

Addition retraction and the surviving multiplicative core

24 September 2026. The user explicitly retracts $\tau + \tau = \tau$ and says $\tau + \tau$ has no meaning. Accordingly this expression is no longer a defined source operation. This is stronger than changing its value. Earlier complete-operation tables remain historical constructions; they are not the current specification. No decision about every other previously supplied mixed sum is inferred from this one retraction.

R1. Exact dependency correction

The equation-based identification of $\{0, \tau\}$ as the additive idempotents no longer applies: $\tau + \tau$ cannot be evaluated. The previous statement that integer 1 is the unique multiplicative but not additive idempotent likewise cannot be used as an intrinsic predicate on this whole carrier. The old distributive-multiplier test at $x = y = \tau$ has an undefined input sum. The basis-relation defect $[\tau + \tau] - [\tau] - [\tau]$ is no longer a relation of the source. The earlier Boolean addition and repeated- τ -sum arguments are withdrawn as descriptions of the current source.

The full congruence and quotient statements require a specified domain of addition and an explicit notion of morphism for that domain. They must not be carried from the old total algebra into the current partial specification without reconstruction. This does not invalidate proofs about the former explicitly defined total algebra; it changes which algebra the user is asking to study.

R2. Addition can exist without a nontrivial parity assignment

Consider the three-element set $C = \{0, a, b\}$ with the complete operation table:

+	0	a	b
0	0	a	b
a	a	b	0
b	b	0	a

This operation is associative and commutative, has identity 0, and every element has an additive inverse. For a direct proof, take a permutation σ cycling three positions, so σ cubed is the identity. Identify 0 with the identity permutation, a with σ and b with σ squared. The displayed table is exactly composition of these three permutations. Composition is associative; the three powers commute; σ and σ squared are mutual inverses. These facts prove all the additive-group axioms for the table without assigning even or odd states.

In fact no nontrivial ordinary two-state parity assignment is compatible with these sums. Suppose the nonzero elements were assigned even or odd, with the usual rule that two odd states sum to an even state and two even states sum to an even state when the result is nonzero. If a were odd, $a + a = b$ would force b even, and $b + b = a$ would then force a even, a contradiction. Exchanging a and b gives the same contradiction for b odd. Thus the only such assignment labels both nonzero elements even; no odd state can occur. This proof does not assign parity to zero.

The example establishes the general mathematical point: existence of addition does not entail a nontrivial even/odd distinction. It does not define an addition on τ . The source retraction in R1 is respected independently of this counterexample. C is not renamed using the user's Z_n layer notation.

R3. The global unit and the requested multiplicative quotient survive

Retain $M = Z$ disjoint union $\{\tau\}$, ordinary multiplication on integers, and $\tau x = x \tau = x$. The integer zero is absorbing. Products associate by removal of τ unit factors followed by integer associativity. τ is the unique global multiplicative unit. Its distinction from integer 1 follows from $1 \tau = 1$ different from τ , so integer 1 is not a unit on the whole carrier.

Let $B_{\text{times}}=\{0,\tau\}$ carry only its specified multiplication: zero absorbs, τ is the unit. No addition on τ is introduced. Define

$$\chi : M \rightarrow B_{\times}, \quad \chi(\tau) = \tau, \quad \chi(n) = 0 \quad (n \in \mathbb{Z}). \quad (\text{R3.1})$$

A product equals τ exactly when both factors are τ . Every other product is an integer. These two cases prove $\chi(xy)=\chi(x)\chi(y)$, including zero and unit. Its exact fibres are

$$\chi^{-1}(\tau) = \{\tau\}, \quad \chi^{-1}(0) = \mathbb{Z}. \quad (\text{R3.2})$$

Thus the requested ‘it and only it maps to it; everything else maps to zero’ quotient remains valid multiplicatively and uses no τ addition.

The second map $r:M \rightarrow (\mathbb{Z},\text{multiplication},1)$, with $r(\tau)=1$ and $r(n)=n$, is multiplicative by the same pairwise check. Pairing χ and r gives the injective map

$$\tau \mapsto (\tau, 1), \quad n \mapsto (0, n), \quad (\text{R3.3})$$

whose image is $(\{0\} \text{ times } \mathbb{Z}) \text{ union } \{(\tau,1)\}$. Its inverse sends the displayed pairs to their source elements. Componentwise multiplication preserves the image and the inverse preserves multiplication. This retains every integer together with the distinction of τ . It does not use componentwise addition or assign a value to $\tau+\tau$.

R4. Algebra-preserving holonomy can still be proved trivial

Consider automorphisms of the multiplicative carrier M that also preserve the ordinary integer addition. The integer part is the complement of the unique global unit τ , so such a bijection maps the integer part to itself. The addition requirement concerns integer pairs only; it assigns no sum to a pair of τ s.

Every such automorphism f is the identity. First f fixes τ , the unique multiplicative unit, and 0, the unique multiplicative absorber. The multiplicative idempotents are exactly $0,1,\tau$: the integer equation $n^2=n$ is $n(n-1)=0$, giving 0 or 1. Hence f fixes the remaining idempotent 1 as well. Integer addition then fixes every positive integer inductively. The only elements satisfying $x^2=1$ are integer 1 and -1, so injectivity and $f(1)=1$ force $f(-1)=-1$. Multiplication fixes every negative integer as $(-1)n$. This exhausts M .

Therefore transport by isomorphisms preserving this multiplicative carrier and its integer addition has identity loop transport. This is a reconstructed proof of the holonomy conclusion that does not use the retracted $\tau+\tau$ law, the former congruence theorem, or the old additive-idempotent characterization. Multiplicative-only transport can still permute primes, as the explicit 3/5 interchange in GS1 demonstrates; that interchange fixes τ and preserves parity availability but fails integer addition.

R5. A global τ section and an integer subsheaf without τ addition

For a topological space X , let $M_X(U)$ be the locally constant functions $U \rightarrow M$, with pointwise multiplication. Compatible functions glue uniquely and the glued function remains locally constant, proving the sheaf property. Let $I_X(U)$ be the integer-valued locally constant functions. This is a subsheaf, is a multiplicative ideal, and carries its ordinary pointwise integer addition. It includes the zero section. No addition on sections involving τ has been defined by this construction.

The constant τ function is the unique global unit section: evaluating any unit section at a point gives the unique unit of M . The restrictions of these unit sections agree on all overlaps, so their globality follows by the sheaf gluing axiom. At the empty open there is the unique empty section, which does not identify distinct source elements on any nonempty open.

The pointwise map $\chi_X: M_X \rightarrow (B \times) X$ is multiplicative, is onto by the literal inclusion of $\{0, \tau\}$, and satisfies χ_X inverse of the constant τ section = {that same τ section}; its zero fibre is $I_X(U)$. Pairing it with r pointwise reconstructs each original section by R3.3. This proves the global-section and ‘everything else’ sheaf construction with no use of $\tau + \tau$.

The pointed prime spectrum and the multiplicative structure-sheaf calculations in GS5-GS8 likewise use the unchanged multiplication. The properties of the closed point $c=Z$, its whole-space-only open neighbourhood, global multiplicative sections, and the closed quotient remain available. Any former Boolean-addition or total-algebra assertion in those notes must instead be read with the correction R1. The prime point c is an ideal; it is not the element τ .

R6. Parity-bearing fractions and their exact quotient map

The usual construction of nonzero rational numbers starts from pairs (a,b) of nonzero integers. Those two integers have parity. This is a fact about the input presentation. It is different from a compatible parity assignment on the represented rational value. The following maps retain the actual input data and prove their relation.

Let F be the set of such pairs and put (a,b) equivalent to (c,d) exactly when $ad=bc$. The rational quotient map is $q(a,b)=a/b$. The numerator/denominator parity record is the pair of their ordinary even/odd states. This record is not constant on fibres of q :

$$q(1, 2) = q(2, 4), \quad (\text{odd, even}) \neq (\text{even, even}). \quad (\text{R6.1})$$

More generally every input (a,b) is equivalent to $(2a, 2b)$, whose two parity entries are even. Therefore any function of only the raw two parity entries that is invariant under the fraction equivalence is constant on the rational quotient: every quotient element admits that same even/even pair of entries. This proves precisely why the raw labels do not descend. It does not remove the labels from the original input pair, and the pair remains retained in F .

A stronger observable does descend. For a nonzero integer a , let $v_2(a)$ be the largest nonnegative integer j for which 2^j divides a . Existence follows because $2^j \leq |a|$ whenever it divides a . Writing $a=2^j u$ with u an odd signed integer proves $v_2(ac)=v_2(a)+v_2(c)$: the product of two odd integers is odd. For the original pair define

$$v(a, b) = v_2(a) - v_2(b). \quad (\text{R6.2})$$

If $ad=bc$, taking integer valuations gives $v_2(a)+v_2(d)=v_2(b)+v_2(c)$, so the two differences agree. Thus there is a well-defined observable v on nonzero rational values with $v(q(a,b))=v(a,b)$. No reduction of the fraction is required, and the full pair remains available separately. Multiplication of pairs proves $v(xy)=v(x)+v(y)$.

For addition use the larger pair carrier $F_0 = Z \times (Z \setminus \{0\})$, so a zero numerator is also allowed. The quotient map $q_0: F_0 \rightarrow Q$ sends (a,b) to a/b and restricts to q on F . The addition below has domain $F_0 \times F_0$ and codomain F_0 ; its denominator is nonzero because b and d are. Retain the complete formula

$$(a, b) + (c, d) = (ad + bc, bd). \quad (\text{R6.3})$$

This formula descends to the rational quotient. Indeed, if $ab'=a'b$ and $cd'=c'd$, then

$$\begin{aligned} (ad + bc)b'd' &= ab'dd' + bb'cd' \\ &= a'bdd' + bb'c'd \\ &= (a'd' + b'c')bd. \end{aligned} \quad (\text{R6.3a})$$

These are exactly the cross-products asserting equivalence of the two output presentations. The original presentations and their parity records remain in F_0 ; the quotient map identifies rational values without asserting that their presentation records coincide.

When $ad+bc$ is nonzero, its output observable is exactly $v_2(ad+bc)-v_2(bd)$. When the numerator is zero, the result is the zero rational and lies outside the domain of this nonzero-value observable. In particular $x+x=2x$ for nonzero rational x gives

$$v(x+x) = 1 + v(x), \quad v(1/2) = -1, \quad v(1) = 0. \quad (\text{R6.4})$$

This retains an exact connection to the integer parity data: a nonzero integer n is odd precisely when $v(n)=0$ and even precisely when $v(n)\geq 1$. Nonzero rationals also admit negative v values. The observable has not been relabeled as one of the user's Z_n layers or identified with a two-state grading.

The earlier half-plus-half argument correctly rules out a compatible two-state parity on rational values extending integer parity. It does not establish that no parity-bearing ingredients were used to construct those values. The claim that every possible construction of addition needs parity is a broader claim about allowed constructions; it does not follow from the usual integer-pair presentation. R2 gives a complete additive structure without a nontrivial compatible two-state grading, while R6 records exactly the integer-derived data the user asks to retain. Neither calculation defines $\tau+\tau$ or restores its retracted law.

Current status

The active common core consists of multiplication on M , ordinary addition on its integer part, and the displayed partial-parity domain on nonzero integers. Previously written additional mixed sums are not promoted to a new completion here. The user's latest explicit change removes $\tau+\tau$ from the domain. R2 disproves the requirement of a compatible nontrivial two-state parity on the elements of every additive structure. It does not assert that an arbitrary construction uses no input with parity data. R6 proves the exact distinction for integer fraction presentations. R3-R5 construct the requested global τ and multiplicative quotient without defining the retracted sum.

The reproducible source is `draw_global_tau_sheaf.py`. The diagram illustrates R3-R5; every arrow preserves multiplication, and neither it nor those proofs defines addition on τ . The image was rendered and visually inspected.

The fixed τ and integer torsion: construction from the retained source

24 September 2026. This proof starts with the user's primitive τ and existing integers. It does not derive integer arithmetic from presentness alone. It introduces no metric, midpoint, vector, pairing, or τ addition. The exact user quotations are in `VERBATIM_USER_DEFINITIONS_AND_CORRECTIONS.md`; V01-V08 concern the source, and V09-V11 direct continued work from that source. Integer translations TRZ1-TRZ4 were read in full before this receiving calculation. Their exact source is recorded below.

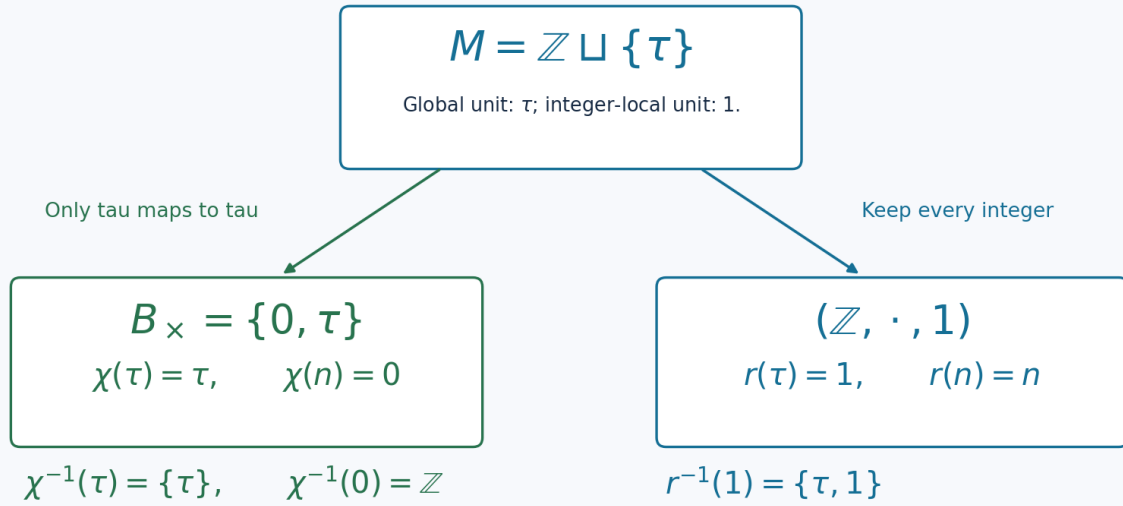
F0. Supplied definitions, followed by derived distinctions

The supplied starting data are Z_1 presentness, τ with no Z_2 parity (V16), existing integer arithmetic with its complete amounts (V03), and the retained Z_2 grading on nonzero integers (V17). The grading is part of the structured elements, so equality preserves its availability and value. The supplied multiplicative rules include $\tau 1=1$ and the *global identity declaration* (V13-V14). *Ordinary integer arithmetic includes* $01=0$ and 1 different from 0. No statement of τ 's distinctness is added as an independent axiom.

First derive τ different from each nonzero integer n . Such an n carries Z_2 parity; τ does not. Equality would preserve that retained structure, giving a contradiction. Next derive τ different from zero: if $\tau=0$,

Global presence and the integer complement

Multiplicative maps only. No value of $\tau + \tau$ is defined or used.



Both maps together give an exact return:

$$\tau \mapsto (\tau, 1), \quad n \mapsto (0, n) \quad (n \in \mathbb{Z})$$

Apply these maps pointwise to locally constant sections: τ is the unique global unit section.

Complete proofs: ADDITION_RETRACTION_AND_MULTIPLICATIVE_CORE.md, R3-R5. The integer subsheaf is the zero fibre of χ .

Figure 1: The two multiplicative receivers and their exact joint return

substituting in $\tau=1$ and using integer multiplication gives $1=\tau=0*1=0$, a contradiction. These cases prove τ is not any integer, without assigning a new parity to zero.

The intrinsic integer-amount map has domain Z and value n at integer n . Since the preceding proof places τ outside Z , this map has no value at τ . Thus the absence of an intrinsic integer amount at τ is a consequence of that domain and the proved distinction, not a second assumption used to prove the distinction. An independently added comparison map, such as the earlier multiplicative $r(\tau)=1$, is not this intrinsic amount map.

Let S denote the part of the programme consisting of these integer elements and τ . The presentation $S=Z$ disjoint union $\{\tau\}$ now records the proved distinction. This is the specified stage under study, not an assertion that the user's higher Z_n layers have no further elements.

On integer inputs addition, subtraction and multiplication have their usual integer values. Addition and subtraction have no input involving τ in the latest source transcription. Multiplication on all S has the supplied global identity law $\tau x = x \tau = x$ (V13-V14). The original integer values, signs and Z_n labels remain in S . The earlier convention that integer zero has no intrinsic Z_2 label is not changed by an auxiliary parity calculation below.

On Z define $T_k(n)=n+k$, for integer k . This uses already supplied integer addition; it is not a formula $\tau+k$ and is not a distance from τ . T_1 represents the step between the existing integer amounts by the supplied arithmetic; no identity with an unconstructed geometric involution is presumed. The question is how these integer bijections extend to S , with no geometric interpretation imposed.

F1. Unique extension and derived fixedness

Every bijection $f:Z \rightarrow Z$ has exactly one bijection $f_{\text{hat}}:S \rightarrow S$ whose restriction to Z is f . To prove necessity, take any such extension. The image of Z is all Z . If $f_{\text{hat}}(\tau)$ were an integer m , there would exist n in Z with $f(n)=m$. Injectivity would then give $\tau=n$, contradicting the distinction proved in F0 from the source grading and multiplication laws. Thus $f_{\text{hat}}(\tau)$ must be τ . Existence follows by setting the value at τ to τ and using f on integers; the resulting function is bijective, with inverse the same extension of f inverse. This proves both assertions.

Applying this construction to T_k gives a unique extension, written T_{hat_k} . Fixedness of τ follows from the extension theorem, rather than being a geometric midpoint premise. Its values, after this proof, are

$$\widehat{T}_k(\tau) = \tau, \quad \widehat{T}_k(n) = n + k \quad (n \in \mathbb{Z}). \quad (\text{F1.1})$$

For k nonzero, no integer is fixed, because $n+k=n$ would imply $k=0$ by integer cancellation. Consequently

$$\text{Fix}(\widehat{T}_k) = \{\tau\} \quad (k \neq 0), \quad \text{Fix}(\widehat{T}_0) = S. \quad (\text{F1.2})$$

This is the unique fixed element for these actual transformations. It proves no distance or radius statement. The derivation uses the exact source carrier through its integer layer; adding further source elements would require a new extension calculation.

F2. The full amount is retained by composition

For an integer n ,

$$(\widehat{T}_k \circ \widehat{T}_l)(n) = (n + l) + k = n + (k + l).$$

Both sides fix τ by F1. Therefore, as functions on all S ,

$$\widehat{T}_k \circ \widehat{T}_l = \widehat{T}_{k+l}, \quad \widehat{T}_k^{-1} = \widehat{T}_{-k}, \quad \widehat{T}_0 = \text{id}_S. \quad (\text{F2.1})$$

The map $k \rightarrow T_hat_k$ is an isomorphism from the integer additive group onto this transformation family. Surjectivity is the definition of the family; injectivity follows by evaluation at integer zero, $T_hat_k(0)=k$. Evaluation there is the inverse. It is not evaluation at tau: evaluation at tau gives tau for every member, and therefore cannot recover k .

In particular T_hat_1 composed with itself sends integer zero to 2 and still fixes tau. The source stage S has retained the integer amount two. Nothing has assigned two copies, two units of amount, or a changed parity to tau itself. The source records the amount in the integer and in its corresponding transformation. Opposite signs give inverse transformations; they do not assign negative distances.

F3. The two-state quotient and the retained integer amount

On integer amounts put n equivalent to m exactly when $n-m=2j$ for some integer j . This is the orbit relation of the subgroup consisting of T_2j : reflexivity uses $j=0$, symmetry uses $-j$, and transitivity adds the two integer j values. Integer division by 2 writes each n uniquely as $2j+r$ with r in $\{0,1\}$. Thus there are exactly two equivalence classes, denoted E and O , containing 0 and 1 respectively.

Translation by one induces a well-defined bijection σ on these classes. Well-definedness follows because $(n+1)-(m+1)=n-m$. Its values are $\sigma(E)=O$ and $\sigma(O)=E$, since $2j+1$ is in O and $(2j+1)+1=2(j+1)$ is in E . Hence σ squared is the identity, while T_1 squared is T_2 and is not the identity on integers. The exact map $\rho:Z \rightarrow \{E,O\}$ obeys

$$\rho(T_1(n)) = \sigma(\rho(n)), \quad \rho(T_2(n)) = \rho(n), \quad T_2(n) = n + 2 \neq n. \quad (\text{F3.1})$$

The last inequality follows from 2 nonzero. These identities retain both the original amount and the quotient; they do not substitute the two classes for the integer source. The graph map $n \rightarrow (n, \rho(n))$ has inverse projection to its first coordinate, so recording the class alongside the integer loses no integer information.

On nonzero integers E/O agrees exactly with the user's Z_2 even/odd labels. If intrinsic parity is undefined at zero under the retained Z_0 convention, its compatibility with T_1 has the exact domain Z minus $\{-1,0\}$, so both input and output are nonzero. The integer orbit map ρ is a separate total map; its value at zero is not a change to that intrinsic convention. No $\rho(\text{tau})$ is defined. No cyclic group is substituted for the user's Z_2 notation.

F4. Exact comparison with the previous rigidity result

The maps T_hat_k are bijections describing a changing integer amount. They are not asserted to preserve all arithmetic operations. For integer m, n ,

$$\begin{aligned} \widehat{T}_k(m+n) &= m+n+k, \\ \widehat{T}_k(m) + \widehat{T}_k(n) &= m+n+2k, \\ \widehat{T}_k(mn) &= mn+k, \\ \widehat{T}_k(m)\widehat{T}_k(n) &= mn+k(m+n)+k^2. \end{aligned} \quad (\text{F4.1})$$

The additive right-minus-left difference in the integer target is k ; the multiplicative right-minus-left difference is $k(m+n)+k^2-k$. All terms and domains are retained. On a product involving tau, the multiplicative comparison instead agrees: $T_hat_k(\text{taun})=n+k=T_hat_k(\text{tau})T_hat_k(n)$. For $\text{tau}*\text{tau}$ both sides equal tau.

Thus the nontrivial translation family is compatible with the earlier theorem R4 that the only automorphism preserving multiplication and integer addition is the identity: the operation domains preserved by these two

transformation notions differ, and F4.1 is their exact comparison. No conclusion about arbitrary analytic holonomy is inferred.

F5. What is actually excluded by the source operation language

In the partial algebra of F0, $n\text{-}\tau$ and $\tau+n$ are outside the respective operation domains. There is no source distance symbol. Integer expressions formed using integer constants and integer addition, subtraction and multiplication remain integers by induction on the expression: leaves are integers and every operation on integer inputs has integer output. They cannot evaluate to τ by F0.

These statements prove an exclusion for the given source operations. They neither introduce a positive scalar distance nor assert a universal theorem forbidding added mathematical structures. The attachment's unproved conversion from difference of structure to numerical positivity has not been used anywhere in F0-F4. The construction here concerns the actual retained integer amounts and the uniquely fixed τ .

Sources, reading and scope

User source: VERBATIM_USER_DEFINITIONS_AND_CORRECTIONS.md V01-V17, with exact quotation locators and receipt. Every interpretation above is stated separately from those verbatim inputs.

Programme source: work/rh_counterfactual_20260913/tau_prime_weil_20260922/cc_totality_20260923/theorems/integer_translations_20260924/TRANSLATION_THEOREM.md, TRZ1-TRZ4, read in full. TRZ1 supplies the existing integer action; F1 uniquely extends it to the current τ source. TRZ3 supplies the partial parity domain retained in F3. The existing RATIONAL_PRESENTATIONS_AND_PARITY.md in the retrospective task was also read in full, including its receiving comparison. No historical paper cited by either note is claimed to have been read here.

The corpus queries are in TRANSLATION_PIVOT_ROUTING.json and are routing only. No new external source theorem is imported. These are elementary constructions with their complete proofs, not a novelty claim, a construction of integers from τ alone, or a proof of RH.

The diagram displays mappings, not a metric or source coordinate space. Drawing positions have no mathematical distance meaning. Reproducible source: draw_tau_translation_fixedness.py. Proof locators F1-F4. The rendered image was inspected, and the caption was moved clear of the map arrows before the final inspection.

The Z_1 base, the twistor involution, and the original-zeta receiver

24 September 2026. This is a source-directed calculation on the supplied author TeX. It retains the user's notation: τ carries Z_1 presentness without Z_2 parity; integer 1 is the identity on the integer part; τ is the global multiplicative identity. Addition involving τ is not used. An operation on an auxiliary coefficient ring or on a space of functions is not an additional operation on τ . Every comparison below states its domain and what it retains.

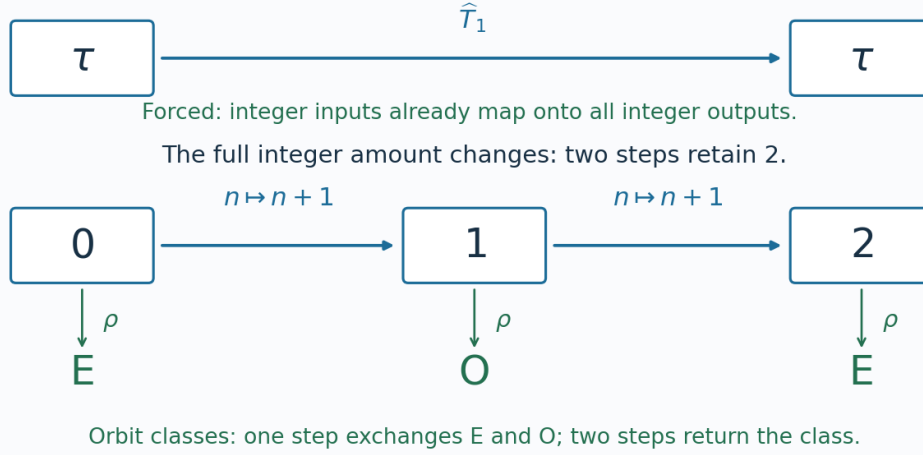
The human sources are Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of Spec Z* (2609.00299v1, CC.tex); the same authors, *On the Absolute Geometry of Spec Z* (2606.06604v1, FF.tex); their *On the Jacobian of Spec Z* (2602.15941v1, Jacobian.tex); and Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time* (2602.04022v1, rhready.tex). All four supplied TeX files are byte-identical to the respective original arXiv sources. Source archives and the exact reading ledger are retained. The calculations marked C1-C10 below are this note's derivations, not claims of literature novelty.

C1. The multiplicative base comparison retains τ and integer 1

Let M be the multiplicative integer-plus- τ stage. Its products are the ordinary integer products and $\tau x = x\tau = x$ for every x in M . τ differs from each nonzero integer by the retained Z_2 availability;

Fixedness from the source bijections

An operation diagram. No radius, displacement, or pairing is assigned.



E and O are the integer orbit classes modulo steps of two.

This auxiliary quotient does not change the intrinsic Z_0 label retained at integer zero.

Proofs F0-F4: TAU_FIXEDNESS_FROM_INTEGER_TRANSLATIONS.md. All source integer amounts remain present.

Figure 2: Operation diagram: the unique fixed tau and the full integer sequence

$\tau=0$ would imply $1=\tau 1=0 1=0$. Thus the disjoint presentation $M=\mathbb{Z}$ disjoint-union $\{\tau\}$ records a consequence. Higher stages of the user's hierarchy are not declared absent.

Put $B_\tau=\{0,\tau\}$ with these products. The map

$$b : B_\tau \longrightarrow \{0, 1_{CC}\}, \quad b(0) = 0, \quad b(\tau) = 1_{CC} \quad (C1.1)$$

is an isomorphism of pointed multiplicative monoids: 0 is absorbing and tau is the unit, exactly matching the two source products. Its inverse sends the source unit to tau. This is the pointed-monoid part of the source's F_1 base, not a claim that an unspecified additive structure or every framework called F_1 has been identified.

The integer receiver $r:M \rightarrow (\mathbb{Z}, \text{multiplication})$, $r(n)=n$, $r(\tau)=1$ is multiplicative. It identifies the images of tau and integer 1. Retain also

$$\chi : M \longrightarrow B_\tau, \quad \chi(\tau) = \tau, \quad \chi(n) = 0 \quad (n \in \mathbb{Z}). \quad (C1.2)$$

For an integer product both sides of multiplicativity are 0; for a product with tau both sides equal the value of the other factor. Hence chi is multiplicative. The pair (χ, r) is injective and has image

$$(\{0\} \times \mathbb{Z}) \cup \{(\tau, 1)\}. \quad (C1.3)$$

Indeed an integer n has the unique pair $(0, n)$, whereas tau has $(\tau, 1)$. In particular using the complex-valued points of the authors' geometry does not make the scalar value 1 a faithful record of the original source element. Equations C1.2–C1.3 give an explicit faithful receiver. No integer amount or parity is assigned to tau.

C2. Signed extension and the exact generic involution

CC.tex, Definition `def:alpha_symmetry`, uses $\epsilon^2=1$ and $J^2=\epsilon$. Retain its sign as a new multiplicative datum. It is not automatically the user's Z_2 parity.

Here is an extension preserving the full M before taking geometric points. Take the disjoint union of the Gaussian integers $Z[i]$ and four new units u_0, u_1, u_2, u_3 . Set $u_0=\tau$, $u_2=\epsilon$, $u_1=J$; define

$$u_r u_t = u_{r+t \bmod 4}, \quad u_r a = a u_r = i^r a \quad (a \in Z[i]), \quad (C2.1)$$

and use ordinary products on $Z[i]$. Associativity follows separately for three units from addition modulo 4, for two units and a Gaussian integer from $i^{(r+t)=i_r} i^t$, and for one unit and two Gaussian integers from $i^{r(ab)=(i_r a)b}$. Three Gaussian integers use their usual associativity. Commutativity reduces all orders to these cases. Zero is absorbing and τ is the global unit. Inclusion embeds M multiplicatively. Local Gaussian i and global J remain different, as do local -1 and ϵ . $J^2=\epsilon$ and $\epsilon^2=\tau$ hold without identifying local integer 1 with τ .

For the generic source chart itself let

$$A_\eta = \{0\} \cup \{J^r T^k : r \in Z/4Z, k \in Z\}. \quad (C2.2)$$

Its product adds the exponent k and adds r modulo 4; 0 is absorbing, $J^0 T^0=\tau$, and $\epsilon=J^2 T^0$. This is the explicitly presented multiplicative monoid of the source chart. Its isomorphism to the authors' monomial presentation sends τ to their unit and retains J, T, ϵ . The inverse recovers the two exponents. The infinite exponent k is never reduced modulo 2 or 4.

The source's signed inversion has the exact coordinate form

$$\alpha^*(J^r T^k) = J^{-r+2k} T^{-k}, \quad \alpha^*(0) = 0. \quad (C2.3)$$

This follows by multiplying $(J^{(-1)})^r$ and $(\epsilon^{(1)})^k$. The map on the exponent pair is additive, so it preserves products. Applying it twice sends (r, k) to $(-(-r+2k)+2(-k), -(-k))=(r, k)$; hence it is an involution. Fixedness requires $k=-k$ in Z and $r=-r$ modulo 4. Consequently its entire fixed submonoid is

$$\text{Fix}(\alpha^*) = \{0, \tau, \epsilon\}. \quad (C2.4)$$

τ 's fixedness is thus proved in the actual generic algebra. ϵ is also fixed; it must not be discarded. For every positive odd n , the power map $F_n: (r, k) \rightarrow (nr, nk)$ commutes with α . The joint fixed submonoid of all these maps is again $\{0, \tau, \epsilon\}$: $n=3$ forces $k=0$ and $2r=0$, and those three elements are fixed for every odd n .

CC's underlying three-point space has $\alpha(+)=-, \alpha(-)=+, \alpha(\epsilon)=\epsilon$. The earlier integer-step orbit receiver $Q=\{\tau\text{-marker}, E, O\}$ is isomorphic as an involutive set to it by $\tau\text{-marker} \rightarrow \epsilon, E \rightarrow +, O \rightarrow -$. Checking the three values proves equivariance and the inverse. This is a map of involutive sets. It does not identify a base element τ with a point carrying the entire generic stalk A_ϵ , and it does not discard the full integers in M .

C3. The sign distinguishing two real structures

Over C , source points have J -coordinate i or $-i$. α sends (z, i) to $(-1/z, -i)$, while complex conjugation sends it to $(\bar{z}, -i)$. Returning to the i branch after conjugation gives exactly

$$\sigma(z) = -\frac{1}{\bar{z}}, \quad \sigma(0) = \infty, \quad \sigma(\infty) = 0. \quad (C3.1)$$

This reconstructs CC.tex, Theorem `thm:twistor_space_scheme`, without dropping the sign. For finite nonzero z , fixedness would give $|z|^2 = -1$, which is impossible; the endpoints are exchanged. Thus σ has no fixed points on $P^1(C)$.

There is also an explicitly related involution. The source automorphism R has $R(T) = \epsilon T$ and $R(J) = J$; it extends to both charts, preserves their restrictions, and commutes with α . For example $\alpha R(T) = T^{-1} = R \alpha(T)$, and both composites take J to J^{-1} . Its induced complex map is $R(z) = -z$. Therefore

$$\kappa = R \circ \sigma, \quad \kappa(z) = \frac{1}{\bar{z}}, \quad \text{Fix}(\kappa) = \{|z| = 1\} \subset \mathbb{C}^\times. \quad (\text{C3.2})$$

The endpoints are again exchanged; the displayed fixed locus follows by multiplying $z = 1/\bar{z}$ by $\bar{z}$. $R^2 = \text{id}$ and R commutes with σ , so $\kappa^2 = \text{id}$. This is a proved sign twist of the source involution, not the same real structure silently renamed. No bijective conjugacy can turn σ into κ , since a conjugacy maps fixed sets bijectively and their fixed sets are empty and nonempty respectively.

The sign remains visible in homogeneous coordinates. With $z = x/y$, put

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (\text{C3.3})$$

Then $A \text{bar}(x, y)$ induces σ , $B \text{bar}(x, y)$ induces κ , $B = DA$, $A \text{bar}(A) = -I$ and $B \text{bar}(B) = I$. Projectivization makes both induced squares the identity but retains different real structures. The unit comparison $\tau \rightarrow I$ and $\epsilon \rightarrow -I$ uses the source multiplicative sign; it does not define τ as an additive zero vector.

For comparison with the user's radius proposal, the complete sphere map is

$$\Phi(z) = \frac{(2\Re z, 2\Im z, |z|^2 - 1)}{1 + |z|^2}, \quad \Phi(\infty) = (0, 0, 1). \quad (\text{C3.4})$$

Its numerator has squared norm $4|z|^2 + (|z|^2 - 1)^2 = (1 + |z|^2)^2$. Thus its radius is 1 for every z . Direct substitution gives $\Phi(\sigma(z)) = -\Phi(z)$. Away from the north pole its inverse is $z = (X + iY)/(1 - Z)$; the north pole returns infinity. Equal sphere radius therefore coexists with a retained, recoverable coordinate $|z|$. It is a property of this target geometry, not an assigned metric at the user's τ .

The actual odd Frobenius in CC.tex, Proposition `prop:twistor_frobenius`, is z^n for $n \equiv 1 \pmod{4}$ and $-1/\bar{z}^n$ for $n \equiv 3 \pmod{4}$. Consequently $|z|$ becomes $|z|^n$ or $|z|^{-n}$, respectively, while Φ of either result still has radius 1. This is an exact calculation within the supplied geometry; it retains the spectral size that a sphere-radius observation alone does not record.

C4. Descent to the supplied Tate curve and its two fixed circles

Figure 1. The left drawing uses $p=3$ solely to display the annulus. C4 proves the circles for every prime p . The right drawing displays the actual quotient maps; the marked point is an arbitrary geometric point, not an asserted zeta zero. Reproducible source: `draw_tate_fixed_locus.py`. Proofs: C4–C5. The two radial boundary circles are identified by the deck factor p .

FF.tex, The Tate curve E_p and `thm:tate_decomposition`, supplies

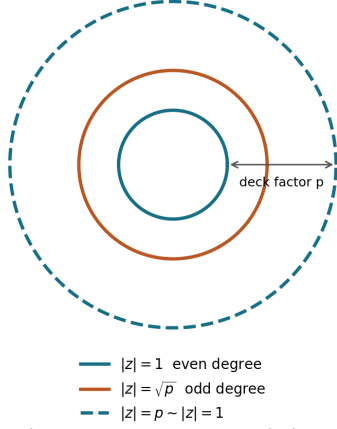
$$E_p = \mathbb{C}^\times / p^{\mathbb{Z}}, \quad p \text{ a positive rational prime.} \quad (\text{C4.1})$$

Both σ and κ descend to E_p , since $\sigma(p^k z) = p^{-k} \sigma(z)$ and $\kappa(p^k z) = p^{-k} \kappa(z)$. A σ -fixed class would satisfy $-1/\bar{z} = p^k z$, hence $-1 = p^k |z|^2$. No such class exists.

The sign-adjusted involution on the paper's Tate curve

$$E_p = \mathbb{C}^\times / p^{\mathbb{Z}}, \quad \kappa[z] = [1/\bar{z}], \quad \sigma[z] = [-1/\bar{z}]$$

Annulus representatives (drawing: $p = 3$)



Proof: C4-C5 in TAU_TWISTOR_TATE_AND_WEIL.md. The even/odd labels are the parity of the lifted integer n in $|z|^2 = p^n$.

For $z = p^s$ with $0 < \text{Re}(s) < 1$, κ -fixedness forces $n = 1$ and hence $\text{Re}(s) = 1/2$. This does not assert fixedness of all zeta zeros.

Sources: Connes-Consani, arXiv:2609.00299v1 (twistor sign) and 2606.06604v1 (Tate quotient). No metric at τ is assumed.

Exact maps on the quotient coordinates

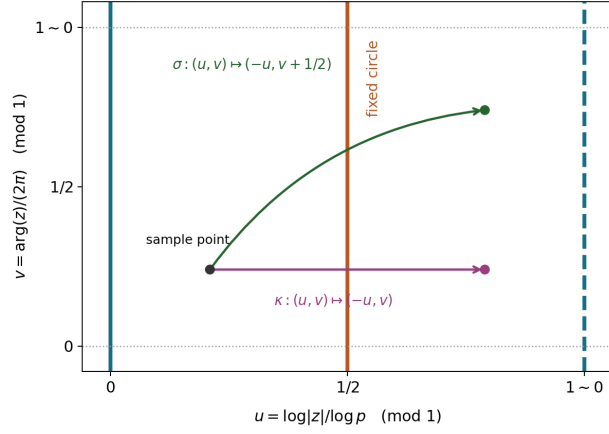


Figure 3: The Tate fixed circles and the retained signed action

A kappa-fixed class instead satisfies

$$\frac{1}{\bar{z}} = p^k z \iff |z|^2 = p^{-k}. \quad (\text{C4.2})$$

Retain $n=-k$. Before quotienting, the full solution set with its integer degree is

$$\tilde{F}_p = \{(z, n) \in \mathbb{C}^\times \times \mathbb{Z} : |z|^2 = p^n\}. \quad (\text{C4.3})$$

Changing representative z to $p^j z$ changes n to $n+2j$. Thus n modulo 2 is independent of representative, while n itself is retained on the lifted set. Multiplication is $(z, n)(w, m) = (zw, n+m)$, which preserves the equation because $|zw|^2 = p^{n+m}$. Its identity is $(1, 0)$, and its inverse is $(1/\bar{z}, -n)$.

Every even n has a unique representative with radius 1 after division by $p^{n/2}$; every odd n has a unique representative with radius $\sqrt{p}$ after division by $p^{(n-1)/2}$. Their phases do not change. Explicitly,

$$S^1 \times \{0, 1\} \longrightarrow \text{Fix}(\kappa : E_p \rightarrow E_p), \quad (v, a) \longmapsto [p^{a/2} v] \quad (\text{C4.4})$$

is bijective. Equality of two images implies $p^{((a-b)/2)*v/w} = p^j$. Taking moduli forces $a=b$ and $j=0$, then $v=w$. Surjectivity was proved by the representative reduction. On the source of C4.4 the group law is $(v, a)(w, b) = (vw, a+b \bmod 2)$: the factor p arising when $a=b=1$ is precisely the retained deck multiplication. Hence the fixed locus is a group isomorphic to S^1 times the two-element degree group.

This proves the two-circle classification, including its lifted integer data. Its two-state label is the parity of n . It is not silently identified with the intrinsic $\mathbb{Z}_2$ state of an arbitrary integer element in the user's base. The exact comparison is $n \rightarrow n \bmod 2$ on the displayed lifted integer coordinate.

C5. Original zeta coordinates and the weight-one circle

For every complex s retain the unquotiented value $\Lambda_p(s) = p^s = \exp(s \log p)$, with the real positive $\log p$. Define $j_p(s) = [p^s]$ in E_p . No zeta completion or rescaling of the zeta function is involved.

In the open critical strip $S = \{s: 0 < \text{Re}(s) < 1\}$, C4.2 gives

$$\begin{aligned} j_p(s) \in \text{Fix}(\kappa) &\iff |p^s|^2 = p^n \text{ for some } n \in \mathbb{Z} \\ &\iff 2\Re s = n \in \mathbb{Z} \\ &\iff \Re s = \frac{1}{2}. \end{aligned} \tag{C5.1}$$

The last implication uses $0 < 2\text{Re}(s) < 2$, so $n=1$, not a choice of a radius. The full unquotiented radius on this locus is $|p^s| = \sqrt{p}$. Thus the strip selects exactly the odd-degree circle. Endpoints of the strip were not included; their real parts 0 and 1 land on the even-degree circle and are not removed from the global map.

Let $R_\text{zetaeta}(s) = 1 - \bar{s}$. The exact equivariance is

$$\Lambda_p(R_\zeta s) = \frac{p}{\Lambda_p(s)}, \quad j_p(R_\zeta s) = \kappa(j_p(s)). \tag{C5.2}$$

The factor p in the first equality becomes a deck transformation in the second, and is retained explicitly. The involution on the unquotiented value is $\iota_p(z) = p/\bar{z} = -p^* \sigma(z)$, whose fixed circle is $|z|^2 = p$. This is the precise weight-one comparison with the supplied twistor map, including the sign and prime factor.

The arithmetic fixedness defect has a full scalar representative

$$\Delta_p(s) = |p^s|^2 - p = p^{s+\bar{s}} - p. \tag{C5.3}$$

It is zero, positive or negative according as $\text{Re}(s)$ is $1/2$, greater than $1/2$ or less than $1/2$, since exponentiation by $p > 1$ is strictly increasing. Its reflected value is

$$\Delta_p(R_\zeta s) = -\frac{p}{|p^s|^2} \Delta_p(s), \tag{C5.4}$$

by substituting $|\Lambda_p(R_\zeta s)|^2 = p^2/|p^s|^2$. A paired sign change thus preserves the precise nonzero defect when the point is not fixed.

The quotient has an exact faithful two-prime receiver on S . For distinct primes p, q the map $(j_p, j_q): S \rightarrow E_p \times E_q$ is injective. To prove this, $j_p(s) = j_p(t)$ implies $s-t = m + 2\pi i k / \log(p)$, m, k integers. Since $\text{Re}(s-t)$ lies strictly between -1 and 1 , $m=0$. Equality for q similarly gives $\text{Im}(s-t) = 2\pi i l / \log(q)$. If this common number is nonzero, k, l are nonzero and $k \log(q) = l \log(p)$, contradicting unique prime factorization after exponentiation, also for negative integer exponents. Therefore $k=l=0$ and $s=t$. This provides an exact way to retain the original strip coordinate rather than treating its one-prime quotient as injective.

In particular RH is exactly the assertion that every nontrivial zero ρ of the ORIGINAL zeta has $j_p(\rho)$ in the odd fixed circle, for one (hence every) prime p . This is a proved equivalence, not an assumption inserted to prove RH. Multiplicity is retained by attaching the original integer m_ρ to each original zero and using the injective two-prime receiver. No infinite pushforward is asserted to be a locally finite divisor on a compact torus.

The functional equation supplies the paired zero $1-\bar{\rho}$; C5.2 proves the corresponding symmetry of their images. It does not by itself identify the two original zeros. The injective two-prime receiver makes the distinction testable on the actual source zeros, without changing their equation to the zeros of an auxiliary function.

C6. The original-zeta multiplier and an exact factor-four correction

The supplied rhready.tex, lines 1062–1080, defines

$$\begin{aligned}\xi(s) &= \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), & \mathcal{E}(f)(u) &= u^{1/2}\sum_{n=1}^{\infty}f(nu), \\ h(u) &= \frac{\pi}{2}u^2(2\pi u^2 - 3)e^{-\pi u^2}, & k &= \mathcal{E}(h).\end{aligned}\tag{C6.1}$$

Compute from these actual definitions. For $\text{Re}(s) > 1$, absolute convergence permits the change $v=nu$ and interchange of integral and sum:

$$\int_0^{\infty} \mathcal{E}(h)(u)u^{s-1/2}\frac{du}{u} = \zeta(s) \int_0^{\infty} h(v)v^s\frac{dv}{v}.\tag{C6.2}$$

Indeed the sum of absolute integrals is bounded by $\zeta(\text{Re}(s))$ times the finite integral of $|h(v)|v^{\widehat{\text{Re}(s)}}dv/v$. The Gaussian controls infinity and $h(v)=O(v^2)$ controls zero. Substitution $t=\pi^*v^2$ yields the complete two-term integral

$$\begin{aligned}\int_0^{\infty} h(v)v^s\frac{dv}{v} &= \frac{\pi}{2} \left[2\pi \frac{1}{2}\pi^{-(s+4)/2}\Gamma\left(\frac{s+4}{2}\right) - 3\frac{1}{2}\pi^{-(s+2)/2}\Gamma\left(\frac{s+2}{2}\right) \right] \\ &= \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2).\end{aligned}\tag{C6.3}$$

The second displayed expression is an evaluated identity, not a replacement of the two original contributions; both remain in C6.3. The Gamma recurrence proves the identity by extracting $\Gamma(s/2)$ and evaluating $(1/2)(s/2)((s/2+1)-3/2)$.

Consequently the correct exact comparison is

$$\boxed{\int_0^{\infty} k(u)u^{s-1/2}\frac{du}{u} = \frac{1}{8}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \frac{1}{4}\xi(s).}\tag{C6.4}$$

The equality to ξ asserted for this displayed k in the source is therefore missing a factor 4. At $s=2$ the left side is $\pi/24$ and $\xi(2)=\pi/6$. This also supplies a direct exact check independent of numerical plotting.

For completeness the equality continues to the critical line without discarding boundary terms. Under the source Fourier convention $\exp(2\pi ixy)$, the Gaussian $g=\exp(-\pi x^2)$ transforms to itself; differentiating the Gaussian transform gives $F(x^{2g})=(1/(2^* \pi i)-y/2)g$ and $F(x^{4g})=(y/4-3y^2/\pi+3/(4^* \pi i^2))g$. Since $h=\pi^{2^*}x^4g-(3\pi/2)x^2g$, these identities give $Fh=h$, $h(0)=0$ and $\text{integral } h=0$. Poisson summation then gives $k(u)=k(1/u)$. Gaussian decay of k at infinity and this symmetry give rapid decay at both ends in logarithmic coordinates. Thus its Mellin integral is entire in s , and analytic continuation extends C6.4. For $\Xi(t)=\xi(1/2+it)$ the corrected equality is $\Xi(t)=4^*\text{integral } k(u)u^{\widehat{it}}du/u$, keeping exactly the source kernel.

The original zeta remains recovered on every domain where the displayed inverse is defined:

$$\zeta(s) = \frac{8\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^{\infty} k(u)u^{s-1/2}\frac{du}{u}.\tag{C6.5}$$

In the open critical strip the multiplier $C(s)=s(s-1)\pi^{\widehat{-(s/2)}}\Gamma(s/2)$ is holomorphic and nonzero, so C6.4–C6.5 preserve all zero multiplicities there. Globally keep

$$\frac{\xi'}{\xi} = \frac{\zeta'}{\zeta} + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}. \quad (\text{C6.6})$$

At 1 the factor C has a simple zero and zeta a simple pole. At 0, $s\Gamma(s/2)$ has limit 2, so $C(0)=-1$ and $\zeta(0)=-1/2$ yields $\xi(0)=1/2$. At $-2m$, $m \geq 1$, C has leading coefficient $2m(2m+1)(-1)^m m/m!$ times $(s+2m)^{-1}$; the original zeta has a simple trivial zero there. Retain the local pair $\zeta(s)=(s+2m)a_m(s)$, $C(s)=(s+2m)^{-1}b_m(s)$, so $\xi=a_m b_m$ and $\zeta=(s+2m)\xi/b_m$. The nonzero $b_m(-2m)$ is the displayed coefficient. These are local meromorphic identities; no canceled divisor is declared absent from the original zeta.

C7. The plotted projection defect and its actual involution

Use exactly $H=L^2(\mathbb{R})$ even, the Fourier transform F with kernel $\exp(2\pi i xy)$, $P=P_\lambda$ the projection to $[-\lambda, \lambda]$, and $Q=F P F^{-1}$. On H , $F^2=I$, so $Q=F P F$. For the source prolate eigenfunction $h=h(2m, \lambda)$, $P h=h$ and $P F P h=\chi_m h$. Therefore

$$P Q P h = \chi_m^2 h, \quad \|(I - Q)h\|^2 = (1 - \chi_m^2)\|h\|^2. \quad (\text{C7.1})$$

Proof: $PQP=(FPF)^2$ on PH . Since Q is an orthogonal projection, $\langle Qh, h \rangle = \|Qh\|^2 = \chi_m^2 \|h\|^2$, and orthogonal Pythagoras gives the second identity. For $m=2$, the source χ_2 is positive and strictly below 1. Hence the precise orange quantity in the supplied plot is

$$1 - \chi_2 = \frac{\|(I - Q)h_{4,\lambda}\|^2}{(1 + \chi_2)\|h_{4,\lambda}\|^2}. \quad (\text{C7.2})$$

Every norm and factor remains displayed.

The exact reflection product behind this defect is $U=(2P-I)(2Q-I)$. Put $a=|\chi_m|$, $b=\sqrt{1-a^2}$, with $0 < a < 1$, and choose $e=h/\|h\|$, $f=(Qe - a^2 e)/(ab)$. Then $P f=0$, $\langle f, e \rangle = 0$, $\|f\|=1$. Applying $Q^2=Q$ gives $Q f=ab e + b^2 f$. Thus the span of e, f is invariant under both projections and their matrices are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} a^2 & ab \\ ab & b^2 \end{pmatrix}, \quad U = \begin{pmatrix} 2a^2 - 1 & 2ab \\ -2ab & 2a^2 - 1 \end{pmatrix}. \quad (\text{C7.3})$$

The relation $a^2 + b^2 = 1$ gives $U^*U=I$. Writing $a=\cos(\alpha)$, $b=\sin(\alpha)$, $0 < \alpha < \pi/2$, its eigenvalues are $\exp(2i\alpha)$ and $\exp(-2i\alpha)$. This establishes the nontrivial phase of the product of the two involutions from the source projections; no physical holonomy is assumed. The leakage $1-a^2=\sin^2(\alpha)$ remains recoverable from it.

C8. The same projection block retains both trace signs

The source's archimedean and semilocal trace expressions use $D=I-P-Q$, not just $I-PQP$. In the exact block C7.3,

$$D = \begin{pmatrix} -a^2 & -ab \\ -ab & a^2 \end{pmatrix}, \quad D^2 = a^2 I, \quad \text{Tr } D = 0. \quad (\text{C8.1})$$

Multiplication of this displayed matrix proves the square identity; its characteristic polynomial is $t^2 - a^2$. Thus its two eigenvalues are $+a$ and $-a$. A unitary product of reflections and a positive leakage quantity coexist with these two trace signs on the same source block. For any positive operator B on this block, in an orthonormal D -eigenbasis, $\text{Tr}(BD)=a(\langle Be_+, e_+ \rangle - \langle Be_-, e_- \rangle)$. Both terms must be retained. This does not assert that an arbitrary B is the image of a global admissible Weil test.

The full source formula, `rhready.tex weilqb`, is

$$-\sum_{v \in S} W_v(f) = \log(TW)f(1) + \text{Trace}(\vartheta(f)(I - P_T^S - \widehat{P}_W^S)). \quad (\text{C8.2})$$

The cutoff term, the admissible test representation, the prime contributions and the trace's meaning in the source are not removed. C8.1 computes its projection factor only; it is not a claim that the individual unsmoothed factors are globally trace class. The blue plot quantity `epsilon(lambda)` is the least eigenvalue of the full Weil operator `A_lambda` in the different space $L^2([\text{lambda}(-1), \text{lambda}], du/u)$. No equality between that operator and the positive leakage operator is supplied by C7.2. The source itself describes the comparison in `poisson` and `miss` as numerical and leaves the eigenfunction comparison unfinished.

C9. The two prolate modes have an exact nonzero endpoint defect

Fix $\text{lambda} > 1$. Let $h_0 = h_-(0, \text{lambda})$, $h_4 = h_-(4, \text{lambda})$, $c_0 = h_0(0)$, $c_4 = h_4(0)$, with the source support extension by zero. Both c_0 and c_4 are nonzero: the functions are even, so their first derivatives at 0 vanish; their second-order prolate differential equation is regular at 0. If their values vanished too, uniqueness for the initial-value problem would make the eigenfunction identically zero, a contradiction.

The compressed Fourier identities give $\text{integral } h_0 = \text{chi}_0 c_0$ and $\text{integral } h_4 = \text{chi}_2 c_4$ by evaluation at Fourier coordinate 0. The source's strictly ordered squared eigenvalues and positive signs at indices 0,2 give $1 > \text{chi}_0 > \text{chi}_2 > 0$. Thus the exact endpoint map on their two-dimensional span is

$$L(a_0 h_0 + a_4 h_4) = \begin{pmatrix} f(0) \\ \hat{f}(0) \end{pmatrix} = \begin{pmatrix} c_0 & c_4 \\ \chi_0 c_0 & \chi_2 c_4 \end{pmatrix} \begin{pmatrix} a_0 \\ a_4 \end{pmatrix}, \quad \det L = c_0 c_4 (\chi_2 - \chi_0) \neq 0. \quad (\text{C9.1})$$

No nonzero combination of these two modes kills both endpoints. This is an invertible map onto the two endpoint values, so the obstruction also constructs their complete two-mode parameterization: for prescribed values (u, v) ,

$$a_0 = \frac{\chi_2 u - v}{c_0 (\chi_2 - \chi_0)}, \quad a_4 = \frac{v - \chi_0 u}{c_4 (\chi_2 - \chi_0)}. \quad (\text{C9.2})$$

For the source's integral-zero choice retain any scalar A and write

$$h_\lambda = A(\chi_2 c_4 h_0 - \chi_0 c_0 h_4), \quad \hat{h}_\lambda(0) = 0, \quad h_\lambda(0) = A c_0 c_4 (\chi_2 - \chi_0). \quad (\text{C9.3})$$

The last value is nonzero for $A \neq 0$; it tends toward a small value only under additional estimates, not by an exact cancellation. Conversely choosing $f(0) = 0$ leaves the Fourier endpoint $c_0 c_4 (\text{chi}_0 - \text{chi}_2)$ times the chosen scalar. These formulas retain the distinction between near cancellation and actual vanishing.

The next source mode constructs an exact repair of both endpoint equations. Put $h_8 = h_-(8, \text{lambda})$, $c_8 = h_8(0)$, and let chi_4 be its compressed Fourier eigenvalue. As above $c_8 \neq 0$, and the source ordering gives $1 > \text{chi}_0 > \text{chi}_2 > \text{chi}_4 > 0$. Define the actual function

$$g_\lambda = \frac{\chi_2 - \chi_4}{c_0} h_0 + \frac{\chi_4 - \chi_0}{c_4} h_4 + \frac{\chi_0 - \chi_2}{c_8} h_8. \quad (\text{C9.4})$$

Its value at zero is the sum of the three displayed numerators, which is 0. Its integral is $\text{chi}_0(\text{chi}_2 - \text{chi}_4) + \text{chi}_2(\text{chi}_4 - \text{chi}_0) + \text{chi}_4(\text{chi}_0 - \text{chi}_2) = 0$ by direct expansion. The coefficient of h_0 is nonzero; the distinct prolate eigenfunctions are linearly independent, so g_λ is nonzero. The endpoint map on the three-mode span has rank 2 by its already invertible first two columns. Its kernel is therefore exactly the one-dimensional space spanned by g_λ . This proves existence, uniqueness up to its retained scalar, and the exact map to both endpoint values.

The remaining Fourier defect is also explicit. Set $d_0=(\chi_2-\chi_4)/c_0$, $d_2=(\chi_4-\chi_0)/c_4$, $d_4=(\chi_0-\chi_2)/c_8$, so $g_\lambda=\sum_{j=0,2,4} d_j h_{2j,\lambda}$. Orthogonality of the distinct selfadjoint prolate eigenfunctions and the compressed Fourier equation give

$$\|(F-I)g_\lambda\|^2 = 2 \sum_{j \in \{0,2,4\}} |d_j|^2 (1 - \chi_j) \|h_{2j,\lambda}\|^2 > 0. \quad (C9.5)$$

Indeed F is a unitary involution on the even Hilbert space, so the left side equals $2\|g_\lambda\|^2 - 2\operatorname{Re}\langle Fg_\lambda, g_\lambda \rangle$. Since every $h_{2j,\lambda}$ is supported in P , its pairing with $Fh_{2k,\lambda}$ equals its pairing with $\chi_k h_{2k,\lambda}$. Expanding gives exactly C9.5, including every coefficient and norm. Strict positivity follows from $\chi_j < 1$ and the nonzero coefficients. The endpoint repair thus retains a nonzero Fourier defect; no finite-cutoff function has been declared exactly Fourier invariant.

For an even Schwartz function f , Poisson summation gives the full endpoint formula

$$\mathcal{E}(\hat{f})(x) = \mathcal{E}(f)(x^{-1}) + \frac{f(0)}{2\sqrt{x}} - \frac{\sqrt{x}}{2} \hat{f}(0). \quad (C9.6)$$

To prove it, separate $n=0$ in $\sum_{n \in \mathbb{Z}} f(n/x) = x \sum_{n \in \mathbb{Z}} \hat{f}(nx)$, pair n with $-n$, and multiply by $x^{(-1/2)/2}$. This retains both half-factors and both endpoints. The source's equation `poissonintro` is its restriction to both endpoint values zero.

The compactly truncated prolate functions are not automatically Schwartz; C9.1–C9.5 require only their integrability and the source compressed Fourier equations, and are already exact. C9.6 is stated on its proved Schwartz domain and is not silently applied to nonsmooth extensions. Its endpoint receiver `L` identifies precisely the values that an extension of that identity or a smoothing construction must retain. The nonzero two-mode endpoint and the three-mode repair are therefore explicit data, not a discarded obstacle and not a claimed negative Weil value.

C10. Return to the existing supported original-zeta form

The existing coefficient receiver is used rather than replacing the programme by the authors' scalar form. Write $U=[\tau]$, $f=[1]$, $Q_0=U-f$ in the auxiliary coefficient algebra of `WEIL_UNIT_DIFFERENCE_LIFT.md`, `WU1–WU8`. These are formal linear coefficients; no source subtraction `tau-1` is asserted. Products give $f^2=f$, $Q_0^2=Q_0$, $fQ_0=0$, $U=f+Q_0$. Arithmetic evaluation sends both U and f to scalar 1 and Q_0 to scalar 0.

For an original compact smooth test $h(v)$, retain $H(s)=\int h(v)\exp(-(s-1/2)v)dv$, the full prime-power term $P(h)$, and the Gamma term $A_\infty(h)$. The programme's exact original-zeta receiver is

$$\begin{aligned} W_\zeta(h) &= H(0) + H(1) + A_\infty(h) - P(h) = \sum_\rho m_\rho H(\rho), \\ P(h) &= \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (h(\log n) + h(-\log n)), \\ A_\infty(h) &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt. \end{aligned} \quad (C10.1)$$

Every finite trivial-zero cutoff is retained in the original identity $V_\zeta(\zeta, N) = Z(H) + \sum_{m=1}^N H(-2m) - H(1) = G_N - P(h)$, $G_N = A_\infty + H(0) + \sum_{m=1}^N H(-2m)$. The cancellation in the displayed W_ζ is an explicitly recorded map from this triple, not removal of its components. Its full contour derivation is `OZC1–OZC15` in the retained programme source; `WU2` states the boundary domains and convergence.

The source test identification is $a(x)=h(\log x)$. It preserves convolution and the involution $a^*(x)=\bar{a}(x^{-1})$ because $dx/x=dv$. The source Fourier integral of a at t is $H(1/2+it)$. Its endpoint values at $\pm i/2$ are respectively $H(0), H(1)$, and the prime terms are exactly $P(h)$. Thus the compared source form is reached by an explicit change of variable with all half-powers retained.

The archimedean comparison also retains its constant. The digamma integral, for $\text{Re}(z)>0$, is $\psi(z)=-\gamma+\int_0^\infty (\exp(-t)-\exp(-zt))/(1-\exp(-t)) dt$. Apply it to $z=1/4+iy/2$ and Fourier inversion of h . First perform the t integral on $[\delta, R]$; then send δ to 0 and R to infinity. The paired numerator at 0 vanishes to first order, and the Schwartz decay of $\hat{h}$ controls the polynomial y factors in its Taylor estimate. At infinity the exponential factors dominate. These bounds justify the passage to the limit and yield, after $t=2v$,

$$A_\infty(h) = -(\gamma + \log \pi)h(0) + \int_0^\infty \frac{2h(0)e^{-2v} - (h(v) + h(-v))e^{-v/2}}{1 - e^{-2v}} dv. \quad (\text{C10.3})$$

The supplied source's archimedean expression, after $x=\exp(v)$, is exactly

$$W_{\mathbb{R}}(a) = (\log(4\pi) + \gamma)h(0) + \int_0^\infty \frac{(h(v) + h(-v))e^{-v/2} - 2h(0)e^{-v}}{1 - e^{-2v}} dv. \quad (\text{C10.4})$$

Adding C10.3 and C10.4 gives $\log(4)h(0)$ minus $2h(0)\int_0^\infty \exp(-v)/(1+\exp(-v))dv$. The last integral equals $\log(2)$, by the substitution $w=\exp(-v)$. Hence $A_\infty(h)=-W_{\mathbb{R}}(a)$, with the exact $\log(4\pi)$, γ and $\log(\pi)$ constants accounted for. Together with the already proved prime and endpoint maps this gives $W_\zeta(h)=H(0)+H(1)-\sum_v W_v(a)$ without dropping an archimedean summand.

For the programme's support lattice L retain the vectors

$$B = \mathbf{e}_{1_L}(H(0) + H(1)) + \sum_{\ell \neq 1_L} \mathbf{e}_\ell H(0), \quad Z = \mathbf{e}_{1_L} Z(H), \\ D = \mathbf{e}_{1_L}(P(h) - A_\infty(h)) + \sum_{\ell \neq 1_L} \mathbf{e}_\ell H(0). \quad (\text{C10.2})$$

C10.1 proves $\text{bold}(B)-\text{bold}(Z)=\text{bold}(D)$ on each coordinate. Tensoring each displayed term with U or f retains the two source records; their difference is the same complete vector tensored with Q_0 . Arithmetic evaluation returns C10.2 without deleting a support coordinate. This is the exact comparison from WU3-WU5, and does not turn Q_0 into an extra classical Euler factor at 0.

The support trace proof is SZW33-SZW38 at the pinned programme edition. The complete local return and full original-zeta reconstruction are retained with their proof locators in `WEIL_UNIT_DIFFERENCE_L` `IFT.md`, WU2-WU5, and the companion original source copies identified in the source ledger.

What has been established here is the actual base comparison, the sign-adjusted Tate fixed locus with all winding data, its faithful two-prime map from the original critical strip, the factor-four repair, the two-mode endpoint isomorphism, its three-mode repair with the complete Fourier defect, and the two-sign trace block. None of these statements assumes every zeta zero is fixed. The supplied geometry's involution and equal sphere radius have been carried through to these exact arithmetic and trace expressions; their values have not been assigned a positive sign by definition.

Prime monodromy, complete clock observations, and the original zeta weights

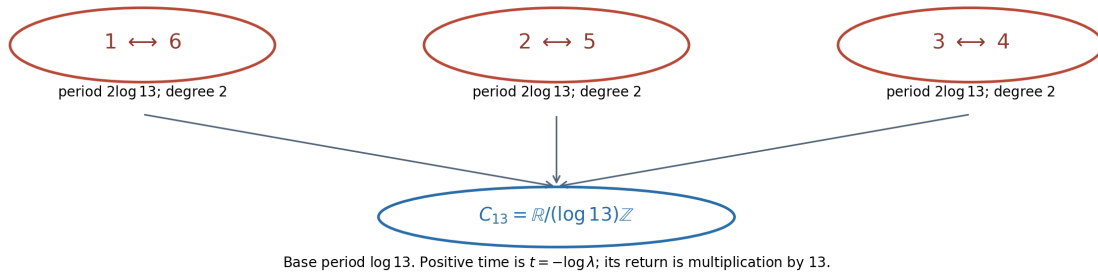
24 September 2026. This receives the two identical pasted transcripts accompanying arxiv-pisa.tex and stack2.png. The transcripts are retained verbatim, including their tentative formulations; assistant text within them is not promoted to a user definition. The question investigated here is the user's proposed common global requirement of infinite observation, and its relation to arithmetic weight.

The notation $Z_0, Z_1, Z_2, \dots$ continues to mean the user's absence/presence and quantum-number layers. The symbols Q_d below name explicitly constructed d-state winding clocks; they do not silently redefine the user's entire Z_d algebra. The distinction between an absent datum and an existing datum whose observed count is zero remains throughout. Addition at tau is not restored.

The primary source is Alain Connes and Caterina Consani, *Knots, primes and class field theory*, arXiv:2501.06560v1, original arxiv-pisa.tex. The supplied file is byte-identical to the original source. Its mapping-torus propositions, finite-cover theorem and class-field comparison are read directly in TeX. The picture's three upper components are over ONE prime p. Their labels are primes above p in an extension; their vertical drawing positions are not three successive quantum-number levels. M2 constructs an exact example with those three components and their winding data.

Prime monodromy and the complete stack of winding observations Three covering circles over ONE prime orbit

Exact example: $p = 13$, $Q(\zeta_7)$, $G = (\mathbb{Z}/7\mathbb{Z})^\times$, return $g \mapsto 6g$



Retain every earlier winding clock: exact refinement data

New clock N	1	2	3	4	5	6	7	8	9
Total states L_N	1	2	6	12	60	60	420	840	2520
L_N/L_{N-1}	—	2	3	2	5	1	7	2	3
$\log L_N - \log L_{N-1}$	—	$\log 2$	$\log 3$	$\log 2$	$\log 5$	0	$\log 7$	$\log 2$	$\log 3$

$$S_N \cong \mathbb{Z}/L_N\mathbb{Z}, \quad L_N = \text{lcm}(1, \dots, N), \quad \log L_N - \log L_{N-1} = \Lambda(N)$$

Green: first prime channel. Gold: refinement of an existing prime channel. Gray: no new independent state.

All clocks remain recorded. The compatible infinite system is $\widehat{\mathbb{Z}}$; no finite prefix isolates an unbounded integer.

Figure 4: Exact prime cover and complete winding-clock refinements

Figure M. Above: the three components for $p=13$ in $Q(\zeta_7)$, with return multiplication by 6 modulo 7, covering degree 2 and both real periods retained. Below: the exact first nine stages of the all-clock receiver. The all-N argument is M4–M6, not an extrapolation from this table. The supported original-zeta comparison is M7. Human source: Connes–Consani, original arxiv-pisa.tex, Proposition mappingtorus and its finite-cover calculation. Reproducible figure source: draw_prime_monodromy_history.py; exact displayed data: MONODROMY_FIGURE_CHECKS.json. The drawing is schematic; ellipse sizes are not

a metric on the source base.

M1. Presence and a zero reading have an explicit comparison

Use the actual generic monoid retained in C2 of TAU_TWISTOR_TATE_AND_WEIL.md:

$$A_\eta = \{0\} \cup \{J^r T^m : r \in \mathbb{Z}/4\mathbb{Z}, m \in \mathbb{Z}\}, \quad J^2 = \epsilon, \quad J^0 T^0 = \tau. \quad (\text{M1.1})$$

The multiplication adds m, adds r modulo 4, and has absorbing element 0. Define the presence observation

$$P : A_\eta \longrightarrow \{0, \tau\}, \quad P(0) = 0, \quad P(J^r T^m) = \tau. \quad (\text{M1.2})$$

It is multiplicative because a product of two nonabsorbing elements is nonabsorbing, and a product with 0 is 0. It is invariant under the source mirror $J^r T^m \rightarrow J^{(-r+2m)T}(-m)$. Thus zero accumulated winding $m=0$ does not make a present datum absent: the whole fibre $m=0$ contains τ, J, ϵ, J^3 . The winding projection is defined on the nonabsorbing group; P is defined on the pointed monoid. These are two proved maps on the same structure.

This comparison does not settle the user's tentative proposal to rename eta or the pair Z_0/Z_1 . The source generic point eta, its stalk A_η and the stalk unit tau have different types, connected by the stalk's projection to its supporting point and by M1.2. The unit in M1.1 is not identified with integer zero. When an integer count is represented by an exponent m, retain its count label separately from the base label tau, including at $m=0$. Winding-return parity is an observation on this exponent; it is not an assignment of intrinsic integer parity to tau.

M2. What the supplied stacked picture represents

Fix a rational prime p and put $\ell_p = \log(p)$. To avoid collision with the source's additive group denoted H_p in FF.tex, write

$$\mathcal{H}_p = \prod_{q \neq p} \mathbb{Z}_q^\times. \quad (\text{M2.1})$$

The original Proposition `mappingtorus` identifies the inverse image of the prime orbit with

$$\pi^{-1}(C_p) \cong (\mathcal{H}_p \times \mathbb{R}_{>0}) / ((h, \lambda) \sim (ph, p\lambda)), \quad C_p = \mathbb{R}_{>0} / p^\mathbb{Z}. \quad (\text{M2.2})$$

The retained adelic representative has coordinate h_q at every $q \neq p$, coordinate 0 at p, and archimedean coordinate lambda. Multiplication by a positive rational preserves this form exactly when that rational is a power of p, because all its valuations outside p must be zero. This proves the orbit equivalence in M2.2 directly. The full existence and equivariance are the source proposition, with its representative map kept here.

For orientation, let $t = -\log(\lambda)$. Then M2.2 becomes

$$(h, t + \ell_p) \sim (ph, t). \quad (\text{M2.3})$$

Thus a positive t-circuit returns h to ph. A positive $\log(\lambda)$ -circuit has inverse return. Both orientations and the original length ℓ_p remain displayed.

For a finite unramified cover in the source, let G be its finite abelian Galois group and c the image of p, its arithmetic Frobenius. Replacing H_p by G in M2.3 gives

$$\mathcal{M}(G, c) = (G \times \mathbb{R}) / ((g, t + \ell_p) \sim (cg, t)). \quad (\text{M2.4})$$

Let f be the order of c . Its connected components are indexed by the cosets $G/$. For a representative g the explicit map

$$\mathbb{R}/(f\ell_p\mathbb{Z}) \longrightarrow \mathcal{M}(G, c), \quad [u] \mapsto [g, u] \quad (\text{M2.5})$$

has image that component and is a homeomorphism onto it. Indeed $[c^j g, t] = [g, t + j \ell_p]$, so it is surjective there. Equality of $[g, u]$ and $[g, v]$ requires $u - v = k \ell_p$ and $c^k = 1$, equivalently $u - v$ is in $f \ell_p \mathbb{Z}$. The local covering charts prove continuity and the inverse continuity. The projection to C_p is u modulo ℓ_p , so its degree is f and its length is $f \log(p)$. There are $|G|/f$ such components. In the abelian unramified field extension these cosets are exactly the primes over p , by the source's decomposition-group calculation.

A concrete instance of the three-component picture is the cyclotomic extension $\mathbb{Q}(\zeta_7)$ and $p=13$. Here $G=(\mathbb{Z}/7\mathbb{Z})^\wedge$ has six elements, $c=13 \bmod 7=6=-1$, and c has order 2. The components are

$$\{1, 6\}, \quad \{2, 5\}, \quad \{3, 4\}. \quad (\text{M2.6})$$

Each component is a two-sheeted circle of length $2 \log(13)$ over the circle of length $\log(13)$. Multiplication by 6 exchanges its two states. This is an exact example of the drawing, not a claim that the drawing itself specified 13 or this field.

M3. The full monodromy closure is the complete winding group

Define $K_p = \text{closure of } \{p^n : n \text{ in } \mathbb{Z}\} \text{ inside } H_p$. The map $n \mapsto p^n$ extends canonically to a continuous surjection

$$\Theta_p : \widehat{\mathbb{Z}} \longrightarrow K_p. \quad (\text{M3.1})$$

Here $\widehat{\mathbb{Z}}$ is the inverse limit of ALL finite integer residue groups. To construct the map, for every integer m coprime to p reduce p modulo m and exponentiate using a modulo $\text{ord}_m(p)$. These coordinates are compatible under divisibility of m , and give an element of H_p . They agree with p^n on ordinary integers. Continuity follows coordinatewise. The image is compact and contains the dense cyclic subgroup; it is contained in K_p by density of $\mathbb{Z}$ in $\widehat{\mathbb{Z}}$, proving surjectivity onto K_p .

There is no kernel. For every $d \geq 2$ take

$$m_d = p^d - 1, \quad \gcd(m_d, p) = 1. \quad (\text{M3.2})$$

Modulo m_d the element p has EXACT order d . Its d -th power is 1. If $1 \leq k < d$, then $0 < p^{k-1} < p^d - 1$, so m_d cannot divide p^{k-1} . This proves the asserted order. Consequently $\Theta_p(a) = 1$ implies $a = 0$ modulo d for every $d \geq 2$, hence $a = 0$ in $\widehat{\mathbb{Z}}$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism; all maps here are group homomorphisms. We have proved

$$\boxed{\Theta_p : \widehat{\mathbb{Z}} \simeq K_p.} \quad (\text{M3.3})$$

This identifies the exact completed image of the source's integer powers. It also proves that the source map from the Frobenius group $\widehat{\text{pi}_1 \text{et}(\text{Spec } F_p)}$ to H_p is injective, using its specified generator p ; no injectivity is deduced merely from injectivity on a dense subgroup. No literature-priority claim is made for this elementary completed-group calculation.

For two primes p, r the generator-preserving isomorphism is $\text{Theta}_r \text{Theta}_p^{-1}(-1): K_p \rightarrow K_r$. It sends p^a to r^a for every profinite exponent a . The embeddings into H_p, H_r and the real periods $\log(p), \log(r)$ remain separate retained data. On their mapping tori the corresponding isomorphism is

$$[h, t] \mapsto [\Theta_r \Theta_p^{-1}(h), (\log r / \log p)t]. \quad (\text{M3.4})$$

Its well-definedness follows from M2.3; a shift $\log(p)$ becomes $\log(r)$ and multiplication by p becomes multiplication by r . Its inverse interchanges p and r . Thus the clocks have the same completed group type, with an explicitly retained change in flow parameter.

M4. The stack keeps every earlier finite-state observation

For each $d \geq 1$ let Q_d be the actual d -state winding quotient Z/dZ , obtained within K_p by reduction modulo p^{d-1} and discrete exponent as in M3. For $d=1$ it is the singleton quotient. This construction assigns a cyclic transition to the source winding clock; it does not assert that every object with the user's label Z_d must be this cyclic group.

To keep all previous floors, take the simultaneous receiver

$$S_N = \text{im} \left(\widehat{\mathbb{Z}} \rightarrow \prod_{d=1}^N Q_d \right), \quad L_N = \text{lcm}(1, \dots, N). \quad (\text{M4.1})$$

Two residues define the same tuple precisely when their difference is divisible by every $d \leq N$, equivalently by L_N . Every tuple in this image comes from an ordinary residue modulo L_N . Hence the explicit map

$$\mathbb{Z}/L_N\mathbb{Z} \simeq S_N, \quad [n] \mapsto ([n]_1, \dots, [n]_N) \quad (\text{M4.2})$$

is bijective and preserves the winding step. The transition $S_N \rightarrow S_{N-1}$ forgets only the last coordinate and corresponds to reduction modulo L_{N-1} , since L_{N-1} divides L_N . The entire compatible system has inverse limit $\widehat{\mathbb{Z}}$: each modulus d occurs as a coordinate, so a compatible system determines, and is determined by, all integer residues. These two assignments are explicit inverse maps.

This gives a stack in which the d -state observation comes with every earlier observation. For example S_3 retains both parity and the three-state reading and has SIX compatible states; S_4 has twelve. A state labelled 0 at any of these coordinates is still a state. To include the user's Z_0 absence, use an additional absent symbol outside the set of present tuples; the absence observation is distinct from the all-zero tuple. The construction imports no tau addition and assigns no intrinsic integer parity to tau.

All source sign data can also be retained in this completion. For the generic nonabsorbing group in M1, the map

$$J^r T^n \mapsto (J^r, \Theta_p(n)) \quad \text{into} \quad (\mathbb{Z}/4\mathbb{Z}) \times K_p \quad (\text{M4.3})$$

is injective: equality fixes r modulo 4 and n by M3.3. On its closure, the source mirror is $(J^r, \text{Theta}_p(a)) \rightarrow (J^{(-r+2a)}, \text{Theta}_p(-a))$, where $2a$ modulo 4 is defined by a modulo 2. This is a continuous multiplicative involution by direct composition. The full quartic sign and branch data therefore survive the completed winding observations.

M5. The exact infinite-observation statement holds uniformly

Fix an integer n . Its fibre under observation through level N is

$$n + L_N\mathbb{Z}. \quad (\text{M5.1})$$

This follows from M4.2. It is infinite for every finite N , while

$$\bigcap_{N \geq 1} (n + L_N \mathbb{Z}) = \{n\}. \quad (\text{M5.2})$$

Indeed a difference belonging to every $L_N \mathbb{Z}$ is divisible by every positive integer d . A nonzero integer cannot be divisible by an integer larger than its absolute value. This proves M5.2 without a numerical cutoff or an assumed prime list. Translation by n carries the entire fibre system at 0 to the fibre system at n , proving the same infinite refinement requirement for EVERY integer in this observation model.

This is the exact global assertion supported by the construction: no finite family of these residue observations singles out an unbounded integer, while the complete family does. It concerns recovery FROM these observations. It does not change the original step generating n after n positive steps. The typed maps relating generation and observation are the prefix inclusions of the integer winding history into Z and the diagonal injection $Z \rightarrow \lim S_N$. They commute with the step $n \rightarrow n+1$; one is a direct system of generated histories and the other is an inverse system of observations of those same counts.

The completion also contains coherent states that are not ordinary integers. For example, by the Chinese remainder construction choose residues that converge to 0 in Z_2 and to 1 in every Z_q for q odd. They define an element of $\hat{Z}$. It cannot be an integer, since an integer that is 0 in Z_2 is 0, and 0 is not 1 in Z_3 . Thus the completion is not silently substituted for the source integers; M5.2 recovers the original integer within its image.

M6. The retained growth is precisely the original prime-power weight

The total stack S_N has L_N states. Retain its exact growth ratio and logarithmic increment

$$r_N = L_N / L_{N-1}, \quad w_N = \log L_N - \log L_{N-1} \quad (N \geq 2), \quad L_1 = 1. \quad (\text{M6.1})$$

For each prime p , unique factorization gives $v_p(L_N) = \max\{a \geq 0 : p^a \leq N\}$. This is because lcm takes the maximum valuation of its arguments. Passing from $N-1$ to N increases this maximum exactly when $N = p^a$ for some $a \geq 1$; the increase is then 1. At most one prime has this property for a given N . Therefore

$$r_N = \begin{cases} p & N = p^a, \ a \geq 1, \\ 1 & N \text{ is not a prime power,} \end{cases} \quad \boxed{w_N = \Lambda(N)}. \quad (\text{M6.2})$$

Λ is the ordinary von Mangoldt function, with every prime power retained. The new prime line at p is its first occurrence. At $p^2, p^3, \dots$ the same prime contributes further resolution. A count such as 6 introduces a floor whose observation is already determined by earlier floors; that floor is still present. Explicitly $L_2=2$, $L_3=6$, $L_4=12$, $L_5=L_6=60$. The step at 4 refines the 2-channel; the step at 6 adds no independent clock state.

This relates the user's first-appearance observation to the higher repeated data rather than replacing one by the other. The earlier prime-activation sieve recovers the set of first primes. The present all-floor receiver recovers their complete prime-power weights. Finite resolution and multiplicity remain available before the infinite limit.

For $\text{Re}(s) > 1$ the resulting exact analytic identity is

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{N=2}^{\infty} (\log L_N - \log L_{N-1}) N^{-s} = \sum_p \sum_{a=1}^{\infty} (\log p) p^{-as}. \quad (\text{M6.3})$$

Proof: finite Euler products expand into the sum of n^{-s} over integers using only those primes. Absolute convergence of $\sum n^{-\text{Re}(s)}$ permits passage to all primes and gives the original zeta Euler product.

Expanding each logarithm gives $\log(\zeta(s)) = \sum_p \sum_{a \geq 1} p^{-as}/a$, with the branch tending to 0 on the positive real axis at infinity. Differentiation on compact subsets of $\text{Re}(s) > 1$ is justified by the bound $\sum_{N \geq 2} (\log N) N^{-\sigma} 1$. It yields the last series in M6.3. Substitution of M6.2 yields the first. All factors, repetitions and signs are retained.

Thus the uniform need for the whole observation system coexists with an exact, nonconstant growth record. The latter is already the prime-power term entering the original zeta function; replacing it by the single statement that all final histories are infinite would remove this computed datum.

M7. Substitute the complete history into the supported Weil receiver

For h in $C_c^\infty(\mathbb{R})$, put $H(s) = \int h(v) \exp(-(s-1/2)v) dv$ and $\hat{h}(t) = \int h(v) \exp(-itv) dv$, with the displayed conventions fixed. In the existing original-zeta receiver of C10, the complete prime term becomes exactly

$$P_{\text{hist}}(h) = \sum_{N=2}^{\infty} \frac{\log L_N - \log L_{N-1}}{\sqrt{N}} (h(\log N) + h(-\log N)). \quad (\text{M7.1})$$

This sum is finite for each compactly supported h , but its formula is valid on the entire test space; it is not a fixed-range numerical approximation. It equals $P(h)$ term by term by M6.2. Keep

$$A_{\infty}(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt.$$

$$W_{\zeta}(h) = H(0) + H(1) + A_{\infty}(h) - P_{\text{hist}}(h) = \sum_{\rho} m_{\rho} H(\rho). \quad (\text{M7.2})$$

This is an exact substitution into the proved original-zeta contour formula, retaining its Gamma, endpoint and multiplicity contributions. The finite trivial-zero cutoff triple remains $V_{\zeta}(B) = Z(H) + \sum_{m=1}^B H(-2m) - H(1) = G_B - P_{\text{hist}}(h)$, $G_B = A_{\infty} + H(0) + \sum_{m=1}^B H(-2m)$; the integer cutoff B is not sent to infinity without convergence. The complete contour proof is retained with C10 and WU2 in the programme sources.

For the existing support lattice $\mathcal{L}$ keep the source vectors

$$B(h) = \mathbf{e}_{1_{\mathcal{L}}}(H(0) + H(1)) + \sum_{\ell \neq 1_{\mathcal{L}}} \mathbf{e}_{\ell} H(0),$$

$$Z(h) = \mathbf{e}_{1_{\mathcal{L}}} \sum_{\rho} m_{\rho} H(\rho), \quad (\text{M7.3})$$

$$D_{\text{hist}}(h) = \mathbf{e}_{1_{\mathcal{L}}}(P_{\text{hist}}(h) - A_{\infty}(h)) + \sum_{\ell \neq 1_{\mathcal{L}}} \mathbf{e}_{\ell} H(0).$$

M7.2 proves $\text{bold}(B) - \text{bold}(Z) = \text{bold}(D_{\text{hist}})$ in every support coordinate. Tensoring all terms with $[\tau]$ or with $[1]$ gives the two distinct programme coefficient records exactly as in WU1–WU8; their full difference remains in $[\tau] - [1]$ before arithmetic evaluation. The support-zero contributions have not been replaced by a classical prime sum, and the new history formula has not silently removed them.

The proof source for the support vectors is SZW33–SZW38 at the pinned programme edition. Their complete original-zeta comparison is C10 and the retained WU2–WU5. M6.2 proves the present substitution without a new analytic assumption.

M8. What equal modulus proves for this actual monodromy

Translation by p preserves Haar measure on the compact group K_p . Its operator $U_p f(h) = f(ph)$ is therefore unitary on $L^2(K_p)$: substitution $h' = ph$ proves equality of norms and $f(h) \rightarrow f(p^{-1}h)$ is its inverse. Under M3.3 a finite character has the form

$$\chi_{j,d}(\Theta_p(a)) = \exp(2\pi i j a / d), \quad U_p \chi_{j,d} = \exp(2\pi i j / d) \chi_{j,d}. \quad (\text{M8.1})$$

These are characters on Q_d pulled back to K_p . Every continuous character $K_p \rightarrow \hat{S}^1$ factors through a finite quotient: continuity puts an open subgroup into a sufficiently short arc about 1; the only subgroup of $\hat{S}^1$ contained in that arc is $\{1\}$, so that open subgroup lies in the kernel. Hence every such character has the displayed form. This proves exact modulus 1 for the entire character family, including after the infinite observation limit.

For the mapping torus in the t -orientation of M2.3, the function $\chi_{j,d}(h) \exp(i \omega t)$ is well defined precisely when $\exp(i \omega \ell_p) = \exp(2\pi i j / d)$. Its full frequencies are therefore

$$\omega = \frac{2\pi}{\log p} \left(k + \frac{j}{d} \right), \quad k \in \mathbb{Z}. \quad (\text{M8.2})$$

This proves a common modulus for the return eigenvalues while retaining the prime-dependent real scale. It is the spectrum of this explicitly defined monodromy representation; no identification with zeta zeros is hidden in the word spectrum.

The factor $1/2$ used in the original-zeta Mellin receiver also has an exact source comparison. For the raw dilation $D_a f(x) = f(x/a)$ on $L^2(\mathbb{R}, dx)$, substitution $x = ay$ gives $\|D_a f\|^2 = a^{-1} \|f\|^2$. Thus a positive scalar multiple $c(a)D_a$ is unitary exactly when $c(a) = a^{1/2}$. The half-power follows from this measure and this dilation, rather than from assigning all integer histories an infinite weight. Define $U_a = a^{1/2} D_a$; the full map to the multiplicative receiver $V_a k(u) = k(u/a)$ is

$$\mathcal{E}(U_a f)(u) = u^{1/2} \sum_{n \geq 1} a^{-1/2} f(nu/a) = \mathcal{E}(f)(u/a) = V_a \mathcal{E}(f)(u). \quad (\text{M8.3})$$

All square roots and the original summation are retained. The next calculation identifies precisely the spectral information in this map.

M9. The spectral quotient retaining the original zeta zeros

Let S_0^{even} consist of even Schwartz functions f on $\mathbb{R}$ with $f(0) = \int_{-\infty}^{\infty} f = 0$. Let A be the smooth functions k on $\mathbb{R}_{>0}$ such that, for all nonnegative integers N, j , $\sup_{u > 0} (u^{N+u} (-N)) | (u d/du)^j k(u) |$ is finite. This is a locally convex space with these seminorms; its elements and all logarithmic derivatives decay faster than every power at both endpoints. It sits in $L^2(\mathbb{R}_{>0}, du/u)$.

The map E from M8.3 sends S_0^{even} into A . At infinity this follows by summing the Schwartz bounds for $f(nu)$ and its differentiated series with an exponent greater than 1. At zero, Poisson summation with both endpoint values zero gives $E(f)(u) = E(\hat{f})(1/u)$; the same argument at infinity for $\hat{f}$ supplies the zero-end estimates, including derivatives. The maps U_a and V_a preserve these domains and have the exact covariance M8.3.

Define the full centered Mellin observation

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{M9.1})$$

It is entire. For every fixed s , including its derivatives in s , it is continuous on A : choose $N > |\operatorname{Re}(s) - 1/2|$ and use the displayed weighted seminorm to bound the integrand at 0 and infinity; any power of $|\log u|$ is still

integrable with a slightly larger N . Locally uniform domination also proves complex differentiation under the integral.

For $\text{Re}(s) > 1$, absolute convergence and $v = nu$ give

$$\mathcal{ME}(f)(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{M9.2})$$

The right-hand Mellin integral is holomorphic on $\text{Re}(s) > -2$ because even f with $f(0)=0$ has $f(v)=O(v^2)$ at zero. At $s=1$ the zero integral condition cancels the simple zeta pole. Analytic continuation therefore gives M9.2 throughout the open critical strip, retaining the ORIGINAL zeta. A zero ρ of multiplicity m_ρ makes every derivative of the left side of orders 0 through $m_\rho-1$ vanish there. Thus these continuous, nonzero Mellin jet functionals descend to

$$\mathcal{Q} = \mathcal{A} / \overline{\mathcal{E}(S_0^{\text{even}})}^{\mathcal{A}}. \quad (\text{M9.3})$$

The evaluation functional at ρ is nonzero on $\mathcal{A}$: choose a nonzero nonnegative compactly supported smooth bump $b(u)$ away from 0 and set $k(u) = u^{-i \text{Im}(\rho)} b(u)$; then $\mathcal{M}k(\rho)$ is a strictly positive real integral. The jet functionals of all orders are linearly independent before quotient: a linear combination integrates $k(u)u^{(\rho-1/2)}$ times a polynomial in $\log u$; vanishing on every compact bump forces that polynomial to vanish. Continuity and M9.2 prove that the jets up to $m_\rho-1$ remain nonzero independent functionals on $\mathcal{Q}$. This retains multiplicity information without assuming RH.

They have the exact scaling action

$$\mathcal{M}(V_a k)(s) = a^{s-1/2} \mathcal{M}k(s). \quad (\text{M9.4})$$

Differentiation retains all jet terms: $(\mathcal{M} V_a k)^{(j)}(\rho) = a^{(\rho-1/2)} \sum_{l=0}^j \binom{j}{l} (\log a)^{j-l} (\mathcal{M} k)^{(l)}(\rho)$. In particular the evaluation eigenvalue has modulus $a^{\text{Re}(\rho)-1/2}$. This is the exact place where critical-line modulus is asked of the ACTUAL zeros, rather than of the finite monodromy characters in M8.

There is a concrete reason the positive Haar inner product does not simply descend as a nontrivial Hilbert quotient: $E(S_0^{\text{even}})$ is dense in $L^2(\mathbb{R}_{>0}, du/u)$. Here is a complete proof using the programme's retained kernel. The nonzero Gaussian polynomial $f_0(x) = (\pi/2) x^{2(2^{p_{ix}}-3)} \exp(-\pi x^2)$ is in S_0^{even} and has Fourier transform itself, as computed in C6. Put $k_0 = E(f_0)$. Its Mellin transform is the entire function

$$\mathcal{M}k_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad (\text{M9.5})$$

with the factor four relative to ξ retained. It is not identically zero: at $s=2$ its value is $\pi/24$. As an entire nonzero function it has isolated zeros. With the Fourier convention $\widehat{K}(t) = \int_{\mathbb{R}} K(v) e^{-itv} dv$, the logarithmic function $K_0(v) = k_0(e^v)$ has transform $\widehat{K}_0(t) = \mathcal{M}k_0(1/2 - it)$. This transform is consequently nonzero almost everywhere. The minus sign in this comparison is retained.

Suppose F in L^2 is orthogonal to every $V_a k_0$. Set $F_{\log}(v) = F(e^v)$ and $b = \log a$. In logarithmic coordinate the translate is $K_0(v-b)$, whose Fourier transform is $e^{-itb} \widehat{K}_0(t)$. Plancherel, with its full constant, gives

$$0 = \langle F, V_a k_0 \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{F}_{\log}(t) \overline{\widehat{K}_0(t)} e^{itb} dt \quad (b \in \mathbb{R}). \quad (\text{M9.6})$$

The product in this integral is L^1 by Cauchy–Schwarz. Uniqueness of the Fourier transform on L^1 gives product=0 almost everywhere. Since the second factor is nonzero almost everywhere, the transform of $F_{\log}$ is zero almost everywhere, and $F=0$. This proves density of the translates of k_0 . M8.3 puts every such translate in $E(S_0^{\text{even}})$, proving the density claim.

Accordingly the L^2 quotient is zero, whereas Q in M9.3 has the proved nonzero zero-evaluation functionals. The identity inclusion $A \rightarrow L^2$ induces the zero map from Q to that Hilbert quotient. Every original zero evaluation annihilates $E(S_0^{\text{even}})$ but is nonzero, so it cannot extend to a continuous L^2 functional. This proves the exact relation between the two receiving spaces instead of treating their different behaviour as absence of a relation. The richer quotient is constructed explicitly and keeps the original zeros, their multiplicities and the full scaling defect.

The programme’s linear coefficient spaces pass through this quotient by the following exact tensor isomorphism. Support records at zero require the separate record-retaining map proved below and in `SUPPORT_ZERO_RECEIVER_REPAIR.md`; they cannot be recovered from a zero tensor. Let C_{rec} be the coefficient algebra of $WU1$: its independent vector-space basis is $U=[\tau]$ and $N_n=[n]$ for nonzero integers n , with U the unit, $N_n N_m = N_{nm}$, and $N_0=0$. Put $f=N_1$. Let $C^{\text{mathcal L}}$ be the free vector space on all retained support labels. All tensor products in the following formula are algebraic, so each element uses finitely many coefficient directions:

$$\frac{\mathcal{A} \otimes C_{\text{rec}} \otimes \mathbb{C}^{(\mathcal{L})}}{\mathcal{E}(S_0^{\text{even}})^{\mathcal{A}} \otimes C_{\text{rec}} \otimes \mathbb{C}^{(\mathcal{L})}} \simeq Q \otimes C_{\text{rec}} \otimes \mathbb{C}^{(\mathcal{L})}. \quad (\text{M9.7})$$

The map sends the class of k tensor c tensor e_{ell} to $[k]$ tensor c tensor e_{ell} . Choose a basis of the two coefficient spaces. Each element then has a unique finite coordinate expansion. Its image is zero exactly when each A -coordinate lies in the closed subspace in the denominator. This proves well-definedness, its exact kernel and surjectivity, hence the displayed isomorphism of the linear spaces. It does not assert that the preceding map from labelled states to tensors retains a label at coefficient zero.

The two sections on Q tensor $C^{\text{mathcal L}}$ send q tensor e_{ell} to q tensor U tensor e_{ell} and q tensor f tensor e_{ell} . Arithmetic evaluation sends U to 1 and N_n to n , and composes with each section to the identity. Their difference is q tensor $(U-f)$ tensor e_{ell} ; it is injective because the coefficient functional taking U to 1 and every N_n to 0 recovers q tensor e_{ell} . Multiplication gives $(U-f)^2=U-f$ and $(U-f)f=0$, retaining the programme’s unit-difference idempotent. The Mellin jets and scaling act on Q . To retain every original support record, including a selected section at $q=0$, use the disjoint family of pairs (record, q) , with record in mathcal L or in $\text{mathcal L} \times \{U, f\}$ when the section selection is retained. Every operator acts only on q . Equality of paired observations requires equality of the record and then equality of q ; the injective synthesis map proves the latter. The exact maps to the previous tensors, their zero fibres, and the full pairing are proved in ZR1-ZR5 of `SUPPORT_ZERO_RECEIVER_REPAIR.md`. This is a formal coefficient construction, not addition or subtraction of source τ , and it is not a new assignment of parity to τ .

The results establish a faithful infinite-observation completion, the same refinement requirement for every integer, the full prime-power growth in the original-zeta Weil term, and an exact spectral receiving quotient. They do not establish nonnegativity of M7.2 on all Weil tests or modulus one for every eigenvalue in M9.4. No equality of those moduli has been inserted as a definition or inferred from a shared infinite cardinality.

Complete history, integral adjoints, and the original zeta

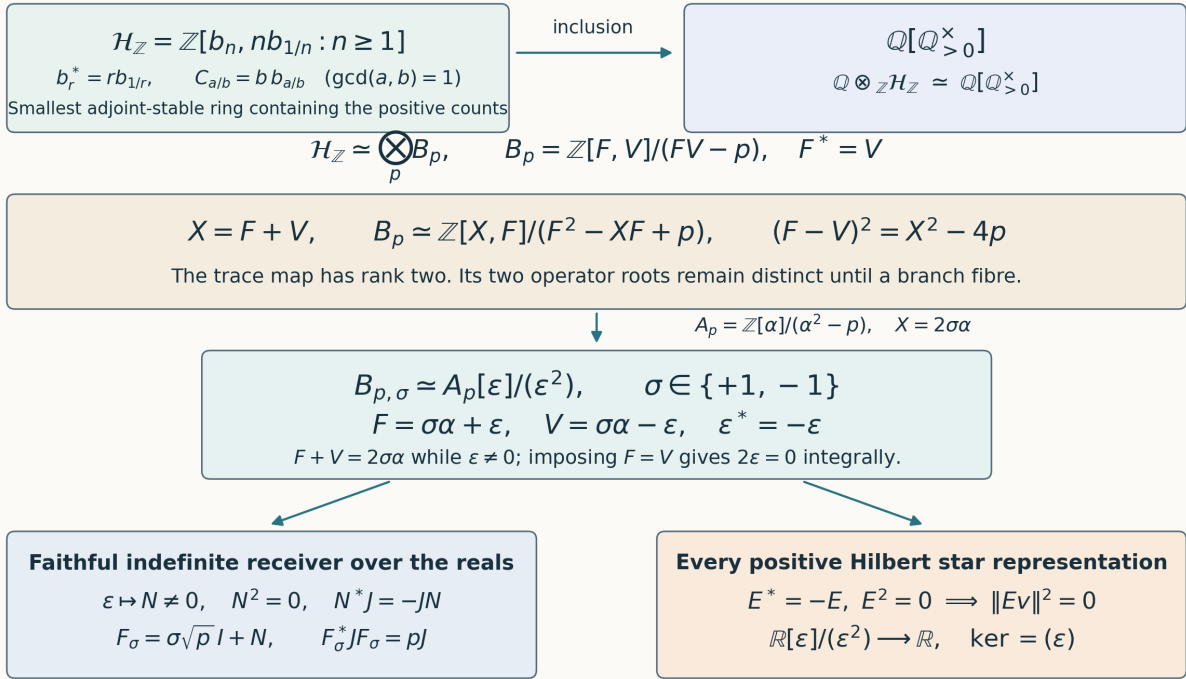
24 September 2026. This reader retains the complete inverse-return proofs and extends them to the integral adjoint algebra, its prime fibres, its exact dual-number trace fibres, and the joint prime-trace observation of the actual original-zeta jets. The original count measure, arithmetic unit atom, support labels, and every prime-power repetition remain explicit. τ is the separately retained parityless supporting datum; addition at τ is not introduced.

The smallest adjoint-stable integral algebra generated by positive counts is proved, including its full basis and the relation $F_p V_p = p$ at every prime. Its trace branch fibres are actual dual-number algebras, with a faithful indefinite representation and a precisely calculated positive quotient. The joint trace calculation retains the full zero jets and states its exact reflection fibres. None of these results establishes positivity of the complete Weil form or RH.

The mathematical reviews are POLARIZED_RETURN_INVERSE_REVIEW.md and ADJOINT_TRACE_AND_DUAL_FIBRES_REVIEW.md. Their corrections have been incorporated. Complete receiving-source proofs are appended as programme dependencies, with their authorship and original notation retained; they are distinct from original-author papers. Source versions and actual reading coverage are in INTEGRAL_ADJOINT_SOURCE_LEDGER.json. The Bhatt-Lurie author-source comparison is confined to the exact spans in INTEGRAL_ADJOINT_APC_SOURCE_RECEIPT.json and does not assert a prismatic identification.

Integral counting, the adjoint, and a trace that retains an infinitesimal fibre

Every prime p is retained. These are operator coordinates; no coordinate or addition is assigned to τ .



IAR1-IAR3; AT2-AT5. Actual zeta operators and every jet are retained in AT5. Positivity of the full Weil form is not inferred.

Figure 5: Exact integral and infinitesimal maps. The two lower receivers use the real base extension alpha to sqrt(p). IAR1-IAR3 and AT2-AT5 prove every displayed arrow; all integral coefficients and both signs are retained.

Global inverse reconstruction of the original zeta return measures

24 September 2026. Independent derivation. Result locators RMR0–RMR10.

This note proves the converse to the programme's timed-prime reconstruction: the original Riemann zeta function determines the complete scalar count measure, its primitive periods, all repeated returns, their coefficients, and the time origin and scale. Every proof applies to the complete measure. No finite range of numerical tests enters an argument.

The supporting datum remains

$$\tau\langle Z_1; \text{no intrinsic } Z_2 \text{ parity} \rangle.$$

The time-zero atom below records the arithmetic unit. It does not identify that atom, the real time coordinate zero, integer zero, or the supporting datum with one another. Addition and convolution below act on measures and counts; no addition on tau is introduced. In particular the retracted equation involving the sum of tau with itself is not used.

RMR0. Sources, order of reconstruction, and the exact input

Read for this calculation:

- work/tau_weight_cohomology_20260924/TIMED_PRIME_PERIOD_DERIVATION.md: TP9–TP14 read in full for the measures and transforms; TP0–TP5 read for reconstruction order. The prior public proof is TP0–TP14.
- SEQUENTIAL_ARGUMENT_RECONSTRUCTION.md, A18–A25: complete-history certification, joint reconstruction of the supporting viewpoint and unit, fixed-point and signed-norm arguments, and the proposed uniqueness-to-purity inference.

The inverse calculation starts with the full original analytic function on its original half-plane

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s}, \quad \Re s > 1. \quad (\text{RMR0.1})$$

It does not obtain the unit by subtracting two previously named prime events. Once a measure has been recovered, its additive support determines its primitive elements intrinsically. The classical labels in the proof verify which recovered objects these are. The construction of the complete source before this analytic comparison is the subject of TP0 and the separate generic-reconstruction proof; this note does not derive that source from a bare supporting point.

No completed zeta function is substituted for (RMR0.1). Gamma factors, powers of pi and endpoint multipliers are absent because none is introduced. Later meromorphic continuation is that of this same original function.

RMR1. The three complete measures and their transforms

On $[0, \infty)$, retain

$$\begin{aligned} \mathcal{D} &= \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \\ \mathcal{R} &= \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \\ \mathcal{W} &= t\mathcal{R} = \sum_p \sum_{k \geq 1} \frac{k \log p}{k} \delta_{k \log p} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p}. \end{aligned} \quad (\text{RMR1.1})$$

The last two measures have mass zero at time zero. On any compact interval $[0, T]$, an atom of $\mathcal{D}$ has $n \leq e^T$; an atom of $\mathcal{R}$ or $\mathcal{W}$ has $p^k \leq e^T$. Consequently all three measures are locally finite.

For $\sigma \geq 1 + \delta > 1$,

$$\sum_p \sum_{k \geq 1} \frac{p^{-k\sigma}}{k} \leq \frac{1}{1 - 2^{-1-\delta}} \sum_{n \geq 2} n^{-1-\delta} < \infty,$$

$$\sum_p \sum_{k \geq 1} (\log p) p^{-k\sigma} \leq \frac{1}{1 - 2^{-1-\delta}} \sum_{n \geq 2} (\log n) n^{-1-\delta} < \infty. \quad (\text{RMR1.2})$$

The second last series converges by the integral test, or by comparing its tail with $n^{-1-\delta/2}$. These majorants give absolute and locally uniform convergence. Define the branch

$$L(s) := \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k}. \quad (\text{RMR1.3})$$

The power-series identity $\exp(\sum_{k \geq 1} z^k/k) = (1 - z)^{-1}$, followed by unique factorization and the convergent Euler product, gives

$$\mathcal{LD}(s) = \zeta(s), \quad \mathcal{LR}(s) = L(s), \quad e^{L(s)} = \zeta(s), \quad \mathcal{LW}(s) = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{RMR1.4})$$

Here $\mathcal{L}\mu(s) = \int e^{-st} d\mu(t)$, and all four identities hold on $\Re s > 1$. Differentiation in the last identity retains the factor $k \log p$ multiplying the coefficient $1/k$, as shown in (RMR1.1). The logarithm is specified by (RMR1.3), equivalently by its real values for real $s > 1$ and its limit zero as $s \rightarrow +\infty$. It is not an unspecified branch.

For completeness, the needed Euler product can be checked globally without a numerical approximation. A product over the primes at most N expands into the Dirichlet terms whose prime factors are all at most N . Unique factorization supplies coefficient one to each such term. Every omitted integer exceeds N , so on $\Re s \geq 1 + \delta$ the absolute remainder is at most $\sum_{n > N} n^{-1-\delta}$, which tends to zero. This proves the equality for the entire infinite product. The bound is a convergence proof, not a finite verification of a global assertion.

RMR2. Injectivity of the complete Laplace observation

Let $\mathcal{M}_1^+$ be the set of positive Borel measures μ on $[0, \infty)$ which are locally finite and satisfy

$$\int_0^\infty e^{-ct} d\mu(t) < \infty \quad \text{for every real } c > 1. \quad (\text{RMR2.1})$$

Theorem. The map $\mu \mapsto \mathcal{L}\mu$ is injective on $\mathcal{M}_1^+$. More strongly, for any one fixed $c > 1$, the values $\mathcal{L}\mu(c + j)$, $j = 0, 1, 2, \dots$, determine μ .

Proof. Define a finite positive measure on $[0, 1]$ by

$$\nu_{\mu,c} := (t \mapsto e^{-t})_*(e^{-ct}\mu), \quad \nu_{\mu,c}(\{0\}) = 0. \quad (\text{RMR2.2})$$

For every nonnegative integer j ,

$$\int_{[0,1]} x^j d\nu_{\mu,c}(x) = \mathcal{L}\mu(c + j). \quad (\text{RMR2.3})$$

Equal Laplace values therefore give equal integrals of every polynomial. These imply equal integrals of every continuous function. Here is the approximation argument explicitly. For $f \in C([0, 1])$, its Bernstein polynomial is

$$B_N f(x) = \sum_{k=0}^N f(k/N) \binom{N}{k} x^k (1-x)^{N-k}.$$

If K has binomial law with parameters N, x , this is $\mathbb{E}f(K/N)$, while $\mathbb{E}(K/N - x)^2 = x(1-x)/N \leq 1/(4N)$. Given $\varepsilon > 0$, uniform continuity provides $\eta > 0$ such that $|f(y) - f(x)| < \varepsilon$ when $|y - x| < \eta$. Splitting the expectation at that event and using the second-moment bound gives uniformly in x

$$|B_N f(x) - f(x)| \leq \varepsilon + \frac{2\|f\|_\infty}{4N\eta^2}.$$

Thus the polynomials converge uniformly. Finite measures with equal polynomial integrals have equal continuous-function integrals. To see the last measure-identification step directly, continuous ramps increasing pointwise to the indicator of any open interval give its measure by monotone convergence; intervals determine the Borel measure. Hence $\nu_{\mu,c} = \nu_{\lambda,c}$.

The transformation (RMR2.2) has the exact inverse

$$\mu(E) = \int_{\{e^{-t}: t \in E\}} x^{-c} d\nu_{\mu,c}(x) \quad (\text{RMR2.4})$$

for every Borel $E \subseteq [0, \infty)$. The integrand is bounded on the image of any bounded E ; the identity for unbounded sets follows by monotone convergence. Equality of the ν 's therefore proves equality of the original measures. $\square$

This also gives an explicit inverse on continuous test functions, without selecting atom locations in advance:

$$\int f d\nu_{\mu,c} = \lim_{N \rightarrow \infty} \sum_{k=0}^N f(k/N) \binom{N}{k} \sum_{j=0}^{N-k} (-1)^j \binom{N-k}{j} \mathcal{L}\mu(c+k+j). \quad (\text{RMR2.5})$$

Indeed, expanding each $(1-x)^{N-k}$ in $B_N f$ gives the finite sum, and uniform convergence gives its limit. Equations (RMR2.4)–(RMR2.5) reconstruct the full measure from the full observation.

RMR3. The converse for the original zeta and its unit

Theorem. Among all measures in $\mathcal{M}_1^+$, precisely one has Laplace transform the original $\zeta(s)$ on $\Re s > 1$, namely

$$\boxed{\mathcal{D} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}.} \quad (\text{RMR3.1})$$

Proof. RMR1 establishes existence and RMR2 establishes uniqueness. In fact equality only at $c, c+1, c+2, \dots$ already suffices. This is an infinite exact observation, not a finite event prefix. $\square$

The unit atom is itself recoverable. For any $\mu \in \mathcal{M}_1^+$, fix $c > 1$. As real $s \rightarrow +\infty$, the functions e^{-st} , for $s \geq c$, are dominated by the integrable e^{-ct} and tend pointwise to $\mathbf{1}_{\{0\}}(t)$. Consequently

$$\mu(\{0\}) = \lim_{s \rightarrow +\infty} \mathcal{L}\mu(s). \quad (\text{RMR3.2})$$

For the original zeta that limit is exactly one: its terms with $n \geq 2$ tend to zero and are dominated by the convergent series at any fixed real $c > 1$. Thus the complete analytic observation fixes both the unit's time and its multiplicity. This is a statement about δ_0 ; tau remains the separately retained supporting datum.

The conclusion is a reconstruction of the actual infinite measure. It does not certify that measure from finitely many observations. For example, for any $T > 0$ and any $u > T$, the positive measure $\mathcal{D} + \delta_u$ has exactly the same restriction to $[0, T]$ and has all the required exponential moments, but its transform is $\zeta(s) + e^{-su}$. This is the exact map that retains the user's later-event possibility: a finite time restriction has a nontrivial fibre, whereas the complete Laplace observation does not.

RMR4. Recovering repeated returns by a convolution logarithm

Convolution is taken under addition of nonnegative real times. Thus

$$\delta_a * \delta_b = \delta_{a+b}, \quad \delta_0 * \mu = \mu. \quad (\text{RMR4.1})$$

For $\mathcal{M} := \mathcal{D} - \delta_0$, the support lies in $[h, \infty)$ with $h = \log 2 > 0$. Define

$$\log_* \mathcal{D} = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} \mathcal{M}^{*r}. \quad (\text{RMR4.2})$$

This signed sum is locally finite: $\mathcal{M}^{*r}$ is supported in $[rh, \infty)$, so only finitely many r contribute on any fixed compact interval. Each of these convolution measures has finitely many atoms there. No convergence assertion or sign inference is hidden in the definition.

Theorem. The logarithm (RMR4.2) is positive and is exactly $\mathcal{R}$ in (RMR1.1). Its inverse map is

$$\exp_* \mathcal{R} = \delta_0 + \sum_{r \geq 1} \frac{\mathcal{R}^{*r}}{r!} = \mathcal{D}. \quad (\text{RMR4.3})$$

Proof. On $[0, T]$, discard contributions supported strictly after T . This is the quotient of the convolution algebra by the ideal of such contributions. Its positive-time part generated by measures with support at least h is nilpotent: a product of more than T/h factors is zero there. Therefore exponential and logarithm are finite polynomial operations in this quotient. The formal identities $\log(\exp X) = X$, $\exp(\log(1 + X)) = 1 + X$, and $\exp(X + Y) = \exp X \exp Y$ hold in it. They follow by differentiation of the formal series, their constant terms, and the product rule; all coefficients have rational denominators and commute.

For a fixed prime the identity

$$\exp_* \left(\sum_{k \geq 1} \frac{1}{k} \delta_{k \log p} \right) = \sum_{j \geq 0} \delta_{j \log p} \quad (\text{RMR4.4})$$

is obtained from $\exp(\sum_{k \geq 1} X^k/k) = (1 - X)^{-1}$ by substituting $X^k \mapsto \delta_{k \log p}$. Products over prime factors give exactly one atom at each $\log n$, including the exponent-zero tuple giving δ_0 . On $[0, T]$, only primes at most e^T enter. This is a finite product in the quotient just described. It proves (RMR4.3) on every compact interval, hence globally. Applying its polynomial inverse gives (RMR4.2) equal to $\mathcal{R}$. Its coefficients $1/k$ are positive, proving positivity rather than assuming it for a convolution logarithm. $\square$

In particular, the inverse formula yields the exact coefficient identity, for every integer $n \geq 2$,

$$\sum_{r \geq 1} \frac{(-1)^{r+1}}{r} \# \{ (d_1, \dots, d_r) : d_i \geq 2, d_1 \cdots d_r = n \} = \begin{cases} 1/k, & n = p^k, \\ 0, & n \text{ has at least two distinct prime factors.} \end{cases} \quad (\text{RMR4.5})$$

The sum is finite because $2^r \leq n$. The proof above verifies this identity for every n ; no range of values is substituted for it.

RMR5. Primitive periods, repetitions and arithmetic are recovered intrinsically

Let $M = \text{supp } \mathcal{D}$, with its addition inherited from time. This is the additive monoid $\{\log n : n \geq 1\}$, with neutral element zero. Define its positive irreducibles without using prime labels:

$$P(M) = \{t \in M \setminus \{0\} : t \neq a + b \text{ for all } a, b \in M \setminus \{0\}\}. \quad (\text{RMR5.1})$$

Theorem. The recovered set $P(M)$ is precisely $\{\log p : p \text{ prime}\}$. Every element of M has a unique expression as a finite sum of these primitive elements with nonnegative integral multiplicities.

Proof. If $n = ab$ with $a, b > 1$, then $\log n = \log a + \log b$, so $\log n$ is reducible. Conversely an equality $\log n = \log a + \log b$ exponentiates to $n = ab$; if n is prime, one factor must be one and the corresponding summand zero.

Here is the arithmetic existence and uniqueness argument used throughout this note. Every integer greater than one has a least divisor greater than one, and that divisor is prime: a proper factor of the divisor would contradict its minimality. Induction on the integer gives a finite prime factorization by factoring a composite into two smaller positive integers. To prove uniqueness, Euclidean division gives the Euclidean algorithm; the successive nonzero remainders strictly decrease, and tracing its equations backwards expresses their last nonzero member, the greatest common divisor, as an integral linear combination of the two inputs. If a prime p does not divide a , its greatest common divisor with a is one, so $up + va = 1$ for some integers u, v . Multiplying this equation by b proves that $p \mid ab$ implies $p \mid b$. Applying this lemma repeatedly to a product shows that a prime in either factorization must equal a prime in the other; cancelling that equal factor and inducting proves uniqueness of every multiplicity. Taking real logarithms supplies exactly the asserted unique finite sum. This proves the intrinsic classification and its arithmetic comparison. $\square$

An entirely equivalent recovery uses $\mathcal{R}$: its primitive positive atoms are the atoms not equal to an integer multiple $k \geq 2$ of another positive atom. Indeed, $k \log p$ with $k \geq 2$ is visibly a repetition, while $\log p = j\ell \log q$ would give $p = q^{j\ell}$, impossible for $j\ell \geq 2$. Thus the time and repetition number are reconstructed before their conventional prime-power names are attached.

The exact monoid isomorphism is

$$\bigoplus_{a \in P(M)} \mathbb{N}_0 \longrightarrow M, \quad (k_a)_a \mapsto \sum_a k_a a. \quad (\text{RMR5.2})$$

Its inverse records the unique multiplicities. The empty tuple maps to time zero, retaining the unit. The exponential map $t \mapsto e^t$ then identifies this with the positive-integer multiplicative monoid, including integer one. The order inherited from time orders its countable elements and identifies the complete sequence of counts. This comparison uses all of M ; it does not claim that a finite prefix certifies the whole source.

The measure $\mathcal{W}$ has two exact reconstructions:

$$\mathcal{W} = t \log_* \mathcal{D}, \quad \mathcal{W} = (t\mathcal{D}) * \mathcal{D}^{-1}, \quad (\text{RMR5.3})$$

where

$$\mathcal{D}^{-1} = \sum_{r \geq 0} (-1)^r \mathcal{M}^{*r} = \sum_{n \geq 1} \mu_{\text{ar}}(n) \delta_{\log n}. \quad (\text{RMR5.4})$$

Here μ_{ar} is the integer Möbius function: coefficient zero when a prime is repeated and coefficient $(-1)^r$ for a product of r distinct primes; its value at one is one. To prove (RMR5.4), invert each prime factor $\sum_{j \geq 0} \delta_{j \log p}$ by $\delta_0 - \delta_{\log p}$; multiply, and use unique factorization on each compact interval. The alternating geometric sum gives the same inverse.

Finally, multiplication by time is a derivation of convolution:

$$t(\mu * \nu) = (t\mu) * \nu + \mu * (t\nu),$$

because the product time is the sum of its two input times. Apply this to the finite local exponential expansion to obtain $t \exp_* \mathcal{R} = (t\mathcal{R}) * \exp_* \mathcal{R}$. This proves the second formula in (RMR5.3). In coefficients it retains

$$\sum_{d|n} (\log d) \mu_{\text{ar}}(n/d) = \Lambda(n), \quad \Lambda(p^k) = \log p, \quad \Lambda(1) = 0. \quad (\text{RMR5.5})$$

RMR6. Converse uniqueness for both repeated-return transforms

Theorem. The following three sets each contain exactly the displayed measure:

$$\begin{aligned} \{\nu \in \mathcal{M}_1^+ : \mathcal{L}\nu = L\} &= \{\mathcal{R}\}, \\ \{\nu \in \mathcal{M}_1^+ : \exp(\mathcal{L}\nu) = \zeta\} &= \{\mathcal{R}\}, \\ \{\nu \in \mathcal{M}_1^+ : \mathcal{L}\nu = -\zeta'/\zeta\} &= \{\mathcal{W}\}. \end{aligned} \quad (\text{RMR6.1})$$

The transform identities in each set are on $\Re s > 1$.

Proof. Existence was proved in RMR1; the first and third uniqueness statements are RMR2. In the second statement, for every real $s > 1$, $\mathcal{L}\nu(s)$ is a finite nonnegative real number. The real exponential is injective and $L(s)$ is the real logarithm of $\zeta(s)$. Therefore $\mathcal{L}\nu(s) = L(s)$ on that real half-line. RMR2 applies again. In particular no atom at time zero can be added: it would multiply the real exponential by a positive constant different from one. $\square$

The precise inverse between the last two measures is multiplication by t and multiplication by $1/t$ on $(0, \infty)$, together with their established zero masses at zero. These are genuine mutually inverse maps on the measures in question. They do not add a time-zero contribution to a logarithmic derivative whose coefficient there is zero. The arithmetic unit is recovered in passing back to $\exp_* \mathcal{R}$.

RMR7. Global rigidity of time origin and scale

For $a > 0$ and $b \in \mathbb{R}$, push $\mathcal{D}$ along the affine clock change $A_{a,b}(t) = at + b$. Allow a target real half-line bounded below, so the proof does not silently exclude shifts of either sign. Its transform, on the common half-plane of convergence, is exactly

$$\mathcal{L}((A_{a,b})_* \mathcal{D})(s) = e^{-bs} \zeta(as). \quad (\text{RMR7.1})$$

Theorem. Equality of (RMR7.1) with the original $\zeta(s)$ for all sufficiently large real s forces $b = 0$ and $a = 1$.

Proof. Both $\zeta(as)$ and $\zeta(s)$ tend to one as real $s \rightarrow +\infty$. Therefore $e^{-bs} = \zeta(s)/\zeta(as)$ tends to one, forcing $b = 0$. For real $x > 1$, the original Dirichlet series is strictly decreasing: if $y > x$, every term n^{-y} is at most n^{-x} , and the inequality is strict for every $n \geq 2$; the absolutely convergent positive sums preserve strictness. Hence $\zeta(as) = \zeta(s)$, at any real s with $s > 1$ and $as > 1$, implies $as = s$, so $a = 1$. $\square$

The same conclusion holds for every order-preserving additive bijection of real nonnegative time. Such a map h satisfies $h(t) = ct$, $c > 0$. To prove this, additivity gives the formula on nonnegative rational arguments from $h(1) = c$; rational sequences increasing and decreasing to any real t , together with monotonicity, give the formula there. RMR7 then fixes $c = 1$.

For the intrinsic return-time monoid M , any order-preserving monoid automorphism is already the identity: it must fix the least element, the next element, and, inductively, every element in the increasing sequence, since the increasing bijection of this sequence cannot skip a position. This induction proves a statement about the whole automorphism, rather than certifying an unknown whole history from a finite record.

RMR8. What a scalar return observation retains about time between events

The preceding rigidity concerns the actual additive clock and every event time. To specify its precise strength, consider only an increasing homeomorphism of ambient time, without assuming it respects addition. There exist such maps preserving the entire count measure but changing the intervals between its atoms.

Here is one complete construction. For every $n \geq 1$, put $a_n = \log n$, $d_n = \log(n+1) - \log n$, and define on $[a_n, a_{n+1}]$

$$h(t) = a_n + d_n \psi\left(\frac{t - a_n}{d_n}\right), \quad \psi(x) = x + \frac{1}{4\pi} \sin(2\pi x). \quad (\text{RMR8.1})$$

The endpoints agree between adjacent intervals, $\psi(0) = 0$, $\psi(1) = 1$, and $\psi'(x) = 1 + \frac{1}{2} \cos(2\pi x) \geq \frac{1}{2}$. Hence h is an increasing homeomorphism of $[0, \infty)$, fixes every $\log n$, and satisfies $h_* \mathcal{D} = \mathcal{D}$. It is not the identity inside these intervals and cannot be additive by RMR7's classification. This preserves rather than changes every event time and coefficient.

Thus the scalar original zeta recovers the event measure exactly, and the additive structure fixes its clock scale. The ambient interpolation of time is additional structure. Formula (RMR8.1) identifies exactly the observation fibre exposed by omitting that structure; it says nothing against the source's already supplied additive flow.

RMR9. The exact map that forgets phase and support labels

The scalar measure reconstructs scalar arithmetic. Here is the explicit comparison with retained labels.

Let S be a finite label set, and let

$$\pi : [0, \infty) \times S \longrightarrow [0, \infty), \quad \pi(t, \ell) = t. \quad (\text{RMR9.1})$$

For a positive labelled measure $\widetilde{\mathcal{D}}$, the scalar observation is $\pi_* \widetilde{\mathcal{D}}$. The positive fibre over $\mathcal{D}$ consists exactly of

$$\widetilde{\mathcal{D}} = \sum_{n \geq 1} \delta_{\log n} \otimes \nu_n, \quad \nu_n \text{ a probability measure on } S. \quad (\text{RMR9.2})$$

Proof. Positivity and the zero scalar mass outside the count times imply zero labelled mass there. Above each count time, the total labelled mass must be the scalar coefficient one. That is exactly a probability measure on the finite fibre. Conversely every such family pushes to $\mathcal{D}$, and local finiteness follows from that of $\mathcal{D}$. $\square$

For the retained quarter-phase group

$$H = \{1, J, \epsilon, \epsilon J\}, \quad J^2 = \epsilon, \quad \epsilon^2 = 1,$$

there are labelled lifts which retain multiplication and the complete convolution exponential as well. Choose any monoid homomorphism $\chi : (\mathbb{N}_{>0}, \cdot) \rightarrow H$. Its value at one is the identity. Define

$$\begin{aligned} \mathcal{D}_\chi &= \sum_{n \geq 1} \delta_{(\log n, \chi(n))}, \\ \mathcal{R}_\chi &= \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{(k \log p, \chi(p)^k)}. \end{aligned} \quad (\text{RMR9.3})$$

Convolution on this labelled space uses $(t, h)(u, j) = (t + u, hj)$. Unique factorization and the proof of RMR4, with this multiplication retained, give

$$\mathcal{D}_\chi = \exp_* \mathcal{R}_\chi, \quad \pi_* \mathcal{D}_\chi = \mathcal{D}, \quad \pi_* \mathcal{R}_\chi = \mathcal{R}. \quad (\text{RMR9.4})$$

For every primitive prime period, its assigned phase can be specified in H and then extended uniquely by multiplication. Assigning a nonidentity phase to one primitive period produces a measure different from the all-identity assignment, with the same scalar transform and the same unit atom. This establishes the precise information lost by scalarization. It does not assert that every such assignment meets every additional condition of the Connes–Consani geometry.

In coefficient notation, the projection is the augmentation

$$\begin{aligned} \text{aug} : \mathbb{Z}[H] &\longrightarrow \mathbb{Z}, \quad \sum_{h \in H} a_h [h] \longmapsto \sum_{h \in H} a_h, \\ \ker(\text{aug}) &= \bigoplus_{h \in H \setminus \{1\}} \mathbb{Z}([h] - [1]). \end{aligned} \quad (\text{RMR9.5})$$

The kernel formula follows by rewriting a sum with total coefficient zero as $\sum_{h \neq 1} a_h ([h] - [1])$; the three displayed generators are linearly independent by inspecting their nonidentity coefficients. The map respects multiplication because each basis group element maps to one. The same formula over rational coefficients accommodates the retained return coefficient $1/k$.

If a support label is retained alongside the phase, use $[0, \infty) \times S \times H$ and the same projection to time. The resulting scalar formula does not identify distinct support labels with one another; it observes them through a map having the fibre (RMR9.2). A prescribed supporting point can be held fixed in every term, so the scalar reconstruction is fully compatible with retaining that point. Recovering the whole support-labelled programme requires its additional labelled observations or the source’s proven reconstruction maps, not a silent replacement of those objects by their scalar image.

RMR10. The proved consequence for the complete-history argument

The following is now an exact chain of mutually recovering arithmetic data:

$$\begin{array}{ccccc} \mathcal{D} & \xrightarrow{\log_*} & \mathcal{R} & \xrightarrow{t \cdot} & \mathcal{W} \\ \mathcal{L} \downarrow & & \mathcal{L} \downarrow & & \mathcal{L} \downarrow \\ \zeta & \xrightarrow{\text{the branch } L(+\infty)=0} & L & \xrightarrow{-\partial_s} & -\zeta'/\zeta. \end{array} \quad (\text{RMR10.1})$$

Inverses are respectively convolution exponential, division by positive time with the zero atom prescribed as above, and the injective Laplace reconstruction RMR2. The left-hand unit atom has coefficient one. Additive irreducibles of its complete support recover every primitive period; RMR4 recovers every repetition coefficient. The original zeta also fixes every affine origin and scale by RMR7.

This proves a global uniqueness statement stronger than uniqueness of an identity inside a previously chosen algebra: no different positive scalar measure with the stated exponential moments has the same complete original-zeta observation. In particular a proposed later event outside that measure changes the original function, by RMR2, even when it leaves every earlier time restriction unchanged.

It also states the remaining comparison exactly. The scalar measure is an image of the full support- and phase-labelled history. Its inverse is unique in the scalar measure category; (RMR9.2)–(RMR9.5) describe the fibre of the labelled-to-scalar map. These formulas retain the supporting datum, the quarter phases, and their relationship to the unique scalar arithmetic observation, without asserting that equality of one scalar transform identifies arbitrary geometries.

Once the original zeta is recovered on $\Re s > 1$, its meromorphic continuation, wherever supplied, is unique: two meromorphic continuations on a connected common domain that agree on this open half-plane agree

everywhere there by the identity theorem. Thus the continuation has the same original zeros, poles and multiplicities. No completion multiplier is introduced, and none of its exceptional loci is cancelled by this argument. The inverse theorem does not place those zeros on a line; its proved contribution is the complete global rigidity of the arithmetic return data that any subsequent purity argument must use.

Recovering the original arithmetic history from the polarized cohomology

24 September 2026. Complete global derivation, PRI0–PRI8. This continues the user’s complete-history and retrospective-uniqueness argument through the actual cohomology constructed in the linked task.

The supporting datum remains $\tau\langle Z_1; \text{no intrinsic } Z_2 \rangle$. Every addition below is addition of measures, cohomology classes, or arithmetic counts. No addition, numerical distance, or parity is assigned to tau. The arithmetic unit atom is retained separately.

PRI0. Exact inputs and provenance

The source complex curves are those of Alain Connes and Caterina Consani, On the Absolute Geometry of Spec Z, arXiv:2606.06604v1, author `FF.tex`, Proposition `prop:archimedean_points_C` and Theorem `thm:tate_decomposition`, lines 1465–1499 and 1548–1635. The author’s original periods are retained.

The programme calculations used here are the linked task’s POLARIZED_DETERMINANT_AND_ORIGINAL_ZETA.md, PD1–PD8, read in full; CANONICAL_GENERIC_RECONSTRUCTION.md, CG0–CG8, read in full; and the published complete timed-prime proof TP0–TP14. The exact inverse theorem is RETURN_MEASURE_RIGIDITY.md, RMR0–RMR10. The retained original-zeta Weil receiver is PRIME_MONODROMY_STACKED_HISTORY.md, M7, and its contour comparison C10. This note proves the measure comparison and its inverse; it does not attribute this constructed product isogeny to the source as its Frobenius.

PRI1. The polarized operator and all its return traces

For every prime p , including two, retain

$$A = \log p, \quad B = 2\pi, \quad E_A = \mathbb{C}/(A\mathbb{Z} + iB\mathbb{Z}), \quad E_{pA} = \mathbb{C}/(pA\mathbb{Z} + iB\mathbb{Z}). \quad (\text{PRI1.1})$$

The exponential map identifies these with $\mathbb{C}^\times/p^\mathbb{Z}$ and $\mathbb{C}^\times/(p^p)^\mathbb{Z}$, respectively. The maps $u : E_{pA} \rightarrow E_A, [w] \mapsto [w]$ and $v : E_A \rightarrow E_{pA}, [w] \mapsto [pw]$ each have degree p ; their respective kernels are $[jA]$ and $[ijB/p]$, $0 \leq j < p$. Both composites are multiplication by p .

On $X_p = E_A \times E_{pA}$, the map $\Phi_p(x, y) = (u(y), v(x))$ is therefore an actual self-isogeny. In the invariant degree-one bases $(dx_1, d\theta_1, dx_2, d\theta_2)$, its pullback and the full Hermitian matrix are

$$F_p = \begin{pmatrix} 0 & pI_2 \\ I_2 & 0 \end{pmatrix}, \quad H_p = \begin{pmatrix} ABI_2 & 0 \\ 0 & pABI_2 \end{pmatrix}. \quad (\text{PRI1.2})$$

The pairing is obtained by integrating $\alpha \wedge \bar{*}\beta$ on the two curves: their areas are exactly AB and pAB . Direct multiplication gives

$$F_p^2 = pI_4, \quad F_p^* H_p F_p = pH_p. \quad (\text{PRI1.3})$$

The vectors $(\sqrt{p}b, b)$ and $(-\sqrt{p}b, b)$, $b \in \mathbb{C}^2$, are the complete eigenspaces. Each has dimension two. In particular this is a proved positive weight-one calculation, retaining both signs. For every positive integer r , it gives

$$\text{Tr}(F_p^r) = \begin{cases} 0, & r \text{ odd}, \\ 4p^k, & r = 2k. \end{cases} \quad (\text{PRI1.4})$$

The odd return trace vanishes because its two sign contributions cancel, not because either eigenspace is absent.

The other degrees are also retained. On $H^j(X_p) = \bigwedge^j H^1(X_p)$ the spectra, with multiplicities, are

j	eigenvalues	
0	1 (1)	
1	$\sqrt{p}$ (2), $-\sqrt{p}$ (2)	
2	p (2), $-p$ (4)	
3	$p^{3/2}$ (2), $-p^{3/2}$ (2)	
4	p^2 (1).	(PRI1.5)

Indeed wedge products multiply the selected eigenvalues, with multiplicities $\binom{2}{a}\binom{2}{j-a}$. The complete local alternating determinant is consequently

$$Z_p(T) = \frac{(1 - \sqrt{p}T)^2(1 + \sqrt{p}T)^2(1 - p^{3/2}T)^2(1 + p^{3/2}T)^2}{(1 - T)(1 - pT)^2(1 + pT)^4(1 - p^2T)}. \quad (\text{PRI1.6})$$

Subsequent use of the degree-one trace refers to that named component of this retained object.

PRI2. An invertible map of the complete measure algebras

Let $\mathcal{M}$ be the real signed or complex Borel measures on $[0, \infty)$ whose total variation is finite on compact sets. Convolution uses addition of times. It is well defined locally: the preimage of $[0, T]$ under addition lies in $[0, T]^2$, where the product of the two total variations is finite. Put $d_2(t) = 2t$. Define

$$\mathcal{P}\mu = (d_2)_*(e^t\mu), \quad \mathcal{P}^{-1}\nu = e^{-t}(d_{1/2})_*\nu = (d_{1/2})_*(e^{-u/2}\nu). \quad (\text{PRI2.1})$$

The letters t and u denote source and target time, respectively. The exact test-function formulas are

$$\int f(u) d(\mathcal{P}\mu)(u) = \int e^t f(2t) d\mu(t), \quad \int g(t) d(\mathcal{P}^{-1}\nu)(t) = \int e^{-u/2} g(u/2) d\nu(u). \quad (\text{PRI2.2})$$

For compactly supported tests all integrals are finite. Substituting one formula into the other proves both inverse identities. Each map preserves and reflects positivity, since its weight is strictly positive and its time map is a homeomorphism.

Moreover $\mathcal{P}$ is a unital convolution-algebra isomorphism. For two input times, $e^{t+v} = e^t e^v$ and $2(t+v) = 2t + 2v$. Applying Fubini on the bounded preimage of a compact test's support therefore gives

$$\mathcal{P}(\mu * \lambda) = (\mathcal{P}\mu) * (\mathcal{P}\lambda), \quad \mathcal{P}\delta_0 = \delta_0. \quad (\text{PRI2.3})$$

For a measure μ supported in $[a, \infty)$, $a > 0$, with no atom at zero, the exponential $\exp_* \mu = \delta_0 + \sum_{r \geq 1} \mu^{*r}/r!$ has only finitely many contributing terms on each compact interval. For $\delta_0 + \mu$, whose atom at zero is exactly one, the logarithm $\sum_{r \geq 1} (-1)^{r+1} \mu^{*r}/r$ has the same property. Thus PRI2.3 also proves that $\mathcal{P}$ commutes with these operations on their stated inputs. Every use below satisfies these zero-atom conditions. No infinite interchange without a local bound is needed.

PRI3. Exact forward and inverse maps for the cohomological return signal

Retain the original arithmetic measures, including their unit,

$$\mathcal{D} = \sum_{n \geq 1} \delta_{\log n}, \quad \mathcal{R} = \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{W} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p} = t\mathcal{R}. \quad (\text{PRI3.1})$$

Their complete original comparison is $\mathcal{D} = \exp_* \mathcal{R}$, $\mathcal{L}\mathcal{D} = \zeta$, $\mathcal{L}\mathcal{W} = -\zeta'/\zeta$ on $\Re s > 1$. TP9–TP13 and RMR1–RMR6 prove these statements and the inverse maps.

The degree-one operator defines actual return measures

$$\begin{aligned} \mathcal{R}_1 &= \sum_p \sum_{r \geq 1} \frac{\text{Tr}(F_p^r)}{r} \delta_{r \log p} = \sum_p \sum_{k \geq 1} \frac{2p^k}{k} \delta_{2k \log p}, \\ \mathcal{W}_1 &= u\mathcal{R}_1 = \sum_p \sum_{k \geq 1} 4p^k \log p \delta_{2k \log p}. \end{aligned} \quad (\text{PRI3.2})$$

PRI1.4 proves both equalities, including the vanished odd traces and the factor four. These are positive locally finite measures; on a bounded target interval only $p^k \leq e^{u/2}$ occur. Termwise comparison gives the global identities

$$\boxed{\mathcal{R}_1 = 2\mathcal{P}\mathcal{R}, \quad \mathcal{W}_1 = 4\mathcal{P}\mathcal{W}.} \quad (\text{PRI3.3})$$

The differing factors are forced by multiplication by time: $u\mathcal{P}\mu = 2\mathcal{P}(t\mu)$. In particular the original arithmetic is recovered by the explicit positive inverses

$$\boxed{\mathcal{R} = \tfrac{1}{2}\mathcal{P}^{-1}\mathcal{R}_1, \quad \mathcal{W} = \tfrac{1}{4}\mathcal{P}^{-1}\mathcal{W}_1, \quad \mathcal{D} = \exp_*(\tfrac{1}{2}\mathcal{P}^{-1}\mathcal{R}_1).} \quad (\text{PRI3.4})$$

At an arbitrary original return $t = k \log p$, the inverse to the second formula replaces time $2t$ by t and its mass $4p^k \log p$ by $\frac{1}{4}e^{-t}4p^k \log p = \log p$. This is a proof for every prime and every repetition. The exponential restores precisely δ_0 with coefficient one.

Consequently the absence of an odd-iterate trace has not made this named scalar arithmetic receiver noninjective: every original primitive and repeated return is still present at its doubled time, with a known nonzero coefficient. The two signed eigenspaces remain in PRI1.2–PRI1.6. Scalar trace alone does not reconstruct arbitrary labelled cohomology; RMR9 gives the exact scalar-forgetting fibres.

PRI4. The counting measure and the analytic inverse, with multiplicities

Let $\mathcal{D}_1 = \exp_* \mathcal{R}_1$. PRI2.3 and PRI3.3 imply

$$\mathcal{D}_1 = \mathcal{P}(\mathcal{D} * \mathcal{D}) = \sum_{n \geq 1} n d(n) \delta_{2 \log n}, \quad (\text{PRI4.1})$$

where $d(n) = \#\{(a, b) \in \mathbb{N}_{>0}^2 : ab = n\}$. The last equality follows by multiplying the two counting measures: their coefficient at $\log n$ is the number of ordered factor pairs. In particular the atom at zero still has coefficient one. Conversely,

$$\mathcal{D} = \exp_*(\tfrac{1}{2}\mathcal{P}^{-1} \log_* \mathcal{D}_1). \quad (\text{PRI4.2})$$

All logarithms here have unit atom one and a positive support gap, so RMR4's local polynomial proof applies. The convolution square root in PRI4.2 is therefore fixed by its unit coefficient and logarithm; it is not an arbitrary sign choice.

For $\Re s > 1$, absolute convergence of $\sum n^{-(2\Re s-1)}$ and its square yields

$$L_1(s) := \mathcal{L}\mathcal{D}_1(s) = \prod_p \frac{1}{(1 - \sqrt{p}p^{-s})^2(1 + \sqrt{p}p^{-s})^2} = \zeta(2s-1)^2. \quad (\text{PRI4.3})$$

The two sign factors remain displayed. The complete original-zeta inverse is, on $\Re z > 1$,

$$\boxed{\zeta(z) = \exp\left(\frac{1}{2} \int_{(0,\infty)} e^{-(z+1)u/2} d\mathcal{R}_1(u)\right) = \exp\left(\frac{1}{2} \int_{(0,\infty)} \frac{e^{-(z+1)u/2}}{u} d\mathcal{W}_1(u)\right).} \quad (\text{PRI4.4})$$

Substitution of PRI3.2 gives exactly $\sum_{p,k} p^{-kz}/k$ in the exponent. The integrals converge absolutely there by this series. They tend to zero on the positive real axis at infinity, so the resulting zeta tends to one. This fixes the original unit and the logarithm branch. Equivalently its square is $L_1((z+1)/2)$, with the branch selected by PRI4.4. Original meromorphic continuation then retains its original divisor.

More explicitly, at every finite s_0 ,

$$\text{ord}_{s_0} L_1 = 2 \text{ord}_{2s_0-1} \zeta. \quad (\text{PRI4.5})$$

Indeed the coordinate $z = 2s-1$ has nonzero derivative two, and squaring doubles the leading Laurent order. Thus a zero ρ of multiplicity m goes to $(\rho+1)/2$ with multiplicity $2m$; the pole at one becomes a double pole at one; each trivial zero $-2j$ goes to $(1-2j)/2$ with twice its order. These contributions are recorded, not cancelled away.

PRI5. The induced reflection and its full original-zeta factor

Under $z = 2s - 1$, the original reflection $z \mapsto 1 - \bar{z}$ becomes

$$s \mapsto \frac{3}{2} - \bar{s}. \quad (\text{PRI5.1})$$

This follows by solving $2s' - 1 = 1 - \overline{2s - 1}$. Its fixed line is $\Re s = 3/4$. It is the image of the original critical line under this receiver's proved affine map.

For completeness retain the full original functional-equation multiplier

$$C(z) = 2^z \pi^{z-1} \sin(\pi z/2) \Gamma(1-z), \quad \zeta(z) = C(z) \zeta(1-z) \quad (\text{PRI5.2})$$

as an identity of meromorphic functions, with products at exceptional points interpreted by their Laurent expansions. Substitution into PRI4.3 gives

$$L_1(s) = \underbrace{2^{4s-2} \pi^{4s-4} \sin^2(\pi(2s-1)/2) \Gamma(2-2s)^2}_{C(2s-1)^2} L_1\left(\frac{3}{2} - s\right). \quad (\text{PRI5.3})$$

No Gamma, pi, sine or power-of-two factor has been suppressed. Local cancellations follow the signed divisor identity associated to this full product. The global reflection centre therefore follows from the original zeta equation and the time-and-amplitude map. The local eigenvalue modulus $\sqrt{p}$ and this global divisor reflection are both retained; one is not substituted for the other.

The other cohomological degrees give, on $\Re s > 3$, the uncanceled complete Euler determinant

$$\prod_p Z_p(p^{-s}) = \frac{1}{\zeta(2s-1)^2} \frac{1}{\zeta(2s-3)^2} \zeta(s) \zeta(s-1)^2 \left(\frac{\zeta(2s-2)}{\zeta(s-1)} \right)^4 \zeta(s-2). \quad (\text{PRI5.4})$$

The factors are respectively from degrees one, three, zero, positive degree two, negative degree two, and four. For the negative degree-two part, $(1+x)^{-1} = (1-x)/(1-x^2)$, with $x = p^{1-s}$, gives the displayed numerator and denominator. All products converge absolutely on this half-plane; in particular $\sum_p p^{2-\Re s} < \infty$. This proves the comparison from PRI1.6 without discarding another degree.

PRI6. Transport into the full original Weil formula

Let $h \in C_c^\infty(\mathbb{R})$, with

$$H_h(z) = \int_{\mathbb{R}} h(v) e^{-(z-1/2)v} dv, \quad \hat{h}(y) = \int_{\mathbb{R}} h(v) e^{-iyv} dv. \quad (\text{PRI6.1})$$

The original prime-power term is exactly

$$P(h) = \int_{(0,\infty)} e^{-t/2} (h(t) + h(-t)) d\mathcal{W}(t).$$

Apply the inverse in PRI3.4. Its exponential contributes $e^{-u/2}$; the original square-root prime weight contributes $e^{-u/4}$. Thus

$$\boxed{P(h) = \frac{1}{4} \int_{(0,\infty)} e^{-3u/4} (h(u/2) + h(-u/2)) d\mathcal{W}_1(u).} \quad (\text{PRI6.2})$$

At $u = 2k \log p$, the complete coefficient is $\frac{1}{4} p^{-3k/2} 4p^k \log p = p^{-k/2} \log p$, exactly the original coefficient. Compact support makes each test's prime sum finite, but the identity holds for the entire test space and all return times. It is not a bounded calculation of zeros.

Retain the archimedean term

$$A_\infty(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(y) \left(\Re \frac{\Gamma'(1/4 + iy/2)}{\Gamma(1/4 + iy/2)} - \log \pi \right) dy. \quad (\text{PRI6.3})$$

Substitution in the existing original-zeta contour identity gives the exact global receiver

$$W_\zeta(h) = H_h(0) + H_h(1) + A_\infty(h) - \frac{1}{4} \int e^{-3u/4} (h(u/2) + h(-u/2)) d\mathcal{W}_1(u) = \sum_\rho m_\rho H_h(\rho). \quad (\text{PRI6.4})$$

Here ρ ranges over distinct nontrivial zeros of the original zeta and m_ρ is their multiplicity. The integral excludes zero, as in PRI6.2. The zero sum is absolutely convergent for this test space by the contour formula and the rapid vertical decay of H_h . Endpoints and every archimedean factor are unchanged because PRI6.2 changes only the prime term by an equality.

The original trivial-zero bookkeeping is retained as well: for every finite positive integer B ,

$$\begin{aligned} V_{\zeta,B}(h) &= \sum_\rho m_\rho H_h(\rho) + \sum_{j=1}^B H_h(-2j) - H_h(1) = G_B(h) - P(h), \\ G_B(h) &= A_\infty(h) + H_h(0) + \sum_{j=1}^B H_h(-2j). \end{aligned} \quad (\text{PRI6.5})$$

The finite cutoff triple is not replaced by a possibly divergent infinite sum.

PRI7. Support and positivity data remain explicit

For the programme's support lattice $\mathcal{L}$, use the existing retained basis $\mathbf{e}_\ell$. Define $P_1(h)$ to be the entire integral in PRI6.2, including its factor $1/4$. The supported formula becomes

$$\begin{aligned} B(h) &= \mathbf{e}_{1_\mathcal{L}}(H_h(0) + H_h(1)) + \sum_{\ell \neq 1_\mathcal{L}} \mathbf{e}_\ell H_h(0), \\ Z(h) &= \mathbf{e}_{1_\mathcal{L}} \sum_\rho m_\rho H_h(\rho), \\ D_1(h) &= \mathbf{e}_{1_\mathcal{L}}(P_1(h) - A_\infty(h)) + \sum_{\ell \neq 1_\mathcal{L}} \mathbf{e}_\ell H_h(0), \\ B(h) - Z(h) &= D_1(h). \end{aligned} \quad (\text{PRI7.1})$$

For infinitely many labels, these equations are interpreted coordinatewise in the product of scalar coefficient spaces, not as unsupported infinite sums in a direct sum. Each coordinate follows from PRI6.4. The support-prime source is SZW33–SZW38 in the pinned derivation. Tensoring each coordinate with either formal coefficient $[\tau]$ or $[1]$ retains the two earlier coefficient sections; their formal difference is not subtraction of source elements. Neither supported zero nor the other support coordinates have been replaced by an ordinary prime sum.

To exhibit the exact form whose sign is required, let $h = f^\# * f$, where $f \in C_c^\infty(\mathbb{R})$, $f^\#(v) = \overline{f(-v)}$, and convolution uses Lebesgue measure. Fubini and substitution give

$$\hat{h}(y) = |\hat{f}(y)|^2, \quad H_h(z) = \overline{H_f(1-\bar{z})} H_f(z), \quad h(-v) = \overline{h(v)}.$$

Consequently the full scalar Weil value in the retained support coordinate is

$$\begin{aligned} W_\zeta(f^\# * f) &= 2\Re(\overline{H_f(1)} H_f(0)) \\ &\quad + \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}(y)|^2 \left(\Re \frac{\Gamma'(1/4 + iy/2)}{\Gamma(1/4 + iy/2)} - \log \pi \right) dy \\ &\quad - \frac{1}{2} \int_{(0,\infty)} e^{-3u/4} \Re(f^\# * f)(u/2) d\mathcal{W}_1(u). \end{aligned} \quad (\text{PRI7.2})$$

This is the original form expressed through the actual polarized traces. The positivity of the cohomological matrix H_p , the positivity of the measure $\mathcal{W}_1$, and the sign of PRI7.2 are statements about their displayed respective objects. PRI7.2 gives their exact receiving formula, including the subtractive prime term and the endpoint and Gamma terms. Its nonnegativity for every test has not been established by the calculation.

PRI8. Consequence for the user's complete-history argument

The proved composition is reversible on its named arithmetic data:

$$(\mathcal{D}, \mathcal{R}, \mathcal{W}) \longleftrightarrow (\mathcal{D}_1, \mathcal{R}_1, \mathcal{W}_1) \longrightarrow W_\zeta \text{ with every support coordinate retained.} \quad (\text{PRI8.1})$$

PRI3.4 and PRI4.2 are its inverse maps, and RMR2–RMR7 prove that the original zeta fixes the first triple uniquely, including the time origin and additive scale. Thus the complete polarized trace signal does recover the full original arithmetic history. Recovering it does not require discarding either eigenvalue sign or assigning arithmetic structure to tau.

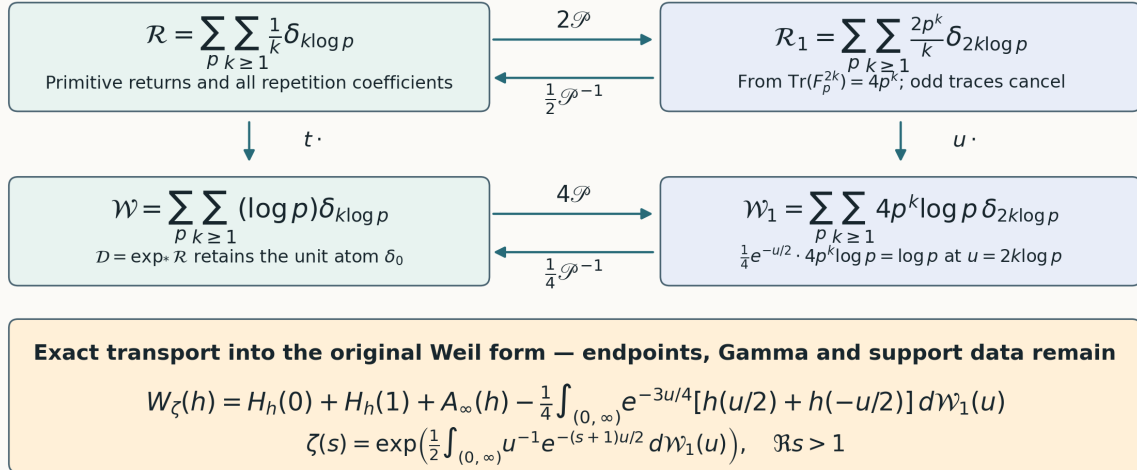
The resulting exact purity test is PRI7.2. The linked task's local weight-one theorem is retained as a proved result; its use in an RH proof requires the sign of this complete global expression. This note has supplied and proved the inverse transport and the full receiving expression, without inserting that sign as an assumption or presenting an RH proof.

The polarized return trace retains the complete original arithmetic history

$$\mathcal{P}\mu = (d_2)_*(e^t\mu), \quad d_2(t) = 2t, \quad \mathcal{P}^{-1}\nu = (d_{1/2})_*(e^{-u/2}\nu)$$

Original arithmetic: every prime p , every $k \geq 1$

Polarized cohomology: both eigenvalue signs retained



PRI2–PRI7: all arrows are proved for complete locally finite measures. No finite range is substituted. The unit atom records integer 1; it does not identify τ with that atom. Positivity of the full Weil form remains unproved.

Figure 6: Exact global return comparison, PRI2–PRI7. The upper and lower arrows retain their different factors two and four; multiplication by source and target time makes the square commute. The bottom formulas give the original-zeta inverse and the full Weil transport. The original periods and both eigenvalue signs come from the Connes–Consani curve and the constructed operator in PRI1.

The reproducible figure source is `draw_return_inverse.py`; the SVG is `return_inverse.svg`. The diagram represents identities for every prime and every repetition, with no numerical sample of zeros. Independent mathematical review is recorded in `POLARIZED_RETURN_INVERSE_REVIEW.md`; its zero-atom wording correction has been incorporated in PRI2.

The signed source norm and the original zeta counting measure

24 September 2026. Exact comparison PRZ1–PRZ4. The user's Z_2 denotes the retained parity observation; this note does not redefine the user's entire Z_n hierarchy as a family of groups. No addition at tau is used.

PRZ1. The source operation actually supplying parity

The signed generic monoid in the author's `CC.tex`, Connes–Consani, arXiv:2609.00299v1, has nonabsorbing group

$$G = \{T^m J^k : m \in \mathbb{Z}, k \in \mathbb{Z}/4\mathbb{Z}\}, \quad J^2 = \epsilon, \quad \epsilon^2 = 1,$$

and involution

$$A(T) = \epsilon T^{-1}, \quad A(J) = \epsilon J = J^{-1}. \quad (\text{PRZ1.1})$$

Here the cyclic four-element factor is the author's quarter-phase structure, not an imposed replacement for the user's quantum-number notation. The full source, its absorbing element and the supporting point remain separate objects. Multiplying the two source elements gives

$$A(T^m J^k) = T^{-m} J^{2m-k}, \quad (T^m J^k)A(T^m J^k) = J^{2m} = \epsilon^m. \quad (\text{PRZ1.2})$$

Thus the norm observation is independent of the quarter phase and descends to winding $L = G/\langle J \rangle$. It records even and odd winding while retaining the full original element alongside the observation. This is the earlier W2 proof, reproduced here to prove the exact source of the following labels.

The complete chart construction recovers arithmetic endomorphisms $[n] : L \rightarrow L$; evaluation on the positive chart's intrinsic generator identifies the positive count n with the winding class of T^n . Reversing the chart changes n to $-n$, which has the same norm ϵ^n . No exchanged parity label is thereby assigned to the fixed supporting point τ .

PRZ2. Retaining the two parity components, the unit, and all four lifts

For the original count measure $\mathcal{D} = \sum_{n \geq 1} \delta_{\log n}$, define

$$\mathcal{D}_{\text{even}} = \sum_{m \geq 1} \delta_{\log(2m)}, \quad \mathcal{D}_{\text{odd}} = \sum_{m \geq 0} \delta_{\log(2m+1)}. \quad (\text{PRZ2.1})$$

These are the two fibres of PRZ1.2 on positive counts. The integer-one atom δ_0 belongs to the odd component. It is neither integer zero nor tau. The maps are exactly

$$\mathcal{D}_{\text{even}} = \delta_{\log 2} * \mathcal{D}, \quad \mathcal{D}_{\text{odd}} = \mathcal{D} - \mathcal{D}_{\text{even}}, \quad \mathcal{D} = \mathcal{D}_{\text{even}} + \mathcal{D}_{\text{odd}}. \quad (\text{PRZ2.2})$$

The first identity follows from $\log(2m) = \log 2 + \log m$. The second is a disjoint partition of the positive counts, so both measures are positive despite the displayed subtraction. The ordered pair is retained, rather than only its sum.

One may also retain every source lift explicitly as

$$\widetilde{\mathcal{D}} = \sum_{n \geq 1} \sum_{k=0}^3 \delta_{(\log n, T^n J^k)}. \quad (\text{PRZ2.3})$$

For the map $\pi(t, g) = t$, the exact pushforwards are $\pi_* \widetilde{\mathcal{D}} = 4\mathcal{D}$ and, on each parity component, $4\mathcal{D}_{\text{even}}$ or $4\mathcal{D}_{\text{odd}}$. The factor four has not been absorbed. The arithmetic count measure counts winding classes once; the full lift measure counts the four representatives once each. These are the stated map and its multiplicity, not an identification of the two measures.

PRZ3. A faithful meromorphic receiver of the original divisor

On $\Re s > 1$, absolute convergence permits the partition of the original series, giving

$$E(s) = \mathcal{L}\mathcal{D}_{\text{even}}(s) = 2^{-s} \zeta(s), \quad O(s) = \mathcal{L}\mathcal{D}_{\text{odd}}(s) = (1 - 2^{-s}) \zeta(s). \quad (\text{PRZ3.1})$$

The ordered receiver $\zeta \mapsto (E, O)$ has the explicit inverse $(E, O) \mapsto E + O$. More strongly, $2^{-s} = \exp(-s \log 2)$ is entire and nowhere zero, with entire inverse 2^s . Therefore $\zeta = 2^s E$ is a global equality of meromorphic functions, and

$$\text{ord}_{s_0} E = \text{ord}_{s_0} \zeta \quad \text{at every finite } s_0. \quad (\text{PRZ3.2})$$

Every original zero, multiplicity, and pole is retained. The odd component has the additional explicitly recorded divisor

$$\text{ord}_{s_0} O = \text{ord}_{s_0} \zeta + \text{ord}_{s_0} (1 - 2^{-s}). \quad (\text{PRZ3.3})$$

The extra factor's zeros are $s_0 = 2\pi i k / \log 2$, $k \in \mathbb{Z}$, each simple because its derivative there is $\log 2 \neq 0$. In particular no zero or pole of the original zeta is silently deleted by using the ordered pair.

PRZ4. What the signed trace does to the endpoint

Evaluating the norm sign by $\epsilon \mapsto -1$ gives the signed scalar

$$S(s) = E(s) - O(s) = (2^{1-s} - 1)\zeta(s) = \sum_{n \geq 1} (-1)^n n^{-s} \quad (\Re s > 1). \quad (\text{PRZ4.1})$$

This is the negative of the usual alternating Dirichlet eta function. The letter S avoids conflating it with the source's generic point η . The cancellation can be traced without losing either parity component. The paired series

$$S(s) = \sum_{m \geq 1} ((2m)^{-s} - (2m-1)^{-s}) = -s \sum_{m \geq 1} \int_{2m-1}^{2m} x^{-s-1} dx \quad (\text{PRZ4.2})$$

converges absolutely and locally uniformly on $\Re s > 0$: on a compact set there, its terms are bounded by a constant times $(2m-1)^{-1-\delta}$ for some $\delta > 0$. It agrees with PRZ4.1 on the original convergence half-plane, hence gives its analytic continuation to this larger domain.

Put $c(s) = 2^{1-s} - 1$. It has simple zeros at $1 + 2\pi i k / \log 2$. Globally the exact meromorphic return is $\zeta = S/c$, with Laurent division at these points. At $s = 1 + h$, the original pole and the retained multiplier give

$$c(1+h) = e^{-h \log 2} - 1 = -h \log 2 + O(h^2), \quad \zeta(1+h) = h^{-1} + O(1), \quad S(1) = -\log 2. \quad (\text{PRZ4.3})$$

The pole has cancelled in this signed scalar, while remaining in E , O , their sum, and the displayed inverse. At every finite point the full order relation is $\text{ord } S = \text{ord } c + \text{ord } \zeta$. Inside $0 < \Re s < 1$, $c \neq 0$, so this signed observation preserves all original zeros and their multiplicities there. The source norm supplies an actual parity-resolved zeta map; none of these identities determines their real parts.

An integral adjoint algebra for the complete positive counting action

24 September 2026. Independent derivation. Proof locators IAR0–IAR9.

This note constructs the smallest unital subring of the original rational group algebra that contains every positive counting operator and is preserved by the actual weighted adjoint. It retains the original rational group algebra as an ambient object and recovers it by scalar extension. No quotient replaces it. The supporting datum remains $\tau\langle Z_1; \text{no intrinsic } Z_2 \rangle$; no addition or arithmetic value is assigned to tau.

IAR0. Input and exact scope of the correction

Read work/tau_weight_cohomology_20260924/INTEGRAL_COUNTING_AND_WEIGHT_COMPARISON.md, IC1–IC9, particularly the complete IC3 and IC8 arguments.

Retain the original rational group algebra

$$K = \mathbb{Q}[\mathbb{Q}_{\geq 0}^\times] = \bigoplus_{r \in \mathbb{Q}_{\geq 0}^\times} \mathbb{Q} b_r, \quad b_r b_s = b_{rs}, \quad 1_K = b_1, \quad (\text{IAR0.1})$$

with the specified involution

$$b_r^* = rb_{1/r}, \quad c^* = c \quad (c \in \mathbb{Q}). \quad (\text{IAR0.2})$$

All sums have finite support. The multiplication is commutative, so its involution can equivalently be described as multiplicative or antimultiplicative. Involution follows directly from $(rb_{1/r})^* = r(1/r)b_r$.

IC8 proves that imposing this involution on a ring already containing every bare $b_{1/n}$ forces rational scalars. That theorem is correct. Here the input is the actual positive counting monoid $\{b_n : n \geq 1\}$; its inverses are not assumed to be integral. The resulting ring is

$$\mathcal{H}_{\mathbb{Z}} = \mathbb{Z}[b_n, nb_{1/n} : n \geq 1] \subseteq K. \quad (\text{IAR0.3})$$

The word integral here refers to an explicit free $\mathbb{Z}$ -form of the algebra. It does not mean that its generators are integral over $\mathbb{Z}$, and it does not assert a finite-rank integral lattice for a complex zero block.

IAR1. A complete additive basis, including every denominator

Write a positive rational uniquely in lowest terms as $r = a/b$, where $a, b \geq 1$ and $\gcd(a, b) = 1$. Set

$$\text{den}(r) = b, \quad \text{num}(r) = a, \quad C_r = bb_r. \quad (\text{IAR1.1})$$

Theorem. The elements C_r , for all positive rationals r , form a free additive $\mathbb{Z}$ -basis of $\mathcal{H}_{\mathbb{Z}}$. Their exact multiplication and involution are

$$C_r C_s = \frac{\text{den}(r) \text{den}(s)}{\text{den}(rs)} C_{rs}, \quad C_r^* = C_{1/r}. \quad (\text{IAR1.2})$$

Proof. Put $r = a/b$, $s = c/d$ in lowest terms and let $g = \gcd(ac, bd)$. Then

$$rs = \frac{ac/g}{bd/g}, \quad \text{den}(rs) = bd/g,$$

so the coefficient in (IAR1.2) is the positive integer g . In the original basis,

$$C_r C_s = bd b_{rs} = \frac{bd}{bd/g} C_{rs} = g C_{rs}.$$

Every denominator factor has therefore been included. Also

$$C_r^* = b \frac{a}{b} b_{b/a} = ab_{b/a} = C_{1/r}.$$

The additive span $L = \bigoplus_r \mathbb{Z} C_r$ is a unital subring, since $C_1 = b_1$, and is preserved by $*$. It contains $b_n = C_n$ and $nb_{1/n} = C_{1/n}$, so $\mathcal{H}_{\mathbb{Z}} \subseteq L$. Conversely,

$$b_a (bb_{1/b}) = bb_{a/b} = C_{a/b},$$

which puts every basis element of L in $\mathcal{H}_{\mathbb{Z}}$. Finally, any finite integral relation $\sum_r m_r C_r = 0$ has b_r -coefficient $m_r \text{den}(r)$; freeness of the original rational group basis forces every $m_r = 0$. Thus the span is free and equals the claimed ring. $\square$

This proves the exact chain of inclusions

$$\mathbb{Z}[b_n : n \geq 1] \subseteq \mathcal{H}_{\mathbb{Z}} \subseteq \Lambda := \mathbb{Z}[\mathbb{Q}_{\geq 0}^{\times}] \subseteq K. \quad (\text{IAR1.3})$$

For $n > 1$, the bare basis vector $b_{1/n}$ does not lie in $\mathcal{H}_{\mathbb{Z}}$: its coefficient one is not a multiple of the required denominator n . Also

$$\mathcal{H}_{\mathbb{Z}} \cap \mathbb{Q}b_1 = \mathbb{Z}b_1. \quad (\text{IAR1.4})$$

Thus stability under this adjoint has not introduced any nonintegral scalar.

IAR2. Minimality and rational scalar extension

Theorem. $\mathcal{H}_{\mathbb{Z}}$ is the unique smallest unital subring of K that contains all b_n , $n \geq 1$, and is preserved by $*$.

Proof. A unital subring contains the integral scalar multiples of its unit. If it contains b_n , involution stability also gives $b_n^* = nb_{1/n}$. It therefore contains every generator in (IAR0.3), hence the entire displayed ring. Conversely IAR1 proves that this ring is involution-stable and contains the required counting elements. $\square$

Scalar extension is an isomorphism onto the original ambient algebra:

$$\boxed{\mathbb{Q} \otimes_{\mathbb{Z}} \mathcal{H}_{\mathbb{Z}} \xrightarrow{\sim} K, \quad q \otimes C_r \mapsto q \operatorname{den}(r) b_r.} \quad (\text{IAR2.1})$$

Surjectivity follows from $b_r = \operatorname{den}(r)^{-1} C_r$; injectivity follows from the same free-basis argument as IAR1. This map preserves products, units and involutions by their explicit formulas.

There is also an exact additive record of the difference from the bare integral group ring:

$$\Lambda / \mathcal{H}_{\mathbb{Z}} \cong \bigoplus_{r \in \mathbb{Q}_{\geq 0}^{\times}} \mathbb{Z} / \operatorname{den}(r) \mathbb{Z}. \quad (\text{IAR2.2})$$

This is an additive quotient, not a quotient ring: $\mathcal{H}_{\mathbb{Z}}$ contains the unit and is not an ideal of Λ . Formula (IAR2.2) follows coordinate by coordinate from (IAR1.1). In particular the discrepancy is torsion, exactly as the scalar-extension isomorphism records.

IAR3. The prime-by-prime presentation

For every prime p , define in the original ambient ring

$$F_p = b_p, \quad V_p = pb_{1/p}. \quad (\text{IAR3.1})$$

They obey

$$F_p V_p = V_p F_p = pb_1, \quad F_p^* = V_p, \quad V_p^* = F_p. \quad (\text{IAR3.2})$$

Elements associated to distinct primes commute. The letter V denotes this particular generator and is not an assertion identifying it with a Frobenius or Verschiebung functor from another theory.

Theorem. There is a unital, involution-preserving isomorphism

$$\boxed{\mathcal{H}_{\mathbb{Z}} \cong \bigotimes_{p \text{ prime}, \mathbb{Z}} \mathbb{Z}[F_p, V_p] / (F_p V_p - p).} \quad (\text{IAR3.3})$$

The infinite tensor product means the filtered union of its finite-prime tensor products under their unital inclusions. Equivalently its presentation is the commutative polynomial ring in all displayed variables, divided by the ideal generated by all $F_p V_p - p$.

Proof. In each monomial, replace a simultaneous factor $F_p^a V_p^b$ by

$$p^{\min(a,b)} \begin{cases} F_p^{a-b}, & a \geq b, \\ V_p^{b-a}, & b \geq a. \end{cases} \quad (\text{IAR3.4})$$

Thus every polynomial is an integral linear combination of monomials that use, for each prime, only a power of F_p , only a power of V_p , or neither. The monomial indexed by finitely supported integers $(e_p)_p$ is

$$M_e = \prod_{e_p > 0} F_p^{e_p} \prod_{e_p < 0} V_p^{-e_p}.$$

Its image in K is

$$\left(\prod_{e_p < 0} p^{-e_p} \right) b_{\prod_p p^{e_p}} = C_{\prod_p p^{e_p}}. \quad (\text{IAR3.5})$$

Unique prime factorization makes these rational indices distinct for distinct exponent tuples. IAR1 proves linear independence of their images. Hence the reduction has no further relation, the presented ring embeds in K , and its image is the entire basis span $\mathcal{H}_{\mathbb{Z}}$. Swapping F_p, V_p preserves each relation and agrees with the computed involution. This proves every assertion. $\square$

In particular the ring is a domain: any two elements of its ambient algebra belong to a Laurent polynomial ring over $\mathbb{Q}$ on a finite set of prime variables, which is a domain. It is also free, hence flat, as a $\mathbb{Z}$ -module; tensoring its direct sum of copies of $\mathbb{Z}$ preserves every injection. No finiteness over $\mathbb{Z}$ is claimed: already the powers of a single F_p are linearly independent. In fact F_p satisfies no monic polynomial over $\mathbb{Z}$, since such a polynomial would be a relation among the distinct basis elements C_{p^j} . The integral adjoint algebra therefore does not itself make every represented eigenvalue an algebraic integer.

IAR4. How the earlier rationalization obstruction fits exactly

Localizing the integral adjoint algebra at all positive generators gives the previously used bare integral group ring:

$$\mathcal{H}_{\mathbb{Z}}[F_p^{-1} : p \text{ prime}] \cong \mathbb{Z}[b_p, b_p^{-1} : p \text{ prime}] = \Lambda. \quad (\text{IAR4.1})$$

Indeed $F_p^{-1} = b_{1/p}$, and all generators in either direction are then present, with the same injection into K . This localization is not stable under the weighted involution over its integral scalars, precisely as IC8 proves.

Inverting both adjoint partners gives

$$\mathcal{H}_{\mathbb{Z}}[F_p^{-1}, V_p^{-1} : p \text{ prime}] \cong K. \quad (\text{IAR4.2})$$

To see the rational scalars explicitly, $p = F_p V_p$ becomes a unit for every prime, so $1/p = F_p^{-1} V_p^{-1}$ belongs to this localization. It also contains all group basis elements and hence every finite rational combination. Conversely all these inverses already exist in K .

Thus the IC8 obstruction is retained, with its exact input identified. It concerns adjoining the bare inverse operators as integral elements and then imposing the adjoint. Starting from the actual positive counting operators and adjoining their adjoints instead produces the smaller involution-stable integral form proved above. The connecting localizations (IAR4.1)–(IAR4.2) give their complete relation.

IAR5. Every prime special fibre, with both components retained

Fix any prime ℓ and write $k = \mathbb{F}_\ell$. Let

$$B_\ell = k[U_p, U_p^{-1} : p \text{ prime}, p \neq \ell]. \quad (\text{IAR5.1})$$

Then base change of the proved presentation yields

$$\boxed{\mathcal{H}_\mathbb{Z} \otimes_\mathbb{Z} k \cong B_\ell[F, V]/(FV)}, \quad F = F_\ell \bmod \ell, \quad V = V_\ell \bmod \ell. \quad (\text{IAR5.2})$$

Proof. For $p \neq \ell$, the relation is $F_p V_p = \bar{p} \neq 0$. Thus F_p is a unit and $V_p = \bar{p} F_p^{-1}$; this factor is exactly the Laurent polynomial ring with coordinate $U_p = F_p$. At $p = \ell$ the relation becomes $FV = 0$. Polynomial presentations commute with scalar base change, proving (IAR5.2). Alternatively the free integral basis in IAR3 reduces directly to its stated basis over k . $\square$

This fibre is reduced and has exactly two minimal prime ideals, (F) and (V) . Its two components are $\text{Spec } B_\ell[V]$ and $\text{Spec } B_\ell[F]$, intersecting in $\text{Spec } B_\ell$, the locus $F = V = 0$. Here is a proof that includes all scheme-theoretic information in these claims. The map

$$B_\ell[F, V]/(FV) \longrightarrow B_\ell[F] \times B_\ell[V], \quad q \longmapsto (q(F, 0), q(0, V)) \quad (\text{IAR5.3})$$

is injective, since each class has a unique form consisting of a constant, positive powers of F , and positive powers of V . Its image is precisely the pairs with equal constant term. Each factor is a domain, so the fibre has no nonzero nilpotents. Every prime ideal contains F or V , since their product is zero, and the quotients by either displayed ideal are domains. Neither ideal contains the other. They are therefore exactly the minimal primes. Their intersection in the fibre is zero, their sum is (F, V) , and the quotient by their sum is B_ℓ .

Both generators are nonzero, although their product is zero. More precisely

$$\text{Ann}(F) = (V), \quad \text{Ann}(V) = (F). \quad (\text{IAR5.4})$$

This follows immediately from the same unique normal form and the domain property of B_ℓ .

The involution descends to every special fibre, and its exact formulas are

$$F^* = V, \quad V^* = F, \quad U_p^* = \bar{p} U_p^{-1} \quad (p \neq \ell). \quad (\text{IAR5.5})$$

It exchanges the two components and preserves their intersection. In particular the characteristic-two fibre has the same two-component description; no prime has been omitted.

For comparison, the inclusion into the bare integral group ring induces

$$B_\ell[F, V]/(FV) \longrightarrow k[b_p, b_p^{-1} : p \text{ prime}], \quad F \longmapsto b_\ell, \quad V \longmapsto 0, \quad (\text{IAR5.6})$$

and $U_p \mapsto b_p$ otherwise. The kernel is exactly (V) . After quotienting by (V) , the map is the injection of the polynomial ring in F into its Laurent localization. Thus adjoining the bare inverse singles out one of the two fibre components and then inverts its surviving coordinate. The integral adjoint algebra retains both, exchanged by its involution. This is a derived description of the map, not a replacement of the full rational action.

IAR6. The actual original-zeta receiver

Retain the receiving space from IC2:

$$\mathcal{B} = \{\text{entire } f : f \text{ decreases faster than every inverse power vertically in each bounded real strip}\},$$

with seminorms

$$b_{A,M}(f) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |f(s)|.$$

Let $\mathcal{Z}$ be the actual distinct nontrivial zeros of the original ζ , with their full multiplicities m_ρ , and retain

$$\mathcal{J} = \{f \in \mathcal{B} : f^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{IAR6.1})$$

The source comparison in IC2 retains its full multiplier $s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/8$, including its endpoint and trivial-zero comparison. The present coefficient construction does not modify that comparison or replace the original zeta. It acts on the already proved quotient.

For $r > 0$, define $T_r[f] = [r^s f(s)]$, using the real logarithm of r . The map from the new integral algebra is

$$\boxed{\Psi : \mathcal{H}_{\mathbb{Z}} \longrightarrow \text{End}_{\mathbb{C}, \text{cont}}(\mathcal{Q}), \quad \Psi(C_{a/b}) = bT_{a/b}.} \quad (\text{IAR6.2})$$

It is the restriction of the full rational group-algebra action, not a newly assigned representation. In particular

$$\Psi(F_p) = T_p, \quad \Psi(V_p) = pT_{1/p}, \quad \Psi(F_p)\Psi(V_p) = pI. \quad (\text{IAR6.3})$$

Proof of well-definedness and continuity. Each multiplier is entire and nowhere zero. Multiplication therefore preserves every required vanishing order. On representatives,

$$b_{A,M}(br^s f) \leq b \max(r^A, r^{-A}) b_{A,M}(f), \quad (\text{IAR6.4})$$

and finite sums satisfy the sum of these bounds. Taking the quotient gives continuous operators on $\mathcal{Q}$. Multiplication of two basis multipliers gives $\text{den}(r) \text{den}(s)(rs)^z$, exactly the image of (IAR1.2), so the map respects every product and the unit. $\square$

The full rational action is recovered by scalar extension of (IAR6.2) through (IAR2.1). In particular every $T_{1/p}$ remains available in the original rational action: it is $p^{-1}\Psi(V_p)$. The new statement is that its integral adjoint partner is $pT_{1/p}$, with the degree factor retained.

IAR7. Every zero jet and its reflected involution

For an actual zero ρ , put $m = m_\rho$. The complete germ receiver is $\mathbb{C}[t]/(t^m)$, with $s = \rho + t$. Its action is

$$\begin{aligned} \Psi(F_p)_\rho &= p^\rho \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} N_\rho^j, \\ \Psi(V_p)_\rho &= p^{1-\rho} \sum_{j=0}^{m-1} \frac{(-\log p)^j}{j!} N_\rho^j, \quad N_\rho^m = 0, \end{aligned} \quad (\text{IAR7.1})$$

where N_ρ is multiplication by t . These are the Taylor expansions of the actual multipliers, so all nilpotent coefficients are retained. Their product is pI : the coefficient of N_ρ^j , for $j \geq 1$, is

$$p(\log p)^j \sum_{a=0}^j \frac{(-1)^{j-a}}{a!(j-a)!} = \frac{p(\log p)^j}{j!} (1-1)^j = 0,$$

while the constant coefficient is p . More generally the basis element $C_{a/b}$ acts by

$$b(a/b)^\rho \sum_{j=0}^{m-1} \frac{(\log(a/b))^j}{j!} N_\rho^j. \quad (\text{IAR7.2})$$

For $A = \sum_r c_r b_r \in K$, $c_r \in \mathbb{Q}$, denote its multiplier by $a_A(s) = \sum_r c_r r^s$. Its algebra involution has multiplier

$$a_{A^*}(s) = \sum_r c_r r^{1-s} = \overline{a_A(1-\bar{s})}. \quad (\text{IAR7.3})$$

The actual zero divisor is invariant under $\rho \mapsto \rho^\# = 1 - \bar{\rho}$, with equal multiplicities. Consequently the full jet comparison is

$$j_\rho(a_{A^*}) = \sigma(j_{\rho^\#}(a_A)), \quad \sigma \left(\sum_{j=0}^{m-1} z_j t^j \right) = \sum_{j=0}^{m-1} (-1)^j \bar{z}_j t^j. \quad (\text{IAR7.4})$$

This formula retains both reflection and the sign on every higher Taylor coefficient. It is not a restriction to simple zeros or to constant jets.

For completeness the represented kernel is involution-stable. If $\Psi(A) = 0$, then $a_A f \in \mathcal{I}$ for every $f \in \mathcal{B}$. For any fixed actual ρ , the function $f(s) = e^{(s-\rho)^2}$ belongs to $\mathcal{B}$ and has value one at ρ . It is an invertible local germ there, so $a_A f$ vanishing to order m_ρ forces the same vanishing order for a_A . Formula (IAR7.4) proves that a_{A^*} has all those orders too, and thus $a_{A^*} f \in \mathcal{I}$ for every test f . Hence $\Psi(A^*) = 0$. The involution therefore descends to the represented image as an algebra involution, even without claiming that Ψ is injective.

IAR8. The weighted adjoint identity, including the trace radical

Retain the actual Weil form

$$W([f], [g]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{f(\rho^\#)} g(\rho), \quad \rho^\# = 1 - \bar{\rho}. \quad (\text{IAR8.1})$$

Its absolute convergence and Hermitian property are supplied by the original-zeta receiver. The finite multipliers here are bounded on the strip containing all these zeros, so the same convergence permits their insertion. For $r = a/b > 0$,

$$\begin{aligned} W(\Psi(C_r)x, y) &= \sum_{\rho} m_\rho \overline{br^{\rho^\#} x(\rho^\#)} y(\rho) \\ &= \sum_{\rho} m_\rho \overline{br^{1-\rho} x(\rho^\#)} y(\rho) \\ &= \sum_{\rho} m_\rho \overline{ar^{-\rho} x(\rho^\#)} y(\rho) \\ &= W(x, \Psi(C_{1/r})y). \end{aligned} \quad (\text{IAR8.2})$$

The integer relation $br = a$ accounts for the coefficient change. Additivity and real integral coefficients now give, for every $A \in \mathcal{H}_{\mathbb{Z}}$,

$$\boxed{W(\Psi(A)x, y) = W(x, \Psi(A^*)y)}. \quad (\text{IAR8.3})$$

In particular T_p and $pT_{1/p}$ are adjoint partners for this exact form, and

$$W(T_p x, T_p y) = W(x, pT_{1/p} T_p y) = pW(x, y). \quad (\text{IAR8.4})$$

This proves the adjoint identity without a positivity assumption. Higher jets remain present in IAR7 even where this value-only trace form has a radical. Thus (IAR8.3) is an exact adjoint identity for the specified algebra involution; it does not assert a unique adjoint for every continuous endomorphism of a possibly degenerate Hermitian space.

The same distinction matters for finite characteristic. The actual receiver $\mathcal{Q}$ is a complex vector space, so multiplication by every prime ℓ is invertible. Therefore

$$\mathcal{Q}/\ell\mathcal{Q} = 0. \quad (\text{IAR8.5})$$

The two nonzero special-fibre components in IAR5 belong to the integral coefficient algebra. They cannot be obtained as nonzero reductions of this complex receiving vector space. A nontrivial reduction of a receiving module would require an actual integral module inside or before it; none is silently supplied by the coefficient-ring construction. Formula (IAR8.5) is the exact existing comparison, while IAR5 retains the nontrivial integral geometry rather than discarding it.

IAR9. Proved repair and the next receiving object

The complete proved diagram is

$$\begin{array}{ccccc} \mathbb{Z}[b_n : n \geq 1] & \hookrightarrow & \mathcal{H}_{\mathbb{Z}} & \hookrightarrow & \mathbb{Q}[\mathbb{Q}_{>0}^{\times}] \\ & & \downarrow \Psi & & \downarrow \Phi_{\mathbb{Q}} \\ & & \text{End}_{\mathbb{C}, \text{cont}}(\mathcal{Q}) & = & \text{End}_{\mathbb{C}, \text{cont}}(\mathcal{Q}). \end{array} \quad (\text{IAR9.1})$$

The middle ring is the minimal integral adjoint closure of the actual positive counting action. Its basis contains all rational labels with their exact denominator coefficients. Its rational scalar extension is the original full group algebra; its prime special fibres have two reduced components exchanged by the involution; and its represented generators retain all original-zeta zero jets and the exact degree-weighted adjoint.

This resolves the coefficient-domain question for the positive counting monoid. It does not contradict IC8's different minimum when bare rational inverses have already been declared integral. The relation between those two inputs and their resulting rings is proved by the inclusions and localizations above.

The equation $F_p V_p = p$, by itself, is not an identification with a prismatic or Dieudonné object. Here F_p, V_p are the explicitly defined elements of a commutative integral subring of a rational group algebra, with the actual complex receiver and special fibres calculated. Any comparison with such cohomological theories must preserve those rings, their coefficient maps, their possible semilinear actions, and the receiving modules. No such further identification, eigenvalue integrality conclusion, global positivity result, or RH proof is asserted in this note.

Integral adjoints, the trace coordinate, and its exact infinitesimal fibres

24 September 2026. Direct derivation AT1–AT6. This continues the linked task’s IC8 integral-adjoint calculation and the present integral return algebra. The variables here are operators and their algebraic coordinates, not coordinates or distances assigned to the parityless supporting datum tau.

AT1. The retained integral algebra at one prime

For each prime p , including two, let

$$B_p = \mathbb{Z}[F, V]/(FV - p), \quad F^\star = V, \quad V^\star = F. \quad (\text{AT1.1})$$

The involution fixes integer scalars. The defining relation is preserved, so it defines an actual involution of the ring. The integral-adjoint construction gives the injection

$$B_p \longrightarrow \mathbb{Q}[\mathbb{Q}_{>0}^\times], \quad F \mapsto b_p, \quad V \mapsto pb_{1/p}. \quad (\text{AT1.2})$$

For completeness its injectivity follows by replacing each mixed monomial $F^a V^b$ by $p^{\min(a,b)} F^{a-b}$ or $p^{\min(a,b)} V^{b-a}$. Thus every element has a representative $c_0 + \sum_{j \geq 1} c_j F^j + \sum_{j \geq 1} d_j V^j$, with finite sums and integer coefficients. Its image is $c_0 b_1 + \sum c_j b_{p^j} + \sum d_j p^j b_{p^{-j}}$. Distinct positive rational indices are independent in the group-algebra basis. Vanishing forces every coefficient to vanish. This proves the asserted injection and the unique representative. In particular neither F nor V has been made an invertible integral generator by a hidden division by p .

The all-prime counterpart and its exact minimality are proved in INTEGRAL_ADJOINT_RETURN_AL_GEBRA.md. The ambient rational group algebra, including its original inverse generators, is retained by scalar extension. Here the calculation uses the displayed integral subalgebra and its specified maps.

AT2. The exact quadratic presentation and its discriminant

Put $X = F + V$. There is a ring isomorphism

$$\boxed{B_p \simeq \mathbb{Z}[X, F]/(F^2 - XF + p), \quad V = X - F.} \quad (\text{AT2.1})$$

The forward substitution gives $F^2 - (F + V)F + p = -FV + p = 0$. The reverse gives $F(X - F) = p$ by its displayed relation. Each substitution fixes F and sends the other generator back to itself, so they are mutually inverse on all elements. Polynomial division by the monic quadratic proves that this is a free rank-two module over $\mathbb{Z}[X]$, with basis $1, F$: existence follows from division, and a remainder of degree less than two cannot be a nonzero multiple of a monic quadratic. No inversion of two or of p is used.

The involution fixes X and sends F to $X - F$. Its exact discriminant identity is

$$\boxed{(F - V)^2 = X^2 - 4p.} \quad (\text{AT2.2})$$

Expand both sides using $FV = p$. Every constant, including four, remains. This identifies the two-sheeted trace map algebraically while retaining the original operator coordinates.

The relative differential module makes the ramification calculation exact:

$$\Omega_{B_p/\mathbb{Z}[X]} = B_p/(2F - X) dF. \quad (\text{AT2.3})$$

Differentiating the single monic relation over the fixed base gives precisely $(2F - X)dF = 0$. This description includes residue characteristic two.

AT3. Its two characteristic-zero branch fibres are dual numbers

Retain the integral extension

$$A_p = \mathbb{Z}[\alpha]/(\alpha^2 - p). \quad (\text{AT3.1})$$

Extend the involution by $\alpha^* = \alpha$; this specifies its action on the new scalar base. For each sign $\sigma \in \{+1, -1\}$, take the actual base change of AT2.1 along $X \mapsto 2\sigma\alpha$. Its coordinate algebra is

$$B_{p,\sigma} = A_p[F]/(F^2 - 2\sigma\alpha F + p) = A_p[F]/((F - \sigma\alpha)^2). \quad (\text{AT3.2})$$

The equalities use $\alpha^2 = p$ and retain the full polynomial before factoring. The mutually inverse maps

$$\boxed{A_p[\varepsilon]/(\varepsilon^2) \longleftrightarrow B_{p,\sigma}, \quad \varepsilon \mapsto F - \sigma\alpha, \quad F \mapsto \sigma\alpha + \varepsilon} \quad (\text{AT3.3})$$

give an exact dual-number isomorphism. They also give

$$F = \sigma\alpha + \varepsilon, \quad V = \sigma\alpha - \varepsilon, \quad F + V = 2\sigma\alpha, \quad FV = p - \varepsilon^2 = p, \quad \varepsilon^* = -\varepsilon. \quad (\text{AT3.4})$$

The nilpotent ε is nonzero: the dual-number ring is free over A_p with basis $1, \varepsilon$, again by division by a monic polynomial. The trace is exactly constant on this nonreduced fibre while the two operators change by opposite nonzero first-order terms. This is the precise vanishing infinitesimal in this construction. It is not a claim that an original zeta zero has this multiplicity or lies over one of these two trace values.

There are additional integral features, which must not be hidden by the characteristic-zero description. The two fibres are defined over A_p ; their base values differ by 4α , not by an invertible constant in general. The relative derivative of the monic quadratic in AT2.1 is $2F - X$. Thus its ramification locus is defined by this derivative, with discriminant $X^2 - 4p$; in residue characteristic two this locus has additional behaviour. AT3.3 is an integral identity valid even above two, without asserting that the two fibres remain disjoint there.

The self-adjoint operator quotient has a different exact map:

$$B_p/(F - V) \simeq \mathbb{Z}[F]/(F^2 - p) \simeq A_p. \quad (\text{AT3.5})$$

Within AT3.3, imposing $F = V$ imposes $2\varepsilon = 0$, not automatically $\varepsilon = 0$ over the integral base. After passage to a real embedding $\alpha = \sqrt{p}$, it imposes $\varepsilon = 0$. The positive cohomological matrix of PRI1 satisfies $F_p^2 = pI$ and has its adjoint equal to itself, so it receives AT3.5. This retains the distinction between the nonreduced trace fibre, its reduced real point, and the integral self-adjoint quotient.

More explicitly, the common quotient after the trace base change is

$$A_p[\varepsilon]/(\varepsilon^2, 2\varepsilon) = A_p \oplus (A_p/2A_p)\varepsilon \quad \text{as an } A_p\text{-module.} \quad (\text{AT3.6})$$

The second summand is nonzero since A_p is free over $\mathbb{Z}$ with basis $1, \alpha$. To check the comparison without identifying two scalar variables prematurely, write the original self-adjoint quotient as $\mathbb{Z}[\beta]/(\beta^2 - p)$. Its trace base change also imposes $2\beta = 2\sigma\alpha$. Substitution $\varepsilon = \beta - \sigma\alpha$ converts these relations into $\varepsilon^2 + 2\sigma\alpha\varepsilon = 0$ and $2\varepsilon = 0$, exactly the two relations displayed. The retained integral infinitesimal is therefore nonzero two-torsion; its involution fixes it because $-\varepsilon = \varepsilon$ in this quotient.

AT4. A faithful indefinite receiver and the exact positive reduction

Base change AT3.3 to $\mathbb{R}$ through $\alpha = \sqrt{p}$. On $\mathbb{C}^2$, retain

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad F_\sigma = \sigma\sqrt{p}I + N, \quad V_\sigma = \sigma\sqrt{p}I - N. \quad (\text{AT4.1})$$

Then $N^2 = 0$ but $N \neq 0$, $J^* = J$, and J has eigenvalues $+1, -1$. Direct multiplication gives

$$N^*J + JN = 0, \quad F_\sigma^*J = JV_\sigma, \quad F_\sigma V_\sigma = pI, \quad F_\sigma^*JF_\sigma = pJ. \quad (\text{AT4.2})$$

The map $a + b\varepsilon \mapsto aI + bN$ is faithful since its diagonal and upper-right entry recover a and b . It therefore retains the complete infinitesimal algebra with its involution in a nondegenerate indefinite Hermitian space. The constant trace has not erased the nilpotent in this receiver.

Conversely, in a representation of this same real dual-number star algebra by bounded operators on a positive-definite Hilbert space, let E be the image of ε . The relations require $E^* = -E$ and $E^2 = 0$. For every vector v ,

$$\|Ev\|^2 = \langle v, E^*Ev \rangle = \langle v, -E^2v \rangle = 0. \quad (\text{AT4.3})$$

Thus $E = 0$. This proves that every such positive star representation factors through the explicit quotient

$$\mathbb{R}[\varepsilon]/(\varepsilon^2) \longrightarrow \mathbb{R}, \quad a + b\varepsilon \longmapsto a, \quad \ker = (\varepsilon). \quad (\text{AT4.4})$$

Evaluation on the one-dimensional real Hilbert space is faithful on the quotient. Consequently (ε) is exactly the intersection of kernels of these positive star representations. The indefinite faithful map and the positive quotient are both constructed; neither justifies deleting the source infinitesimal from other calculations.

AT5. The actual original-zeta receiver and the purity trace

Use the actual zero-jet quotient $\mathcal{Q}$ and its continuous action $T_a[F(s)] = [a^s F(s)]$, as proved in the linked task's CD4–CD6 and RZ8. Its multiplier comparison is with the original ζ , including the full $s(s-1)\pi^{-s/2}\Gamma(s/2)/8$ source factor. On this quotient there is the ring homomorphism

$$B_p \longrightarrow \text{End}_{\text{cont}}(\mathcal{Q}), \quad F \longmapsto T_p, \quad V \longmapsto pT_{1/p}, \quad X \longmapsto T_p + pT_{1/p}. \quad (\text{AT5.1})$$

Composition gives $T_p(pT_{1/p}) = pI$, proving the defining relation. The reflected Weil form makes these two operators adjoints, exactly as IC8.5 proves; no positive metric is presumed.

For every actual nontrivial zero ρ with multiplicity m_ρ , let $N_\rho^{m_\rho} = 0$ be its retained Jordan coordinate. The full receiving formulas are

$$\begin{aligned} F|_\rho &= p^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log p)^j}{j!} N_\rho^j, \\ V|_\rho &= p^{1-\rho} \sum_{j=0}^{m_\rho-1} \frac{(-\log p)^j}{j!} N_\rho^j, \\ X|_\rho &= \sum_{j=0}^{m_\rho-1} \frac{(\log p)^j}{j!} (p^\rho + (-1)^j p^{1-\rho}) N_\rho^j. \end{aligned} \quad (\text{AT5.2})$$

Taylor expansion of the entire multipliers proves each finite sum; no nilpotent term is dropped. The scalar trace-coordinate value is

$$x_p(\rho) = p^\rho + p^{1-\rho}. \quad (\text{AT5.3})$$

The following characterization holds for every complex s , hence also on the entire actual-zero set:

$$\boxed{\Re s = \frac{1}{2} \iff p^s + p^{1-s} \in [-2\sqrt{p}, 2\sqrt{p}] \subset \mathbb{R}.} \quad (\text{AT5.4})$$

To prove the forward direction write $s = 1/2 + it$, giving the retained value $2\sqrt{p} \cos(t \log p)$. For the reverse direction put $x = p^s + p^{1-s}$ in the real interval. Both p^s and p^{1-s} are roots of $Y^2 - xY + p$. Its roots are $(x \pm i\sqrt{4p - x^2})/2$, including the coincident endpoints, and both have squared modulus $(x^2 + 4p - x^2)/4 = p$. Therefore $|p^s|^2 = p^{2\Re s} = p$. Since $p > 1$, taking real logarithms proves the equivalence.

Equivalently the character $F \mapsto p^s, V \mapsto p^{1-s}$ preserves $\star$ as complex conjugation exactly on that same line: the required equality is $p^{1-s} = \overline{p^s}$, which holds there; its absolute values imply $1 - \Re s = \Re s$ in the converse direction. The full phase $\Im s \log p$ remains. Thus the integral trace algebra supplies an exact place to test

purity of the actual zero characters, rather than identifying an integer degree with an unknown algebraic eigenvalue.

AT5.4 is an exact identity for all complex inputs. It is not a finite interval computation of zeros, an established bound for their trace values, or an RH conclusion. The displayed trace interval is the mathematical condition whose global proof is sought.

The scalar trace observation has the following exact fibre for any $s, t \in \mathbb{C}$:

$$x_p(t) = x_p(s) \iff t = s + \frac{2\pi ik}{\log p} \text{ or } t = 1 - s + \frac{2\pi ik}{\log p}, \quad k \in \mathbb{Z}. \quad (\text{AT5.5})$$

Both exponentials are roots of the same quadratic precisely when p^t is either p^s or p^{1-s} . The exponential with real logarithm has exactly the displayed periods, proving both directions. Thus the full receiver retains its two ordered phase factors, while their sum identifies the two roots; the involution and exponential periods are recorded by this observation map.

There is also an exact check of when the actual jet factors through the constant-trace fibre. At a coincident-root value, put $a = p^s = p^{1-s} = \sigma\sqrt{p}$, $q = \log p$, and let N be the retained nilpotent local coordinate. Formula AT5.2 gives

$$X - 2aI = aq^2N^2 \left(I + \frac{q^2}{12}N^2 + \frac{q^4}{360}N^4 + \dots \right). \quad (\text{AT5.6})$$

The sum stops when the powers of N vanish. Its parenthesized factor is invertible: it is $I + K$ with K nilpotent, whose inverse is the finite geometric series $I - K + K^2 - \dots$. Also $aq^2 \neq 0$. Consequently the trace is constant exactly when $N^2 = 0$. On a nonzero first-order jet the exact map from AT3.3 after complex scalar extension is $\varepsilon \mapsto aqN$, with inverse $N \mapsto \varepsilon/(aq)$; it sends $F = a + \varepsilon$ and $V = a - \varepsilon$ to the actual multiplier polynomials. For higher jets the displayed second-order term survives, and the full jet does not factor through that same constant-trace fibre. No existence or multiplicity of an actual zeta zero at these special values is assumed.

AT6. Relation to the earlier infinitesimal and integral questions

The integral adjoint algebra closes the weighted-adjoint operation without making every bare inverse integral. Its trace map then has the exact dual-number fibres AT3.3. The infinitesimal disappears under the trace while surviving faithfully in AT4.1, and positive star representations kill it through the explicit quotient AT4.4. These are completed calculations on specified objects, with their connecting maps.

For the original-zeta quotient, the exact receiving map is AT5.1–AT5.2. No identification of its unknown zero locations with the branch fibres is made. The full receiver retains both phase factors and every nilpotent coefficient; the scalar trace has the exact fibres AT5.5. The criterion AT5.4 retains the original prime norm, and proving it on every actual zero remains the purity problem. The nilpotent computation gives a usable representation and its precise positive reduction, not a sign change of the original Weil form by declaration.

Global joint traces and the complete original-zeta jets

24 September 2026. Independent derivation. Proof locators JTR0–JTR8.

This note proves exactly which information the ordered counting-adjoint operators and their trace coordinates retain, globally and at every actual nontrivial zero of the original zeta. All statements concern the complete indicated sets or spaces. No numerical range of zeros is tested. The supporting datum $\tau\langle Z_1; \text{no intrinsic } Z_2 \rangle$ receives no arithmetic coordinate, addition, or metric.

JTR0. Retained receiving objects and conventions

The source dependencies are ADJOINT_TRACE_AND_DUAL_FIBRES.md, AT1–AT6, read in full, and INTEGRAL_ADJOINT_RETURN_ALGEBRA.md, IAR6–IAR8, independently derived and proved

before this calculation. The latter receives the original quotient and weighted Weil adjoint from the linked task's IC2 and IC8. No additional source discovery or new cohomological identification is used here.

For each prime p , retain its positive real logarithm $q_p = \log p$, and define entire functions on the original complex s -plane:

$$F_p(s) = p^s = e^{q_p s}, \quad V_p(s) = p^{1-s} = p e^{-q_p s}, \quad x_p(s) = F_p(s) + V_p(s). \quad (\text{JTR0.1})$$

The ordered pair is retained; it is not replaced by its sum. Its product is exactly p . On the actual zero-jet quotient, these are the multipliers of T_p , $pT_{1/p}$, and their sum.

The holomorphic reflection and the anti-holomorphic Weil reflection are respectively

$$j(s) = 1 - s, \quad \iota(s) = 1 - \bar{s}. \quad (\text{JTR0.2})$$

They must not be interchanged: the trace functions satisfy

$$x_p(j(s)) = x_p(s), \quad x_p(\iota(s)) = \overline{x_p(s)}. \quad (\text{JTR0.3})$$

The first is a holomorphic two-root identification; the second includes conjugation. Both follow directly from (JTR0.1) and the real logarithm convention.

JTR1. The elementary independence of distinct prime periods

For distinct primes p, q ,

$$\frac{\log p}{\log q} \notin \mathbb{Q}. \quad (\text{JTR1.1})$$

Proof. Both logarithms are positive. A rational ratio would be a/b for positive integers a, b , so $b \log p = a \log q$, hence $p^b = q^a$. Divisibility by the distinct primes contradicts unique prime factorization. $\square$

In particular, for integers k, ℓ ,

$$\frac{k}{\log p} = \frac{\ell}{\log q} \implies k = \ell = 0. \quad (\text{JTR1.2})$$

If one integer is zero this is immediate; otherwise it would give a rational ratio in (JTR1.1). This exact relation is the only independence fact required below. No unproved algebraic independence of logarithms enters.

JTR2. Two complete ordered pairs separate every complex point

For a single prime, the precise fibre of the complete ordered pair is

$$(F_p(t), V_p(t)) = (F_p(s), V_p(s)) \iff t - s \in \frac{2\pi i}{\log p} \mathbb{Z}. \quad (\text{JTR2.1})$$

Proof. Equality of the first coordinates says $e^{q_p(t-s)} = 1$, equivalent to $q_p(t-s) = 2\pi i k$ for an integer k . Conversely such a period preserves the first coordinate and also its reciprocal times p , the second coordinate. $\square$

Consequently the entire map

$$\mathcal{F}_{p,q} : \mathbb{C} \longrightarrow (\mathbb{C}^\times)^4, \quad s \longmapsto (F_p(s), V_p(s), F_q(s), V_q(s)) \quad (\text{JTR2.2})$$

is injective for any two distinct primes. Equality at two points gives

$$t - s = \frac{2\pi i k}{\log p} = \frac{2\pi i \ell}{\log q},$$

and JTR1 forces both integers to be zero. This proof covers every complex s, t . It does not merely distinguish points in a selected strip or bounded height range.

The differential of even one first coordinate is $F'_p(s) = q_p p^s \neq 0$ everywhere. Thus there is also no local loss of an infinitesimal coordinate in the full pair. Its global one-prime periodic ambiguity and its local invertibility are different properties, with the exact fibre (JTR2.1) relating them.

JTR3. Three traces separate precisely the reflection orbits

For a single prime the complete trace fibre is

$$x_p(t) = x_p(s) \iff t = s + \frac{2\pi i k}{\log p} \text{ or } t = 1 - s + \frac{2\pi i k}{\log p}, \quad k \in \mathbb{Z}. \quad (\text{JTR3.1})$$

Proof. Both $F_p(s)$ and $V_p(s)$ are roots of $Y^2 - x_p(s)Y + p$. If the trace values agree, the same equation has $F_p(t)$ as a root. It follows that $F_p(t) = F_p(s)$ or $F_p(t) = V_p(s)$. Their exponential fibres give exactly the two alternatives. Each alternative conversely gives the same trace by substitution. Repeated roots cause no exception: then the two alternatives coincide modulo their displayed period. $\square$

Theorem. For any three distinct primes p, q, r ,

$$(x_p(t), x_q(t), x_r(t)) = (x_p(s), x_q(s), x_r(s)) \iff t = s \text{ or } t = 1 - s. \quad (\text{JTR3.2})$$

Proof. At each of the three primes, choose one of the valid alternatives in (JTR3.1). There are two types and three choices, so two distinct primes have the same type. If they both have the first type, their two periods give $t - s = 2\pi i k / \log p = 2\pi i \ell / \log q$; JTR1 gives $t = s$. If both have the second type, apply the same argument to $t - (1 - s)$ to obtain $t = 1 - s$. The converse follows from (JTR0.3). This argument still applies at coincident roots, because either of their valid types may be chosen. $\square$

Equivalently the trace map factors through $s \mapsto \{s, 1 - s\}$ and is injective on that orbit set. It does not pick one of the two points of an orbit. The two full ordered pairs in JTR2 retain precisely the branch information that this trace observation identifies.

Two traces alone can have further coincidences; the following exact family locates them rather than dropping them. For distinct primes p, q and nonzero integers k, ℓ , put

$$\begin{aligned} s &= \frac{1}{2} + \pi i \left(\frac{\ell}{\log q} - \frac{k}{\log p} \right), \\ t &= \frac{1}{2} + \pi i \left(\frac{\ell}{\log q} + \frac{k}{\log p} \right). \end{aligned} \quad (\text{JTR3.3})$$

Then $t = s + 2\pi i k / \log p$ and $t = 1 - s + 2\pi i \ell / \log q$, so both traces agree. Neither $t = s$ nor $t = 1 - s$ holds because the two integers are nonzero. This is an infinite symbolic family, not a numerical search or a claim about actual zeta zeros.

More generally, any coincidence of two traces outside the two reflection alternatives must use different types for the two primes in (JTR3.1). In that case taking real parts gives both $\Re t = \Re s$ and $\Re t = 1 - \Re s$, so

both points lie on $\Re s = 1/2$. Thus the extra ambiguity of two traces has an exact location. The third trace removes it globally by the proved orientation argument.

JTR4. The only common ramification point of two trace coordinates

Differentiate the complete original trace:

$$x'_p(s) = q_p(p^s - p^{1-s}). \quad (\text{JTR4.1})$$

Since $q_p > 0$, it vanishes exactly when $p^{2s-1} = 1$. Therefore

$$x'_p(s) = 0 \iff s = \frac{1}{2} + \frac{\pi i k}{\log p}, \quad k \in \mathbb{Z}. \quad (\text{JTR4.2})$$

For any distinct primes p, q , simultaneous vanishing requires $k/\log p = \ell/\log q$, so JTR1 gives

$$\boxed{x'_p(s) = x'_q(s) = 0 \iff s = \frac{1}{2}.} \quad (\text{JTR4.3})$$

This is the unique common ramification point of the joint local trace observation. It is a statement about the actual functions on the original s -plane, without assigning a coordinate to the supporting point tau.

JTR5. The original zeta is nonzero at the common ramification point

Retain the exact alternating-series factor, rather than substituting another function for the original zeta. On $\Re s > 0$, define

$$\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}. \quad (\text{JTR5.1})$$

This series defines a holomorphic function there. For a direct convergence proof, the partial sums of the alternating coefficients of any tail have absolute value at most one. Summation by parts and

$$|n^{-s} - (n+1)^{-s}| \leq |s| \int_n^{n+1} x^{-\Re s-1} dx$$

bound the infinite tail after N , on any compact set with $\Re s \geq \delta > 0$ and $|s| \leq C$, by $(C/\delta)(N+1)^{-\delta}$. A terminal finite-sum boundary tends to zero. Thus the convergence is locally uniform, and the terms are holomorphic.

On the half-plane of absolute convergence, separate the original odd and even terms:

$$\begin{aligned} \eta(s) &= \sum_{n \geq 1} n^{-s} - 2 \sum_{n \geq 1} (2n)^{-s} \\ &= (1 - 2^{1-s}) \zeta(s), \quad \Re s > 1. \end{aligned} \quad (\text{JTR5.2})$$

The same equality holds meromorphically on $\Re s > 0$ by uniqueness of continuation of the original zeta. Its entire multiplier $1 - 2^{1-s}$ is retained; this is not an assertion that η and ζ have the same global zero or pole divisor. At $s = 1/2$ the multiplier is $1 - \sqrt{2} \neq 0$, so inversion at that specific point is valid.

Pair consecutive terms at this real argument. Every pair is positive, and the convergence just proved permits the consecutive pairing:

$$\eta(1/2) = \sum_{n \geq 1} ((2n-1)^{-1/2} - (2n)^{-1/2}) \geq 1 - 2^{-1/2} > 0. \quad (\text{JTR5.3})$$

Consequently

$$\boxed{\zeta(1/2) = \frac{\eta(1/2)}{1 - \sqrt{2}} < 0,} \quad (\text{JTR5.4})$$

in particular it is not zero. This proof uses the original factor and no assumption about the nontrivial zeros. The same argument, with exponent any real $x \in (0, 1)$, proves $\zeta(x) < 0$: the paired numerator is positive and $1 - 2^{1-x} < 0$. No general division by the factor at its other zeros is made.

Combining this with JTR4, every actual nontrivial zero ρ has at least one unramified trace coordinate among the fixed primes 2 and 3:

$$\boxed{\text{For every actual } \rho, \quad x'_2(\rho) \neq 0 \text{ or } x'_3(\rho) \neq 0.} \quad (\text{JTR5.5})$$

The assertion is global on the entire actual-zero set and retains all multiplicities.

JTR6. Exact recovery of every finite zero jet

Let ρ be an actual nontrivial zero of multiplicity $m = m_\rho$. Its retained local algebra is

$$A_\rho = \mathbb{C}[h]/(h^m), \quad s = \rho + h. \quad (\text{JTR6.1})$$

This is the full germ receiver, not only a formal upper bound for its possible values. The quotient $\mathcal{Q} = \mathcal{B}/\mathcal{I}$ has the well-defined map $j_\rho[f] = \sum_{j=0}^{m-1} f^{(j)}(\rho)h^j/j!$. For any desired polynomial $q(h)$ of degree less than m , let $P(h)$ be the truncation of $e^{-h^2}q(h)$ modulo h^m . Then

$$f(s) = e^{(s-\rho)^2}P(s-\rho)$$

is entire and rapidly decreasing vertically on every bounded real strip, and $j_\rho[f] = q(h)$. Thus the receiver onto A_ρ is surjective. The joint map over all actual zeros is injective on $\mathcal{Q}$, because its kernel is exactly the defining ideal $\mathcal{I}$; this does not assert unrestricted surjectivity onto an infinite product.

Choose $p \in \{2, 3\}$ for which JTR5.5 gives $x'_p(\rho) \neq 0$. Set

$$u_p(h) = x_p(\rho + h) - x_p(\rho) = \sum_{j=1}^{m-1} a_j h^j, \quad a_j = \frac{q_p^j}{j!} (p^\rho + (-1)^j p^{1-\rho}), \quad a_1 \neq 0. \quad (\text{JTR6.2})$$

All terms are evaluated in A_ρ ; this is the actual trace operator from AT5.2 with its scalar constant subtracted, not an assumed formal surrogate.

Theorem. The substitution map

$$\boxed{\mathbb{C}[u]/(u^m) \longrightarrow A_\rho, \quad u \longmapsto u_p(h),} \quad (\text{JTR6.3})$$

is an algebra isomorphism, with a fully specified truncated inverse.

Proof. For each $0 \leq j < m$, the polynomial $u_p(h)^j$ has first nonzero term $a_1^j h^j$, with the convention $j = 0$ giving one. Relative to the bases $1, u, \dots, u^{m-1}$ and $1, h, \dots, h^{m-1}$, the substitution matrix is triangular with nonzero diagonal

$$1, a_1, a_1^2, \dots, a_1^{m-1}.$$

It is invertible. Substitution preserves products and unit, so this is an algebra isomorphism.

For its exact inverse, put $h = g(u) = \sum_{n=1}^{m-1} b_n u^n$. For $m \geq 2$, define recursively

$$b_1 = a_1^{-1}, \quad b_n = -a_1^{-1} \sum_{k=2}^n a_k \sum_{\substack{j_1, \dots, j_k \geq 1 \\ j_1 + \dots + j_k = n}} b_{j_1} \cdots b_{j_k} \quad (2 \leq n < m). \quad (\text{JTR6.4})$$

Every index in the inner sum is at most $n - 1$, so the recursion uses only coefficients already defined. The coefficient of u^n in $u_p(g(u))$ is $a_1 b_n$ plus precisely the sum cancelled by this formula. Thus $u_p(g(u)) = u$ modulo u^m . The other composition is the identity as well: for nilpotent inputs a, b , the difference $u_p(a) - u_p(b)$ equals $(a - b)$ times a polynomial with constant term a_1 , hence a unit. Applying this to the two compositions proves $g(u_p(h)) = h$ modulo h^m . For $m = 1$, both algebras are just $\mathbb{C}$, and the inverse contains no nonconstant term. $\square$

In operator form, with N_ρ multiplication by h , this is the exact identity

$$\boxed{N_\rho = g(X_p|_\rho - x_p(\rho)I)}. \quad (\text{JTR6.5})$$

The trace operator therefore retains the entire local multiplicity algebra, including every nilpotent coefficient, once an unramified coordinate has been selected. The pair of primes 2, 3 supplies such a selection at every actual zero; there is no common first-order loss on the actual-zero set.

There remains a precise global branch distinction. The trace functions identify ρ and $1 - \rho$. Their local comparison sends the coordinate h at one point to $-h$ at the other:

$$x_p(1 - \rho + h) = x_p(\rho - h). \quad (\text{JTR6.6})$$

Thus trace data retain full jets on each chosen branch, while their global observation identifies the two reflected branches by this explicit isomorphism. The complete ordered pairs distinguish the branches by JTR2. This does not assert that scalar trace values independently prescribe arbitrary data at both identified points.

JTR7. The completed local observation at the common point

Although it is not a zero of the original zeta, the unique common ramification point has an exact local observation algebra. Put $s = 1/2 + h$ and work in the complete local ring $\mathbb{C}[[h]]$. Then

$$\begin{aligned} F_p &= \sqrt{p} e^{q_p h}, \\ V_p &= \sqrt{p} e^{-q_p h}, \\ u_p &:= x_p - 2\sqrt{p} = 2\sqrt{p} \sum_{j \geq 1} \frac{q_p^{2j}}{(2j)!} h^{2j}. \end{aligned} \quad (\text{JTR7.1})$$

All constants and powers of $\log p$ are retained. Define the completed trace-observation algebra to be the closure, in the (h) -adic topology, of the $\mathbb{C}$ -subalgebra generated by the u_p for all primes.

Theorem. This completed observation algebra is exactly

$$\boxed{\mathbb{C}[[h^2]] \hookrightarrow \mathbb{C}[[h]]}. \quad (\text{JTR7.2})$$

In fact the closure of the subalgebra from any one prime already equals $\mathbb{C}[[h^2]]$.

Proof. Each u_p is even, so every polynomial in these elements and every formal limit has only even powers of h . This proves containment in $\mathbb{C}[[h^2]]$, a closed subalgebra. Conversely put $v = h^2$. For each prime,

$$u_p = \sqrt{p} q_p^2 v + 2\sqrt{p} \sum_{j \geq 2} \frac{q_p^{2j}}{(2j)!} v^j. \quad (\text{JTR7.3})$$

Its linear coefficient in v is nonzero. The coefficient recursion in JTR6, now continued at every degree, constructs a unique inverse formal series $v = g_p(u_p)$. Every coefficient requires only finitely many earlier coefficients, so this defines a formal inverse without an analytic convergence assumption. Hence v is in the completed algebra generated by u_p , as is every series in v . This proves equality. $\square$

The holomorphic sign involution is

$$j^* f(h) = f(-h).$$

Its fixed subalgebra is exactly $\mathbb{C}[[h^2]]$, since equality of coefficients forces every odd coefficient to vanish. More completely,

$$\mathbb{C}[[h]] = \mathbb{C}[[h^2]] \oplus h\mathbb{C}[[h^2]], \quad (\text{JTR7.4})$$

with the involution acting by $+1$ and -1 on the two summands. The two projections are exactly $(f(h) + f(-h))/2$ and $(f(h) - f(-h))/2$. This is a decomposition of the local power-series ring, not a parity assignment to the supporting datum τ .

The weighted adjoint involution on complex germs includes conjugation as well:

$$f(h) = \sum_j a_j h^j \mapsto \sum_j (-1)^j \bar{a}_j h^j. \quad (\text{JTR7.5})$$

It exchanges F_p, V_p and fixes the real-coefficient trace. Its real fixed set should not be confused with the complex fixed subalgebra of the holomorphic sign involution in (JTR7.4).

At any finite jet length m , the trace image is the exact even subalgebra

$$\mathbb{C}[v]/(v^{\lceil m/2 \rceil}) \hookrightarrow \mathbb{C}[h]/(h^m), \quad v \mapsto h^2. \quad (\text{JTR7.6})$$

Its kernel before dividing by the displayed power is exactly $(v^{\lceil m/2 \rceil})$, because h^{2j} is nonzero precisely for $2j < m$. This records the entire finite loss at the exceptional point, not only its first-order term.

The odd information is preserved by the full pair. For example

$$F_p - V_p = 2\sqrt{p} \sum_{j \geq 0} \frac{q_p^{2j+1}}{(2j+1)!} h^{2j+1} \quad (\text{JTR7.7})$$

has a nonzero linear coefficient. Even the first coordinate gives the exact formal inverse

$$h = \frac{1}{q_p} \sum_{j \geq 1} \frac{(-1)^{j+1}}{j} \left(\frac{F_p - \sqrt{p}}{\sqrt{p}} \right)^j. \quad (\text{JTR7.8})$$

This is the formal logarithm of the original exponential, with its constant $\sqrt{p}$ and factor q_p explicitly retained. It is well defined because its input has zero constant term. It proves that retaining the full operator coordinate recovers $\mathbb{C}[[h]]$, and reduction modulo h^m gives its finite-jet version.

JTR8. Consequence and exact limits of the joint observation

The global and local maps proved here are:

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{\mathcal{F}_{2,3}} & (\mathbb{C}^\times)^4, \quad \text{injective with all local jets retained;} \\ \mathbb{C}/\{s \sim 1-s\} & \xrightarrow{(x_2, x_3, x_5)} & \mathbb{C}^3, \quad \text{injective on reflection orbits;} \\ \mathbb{C}[h]/(h^{m_\rho}) & \longleftrightarrow & \mathbb{C}[u_p]/(u_p^{m_\rho}), \quad p \in \{2, 3\} \text{ chosen with } x'_p(\rho) \neq 0, \end{array} \quad (\text{JTR8.1})$$

where the third line applies to every actual original-zeta zero and has the exact inverse coefficients (JTR6.4). These statements do not replace the full arithmetic spectrum by three primes; they prove that these particular coordinates separate the indicated parameters within the complete receiving construction. Every prime and its original return data remain in that construction.

At $s = 1/2$, the complete trace observation is the even subalgebra and the full pair retains the odd part. The original zeta is not zero there, so this common ramification does not erase a jet of any actual nontrivial zero. Individual trace coordinates may ramify at other points; another coordinate among the two fixed primes repairs that loss by the proved local isomorphism.

The results provide global point separation and exact preservation of the finite multiplicity algebras. They do not assert that every arbitrary infinite list of jets is represented by a test function, nor that trace values choose a branch within a reflected orbit. They supply the precise maps that retain or identify those data.

Finally, the purity criterion from AT5 remains an additional mathematical statement to establish on the actual spectrum:

$$x_p(\rho) \in [-2\sqrt{p}, 2\sqrt{p}] \subset \mathbb{R}.$$

The joint observation theorems do not assume this inclusion or infer it merely from injectivity. They show that its target contains faithful local arithmetic data and a globally specified reflection identification, while preserving the ordered operator pairs and all phases necessary for the subsequent original Weil calculation. No global positivity theorem or RH conclusion is claimed.

Retaining support at every zero coefficient

24 September 2026. ZR1-ZR5. This corrects the support-retention wording of M9.7 and S7 while retaining their linear isomorphisms and synthesis theorem. The correction was supplied by the receiving task in ITEM26_GLOBAL_RECEIVERS_INTEGRATION.md, I26.10; its complete argument is reproduced and propagated here. The record set below is the programme's existing set of labels, not an arithmetic value assigned to an unsupported element.

ZR1. Exact domain of the linear statement

Let $\mathcal{Q} = \mathcal{B}/\mathcal{J}$ be the actual original-zeta quotient, and let $J : \mathcal{Q} \rightarrow \prod_{\rho} \mathbb{C}^{m_{\rho}}$ be the injective map of all zero jets proved in S7. Let $\mathcal{L}$ be the retained set of support records. Write e_{ℓ} for the basis of the free vector space $\mathbb{C}^{(\mathcal{L})}$. The linear comparison

$$J \otimes \text{id} : \mathcal{Q} \otimes \mathbb{C}^{(\mathcal{L})} \longrightarrow \left(\prod_{\rho} \mathbb{C}^{m_{\rho}} \right) \otimes \mathbb{C}^{(\mathcal{L})} \quad (\text{ZR1.1})$$

is injective: every tensor has a finite expansion $\sum q_{\ell} \otimes e_{\ell}$, and the coordinate functional at each e_{ℓ} makes a zero image imply $Jq_{\ell} = 0$, hence $q_{\ell} = 0$. This proves precisely the linear assertion. However,

$$0_{\mathcal{Q}} \otimes e_{\ell} = 0_{\mathcal{Q}} \otimes e_{\ell'} = 0 \quad (\text{ZR1.2})$$

for every pair of records. The passage from a labelled coefficient to a tensor has already identified its zero sections. Injectivity of ZR1.1 does not undo that earlier map.

ZR2. Exact record-retaining quotient and observation

Retain the disjoint family

$$\widehat{\mathcal{Q}} = \coprod_{\ell \in \mathcal{L}} \{\ell\} \times \mathcal{Q}. \quad (\text{ZR2.1})$$

It is equivalently the quotient of $\mathcal{L} \times \mathcal{B}$ by $(\ell, f) \sim (\ell', g)$ exactly when $\ell = \ell'$ and $f - g \in \mathcal{J}$. This is an equivalence relation since $\mathcal{J}$ is a vector subspace. The map to ZR2.1 is $(\ell, f) \mapsto (\ell, [f])$; its fibres are exactly that relation, proving the comparison. All subtractions occur in the coefficient functions. No addition is defined on $\mathcal{L}$.

The complete observation is

$$\widehat{J}(\ell, q) = (\ell, Jq). \quad (\text{ZR2.2})$$

If its two values agree, their first coordinates give equal records and its second coordinates give equal coefficient classes by injectivity of J . Thus $\widehat{J}$ is injective. The fibre of the coefficient-only zero observation is exactly

$$\coprod_{\ell \in \mathcal{L}} \{(\ell, 0_{\mathcal{Q}})\}. \quad (\text{ZR2.3})$$

It contains every original record, including those that must distinguish absence from a supported zero.

ZR3. The map to the previous tensors, with its full fibres

Define

$$c : \widehat{\mathcal{Q}} \longrightarrow \mathcal{Q} \otimes \mathbb{C}^{(\mathcal{L})}, \quad c(\ell, q) = q \otimes e_{\ell}. \quad (\text{ZR3.1})$$

Its zero fibre is exactly ZR2.3, because the coordinate functional at ℓ recovers q . If its common output is nonzero, equality of two images with different labels is impossible: the coordinate functional at the first label would give $q = 0$. Equality with the same label gives equal coefficients. Thus every nonzero fibre in its image is a singleton; tensors with several nonzero label coordinates are outside its image. This states every fibre without asserting that the map is surjective. The square with $\widehat{J}$, c , and $J \otimes \text{id}$ commutes by evaluation on (ℓ, q) .

ZR4. Arithmetic action, projectors, and the full pairing

Every existing coefficient operator $A : \mathcal{Q} \rightarrow \mathcal{Q}$ lifts as

$$\widehat{A}(\ell, q) = (\ell, Aq). \quad (\text{ZR4.1})$$

It satisfies $\widehat{AB} = \widehat{A}\widehat{B}$, $\widehat{\text{id}} = \text{id}$, and $c\widehat{A} = (A \otimes \text{id})c$ by direct substitution. This applies to all actual prime operators, every IAR coefficient operator, and every zero projector. Addition of operators remains fibrewise

addition on $\mathcal{Q}$; it is not an operation on the primitive records. An operator sending a coefficient to zero sends (ℓ, q) to $(\ell, 0)$, retaining its record.

The full Weil pairing has the record-preserving receiver

$$((\ell, x), (\ell', y)) \mapsto (\ell, \ell', W(x, y)). \quad (\text{ZR4.2})$$

Projection to the last coordinate is exactly the previous form, with all its endpoint, Gamma, prime and original-zero contributions unchanged. Its weighted-adjoint identity is therefore the same equality of last coordinates with the same two retained labels. A vanishing form value deletes neither input record. No sign or positivity is inferred from this recording map.

ZR5. The two coefficient sections at zero

The formal coefficient algebra C_{rec} of M9.7 has independent basis elements $U = [\tau]$ and $f = [1]$. Its two linear sections $q \mapsto q \otimes U$ and $q \mapsto q \otimes f$ are distinct maps, but both take zero to zero. Their linear difference is injective, as M9.7 proves; that still does not distinguish which zero section was selected.

To retain that already specified selection, use the record set $\mathcal{L} \times \{U, f\}$ and apply ZR2. The maps from the two selected families to the previous coefficient receiver are exactly

$$(\ell, U, q) \mapsto q \otimes U \otimes e_\ell, \quad (\ell, f, q) \mapsto q \otimes f \otimes e_\ell. \quad (\text{ZR5.1})$$

For nonzero coefficients their images are distinct by the independent basis coordinates. At zero the original records remain distinct in the domain and their comparison images coincide. This proves both the retained distinction and the specific loss under linear observation. It assigns no source sum, difference, parity or numerical value to τ .

Complete receiving-source dependencies

Global synthesis for the original-zeta Mellin quotient

Independent derivation, 24 September 2026. This note proves the equality left open in M9 of PRIME_MO NODROMY_STACKED_HISTORY.md. It uses no RH assumption, no simplicity assumption, no bounded numerical experiment, and no assumed lower bound for zeta inside the critical strip. Addition at source tau is not used. The user's state-layer notation is unchanged.

Source identity and actual reading

The existing machine-readable corpus index was queried before consulting further literature: `canonicalresearchliteratureindex(privateroutingdatabase)`, table `canonical_units`, for spectral/zeta, spectral synthesis, Hadamard, and explicit-formula routing. Routing records were not treated as proofs. The relevant retained original author source is Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, original TeX sources/2602.04022v1/author/rhready.tex, SHA256 7f1888d82b42263faca4264f3f9fc77e5750c348640f85db0ac4569fe88ebab2. The author identity and version were checked against arXiv. Original TeX lines 501–546 and 1570–1600 were read; the theorem actually used is Hadamard factorization at lines 526–535. The source's subsequent symmetric-product presentation is not substituted for the full genus-one product used below. Its historical attribution is J. Hadamard, *Essai sur l'étude des fonctions données par leur développement de Taylor*, J. Math. Pures Appl. 4 8 1892, 101–186. That historical article was not newly read in this derivation.

The retained programme sources read here are M8–M9 in full, with the preceding M6–M7 context, and the C6 kernel formula cited there. The full original-zeta multiplier is preserved below, including its factor 1/8. Hadamard factorization is the standard complex-analysis theorem used; the division estimate needed for this particular space is proved here instead of assumed.

S1. The exact function spaces and Mellin isomorphism

Write

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) := \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{S1.1})$$

All indices in these seminorms are nonnegative integers. In the logarithmic coordinate $x = \log u$, put $K(x) = k(e^x)$. The topology is equivalently given by

$$\|K\|_{N,j} = \sup_{x \in \mathbb{R}} e^{N|x|} |K^{(j)}(x)|. \quad (\text{S1.2})$$

The equivalence follows from

$$e^{N|x|} \leq e^{Nx} + e^{-Nx} \leq 2e^{N|x|}.$$

Let $\mathcal{B}$ be the space of entire functions F for which

$$b_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty \quad (A, M \geq 0). \quad (\text{S1.3})$$

The centered Mellin map

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u} = \int_{\mathbb{R}} K(x) e^{(s-1/2)x} dx \quad (\text{S1.4})$$

is a linear topological isomorphism $\mathcal{A} \rightarrow \mathcal{B}$.

Here are the estimates and inverse. On each fixed real strip the integral and every derivative in s are absolutely convergent, because an arbitrary polynomial in x times $e^{(\sigma-1/2)x}$ is controlled by a larger seminorm (S1.2). Repeated integration by parts in x , applied to $e^{(\sigma-1/2)x} K(x)$, proves (S1.3), uniformly in the strip. Boundary terms vanish by (S1.2). This proves continuity of the map.

For $F \in \mathcal{B}$, define

$$K(x) = \frac{1}{2\pi} \int_{\mathbb{R}} F(1/2 + it) e^{-itx} dt. \quad (\text{S1.5})$$

For any real c , contour shifting in a finite vertical strip gives

$$K(x) = \frac{e^{-(c-1/2)x}}{2\pi} \int_{\mathbb{R}} F(c + it) e^{-itx} dt. \quad (\text{S1.6})$$

Indeed, the horizontal sides of the rectangle for $F(s)e^{-(s-1/2)x}$ tend to zero by (S1.3), uniformly between its two vertical sides. Differentiation j times inserts the exact factor $-(c-1/2+it)^j$ in the integral. For $x \geq 0$ choose $c = 1/2 + N + 1$, and for $x \leq 0$ choose $c = 1/2 - N - 1$. Integrability with $M > j + 1$ proves (S1.2) and continuity of the inverse. Fourier inversion gives (S1.4) on the line $c = 1/2$; the identity theorem gives it everywhere. The sign and constant in (S1.5) are fixed throughout.

Both spaces are complete metrizable locally convex spaces. For $\mathcal{A}$, a Cauchy sequence in all seminorms converges in C^∞ on each compact interval; the weighted uniform limits give the same limit with every seminorm finite and convergence in every seminorm. The corresponding assertion for $\mathcal{B}$ follows from this proved isomorphism (or from locally uniform holomorphic convergence together with the strip bounds).

S2. One retained source kernel, with every factor

Set

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad k_0(u) = \mathcal{E} f_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu). \quad (\text{S2.1})$$

The even Schwartz function f_0 satisfies $f_0(0) = 0$ and $\int_{\mathbb{R}} f_0 = 0$. The Gaussian moments give the latter directly:

$$\int v^2 e^{-\pi v^2} dv = \frac{1}{2\pi}, \quad \int v^4 e^{-\pi v^2} dv = \frac{3}{4\pi^2}, \quad \frac{\pi}{2} \left(2\pi \frac{3}{4\pi^2} - 3 \frac{1}{2\pi} \right) = 0.$$

For the Fourier convention $\hat{f}(t) = \int f(v)e^{-2\pi i vt} dv$, the Gaussian identity and differentiation give

$$\widehat{v^2 e^{-\pi v^2}}(t) = \left(\frac{1}{2\pi} - t^2\right) e^{-\pi t^2},$$

$$\widehat{v^4 e^{-\pi v^2}}(t) = \left(t^4 - \frac{3}{\pi}t^2 + \frac{3}{4\pi^2}\right) e^{-\pi t^2}.$$

Substituting into (S2.1) proves $\hat{f}_0 = f_0$. Poisson summation, with both endpoint terms zero, consequently proves $k_0(u) = k_0(1/u)$. It also proves $k_0 \in \mathcal{A}$, as in M9.

For $\Re s > 1$, the absolutely convergent Mellin calculation is

$$F_0(s) := \mathcal{M}k_0(s) = \zeta(s) \int_0^\infty f_0(v) v^s \frac{dv}{v} = \boxed{\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s)}. \quad (\text{S2.2})$$

For precision, the integral before the last equality equals

$$\frac{\pi}{2} \left[2\pi \cdot \frac{1}{2} \pi^{-(s+4)/2} \Gamma((s+4)/2) - 3 \cdot \frac{1}{2} \pi^{-(s+2)/2} \Gamma((s+2)/2) \right].$$

Applying $\Gamma(z+1) = z\Gamma(z)$ twice gives exactly the factor in (S2.2). Thus $F_0 = \xi/4$ for the source's definition of ξ ; no factor is absorbed. The Mellin integral makes F_0 entire, and $k_0(u) = k_0(1/u)$ gives $F_0(s) = F_0(1-s)$.

The apparent endpoint and trivial-zero singularities in the displayed multiplier have the following explicit values:

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2m) = F_0(1+2m) = \frac{(1+2m)(2m)}{8} \pi^{-(1+2m)/2} \Gamma((1+2m)/2) \zeta(1+2m) \neq 0 \quad (m \geq 1). \quad (\text{S2.3})$$

At 1, the zeta residue is 1 and $\pi^{-1/2} \Gamma(1/2) = 1$; the value at 0 follows also by the functional equation. At every negative even integer the reflection gives the positive, finite expression in (S2.3). The Gamma factor has no zeros. Consequently the zeros of F_0 , with their orders, are exactly the nontrivial zeros of the ORIGINAL zeta function. Trivial zeros are cancelled in this auxiliary transform, with their values explicitly retained in (S2.3); S6 below recovers their full derivative data in the approximating source functions.

We also need the order of F_0 , for which no zero-distribution hypothesis is required. There exist $B, c, C > 0$ such that

$$|k_0(u)| \leq C u^B e^{-cu^2} \quad (u \geq 1).$$

This follows by summing the Gaussian series (S2.1), retaining a smaller positive Gaussian exponent to absorb its polynomial factors. Reflection supplies the other endpoint. Thus for $|s| \leq R$,

$$|F_0(s)| \leq 2C \int_1^\infty u^{B+R+1/2} e^{-cu^2} \frac{du}{u} \leq \exp(C_1(R+2) \log(R+2)). \quad (\text{S2.4})$$

The last bound follows by the Gamma integral after $y = cu^2$; alternatively its maximum at y proportional to $R+2$ and the remaining exponential tail give the same bound. Thus F_0 has order at most 1. It is not zero, since $F_0(2) = \pi/24$.

Jensen's formula at radius $(2R)$, using $F_0(0) = 1/8 \neq 0$, now gives a constant C_2 such that the zero count, with multiplicities, obeys

$$n(R) \leq C_2(R+2)^{3/2} \quad (R \geq 1). \quad (\text{S2.5})$$

Indeed each zero of modulus at most R contributes at least $\log 2$ to Jensen's sum at $(2R)$, and (S2.4) is bounded by a constant times $(R+2)^{3/2}$. Hadamard's factorization theorem in the retained original TeX then gives

$$F_0(s) = e^{a+bs} \prod_p \left(1 - \frac{s}{\rho}\right) e^{s/\rho}, \quad e^a = \frac{1}{8}, \quad b = \frac{F_0'(0)}{F_0(0)}. \quad (\text{S2.6})$$

The product retains every repeated zero and every exponential factor. The value of b is exactly $\frac{1}{2} \log(4\pi) - 1 - \frac{1}{2}\gamma$, as follows by expanding all factors of (S2.2) at 0; no value of b is needed to discard a term below. Formula (S2.5) implies convergence of $\sum_\rho |\rho|^{-2}$, so the product in (S2.6) converges absolutely on compact subsets after its genus-one factors are grouped as displayed.

S3. Global division with a proved subquadratic growth bound

Let $F \in \mathcal{B}$ vanish at every zero ρ of F_0 to at least the exact multiplicity m_ρ . Then

$$Q(s) = F(s)/F_0(s) \quad (\text{S3.1})$$

extends to an entire function. This follows from Taylor division at each zero; the extensions agree with the quotient on punctured neighborhoods.

For every $A \geq 0$ there are constants $C_{A,F}, C > 0$ such that

$$\sup_{|\sigma| \leq A} |Q(\sigma + it)| \leq C_{A,F} \exp(C(|t| + 2)^{3/2} \log(|t| + 2)), \quad t \in \mathbb{R}. \quad (\text{S3.2})$$

The supremum is over the real coordinate at the fixed imaginary coordinate t . This incorporates the independently verified notation correction in GLOBAL_MELLIN_SYNTHESIS_REVIEW.md R1.1. The exponent is strictly subquadratic. This is a statement on the whole unbounded strip, not a finite interval estimate.

We prove the bound including neighborhoods of arbitrarily close or multiple zeros. Fix a large R . Around each zero with $|\rho| \leq 8R$, take the disk of radius R^{-2} , and let D_R be their union. By (S2.5) the sum of the radii, counted with multiplicity if necessary, is at most $CR^{-1/2}$. For sufficiently large R , each connected component of this finite union has diameter less than 1: a chain of intersecting disks joining two points has total diameter at most twice the sum of the radii in that chain.

For $R/2 \leq |s| \leq 3R$ outside D_R , use (S2.6). For the factors with $|\rho| \leq 8R$,

$$\left| 1 - \frac{s}{\rho} \right| = \frac{|s - \rho|}{|\rho|} \geq \frac{1}{8R^3}. \quad (\text{S3.3})$$

Their logarithms contribute at least $-CR^{3/2} \log R$. The exponential parts contribute at least

$$-|s| \sum_{|\rho| \leq 8R} |\rho|^{-1} \geq -CR^{3/2}. \quad (\text{S3.4})$$

The last inequality follows by partial summation of (S2.5); the finitely many small zeros are harmless because $F_0(0) \neq 0$, so their moduli have a positive lower bound.

For $|\rho| > 8R$, the ratio $w = s/\rho$ has $|w| \leq 3/8$. The absolutely convergent logarithmic series gives

$$\log |(1 - w)e^w| = \Re \left(- \sum_{j \geq 2} \frac{w^j}{j} \right) \geq -C|w|^2.$$

Partial summation again gives

$$\sum_{|\rho| > 8R} |\rho|^{-2} \leq CR^{-1/2}.$$

Therefore the tail contributes at least $-CR^{3/2}$. Finally e^{a+bs} contributes at least $-|a| - 3|b|R$. Altogether,

$$|F_0(s)| \geq \exp(-CR^{3/2} \log R) \quad (R/2 \leq |s| \leq 3R, s \notin D_R). \quad (\text{S3.5})$$

Now take $R \leq |s| \leq 2R$ with $|\Re s| \leq A$. Outside D_R , (S3.5) and the boundedness of F on the strip give the desired bound for Q . Inside D_R , the whole component containing s stays within distance 1 of s . Its boundary lies in the strip $|\Re z| \leq A + 1$, in the larger annulus used in (S3.5), and outside the open disk union. The same bound holds for Q on that boundary. Since Q is holomorphic even at all enclosed zeros, the maximum-modulus principle gives the bound inside. No zero spacing or simplicity has been assumed. Taking dyadic R proves (S3.2); the remaining compact region is bounded by continuity.

S4. Gaussian division produces actual source functions

For every $\varepsilon > 0$, define the entire function

$$Q_\varepsilon(s) = e^{\varepsilon(s-1/2)^2} Q(s). \quad (\text{S4.1})$$

On the strip $|\Re s| \leq A$,

$$|e^{\varepsilon(s-1/2)^2}| \leq e^{\varepsilon(A+1/2)^2} e^{-\varepsilon(\Im s)^2}.$$

Together with (S3.2), this proves $Q_\varepsilon \in \mathcal{B}$: for every fixed strip and every polynomial power, the negative quadratic dominates $C|t|^{3/2} \log |t|$. The source preimage $q_\varepsilon \in \mathcal{A}$ is therefore explicitly

$$q_\varepsilon(e^x) = \frac{1}{2\pi} \int_{\mathbb{R}} Q_\varepsilon(1/2 + it) e^{-itx} dt. \quad (\text{S4.2})$$

Let multiplicative convolution be

$$(k * q)(u) = \int_0^\infty k(u/a) q(a) \frac{da}{a}. \quad (\text{S4.3})$$

All such integrals and their logarithmic derivatives converge in $\mathcal{A}$, because in logarithmic coordinate

$$e^{N|x|} \leq e^{N|x-y|} e^{N|y|}$$

and the second factor is integrable against a seminorm with one additional exponential weight. The Mellin transform of (S4.3) is the product of Mellin transforms, with Fubini justified by the same bounds.

Define, retaining the full unitary factor of the source dilation,

$$U_a f(v) = a^{-1/2} f(v/a), \quad f_\varepsilon(v) = \int_0^\infty q_\varepsilon(a) U_a f_0(v) \frac{da}{a}. \quad (\text{S4.4})$$

This is an actual even Schwartz function. For a Schwartz seminorm $\sup_v |v|^r |\partial_v^j f(v)|$, substitution $v = aw$ bounds the corresponding seminorm of $U_a f_0$ by

$$a^{r-j-1/2} \sup_w |w|^r |f_0^{(j)}(w)|.$$

Every such power is integrable against $q_\varepsilon(a) da/a$, by its arbitrary endpoint decay. This proves convergence in every Schwartz seminorm and permits all required differentiations. Evenness is preserved. Moreover

$$f_\varepsilon(0) = 0, \quad \int_{\mathbb{R}} f_\varepsilon(v) dv = \int_0^\infty q_\varepsilon(a) a^{1/2} \frac{da}{a} \int_{\mathbb{R}} f_0(w) dw = 0. \quad (\text{S4.5})$$

Absolute convergence follows also from $\|U_a f_0\|_1 = a^{1/2} \|f_0\|_1$. Thus $f_\varepsilon \in \mathcal{S}_0^{\text{even}}$, the actual source space in M9.

The continuity $\mathcal{E} : \mathcal{S}_0^{\text{even}} \rightarrow \mathcal{A}$ follows directly from the estimates already used in M9. More explicitly, at $u \geq 1$ the differentiated series is bounded using a sufficiently high Schwartz seminorm and the convergent sum $\sum n^{-M}$, $M > 1$; at $u \leq 1$, Poisson gives $\mathcal{E} f(u) = \mathcal{E} \hat{f}(1/u)$, and Fourier transformation is continuous on Schwartz space. Consequently (S4.4), or direct absolute summation, gives the exact receiver identity

$$\boxed{\mathcal{E} f_\varepsilon = k_0 * q_\varepsilon.} \quad (\text{S4.6})$$

It has Mellin transform

$$\mathcal{M} \mathcal{E} f_\varepsilon(s) = F_0(s) Q_\varepsilon(s) = e^{\varepsilon(s-1/2)^2} F(s). \quad (\text{S4.7})$$

Every source, domain and multiplier in this construction is specified; the argument has not inserted a hypothetical division preimage.

S5. Exact global closure equality

Let

$$\mathcal{J}_\zeta = \{k \in \mathcal{A} : (\mathcal{M}k)^{(j)}(\rho) = 0 \text{ for every nontrivial zero } \rho \text{ and every } 0 \leq j < m_\rho\}. \quad (\text{S5.1})$$

Then

$$\overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})^{\mathcal{A}}} = \mathcal{J}_\zeta. \quad (\text{S5.2})$$

The inclusion from left to right is M9's original-zeta Mellin identity and continuity of every jet. For completeness, on $\Re s > 1$,

$$\mathcal{M}\mathcal{E}f(s) = \zeta(s) \int_0^\infty f(v)v^s \frac{dv}{v}.$$

The last integral is holomorphic on $\Re s > -2$, since f is even and $f(0) = 0$. Its value at 1 is zero, cancelling the simple pole of zeta. Continuation into the critical strip therefore proves vanishing at every original nontrivial-zero jet of the required orders. Those functionals are continuous on $\mathcal{A}$, so the closed image is contained in (S5.1).

For the reverse inclusion, take any $k \in \mathcal{J}_\zeta$, let $F = \mathcal{M}k$, and construct f_ε by S3–S4. Formula (S4.7) is the transform of logarithmic Gaussian convolution. Precisely, put

$$g_\varepsilon(x) = \frac{1}{\sqrt{4\pi\varepsilon}} e^{-x^2/(4\varepsilon)}. \quad (\text{S5.3})$$

The full Gaussian integral gives

$$\int_{\mathbb{R}} g_\varepsilon(x) e^{(s-1/2)x} dx = e^{\varepsilon(s-1/2)^2}.$$

Thus Mellin injectivity proves

$$(\mathcal{E}f_\varepsilon)(e^x) = (g_\varepsilon * K)(x), \quad K(x) = k(e^x). \quad (\text{S5.4})$$

These convolutions tend to K in every seminorm (S1.2). Here is a weighted proof, so that a weaker L^2 convergence is not substituted. For each (N, j) , the fundamental theorem of calculus gives

$$\|K(\cdot - y) - K\|_{N,j} \leq |y| e^{N|y|} \|K\|_{N,j+1}. \quad (\text{S5.5})$$

Indeed write the derivative difference as the integral of $K^{(j+1)}(x-r)$ from $r=0$ to $r=y$, and use $e^{N|x|} \leq e^{N|x-r|} e^{N|r|}$. Therefore

$$\|g_\varepsilon * K - K\|_{N,j} \leq \|K\|_{N,j+1} \int_{\mathbb{R}} g_\varepsilon(y) |y| e^{N|y|} dy \rightarrow 0. \quad (\text{S5.6})$$

After $y = \sqrt{\varepsilon}z$, the last integral is bounded for $0 < \varepsilon \leq 1$ by $\sqrt{\varepsilon}$ times the finite integral of $|z| e^{N|z|} e^{-z^2/4} / \sqrt{4\pi}$. This proves the asserted limit. Every approximant (S5.4) lies in the actual image by (S4.4)–(S4.6), proving the reverse inclusion globally.

The parameter $\varepsilon \downarrow 0$ is an explicit approximation to the identity on the entire space. It is not a restriction to finitely many zeros, a finite prime range, or an assumed convergence of a numerical experiment.

S6. Exceptional points, source jets and endpoint recovery

The full construction retains the information at the places cancelled by the auxiliary factor F_0 . Put

$$F_\varepsilon(s) = e^{\varepsilon(s-1/2)^2} F(s).$$

The original, uncompleted zeta relation is

$$\int_0^\infty f_\varepsilon(v)v^s \frac{dv}{v} = \frac{F_\varepsilon(s)}{\zeta(s)} \quad (\Re s > 1), \quad (\text{S6.1})$$

with meromorphic continuation retaining all terms.

For an even Schwartz function with $f(0) = 0$, subtracting its Taylor polynomial on $0 < v < 1$ proves that its Mellin integral extends meromorphically to the whole plane. The only possible poles are simple poles at $(-2m)$, $m \geq 1$, and their residues are $f^{(2m)}(0)/(2m)!$. Indeed, the subtracted remainder $O(v^{2M+2})$ gives a holomorphic integral on $\Re s > -2M-2$, while each subtracted monomial contributes exactly $f^{(2m)}(0)/((2m)!(s+2m))$. As M increases these continuations agree on their common domains.

At the simple trivial zero $(-2m)$, (S6.1) consequently gives

$$\boxed{\frac{f_\varepsilon^{(2m)}(0)}{(2m)!} = \frac{e^{\varepsilon(-2m-1/2)^2} F(-2m)}{\zeta'(-2m)} \quad (m \geq 1).} \quad (\text{S6.2})$$

This can also be checked directly from (S4.4): its $(2m)$ -th derivative at 0 multiplies the coefficient of f_0 by $Q_\varepsilon(-2m)$; (S2.2) identifies that coefficient as $F_0(-2m)/\zeta'(-2m)$. The trivial zeros are simple because the sine factor in the zeta functional equation has a simple zero there, whereas $\Gamma(1+2m)\zeta(1+2m)$ is finite and nonzero. Equivalently, the finite nonzero value in (S2.3), together with the simple Gamma pole, gives the same conclusion.

At $s = 1$, since $\zeta(s) = (s-1)^{-1} + O(1)$, the Mellin integral in (S6.1) has the exact expansion

$$\int_0^\infty f_\varepsilon(v) v^s \frac{dv}{v} = (s-1)F_\varepsilon(1) + O((s-1)^2). \quad (\text{S6.3})$$

Its value is the required zero integral, and its first derivative records $F_\varepsilon(1)$. At $s = 0$, $\zeta(0) = -1/2$ gives

$$\int_0^\infty f_\varepsilon(v) \frac{dv}{v} = -2F_\varepsilon(0). \quad (\text{S6.4})$$

The integral converges because $f_\varepsilon(v) = O(v^2)$. No endpoint value of the receiver has been discarded merely because the source has zero endpoint constraints.

S7. The resulting faithful receiver and its remaining mathematical role

By (S5.2), the joint actual-zero jet map has exactly the original closed image as its kernel:

$$J : \mathcal{A} \longrightarrow \prod_\rho \mathbb{C}^{m_\rho}, \quad J(k) = ((\mathcal{M}k)^{(j)}(\rho))_{\rho, 0 \leq j < m_\rho},$$

$$\ker J = \overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}. \quad (\text{S7.1})$$

Thus J induces an injective linear map

$$\boxed{\mathcal{Q} \hookrightarrow \prod_\rho \mathbb{C}^{m_\rho}.} \quad (\text{S7.2})$$

Its image is $J(\mathcal{A})$, with the quotient topology if a topological isomorphism is desired. No claim that this image is the entire product, or is closed in the product topology, is made. The conclusion proved here is stronger than the prior existence of some descending zero functionals: **all actual zero jets together now distinguish every class in the quotient**. There is no additional quotient class invisible to every original zeta zero and its retained multiplicities.

The scaling action remains exactly

$$J_{\rho,j}(V_a k) = a^{\rho-1/2} \sum_{\ell=0}^j \binom{j}{\ell} (\log a)^{j-\ell} J_{\rho,\ell}(k). \quad (\text{S7.3})$$

Thus the quotient has been identified using the actual zeta spectral data, including its full triangular multiplicity action. This theorem does not force $\Re \rho = 1/2$: it does not insert a positive definite norm on this

quotient, and it does not assert that (S7.3) is unitary. M9’s proof that the image is dense in the ordinary $L^2(du/u)$ space still holds. The faithful jet receiver (S7.2), its finer topology, and the zero Hilbert quotient are all retained with their exact maps.

Finally, let C_{rec} and the free support space $\mathbb{C}^{(\mathcal{L})}$ be exactly those of M9.7. Tensoring (S7.2) algebraically with these coefficient spaces gives an injective linear map: a finite basis expansion reduces the assertion to injectivity of J in each coordinate. This statement concerns the linear spaces after passage to tensors. That preceding passage identifies $0_Q \otimes e_\ell$ for every label, so it does not retain a support record at coefficient zero. The two coefficient sections are distinct maps, with the same zero value.

The exact repair is the disjoint family $\widehat{Q} = \coprod_{\ell \in \mathcal{L}} \{\ell\} \times \mathcal{Q}$, with $\widehat{J}(\ell, q) = (\ell, Jq)$. This is injective because equality first retains the label and then determines the coefficient through S7.2. Its coefficient-zero fibre retains every $(\ell, 0_Q)$. The map $(\ell, q) \mapsto q \otimes e_\ell$ is the explicit comparison to the old receiver: its zero fibre is that entire labelled family and every nonzero fibre in its image is a singleton. All arithmetic operators and projectors act on q while retaining the label. Section selections through $[\tau]$ and $([1])$ can be included in the existing record as proved in `SUPPORT_ZERO_RECEIVER_REPAIR.md` ZR5. ZR1-ZR5 supply the complete proofs and the exact full-pairing map, incorporating the receiving task’s I26.10 correction. No source addition or integer parity is assigned to τ .

The proof diagram is

$$\begin{array}{ccc}
 F \in \mathcal{B}, J(F) = 0 & \xrightarrow{/F_0} & Q \text{ entire } (\log|Q| = O_A(|t|^{3/2} \log|t|)) \\
 \downarrow \times e^{\varepsilon(s-1/2)^2} & & \downarrow \times e^{\varepsilon(s-1/2)^2} \\
 F_\varepsilon = F_0 Q_\varepsilon & \xleftarrow{\mathcal{M}} & \mathcal{E} f_\varepsilon = k_0 * q_\varepsilon,
 \end{array}$$

with $F_\varepsilon \rightarrow F$ in $\mathcal{B}$ proved through the equivalent $\mathcal{A}$ convergence (S5.6). Every arrow is constructed above on its full domain.

Integral counting, arithmetic operators, and the weight comparison

24 September 2026. Proof labels IC1–IC9. This note receives the proposed implication from the complete global counting unit to integrality and then purity. It calculates the intervening maps. It does not assume an off-critical zero exists, and the projective-line calculation below is not asserted to satisfy the entire tau/Connes–Consani programme.

The supporting point remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition, subtraction, distance, coordinate, vector or numerical value is assigned to that point. Every operation below has as its domain a stalk quotient, an endomorphism ring, a group algebra, a cohomology group or the original-zeta quotient explicitly specified below.

Sources actually used

- `CANONICAL_GENERIC_RECONSTRUCTION.md`, CG0–CG8, read in full: the complete winding quotient, its endomorphism ring, the canonical unit, prime norms, and original-zeta return measure. Its original Connes–Consani source and exact author-source locators are retained there.
- `TWISTOR_DEGREE_DERIVATION.md`, TD3–TD4 and TD8, read in full for the integral sphere cohomology, its degree action, both signed source components, and the finite-characteristic fixed-point calculation. The calculation below restates the relevant proof rather than importing a conclusion called purity.
- `ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`, RZ8 and RZ10–RZ11, read in full: the actual multiplier a^s , its full source factor $a^{1/2}$, all multiplicity jets, and the reflected Weil form. RZ5 supplies its proved zero-isolating representatives. This note uses that proved receiving space, not an assumed spectral object.
- No claim is made here to have read Deligne’s original author TeX or to reconstruct Deligne’s proof. The elementary finite-field comparison has the weight convention $|\lambda| = p^{w/2}$, with its actual weights calculated below.

IC1. The integral ring recovered from the complete winding object

Retain the complete CC generic group and its finite subgroup:

$$G = \langle T, J, \epsilon : TJ = JT, T\epsilon = \epsilon T, J\epsilon = \epsilon J, J^2 = \epsilon, \epsilon^2 = 1 \rangle, \quad H = \langle J \rangle, \quad L_w = G/H. \quad (\text{IC1.1})$$

All source commutation relations remain displayed, including those also implied by $\epsilon = J^2$. CG2–CG3 prove that $G \cong \mathbb{Z} \times C_4$ and L_w is infinite cyclic, retaining $1 \rightarrow H \rightarrow G \rightarrow L_w \rightarrow 1$. The subscript here distinguishes the winding group from the spectral operator below.

Let

$$R = \text{End}_{\text{Ab}}(L_w), \quad (f \oplus g)(x) = f(x)g(x), \quad (f \odot g)(x) = f(g(x)). \quad (\text{IC1.2})$$

An endomorphism is exactly a power map $[n] : x \mapsto x^n$, with $n \in \mathbb{Z}$. Indeed its value on a generator determines every value, and is a unique power of that generator. Choosing the inverse generator leaves the exponent unchanged. Thus

$$R \cong \mathbb{Z}, \quad [a] \oplus [b] = [a + b], \quad [a] \odot [b] = [ab], \quad \mathbf{1}_R = [1] = \text{id}_{L_w}. \quad (\text{IC1.3})$$

The nonnegative submonoid R_+ consists of the successive sums of this identity, including the zero endomorphism. For nonzero $f = [n] \in R_+$, define

$$d(f) = |L_w / f(L_w)| = n. \quad (\text{IC1.4})$$

Representatives are the first n powers of any generator. Division of exponents proves that they exhaust the quotient without repetition. In particular

$$d(f \odot g) = d(f)d(g), \quad d(\mathbf{1}_R) = 1. \quad (\text{IC1.5})$$

This is a proved integrality statement: every counting endomorphism has an integer degree and every positive prime degree is a finite quotient cardinality. It makes no assertion that eigenvalues of a different representation of these degrees are algebraic integers.

IC2. The exact map from the counting operators to the original-zeta action

Use the original-zeta receiving object from RZ. Let $\mathcal{B}$ be the entire functions rapidly decreasing vertically in each bounded real strip, with seminorms

$$b_{A,M}(F) = \sup_{|\text{Re } s| \leq A} (1 + |\text{Im } s|)^M |F(s)|. \quad (\text{IC2.1})$$

Let $\mathcal{Z}$ be the set of distinct actual nontrivial zeros of the original ζ , and m_ρ their full multiplicities. The closed ideal is

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}. \quad (\text{IC2.2})$$

The prior global synthesis proves that this is the quotient of the original arithmetic source by its closed image, with the full multiplier

$$F_0(s) = \frac{s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)}{8} \quad (\text{IC2.3})$$

retained in the source comparison. Formula (IC2.3) is not a replacement for ζ : the full endpoint, Gamma and trivial-zero comparison belongs to that proof. Only the actual nontrivial-zero ideal (IC2.2) is used for the receiving calculation here.

Write $L_\zeta[F] = [sF(s)]$. For every real $a > 0$, RZ8 defines the continuous invertible operator

$$T_a[F] = [a^s F(s)], \quad a^s = \exp(s \log a). \quad (\text{IC2.4})$$

Here $\log a$ is real. Direct multiplication proves

$$T_a T_b = T_{ab}, \quad T_1 = I, \quad T_a^{-1} = T_{1/a}. \quad (\text{IC2.5})$$

The exact continuity estimate before passage to the quotient is

$$b_{A,M}(T_a F) \leq \max(a^A, a^{-A}) b_{A,M}(F). \quad (\text{IC2.6})$$

The multiplier is entire and nowhere zero, so it preserves every vanishing order in (IC2.2). Taking infima over representatives gives the same bound for quotient seminorms. The source Mellin transform and its inverse comparison are

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}, \quad \mathcal{M}^{-1} T_a \mathcal{M}k(u) = a^{1/2} k(u/a). \quad (\text{IC2.7})$$

Substituting $u = av$ into the integral proves this identity with both powers retained. In the multiplier sense, RZ writes $a^{L_\zeta} = T_a$; it does not require convergence of an operator power series.

The requested canonical map out of the positive counting degrees is therefore

$$(R_+ \setminus \{0\}, \odot) \longrightarrow \text{Aut}_{\text{cont}}(\mathcal{Q}), \quad f \longmapsto T_{d(f)}. \quad (\text{IC2.8})$$

Equations (IC1.5) and (IC2.5) prove the homomorphism property, including the unit. No arbitrary orientation or preassigned prime label is needed. The full source ring is retained; (IC2.8) states which of its operations this particular receiving map preserves.

IC3. The integral group algebra carrying the arithmetic action

Let $\mathbb{Q}_{>0}^\times$ be the multiplicative group of positive rational numbers. Define

$$\Lambda = \mathbb{Z}[\mathbb{Q}_{>0}^\times] = \left\{ \sum_{r \in S} a_r b_r : S \subset \mathbb{Q}_{>0}^\times \text{ finite, } a_r \in \mathbb{Z} \right\}, \quad b_r b_t = b_{rt}, \quad 1_\Lambda = b_1. \quad (\text{IC3.1})$$

Different symbols b_r are a free additive basis; coefficients of equal symbols are added. Multiplication is the displayed finite convolution, hence associative, commutative and distributive, with identity b_1 . Negative coefficients are additive inverses. This constructs the ring directly.

Unique prime factorization gives the exact identification

$$\Lambda \cong \mathbb{Z}[b_p, b_p^{-1} : p \text{ prime}], \quad b_r = \prod_p b_p^{v_p(r)}, \quad (\text{IC3.2})$$

where only finitely many integer valuations $v_p(r)$ are nonzero. All primes, including two, occur.

The original arithmetic action extends to the unital ring homomorphism

$$\Phi : \Lambda \longrightarrow \text{End}_{\text{cont}}(\mathcal{Q}), \quad \Phi \left(\sum_{r \in S} a_r b_r \right) = \sum_{r \in S} a_r T_r. \quad (\text{IC3.3})$$

Additivity holds term by term. Multiplying two finite sums and using $T_r T_t = T_{rt}$ proves multiplicativity, and $\Phi(b_1) = I$. Each image is continuous because

$$b_{A,M} \left(\sum_{r \in S} a_r r^s F(s) \right) \leq \left(\sum_{r \in S} |a_r| \max(r^A, r^{-A}) \right) b_{A,M}(F). \quad (\text{IC3.4})$$

The same estimate passes to the quotient. We assert continuity of every represented operator; no topology on the abstract finite-support ring Λ is implicit in this statement. No injectivity of Φ is needed or asserted.

There are now two exact maps, with different domains of preserved operations:

$$\begin{aligned} R \cong \mathbb{Z} &\longrightarrow \Lambda, & [n] &\longmapsto n b_1, & \text{a unital ring homomorphism;} \\ R_+ \setminus \{0\} &\longrightarrow \Lambda^\times, & [n] &\longmapsto b_n, & \text{a multiplicative-monoid homomorphism.} \end{aligned} \quad (\text{IC3.5})$$

The first is injective because the b_1 -coefficient of nb_1 is n . Their represented images are respectively

$$\Phi(nb_1) = nI, \quad \Phi(b_n) = T_n = n^{L\zeta}. \quad (\text{IC3.6})$$

For example $b_2 \neq 2b_1$ already in the free additive basis. The difference is not erased in the actual receiving quotient. On any actual zero block $\mathcal{Q}_\rho$, RZ8 proves the eigenvalue of T_2 is 2^ρ . Since $0 < \text{Re } \rho < 1$,

$$|2^\rho| = 2^{\text{Re } \rho} < 2. \quad (\text{IC3.7})$$

Thus $T_2 \neq 2I$. The nontrivial-zero set is nonempty, independently of RH; the source quotient used here includes its proved nonzero isolating blocks. Consequently (IC2.8) cannot be an additive extension of the identity map on R : such an extension would give $T_2 = T_1 + T_1 = 2I$, contradicting (IC3.7).

This pinpoints the integrality transfer rather than discarding the relation between the two objects. The full map is (IC3.5) followed by (IC3.3), and its scalar and multiplicative inputs are explicitly distinct.

IC4. What integrality of the group algebra does, and does not, assert

An endomorphism of a finite-rank free abelian group has an integer matrix in an integral basis. Expanding $\det(XI - A)$ by permutations yields a monic polynomial with integer coefficients; every complex eigenvalue annihilates this polynomial. This proves that such an eigenvalue is an algebraic integer.

In contrast, the word integral in the coefficient ring $\Lambda = \mathbb{Z}[\mathbb{Q}_{>0}^\times]$ does not assert that each b_p is integral over its scalar subring $\mathbb{Z}b_1$. In fact it is not. If a monic relation of positive degree existed, it would have the form

$$b_p^d + a_{d-1}b_p^{d-1} + \cdots + a_0b_1 = 0, \quad a_j \in \mathbb{Z}. \quad (\text{IC4.1})$$

But $b_p^j = b_{p^j}$, and the positive rationals $1, p, \dots, p^d$ are distinct. The free additive basis in (IC3.1) makes the coefficient of b_{p^d} on the left equal to one, a contradiction. This proves the precise algebraic distinction without a spectral assumption.

On an actual zero block, with full multiplicity $m = m_\rho$, the represented action is

$$T_p|_{\mathcal{Q}_\rho} = p^\rho \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} N_\rho^j, \quad N_\rho^m = 0, \quad (\text{IC4.2})$$

and its characteristic polynomial is exactly

$$\det(XI - T_p|_{\mathcal{Q}_\rho}) = (X - p^\rho)^m. \quad (\text{IC4.3})$$

The nilpotent terms have not been removed; their diagonal is zero and their full coefficients remain (IC4.2). Neither CG's integral counting ring nor the representation (IC3.3) proves that the coefficients of (IC4.3) are integers, or that p^ρ is an algebraic integer. No finite-rank invariant integral lattice for these blocks has been constructed in these calculations. This is a record of the exact established property, not an assertion that such a lattice or a different weight construction is impossible.

IC5. An actual finite-field comparison, with every point count retained

Fix a rational prime p , including $p = 2$, and consider the projective line over $\mathbb{F}_p$. For every $r \geq 1$, its $\mathbb{F}_{p^r}$ -points consist of the p^r affine elements and the point at infinity. Hence

$$\#\mathbb{P}^1(\mathbb{F}_{p^r}) = 1 + p^r. \quad (\text{IC5.1})$$

Equivalently, the morphism $f_p : [x : y] \mapsto [x^p : y^p]$ over $\overline{\mathbb{F}}_p$ has these fixed-point counts for all powers: in its affine chart, f_p^r fixes the roots of $z^{p^r} - z$. The derivative is -1 , giving p^r distinct roots, and infinity is fixed.

Here is an integral constants-and-divisors realization of its degree-zero and degree-two cohomological action. Let

$$M = \mathbb{Z}e_0 \oplus \mathbb{Z}e_2, \quad e_0 \cdot e_0 = e_0, \quad e_0 \cdot e_2 = e_2 \cdot e_0 = e_2, \quad e_2 \cdot e_2 = 0. \quad (\text{IC5.2})$$

The first class is the constant class. The second is the degree-one divisor class $[\infty]$. Over an algebraically closed field every divisor is a finite integer sum of points. The rational function $z - a$ has divisor $[a] - [\infty]$; multiplying its integer powers shows every divisor of degree d is equivalent to $d[\infty]$. Principal divisors have degree zero, by counting numerator and denominator zeros including infinity. Thus the divisor class group is exactly $\mathbb{Z}[\infty]$, proving the integral second summand directly.

The same ring is the full integral singular cohomology of $\mathbb{P}^1(\mathbb{C})$: its sphere cell decomposition has a cell in degrees zero and two and none in degree one, and the square of a degree-two class vanishes because degree four is absent. This gives the integral comparison package conventionally denoted $H^0 \oplus H^2$. In the finite-field cohomological notation the second summand is the Tate class $\mathbb{Z}_\ell(-1)$, for $\ell \neq p$; the Frobenius convention here is the one entering point-count traces and acting by p on that untwisted degree-two class. The elementary calculation uses the displayed constants and divisor pullback; it does not require reconstructing the general étale comparison theorem.

The pullback of a constant is that constant. In the local parameter $w = 1/z$ at infinity, f_p sends w to w^p , so $f_p^*[\infty] = p[\infty]$. Therefore the full action is

$$F_p e_0 = e_0, \quad F_p e_2 = p e_2, \quad F_p = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}. \quad (\text{IC5.3})$$

Its trace on every iterate is exactly $1 + p^r$, matching (IC5.1), and its characteristic polynomial and determinant factor are

$$\det(XI - F_p) = X^2 - (1 + p)X + p, \quad \det(I - tF_p) = (1 - t)(1 - pt). \quad (\text{IC5.4})$$

Both eigenvalues are integers. All extension-field point counts are present. Their full generating function is

$$Z(\mathbb{P}^1/\mathbb{F}_p, t) = \exp \left(\sum_{r \geq 1} \frac{1 + p^r}{r} t^r \right) = \frac{1}{(1 - t)(1 - pt)}. \quad (\text{IC5.5})$$

This is an identity of formal power series and of analytic functions for $|t| < 1/p$: expand separately $-\log(1 - t)$ and $-\log(1 - pt)$, retaining both terms. It is not an identification of (IC5.5) with the original Riemann zeta function.

For connection with the already calculated actual CC signed sphere maps, TD8 has $z \mapsto z^p$ for $p \equiv 1 \pmod{4}$, and $z \mapsto -1/z^p$ for $p \equiv 3 \pmod{4}$. The latter fixed-point equation in the odd reciprocal iterates is $z^{p^r+1} = -1$, with exactly $p^r + 1$ distinct roots. Its degree action and all counts are therefore precisely the same (IC5.3)–(IC5.5). TD8 requires odd p for that signed source lift. The present projective-line Frobenius example itself includes two; it is not an unproved extension of that odd source lift.

IC6. Its degree- p pairing and its exact weights

Let $\text{tr}(e_0) = 0$ and $\text{tr}(e_2) = 1$, and define the integral bilinear form by extracting the top-degree coefficient:

$$B(x, y) = \text{tr}(xy), \quad B(ae_0 + be_2, ce_0 + de_2) = ad + bc. \quad (\text{IC6.1})$$

Its matrix and exact degree relation are

$$J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad F_p^\top J F_p = pJ. \quad (\text{IC6.2})$$

The determinant of J is -1 , so the integral pairing is perfect. Direct evaluation gives

$$B(e_0 + e_2, e_0 + e_2) = 2, \quad B(e_0 - e_2, e_0 - e_2) = -2. \quad (\text{IC6.3})$$

Thus its real signature is $(1, 1)$. The multiplier p pairs the eigenvalues 1 and p ; it does not make both eigenvalues have modulus $\sqrt{p}$.

The complete degree filtration is

$$W_{-1}M = 0, \quad W_0M = W_1M = \mathbb{Z}e_0, \quad W_2M = M. \quad (\text{IC6.4})$$

This records the proved cohomological degrees in (IC5.2), rather than assigning an unknown sector a desired degree. The associated graded degree-zero action is 1, and the degree-two action is p . Their absolute values are exactly

$$|1| = p^{0/2}, \quad |p| = p^{2/2}. \quad (\text{IC6.5})$$

They are pure of weights zero and two, respectively, in the finite-field weight convention. The direct sum is not pure of weight one. Its degree-one cohomology vanishes in this calculation. This is consistent with, and is not a violation of, the purity assertion for each specified cohomological degree.

There is also no positive-definite Hermitian form K on $M \otimes \mathbb{C}$ having this same degree- p similitude. Evaluating $K(F_p e_0, F_p e_0) = pK(e_0, e_0)$ would give $(p-1)K(e_0, e_0) = 0$. Since $p > 1$, this forces $K(e_0, e_0) = 0$, incompatible with positive definiteness on the nonzero vector e_0 . This explicit calculation identifies the role of the cohomological sector and its form. It does not assert that positivity cannot be obtained on a different, correctly constructed weight-one object.

IC7. The exact map from the counting ring to the comparison package

The preceding integral package can be built canonically over the already reconstructed ring R : take $Re_0 \oplus Re_2$ and, for each nonzero $f \in R_+$, put

$$\mathcal{F}_f(ae_0 + be_2) = ae_0 + (f \odot b)e_2. \quad (\text{IC7.1})$$

In the canonical coordinates (IC1.3), $f = [n]$ gives $\text{diag}(1, n)$, not a chosen numerical coordinate at τ . Associativity of composition gives

$$\mathcal{F}_f \mathcal{F}_g = \mathcal{F}_{f \odot g}, \quad \mathcal{F}_{1_R} = I. \quad (\text{IC7.2})$$

The pairing $B_R(ae_0 + be_2, ce_0 + de_2) = a \odot d \oplus b \odot c$ obeys

$$B_R(\mathcal{F}_f x, \mathcal{F}_f y) = f \odot B_R(x, y). \quad (\text{IC7.3})$$

For $f = [p]$, applying $R \cong \mathbb{Z}$ gives exactly the degree- p relation (IC6.2), with no altered multiplier. Thus the same canonical unit and recovered prime norms have an explicit integral cohomological representation whose actual weight pieces are zero and two. The representation supplies a precise comparison of integrality and weight. It is not a model of the full original-zeta receiving quotient, nor is it a claim that the CC programme has only these pieces.

IC8. The actual original-zeta pairing has its own precise integral-indexed degree relation

Let $\rho^\# = 1 - \bar{\rho}$. On $\mathcal{Q}$, the full source-derived Weil form is

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(\rho^\#)} G(\rho). \quad (\text{IC8.1})$$

The previous source calculation proves absolute convergence and Hermitian symmetry. Since $\overline{\rho^\#} + \rho = 1$, the actual unshifted arithmetic action satisfies, for every real $a > 0$,

$$W(T_a x, T_a y) = aW(x, y), \quad W(T_p x, T_p y) = pW(x, y) \quad \text{when } a = p. \quad (\text{IC8.2})$$

Every zero and every multiplicity is retained in this equality. The numerical norm $p = |L_w/[p]L_w|$ is exactly its multiplier. This is the concrete meeting point between the global counting ring and the spectral pairing.

On a reflected pair of distinct actual zeros, put $\lambda = p^\rho$, $\mu = p^{\rho^\#}$. Then

$$\bar{\lambda}\mu = p, \quad |\lambda| = p^{\text{Re } \rho}, \quad |\mu| = p^{1-\text{Re } \rho}. \quad (\text{IC8.3})$$

The exact isolating representatives give matrix $\begin{pmatrix} 0 & m_\rho \\ m_\rho & 0 \end{pmatrix}$ on the two constant-jet coordinates. These statements describe what the actual formula would record at such a zero; they do not invent a zero or discard

any prime. Higher jets remain the full nilpotent structure (IC4.2), even where the ordinary trace pairing has a radical.

The weighted adjoint relation also determines an exact coefficient-domain enlargement. Set

$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[\mathbb{Q}_{\geq 0}^{\times}], \quad \Phi_{\mathbb{Q}} \left(\sum_{r \in S} c_r b_r \right) = \sum_{r \in S} c_r T_r, \quad c_r \in \mathbb{Q}, \quad (\text{IC8.4a})$$

where S is finite. This explicitly extends Φ from (IC3.3). The same finite-convolution proof establishes its unital ring homomorphism property, and (IC3.4) with $|c_r|$ proves continuity of every image operator. On $\Lambda_{\mathbb{Q}}$, the assignment

$$b_r^{\star} = r b_{1/r}, \quad a^{\star} = a \quad (a \in \mathbb{Q}) \quad (\text{IC8.4})$$

is an involution: $(b_r^{\star})^{\star} = r(1/r)b_r = b_r$, and multiplication is respected because $rt b_{1/(rt)} = (rb_{1/r})(tb_{1/t})$. The form satisfies

$$W(\Phi_{\mathbb{Q}}(A)x, y) = W(x, \Phi_{\mathbb{Q}}(A^{\star})y), \quad A \in \Lambda_{\mathbb{Q}}. \quad (\text{IC8.5})$$

For a monomial this is $W(T_r x, y) = W(x, rT_{1/r} y)$, obtained from (IC8.2) and invertibility. Rational coefficients are real, so conjugate-linearity in the first argument and linearity in the second give exactly the displayed equality for finite rational sums.

This rational enlargement is the smallest unital subring of $\Lambda_{\mathbb{Q}}$ containing Λ and preserved by this specified weighted involution. To prove minimality, let A be any such subring. For every prime p , the monomial $b_{1/p}$ belongs to $\Lambda \subseteq A$. Stability under $\star$ and closure under multiplication give

$$(b_{1/p})^{\star} b_{1/p} = \left(\frac{1}{p} b_p \right) b_{1/p} = \frac{1}{p} b_1 \in A. \quad (\text{IC8.6})$$

Every positive integer denominator is a finite product of prime powers. Multiplying the corresponding scalar elements in (IC8.6), and then multiplying by an arbitrary integer scalar, proves $\mathbb{Q}b_1 \subseteq A$. Since A also contains every b_r , multiplication supplies $c_r b_r$ for every rational c_r , and finite addition supplies every element of $\Lambda_{\mathbb{Q}}$. Hence $A = \Lambda_{\mathbb{Q}}$. Conversely, (IC8.4) directly preserves $\Lambda_{\mathbb{Q}}$, so this ring attains the asserted minimum.

In particular Λ itself is not preserved: the b_p -coefficient of $(b_{1/p})^{\star}$ is $1/p$, which is not an integer. Equations (IC8.4a)–(IC8.6) construct and characterize the object required to retain the full weighted adjoint symmetry in this exact group-algebra presentation. This does not rule out integral arithmetic geometry, a different integral realization, or a pairing valued in an integral Tate object; those are not the scalar-valued coefficient presentation classified here.

IC9. What the complete spectrum detects

CG7 proves that the complete normed prime spectrum, including its unit atom and every prime-power repetition, determines

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad \text{Re } s > 1, \quad (\text{IC9.1})$$

with absolute convergence. Its proved meromorphic continuation is unique. Consequently an actual nontrivial zero, wherever located, belongs to the analytic function determined by that same complete prime data. It would not be an additional prime omitted by the counting process. The whole prime spectrum determines the whole analytic function, including a possible off-critical zero; uniqueness alone does not tell us the location of that zero.

This last statement has a direct logical proof. If two meromorphic functions had the same complete series in (IC9.1), they would coincide on the open half-plane and hence throughout the connected continuation domain by the meromorphic identity theorem. Their zero divisors, with multiplicities, would therefore coincide. The argument contains no inequality or equality restricting the real parts of those zeros. The constructed original-zeta receiver strengthens the detection statement: its full joint arithmetic action has

a block of dimension m_ρ for each actual zero, with coefficients (IC4.2). Such a zero is detected there, not missing from the reconstructed function.

The proved chain is therefore complete at the following levels: the source supplies the unique counting identity; its full endomorphism ring has integral degrees and prime norms; those degrees act through the explicit integral group algebra; the actual original-zeta quotient receives that action with every zero and multiplicity; and the Weil pairing has exactly the same prime norm as its similitude multiplier. The projective-line calculation identifies why integrality, complete counts and a degree- p pairing do not by themselves specify a weight-one sector. None of these calculations asserts a counterexample to the user's full geometric programme, or an RH resolution.

What the complete prime record detects, and the resulting arithmetic weight

24 September 2026. Proof locators CD1–CD7. This calculation receives the user's integrality and completeness argument in WU037. It retains the common supporting point $\tau\langle Z_1; \text{no } Z_2 \rangle$, the full signed stalk of CG2–CG3, and the original Riemann zeta function. No operation, coordinate, or distance is introduced on the supporting point.

The completed source reconstruction in CG0–CG8 is an input: its uniqueness is uniqueness of the specified complete Connes–Consani chart diagram and its universal localization, with all chart maps retained. Neither the existence of a cyclic stalk nor a spectral weight is deduced merely from the duration of an observation. The analytic quotient used below is the proved quotient in GLOBAL_MELLIN_SYNTHESIS.md, S1–S7; its complete proof is included in the cumulative reader. This note proves the next receiving calculation, rather than assuming that a zero outside the critical line is an omitted arithmetic event.

CD1. The full return record determines the original logarithmic derivative

Write $R = \text{End}_{\text{Ab}}(L)$, with its orientation-independent identity $\mathbf{1}_R = \text{id}_L$, as constructed in CG4. For each nonzero prime ideal $\mathfrak{p}$, its residue norm is

$$N(\mathfrak{p}) = |R/\mathfrak{p}|. \quad (\text{CD1.1})$$

CG5 identifies these norms with every positive prime integer, including two. Retain the three complete measures

$$\mathcal{D} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad \mathcal{R} = \sum_p \sum_{k \geq 1} \frac{1}{k} \delta_{k \log p}, \quad \mathcal{W} = \sum_p \sum_{k \geq 1} (\log p) \delta_{k \log p}. \quad (\text{CD1.2})$$

The proof of $\mathcal{D} = \exp_* \mathcal{R}$, including each coefficient and the unit atom, is TP9–TP13. For $\text{Re } s > 1$ it gives

$$\zeta(s) = \int e^{-st} d\mathcal{D}(t) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad (\text{CD1.3})$$

$$H(s) := \int e^{-st} d\mathcal{W}(t) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} = -\frac{\zeta'(s)}{\zeta(s)}. \quad (\text{CD1.4})$$

Every displayed identity is on its convergence domain. In particular the prime sum is not asserted to converge at a nontrivial zero. To justify the derivative and all local termwise operations, on a compact set in $\text{Re } s > 1$, choose $\sigma_0 > 1$ below all its real parts. The sum with an additional fixed power of $k \log p$ is bounded by a constant times $\sum_{n \geq 2} (\log n)^j n^{-\sigma_0}$, with a possibly larger nonnegative integer j . This converges by the integral test. The exponential of the absolutely convergent prime logarithm is nonzero on this half-plane, so logarithmic differentiation is justified there.

The complete record therefore determines one holomorphic germ H on that half-plane. Its meromorphic continuation is the indicated logarithmic derivative of the original meromorphic ζ . Two continuations on a connected domain agree by the identity theorem. This is analytic uniqueness from the full record, without a finite-prefix inference.

CD2. Every zero is detected, with its original multiplicity

Let ρ be any actual nontrivial zero of the original ζ , of multiplicity m_ρ . Holomorphic Taylor division gives, on a sufficiently small disk,

$$\zeta(s) = (s - \rho)^{m_\rho} g_\rho(s), \quad g_\rho(\rho) \neq 0. \quad (\text{CD2.1})$$

Shrinking the disk ensures g_ρ has no zero there. Direct differentiation of this original function gives

$$H(s) = -\frac{m_\rho}{s - \rho} - \frac{g'_\rho(s)}{g_\rho(s)}. \quad (\text{CD2.2})$$

Consequently the positive counterclockwise boundary γ_ρ of that disk detects exactly

$$-\frac{1}{2\pi i} \int_{\gamma_\rho} H(s) ds = m_\rho. \quad (\text{CD2.3})$$

The second term in (CD2.2) is holomorphic, so its integral is zero; the first term has residue $-m_\rho$. This proves (CD2.3). No step uses $\text{Re } \rho = 1/2$.

The original pole at $s = 1$ is retained too. Writing $\zeta(s) = (s - 1)^{-1} g_1(s)$, with $g_1(1) = 1$, gives

$$H(s) = \frac{1}{s - 1} - \frac{g'_1(s)}{g_1(s)}. \quad (\text{CD2.4})$$

Every original trivial zero $-2r$, $r \geq 1$, gives residue -1 in H . Thus this scalar detector contains the pole and both types of zero, with their signs.

An off-line zero, if there is one, is consequently a zero detected by the same uniquely reconstructed record. It is not an extra prime, a fractional prime norm, or a prime-power event omitted from (CD1.2). The assertion follows from the actual detector (CD2.3), not from substituting another zeta function. Completeness determines all zero locations; it does not restrict the definition of detection to locations on the critical line.

CD3. The full auxiliary multiplier and its contribution to the detector

The exact Mellin transform used for the source realization is

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CD3.1})$$

It is the Mellin transform of the explicit original kernel

$$k_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu), \quad f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2},$$

under $\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} du/u$. S2 proves this identity, entire continuation, and every exceptional value. In particular

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{CD3.2})$$

This is an auxiliary source transform with an explicit comparison, not a replacement for the original function.

Let $\psi = \Gamma'/\Gamma$. Logarithmic differentiation of the complete product, on the open set where its factors are nonsingular and nonzero, gives

$$\boxed{-\frac{\zeta'(s)}{\zeta(s)} = -\frac{F'_0(s)}{F_0(s)} + \frac{1}{s} + \frac{1}{s-1} - \frac{\log \pi}{2} + \frac{1}{2} \psi(s/2).} \quad (\text{CD3.3})$$

Both sides are meromorphic, so the identity extends meromorphically. The constant $1/8$ has derivative zero and remains in (CD3.1), (CD3.2), and every source projector below. At $s = 0$, the residues of $1/s$ and $\psi(s/2)/2$ cancel. At $s = 1$, the retained endpoint term has residue $+1$. At $s = -2r$, the Gamma logarithmic derivative has residue -1 , while (CD3.2) makes F'_0/F_0 regular. At a nontrivial zero, the elementary and Gamma terms are regular and $-F'_0/F_0$ has residue $-m_\rho$. Thus the quotient detector and the full original detector agree on nontrivial zeros with every discarded-looking term accounted for explicitly.

CD4. Detection inside the actual whole source quotient

For the following stronger detection statement, retain the exact function spaces

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty \text{ for all integers } A, M \geq 0\},$$

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CD4.1})$$

The synthesis proof S5 identifies $\mathcal{J}$ with the closed original source image; zeros have not been freely appended as extra states. The physical source space and full proof of this equality are included in the cumulative reader.

For a particular actual zero, put $m = m_\rho$ and

$$A_\rho(s) = \frac{F_0(s)}{(s - \rho)^m}, \quad B_\rho(s) = \operatorname{Tay}_{\rho, < m}(1/A_\rho)(s), \quad e_\rho = A_\rho B_\rho. \quad (\text{CD4.2})$$

Taylor division gives $A_\rho(\rho) \neq 0$. At ρ the product is 1 modulo $(s - \rho)^m$; at every other zero it has the full required vanishing order because its denominator is nonzero there. Division by a fixed linear factor of a vanishing entire strip-Schwartz function stays in $\mathcal{B}$: outside a unit disk this follows by division, and inside it follows by Cauchy's maximum principle on a radius-two disk. Multiplication by a polynomial also preserves $\mathcal{B}$. These facts establish $e_\rho \in \mathcal{B}$ on the whole plane, not only on a finite test window.

The exact projector lift is

$$\mathcal{P}_\rho F(s) = A_\rho(s) \operatorname{Tay}_{\rho, < m}(F/A_\rho)(s). \quad (\text{CD4.3})$$

Its output has all of the first m jets of F at ρ and all zero jets at every other zero. Taylor coefficients are continuous on $\mathcal{B}$; hence this is a continuous finite-rank operator. A second application does not change its polynomial, so it is idempotent. It kills $\mathcal{J}$ and defines P_ρ on $\mathcal{Q}$. The vectors

$$[e_\rho(s)(s - \rho)^j], \quad 0 \leq j < m, \quad (\text{CD4.4})$$

are independent because their derivative jets at ρ are respectively $j!$ in position j and zero in the other positions. Their quotient classes span $\operatorname{im} P_\rho$, since

$$P_\rho[F] = \sum_{j=0}^{m-1} \frac{F^{(j)}(\rho)}{j!} [e_\rho(s)(s - \rho)^j].$$

At the representative level, the range of (CD4.3) is $A_\rho \operatorname{Pol}_{< m}$; the displayed representatives agree with their projected representatives modulo $\mathcal{J}$. Consequently $\operatorname{rank} P_\rho = m$, and

$$\bigcap_{\rho} \ker P_\rho = \{0\}. \quad (\text{CD4.5})$$

Indeed the intersection forces all required jets to vanish, exactly the definition of $\mathcal{J}$, which the synthesis theorem identifies with the original closed source image. No convergence of an infinite sum of projectors is asserted or needed. The complete resolvent and every source-level correction are proved in RZ3–RZ7.

Thus completeness of the quotient makes every actual zero detectable, including its multiplicity, independently of its location. An off-line zero would not fail (CD4.5); its own nonzero projector would be among the detectors proving that equality.

CD5. The exact arithmetic weight seen by those detectors

Use a different symbol L_ζ for the spectral operator to avoid confusing it with the winding group L . It is induced by $F(s) \mapsto sF(s)$. For every positive real a , the actual action is

$$T_a[F] = [a^s F(s)], \quad \mathcal{M}^{-1} T_a \mathcal{M} k(u) = a^{1/2} k(u/a). \quad (\text{CD5.1})$$

The multiplier preserves $\mathcal{B}$, with seminorm factor $\max(a^A, a^{-A})$, and preserves $\mathcal{J}$, with every order unchanged. Substitution $u = av$ in the Mellin integral proves the second identity including its factor. Composition gives $T_a T_b = T_{ab}$ and $T_1 = I$.

On the range of P_ρ , let $N_\rho = L_\zeta - \rho I$. Multiplication by $s - \rho$ shifts the vectors (CD4.4) to the next one, and kills the final one modulo $\mathcal{J}$. Thus $N_\rho^m = 0$, $N_\rho^{m-1} \neq 0$, and Taylor expansion gives the full operator

$$T_p|_{\text{im } P_\rho} = p^\rho \sum_{j=0}^{m-1} \frac{(\log p)^j}{j!} N_\rho^j, \quad |p^\rho| = p^{\text{Re } \rho}. \quad (\text{CD5.2})$$

Every logarithmic jet term remains present. The weight-one modulus is precisely the further assertion $|p^\rho| = p^{1/2}$ for these actual blocks. Equation (CD5.2) proves that assertion is equivalent to $\text{Re } \rho = 1/2$, for any one prime $p > 1$; it does not assume it from the source's unique supporting point.

CD6. The retained reflection gives an exact degree factor, and exposes the sign

The original functional equation, with all factors retained in (CD3.1), gives the involution $\rho^\# = 1 - \bar{\rho}$ with $m_{\rho^\#} = m_\rho$. On $\mathcal{B}$, $F^\#(s) = \overline{F(1 - \bar{s})}$. The actual reflected form is

$$W([F], [G]) = \sum_\rho m_\rho \overline{F(\rho^\#)} G(\rho). \quad (\text{CD6.1})$$

The sum is over distinct zeros, with multiplicity supplied by m_ρ , and is absolutely convergent and continuous. For example strip decay of order two bounds the summand by a constant times $(1 + |\text{Im } \rho|)^{-4}$; the zero count from S2 is $O((R + 2)^{3/2})$. A dyadic decomposition makes the sum finite. RZ10 also gives the quotient seminorm bound and its exact radical. Direct substitution yields

$$W(T_p x, T_p y) = p W(x, y), \quad (\text{CD6.2})$$

because $\overline{\rho^\#} + \rho = 1$. Thus the similitude factor is already derived on the actual whole arithmetic quotient. It equals the prime norm $p = |L/[p]L|$; no finite geometric morphism with degree p on this quotient is asserted by that equality.

Its sign is not supplied by that factor. The globally constructed vectors e_ρ have value one at ρ and zero at all other zeros. On a distinct reflected pair their actual form is

$$W|_{\langle [e_\rho], [e_{\rho^\#}] \rangle} = \begin{pmatrix} 0 & m_\rho \\ m_\rho & 0 \end{pmatrix}, \quad (\text{CD6.3})$$

as follows by inserting those two exact value functions in (CD6.1). In particular their difference has value $-2m_\rho$. If a zero is fixed by reflection, its corresponding value vector instead has squared value m_ρ . All higher multiplicity jets lie in the radical of this value form and remain nonzero elements of $\mathcal{Q}$; no simplicity assertion is implicit.

The two eigenvalue moduli on a distinct reflected pair are exactly $p^{\text{Re } \rho}$ and $p^{1-\text{Re } \rho}$. Their product is p , and (CD6.2) holds on that pair. The calculated negative direction and the calculated unequal moduli are two descriptions of the same potential obstruction in this particular quotient. The unique complete counting record has supplied the quotient, all these projectors, and this pairing. Its uniqueness has not changed (CD6.3) into a positive matrix.

CD7. What follows for the proposed contradiction

The calculation establishes three connected facts. First, the complete reconstructed prime record determines the original zeta and hence every actual zero and multiplicity. Second, each such zero has an explicit nonzero projector in the source quotient; the joint kernel of all these projectors is zero. Third, the arithmetic action on those projectors has modulus $p^{\text{Re } \rho}$ and satisfies the reflected degree identity (CD6.2), with its sign fully calculated.

The detector identities prove that every actual zero has residue count $m_\rho > 0$ and a nonzero projector, with no restriction on its real part. Thus the proposed contradiction has not been derived: off-line location has not been shown to imply a failure of detection. This identifies the exact inference not established by the reconstruction; it does not discard the supporting geometry or the programme of deriving a weight theorem over it.

The integrality comparison is given in INTEGRAL_COUNTING_AND_WEIGHT_COMPARISON.md. Its exact connecting object is the integral group algebra of the multiplicative positive rationals, acting through (CD5.1). This retains integer arithmetic and the full prime scaling representation without falsely treating that representation as additive. No theorem of positivity or purity is introduced as a missing hypothesis.

Sources and proof provenance

- Alain Connes and Caterina Consani, *The Absolute Twistor Line and the Geometry of $\overline{\text{Spec } \mathbb{Z}}$* , arXiv:2609.00299v1, original CC.tex, especially lines 494–615 and 772–904. The common generic support and actual signed stalk are inputs from this source, with the comparison proved in CG1–CG3. Public source: <https://arxiv.org/html/2609.00299v1#S3>.
- Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, original rhready.tex, lines 526–545. The source’s complete multiplier is compared in S2 and (CD3.1); its factorization theorem is used with all genus-one factors in the full synthesis proof. Public source: <https://arxiv.org/abs/2602.04022>.
- Local complete derivations: CG0–CG8 and TP0–TP14; GLOBAL_MELLIN_SYNTHESIS.md S1–S7; GLOBAL_CANONICAL_MELLIN_INVOLUTION.md G1–G5; ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md RZ1–RZ12. These are programme derivations, not quotations or novelty claims. Their complete sources are retained in the cumulative local reader.

The disk index routes Deligne’s original Weil I and Weil II to PDF-only records. They were not newly read as TeX in this calculation; no claim of reconstructing Deligne’s proof is made here. The elementary integral comparison in the companion note is proved directly. A complete derivation of Deligne’s machinery over the present source remains a different mathematical task from the detection result proved above.

The original-zeta resolvent, global projectors, and every multiplicity jet

Independent mathematical derivation, 24 September 2026. Proof locators RZ1–RZ12.

This note works on the entire quotient already constructed in GLOBAL_MELLIN_SYNTHESIS.md, S1–S7, and GLOBAL_CANONICAL_MELLIN_INVOLUTION.md, G1–G5. Those portions were read for this derivation. S5 supplies the equality between the closed original source image and the ideal of all actual nontrivial-zero jets. That synthesis theorem is an input here, not a result reproved or attributed to this note. The source Fourier, Mellin, dilation and reflection conventions are retained exactly.

Every zero below is a zero of the original Riemann zeta function with its actual multiplicity. No simplicity, critical-line location, positivity or RH assumption is made. All operations are on the named function spaces and their quotient; they put no addition, coordinate or metric on the user’s supporting datum.

RZ1. Spaces, full multiplier, and the operator domain

Let

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) := \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j k(u)| < \infty \quad (N, j \geq 0) \right\}, \quad (\text{RZ1.1})$$

and let $\mathcal{B}$ be the entire functions with every seminorm

$$b_{A,M}(F) = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty. \quad (\text{RZ1.2})$$

Integer nonnegative A, M, N, j suffice. The exact topological isomorphism from S1 is

$$\mathcal{M}k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{RZ1.3})$$

Write $\mathcal{Z}$ for the set of distinct original nontrivial zeros, and m_ρ for the full positive multiplicity of $\rho \in \mathcal{Z}$. Define the closed ideal

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \quad (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{RZ1.4})$$

The source quotient $\mathcal{A}/\overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}$ is identified with this quotient by the already proved S5. A bracket $[F]$ denotes its class. We use the quotient Fréchet topology, not an unspecified product topology on jets.

The exact retained element of $\mathcal{J}$ is

$$F_0(s) = \mathcal{M}k_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad (\text{RZ1.5})$$

where

$$k_0(u) = u^{1/2} \sum_{n \geq 1} f_0(nu), \quad f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}. \quad (\text{RZ1.6})$$

The whole product (RZ1.5) has its entire continuation. Its zeros and their orders are exactly (ρ, m_ρ) . Its exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad (\text{RZ1.7})$$

and, for every positive integer r ,

$$\begin{aligned} F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_0(1+2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \end{aligned} \quad (\text{RZ1.8})$$

These are the original endpoint, Gamma-pole and trivial-zero cancellations, with their resulting values retained. They are inputs established in S2 and G4; the function ζ is not replaced by another function in the arithmetic argument.

Define on all of $\mathcal{B}$

$$\mathbf{L}F(s) = sF(s). \quad (\text{RZ1.9})$$

For all A, M ,

$$b_{A,M}(\mathbf{L}F) \leq (A+1)b_{A,M+1}(F). \quad (\text{RZ1.10})$$

Multiplication by s preserves every required zero order, so $\mathbf{L}\mathcal{J} \subseteq \mathcal{J}$. It therefore induces a continuous operator L on all of $\mathcal{Q}$; its domain is the entire quotient.

On $\mathcal{A}$, the exact corresponding operator is

$$\mathbf{D} = \frac{1}{2} - u\partial_u. \quad (\text{RZ1.11})$$

Indeed integration by parts, whose two boundary terms vanish by (RZ1.1), gives

$$\mathcal{M}(u\partial_u k)(s) = -(s - \tfrac{1}{2})\mathcal{M}k(s), \quad \mathcal{M}(\mathbf{D}k)(s) = s\mathcal{M}k(s). \quad (\text{RZ1.12})$$

Moreover

$$p_{N,j}(\mathbf{D}k) \leq \tfrac{1}{2}p_{N,j}(k) + p_{N,j+1}(k). \quad (\text{RZ1.13})$$

The variable of L is thus the original s , not $s - \tfrac{1}{2}$.

RZ2. Estimates for evaluation, multiplication and entire division

For a fixed $z \in \mathbb{C}$, positive radius r , and $j \geq 0$, the Cauchy integral formula gives

$$\frac{|F^{(j)}(z)|}{j!} \leq r^{-j}b_{A,0}(F) \quad \text{when } A \geq |\operatorname{Re} z| + r. \quad (\text{RZ2.1})$$

Consequently every jet functional is continuous and $\mathcal{J}$ is closed. It is an ideal because the Leibniz formula preserves a zero of any specified finite order under multiplication by an entire function.

For completeness, the quotient used here is itself complete. The increasing seminorms $b_{j,j}$, $j \geq 1$, generate the topology of $\mathcal{B}$; their quotient seminorms generate that of $\mathcal{Q}$. From a Cauchy sequence in the quotient choose a subsequence whose j -th consecutive difference has quotient seminorm below 2^{-j} for $b_{j,j}$. Lift that difference to $H_j \in \mathcal{B}$ with $b_{j,j}(H_j) < 2^{1-j}$. The series $\sum_j H_j$ converges in every fixed seminorm: after its index exceeds that seminorm's indices, the displayed geometric bound controls the tail. Completeness of $\mathcal{B}$ therefore gives a limit of the lifted partial sums. Its quotient class is the limit of the selected subsequence and hence of the original Cauchy sequence. Closedness of $\mathcal{J}$ makes the quotient Hausdorff. This proves the stated complete Fréchet domain.

For $F, G \in \mathcal{B}$, and a nonnegative integer r ,

$$b_{A,M}(FG) \leq b_{A,0}(F)b_{A,M}(G), \quad (\text{RZ2.2})$$

$$b_{A,M}((s-z)^r F) \leq (A + |z| + 1)^r b_{A,M+r}(F). \quad (\text{RZ2.3})$$

Thus multiplication by a fixed member of $\mathcal{B}$, and by every fixed polynomial, is continuous on $\mathcal{B}$.

We need entire division, including its local estimate. Suppose that $H \in \mathcal{B}$ vanishes at z to order at least $r \geq 1$. Then

$$D_{z,r}H(s) = \frac{H(s)}{(s-z)^r} \quad (\text{RZ2.4})$$

extends to an entire member of $\mathcal{B}$, continuously on the closed subspace with that vanishing condition. Outside $|s-z| < 1$, its weighted absolute value is at most that of H . Inside $|s-z| \leq 1$, the holomorphic quotient and the maximum principle on $|s-z| \leq 2$ give

$$|D_{z,r}H(s)| \leq 2^{-r} \sup_{|w-z|=2} |H(w)|. \quad (\text{RZ2.5})$$

The weight in that smaller disk is at most $(2 + |\operatorname{Im} z|)^M$. Taking any integer $A' \geq \max(A, |\operatorname{Re} z| + 2)$ proves, for example,

$$b_{A,M}(D_{z,r}H) \leq b_{A,M}(H) + 2^{-r}(2 + |\operatorname{Im} z|)^M b_{A',0}(H). \quad (\text{RZ2.6})$$

The same estimate is uniform as z ranges over a fixed compact set. The quotient at $s = z$ is the corresponding Taylor coefficient, so no singular integral is hidden in this assertion.

All further divisions in this note are applications of this estimate or the first-order version with a constructed vanishing numerator.

RZ3. The continuous global resolvent

For $\lambda \notin \mathcal{Z}$, define on all of $\mathcal{B}$

$$\boxed{\mathcal{R}_\lambda F(s) = \frac{F(s) - F_0(s) F(\lambda)/F_0(\lambda)}{\lambda - s}.} \quad (\text{RZ3.1})$$

The numerator is in $\mathcal{B}$, vanishes at $s = \lambda$, and $F_0(\lambda) \neq 0$. The removable value is exactly

$$\mathcal{R}_\lambda F(\lambda) = -F'(\lambda) + F_0'(\lambda) \frac{F(\lambda)}{F_0(\lambda)}. \quad (\text{RZ3.2})$$

The negative sign is caused by the denominator $\lambda - s$. RZ2 proves that $\mathcal{R}_\lambda F \in \mathcal{B}$.

Here is the needed uniform continuity statement. Let K be nonempty and compact in $\mathbb{C} \setminus \mathcal{Z}$. Set

$$c_K = \min_{\lambda \in K} |F_0(\lambda)| > 0, \quad B_K = \sup_{\lambda \in K} |\operatorname{Re} \lambda|, \quad T_K = \sup_{\lambda \in K} |\operatorname{Im} \lambda|.$$

For each A, M , choose an integer $A' \geq \max(A, B_K + 2)$. Evaluation is bounded by $b_{A',0}(F)$, so the numerator in (RZ3.1) satisfies

$$b_{A,M} \left(F - F_0 \frac{F(\lambda)}{F_0(\lambda)} \right) \leq b_{A,M}(F) + c_K^{-1} b_{A,M}(F_0) b_{A',0}(F). \quad (\text{RZ3.3})$$

Write $G_\lambda = F - F_0 F(\lambda)/F_0(\lambda)$ for the numerator. Applying (RZ2.6) once, with the output strip A , gives

$$b_{A,M}(\mathcal{R}_\lambda F) \leq b_{A,M}(G_\lambda) + \frac{1}{2}(2 + T_K)^M b_{A',0}(G_\lambda).$$

The second term requires only a bound on the numerator itself, not another division. Directly,

$$b_{A',0}(G_\lambda) \leq (1 + c_K^{-1} b_{A',0}(F_0)) b_{A',0}(F).$$

Combining this with (RZ3.3) gives the explicit finite constant

$$C_{K,A,M} = 1 + c_K^{-1} b_{A,M}(F_0) + \frac{1}{2}(2 + T_K)^M (1 + c_K^{-1} b_{A',0}(F_0))$$

and proves

$$\sup_{\lambda \in K} b_{A,M}(\mathcal{R}_\lambda F) \leq C_{K,A,M} b_{A',M}(F). \quad (\text{RZ3.4})$$

The condition $A' \geq \max(A, B_K + 2)$ already contains every radius-two disk centered at a point of K . Thus the same A' controls the division disk and both numerator bounds; no further enlargement to $A' + 2$ is needed. Only this stated increase of the strip index has been used.

If $F \in \mathcal{I}$, its numerator in (RZ3.1) vanishes to order at least m_ρ at every ρ . Since $\lambda - \rho \neq 0$, division preserves each such order. Thus

$$\mathcal{R}_\lambda \mathcal{I} \subseteq \mathcal{I}. \quad (\text{RZ3.5})$$

The induced operator R_λ is continuous on $\mathcal{Q}$. More explicitly, with quotient seminorms

$$q_{A,M}([F]) = \inf_{H \in \mathcal{I}} b_{A,M}(F + H),$$

equation (RZ3.4) gives

$$\sup_{\lambda \in K} q_{A,M}(R_\lambda[F]) \leq C_{K,A,M} q_{A',M}([F]). \quad (\text{RZ3.6})$$

For the supremum, this follows by using the same representative $F + H$ for every λ , then taking the infimum.

The exact identities before taking the quotient are

$$(\lambda - \mathbb{L}) \mathcal{R}_\lambda F = F - F_0 \frac{F(\lambda)}{F_0(\lambda)}, \quad \mathcal{R}_\lambda((\lambda - \mathbb{L})F) = F. \quad (\text{RZ3.7})$$

The correction in the first identity belongs to $\mathcal{J}$. Therefore on the actual quotient

$$\boxed{R_\lambda = (\lambda - L)^{-1} \quad (\lambda \notin \mathcal{Z}).} \quad (\text{RZ3.8})$$

Although $F(\lambda)$ by itself generally does not descend to this quotient, the full expression (RZ3.1) does, by (RZ3.5). In fact $F_0(\lambda) \neq 0$ already proves that evaluation at such a λ does not annihilate $\mathcal{J}$. The correction term is essential.

For example the endpoint resolvents use exactly

$$\mathcal{R}_0 F(s) = \frac{F(s) - 8F_0(s)F(0)}{-s}, \quad \mathcal{R}_1 F(s) = \frac{F(s) - 8F_0(s)F(1)}{1-s}. \quad (\text{RZ3.9})$$

At $\lambda = -2r$, (RZ3.1) uses the nonzero value (RZ1.8). Thus 0, 1 and the original trivial zeros are resolvent points of this specified nontrivial-zero quotient. Their full original multiplier contributions have been evaluated, not dropped.

RZ4. Parameter holomorphy and the source resolvent

The numerator

$$F(s)F_0(\lambda) - F_0(s)F(\lambda) \quad (\text{RZ4.1})$$

vanishes on the diagonal $s = \lambda$. Its division by $\lambda - s$ is holomorphic jointly in the two variables: locally expand both entire functions into power series, or factor each difference by the integral of its derivative along the segment from λ to s . Dividing additionally by $F_0(\lambda)$ is holomorphic when $\lambda \notin \mathcal{Z}$. Hence $\lambda \mapsto \mathcal{R}_\lambda F(s)$ is pointwise holomorphic.

Together with (RZ3.4), this is holomorphy in the topology of $\mathcal{B}$, and in the topology of uniform convergence on bounded subsets for the operator family. Here are the details needed for that conclusion. On a small closed parameter disk contained in the resolvent set, the family is uniformly bounded in each output seminorm by one input seminorm. Cauchy's formula in λ can therefore be integrated in each seminorm; completeness of $\mathcal{B}$ puts the integral in $\mathcal{B}$. Pointwise Cauchy equality identifies this integral with the given entire function of s . The Cauchy remainder estimate on a slightly larger disk bounds the difference-quotient remainder in any output seminorm by a constant times $|h|$ and one input seminorm. Taking the supremum over any bounded input set proves operator holomorphy in that topology. The identical argument, using (RZ3.6), applies to the quotient.

In particular contour integrals of R_λ on compact contours in the resolvent set are well defined continuous operators on the complete Fréchet quotient. The usual resolvent identity follows directly from the two inverse identities:

$$R_\lambda - R_\mu = (\mu - \lambda)R_\lambda R_\mu. \quad (\text{RZ4.2})$$

The full source representative is also explicit. If $F = \mathcal{M}k$, then

$$(r_\lambda k)(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(\frac{1}{2} + it) - F_0(\frac{1}{2} + it)F(\lambda)/F_0(\lambda)}{\lambda - \frac{1}{2} - it} u^{-it} dt. \quad (\text{RZ4.3})$$

If the denominator vanishes on this line, the integrand has the removable value (RZ3.2). Its Mellin transform is (RZ3.1).

For precision, every $H \in \mathcal{B}$ has inverse

$$k_H(e^x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(\frac{1}{2} + it) e^{-itx} dt. \quad (\text{RZ4.4})$$

Take $c = \frac{1}{2} + N + 1$ for $x \geq 0$, and $c = \frac{1}{2} - N - 1$ for $x \leq 0$. The proven strip contour shift gives

$$\frac{d^j}{dx^j} k_H(e^x) = \frac{e^{-(c-1/2)x}}{2\pi} \int_{\mathbb{R}} [-(c - \frac{1}{2} + it)]^j H(c + it) e^{-itx} dt. \quad (\text{RZ4.5})$$

Since $\int_{\mathbb{R}} (1 + |t|)^{-2} dt = 2$, this yields the explicit estimate

$$\sup_x e^{N|x|} \left| \frac{d^j}{dx^j} k_H(e^x) \right| \leq \frac{(N+2)^j}{\pi} b_{N+2,j+2}(H), \quad (\text{RZ4.6})$$

and consequently

$$p_{N,j}(k_H) \leq \frac{2(N+2)^j}{\pi} b_{N+2,j+2}(H). \quad (\text{RZ4.7})$$

The notation $d^j/dx^j k_H(e^x)$ denotes derivatives of the composite function. These estimates prove that (RZ4.3) belongs to $\mathcal{A}$, continuously and uniformly on compact resolvent-parameter sets.

The exact source identities are

$$(\lambda - \mathbf{D})r_\lambda k = k - k_0 \frac{\mathcal{M}k(\lambda)}{F_0(\lambda)}, \quad r_\lambda((\lambda - \mathbf{D})k) = k. \quad (\text{RZ4.8})$$

The first correction lies in the original source image because $k_0 = \mathcal{E}f_0$. The whole closed source image is preserved by r_λ , by S5 and (RZ3.5).

RZ5. Global representatives of every actual zero jet

Fix an actual zero ρ , and write $m = m_\rho$. Put

$$A_\rho(s) = \frac{F_0(s)}{(s - \rho)^m}. \quad (\text{RZ5.1})$$

It is entire, belongs to $\mathcal{B}$ by RZ2, and

$$a_r := \frac{A_\rho^{(r)}(\rho)}{r!} = \frac{F_0^{(m+r)}(\rho)}{(m+r)!}, \quad a_0 \neq 0. \quad (\text{RZ5.2})$$

Define the coefficients of the reciprocal Taylor polynomial by

$$b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{h=1}^r a_h b_{r-h} \quad (1 \leq r < m), \quad (\text{RZ5.3})$$

and set

$$B_\rho(s) = \sum_{r=0}^{m-1} b_r (s - \rho)^r, \quad e_\rho(s) = A_\rho(s) B_\rho(s). \quad (\text{RZ5.4})$$

The recursion says exactly

$$e_\rho(s) = 1 + O((s - \rho)^m) \quad (s \rightarrow \rho). \quad (\text{RZ5.5})$$

Every function in (RZ5.4) after multiplication by A_ρ belongs to $\mathcal{B}$, by the polynomial estimate (RZ2.3).

At each other actual zero $\sigma \neq \rho$, the denominator in (RZ5.1) is nonzero. Hence A_ρ , e_ρ , and $e_\rho(s)(s - \rho)^j$ vanish there to order at least m_σ . This holds at every other zero simultaneously, without a finite interpolation range.

Define, for $0 \leq j < m$,

$$E_{\rho,j}(s) = e_\rho(s)(s - \rho)^j, \quad D_{\rho,j}(s) = \frac{1}{j!} E_{\rho,j}(s). \quad (\text{RZ5.6})$$

The complete exact jet statement is

$$E_{\rho,j}^{(k)}(\sigma) = j! \mathbf{1}_{\{\sigma=\rho, k=j\}} \quad (\sigma \in \mathcal{Z}, 0 \leq k < m_\sigma), \quad (\text{RZ5.7})$$

and $D_{\rho,j}^{(k)}(\sigma)$ is the same expression without $j!$. Thus these are global representatives for each individual derivative coordinate, including its factorial.

They have actual prequotient source representatives

$$k_{\rho,j}(u) = \frac{1}{2\pi} \int_{\mathbb{R}} e_{\rho}(\frac{1}{2} + it)(\frac{1}{2} + it - \rho)^j u^{-it} dt, \quad \mathcal{M}k_{\rho,j} = E_{\rho,j}. \quad (\text{RZ5.8})$$

The estimate (RZ4.7), with $H = E_{\rho,j}$, proves every source seminorm is finite. In particular these representatives belong to the whole $\mathcal{A}$, rather than merely defining distributions or a finite set of prescribed values. They do not belong to the annihilated source image when the indicated jet is nonzero; that would contradict (RZ5.7) and S5.

All multiplier derivatives in the recursion can be retained explicitly in terms of the original zeta. Set only for displaying these derivatives

$$C(s) = \frac{1}{8} s(s-1) \pi^{-s/2} \Gamma(s/2). \quad (\text{RZ5.9})$$

At the nontrivial zero ρ , C is holomorphic and nonzero. The exact coefficient formula is

$$a_r = \sum_{h=0}^r \frac{C^{(h)}(\rho)}{h!} \frac{\zeta^{(m+r-h)}(\rho)}{(m+r-h)!}. \quad (\text{RZ5.10})$$

No derivative term has been suppressed: derivatives of ζ of order less than m vanish by the definition of its actual multiplicity. For $P(s) = s(s-1)$,

$$C^{(h)}(s) = \frac{\pi^{-s/2}}{8} \sum_{\substack{a+b+c=h \\ 0 \leq a \leq 2}} \frac{h!}{a! b! c!} P^{(a)}(s) \left(-\frac{\log \pi}{2} \right)^b 2^{-c} \Gamma^{(c)}(s/2), \quad (\text{RZ5.11})$$

where $P^{(0)} = s(s-1)$, $P' = 2s-1$, and $P'' = 2$.

At an exceptional point $z \in \{0, 1, -2, -4, \dots\}$, the same entire representatives have the full value

$$E_{\rho,j}(z) = F_0(z) B_{\rho}(z) (z - \rho)^{j-m}, \quad (\text{RZ5.12})$$

with $F_0(z)$ given by (RZ1.7)–(RZ1.8). These values have not been forced to vanish or removed from the source functions. Evaluation at these points is not a quotient character, as RZ3 already proves.

RZ6. Continuous projectors and exact generalized eigenspaces

For a holomorphic germ H at ρ , write

$$\text{Tay}_{\rho, < r} H(s) = \sum_{j=0}^{r-1} \frac{H^{(j)}(\rho)}{j!} (s - \rho)^j \quad (r \geq 1). \quad (\text{RZ6.1})$$

The promised finite-rank lift on the entire function space is

$$\boxed{\mathcal{P}_{\rho} F(s) = A_{\rho}(s) \text{Tay}_{\rho, < m} \left(\frac{F}{A_{\rho}} \right) (s).} \quad (\text{RZ6.2})$$

The quotient F/A_{ρ} is used only as a germ at ρ , where $A_{\rho}(\rho) \neq 0$. The displayed output is a globally defined entire function. It belongs to $\mathcal{B}$ as a finite polynomial times A_{ρ} . Its coefficients are fixed finite linear combinations of the jets of F at ρ , so (RZ2.1) proves continuity.

The output has exactly the same first m Taylor coefficients as F at ρ , and has every required vanishing jet at all other zeros. Moreover

$$\mathcal{P}_{\rho}^2 = \mathcal{P}_{\rho}, \quad \mathcal{P}_{\rho} \mathcal{J} = \{0\}, \quad \mathcal{P}_{\rho} \mathcal{P}_{\sigma} = 0 \quad (\rho \neq \sigma). \quad (\text{RZ6.3})$$

For the first equality, divide $\mathcal{P}_\rho F$ by A_ρ ; its quotient is already a polynomial of degree below m , so its Taylor truncation is unchanged. For the second, F/A_ρ has a zero of order at least m at ρ . For the third, $\mathcal{P}_\sigma F$ has a zero of order at least m_ρ at ρ , and the same Taylor argument applies.

Consequently the quotient projector

$$P_\rho[F] = [\mathcal{P}_\rho F] = [e_\rho F] \quad (\text{RZ6.4})$$

is well defined and continuous. To verify the second equality, both functions have the jets of F at ρ and zero jets at every other zero; their difference is in $\mathcal{I}$. Equivalently

$$P_\rho[F] = \sum_{j=0}^{m-1} \frac{F^{(j)}(\rho)}{j!} [E_{\rho,j}] = \sum_{j=0}^{m-1} F^{(j)}(\rho) [D_{\rho,j}]. \quad (\text{RZ6.5})$$

The range $\mathcal{Q}_\rho = P_\rho \mathcal{Q}$ has dimension exactly m , since (RZ5.7) proves independence and (RZ6.5) proves spanning.

The lift $\mathcal{P}_\rho : \mathcal{B} \rightarrow \mathcal{B}$ annihilates $\mathcal{I}$, so it also defines a continuous map $\mathcal{Q} \rightarrow \mathcal{B}$, by the defining quotient topology. On $\mathcal{Q}_\rho$ it is a continuous choice of source representative. In addition $\mathcal{Q} = \mathcal{Q}_\rho \oplus \ker P_\rho$ is a topological direct sum: the forward map sends x to $(P_\rho x, x - P_\rho x)$, and its continuous inverse adds the two components. This is a split finite-dimensional spectral block of the full quotient.

This range is exactly the actual generalized eigenspace:

$$\mathcal{Q}_\rho = \ker(L - \rho)^m. \quad (\text{RZ6.6})$$

The inclusion from left to right follows from the required local orders, or from $(s - \rho)^m e_\rho = F_0 B_\rho$. Conversely, if $(s - \rho)^m F \in \mathcal{I}$, then at every other zero σ the factor $(s - \rho)^m$ is an invertible holomorphic germ, so F has all required zero jets there. At ρ , $F - \mathcal{P}_\rho F$ has all required zero jets by construction. Hence $[F] = P_\rho[F]$, proving the reverse inclusion.

Put $N_\rho = (L - \rho)|_{\mathcal{Q}_\rho}$. Its complete action is

$$N_\rho[E_{\rho,j}] = \begin{cases} [E_{\rho,j+1}], & j < m-1, \\ 0, & j = m-1. \end{cases} \quad (\text{RZ6.7})$$

In derivative-coordinate representatives,

$$N_\rho[D_{\rho,j}] = (j+1)[D_{\rho,j+1}] \quad (j < m-1). \quad (\text{RZ6.8})$$

Thus $N_\rho^m = 0$ and $N_\rho^{m-1} \neq 0$; its ranks are $\text{rank } N_\rho^r = m - r$ for $0 \leq r < m$. In particular $[E_{\rho,m-1}] \neq 0$ is an eigenvector with eigenvalue ρ .

It now follows from RZ3 and this eigenvector that the finite continuous spectrum is exactly

$$\boxed{\text{Spec}(L) = \mathcal{Z}.} \quad (\text{RZ6.9})$$

Here the resolvent set means those $\lambda \in \mathbb{C}$ for which $\lambda - L$ has an everywhere-defined continuous inverse. No Banach-space spectral convention is being silently substituted.

The family of projectors also separates the complete quotient:

$$\bigcap_{\rho \in \mathcal{Z}} \ker P_\rho = \{0\}. \quad (\text{RZ6.10})$$

Indeed every vanishing projector forces all the corresponding actual jets to vanish, and S5 identifies their common kernel with $\mathcal{I}$. This is a statement on every quotient class. It does not assume convergence of an infinite sum of projectors.

RZ7. Riesz contour formula, every Laurent coefficient, and signs

Choose a positively oriented circle γ_ρ enclosing ρ and no other zero. Its radius can be chosen positive because the zeros of the nonzero entire F_0 are isolated. On that contour, RZ3–RZ4 give a holomorphic, locally equicontinuous resolvent.

We first compute its integral at the level of $\mathcal{B}$:

$$\frac{1}{2\pi i} \int_{\gamma_\rho} \mathcal{R}_\lambda F d\lambda = \mathcal{P}_\rho F. \quad (\text{RZ7.1})$$

For s outside the closed circle, the first term $F(s)/(\lambda - s)$ in (RZ3.1) has integral zero. Put $t = \lambda - \rho$, $d = s - \rho$, and write locally

$$\frac{F(\rho + t)}{A_\rho(\rho + t)} = \sum_{\ell \geq 0} h_\ell t^\ell. \quad (\text{RZ7.2})$$

The remaining term is

$$-F_0(s) \frac{t^{-m} \sum_{\ell \geq 0} h_\ell t^\ell}{t - d}.$$

Since

$$\frac{1}{t - d} = - \sum_{r \geq 0} \frac{t^r}{d^{r+1}}, \quad (\text{RZ7.3})$$

its residue is

$$F_0(s) \sum_{\ell=0}^{m-1} \frac{h_\ell}{(s - \rho)^{m-\ell}} = A_\rho(s) \sum_{\ell=0}^{m-1} h_\ell (s - \rho)^\ell. \quad (\text{RZ7.4})$$

The two negative signs cancel. This is (RZ6.2). Both sides of (RZ7.1) are entire functions of s , so equality outside the circle proves equality everywhere. This also handles s on or inside the contour without separating singular terms there.

Passing to the quotient proves the actual Riesz formula

$$\boxed{P_\rho = \frac{1}{2\pi i} \int_{\gamma_\rho} (\lambda - L)^{-1} d\lambda.} \quad (\text{RZ7.5})$$

Thus the explicit projectors are the Riesz projectors of the actual continuous operator, not merely finite interpolation maps.

Every principal Laurent coefficient has an equally explicit global lift. For $0 \leq r < m$,

$$\mathcal{C}_{\rho,r} F(s) = A_\rho(s) (s - \rho)^r \text{Tay}_{\rho, < m-r} \left(\frac{F}{A_\rho} \right) (s), \quad (\text{RZ7.6})$$

and

$$\frac{1}{2\pi i} \int_{\gamma_\rho} (\lambda - \rho)^r \mathcal{R}_\lambda F d\lambda = \mathcal{C}_{\rho,r} F. \quad (\text{RZ7.7})$$

For $r \geq m$ this integral is zero. The proof repeats (RZ7.3): the residue condition becomes $\ell + \text{summation index} = m - r - 1$, giving exactly (RZ7.6).

The induced coefficient is

$$[\mathcal{C}_{\rho,r} F] = (L - \rho)^r P_\rho[F]. \quad (\text{RZ7.8})$$

Indeed replacing the Taylor truncation of length $m-r$ by the one of length m changes the result by $F_0(s)P(s)$, where P is a polynomial. This product belongs to $\mathcal{B}$ by (RZ2.3) and to $\mathcal{I}$ by its full zero orders; no claim that P itself belongs to $\mathcal{B}$ is needed.

There is also a one-step identity retaining the exact source correction. With the coefficients h_j of (RZ7.2), and with $\mathcal{C}_{\rho,m} = 0$,

$$(s - \rho)\mathcal{C}_{\rho,r}F(s) - \mathcal{C}_{\rho,r+1}F(s) = F_0(s)h_{m-r-1} \quad (0 \leq r < m). \quad (\text{RZ7.11})$$

It follows by subtracting the two Taylor polynomials, whose sole remaining term has degree $m - r - 1$. These corrections are present on $\mathcal{B}$, even though their quotient classes vanish. In particular

$$s\mathcal{P}_\rho F - \mathcal{P}_\rho(sF) = F_0 h_{m-1}, \quad (\text{RZ7.12})$$

so $P_\rho L = LP_\rho$ is exact on $\mathcal{Q}$ with the displayed correction at the representative level.

There is a meromorphic operator family on the whole finite λ -plane. To see this directly before quotienting, the jointly holomorphic divided expression in RZ4 is divided only by $F_0(\lambda)$. Near ρ , its pole order is at most m ; the uniform seminorm estimates on a surrounding circle and the parameter Cauchy formula give its Laurent expansion in continuous operators, uniformly on bounded input sets. Consequently on $\mathcal{Q}$,

$$R_\lambda = \sum_{r=0}^{m-1} \frac{(L - \rho)^r P_\rho}{(\lambda - \rho)^{r+1}} + \text{a holomorphic operator family near } \rho. \quad (\text{RZ7.9})$$

The pole order is exactly m , because its last coefficient has rank one and is nonzero by (RZ6.7). More generally its coefficient of order $(\lambda - \rho)^{-r-1}$ has rank $m - r$.

On the actual block, the resolvent is the finite algebraic identity

$$R_\lambda|_{\mathcal{Q}_\rho} = \sum_{r=0}^{m-1} \frac{N_\rho^r}{(\lambda - \rho)^{r+1}}. \quad (\text{RZ7.10})$$

Multiplication by $(\lambda - \rho)I - N_\rho$ telescopes to the identity because $N_\rho^m = 0$, verifying every sign independently of the contour calculation.

RZ8. The entire unshifted arithmetic action

For every positive real a , define

$$T_a F(s) = a^s F(s), \quad a^s = e^{s \log a}, \quad (\text{RZ8.1})$$

where $\log a$ is the real logarithm. This preserves $\mathcal{B}$ and its ideal $\mathcal{J}$, since the multiplier is entire and nowhere zero. The exact estimates are

$$b_{A,M}(T_a F) \leq \max(a^A, a^{-A}) b_{A,M}(F), \quad (\text{RZ8.2})$$

and on the source,

$$\mathcal{M}^{-1} T_a \mathcal{M} k(u) = a^{1/2} k(u/a) = a^{1/2} V_a k(u), \quad (\text{RZ8.3})$$

$$p_{N,j}(a^{1/2} V_a k) \leq a^{1/2} \max(a^N, a^{-N}) p_{N,j}(k). \quad (\text{RZ8.4})$$

Both the factor $a^{1/2}$ and the original Mellin variable s are retained.

These operators obey $T_a T_b = T_{ab}$ and $T_1 = I$. In the real parameter $t = \log a$, they form a strongly smooth group whose generator is L :

$$\frac{d^r}{dt^r} T_{e^t} [F] = L^r T_{e^t} [F]. \quad (\text{RZ8.5})$$

For a direct topological justification, on a bounded real t -interval the derivative multipliers $s^r e^{ts}$ satisfy (RZ8.2) followed by (RZ1.10) repeatedly. Taylor's integral remainder for real t gives difference-quotient convergence in every seminorm, using one additional power of s . Equivalently (RZ8.4) and the logarithmic derivative bounds give the same assertion on $\mathcal{A}$.

In this explicit real-parameter multiplier sense we write

$$\boxed{n^L = T_n = \sqrt{n} V_n \quad (n = 1, 2, 3, \dots)}. \quad (\text{RZ8.6})$$

This notation is justified by the multiplier and group formulas just proved. It does not assume convergence of the exponential power series of a continuous operator in a nonnormable Fréchet space. Such a series convergence is unnecessary here.

On every actual multiplicity block,

$$T_n[E_{\rho,j}] = n^\rho \sum_{r=0}^{m_\rho-1-j} \frac{(\log n)^r}{r!} [E_{\rho,j+r}]. \quad (\text{RZ8.7})$$

This follows by taking the full Taylor series of $n^s = n^\rho e^{(s-\rho)\log n}$ at ρ , retaining all terms that survive modulo the order m_ρ . All other-zero jets still vanish. Thus

$$T_n|_{\mathcal{Q}_\rho} = n^\rho \sum_{r=0}^{m_\rho-1} \frac{(\log n)^r}{r!} N_\rho^r. \quad (\text{RZ8.8})$$

On this block the eigenvalue is n^ρ , with algebraic multiplicity m_ρ ; its modulus is exactly $n^{\text{Re } \rho}$. No conclusion that this modulus is $\sqrt{n}$ has been assumed.

For the original unscaled derivative jets, the complete triangular formula is

$$(T_n F)^{(j)}(\rho) = n^\rho \sum_{\ell=0}^j \binom{j}{\ell} (\log n)^{j-\ell} F^{(\ell)}(\rho). \quad (\text{RZ8.9})$$

Equivalently, on the representatives of individual derivative coordinates,

$$T_n[D_{\rho,j}] = n^\rho \sum_{r=0}^{m_\rho-1-j} \binom{j+r}{r} (\log n)^r [D_{\rho,j+r}]. \quad (\text{RZ8.10})$$

These formulas also cover $n = 1$ with its identity action. The trace of this finite block is $m_\rho n^\rho$, since every positive power of N_ρ has zero diagonal.

For $n > 1$, the arithmetic operator itself retains the full Jordan length m_ρ . Indeed

$$n^{-\rho} T_n - I = N_\rho H_n(N_\rho), \quad H_n(z) = \sum_{r=1}^{m_\rho} \frac{(\log n)^r}{r!} z^{r-1}, \quad H_n(0) = \log n \neq 0. \quad (\text{RZ8.11})$$

The last possible term is harmless because $N_\rho^{m_\rho} = 0$. The polynomial $H_n(N_\rho)$ is invertible: its nonzero constant term times the identity has a nilpotent correction, whose inverse is its finite geometric series. Hence $(T_n - n^\rho I)^r$ has the same rank as N_ρ^r , including the nilpotence index m_ρ .

Different zero blocks are never identified merely because one arithmetic operator has coincident eigenvalues. For a fixed $n > 1$,

$$n^\rho = n^\sigma \iff \rho - \sigma = \frac{2\pi i k}{\log n} \text{ for some } k \in \mathbb{Z}. \quad (\text{RZ8.12})$$

The full family separates the original s -values. Already equality at both $n = 2$ and $n = 3$ implies $k \log 3 = l \log 2$, hence $3^k = 2^l$ for integers k, l . Unique prime factorization forces $k = l = 0$, so $\rho = \sigma$. This statement retains the exact phase periods of each individual operator.

RZ9. Reflection acts on every jet, including its sign

The exact source involution is $k^\#(u) = \overline{k(1/u)}$. On $\mathcal{B}$ it is

$$F^\#(s) = \overline{F(1 - \bar{s})}. \quad (\text{RZ9.1})$$

It is conjugate-linear and continuous:

$$b_{A,M}(F^\#) \leq b_{A+1,M}(F). \quad (\text{RZ9.2})$$

The original functional equation and conjugation preserve zero multiplicities under

$$\rho^\# = 1 - \bar{\rho}, \quad m_{\rho^\#} = m_\rho. \quad (\text{RZ9.3})$$

Thus $\mathcal{J}$ is invariant and the involution descends to the whole quotient. Its derivative formula is

$$(F^\#)^{(j)}(\rho^\#) = (-1)^j \overline{F^{(j)}(\rho)}. \quad (\text{RZ9.4})$$

To prove it, put $s = \rho^\# + h$; then $1 - \bar{s} = \rho - \bar{h}$, and conjugate the full local Taylor series.

Applying (RZ9.4) to the isolating jets gives

$$[E_{\rho,j}]^\# = (-1)^j [E_{\rho^\#,j}], \quad [D_{\rho,j}]^\# = (-1)^j [D_{\rho^\#,j}]. \quad (\text{RZ9.5})$$

These are equalities of actual quotient classes because all their zero jets agree globally. In particular

$$(P_\rho x)^\# = P_{\rho^\#}(x^\#), \quad (Lx)^\# = (1 - L)x^\#, \quad (\text{RZ9.6})$$

$$(T_a x)^\# = a^{1-L} x^\# = a T_a^{-1} x^\#. \quad (\text{RZ9.7})$$

For the nilpotent parts, (RZ9.6) reads

$$(N_\rho x)^\# = -N_{\rho^\#} x^\# \quad (x \in \mathcal{Q}_\rho). \quad (\text{RZ9.8})$$

The sign is essential and does not disappear when a zero happens to be on the critical line.

For the specific source representatives constructed here, (RZ9.5) holds even before passage to the quotient.

The source symmetry $F_0^\# = F_0$ gives

$$A_\rho^\# = (-1)^{m_\rho} A_{\rho^\#}. \quad (\text{RZ9.9})$$

Applying reflection to the uniquely determined reciprocal Taylor polynomial gives

$$B_\rho^\# = (-1)^{m_\rho} B_{\rho^\#}. \quad (\text{RZ9.10})$$

Indeed the reflected product is one modulo the required order, and the reciprocal polynomial of degree below m_ρ is unique by the recursion (RZ5.3). The two signs cancel in the product $e_\rho^\# = e_{\rho^\#}$. Multiplying by the reflected factor $(s - \rho)^j$ supplies exactly the remaining $(-1)^j$ in (RZ9.5). Thus the corresponding source functions in (RZ5.8) also satisfy the exact reflected identity.

RZ10. The actual reflected Weil form and its block trace

Use the existing source-derived form from G5, written on this exact quotient:

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(\rho^\#)} G(\rho). \quad (\text{RZ10.1})$$

This is conjugate-linear in the first input and linear in the second. Its absolute convergence and continuity follow already from the weaker zero-count bound proved in S2,

$$\sum_{|\rho| \leq R} m_\rho \leq C(R + 2)^{3/2}. \quad (\text{RZ10.2})$$

Indeed $0 < \text{Re } \rho < 1$, the reflected zero has the same imaginary part, and

$$|F(\rho^\#)G(\rho)| \leq b_{1,2}(F)b_{1,2}(G)(1 + |\text{Im } \rho|)^{-4}.$$

The sum of the last weights with multiplicities is finite by grouping zeros into dyadic height intervals and applying (RZ10.2). Therefore

$$|W([F], [G])| \leq C_W b_{1,2}(F)b_{1,2}(G). \quad (\text{RZ10.3})$$

It descends because $\mathcal{J}$ annihilates every displayed value. Taking infima over representatives also proves a bound by $C_W q_{1,2}([F]) q_{1,2}([G])$. Reflection and multiplicity preservation prove Hermitian symmetry by relabelling the absolutely convergent sum.

The word trace here has an exact interpretation on the constructed blocks. Multiplication by any $H \in \mathcal{B}$ preserves $\mathcal{Q}_\rho$; in the basis $[E_{\rho,j}]$, its diagonal entries are all $H(\rho)$. Hence

$$\mathrm{Tr}_{\mathcal{Q}_\rho}(M_H) = m_\rho H(\rho), \quad (\text{RZ10.4})$$

and

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} \mathrm{Tr}_{\mathcal{Q}_\rho}(M_{F^\# G}). \quad (\text{RZ10.5})$$

This is the absolutely convergent sum of the specified finite-dimensional block traces. It does not assert a trace-class operator on an unstated Hilbert completion.

Every block entry is explicit:

$$\boxed{W([E_{\rho,j}], [E_{\sigma,k}]) = m_\sigma \mathbf{1}_{\{\rho=\sigma^\#\}} \mathbf{1}_{\{j=0\}} \mathbf{1}_{\{k=0\}}.} \quad (\text{RZ10.6})$$

The same formula holds for $D_{\rho,j}$, since only the zeroth representative contributes. This is immediate from all-zero isolation (RZ5.7).

If $\rho = \rho^\#$, the form on $\mathcal{Q}_\rho$ has one positive direction of squared length m_ρ and radical of dimension $m_\rho - 1$. If $\rho \neq \rho^\#$, put $m = m_\rho = m_{\rho^\#}$. On the two constant-jet representatives its matrix is exactly

$$\begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}. \quad (\text{RZ10.7})$$

Its values on the sum and difference of those representatives are $2m$ and $-2m$, respectively. The entire paired space $\mathcal{Q}_\rho \oplus \mathcal{Q}_{\rho^\#}$ has one positive direction, one negative direction, and radical dimension $2(m - 1)$.

Globally the radical is precisely

$$\mathrm{rad} W = \{[F] \in \mathcal{Q} : F(\rho) = 0 \text{ for all } \rho \in \mathcal{Z}\}. \quad (\text{RZ10.8})$$

One inclusion follows from (RZ10.1). Conversely, pairing with $[E_{\sigma,0}]$ gives $m_\sigma \overline{F(\sigma^\#)}$; if every such pairing vanishes, every zero value is zero. Higher multiplicity jets therefore remain real nonzero quotient data even though this ordinary trace form cannot detect them.

RZ11. Full arithmetic similitude, reflected adjoints, and the exact sign question

For every $a > 0$, with no restriction on the real parts of the actual zeros,

$$\begin{aligned} W(T_a x, T_a y) &= \sum_{\rho} m_\rho \overline{a^{\rho^\#} F(\rho^\#)} a^\rho G(\rho) \\ &= \sum_{\rho} m_\rho a^{\overline{\rho^\#} + \rho} \overline{F(\rho^\#)} G(\rho) \\ &= a W(x, y), \end{aligned} \quad (\text{RZ11.1})$$

because $\overline{\rho^\#} = 1 - \rho$. In particular

$$\boxed{W(n^L x, n^L y) = n W(x, y), \quad n^L = \sqrt{n} V_n.} \quad (\text{RZ11.2})$$

This preserves the full unshifted arithmetic factor. Dividing both arguments by the displayed $\sqrt{n}$ recovers the earlier exact invariance of V_n , but the original n^L is the operator in (RZ11.2).

The corresponding identities for the pairing are

$$W(Lx, y) = W(x, (1 - L)y), \quad (\text{RZ11.3})$$

$$W(T_a x, y) = W(x, aT_a^{-1}y), \quad W(P_\rho x, y) = W(x, P_{\rho^\#}y). \quad (\text{RZ11.4})$$

For the first equation, conjugation of the first factor changes the spectral scalar $\rho^\#$ into $1 - \rho$. The second follows by the same direct scalar calculation, and the projector identity follows by its global zero-jet support. These are identities for a possibly indefinite and degenerate form, not adjoint assertions for an unconstructed positive Hilbert norm.

For the resolvent there is a further retained sign:

$$W(R_\lambda x, y) = -W(x, R_{1-\bar{\lambda}}y) \quad (\lambda \notin \mathcal{Z}). \quad (\text{RZ11.5})$$

The reflected parameter is also outside $\mathcal{Z}$ by (RZ9.3). Directly,

$$\frac{1}{\bar{\lambda} - \rho^\#} = -\frac{1}{1 - \bar{\lambda} - \rho},$$

which proves (RZ11.5) term by term in the convergent form. Equivalently it follows by taking the pairing adjoint of $\lambda - L$ using (RZ11.3).

On an off-line pair the actual eigenvalues obey

$$n^{\rho^\#} = \frac{n}{n^\rho}, \quad |n^\rho| = n^{\text{Re } \rho}, \quad |n^{\rho^\#}| = n^{1-\text{Re } \rho}. \quad (\text{RZ11.6})$$

Their reflected pairing therefore has the exact factor n even when neither modulus is $\sqrt{n}$. The complete nilpotent coefficients of their action remain (RZ8.7)–(RZ8.10); their absence from the value trace is accounted for by the radical (RZ10.8), not by removal of the corresponding vectors.

These explicit representatives also give the precise positivity equivalences for this specified form. The form W is positive semidefinite exactly when every actual zero satisfies $\rho = \rho^\#$: the forward implication follows from the negative vector in (RZ10.7), and the reverse implication follows from the convergent sum $\sum_\rho m_\rho |F(\rho)|^2$. It is positive definite exactly when, in addition, all multiplicities are one. If a multiplicity exceeds one, (RZ5.7) gives a nonzero higher-jet radical vector. If all multiplicities are one, zero values at every zero mean membership in $\mathcal{J}$, so no nonzero quotient vector can have zero squared length in the positive case.

These are proved equivalences and explicit block signatures. No positive-semidefiniteness assertion or RH conclusion has been inserted.

RZ12. Exact scope and retained source information

The resolvent acts continuously on the entire Fréchet quotient and has exactly the actual original nontrivial zeros as its finite spectrum. Its poles have exactly their original multiplicities. The projectors are simultaneously actual contour projectors, explicit continuous finite-rank lifts on the full source function space, and isolators of the full jet at one zero with every other-zero jet zero. Every source representative has all endpoint and derivative seminorms controlled by (RZ4.7).

The full multiplier $s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/8$, its endpoint values, the finite values at the original trivial zeros, and all local derivative contributions remain explicit. The arithmetic action is the original-variable operator $n^L = \sqrt{n}V_n$, with every logarithmic multiplicity term and the exact reflected form. The independent source synthesis S5 is used on its actual domain; no finite numerical experiment, unrestricted jet-product identification, positive completion, or RH assumption replaces it.

Source-availability update. Earlier sections record the sources then available. The entire original published Deligne article, printed 137–252, has now been read, following the complete French transcription used in the initial reconstruction. Individual original-formula checks and external-reference coverage remain separately recorded. The main weight argument is reconstructed below. This is a new source reconstruction; it does not retroactively identify earlier elementary comparisons with Deligne’s full cohomological proof.

Original-source reading update: every printed page 137–252 of Deligne’s published article has now been read. The reading ledger distinguishes this from individual formula checks and unread external references. The current corpus and operation instruction governs all receiving calculations. The actual tau-supported lifting target remains active.

Deligne’s weight control and the complete return measure

24 September 2026. Proof reconstruction and programme comparison. Locators DWR0–DWR14.

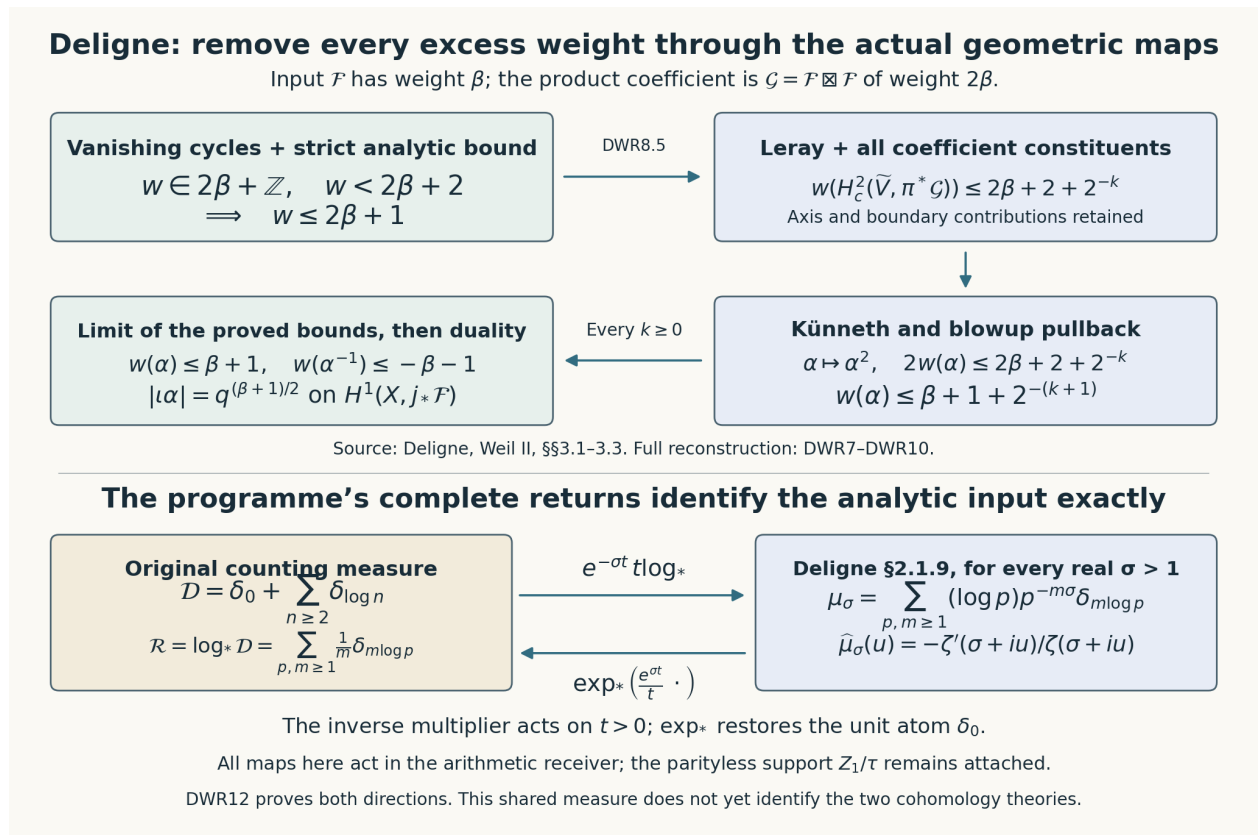


Figure 7: The top diagram shows the complete global weight-tightening mechanism, including the integral congruence class, strict bound, product map, and duality. The bottom diagram shows the proved invertible map between the complete counting measure and Deligne’s original-zeta return measure. Proofs: DWR7–DWR12. The figure contains no finite numerical verification or claim of an established cohomology equivalence.

DWR0. Sources, scope, and the user’s proposed comparison

Pierre Deligne, *La conjecture de Weil. II*, Publications mathématiques de l’IHÉS 52 (1980), 137–252, published source identity. The initial reconstruction used the complete French S20 LaTeX transcription, with selected comparisons to the historical English math-mode transcription. The entire French text has been read, including chapters IV–VI. This is not a claim to have verified every transcription against the original

scan, or to have independently reconstructed every cited SGA theorem. The exact coverage and discrepancies are retained in the private reading ledger. Neither transcription is the author's original TeX. The subsequent reading covered the entire original published article, printed 137–252, using its original PDF and direct visual checks. The ledger records the exact original source hash, coverage and independently justified corrections. That original reading supersedes the earlier transcription-only coverage.

This note reconstructs the proof of the fundamental weight bound, its local-monodromy input, and the passage to purity. It identifies the foundational theorems being used instead of disguising them as new proofs. Applications to root systems and rational homotopy were read but are not new theorems claimed here.

The receiving datum is the user's parityless Z_1/τ support with the complete arithmetic history retained in its receiver. No addition at τ is defined here. No count of copies of τ is taken. Integer labels below refer to the already reconstructed arithmetic receiver; these calculations do not replace the requested preceding construction of that receiver from the complete history.

The proposed replacement of a source zero by Z_1/τ is investigated through the actual maps. It is not installed as an axiom. In particular, weight zero in Deligne means a modulus condition on nonzero Frobenius eigenvalues, not an unsupported arithmetic element.

DWR1. The exact weight and operator conventions

Let X_0 be of finite type over $\mathbf{F}_q$, let $X = X_0 \otimes_{\mathbf{F}_q} \overline{\mathbf{F}}_q$, and let F be geometric Frobenius. A closed point x has degree $d_x = [k(x) : \mathbf{F}_q]$ and norm $N(x) = q^{d_x}$. For a fixed coefficient embedding $\iota : \overline{\mathbf{Q}}_\ell \rightarrow \mathbf{C}$, define

$$w_Q(\alpha) = \frac{2 \log |\iota \alpha|}{\log Q}, \quad Q > 1, \quad \alpha \neq 0. \quad (\text{DWR1.1})$$

A lisse sheaf is pointwise ι -pure of weight β when each eigenvalue at each closed point satisfies $|\iota \alpha| = N(x)^{\beta/2}$. Mixed means a finite filtration with pure successive quotients. Without ι , purity requires algebraicity and this modulus for every complex embedding, with integer weight. Source: §§1.2.1–1.2.6.

Tensor products multiply eigenvalues and add weights; the dual replaces α by α^{-1} . The Tate object $\mathbf{Q}_\ell(1)$ has geometric Frobenius Q^{-1} , hence weight -2 . Thus a twist by (a) multiplies eigenvalues by Q^{-a} , retaining the shift $-2a$.

For a rank-one Weil sheaf on $\text{Spec } \mathbf{F}_p$ with Frobenius b , its pullback at x multiplies an eigenvalue by

$$b^{[k(x):\mathbf{F}_p]} = b^{[\mathbf{F}_q:\mathbf{F}_p]d_x}. \quad (\text{DWR1.2})$$

The inverse tensor functor uses b^{-1} , and evaluation gives the natural isomorphism between their composite and the original sheaf. No factor in (DWR1.2) is discarded. A numerical shift of weight is therefore a specified change of coefficient object and Frobenius, not a relocation of the supporting point. The reconstruction below keeps arbitrary β in its formulas.

DWR2. Why determinant weights are controlled

Sources: §§1.3.1–1.3.15 and 1.11.4. For a normal geometrically connected X_0 , the Weil group fits into

$$1 \longrightarrow \pi_1(X, \bar{x}) \longrightarrow W(X_0, \bar{x}) \xrightarrow{\deg} \mathbf{Z} \longrightarrow 0. \quad (\text{DWR2.1})$$

In the abelianization of the Weil group, the geometric image is the product of a finite group of order prime to p and a pro- p group. Here is the source's reduction: for a curve, class field theory describes it through degree-zero divisor class groups with moduli at the removed points. The ordinary degree-zero class group is finite because it is the rational-point group of the finite-type Picard variety over a finite field. The additional local units are finite groups times pro- p groups. For general normal X_0 , pass to a dense quasi-projective

open and a sufficiently general curve section. Bertini surjectivity on fundamental groups and specialization of the prime-to- p abelian quotient reduce the assertion to the curve. These are the precise class-field, Picard, Bertini and specialization inputs of §1.3.1, completed in §1.11.4.

For a rank-one ℓ -adic character χ , the geometric image also lies in a compact subgroup of E^* for a finite extension $E/\mathbf{Q}_\ell$. Such a compact subgroup is finite times pro- ℓ . Because $p \neq \ell$, the geometric image is finite. A power of χ is therefore trivial geometrically and has the form $b^{\deg}$. Taking a root of b proves

$$\chi(w) = c^{\deg(w)} \epsilon(w), \quad \epsilon \text{ of finite order.} \quad (\text{DWR2.2})$$

Roots of unity have complex modulus one, so the rank-one weight is the constant $w_q(c)$.

For an irreducible constituent V of rank r , its determinant is rank one. Its determinantal weight is $w(\det V)/r$. To control tensor and exterior powers, Deligne forms G^0 , the Zariski closure of geometric monodromy, and its Weil extension G . A degree-one Weil element acts on G^0 by conjugation, giving the exact semidirect-product construction of §1.3.7.

For a semisimple sheaf the connected geometric group is semisimple. To see the source's argument, a central torus has finitely many weights in a faithful representation. After a finite cover and a finite extension of constants the Weil action fixes those weights, acts on the connected reductive group by inner automorphisms, and the extension splits as a product with $\mathbf{Z}$. Any remaining torus quotient would then supply a rank-one character with Zariski-dense geometric image. Equation (DWR2.2) makes that image finite, forcing the torus quotient to be trivial. For a nonsemisimple sheaf, passage to its Jordan–Hölder graded object has unipotent kernel. This proves the unipotent-radical assertion used in the source.

For the following central-element calculation use the semisimplified fibre: its constituent determinant weights are unchanged, while extensions in the original sheaf remain part of the later cohomology calculation. Write G^{00} for the identity component of G^0 , and w for the degree-one lift. A positive power w^a acts by an inner automorphism on G^{00} , and trivially on the finite group G^0/G^{00} . Choose $h_0 \in G^{00}$ so that $y = h_0 w^a$ centralizes G^{00} . For each representative h of G^0/G^{00} , the commutator $[y, h]$ lies in G^{00} and centralizes it; it is therefore in the finite centre $Z(G^{00})$. The same is true of $[y, w]$: membership in G^{00} follows from the expression for y , and centralization follows because both y and wyw^{-1} centralize G^{00} . Since y centralizes this finite centre, a common positive power kills all these commutators. It centralizes the identity component, all component representatives, and the degree generator. Thus it gives $g \in Z(G)$ of degree $m > 0$. The centre's degree kernel is finite and its image has finite index in $\mathbf{Z}$.

On an irreducible representation V , g is scalar. The rank-one determinant identity extends from the Weil group to G : a power of the determinant character is trivial on geometric monodromy, hence on its Zariski closure G^0 . It therefore factors through degree. Taking the corresponding scalar root expresses the determinant as $c^{\deg} \epsilon$ with ϵ of finite order on G itself. Evaluation at g is legitimate even when g is not in the Weil-group image. If V has rank r and central scalar α , then $\alpha^r = c^m \epsilon(g)$, whence $|\iota \alpha|^r = |\iota c|^m$. Consequently

$$V \text{ has determinantal weight } \beta \iff |\iota \alpha| = q^{m\beta/2} \text{ for every eigenvalue of } g \text{ on } V. \quad (\text{DWR2.3})$$

Multiplication of these central eigenvalues proves addition of determinantal weights under tensor product. If $n(\beta)$ is the total rank of constituents of weight β , the weights in $\bigwedge^a V$ are exactly

$$\sum_{\beta} a(\beta) \beta, \quad \sum_{\beta} a(\beta) = a, \quad 0 \leq a(\beta) \leq n(\beta). \quad (\text{DWR2.4})$$

The coefficients in (DWR2.4) are integers; this is exterior algebra on the complete representation, not a finite numerical test of primes.

DWR3. Trace identity and all even tensor powers

Sources: §§1.4–1.5. Grothendieck's trace identity is

$$\prod_{x \in |X_0|} \det(1 - F_x t^{d_x}, \mathcal{F}_x)^{-1} = \prod_i \det(1 - Ft, H_c^i(X, \mathcal{F}))^{(-1)^{i+1}}. \quad (\text{DWR3.1})$$

It first holds as a formal power series. If the local weights are at most β , and $\dim X_0 = d$, the bound on the number of closed points of degree n by Cq^{dn} gives absolute convergence for $|t| < q^{-\beta/2-d}$. This also proves the product has no zero there: the sum of the absolutely convergent logarithmic series defines its logarithm.

For a smooth connected curve, H^0 consists of geometric invariants and H_c^2 consists of geometric coinvariants twisted by (-1) . Their weights are respectively determinantal weights γ and $\gamma + 2$. The full curve formula is

$$L(\mathcal{F}, t) = \frac{\det(1 - Ft, H_c^1)}{\det(1 - Ft, H_c^0) \det(1 - Ft, H_c^2)}. \quad (\text{DWR3.2})$$

On an affine curve with lisse coefficients, $H_c^0 = 0$; its determinant is exactly one. Its removal from the displayed quotient is justified by that vanishing, not an arbitrary cancellation.

Suppose now that the local characteristic polynomials of $\mathcal{F}$, after applying ι , have real coefficients. Write r for its largest determinantal weight. For every positive integer k , every local factor for $\mathcal{F}^{\otimes 2k}$ has nonnegative real coefficients. Indeed,

$$\det(1 - At^{d_x})^{-1} = \exp \left(\sum_{m \geq 1} \frac{\text{Tr}(A^m)}{m} t^{md_x} \right), \quad \text{Tr}((F_x^{\otimes 2k})^m) = \text{Tr}(F_x^m)^{2k} \geq 0. \quad (\text{DWR3.3})$$

The trace is real because the characteristic polynomial is real; Newton identities or conjugate eigenvalue multisets prove this for every power. Exponentiating a series of nonnegative coefficients preserves nonnegativity. Each local factor has constant coefficient one. Thus each local series is coefficientwise bounded by the complete product series.

The denominator of the cohomological product has no zero in

$$|t| < q^{-(2kr+2)/2}, \quad (\text{DWR3.4})$$

by (DWR2.4) and the H_c^2 calculation. Hence the product is analytic there. Coefficientwise domination makes every local series converge in that disk too. If α is a local eigenvalue, α^{2k} is a tensor-power eigenvalue, and its local reciprocal determinant has a pole at every solution of $t^{d_x} = \alpha^{-2k}$. There is no numerator in this local factor to cancel the pole. Comparing its modulus with (DWR3.4) yields

$$|\iota\alpha|^{2k/d_x} \leq q^{(2kr+2)/2}, \quad w_{N(x)}(\alpha) \leq r + \frac{1}{k}. \quad (\text{DWR3.5})$$

This holds for every k , so $w_{N(x)}(\alpha) \leq r$.

For each determinant-weight class β , write its local eigenvalues as α_i^β . The determinant gives the exact equality

$$\sum_i w_{N(x)}(\alpha_i^\beta) = n(\beta)\beta. \quad (\text{DWR3.6})$$

Let $a = 1 + \sum_{\gamma > \beta} n(\gamma)$. The largest determinant weight of $\bigwedge^a \mathcal{F}$ is $\beta + \sum_{\gamma > \beta} n(\gamma)\gamma$, by (DWR2.4). This exterior power is also real: its eigenvalue multiset is made of products of distinct indices from a conjugation-stable multiset. Apply (DWR3.5), for all k , to the eigenvalue $\alpha_i^\beta \prod_{\gamma > \beta} \prod_j \alpha_j^\gamma$. Subtract (DWR3.6) for every $\gamma > \beta$. The result is $w(\alpha_i^\beta) \leq \beta$. Summing those inequalities gives equality (DWR3.6), so every inequality is equality. This proves §1.5.1: every constituent of a real lisse sheaf on the curve is pure.

DWR4. Boundary weights with every boundary factor retained

Source: §1.8.1. Let $j : U_0 \hookrightarrow X_0$ be the complement of a finite set S_0 in a smooth curve, and let $\mathcal{F}_0$ be lisse pure of weight β on U_0 . Remove a point of U_0 , if necessary, to make the ambient curve affine without removing the boundary point under consideration. Then

$$\prod_{x \in |U_0|} \det(1 - F_x t^{d_x}, \mathcal{F})^{-1} \prod_{x \in |S_0|} \det(1 - F_x t^{d_x}, j_* \mathcal{F})^{-1} = \frac{\det(1 - Ft, H_c^1(X, j_* \mathcal{F}))}{\det(1 - Ft, H_c^2(X, j_* \mathcal{F}))}. \quad (\text{DWR4.1})$$

The interior product is nonzero and holomorphic in $|t| < q^{-(\beta+2)/2}$. The right denominator has no zero in that disk. Therefore the boundary product has no pole there. Its reciprocal-polynomial factors cannot cancel each other's poles. This proves each boundary eigenvalue has weight at most $\beta + 2$.

At a boundary stalk, the inclusion

$$(V^I)^{\otimes k} \hookrightarrow (V^{\otimes k})^I \quad (\text{DWR4.2})$$

is the literal inclusion of tensors of inertia-invariant vectors into invariant tensors. Applying the preceding bound to $\mathcal{F}^{\otimes k}$ gives

$$kw_{N(x)}(\alpha) \leq k\beta + 2, \quad w_{N(x)}(\alpha) \leq \beta + 2/k. \quad (\text{DWR4.3})$$

All positive integers k give the sharp boundary bound $w \leq \beta$. The zero-free comparison of the same complete identity bounds H_c^1 by $\beta + 2$. Extension to general open immersions is by sheaf extensions, support reduction, finite normalization, and restriction to generic points of boundary divisors, as in §1.8.9; transverse curves supply the boundary calculation. The dual gives the opposite bound wherever a lisse sheaf extends across the boundary.

DWR5. The local nilpotent operator and its exact purity calculation

Sources: §§1.6–1.8. For a nilpotent endomorphism N , its monodromy filtration M is characterized by

$$NM_i \subset M_{i-2}, \quad N^j : \text{Gr}_j^M V \xrightarrow{\sim} \text{Gr}_{-j}^M V. \quad (\text{DWR5.1})$$

For a Jordan chain of length $d+1$, place its vectors in degrees $d, d-2, \dots, -d$; N lowers degree by two. Their sums construct M . Intrinsically, if $N^{d+1} = 0$, take $M_d = V$, $M_{d-1} = \ker N^d$, $M_{-d} = \text{im } N^d$, $M_{-d-1} = 0$, and recurse on $\ker N^d / \text{im } N^d$. The same forced extreme terms and induction prove uniqueness. For a relative filtration, §1.6.13 uses induction on the pre-existing filtration; no existence is asserted solely from uniqueness.

In the geometric application the operator is typed as

$$N : V(1) \longrightarrow V, \quad \rho(\sigma) = \exp(Nt_\ell(\sigma)) \quad (\text{DWR5.2})$$

on a finite-index inertia subgroup. After a finite cover, this subgroup is all inertia. The cover changes neither the weights nor the monodromy filtration: it raises Frobenius powers and multiplies the chosen logarithm by a nonzero scalar. Thus $V^I = \ker N$ is the boundary stalk.

Put $P_{-j} = \ker(N : \text{Gr}_{-j}^M V \rightarrow \text{Gr}_{-j-2}^M V(-1))$. Keeping the one-dimensional space of logarithms gives

$$\text{Gr}_i^M V \cong \bigoplus_{j \geq |i|, j \equiv i \pmod{2}} P_{-j}(-(i+j)/2). \quad (\text{DWR5.3})$$

The sign is forced: $N^{(i+j)/2}$ maps the corresponding summand of $\text{Gr}_i^M V((i+j)/2)$ to P_{-j} . Solving for the untwisted source gives (DWR5.3). The local transcription's positive sign in §1.6.14.3 fails this test and its own weight check; the original file is preserved and the correction is recorded separately.

The $\mathfrak{sl}_2$ decomposition of Jordan chains gives the primitive tensor and dual maps

$$P_{-j'}(V) \otimes P_{-j''}(W)((j-j'-j'')/2) \longrightarrow P_{-j}(V \otimes W) \quad (\text{DWR5.4})$$

as direct summands for $|j' - j''| \leq j \leq j' + j''$ with the matching parity, and

$$P_{-j}(V^\vee) \cong P_{-j}(V)^\vee(j). \quad (\text{DWR5.5})$$

For specificity, the tensor decomposition is $S_{j'} \otimes S_{j''} = \bigoplus_{j=|j'-j''|, \text{ step } 2}^{j'+j''} S_j$, where $S_j = \text{Sym}^j$ of the two-dimensional representation. Comparing the weights $j, j-2, \dots, -j$ proves its character identity; complete reducibility in characteristic zero gives the decomposition. Choosing lowest-weight vectors determines the maps, and tracking their rescaling by the choice of N gives exactly the twist in (DWR5.4). Applying the dual to a chain gives (DWR5.5).

Now retain arbitrary input weight β , and write $Q = N(s)$ at the boundary point. A primitive eigenvalue α in $P_{-j}(V)$ gives an eigenvalue $\alpha^2 Q^j$ in the summand $P_{-j}(V)^{\otimes 2}(-j) \subset P_0(V^{\otimes 2})$. By DWR4 applied to the sheaf of weight 2β ,

$$|\iota\alpha|^2 Q^j \leq Q^\beta. \quad (\text{DWR5.6})$$

Apply this same result to the dual sheaf, of weight $-\beta$. Equation (DWR5.5) supplies its primitive eigenvalue $\alpha^{-1} Q^{-j}$, hence

$$|\iota\alpha|^{-2} Q^{-2j} Q^j \leq Q^{-\beta}. \quad (\text{DWR5.7})$$

Equations (DWR5.6)–(DWR5.7) force

$$|\iota\alpha| = Q^{(\beta-j)/2}. \quad (\text{DWR5.8})$$

Finally (DWR5.3) multiplies it by $Q^{(i+j)/2}$, so every eigenvalue on $\text{Gr}_i^M V$ has modulus $Q^{(\beta+i)/2}$. This is the full local-monodromy purity theorem §1.8.4, with its original weight and every Tate factor retained. Boundary invariants occupy only indices $i \leq 0$.

DWR6. Why the preliminary cohomological bound is strict

Sources: chapter II, especially §§2.1.4–2.1.9 and 2.2.10. Let a group G be an extension of $\mathbf{Z}$ or $\mathbf{R}$ by a compact group; let x_v be conjugacy classes with $Nv > 1$. Suppose the Euler products converge for $\Re s > 1$. For a unitary representation ρ ,

$$-\frac{L'}{L}(\rho, s) = \sum_{v, m \geq 1} (\log Nv)(Nv)^{-ms} \text{Tr } \rho(x_v^m). \quad (\text{DWR6.1})$$

For real $\sigma > 1$, the weights on the right define a finite positive measure. Since $\text{Tr}(\rho \otimes \bar{\rho})(g) = |\text{Tr } \rho(g)|^2$, its transform is nonnegative on every such tensor square, including virtual representations.

Assume now the source's specified meromorphic continuation with only the trivial-character simple pole on the boundary $\Re s \geq 1$. Multiplying the preceding inequality by $\sigma - 1 > 0$ and taking $\sigma \downarrow 1$ gives

$$\nu(\rho \otimes \bar{\rho}) \geq 0, \quad (\text{DWR6.2})$$

where ν is pole order minus zero order at the boundary. Thus $\nu(1) = 1$, $\nu(\chi) = \nu(\bar{\chi})$, and $\nu(\chi) \leq 0$ for nontrivial irreducibles.

Here is the global character argument that uses all representations. First take the closure K of the image of G in the product of the unitary groups of all its finite-dimensional unitary representations. This is compact. Each such representation extends by its coordinate projection; density preserves irreducibility, intertwiners, tensor products and multiplicities because each intertwining equation is a closed condition. Conversely every continuous representation of K restricts to one of G . The character calculation can therefore be made on K , with Haar measure of total mass one.

Fix a finite set of irreducibles and $0 < \epsilon < 1$. Put $e_1 = e_2 = \epsilon/4$, and choose a central inversion-invariant neighbourhood U of the identity on which $|\chi(g) - \dim \chi| < e_1 \dim \chi$ for every chosen character. A continuous central inversion-invariant function supported in U is obtained by averaging a bump function over conjugation and inversion. Add a sufficiently small positive constant and approximate uniformly by a finite character combination using the central Peter–Weyl density theorem. Taking its real part and inversion average gives a real inversion-invariant combination. Its character coefficients are real: substitution $g \mapsto g^{-1}$ in their Haar integrals identifies each coefficient with its complex conjugate. Approximate these coefficients by rationals, preserving conjugate pairs, and multiply by a common positive denominator. The resulting integral virtual character f can still be chosen with $M = \int_K |f|^2 > 0$ and $\int_{K \setminus U} |f|^2 \leq e_2 M$, since both strict concentration and positivity persist under sufficiently small uniform perturbations.

Write $f = \rho^+ - \rho^-$ with disjoint irreducible supports and put $\rho = \rho^+ + \rho^-$. The exact difference

$$\rho \otimes \bar{\rho} - f \otimes \bar{f} = 2(\rho^+ \otimes \bar{\rho}^- + \rho^- \otimes \bar{\rho}^+) \quad (\text{DWR6.2a})$$

has nonnegative multiplicities and trivial multiplicity zero. The exterior part of the Haar integral cannot be dropped: $\operatorname{Re} \overline{\chi(g)} \geq -\dim \chi$ everywhere, so

$$\begin{aligned} [f \otimes \bar{f} : \chi] &= \operatorname{Re} \int_K |f|^2 \bar{\chi} dg \\ &\geq (\dim \chi) [(1 - e_1)(1 - e_2) - e_2] M \\ &= (\dim \chi) (1 - 3\epsilon/4 + \epsilon^2/16) M \geq (1 - \epsilon)(\dim \chi) M. \end{aligned} \tag{DWR6.2b}$$

Since $[\rho \otimes \bar{\rho} : 1] = M$, this proves

$$[\rho \otimes \bar{\rho} : \chi] \geq (1 - \epsilon) \dim \chi [\rho \otimes \bar{\rho} : 1]. \tag{DWR6.3}$$

Use (DWR6.3) in (DWR6.2), let $\epsilon \downarrow 0$, and enlarge the finite set. Because all nontrivial contributions are nonpositive integers,

$$1 + \sum_{\chi \neq 1} \dim \chi \nu(\chi) \geq 0. \tag{DWR6.4}$$

At most one nontrivial irreducible can have negative order; it must have dimension one, order -1 , and be self-conjugate. It is a character of order two. This is Deligne's precise exceptional possibility, not an unmentioned deletion.

For the curve's monodromy representation, DWR2 supplies a semisimple geometric group. Its complexification has a compact form after retaining the degree direction. Twisting each pure constituent by its inverse central scalar makes its geometric semisimple Frobenius powers relatively compact; conjugacy of maximal compact subgroups places those classes in the compact form. The tensor functor and all inverse twists remain those of (DWR1.2). The trace formula supplies the meromorphic continuation required above. A nontrivial irreducible on the geometric subgroup has neither invariant nor coinvariant and hence no denominator in (DWR3.2). The geometrically trivial one supplies the curve's zeta pole. The possible quadratic exception is removed by the exact double-cover identity

$$\zeta_{X'_0}(s) = \zeta_{X_0}(s) L(\epsilon, s) : \tag{DWR6.5}$$

both curve zeta functions have a simple pole at $s = 1$, so the second factor cannot vanish there. These statements concern the actual connected cover; extensions of the constant field retain a simple pole at this real point as well.

Consequently the cohomological numerator has no zero on its outer weight boundary. Together with DWR4 this proves

$$\alpha \in \operatorname{Spec}(F|H_c^1(U, \mathcal{F})) \implies w_q(\alpha) < \beta + 2. \tag{DWR6.6}$$

At a trivial-character pole there is also no hidden numerator cancellation: the actual pole is simple and the denominator has exactly that simple zero. Arbitrary β follows with the explicit eigenvalue multiplier and inverse from (DWR1.2).

DWR7. Vanishing cycles retain the orientation data

Source: §3.1. Embed a smooth projective surface S , with normal-crossings boundary D , in projective space and choose a Lefschetz pencil transverse to its strata. Blow up its axis: $\pi : \tilde{S} \rightarrow S$, and write $f : \tilde{S} \rightarrow \mathbf{P}^1$. The three exceptional fibres have an ordinary node, a tangency to one boundary branch, or a crossing of two boundary branches. Assume unipotent local monodromy of the coefficient sheaf, obtained later by a finite cover.

For a two-element set B , the orientation object is

$$\epsilon(B) = \mathbf{Z}^B / \mathbf{Z}(1, 1). \tag{DWR7.1}$$

Exchanging the two labels acts by minus one. This actual source orientation is retained; no identification with a user's information layer is assumed. The only nonzero vanishing-cycle degree is one. At a node,

$$\Phi_x^1 = \mathcal{G}_x(-1) \otimes \epsilon(B). \tag{DWR7.2}$$

At a boundary tangency or crossing, a local filtration A with constant quotients gives

$$\mathrm{Gr}_A^i \Phi_x^1 = \mathrm{Gr}_A^i(\mathcal{G})_x \otimes \epsilon(B). \quad (\text{DWR7.3})$$

At a node $uv = t$, Picard–Lefschetz retains the two Kummer branch classes $[u], [v]$. Over the geometric generic fibre $[u] + [v] = [t] = 0$, and exchanging branches sends $[u]$ to $-[u]$. Thus the quotient of the two labelled generators by their sum gives exactly the orientation object in (DWR7.1); the Kummer coefficient twist supplies (-1) . The class $[u/v] = [u] - [v] = 2[u]$ must not replace the labelled branch generator in integral or mod-two coefficients. For tangency, apply vanishing cycles to $0 \rightarrow j_! \mathbf{Q}_\ell \rightarrow \mathbf{Q}_\ell \rightarrow \mathbf{Q}_{\ell,D} \rightarrow 0$; the ambient smooth contribution vanishes and the boundary’s degree-zero difference moves into degree one. For a crossing use the two-branch normalization sequence $0 \rightarrow j_! \mathbf{Q}_\ell \rightarrow \mathbf{Q}_\ell \rightarrow i_* \mathbf{Q}_\ell \rightarrow \mathbf{Q}_{\ell,x} \otimes \epsilon(B) \rightarrow 0$. Its last term has vanishing cycles in degree -1 , so the two shifts again place the answer in degree one. Applying the local filtration proves (DWR7.3) for the original sheaf.

The axis correction and the receiving spectral sequence are

$$H_c^*(\widetilde{V}, \pi^* \mathcal{G}) \cong H_c^*(V, \mathcal{G}) \oplus H^0(V \cap A, \mathcal{G})(-1)[-2], \quad (\text{DWR7.4})$$

$$E_2^{ab} = H^a(\mathbf{P}^1, R^b f_! \pi^* \mathcal{G}) \implies H_c^{a+b}(\widetilde{V}, \pi^* \mathcal{G}). \quad (\text{DWR7.5})$$

The extra summand in (DWR7.4) is in cohomological degree two. The pullback injection of the original cohomology has a retraction by proper pushforward/Poincaré duality; no exceptional divisor contribution is discarded.

DWR8. The global product argument, with the original input weight

Source: §§3.2.1–3.2.14. We prove, for every integer $k \geq 0$, every smooth curve U_0 , and every lisse pure coefficient sheaf of weight β ,

$$w_q(\alpha) \leq \beta + 1 + 2^{-k} \quad (\alpha \text{ on } H_c^1(U, \mathcal{F})). \quad (\text{DWR8.1})$$

The case $k = 0$ is DWR4. A sheaf on a curve that is lisse pure on a dense open has its H_c^1 as a quotient of that open’s H_c^1 : the complementary finite support has no H_c^1 . Thus (DWR8.1) applies to such intermediate extensions too.

Assume (DWR8.1) for a fixed k . Let X_0 compactify U_0 , put $V_0 = U_0 \times U_0$, and retain the actual coefficient

$$\mathcal{G}_0 = \mathcal{F}_0 \boxtimes \mathcal{F}_0, \quad w(\mathcal{G}_0) = 2\beta. \quad (\text{DWR8.2})$$

Choose the pencil of DWR7 on $X_0 \times X_0$. For arbitrary real β , take a rank-one Weil object L whose eigenvalue at a point of norm Q , after ι , is $Q^{2\beta}$. Such an object is obtained on $\mathbf{F}_p$ by $\iota(b) = p^{2\beta}$, as in (DWR1.2). Then

$$\mathcal{G}_0 \oplus (\mathcal{G}_0' \otimes L) \quad (\text{DWR8.3})$$

is pure of weight 2β and real: if $|\lambda| = Q^\beta$, its paired eigenvalue is $Q^{2\beta}/\lambda = \bar{\lambda}$. This keeps the companion and its factor explicitly, without replacing $\mathcal{G}_0$.

The fibre trace formula and strict bound DWR6 separate the H_c^2 poles from H_c^1 zeros; their conjugate multisets are therefore separately invariant. Duality handles H_c^0 for a proper fibre. Thus the H_c^1 characteristic polynomial of a real pure coefficient sheaf is real (§3.2.1). Applied fibrewise to (DWR8.3), this makes $R^1 f_! \pi^* \mathcal{G}_0$ on the smooth pencil locus a direct summand of a real sheaf. DWR3 proves that all its constituents are pure.

The vanishing-cycle exact sequence is

$$0 \rightarrow (R^1 f_! \pi^* \mathcal{G})_{\bar{t}} \rightarrow (R^1 f_! \pi^* \mathcal{G})_{\bar{\eta}} \rightarrow \Phi_x^1 \rightarrow (R^2 f_! \pi^* \mathcal{G})_{\bar{t}} \rightarrow (R^2 f_! \pi^* \mathcal{G})_{\bar{\eta}} \rightarrow 0. \quad (\text{DWR8.4})$$

There are no punctual sections in R^1 . Extend a constituent filtration by intersection in the direct image from the smooth locus. The defect of a constituent’s specialization is a subquotient of Φ_x^1 .

DWR5 at the boundary of U_0 gives weights in $\beta + \mathbf{Z}$; tensoring the two factors gives $2\beta + \mathbf{Z}$. Equations (DWR7.2)–(DWR7.3), including the twist (-1) and the finite orientation character, keep vanishing-cycle weights in $2\beta + \mathbf{Z}$. A constituent with nonzero specialization defect therefore has weight in this same class modulo $\mathbf{Z}$. If it has no defect anywhere, it is lisse on all of $\mathbf{P}^1$, hence geometrically constant, and its H^1 is zero. DWR6 applied to every smooth fibre gives weight strictly below $2\beta + 2$. Every contributing constituent thus has

$$w \in 2\beta + \mathbf{Z}, \quad w < 2\beta + 2, \quad \text{hence } w \leq 2\beta + 1. \quad (\text{DWR8.5})$$

This is the exact place where strictness and integrality work together.

Apply the induction bound to these constituents on the pencil. It gives weight at most $2\beta + 2 + 2^{-k}$ on E_2^{11} . The invariant/coinvariant computations of DWR3 give at most $2\beta + 2$ on E_2^{02} and E_2^{20} . The spectral sequence (DWR7.5), whose later terms are subquotients, therefore bounds the whole $H_c^2(\tilde{V}, \pi^* \mathcal{G})$ by $2\beta + 2 + 2^{-k}$.

Künneth and the actual pullback give the Frobenius-equivariant injections

$$H_c^1(U, \mathcal{F}) \otimes H_c^1(U, \mathcal{F}) \hookrightarrow H_c^2(V, \mathcal{G}) \hookrightarrow H_c^2(\tilde{V}, \pi^* \mathcal{G}). \quad (\text{DWR8.6})$$

An eigenvalue α therefore supplies α^2 in that bounded group. Consequently

$$2w_q(\alpha) \leq 2\beta + 2 + 2^{-k}, \quad w_q(\alpha) \leq \beta + 1 + 2^{-(k+1)}. \quad (\text{DWR8.7})$$

To remove the auxiliary pencil and ramification assumptions, a finite cover makes boundary monodromy unipotent, and a finite extension of the field defines a sufficiently general pencil and its exceptional points. Pullback on cohomology is injective because trace divided by the nonzero covering degree is a retraction in $\mathbf{Q}_\ell$. Extension $\mathbf{F}_q \subset \mathbf{F}_{q^d}$ replaces F by F^d , and $w_{q^d}(\alpha^d) = w_q(\alpha)$; the bound descends. This completes induction for every k . Taking all k proves $w_q(\alpha) \leq \beta + 1$, globally.

DWR9. The reverse inequality and purity

Source: §3.2.15. For $j : U \hookrightarrow X$ with X a smooth proper curve, $H^1(X, j_* \mathcal{F})$ is the image of $H_c^1(U, \mathcal{F}) \rightarrow H^1(U, \mathcal{F})$. Its upper bound is DWR8. Poincaré duality supplies the perfect pairing with $H^1(X, j_* \mathcal{F}^\vee(1))$. The latter sheaf has input weight $-\beta - 2$, so its eigenvalue paired with α is α^{-1} , and DWR8 gives

$$w_q(\alpha^{-1}) \leq (-\beta - 2) + 1 = -\beta - 1. \quad (\text{DWR9.1})$$

Thus $w_q(\alpha) \geq \beta + 1$. Combining the two proved bounds gives equality. Degrees zero and two are the invariant and twisted-coinvariant calculations. This proves

$$w_q(\alpha) = \beta + i \quad \text{on } H^i(X, j_* \mathcal{F}). \quad (\text{DWR9.2})$$

The source's arithmetic zeros in kernels and exact sequences remain zero vectors throughout. Neither inequality is supplied by relabelling them.

DWR10. General direct images and every boundary term

Source: §3.3. For $f : X \rightarrow Y$ of finite type and mixed $\mathcal{F}$ of weights at most n , the conclusion is that $R^i f_! \mathcal{F}$ has weights at most $n + i$. The source reductions are: sheaf extensions by long exact sequences; open/closed decompositions in X and Y ; composition by the Leray spectral sequence; invariance under universal homeomorphisms; and relative dimension zero, where only $f_!$ remains and preserves weight. An affine-coordinate factorization followed by these reductions reduces relative dimension to one. Dense-open reduction and a finite cover then give a smooth proper curve family $\bar{f} : \bar{X} \rightarrow Y$, a finite étale boundary $i : D \hookrightarrow \bar{X}$, and $j : X \hookrightarrow \bar{X}$, with tame lisse pure coefficients of weight n .

Retain the exact sequence

$$0 \rightarrow j_! \mathcal{F} \rightarrow j_* \mathcal{F} \rightarrow i_* i^* j_* \mathcal{F} \rightarrow 0. \quad (\text{DWR10.1})$$

DWR9 on each fibre makes $R^i \bar{f}_* j_* \mathcal{F}$ pure of weight $n + i$. DWR5 filters the boundary term into weights $n + a$, $a \leq 0$. Since $\bar{f}i$ is finite, its higher direct images vanish and its degree-zero direct image preserves

those weights. The long exact sequence of (DWR10.1) proves the claimed bound for $R^i f_! \mathcal{F}$, including the boundary injection into degree one. This is the final calculation behind the general theorem, rather than an assumption of purity for the boundary.

For X smooth of pure dimension d and $\mathcal{F}$ lisse of weights at least n , apply the upper bound to $\mathcal{F}^\vee$. The group

$$H_c^{2d-i}(X, \mathcal{F}^\vee)(d) \quad (\text{DWR10.2})$$

has weights at most $-n + (2d - i) - 2d = -n - i$. Its dual is $H^i(X, \mathcal{F})$, giving the lower bound $n + i$. Hence the image of $H_c^i \rightarrow H^i$, for pure coefficients, is pure of weight $n + i$. Properness makes that map an isomorphism.

DWR11. What the later mixed-sheaf formalism adds

Sources: §§3.4, 3.6 and 6.1–6.2. Initially let $\mathcal{F}_0, \mathcal{G}_0$ be lisse pure of weights β, γ on a smooth finite-type $X_0/\mathbf{F}_q$. The extension exact sequence

$$0 \rightarrow H^0(X, \mathcal{H}om(\mathcal{F}, \mathcal{G}))_F \rightarrow \text{Ext}^1(\mathcal{F}_0, \mathcal{G}_0) \rightarrow H^1(X, \mathcal{H}om(\mathcal{F}, \mathcal{G}))^F \quad (\text{DWR11.1})$$

comes from comparing a geometric splitting with its Frobenius translate: its difference is a homomorphism, changed by $(F - 1)h$ when the splitting changes. A geometric extension needs eigenvalue one in the final group. DWR10 bounds that group's weights below by $\gamma - \beta + 1$, in the congruence class $\gamma - \beta + \mathbf{Z}$. Thus a geometrically nontrivial extension of weights β, γ requires $\beta > \gamma$ and equal classes modulo $\mathbf{Z}$. This proves geometric semisimplicity for pure sheaves, and permits induction constructing the unique weight filtration and decomposition modulo $\mathbf{Z}$. The stalks are sums of the corresponding generalized Frobenius eigenspaces; this gives uniqueness and strict compatibility of maps.

For a bounded constructible complex K , the dual is

$$D_X K = R\mathcal{H}om(K, K_X), \quad K_X = Ra^! \mathbf{Q}_\ell. \quad (\text{DWR11.2})$$

For smooth d -dimensional X , $K_X = \mathbf{Q}_\ell(d)[2d]$. Weight at most n means $\mathcal{H}^i K$ has weight at most $n + i$. The spectral sequence with terms $R^p f_! \mathcal{H}^q K$ then proves preservation of the complex bound by $Rf_!$. Purity requires the upper bound for K and the opposite upper bound for $D_X K$. Proper duality $D_Y Rf_* K \cong Rf_* D_X K$ proves proper pushforward preserves this purity. The operator, shift, and Tate factor all belong to that argument.

For the local invariant-cycle calculation of §3.6, take a proper morphism $f : X \rightarrow S$, with S a henselian trait with algebraically closed residue field k , X essentially smooth over k , and $X_{\bar{\eta}}$ smooth over its geometric generic point. All cohomology in the conclusion has $\mathbf{Q}_\ell$ coefficients, with ℓ invertible on S . Spreading out and specialization reduce to S the henselization at a point of a smooth curve, and X the corresponding localization of a smooth variety X' . Descent to a finite field supplies the arithmetic Frobenius. Work on a connected component of X' , and retain $N = \dim X'$, the total dimension, not the dimension of the special fibre.

Write $I = \text{Gal}(\bar{\eta}/\eta)$. Its prime-to- ℓ part has no higher ℓ -primary cohomology; the remaining procyclic quotient $\mathbf{Z}_\ell(1)$ has cohomology only in degrees zero and one, namely invariants and coinvariants with twist (-1) . The Hochschild–Serre sequence therefore gives the complete exact row

$$0 \rightarrow H^{i-1}(X_{\bar{\eta}})_I(-1) \rightarrow H^i(X_{\bar{\eta}}) \rightarrow H^i(X_{\bar{\eta}})^I \rightarrow 0. \quad (\text{DWR11.3})$$

Proper base change identifies $H^i(X)$ with $H^i(X_s)$. Localization and the duality for the smooth total space give the exact portion

$$H^i(X_s) \rightarrow H^i(X_{\bar{\eta}}) \xrightarrow{\partial} H_{X_s}^{i+1}(X) \cong H^{2N-i-1}(X_s)^\vee(-N). \quad (\text{DWR11.4})$$

The duality assertion here is made after passage to $\mathbf{Q}_\ell$; integral torsion has not been asserted absent. The specialization map is the composite of the first arrow of (DWR11.4) and the surjection in (DWR11.3).

The generic H^i arises from the pure weight- i sheaf of the smooth proper family away from the special point. DWR5 bounds its inertia invariants above by i . The proper special fibre has H^{2N-i-1} of weights at most $2N - i - 1$, by DWR10. Taking its dual and the displayed twist gives the opposite bound

$$w(H^i(X_{\bar{\eta}})^I) \leq i, \quad w(H^{2N-i-1}(X_s)^\vee(-N)) \geq -(2N - i - 1) + 2N = i + 1. \quad (\text{DWR11.5})$$

Let W_i select all generalized Frobenius eigenspaces of weight at most i . This is an exact functor on the finite-dimensional mixed Frobenius modules: an equivariant map respects the generalized-eigenvalue decomposition, and taking any chosen set of those summands is exact. Equation (DWR11.3) thus makes $W_i H^i(X_{\bar{\eta}})$ surject onto the whole invariant target. Equation (DWR11.5) makes ∂ vanish on $W_i H^i(X_{\bar{\eta}})$, since its target has no such weights. Exactness of (DWR11.4) lifts each of those classes to $H^i(X_s)$. Consequently

$$\text{sp}^* : H^i(X_s) \twoheadrightarrow H^i(X_{\bar{\eta}})^I \quad (\text{DWR11.6})$$

is surjective. This is the source's actual quotient-and-lift calculation: the obstruction map vanishes on the required weight part, while the entire obstruction group, the coinvariant term, and both Tate twists remain in the diagram.

DWR12. Exact map from complete returns to Deligne's measure

Receiving source: RETURN_MEASURE_RIGIDITY.md, RMR0–RMR5, read for this comparison, and GLOBAL_HISTORY_WEIL_ENERGY.md, GHE1–GHE9. The preceding complete-history construction is required before attaching the conventional integer labels. In its arithmetic receiver retain

$$\mathcal{D} = \delta_0 + \sum_{n \geq 2} \delta_{\log n}, \quad \mathcal{R} = \sum_p \sum_{m \geq 1} \frac{1}{m} \delta_{m \log p}, \quad \mathcal{W} = t\mathcal{R} = \sum_p \sum_{m \geq 1} \frac{m \log p}{m} \delta_{m \log p}. \quad (\text{DWR12.1})$$

Each measure is locally finite. To specify the domains, let $\mathfrak{I}$ be the additive group of locally finite complex measures of locally finite total variation on $[0, \infty)$, each supported in $[a, \infty)$ for some $a > 0$. The value of a may depend on the measure. Convolution is defined by addition of times. It is locally finite because the inverse image of $[0, T]$ is contained in $[0, T]^2$, where both total variations are finite. The convolution of r copies of a measure supported in $[a, \infty)$ is supported in $[ra, \infty)$. Accordingly $\delta_0 + \mathfrak{I}$ is a group under convolution, with inverse $\sum_{r \geq 0} (-h)^{*r}$ for $\delta_0 + h$; only finitely many terms contribute on each $[0, T]$.

In this same group define

$$\log_* (\delta_0 + h) = \sum_{r \geq 1} \frac{(-1)^{r+1}}{r} h^{*r}, \quad \exp_* h = \delta_0 + \sum_{r \geq 1} \frac{h^{*r}}{r!}. \quad (\text{DWR12.1a})$$

The formal power-series identities for exponential and logarithm become finite identities after restriction to any $[0, T]$. Therefore they are inverse group isomorphisms between $(\delta_0 + \mathfrak{I}, *)$ and $(\mathfrak{I}, +)$. This proves the identities for the complete measures, not just a finite set of return times. The exact arithmetic convolution maps are

$$\mathcal{R} = \log_* \mathcal{D}, \quad \mathcal{D} = \exp_* \mathcal{R} = \delta_0 + \sum_{r \geq 1} \frac{\mathcal{R}^{*r}}{r!}. \quad (\text{DWR12.2})$$

Here convolution adds real times in the receiver. No operation on τ is implied. On every compact time interval, the positive-time part is nilpotent under truncated convolution since its support starts at $\log 2 > 0$. Exponential and logarithm are therefore finite polynomial identities on that interval. For each primitive period, $\exp_*(\sum_{m \geq 1} \delta_{m \log p}/m) = \sum_{a \geq 0} \delta_{a \log p}$. Multiplying these factors gives exactly $\mathcal{D}$ by unique factorization. This proves (DWR12.2) globally by equality on every compact interval, with its unit atom intact.

For real $\sigma > 1$, set

$$\mu_\sigma = e^{-\sigma t} \mathcal{W} = \sum_p \sum_{m \geq 1} (\log p) p^{-m\sigma} \delta_{m \log p}. \quad (\text{DWR12.3})$$

This is **exactly** the measure in Deligne §2.1.9, not merely an analogous shape. The morphism on the original complete measures and its inverse are

$$\mathcal{D} \mapsto e^{-\sigma t} t \log_* \mathcal{D} = \mu_\sigma, \quad \mu_\sigma \mapsto \exp_* \left(\frac{e^{\sigma t}}{t} \mu_\sigma \right) = \mathcal{D}. \quad (\text{DWR12.4})$$

More generally $\Phi_\sigma : \delta_0 + \mathfrak{J} \rightarrow \mathfrak{J}$, given by the first formula in (DWR12.4), is a group isomorphism from convolution to addition. Indeed $\log_*(D_1 * D_2) = \log_* D_1 + \log_* D_2$, and multiplication of a measure by $e^{-\sigma t}$ is a linear automorphism of $\mathfrak{J}$, with inverse multiplication by $e^{\sigma t}/t$. Both multipliers and their inverses are bounded on every compact subset of the positive support. This proves both the operation law and the displayed inverse. For the arithmetic $\mathcal{D}$, its image is also a finite positive measure when $\sigma > 1$; general elements of $\mathfrak{J}$ need not be positive or have finite total mass.

The inverse multiplier is only applied on $t > 0$, where the return measure is supported; it has no division at a zero atom. The unit δ_0 returns as the constant term of $\exp_*$. Thus all prime-power repetitions and the unit are recovered. The absence of an atom in $\mathcal{R}$ at time zero is distinct from the parityless supporting datum. The marked comparison $(\tau, D) \mapsto (\tau, \Phi_\sigma(D))$ fixes that supporting datum in both directions and adds no operation on it.

For $\Re s > 1$, absolute convergence follows from comparison with $\sum_{n \geq 2} (\log n) n^{-1-\delta}$ on every $\Re s \geq 1 + \delta$. Consequently

$$\zeta(s) = \int e^{-st} d\mathcal{D}(t) = \exp \left(\int_{t>0} \frac{e^{-(s-\sigma)t}}{t} d\mu_\sigma(t) \right), \quad -\frac{\zeta'}{\zeta}(\sigma + iu) = \int e^{-iut} d\mu_\sigma(t). \quad (\text{DWR12.5})$$

The original ζ , with constant term one, is used throughout. There is no Gamma multiplier or completed replacement in (DWR12.5).

Positive definiteness is now a calculation for every finite family of real u_j and complex c_j :

$$\sum_{j,k} \bar{c}_j c_k \left[-\frac{\zeta'}{\zeta}(\sigma + i(u_k - u_j)) \right] = \int \left| \sum_k c_k e^{-iu_k t} \right|^2 d\mu_\sigma(t) \geq 0. \quad (\text{DWR12.6})$$

This is the positivity that enters chapter II. Its stated domain is $\sigma > 1$. Analytic continuation of the scalar transform does not continue a finite positive measure by itself; the source's boundary argument explicitly supplies the additional meromorphic and pole information before taking its residue limit.

DWR13. The base-point map and the weight datum in its stalk

The following comparison uses the already constructed pointed receiver; it does not define τ by a complex coordinate. On a singleton supporting space named τ , a sheaf of coefficient vector spaces is specified by its stalk V , with its operator F , filtration and any pairing. The unique map from a singleton named b to this singleton induces mutually inverse sheaf functors: keep the same stalk and each of its maps. Explicitly,

$$(\{b\}, V, F, M, \langle \cdot, \cdot \rangle) \mapsto (\{\tau\}, V, F, M, \langle \cdot, \cdot \rangle), \quad (\text{DWR13.1})$$

with inverse changing only the support name back. Compositions fix every stalk element and every operator. This proves the exact base-renaming isomorphism. It assigns no parity or arithmetic value to τ . It also leaves every eigenvalue unchanged, because $\det(TI - F)$ is literally the same polynomial before and after the functor.

An actual Frobenius-equivariant stalk isomorphism $f : V \rightarrow W$ likewise has $F_W = f F_V f^{-1}$, so

$$\det(TI - F_W) = \det(f(TI - F_V) f^{-1}) = \det(TI - F_V). \quad (\text{DWR13.2})$$

These are useful faithful maps, but neither creates the weight estimates DWR5–DWR10. The parityless supporting point can carry the coefficient calculation while remaining distinct from all its arithmetic elements.

The role of the coefficient zero is different: any additive coefficient isomorphism sends zero vector to zero vector, since $f(0) + f(v) = f(v)$. This uses addition in the coefficient stalk and makes no statement about addition on the supporting τ .

For a Frobenius-equivariant quotient $V \twoheadrightarrow Q$ with kernel K , choose a basis adapted to K . The matrix is block triangular, proving

$$\det(1 - tF_V) = \det(1 - tF_K) \det(1 - tF_Q), \quad \text{Tr}(F_V^m) = \text{Tr}(F_K^m) + \text{Tr}(F_Q^m). \quad (\text{DWR13.3})$$

Thus the user's global quotient proposal has an exact trace receiver: its kernel contributes a specified factor, which has to be retained in the complete return comparison. Deligne's local invariant-cycle argument DWR11 controls a quotient by proving that its obstruction lies in disjoint weights. That is a closer mathematical target than identifying a support point with a numerical zero.

DWR14. What has actually transferred

The complete-return measure has now been identified with the measure used in Deligne's own original-zeta example, with the inverse (DWR12.4) restoring the unit and every repetition. This identifies the scalar analytic input to his chapter-II method exactly. It does not assert that his finite-field coefficient sheaves and the programme's entire cohomology are already isomorphic.

The global weight-control proof itself has been reconstructed through explicit maps: tensor and exterior powers, inertia invariants, nilpotent primitive pieces and their duals, vanishing cycles, blowup pullback and trace, Künneth, Leray, and the dual cohomology pairing. Its removal of all excess weight is (DWR8.5)–(DWR8.7), followed by (DWR9.1). Neither a selected pair of primes nor a bounded calculation is used to infer the result.

For the programme, the remaining unconstructed comparison is the coefficient object and its geometric operations realizing those particular maps for the original zeta spectrum. This note has not proved their existence, assumed them as purity axioms, or declared the RH problem solved. It has proved the return-measure map, reconstructed the source proof that the proposed transfer must carry, and identified the quotient-kernel determinant and duality data that must be computed on the existing programme objects next.

The CC chart realization, its Tate class, and its ramified specialization

Independent derivation, 24 September 2026. Proof locators CGT0–CGT12.

CGT0. Objects, prerequisites, and source coverage

The supporting datum remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition on τ is used. The ring $\mathbf{Z}$ in this calculation is the already recovered arithmetic coefficient ring. It is not defined by selecting the primes 2, 3, or by a finite part of the timing history. Every ring operation, scheme, cohomology group, and quotient below is downstream of that reconstruction.

The specific object evaluated here is Connes–Consani's supplied chart construction. The Gaussian coefficient ring appearing in its evaluation is a receiver for those charts. It is not an alternative prime spectrum offered as a counterexample to the user's global reconstruction. In particular, this calculation makes no identification of τ with a Gaussian integer, a zero, a unit, or a cohomology class.

The instruction CORPUS_AND_OPERATION_RULE.md was read before the calculation. The root task additionally supplied the relevant correction: the user has already rejected replacing the reconstruction by a Gaussian-integer counterexample. That rejected use is not made here.

Original-author source actually read: Alain Connes and Caterina Consani, The Absolute Twistor Line and the Geometry of the compactified Spec $\mathbf{Z}$, supplied author file CC.tex, especially lines 430–460, 526–542, 576–602, 617–758 and 765–851. These are the signed base, chart and overlap presentations, field-valued point comparison, and involution. SHA-256 of that file is 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4.

The Stacks Project original TeX was read locally at commit d3e53496bbd24b82a1d87ce513c913923ebbb2:

- etale-cohomology.tex, lines 11071–11244: Kummer calculation and the pullback degree formula, 03RQ, 0AMB.
- The same file, lines 6547–6570: invariance under a nilpotent closed immersion, 03SI.
- The same file, lines 18610–18704: proper base change and its proof, 095T.
- The same file, lines 19523–19585: proper specialization and the smooth specialization consequence, 0GKD.
- trace.tex, lines 196–354: arithmetic/geometric Frobenius convention and the degree computation on the projective line.
- more-etale.tex, lines 4278–4362: the smooth proper curve consequence and its references.

This is reading coverage of the stated passages, not a claim to have reread the entire Stacks Project or CC paper. The two source hashes are etale-cohomology.tex: 015f27b1d332cc096b1f323daeeb855b299bb5a08f43c05131f196d4c183e7bf; trace.tex: 642cb0f2314fc73bef27542c79f55af87dcc0b0bf87d7cecf29b9b61860e4bfc.

CGT1. The ring realization and its exact twisted overlap

For a commutative coefficient ring R , define

$$\begin{aligned} A_R &= R[J]/(J^2 + 1), & A_+ &= R[J_+]/(J_+^2 + 1)[T], & A_- &= R[J_-]/(J_-^2 + 1)[U], \\ A_o &= R[J]/(J^2 + 1)[T, T^{-1}]. \end{aligned} \tag{CGT1.1}$$

The overlap homomorphisms, with all signs retained, are

$$\rho_+(J_+) = J, \quad \rho_+(T) = T, \quad \rho_-(J_-) = -J, \quad \rho_-(U) = T^{-1}. \tag{CGT1.2}$$

These are precisely the ring presentations representing the signed chart evaluations: for every commutative R -algebra D , a homomorphism $A_+ \rightarrow D$ is uniquely a pair (t, j_+) with $j_+^2 = -1$; the other chart and overlap have the same universal property, with t invertible on the overlap. The signed source element ϵ is sent to -1_D .

Thus this realization is specified by a universal mapping property, not merely by agreement of geometric points over fields.

Let X_R be the scheme obtained by gluing these affine spectra along their displayed localizations. Define the global coefficient J to be J_+ on the plus chart and $-J_-$ on the minus chart. The two definitions have the same restriction, because $\rho_-(-J_-) = J$.

The standard two charts of $\mathbf{P}_{A_R}^1$, with $T = X/Y$ and $U = Y/X$, now give isomorphisms to the two charts of X_R : their coefficient maps are respectively $J \mapsto J_+$ and $J \mapsto -J_-$, and their coordinate maps are $T \mapsto T$, $U \mapsto U$. On the overlap both become the same homomorphism $A_R[T, T^{-1}] \rightarrow A_\circ$. The inverse homomorphisms send $J_+ \mapsto J$, $J_- \mapsto -J$. Both composites fix every generator. The gluing universal property therefore proves

$$\boxed{X_R \simeq \mathbf{P}_{A_R}^1} \quad (\text{CGT1.3})$$

with the original twisted chart identifications explicitly retained. This does not replace J_- by J_+ : it records their comparison $J_- = -J$ in the global coefficient chart.

For $R = \mathbf{Z}$, write $A = \mathbf{Z}[J]/(J^2 + 1)$. The evaluation $J \mapsto i$ is a ring isomorphism $A \rightarrow \mathbf{Z}[i]$, with inverse $a + bi \mapsto a + bJ$. Uniqueness follows from the monic quadratic relation, which gives the free $\mathbf{Z}$ -basis $1, J$.

CGT2. The involution on the entire scheme

Let $c : A_R \rightarrow A_R$ be $c(J) = -J$, $c(r) = r$ for $r \in R$. Under CGT1 the supplied CC involution is

$$\alpha([X : Y], J) = ([-Y : X], -J), \quad \alpha^*(T) = -T^{-1}, \quad \alpha^*(J) = -J. \quad (\text{CGT2.1})$$

This notation describes the semilinear scheme morphism over R ; it does not assert that α is over A_R .

Its plus-to-minus chart map is

$$A_- \longrightarrow A_+, \quad U \mapsto -T, \quad J_- \mapsto J_+, \quad (\text{CGT2.2})$$

and its minus-to-plus map is

$$A_+ \longrightarrow A_-, \quad T \mapsto -U, \quad J_+ \mapsto J_-. \quad (\text{CGT2.3})$$

For example, restricting (CGT2.2) sends J_- to J , while applying the generic formula to $\rho_-(J_-) = -J$ also gives J . The coordinate comparison sends U to $-T$ on both routes. The other comparison is identical on its stated generators. Thus these are morphisms of the glued scheme.

Applying (CGT2.1) twice gives $[-X : -Y] = [X : Y]$ and $J \mapsto J$. Hence $\alpha^2 = 1$. The projective transformation has degree 1, although its affine formula contains a minus sign and inversion. This degree will determine its action on H^2 .

CGT3. All finite-field coefficient fibres, including the nonreduced fibre

Let $q = p^f$, $k = \mathbf{F}_q$, and $\bar{k}$ an algebraic closure.

For $p \neq 2$, choose once a root $j \in \bar{k}$ of $J^2 + 1$. The Chinese remainder map is

$$A_{\bar{k}} \longrightarrow \bar{k} \oplus \bar{k}, \quad F(J) \mapsto (F(j), F(-j)). \quad (\text{CGT3.1})$$

It is an isomorphism because $2j \neq 0$. Its inverse on a pair (a, b) is

$$\frac{a+b}{2} + \frac{a-b}{2j}J. \quad (\text{CGT3.2})$$

Both branches remain labeled by their actual coefficient values $j, -j$. Consequently

$$X_{\bar{k}} = \mathbf{P}_{\bar{k},+}^1 \sqcup \mathbf{P}_{\bar{k},-}^1. \quad (\text{CGT3.3})$$

The involution interchanges these components and applies $[X : Y] \mapsto [-Y : X]$ on their coordinates.

For $p = 2$, set $\nu = J - 1$. Then

$$A_{\bar{k}} = \bar{k}[\nu]/(\nu^2), \quad X_{\bar{k}} = \mathbf{P}_{\bar{k}[\nu]/(\nu^2)}^1. \quad (\text{CGT3.4})$$

This is a nonreduced scheme. Its closed immersion

$$\iota : \mathbf{P}_{\bar{k}}^1 \hookrightarrow X_{\bar{k}} \quad (\text{CGT3.5})$$

is defined by the nonzero square-zero ideal (ν) . The scheme and its reduction are not identified in this calculation. Instead, topological invariance gives the actual equivalence of étale topoi ι^{-1} , and hence the cohomology isomorphisms ι^* . The nilpotent coefficient ν is retained in (CGT3.4), even though the constant étale-cohomology receiver does not distinguish it.

Here $\alpha^*(J) = -J = J$, so $\alpha^*(\nu) = \nu$. The coordinate formula becomes $T \mapsto T^{-1}$; it is still the reduction of the full formula $-T^{-1}$.

CGT4. Cohomology and the geometric origin of the Tate factor

Fix a coefficient prime $\ell \neq p$, put $E = \mathbf{Q}_\ell$, and first work with coefficients $\mathbf{Z}/\ell^r\mathbf{Z}$. On $\mathbf{P}_{\bar{k}}^1$, the Kummer sequence gives

$$\begin{aligned} H^1(\mathbf{P}_{\bar{k}}^1, \mu_{\ell^r}) &= \text{Pic}(\mathbf{P}_{\bar{k}}^1)[\ell^r] = 0, \\ H^2(\mathbf{P}_{\bar{k}}^1, \mu_{\ell^r}) &= \text{Pic}(\mathbf{P}_{\bar{k}}^1)/\ell^r \text{Pic}(\mathbf{P}_{\bar{k}}^1) = \mathbf{Z}/\ell^r\mathbf{Z}. \end{aligned} \quad (\text{CGT4.1})$$

Here $\text{Pic}(\mathbf{P}_{\bar{k}}^1) = \mathbf{Z}[\mathcal{O}(1)]$. The other possible terms vanish because $\bar{k}^\times$ is ℓ^r -divisible and the higher $\mathbf{G}_m$ -cohomology of this smooth curve vanishes, as in the cited proof of 03RQ. The same source gives vanishing in degrees above 2. Passing through the compatible surjective coefficient transition maps, then tensoring with E , yields

$$H^0 = E, \quad H^1 = 0, \quad H^2 = E(-1). \quad (\text{CGT4.2})$$

The isomorphism $E \rightarrow H^2(\mathbf{P}_{\bar{k}}^1, E(1))$ sends 1 to $c_1(\mathcal{O}(1))$, not to an arbitrarily chosen eigenvector.

Here is the actual geometric multiplication factor. For every integer $n \geq 1$, the morphism

$$b_n : \mathbf{P}_{\bar{k}}^1 \rightarrow \mathbf{P}_{\bar{k}}^1, \quad [X : Y] \mapsto [X^n : Y^n] \quad (\text{CGT4.3})$$

has

$$b_n^* \mathcal{O}(1) = \mathcal{O}(n), \quad b_n^* c_1(\mathcal{O}(1)) = n c_1(\mathcal{O}(1)). \quad (\text{CGT4.4})$$

The first equality follows by pulling back the homogeneous transition function X/Y , which becomes $(X/Y)^n$; the second follows from the tensor-product additivity and functoriality of the Kummer boundary. Thus the factor n comes from a morphism and its divisor pullback.

For $n = q$ the coordinate morphism is the relative q -power Frobenius of the projective line. With geometric Frobenius defined as inverse arithmetic Galois Frobenius, its action on untwisted H^2 is multiplication by q . Equivalently, arithmetic Frobenius acts by q on $\mathbf{Z}_\ell(1)$, geometric Frobenius by q^{-1} , and the class $c_1(\mathcal{O}(1))$ is Galois-fixed in $H^2(E(1))$. Untwisting therefore gives q , with no rescaling.

Applying α 's coordinate morphism to the same transition function gives degree 1, so its pullback on H^2 is $+1$. The minus sign in $-T^{-1}$ is not a negative cohomological degree.

CGT5. Full Frobenius matrices and all repetition traces

For odd p , in the ordered component basis $(+, -)$, let

$$P_q = \begin{cases} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, & q \equiv 1 \pmod{4}, \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & q \equiv 3 \pmod{4}. \end{cases} \quad (\text{CGT5.1})$$

Indeed arithmetic Frobenius sends j to j^q ; since $j^2 = -1$, this is j or $-j$ according to $q \bmod 4$. The inverse permutation for geometric Frobenius is the same, because the permutation has order at most two. Combining this permutation with CGT4 gives

$$H^0(X_{\bar{k}}, E) = E^2, \quad F_q = P_q; \quad H^1 = 0; \quad H^2 = E(-1)^2, \quad F_q = qP_q. \quad (\text{CGT5.2})$$

Thus the eigenvalues are $1, 1$ and q, q when $q \equiv 1 \pmod{4}$, and $1, -1$ and $q, -q$ when $q \equiv 3 \pmod{4}$.

The matrix description is valid already at integral coefficient level. The decomposition into plus and minus eigenlines uses $1/2$, so it is asserted here over E , including $E = \mathbf{Q}_2$, not indiscriminately over $\mathbf{Z}_2$.

For $p = 2$, the pullback along (CGT3.5) and CGT4 give

$$H^0(X_{\bar{k}}, E) = E, \quad F_q = 1; \quad H^1 = 0; \quad H^2(X_{\bar{k}}, E) = E(-1), \quad F_q = q. \quad (\text{CGT5.3})$$

The q -power map sends ν to 0 , whereas α fixes ν . They are different scheme maps. They nevertheless give the stated cohomology actions through the explicitly proved reduction map.

For every repetition number $m \geq 1$, the full alternating Frobenius trace is

$$\text{Tr}(F_q^m \mid H^\bullet) = \begin{cases} (1 + q^m) \text{Tr}(P_q^m), & p \neq 2, \\ 1 + q^m, & p = 2. \end{cases} \quad (\text{CGT5.4})$$

In the nonsplit odd case this is zero for odd m and $2(1 + q^m)$ for even m . In the split odd case it is $2(1 + q^m)$ for every m . These agree directly with the number of $\mathbf{F}_{q^m}$ -points: the projective coordinate gives $q^m + 1$ for each available root of $J^2 + 1$. No repetition or coefficient branch is dropped.

The cohomological weights here are 0 in degree 0 and 2 in degree 2 , since the complex absolute values of the eigenvalues are $1 = q^{0/2}$ and $q = q^{2/2}$. There is no degree-one cohomology in this particular scheme.

CGT6. The signed absolute powers and the prime 2

The supplied signed absolute power map has $T \mapsto T^n$, $J \mapsto J^n$, and must preserve $\epsilon = -1$. It extends over the ring realization for odd n , because

$$(J^n)^2 = (-1)^n = -1. \quad (\text{CGT6.1})$$

The minus-chart compatibility is

$$(-J)^n = -J^n. \quad (\text{CGT6.2})$$

Its coordinate degree is n , and its coefficient action is identity for $n \equiv 1 \pmod{4}$ and conjugation for $n \equiv 3 \pmod{4}$. Hence its action on the two geometric component cohomologies is P_n in degree 0 and nP_n in degree 2 . It commutes with α : on T , both pullback composites give $-T^{-n}$; on J , both give $-J^n$.

For even n , $J \mapsto J^n$ is not a homomorphism of this global signed presentation over $\mathbf{Z}$, since its squared image is 1 , not -1 . The morphism b_n from CGT4 does exist with the coefficient J fixed, but it is not that signed source power map. In characteristic 2 , the Frobenius of the fibre also exists, including its action $\nu \mapsto 0$. These maps have different stated domains and coefficient actions. The cohomology at 2 in CGT5 is obtained from the fibre Frobenius, not by inventing an even signed source endomorphism.

CGT7. The actual wild specialization at 2

Let

$$S = \text{Spec } \mathbf{Z}_2, \quad S' = \text{Spec } \mathbf{Z}_2[i], \quad Y = \mathbf{P}_{S'}^1, \quad g : Y \rightarrow S. \quad (\text{CGT7.1})$$

The polynomial of $\pi = 1 - i$ is

$$\pi^2 - 2\pi + 2 = 0. \quad (\text{CGT7.2})$$

It is Eisenstein at 2 , so $\mathbf{Q}_2(i)/\mathbf{Q}_2$ is a totally ramified quadratic extension, with uniformizer π , residue field $\mathbf{F}_2$, and

$$2 = i\pi^2, \quad c(\pi) = 1 + i = i\pi. \quad (\text{CGT7.3})$$

It is wild because its ramification degree is the residue characteristic. The quotient of inertia acting on its two embeddings is the order-two group generated by $i \mapsto -i$. This action is retained.

Let $\ell \neq 2$. Over a geometric generic point, $Y_{\bar{\eta}}$ is the disjoint union of two projective lines, indexed by the embeddings $i \mapsto i$ and $i \mapsto -i$. Over a geometric special point its coefficient ring is $\overline{\mathbf{F}}_2[\nu]/(\nu^2)$. Hence

$$\begin{aligned} H^0(Y_{\bar{s}}, E) &= E, & H^2(Y_{\bar{s}}, E) &= E(-1), \\ H^0(Y_{\bar{\eta}}, E) &= E^2, & H^2(Y_{\bar{\eta}}, E) &= E(-1)^2. \end{aligned} \quad (\text{CGT7.4})$$

All degree-one groups vanish.

The actual specialization maps are

$$\text{sp}_0 : a \mapsto (a, a), \quad \text{sp}_2 : a h_s \mapsto (a h_+, a h_-), \quad (\text{CGT7.5})$$

where the notation $h = c_1(\mathcal{O}(1))$ is interpreted in the twisted group $H^2(E(1))$ before untwisting. To prove these formulas, use the global unit section for degree zero and the global line bundle $\mathcal{O}_Y(1)$ for degree two. Their restrictions to each geometric generic component are respectively the unit and the degree-one hyperplane class. These generate all groups by CGT4. Proper base change defines the special-fibre stalk, and smooth proper specialization for $Y \rightarrow S'$ proves that the restriction of each generator to each generic branch is an isomorphism. This proves (CGT7.5), including its coefficient 1 on each branch.

For $\ell \neq 2$, inertia acts trivially on $\mathbf{Q}_\ell(1)$: every ℓ^r -th root of unity belongs to an unramified extension of $\mathbf{Q}_2$, by lifting the simple roots in the algebraic closure of the residue field. It therefore acts on (CGT7.4) solely by the wild branch swap

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{CGT7.6})$$

The invariant subspaces are exactly the diagonals. Consequently the actual maps

$$\boxed{\text{sp}_j : H^j(Y_{\bar{s}}, E) \xrightarrow{\sim} H^j(Y_{\bar{\eta}}, E)^I, \quad j = 0, 1, 2} \quad (\text{CGT7.7})$$

are isomorphisms. In degree 1 both sides are zero. Their cokernels, the obstruction spaces for this specified specialization problem, are zero.

The anti-diagonal generic class $(a, -a)$ is not discarded. It is the sign representation of the wild inertia quotient and is not invariant unless $a = 0$. Over E it is the kernel of the sum map, complementary to the image in (CGT7.5). The factor 2 in the sum of a diagonal vector, $(a, a) \mapsto 2a$, is retained. Thus the disappearance of a second geometric special component is represented by the nonreduced coefficient fibre and this explicit ramified representation, not by silently merging two generic basis vectors.

Residual geometric Frobenius acts on the invariant degree-zero line by 1 and on the invariant degree-two line by 2. Any two lifts of residual Frobenius differ by inertia, so these values are independent of the chosen lift. On the full generic two-component space the branch permutation of a lift may differ; we make no choice-independent assertion about that extra permutation.

This is a calculation in a mixed-characteristic model smooth over S' , not an assertion that $g : Y \rightarrow S$ is smooth. Indeed the nonreduced special fibre shows that g is not smooth. The equal-characteristic hypotheses of Deligne's §3.6 are not silently substituted for the actual hypotheses here.

CGT8. The full proper pushforward before taking invariants

Fix ℓ and put

$$B = \text{Spec } \mathbf{Z}[1/\ell], \quad C = \text{Spec}(\mathbf{Z}[1/\ell, J]/(J^2 + 1)), \quad f : C \rightarrow B, \quad p : X = \mathbf{P}_C^1 \rightarrow C, \quad g = f \circ p. \quad (\text{CGT8.1})$$

The unit and hyperplane class define a map in the étale derived category

$$\mathbf{Q}_{\ell, C} \oplus \mathbf{Q}_{\ell, C}(-1)[-2] \longrightarrow R p_* \mathbf{Q}_{\ell, X}. \quad (\text{CGT8.2})$$

The first summand is the adjunction unit. The second is the morphism represented by $c_1(\mathcal{O}_X(1)) \in H^2(X, \mathbf{Q}_\ell(1))$, with the twist and cohomological shift explicitly displayed.

At every geometric point of C , proper base change identifies (CGT8.2) with the unit and hyperplane maps of CGT4. They are isomorphisms in degrees 0 and 2, and both sides have no other cohomology. A map of complexes inducing an isomorphism on all cohomology stalks is a quasi-isomorphism. Therefore (CGT8.2) is an isomorphism. One may first carry out the same construction with $\mathbf{Z}/\ell^r$ coefficients; the computed groups have surjective transitions, so passage to $\mathbf{Z}_\ell$ and then $\mathbf{Q}_\ell$ preserves these computations.

The finite morphism f has no higher direct images on these coefficients. Proper base change computes a stalk as cohomology of a finite geometric fibre; each such fibre is a disjoint union of spectra of Artinian local algebras over an algebraically closed field. Reduction and étale topological invariance identify its cohomology with that of a finite discrete set, which has no positive cohomology. This proves the vanishing by stalks, including the ramified fibre.

Applying $Rf_* = f_*$ to (CGT8.2) yields the full comparison

$$\boxed{Rg_* \mathbf{Q}_\ell \simeq f_* \mathbf{Q}_\ell \oplus f_* \mathbf{Q}_\ell(-1)[-2].} \quad (\text{CGT8.3})$$

No branch has been removed in (CGT8.3). The sheaf $f_* \mathbf{Q}_\ell$ has geometric stalk dimension 2 away from 2, and dimension 1 at 2 when $2 \in B$; its specialization is the diagonal map just calculated.

The involution acts by coefficient conjugation on the first summand. It has the same coefficient action on the second, because the projective part pulls back $\mathcal{O}(1)$ to $\mathcal{O}(1)$. Thus the decomposition is α -equivariant.

CGT9. The invariant sheaf, with the ramified stalk retained

Define the unit map

$$d : \mathbf{Q}_{\ell,B} \longrightarrow f_* \mathbf{Q}_{\ell,C}. \quad (\text{CGT9.1})$$

At an odd geometric residue characteristic its stalk is $a \mapsto (a, a)$. Conjugation interchanges the two entries, so its image is exactly the invariant subspace. At residue characteristic 2, its stalk is $a \mapsto a$ on the single-point étale receiver of the nonreduced ring. Conjugation acts trivially there. The same two-entry calculation applies at a geometric characteristic-zero point.

It follows on every geometric stalk that

$$\boxed{\mathbf{Q}_{\ell,B} \xrightarrow{\sim} (f_* \mathbf{Q}_{\ell,C})^\alpha.} \quad (\text{CGT9.2})$$

This is a sheaf isomorphism, not solely a fibrewise dimension comparison, because it is induced by the single map (CGT9.1).

Since 2 is invertible in $\mathbf{Q}_\ell$, the projector

$$e_+ = (1 + \alpha)/2 \quad (\text{CGT9.3})$$

is defined on rational ℓ -adic sheaves and makes taking invariants exact. This includes $\ell = 2$ with rational coefficients; it is not a statement about exact invariants of integral 2-adic modules. Apply the projector to the equivariant decomposition (CGT8.3). The result is the proved derived comparison

$$\boxed{(Rg_* \mathbf{Q}_\ell)^\alpha \simeq \mathbf{Q}_{\ell,B} \oplus \mathbf{Q}_{\ell,B}(-1)[-2].} \quad (\text{CGT9.4})$$

The complementary sheaf $(f_* \mathbf{Q}_\ell)^-$ is retained as the kernel of $1 + \alpha$. Its geometric stalk is the anti-diagonal at odd primes and at the generic point, and zero at 2. Thus (CGT9.4) records a projection with an explicitly described complementary receiver. It does not identify the original scheme or the full pushforward with the invariant part.

CGT10. The resulting full Euler products in their convergence domain

For every prime $p \neq \ell$ and every $m \geq 1$, (CGT9.4) gives the invariant alternating trace

$$1 + p^m. \quad (\text{CGT10.1})$$

With a formal variable u , its local factor is

$$\exp \left(\sum_{m \geq 1} \frac{1 + p^m}{m} u^m \right) = \frac{1}{(1 - u)(1 - pu)}. \quad (\text{CGT10.2})$$

This retains both cohomological degrees and every repetition. The equality follows from the convergent power series for $-\log(1 - z)$ near zero, or as a formal power-series identity with constant term 1.

To include $p = \ell$, compute that fibre with a different coefficient prime $\ell' \neq p$. CGT4–CGT5 give the same integer traces $1 + p^m$; hence the local polynomial is independent of this choice. This procedure retains the entire prime spectrum and does not place a missing-prime factor into an unspecified unit.

On the explicit domain $\Re(s) > 2$, all prime-repetition series are absolutely convergent, and $u = p^{-s}$ gives

$$\begin{aligned} \prod_p \frac{1}{(1 - p^{-s})(1 - p^{1-s})} &= \exp \left(\sum_p \sum_{m \geq 1} \frac{p^{-ms} + p^{m(1-s)}}{m} \right) \\ &= \left(\sum_{a \geq 1} a^{-s} \right) \left(\sum_{b \geq 1} b^{1-s} \right) \\ &= \boxed{\zeta(s)\zeta(s-1)}. \end{aligned} \quad (\text{CGT10.3})$$

The second equality is the Euler product using the already recovered arithmetic and unique factorization; it is not a construction of arithmetic from a finite prime selection. The unit terms $a = b = 1$ and the factor at 2 are explicit. These are the original zeta functions; no completion, Gamma multiplier, heat rescaling, or endpoint cancellation has been introduced.

For comparison, the full unprojected object of CGT8 has local factors

$$\begin{cases} ((1 - p^{-s})^2(1 - p^{1-s})^2)^{-1}, & p \equiv 1 \pmod{4}, \\ ((1 - p^{-2s})(1 - p^{2(1-s)}))^{-1}, & p \equiv 3 \pmod{4}, \\ ((1 - 2^{-s})(1 - 2^{1-s}))^{-1}, & p = 2. \end{cases} \quad (\text{CGT10.4})$$

These follow directly by taking determinants of the matrices in CGT5. They are recorded alongside the invariant factors so that projection has a visible effect on the Euler product.

CGT11. The exact lifting result and the boundary of this calculation

The specialization problem solved here is (CGT7.7): lift every inertia-invariant geometric generic class of the realized CC projective line from its geometric special fibre at 2. The maps are written in (CGT7.5), their images are computed in (CGT7.6), and their cokernels are zero. This proof retains the wild branch permutation and the nilpotent coefficient fibre.

The Tate factor in CGT4–CGT10 is obtained from the actual class $c_1(\mathcal{O}(1))$, its Kummer boundary, and pullback of the degree- q morphism. No character is assigned to an arbitrary analytic vector or to τ .

There is no proved identification here of the original-zeta zero-jet sheaf, the full timing reconstruction, or the user's desired lifting map with this $Rg_*\mathbf{Q}_\ell$. In particular, the weight-0 and weight-2 summands and the vanishing H^1 of this concrete projective line are exactly what the calculation gives. Calling that a weight-1 zero space or a proof of RH would add a conclusion not established by any map in this file.

The comparison is nevertheless explicit: it evaluates the supplied signed CC charts, retains their involution, derives a geometric Tate factor, proves the ramified specialization lift, and recovers the full original product $\zeta(s)\zeta(s-1)$ after the specified invariant projection. The factors and the complementary branch sheaf remain available for any further comparison.

CGT12. Diagram of the actual maps

The following diagram records the maps of CGT7. The left column is the geometric special fibre's cohomology; the middle column is the full geometric generic cohomology; the right column records its actual inertia invariants. In degree 2, the displayed $E(-1)$ is produced by CGT4.

$$\begin{array}{c|ccc}
 \text{degree} & H^j(Y_s, E) & H^j(Y_{\bar{\eta}}, E) & H^j(Y_{\bar{\eta}}, E)^I \\
 \hline
 0 & E \xrightarrow{a \mapsto (a, a)} & E^2 & \{(a, a) : a \in E\} \\
 2 & E(-1) \xrightarrow{a \mapsto (a, a)} & E(-1)^2 & \{(a, a) : a \in E(-1)\}.
 \end{array} \tag{CGT12.1}$$

The unit and hyperplane comparisons have the respective derived types $E \rightarrow R\Gamma(\mathbf{P}^1, E)$ and $E(-1)[-2] \rightarrow R\Gamma(\mathbf{P}^1, E)$, as in (CGT8.2). There is no asserted unshifted equivariant map $E \rightarrow E(-1)$. The horizontal arrows in (CGT12.1) are the specialization morphisms; their images are exactly the invariant subspaces in the right column.

The CC hyperplane class and the continuous global zeta lift

24 September 2026. Complete comparison proof; locators GTZ0–GTZ9.

GTZ0. Prerequisites, notation, and the object being compared

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$, in the user’s notation. This note introduces no addition at τ , numerical coordinate on it, or identification with an integer. The complete arithmetic reconstruction precedes every integer, prime, complex scalar, function space, and cohomology group used below.

The full relevant user corrections were read in the private corpus, including the complete-history-before-unit argument, the rejection of a Gaussian-integer example as a competing foundation, and the instruction to quotient branches separately. The governing route is `argument_reconstruction/CORPUS_AND_OPERATION_RULE.md`. Those private source records are not part of a public mathematical edition.

The input arithmetic is the complete reconstructed arithmetic of CG0–CG8 and BQC0–BQC6. Its original Riemann zeta function is denoted ζ . We do not reconstruct it from two selected primes. The supplied Connes–Consani charts are evaluated after that input is present. Their Gaussian coefficient ring in CGT1 is this particular chart receiver, not an alternative origin for the arithmetic.

We compare two already constructed objects:

1. The continuous quotient of the full Mellin source constructed in GSL0–GSL11, retaining all actual nontrivial zeros and their multiplicities.
2. The cohomology of the actual complex CC twistor line with that quotient as constant coefficients. Its hyperplane class is geometrically constructed.

The resulting isomorphism is in continuous complex linear representations of the CC odd monoid. It is not asserted to be an isomorphism of algebras, a complex-to- ℓ -adic coefficient identification, or a construction of the original coefficient space from the projective line alone. Those types will be visible in every map.

Source reading:

- Connes–Consani, *The Absolute Twistor Line and the Geometry of the compactified Spec Z*, arXiv:2609.00299v1.

The original CC.tex was read especially for the chart restrictions, geometric involution, and Proposition “The odd arithmetic monoid acts on the twistor line,” lines1159–1195. The supplied file has SHA256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4.

- Deligne, *La conjecture de Weil. II*, IHÉS52 (1980), printed213–214, §3.6; printed217 and221, chapterIV’s coefficient choices; printed248–249, §6.2.9. The entire original article, printed137–252, has now been read, with coverage recorded separately from checks of individual formulas. Original article.
- GLOBAL_SHIFTED_ZETA_LIFT.md, GSL0–GSL11, including the full continuous CRT proof and the higher-weight addition GSL6.5–GSL6.11, read completely.
- CC_GEOMETRIC_TATE_INDEPENDENT.md, CGT0–CGT12, read completely; it proves the signed chart realization, all finite-field fibres, Kummer class, ramified specialization, proper pushforward and invariant Euler factors.

These source comparisons do not ascribe the new map below to Deligne or Connes–Consani.

GTZ1. The complete coefficient spaces and their operations

Let $\mathcal{B}$ be the Fréchet space of entire functions with all seminorms

$$b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty, \quad A, M \in \mathbf{Z}_{\geq 0}. \quad (\text{GTZ1.1})$$

Let $\mathcal{Z}$ be the distinct nontrivial zeros of the original ζ , with actual multiplicities m_ρ . The already proved boundary nonvanishing places them in $0 < \Re \rho < 1$. Define closed ideals

$$\begin{aligned} \mathcal{J} &= \{F : F^{(j)}(\rho) = 0 \text{ for every } \rho, 0 \leq j < m_\rho\}, \\ \mathcal{J}_+ &= \{F : F^{(j)}(\rho + 1) = 0 \text{ for every } \rho, 0 \leq j < m_\rho\}, \\ \mathcal{Q} &= \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+, \quad \mathcal{E}_+ = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+). \end{aligned} \tag{GTZ1.2}$$

All quotients have their actual quotient Fréchet topologies. No unrestricted product of zero jets replaces them.

For every already reconstructed $a > 0$, the continuous coefficient operator is

$$T_a F(s) = a^s F(s), \quad L F(s) = s F(s). \tag{GTZ1.3}$$

Multiplication by a^s preserves each ideal and is continuous, since its modulus is $a^{\Re s}$, bounded on each fixed vertical strip. Multiplication by s is continuous with one additional polynomial seminorm. They consequently induce the indicated operators on all quotients.

Write $U F(s) = F(s + 1)$ and $V F(s) = F(s - 1)$. Their continuity follows from

$$b_{A,M}(U F), b_{A,M}(V F) \leq b_{A+1,M}(F). \tag{GTZ1.4}$$

The map V sends $\mathcal{J}$ to $\mathcal{J}_+$, and U reverses this map. Thus they induce mutually inverse maps $\bar{V} : \mathcal{Q} \rightarrow \mathcal{Q}_+$ and $\bar{U} : \mathcal{Q}_+ \rightarrow \mathcal{Q}$.

GSL4–GSL6 construct an entire multiplier E_+ of $\mathcal{B}$ with the full jet congruences

$$E_+ \equiv 1 \pmod{\mathfrak{m}_\rho^{m_\rho}}, \quad E_+ \equiv 0 \pmod{\mathfrak{m}_{\rho+1}^{m_{\rho+1}}}. \tag{GTZ1.5}$$

Here congruences are in the local holomorphic rings at the named points. The construction uses all zeros simultaneously and proves polynomial strip growth, so multiplication by E_+ and $1 - E_+$ is continuous on $\mathcal{B}$. In particular these are not formal infinite sums of local projectors.

The resulting continuous maps are

$$\begin{aligned} \pi[F] &= [F]_{\mathcal{J}}, \\ j[h] &= [(1 - E_+(s))h(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}, \\ \sigma[f] &= [E_+(s)f(s)]_{\mathcal{J} \cap \mathcal{J}_+}. \end{aligned} \tag{GTZ1.6}$$

They give the split exact row

$$0 \longrightarrow \mathcal{Q}(-1) \xrightarrow{j} \mathcal{E}_+ \xrightarrow{\pi} \mathcal{Q} \longrightarrow 0. \tag{GTZ1.7}$$

The notation $\mathcal{Q}(-1)$ here specifies $T_a^{(-1)} = a T_a$, $L^{(-1)} = L + 1$, on the same underlying Fréchet vector space. Its compatibility with an actual geometric Tate class, rather than a declaration about a class on τ , is what is proved next.

GTZ2. The actual complex line and its degree action

The CC object has complex points

$$X(\mathbf{C}) = \mathbf{P}^1(\mathbf{C})_+ \sqcup \mathbf{P}^1(\mathbf{C})_-, \quad \alpha(z, J) = (-1/z, -J), \quad J \in \{i, -i\}. \tag{GTZ2.1}$$

The full signed coordinate formula is retained. Its quotient by α is the complex line $M = \mathcal{M}_\infty(\mathbf{C})$. Each orbit has exactly one point with $J = i$, giving an isomorphism $M \simeq \mathbf{P}^1(\mathbf{C})$; the other representative is explicitly obtained by α .

For odd $n \geq 1$, the descended source morphism is

$$f_n(z) = \begin{cases} z^n, & n \equiv 1 \pmod{4}, \\ -1/z^n, & n \equiv 3 \pmod{4}. \end{cases} \quad (\text{GTZ2.2})$$

In homogeneous coordinates these are respectively $[X : Y] \mapsto [X^n : Y^n]$ and $[X : Y] \mapsto [-Y^n : X^n]$. Neither pair has a common projective zero. The first pulls the transition function of $\mathcal{O}(1)$ to its n -th power; the second is its composition with a projective linear automorphism. Hence

$$f_n^* \mathcal{O}(1) \simeq \mathcal{O}(n). \quad (\text{GTZ2.3})$$

Give M its complex orientation and let $h = c_1(\mathcal{O}(1)) \in H^2(M, \mathbf{C})$. This is the image of the integral first Chern class; its integral on M is 1, by the clutching function of degree one. Thus it retains a specified integral generator before scalar extension. The projective line has a CW decomposition with one cell in degree zero and one in degree two. Its cellular cochain complex is $\mathbf{C}$ in those two degrees and zero otherwise, with zero differential. It follows that

$$H^0(M, \mathbf{C}) = \mathbf{C} 1, \quad H^1(M, \mathbf{C}) = 0, \quad H^2(M, \mathbf{C}) = \mathbf{C} h. \quad (\text{GTZ2.4})$$

Naturality of c_1 and (GTZ2.3) prove

$$f_n^* 1 = 1, \quad f_n^* h = n h. \quad (\text{GTZ2.5})$$

The coordinate minus sign in (GTZ2.2) has not become a negative degree: its projective automorphism has degree +1.

The quotient and unquotiented calculation agree by actual pullback. If $q : X(\mathbf{C}) \rightarrow M$, then

$$q^* 1 = (1, 1), \quad q^* h = (h_+, h_-). \quad (\text{GTZ2.6})$$

These follow because the restriction of q to either component is a holomorphic projective automorphism. They identify $H^\bullet(M, \mathbf{C})$ with the α -invariant cohomology. The anti-diagonal component remains the complementary eigenspace; it has not been identified with zero.

GTZ3. The geometric receiver with the full original coefficients

Take constant cellular coefficients in the already constructed $\mathcal{Q}$. The cellular cochain complex is $\mathcal{Q}$ in degrees zero and two, with zero differential; all its maps are continuous. Its cohomology, with the finite product topology, is

$$\mathcal{H} = H^0(M; \mathcal{Q}) \oplus H^2(M; \mathcal{Q}) = \mathcal{Q} 1 \oplus \mathcal{Q} h. \quad (\text{GTZ3.1})$$

There is no completion of an infinite tensor product here: the geometric cochain complex has two cells.

Precisely, $x1 + yh$ denotes $x \otimes 1 + y \otimes h$ in $\mathcal{Q} \otimes_{\mathbf{C}} H^{\text{even}}(M, \mathbf{C})$, with its finite-product topology. Both 1 and h belong to the scalar geometric cohomology; neither notation presupposes a multiplicative identity in $\mathcal{Q}$. All coefficient products below are the products of the actual quotient algebra.

The joint coefficient and geometric action is

$$\mathcal{T}_n = T_n \otimes f_n^* : \quad x1 + yh \mapsto (T_n x) 1 + (n T_n y) h, \quad n \text{ odd}. \quad (\text{GTZ3.2})$$

It defines a continuous monoid representation. Indeed the source maps compose by $f_m f_n = f_{mn}$, and $T_m T_n = T_{mn}$; the displayed formula also verifies composition directly.

Define the geometric inclusion and projection

$$\iota_h : \mathcal{Q}(-1) \rightarrow \mathcal{H}, \quad y \mapsto yh, \quad \epsilon_0 : \mathcal{H} \rightarrow \mathcal{Q}, \quad x1 + yh \mapsto x. \quad (\text{GTZ3.3})$$

Their types retain degree two and its degree character; ι_h is equivariant by (GTZ2.5). The section $x \mapsto x1$ is equivariant. The class h is also the Gysin image of the oriented point class in M : its Poincaré dual integrates to the point's multiplicity one, and the cellular H^2 has precisely that integral generator. In the arithmetic étale formulation the corresponding map is $\mathbf{Q}_\ell(-1)[-2] \rightarrow R p_* \mathbf{Q}_\ell$, constructed by the Kummer Chern class in CGT4 and CGT8. Thus the degree and twist have an actual geometric source.

GTZ4. The exact map and its inverse

Define

$$\Theta : \mathcal{E}_+ \longrightarrow \mathcal{H}, \quad [F] \longmapsto [F]_{\mathcal{J}} 1 + [F(s+1)]_{\mathcal{J}} h. \quad (\text{GTZ4.1})$$

This is well defined. Changing F by $D \in \mathcal{J} \cap \mathcal{J}_+$ changes the first component by an element of $\mathcal{J}$; since D has all required jets zero at $\rho + 1$, $D(s+1)$ has those jets zero at ρ , so the second component is unchanged as well. Quotient maps and translation are continuous, hence so is Θ .

For $x = [f]_{\mathcal{J}}$, $y = [g]_{\mathcal{J}}$, define

$$\Xi(x1 + yh) = [E_+(s)f(s) + (1 - E_+(s))g(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{GTZ4.2})$$

This is independent of both representatives. If f changes by an element of $\mathcal{J}$, its product with E_+ vanishes to the required orders on the original divisor and, by (GTZ1.5), on the shifted divisor. If g changes by an element of $\mathcal{J}$, its translate by V vanishes on the shifted divisor; multiplication by $1 - E_+$ supplies vanishing on the original divisor. The induced map is continuous by the quotient property, the continuous multiplier bounds, and (GTZ1.4).

At every original zero the expression in (GTZ4.2) has exactly the full jet of f . At every shifted zero $\rho + 1$ it has exactly the full jet of $g(s-1)$. Hence $\Theta\Xi(x1 + yh) = x1 + yh$. Conversely, the expression $\Xi\Theta[F] - [F]$ has all required jets zero at both complete divisors, and therefore belongs to the defining ideal $\mathcal{J} \cap \mathcal{J}_+$. This proves $\Xi\Theta = 1$.

Thus Θ is a continuous linear isomorphism with the stated continuous inverse. The argument compares the actual quotient classes, not merely a finite set of their observations.

GTZ5. Equivariance, the entire lifting row, and every jet

For every odd n , its first component satisfies $[n^s F] = T_n[F]$. Its second component is

$$[(n^s F)(s+1)]_{\mathcal{J}} = [n^{s+1} F(s+1)]_{\mathcal{J}} = n T_n[F(s+1)]_{\mathcal{J}}. \quad (\text{GTZ5.1})$$

Equations (GTZ3.2) and (GTZ5.1) prove

$$\Theta T_n = \mathcal{T}_n \Theta. \quad (\text{GTZ5.2})$$

The factor n is exactly the degree factor already proved on h .

The maps of the entire extension agree:

$$\Theta j(y) = yh, \quad \epsilon_0 \Theta = \pi, \quad \Theta \sigma(x) = x1. \quad (\text{GTZ5.3})$$

The first identity uses both congruences in (GTZ1.5), with the translation $UV = 1$; the other two use the same congruences. Thus the following is a diagram of split exact continuous equivariant rows:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathcal{Q}(-1) & \xrightarrow{j} & \mathcal{E}_+ & \xrightarrow{\pi} & \mathcal{Q} \rightarrow 0 \\ & & \parallel & & \downarrow \Theta & & \parallel \\ 0 & \rightarrow & \mathcal{Q}(-1) & \xrightarrow{\iota_h} & \mathcal{H} & \xrightarrow{\epsilon_0} & \mathcal{Q} \rightarrow 0. \end{array} \quad (\text{GTZ5.4})$$

In particular the analytic continuous section is sent to the geometric degree-zero section, while the kernel is sent to the hyperplane summand.

Every multiplicity remains visible. At an original zero ρ , write its full jet coordinate as $t = s - \rho$, taken modulo t^{m_ρ} . The coefficient action is

$$n^{\rho+t} = n^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j. \quad (\text{GTZ5.5})$$

The shifted jet has coordinate $t = s - (\rho + 1)$. Translation U identifies these coordinates with the same t , without a sign or factorial change. Its action is

$$n^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j = n \left(n^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j \right). \quad (\text{GTZ5.6})$$

This is precisely (GTZ5.1) on the entire jet. No zero simplicity or RH has been used.

The analytic row has operators for all positive real a . Its geometric identification here is asserted for the actual CC odd monoid. The absence of an even signed endomorphism in that monoid does not remove the prime 2 from $\mathcal{Q}$: the coefficient object was constructed from the full original ζ . The fibre Frobenius at 2 and its separate geometric Tate factor are explicitly constructed in CGT5–CGT7.

GTZ6. The original zeta product and all exceptional contributions

The finite-prime calculation of CGT8–CGT10 gives an independent exact comparison to the same shift. The α -invariant proper pushforward is

$$(Rg_* \mathbf{Q}_\ell)^\alpha \simeq \mathbf{Q}_\ell \oplus \mathbf{Q}_\ell(-1)[-2]. \quad (\text{GTZ6.1})$$

Its full trace at prime p , repetition m , is $1 + p^m$. At $p = 2$ this is obtained from the actual nonreduced fibre and its reduction map; at $p = \ell$ the compatible local polynomial is calculated with a different coefficient prime. Therefore

$$\prod_p \frac{1}{(1 - p^{-s})(1 - p^{1-s})} = \exp \left(\sum_p \sum_{m \geq 1} \frac{p^{-ms} + p^{m(1-s)}}{m} \right) \quad (\text{GTZ6.2})$$

$$= \zeta(s)\zeta(s-1), \quad \Re s > 2.$$

Both original factors, all repetitions, and both unit terms remain. This geometric Euler product is not identified with a determinant of $\mathcal{Q}$; such a determinant has not been constructed in this note.

The analytic full source functions used to implement the two nontrivial zero divisors are

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_{-1}(s) = \frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1). \quad (\text{GTZ6.3})$$

Their product is the product of both displayed multipliers times both original zeta functions. It is not a replacement definition of those original functions. The meromorphic product $\zeta(s)\zeta(s-1)$ has:

- nontrivial zero divisors (ρ, m_ρ) and $(\rho + 1, m_\rho)$;
- simple trivial zeros at every $-2r$ and $1 - 2r$, $r \geq 1$;
- simple poles at $s = 1$ and $s = 2$.

There are no cancellations within this list: at ρ , the factor $\zeta(s-1)$ is a unit, by the functional equation and Euler nonvanishing in the reflected half-plane; at $\rho + 1$, $\zeta(s)$ is a unit by Euler nonvanishing. At the listed trivial zeros the other zeta factor is nonzero; at the poles the other factor is respectively $\zeta(0) = -1/2$ and $\zeta(2) \neq 0$. The trivial zeros are simple by the full functional equation and the nonzero positive-half-plane value.

The full comparison retains

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_{-1}(1) = F_{-1}(2) = \frac{1}{8}, \quad (\text{GTZ6.4})$$

and, for every $r \geq 1$,

$$\begin{aligned} F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \\ F_{-1}(1-2r) &= F_0(-2r), \quad F_{-1}(2+2r) = F_0(1+2r). \end{aligned} \quad (\text{GTZ6.5})$$

These nonzero values record the cancellation by the displayed multipliers; they are not zeros of the quotient divisor. For every derivative order d , if $K(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$, then

$$F_0^{(d)}(s) = \sum_{j=0}^d \binom{d}{j} K^{(d-j)}(s) \zeta^{(j)}(s), \quad F_{-1}^{(d)}(s) = F_0^{(d)}(s-1). \quad (\text{GTZ6.6})$$

At exceptional points these identities are interpreted by the meromorphic product's holomorphic continuation, as (GTZ6.4)–(GTZ6.5) explicitly compute. Thus no endpoint, Gamma derivative, multiplicity or constant is suppressed by the map.

GTZ7. The multiplication transported by the comparison

The map Θ is not claimed to respect the usual cup product on $H^\bullet(M; \mathcal{Q})$. There is an exact relation to it.

The quotients of $\mathcal{B}$ are commutative algebras under function multiplication. Translation U respects multiplication. Therefore the product transported from $\mathcal{E}_+$ to the coordinates of $\mathcal{H}$ is

$$(x1 + yh) \diamond (x'1 + y'h) = (xx')1 + (yy')h. \quad (\text{GTZ7.1})$$

This follows directly by applying Θ to representatives FG , at the two divisors. With $\diamond$, Θ is an algebra isomorphism.

The geometric cup product is instead

$$(x1 + yh) \smile (x'1 + y'h) = (xx')1 + (xy' + yx')h, \quad (\text{GTZ7.2})$$

because $h \smile h = 0$ in degree four of the two-dimensional sphere and 1 is the degree-zero identity in its geometric cohomology. The exact difference, in this already additive coefficient space, is

$$(u \diamond v) - (u \smile v) = (yy' - xy' - yx')h. \quad (\text{GTZ7.3})$$

Thus the continuous representation comparison is fully specified, including the extra multiplicative structure it does and does not preserve. It does not identify an idempotent divisor projector with a square-zero cohomology class as algebras.

GTZ8. What is proved about lifting, and what remains active

GTZ4–GTZ5 construct an actual continuous isomorphism of the whole original-zeta shifted extension with the cohomology of the CC complex line carrying the already constructed $\mathcal{Q}$. Its kernel's factor n is the pullback degree of the geometric hyperplane class, and its entire section corresponds to degree zero. CGT7 separately proves the actual ramified invariant-cycle lift at 2, by $a \mapsto (a, a)$.

Deligne's local invariant-cycle proof in §3.6, and its pure-complex version §6.2.9, involve a localization cross and a dual support-cohomology obstruction. GTZ5.4 is a proved comparison to this particular cohomology receiver; it does not yet identify that localization cross with the user's desired τ -supported lifting construction. In particular $\mathcal{Q}$ is an input coefficient space here, and its spectral purity has not been obtained from the purity of the projective line's geometric Tate class.

No conditional theorem replaces that calculation. The concrete result is the map Θ , its inverse, all equivariance and full jet identities, and the two geometric rows just proved. The active research goal remains to construct and calculate the actual receiving localization/support maps for the programme, using the complete source and the user's prerequisite order.

GTZ9. Proof dependencies and illustration

The complete continuous multiplier and CRT estimates are in GSL4–GSL6; this note uses that proved multiplier, not an assumed separator. The original Mellin source identification and its topological quotient are S5 and RZ1–RZ4, with source formulas retained in GSL8. The geometric integral Chern class calculation

and finite-prime realizations are CGT4, CGT7–CGT10. The complete comparison between these inputs is GTZ1–GTZ7 above.

The accompanying figure is a diagram of these exact maps and the ramified specialization. Its labels retain the shift, degree, twist and coefficient spaces. It does not depict a proved purity statement for $\mathcal{Q}$.

The geometric Tate factor in the global zeta lift

All arithmetic is already reconstructed. The supporting datum $\tau(Z_1; \text{no } Z_2)$ receives no new operation.

A. The entire continuous lifting row

$$\begin{array}{ccccccc}
 0 \rightarrow \mathcal{Q}(-1) & \xrightarrow{j} & \mathcal{E}_+ = B/(I \cap I_+) & \xrightarrow{\pi} & \mathcal{Q} = B/I & \rightarrow & 0 \\
 \parallel & & \downarrow \Theta[F] = [F]_I 1 + [F(s+1)]_I h & & \parallel & & \\
 0 \rightarrow \mathcal{Q}(-1) & \xrightarrow{y \mapsto yh} & \mathcal{Q} 1 \oplus \mathcal{Q} h & \xrightarrow{\varepsilon_0} & \mathcal{Q} & \rightarrow & 0
 \end{array}$$

$$M = \mathcal{M}_\infty(\mathbf{C}) \simeq \mathbf{P}^1(\mathbf{C}), \quad h = c_1(\mathcal{O}(1)), \quad f_n^* h = nh \quad (n \text{ odd}).$$

GTZ4–GTZ5: continuous inverse; exact section; every zero jet and multiplicity retained.

B. Ramified specialization at 2

$$\bar{\mathbf{F}}_2[\nu]/(\nu^2)$$

The nonreduced fibre remains explicit.

$$E(-1) \xrightarrow{\text{sp}_2} \{(a, a)\} \subset E(-1)^2$$

$$a \mapsto (a, a), \quad I: (x, y) \mapsto (y, x)$$

$$\text{image}(\text{sp}_2) = H_{\eta}^{2I}$$

CGT7. $E = \mathbf{Q}_\ell$, $\ell \neq 2$. The cokernel is zero.

C. Every finite prime and repetition

$$\text{Tr}(F_p^m | H_{\alpha=1}^\bullet) = 1 + p^m$$

$$\prod_p \frac{1}{(1-p^{-s})(1-p^{1-s})} = \zeta(s)\zeta(s-1)$$

$$\Re(s) > 2, \quad m \geq 1, \quad \text{including } p=2.$$

CGT8–CGT10 retain both degrees and all local factors.

The global analytic and geometric maps use the same +1 shift.

Proofs: GTZ0–GTZ9 and CGT0–CGT12. Sources: Connes–Consani (2609.00299v1), Deligne, Weil II §3.6 and §6.2.9.

The coefficient space $\mathcal{Q}$ is an input here. Its purity and the identification with the programme's full localization cross remain unproved.

Figure 8: The full continuous comparison and the geometric source of its degree factor. Panel A is GTZ5.4, with Theta and its inverse proved in GTZ4. Panel B is the actual ramified specialization of CGT7. Panel C is the all-prime Euler product of CGT10. $\mathcal{Q}$ is already constructed input data.

The independent mathematical check is GTZ_INDEPENDENT_MATHEMATICAL_CHECK.md, GTC0–GTC5. It checked the entire map, its topology and multiplication types, all exceptional-value factors, and the full GSL multiplier mechanism. Its tensor-notation clarification is incorporated in GTZ3.

Independent check of the CC hyperplane and shifted-zeta comparison

24 September 2026. Mathematical checking only. Locators GTC0–GTC5.

GTC0. Exact coverage

The file `CC_GYSIN_TO_GLOBAL_SHIFTED_ZETA.md`, GTZ0–GTZ9, was read completely. The dependency `GLOBAL_SHIFTED_ZETA_LIFT.md` was read for GSL0–GSL11; the Cauchy-kernel separator, strip bounds, full CRT inverse and higher-shift formulas were checked directly. The minor output truncation of GSL5.5 was corrected by rereading GSL4.12–GSL6.6 from the source file. The CC author source was additionally read at lines 1144–1205 for the odd-monoid quotient maps. CGT0–CGT12 is the independent geometric derivation by this checker.

No mathematical defect was found in the map Θ or its stated continuous linear, equivariant type. The coefficient quotient $\mathcal{Q}$ is an input throughout this check. It is not newly obtained from the geometric line, and no purity or RH conclusion is inferred.

GTC1. The full quotient map and topology

With the actual ideals $\mathcal{J}, \mathcal{J}_+$ and $\mathcal{J}_+ = \mathcal{J} \cap \mathcal{J}_+$, the canonical continuous CRT map is

$$C : \mathcal{B}/\mathcal{J}_+ \rightarrow (\mathcal{B}/\mathcal{J}) \oplus (\mathcal{B}/\mathcal{J}_+).$$

Translation $U : F(s) \mapsto F(s+1)$ induces the topological isomorphism $\bar{U} : \mathcal{Q}_+ \rightarrow \mathcal{Q}$. The proposed map is exactly

$$\Theta = (1 \oplus \bar{U})C$$

followed by the stated two-cell geometric coordinate identification. Its inverse is

$$(x, y) \mapsto [E_+(s)f(s) + (1 - E_+(s))g(s-1)]_{\mathcal{J}_+}$$

for any representatives $f, g \in \mathcal{B}$ of x, y .

If f changes by $\mathcal{J}$, its multiplied difference vanishes on the original divisor by that ideal and on the shifted divisor by E_+ . If g changes by $\mathcal{J}$, its translate vanishes on the shifted divisor, while $1 - E_+$ supplies the original divisor. Thus the inverse is defined on both quotient classes. The two jet identities for E_+ prove both compositions are identities, including every derivative through $m_\rho - 1$.

For an explicit topology check, let $q_{A,M}^{\mathcal{K}}$ be the quotient seminorm induced by $b_{A,M}$, and let multiplication by $E_+, 1 - E_+$ have bound $C_A b_{A,M+d_A}$. Then

$$q_{A,M}^{\mathcal{J}_+}(\Theta^{-1}(x, y)) \leq C_A \left(q_{A,M+d_A}^{\mathcal{J}}(x) + q_{A+1,M+d_A}^{\mathcal{J}}(y) \right).$$

This follows by choosing representatives independently, applying the multiplier and translation estimates, and taking their two infima. The forward map is bounded by the quotient seminorms with strip indices A and $A+1$. Hence the claimed inverse is continuous in the actual quotient topology.

GTC2. The multiplier dependency

The GSL argument supplies an entire multiplier, not an element of $\mathcal{B}$. Its proof has the following checked ingredients:

1. The original prime-power identity with coefficients 3, 4, 2 has summand $(1 + 2 \cos \theta)^2$. Its lower bound, combined with the horizontal derivative estimate and $\delta = c(1 + |t|)^{-6}$, gives the stated $(1 + |t|)^{-5}$ zeta lower bound.

2. The full Gamma reflection product for $F_0(s)F_0(s+1)$ has the displayed denominator $64 \cos(\pi s/2)$ and all four endpoint factors. Their magnitude gives the polynomial factor $(1 + |t|)^{-6}$ multiplying $e^{-\pi|t|/2}$ in the corridor lower bound.
3. The kernel $e^{(s-w)^2}/(\pi(s-w))$ has $\bar{\partial}$ -derivative the point mass of mass one. Its absolute value is $e^{(x-u)^2-(t-v)^2}/(\pi|s-w|)$; the real-coordinate integration creates only a logarithmic singularity, which is integrable.
4. The resulting solution u grows at most $e^{\pi|t|/2}(1 + |t|)^{12}$ on each fixed strip. Multiplication by the full G cancels this exponential estimate, proving polynomial strip growth of $E = \chi - Gu$.
5. At every zero of either divisor the cutoff is locally constant, u is holomorphic, and the complete multiplicity of G proves the full jet congruences. Translation gives $E_+(s) = 1 - E(s-1)$ with the correct direction.

These are global estimates; there is no finite-zero truncation, no unrestricted-jet-product replacement, and no zero-simplicity premise.

GTC3. The geometric degree and representation

Write $b_n(z) = z^n$, $I(z) = -1/z$, and $c(n) = (n-1)/2 \bmod 2$ for odd n . Then $I^2 = 1$, $b_n I = I b_n$, and $c(mn) = c(m) + c(n) \bmod 2$. Consequently the CC maps $f_n = I^{c(n)} b_n$ satisfy $f_m f_n = f_{mn}$, including all four congruence cases.

Their degree is n : the polynomial map pulls $\mathcal{O}(1)$ back to $\mathcal{O}(n)$, and I is a projective linear automorphism of degree 1. Thus $f_n^* h = nh$, with positive sign and the specified integral orientation.

On the shifted quotient,

$$U(T_n F)(s) = n^{s+1} F(s+1) = n T_n(UF)(s).$$

It follows that Θ intertwines T_n with $T_n \oplus n T_n$. At a multiplicity- m_ρ point this identity reads

$$n^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j = n \left(n^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} t^j \right).$$

The coordinates on the two germs are identified by translation with no sign change. This verifies all jets and every factorial. It also proves the three identities for the kernel inclusion, quotient projection, and section in GTZ5.3.

GTC4. Original factors and exceptional divisor

The factors in GTZ6 are $F_0(s)$ and $F_0(s-1)$, with constants $1/8$, endpoint products, powers of π , and Gamma factors retained. Their zero divisors are precisely the original nontrivial divisor and its shift by $+1$, while the meromorphic original product $\zeta(s)\zeta(s-1)$ additionally has every negative-integer trivial zero and its two simple poles at $1, 2$.

The two nontrivial divisors have disjoint real-part strips. At an original zero ρ , the other argument $\rho-1$ is in $-1 < \Re s < 0$, where the full functional equation relates it to a nonvanishing Euler-product value. At $\rho+1$, the other zeta factor is already in $\Re s > 1$. The trivial-zero and pole evaluations in GTZ6 therefore involve no cross-factor cancellation.

At $s = -2r + h$, the Gamma residue is

$$\Gamma(s/2) = \frac{2(-1)^r}{r! h} + O(1).$$

Multiplying it by $s(s-1)/8$, $\pi^{-s/2}$, and $\zeta'(-2r)h + O(h^2)$ gives exactly

$$\frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r).$$

This checks the constant in GTZ6.5. The endpoint values $1/8$ and the translation identities are also correct. The derivative identity is the full Leibniz rule, with meromorphic continuation at exceptional points as stated.

GTC5. Products and the coefficient-unit distinction

Both $\mathcal{B}$ and its quotient are being used as algebras without an assumed identity. The geometric notation must be read as

$$x1 + yh := x \otimes 1 + y \otimes h,$$

where $1, h$ are the scalar geometric cohomology basis. It must not mean that $\mathcal{Q}$ has an element 1 supplied by an entire constant function: that constant is not in $\mathcal{B}$.

Indeed the actual unbounded zero set also precludes a multiplicative identity in $\mathcal{Q}$. If $[u]$ were such an identity, evaluate its product with $e^{(s-\rho)^2} \in \mathcal{B}$ at any zero ρ . Since this test function has value 1 there, $u(\rho) = 1$. This holds for every zero, contradicting the rapid decay of u on a strip containing $0 < \Re s < 1$ along the unbounded zero ordinates. This observation does not affect any of the continuous linear comparison maps.

The multiplication in the quotient is transported to coordinatewise multiplication because both reduction and translation preserve products:

$$(x, y) \diamond (x', y') = (xx', yy').$$

The geometric cup product with these coefficients is instead

$$(x, y) \smile (x', y') = (xx', xy' + yx').$$

Their difference is the second-coordinate class $yy' - xy' - yx'$, exactly GTZ7.3. The identities remain valid for nonunital coefficient algebras. The monoid operators T_n are linear representation operators; they are not asserted to be algebra homomorphisms for either product.

With the explicit tensor interpretation of the geometric basis, the comparison is mathematically valid in the stated category. Its geometric Tate factor is proved; the original spectral coefficient purity is not inferred from that factor.

A continuous lift for the original and shifted zeta quotients

Independent mathematical derivation, 24 September 2026. Stable proof locators GSL0–GSL11.

The result is a continuous, arithmetic-equivariant isomorphism on the actual Fréchet spaces:

$$\mathcal{B}/(\mathcal{I} \cap \mathcal{I}_-) \xrightarrow{\sim} (\mathcal{B}/\mathcal{I}) \oplus (\mathcal{B}/\mathcal{I}_-). \quad (\text{GSL0.1})$$

Its inverse is explicitly constructed below. The proof retains the original zeta multiplier, all zero multiplicities, and the original strip seminorms. It makes no identification of either quotient with an unrestricted jet product or an algebraic direct sum.

GSL0. Input stage, source reading, and exact scope

The calculation takes place after the complete arithmetic and the original Riemann zeta function have been recovered. The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$; no arithmetic, coordinate, norm or addition is assigned to that support. All sums and integrals below occur in the existing coefficient function spaces. No selected prime determines the earlier arithmetic, and separate branches are not pooled.

The prerequisite correction audit is recorded in ZTM0 of ZETA_TATE_MONODROMY_DERIVATION.md. It includes READ_FIRST_USER_CONSTRUCTION.md, the full detailed reconstruction, USER_ARGUMENT_CONNECTIONS.md, and the pertinent verbatim passages U123–U138, WU042–WU055 and the complete ad55fb5f attachment in USER_CONSTRUCTION_FULL_LOGBOOK.md. The present operations use the same corrected input stage. In particular this is a calculation on a specified extension of the actual original-zeta coefficient quotient, not a replacement definition of the user’s whole global quotient.

The actual mathematical sources read for this derivation are:

- GLOBAL_MELLIN_SYNTHESIS.md, S1–S7, with particular attention to the full proof of S5 and the precise range statement S7. SHA256 334856FB8C568A7D1FD7CFC0326CD6E30298639E3A107201C9A196E2019E8730.
- ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md, RZ1–RZ12, including all finite-source representatives, estimates, and its actual quotient topology. SHA256 F3553E1362B4C2C1C1556098B55EE153BE9FC716A9D4C35BAEDCF39BD28AD755.
- The independent S3–S5 review, read for its division and topology checks, and DB9 of DELIGNE_BOUNDARY_NONVANISHING_RECONSTRUCTION.md, including its complete original-prime-power measure and coefficients 3, 4, 2.
- The original author source Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, `rhready.tex`, full multiplier and functional equation at local lines 526–546, with author-source identity recorded in OZR0 and ZTM0. No PDF is used. Its canonical IDs are PUBUNIT-171DAC50D1C3B7A2438B39AE and PUBUNIT-90A9C2A0907A753437719FD9.

The complete source-image equality S5 is an input to identifying the quotient used here with the original Mellin source. It is not newly attributed to this note. The global splitting and its continuous inverse are proved here, rather than deduced from the algebraic splitting in ZTM4–ZTM5. The estimates below use the original Euler product, elementary continued integrals, and the exact Gamma reflection identity; no zero spacing, zero simplicity, RH, or complex-to- ℓ -adic coefficient map is assumed.

GSL1. The Fréchet spaces, the full original multiplier, and translation

Let $\mathcal{B}$ consist of entire functions for which

$$b_{A,M}(F) = \sup_{|\operatorname{Re} s| \leq A} (1 + |\operatorname{Im} s|)^M |F(s)| < \infty \quad (A, M \in \mathbb{Z}_{\geq 0}). \quad (\text{GSL1.1})$$

These seminorms give the complete Fréchet topology proved in S1 and RZ2. Let $\mathcal{Z}$ be the actual distinct nontrivial zeros of the original ζ , with full multiplicities m_ρ . They lie in $0 < \operatorname{Re} \rho < 1$; the boundary

nonvanishing used here is the already proved DB9 statement. Set

$$\begin{aligned}\mathcal{J} &= \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for all } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \\ \mathcal{J}_- &= \{F \in \mathcal{B} : F^{(j)}(\rho - 1) = 0 \text{ for all } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \\ \mathcal{J} &= \mathcal{J} \cap \mathcal{J}_-, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_- = \mathcal{B}/\mathcal{J}_-, \quad \mathcal{Q}_\cup = \mathcal{B}/\mathcal{J}.\end{aligned}\tag{GSL1.2}$$

Every jet evaluation is continuous by Cauchy's formula, so these ideals are closed. The three quotients carry their actual quotient Fréchet topologies.

The retained source element is

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s).\tag{GSL1.3}$$

It is entire and belongs to $\mathcal{B}$, by the original Mellin calculation S2. Its zeros are exactly $\mathcal{Z}$, with their original multiplicities. The full exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8},\tag{GSL1.4}$$

$$\begin{aligned}F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_0(1+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1).\end{aligned}\tag{GSL1.5}$$

These account for the original endpoint factors, Gamma poles and trivial zeros; none is declared absent from the original function.

Translation defines a topological automorphism of $\mathcal{B}$:

$$(UF)(s) = F(s+1), \quad (U^{-1}F)(s) = F(s-1), \quad b_{A,M}(U^{\pm 1}F) \leq b_{A+1,M}(F).\tag{GSL1.6}$$

It satisfies $UJ = J_-$. Consequently it induces a topological isomorphism $\bar{U} : \mathcal{Q} \rightarrow \mathcal{Q}_-$, with the same inverse translation.

In particular the shifted full source multiplier is

$$F_1(s) = F_0(s+1) = \frac{(s+1)s}{8} \pi^{-(s+1)/2} \Gamma((s+1)/2) \zeta(s+1).\tag{GSL1.7}$$

Its endpoint values are $F_1(-1) = F_1(0) = 1/8$; its values at $-2r-1$ and $2r$ are precisely the two original expressions in (GSL1.5). Its zero divisor is $(\rho-1, m_\rho)$. The original and shifted divisors are disjoint because their real parts lie respectively in $(0, 1)$ and $(-1, 0)$.

We retain throughout the proof the entire function

$$G(s) = F_0(s)F_1(s),\tag{GSL1.8}$$

with both full factors (GSL1.3), (GSL1.7). It belongs to $\mathcal{B}$, since multiplication by a fixed member of $\mathcal{B}$ is continuous there. Its zero divisor is exactly the union with all multiplicities, and $G(0) = 1/64$.

GSL2. Polynomial bounds from the original zeta integral

The original identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx \quad (\operatorname{Re} s > 0)\tag{GSL2.1}$$

has the simple pole at one retained. It follows from the Dirichlet series by integration of the counting function and then by locally uniform continuation. On $3/4 \leq \operatorname{Re} s \leq 2$, $|\operatorname{Im} s| \geq 1$, it and its derivative give constants C_0, C_1 with

$$|\zeta(s)| \leq C_0(1 + |\operatorname{Im} s|), \quad |\zeta'(s)| \leq C_1(1 + |\operatorname{Im} s|).\tag{GSL2.2}$$

For the derivative, the complete formula is

$$\zeta'(s) = -\frac{1}{(s-1)^2} - \int_1^\infty \{x\}x^{-s-1} dx + s \int_1^\infty \{x\} \log x x^{-s-1} dx. \quad (\text{GSL2.3})$$

The two integrals are bounded by $1/\operatorname{Re} s$ and $1/(\operatorname{Re} s)^2$, respectively. This proves (GSL2.2) with every derivative term retained.

We also need polynomial growth on any fixed vertical strip. It can be proved without importing a bound in the critical strip. Define $P_1(x) = \{x\} - 1/2$ away from integers, and recursively let P_{j+1} be its periodic antiderivative of mean zero. Each P_j is bounded, P_j is continuous for $j \geq 2$, and $P'_{j+1} = P_j$ almost everywhere. To construct it, integrate P_j over a period and subtract the mean of the resulting primitive; the zero mean of P_j makes the primitive periodic.

Repeated integration by parts in (GSL2.1) gives, for every integer $K \geq 2$,

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \sum_{j=2}^K (s)_{j-1} P_j(1) - (s)_K \int_1^\infty P_K(x) x^{-s-K} dx, \quad \operatorname{Re} s > 1-K. \quad (\text{GSL2.4})$$

Here $(s)_j = s(s+1)\cdots(s+j-1)$. For the first step, $\int_1^\infty P_1(x)x^{-s-1}dx = -P_2(1) + (s+1) \int_1^\infty P_2(x)x^{-s-2}dx$. Each following integration gives the next displayed boundary constant with its positive sign and the final negative remainder. All boundary terms at infinity vanish initially where the integrals converge absolutely; the successively enlarged half-planes follow by the new convergent remainder and uniqueness of continuation. The constants, pole and $1/2$ are retained even when some periodic boundary constants are zero.

For $|\operatorname{Re} s| \leq A$, choose $K > A+2$. The last integral is bounded by $\|P_K\|_\infty/(K-1-A)$. For $|\operatorname{Im} s| \geq 1$ the pole term is bounded as well, so

$$|\zeta(s)| \leq C_A(1 + |\operatorname{Im} s|)^K \quad (|\operatorname{Re} s| \leq A, |\operatorname{Im} s| \geq 1). \quad (\text{GSL2.5})$$

This estimate is used only as an inequality for the original function; no new working zeta function has been defined.

GSL3. A quantitative zero-free corridor around the separating line

For $\delta > 0$, the absolutely convergent original Euler product gives

$$\begin{aligned} & 3 \log \zeta(1+\delta) + 4 \log |\zeta(1+\delta+it)| + 2 \log |\zeta(1+\delta+2it)| \\ &= \sum_p \sum_{n \geq 1} \frac{p^{-n(1+\delta)}}{n} (3 + 4 \cos(nt \log p) + 2 \cos(2nt \log p)) \geq 0. \end{aligned} \quad (\text{GSL3.1})$$

Every prime and positive prime power is present. The summand's trigonometric factor is exactly $(1 + 2 \cos(nt \log p))^2$. Exponentiating gives

$$\zeta(1+\delta)^3 |\zeta(1+\delta+it)|^4 |\zeta(1+\delta+2it)|^2 \geq 1. \quad (\text{GSL3.2})$$

For $0 < \delta \leq 1/4$, the Dirichlet-series integral bound gives $\zeta(1+\delta) \leq 1 + 1/\delta \leq 2/\delta$. Put $T = 1 + |t|$, with $|t| \geq 1$. By (GSL2.2), the last zeta factor is at most a constant times T . Consequently there is $c_0 > 0$, independent of δ, t , such that

$$|\zeta(1+\delta+it)| \geq c_0 \delta^{3/4} T^{-1/2}. \quad (\text{GSL3.3})$$

Choose a positive constant $c \leq 1/4$ small enough that $2C_1 c \leq (c_0/2)c^{3/4}$, enlarging C_1 in (GSL2.2) if required to cover the same horizontal segments. Set $\delta = cT^{-6}$. For every real x with $|x| \leq cT^{-6}$, (GSL2.3) integrated along that segment gives

$$\begin{aligned} |\zeta(1+x+it)| &\geq |\zeta(1+\delta+it)| - 2C_1 T \delta \\ &\geq \frac{c_0 c^{3/4}}{2} T^{-5}. \end{aligned} \quad (\text{GSL3.4})$$

Thus the width and lower bound have been derived from the original function, not assumed as an RH consequence.

The full functional equation gives $F_0(s) = F_0(1-s)$. For an exact growth comparison, retain both complete factors after that substitution:

$$G(s) = \left[\frac{(1-s)(-s)}{8} \pi^{-(1-s)/2} \Gamma((1-s)/2) \zeta(1-s) \right] \cdot \left[\frac{(1+s)s}{8} \pi^{-(1+s)/2} \Gamma((1+s)/2) \zeta(1+s) \right]. \quad (\text{GSL3.5})$$

The exact Gamma reflection identity gives the comparison

$$\Gamma((1-s)/2) \Gamma((1+s)/2) = \frac{\pi}{\cos(\pi s/2)}. \quad (\text{GSL3.6})$$

This identity follows from $\Gamma(z) \Gamma(1-z) = \pi / \sin(\pi z)$, with $z = (1-s)/2$. For example the latter follows from the Euler products for $1/\Gamma$ and $\sin$, multiplying the factors at $z, -z$ and then using $\Gamma(1-z) = -z\Gamma(-z)$. It is a meromorphic identity retaining both Gamma factors, not a change to the function in (GSL1.8).

Combining the two powers of π with (GSL3.6) gives the exact additional expression

$$G(s) = \frac{(1-s)(-s)(1+s)s}{64 \cos(\pi s/2)} \zeta(1-s) \zeta(1+s). \quad (\text{GSL3.7})$$

All four endpoint factors, the denominator 64, both original zeta factors and the Gamma comparison remain explicit. At apparent exceptional points the expression has the entire continuation specified by (GSL1.3)–(GSL1.8); it is used for estimates only where $|t| \geq 1$.

For $s = x + it$, $|t| \geq 1$, $|x| \leq cT^{-6}$, the product of the four endpoint magnitudes is bounded below by a positive constant times T^4 . Further,

$$|\cos(\pi(x+it)/2)|^2 = \cos^2(\pi x/2) + \sinh^2(\pi t/2). \quad (\text{GSL3.8})$$

The reciprocal cosine has magnitude comparable to $e^{-\pi|t|/2}$ in this region. Apply (GSL3.4) to $1+x+it$ and to $1-x-it$; conjugation preserves the same bound. Therefore

$$|G(x+it)| \geq c_2 T^{-6} e^{-\pi|t|/2} \quad (|t| \geq 1, |x| \leq cT^{-6}). \quad (\text{GSL3.9})$$

On every fixed vertical strip, (GSL2.5), (GSL3.7) and (GSL3.8) also give an integer $k_A \geq 0$ and a constant C_A such that

$$|G(x+it)| \leq C_A (1+|t|)^{k_A} e^{-\pi|t|/2} \quad (|x| \leq A, t \in \mathbb{R}). \quad (\text{GSL3.10})$$

For bounded t , this follows by the entire continuation and compactness; no pole in (GSL3.7) is retained spuriously.

The line $x = 0$ has no zero of G . For $t \neq 0$, this follows from DB9 at $1 \pm it$, (GSL3.5), and the absence of a Gamma zero or pole there. At $t = 0$, it is the exact value $G(0) = 1/64$. Compactness gives a positive-width zero-free neighborhood of the bounded segment $|t| \leq 1$. Choose $\varepsilon > 0$ sufficiently small, with $\varepsilon \leq 1/4$, so that

$$r(t) = \varepsilon(1+t^2)^{-3} \quad (\text{GSL3.11})$$

lies within that compact neighborhood and within the corridor in (GSL3.9) for $|t| \geq 1$. This is possible because $(1+t^2)^{-3} \leq 8(1+|t|)^{-6}$. Shrinking the constants to include the compact segment proves

$$G(x+it) \neq 0, \quad \frac{1}{|G(x+it)|} \leq C(1+|t|)^6 e^{\pi|t|/2} \quad (|x| \leq r(t)). \quad (\text{GSL3.12})$$

The entire original and shifted zero sets are outside this corridor, on its right and left respectively. No separation estimate between zeros within either set was needed.

GSL4. An explicit entire separator with controlled strip growth

Choose a real smooth function η with $0 \leq \eta \leq 1$, $\eta(v) = 0$ for $v \leq -1$, and $\eta(v) = 1$ for $v \geq 1$. For an explicit choice let $b(v) = 0$ for $v \leq 0$, $b(v) = e^{-1/v}$ for $v > 0$, and take

$$\eta(v) = \frac{b(v+1)}{b(v+1) + b(1-v)}. \quad (\text{GSL4.1})$$

The denominator is positive everywhere, and the vanishing of all right derivatives of b at zero proves the stated smoothness. Define on the complex plane

$$\chi(x+it) = \eta(x/r(t)), \quad \bar{\partial} = \frac{1}{2}(\partial_x + i\partial_t). \quad (\text{GSL4.2})$$

Its derivative is supported in the corridor. There,

$$\bar{\partial}\chi(x+it) = \frac{1}{2}\eta'(x/r(t)) \left(\frac{1}{r(t)} - i \frac{xr'(t)}{r(t)^2} \right). \quad (\text{GSL4.3})$$

Since $|x| \leq r(t)$, $r'/r = -6t/(1+t^2)$, and $1/r \leq C(1+|t|)^6$, we have

$$|\bar{\partial}\chi(x+it)| \leq C(1+|t|)^6. \quad (\text{GSL4.4})$$

Define the smooth function

$$h(w) = \frac{\bar{\partial}\chi(w)}{G(w)} \quad (\text{GSL4.5})$$

on the corridor, extended by zero outside it. Smoothness across its edges follows because the derivative of η vanishes to every order there. There is no division at a zero of G . With $\alpha = \pi/2$, (GSL3.12) gives

$$\text{supp } h \subset \{| \text{Re } w | \leq \varepsilon\}, \quad |h(u+iv)| \leq C e^{\alpha|v|} (1+|v|)^{12}. \quad (\text{GSL4.6})$$

Use the explicit Cauchy kernel with its Gaussian factor:

$$u(s) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{e^{(s-w)^2}}{s-w} h(w) dA(w), \quad (\text{GSL4.7})$$

where dA is ordinary planar area. This integral is absolutely convergent. Its local singularity has order $1/|s-w|$, which is integrable in two real dimensions. For the tails, writing $s = x+it$, $w = u+iv$, its exponential modulus is exactly

$$|e^{(s-w)^2}| = e^{(x-u)^2 - (t-v)^2}. \quad (\text{GSL4.8})$$

The negative quadratic in $t-v$ dominates the fixed exponential and polynomial in (GSL4.6), while $|u| \leq \varepsilon$.

Here is a uniform strip estimate, including the local singularity. For $y \neq 0$, integration in u gives

$$\sup_{x \in \mathbb{R}} \int_{-\varepsilon}^{\varepsilon} \frac{du}{\sqrt{(x-u)^2 + y^2}} \leq C_{\varepsilon} \left(1 + \log_+ \frac{1}{|y|} \right). \quad (\text{GSL4.9})$$

One can see this by translating the integration interval towards zero, which increases the integral of the even decreasing function of $|x-u|$, and integrating $(v^2 + y^2)^{-1/2}$ there. The result is bounded by a constant plus $2 \log(1/|y|)$ when $|y| \leq 1$, and by $2\varepsilon/|y|$ otherwise. The infinite value at $y = 0$ is on a set of one-dimensional measure zero; the displayed logarithm is integrable near zero.

For $|x| \leq A$, use $(1+|v|) \leq (1+|t|)(1+|t-v|)$ and $e^{\alpha|v|} \leq e^{\alpha|t|} e^{\alpha|t-v|}$. Equations (GSL4.6)–(GSL4.9) yield

$$|u(x+it)| \leq C_A e^{\alpha|t|} (1+|t|)^{12}, \quad (\text{GSL4.10})$$

because the remaining integral

$$\int_{\mathbb{R}} e^{-y^2 + \alpha|y|} (1+|y|)^{12} \left(1 + \log_+ \frac{1}{|y|} \right) dy \quad (\text{GSL4.11})$$

is finite, both near zero and at infinity.

The fundamental-solution identity is $\bar{\partial}(1/(\pi s)) = \delta_0$. It follows directly by integrating a compactly supported smooth test function over the punctured plane and taking the boundary integral on a shrinking circle; the factor $1/\pi$ and the convention in (GSL4.2) give mass one. Multiplication by the holomorphic function e^{s^2} , whose value at zero is one, preserves that identity. Thus (GSL4.7) satisfies

$$\bar{\partial}u = h. \quad (\text{GSL4.12})$$

For precision the identity may first be applied to compactly supported truncations of h ; Gaussian domination makes their integrals and distributional derivatives converge on each compact set. Locally the compact Cauchy transform of a smooth function is smooth: transfer its derivatives to that function using a smooth cutoff, while the complementary kernel has no singularity. The same Gaussian tail estimates apply to the complementary part. Hence u is smooth, and (GSL4.12) is also a pointwise identity.

Define

$$E(s) = \chi(s) - G(s)u(s). \quad (\text{GSL4.13})$$

Its $\bar{\partial}$ derivative is $\bar{\partial}\chi - Gh = 0$, so it is entire. Combining (GSL3.10) and (GSL4.10) proves, for each strip, an integer $d_A \geq 0$ and constant C'_A with

$$|E(x + it)| + |1 - E(x + it)| \leq C'_A(1 + |t|)^{d_A} \quad (|x| \leq A). \quad (\text{GSL4.14})$$

The exponential factors $e^{-\pi|t|/2}$ and $e^{\pi|t|/2}$ have both been retained in the proof before this estimate is taken.

Every original zero is to the right of the corridor, where $\chi = 1$; every shifted zero is to the left, where $\chi = 0$. In a neighborhood of any such zero, $h = 0$, so u is holomorphic and (GSL4.13) gives the complete jet identities

$$\begin{aligned} E(s) - 1 &\in (s - \rho)^{m_\rho} \mathcal{O}_\rho, \\ E(s) &\in (s - (\rho - 1))^{m_\rho} \mathcal{O}_{\rho-1} \quad (\rho \in \mathcal{Z}). \end{aligned} \quad (\text{GSL4.15})$$

Every multiplicity is preserved. The constructed function E is a multiplier of $\mathcal{B}$; it is not assumed to be a member of $\mathcal{B}$. This distinction matters because it takes the value one at the entire original zero set.

GSL5. Continuous multiplication and the full CRT inverse

Equation (GSL4.14) gives the explicit seminorm estimates

$$b_{A,M}(EF) \leq C'_A b_{A,M+d_A}(F), \quad b_{A,M}((1-E)F) \leq C'_A b_{A,M+d_A}(F). \quad (\text{GSL5.1})$$

Thus both are continuous linear operators $\mathcal{B} \rightarrow \mathcal{B}$. Their full ideal behavior is

$$\begin{aligned} E\mathcal{B} &\subset \mathcal{J}_-, & (1-E)\mathcal{B} &\subset \mathcal{J}, \\ E\mathcal{J} &\subset \mathcal{J}, & (1-E)\mathcal{J}_- &\subset \mathcal{J}. \end{aligned} \quad (\text{GSL5.2})$$

Each inclusion follows from the appropriate two sets of jets in (GSL4.15) and the full Leibniz formula. In particular

$$F = EF + (1-E)F \quad \text{proves} \quad \mathcal{J} + \mathcal{J}_- = \mathcal{B}, \quad (\text{GSL5.3})$$

with an explicit continuous choice of the two summands.

The canonical map and its constructed inverse are

$$\Phi : \mathcal{Q}_\cup \longrightarrow \mathcal{Q} \oplus \mathcal{Q}_-, \quad [F]_{\mathcal{J}} \longmapsto ([F]_{\mathcal{J}}, [F]_{\mathcal{J}_-}), \quad (\text{GSL5.4})$$

$$\Psi([f]_{\mathcal{J}}, [g]_{\mathcal{J}_-}) = [Ef + (1-E)g]_{\mathcal{J}}. \quad (\text{GSL5.5})$$

The final two inclusions in (GSL5.2) prove independence of the representatives in (GSL5.5). At every original zero its representative has exactly the jets of f ; at every shifted zero it has exactly the jets of g . Thus $\Phi\Psi = 1$. For a common representative F , its image under the opposite composition is $[EF + (1-E)F]_{\mathcal{J}} = [F]_{\mathcal{J}}$, so

$\Psi\Phi = 1$. This proves surjectivity onto the full direct sum of the two actual quotient spaces, not merely a dense range.

For explicit topological control, let $q_{A,M}^{\mathcal{K}}$ be the quotient seminorm obtained from $b_{A,M}$ by a closed subspace $\mathcal{K}$. The quotient maps in (GSL5.4) are continuous with

$$q_{A,M}^{\mathcal{J}}([F]), \quad q_{A,M}^{\mathcal{J}_-}([F]) \leq q_{A,M}^{\mathcal{J}}([F]). \quad (\text{GSL5.6})$$

Taking the two independent infima over representatives in (GSL5.1) proves

$$q_{A,M}^{\mathcal{J}}(\Psi(x, y)) \leq C'_A \left(q_{A,M+d_A}^{\mathcal{J}}(x) + q_{A,M+d_A}^{\mathcal{J}_-}(y) \right). \quad (\text{GSL5.7})$$

This is the required continuous inverse estimate on the actual quotient topology. Neither a blockwise formal inversion nor an open-mapping assertion is being used as a substitute for it.

The lift and the shifted inclusion are consequently

$$\iota_+(x) = \Psi(x, 0), \quad \iota_-(y) = \Psi(0, y), \quad \pi_+([F]) = [F]_{\mathcal{J}}. \quad (\text{GSL5.8})$$

They give the continuously split exact sequence

$$0 \longrightarrow \mathcal{Q}_- \xrightarrow{\iota_-} \mathcal{Q}_{\cup} \xrightarrow{\pi_+} \mathcal{Q} \longrightarrow 0, \quad \pi_+ \iota_+ = 1. \quad (\text{GSL5.9})$$

The corresponding projection is multiplication by E modulo $\mathcal{J}$. It is idempotent there because $E^2 - E$ vanishes to every required jet at both sets. No assertion that $E^2 = E$ as an entire function is made.

GSL6. Original arithmetic action, degree shift, and uniqueness of the lift

On $\mathcal{B}$ use the actual operators

$$\mathbf{L}F(s) = sF(s), \quad T_a F(s) = a^s F(s), \quad a > 0. \quad (\text{GSL6.1})$$

Their continuity estimates are exactly RZ1.10 and RZ8.2. Both preserve all ideals in (GSL1.2), so they descend to every displayed quotient. Multiplication by E commutes with both already on $\mathcal{B}$. Hence $\Phi, \Psi, \iota_+, \iota_-$ commute with L and every T_a in their specified domains and codomains.

Translation keeps its exact character and degree factors:

$$T_{a,-} \bar{U} = a^{-1} \bar{U} T_a, \quad L_- \bar{U} = \bar{U} (L - 1). \quad (\text{GSL6.2})$$

For example $U(sF)(s) = (s+1)F(s+1)$, proving the second identity including its sign. Thus (GSL5.4) also gives a topological representation isomorphism

$$\mathcal{Q}_{\cup} \simeq \mathcal{Q} \oplus \mathcal{Q}(1), \quad (\text{GSL6.3})$$

where the second summand is realized as the actual translated quotient, and its character twist means $T_a \mapsto a^{-1} T_a$. This is an analytic character twist, not an asserted étale Tate realization. On each actual shifted jet its full operator is

$$T_{a,-} = a^{\rho-1} \sum_{j=0}^{m_{\rho}-1} \frac{(\log a)^j}{j!} M_w^j, \quad w = s - (\rho - 1). \quad (\text{GSL6.4})$$

The original block retains a^{ρ} with the same complete nilpotent exponential. For the weight statement fix a recovered $a > 1$ and define $w_a(\lambda) = 2 \log |\lambda| / \log a$ for $\lambda \neq 0$. Then $w_a(a^{\rho-1}) = 2 \operatorname{Re} \rho - 2$ and $w_a(a^{\rho}) = 2 \operatorname{Re} \rho$, so their logarithmic-modulus weights occupy $(-2, 0)$ and $(0, 2)$, respectively, without any critical-line assumption. The action itself remains defined for every $a > 0$, including its identity at $a = 1$; no weight is assigned using a denominator $\log 1$.

There is also an exact version with the higher-weight term as the kernel, obtained by translating the proved construction, rather than positing another splitting. Set

$$\mathcal{J}_+ = U^{-1}\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0 \ (0 \leq j < m_\rho)\}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+, \quad \mathcal{Q}_{\cup,+} = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+). \quad (\text{GSL6.5})$$

Translation by -1 sends $\mathcal{J}$ onto $\mathcal{J}_+ \cap \mathcal{J}$ and gives a topological character comparison from $\mathcal{Q}_{\cup}(-1)$ to $\mathcal{Q}_{\cup,+}$. Its full translated multiplier remains

$$G(s-1) = \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right] \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right]. \quad (\text{GSL6.6})$$

The multiplier $E_+(s) = 1 - E(s-1)$ has jet one on the original divisor and jet zero on the divisor shifted by $+1$. Its strip-growth estimate follows from (GSL4.14) with $A+1$ in place of A . Applying the already proved inverse with this multiplier gives the exact continuously split row

$$0 \longrightarrow \mathcal{Q}_+ \simeq \mathcal{Q}(-1) \longrightarrow \mathcal{Q}_{\cup,+} \longrightarrow \mathcal{Q} \longrightarrow 0. \quad (\text{GSL6.7})$$

The section is $[f]_{\mathcal{J}} \mapsto [E_+ f]_{\mathcal{J} \cap \mathcal{J}_+}$; the inclusion of the kernel uses $1 - E_+$. On its kernel, translation gives $a^{\rho+1} \exp((\log a) M_w)$, retaining every nilpotent coefficient. The original scalar weights are in $(0, 2)$, the kernel weights in $(2, 4)$. Thus the higher-weight version of the separated lift is also realized on a fully specified Fréchet quotient of the original source, with both original and translated full factors retained.

Here are the full maps and exceptional values for that counterpart. Write $VF(s) = F(s-1)$, so $V = U^{-1}$, and put

$$F_{-1}(s) = F_0(s-1) = \frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1). \quad (\text{GSL6.8})$$

Its endpoint values are $F_{-1}(1) = F_{-1}(2) = 1/8$. The entire values at the translated trivial-zero and reflected locations are, with every original coefficient retained,

$$\begin{aligned} F_{-1}(1-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_{-1}(2+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{GSL6.9})$$

The apparent Gamma and zeta singularities in (GSL6.8) have exactly these continuations; its entire derivatives are $F_{-1}^{(j)}(s) = F_0^{(j)}(s-1)$.

Translation induces $\bar{V} : \mathcal{Q} \rightarrow \mathcal{Q}_+$, with inverse $\bar{U}$ between these two specified quotients. Substituting the functions gives

$$T_{a,+} \bar{V} = a \bar{V} T_a, \quad L_+ \bar{V} = \bar{V}(L+1). \quad (\text{GSL6.10})$$

For instance $V(sF)(s) = (s-1)F(s-1)$, so $sVF = V(sF) + VF$; also $V(a^s F)(s) = a^{s-1} F(s-1)$, giving exactly the first factor in (GSL6.10). The map $\bar{V}$ is thus an equivariant isomorphism from $\mathcal{Q}(-1)$ to $\mathcal{Q}_+$. The high-weight row, including its injection from that domain, its quotient map and its section, is

$$\begin{aligned} j_{\text{high}}([h]_{\mathcal{J}}) &= [(1 - E_+(s))h(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}, \\ \pi_{\text{high}}([F]_{\mathcal{J} \cap \mathcal{J}_+}) &= [F]_{\mathcal{J}}, \\ s_{\text{high}}([f]_{\mathcal{J}}) &= [E_+(s)f(s)]_{\mathcal{J} \cap \mathcal{J}_+}. \end{aligned} \quad (\text{GSL6.11})$$

The first displayed domain is the underlying vector space of $\mathcal{Q}(-1)$, with its specified twisted action. The ideal identities follow by the same two full jet conditions on E_+ , proving independence of every representative. The CRT inverse applied to $([f]_{\mathcal{J}}, \bar{V}[h]_{\mathcal{J}})$ is the sum of the first and third maps. It proves exactness, surjectivity, and $\pi_{\text{high}} s_{\text{high}} = 1$. Continuity follows from (GSL1.6) and the multiplier bound for E_+ . Equations (GSL6.10) and commutation of the multipliers prove equivariance, with no dropped factor of a .

The equivariant section is unique in a precise continuous linear category. Let $H : \mathcal{Q} \rightarrow \mathcal{Q}_-$ be a continuous linear map commuting with L . For $\lambda = \rho - 1$ and $m = m_\rho$, the first m jets of $(L_- - \lambda)^m H(x)$ vanish at λ . By

linearity and intertwining these are the same jets as those of $H((L - \lambda)^m x)$. But $\lambda \notin \mathcal{Z}$, and RZ3 constructs an everywhere-defined continuous inverse for $\lambda - L$. Hence $(L - \lambda)^m$ is onto $\mathcal{Q}$, so every jet of order less than m of every $H(x)$ vanishes at λ . This holds for all shifted zeros. The definition of $\mathcal{J}_-$ implies $H = 0$. The identical argument with $\lambda = \rho + 1 \notin \mathcal{Z}$ proves that every continuous linear L -intertwiner $\mathcal{Q} \rightarrow \mathcal{Q}_+$ is zero.

Consequently two continuous linear L -equivariant sections of π_+ must coincide: their difference lands in $\ker \pi_+ = \iota_- \mathcal{Q}_-$, and the preceding argument applies. The same proof gives uniqueness for the high-weight row. A continuous linear map commuting with every recovered positive rational T_a also commutes with the real group and its generator L . Indeed the positive rationals are dense in the positive reals after the full arithmetic is identified, RZ8.5 proves strong continuity and differentiability, and continuity of the map permits passage first to the real parameter limit and then to its derivative. The same uniqueness therefore holds for continuous linear sections respecting the full recovered arithmetic action. This proof uses the actual global resolvent and the separating jet family, not a claimed convergence of an infinite sum of finite projectors.

Different allowed cutoffs or corridor constants in GSL4 may produce different functions E , but their quotient inverses (GSL5.5) coincide. Their difference has zero jets at both divisors, and its product with any member of $\mathcal{B}$ belongs to $\mathcal{J}$. Equivalently both are the inverse of the canonical map Φ . The continuous quotient lift is therefore independent of those auxiliary choices.

GSL7. Finite-source projectors inside the actual global extension

The construction is compatible with the RZ finite-source projectors. For a point λ in the union divisor, with its full multiplicity m , put

$$A_\lambda^\cup(s) = \frac{G(s)}{(s - \lambda)^m}, \quad \mathcal{P}_\lambda^\cup F(s) = A_\lambda^\cup(s) \text{Tay}_{\lambda, < m} \left(\frac{F}{A_\lambda^\cup} \right) (s). \quad (\text{GSL7.1})$$

Division is entire and continuous on the relevant vanishing subspace by RZ2. The local Taylor quotient is taken only where $A_\lambda^\cup(\lambda) \neq 0$; its output is a global polynomial times that member of $\mathcal{B}$. Thus it is a continuous finite-rank map $\mathcal{B} \rightarrow \mathcal{B}$, annihilates $\mathcal{J}$, and has exactly the original jet at λ and zero jets at every other union point.

Let $P_\lambda^\cup$ be its induced quotient projector. Equality of all retained jets proves

$$P_\rho^\cup \iota_+ = \iota_+ P_\rho, \quad P_{\rho-1}^\cup \iota_+ = 0, \quad (\text{GSL7.2})$$

$$P_{\rho-1}^\cup \iota_- \bar{U} = \iota_- \bar{U} P_\rho, \quad P_\rho^\cup \iota_- \bar{U} = 0. \quad (\text{GSL7.3})$$

These are exact global quotient identities. They retain every Taylor derivative and factorial of RZ5–RZ6. No assertion about convergence of $\sum_\rho P_\rho$ is used.

The full union resolvent also has the direct representative

$$\mathcal{R}_\lambda^\cup F(s) = \frac{F(s) - G(s)F(\lambda)/G(\lambda)}{\lambda - s} \quad (\lambda \notin \mathcal{Z} \cup (\mathcal{Z} - 1)). \quad (\text{GSL7.4})$$

The numerator vanishes at $s = \lambda$, and the RZ2 division estimate applies with the full retained G . It preserves $\mathcal{J}$ and induces $(\lambda - L_\cup)^{-1}$, by the same two exact multiplication identities as RZ3.7. The pole orders and all Laurent coefficients are consequently the full multiplicities at the two divisors, with the projectors (GSL7.1). This verifies the global degree operator on the actual extension directly.

GSL8. The original Mellin source and a continuous source representative formula

The source is the original space

$$\mathcal{A} = \left\{ k \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(k) = \sup_{u>0} (u^N + u^{-N}) |(u \partial_u)^j k(u)| < \infty \right\}, \quad (\text{GSL8.1})$$

with its exact Mellin isomorphism

$$\mathcal{M}k(s) = \int_0^\infty k(u)u^{s-1/2}\frac{du}{u}. \quad (\text{GSL8.2})$$

Let $\mathcal{C} = \overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}$. S5 proves $\mathcal{M}\mathcal{C} = \mathcal{I}$, with its full source-image equality and topology. Multiplication by u is a topological automorphism of $\mathcal{A}$, and

$$\mathcal{M}(uk)(s) = \mathcal{M}k(s+1), \quad \mathcal{M}(u\mathcal{C}) = \mathcal{I}_-. \quad (\text{GSL8.3})$$

For its continuity, expand $(u\partial_u)^j(uk) = u \sum_{h=0}^j \binom{j}{h} (u\partial_u)^h k$ and use $u(u^N + u^{-N}) \leq u^{N+1} + u^{-(N+1)} + u^{N+1} + u^{-(N+1)}$; the harmless factor two supplies an explicit bound by finitely many $p_{N+1,h}$. The inverse uses u^{-1} and the corresponding signs. Thus (GSL0.1) is transported to

$$\mathcal{A}/(\mathcal{C} \cap u\mathcal{C}) \xrightarrow{\sim} (\mathcal{A}/\mathcal{C}) \oplus (\mathcal{A}/u\mathcal{C}) \quad (\text{GSL8.4})$$

by constructed continuous maps on the original source spaces.

For $F = \mathcal{M}k$, a representative of the first lifted quotient class has the explicit source function

$$k_+(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E(\tfrac{1}{2} + it) F(\tfrac{1}{2} + it) u^{-it} dt. \quad (\text{GSL8.5})$$

It lies in $\mathcal{A}$, because $EF \in \mathcal{B}$. The exact RZ4.7 estimate and (GSL5.1) give

$$p_{N,j}(k_+) \leq \frac{2(N+2)^j}{\pi} C'_{N+2} b_{N+2,j+2+d_{N+2}}(F). \quad (\text{GSL8.6})$$

The other summand uses $1-E$ with the same type of estimate. Their sum is k by Mellin inversion. Changing the representative of a quotient class changes its lifted representative by $\mathcal{C} \cap u\mathcal{C}$, exactly as (GSL5.2) proves. Thus (GSL8.5) supplies a continuous prequotient operator and the stated quotient lift; it is not an assertion of a separately chosen linear section $\mathcal{Q} \rightarrow \mathcal{B}$.

On the source the original T_a is $k(u) \mapsto a^{1/2}k(u/a)$, and the original degree operator is $1/2 - u\partial_u$, as proved in RZ1 and RZ8. Both factors are retained when transporting (GSL6.1)–(GSL6.2). The variable u belongs to the existing Mellin source; it is not a coordinate assigned to τ .

For the high-weight counterpart the exact source identities are

$$\mathcal{M}(u^{-1}k)(s) = \mathcal{M}k(s-1), \quad \mathcal{M}(u^{-1}\mathcal{C}) = \mathcal{I}_+, \quad (\text{GSL8.7})$$

and its full continuous source quotient is $\mathcal{A}/(\mathcal{C} \cap u^{-1}\mathcal{C})$. The source injection corresponding to j_{high} first multiplies its representative by u^{-1} and then applies the Mellin multiplier $1 - E_+$. The lifted original class is represented by

$$k_{\text{high}}(u) = \frac{1}{2\pi} \int_{\mathbb{R}} E_+(\tfrac{1}{2} + it) F(\tfrac{1}{2} + it) u^{-it} dt. \quad (\text{GSL8.8})$$

Multiplication by u^{-1} has the finite derivative expansion $(u\partial_u)^j(u^{-1}k) = u^{-1} \sum_{h=0}^j \binom{j}{h} (-1)^{j-h} (u\partial_u)^h k$, giving the same endpoint-seminorm control as above. For (GSL8.8), the translated strip estimate and RZ4.7 give

$$p_{N,j}(k_{\text{high}}) \leq \frac{2(N+2)^j}{\pi} C'_{N+3} b_{N+2,j+2+d_{N+3}}(F). \quad (\text{GSL8.9})$$

These are continuous maps on the full original source, with the stated quotient kernels. All shifted endpoint and trivial-zero contributions are the retained values in (GSL6.8)–(GSL6.9).

GSL9. Exact range, shifted ideals, and the original functional equation

The constructed CRT map has full range, with the continuous inverse (GSL5.5). The zero-germ data used by either quotient remain the actual image of $\mathcal{B}$ under its corresponding jet map. The proof never enlarges that image to arbitrary sequences and never substitutes a product topology for the quotient topology.

At a shifted center $\rho - 1$, the original $\zeta(s)$ is a unit while $\zeta(s + 1)$ has the original zero multiplicity. This follows from the full functional equation with

$$C(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \zeta(s) = \frac{C(1-s)}{C(s)}\zeta(1-s), \quad (\text{GSL9.1})$$

because $-1 < \operatorname{Re} s < 0$ maps into $\operatorname{Re}(1-s) > 1$, where the original Euler product is nonzero, and both displayed factors are holomorphic units on those open strips. The shifted original-zero algebra is exactly $\mathcal{O}_{\rho-1}/(\zeta(s+1))$, not $\mathcal{O}_{\rho-1}/(\zeta(s))$. The full multipliers (GSL1.3) and (GSL1.7) implement these precise divisors globally without erasing the exceptional values (GSL1.4)–(GSL1.5).

For every derivative order, the multiplier comparison remains the full product rule. Writing $K(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$ only to display its terms,

$$F_0^{(d)}(s) = \sum_{j=0}^d \binom{d}{j} K^{(d-j)}(s) \zeta^{(j)}(s), \quad F_1^{(d)}(s) = F_0^{(d)}(s+1). \quad (\text{GSL9.2})$$

The finite projector formulas in GSL7 use these full derivatives. The constant $1/8$, both endpoint factors, the power of π , and every Gamma derivative are not absorbed into a different original function.

The extension class of (GSL5.9) vanishes in the category of Fréchet spaces with the stated continuous arithmetic action, because (GSL5.8) is its explicit continuous equivariant section. This is an actual split extension of the original global quotient $\mathcal{Q}$. It is more than a finite-zero calculation, and more than an algebraic direct-sum splitting. It does not assert that every abstract extension of $\mathcal{Q}$ has a continuous section, nor identify this explicitly constructed extension with a geometric localization cross that has different specified maps.

GSL10. Branch transport and label retention

An admissible branch comparison at this stage preserves the recovered arithmetic, its original function ζ , and the specified Mellin receiver. Under that comparison the definitions of $\mathcal{I}$, $\mathcal{I}_-$, the full multipliers, translation and Φ agree. The inverse Ψ , being the proved inverse of this canonical map, therefore agrees as well. Each branch can carry out the construction using its own recovered counter and then apply its isomorphism; no pooled measure or comparison with an arithmetic-destroying candidate appears.

If the existing source record retains a separate support label ℓ , apply the maps to the coefficient in $\{\ell\} \times \mathcal{Q}$ and keep ℓ fixed. In particular a zero coefficient is still the labelled pair $(\ell, 0)$. This respects the S7 support repair and introduces no addition between support labels or at τ .

GSL11. Independent check

An independent mathematical derivation checked the entire construction mechanism: the 3, 4, 2 lower bound with $\delta = cT^{-6}$; the full Gamma reflection comparison and corridor bounds; the sign and factor of the Gaussian Cauchy kernel; the logarithmic-integrability estimate after real-coordinate integration; and the resulting continuous CRT inverse. It confirmed that the universal separator need only lie in the multiplier algebra, that every multiplicity is retained, and that the construction proves the full range without identifying the quotient with arbitrary jets. The complete written note is also made available to the parent for its independent reading.

The checker subsequently read GSL0–GSL11 and the receiving RZ3–RZ4 and RZ8 passages. It verified the all-strip integration formula, every corridor factor, the quotient-seminorm estimate, the global-resolvent uniqueness argument, and the full source inversion constant. Its two detected TeX spacing errors were corrected. It then separately checked the added higher-weight maps (GSL6.5)–(GSL6.11) and source formulas

(GSL8.7)–(GSL8.9), including the asymmetric strip indices in the last bound. No mathematical correction was needed; the positive-integer range in (GSL6.9) was made explicit.

The proved output is the explicit global shifted extension, its full continuous equivariant splitting, its source maps and its finite-jet compatibility. The checking coverage does not identify this extension with a different geometric localization obstruction or supply an RH conclusion.

Exact Schwartz summation image and the original-zeta cohomology

24 September 2026. Complete global calculation SSI0–SSI10. This strengthens the closure result S5 in GLOBAL_MELLIN_SYNTHESIS.md to an equality with the actual image. It uses the original zeta function throughout. All zero multiplicities, trivial-zero residues, endpoint conditions, the two signs in rational summation, and the two chart orientations are retained.

SSI0. Prerequisites and the precise operation

The governing user arguments and corrections are in argument_reconstruction/CORPUS_AND_OPERATION_RULE.md, LATEST_DIRECT_ARGUMENTS.md, CORRECTION_CHAINS.md, and the complete verbatim corpus. They require global reconstruction before arithmetic operations. The integers, their unit, primes, prime powers, complex coefficient field, Schwartz spaces and Mellin maps in the present calculation are the already reconstructed receivers in CANONICAL_GENERIC_RECONSTRUCTION.md, CG0–CG4, and GLOBAL_MELLIN_SYNTHESIS.md, S1–S6. No operation here constructs them by selecting a few primes. In particular, the Möbius inverse in SSI7 uses the entire reconstructed arithmetic.

The support notation remains $Z_0, Z_1, Z_2, \dots, \tau$. No addition at τ , numerical value of τ , distance to τ , or purity assumption is introduced. The theorem concerns an actual linear map on the coefficient spaces carried at the support. Its receiving cohomology maps are given in SSI8; no arithmetic weight separation is inferred from closedness.

The author source is Alain Connes and Caterina Consani, *Schemes over F_1 and zeta functions*, arXiv:0903.2024v3, original `announc3.tex`, §5, lines1376–1667, read in full. Source SHA256: 8254a72d27b082e0a3550e6be9398bae1c7e267303dc2b86a904703f58569fd4. The retained archive has SHA256 f09e4f8a917de991a84b508023598259e9d79694425ce5bd1ddd33d7c729e53f. This note proves a real even-Schwartz summation theorem and applies it to the explicitly defined coefficient sheaf CSL1. Its further identification with a particular convention for adelic coinvariants is a separate comparison; no missing coinvariant proof is supplied by a name.

The full dependency for the growth estimate is S2–S3 and its independent proof audit R1–R3 in GLOBAL_MELLIN_SYNTHESIS_REVIEW.md. The proof below uses the subquadratic division bound proved there, rather than a conjectural lower bound on zeta in the critical strip. Poisson summation and ordinary Fourier inversion use the same convention as that source. The standard Gamma recurrence and reflection identities are used with their factors displayed.

SSI1. Spaces, maps and the claimed equality

Let

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), \quad f(0) = 0, \quad \int_{\mathbb{R}} f(v) dv = 0\}. \quad (\text{SSI1.1})$$

It is a closed subspace of Schwartz space. Its seminorms may be taken to be

$$q_{r,j}(f) = \sup_{v \in \mathbb{R}} (1 + |v|)^r |f^{(j)}(v)|, \quad r, j \geq 0. \quad (\text{SSI1.2})$$

Let

$$A = \{b \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(b) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for all } N, j \geq 0\}, \quad (\text{SSI1.3})$$

and let $\mathcal{B}$ consist of entire functions with seminorms

$$b_{A_0,M}(F) = \sup_{|\sigma| \leq A_0, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty \quad (A_0, M \geq 0). \quad (\text{SSI1.4})$$

Integer indices suffice in these seminorms. The raw Mellin map and its inverse are

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}, \quad b(e^x) = \frac{e^{-cx}}{2\pi} \int_{\mathbb{R}} (\mathcal{M}_0 b)(c + it) e^{-itx} dt. \quad (\text{SSI1.5})$$

They are mutually inverse continuous maps $A \leftrightarrow B$, for any real c . This is S1 with its exact change of exponent. Specifically, for its centered map $\mathcal{M}$ and $\mathcal{T}k(u) = 2u^{-1/2}k(u)$, one has $\mathcal{M}_0\mathcal{T} = 2\mathcal{M}$. Multiplication by these displayed powers preserves all the spaces with continuous inverse.

For every actual nontrivial zero ρ of the original ζ , let m_ρ be its original multiplicity. Define the closed subspace

$$I = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \text{ and } 0 \leq j < m_\rho\}. \quad (\text{SSI1.6})$$

Jet evaluation is continuous by Cauchy's integral formula on a fixed small circle, so this intersection is closed. Define

$$\Sigma : S \longrightarrow A, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad J = \Sigma S. \quad (\text{SSI1.7})$$

The factor 2 is the contribution of both signs in the even sector. The theorem proved below is the topological isomorphism

$$\boxed{\mathcal{M}_0 \Sigma : S \xrightarrow{\sim} I.} \quad \boxed{J = \mathcal{M}_0^{-1} I = \overline{J}^A.} \quad (\text{SSI1.8})$$

The inverse is actual Schwartz inversion, with no limiting class substituted for a source function.

SSI2. Continuity, inclusion, injectivity and full meromorphic division

On $u \geq 1$, every logarithmic derivative of the series in (SSI1.7) is bounded by a finite sum of Schwartz seminorms times $u^{-L} \sum n^{-L}$, for any sufficiently large integer $L > 1$. For $0 < u \leq 1$, the Fourier convention

$$\hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv \quad (\text{SSI2.1})$$

and Poisson summation give the full identity

$$\Sigma \hat{f}(u) = u^{-1} \Sigma f(u^{-1}) + u^{-1} f(0) - \int_{\mathbb{R}} f(v) dv. \quad (\text{SSI2.2})$$

For $f \in S$, the last two terms vanish by the two stated moment conditions; $\hat{f} \in S$. This proves the required estimates at zero and the continuity $\Sigma : S \rightarrow A$.

For $\Re s > 1$, absolute integration and summation give

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{SSI2.3})$$

The positive-axis Mellin integral is holomorphic on $\Re s > -2$, since evenness and $f(0) = 0$ give $f(v) = O(v^2)$. It vanishes at $s = 1$, because $\int_{\mathbb{R}} f = 0$. Thus (SSI2.3) continues through the zeta pole at 1 and proves full-order vanishing at every nontrivial zero. This proves $\mathcal{M}_0 \Sigma S \subset I$.

If $\Sigma f = 0$, the nonvanishing Euler product for ζ on $\Re s > 1$ implies that the positive-axis Mellin transform is zero on that half-plane. Fourier uniqueness on a vertical line gives $f = 0$ on $\mathbb{R}_{>0}$, and evenness and continuity give $f = 0$ on all of $\mathbb{R}$. Hence Σ is injective.

Conversely, let $F \in I$ and put

$$H(s) = \frac{F(s)}{2\zeta(s)}. \quad (\text{SSI2.4})$$

At a nontrivial zero ρ , write $\zeta(s) = (s-\rho)^{m_\rho} u(s)$, with $u(\rho) \neq 0$, and $F(s) = (s-\rho)^{m_\rho} v(s)$. Then $H = v/(2u)$ extends holomorphically there. No simplicity or separation of these zeros is assumed. The only possible poles of H are the simple trivial-zero poles

$$\text{Res}_{s=-2r} H(s) = c_r(F) := \frac{F(-2r)}{2\zeta'(-2r)}, \quad r \geq 1. \quad (\text{SSI2.5})$$

The original functional equation proves that these zeros are simple: its sine factor has a simple zero at $-2r$, whereas its Gamma factor and $\zeta(1+2r)$ are finite and nonzero. At zero and one the exact values are

$$H(0) = -F(0), \quad H(1) = 0, \quad H'(1) = \frac{F(1)}{2}. \quad (\text{SSI2.6})$$

Here $\zeta(0) = -1/2$ and the residue of ζ at 1 is 1. The endpoint pole and each trivial zero have been used, not discarded.

SSI3. Growth and exact estimates on the two vertical edges

Keep the auxiliary source transform of S2 with all factors:

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{SSI3.1})$$

It is the function denoted F_0 in S2. Its zeros are exactly the nontrivial zeros with their multiplicities, and $F_*(0) = F_*(1) = 1/8$. It is used only for the proved division estimate. S3 gives

$$\sup_{|\sigma| \leq A_0} |F/F_*(\sigma + it)| \leq C_{A_0, F} \exp\{C_{A_0} (|t| + 2)^{3/2} \log(|t| + 2)\}. \quad (\text{SSI3.2})$$

Retaining the comparison rather than replacing ζ , one has

$$H(s) = \frac{s(s-1)}{16} \pi^{-s/2} \Gamma(s/2) \frac{F(s)}{F_*(s)}. \quad (\text{SSI3.3})$$

On every bounded real strip and $|t| \geq 1$, the Gamma factor has an upper bound by a polynomial in $1 + |t|$. Indeed choose an integer L for which $\sigma/2 + L \geq 1$ throughout the strip. The recurrence and Euler integral give

$$|\Gamma(s/2)| = \frac{|\Gamma(s/2 + L)|}{\prod_{j=0}^{L-1} |s/2 + j|} \leq \frac{\Gamma(\sigma/2 + L)}{(|t|/2)^L}. \quad (\text{SSI3.4})$$

The numerator is bounded on the compact interval of its real arguments. Equations (SSI3.2)–(SSI3.4) therefore give subquadratic growth for H , away from its finitely many poles in each strip. This is sufficient for the maximum-principle argument; no conjectural reciprocal-zeta estimate is used.

Choose an integer $N \geq 0$, let $a = -2N - 1$, and choose a real $b \geq 2$. On the right edge, the absolutely convergent reciprocal Euler series yields

$$|\zeta(b + it)^{-1}| \leq \sum_{n \geq 1} |\mu(n)| n^{-b} \leq \zeta(b). \quad (\text{SSI3.5})$$

Here μ is the Möbius function of the complete reconstructed integers; its use does not select a finite set of primes.

On the left edge retain the functional equation in its full multiplier form

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)} = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s). \quad (\text{SSI3.6})$$

Writing $y = t/2$, Gamma recurrence and reflection give the exact identity

$$|\chi(-2N - 1 + it)|^2 = \pi^{-4N-3} y \coth(\pi y) \prod_{k=1}^N (k^2 + y^2) \prod_{j=0}^N ((j + \frac{1}{2})^2 + y^2). \quad (\text{SSI3.7})$$

The empty product is 1 and $y \coth(\pi y)$ has the value $1/\pi$ at zero. To verify (SSI3.7), apply the recurrence to $\Gamma(N+1-iy)$, and to $\Gamma(-N-1/2+iy)$ up to $\Gamma(1/2+iy)$. The remaining exact quotient of squared moduli is

$$\frac{|\Gamma(1-iy)|^2}{|\Gamma(1/2+iy)|^2} = \frac{\pi y / \sinh(\pi y)}{\pi / \cosh(\pi y)} = y \coth(\pi y). \quad (\text{SSI3.8})$$

These two modulus identities follow from $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ and $\Gamma(1+z) = z\Gamma(z)$; their values at zero are their continuous limits. All factors on the right of (SSI3.7) are positive for real y , and their growth at infinity gives constants $c_N, C_N > 0$ with

$$c_N(1+|t|)^{4N+3} \leq |\chi(a+it)|^2 \leq C_N(1+|t|)^{4N+3}. \quad (\text{SSI3.9})$$

For completeness, this follows on $|t| \geq 1$ by bounding each quadratic between positive multiples of $1+t^2$, and bounding $y \coth(\pi y)$ between positive multiples of $1+|y|$. On $|t| \leq 1$, positivity and continuity give the same inequalities after changing the constants. Since $\Re(1-a-it) = 2N+2 > 1$, (SSI3.5) applies to $\zeta(1-a-it)$. Thus $1/\zeta$ is polynomially bounded on the left edge too, with no zero on either edge.

It follows that H decreases faster than every inverse power on each edge. Every such bound is bounded by a fixed constant times a finite seminorm of F in $\mathcal{B}$. The constants can depend on N, b and the requested power, but not on F .

SSI4. The maximum principle removes the Gaussian approximation

Define

$$P_N(s) = \prod_{r=1}^N (s+2r), \quad P_0 = 1, \quad V_N = P_N H. \quad (\text{SSI4.1})$$

This product cancels exactly the possible poles inside the strip $a \leq \Re s \leq b$. Hence V_N is holomorphic on a neighborhood of that closed strip. For an integer $M \geq 0$ set $\kappa = 2N+2$ and

$$W(s) = (s+\kappa)^M V_N(s). \quad (\text{SSI4.2})$$

The estimates in SSI3 give, on the two vertical edges,

$$|W(s)| \leq C_{N,b,M} b_{A_0, M+N}(F), \quad A_0 \geq \max(2N+1, b). \quad (\text{SSI4.3})$$

Indeed the polynomial factors have total degree $M+N$, and the reciprocal-zeta bounds on each edge are bounded, in addition to having the more precise polynomial behavior already computed.

For $\varepsilon > 0$, apply the maximum-modulus principle to $e^{\varepsilon s^2} W(s)$ on the rectangle with vertical edges a, b and horizontal edges $\Im s = \pm T$. On the horizontal edges, (SSI3.2)–(SSI3.4) give a bound of the form

$$C_{F,N,b,M,\varepsilon} (1+T)^L \exp\{-\varepsilon T^2 + C_{N,b}(T+2)^{3/2} \log(T+2)\}, \quad (\text{SSI4.4})$$

which tends to zero as $T \rightarrow \infty$, uniformly in the real coordinate. On the vertical edges, (SSI4.3) gives

$$|e^{\varepsilon s^2} W(s)| \leq e^{\varepsilon \max(a^2, b^2)} C_{N,b,M} b_{A_0, M+N}(F). \quad (\text{SSI4.5})$$

Fix any interior s , let $T \rightarrow \infty$, and then let $\varepsilon \downarrow 0$. The resulting estimate is

$$|W(s)| \leq C_{N,b,M} b_{A_0, M+N}(F). \quad (\text{SSI4.6})$$

This also holds on the edges by continuity. Since $\Re(s+\kappa) \geq 1$ throughout the strip,

$$\boxed{\sup_{a \leq \sigma \leq b, t \in \mathbb{R}} (1+|t|)^M |P_N(\sigma+it)H(\sigma+it)| \leq C'_{N,b,M} b_{A_0, M+N}(F).} \quad (\text{SSI4.7})$$

The growth constant depending on F in (SSI4.4) is used only to remove the horizontal edges for a fixed F . It does not enter the final bound (SSI4.7). Thus (SSI4.7) proves continuous seminorm control as well as existence of rapid decrease. There is no limit of source classes left in the assertion.

On a vertical line avoiding the finitely many trivial poles, division by P_N gives corresponding rapid bounds for H . On horizontal lines with $|t| \geq 1$, each $|s+2r| \geq |t|$, so the same bounds justify all contour shifts below. Cauchy's integral formula on small disks away from the poles also gives rapid bounds for every holomorphic derivative of H .

SSI5. Actual inverse Mellin and all even Taylor coefficients

Define for $x > 0$

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(2+it)x^{-2-it} dt. \quad (\text{SSI5.1})$$

This integral is absolutely convergent and can be differentiated any finite number of times. On every compact interval of $x > 0$, the integrable majorant follows from (SSI4.7). The exact derivative factor is

$$\frac{d^j}{dx^j} x^{-s} = (-1)^j (s)_j x^{-s-j}, \quad (s)_j = s(s+1)\cdots(s+j-1), \quad (s)_0 = 1. \quad (\text{SSI5.2})$$

Contour shifting to any $c \geq 2$ crosses no pole and gives

$$f_F^{(j)}(x) = \frac{(-1)^j}{2\pi} \int_{\mathbb{R}} (c+it)_j H(c+it)x^{-c-it-j} dt. \quad (\text{SSI5.3})$$

The horizontal contributions tend to zero by SSI4. Taking c arbitrarily large proves faster-than-any-power decay of every derivative at infinity.

Shifting the contour to $a = -2N - 1$ crosses precisely the poles in (SSI2.5) for $1 \leq r \leq N$, with their positive residue sign. Thus

$$f_F(x) = \sum_{r=1}^N c_r(F)x^{2r} + \frac{1}{2\pi} \int_{\mathbb{R}} H(a+it)x^{-a-it} dt. \quad (\text{SSI5.4})$$

For every integer $j \geq 0$, the remainder after differentiating this equality j times is bounded, on $0 < x \leq 1$, by

$$\frac{x^{2N+1-j}}{2\pi} \int_{\mathbb{R}} |(a+it)_j H(a+it)| dt. \quad (\text{SSI5.5})$$

This integral is finite and bounded by a finite seminorm of F . For any prescribed derivative order choose N large enough that $2N+1-j > 0$. Each derivative then has a finite limit as $x \downarrow 0$, equal to the derivative at zero of the even polynomial in (SSI5.4). These limits consistently extend f_F to a smooth function on the half-line: the fundamental theorem of calculus applied to successive derivatives, followed by their continuous limits, proves that the extended derivative is indeed the derivative of the extended preceding function. Extending by $f_F(-x) = f_F(x)$ is smooth because every odd one-sided derivative is zero. In particular,

$$f_F(0) = 0, \quad f_F^{(2r-1)}(0) = 0, \quad \boxed{\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)}} \quad (r \geq 1). \quad (\text{SSI5.6})$$

Equations (SSI5.3) and (SSI5.4) prove $f_F \in \mathcal{S}(\mathbb{R})$, not merely a function on the punctured positive axis.

On the line $\Re s = 2$, $t \mapsto H(2+it)$ is Schwartz by SSI4 and Cauchy's formula. Ordinary Fourier inversion applied to $x = e^y$ in (SSI5.1) therefore gives

$$\int_0^\infty f_F(x)x^s \frac{dx}{x} = H(s) \quad (\Re s = 2). \quad (\text{SSI5.7})$$

Both sides are holomorphic on $\Re s > -2$, since $f_F(x) = O(x^2)$ at zero, and have no poles there. The identity theorem extends (SSI5.7) throughout that half-plane. At $s = 1$, (SSI2.6) gives

$$\int_{\mathbb{R}} f_F(x) dx = 2H(1) = 0. \quad (\text{SSI5.8})$$

Thus $f_F \in S$. By (SSI2.3), $\mathcal{M}_0 \Sigma f_F = 2\zeta H = F$ on $\Re s > 1$; both sides are entire, so equality holds on the whole plane. This proves the surjectivity in (SSI1.8).

SSI6. Continuity of the inverse and endpoint information

For any Schwartz seminorm $q_{r,j}$, split its domain into $|x| \geq 1$ and $|x| \leq 1$. On the first part use (SSI5.3) with $c \geq \max(2, r+2)$, and a rapid bound for $H(c+it)$ of order greater than $j+1$. It yields a constant times one finite seminorm of F . On the second part choose $N \geq j+1$ in (SSI5.4). The finitely many polynomial coefficients satisfy

$$|c_k(F)| \leq \frac{b_{A_0,0}(F)}{2|\zeta'(-2k)|} \quad (A_0 \geq 2N), \quad (\text{SSI6.1})$$

and the remainder is bounded by (SSI5.5) and (SSI4.7). Consequently, for suitable finite indices A_0, M depending on r, j ,

$$q_{r,j}(f_F) \leq C_{r,j} b_{A_0,M}(F). \quad (\text{SSI6.2})$$

This proves continuity of the inverse. Together with SSI2 and SSI5 it proves the topological isomorphism (SSI1.8). Because I is closed, the image J is closed in A .

The endpoint information is more than the two vanishing conditions. Differentiating (SSI5.7), with dominated convergence, and using (SSI2.6) gives

$$\int_0^\infty f_F(x) \frac{dx}{x} = -F(0), \quad \int_0^\infty f_F(x) \log x \, dx = \frac{F(1)}{2}. \quad (\text{SSI6.3})$$

All Taylor coefficients at the original trivial zeros are retained in (SSI5.6). Higher coefficients at these points are obtained by Laurent division of the full original $F/(2\zeta)$; no substitution by the completed function is required. In particular, the Gamma cancellation in (SSI3.1) did not erase the information that reconstructs the source jet.

In the centered convention of S1–S6, $\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu)$. Hence $\Sigma = \mathcal{T}\mathcal{E}$, and the strengthened centered statement is

$$\mathcal{M}\mathcal{E} : S \xrightarrow{\sim} I, \quad \int_0^\infty f_F^{\text{centered}}(x) x^s \frac{dx}{x} = \frac{F(s)}{\zeta(s)}, \quad f_F^{\text{centered}} = 2f_F. \quad (\text{SSI6.4})$$

The original Gaussian approximants in S4–S5 therefore converge in Schwartz space to this actual source function: their transforms converge in I , and the continuous inverse just proved carries that convergence to every Schwartz seminorm. This strengthens the older closure statement without invalidating its proof.

SSI7. An inverse on the counting side and the full Fourier map

For $b \in J$, the exact inverse can also be written directly in the original counting variable:

$$\boxed{\Sigma^{-1}b(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) b(nx)} \quad (x > 0). \quad (\text{SSI7.1})$$

For fixed $x > 0$, the sum and all its derivatives converge absolutely and locally uniformly, using the arbitrary decay of b at infinity. To prove the identity, write $b = \Sigma f$ by SSI5. Then, for any $L > 1$,

$$\sum_{n,m \geq 1} |f(mnx)| \leq C_{x,L} \sum_{n,m \geq 1} (mn)^{-L} < \infty. \quad (\text{SSI7.2})$$

The same bound with a sufficiently higher decay order applies to derivatives. Thus sums can be rearranged, and

$$\frac{1}{2} \sum_n \mu(n) b(nx) = \sum_{k \geq 1} f(kx) \sum_{n|k} \mu(n) = f(x). \quad (\text{SSI7.3})$$

The last finite divisor sum is 1 at $k = 1$ and 0 otherwise, as follows by expanding $\prod_{p|k} (1-1)$. All prime divisors are those already present in the complete arithmetic. This calculation does not posit independent meanings for a few labels before that reconstruction. Formula (SSI7.1) is not asserted to be termwise differentiable at $x = 0$; its smooth extension and its complete jet there are supplied by (SSI5.4)–(SSI5.6).

Let $Rb(u) = u^{-1}b(u^{-1})$ and $jF(s) = F(1-s)$. The functional equation, with its nonzero multiplier on the critical strip, pairs all actual nontrivial-zero jets, so $jI = I$. The exact inverse intertwines the full reflection with Fourier transform:

$$\Sigma^{-1}Rb = \widehat{\Sigma^{-1}b}, \quad f_{jF} = \widehat{f_F}. \quad (\text{SSI7.4})$$

Proof: (SSI2.2) gives $R\Sigma f = \Sigma \widehat{f}$ on S ; injectivity proves the first equality. Raw Mellin gives $\mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1-s)$, proving the second. Equivalently their positive-axis Mellin transforms satisfy

$$\int_0^\infty \widehat{f_F}(x) x^s \frac{dx}{x} = \frac{F(1-s)}{2\zeta(s)} = \frac{\zeta(1-s)}{\zeta(s)} H(1-s), \quad (\text{SSI7.5})$$

as meromorphic identities. The complete functional multiplier is retained in the last quotient and in (SSI3.6); it has not been set equal to 1.

For $a > 0$, let $T_a b(u) = b(u/a)$ and $D_a f(x) = f(x/a)$. Direct summation gives $\Sigma D_a = T_a \Sigma$, and direct Mellin transformation gives multiplication by a^s . Therefore the inverse is continuously equivariant for every positive dilation. Its derivative $-x\partial_x$ on S corresponds to multiplication by s on I ; this follows by differentiating in $\log a$, or by integration by parts with the endpoint decay already proved. Every multiplicity and nilpotent zero-jet action in the quotient consequently remains unchanged.

SSI8. Actual cohomology, strict support maps and the vanishing closure quotient

Apply the theorem to the sheaf explicitly constructed in CC_SUPPORTED_LOCALIZATION_AND_WEIGHT_MAPS.md, CSL1–CSL9. Its underlying space has opens $\emptyset, U, U_+, U_-, X$, with $U = \{\eta\}$, $U_\pm = \{x_\pm, \eta\}$; its chart spaces are $V_\pm = S \oplus \mathbb{C}^2$, overlap A , and restrictions

$$r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h. \quad (\text{SSI8.1})$$

Each map is a continuous map onto J , with continuous right inverse into its Schwartz summand, and kernel exactly its two endpoint lines. Equation (SSI7.4) proves that the two images are the same closed subspace, with their Fourier comparison retained.

The ordinary Čech complex is

$$V_+ \oplus V_- \xrightarrow{r_+ - r_-} A. \quad (\text{SSI8.2})$$

Its differential has closed image J ; the inverse of Σ gives a continuous right inverse from that image to the first chart summand. Thus this differential is strict onto its image. Ordinary sheaf cohomology, with its quotient topology, is already Hausdorff, and the exact isomorphism is

$$\boxed{H^1(X, \Omega) = A/J \xrightarrow{\sim} \mathcal{B}/I, \quad [b] \mapsto [\mathcal{M}_0 b].} \quad (\text{SSI8.3})$$

In the notation $C = \overline{J}$, $Q_{\text{alg}} = A/J$, $Q_H = A/C$, $N = C/J$, the previously retained comparison kernel is now calculated:

$$\boxed{C = J, \quad N = 0, \quad Q_{\text{alg}} \xrightarrow{\sim} Q_H.} \quad (\text{SSI8.4})$$

This is an unconditional vanishing of this particular closure quotient. It is not identified with Deligne's lifting-obstruction group.

All supported rows in CSL therefore act on the actual ordinary cohomology, with no additional Hausdorffification step. For one closed point, $H^1_{\{x_+\}}(X, \Omega) \rightarrow H^1(X, \Omega)$ is the identity on A/J ; the other chart has the retained orientation minus sign. For both closed points the exact row is

$$0 \longrightarrow Q \xrightarrow{q \mapsto (q, q)} Q \oplus Q \xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q \longrightarrow 0, \quad Q = \mathcal{B}/I. \quad (\text{SSI8.5})$$

The quotient of the generic restriction by its kernel J is the first Q . The section $q \mapsto \frac{1}{2}(q, -q)$ is continuous and dilation-equivariant. It is also mirror-equivariant with j on the first Q , $(q_+, q_-) \mapsto (jq_-, jq_+)$ in the

middle, and $-j$ on the last Q . Substituting into the difference verifies both the section identity and the orientation sign. The scalar $1/2$ belongs to the already reconstructed coefficient field; it is not an operation on τ .

The full zeroth cohomology remains the Fourier graph

$$\{((\hat{h}, c_0, c_1), (h, d_0, d_1)) : h \in S, c_0, c_1, d_0, d_1 \in \mathbb{C}\}. \quad (\text{SSI8.6})$$

Its four endpoint lines retain the unshifted dilation characters $1, a, a, 1$. None is removed by the closed-image theorem. The proof affects the topological comparison kernel in (SSI8.4), not the original nontrivial-zero jets of Q .

SSI9. Exact scope of the new vanishing result

Every $F \in I$ now has a unique actual source function, given by (SSI5.1), with its complete expansion (SSI5.4), continuity (SSI6.2), counting inverse (SSI7.1), and Fourier comparison (SSI7.4). This resolves the old image-versus-closure question globally, for every zero and every multiplicity at once. It strengthens S5 and allows the algebraic cohomology and supported maps of CSL to receive the exact quotient $\mathcal{B}/I$ directly.

The target requested from Deligne's theorem remains the weight-separated vanishing for the actual τ -supported geometric lifting map. In Deligne §3.6, the obstruction receiver is $H^{2N-i-1}(X_s, E)^\vee(-N)$, and the inertia term is $H^{i-1}(X_{\bar{\eta}}, E)_I(-1)$. Equations (SSI8.3)–(SSI8.5) construct the analytic receiver and its support maps; they do not insert an identification with those two geometric terms. The present proof does not move any original zero or show $\Re\rho = 1/2$. Its proved new vanishing is precisely $\overline{\Sigma S}/\Sigma S = 0$, with the displayed inverse and preserved endpoint data.

SSI10. The adelic sector and the alternative source's retained prime directions

The linked source comparison CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md, CS0–CS10 including CS2A, has now been read in full. It proves the adelic comparison that SSI0 kept separate. Here are its receiving maps and the additional Fourier and dilation calculation, retaining the source-domain distinction.

The full companion CC_ENDPOINT_DEFECT_ACTIONS.md, CSB0–CSB10, has also been read. Its CSB1–CSB9 independently proves these endpoint actions and supplies an exactly Fourier-equivariant Schwartz representative. CSB10 proves the coordinate comparison to the prior FR15–FR25 calculation, whose earlier public source is Prime boundary classes in the actual adelic theta receiver, PM4–PM15. That public source identity was verified; a new complete reading of its earlier base conventions is not claimed. The prime-boundary component is retained programme mathematics. SSI10 receives and checks its maps without claiming discovery of that component.

Put $K = \widehat{\mathbb{Z}}^\times$. On the moment-null adelic Schwartz space let D be the Hausdorff quotient by rational differences generated by arbitrary adelic Schwartz functions, and let D_{null} instead restrict the generators themselves to moment-null functions. Both kinds of difference are moment-null because the adelic modulus of a rational number is 1. CS2 proves the continuous inverse maps

$$j : S \longrightarrow D^K, \quad f \longmapsto [f \otimes 1_{\widehat{\mathbb{Z}}}], \quad \bar{B} \left[\sum_d f_d \otimes 1_{d\widehat{\mathbb{Z}}} \right] (v) = \sum_d f_d^{\text{ev}}(dv). \quad (\text{SSI10.1})$$

The radial-ball expansion is finite after K -averaging. Its uniqueness and the preservation of both endpoint functionals are proved in CS2. Rational scaling takes the ball $d\widehat{\mathbb{Z}}$ to $(d/|q|)\widehat{\mathbb{Z}}$, while scaling the real function by $|q|$; the two changes cancel in $\bar{B}$. This proves that the formula descends to the quotient. The difference between a ball tensor and its displayed real representative is a sum of two full rational differences, as proved in CS2.7a. Thus $\bar{B}j = 1$ and $j\bar{B} = P_K$, with both inverses continuous.

On the overlap, summation of jf over $\mathbb{Q}^\times$ is exactly Σf : a rational number in $\widehat{\mathbb{Z}}$ is an integer, and both signs contribute. Therefore the actual full-difference CC source sector has the continuous isomorphism

$$D^K \xrightarrow{\bar{B}} S \xrightarrow{\Sigma} J \xrightarrow{\mathcal{M}_0} I. \quad (\text{SSI10.2})$$

The middle isomorphism is SSI1–SSI6; thus this statement now concerns the proved adelic source sector as well as its explicit real presentation.

For the second source convention CS2A constructs

$$V = \bigoplus_{p \text{ prime}} \mathbb{C}^2, \quad D_{\text{null}} \xrightarrow{\sim} D \oplus V, \quad [F]_{\text{null}} \mapsto ([F], \mathcal{L}F), \quad (\text{SSI10.3})$$

with its continuous inverse and locally convex direct-sum topology. On a ball tensor the components are

$$\ell_p(f \otimes 1_{d\widehat{z}}) = v_p(d) \left(f(0), d^{-1} \int_{\mathbb{R}} f(v) dv \right), \quad \mathcal{L} = (\ell_p)_p. \quad (\text{SSI10.4})$$

For nonradial inputs average over K first. All operations in (SSI10.3)–(SSI10.4) concern the complete reconstructed prime system. Each test function has a finite radial-ball expansion, so its image has finite prime support; this property of individual sections is not a truncation of the prime system. CS2A proves that the extra V is precisely the kernel of the source comparison, not an unspecified possible defect.

Let $C(a, b) = (b, a)$ on each $\mathbb{C}^2$. Under the self-dual adelic Fourier convention, the finite Fourier transform satisfies

$$\widehat{1_{d\widehat{z}}} = d^{-1} 1_{d^{-1}\widehat{z}}. \quad (\text{SSI10.5})$$

Indeed the annihilator of the ball is the displayed reciprocal ball, and its Haar volume is d^{-1} . For a real Schwartz factor the two Fourier endpoint identities are $\widehat{f}(0) = \int f$ and $\int \widehat{f} = f(0)$. Substitute all these values into (SSI10.4). Since $v_p(d^{-1}) = -v_p(d)$, one obtains

$$\ell_p(\widehat{F}) = -C\ell_p(F), \quad \boxed{\mathcal{L}\widehat{F} = -C\mathcal{L}F}. \quad (\text{SSI10.6})$$

The same identity holds before averaging: Fourier conjugates the action of a finite unit to its inverse, and Haar averaging is unchanged by inversion. Linearity and continuity then extend the calculation from ball tensors to the stated source. Thus (SSI10.3) intertwines Fourier transform with $(\widehat{}_D, -C)$. The minus sign records the inverted finite valuation; it cannot be suppressed by only retaining the two unordered endpoint coordinates.

For positive real dilation, define

$$\rho_+(a)F(v, x_f) = F(v/a, x_f), \quad \rho_-(a)F(v, x_f) = aF(av, x_f). \quad (\text{SSI10.7})$$

Evaluation at zero and change of variables in the real integral give

$$\mathcal{L}\rho_+(a) = \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix} \mathcal{L}, \quad \mathcal{L}\rho_-(a) = \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \mathcal{L}. \quad (\text{SSI10.8})$$

These formulas hold in every prime summand of V . Consequently the exact splitting (SSI10.3) is equivariant for these displayed actions, without requiring its chosen endpoint section to be equivariant. This also verifies Fourier interchange of the two actions, with the retained $-C$.

For the sheaves in this note, take the K -fixed sector explicitly. The group K acts trivially on V , and (SSI10.3) is K -equivariant, so it restricts to the topological isomorphism $D_{\text{null}}^K \cong D^K \oplus V$. Define Ω_{null} with chart source D_{null}^K , its two endpoint lines on each chart, and the same K -fixed overlap A as Ω . This is not the sheaf containing every nontrivial finite-unit character. Each extra V class has the representative $\mathcal{T}(v)$ constructed in CS2A.7, a finite sum of full rational differences. Its sum vanishes by reindexing. The relation closure also sums to zero by continuity, so the whole V summand restricts to zero on the overlap. Let $i_{\pm} : \{x_{\pm}\} \hookrightarrow X$. The source comparison therefore gives the actual sheaf decomposition

$$\Omega_{\text{null}} \cong \Omega \oplus (i_+)_* V \oplus (i_-)_* V. \quad (\text{SSI10.9})$$

This is verified on both charts, the overlap and their restrictions; hence it is an isomorphism of sheaves. In the two-term Čech complex it adds $V \oplus V$ in degree 0 with zero differential, and adds nothing in degree 1. It follows directly that

$$H^0(X, \Omega_{\text{null}}) \cong H^0(X, \Omega) \oplus V \oplus V, \quad \boxed{H^1(X, \Omega_{\text{null}}) \cong H^1(X, \Omega) \cong \mathcal{B}/I}. \quad (\text{SSI10.10})$$

The same calculation retains one additional V at the corresponding single closed support. The chart-exchanging sheaf mirror swaps the two V copies; this is a different typed map from the adelic Fourier map $-C$ within a source copy in (SSI10.6). Formula (SSI10.8) gives their separate endpoint characters. Thus the extra prime directions are preserved, while the degree-one original-zeta quotient remains exactly the same. This identifies the consequence of the source-domain choice without confusing it with the vanished closure quotient or the still-unproved geometric weight separation.

Exact Schwartz image of the raw Connes–Consani summation map

Independent mathematical derivation and check, 24 September 2026.

Conclusion. For the spaces specified below, the actual image of the raw summation map is closed. Its image equals the complete nontrivial-zeta-zero jet ideal under the raw Mellin transform. Consequently no closure needs to be added to this particular image. The proof concerns the explicitly specified even real Schwartz source; it does not identify that source with an adelic coinvariant space under an unspecified convention, and it does not prove a purity or RH statement.

The complete files `GLOBAL_MELLIN_SYNTHESIS.md` and `GLOBAL_MELLIN_SYNTHESIS_REVIEW.md` were read for this check. The dependencies used are S1’s Mellin isomorphism, S2’s full auxiliary kernel and genus-one product, and the proved S3 strip division estimate, independently checked in R1–R3. The construction below removes the Gaussian approximation used in S4–S5. No zero spacing, zero simplicity, RH, source addition at tau, or identification of a support label with a coefficient is assumed. All arithmetic here is in the already reconstructed arithmetic coefficient space.

ESI0. Spaces, conventions, and the exact statement

Let

$$\mathcal{S} = \{f \in \mathcal{S}(\mathbb{R}) : f(-x) = f(x), f(0) = 0, \int_{\mathbb{R}} f(x) dx = 0\}. \quad (\text{ESI0.1})$$

Its topology is induced by the Schwartz seminorms

$$q_{R,j}(f) = \sup_{x \in \mathbb{R}} (1 + |x|)^R |f^{(j)}(x)|, \quad R, j \in \mathbb{Z}_{\geq 0}. \quad (\text{ESI0.2})$$

Let

$$\mathcal{A} = \left\{ b \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for all } N, j \geq 0 \right\}, \quad (\text{ESI0.3})$$

and let $\mathcal{B}$ consist of entire functions with seminorms

$$\beta_{A,M}(F) = \sup_{|\sigma| \leq A, t \in \mathbb{R}} (1 + |t|)^M |F(\sigma + it)| < \infty. \quad (\text{ESI0.4})$$

The raw Mellin map and raw summation map are

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Sigma f(u) = 2 \sum_{n=1}^\infty f(nu). \quad (\text{ESI0.5})$$

The comparison with S1’s centered transform is the exact identity $\mathcal{M}_0 b(s) = \mathcal{M} b(s + \frac{1}{2})$. Thus S1 gives a topological isomorphism $\mathcal{M}_0 : \mathcal{A} \rightarrow \mathcal{B}$, with inverse, for every real c ,

$$b(u) = \frac{1}{2\pi} \int_{\mathbb{R}} F(c + it) u^{-c-it} dt. \quad (\text{ESI0.6})$$

The shift in this comparison changes the transform convention; it does not change the original zeta variable in the formulas below.

Let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every nontrivial zero } \rho \text{ of } \zeta, 0 \leq j < m_\rho\}. \quad (\text{ESI0.7})$$

Every zero is included with its actual multiplicity. The closedness of $\mathcal{J}$ follows from continuity of holomorphic jet evaluations: Cauchy’s formula on any fixed circle about the evaluation point bounds a jet by a finite $\beta_{A,0}$ seminorm.

The exact result is the topological linear isomorphism

$$\boxed{\Sigma : \mathcal{S} \xrightarrow{\sim} \mathcal{M}_0^{-1}(\mathcal{J})}. \quad (\text{ESI0.8})$$

In particular, with $J = \Sigma \mathcal{S} \subset \mathcal{A}$,

$$J = \overline{J}^{\mathcal{A}} = \mathcal{M}_0^{-1}(\mathcal{J}). \quad (\text{ESI0.9})$$

ESI1. Forward map, all factors, and injectivity

For $f \in \mathcal{S}$, the differentiated series defining Σf converges locally uniformly with all derivatives on $u > 0$. At $u \geq 1$, for each $L > 1$, a finite collection of Schwartz seminorms bounds its logarithmic derivatives by a constant times $u^{-L} \sum_{n \geq 1} n^{-L}$.

Use the Fourier convention

$$\hat{f}(y) = \int_{\mathbb{R}} f(x) e^{-2\pi i x y} dx.$$

Evenness and the two source constraints give $\hat{f}(0) = 0$, $\int \hat{f} = f(0) = 0$, and $\hat{f} \in \mathcal{S}$. Poisson summation gives, retaining the raw factor of two,

$$\Sigma f(u) = u^{-1} \Sigma \hat{f}(1/u). \quad (\text{ESI1.1})$$

Indeed both complete integer sums have vanishing zero terms. If $D = u \partial_u$, differentiation of the right side gives

$$D^j(u^{-1} \Sigma \hat{f}(1/u)) = u^{-1} (-1 - D_v)^j \Sigma \hat{f}(v) \big|_{v=1/u}. \quad (\text{ESI1.2})$$

The estimates at infinity therefore give arbitrary power decay at zero too. Fourier transformation is continuous on Schwartz space, so these estimates prove continuity $\Sigma : \mathcal{S} \rightarrow \mathcal{A}$.

For $\Re s > 1$, absolute convergence and the substitution $v = nu$ give

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{ESI1.3})$$

The last integral is holomorphic on $\Re s > -2$, since an even smooth function with $f(0) = 0$ satisfies $f(v) = O(v^2)$. Its value at $s = 1$ is zero. Consequently (ESI1.3), continued to the nontrivial zeros, proves $\mathcal{M}_0 \Sigma f \in \mathcal{I}$, including every required jet order.

If $\Sigma f = 0$, then on $s = 2 + it$ the Euler product gives $\zeta(s) \neq 0$, so the Mellin integral of f there vanishes identically. Fourier uniqueness in the variable v gives $f = 0$. Thus Σ is injective.

ESI2. The quotient and its possible poles

Fix $F \in \mathcal{I}$ and put

$$H(s) = \frac{F(s)}{2\zeta(s)}. \quad (\text{ESI2.1})$$

Local Taylor division at every nontrivial zero gives a holomorphic extension there. The pole of ζ at 1 makes H holomorphic with

$$H(1) = 0, \quad H'(1) = \frac{F(1)}{2}. \quad (\text{ESI2.2})$$

The only possible poles of H are simple poles at the trivial zeros $-2r$, $r \geq 1$, with residues

$$c_r := \operatorname{Res}_{s=-2r} H(s) = \frac{F(-2r)}{2\zeta'(-2r)}. \quad (\text{ESI2.3})$$

If the numerator also vanishes, this residue is zero and the apparent pole is removable. At zero,

$$H(0) = -F(0), \quad (\text{ESI2.4})$$

because $\zeta(0) = -\frac{1}{2}$.

The simplicity and nonvanishing derivatives of the trivial zeros follow from the full functional equation

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{ESI2.5})$$

At $-2r$, the sine has a simple zero, while the other factors and $\zeta(1 + 2r)$ are finite and nonzero. More precisely, differentiating its simple sine zero gives

$$\zeta'(-2r) = (-1)^r 2^{-2r-1} \pi^{-2r} (2r)! \zeta(1 + 2r). \quad (\text{ESI2.6})$$

Thus no completion cancellation is erased: its exact coefficient remains in (ESI2.3) and (ESI2.6).

The auxiliary function used only to prove a growth estimate is the complete retained S2 expression

$$F_0(s) = K(s)\zeta(s), \quad K(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2). \quad (\text{ESI2.7})$$

By S3, $Q = F/F_0$ is entire and, on each bounded real strip,

$$|Q(\sigma + it)| \leq C_{A,F} \exp(C(|t| + 2)^{3/2} \log(|t| + 2)). \quad (\text{ESI2.8})$$

The exact relationship is

$$H(s) = \frac{1}{2} Q(s) K(s) = Q(s) \frac{s(s-1)}{16} \pi^{-s/2} \Gamma(s/2). \quad (\text{ESI2.9})$$

It holds meromorphically at all exceptional points too.

For later use, on a fixed real strip and $|t| \geq 1$, K has at most polynomial growth. To verify this without an interior lower bound for zeta, choose an integer m making $\Re(s/2) + m \geq 1$ throughout the strip. Gamma recurrence and its defining integral give

$$\Gamma(s/2) = \frac{\Gamma(s/2 + m)}{\prod_{j=0}^{m-1} (s/2 + j)}, \quad |\Gamma(s/2 + m)| \leq \Gamma(\Re(s/2) + m). \quad (\text{ESI2.10})$$

The numerator is bounded on the resulting compact interval of positive real parts; each denominator factor has modulus at least $|t|/2$. Including the polynomial $s(s-1)$ and the bounded modulus of $\pi^{-s/2}$ proves the assertion. Together with (ESI2.8), this proves strictly subquadratic logarithmic growth of H , and of any fixed polynomial times H , on the horizontal parts $|t| \geq 1$ of each strip.

ESI3. Exact boundary estimates

On a right boundary $\Re s = b > 1$, the absolutely convergent Euler product, or its reciprocal Dirichlet series, gives

$$\frac{1}{|\zeta(b + it)|} \leq \sum_{n \geq 1} \frac{|\mu(n)|}{n^b} \leq \zeta(b). \quad (\text{ESI3.1})$$

Thus H has rapid vertical decay there, bounded continuously by the corresponding seminorms of F .

For a left boundary choose $a = -2N - 1$, $N \geq 0$, and write $y = t/2$. The exact Gamma identity is

$$|\chi(a + it)|^2 = \pi^{-4N-3} y \coth(\pi y) \prod_{k=1}^N (k^2 + y^2) \prod_{j=0}^N ((j + \frac{1}{2})^2 + y^2). \quad (\text{ESI3.2})$$

Here the first product is 1 when $N = 0$, and $y \coth(\pi y)$ has value $1/\pi$ at $y = 0$.

For completeness, the Gamma numerator in (ESI2.5) is $\Gamma(N + 1 - iy)$; recurrence gives its squared modulus as $\prod_{k=1}^N (k^2 + y^2) |\Gamma(1 - iy)|^2$. The denominator is $\Gamma(-N - \frac{1}{2} + iy)$; recurrence to $\Gamma(\frac{1}{2} + iy)$ contributes the second product in the numerator of the modulus ratio. The reflection formula gives

$$|\Gamma(1 - iy)|^2 = \frac{\pi y}{\sinh(\pi y)}, \quad |\Gamma(\frac{1}{2} + iy)|^2 = \frac{\pi}{\cosh(\pi y)}. \quad (\text{ESI3.3})$$

Their ratio is $y \coth(\pi y)$, and the squared modulus of $\pi^{s-1/2}$ is π^{-4N-3} . This proves every factor in (ESI3.2).

This expression is strictly positive for every real t . It yields the explicit lower estimate

$$|\chi(a+it)|^2 \geq \frac{\pi^{-4N-4}}{2 \cdot 8^{2N+1}} (1+|t|)^{4N+3}. \quad (\text{ESI3.4})$$

Indeed $y \coth(\pi y) \geq \max(|t|/2, 1/\pi) \geq (1+|t|)/(2\pi)$. Each of the $2N+1$ quadratic factors is at least $(1+|t|)^2/8$. The first inequality follows from $\coth x \geq 1$ and $\tanh x \leq x$ for $x > 0$, with continuity at zero.

Because $1-a = 2N+2 > 1$, (ESI2.5) and (ESI3.1) give

$$\frac{1}{|\zeta(a+it)|} \leq \zeta(2N+2) \sqrt{2 \cdot 8^{2N+1}} \pi^{2N+2} (1+|t|)^{-2N-3/2}. \quad (\text{ESI3.5})$$

In particular H is rapidly decreasing on this left boundary too, with continuous bounds in F . This proves the boundary estimates on complete unbounded lines, not only on a finite height range.

ESI4. The strip argument and its uniformity

Fix $N \geq 0$, $a = -2N-1$, and any $b > 1$. Put

$$P_N(s) = \prod_{r=1}^N (s+2r), \quad V(s) = P_N(s)H(s). \quad (\text{ESI4.1})$$

For $N=0$, $P_0=1$. The poles described in ESI2 show that V is holomorphic on a neighborhood of the closed strip $a \leq \Re s \leq b$. No poles lie on either boundary.

Choose a real number L with $a+L \geq 1$. For any integer $M \geq 0$, set

$$W_M(s) = (s+L)^M V(s), \quad B_M(F) = \max_{c \in \{a,b\}} \sup_{t \in \mathbb{R}} |W_M(c+it)|. \quad (\text{ESI4.2})$$

ESI3 proves that $B_M(F)$ is finite and at most a constant times a finite $\beta_{A,M'}(F)$ seminorm. This constant depends on the fixed strip, L , and M , not on F .

For $\varepsilon > 0$, apply the maximum-modulus principle on the rectangle $[a, b] + i[-T, T]$ to

$$e^{\varepsilon s^2} W_M(s). \quad (\text{ESI4.3})$$

On either vertical side its modulus is at most $e^{\varepsilon \max(a^2, b^2)} B_M(F)$, since $|e^{\varepsilon(c+it)^2}| = e^{\varepsilon c^2 - \varepsilon t^2}$. On the two horizontal sides ESI2 gives a bound of the form

$$C_{F,N,b,M,\varepsilon} (1+T)^d \exp(-\varepsilon T^2 + C(T+2)^{3/2} \log(T+2)), \quad (\text{ESI4.4})$$

which tends to zero as $T \rightarrow \infty$, uniformly in the real coordinate. The exponent's second term divided by T^2 tends to zero. No growth constant uniform in F is needed for this limiting assertion.

For any fixed s in the strip, first let $T \rightarrow \infty$. This yields

$$|e^{\varepsilon s^2} W_M(s)| \leq e^{\varepsilon \max(a^2, b^2)} B_M(F).$$

Now let $\varepsilon \downarrow 0$, with s fixed. The exact consequence is

$$|W_M(s)| \leq B_M(F) \quad (a \leq \Re s \leq b). \quad (\text{ESI4.5})$$

Since $|s+L| \geq \sqrt{1+t^2} \geq (1+|t|)/\sqrt{2}$, this proves

$$\sup_{a \leq \sigma \leq b} (1+|t|)^M |P_N(\sigma+it)H(\sigma+it)| \leq 2^{M/2} B_M(F). \quad (\text{ESI4.6})$$

This is an interior rapid-decay estimate with constants controlled by boundary seminorms of F . The use of a potentially uncontrolled constant in (ESI2.8) does not compromise (ESI4.6); that constant has disappeared after taking the two limits in their stated order.

For $|t| \geq 1$, $|P_N(\sigma+it)| \geq |t|^N$. Thus H itself has arbitrarily rapid vertical decay there uniformly on the strip. On compact regions away from its finitely many poles it is bounded by holomorphicity. Cauchy's formula applied on small circles in a slightly larger strip gives the same rapid bounds for all complex derivatives on any vertical line avoiding the poles. All contour shifts and differentiations used next are therefore justified on complete lines.

ESI5. The actual inverse source and every endpoint derivative

Define, for $x > 0$,

$$f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(2 + it) x^{-2-it} dt. \quad (\text{ESI5.1})$$

ESI4 proves absolute convergence and permits all ordinary derivatives in x . Define the rising product $(s)_0 = 1$, $(s)_j = s(s+1)\cdots(s+j-1)$ for $j \geq 1$; then

$$\partial_x^j x^{-s} = (-1)^j (s)_j x^{-s-j}. \quad (\text{ESI5.2})$$

For every real $c > 1$, shifting between 2 and c crosses no poles. The horizontal integrals tend to zero by ESI4, giving

$$f^{(j)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-1)^j (c + it)_j H(c + it) x^{-c-it-j} dt. \quad (\text{ESI5.3})$$

For $x \geq 1$, this is at most $C_{c,j,F} x^{-c-j}$ in modulus. Since c can be chosen arbitrarily large, every derivative has arbitrary power decay at infinity.

Now shift the original contour to $a = -2N - 1$, crossing exactly the possible poles $-2, -4, \dots, -2N$. Residue calculus with the original orientation gives

$$f(x) = \sum_{r=1}^N c_r x^{2r} + R_N(x), \quad R_N(x) = \frac{1}{2\pi} \int_{\mathbb{R}} H(a + it) x^{-a-it} dt. \quad (\text{ESI5.4})$$

Every residue has the positive sign displayed. For every $j \geq 0$, differentiation under this last integral yields

$$|R_N^{(j)}(x)| \leq C_{N,j,F} x^{2N+1-j} \quad (0 < x \leq 1). \quad (\text{ESI5.5})$$

For a given derivative order j , choose $2N \geq j$. The remainder derivative then tends to zero at 0. The polynomial in (ESI5.4) proves that every right derivative has a finite limit, equal to zero for odd j , equal to $(2r)!c_r$ for $j = 2r \geq 2$, and zero for $j = 0$.

Here finite right derivative limits really give a smooth extension. For each j , $f^{(j+1)}$ is bounded near zero and has a finite limit. The fundamental theorem of calculus on $[\delta, x]$, followed by $\delta \downarrow 0$, gives

$$f^{(j)}(x) - \lim_{v \downarrow 0} f^{(j)}(v) = \int_0^x f^{(j+1)}(v) dv. \quad (\text{ESI5.6})$$

This identity proves differentiability at zero with the asserted next derivative. Iterating in j proves smoothness on the half-line. All its odd derivatives there are zero, so the extension $f(-x) = f(x)$, $f(0) = 0$, is smooth on $\mathbb{R}$. Together with the infinity estimates, it belongs to Schwartz space.

The complete even Taylor data are

$$\boxed{\frac{f^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1),} \quad f^{(2r+1)}(0) = 0, \quad f(0) = 0. \quad (\text{ESI5.7})$$

Thus the cancelled trivial-zero locations return as actual source derivatives; they have not been discarded by the inverse construction.

ESI6. Mellin recovery, source moment, and surjectivity

ESI4 and Cauchy's formula show that $t \mapsto H(2 + it)$ is a Schwartz function of t . Formula (ESI5.1) says that $y \mapsto e^{2y} f(e^y)$ is its inverse Fourier transform with phase e^{-ity} and factor $1/(2\pi)$. Fourier inversion therefore gives

$$\int_0^\infty f(x) x^{2+it} \frac{dx}{x} = H(2 + it). \quad (\text{ESI6.1})$$

By ESI5, the Mellin integral of f is holomorphic on $\Re s > -2$. There are no possible poles of H in that open half-plane. The identity theorem applied to (ESI6.1) gives

$$\int_0^\infty f(x)x^s \frac{dx}{x} = H(s) = \frac{F(s)}{2\zeta(s)} \quad (\Re s > -2), \quad (\text{ESI6.2})$$

where the quotient uses its holomorphic extensions at the nontrivial zeros and at 1.

Taking $s = 1$, ESI2 gives $\int_0^\infty f(x)dx = 0$. Evenness then gives $\int_{\mathbb{R}} f = 0$. Hence the constructed function is in the precise source $\mathcal{S}$, including both constraints.

The endpoint identities are

$$\int_0^\infty f(x) \frac{dx}{x} = -F(0), \quad \int_0^\infty f(x) \log x \, dx = \frac{F(1)}{2}. \quad (\text{ESI6.3})$$

The logarithmic integral is the derivative of (ESI6.2) at 1, justified by absolute convergence. These factors differ from S6's centered, half-sum source because the present map is the raw sum $2 \sum f(nu)$.

By ESI1,

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s)H(s) = F(s) \quad (\Re s > 1). \quad (\text{ESI6.4})$$

Both sides are entire, so the identity holds everywhere. Mellin injectivity now gives $\Sigma f = \mathcal{M}_0^{-1}F$. Together with ESI1's injectivity, this proves the algebraic bijection in (ESI0.8).

ESI7. Continuity of the inverse and closedness

The inverse constructed above is linear. Its continuity follows directly from the two boundary integrals and the retained residues, without an unproved uniform estimate for the quotient F/F_0 .

Fix a source seminorm $q_{R,j}$. On $x \geq 1$, choose a real $c > \max(1, R)$. Formula (ESI5.3), the polynomial bound on $(c + it)_j$, and (ESI3.1) bound

$$(1+x)^R |f^{(j)}(x)| \leq C_{R,j,c} \beta_{A,j+2}(F) \quad (x \geq 1) \quad (\text{ESI7.1})$$

for any $A \geq c$. Indeed $(1+x)^R \leq 2^R x^R$ there, $x^{R-c-j} \leq 1$, and $\int_{\mathbb{R}} (1+|t|)^{-2} dt < \infty$.

On $0 < x \leq 1$, choose N with $2N+1 \geq j$. Formula (ESI5.5) follows directly from the boundary $a = -2N-1$, where (ESI3.5) bounds the reciprocal zeta factor by a fixed constant, and in fact by a decaying power. Thus

$$\sup_{0 < x \leq 1} |R_N^{(j)}(x)| \leq C_{N,j} \beta_{A,j+2}(F) \quad (A \geq 2N+1). \quad (\text{ESI7.2})$$

The remaining polynomial derivatives in (ESI5.4) are controlled by the finitely many continuous evaluations

$$|c_r| \leq \frac{\beta_{A,0}(F)}{2|\zeta'(-2r)|}, \quad 1 \leq r \leq N, \quad (\text{ESI7.3})$$

with $A \geq 2N$. Terms whose degree is below j have zero j -th derivative. All other powers of x are bounded by 1 on this interval. The same estimates hold at 0 by the proved smooth extension, and on negative x by evenness. Consequently there exist A , M , and a constant $C_{R,j}$, independent of F , such that

$$q_{R,j}(f) \leq C_{R,j} \beta_{A,M}(F). \quad (\text{ESI7.4})$$

This proves continuity $\mathcal{I} \rightarrow \mathcal{S}$. Combining it with the continuous Mellin isomorphism and the continuity of Σ proves (ESI0.8) as a topological isomorphism. Since $\mathcal{I}$ is closed, its inverse Mellin image is closed in $\mathcal{A}$. This proves (ESI0.9).

In particular, if $C = \overline{\Sigma \mathcal{S}}^{\mathcal{A}}$, then the previously possible difference is exactly

$$C/\Sigma \mathcal{S} = 0. \quad (\text{ESI7.5})$$

The ordinary quotient $\mathcal{A}/\Sigma \mathcal{S}$ is already Hausdorff and maps isomorphically to $\mathcal{A}/C$ and then to $\mathcal{B}/\mathcal{I}$. This assertion concerns the specific source map in (ESI0.5). Any further identification with a differently defined adelic source must still use its actual proved source comparison.

ESI8. The exact Möbius inverse and its relation to the analytic inverse

For $b = \Sigma f$, the inverse is also, for every $x > 0$,

$$f(x) = \frac{1}{2} \sum_{n=1}^{\infty} \mu(n) b(nx). \quad (\text{ESI8.1})$$

For every compact interval $0 < \delta \leq x \leq R$, the series and each ordinary derivative converge absolutely and uniformly. This follows from arbitrary power decay of the derivatives of b at infinity; its logarithmic seminorms imply those derivative bounds by the identities expressing $x^j \partial_x^j$ as a polynomial in $x \partial_x$. In particular a sufficiently large exponent controls $n^j b^{(j)}(nx)$ by a summable power of n .

Substituting $b = \Sigma f$ into the right side of (ESI8.1) gives the absolutely convergent double series

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) 2 \sum_{m \geq 1} f(mnx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) f(kx) = f(x). \quad (\text{ESI8.2})$$

Absolute convergence follows, for each fixed $x > 0$, by choosing $L > 1$ and bounding $|f(mnx)| \leq C_x (mn)^{-L}$; the sum is then at most $C_x \zeta(L)^2$. The divisor sum is 1 for $k = 1$, and 0 for $k > 1$, by the full prime factorization identity $\prod_{p|k} (1 - 1)$. This uses the globally reconstructed prime system, not a truncated list of primes.

For an arbitrary $b \in \mathcal{M}_0^{-1}(\mathcal{J})$, ESI5–ESI7 already produce the unique Schwartz f with $b = \Sigma f$; (ESI8.2) therefore identifies its pointwise inverse with (ESI8.1). No termwise evaluation of the Möbius series at zero is asserted. The smooth extension and all its derivatives at zero are instead proved by (ESI5.4)–(ESI5.7), which retain the complete trivial-zero residue data.

ESI9. Audit scope and receiving consequence

The candidate mechanism succeeds. Its essential points are the already proved subquadratic strip division estimate, the exact nonvanishing boundary factors of the original zeta, the order of limits $T \rightarrow \infty$ followed by $\varepsilon \downarrow 0$, and the full Mellin residue reconstruction at zero. The finite collection of poles in any one contour shift is used for that exact contour computation; all negative even points and every derivative are covered by allowing arbitrary N .

Every nontrivial zero and its multiplicity remains in $\mathcal{J}$. The theorem neither relocates those zeros nor supplies a positive definite form or a Deligne weight gap. It closes the concrete source-image versus closure distinction: for this source the ordinary cohomological coefficient quotient already is the full Hausdorff zeta-jet quotient. Its exact maps are the raw Σ , $\mathcal{M}_0$, and inverse (ESI5.1), equivalently (ESI8.1).

ESI10. Comparison with the independently written SSI proof

After completing ESI0–ESI9, the complete file EXACT_SCHWARTZ_SUMMATION_IMAGE.md, SSI0–SSI9, was read. The checked file SHA256 is 598560088859a90181e444af2ceaf49bdd9b8c4fe2f265ddd2270bc14b324f22. The requested checks of SSI4–SSI7 require no mathematical correction.

In SSI4.3, the polynomial multiplying H has degree $M + N$. ESI3 shows that the reciprocal zeta on both boundary lines is bounded, including the left line where it in fact decreases. Thus the specific seminorm $b_{A_0, M+N}(F)$, rather than an unspecified infinite collection, suffices there. The use of $\kappa = 2N + 2$ gives $\Re(s + \kappa) \geq 1$ exactly; conversion to $(1 + |t|)^M$ costs only $2^{M/2}$. SSI4.7 therefore follows with the stated continuity and order of limits.

In SSI5, the rising product $(s)_j$, positive signs of the crossed residues, and the exponent $2N + 1 - j$ of the derivative remainder agree with ESI5. The arbitrarily large choice of N supplies every derivative limit; the fundamental theorem of calculus proves that these are derivatives of the extension. Odd derivatives vanish, making the even extension smooth. On the line $\Re s = 2$, multiplication by e^{2y} after $x = e^y$ gives precisely the Fourier inversion pair with no missing scale or 2π factor. The integral condition is $2H(1) = 0$, as stated.

The continuity in SSI6 also follows from ESI7's direct boundary estimates. Its centered source is exactly twice the raw source for the same entire target F : $\mathcal{M}_0\Sigma = 2\mathcal{M}\mathcal{E}$. Thus the centered quotient is F/ζ , the raw quotient is $F/(2\zeta)$, and the endpoint and derivative factors all agree.

The Möbius proof in SSI7 uses the same absolutely convergent double sum as (ESI8.2). Its Fourier and dilation comparisons can additionally be checked directly. Define

$$Rb(u) = u^{-1}b(u^{-1}), \quad \jmath F(s) = F(1-s), \quad T_a b(u) = b(u/a), \quad D_a f(x) = f(x/a). \quad (\text{ESI10.1})$$

The exact Poisson identity on the source is $R\Sigma f = \Sigma \hat{f}$. Since Σ is injective and surjective onto its image, this proves

$$\Sigma^{-1}R\Sigma f = \hat{f}. \quad (\text{ESI10.2})$$

Substitution $v = 1/u$ gives $\mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1-s)$; hence the source with target $\jmath F$ is $\widehat{f_F}$. Its Mellin formula is exactly

$$\int_0^\infty \widehat{f_F}(x) x^s \frac{dx}{x} = \frac{F(1-s)}{2\zeta(s)} = \frac{\zeta(1-s)}{\zeta(s)} H(1-s), \quad (\text{ESI10.3})$$

first on a common regular domain and then meromorphically. The functional-equation factor is the reciprocal of the full $\chi(s)$ in (ESI2.5), including its Gamma, powers, and trigonometric contributions. No unit-modulus conclusion follows from this identity alone.

Direct summation gives $\Sigma D_a = T_a \Sigma$, and substitution in the raw Mellin integral gives $\mathcal{M}_0 T_a b(s) = a^s \mathcal{M}_0 b(s)$. Differentiation with respect to $\log a$ at zero gives the source operator $-x\partial_x$; integration by parts, with both endpoint terms vanishing by the established source estimates, gives its Mellin operator s . These are the signs and factors in SSI7. They preserve the existing full zero multiplicities and do not assert new restrictions on their real parts.

The receiving support rows in SSI8 use the separately constructed sheaf CSL and its orientation conventions; this check supplies their claimed input $J = \bar{J}$. It does not replace the independent verification of those sheaf and support complexes. SSI9 correctly leaves the Deligne lifting obstruction and purity distinct from the proved vanishing $\bar{J}/J = 0$.

ESI11. The full and moment-null adelic difference quotients

The complete linked file `../tau_weight_cohomology_20260924/CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md`, CS0–CS11 including CS2A, was subsequently read; its checked SHA256 is 6cb067ddffae3b15bb0d19b32d30e1df90bf0c7b4214f2ac7f5b9118a085279. The appended SSI10 was also read. The adelic source proof and the claimed actions are valid, with the finite-unit sector explicitly retained in the receiving sheaf as below.

Write $E = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$, $\epsilon(F) = (F(0), \int_{\mathbb{A}} F)$, and $E_0 = \ker \epsilon$. Give these spaces the stated inductive-limit Schwartz topology. Let

$$\begin{aligned} \mathcal{R} &= \text{span}\{F - F(q\cdot) : F \in E, q \in \mathbb{Q}^\times\}, & \mathcal{R}_{\text{null}} &= \text{span}\{F - F(q\cdot) : F \in E_0, q \in \mathbb{Q}^\times\}, \\ D &= E_0/\overline{\mathcal{R}}, & D_{\text{null}} &= E_0/\overline{\mathcal{R}_{\text{null}}}. \end{aligned} \quad (\text{ESI11.1})$$

The adelic product formula $|q|_{\mathbb{A}} = 1$ ensures that both displayed relation spaces lie in E_0 . Their difference is therefore a genuine choice of the generating domain, not a failure of the endpoint constraints.

Let $K = \widehat{\mathbb{Z}}^\times$. On each test-function stage, averaging over K is a finite averaging operation after taking the stabilizer of that stage. A K -invariant finite Schwartz function is a finite linear combination of the indicators $1_{d\widehat{\mathbb{Z}}}$, $d \in \mathbb{Q}_{>0}$. At each exceptional prime, compact support gives a lower valuation bound and local constancy gives a terminal ball; their shells are differences of nested balls. Finite tensor products of these expansions give the global statement. These ball indicators are linearly independent: evaluations on successive local shells give an invertible triangular evaluation matrix, and tensor products preserve independence.

Thus the following maps have unique well-defined formulas:

$$jf = [f \otimes 1_{\widehat{\mathbb{Z}}}], \quad B \left(\sum_d f_d \otimes 1_{d\widehat{\mathbb{Z}}} \right) (v) = \sum_d f_d^{\text{ev}}(dv), \quad B(F) = B(P_K F). \quad (\text{ESI11.2})$$

They are continuous on every finite stage and hence on the given inductive limit. The two exact endpoint identities are $B(F)(0) = F(0)$ and $\int_{\mathbb{R}} B(F) = \int_{\mathbb{A}} F$, using $\text{vol}(d\widehat{\mathbb{Z}}) = d^{-1}$. Rational scaling sends the ball index to $d/|q|$ and its even real function to $f^{\text{ev}}(|q|v)$, so $B(F(q\cdot)) = B(F)$. Consequently B descends to $\bar{B} : D \rightarrow \mathcal{S}$.

For a ball tensor $G = f \otimes 1_{d\widehat{\mathbb{Z}}}$, put $H_d = G(d\cdot)$. The complete representative identity is

$$G - \tilde{j}(BG) = (G - G(d\cdot)) + \frac{1}{2}(H_d - H_d(-\cdot)) \in \mathcal{R}. \quad (\text{ESI11.3})$$

Although individual tensors need not lie in E_0 , both differences in this equation do; summing the equation for a moment-null input before taking its class is legitimate. It follows that $\bar{B}j = 1$ and $j\bar{B} = P_K$ on D . A K -fixed class is equal to its average because its representative has a finite averaging orbit. Thus

$$\boxed{D^K \xleftarrow[\bar{j}]{\bar{B}} \mathcal{S}} \quad (\text{ESI11.4})$$

is an actual topological isomorphism. Its overlap sum is precisely $2 \sum_{n \geq 1} f(nu)$, since $\mathbb{Q} \cap \widehat{\mathbb{Z}} = \mathbb{Z}$; there is no lost negative-integer contribution.

To check the entire alternative quotient, let

$$V = \bigoplus_{p \text{ prime}} \mathbb{C}^2, \quad \ell_p(F) = \sum_d v_p(d) \left(f_d(0), d^{-1} \int f_d \right), \quad \mathcal{L} = (\ell_p)_p, \quad (\text{ESI11.5})$$

with averaging understood for general inputs. Each fixed test stage has only finitely many relevant primes; therefore $\mathcal{L} : E \rightarrow V$ is continuous for the locally convex direct-sum topology. Its rational-scaling identity is

$$\ell_p(F(q\cdot)) = \ell_p(F) - v_p(|q|)\epsilon(F). \quad (\text{ESI11.6})$$

Hence $\mathcal{L}$ vanishes on $\overline{\mathcal{R}_{\text{null}}}$.

The section used in CS2A has real factor $\sigma(a, b)(v) = [a(1 - 2\pi v^2) + b 2\pi v^2]e^{-\pi v^2}$ and finite factor $1_{\widehat{\mathbb{Z}}}$. The Gaussian integrals give $\epsilon\sigma(a, b) = (a, b)$. Define

$$\mathcal{T}(v) = \sum_p (\sigma(v_p) - \sigma(v_p)(p\cdot)). \quad (\text{ESI11.7})$$

The sum is finite for every $v \in V$, its map is continuous by the direct-sum universal property, and (ESI11.6) gives $\mathcal{L}\mathcal{T} = 1$.

Put $r_q(F) = F - F(q\cdot)$. Modulo $\mathcal{R}_{\text{null}}$, its class depends only on $\epsilon(F)$. Also $r_{qr}(F) - r_q(F) - r_r(F) = r_r(F(q\cdot) - F)$, whose generating function is moment-null. The sign relation is $r_{-1}(F) = \frac{1}{2}r_{-1}(F - F(-\cdot))$. Full rational factorization therefore proves

$$r_q(F) \equiv \sum_p v_p(|q|)r_p(\sigma(\epsilon F)) = \mathcal{T}\mathcal{L}r_q(F) \pmod{\mathcal{R}_{\text{null}}}. \quad (\text{ESI11.8})$$

Thus $(1 - \mathcal{T}\mathcal{L})\overline{\mathcal{R}} \subset \overline{\mathcal{R}_{\text{null}}}$ by continuity. It proves the mutually inverse continuous maps

$$\begin{aligned} \Psi : D_{\text{null}} &\longrightarrow D \oplus V, & [F]_{\text{null}} &\longmapsto ([F], \mathcal{L}F), \\ \Psi^{-1}([F], v) &= [F - \mathcal{T}\mathcal{L}F + \mathcal{T}v]_{\text{null}}. \end{aligned} \quad (\text{ESI11.9})$$

In particular the second formula is independent of the representative by (ESI11.8); continuity follows by factoring a continuous map through the defining quotient. This checks both the algebra and topology of CS2A.10.

The map $\mathcal{L}$ is K -invariant, and $\mathcal{T}$ takes values in K -fixed functions, since all its finite factors are radial balls. Thus Ψ is K -equivariant with trivial K -action on V , and the receiving sector is explicitly

$$\boxed{D_{\text{null}}^K \cong D^K \oplus V \cong \mathcal{S} \oplus V.} \quad (\text{ESI11.10})$$

This sector restriction is necessary for the subsequent identification of degree-one cohomology with the original Riemann-zeta quotient. The full sheaf formed from all of D has further finite-unit character sectors; they have not been identified with $\mathcal{B}/\mathcal{I}$.

ESI12. Retained Fourier, dilation, and chart-mirror maps

For self-dual adelic Fourier measures,

$$f \otimes \widehat{1_{d\widehat{z}}} = d^{-1} \widehat{f} \otimes 1_{d^{-1}\widehat{z}}. \quad (\text{ESI12.1})$$

Its finite factor comes from the reciprocal annihilator ball and the original volume d^{-1} . For $C(a, b) = (b, a)$, substitution in (ESI11.5) gives, without dropping that volume,

$$\ell_p(\widehat{F}) = -v_p(d) \left(d^{-1} \int f, f(0) \right) = -C\ell_p(F) \quad (\text{ESI12.2})$$

on each ball tensor. Fourier conjugates the finite-unit action to its inverse and hence commutes with averaging. It preserves both relation closures because it takes a rational difference at q to one at q^{-1} , the full adelic modulus being 1. Thus (ESI12.2) holds on the actual quotient and proves

$$\widehat{\Psi}_{D_{\text{null}}} \Psi^{-1} = (\widehat{}, -C). \quad (\text{ESI12.3})$$

The sign is required: it is the sign of the inverted finite valuation.

For the two real-place actions

$$\rho_+(a)F(v, x_f) = F(v/a, x_f), \quad \rho_-(a)F(v, x_f) = aF(av, x_f), \quad a > 0, \quad (\text{ESI12.4})$$

evaluation and the change of integration variable give

$$\begin{aligned} \mathcal{L}\rho_+(a) &= A_+(a)\mathcal{L}, & A_+(a) &= \begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}, \\ \mathcal{L}\rho_-(a) &= A_-(a)\mathcal{L}, & A_-(a) &= \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \quad (\text{ESI12.5})$$

These act in every prime summand. The two actions commute with rational dilation, so descend to the source quotients. The canonical map Ψ is equivariant for (ESI12.5) because both its coordinates intertwine the specified actions; this conclusion does not require its auxiliary endpoint section σ to be equivariant. Fourier interchange is explicitly consistent:

$$(-C)A_+(a) = A_-(a)(-C), \quad \widehat{\rho_+(a)F} = \rho_-(a)\widehat{F}. \quad (\text{ESI12.6})$$

Let Ω denote the two-chart K -sector sheaf already used in SSI8, with chart spaces $D^K \oplus \mathbb{C}^2$, overlap $\mathcal{A}$, and the specified positive and reflected summation restrictions. Define Ω_{null} with $D_{\text{null}}^K \oplus \mathbb{C}^2$ in those charts. The V -coordinates of (ESI11.10) restrict to zero: they are represented by the finite sums $\mathcal{T}(v)$ of full rational differences, whose summations vanish by rational reindexing. Therefore (ESI11.10), the identity on both endpoint spaces, and the identity on the overlap give the actual sheaf isomorphism

$$\boxed{\Omega_{\text{null}} \cong \Omega \oplus (i_+)_* V \oplus (i_-)_* V.} \quad (\text{ESI12.7})$$

This is checked on each of the two charts, their overlap, and all restriction arrows. On global sections it follows by the defining fiber product. Its additional Čech terms are $V \oplus V$ in degree zero and zero in degree one, with zero differential. Hence

$$H^0(X, \Omega_{\text{null}}) \cong H^0(X, \Omega) \oplus V \oplus V, \quad H^1(X, \Omega_{\text{null}}) \cong H^1(X, \Omega) \cong \mathcal{B}/\mathcal{I}. \quad (\text{ESI12.8})$$

The single closed support receives exactly its own degree-zero V ; its degree-one quotient is unchanged. These statements are also continuous for the indicated section and quotient topologies.

Finally the chart-exchanging sheaf mirror is not the Fourier map on a single source. On the new degree-zero coordinates it is

$$W_V : V_+ \oplus V_- \longrightarrow V_+ \oplus V_-, \quad (v_+, v_-) \longmapsto (v_-, v_+), \quad (\text{ESI12.9})$$

whereas Fourier inside either typed source copy acts by $-C$ from (ESI12.3). With $\rho_V(a) = \text{diag}(A_+(a), A_-(a))$, the unshifted mirror identity is

$$\rho_V(a)W_V = aW_V\rho_V(a^{-1}), \quad (\text{ESI12.10})$$

as follows from $A_+(a) = aA_-(a^{-1})$ and $A_-(a) = aA_+(a^{-1})$. On the pre-existing degree-one Čech quotient the mirror remains $-R$, or $-j$ after the raw Mellin map, from its chart-order sign. Thus all three typed maps and their signs remain distinct: the source Fourier map $-C$, the chart swap W_V , and the degree-one orientation action $-j$.

This verifies the retained prime directions and their actions. The sheaf formula is understood with the explicit sector typing of (ESI11.10), and no purity conclusion is added.

Localization on the three-point CC space, including full nilpotent jets

Independent mathematical derivation, 24 September 2026. Proof locators TPL0–TPL13.

TPL0. Source, prerequisites, and signs

The space is the one in Connes–Consani’s supplied author source CC.tex, lines 496–541, read directly: two closed points $+$, $-$, one generic point η , and opens

$$\emptyset, \quad U_\eta = \{\eta\}, \quad U_+ = \{+, \eta\}, \quad U_- = \{-, \eta\}, \quad X. \quad (\text{TPL0.1})$$

The source is Alain Connes and Caterina Consani, The Absolute Twistor Line and the Geometry of the compactified Spec $\mathbb{Z}$, supplied-file SHA-256 57a10dcef5cd758e6b2f1c54b1c0a11b0fb093638f74fb1be515ec3bc7080ba4.

The instruction argument_reconstruction/CORPUS_AND_OPERATION_RULE.md was reread before this derivation. These are sheaves of complex vector spaces after their coefficients and arithmetic have been constructed. No finite prime prefix defines those coefficients. No equality $\tau = \eta$, addition on τ , numerical coordinate on τ , weight separation, or RH conclusion is asserted.

Complexes are cohomological, with $C[k]^n = C^{n+k}$, $d_{C[k]} = (-1)^k d_C$. For $f : C \rightarrow D$, use $\text{Cone}(f)^n = D^n \oplus C^{n+1}$, with differential $(y, x) \mapsto (d_D y + f x, -d_C x)$. Thus $\text{Cone}(r : A[0] \rightarrow B[0])[-1]$ is $A \xrightarrow{-r} B$ in degrees 0, 1. For the cone triangle take the final arrow to be projection to $C[1]$. Under the cone-to-quotient identification for a short exact sequence, its connecting representative is minus the differential of a lift: a cone cocycle (y, x) satisfies $d_D y + f x = 0$. This records the connecting-arrow convention explicitly; the opposite lift-and-differentiate convention changes those arrows by a minus sign. The ordinary vector-space calculations come first; TPL11–TPL12 specify the topological qualifications.

TPL1. The diagram describing every sheaf

A sheaf F is specified by

$$A_+ = F(U_+), \quad A_- = F(U_-), \quad B = F(U_\eta), \quad r_\pm : A_\pm \rightarrow B. \quad (\text{TPL1.1})$$

Its global sections are

$$\Gamma(X, F) = A_+ \times_B A_- = \{(a_+, a_-) : r_+ a_+ = r_- a_-\}. \quad (\text{TPL1.2})$$

Indeed the sheaf condition for $X = U_+ \cup U_-$ is exactly (TPL1.2). Every cover of U_+ contains U_+ , because no smaller open contains $+$; similarly for U_- , and a cover of U_η contains U_η . No additional gluing condition occurs. Conversely, (TPL1.2) and the three assigned spaces therefore define a sheaf on every open.

The stalks are $F_+ = A_+$, $F_- = A_-$, $F_\eta = B$, since each point has the displayed smallest open neighborhood. Morphisms are triples commuting with $r_\pm$. Kernels and cokernels are computed at these stalks. This proves the equivalence with the category of diagrams $A_+ \rightarrow B \leftarrow A_-$.

TPL2. An explicit resolution and complete Čech cohomology

For any vector space V , put

$$I_+(V) = (V \rightarrow 0 \leftarrow 0), \quad I_-(V) = (0 \rightarrow 0 \leftarrow V), \quad I_\eta(V) = (V \xrightarrow{1} V \xleftarrow{1} V). \quad (\text{TPL2.1})$$

The displayed order is plus stalk, generic stalk, minus stalk. A morphism $F \rightarrow I_+(V)$ is uniquely a map $A_+ \rightarrow V$; similarly for minus. A morphism $F \rightarrow I_\eta(V)$ is uniquely a map $B \rightarrow V$, its other components being obtained by restriction. These functors are right adjoint to the three exact stalk evaluations. Since linear maps from a subspace extend after choosing a basis, every vector space is injective; the adjunctions therefore prove that each sheaf in (TPL2.1) is injective.

There is the injective resolution

$$0 \rightarrow F \rightarrow I^0 = I_+(A_+) \oplus I_-(A_-) \oplus I_\eta(B) \xrightarrow{D} I^1 = I_+(B) \oplus I_-(B) \rightarrow 0. \quad (\text{TPL2.2})$$

At plus, the injection is $a \mapsto (a, r_+ a)$ into $A_+ \oplus B$, and $D_+(a, b) = b - r_+ a$. At minus the formulas use r_- . At η the injection is the identity $B \rightarrow B$, followed by zero. Restrictions in I^0 project onto the B coordinate. These formulas commute with restriction and are exact at every stalk, proving (TPL2.2).

Global sections of this resolution give

$$R\Gamma(X, F) = [A_+ \oplus A_- \oplus B \xrightarrow{D} B^2], \quad D(a_+, a_-, b) = (b - r_+ a_+, b - r_- a_-). \quad (\text{TPL2.3})$$

It maps onto the Čech complex

$$C(F) = [A_+ \oplus A_- \xrightarrow{d} B], \quad d(a_+, a_-) = r_+ a_+ - r_- a_-, \quad (\text{TPL2.4})$$

by forgetting b in degree zero and by

$$(v_+, v_-) \mapsto v_- - v_+ \quad (\text{TPL2.5})$$

in degree one. Applying (TPL2.5) to D gives d , so this is a chain map. Its kernel is the acyclic complex $B \xrightarrow{b \mapsto (b, b)} \text{diag } B$. Hence it is a quasi-isomorphism. It follows that

$$H^0(X, F) = A_+ \times_B A_-, \quad H^1(X, F) = B/(r_+ A_+ + r_- A_-), \quad H^q(X, F) = 0 \ (q \geq 2). \quad (\text{TPL2.6})$$

The image here is the algebraic sum of the actual images, with no closure.

TPL3. Open and closed functors

Let $j : \{\eta\} \hookrightarrow X$, $i : Z = \{+, -\} \hookrightarrow X$. The topology on Z is discrete. Directly on diagrams,

$$j^* F = B, \quad j_* V = (V \xrightarrow{1} V \xleftarrow{1} V), \quad j_! V = (0 \rightarrow V \leftarrow 0), \quad (\text{TPL3.1})$$

$$i^* F = (A_+, A_-), \quad i_*(W_+, W_-) = (W_+ \rightarrow 0 \leftarrow W_-), \quad i^! F = (\ker r_+, \ker r_-). \quad (\text{TPL3.2})$$

Every nonempty open meets η , giving the formula for j_* . Extension by zero has zero closed stalks, giving $j_!$. The adjunctions $j_! \dashv j^* \dashv j_*$ follow by writing the three components of a diagram morphism. A map $i_* W \rightarrow F$ must land in the kernels of $r_\pm$, proving $i_* \dashv i^!$; the other adjunction is $i^* \dashv i_*$.

All these functors except $i^!$ are exact, as their stalk formulas show. In particular $Rj_* V = j_* V$. The higher functor $R^1 i^!$ will retain a cokernel.

TPL4. Both costalks and the first localization triangle

Apply $i^!$ to (TPL2.2). The plus-supported part of I^0 is $\{(a_+, 0)\}$, whose differential is $-r_+ a_+$; the minus case is identical. Therefore

$$Ri^! F = (K_+, K_-), \quad K_\pm = [A_\pm \xrightarrow{-r_\pm} B] = \text{Cone}(r_\pm)[-1]. \quad (\text{TPL4.1})$$

In particular

$$H^0 K_\pm = \ker r_\pm, \quad H^1 K_\pm = B/r_\pm A_\pm, \quad H^q K_\pm = 0 \ (q \neq 0, 1). \quad (\text{TPL4.2})$$

There is a short exact sequence of sheaf complexes

$$0 \rightarrow i_* Ri^! F \rightarrow I^\bullet \rightarrow j_* B[0] \rightarrow 0. \quad (\text{TPL4.3})$$

The last map in degree zero projects $A_\pm \oplus B$ to B and is the identity at η ; in degree one it is zero. Its kernel is exactly (TPL4.1) at the two closed points. Consequently the actual localization triangle is

$$\boxed{i_* Ri^! F \rightarrow F \rightarrow Rj_* j^* F \rightarrow (i_* Ri^! F)[1].} \quad (\text{TPL4.4})$$

This follows from displayed complexes, rather than identification with a different geometric lifting problem.

TPL5. The connecting map on representatives

Global sections of (TPL4.3) give

$$0 \rightarrow K \rightarrow [A_+ \oplus A_- \oplus B \xrightarrow{D} B^2] \xrightarrow{(a_+, a_-, b) \mapsto b} B[0] \rightarrow 0, \quad K = [A_+ \oplus A_- \xrightarrow{(-r_+, -r_-)} B^2]. \quad (\text{TPL5.1})$$

Lift $b \in B$ to $(0, 0, b)$. Its differential is (b, b) , so with the cone convention of TPL0 the boundary is

$$\boxed{\delta : B \rightarrow (B/r_+ A_+) \oplus (B/r_- A_-), \quad b \mapsto ([-b], [-b])}. \quad (\text{TPL5.2})$$

Changing each costalk differential from $-r_\pm$ to $+r_\pm$ uses the chain isomorphism which is identity in degree zero and minus identity in degree one; it changes the boundary to $([b], [b])$. With the alternative lift-and-differentiate convention the signs of both boundary formulas are reversed. Their kernels and images are unchanged.

The complete long exact sequence is

$$\begin{aligned} 0 \rightarrow \ker r_+ \oplus \ker r_- \rightarrow A_+ \times_B A_- \xrightarrow{\text{res}} B \\ \xrightarrow{\delta} (B/r_+ A_+) \oplus (B/r_- A_-) \xrightarrow{q} B/(r_+ A_+ + r_- A_-) \rightarrow 0, \end{aligned} \quad (\text{TPL5.3})$$

where

$$\text{res}(a_+, a_-) = r_+ a_+ = r_- a_-, \quad q([v_+], [v_-]) = [v_- - v_+]. \quad (\text{TPL5.4})$$

The last map is induced by (TPL2.5), retaining the cover ordering.

In particular

$$\ker \delta = r_+ A_+ \cap r_- A_- = \text{im}(\text{res}). \quad (\text{TPL5.5})$$

Indeed a zero boundary means the existence of actual $a_\pm$ with the same image b ; these form a global section. Any two lifts differ by $\ker r_+ \oplus \ker r_-$.

Exactness at q can be checked without derived terminology. If $v_- - v_+ = r_+ a_+ + r_- a_-$, then $b = v_+ + r_+ a_+ = v_- - r_- a_-$ represents both residue classes, so the pair is $\delta(-b)$. Conversely every diagonal pair has zero difference. Thus the obstruction to lifting a prescribed generic section is the pair (TPL5.2), while Čech H^1 is the receiving quotient under the difference map. Neither replaces the other.

TPL6. The second localization triangle and single-point versions

There is a stalkwise short exact sequence

$$0 \rightarrow j_! j^* F \rightarrow F \rightarrow i_* i^* F \rightarrow 0. \quad (\text{TPL6.1})$$

At η its nonzero map is $B \xrightarrow{1} B$; at each closed point it is $A_\pm \xrightarrow{1} A_\pm$. The maps commute with restriction. Hence

$$\boxed{j_! j^* F \rightarrow F \rightarrow i_* i^* F \rightarrow (j_! j^* F)[1]}. \quad (\text{TPL6.2})$$

TPL2 gives $R\Gamma(X, j_! B) = B[-1]$ and $R\Gamma(X, i_* i^* F) = (A_+ \oplus A_-)[0]$. Under (TPL2.5) and the cone convention TPL0 its boundary is $-r_+ a_+ + r_- a_-$: a lift $(a_+, a_-, 0)$ has differential $(-r_+ a_+, -r_- a_-)$, whose negative has difference $-r_+ a_+ + r_- a_-$. Replacing the identification of $H^1(j_! B)$ with B by its negative makes this boundary d . Thus the Čech kernel/cokernel sequence is recovered with an explicit degree-one sign change.

For $i_+ : \{+\} \hookrightarrow X$, the open complement is U_- , whose sections are the exact evaluation A_- . The corresponding sequence is

$$0 \rightarrow \ker r_+ \rightarrow H^0(X, F) \rightarrow A_- \xrightarrow{a_- \mapsto [-r_- a_-]} B/r_+ A_+ \rightarrow H^1(X, F) \rightarrow 0. \quad (\text{TPL6.3})$$

The representative lift $(0, a_-, r_- a_-)$ has differential $(r_- a_-, 0)$, proving the boundary with the negative prescribed in TPL0. Its costalk-to-Čech map is $[v_+] \mapsto [-v_+]$ by (TPL2.5). For i_- , interchange the two stalks; the boundary is $a_+ \mapsto [-r_+ a_+]$, and its costalk-to-Čech map is $[v_-] \mapsto [v_-]$. All signs are therefore explicit.

TPL7. Operators and exchange of branches

A coefficient or dilation operator is a triple

$$S = (S_+, S_-, S_\eta), \quad r_\pm S_\pm = S_\eta r_\pm. \quad (\text{TPL7.1})$$

It acts on (TPL2.3) diagonally in each degree, on the costalks by $S_\pm, S_\eta$, and on the Čech complex by $S_+ \oplus S_-, S_\eta$. Equation (TPL7.1) proves commutation with every differential, and direct substitution proves

$$\delta S_\eta = (\bar{S}_\eta \oplus \bar{S}_\eta) \delta, \quad q(\bar{S}_\eta \oplus \bar{S}_\eta) = \bar{S}_\eta q. \quad (\text{TPL7.2})$$

The bars denote the operators on the stated quotients. Thus all sequences are equivariant for any given commuting family satisfying the actual restriction equations; no weights are assigned to make them commute.

A symmetry covering $+\leftrightarrow -$ has maps $s_+ : A_+ \rightarrow A_-, s_- : A_- \rightarrow A_+, s_\eta : B \rightarrow B$, with

$$r_- s_+ = s_\eta r_+, \quad r_+ s_- = s_\eta r_-. \quad (\text{TPL7.3})$$

Its Čech action is

$$(a_+, a_-) \mapsto (s_- a_-, s_+ a_+), \quad b \mapsto -s_\eta b. \quad (\text{TPL7.4})$$

Indeed its degree-zero action changes d to $-s_\eta d$. On the costalk sum the degree-one action is instead

$$(b_+, b_-) \mapsto (s_\eta b_-, s_\eta b_+). \quad (\text{TPL7.5})$$

It intertwines the diagonal boundary with the generic action s_η , and (TPL2.5) intertwines (TPL7.5) with (TPL7.4). This Čech cover-orientation sign is not identified with the degree of a projective-line map in the different ring-realized scheme calculation.

TPL8. Full nilpotent jets

Let $J_m = \mathbf{C}[t]/(t^m)$, $m \geq 1$, and $N(v) = tv$. Then $N^m = 0$, and $N^{m-1}(1) = t^{m-1} \neq 0$. These formulas specify an actual nilpotent operator on the full multiplicity jet.

Tensor any diagram F^0 over $\mathbf{C}$ with J_m , using restrictions $r_\pm^0 \otimes 1$. Tensoring over a field is exact, as is seen by choosing a basis and taking copies of the original sequence. Consequently all complexes and exact sequences above tensor with J_m , and

$$H^q(X, F^0 \otimes J_m) = H^q(X, F^0) \otimes J_m. \quad (\text{TPL8.1})$$

The full jet operator has

$$\ker N = H^q(X, F^0) \otimes \mathbf{C}t^{m-1}, \quad \text{coker } N = H^q(X, F^0) \otimes (J_m/tJ_m). \quad (\text{TPL8.2})$$

These are named subquotients of the retained jet factor, not replacements for it.

For a recovered $a > 0$ and specified exponent λ ,

$$T_a = a^\lambda \exp((\log a)N) = a^\lambda \sum_{j=0}^{m-1} \frac{(\log a)^j}{j!} N^j. \quad (\text{TPL8.3})$$

The inverse uses $a^{-\lambda}$ and $-\log a$; multiplication of the polynomials cancels all positive-degree coefficients below m , and higher terms vanish. With this action on all three stalks, the restrictions commute with T_a . With different stalk actions, the requirement remains (TPL7.1). Nilpotence supplies no unproved weight relation.

TPL9. Restriction maps given by actual nilpotent powers

Take all three stalks to be J_m , with

$$r_+ = N^a, \quad r_- = N^b, \quad 0 \leq a, b \leq m. \quad (\text{TPL9.1})$$

Put $k = \min(a, b)$, $h = \max(a, b)$. Directly in the basis $1, t, \dots, t^{m-1}$,

$$H^0 K_+ = t^{m-a} J_m, \quad H^1 K_+ = J_m / t^a J_m, \quad H^0 K_- = t^{m-b} J_m, \quad H^1 K_- = J_m / t^b J_m. \quad (\text{TPL9.2})$$

These include zero and full-power cases. For example $a = 0$ gives zero kernel and cokernel, while $a = m$ gives J_m for both.

The global groups and boundary are

$$H^0(X, F) = \{(x, y) : t^a x = t^b y\}, \quad H^1(X, F) = J_m / t^k J_m, \quad (\text{TPL9.3})$$

$$\delta(z) = (-z \bmod t^a J_m, -z \bmod t^b J_m), \quad \ker \delta = t^h J_m. \quad (\text{TPL9.4})$$

The image-sum and intersection identities follow coefficient by coefficient: $(t^a) + (t^b) = (t^k)$, $(t^a) \cap (t^b) = (t^h)$.

For $a \leq b$, the map

$$J_m \oplus t^{m-a} J_m \rightarrow H^0(X, F), \quad (y, u) \mapsto (t^{b-a} y + u, y) \quad (\text{TPL9.5})$$

is an N -equivariant isomorphism. Its inverse is $(x, y) \mapsto (y, x - t^{b-a} y)$; the defining equation puts the second entry in $\ker N^a$. For $b < a$, interchange the two stalks. In particular

$$\dim H^0 = m + k, \quad \dim H^1 = k, \quad \dim \text{im}(\text{res}) = m - h. \quad (\text{TPL9.6})$$

For $a = b = 1$, the generic lift exists exactly for $z \in tJ_m$, and its boundary is minus the diagonal pair of constant terms in $\mathbf{C}^2$. Both costalk quotients and Čech H^1 are $\mathbf{C}$, with their difference map as in TPL5. This is a proved family for arbitrary multiplicity m , not a search over finitely many zeros.

TPL10. Unipotent invariants without dropping kernels or cokernels

Let a general diagram have nilpotent operators $N_\pm, N_B$, with $r_\pm N_\pm = N_B r_\pm$. Put $A = A_+ \oplus A_-$, $N_A = N_+ \oplus N_-$, $d = r_+ - r_-$. The actual unipotent automorphism is $M = \exp N$. The finite polynomial

$$H(N) = \sum_{j \geq 0} \frac{N^j}{(j+1)!} \quad (\text{TPL10.1})$$

has constant term one and therefore a finite-polynomial inverse. Since $M - 1 = NH(N)$,

$$\ker(M - 1) = \ker N, \quad \text{im}(M - 1) = \text{im} N, \quad \text{coker}(M - 1) = \text{coker} N. \quad (\text{TPL10.2})$$

The chain isomorphism $[V \xrightarrow{M-1} V] \rightarrow [V \xrightarrow{N} V]$ is identity in degree zero and $H(N)^{-1}$ in degree one.

If an actual infinite cyclic action is included with generator M , its derived invariants are computed by that two-term complex. This follows by applying $\text{Hom}_{\mathbf{C}[z, z^{-1}]}(-, V)$ to

$$0 \rightarrow \mathbf{C}[z, z^{-1}] \xrightarrow{z-1} \mathbf{C}[z, z^{-1}] \rightarrow \mathbf{C} \rightarrow 0.$$

This resolution is exact because evaluation at $z = 1$ has kernel $(z - 1)$ and multiplication by $z - 1$ is injective. The boundary on V is $M - 1$.

Combining the nilpotent form with the Čech complex gives

$$\boxed{A \xrightarrow{d_0} B \oplus A \xrightarrow{d_1} B, \quad d_0(a) = (da, N_A a), \quad d_1(b, a') = da' - N_B b.} \quad (\text{TPL10.3})$$

The composite is zero because $dN_A = N_B d$. Its complete cohomology is

$$H_{\text{tot}}^0 = \ker d \cap \ker N_A,$$

$$H_{\text{tot}}^1 = \{(b, a') : da' = N_B b\} / \{(da, N_A a) : a \in A\}, \quad H_{\text{tot}}^2 = B / (dA + N_B B). \quad (\text{TPL10.4})$$

Degree two comes from the added cyclic-invariants direction, not from ordinary sheaf $H^2(X, F)$.

The exact invariant/coinvariant sequence is

$$0 \rightarrow \text{coker}(N : H^0(X, F) \rightarrow H^0(X, F)) \xrightarrow{z \mapsto [(0, z)]} H_{\text{tot}}^1 \xrightarrow{[(b, a')] \mapsto [b]} \ker(N : H^1(X, F) \rightarrow H^1(X, F)) \rightarrow 0. \quad (\text{TPL10.5})$$

If $N[b] = 0$ in B/dA , an actual a' with $N_B b = da'$ supplies a cocycle, proving surjectivity. If $b = dv$, subtract the coboundary $(dv, N_A v)$; the result is $(0, a' - N_A v)$, whose second entry lies in $\ker d$. Two such representatives differ by $N_A w$ for $dw = 0$, precisely the displayed cokernel. This proves exactness. Also H_{tot}^2 is the cokernel of N on $H^1(X, F)$.

Apply the same two-term construction to the localization short exact sequence (TPL5.1). All its maps commute with N , so it gives a short exact sequence of total complexes and hence its derived localization triangle. Its boundary is obtained by taking minus the differential of a lift, as fixed in TPL0. No underived invariant kernel is interchanged with cohomology without this total complex.

TPL11. Fréchet qualifications and strictness

Suppose now that $A_{\pm}, B$ are Fréchet spaces and $r_{\pm}$ are continuous. Finite products, kernels, and all the displayed maps are continuous. The space $H^0(X, F) = \ker d$ is closed in $A_+ \oplus A_-$, hence Fréchet. Forgetting topology gives all the already proved algebraic exact sequences.

The quotient topology on

$$H_{\text{alg}}^1(X, F) = B / (r_+ A_+ + r_- A_-) \quad (\text{TPL11.1})$$

is Hausdorff exactly when the actual sum is closed; it is then Fréchet. Each costalk quotient $B/r_{\pm} A_{\pm}$ is Hausdorff and Fréchet exactly when $r_{\pm} A_{\pm}$ is closed.

If both images and their sum are closed, every term of (TPL5.3) is Fréchet and each map is continuous. Algebraic exactness makes all its images kernels of continuous maps to Hausdorff spaces, hence closed. The induced maps from quotient by kernel onto image are topological isomorphisms by the open mapping theorem. Thus the sequence is strict in this case.

Without those facts, Hausdorff reduction changes the object:

$$H_{\text{red}}^1(X, F) = B / \overline{r_+ A_+ + r_- A_-}. \quad (\text{TPL11.2})$$

The natural map from (TPL11.1) has kernel

$$\overline{r_+ A_+ + r_- A_-} / (r_+ A_+ + r_- A_-). \quad (\text{TPL11.3})$$

The reduced branch boundary has kernel

$$\overline{r_+ A_+} \cap \overline{r_- A_-}, \quad (\text{TPL11.4})$$

not necessarily the actual liftable intersection (TPL5.5).

Nilpotence alone does not imply closed image. On $\ell^2 \oplus \ell^2$, set $S(x_n) = (x_n/n)$ and $N(x, y) = (Sy, 0)$. Then $N^2 = 0$, while its image is $S(\ell^2) \oplus 0$, proper and dense in $\ell^2 \oplus 0$. Every finite sequence belongs to $S(\ell^2)$, proving density; the sequence $(1/n)$ has the non-square-summable putative preimage $(1, 1, \dots)$, proving properness.

For a finite jet factor J_m , however, tensoring a Fréchet space with J_m is a finite product. Multiplication by t^a has a closed coordinate image and a continuous coordinate section onto that image. Thus the special

family TPL9 remains strict when each scalar coefficient is replaced by a fixed Fréchet space. In a general diagram tensored with J_m , the original restriction-map strictness questions remain.

For the total complex TPL10, the displayed formulas always compute algebraic kernels and quotients. Its Hausdorff cohomology requires closed images of the actual total differentials. Closedness of all cohomological maps used in an invariant long exact sequence must likewise be checked; neither the existence of a nilpotent operator nor the name “jet” proves it.

TPL12. An exact demonstration of the closure distinction

Take $A_+ = A_- = B = \ell^2$, $r_+ = S$ as above, and $r_- = 1$. Then

$$H^0(X, F) = \{(x, Sx) : x \in \ell^2\}, \quad H_{\text{alg}}^1(X, F) = 0, \quad (\text{TPL12.1})$$

because $r_- A_- = B$. The actual generic restriction image is $S(\ell^2)$, and

$$\delta(b) = ([-b]_{B/S(\ell^2)}, 0). \quad (\text{TPL12.2})$$

For $b = (1/n)$, the class is nonzero: its plus lift would be $(1, 1, \dots) \notin \ell^2$, although its minus lift exists. But $\overline{S(\ell^2)} = B$, so both reduced costalk quotients are zero and the reduced boundary vanishes on every b . Replacing image by closure has therefore lost the actual lifting question, even though ordinary Čech H^1 is already zero.

This example proves a statement about these explicitly specified continuous maps. It is not an obstruction asserted for τ , the user’s global arithmetic, or the original zeta quotient. The programme’s actual restrictions must be inserted into TPL5 with their domains, images and topology before making such a conclusion.

TPL13. Exact comparison with the actual CSL restriction sheaf

After the independent derivation, the complete file CC_SUPPORTED_LOCALIZATION_AND_WEIGHT_MAPS.md, CSL0–CSL9, in the companion tau_weight_cohomology_20260924 directory was read. Its actual substitution into the general diagram is

$$\begin{aligned} A_+ = V_+ = S \oplus \mathbf{C}^2, \quad A_- = V_- = S \oplus \mathbf{C}^2, \quad B = A, \\ r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h. \end{aligned} \quad (\text{TPL13.1})$$

The maps Σ, R , all factors, moments and coefficient actions remain those in CSL1; the substitution does not replace their algebraic image by its closure.

CSL uses positive costalk differentials. The precise comparison of its two-support complex with this note is

$$\begin{aligned} S_K : [V_+ \oplus V_- \xrightarrow{(r_+, r_-)} A^2] \longrightarrow [V_+ \oplus V_- \xrightarrow{(-r_+, -r_-)} A^2], \\ S_K^0 = 1, \quad S_K^1 = -1. \end{aligned} \quad (\text{TPL13.2})$$

It is a chain isomorphism: $S_K^1(r_+, r_-) = (-r_+, -r_-)S_K^0$; its inverse is itself. The one-support versions are the same map on $[V_{\pm} \rightarrow A]$.

Both notes use the Čech complex with $d = r_+ - r_-$. CSL’s support-to-Čech map is $b_+ - b_-$; ours is $b_- - b_+$. They commute with (TPL13.2), because

$$(-b_-) - (-b_+) = b_+ - b_-. \quad (\text{TPL13.3})$$

Likewise the CSL boundary $b \mapsto ([b], [b])$ is sent by S_K^1 to $b \mapsto ([-b], [-b])$, exactly the boundary in TPL5 under its explicitly fixed cone convention. Thus the entire localization triangle comparison retains both the costalk differential and the connecting-arrow sign. Merely changing the differential while retaining the same numerical boundary would not be this comparison.

CSL1 proves $R\Sigma h = \Sigma\hat{h}$, hence both restriction images equal the actual subspace $J = \Sigma(S)$. Substituting into TPL5 gives kernel J , the supported quotients $(A/J)^2$, and the last quotient A/J . Under (TPL13.2) this is exactly CSL4.3–CSL4.5, including its positive diagonal and positive-first-coordinate difference. The four endpoint lines are the kernels of r_+ and r_- , each of dimension two, by CSL1’s injectivity of Σ . The graph of Fourier transform in CSL2.4 is precisely the fibre product (TPL1.2).

The explicit lift in CSL3.4 agrees with TPL6: its boundary is represented by $r_-(h, d_0, d_1) = R\Sigma h = \Sigma\hat{h}$, which belongs to the plus restriction image. Both the negative representative in our costalk and the positive representative in CSL therefore vanish. The Fourier formula constructs a continuous lift, rather than inferring it from a reduced cokernel.

The involution is $s_\eta = R$ and swaps the two chart coefficient copies. Formula (TPL7.4) is therefore exactly $w_D^1 = -R$, and (TPL7.5) is exactly the swapped R -action on the two costalk factors. The degree-one scalar -1 in (TPL13.2) commutes with those actions and with every coefficient dilation. Thus the two equivariant localization calculations agree as well.

Finally, the exact algebraic-to-Hausdorff map retains its kernel C/J , where $C = \overline{J}$. TPL11 neither proves nor assumes $J = C$. A subsequent proof of that equality for the actual Schwartz summation image can be inserted here and then makes the corresponding quotient comparison an isomorphism. Until such a proof is checked, the two receivers remain the distinct objects and exact map already specified by CSL2.

The Connes–Consani sheaf and the original-zeta Hausdorff cohomology

Independent derivation, 24 September 2026. Proof locators CS0–CS11, including CS2A.

CS0. Construction stage, source, and precise conclusion

This calculation constructs a sheaf on the three-point geometric realization used by Connes–Consani, identifies its trivial finite-unit sector over the already specified field $\mathbb{Q}$, and proves that the Hausdorff quotient of its first ordinary sheaf cohomology is the existing original-zeta quotient $\mathcal{Q}$. It first retains the exact comparison with ordinary cohomology. The subsequent complete image theorem in CS11 proves that this cohomology is already Hausdorff. It is a construction on arithmetic coefficient spaces after that arithmetic is available. It does not define arithmetic operations, coordinates, a midpoint, or a metric on $\tau\langle Z_1; \text{no } Z_2 \rangle$, and does not identify the authors’ generic point with the user’s entire supported construction.

The connected user reconstruction and its correction-precedence table were read before this calculation, together with the full governing passages U126–U128, U135–U141, WU050–WU055, WU057–WU063 and the pertinent branch-comparison attachment. Those passages require retaining the complete-history dependency, the retraction of addition at τ , and the separate branch counters. Here the two charts remain separately typed; their restrictions form a difference in a sheaf complex, not a pooled arithmetic return measure. No implication from this difference to common weights is inserted.

The original author source actually read is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, retained as sources/CC_0903_2024_v3/author_source/announc3.tex. Section 5, source lines 1376–1667, was read in full. Its exact source labels used here are:

- `cechprop`, `vanish`, `cechequal`, `longexact1`: affine acyclicity and supported cohomology;
- `projspace1`, `projspace2`, `projback`: the three-point base and the projective adèle-class projection;
- `fonction1`, `fonction3`, `sheaf1`, `SCK`, `restmaps`, `Smap`: the source spaces and restrictions;
- `bord`, `lemsurject`, `hzero`, `fourier`, `poisson2`, `poisson3`, `coinv`: the Čech differential and Fourier graph;
- `rep1`, `liftw`, `compinv`, `court`, `surject`: the unshifted action, Weyl lift, and single-support comparison.

The original text invokes Meyer’s Lemma 5.4 to prove injectivity of its full adèlic summation map on Hausdorff coinvariants. This note does not claim to have reconstructed that general theorem. CS2–CS4 prove the needed injectivity directly for the trivial finite-unit sector over $\mathbb{Q}$. A domain qualification matters: source `fonction3` does not explicitly type its generating f . We adopt and state the full-difference reading, in which arbitrary adèlic Schwartz functions generate differences already lying in the moment-null subspace by invariance of both endpoint functionals. CS2–CS3 prove its compatibility with the source’s stated Fourier-graph result. Restricting the generating functions themselves to the smaller moment-null subspace changes the quotient; CS2A proves that this alternative contradicts that graph result for the specified topology. This is a proved interpretation and comparison, not a claim that the source writes an omitted domain explicitly.

The existing programme theorem used below is GLOBAL_MELLIN_SYNTHESIS.md, S1–S7, in the adjacent quantum_tau_programme_bridge_20260924 directory: the topological Mellin isomorphism and the equality of the closed original source image with the ideal of every actual nontrivial-zero jet. Its full formula, exceptional values, and exact role are restated in CS6–CS7. The global synthesis proof remains its own proof, rather than being attributed to Connes–Consani.

CS1. The adèlic source and its compact unit projection

Write

$$\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \mathbb{A}_f, \quad \widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p, \quad K = \widehat{\mathbb{Z}}^{\times}. \quad (\text{CS1.1})$$

Use Lebesgue measure at the real place and finite additive Haar measure with $\text{vol}(\widehat{\mathbb{Z}}) = 1$. The adèlic Bruhat–Schwartz space is the locally convex inductive limit of the spaces of finite sums $f_i(v)\phi_i(x_f)$, where $f_i \in \mathcal{S}(\mathbb{R})$

and the ϕ_i belong to one finite-dimensional space of locally constant, compactly supported functions on $\mathbb{A}_f$. On each such stage the topology is a finite direct sum of the ordinary real Schwartz topology.

Put

$$\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 = \left\{ F : F(0) = 0, \int_{\mathbb{A}_{\mathbb{Q}}} F(x) dx = 0 \right\}, \quad \mathcal{R} = \text{span}_{\mathbb{C}}\{F - F(q \cdot) : F \in \mathcal{S}(\mathbb{A}_{\mathbb{Q}}), q \in \mathbb{Q}^{\times}\},$$

$$D = \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 / \overline{\mathcal{R}}. \quad (\text{CS1.2})$$

The closure is taken in $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ with its subspace topology. Rational scaling preserves both endpoint functionals, since $|q|_{\mathbb{A}} = 1$, so every displayed difference belongs to that moment-null space even when its generating F does not. This is the full-difference reading of the source's "subspace of $\mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0$ generated by the differences." The source line does not separately write a domain after its f ; the distinction and its mathematically forced consequence are recorded in CS2A.

The group K acts on finite coordinates. Its normalized Haar average P_K is continuous on this source: each test function is fixed by an open subgroup of K , so its average is a finite average on that test-function stage. The average commutes with rational scaling and preserves both endpoint conditions. It consequently descends to a continuous idempotent on D .

Every idèle class has a unique representative (u, k) , with $u > 0$ and $k \in K$. To see this, multiply its finite component by the positive rational with opposite finite valuations, and then use the sign in $\mathbb{Q}^{\times}$ to make its real component positive. Uniqueness follows because a rational number with every finite valuation zero is ± 1 , and positivity excludes the minus sign. Thus

$$C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^{\times} / \mathbb{Q}^{\times} \cong \mathbb{R}_{>0}^{\times} \times K, \quad |(u, k)| = u. \quad (\text{CS1.3})$$

This is a coordinate on the authors' already constructed idèle-class group.

CS2. An exact topological identification of the source sector

Define

$$\mathcal{S}_0^{\text{even}} = \left\{ f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), \quad f(0) = 0, \quad \int_{\mathbb{R}} f(v) dv = 0 \right\},$$

$$j : \mathcal{S}_0^{\text{even}} \longrightarrow D, \quad j(f) = [f \otimes 1_{\widehat{\mathbb{Z}}}]. \quad (\text{CS2.1})$$

This is continuous, and its image is fixed by K .

We construct its continuous inverse on $P_K D$. A K -invariant finite Schwartz function is a finite linear combination of

$$1_{d\widehat{\mathbb{Z}}}, \quad d \in \mathbb{Q}_{>0}. \quad (\text{CS2.2})$$

Here $d\widehat{\mathbb{Z}}$ is the product of the balls $p^{v_p(d)}\mathbb{Z}_p$. To prove the claim, choose the finitely many primes on which the function is nonstandard. At each such prime, compact support bounds valuations below and local constancy makes the function constant on one sufficiently small ball around zero. Unit invariance leaves a finite list of valuation shells and that final ball. Each shell is the difference of two nested ball indicators. Taking tensor products gives (CS2.2).

These ball indicators are linearly independent. In one local factor, evaluation successively on the shells of largest and then smaller absolute values proves independence of a finite list of nested balls. Finite tensor products of independent lists are independent, and every proposed relation involves such a finite product. This also proves that the following definition is independent of its expansion.

For $f^{\text{ev}}(v) = (f(v) + f(-v))/2$, define on K -averaged finite sums

$$B(f \otimes 1_{d\widehat{\mathbb{Z}}})(v) = f^{\text{ev}}(dv), \quad B(F) := B(P_K F) \quad \text{on the full source.} \quad (\text{CS2.3})$$

On each finite stage this is a finite sum of fixed real dilations and the even projection, hence is continuous into $\mathcal{S}^{\text{even}}(\mathbb{R})$. The defining inductive-limit topology proves continuity on the full source.

The endpoint identities are exact:

$$B(F)(0) = F(0), \quad \int_{\mathbb{R}} B(F)(v) dv = \int_{\mathbb{A}_{\mathbb{Q}}} F(x) dx. \quad (\text{CS2.4})$$

For a ball tensor the second identity follows from

$$\text{vol}(d\widehat{\mathbb{Z}}) = \prod_p p^{-v_p(d)} = d^{-1}, \quad \int_{\mathbb{R}} f^{\text{ev}}(dv) dv = d^{-1} \int_{\mathbb{R}} f(v) dv. \quad (\text{CS2.5})$$

The first identity is immediate; averaging does not change either endpoint functional. Thus B maps the source in (CS1.2) into $\mathcal{S}_0^{\text{even}}$.

For $q \in \mathbb{Q}^\times$, direct substitution on a ball tensor gives

$$B((f \otimes 1_{d\widehat{\mathbb{Z}}})(q \cdot))(v) = f^{\text{ev}}(dv) = B(f \otimes 1_{d\widehat{\mathbb{Z}}})(v). \quad (\text{CS2.6})$$

Indeed the new finite ball is $(d/|q|)\widehat{\mathbb{Z}}$, while the new even real function is $f^{\text{ev}}(|q|v)$. Since P_K commutes with this scaling, (CS2.6) holds on the full source. Continuity makes B vanish on $\overline{\mathcal{R}}$, so it descends to a continuous map $\bar{B} : D \rightarrow \mathcal{S}_0^{\text{even}}$.

The two compositions are

$$\bar{B}j = \text{id}, \quad j\bar{B} = P_K \quad \text{on } D. \quad (\text{CS2.7})$$

For the second identity, individual tensors need not themselves be moment-null, so first work with differences before taking a quotient class. Put $G = f \otimes 1_{d\widehat{\mathbb{Z}}}$ and $H_d = G(d \cdot) = f(dv) \otimes 1_{\widehat{\mathbb{Z}}}$. Let $\tilde{j}$ denote tensoring with $1_{\widehat{\mathbb{Z}}}$ on all even Schwartz functions. The exact identity is

$$G - \tilde{j}(BG) = (G - G(d \cdot)) + \frac{1}{2}(H_d - H_d(-\cdot)) \in \mathcal{R}. \quad (\text{CS2.7a})$$

Both parenthesized differences have zero endpoint moments, whether or not their generating functions do. Sum (CS2.7a) over a radial-ball expansion of $P_K F$, with F moment-null. Only then take its class in D . This proves $j\bar{B}[F] = P_K[F]$, with every class taken on its actual domain.

Finally $P_K D = D^K$. One inclusion follows from averaging. For a fixed quotient class, choose a representative F . Its K -orbit lies in a finite averaging orbit modulo an open subgroup. Averaging its equal quotient classes therefore gives the same class. We have proved the actual topological isomorphism

$$\boxed{j : \mathcal{S}_0^{\text{even}} \xrightarrow{\sim} D^K, \quad j^{-1} = \bar{B}|_{D^K}.} \quad (\text{CS2.8})$$

In particular, this identification neither discards the closed coinvariant relation nor assumes it has no kernel.

CS2A. Why the generating domain must be retained

Let $\mathcal{R}_{\text{null}}$ denote instead the span of $F - F(q \cdot)$ with F itself moment-null. Then $\mathcal{R}_{\text{null}} \subset \mathcal{R}$, and there is a natural continuous surjection

$$D_{\text{null}} := \mathcal{S}(\mathbb{A}_{\mathbb{Q}})_0 / \overline{\mathcal{R}_{\text{null}}} \longrightarrow D. \quad (\text{CS2A.1})$$

This alternative quotient is genuinely larger. Here is a direct proof which also identifies the source of the difference.

For a prime p , expand a K -averaged Schwartz function as $\sum_d f_d \otimes 1_{d\widehat{\mathbb{Z}}}$, and define the two-component continuous functional

$$\ell_p(F) = \sum_d v_p(d) \left(f_d(0), d^{-1} \int_{\mathbb{R}} f_d(v) dv \right), \quad \ell_p(F) := \ell_p(P_K F) \text{ before averaging.} \quad (\text{CS2A.2})$$

The sum is finite, its expansion is unique by CS2, and continuity follows on each finite test-function stage. Put $\epsilon(F) = (F(0), \int_{\mathbb{A}} F)$. On ball tensors, scaling changes the index d to $d/|q|$ and leaves the two displayed endpoint values unchanged. Hence

$$\ell_p(F(q \cdot)) = \ell_p(F) - v_p(|q|)\epsilon(F). \quad (\text{CS2A.3})$$

It follows that ℓ_p vanishes on $\overline{\mathcal{R}_{\text{null}}}$, but not on all of $\mathcal{R}$. Indeed, take any even Schwartz f and

$$G = f \otimes 1_{\widehat{\mathbb{Z}}} - (f \otimes 1_{\widehat{\mathbb{Z}}})(p \cdot).$$

Then G is moment-null, its summation is zero by rational reindexing, and

$$\ell_p(G) = \left(f(0), \int_{\mathbb{R}} f \right). \quad (\text{CS2A.4})$$

The endpoint map on even Schwartz functions is onto $\mathbb{C}^2$: for example the functions $e^{-\pi v^2}$ and $v^2 e^{-\pi v^2}$ have endpoint pairs $(1, 1)$ and $(0, 1/(2\pi))$. Thus these are nonzero classes in the alternative quotient whenever that pair is nonzero.

In particular, the injective summation and Fourier-graph claim in the original source cannot hold on D_{null} with this topology: (CS2A.4) supplies an explicit kernel. The full-difference reading in CS1 removes exactly these displayed differences, and CS2–CS3 prove the needed sector result for it. Even under the alternative reading, the map (CS2A.1) and the commuting summation maps give the same summation image and the same Hausdorff first-cohomology receiver. Its degree-zero kernel contains additional classes, so its H^0 must not be described by (CS5.3).

The complete difference between the two quotient spaces can also be computed. Give

$$V = \bigoplus_{p \text{ prime}} \mathbb{C}^2 \quad (\text{CS2A.5})$$

the locally convex direct-sum topology, and put $\mathcal{L}(F) = (\ell_p(F))_p$. A fixed test-function stage has only finitely many relevant primes in its radial-ball expansion. Thus $\mathcal{L}$ has finite support on every test function and is continuous into V .

Choose the continuous endpoint section $\sigma : \mathbb{C}^2 \rightarrow \mathcal{S}(\mathbb{A}_{\mathbb{Q}})$ given by tensoring

$$\sigma(a, b)_{\infty}(v) = a(1 - 2\pi v^2)e^{-\pi v^2} + b 2\pi v^2 e^{-\pi v^2} \quad (\text{CS2A.6})$$

with $1_{\widehat{\mathbb{Z}}}$. The Gaussian moments give $\epsilon(\sigma(a, b)) = (a, b)$. Define the continuous finite-sum map

$$\mathcal{T}(v) = \sum_p (\sigma(v_p) - \sigma(v_p)(p \cdot)) \in \mathcal{R}, \quad \mathcal{L}\mathcal{T} = \text{id}_V. \quad (\text{CS2A.7})$$

The second identity is (CS2A.3) at every prime.

Write $r_q(F) = F - F(q \cdot)$. Modulo $\mathcal{R}_{\text{null}}$, this depends only on $\epsilon(F)$, because subtracting $\sigma(\epsilon F)$ makes its argument moment-null. Also

$$r_{qr}(F) \equiv r_q(F) + r_r(F) \pmod{\mathcal{R}_{\text{null}}}. \quad (\text{CS2A.8})$$

Indeed their difference is a rational difference with generating function $F(q \cdot) - F$, which is moment-null. The sign $q = -1$ contributes zero in this quotient, since $r_{-1}(F) = \frac{1}{2}r_{-1}(F - F(-\cdot))$. Unique factorization, including negative prime exponents, now gives

$$r_q(F) \equiv \sum_p v_p(|q|) r_p(\sigma(\epsilon F)) = \mathcal{T}\mathcal{L}(r_q(F)) \pmod{\mathcal{R}_{\text{null}}}. \quad (\text{CS2A.9})$$

Thus $(I - \mathcal{T}\mathcal{L})\mathcal{R} \subset \mathcal{R}_{\text{null}}$. Continuity gives the same inclusion for closures. It follows that the exact quotient map has the topological splitting

$$\boxed{D_{\text{null}} \xrightarrow{\sim} D \oplus V, \quad [F]_{\text{null}} \mapsto ([F], \mathcal{L}F).} \quad (\text{CS2A.10})$$

For injectivity, if $F \in \overline{\mathcal{R}}$ and $\mathcal{L}F = 0$, the closure inclusion puts F in $\overline{\mathcal{R}_{\text{null}}}$. For surjectivity, start with any representative F of the required D -class and replace it by $F + \mathcal{T}(v - \mathcal{L}F)$. A continuous inverse sends $([F], v)$ to $[F - \mathcal{T}\mathcal{L}F + \mathcal{T}v]_{\text{null}}$. Independence of the representative and continuity follow from the closure inclusion and the defining quotient topology. This proves the full alternative receiver and its connecting maps; the additional summand records two endpoint directions at every prime.

CS3. The overlap, Fourier transform, and all rational summands

On the trivial K -sector, the source overlap space $\bigcap_{\beta \in \mathbb{R}} \mu^\beta \mathcal{S}(C_{\mathbb{Q}})$ is the Fréchet space

$$H = \left\{ h \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(h) := \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j h(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{CS3.1})$$

To verify both the set and topology, use $x = \log u$. The ordinary multiplicative Schwartz seminorms are the polynomially weighted derivative seminorms in x ; the source additionally requires every factor $e^{\beta x}$. Derivatives of $e^{\beta x} h(e^x)$ are finite sums of $e^{\beta x} \partial_x^\ell h(e^x)$. Polynomial factors in x are bounded by a constant times $e^{|x|}$, and $e^{N|x|} \leq e^{Nx} + e^{-Nx} \leq 2e^{N|x|}$. These identities bound every source seminorm by finitely many (CS3.1) seminorms and conversely. Integer N suffice for every real β .

For $f \in \mathcal{S}_0^{\text{even}}$, summation of its actual adèlic representative at $(u, 1)$ gives

$$Sf(u) := \Sigma(jf)(u) = \sum_{q \in \mathbb{Q}^\times} f(qu) 1_{\widehat{\mathbb{Z}}}(q) = \sum_{n \in \mathbb{Z} \setminus \{0\}} f(nu) = 2 \sum_{n \geq 1} f(nu). \quad (\text{CS3.2})$$

The criterion in the middle step is $q \in \widehat{\mathbb{Z}} \cap \mathbb{Q} = \mathbb{Z}$. Both rational signs are present, and the final factor two is retained.

Use the real Fourier convention

$$\hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv. \quad (\text{CS3.3})$$

Choose the corresponding character of $\mathbb{A}_{\mathbb{Q}}/\mathbb{Q}$ and self-dual measures; at finite places the annihilator of $\widehat{\mathbb{Z}}$ is itself, with volume one. Direct Fourier integration of its indicator is therefore the same indicator. Consequently the adèlic Fourier transform of $f \otimes 1_{\widehat{\mathbb{Z}}}$ is $\hat{f} \otimes 1_{\widehat{\mathbb{Z}}}$. Fourier transformation preserves both vanishing endpoint conditions and evenness, and $\hat{\hat{f}} = f$ on this even space.

Set

$$(Rh)(u) = u^{-1} h(1/u), \quad R^2 = \text{id}. \quad (\text{CS3.4})$$

For an arbitrary even real Schwartz function, Poisson summation retains the exact endpoint identity

$$Sf(u) - u^{-1} S\hat{f}(1/u) = u^{-1} \int_{\mathbb{R}} f(v) dv - f(0). \quad (\text{CS3.5})$$

Indeed the full lattice sums have their zero summands $f(0)$ and $u^{-1} \hat{f}(0)$, and $\hat{f}(0) = \int f$. For our actual moment-null source both terms on the right vanish, so

$$S\hat{f} = RSf, \quad RS\hat{f} = Sf. \quad (\text{CS3.6})$$

This is the exact plus/minus gluing identity, including the factor u^{-1} .

The map $S : \mathcal{S}_0^{\text{even}} \rightarrow H$ is continuous. At $u \geq 1$, each logarithmic derivative of the series is bounded by a sufficiently high Schwartz seminorm times a convergent sum $\sum_{n \geq 1} n^{-M}$, $M > 1$. For $u \leq 1$, (CS3.6) expresses the answer through the Fourier transform at $1/u \geq 1$, multiplied by u^{-1} . Fourier continuity and one additional endpoint weight prove the required bounds. The same arguments justify termwise differentiation.

It is also injective. For $\Re s > 1$, absolute convergence and the substitution $v = nu$ give

$$\int_0^\infty Sf(u) u^s \frac{du}{u} = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CS3.7})$$

If $Sf = 0$, choose any real $c > 1$. Since $\zeta(c + it) \neq 0$ on this half-plane, the Mellin transform of $f|_{\mathbb{R}_{>0}}$ vanishes on the line $c + it$. It is the Fourier transform of the integrable function $x \mapsto e^{cx} f(e^x)$, with Fourier frequency sign fixed by e^{itx} . Fourier injectivity gives $f(e^x) = 0$; evenness gives $f = 0$ on $\mathbb{R}$. This proves the source-sector injectivity independently of the full adèlic coinvariant theorem.

CS4. The actual three-point sheaf

Let

$$X = \{0, \eta, \infty\}, \quad U_+ = \{0, \eta\}, \quad U_- = \{\eta, \infty\}, \quad U = \{\eta\}. \quad (\text{CS4.1})$$

The nonempty opens are U, U_+, U_-, X . The point η is generic; $0, \infty$ are closed. These are the authors' base labels.

For each chart retain its two separate endpoint lines,

$$E_+ = \mathbb{C}e_{+,0} \oplus \mathbb{C}e_{+,1}, \quad E_- = \mathbb{C}e_{-,0} \oplus \mathbb{C}e_{-,1}, \quad A_{\pm} = \mathcal{S}_0^{\text{even}} \oplus E_{\pm}. \quad (\text{CS4.2})$$

The index 0 labels the value-at-zero endpoint functional, and index 1 labels the integral endpoint functional. These are the two explicit summands in the source definition of $\mathcal{S}(M)$, separately on each chart.

Define

$$\begin{aligned} \mathcal{F}(U_+) &= A_+, \quad \mathcal{F}(U_-) = A_-, \quad \mathcal{F}(U) = H, \\ r_+(f, e_+) &= Sf, \quad r_-(h, e_-) = RSh = \hat{S}h. \end{aligned} \quad (\text{CS4.3})$$

Every endpoint restriction is zero, exactly as in restmaps. Define

$$\mathcal{F}(X) = \{(a, b) \in A_+ \oplus A_- : r_+a = r_-b\}, \quad \mathcal{F}(\emptyset) = 0. \quad (\text{CS4.4})$$

Use the evident restrictions from X . The only nontrivial sheaf gluing condition is the cover $X = U_+ \cup U_-$, and it is precisely (CS4.4). A cover of U_+ must contain U_+ , since no smaller open contains 0; the same holds for U_- and ∞ . The overlap is a singleton. Thus these formulas construct a genuine sheaf, including its topology on every section space.

Under j on the two source coordinates and (CS1.3) on the overlap, this is the P_K -image sheaf of the actual Connes–Consani sheaf. Averaging commutes with summation, restrictions, and Fourier transform, and is the identity on the endpoint lines. It is therefore a sheaf idempotent and gives a direct-summand sheaf. This precise use of “trivial finite-unit sector” does not silently identify fixed classes in an arbitrary non-Hausdorff quotient with quotients of fixed representatives.

CS5. Ordinary cohomology, including every endpoint line

Fix the ordered cover (U_+, U_-) . Its complex is

$$C^0 = A_+ \oplus A_-, \quad C^1 = H, \quad d(a, b) = r_+a - r_-b. \quad (\text{CS5.1})$$

This computes ordinary sheaf cohomology in the category of complex vector-space sheaves, before taking closures. Here is an elementary derived-functor verification. A sheaf on this space is exactly a diagram $D_+ \rightarrow H \leftarrow D_-$. Global sections are the kernel of the displayed difference. Short exact sequences of diagrams give short exact sequences of these two-term complexes and their cohomology long exact sequences. The degree-one functor is effaceable: embed the diagram into one with chart spaces $D_+ \oplus H, D_- \oplus H$, overlap H , and restriction maps the projections to H ; embed a as (a, r_+a) , b as (b, r_-b) , and act identically on H . The new difference map is onto H , so its degree-one cohomology vanishes. Thus this effaceable cohomology functor is the right-derived global-section functor. There are no higher terms, and

$$H^i(X, \mathcal{F}) = 0 \quad (i \geq 2). \quad (\text{CS5.2})$$

By (CS3.6) and injectivity of S , $d((f, e_+), (h, e_-)) = 0$ if and only if $f = \hat{h}$, equivalently $h = \hat{f}$. Consequently

$$H^0(X, \mathcal{F}) = \{((f, e_+), (\hat{f}, e_-)) : f \in \mathcal{S}_0^{\text{even}}, e_+ \in E_+, e_- \in E_-\}. \quad (\text{CS5.3})$$

This is a topological isomorphism from $\mathcal{S}_0^{\text{even}} \oplus E_+ \oplus E_-$ onto the kernel, because Fourier transformation and the coordinate projections are continuous. There are four independent endpoint lines, rather than two endpoint lines identified across the charts.

Write

$$J = S(\mathcal{S}_0^{\text{even}}) \subset H. \quad (\text{CS5.4})$$

The minus restriction has the same image J , since Fourier transformation is a bijection of the source. Moreover $d((f, 0), (0, 0)) = Sf$. Therefore

$$\text{im } d = J, \quad \boxed{H_{\text{alg}}^1(X, \mathcal{F}) = H/J}. \quad (\text{CS5.5})$$

The subscript emphasizes that the ordinary cokernel has not been replaced by a closed-image quotient. Give it the quotient locally convex topology. Its closure of zero is $\bar{J}/J$, and its maximal Hausdorff quotient is

$$H_{\text{Haus}}^1(X, \mathcal{F}) := H/\bar{J}, \quad 0 \longrightarrow \bar{J}/J \longrightarrow H/J \longrightarrow H/\bar{J} \longrightarrow 0. \quad (\text{CS5.6})$$

This sequence is algebraically exact without an additional image theorem. CS11 subsequently proves $J = \bar{J}$, so its left term is zero for this particular source.

CS6. The exact comparison with the existing source image

The earlier real source map, with its original factors, is

$$\mathcal{E}f(u) = u^{1/2} \sum_{n \geq 1} f(nu), \quad \mathcal{C} = \overline{\mathcal{E}(\mathcal{S}_0^{\text{even}})}^{\mathcal{A}}, \quad (\text{CS6.1})$$

where $\mathcal{A}$ has exactly the weighted derivative seminorms (CS3.1). Although H and $\mathcal{A}$ have the same underlying class of functions, their receiving formulas differ. Define the comparison explicitly:

$$U : H \longrightarrow \mathcal{A}, \quad (Uh)(u) = \frac{1}{2}u^{1/2}h(u), \quad (U^{-1}k)(u) = 2u^{-1/2}k(u). \quad (\text{CS6.2})$$

For $D_u = u\partial_u$,

$$D_u^j(u^a h) = u^a \sum_{\ell=0}^j \binom{j}{\ell} a^{j-\ell} D_u^\ell h. \quad (\text{CS6.3})$$

Since $(u^N + u^{-N})u^{\pm 1/2} \leq 2(u^{N+1} + u^{-(N+1)})$, these give the concrete continuity estimates

$$\begin{aligned} p_{N,j}(Uh) &\leq \sum_{\ell=0}^j \binom{j}{\ell} (1/2)^{j-\ell} p_{N+1,\ell}(h), \\ p_{N,j}(U^{-1}k) &\leq 4 \sum_{\ell=0}^j \binom{j}{\ell} (1/2)^{j-\ell} p_{N+1,\ell}(k). \end{aligned} \quad (\text{CS6.4})$$

Both compositions in (CS6.2) are the identity, so this is a topological isomorphism. Formula (CS3.2) now gives, without removing its original scalar or power,

$$Sf = 2u^{-1/2}\mathcal{E}f, \quad USf = \mathcal{E}f. \quad (\text{CS6.5})$$

Topological isomorphisms commute with closure. It follows that

$$\boxed{\bar{J}^H = U^{-1}\mathcal{C} = \{2u^{-1/2}k(u) : k \in \mathcal{C}\}}. \quad (\text{CS6.6})$$

Thus the requested source-image comparison includes both topologies and both inverses, rather than only a formal equality of summation expressions.

CS7. Raw Mellin transform, original zeta, and the Hausdorff quotient

Retain the two Mellin maps separately:

$$\mathcal{M}_0 h(s) = \int_0^\infty h(u) u^s \frac{du}{u}, \quad \mathcal{M} k(s) = \int_0^\infty k(u) u^{s-1/2} \frac{du}{u}. \quad (\text{CS7.1})$$

They satisfy

$$\mathcal{M}_0 h = 2\mathcal{M}(Uh), \quad \Theta := \mathcal{M}U = \frac{1}{2}\mathcal{M}_0. \quad (\text{CS7.2})$$

The previous theorem S1 proves $\mathcal{M} : \mathcal{A} \rightarrow \mathcal{B}$ is a topological isomorphism, where

$$\mathcal{B} = \left\{ F \text{ entire} : b_{A,M}(F) := \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for every } A, M \geq 0 \right\}. \quad (\text{CS7.3})$$

Consequently Θ is a topological isomorphism. For example its full inverse, on the real line $c = 1/2$, is

$$(\Theta^{-1}F)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CS7.4})$$

This is $2u^{-1/2}$ times S1's inverse with coefficient $1/(2\pi)$; both factors remain explicit.

Equation (CS3.7) is the raw arithmetic identity

$$\mathcal{M}_0 S f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}, \quad \Theta S f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{CS7.5})$$

For the retained source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad (\text{CS7.6})$$

S2's exact calculation gives

$$\Theta S f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 S f_0(s) = 2F_0(s) = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CS7.7})$$

The entire continuation and exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad \mathcal{M}_0 S f_0(0) = \mathcal{M}_0 S f_0(1) = \frac{1}{4}, \quad (\text{CS7.8})$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{CS7.9})$$

The raw Mellin value at every such point is $2F_0(-2r)$. This retains the endpoint contributions and the Gamma-pole/trivial-zero cancellation. It does not make the completed product a replacement for the original zeta function.

For additional endpoint precision, let $a_f(s) = \int_0^\infty f(v) v^s dv/v$. Its meromorphic continuation has possible simple poles at $-2r$, $r \geq 1$, with residues $f^{(2r)}(0)/(2r)!$, obtained by subtracting the real Taylor polynomial on $0 < v < 1$. At the original trivial zeros,

$$\mathcal{M}_0 S f(-2r) = 2\zeta'(-2r) \frac{f^{(2r)}(0)}{(2r)!}. \quad (\text{CS7.10})$$

At the two endpoints, $a_f(1) = \frac{1}{2} \int_{\mathbb{R}} f = 0$, the zeta residue at 1 is one, and $\zeta(0) = -1/2$, so

$$\mathcal{M}_0 S f(1) = 2a'_f(1), \quad \mathcal{M}_0 S f(0) = -a_f(0). \quad (\text{CS7.11})$$

Thus the source's two zero moment conditions do not erase these receiver values.

Let $\mathcal{Z}$ be the set of distinct actual nontrivial zeros of the original ζ , with their full multiplicities m_ρ , and let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CS7.12})$$

S5 proves $\mathcal{MC} = \mathcal{J}$ on these exact spaces. Combining this established theorem with (CS6.6) proves

$$\boxed{\bar{\mathcal{J}} = \Theta^{-1}\mathcal{J}, \quad H_{\text{Haus}}^1(X, \mathcal{F}) \xrightarrow[\downarrow h]{\mapsto} [\tfrac{1}{2}\mathcal{M}_0 h] \sim \mathcal{Q}.} \quad (\text{CS7.13})$$

The map and its inverse are continuous quotient maps induced by the homeomorphism Θ , because it takes exactly one closed subspace to the other. On ordinary cohomology it is the surjection

$$H_{\text{alg}}^1(X, \mathcal{F}) = H/J \longrightarrow \mathcal{Q}, \quad \ker = \bar{\mathcal{J}}/J. \quad (\text{CS7.14})$$

This proves the Hausdorff identification without presupposing equality with ordinary cohomology. CS11 supplies that equality by constructing the actual inverse source function. Every original-zero jet and its full order is retained by (CS7.12).

CS8. Unshifted scaling, all endpoint characters, and the Čech sign

For $a > 0$, the source actions on the main chart coordinates and the overlap are

$$\rho_+(a)f(v) = f(v/a), \quad \rho_-(a)h(v) = a h(av), \quad \rho_H(a)g(u) = g(u/a). \quad (\text{CS8.1})$$

They preserve the moment-null even Schwartz space and are continuous. On the four endpoint lines their exact characters are

$$\begin{array}{c|cccc} \text{line} & e_{+,0} & e_{+,1} & e_{-,0} & e_{-,1} \\ \hline \text{multiplier} & 1 & a & a & 1. \end{array} \quad (\text{CS8.2})$$

These follow respectively by evaluation at zero and change of variables in the real integral. In particular the minus chart uses the source's extra scalar a ; it does not have the same ordered endpoint characters as the plus chart.

Direct substitution proves

$$r_\pm \rho_\pm(a) = \rho_H(a)r_\pm, \quad \widehat{\rho_+(a)f} = \rho_-(a)\hat{f}. \quad (\text{CS8.3})$$

Thus the Fourier graph and all four endpoint lines carry the actual source action. Under (CS6.2)–(CS7.2),

$$U\rho_H(a)U^{-1}k(u) = a^{1/2}k(u/a), \quad \Theta\rho_H(a)\Theta^{-1}F(s) = a^s F(s). \quad (\text{CS8.4})$$

The infinitesimal operator on raw H is $-u\partial_u$; on $\mathcal{A}$ it is $1/2 - u\partial_u$; on $\mathcal{B}$ it is multiplication by the original variable s . No operator has been silently shifted. On the quotient the arithmetic operator is exactly

$$T_a[F] = [a^s F(s)], \quad T_n = n^L = \sqrt{n} V_n^{\text{centered}}, \quad (\text{CS8.5})$$

where the superscript distinguishes the earlier centered dilation from an integral transfer generator.

The source Weyl lift exchanges chart sections by the identity and acts on overlap sections by R . On the ordered Čech complex (CS5.1), its actual cochain action is

$$W^0(a, b) = (b, a), \quad W^1 g = -Rg. \quad (\text{CS8.6})$$

Indeed

$$dW^0(a, b) = r_+ b - r_- a = -R(r_+ a - r_- b) = -Rd(a, b). \quad (\text{CS8.7})$$

The extra minus sign is the orientation sign from reversing the ordered chart pair. Both degrees square to the identity.

On H^0 , the main Fourier-graph coordinate transforms as $f \mapsto \hat{f}$, and the endpoint lines are exchanged by their chart labels. On the Hausdorff H^1 quotient, changing variables $v = 1/u$ gives

$$\mathcal{M}_0(Rg)(s) = \mathcal{M}_0 g(1-s), \quad \boxed{W^1[F(s)] = [-F(1-s)]}. \quad (\text{CS8.8})$$

The ideal is preserved either by the source identity $RJ = J$ and closure, or by the original functional equation with its multiplicities. The source's unshifted compatibility remains

$$\rho_H(a)R = aR\rho_H(a^{-1}), \quad T_a W^1 = aW^1 T_{a^{-1}}. \quad (\text{CS8.9})$$

This proves the spectral reflection and its chain sign without assuming critical-line location or positivity.

CS9. The single-closed-point support maps

For the closed subset $\{0\}$, the ordinary support functor is $\ker(r_+ : A_+ \rightarrow H)$. Its derived complex is

$$L_0 = [A_+ \xrightarrow{r_+} H]. \quad (\text{CS9.1})$$

For $\{\infty\}$ use

$$L_\infty = [A_- \xrightarrow{r_-} H]. \quad (\text{CS9.2})$$

These derived identifications follow by the same short-exact-complex and effacement proof as CS5 for the single arrow. Source injectivity and (CS3.6) yield

$$\begin{aligned} H_{\{0\}}^0(X, \mathcal{F}) &= E_+, & H_{\{\infty\}}^0(X, \mathcal{F}) &= E_-, \\ H_{\{0\}}^1(X, \mathcal{F}) &= H/J, & H_{\{\infty\}}^1(X, \mathcal{F}) &= H/J, & H_{\{0\}}^i &= H_{\{\infty\}}^i = 0 \quad (i \geq 2). \end{aligned} \quad (\text{CS9.3})$$

The two support-to-global chain maps, with positive restriction differentials in both local presentations, are

$$\begin{aligned} L_0 &\longrightarrow C^\bullet : & a &\longmapsto (a, 0), & g &\longmapsto g, \\ L_\infty &\longrightarrow C^\bullet : & b &\longmapsto (0, b), & g &\longmapsto -g. \end{aligned} \quad (\text{CS9.4})$$

Their chain identities follow from $d(a, 0) = r_+ a$ and $d(0, b) = -r_- b$. Hence both induce first-cohomology isomorphisms, with respectively $+\text{id}$ and $-\text{id}$ in the displayed quotient coordinates.

For completeness, the restriction $H^0(X, \mathcal{F}) \rightarrow A_-$ is onto: given (h, e_-) , choose the global section with plus coordinate $\hat{h}$, arbitrary plus endpoints, and minus coordinate h . Its kernel is E_+ . This proves directly that the connecting map $A_- \rightarrow H_{\{0\}}^1$ is zero and gives the exact sequence

$$0 \longrightarrow E_+ \longrightarrow H^0(X, \mathcal{F}) \longrightarrow A_- \xrightarrow{0} H/J \xrightarrow{\text{id}} H/J \longrightarrow 0. \quad (\text{CS9.5})$$

The analogous sequence for the other closed point has the support-to-global sign from (CS9.4). Passing the individual first-cohomology spaces to their Hausdorff quotients gives the same isomorphisms with $\mathcal{Q}$, with exactly those signs. This does not assert that replacing every term of an arbitrary ordinary localization sequence by its Hausdorff quotient preserves exactness.

The Weyl map from L_0 to L_∞ is the identity on chart coordinates and $+R$ on degree one, because $r_- = Rr_+$. Together with (CS9.4), this gives precisely the global $-R$ in (CS8.6), reconciling the local and global sign conventions.

CS10. Result and limits of this comparison

The constructed object is the explicit $K = \widehat{\mathbb{Z}}^\times$ summand of the original Connes–Consani sheaf. Its source chart coordinate is topologically $\mathcal{S}_0^{\text{even}}(\mathbb{R})$, as proved by the two inverse maps in (CS2.7). Its overlap restriction is the full rational sum $2 \sum_{n \geq 1} f(nu)$. Its ordinary H^0 is the Fourier graph with four separate endpoint lines, and its ordinary H^1 is H/J . The exact topological comparison is

$$\begin{array}{ccc} H & \xrightarrow{U: h \mapsto \frac{1}{2} u^{1/2} h} & \mathcal{A} \\ \downarrow \text{quotient} & & \downarrow \text{quotient} \\ H/\bar{J} & \xrightarrow{\sim} & \mathcal{A}/\mathcal{C} \xrightarrow[\mathcal{M}]{\sim} \mathcal{B}/\mathcal{I} = \mathcal{Q}. \end{array} \quad (\text{CS10.1})$$

Every map in this diagram has been specified on its actual topology, and the ordinary-to-Hausdorff kernel is $\bar{J}/J$. The comparison carries unshifted idèle scaling to multiplication by a^s , and the ordered Čech Weyl action to $-F(1-s)$.

This proves an actual source-sheaf realization and its exact connection to all original-zeta zero jets. It does not prove a finite-field Frobenius structure, positivity of the original Weil form, separation of the weights of a new lifting obstruction, or RH. Such a conclusion is not one of the hypotheses used in any map or quotient above.

CS11. The actual image is closed, with its complete inverse

The later complete derivation SSI0–SSI9 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md, independently checked in ESI0–ESI9 of EXACT_SCHWARTZ_IMAGE_INDEPENDENT_CHECK.md, strengthens the source-image statement used above. Both full proofs are retained in this edition. It gives

$$\mathcal{M}_0 S : \mathcal{S}_0^{\text{even}} \xrightarrow{\sim} \mathcal{J} \quad (\text{CS11.1})$$

as a topological isomorphism. For $b \in H$ with $F = \mathcal{M}_0 b \in \mathcal{J}$, its exact inverse is

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0), \quad S f_F = b. \quad (\text{CS11.2})$$

It extends to an even Schwartz function on the whole real line, with

$$f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1). \quad (\text{CS11.3})$$

The derivation cancels the nontrivial-zero poles with all their actual multiplicities, retains the simple trivial-zero poles of $F/(2\zeta)$, proves rapid bounds on every pole-cleared vertical strip, and shifts the inverse Mellin contour with every residue retained. SSI4 proves continuous bounds from the two vertical boundary lines; SSI5–SSI6 prove all source derivatives and endpoint moments. Thus the inverse is a map into the actual source topology, not merely a limiting representative in a quotient.

Since $\mathcal{J}$ is closed and $\mathcal{M}_0$ is a homeomorphism, (CS11.1) gives

$$J = \bar{J} = \mathcal{M}_0^{-1} \mathcal{J} = \Theta^{-1} \mathcal{J}, \quad \bar{J}/J = 0. \quad (\text{CS11.4})$$

The last equality of inverse images uses the exact scalar relation $\Theta = \frac{1}{2}\mathcal{M}_0$ and the fact that $\mathcal{J}$ is a complex vector subspace. Consequently (CS7.13) identifies the ordinary first cohomology itself with $\mathcal{Q}$, using exactly its displayed map $[b] \mapsto [\frac{1}{2}\mathcal{M}_0 b]$. The four independent endpoint lines, both support-to-global signs, the Weyl action, all actual zero jets and the unshifted arithmetic action remain as calculated in CS5 and CS8–CS9.

This theorem kills precisely the comparison kernel $\bar{J}/J$. It does not kill the prime-indexed endpoint summand of CS2A: that summand occurs in a different quotient of the source and is already in the summation kernel before taking its image. The two quotients and their topologies have not been identified. Neither vanishing is presented as Deligne’s weight-separated lifting obstruction.

The supported localization maps of the original-zeta sheaf

Complete derivation, 24 September 2026. Proof locators CSL0–CSL10.

CSL0. Construction stage and sources

The supporting datum is $\tau\langle Z_1; \text{no } Z_2 \rangle$. Its comparison with the unique generic point uses the complete chart reconstruction CG1–CG3, after the complete arithmetic has been recovered. All vector spaces, sums, scalar factors and quotient operations in this note occur in the coefficient sheaf over that support. None is an arithmetic operation on τ , a metric on it, or a definition of the earlier counting process. The two chart summation maps remain separately labelled.

The corrected argument and relevant verbatim passages were recalled before this calculation; PC04 in the private prerequisite ledger records that reading. In particular, the retracted addition at τ is not used, and fixed support is not substituted for trivial action on its stalk.

The original author source is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. Its unchanged author file is `sources/CC_0903_2024_v3/author_source/anno_unc3.tex`. Root read lines 1376–1667 in full, including the sheaf, both restriction maps, the Fourier proof and the localization sequence. The source archive is retained unchanged. The spectral comparison below uses the complete construction in `CC_SHEAF_ORIGINAL_ZETA_COHOMOLOGY.md` and S1–S7 in `GLOBAL_MELLIN_SYNTHESES.md`; the former proves the exact finite-unit sector of the adelic sheaf and the latter its closed summation image. These are named derivations, not original-author claims.

The weight comparison is with Pierre Deligne, *La conjecture de Weil. II*, §3.6, as reconstructed in DC1–DC8 of `DELIGNE_INVARIANT_CYCLE_QUOTIENT.md`. The present calculation gives the actual supported-cohomology maps of the CC coefficient sheaf. It does not replace Deligne’s inertia and geometric duality terms with similarly named analytic spaces.

CSL1. Exact sheaf and unshifted actions

Let $X = \{x_+, \eta, x_-\}$, with opens

$$\emptyset, \quad U = \{\eta\}, \quad U_+ = \{x_+, \eta\}, \quad U_- = \{x_-, \eta\}, \quad X. \quad (\text{CSL1.1})$$

The notation x_+, x_- distinguishes the two closed chart points from the integer zero and from τ . In the author’s notation these are $0, \infty$; that is a chart notation, not an equation identifying τ with zero.

There is an exact topological comparison with the later signed CC construction: send its points $(+, \eta, -)$ to (x_+, η, x_-) . The open sets of `CC.tex` lines 499–505 are exactly (CSL1.1), so this map and its inverse are continuous. The underlying map of the full involution exchanges the two closed points and fixes the generic point. Therefore this homeomorphism intertwines the underlying involutions, and pullback transports the coefficient sheaf below to that space, with its stalks and all cohomology unchanged. The original signed stalk formulas remain $\alpha^*T = \epsilon T^{-1}$ and $\alpha^*J = \epsilon J$, with $J^2 = \epsilon$. The topological pullback is not asserted to be a comparison of the two structure sheaves or an action of those full signed stalk generators on the new coefficient sheaf. Its domain is precisely the underlying topological space, which is sufficient for the sheaf and support calculations being made here.

Let S be the even real Schwartz space with complex coefficients and with both moments zero:

$$S = \{f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), \quad f(0) = 0, \quad \int_{\mathbb{R}} f(v) dv = 0\}. \quad (\text{CSL1.2})$$

Set $V_+ = S \oplus \mathbb{C}^2$, $V_- = S \oplus \mathbb{C}^2$. Denote the additional coordinates by (c_0, c_1) , retaining the value and integral labels. The overlap is

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\}. \quad (\text{CSL1.3})$$

The exact comparison with S1's centred source is the topological isomorphism

$$\mathcal{T} : \mathcal{A} \longrightarrow A, \quad (\mathcal{T}k)(u) = 2u^{-1/2}k(u), \quad \mathcal{T}^{-1}a(u) = \frac{1}{2}u^{1/2}a(u). \quad (\text{CSL1.4})$$

Multiplication by either displayed power shifts each seminorm by at most one integer weight; Leibniz's formula gives a finite sum of derivative seminorms. Thus both maps are continuous and preserve exactly the stated spaces.

Define

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h. \quad (\text{CSL1.5})$$

These are the original CC restriction maps in the finite-unit sector. The factor 2 comes from both signs in $\mathbb{Q}^\times$; it is not discarded. Poisson summation gives, for general even Schwartz h ,

$$\Sigma \hat{h}(u) = u^{-1}\Sigma h(u^{-1}) + u^{-1}h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{CSL1.6})$$

Here $\hat{h}(t) = \int h(v)e^{-2\pi i vt} dv$. Therefore on S

$$R^2 = 1, \quad \hat{\hat{h}} = h, \quad R\Sigma h = \Sigma \hat{h}. \quad (\text{CSL1.7})$$

The two omitted terms in the restricted identity are zero by the two specified moment conditions; their formula and their independent endpoint coordinates are retained. The convergence and continuity of $\Sigma : S \rightarrow A$ follow by Schwartz summation at infinity and (CSL1.6) at zero, with logarithmic derivatives. Its injectivity also has an elementary proof: for $\Re s > 1$,

$$\int_0^\infty \Sigma f(u) u^s \frac{du}{u} = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CSL1.8})$$

If $\Sigma f = 0$, the nonvanishing Euler product on this half-plane makes the second Mellin transform zero there. Fourier uniqueness on any vertical line in that half-plane gives $f = 0$ on the positive axis, and evenness gives $f = 0$ on $\mathbb{R}$.

For $a > 0$, retain the unshifted actions

$$\begin{aligned} T_a b(u) &= b(u/a), \\ \rho_+(a)(f, c_0, c_1) &= (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a\cdot), ad_0, d_1). \end{aligned} \quad (\text{CSL1.9})$$

Direct substitution shows $r_\pm \rho_\pm(a) = T_a r_\pm$ and

$$T_a R = a R T_{a^{-1}}, \quad \widehat{ah(a\cdot)}(v) = \hat{h}(v/a). \quad (\text{CSL1.10})$$

Thus these are actions on the specified sheaf. No $a^{-1/2}$ character has been used to replace them.

CSL2. The whole Cech complex and its topology

Each of U_+, U_-, U has a point whose only neighbourhood in that open is the whole open. Sections there are the corresponding stalk functor, hence exact. These opens are acyclic for every sheaf of vector spaces. The Cech complex for this two-open cover therefore computes ordinary sheaf cohomology:

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad d(v_+, v_-) = r_+ v_+ - r_- v_-. \quad (\text{CSL2.1})$$

This sign convention is fixed below. Let

$$J = \Sigma(S), \quad C = \overline{J}^A, \quad Q_{\text{alg}} = A/J, \quad Q_H = A/C, \quad N = C/J. \quad (\text{CSL2.2})$$

By (CSL1.7), $r_+(V_+) = r_-(V_-) = J$, so the sum of the chart images is exactly J , without a doubling of its arithmetic spectrum. Hence

$$H^1(X, \Omega) = Q_{\text{alg}}, \quad H^j(X, \Omega) = 0 \ (j \geq 2), \quad 0 \longrightarrow N \longrightarrow Q_{\text{alg}} \longrightarrow Q_H \longrightarrow 0. \quad (\text{CSL2.3})$$

The last arrow is precisely Hausdorffization for the quotient locally convex topology. The construction of this sequence does not presuppose $J = C$. CSL10 now proves that equality by applying the complete inverse-source theorem.

Injectivity of Σ and (CSL1.7) give the full zeroth cohomology:

$$H^0(X, \Omega) = \{((\hat{h}, c_0, c_1), (h, d_0, d_1)) : h \in S, \ c_0, c_1, d_0, d_1 \in \mathbb{C}\}. \quad (\text{CSL2.4})$$

In particular, all four labelled endpoint lines remain in the kernel. The restriction to U has image J , and its kernel consists exactly of these four lines.

CSL3. One closed support and an actual lifting map

Let $Y_+ = \{x_+\}$. Its complement is U_- . The complex with support on Y_+ is represented by

$$K_+ = [V_+ \xrightarrow{r_+} A] \quad \text{in degrees } 0, 1. \quad (\text{CSL3.1})$$

To prove this, use the localization cone of the chain map $D \rightarrow V_-[0]$, $(v_+, v_-) \mapsto v_-$. Its degree0 is $V_+ \oplus V_-$, its degree1 is $A \oplus V_-$, and its differential is $(v_+, v_-) \mapsto (r_+v_+ - r_-v_-, v_-)$. Changing degree1 coordinates to $(b + r_-v_-, v_-)$ splits it as (CSL3.1) plus the contractible identity complex on V_- . This explicitly proves the claimed quasi-isomorphism.

Consequently

$$H_{Y_+}^0(X, \Omega) = \mathbb{C} \oplus \mathbb{C}[1], \quad H_{Y_+}^1(X, \Omega) = Q_{\text{alg}}, \quad H_{Y_+}^j = 0 \ (j \geq 2), \quad (\text{CSL3.2})$$

where $\mathbb{C}[1]$ means the character $a \mapsto a$, not a cohomological shift. The map $K_+ \rightarrow D$ is $v \mapsto (v, 0)$ in degree0 and the identity on A in degree1. It induces the identity

$$H_{Y_+}^1(X, \Omega) \xrightarrow{\sim} H^1(X, \Omega). \quad (\text{CSL3.3})$$

The connecting map from $H^0(U_-, \Omega)$ into this supported H^1 is zero. This is proved by the continuous equivariant lift

$$\ell_- : V_- \longrightarrow H^0(X, \Omega), \quad (h, d_0, d_1) \longmapsto ((\hat{h}, 0, 0), (h, d_0, d_1)). \quad (\text{CSL3.4})$$

Its restriction to U_- is the identity. Equation (CSL1.10) proves equivariance, and the Fourier transform is continuous in the Schwartz topology. Thus the exact lifting is actually constructed, including the endpoint data. Its mechanism is Poisson summation and Fourier transform; it is not presented as a new proof of weight separation.

For $Y_- = \{x_-\}$, the analogous supported complex is $[V_- \xrightarrow{r_-} A]$. Under the fixed sign of (CSL2.1), its map to D is $v \mapsto (0, v)$ in degree0 and multiplication by -1 in degree1. Both support choices are thereby retained with their correct sign.

CSL4. Both closed supports and the diagonal/difference sequence

Put $Y = \{x_+, x_-\}$, whose complement is the generic open U . A complex representing $R\Gamma_Y(X, \Omega)$ is

$$K^0 = V_+ \oplus V_-, \quad K^1 = A \oplus A, \quad d_K(v_+, v_-) = (r_+v_+, r_-v_-). \quad (\text{CSL4.1})$$

Here is an explicit localization-cone proof. Represent the restriction to U by the chain map $D \rightarrow A[0]$ which is r_+ on the first summand of D^0 . Choosing r_- instead gives a homotopic map: their difference is the differential (CSL2.1). The shifted cone has differential

$$(v_+, v_-) \longmapsto (r_+v_+ - r_-v_-, r_+v_+) \quad \text{in } A \oplus A. \quad (\text{CSL4.2})$$

The invertible degree1 change $(k, b) \mapsto (b, b - k)$, with inverse $(b_+, b_-) \mapsto (b_+ - b_-, b_+)$, gives (CSL4.1). Its map to D is the identity in degree0 and $(b_+, b_-) \mapsto b_+ - b_-$ in degree1. The boundary from A into supported degree1 is $b \mapsto (b, b)$.

It follows, with all maps specified, that the complete nonzero portion of the localization sequence is

$$0 \rightarrow \mathbb{C}^4 \rightarrow H^0(X, \Omega) \xrightarrow{\Sigma^\wedge} A \xrightarrow{\beta} Q_{\text{alg}} \oplus Q_{\text{alg}} \xrightarrow{\pi} Q_{\text{alg}} \rightarrow 0, \quad (\text{CSL4.3})$$

where the middle map on the graph (CSL2.4) sends it to $\Sigma^\wedge h$, and

$$\beta(b) = ([b], [b]), \quad \pi([b_+], [b_-]) = [b_+ - b_-]. \quad (\text{CSL4.4})$$

Exactness can also be checked without derived notation. The kernel of β is J . If $[b_+] = [b_-]$, the pair is diagonal and in its image. Every class is the image of $([b], 0)$, and all earlier kernels have already been computed in (CSL2.4). This proves each arrow of (CSL4.3).

After factoring out J in the middle, one obtains

$$0 \longrightarrow Q_{\text{alg}} \xrightarrow{q \mapsto (q, q)} Q_{\text{alg}} \oplus Q_{\text{alg}} \xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q_{\text{alg}} \longrightarrow 0. \quad (\text{CSL4.5})$$

This row is split by $q \mapsto (q, 0)$. Hausdorffization here gives exactly the same row with Q_H in all three places: this follows directly from the explicit maps and the closed subspaces C , $C \oplus C$, rather than from a general exactness claim about Hausdorffization.

CSL5. The mirror sign on cochains and its retained action

The mirror exchanges x_+, x_- and fixes η , since the generic point is unique. A carried exchanged Z_2 -label at that fixed point would have to equal its flip, which is impossible for either of its two states. This derives the relevant absence of that label at $\tau\langle Z_1; \text{no } Z_2 \rangle$. It does not make the stalk involution the identity.

On the sheaf the mirror swaps the two chart sections and uses R on the overlap. The induced map on the ordered Cech complex is

$$w_D^0(v_+, v_-) = (v_-, v_+), \quad w_D^1(b) = -Rb. \quad (\text{CSL5.1})$$

Indeed, $d(v_-, v_+) = \Sigma h - R\Sigma f = -R(\Sigma f - R\Sigma h)$. The minus sign is the orientation sign from reversing the ordered cover; omitting it would not give a chain map. This does not change the author's assertion of spectral symmetry, but it fixes the actual action under our displayed identification of H^1 .

On (CSL4.1) the induced map is

$$w_K^0(v_+, v_-) = (v_-, v_+), \quad w_K^1(b_+, b_-) = (Rb_-, Rb_+). \quad (\text{CSL5.2})$$

The identity $Rr_- = r_+$, with the two chart labels exchanged, verifies the chain-map identity. Consequently (CSL4.5) is equivariant with mirror R on its first term, $(q_+, q_-) \mapsto (Rq_-, Rq_+)$ on its middle term, and $-R$ on its last term. Its continuous arithmetic- and mirror-equivariant section is

$$s(q) = \frac{1}{2}(q, -q). \quad (\text{CSL5.3})$$

The calculation is exact: $\pi s(q) = q$, while $w_K s(q) = \frac{1}{2}(-Rq, Rq) = s(-Rq)$. The scalar $1/2$ is in the already constructed complex coefficient field. It has no interpretation as halving τ .

CSL6. Exact comparison with the original-zeta receiver

Use the raw Mellin map

$$\mathcal{M}_0 b(s) = \int_0^\infty b(u) u^s \frac{du}{u}. \quad (\text{CSL6.1})$$

From (CSL1.4) and S1,

$$\mathcal{M}_0 \mathcal{F} = 2\mathcal{M}, \quad \mathcal{M}_0 T_a = a^s \mathcal{M}_0, \quad \mathcal{M}_0 Rb(s) = \mathcal{M}_0 b(1 - s). \quad (\text{CSL6.2})$$

All integrals converge absolutely in every fixed strip; the inverse is

$$b(e^x) = \frac{e^{-cx}}{2\pi} \int_{\mathbb{R}} (\mathcal{M}_0 b)(c + it) e^{-itx} dt \quad (c \in \mathbb{R}). \quad (\text{CSL6.3})$$

The estimates and contour shifts are those of S1 with the exact change (CSL6.2). Thus $\mathcal{M}_0$ is a topological isomorphism $A \rightarrow \mathcal{B}$.

Let $\mathcal{I}$ be the closed ideal of full-order vanishing at every original nontrivial zero, with its original multiplicity. S5 gives $\mathcal{M}_0 C = \mathcal{I}$, so

$$Q_H \xrightarrow{\sim} \mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad [b] \mapsto [\mathcal{M}_0 b]. \quad (\text{CSL6.4})$$

Under this exact isomorphism, the entire supported row (CSL4.5) is the diagonal/difference row of the actual $\mathcal{Q}$, with the same full multiplicities and quotient topology. It is not an unrestricted product of jets or an algebraic direct sum.

For the retained source function

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2},$$

the complete auxiliary transform is

$$\mathcal{M}_0 \Sigma f_0(s) = \frac{s(s-1)}{4} \pi^{-s/2} \Gamma(s/2) \zeta(s) = 2F_0(s). \quad (\text{CSL6.5})$$

Its endpoint values are $1/4$ at both 0 and 1. At each negative even integer, with $r \geq 1$, its retained cancellation value is

$$(2F_0)(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{4} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0. \quad (\text{CSL6.6})$$

These are values of the transform; the original ζ still has its pole at 1 and trivial zeros at $-2r$. Its unit and prime-power repetitions remain in

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{CSL6.7})$$

All these arithmetic labels belong to the recovered complete arithmetic, not to the primitive support.

CSL7. What the computed action says about weights

On an actual original-zero block at ρ of multiplicity m_ρ , multiplication by s is $\rho I + N_\rho$, where $N_\rho^{m_\rho} = 0$. Consequently the unshifted action in every Q_H term above is

$$T_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j. \quad (\text{CSL7.1})$$

It follows from the original strip $0 < \Re \rho < 1$, for $a > 1$, that its character weight $2 \log_a |a^\rho|$ lies in $(0, 2)$. The kernel, middle terms and quotient of (CSL4.5) have these same character weights. The diagonal and difference are explicit nonzero equivariant maps; they are not forced to vanish by separated weights.

The endpoint kernel has the separately computed weights. In V_+ , value has character 1 and integral has character a ; in V_- , value has character a and integral has character 1. Thus the four endpoint lines have weights 0, 2, 2, 0, in that order. This follows from (CSL1.9), not from a weight assigned to a support point.

This weight computation states the exact analytic character action. It does not assert that these spaces are finite-dimensional ℓ -adic cohomology, that their characters are algebraic Weil numbers, or that an arithmetic weight filtration has been imported without a comparison.

CSL8. The precise comparison with Deligne’s localization cross

In Deligne’s theorem, the top obstruction group and the inertia term are

$$O_i = H^{2N-i-1}(X_s, E)^\vee(-N), \quad K_i = H^{i-1}(X_{\bar{\eta}}, E)_I(-1), \quad (\text{CSL8.1})$$

and the obstruction to specialization is

$$\text{im}(\partial_i)/\partial_i j_i(K_i). \quad (\text{CSL8.2})$$

DC3–DC6 derives the Tate factors from tame inertia and the dualizing complex of the smooth total space. Those operations give the weights $\leq i$ on the classes and $\geq i + 1$ on the obstruction receiver. They are additional geometric maps, not consequences of the name “localization.”

For the CC sheaf calculated here, the exact specialization-like map actually constructed by removing one closed point is (CSL3.3). It is an isomorphism, with inverse the identity on the displayed quotient. The restriction lift (CSL3.4) is explicit. For removal of both closed points, the exact receiver and its maps are (CSL4.3)–(CSL4.5), and the equivariant section is (CSL5.3). These are genuine applications of localization to this coefficient sheaf, and every map has now been calculated on its source.

In particular, neither supported calculation produces an uncomputed Tate shift: the character identity (CSL7.1) determines its action. The constructed comparison to the Mellin receiver is (CSL6.4), not an identification with (CSL8.1). A geometric extension carrying an additional normal direction can have a different dualizing or inertia term; the preceding calculation makes no claim against such an extension. Its exact current contribution is the original-zeta cohomological receiver, its support maps, its Fourier lift, and its mirror-equivariant split supported row. It does not certify completion of the requested Deligne weight-lifting application.

CSL9. All comparison arrows and retained information

The following commuting formulas describe the full receiver used here:

$$\begin{aligned} \mathcal{M}_0 \circ r_+ &= 2 \mathcal{M} \circ \mathcal{E}, \\ \mathcal{M}_0 \circ r_- &= 2 (F(s) \mapsto F(1-s)) \circ \mathcal{M} \circ \mathcal{E}, \\ \overline{\mathcal{M}_0}^{\oplus 2} \circ \beta &= \text{diag} \circ \overline{\mathcal{M}_0}, \\ \overline{\mathcal{M}_0} \circ \pi &= \text{diff} \circ \overline{\mathcal{M}_0}^{\oplus 2}, \\ \overline{\mathcal{M}_0}^{\oplus 2} \circ s(q) &= \tfrac{1}{2}(\overline{\mathcal{M}_0}q, -\overline{\mathcal{M}_0}q). \end{aligned} \quad (\text{CSL9.1})$$

The first two formulas act on the Schwartz component and send each explicitly retained endpoint summand to zero. The last three use the quotient of β by its kernel J , followed by the stated Hausdorffization. Hence their domains and codomains are exactly the displayed Q_H spaces. Every original-zero jet survives in $\mathcal{Q}$; CSL10 calculates the separately retained closure difference as $N = C/J = 0$. The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$, while the full coefficient action, both charts and their orientation sign remain visible.

CSL10. Exact image, strict maps, and the calculated comparison kernel

The complete proof SSI0–SSI9 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md constructs a continuous inverse to

$$\mathcal{M}_0 \Sigma : S \xrightarrow{\sim} \mathcal{I}. \quad (\text{CSL10.1})$$

Its independently checked proof ESI0–ESI9 is retained in EXACT_SCHWARTZ_IMAGE_INDEPENDENT_CHECK.md. With $F = \mathcal{M}_0 b$, the inverse has the full formula

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0). \quad (\text{CSL10.2})$$

The pole-cleared strip estimate, contour residues and smooth even extension in SSI4–SSI6 prove $f_F \in S$, continuity in all Schwartz seminorms, and $\Sigma f_F = b$. Its even derivatives at zero are exactly

$(2r)!F(-2r)/(2\zeta'(-2r))$; both vanishing moments and every nontrivial multiplicity are proved there. Thus the inverse does not discard the trivial-zero data of the original function.

It follows, on the actual spaces already used in CSL2, that

$$J = \mathcal{M}_0^{-1}\mathcal{J} = C, \quad N = 0, \quad Q_{\text{alg}} \xrightarrow{\sim} Q_H \xrightarrow{\sim} \mathcal{Q}. \quad (\text{CSL10.3})$$

Each map is a homeomorphism with the quotient topologies. Both chart restrictions are strict onto J : the plus chart has right inverse $b \mapsto (\Sigma^{-1}b, 0, 0)$, and the minus chart has right inverse $b \mapsto (\widehat{\Sigma^{-1}b}, 0, 0)$. Poisson summation proves the second identity. The Čech differential has right inverse from its image $b \mapsto ((\Sigma^{-1}b, 0, 0), 0)$, so its image is closed and its ordinary first cohomology is already Hausdorff.

Therefore the one-point localization maps and the two-point diagonal/difference row of CSL3–CSL5 are maps on the actual original-zero quotient without any extra closure operation. Their signs, common arithmetic action and section $q \mapsto \frac{1}{2}(q, -q)$ are unchanged. This is the vanishing of the exact comparison kernel N , not a deduction that the source has pure weight or that the independently calculated localization row is Deligne’s cross.

Original-zeta residue duality and the actual support receiver

24 September 2026. Complete derivation RD0–RD9.

RD0. Prerequisites and the specific comparison

The support remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$. The complete arithmetic reconstruction precedes every coefficient, integer, prime, analytic variable and linear operation below. None of these operations is addition at τ , a coordinate on that point, or a replacement of the user's information layers by cyclic groups. The governing corpus rule, correction chains, latest direct global-quotientability arguments, and the complete source passages for the distance, separate-branch and global-admissibility corrections were read before this calculation.

The current receiving objects are the actual closed summation image of SSI0–SSI10 and ESI0–ESI12 and the sphere sheaf calculated in CSP0–CSP13, `../tau_weight_cohomology_20260924/CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`. All of CSP was read. The original-zeta projector proof RZ1–RZ2, RZ5 and RZ8–RZ12 was read directly in `../tau_weight_cohomology_20260924/ORIGINAL_ZETA_RE_SOLVENT_AND_PROJECTORS.md`; its other sections are dependencies with their existing proofs, not newly claimed reading coverage.

Deligne's actual lifting argument is *La conjecture de Weil. II*, §3.6, printed 213–214, particularly the support-dual term in the cross on 214. The full original article has been read with the separate original-source ledger. That term is $H^{2N-i-1}(X_s)^\vee(-N)$; identifying a dual coefficient without its support, degree and twist would not reproduce that term. The calculation here constructs a dual receiver for the programme's actual coefficient spaces. It does not identify these infinite-dimensional complex sheaves with Deligne's constructible étale category.

The operations to construct are: continuous linear functionals on the already formed quotient; residues of the original ζ ; their embedding into the continuous dual of actual sheaf cohomology; and their arithmetic action. The degree character used in the comparison is independently constructed by the sphere cover in CSP8, rather than assigned to the supporting point.

RD1. The complete quotient and its finite-support subspace

Retain the Fréchet space

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for all integers } A, M \geq 0\}. \quad (\text{RD1.1})$$

Let $\mathcal{Z}$ denote all distinct nontrivial zeros of the original $\zeta(s)$, with their full multiplicities m_ρ . Set

$$\mathcal{I} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ } (\rho \in \mathcal{Z}, 0 \leq j < m_\rho)\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}. \quad (\text{RD1.2})$$

This is the actual quotient topology. Cauchy's formula gives $|F^{(j)}(\rho)|/j! \leq r^{-j} b_{A,0}(F)$ whenever $A \geq |\Re \rho| + r$. Therefore every required jet is continuous and $\mathcal{I}$ is closed. All zero multiplicities are retained in the quotient.

The already proved global isolators can also be constructed explicitly here. Keep the entire source transform with its entire multiplier:

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad A_\rho(s) = \frac{F_0(s)}{(s-\rho)^{m_\rho}}. \quad (\text{RD1.3})$$

Division is removable at ρ . It preserves $\mathcal{B}$: away from a disk the denominator is bounded below by a polynomial, and on the disk the maximum principle bounds the holomorphic quotient by a larger boundary circle. Put $a_j = A_\rho^{(j)}(\rho)/j!$, and define

$$b_0 = a_0^{-1}, \quad b_r = -a_0^{-1} \sum_{h=1}^r a_h b_{r-h}, \quad e_\rho(s) = A_\rho(s) \sum_{r=0}^{m_\rho-1} b_r (s-\rho)^r. \quad (\text{RD1.4})$$

The nonzero number a_0 is determined by the actual multiplicity, not by simplicity. The recursion proves that $e_\rho = 1$ modulo $(s - \rho)^{m_\rho}$ and has every required vanishing jet at every other zero. Thus

$$e_{\rho,j} = [e_\rho(s)(s - \rho)^j], \quad 0 \leq j < m_\rho, \quad \mathcal{Q}_{\text{fin}} = \bigoplus_{\rho \in \mathcal{Z}} \text{span}_{\mathbb{C}} \{e_{\rho,j} : 0 \leq j < m_\rho\} \subset \mathcal{Q}. \quad (\text{RD1.5})$$

The sum is direct because evaluation of the full jet at any one zero recovers its coefficients. A member has finitely many nonzero zero blocks, but each block's source representative satisfies the conditions at **all** other zeros simultaneously. This is a subspace of the globally constructed quotient, not a bounded-height replacement for it. No density of $\mathcal{Q}_{\text{fin}}$ in $\mathcal{Q}$ is assumed below.

RD2. The original-zeta local residue dual

Write $\mathcal{Q}'$ for the continuous complex-linear dual. On a zero block put

$$t = s - \rho, \quad \zeta(\rho + t) = t^m u_\rho(t), \quad m = m_\rho, \quad u_\rho(0) = \zeta^{(m)}(\rho)/m! \neq 0. \quad (\text{RD2.1})$$

For a polynomial P of degree below m , define

$$\lambda_{\rho,P}([F]) = \text{Res}_{s=\rho} \frac{P(s - \rho)F(s)}{\zeta(s)} ds = \frac{1}{2\pi i} \int_{|s-\rho|=r} \frac{P(s - \rho)F(s)}{\zeta(s)} ds. \quad (\text{RD2.2})$$

The circle is positively oriented and sufficiently small to contain no other zero or pole. No choice of its admissible radius changes the residue. Changing F by $\mathcal{I}$ makes the integrand holomorphic at ρ , so the functional is well defined. It is a finite linear combination of continuous jets and hence is continuous on the quotient.

Every Taylor-coefficient functional $\delta_{\rho,j}([F]) = F^{(j)}(\rho)/j!$, $0 \leq j < m$, is one of these residues. Specifically set

$$P_{\rho,j}(t) = [t^{m-1-j}u_\rho(t)]_{<m}. \quad (\text{RD2.3})$$

Here the brackets retain exactly the Taylor terms of degrees below m . The discarded remainder is divisible by t^m , so $P_{\rho,j}(t)/(t^m u_\rho(t)) = t^{-j-1} + \text{holomorphic germ}$. Taking the residue after multiplication by $F(\rho + t)$ proves

$$\lambda_{\rho,P_{\rho,j}} = \delta_{\rho,j}. \quad (\text{RD2.4})$$

Unscaled derivatives are $j!\lambda_{\rho,P_{\rho,j}}$; the factorial is not suppressed. Conversely (RD2.2) is a linear combination of these m functionals. Consequently the original local residue, including every derivative of its unit u_ρ , gives the entire dual of the full multiplicity block.

RD3. A separating pairing using every reflected block

The original functional equation preserves zero multiplicities under $\rho \mapsto 1 - \rho$. Define, for $x = [F] \in \mathcal{Q}$ and $g = [G] \in \mathcal{Q}_{\text{fin}}$,

$$\mathcal{R}(x, g) = \sum_{\rho \in \mathcal{Z}} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{RD3.1})$$

This sum has finite support because every zero jet of G outside its finite support vanishes to the required order. It is independent of both representatives by exactly that order of vanishing. For each fixed g , it is a continuous functional on the **entire** $\mathcal{Q}$. It is complex bilinear; no Hermitian or positivity assertion is part of its definition.

For $\sigma = 1 - \rho$, take the exact basis classes in (RD1.5) at ρ and σ . Formula (RD2.1) gives the full block matrix

$$\mathcal{R}(e_{\rho,i}, e_{\sigma,j}) = (-1)^j [t^{m-1-i-j}] u_\rho(t)^{-1}, \quad 0 \leq i, j < m, \quad (\text{RD3.2})$$

where a coefficient with negative index is zero. Blocks not paired by $\sigma = 1 - \rho$ give zero. The entries on $i + j = m - 1$ are $(-1)^j / u_\rho(0)$. Reversing the order of the columns introduces $(-1)^{m(m-1)/2}$; the product of the displayed signs introduces the same sign. Therefore

$$\det(\mathcal{R}(e_{\rho,i}, e_{1-\rho,j}))_{i,j=0}^{m-1} = u_\rho(0)^{-m} \neq 0. \quad (\text{RD3.3})$$

All lower coefficients of u_ρ^{-1} remain in (RD3.2). The determinant does not replace that matrix.

Let

$$\mathcal{D} = \text{span}_{\mathbb{C}}\{\delta_{\rho,j} : \rho \in \mathcal{Z}, 0 \leq j < m_\rho\} \subset \mathcal{Q}'. \quad (\text{RD3.4})$$

Then

$$\iota : \mathcal{Q}_{\text{fin}} \xrightarrow{\sim} \mathcal{D}, \quad \iota(g)(x) = \mathcal{R}(x, g) \quad (\text{RD3.5})$$

is a linear isomorphism. An explicit inverse on (RD2.2) is the class supported at $\sigma = 1 - \rho$ whose local polynomial is $P(-t)$: its reflected germ is $P(t)$. Construct it by the finite linear combination of the $e_{\sigma,j}$. Extend over finite sums. This proves surjectivity; (RD3.3) proves injectivity. It also proves that the pairing separates both arguments: a nonzero class $[F]$ has a nonzero required jet somewhere by (RD1.2), and (RD2.4) detects it.

RD4. The exact global dual closure, without a topology substitution

Give $\mathcal{Q}'$ its weak topology $\sigma(\mathcal{Q}', \mathcal{Q})$, namely pointwise convergence on all $x \in \mathcal{Q}$. Then

$$\overline{\mathcal{D}}^{\sigma(\mathcal{Q}', \mathcal{Q})} = \mathcal{Q}'. \quad (\text{RD4.1})$$

Here is a proof that also specifies its precise strength. Fix $\lambda \in \mathcal{Q}'$ and any finite collection $x_1, \dots, x_k \in \mathcal{Q}$. Let $E : \mathcal{D} \rightarrow \mathbb{C}^k$, $E(d) = (d(x_1), \dots, d(x_k))$. If $(\lambda(x_i))_i$ were not in $E(\mathcal{D})$, finite-dimensional linear algebra would supply $c_1, \dots, c_k$ such that $\sum_i c_i d(x_i) = 0$ for every $d \in \mathcal{D}$ but $\sum_i c_i \lambda(x_i) \neq 0$. The first assertion and separation in RD3 imply $\sum_i c_i x_i = 0$ in $\mathcal{Q}$, contradicting the second. Thus some $d \in \mathcal{D}$ matches λ exactly on this finite collection. Direct the finite subsets of $\mathcal{Q}$ by inclusion and select one such matching functional at each stage. The resulting net converges pointwise to λ , proving (RD4.1).

Every element of the full dual is therefore recovered in this weak topology from the full residue system. This statement does not claim convergence in the strong dual topology, equicontinuity of this chosen net, a continuous extension of ι to all of $\mathcal{Q}$, or replacement of $\mathcal{Q}$ by an unrestricted product of jets. Those are different assertions; none is used in this proof.

RD5. The full arithmetic action and its dual degree factor

On the original quotient let

$$T_a[F] = [a^s F(s)], \quad L[F] = [sF(s)], \quad a > 0. \quad (\text{RD5.1})$$

The operations are continuous since $a^{\Re s}$ is bounded on any fixed vertical strip, and $|s|$ adds at most one polynomial seminorm. They preserve the ideal and every finite block. For a continuous map U , denote its ordinary transpose by $U'\lambda = \lambda \circ U$; its contragredient action is $(U^{-1})'$ when U is invertible. Directly in the original residue integrand,

$$\mathcal{R}(T_a x, T_a g) = a \mathcal{R}(x, g), \quad \mathcal{R}(Lx, g) + \mathcal{R}(x, Lg) = \mathcal{R}(x, g). \quad (\text{RD5.2})$$

The first identity uses the complete product $a^s a^{1-s} = a$, the second $s + (1-s) = 1$, before taking any residue. It follows that

$$\boxed{\iota T_a = a(T_{a^{-1}})'\iota, \quad \iota L = (1 - L')\iota.} \quad (\text{RD5.3})$$

Thus the same factor n produced geometrically by the degree of $z \mapsto z^n$ in CSP8 is present in the exact original-zeta dual receiver. For nonintegral a , (RD5.3) is a coefficient identity, not a claimed single-valued cover $z \mapsto z^a$.

For the Taylor functionals the entire triangular formulas are

$$T'_a \delta_{\rho,j} = a^\rho \sum_{h=0}^j \frac{(\log a)^{j-h}}{(j-h)!} \delta_{\rho,h},$$

$$a(T_{a^{-1}})' \delta_{\rho,j} = a^{1-\rho} \sum_{h=0}^j \frac{(-\log a)^{j-h}}{(j-h)!} \delta_{\rho,h}, \quad L' \delta_{\rho,j} = \rho \delta_{\rho,j} + \delta_{\rho,j-1}, \quad (\text{RD5.4})$$

where $\delta_{\rho,-1} = 0$. They follow by multiplying the original Taylor series, so every factorial and every nilpotent coefficient is accounted for. They are compatible with (RD3.2), including its signs. The ordinary transpose of the actual degree-two action aT_a is instead aT'_a , and its contragredient is $a^{-1}(T_{a^{-1}})'$. Neither is silently replaced by the degree-twisted dual $a(T_{a^{-1}})'$ in (RD5.3).

More generally, for the actual degree-character twists indexed by integers k, ℓ , the same integrand proves

$$\mathcal{R}(a^k T_a x, a^\ell T_a g) = a^{k+\ell+1} \mathcal{R}(x, g). \quad (\text{RD5.5})$$

In particular two normal-direction inputs carry the full factor a^3 , and one normal-direction input together with one unshifted input carries a^2 . Equivalently the map ι intertwines the unshifted T_a with a^2 times the contragredient of the normal action aT_a , since that contragredient is $a^{-1}(T_{a^{-1}})'$. This retains the difference between the coefficient degree-character dual and the dual of an already shifted normal term.

RD6. Functional equation and the retained symmetry defect

The original functional equation in this local comparison is

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{RD6.1})$$

At each nontrivial zero both local multiplier expressions are holomorphic and nonzero. For $\sigma = 1 - \rho$, the full local units satisfy

$$u_\rho(t) = (-1)^m \chi(\rho + t) u_\sigma(-t). \quad (\text{RD6.2})$$

This follows by substituting $s = \rho + t$ into (RD6.1) and retaining the factor $(-t)^m$. For F, G with finite zero support, substitution $s = 1 - z$, including $ds = -dz$, proves the exact reversal law

$$\mathcal{R}([F], [G]) = - \sum_{\sigma \in \mathcal{Z}} \text{Res}_{z=\sigma} \frac{\chi(z) G(z) F(1-z)}{\zeta(z)} dz. \quad (\text{RD6.3})$$

The sum is finite; the formula uses the germs of χ there. It does not assert that multiplication by its meromorphic global formula preserves $\mathcal{B}$. Derivatives of χ in (RD6.2)–(RD6.3) are retained. In particular a symmetric or Hermitian global form has not been obtained by deleting this unit or its differential sign.

The ordinary reflected trace form previously constructed in RZ10 is

$$W([F], [G]) = \sum_{\rho \in \mathcal{Z}} m_\rho \overline{F(1-\bar{\rho})} G(\rho). \quad (\text{RD6.4})$$

Its multiplicity-block trace has no higher-jet entries. The residue pairing (RD3.2) does: its matrix is invertible for every positive multiplicity. This proves an exact additional dual observation of the retained classes. It does not replace (RD6.4) in Weil's positivity criterion, and it makes no sign claim about that form.

RD7. Passage to the actual sheaf cohomology and support maps

CSP gives $H^1(Y, \mathcal{F}) = Q_A = A/J$ with

$$\kappa : Q_A \xrightarrow{\sim} \mathcal{Q}, \quad \kappa([b]) = [\Theta b], \quad \Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}. \quad (\text{RD7.1})$$

This is the CS convention, retaining the factor $1/2$. The original restriction is still $\Sigma f = 2 \sum_{k \geq 1} f(ku)$; SSI's raw transform uses $\mathcal{M}_0 = 2\Theta$. The isomorphism in (RD7.1) follows from the proved actual equality $\Sigma S = J = \Theta^{-1}\mathcal{J}$. It retains the quotient topology, not a closure surrogate.

Transpose (RD7.1) to get an actual isomorphism of continuous duals $\kappa' : \mathcal{Q}' \rightarrow \mathcal{Q}'_A$. Combining it with (RD3.5) gives

$$\mathcal{Q}_{\text{fin}} \xrightarrow{\kappa' \iota} H^1(Y, \mathcal{F})', \quad g \mapsto \left([b] \mapsto \sum_{\rho} \text{Res}_{s=\rho} \frac{\frac{1}{2} \mathcal{M}_0 b(s) G(1-s)}{\zeta(s)} ds \right). \quad (\text{RD7.2})$$

Its image is weakly dense by (RD4.1) and the topological isomorphism. Formula (RD5.3) gives its arithmetic action with the same complete factor. No coefficient arithmetic has been assigned to τ .

The degree-one support-to-global map from CSP7 is

$$k^1 : \mathcal{Q}_A \oplus \mathcal{Q}_A \longrightarrow \mathcal{Q}_A, \quad (x_+, x_-) \mapsto x_+ - x_-. \quad (\text{RD7.3})$$

Therefore its exact transpose on these residue functionals is

$$(k^1)'(\kappa' \iota g) = (\kappa' \iota g, -\kappa' \iota g). \quad (\text{RD7.4})$$

The degree-two support map is instead the sum $A \oplus A \rightarrow A$, whose transpose is $\lambda \mapsto (\lambda, \lambda)$. The annular boundary $a \mapsto (-a, a)$ transposes to $(\lambda_+, \lambda_-) \mapsto -\lambda_+ + \lambda_-$. These three identities follow by evaluating the maps on arbitrary input pairs; they preserve both branches and every orientation sign.

This embeds the residue receiver in the actual dual localization sequence. Its terms are continuous duals of CSP's coefficient complexes. Calling them a Verdier-dual sheaf, or Deligne's étale support-dual cross, would require an additional constructed comparison; that assertion is not used here.

RD8. Endpoint and original-function data remain in the source

The quotient in RD1 records the nontrivial-zero divisor. Its original source retains the other contributions. In the raw convention SSI proves for every $F \in \mathcal{J}$

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1), \quad (\text{RD8.1})$$

$$f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \int_0^\infty f_F(x) \frac{dx}{x} = -F(0), \quad \int_0^\infty f_F(x) \log x \, dx = \frac{F(1)}{2}. \quad (\text{RD8.2})$$

The four independent endpoint lines $E_+ \oplus E_-$ remain in $H^0(Y, \mathcal{F})$ and their full duals remain in the degree-zero transposed cohomology. They have not been folded into (RD7.2) or declared absent.

For the isolator source (RD1.3), the full retained special values are, with $r \geq 1$ in the second formula,

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r). \quad (\text{RD8.3})$$

Every unit coefficient a_j in (RD1.4) is obtained from the full derivatives of (RD1.3). At nontrivial ρ , writing its displayed multiplier as $C(s)$, the exact expression is

$$a_j = \sum_{h=0}^j \frac{C^{(h)}(\rho)}{h!} \frac{\zeta^{(m+j-h)}(\rho)}{(m+j-h)!}. \quad (\text{RD8.4})$$

In contrast, (RD2.1)–(RD3.3) use the original local unit of ζ itself. The comparison has not substituted F_0 for the original arithmetic function.

RD9. What this calculation establishes for the lifting programme

The proved map (RD7.2) supplies the complete finite-jet residue receiver inside the continuous dual of the actual geometric degree-one cohomology. It sees every nilpotent multiplicity coordinate; (RD4.1) recovers the entire continuous dual in its specified weak topology. Its arithmetic action is the degree-character contragredient (RD5.3), and its support maps have the exact transposed signs (RD7.4). These are full-divisor identities, with no bounded numerical experiment and no assumed purity.

The support-dual target used in Deligne §3.6 additionally has geometrically derived weight bounds and a particular specialization/localization cross. Neither a residue similitude nor the positive degree of a sphere cover proves those bounds for this coefficient sheaf. The current target remains to construct and calculate that comparison. The present calculation contributes its actual dual objects and maps; it does not turn the desired weight separation into an input.

Continuous duals of the actual sphere coefficient and support complexes

Independent mathematical derivation, 24 September 2026. Proof locators CDI0–CDI12.

The complete source `../tau_weight_cohomology_20260924/CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`, CSP0–CSP13, was read, including the pole orientation calculation and its localization cone. The exact source-image theorem SSI1–SSI7 and independent ESI0–ESI12 supply the closedness and continuous source inverse used below. This calculation takes the stated coefficient complexes as actual inputs; it constructs their continuous duals and all transposed maps. It does not identify the resulting complexes with an unconstructed Verdier dual sheaf or assume arithmetic purity.

Every vector-space operation below is on the reconstructed arithmetic coefficient spaces. The support notation $Z_0, Z_1, \dots, \tau$ is unchanged. No addition, numerical coordinate, distance, or numerical weight is assigned to τ .

CDI0. Exact input and dual conventions

Retain the CSP coefficient spaces

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}) : f(-x) = f(x), f(0) = 0, \int_{\mathbb{R}} f = 0\}, & V_{\pm} &= S \oplus E_{\pm}, & E_{\pm} &= \mathbb{C}^2, \\ A &= \left\{a \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j \geq 0\right\}, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), & Ra(u) &= u^{-1} a(u^{-1}), & r_+(f, e) &= \Sigma f, & r_-(h, e) &= R \Sigma h = \hat{\Sigma} h. \end{aligned} \quad (\text{CDI0.1})$$

The original Fourier convention is $\hat{h}(t) = \int h(x) e^{-2\pi i x t} dx$. Let

$$J = \Sigma S \subset A, \quad Q = A/J, \quad q : A \rightarrow Q. \quad (\text{CDI0.2})$$

The exact-image theorem proves that J is closed and that $\Sigma : S \rightarrow J$ is a topological isomorphism. All spaces in (CDI0.1) and (CDI0.2) are therefore Hausdorff Fréchet spaces with their stated subspace, finite-product, and quotient topologies. In particular no closure of zero remains in Q .

For a Hausdorff locally convex space E , write E' for its continuous complex-linear dual. Two topologies will remain explicit:

$$E'_{\sigma} = (E', \sigma(E', E)), \quad E'_{\beta} = (E', \beta(E', E)). \quad (\text{CDI0.3})$$

The first is convergence on every finite set of vectors; the second is uniform convergence on every bounded subset of E . No algebraic dual of discontinuous functionals is substituted for either. For a continuous map $f : E \rightarrow F$, its ordinary transpose is $f' : F' \rightarrow E'$, $f' \lambda = \lambda \circ f$. It is continuous for both topologies: finite evaluation sets pull back to finite sets, and continuous linear maps take bounded sets to bounded sets.

For a cochain complex C , define

$$(C^{\vee})^n = (C^{-n})', \quad d_{C^{\vee}}^n \lambda = (-1)^{n+1} \lambda \circ d_C^{-n-1}. \quad (\text{CDI0.4})$$

This is the usual Hom differential. The evaluation map in the order $C^{\vee} \otimes C \rightarrow \mathbb{C}$ satisfies the cochain identity: the two terms $(-1)^{n+1} \lambda(dc)$ and $(-1)^n \lambda(dc)$ cancel. A degree-zero chain map transposes without an additional sign. Weak or strong topologies on each dual coefficient produce the same underlying vector complex; their cohomology topologies will be distinguished below.

CDI1. Explicit source splittings and the exact continuous transpose

Define continuous sections onto the actual closed image:

$$s_+(j) = (\Sigma^{-1}j, 0) \in V_+, \quad s_-(j) = (\widehat{\Sigma^{-1}j}, 0) \in V_-. \quad (\text{CDI1.1})$$

Poisson and $\hat{f} = f$ on S give $r_{\pm}s_{\pm} = 1_J$. Hence

$$V_{\pm} \xrightarrow{\sim} E_{\pm} \oplus J, \quad v \mapsto (v - s_{\pm}r_{\pm}v, r_{\pm}v) \quad (\text{CDI1.2})$$

are continuous isomorphisms. Their inverses are $(e, j) \mapsto e + s_{\pm}j$; the first coordinate on the right of (CDI1.2) lies in the actual endpoint kernel.

Put $W = V_+ \oplus V_-$, $d(v_+, v_-) = r_+v_+ - r_-v_-$, and $Z = \ker d$. The common-restriction description is

$$Z \xrightarrow{\sim} J \oplus E_+ \oplus E_-, \quad (v_+, v_-) \mapsto (r_+v_+, e_+, e_-). \quad (\text{CDI1.3})$$

Its inverse sends (j, e_+, e_-) to $(s_+j + e_+, s_-j + e_-)$. These are the exact Fourier-graph coordinates of CSP4.5, with all four endpoint lines retained.

The map $t : J \rightarrow W$, $t(j) = (s_+j, 0)$, satisfies $dt = 1$. Therefore

$$W \xrightarrow{\sim} Z \oplus J, \quad w \mapsto (w - t(dw), dw), \quad (z, j) \mapsto z + t(j). \quad (\text{CDI1.4})$$

This explicit splitting will be used on the domain side. It does not assert that the closed inclusion $J \hookrightarrow A$ has a continuous projection or that q has a continuous section.

The continuous transpose of $0 \rightarrow J \rightarrow A \xrightarrow{q} Q \rightarrow 0$ is algebraically exact:

$$0 \rightarrow Q' \xrightarrow{q'} A' \xrightarrow{\text{res}_J} J' \rightarrow 0. \quad (\text{CDI1.5})$$

Indeed q' is injective because q is onto. Its image is

$$J^{\perp} = \{\lambda \in A' : \lambda(j) = 0 \text{ for every } j \in J\}, \quad (\text{CDI1.6})$$

since a continuous functional constant on quotient fibers factors continuously through the quotient topology. Finally every continuous complex-linear functional on the subspace J extends continuously to A by the complex Hahn–Banach theorem: continuity bounds it by a continuous seminorm induced from A , and Hahn–Banach gives an extension with that bound. This proves surjectivity in (CDI1.5), not merely density of its image.

The corresponding exact identities are

$$\ker d' = J^{\perp} = q'Q', \quad \text{im } d' = Z^{\perp} \subset W', \quad (\text{CDI1.7})$$

where $d'\lambda = (r'_+\lambda, -r'_-\lambda)$. To prove the second equality completely, a continuous $\phi \in W'$ vanishing on Z gives the continuous functional $j \mapsto \phi(tj)$ on J . Extend it to $\lambda \in A'$ by (CDI1.5); (CDI1.4) gives $\phi = d'\lambda$. Conversely every $d'\lambda$ vanishes on Z . Likewise

$$\ker r'_{\pm} = q'Q', \quad \text{im } r'_{\pm} = E_{\pm}^{\perp} \subset V'_{\pm}. \quad (\text{CDI1.8})$$

Thus every transpose image needed for the coefficient cohomology is calculated exactly.

CDI2. The dual global and costalk complexes

The global CSP complex is

$$D^0 = W \xrightarrow{d} D^1 = A \xrightarrow{0} D^2 = A. \quad (\text{CDI2.1})$$

Its continuous dual, with the convention (CDI0.4), is

$$(D^\vee)^{-2} = A' \xrightarrow{0} (D^\vee)^{-1} = A' \xrightarrow{\lambda \mapsto (r'_+ \lambda, -r'_- \lambda)} (D^\vee)^0 = V'_+ \oplus V'_-. \quad (\text{CDI2.2})$$

The minus sign in the second chart comes from the original chart difference, not from an added dualization convention: the Hom differential at degree -1 has sign $+1$.

Equations (CDI1.7) give all groups and their evaluation pairings:

$$H^{-2}(D^\vee) = A', \quad H^{-1}(D^\vee) = q'Q' \cong Q', \quad H^0(D^\vee) = W'/Z^\perp \cong Z'. \quad (\text{CDI2.3})$$

The last map is restriction to Z ; it is onto because the projection in (CDI1.4) supplies a continuous extension of every functional on Z . For representatives the exact pairings are

$$\begin{aligned} \langle \lambda, a \rangle &= \lambda(a) \quad (H^{-2}, H^2), \\ \langle q'\mu, [a] \rangle &= \mu([a]) \quad (H^{-1}, H^1), \\ \langle [\phi], z \rangle &= \phi(z) \quad (H^0, H^0). \end{aligned} \quad (\text{CDI2.4})$$

The middle expression is independent of a representative because $q'\mu$ vanishes on J ; the last is independent because boundaries are exactly $Z^\perp$.

At each pole, the costalk complex is

$$L_p^0 = V_p \xrightarrow{r_p} L_p^1 = A \xrightarrow{0} L_p^2 = A, \quad p \in \{+, -\}. \quad (\text{CDI2.5})$$

Its dual is

$$(L_p^\vee)^{-2} = A' \xrightarrow{0} (L_p^\vee)^{-1} = A' \xrightarrow{r'_p} (L_p^\vee)^0 = V'_p, \quad (\text{CDI2.6})$$

Its coefficient degrees are $-2, -1, 0$, and its cohomology is

$$H^{-2}(L_p^\vee) = A', \quad H^{-1}(L_p^\vee) = q'Q' \cong Q', \quad H^0(L_p^\vee) = V'_p/E_p^\perp \cong E'_p. \quad (\text{CDI2.7})$$

The pairings are evaluation on the original costalk groups A, Q, E_p , with both positive local complex orientations retained as in CSP6.

If one wants a complex indexed in degrees $0, 1, 2$, the precisely specified shift is $D^\vee[-2]$: its term of degree i is $(D^{2-i})'$, and its differential has no additional sign because the shift is even. It pairs with D in complementary degrees summing to two. This is a shift of the constructed coefficient complex; calling it a shifted complex does not establish a sheaf-duality comparison.

CDI3. Separated perfect continuous-dual pairings and their topologies

For each Hausdorff locally convex space E in the cohomology list $Z, Q, A, E_\pm$, the evaluation pairing $E' \times E \rightarrow \mathbb{C}$ is separated in both arguments. A nonzero vector is separated by a continuous functional using Hahn-Banach and a continuous seminorm nonzero on that vector; a nonzero functional is nonzero on some vector by its definition. The dual groups in (CDI2.3) and (CDI2.7) are the **entire continuous duals** of the respective primal groups. This is the precise sense of a perfect continuous-dual pairing used here. It does not mean that every discontinuous algebraic functional is continuous, and does not assert a Hilbert inner product or positivity.

Equipped with the weak dual topology, the second dual in the opposite direction is also exactly the original vector space. Indeed a continuous linear $\Lambda : E'_\sigma \rightarrow \mathbb{C}$ is bounded by finitely many evaluation seminorms $|\lambda(e_i)|$. It vanishes on the common kernel of those evaluations, factors through their finite-dimensional image, and is a linear combination of them. Hence $\Lambda(\lambda) = \lambda(e)$ for one $e \in E$, unique by separation. Thus

$$(E'_\sigma)' = E \quad (\text{CDI3.1})$$

as vector spaces with the specified evaluation map. The pairing is separately continuous. No claim of joint continuity or an isomorphism between the original topology on E and a bidual topology is required by this argument.

For weak dual coefficient topologies, all cohomology identifications in CDI2 are topological identifications with the respective weak duals. For $q'Q' \subset A'$, evaluation at a is exactly evaluation at $q(a)$, and every element of Q has a representative, so the induced weak topology is exactly $\sigma(Q', Q)$. For the degree-zero quotients, the explicit topological splittings (CDI1.2) and (CDI1.4) transpose to finite direct-product splittings of weak duals; quotienting by the entire complementary dual factor gives precisely $(E_p)'_\sigma$ or Z'_σ .

For clarity, the surjection $A'_\sigma \rightarrow J'_\sigma$ in (CDI1.5) is also a quotient map. A basic weak neighborhood constrains a functional on a finite-dimensional subspace $L \subset A$. The possible restrictions to J are determined solely by its values on $L \cap J$. Restriction $L' \rightarrow (L \cap J)'$ is an open surjective finite-dimensional map. Any two compatible continuous functionals on J and L combine to a continuous functional on $J + L$: a finite-dimensional complement of $J \cap L$ in L maps homeomorphically onto its finite-dimensional image in the Hausdorff quotient A/J , yielding a topological direct sum with J . Hahn–Banach then extends the combined functional to A . This proves that images of basic weak neighborhoods are weak neighborhoods. Thus (CDI1.5) is exact with its stated weak subspace and quotient topologies.

For strong duals the following statements are proved without any bounded-lifting assumption. The decompositions (CDI1.2) and (CDI1.4) are topological finite direct sums, so their strong duals also split. In particular the degree-zero dual cohomology is exactly Z'_β , respectively $(E_p)'_\beta$, and the degree-minus-two group is A'_β . All dual differential images are closed in both weak and strong coefficient topologies, since they are the annihilators of the displayed closed kernels and those annihilators are intersections of evaluation kernels.

The degree-minus-one group with its topology inherited from A'_β is the vector space Q' with the exact seminorm family

$$p_B^{\text{induced}}(\mu) = \sup_{a \in B} |\mu(q(a))|, \quad B \subset A \text{ bounded.} \quad (\text{CDI3.2})$$

The canonical map

$$Q'_\beta \longrightarrow (J^\perp, \text{subspace topology from } A'_\beta), \quad \mu \longmapsto q' \mu \quad (\text{CDI3.3})$$

is a continuous bijection. Each image $q(B)$ is bounded in Q , proving continuity. This proof does not identify (CDI3.2) with uniform convergence on **all** bounded subsets of Q . A proof that bounded subsets of Q have bounded lifts would give that additional identification, but no such unproved assertion is used. Likewise the strong-topology surjection $A'_\beta \rightarrow J'_\beta$ is algebraically onto and continuous by the proved extension theorem; no strong open-mapping claim is inserted. These precise strong seminorms coexist with the full weak-topology duality already proved.

CDI4. Transpose of the two-support localization map

Write $L = L_+ \oplus L_-$, with

$$L^0 = W, \quad L^1 = A^2, \quad L^2 = A^2, \quad d_L^0(v_+, v_-) = (r_+ v_+, r_- v_-), \quad d_L^1 = 0.$$

The actual CSP support-to-global map has

$$k^0 = 1_W, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CDI4.1})$$

Its continuous transpose is the chain map $k^\vee : D^\vee \rightarrow L^\vee$ with

$$(k^\vee)^0 = 1_{W'}, \quad (k^\vee)^{-1} \lambda = (\lambda, -\lambda), \quad (k^\vee)^{-2} \lambda = (\lambda, \lambda). \quad (\text{CDI4.2})$$

The dual supported differential is

$$d_{L^\vee}^{-1}(\lambda_+, \lambda_-) = (r'_+ \lambda_+, r'_- \lambda_-). \quad (\text{CDI4.3})$$

Substitution gives $d_{L^\vee}^{-1}(\lambda, -\lambda) = d_{D^\vee}^{-1} \lambda$, proving the only nonzero chain compatibility directly.

The induced maps on cohomology are therefore

$$\begin{aligned}
A' &\longrightarrow A'^2 : \lambda \mapsto (\lambda, \lambda) \quad \text{in degree } -2, \\
Q' &\longrightarrow Q'^2 : \mu \mapsto (\mu, -\mu) \quad \text{in degree } -1, \\
Z' &\longrightarrow E'_+ \oplus E'_- : \phi \mapsto (\phi|_{E'_+}, \phi|_{E'_-}) \quad \text{in degree } 0.
\end{aligned} \tag{CDI4.4}$$

Each is precisely adjoint to its primal support-to-global map under CDI2's evaluation. For one pole alone, the transpose map from the global dual has degrees $(0, -1, -2)$:

$$k_+^\vee = (\text{pr}_{V'_+}, +1, +1), \quad k_-^\vee = (\text{pr}_{V'_-}, -1, +1). \tag{CDI4.5}$$

Thus the two degree-one signs differ, while both positive local orientation classes have the same degree-two sign.

CDI5. The full dual localization row and cone signs

Let $O = A[\text{degree } 0] \oplus A[\text{degree } 1]$ be the annular model, with zero differential. Reversing the CSP7.6 localization row by the continuous transpose gives the following algebraically exact row:

$$\begin{aligned}
0 \rightarrow A' &\xrightarrow{\lambda \mapsto (\lambda, \lambda)} A'^2 \xrightarrow{(\lambda_+, \lambda_-) \mapsto -\lambda_+ + \lambda_-} A' \xrightarrow{0} Q' \\
&\xrightarrow{\mu \mapsto (\mu, -\mu)} Q'^2 \xrightarrow{(\mu_+, \mu_-) \mapsto q'(\mu_+ + \mu_-)} A' \xrightarrow{r'} Z' \xrightarrow{\text{res}} E'_+ \oplus E'_- \rightarrow 0.
\end{aligned} \tag{CDI5.1}$$

Here $r : Z \rightarrow A$ is the common restriction, and in (CDI1.3) its transpose is

$$r'\lambda = (\lambda|_J, 0, 0). \tag{CDI5.2}$$

Exactness can be checked without a general topological exactness axiom. The first two nonzero maps are diagonal and signed difference, hence have the displayed kernels and images. The next zero map has kernel all of A' , exactly the preceding image. The kernel of the sum on Q'^2 is its anti-diagonal. The image of that sum composed with q' is $J^\perp$, the kernel of r' . Hahn–Banach makes r' onto the J' -summand of Z' , which is exactly the kernel of endpoint restriction. That restriction is onto by the direct-sum decomposition of Z . All arrows are continuous for both dual topologies; the weak topologies are the exact subspace and quotient topologies proved in CDI3. Strong algebraic exactness does not assert the additional strong quotient identifications excluded in CDI3.

One can check every cone sign directly too. Put $E = \text{Cone}(k)$ with the standard differential $d_E(x, y) = (d_Dx + ky, -d_Ly)$. Its coefficient degrees from CSP7.3 are

$$E^{-1} = W, \quad E^0 = W \oplus A^2, \quad E^1 = A \oplus A^2, \quad E^2 = A.$$

By (CDI0.4), its dual differentials are exactly

$$\begin{aligned}
(E^\vee)^{-2} &= A' \longrightarrow (E^\vee)^{-1} = A' \oplus A'^2, \quad \lambda \mapsto (0, -\lambda, -\lambda), \\
(\lambda, \mu_+, \mu_-) &\mapsto (d'\lambda, \lambda, -\lambda) \quad \text{from degree } -1 \text{ to } 0, \\
(\phi, \nu_+, \nu_-) &\mapsto -\phi + (r'_+\nu_+, r'_-\nu_-) \quad \text{from degree } 0 \text{ to } 1.
\end{aligned} \tag{CDI5.3}$$

The last target is W' . In particular the diagonal in the first line has a negative sign from the Hom differential in degree -2 , while the diagonal cohomology map in (CDI4.2) has a positive sign; these are different maps.

The transpose of the CSP7.4 annular inclusion is a quasi-isomorphism $E^\vee \rightarrow O^\vee$. In degree zero it sends $(\phi, \nu_+, \nu_-) \mapsto \nu_+ + \nu_-$; in degree minus one it sends $(\lambda, \mu_+, \mu_-) \mapsto -\mu_+ + \mu_-$. To prove this, a degree-zero cocycle has $\phi = (r'_+\nu_+, r'_-\nu_-)$. Its displayed image is zero precisely when $\nu_- = -\nu_+$, in which case it is the boundary of $(\nu_+, 0, 0)$ in degree minus one. Every target value has a cocycle representative by taking $\nu_+ = \nu$, $\nu_- = 0$, and the required ϕ . A degree-minus-one cocycle has $\lambda = 0$, while the other coordinates are

arbitrary. Its displayed image vanishes precisely on the diagonal, which is the image of degree minus two in (CDI5.3). It is onto by taking $\mu_+ = 0$. The other two degrees have zero cohomology by injectivity of the first differential and surjectivity of the last. These choices are continuous finite operations, so the annular dual comparison is valid with either topology.

For a general chain map with the present Hom convention, the exact relation between the two cone presentations is

$$\text{Cone}(k)^\vee \xrightarrow{\sim} \text{Cone}(k^\vee)[-1], \quad (\phi, \psi) \mapsto ((-1)^{n+1}\psi, \phi) \text{ in degree } n. \quad (\text{CDI5.4})$$

Indeed the differential of the left side is $(d_{D^\vee}\phi, d_{L^\vee}\psi + (-1)^{n+1}k^\vee\phi)$; the target shifted-cone differential is $(\alpha, \beta) \mapsto (-d_{L^\vee}\alpha - k^\vee\beta, d_{D^\vee}\beta)$. Substitution of the displayed sign proves equality. This accounts for the localization-cone shift without suppressing a boundary sign.

CDI6. Ordinary transposes, contragredients, and original spectral jets

For $a > 0$, use the CSP actions

$$T_a b(u) = b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \quad \rho_-(a)(h, d_0, d_1) = (ah(a \cdot), ad_0, d_1). \quad (\text{CDI6.1})$$

The combined sphere-cover action on D , extended from the recovered integer covers to the indicated real coefficient action as in CSP8, is

$$\mathbf{B}_a^0 = \rho_+(a) \oplus \rho_-(a), \quad \mathbf{B}_a^1 = T_a, \quad \mathbf{B}_a^2 = aT_a. \quad (\text{CDI6.2})$$

The same degree factors occur on each costalk, and on O the degree-zero and degree-one actions are T_a and aT_a . A nonintegral a here does not designate a holomorphic map $z \mapsto z^a$.

The ordinary transpose $(\mathbf{B}_a^i)'$ satisfies the adjoint identity $\langle (\mathbf{B}_a^i)' \lambda, x \rangle = \langle \lambda, \mathbf{B}_a^i x \rangle$. For an invariant evaluation pairing one must instead use the contragredient

$$\mathbf{B}_a^\vee|_{(D^\vee)^{-i}} = (\mathbf{B}_{a^{-1}}^i)'. \quad (\text{CDI6.3})$$

Then $\langle \mathbf{B}_a^\vee \lambda, \mathbf{B}_a x \rangle = \langle \lambda, x \rangle$. Its exact coefficient actions, in degrees 0, -1 , -2 , are

$$\rho_+(a^{-1})' \oplus \rho_-(a^{-1})', \quad (T_{a^{-1}})', \quad a^{-1}(T_{a^{-1}})'. \quad (\text{CDI6.4})$$

In particular the four dual endpoint characters are $1, a^{-1}, a^{-1}, 1$. In the common-restriction coordinates (CDI1.3), the other degree-zero dual summand J' carries $(T_{a^{-1}}|_J)'$. Every transposed support and localization map intertwines these actions because its primal map intertwines (CDI6.2); direct substitution in CDI4–CDI5 verifies the factors.

Retain the CSP Mellin comparison including its scalar:

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta T_a b(s) = a^s \Theta b(s), \quad \Theta J = \mathcal{J}. \quad (\text{CDI6.5})$$

Thus $[b] \mapsto [\Theta b]$ identifies Q with $\mathcal{B}/\mathcal{J}$. For every actual nontrivial zero ρ and $0 \leq j < m_\rho$, its continuous jet functional is

$$\delta_{\rho,j}([b]) = (\Theta b)^{(j)}(\rho). \quad (\text{CDI6.6})$$

Using the raw transform instead would multiply this functional by 2; that factor is not suppressed in (CDI6.5).

Leibniz's rule gives the full triangular matrices

$$T_a' \delta_{\rho,j} = a^\rho \sum_{\ell=0}^j \binom{j}{\ell} (\log a)^{j-\ell} \delta_{\rho,\ell}, \quad (T_{a^{-1}})' \delta_{\rho,j} = a^{-\rho} \sum_{\ell=0}^j \binom{j}{\ell} (-\log a)^{j-\ell} \delta_{\rho,\ell}. \quad (\text{CDI6.7})$$

The contragredient of the normal action aT_a has instead

$$a^{-\rho-1} \sum_{\ell=0}^j \binom{j}{\ell} (-\log a)^{j-\ell} \delta_{\rho,\ell}. \quad (\text{CDI6.8})$$

These formulas are valid for every actual zero with its full multiplicity. They retain the extra geometric degree factor and do not assume $\Re \rho = 1/2$.

CDI7. Mirrors, orientation, and the reflected degree characters

Let $\varepsilon = +1$ for the holomorphic sphere swap $w(z) = -1/z$, and $\varepsilon = -1$ for the antiholomorphic swap $\alpha(z) = -1/\bar{z}$. Both use the actual source coefficient comparison of CSP9. In degrees 0, 1, 2, their global maps are

$$M_\varepsilon = (\text{chart swap}, -R, \varepsilon R). \quad (\text{CDI7.1})$$

Since these are involutions, their ordinary transpose and contragredient coincide. On $D^\vee$, in degrees 0, -1, -2, their exact maps are

$$M_\varepsilon^\vee = (\text{dual chart swap}, -R', \varepsilon R'). \quad (\text{CDI7.2})$$

The two supported copies have degree-minus-one action $(\lambda_+, \lambda_-) \mapsto (R'\lambda_-, R'\lambda_+)$ and degree-minus-two action $(\lambda_+, \lambda_-) \mapsto (\varepsilon R'\lambda_-, \varepsilon R'\lambda_+)$. The annular dual has degree-zero action R' and degree-minus-one action $-\varepsilon R'$. Thus (CDI4.2)'s anti-diagonal and diagonal intertwine these maps: in degree minus one, the swap sends $(\lambda, -\lambda)$ to $(-R'\lambda, R'\lambda)$, precisely the image of $-R'\lambda$; in degree minus two, the image of $\varepsilon R'\lambda$ is the required diagonal. The signed annular maps in CDI5 agree too.

Raw Mellin reflection, or its retained scalar version, gives $\Theta Rb(s) = \Theta b(1-s)$. Hence

$$R'\delta_{\rho,j} = (-1)^j \delta_{1-\rho,j}. \quad (\text{CDI7.3})$$

The actual global degree-one mirror is its negative, so its dual acts by $(-1)^{j+1} \delta_{1-\rho,j}$. The degree-two mirror has the factor $\varepsilon(-1)^j$. The functional equation preserves the exact multiplicity between ρ and $1-\rho$, since its multiplier is holomorphic and nonzero there.

The degree character changes the reflection center of the represented exponents. For a coefficient carrying $a^k T_a$, the exact identity is

$$(a^k T_a)R = a^{2k+1} R(a^{-k} T_{a^{-1}}). \quad (\text{CDI7.4})$$

It follows immediately from $T_a R = a R T_{a^{-1}}$, retaining both powers a^k . The exponents paired by this reflection are $\rho + k$ and $1 - \rho + k$, whose sum is $2k + 1$. In the unshifted coefficient this is 1; in the actual normal direction $k = 1$, it is 3. For the invariant-pairing contragredient, transposition at the inverse parameter gives

$$B_a^\vee M_\varepsilon^\vee = a^{-(2k+1)} M_\varepsilon^\vee B_{a^{-1}}^\vee \quad (\text{CDI7.5})$$

on a degree with that factor. Its exponent pair sums to $-(2k+1)$. On the degree-zero graph and endpoint coordinates $k = 0$; the source formulas (CDI6.1) verify the same factor a directly before transposing. Neither mirror allows the normal factor to be omitted, and the antiholomorphic sign $\varepsilon = -1$ remains independent of these positive dilation characters.

CDI8. Finite full-zero blocks inside the actual quotient

An exact finite-block comparison with the dual can be constructed without asserting global self-duality. Write $\mathcal{Q} = \mathcal{B}/\mathcal{J}$ using (CDI6.5). The full zero-jet map is injective by the definition of $\mathcal{J}$, with the closed-image theorem already identifying it with the actual source quotient.

For completeness, each individual full jet block has an actual representative. Use the retained auxiliary entire function

$$F_*(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \in \mathcal{B}. \quad (\text{CDI8.1})$$

It has precisely the nontrivial-zero divisor and all its multiplicities; its endpoint values are $F_*(0) = F_*(1) = 1/8$, and its values at the cancelled trivial-zero locations are those retained in SSI3 and CSP11. For $m = m_\rho$, let

$$A_\rho(s) = F_*(s)/(s - \rho)^m. \quad (\text{CDI8.2})$$

Taylor division makes it entire with $A_\rho(\rho) \neq 0$. It belongs to $\mathcal{B}$: outside a compact neighborhood of ρ , division costs at most a fixed polynomial factor on any strip; inside, holomorphicity gives a finite bound. For $0 \leq j < m$, let $P_{\rho,j}(t)$ be the Taylor polynomial through degree $m - 1$ of $t^j/A_\rho(\rho + t)$, and set

$$\psi_{\rho,j}(s) = A_\rho(s)P_{\rho,j}(s - \rho). \quad (\text{CDI8.3})$$

This is in $\mathcal{B}$, has jet t^j modulo t^m at ρ , and vanishes with the full required order at every other nontrivial zero. This is a comparison construction using the full factor in (CDI8.1), not a replacement for original zeta in the forthcoming residue formula.

Define $\mathcal{Q}_{\text{fin}}$ as those classes whose full zero-jet family has finite support. Equations (CDI8.2)–(CDI8.3), together with injectivity of the joint jet map, prove the exact vector-space identification

$$\mathcal{Q}_{\text{fin}} \cong \bigoplus_{\rho} \mathbb{C}[t]/(t^{m_\rho}). \quad (\text{CDI8.4})$$

Let

$$\mathcal{D}_{\text{fin}} = \text{span}_{\mathbb{C}}\{\delta_{\rho,j} : \rho \text{ actual nontrivial zero, } 0 \leq j < m_\rho\} \subset \mathcal{Q}'. \quad (\text{CDI8.5})$$

The individual jets are continuous, so the displayed inclusion really is in the continuous dual. No statement that $\mathcal{Q}_{\text{fin}}$ is dense in $\mathcal{Q}$ is used.

CDI9. The original-zeta residue map and its full determinant

For $[G] \in \mathcal{Q}_{\text{fin}}$, define a functional on $\mathcal{Q}$ by

$$\iota([G])([F]) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \quad (\text{CDI9.1})$$

Only finitely many residues are nonzero: the potentially nonzero poles are at the reflections of the finite support of the zero jets of G . Changing either representative by an element of $\mathcal{J}$ makes every residue zero. Each residue is a finite linear combination of the continuous jets of F ; consequently (CDI9.1) is a well-defined continuous functional. Its denominator is the original zeta, not (CDI8.1).

Write $\sigma = 1 - \rho$, $m = m_\rho = m_\sigma$, and

$$\zeta(\rho + t) = t^m u(t), \quad u(0) \neq 0. \quad (\text{CDI9.2})$$

Pair the monomial jet t^i at ρ with the monomial jet $(s - \sigma)^j$ at σ . Reflection substitutes $-t$ in the latter. The exact local matrix is

$$P_{ij} = (-1)^j [t^{m-1-i-j}] \frac{1}{u(t)}, \quad 0 \leq i, j < m, \quad (\text{CDI9.3})$$

where a coefficient with negative index is zero. The matrix is anti-triangular. Its anti-diagonal entries are $(-1)^j/u(0)$; the reversal permutation has sign $(-1)^{m(m-1)/2}$, while the product of those entry signs has the same sign. Hence

$$\boxed{\det(P) = u(0)^{-m} \neq 0.} \quad (\text{CDI9.4})$$

Every nilpotent order is included. Equations (CDI8.3) and (CDI9.4) prove the vector-space isomorphism

$$\boxed{\iota : \mathcal{Q}_{\text{fin}} \xrightarrow{\sim} \mathcal{D}_{\text{fin}}.} \quad (\text{CDI9.5})$$

This statement does not choose a topology on $\mathcal{Q}_{\text{fin}}$ or assert continuity of a map extending to all of $\mathcal{Q}$.

There is, however, a proved complete weak-dual detection statement:

$$\overline{\mathcal{D}_{\text{fin}}^{\sigma(Q', Q)}} = Q'. \quad (\text{CDI9.6})$$

To prove it without reflexivity, fix $\lambda \in Q'$ and finitely many classes $q_1, \dots, q_r$. Consider the linear image $L = \{(d(q_1), \dots, d(q_r)) : d \in \mathcal{D}_{\text{fin}}\} \subset \mathbb{C}^r$. Any coefficient vector c annihilating L makes $\sum c_i q_i$ annihilated by every full zero jet, hence zero in Q . It follows that $\sum c_i \lambda(q_i) = 0$. Finite-dimensional linear algebra now puts $(\lambda(q_i))_i$ in L , so some $d \in \mathcal{D}_{\text{fin}}$ agrees **exactly** with λ on all the chosen vectors. This proves density for every weak neighborhood. It proves no strong-dual density assertion.

CDI10. The residue factor and the normal-dual factor remain distinct

The residue comparison has the exact covariance

$$\boxed{\iota(T_a G) = a(T_{a^{-1}})' \iota(G)}. \quad (\text{CDI10.1})$$

Indeed its integrand changes by a^{1-s} when $G(1-s)$ is replaced by $(T_a G)(1-s)$. Pulling out the scalar a leaves $a^{-s} F(s)$, which is exactly the contragredient transpose in (CDI10.1). The equality holds for the full finite residue sum and all its jets, not only for simple zeros.

More generally, with $P(F, G) = \iota(G)(F)$, the full character law is

$$P(a^k T_a F, a^\ell T_a G) = a^{k+\ell+1} P(F, G) \quad (\text{CDI10.2})$$

whenever the second argument has finite zero-jet support. This follows by multiplying the integrand by $a^{k+s} a^{\ell+1-s}$.

The contragredient of the actual normal representation $Q(-1)$, whose action is aT_a , was independently calculated in (CDI6.4) as

$$\mathbf{N}_a^\vee = a^{-1}(T_{a^{-1}})'. \quad (\text{CDI10.3})$$

Therefore the same residue map from the unshifted coefficient has the precise comparison

$$\iota(T_a G) = a^2 \mathbf{N}_a^\vee \iota(G). \quad (\text{CDI10.4})$$

The coefficient-dual character $a(T_{a^{-1}})'$ in (CDI10.1) and the normal contragredient in (CDI10.3) differ by a^2 . They cannot be identified by dropping the geometric degree or the inverse in the dual action. Equations (CDI7.4) and (CDI10.2) are consistent: unshifted paired exponents sum to 1; two normal exponents sum to 3. These are exact morphism and pairing identities. They are not positivity statements, and the bilinear residue pairing has not been declared Hermitian.

CDI11. Actual receiving result

The continuous duals of the specified global, pole-costalk, and localization complexes now have full formulas, all their images and kernels, and perfect separated continuous-dual pairings on every cohomology degree. Closedness of J and the actual inverse Σ^{-1} enter exactly through CDI1; they ensure that the original quotient and its continuous dual are separated and that no image closure is substituted in the transpose calculation.

The transpose localization map has the anti-diagonal in degree minus one and diagonal in degree minus two; the complete cone and long-row signs are CDI5. The antiholomorphic sphere operation retains its negative degree-two orientation. The ordinary transpose, invariant-pairing contragredient, and original-zeta residue correspondence remain separate exact maps with their full degree characters.

This is a completed duality calculation for the actual finite coefficient models. It does not identify the continuous-dual coefficient complex with the derived sheaf Hom into a dualizing complex, and does not supply an unproved comparison to Deligne's finite-rank étale obstruction receiver. The concrete comparison now available for that further calculation is the entire weak continuous dual in CDI2–CDI5, together with the explicitly reflected finite-jet subspace, nonzero local determinant, and weak-dual density in CDI8–CDI10. No operation or metric on the support τ was used.

CDI12. Independent verification of RD0–RD9

After completing CDI0–CDI11, the full independent root derivation `ORIGINAL_ZETA_RESIDUE_DUAL_AND_SUPPORT_MAPS.md`, RD0–RD9, was read and checked. The reviewed file SHA256 is 65ae37a4303a76dae2fa5d79820201858bce8773afae7138e5b7175586671641. Its local determinant, explicit inverse, weak-density argument, original-zeta reflection law, and complete normal-action factor require no mathematical correction.

RD1’s recursion is the Taylor inversion of the nonvanishing local unit A_ρ . Its isolator has jet 1 at precisely the chosen full block and the required zero jet at all other blocks. Multiplication by the stated finite polynomials preserves $\mathcal{B}$. This agrees with CDI8, which supplies the same global block construction independently.

RD2 uses Taylor-coefficient functionals $\delta_{\rho,j}^{\text{coeff}} = F^{(j)}(\rho)/j!$, whereas (CDI6.6) used unscaled derivatives. Its formula

$$P_{\rho,j}(t) = [t^{m-1-j}u_\rho(t)]_{<m}$$

does give the exact coefficient functional: subtraction of its omitted Taylor terms leaves a multiple of t^m ; division by $t^m u_\rho(t)$ therefore gives t^{-j-1} plus a holomorphic germ. The residue is $F^{(j)}(\rho)/j!$. Thus the factorial conventions in RD2 and RD5.4 agree, and the unscaled formulas in CDI6.7 are their exact multiples by $j!$.

For RD3, a class supported at $\sigma = 1 - \rho$ with local polynomial $P(-t)$ has reflected germ $P(t)$ at ρ . It therefore maps under ι to the specified original-zeta functional $\lambda_{\rho,P}$, giving an actual inverse on every finite sum of block functionals. The determinant is exactly (CDI9.4), including cancellation of the two anti-diagonal signs, rather than a determinant modulo a nonzero unspecified constant. RD4 is exactly the finite-evaluation interpolation argument proved in CDI9; the resulting net is pointwise convergent, with no strong-topology or equicontinuity claim.

The second identity in RD5.2 retains the original variable before taking residues: multiplication of the first argument by s and of the reflected second argument by $1-s$ sums to multiplication by 1. Hence $\iota L = (1-L')\iota$. The full factor in RD5.5 is $a^{k+\ell+1}$, as independently proved in CDI10. For two normal-direction arguments it is a^3 ; for one normal and one unshifted argument it is a^2 . In particular RD5.5 correctly preserves the a^2 comparison with the contragredient of an already normal-shifted coefficient. The cochain degree-character distinction is not lost.

RD6’s original-zeta symmetry factor and its minus sign can be checked before any simplification. Set $\sigma = 1 - \rho$. The functional equation gives

$$t^m u_\rho(t) = \chi(\rho + t)(-t)^m u_\sigma(-t),$$

which is precisely its local-unit identity. In a residue, substitute $s = 1 - z$, so $ds = -dz$. The denominator becomes $\zeta(1 - z) = \zeta(z)/\chi(z)$. Thus the reflected integrand is

$$-\frac{\chi(z)G(z)F(1-z)}{\zeta(z)}dz,$$

with all derivatives of the displayed χ -germ retained. This is RD6.3, including the orientation sign. No claim that its global meromorphic multiplier preserves $\mathcal{B}$ is needed. The separately recalled trace form in RD6.4 sees only the constant Taylor coefficient on each block: multiplication by a germ $h(t)$ on $\mathbb{C}[t]/(t^m)$ is triangular with diagonal $h(0)$, so its trace is $mh(0)$. In contrast the residue matrix retains all the jets. For the displayed full trace sum, absolute convergence follows from rapid decay of the two $\mathcal{B}$ factors on the strip of zeros and the retained zero-count bound $n(R) = O(R^{3/2})$ from S2. Taking a decay exponent with the product power greater than $3/2$ proves convergence by dyadic shells. These statements add no positivity assertion.

RD7 uses the actual topological comparison $\kappa([b]) = [\frac{1}{2}\mathcal{M}_0 b]$; its transpose preserves that same factor $1/2$. Its support-map transposes coincide with CDI4–CDI5: anti-diagonal for the degree-one difference, diagonal for the degree-two sum, and $-\lambda_+ + \lambda_-$ for the angular boundary. The weak density passes through κ' because a topological isomorphism induces a weak-dual topological isomorphism.

Finally RD8 retains the correct raw inverse factor $1/(2\zeta)$, the full trivial-zero derivatives, and both endpoint values. Its expansion of A_ρ is the Taylor convolution of the complete multiplier in F_0 with $\zeta(s)/(s-\rho)^m$, so $a_j = \sum_{h=0}^j C^{(h)}(\rho)\zeta^{(m+j-h)}(\rho)/(h!(m+j-h)!)$ is exact. The original local residue uses u_ρ of ζ , not this auxiliary multiplier. RD9 keeps the continuous-dual receiver distinct from the not-yet-constructed Deligne support-duality comparison. This matches the exact scope of CDI11.

Pullback of the original-zeta coefficient sheaf to the complex projective line

Complete derivation, 24 September 2026. Proof locators CSP0–CSP13.

CSP0. Input stage, source coverage, and the map being constructed

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. This calculation starts after the complete arithmetic reconstruction, on the actual coefficient sheaf of CS and CSL. All complex vector spaces, scalar actions, analytic coordinates and quotient maps below belong to this receiving construction. They supply no addition, subtraction, metric, numerical coordinate, or parity on τ .

The complete corrected argument in `USER_ARGUMENT_RECONSTRUCTION.md` and the prerequisite checks PC01–PC04 govern this use. In particular, fixed support does not imply a trivial stalk; the retracted addition at τ is not used; neither the counting unit nor selected primes initialize this construction.

The original author source is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. For this derivation the unchanged author LaTeX file `sources/CC_0903_2024_v3/author_source/announc3.tex` was read at lines1471–1670, including `projback`, `fonction3`, `restmaps`, `bord`, `hzero`, `rep1`, `liftw` and `court`. Its source identity and preserved archive are recorded in `CC_SHEAF_SOURCE_READING_PRIVATE.json`. CS0–CS10 and CSL0–CSL9 were read in full. These local derivations prove the precise finite-unit sector, endpoint conventions, and raw Mellin comparison used here.

The reading of the later Connes–Consani author file `CC.tex` covered lines388–423 and485–608: the field-valued point construction, its three-point support, and the stated antiholomorphic real structure. That bounded reading is not described as a new reading of the entire paper. The source is arXiv:2609.00299v1. The two mirrors will remain distinct: the holomorphic $z \mapsto -1/z$ and the actual antiholomorphic $z \mapsto -1/\bar{z}$ have different orientation degrees.

The subsequently completed exact-image derivations `../quantum_tau_programme_bridge_20260924/EXACT_SCHWARTZ_SUMMATION_IMAGE.md`, `SSI0–SSI10`, and `EXACT_SCHWARTZ_IMAGE_INDEPENDENT_CHECK.md`, `ESI0–ESI12`, were read in full. Their proved equality $J = \bar{J}$, continuous inverse, and source-domain qualification are incorporated below, rather than left as a pending condition.

Let

$$X = \{x_+, \eta, x_-\}, \quad U_+ = \{x_+, \eta\}, \quad U_- = \{\eta, x_-\}, \quad U = \{\eta\}. \quad (\text{CSP0.1})$$

Let $Y = \mathbb{P}^1(\mathbb{C})$ with its usual topology, $D = \{0, \infty\}$, and $Y^\circ = \mathbb{C}^\times$. Define

$$\pi : Y \longrightarrow X, \quad \pi(0) = x_+, \quad \pi(\infty) = x_-, \quad \pi(z) = \eta \quad (z \neq 0, \infty). \quad (\text{CSP0.2})$$

Its inverse images of the three proper nonempty opens are $\mathbb{C}$, $Y \setminus \{0\}$, and $\mathbb{C}^\times$; hence it is continuous. It is the evaluation/specialization map for the projective-line charts: a nonzero field element is a unit and belongs to their common localization.

In particular, $\pi^{-1}(\eta) = \mathbb{C}^\times$. No point of the complex coordinate plane is identified with τ , and the complex chart origin 0 maps to x_+ , not to η . The sphere is a receiving space over the three-point support, not a replacement definition of that support.

The object to compute is

$$\mathcal{F} = \pi^{-1}\Omega \quad (\text{CSP0.3})$$

in the category of sheaves of complex vector spaces. The coefficient spaces and every finite complex below retain their specified locally convex topologies. Derived sheaf operations are taken in that abelian category; no unproved claim about derived categories of locally convex spaces is needed.

CSP1. Coefficients, endpoint lines, and restrictions

Retain the actual source spaces

$$S = \left\{ f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0 \right\},$$

$$V_+ = S \oplus E_+, \quad V_- = S \oplus E_-, \quad E_{\pm} = \mathbb{C}e_{\pm,0} \oplus \mathbb{C}e_{\pm,1}, \quad (\text{CSP1.1})$$

and the Fréchet overlap

$$A = \left\{ a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty, N, j \geq 0 \right\}. \quad (\text{CSP1.2})$$

The endpoint indices retain value at zero and real integral, separately on both charts. Put

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}),$$

$$r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h = \Sigma \hat{h}, \quad \hat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i vt} dv. \quad (\text{CSP1.3})$$

The full Poisson formula, before restricting to S , is

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{CSP1.4})$$

CS3 proves continuity and injectivity of $\Sigma : S \rightarrow A$; Fourier transformation preserves S , squares to one there, and gives $R\Sigma = \Sigma^\wedge$. Consequently

$$J = \Sigma(S) = r_+(V_+) = r_-(V_-), \quad \ker r_{\pm} = E_{\pm}. \quad (\text{CSP1.5})$$

To retain the exact comparison maps used in the earlier source calculations, write

$$C = \bar{J}^A, \quad Q_{\text{alg}} = A/J, \quad Q_H = A/C, \quad N = C/J. \quad (\text{CSP1.6})$$

The exact row $0 \rightarrow N \rightarrow Q_{\text{alg}} \rightarrow Q_H \rightarrow 0$ has the actual quotient topologies. The completed SSI1–SSI7 theorem, independently derived in ESI0–ESI8, now proves

$J = C, \quad N = 0, \quad Q_{\text{alg}} = Q_H = A/J, \quad \Sigma : S \xrightarrow{\sim} J \text{ continuously with continuous inverse.}$

(CSP1.7)

Here is the exact content being imported, including its factors. For every entire F in the full original nontrivial-zero jet ideal, $H(s) = F(s)/(2\zeta(s))$ is regular at those zeros and at 1, and has possible residues $F(-2r)/(2\zeta'(-2r))$ at the trivial zeros. SSI4 proves rapid vertical bounds for $P_N(s)H(s)$, $P_N(s) = \prod_{r=1}^N (s+2r)$, by the full functional equation and its subquadratic-strip maximum-principle estimate. SSI5–SSI6 then construct the actual Schwartz inverse

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt, \quad x > 0, \quad f_F(-x) = f_F(x),$$

$$\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)}, \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \mathcal{M}_0 \Sigma f_F = F. \quad (\text{CSP1.8})$$

The derivative remainder and finite-seminorm bounds there prove that this is a continuous inverse on precisely S , not an approximation in a closure. Equivalently, for $a \in J$ and $x > 0$, $\Sigma^{-1}a(x) = \frac{1}{2} \sum_{k \geq 1} \mu(k)a(kx)$, with the zero-endpoint extension supplied by (CSP1.8), not by termwise evaluation. Those complete proofs supply (CSP1.7). Ordinary cohomology below is therefore already Hausdorff; retaining both earlier quotient names does not leave an unresolved extra kernel.

The sheaf Ω is the diagram $V_+ \xrightarrow{r_+} A \xleftarrow{r_-} V_-$, with global sections the fibre product. CS0–CS2A specify the full-difference interpretation of the original adelic coinvariants used to identify this source sector. Under the alternative moment-null generating-domain reading, the extra kernel computed there must be retained; this note uses the stated CS source, not that alternative.

CSP2. An explicit construction of the inverse-image sheaf

Write $j : Y^\circ \hookrightarrow Y$, $i : D \hookrightarrow Y$, and $\underline{A}_Y, \underline{A}_{Y^\circ}$ for constant sheaves, whose sections are locally constant functions with values in the vector space A . The coefficient topology does not change the meaning of “locally constant.”

The natural map

$$\underline{A}_Y \longrightarrow j_* \underline{A}_{Y^\circ} \quad (\text{CSP2.1})$$

is an isomorphism. Away from the poles this is immediate. At either pole, arbitrarily small punctured disks are connected, so the right-hand stalk is A , and the map of stalks is the identity. A stalkwise isomorphism of sheaves is an isomorphism. This is an assertion about j_* , not an assertion that Rj_* has no higher terms.

Define $i_*(V_+ \oplus V_-)$ as the direct sum of the two skyscraper sheaves, and let

$$\epsilon : \underline{A}_Y \longrightarrow i_*(A \oplus A)$$

be restriction to the two pole stalks. Then

$$\boxed{\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-)}, \quad (a, v_+, v_-) \mapsto (a|_0 - r_+ v_+, a|_\infty - r_- v_-). \quad (\text{CSP2.2})$$

Here and below a pole component is omitted on an open set not containing that pole.

To prove that (CSP2.2) is (CSP0.3), first construct a map $\Omega \rightarrow \pi_* \mathcal{F}$. A section v_+ on U_+ gives the locally constant section $r_+ v_+$ on $\pi^{-1}U_+$, together with pole value v_+ . Use $r_- v_-$ on the other chart and the identity on their overlap. The fibre-product condition makes these maps compatible, so adjunction gives $\pi^{-1}\Omega \rightarrow \mathcal{F}$. Its pole stalks are the identities on $V_\pm$, and its other stalks are the identity on A . It is therefore an isomorphism.

In particular, on a connected disk containing only one pole, sections are exactly $V_\pm$, with their original Fréchet topology. On a connected open avoiding both poles, sections are A . The restrictions into the punctured disk are exactly $r_\pm$, not injections invented from a coordinate picture.

Formula (CSP2.2) gives a short exact sequence of vector-space sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \underline{A}_Y \oplus i_*(V_+ \oplus V_-) \xrightarrow{\epsilon - r} i_*(A \oplus A) \longrightarrow 0. \quad (\text{CSP2.3})$$

Surjectivity is stalkwise: at each pole the constant-sheaf A -coordinate supplies any prescribed value, independently of the restriction map $r_\pm$. All displayed local maps are continuous. This construction does not assume that J is closed or that $\Sigma^{-\frac{1}{2}}$ is continuous.

CSP3. Disk, annulus, and sphere cohomology with these coefficients

Here is the local resolution used in the calculation. For a complex vector space B , let $\mathcal{E}_B^q$ be the sheaf of differential forms locally expressible as a finite sum $\sum_\nu \omega_\nu b_\nu$, with ordinary smooth complex forms ω_ν and vectors $b_\nu \in B$. Its differential acts on the forms. On a star-shaped coordinate ball, the radial homotopy

$$(K\omega)_x = \int_0^1 t^{q-1} \iota_x \omega_{tx} dt \quad (q > 0) \quad (\text{CSP3.1})$$

acts on those finitely many scalar coefficients and satisfies $dK + Kd = 1$ in positive degree and $Kd = 1 - \text{ev}_0$ in degree zero. Thus

$$0 \rightarrow \underline{B} \rightarrow \mathcal{E}_B^0 \xrightarrow{d} \mathcal{E}_B^1 \xrightarrow{d} \cdots \quad (\text{CSP3.2})$$

is a resolution. Partitions of unity act by multiplication, so these sheaves are fine. A locally finite partition produces locally finite sums, which are legitimate sections of the sheaf. Hence this computes sheaf cohomology. It also applies to a possibly noncomplete subspace J or quotient Q_{alg} : only finitely many coefficient vectors occur locally.

For the complete Fréchet spaces $A, V_{\pm}$, one may equivalently use smooth Fréchet-valued forms. The same integral exists as a limit of Riemann sums in every seminorm, and

$$p\left(\int_a^b f(t) dt\right) \leq \int_a^b p(f(t)) dt \quad (\text{CSP3.3})$$

proves continuity of each contraction on compact sets. Finite good covers and their contractions consequently give the same finite coefficient complexes with continuous maps. No completed infinite tensor product is being substituted for a coefficient space.

A disk, or a plane chart with its pole included, is acyclic for a constant sheaf: (CSP3.1) contracts its positive-degree complex. On Y° , put $z = e^{t+i\theta}$. A closed one-form can be written on the covering $\mathbb{R}^2$ as

$$\omega = P(t, \theta) dt + Q(t, \theta) d\theta, \quad \partial_{\theta} P = \partial_t Q,$$

with 2π -periodic coefficients. Its period

$$a = \int_0^{2\pi} Q(0, \theta) d\theta \quad (\text{CSP3.4})$$

is independent of t . After subtracting $a d\theta/(2\pi)$, the explicit primitive

$$g(t, \theta) = \int_0^t P(x, 0) dx + \int_0^{\theta} (Q(t, \phi) - a/(2\pi)) d\phi \quad (\text{CSP3.5})$$

is periodic and has differential equal to the adjusted form. Every two-form $G(t, \theta) dt \wedge d\theta$ is the differential of $(\int_0^t G(x, \theta) dx) d\theta$. These formulas prove

$$R\Gamma(Y^{\circ}, \underline{B}) \simeq B[0] \oplus B[-1]. \quad (\text{CSP3.6})$$

The degree-one coordinate is the positive z -circle period (CSP3.4), with representative $a d\theta/(2\pi)$. They also prove continuity of the period and chosen primitives in the coefficient seminorms on compact sets. A punctured pole disk has the same calculation in its local coordinate.

For the sphere, cover by its two plane charts and use (CSP3.2) to form the Čech double complex. Each chart is acyclic for $\underline{B}$; their overlap has (CSP3.6). The constant restriction difference is $B \oplus B \rightarrow B$, $(a, b) \mapsto a - b$, which is onto. Consequently

$$H^0(Y, \underline{B}) = B, \quad H^1(Y, \underline{B}) = 0, \quad H^2(Y, \underline{B}) = B, \quad H^{>2} = 0. \quad (\text{CSP3.7})$$

The degree-two coordinate is integration with the complex orientation. For a concrete generator one can use

$$\omega_Y = \frac{1}{\pi} \frac{dx \wedge dy}{(1 + |z|^2)^2}, \quad z = x + iy, \quad \int_Y \omega_Y = 1. \quad (\text{CSP3.8})$$

It extends smoothly through infinity. The integral is $2 \int_0^{\infty} r(1 + r^2)^{-2} dr = 1$. The preceding two-chart complex shows that its nonzero integral generates the entire degree-two group; a zero-integral form is exact by that same period calculation. Thus (CSP3.7) is established by the resolution and its maps, not inferred solely from a homotopy picture.

CSP4. Global cohomology and a continuous finite complex

A skyscraper sheaf on either pole has no higher cohomology: sections on an open are either its coefficient space or zero, and all restrictions are onto. Apply (CSP3.7) to (CSP2.3). A finite model for $R\Gamma(Y, \mathcal{F})$ is

$$\begin{aligned} K^0 &= A \oplus V_+ \oplus V_-, & K^1 &= A \oplus A, & K^2 &= A, \\ d_K^0(a, v_+, v_-) &= (a - r_+ v_+, a - r_- v_-), & d_K^1 &= 0. \end{aligned} \quad (\text{CSP4.1})$$

The degree-two coordinate comes from the constant-sheaf summand with the orientation (CSP3.8). There are no other degrees.

The following explicit contraction removes the redundant diagonal A . Define

$$D^0 = V_+ \oplus V_-, \quad D^1 = A, \quad D^2 = A, \quad d_D^0 = r_+ - r_-, \quad d_D^1 = 0. \quad (\text{CSP4.2})$$

Maps $p : K \rightarrow D$ and $s : D \rightarrow K$ are

$$\begin{aligned} p^0(a, v_+, v_-) &= (v_+, v_-), & p^1(b_+, b_-) &= b_- - b_+, & p^2 &= \text{id}, \\ s^0(v_+, v_-) &= (r_- v_-, v_+, v_-), & s^1(b) &= (-b, 0), & s^2 &= \text{id}. \end{aligned} \quad (\text{CSP4.3})$$

They are chain maps and $ps = 1$. Let $h^1(b_+, b_-) = (b_-, 0, 0)$, with every other component of h zero. Direct substitution gives

$$1 - sp = d_K h + h d_K. \quad (\text{CSP4.4})$$

Every map is continuous and uses only the original restrictions, additions in coefficient spaces, and finite products. No inverse to Σ is needed.

It follows that

$$\begin{aligned} H^0(Y, \mathcal{F}) &= \{(\hat{h}, c_0, c_1), (h, d_0, d_1) : h \in S, c_0, c_1, d_0, d_1 \in \mathbb{C}\}, \\ H^1(Y, \mathcal{F}) &= A/J = Q_{\text{alg}}, \\ H^2(Y, \mathcal{F}) &= A, \quad H^{>2}(Y, \mathcal{F}) = 0. \end{aligned}$$

(CSP4.5)

The first isomorphism is topological via the Fourier map and coordinate projections. Degree one has its original quotient topology, which is Fréchet and Hausdorff by (CSP1.7); the previously retained comparison subspace C/J is zero. Degree two has the original Fréchet topology of A , and its isomorphism is induced by the sheaf projection $\mathcal{F} \rightarrow \underline{A}_Y$, followed by complex-oriented integration.

Equivalently, two pole charts for $\mathcal{F}$ are acyclic: on a chart, (CSP2.3) reduces to the complex $A \oplus V_{\pm} \rightarrow A$, $(a, v) \mapsto a - r_{\pm} v$, contracted onto $V_{\pm}$. Their annular overlap has $A[0] \oplus A[-1]$. In the derived Mayer-Vietoris cone its degree-zero overlap produces A/J , and its degree-one loop produces the additional A in degree two. This is the exact Čech-loop contribution.

CSP5. Where the additional term lives over the original support

Because each point of X has a smallest open neighborhood, higher direct-image stalks are the cohomology of the corresponding preimages. The preceding chart and overlap calculations give

$$R^0 \pi_* \mathcal{F} = \Omega, \quad R^1 \pi_* \mathcal{F} = j_{X!} \underline{A}_{\{\eta\}}, \quad R^{q>1} \pi_* \mathcal{F} = 0, \quad (\text{CSP5.1})$$

where $j_X : \{\eta\} \hookrightarrow X$. Indeed the degree-one sheaf has zero stalks at $x_{\pm}$ and stalk A at η , with the zero generation maps from the closed stalks.

Its Čech complex on X is simply $0 \rightarrow A$, in degrees zero and one, so

$$H^0(X, R^1 \pi_* \mathcal{F}) = 0, \quad H^1(X, R^1 \pi_* \mathcal{F}) = A. \quad (\text{CSP5.2})$$

The two-column, two-row Čech resolution computing the composite of direct image and global sections now has no possible nonzero differential beyond its displayed maps. It gives the Leray identifications

$$H^0(Y, \mathcal{F}) = H^0(X, \Omega), \quad H^1(Y, \mathcal{F}) = H^1(X, \Omega), \quad H^2(Y, \mathcal{F}) = H^1(X, R^1 \pi_* \mathcal{F}) = A. \quad (\text{CSP5.3})$$

The first two maps are induced by the adjunction unit $\Omega \rightarrow R\pi_* \pi^{-1} \Omega$. This supplies an exact derived comparison with the finite-poset calculation: the extra term is the punctured-disk loop in the actual receiver. It is not inserted into the stalk of τ as an additional parity label.

For comparison, $R^0 j_* \underline{A}_{Y^{\circ}} = \underline{A}_Y$, whereas $R^1 j_* \underline{A}_{Y^{\circ}}$ has one A -stalk at each pole. Formula (CSP2.1) did not discard those higher direct images. No splitting of the whole object $Rj_* \underline{A}$ is inferred from the list of its cohomology sheaves.

CSP6. Exact pole costalks and their orientation convention

Let p be either pole, B_p a small disk with local coordinate z at 0, or $w = 1/z$ at ∞ . Use the positive local-coordinate circle in each case. Local cohomology is the fibre complex of restriction:

$$R\Gamma_{\{p\}}(B_p, \mathcal{F}) = \text{Cone}(R\Gamma(B_p, \mathcal{F}) \rightarrow R\Gamma(B_p \setminus \{p\}, \mathcal{F}))[-1]. \quad (\text{CSP6.1})$$

Using the explicit resolutions above, choose the fibre-complex convention whose differential is the restriction map. It gives

$$L_p^0 = V_p, \quad L_p^1 = A, \quad L_p^2 = A, \quad d^0 = r_p, \quad d^1 = 0. \quad (\text{CSP6.2})$$

In the unreoriented fibre cone the connecting map inserts the punctured-disk coefficient with a positive sign in the next degree. We keep that coordinate in degree one. In degree two we take its negative, so that the coordinate is positive complex-oriented integration, as checked next. The zero differential into degree two is unchanged. Thus

$$H_{\{p\}}^0 = E_p, \quad H_{\{p\}}^1 = Q_{\text{alg}}, \quad H_{\{p\}}^2 = A, \quad H_{\{p\}}^{>2} = 0. \quad (\text{CSP6.3})$$

These compute the costalk $i_p^! \mathcal{F}$; passage to smaller pole disks gives identity maps on these models.

The degree-two coordinate is the positive complex local fundamental class. To check the sign of the connecting map, choose a radial function χ that is one near the pole and zero near the outside of a small disk. If θ_p is its positive local angle, then

$$-d\chi \wedge \frac{d\theta_p}{2\pi} \quad (\text{CSP6.4})$$

is a compactly supported smooth two-form away from the pole. Its integral is $-\int_0^\infty \chi'(r)dr = 1$. Put $\beta = (1 - \chi)d\theta_p/(2\pi)$, a smooth one-form on the disk, so $d\beta$ is (CSP6.4). Restrict the relative cone to an outer annular collar, where $\beta = d\theta_p/(2\pi)$. Its cone differential sends $(\beta, 0)$ to $(d\beta, d\theta_p/(2\pi))$. Hence the boundary class $(0, d\theta_p/(2\pi))$ equals $(-d\beta, 0)$ in that relative cohomology. It is the **negative** of the positive-integral class. Excision identifies the same sign in the point-support group.

Thus the local connecting map from a positive angular period a is $-a$ in our degree-two integration coordinate. The support-to-global map on that coordinate is $+1$ at both poles, because both positive local complex orientations agree with the sphere orientation. At infinity the local circle is the positive w -circle, not the positive z -circle.

CSP7. Both supports, the complete localization triangle, and all signs

For $D = \{0, \infty\}$, take the direct sum of the disjoint costalk models:

$$L_D^0 = V_+ \oplus V_-, \quad L_D^1 = A \oplus A, \quad L_D^2 = A \oplus A, \\ d_L^0(v_+, v_-) = (r_+v_+, r_-v_-), \quad d_L^1 = 0. \quad (\text{CSP7.1})$$

The support-to-global map $k : L_D \rightarrow D^\bullet$ from (CSP4.2) is

$$k^0 = \text{id}, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CSP7.2})$$

Degree one has the ordered-chart difference; degree two has the sum of the two positive local orientation classes. These are different signs in different degrees.

An explicit verification of the entire triangle is useful. The ordinary cone of k has

$$\text{Cone}(k)^{-1} = V_+ \oplus V_-, \\ \text{Cone}(k)^0 = (V_+ \oplus V_-) \oplus A^2, \quad \text{Cone}(k)^1 = A \oplus A^2, \quad \text{Cone}(k)^2 = A,$$

with

$$d^{-1}(v) = (v, -r_+v_+, -r_-v_-), \\ d^0(v, b_+, b_-) = (r_+v_+ - r_-v_- + b_+ - b_-, 0, 0), \quad d^1(b, c_+, c_-) = c_+ + c_-. \quad (\text{CSP7.3})$$

The overlap model $A[0] \oplus A[-1]$ maps into this cone by

$$a \text{ in degree } 0 \mapsto (0, a, a), \quad a \text{ in degree } 1 \mapsto (0, -a, a). \quad (\text{CSP7.4})$$

These are cocycles. Every degree-zero cocycle differs by $d^{-1}(v)$ from a unique diagonal pair $(0, a, a)$. Every degree-one cocycle has $c_- = -c_+$; its b -component is removed by the boundary of $(0, b, 0)$ in degree zero. Degrees minus one and two have zero cohomology by injectivity of d^{-1} and surjectivity of the sum. Thus (CSP7.4) is a quasi-isomorphism, with all reductions given by continuous finite linear operations.

This proves the localization triangle and its boundaries in the chosen coordinates. In particular,

$$\begin{aligned} \partial^0 : A &\rightarrow Q_{\text{alg}}^2, \quad a \mapsto ([a], [a]), \\ \partial^1 : H^1(Y^\circ, \underline{A}) &= A \rightarrow H_D^2(Y, \mathcal{F}) = A^2, \quad a \mapsto (-a, a). \end{aligned} \quad (\text{CSP7.5})$$

The second formula combines the negative localization-boundary sign proved in CSP6 with $w = 1/z$, which gives $d \arg w = -d \arg z$.

The full nonzero portion of the ordinary localization sequence is

$$\begin{aligned} 0 \rightarrow E_+ \oplus E_- \rightarrow H^0(Y, \mathcal{F}) &\xrightarrow{r} A \xrightarrow{\text{diag} \circ q} Q_{\text{alg}}^2 \\ &\xrightarrow{(q_+, q_-) \mapsto q_+ - q_-} Q_{\text{alg}} \xrightarrow{0} A \xrightarrow{a \mapsto (-a, a)} A^2 \xrightarrow{(a, b) \mapsto a + b} A \rightarrow 0. \end{aligned} \quad (\text{CSP7.6})$$

Here r takes a Fourier-graph section to its common restriction and has image J ; $q : A \rightarrow A/J$. Every kernel and image in this row follows directly from the displayed formulas. In particular, global degree-one classes restrict to zero in annular degree one: they are Čech differences of degree-zero overlap data.

For one pole alone the support-to-global maps are

$$\begin{aligned} L_+ \rightarrow D : \quad (v, b, c) &\mapsto ((v, 0), b, c), \\ L_- \rightarrow D : \quad (v, b, c) &\mapsto ((0, v), -b, c). \end{aligned} \quad (\text{CSP7.7})$$

They induce degree-one isomorphisms with signs $+1, -1$ and degree-two isomorphisms with sign $+1$ in both cases. The complementary chart is acyclic for $\mathcal{F}$, as required by its localization sequence.

CSP8. Holomorphic power maps and the geometric normal factor

For each recovered positive integer n , the actual holomorphic map

$$b_n : Y \rightarrow Y, \quad b_n(z) = z^n, \quad b_n(0) = 0, \quad b_n(\infty) = \infty \quad (\text{CSP8.1})$$

satisfies $\pi b_n = \pi$, so $b_n^{-1}\mathcal{F}$ is canonically identified with $\mathcal{F}$ before applying any coefficient action.

For coefficients retain

$$\begin{aligned} T_a f(u) &= f(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a \cdot), ad_0, d_1), \quad a > 0. \end{aligned} \quad (\text{CSP8.2})$$

CS8 proves $r_\pm \rho_\pm(a) = T_a r_\pm$. These give a continuous sheaf endomorphism over the identity on Y . Combine it with pullback by b_n , using $a = n$, and denote the resulting cohomological operator by B_n .

On an annular angular class,

$$b_n^* \left(a \frac{d\theta}{2\pi} \right) = n a \frac{d\theta}{2\pi}. \quad (\text{CSP8.3})$$

The local coordinate at infinity also maps as $w \mapsto w^n$, so the same positive factor holds for its positive local circle. On degree two, either (CSP6.4) and the boundary give factor n , or one can compute its global degree directly:

$$b_n^* \omega_Y = \frac{1}{\pi} \frac{n^2 r^{2n-2} dx \wedge dy}{(1 + r^{2n})^2}, \quad \int_Y b_n^* \omega_Y = 2n^2 \int_0^\infty \frac{r^{2n-1}}{(1 + r^{2n})^2} dr = n. \quad (\text{CSP8.4})$$

All constants and the branch multiplicities at both poles are included. Therefore the induced chain/cohomology model for the combined operation is

$$\mathbf{B}_n^0 = \rho_+(n) \oplus \rho_-(n), \quad \mathbf{B}_n^1 = T_n, \quad \mathbf{B}_n^2 = nT_n. \quad (\text{CSP8.5})$$

Each pole-support complex has the corresponding actions $(\rho_\pm(n), T_n, nT_n)$. On the overlap, degree zero has T_n and degree one has nT_n ; these make every map in (CSP7.6) equivariant.

This proves, as a representation of the recovered positive-integer monoid,

$$\boxed{H^2(Y, \mathcal{F}) = A(-1), \quad A(-1) \text{ has action } nT_n.} \quad (\text{CSP8.6})$$

The notation records the proved degree character, not a declaration that this infinite-dimensional complex sheaf is an étale Tate sheaf. The coefficient-only action remains T_a in every degree. The additional n in (CSP8.6) is produced by the actual geometric covering, not by changing the coefficient action by definition.

Composition is exact: $b_m b_n = b_{mn}$, $T_m T_n = T_{mn}$, and hence $\mathbf{B}_m \mathbf{B}_n = \mathbf{B}_{mn}$. The induced operators are invertible on all displayed cohomology spaces. On positive rational indices define the cohomological comparison $\mathbf{B}_{n/m} = \mathbf{B}_n \mathbf{B}_m^{-1}$; equality $nm' = n'm$ proves independence of the presentation. In degree two its action is $(n/m)T_{n/m}$. The strongly continuous family aT_a , already defined on the coefficient space, is the unique extension of this rational action to $a > 0$. This does not assert a holomorphic map $z \mapsto z^a$ for a nonintegral real a .

The maps b_n here are receiving-space covers. The full signed CC structure has its own odd-index and chart-swapping rules; even b_n , or all b_n with one uniform chart rule, are not thereby proved to be endomorphisms of that full signed structure.

CSP9. The two mirrors and their different orientation signs

Keep the maps separate:

$$w(z) = -1/z, \quad \alpha(z) = -1/\bar{z}. \quad (\text{CSP9.1})$$

Both exchange the two poles and induce the same swap $x_+ \leftrightarrow x_-$ on X . The former is holomorphic, degree +1 on the sphere; the latter is antiholomorphic, degree -1, and is the actual real structure stated in CC.tex. On the unit circle their angle maps are respectively

$$\theta \mapsto \pi - \theta, \quad \theta \mapsto \pi + \theta. \quad (\text{CSP9.2})$$

Thus their actions on $H^1(Y^\circ, \underline{A})$, before a coefficient map, have signs -1, +1, respectively.

Use the actual source Weyl coefficient map: exchange V_+ and V_- by identity on their correspondingly labelled coordinates, and apply R on A . It is a sheaf map over either swap because $r_- = Rr_+$ and $R^2 = 1$. It is complex linear. It is not being identified without further construction with the full signed structure-sheaf action. If a semilinear coefficient comparison is wanted, composing with complex conjugation gives a separately specified conjugate-linear map.

On the global model (CSP4.2) the two operations are

$$\begin{array}{c|ccc} & D^0 & D^1 & D^2 \\ \hline w & \text{swap} & -R & +R \\ \alpha & \text{swap} & -R & -R. \end{array} \quad (\text{CSP9.3})$$

The degree-one sign is the ordered Čech swap in both cases. Degree two is the product of that Čech sign with the angular sign in (CSP9.2). Equivalently it is the orientation degree of the map, times R . On the pole-support degree-two pair, w acts as $(c_+, c_-) \mapsto (Rc_-, Rc_+)$, whereas α acts as $(c_+, c_-) \mapsto (-Rc_-, -Rc_+)$. Both act on its degree-one pair by $(b_+, b_-) \mapsto (Rb_-, Rb_+)$.

These formulas respect all localization maps. For example,

$$(Rb_- - Rb_+) = -R(b_+ - b_-).$$

For the annular boundary $a \mapsto (-a, a)$, the holomorphic operation sends its value to $(Ra, -Ra)$, which is the boundary of $-Ra$; the antiholomorphic operation sends it to $(-Ra, Ra)$, the boundary of $+Ra$. The degree-two sum is respectively transformed by $+R$ and $-R$.

Thus equality of the finite-poset swap does not erase the sphere's complex-orientation information. In particular, using $+R$ on global degree two for the actual antiholomorphic α would give the wrong map.

CSP10. The exact quotient of the normal direction

For any vector quotient $B = A/J$, there is an actual sheaf morphism

$$q_Y : \mathcal{F} \longrightarrow j_! \underline{Q_{\text{alg}}}_{Y^\circ}. \quad (\text{CSP10.1})$$

On Y° it is $q : A \rightarrow A/J$; at either pole its stalk map is zero. To verify that it is a sheaf morphism into extension by zero, take a section near a pole. Its nearby constant A -value is $r_\pm v \in J$, so its quotient vanishes on a neighborhood of that pole. Thus the quotient section has exactly the vanishing-near-boundary condition required by $j_!$. The map is onto on every stalk.

Its kernel $\mathcal{F}_J$ has pole stalks $V_\pm$, interior stalk J , and the same restrictions with codomain J . Hence

$$0 \rightarrow \mathcal{F}_J \rightarrow \mathcal{F} \xrightarrow{q_Y} j_! \underline{Q_{\text{alg}}} \rightarrow 0 \quad (\text{CSP10.2})$$

is exact. Give J its actual closed-subspace Fréchet topology from (CSP1.7). The restrictions $V_\pm \rightarrow J$ are continuous, with continuous right inverses obtained from Σ^{-1} and Fourier transformation. CSP3 therefore applies with these complete coefficient spaces as well as in the vector-space category.

The extension-by-zero sheaf has the same pole/interior model with zero pole spaces and overlap Q_{alg} . Thus

$$H^0(Y, j_! \underline{Q_{\text{alg}}}) = 0, \quad H^1(Y, j_! \underline{Q_{\text{alg}}}) = Q_{\text{alg}}, \quad H^2(Y, j_! \underline{Q_{\text{alg}}}) = Q_{\text{alg}}. \quad (\text{CSP10.3})$$

For $\mathcal{F}_J$, both restrictions are onto J ; therefore its degree one is zero and its degree two is J . Under these models (CSP10.1) induces the identity $Q_{\text{alg}} \rightarrow Q_{\text{alg}}$ in degree one and $q : A \rightarrow A/J$ in degree two. In particular the normal-direction sequence is exactly

$$\boxed{0 \longrightarrow J(-1) \longrightarrow A(-1) \longrightarrow H^2(Y, j_! \underline{Q_{\text{alg}}}) = Q_{\text{alg}}(-1) \longrightarrow 0.} \quad (\text{CSP10.4})$$

Its kernel is the full original summation image J , not a selected family of zero jets or a finite spectral truncation. The sphere orientation and the covers prove the (-1) action on each term. As locally convex spaces, the quotient topology in (CSP10.4) is precisely the topology of A/J .

There is also the continuous Hausdorff version

$$0 \longrightarrow C(-1) \longrightarrow A(-1) \longrightarrow Q_H(-1) \longrightarrow 0, \quad (\text{CSP10.5})$$

and an exact comparison

$$0 \longrightarrow (C/J)(-1) \longrightarrow Q_{\text{alg}}(-1) \longrightarrow Q_H(-1) \longrightarrow 0. \quad (\text{CSP10.6})$$

The sheaf map in this version is the composition with $Q_{\text{alg}} \rightarrow Q_H$; it does not rely on any general exactness assertion for Hausdorffification. By (CSP1.7), this last map is the identity comparison, $C/J = 0$, and (CSP10.5) is exactly (CSP10.4). Both retained kernel labels are the same actual closed source image $J(-1)$; there is no remaining completion operation.

CSP11. Original zeta, all factors, and the map to the shifted kernel

Use the exact raw transform and its CS comparison

$$\mathcal{M}_0 a(s) = \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta a(s) = \frac{1}{2} \mathcal{M}_0 a(s). \quad (\text{CSP11.1})$$

The factor $1/2$ is the explicitly proved comparison map; the original restriction remains $\Sigma f = 2 \sum_{k \geq 1} f(ku)$. It has not been replaced by a different restriction. The target is

$$\mathcal{B} = \left\{ F \text{ entire} : \sup_{|\Re s| \leq M} (1 + |\Im s|)^N |F(s)| < \infty \text{ for every } M, N \geq 0 \right\}. \quad (\text{CSP11.2})$$

CS7 and S1 give a topological isomorphism $\Theta : A \rightarrow \mathcal{B}$, with inverse

$$(\Theta^{-1}F)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CSP11.3})$$

For $\Re s > 1$, the original identity is

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}, \quad \Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CSP11.4})$$

It retains the original Euler product, unit summand and every repetition:

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{CSP11.5})$$

For the actual retained source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2},$$

$$\Theta \Sigma f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 \Sigma f_0 = 2F_0. \quad (\text{CSP11.6})$$

The original zeta pole at 1, its trivial zeros, and the Gamma poles give

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{CSP11.7})$$

For general $f \in S$, writing $a_f(s) = \int_0^\infty f(v) v^s dv/v$ with its Taylor-subtracted continuation, the exact receiver values are

$$\mathcal{M}_0 \Sigma f(-2r) = 2\zeta'(-2r) \frac{f^{(2r)}(0)}{(2r)!}, \quad \mathcal{M}_0 \Sigma f(1) = 2a'_f(1), \quad \mathcal{M}_0 \Sigma f(0) = -a_f(0). \quad (\text{CSP11.8})$$

These identities keep the endpoint and trivial-zero contributions rather than representing them as absent zeros of a replacement function.

Let $\mathcal{Z}$ be the actual nontrivial zero set of the original ζ , with full multiplicities m_ρ , and put

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CSP11.9})$$

The exact-image theorem SSI1–SSI7 strengthens S5 and CS7 to $\Theta J = \Theta C = \mathcal{J}$. The scalar $1/2$ in Θ leaves this linear subspace unchanged. Hence

$$Q_H \xrightarrow{\sim} \mathcal{Q}, \quad [a] \mapsto [\Theta a], \quad \Theta T_a = a^s \Theta. \quad (\text{CSP11.10})$$

The normal quotient from the actual sphere therefore has the continuous map

$$\nu : A(-1) \rightarrow \mathcal{Q}(-1), \quad a \mapsto [\Theta a]. \quad (\text{CSP11.11})$$

Its kernel is exactly $J(-1) = C(-1)$ by (CSP1.7). Thus its factor through the ordinary geometric quotient (CSP10.4) is a continuous isomorphism $Q_{\text{alg}}(-1) \rightarrow \mathcal{Q}(-1)$. The earlier comparison kernel $(C/J)(-1)$ is zero by the actual inverse calculation, not by an assumed exactness property of Hausdorffification.

The comparison with the already constructed high-weight shifted quotient in GSL6.5–GSL6.11 is now an actual map. Retain

$$\mathcal{J}_+ = \{F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0, 0 \leq j < m_\rho\}, \quad \mathcal{Q}_{\cup,+} = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+).$$

The GSL multiplier E_+ has jet one on the original divisor and jet zero on the divisor shifted by $+1$. Its construction and global strip estimates, not just these jet conditions, are the input guaranteeing a continuous multiplier of $\mathcal{B}$. Define

$$\boxed{\mathbf{n} : A(-1) \longrightarrow \mathcal{Q}_{\cup,+}, \quad \mathbf{n}(a) = [(1 - E_+(s))(\Theta a)(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}} \quad (\text{CSP11.12})$$

This is exactly $j_{\text{high}} \circ \nu$, where GSL proves that j_{high} is injective. Thus its kernel is the full original source $J(-1) = C(-1)$, its image is the entire GSL kernel, and its induced map from the ordinary geometric quotient (CSP10.4) is an isomorphism onto that kernel. Its continuity follows from (CSP11.3), translation continuity and the stated multiplier bounds.

For every recovered n , equivariance includes the full degree factor:

$$(\Theta(nT_n a))(s - 1) = n n^{s-1}(\Theta a)(s - 1) = n^s(\Theta a)(s - 1). \quad (\text{CSP11.13})$$

Multiplication by $1 - E_+$ commutes with n^s ; hence $\mathbf{n}(nT_n a) = T_n \mathbf{n}(a)$. This proves the exact morphism between the geometric normal factor and the existing translated-divisor kernel.

The full analytic multiplier of that extension remains

$$F_0(s)F_0(s - 1) = \left[\frac{s(s - 1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right] \left[\frac{(s - 1)(s - 2)}{8} \pi^{-(s-1)/2} \Gamma((s - 1)/2) \zeta(s - 1) \right]. \quad (\text{CSP11.14})$$

The second factor has values $1/8$ at $s = 1, 2$, and at $s = 1 - 2r$ its value is precisely (CSP11.7). Every derivative is the corresponding derivative of F_0 at $s - 1$. These are translations of the original full formula, not deletions of its poles or trivial-zero cancellations.

On an actual original zero block of multiplicity m_ρ , the normal quotient action is

$$n^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log n)^j}{j!} M_t^j, \quad t = s - \rho. \quad (\text{CSP11.15})$$

Translation identifies this with the block at $\rho + 1$, with local parameter $s - (\rho + 1)$. Every nilpotent coefficient, factorial and multiplicity is retained.

CSP12. A comparison of the whole two-degree receiver with GSL

The geometric calculation has produced the original quotient in degree one and its degree-shifted coefficient in degree two. These degrees must remain visible. Degree one is already Hausdorff by (CSP1.7); degree two is passed through the actual quotient (CSP10.4). Write

$$\mathcal{H}^1 = H^1(Y, \mathcal{F}) = H^1(Y, \mathcal{F})_H \simeq \mathcal{Q}, \quad \mathcal{H}^2 = H^2(Y, j_! \underline{Q}_H) \simeq \mathcal{Q}(-1). \quad (\text{CSP12.1})$$

The combined covering action is T_n on the first and nT_n on the second.

There is a continuous representation isomorphism of their underlying direct sum with the GSL extension:

$$\begin{aligned} \Psi_Y : \mathcal{H}^1 \oplus \mathcal{H}^2 &\longrightarrow \mathcal{Q}_{\cup,+}, \\ \Psi_Y([f]_{\mathcal{J}}, [h]_{\mathcal{J}}) &= [E_+(s)f(s) + (1 - E_+(s))h(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}. \end{aligned} \quad (\text{CSP12.2})$$

The second displayed input has the twisted action in (CSP12.1). Independence of its representatives follows from the full original and shifted jet conditions on E_+ . Its inverse takes a common representative F to

$$([F]_{\mathcal{J}}, [F(s + 1)]_{\mathcal{J}}). \quad (\text{CSP12.3})$$

For the second coordinate, changing F by $\mathcal{I} \cap \mathcal{I}_+$ changes $F(s+1)$ by a function in $\mathcal{I}$. On original jets, (CSP12.2) is f ; on shifted jets it is $h(s-1)$. These two observations prove both inverse identities. The GSL global bounds prove continuity of the forward map, and translation plus quotient seminorms prove continuity of the inverse. Formula (CSP11.13) proves equivariance.

This is a proved comparison after taking the stated quotient and forgetting the distinction between two cohomological degrees in the target direct sum. It does not assert that the short exact GSL row itself is the localization triangle of $\mathcal{F}$, nor that an unconstructed geometric boundary has vanished. The actual localization maps are (CSP7.2)–(CSP7.6).

A diagram of the exact normal-direction maps is:

$$\begin{array}{ccccc}
 J(-1) & \longrightarrow & A(-1) & \longrightarrow & Q_{\text{alg}}(-1) \\
 & & \downarrow \nu & & \downarrow \\
 & & Q(-1) & = & \mathcal{Q}(-1) \\
 & & \downarrow j_{\text{high}} & & \\
 & & \mathcal{Q}_{\cup,+} & &
 \end{array} \tag{CSP12.4}$$

The upper row has kernel $J(-1)$. The middle-left vertical map has the same kernel $C(-1) = J(-1)$; the upper-right vertical map is an isomorphism, its earlier comparison kernel $(C/J)(-1)$ being zero. The bottom map is injective. All arrows intertwine the combined covering actions with their displayed factors.

CSP13. Exact scope of the derived geometric statement

The construction establishes the following concrete facts together:

1. The inverse image of the actual CS coefficient sheaf on the complex projective-line evaluation has the fibre-product presentation (CSP2.2), keeping every source coordinate and all four endpoint lines.
2. Its global groups, costalks and localization maps are computed by (CSP4.2), (CSP6.2) and (CSP7.1)–(CSP7.7). The additional degree-two coefficient comes from annular cohomology, as also identified by $R^1\pi_*$ in (CSP5.1).
3. The positive degree of the holomorphic cover $z \mapsto z^n$ gives the exact factor n . Combined with the original arithmetic coefficient action, this produces nT_n , and therefore the geometric normal representation $A(-1)$.
4. The full-source quotient of that normal coefficient maps by (CSP11.12) onto the entire existing GSL kernel. The exact comparison retains ordinary/Hausdorff kernels, the Mellin factor two, the original zeta, its full multipliers, every zero multiplicity, and the shifted endpoint values.
5. Holomorphic and antiholomorphic swaps act differently on degree two, exactly as in (CSP9.3). The latter is the real structure in the cited later CC construction.

This is an explicit geometric source for the previously translated coefficient character. The coefficient sheaf is infinite-dimensional and complex, with the proved Fréchet/quotient topologies; it has not been made into a finite-rank mixed étale sheaf on a finite-type scheme over a finite field. Deligne’s étale purity theorem is consequently not applied to it by naming its normal factor. The statement proved here is the construction and comparison of the actual objects and maps above.

The source dependencies are CS3 for the exact summation/Fourier source, S1 for Mellin inversion, SSI1–SSI7 and its ESI0–ESI8 independent derivation for the actual closed summation image strengthening S5, and GSL4–GSL6 for its already constructed global continuous separator and shifted quotient. CSP2–CSP10 give the new sheaf, resolution, support and degree calculations in full. No independent norm, coordinate, or parity is assigned to τ .

The actual sheaf lifts and the geometric normal quotient

Complete derivation, 24 September 2026. Stable proof locators CW0–CW11.

CW0. Construction stage and exact question

The support is $\tau\langle Z_1; \text{no } Z_2 \rangle$. All operations below are on the arithmetic coefficient sheaves already constructed in CSP, after the complete arithmetic reconstruction. There is no addition, scalar multiplication, coordinate, distance, or parity operation on τ . The two charts remain separately labelled; a difference in a cochain complex is not a pooled branch return measure.

The connected argument and its correction-precedence table in `USER_ARGUMENT_RECONSTRUCTION.md`, the operative `READ_FIRST_USER_CONSTRUCTION.md`, and the full governing passages WU050–WU055 and WU061–WU063 in `USER_CONSTRUCTION_FULL_LOGBOOK.md` were read for this calculation. In particular, no arithmetic label initializes the supporting construction; the original zeta arithmetic is retained; and a calculation on this receiving sheaf is not promoted into a conclusion about every possible geometric realization of the complete construction.

The actual input is the final `CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`, CSP0–CSP13, read in full. Its version for this derivation has SHA256

EBFF9973E1472EA88E3F88D3350FAD8FFA98364067F485F48D3AF4E119EBAF9D.

The two comparison proofs are `GLOBAL_SHIFTED_ZETA_LIFT.md`, GSL0–GSL11, and `DELIGNE_IN_VARIANT_CYCLE_QUOTIENT.md`, DC0–DC12, previously derived and read in full and checked against their unchanged final hashes

2CC092DD3EBC0F021E9914FC7FD75DA16EA4DE47760B1D12558F33F0A661F529,

3A9AFDAEACD7DB0F154348F6487A204852459B08AE8E0F90017E7755FD627521.

The action and inverse calculations used from those notes are proved again below where they determine the present conclusion.

The source construction is Connes and Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5: the original author file `sources/CC_0903_2024_v3/author_source/announc3.tex`, lines1376–1667, was read in the preceding CSL verification. The relevant source formulas are `fonction3`, `restmaps`, `bord`, `hzero`, `rep1`, `liftw` and `court`. The finite-unit-sector comparison retains CS2A’s full-difference generating-domain qualification. The later signed source `CC.tex`, lines494–610 and773–899, was also read: its topological comparison does not by itself identify its structure sheaf with this complex coefficient sheaf.

The comparison with Deligne concerns *La conjecture de Weil. II*, §3.6. DC reconstructs that argument from the current French transcription, with its actual geometric hypotheses and arithmetic reduction. No new original-author-TeX reading of Deligne is claimed here.

This note computes both actual sheaf short exact sequences in CSP, their connecting maps, and the actual supported localization sequence. It proves the vanishing and existence of the lifts that those sequences define. It also computes exactly which kernel has a separated original-zero character range, and which kernel does not.

CW1. Full coefficients, original zeta, and the source image

Use

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2. \quad (\text{CW1.1})$$

The two endpoint coordinates are labelled value and integral on each chart. Let

$$A = \left\{ a \in C^{\infty}(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \quad (N, j \geq 0) \right\}. \quad (\text{CW1.2})$$

The actual maps are

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, c_0, c_1) = \Sigma f, \quad r_-(h, d_0, d_1) = R\Sigma h = \Sigma \hat{h}. \quad (\text{CW1.3})$$

The factor 2 records both rational signs. The Fourier convention is $\hat{h}(t) = \int_{\mathbb{R}} h(v) e^{-2\pi i vt} dv$. Before imposing the two endpoint conditions, the retained Poisson identity is

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv. \quad (\text{CW1.4})$$

Thus $R^2 = 1$, Fourier transformation preserves S and squares to one there, and

$$J = \Sigma(S) = r_+(V_+) = r_-(V_-), \quad \ker r_{\pm} = E_{\pm}. \quad (\text{CW1.5})$$

Continuity follows by Schwartz summation for $u \geq 1$ and (CW1.4) for $u \leq 1$. Injectivity follows by taking the Mellin transform in $\Re s > 1$, dividing by the nonzero Euler product there, and applying Fourier uniqueness on a vertical line. These are the source maps of CSP1, not new restrictions.

Write

$$\mathcal{B} = \{F \text{ entire} : b_{M,N}(F) = \sup_{|\Re s| \leq M} (1 + |\Im s|)^N |F(s)| < \infty \text{ for all } M, N \geq 0\}.$$

The exact comparison is

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad (\Theta^{-1} F)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{CW1.6})$$

The integrals and their differentiated versions converge by the seminorms. Integration by parts in $\log u$, followed by the inverse Fourier integral and contour shifts in each fixed strip, proves that $\Theta : A \rightarrow \mathcal{B}$ is a topological isomorphism. The $1/2$, $1/\pi$, and $u^{-1/2}$ are retained in these comparison maps.

For $\Re s > 1$, absolute summation gives

$$\Theta \Sigma f(s) = \zeta(s) a_f(s), \quad a_f(s) = \int_0^\infty f(v) v^s \frac{dv}{v}. \quad (\text{CW1.7})$$

The original arithmetic remains

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{CW1.8})$$

In particular, neither the unit summand nor any prime-power repetition is removed.

For the actual source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

the full transform is

$$F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 \Sigma f_0 = 2F_0. \quad (\text{CW1.9})$$

Here $f_0(0) = 0$ and its integral is zero because

$$\frac{\pi}{2} \left(2\pi \frac{3}{4\pi^2} - 3 \frac{1}{2\pi} \right) = 0.$$

Its two Gaussian Mellin integrals, before applying the Gamma recurrence, are

$$\frac{\pi}{2} \left[2\pi \cdot \frac{1}{2} \pi^{-(s+4)/2} \Gamma((s+4)/2) - 3 \cdot \frac{1}{2} \pi^{-(s+2)/2} \Gamma((s+2)/2) \right]. \quad (\text{CW1.10})$$

Multiplication by the original $\zeta(s)$ gives (CW1.9). The endpoint and trivial-zero cancellations have the exact values

$$F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{CW1.11})$$

The original ζ still has its pole at 1 and its trivial zeros at $-2r$. For general $f \in S$, the retained receiver values are

$$\Theta \Sigma f(-2r) = \zeta'(-2r) \frac{f^{(2r)}(0)}{(2r)!}, \quad \Theta \Sigma f(1) = a'_f(1), \quad \Theta \Sigma f(0) = -\frac{1}{2} a_f(0). \quad (\text{CW1.12})$$

Taylor subtraction in the integral defining a_f proves its meromorphic continuation. At $\nu = -2r$, write

$$a_f(\nu + t) = A_{-1}/t + \sum_{k \geq 0} A_k t^k, \quad A_{-1} = f^{(2r)}(0)/(2r)!.$$

Then the coefficient of t^n in $\zeta(\nu + t) a_f(\nu + t)$ is

$$\frac{\zeta^{(n+1)}(\nu)}{(n+1)!} A_{-1} + \sum_{j=1}^n \frac{\zeta^{(j)}(\nu)}{j!} A_{n-j}. \quad (\text{CW1.13})$$

At 1, write $\zeta(1+t) = t^{-1} + \sum_{j \geq 0} c_j t^j$ and $a_f(1+t) = \sum_{k \geq 1} a_k t^k$. The coefficient of t^n is

$$a_{n+1} + \sum_{j=0}^{n-1} c_j a_{n-j}. \quad (\text{CW1.14})$$

Empty sums are zero. Away from these exceptional points every derivative is the full Leibniz sum $\sum_{j=0}^n \binom{n}{j} \zeta^{(j)} a_f^{(n-j)}$. Thus the comparison retains derivative data as well as values.

Let $\mathcal{Z}$ be the actual nontrivial zeros of the original ζ , with their multiplicities m_ρ , and let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CW1.15})$$

CSP1.7 uses the proved SSI1–SSI7 exact-image theorem, independently checked in ESI0–ESI8:

$$J = \overline{J}^A, \quad \Sigma : S \xrightarrow{\sim} J \text{ topologically}, \quad \Theta J = \mathcal{J}. \quad (\text{CW1.16})$$

The actual inverse in that theorem has, for $F = \mathcal{M}_0 a$,

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt, \quad x > 0, \quad f_F(-x) = f_F(x),$$

$$\frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)}, \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0. \quad (\text{CW1.17})$$

Its all-strip estimates and Schwartz remainder estimates are the completed source theorem imported here, not a new assumption of closure. Consequently

$$Q := A/J = Q_{\text{alg}} = Q_H \xrightarrow[\downarrow a]{\mapsto} [\Theta a] \sim \mathcal{Q} \quad (\text{CW1.18})$$

is a topological isomorphism, and the former comparison kernel $\overline{J}/J$ is zero. The chain calculations below also make sense algebraically before that theorem; continuity and the stated Fréchet conclusions use the proved closed image.

CW2. The actual sheaves, finite complexes, and actions

Let $Y = \mathbb{P}^1(\mathbb{C})$, $D = \{0, \infty\}$, $U = \mathbb{C}^\times$, and $j : U \hookrightarrow Y$, $i : D \hookrightarrow Y$. The map $\pi : Y \rightarrow X = \{x_+, \eta, x_-\}$ sends the poles to x_+, x_- and every other point to η . Its inverse-image sheaf is

$$\mathcal{F} = \pi^{-1}\Omega = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-). \quad (\text{CW2.1})$$

The fibre product uses the map $(a, v_+, v_-) \mapsto (a|_0 - r_+v_+, a|_\infty - r_-v_-)$. At the two poles its stalks are $V_\pm$; on U it is the constant sheaf with fibre A .

The actual subsheaf $\mathcal{F}_J$ has the same pole stalks and restrictions, with interior stalk J . Put $\mathcal{G} = j_!Q_U$. Quotienting the interior coefficients and sending the pole stalks to zero gives the actual sheaf map

$$0 \longrightarrow \mathcal{F}_J \longrightarrow \mathcal{F} \xrightarrow{q_Y} \mathcal{G} \longrightarrow 0. \quad (\text{CW2.2})$$

Near a pole, the interior value of a section lies in $r_\pm(V_\pm) = J$. Its quotient is therefore zero near the pole, exactly the extension-by-zero condition. The sequence is exact on every stalk.

CSP2–CSP7 construct fine resolutions and their finite contractions. The resulting finite complexes, with degrees 0, 1, 2, form the degreewise exact row

$$\begin{array}{ccccccccc} 0 & \rightarrow & V_+ \oplus V_- & \xrightarrow{\text{id}} & V_+ \oplus V_- & \rightarrow & 0 & \rightarrow & 0 \\ & & \downarrow r_+ - r_- & & \downarrow r_+ - r_- & & \downarrow 0 & & \\ 0 & \rightarrow & J & \xrightarrow{\iota} & A & \xrightarrow{q} & Q & \rightarrow & 0 \\ & & \downarrow 0 & & \downarrow 0 & & \downarrow 0 & & \\ 0 & \rightarrow & J & \xrightarrow{\iota} & A & \xrightarrow{q} & Q & \rightarrow & 0. \end{array} \quad (\text{CW2.3})$$

The three columns compute $R\Gamma(Y, \mathcal{F}_J)$, $R\Gamma(Y, \mathcal{F})$, and $R\Gamma(Y, \mathcal{G})$. The top row is read as $0 \rightarrow D_J^0 \rightarrow D_F^0 \rightarrow D_G^0 \rightarrow 0$, with $D_G^0 = 0$. All maps are continuous.

For clarity, these are simultaneous models for the sheaf maps, not independently chosen cohomology presentations. For any $B = A, J, Q$, use the fibre-product resolution with its specified pole spaces. It gives the complex

$$B \oplus V_+ \oplus V_- \longrightarrow B^2 \longrightarrow B, \quad d^0(a, v_+, v_-) = (a - r_+v_+, a - r_-v_-), \quad d^1 = 0.$$

For $B = Q$ the pole spaces are zero. The contraction

$$p^0(a, v_+, v_-) = (v_+, v_-), \quad p^1(b_+, b_-) = b_- - b_+, \quad p^2 = \text{id}$$

and the section

$$s^0(v_+, v_-) = (r_-v_-, v_+, v_-), \quad s^1(b) = (-b, 0), \quad s^2 = \text{id} \quad (\text{CW2.4})$$

commute with inclusion $J \rightarrow A$ and quotient $A \rightarrow Q$. With $h^1(b_+, b_-) = (b_-, 0, 0)$, direct substitution gives $ps = 1$ and $1 - sp = dh + hd$. This proves the compatibility asserted in (CW2.3).

The coefficient actions are

$$\begin{aligned} T_a b(u) &= b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \\ \rho_-(a)(h, d_0, d_1) &= (ah(a\cdot), ad_0, d_1), \quad a > 0. \end{aligned} \quad (\text{CW2.5})$$

They preserve every source and satisfy $r_\pm \rho_\pm(a) = T_a r_\pm$. The actual holomorphic covering $b_n(z) = z^n$, for recovered positive integers n , has degree n on the sphere and multiplies an annular angular period by n . Combining that pullback with the coefficient action gives

$$\mathbf{B}_n^0 = \rho_+(n) \oplus \rho_-(n), \quad \mathbf{B}_n^1 = T_n, \quad \mathbf{B}_n^2 = nT_n. \quad (\text{CW2.6})$$

The same formula applies to the three columns where defined. Degree two uses the positive complex orientation. In the notation $B(-1)$, the action is aT_a ; this records the factor just proved from b_n , without asserting an étale realization.

Integer operators are invertible on the computed cohomology. Their inverses define the positive rational action consistently, and strong continuity extends it to all $a > 0$. This is a cohomological action; no nonintegral holomorphic map $z \mapsto z^a$ is asserted.

CW3. The connecting map of the fibre-product presentation

The other actual short exact sequence, CSP2.3, is

$$0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_*(V_+ \oplus V_-) \xrightarrow{\epsilon-r} i_*(A \oplus A) \rightarrow 0. \quad (\text{CW3.1})$$

Constant-sheaf cohomology on the sphere is $A, 0, A$ in degrees $0, 1, 2$; a pole skyscraper has no positive cohomology. Hence its complete nonzero long exact sequence is

$$0 \rightarrow H^0(Y, \mathcal{F}) \rightarrow A \oplus V_+ \oplus V_- \xrightarrow{d} A^2 \xrightarrow{\delta_{\text{fib}}} Q \rightarrow 0, \quad (\text{CW3.2})$$

and the degree-two comparison is the identity

$$H^2(Y, \mathcal{F}) = A(-1) \xrightarrow{\text{id}} H^2(Y, \underline{A}_Y) = A(-1). \quad (\text{CW3.3})$$

Here

$$d(a, v_+, v_-) = (a - r_+v_+, a - r_-v_-), \quad \delta_{\text{fib}}(b_+, b_-) = [b_- - b_+]. \quad (\text{CW3.4})$$

The connecting sign follows from the degree-one contraction p^1 in (CW2.4). Its kernel is exactly

$$\{(b_+, b_-) : b_- - b_+ \in J\}. \quad (\text{CW3.5})$$

Indeed d has differences in J . Conversely, for $b_- - b_+ = r_+v_+$, choose $v_- = 0$ and $a = b_-$. Then $d(a, v_+, 0) = (b_+, b_-)$. Every class $[b]$ is $\delta_{\text{fib}}(0, b)$, so the image is all Q .

This is an actual, generally nonzero connecting map. Its source action is $T_n \oplus T_n$ and its target action is T_n , with no geometric factor n in this degree. Its exact lifting quotient is

$$A^2 / \text{im } d \xrightarrow{\sim} Q, \quad [(b_+, b_-)] \mapsto [b_- - b_+]. \quad (\text{CW3.6})$$

Pairs in (CW3.5) lift through d ; other pairs do not. This statement specifies this sheaf-presentation lifting problem, rather than declaring a failure of a different global construction.

CW4. The actual quotient-sheaf lift and its zero boundary

From (CW2.3),

$$\begin{aligned} H^0(Y, \mathcal{F}_J) &= H^0(Y, \mathcal{F}) = \{(\hat{h}, c_0, c_1), (h, d_0, d_1)\}, \\ H^1(Y, \mathcal{F}_J) &= 0, \quad H^2(Y, \mathcal{F}_J) = J(-1), \\ H^1(Y, \mathcal{F}) &= Q, \quad H^2(Y, \mathcal{F}) = A(-1), \\ H^0(Y, \mathcal{G}) &= 0, \quad H^1(Y, \mathcal{G}) = Q, \quad H^2(Y, \mathcal{G}) = Q(-1), \end{aligned} \quad (\text{CW4.1})$$

and all degree-three groups vanish. The actual long exact sequence is therefore

$$\begin{aligned} 0 \rightarrow H^0(Y, \mathcal{F}_J) &\xrightarrow{\text{id}} H^0(Y, \mathcal{F}) \rightarrow 0, \\ 0 \rightarrow H^1(Y, \mathcal{F}) = Q &\xrightarrow{\text{id}} H^1(Y, \mathcal{G}) = Q \xrightarrow{\delta^1} J(-1) \xrightarrow{\iota} A(-1) \xrightarrow{q} Q(-1) \xrightarrow{\delta^2} 0. \end{aligned} \quad (\text{CW4.2})$$

Every sign and map follows from the simultaneous complexes, including

$$\boxed{\delta^1 = 0, \quad \delta^2 = 0.} \quad (\text{CW4.3})$$

Here is the cochain proof of the first equality, which does not assume a section of $q : A \rightarrow Q$. A class $x \in H^1(Y, \mathcal{G}) = Q$ is represented by the degree-one cocycle x of its zero-differential complex. Choose any $a \in A$ with $q(a) = x$, which exists by the definition of the quotient. Its differential in D_F is $d_F^1 a = 0$. Therefore the defining connecting cocycle in D_J^2 is zero. Changing a by an element of J still gives zero. The degree-one cohomology class $[a] \in H^1(Y, \mathcal{F}) = A/J$ depends only on x and is exactly x .

Thus the actual lift is the continuous equivariant inverse of the identity map in (CW4.2). It is unique at the cohomology-class level because $H^1(Y, \mathcal{F}_J) = 0$. In degree two every $x \in Q(-1)$ also lifts, and its set of lifts is the nonempty affine fibre

$$\text{Lift}_2(x) = q^{-1}(x) = a + J(-1). \quad (\text{CW4.4})$$

The equality describes all representatives, including their complete kernel. It does not select a representative compatible with every arithmetic action.

The degree-one obstruction receiver itself survives:

$$\text{im } \delta^1 = 0, \quad \text{coker}(H^1\mathcal{F} \rightarrow H^1\mathcal{G}) = 0, \quad \ker(H^2\mathcal{F} \rightarrow H^2\mathcal{G}) = J(-1). \quad (\text{CW4.5})$$

In the lifting formula of DC5, there is no additional invariant quotient of $H^1(Y, \mathcal{G})$ in this sequence. Taking its actual target as the classes to be lifted means that the further quotient is the identity and its kernel is zero. The corresponding obstruction quotient is therefore $J(-1)/0$, while the obstruction map into it is zero. This is the complete receiver; it has not been replaced by $Q(-1)$.

CW5. Supported localization: every image, kernel, and lift

For the two pole supports, the finite costalk complex is

$$L_D^0 = V_+ \oplus V_-, \quad L_D^1 = A^2, \quad L_D^2 = A(-1)^2, \quad d^0 = (r_+, r_-), \quad d^1 = 0.$$

The support-to-global maps are

$$k^0 = \text{id}, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CW5.1})$$

The annular positive z -circle gives

$$R\Gamma(U, \underline{A}) = A[0] \oplus A(-1)[-1].$$

The local positive circle at infinity uses $w = 1/z$, so $d \arg w = -d \arg z$. With the positive complex local fundamental classes, the actual localization boundaries are

$$\partial^0(a) = ([a], [a]), \quad \partial^1(a) = (-a, a). \quad (\text{CW5.2})$$

These are precisely the cone maps of CSP7, including the negative local boundary sign and the inverse-coordinate sign at infinity.

The resulting entire nonzero row is

$$\begin{aligned} 0 \rightarrow E_+ \oplus E_- \rightarrow H^0(Y, \mathcal{F}) \xrightarrow{r} A \xrightarrow{\text{diag } q} Q^2 \xrightarrow{\text{diff}} Q \\ \xrightarrow{0} A(-1) \xrightarrow{(-\text{id}, \text{id})} A(-1)^2 \xrightarrow{\text{sum}} A(-1) \rightarrow 0. \end{aligned} \quad (\text{CW5.3})$$

Here r sends a Fourier-graph section to its common restriction and has image J . Consequently:

1. A degree-zero annular section $a \in A$ extends globally exactly when $a \in J$. Its obstruction is $([a], [a])$, with image $\text{diag } Q$, kernel J , and cokernel identified with Q by the difference map.
2. Every global degree-one class in Q lifts from the two supported groups Q^2 . The kernel is $\text{diag } Q$, and a continuous equivariant section is

$$s_1(q) = \frac{1}{2}(q, -q). \quad (\text{CW5.4})$$

The following restriction $Q \rightarrow A(-1)$ is zero.

3. A degree-one annular class $a \in A(-1)$ extends to global degree one exactly when $a = 0$. Its boundary is the injective map $a \mapsto (-a, a)$, with full image the anti-diagonal. On that image its inverse is $(c_+, c_-) \mapsto (c_- - c_+)/2$.
4. Every global degree-two class $a \in A(-1)$ lifts from $A(-1)^2$. The kernel is the anti-diagonal, and a continuous equivariant section is

$$s_2(a) = \frac{1}{2}(a, a). \quad (\text{CW5.5})$$

Each assertion is a direct kernel/image calculation in the displayed row. In particular the quotient after the nonzero annular boundary is

$$A(-1)^2/\{(-a, a) : a \in A(-1)\} \xrightarrow{\sim} A(-1), \quad [(b, c)] \mapsto b + c. \quad (\text{CW5.6})$$

It survives; the existence of the supported degree-two lift does not assert that this receiver is zero.

For a single pole the supported degree-one map to Q is $+\text{id}$ at 0, $-\text{id}$ at infinity, and the supported degree-two map is $+\text{id}$ at both poles. These are the maps of CSP7.7, so both single-support lifting maps are isomorphisms in their displayed degrees.

Every scalar $1/2$ in these sections belongs to the recovered coefficient field. The sections do not divide the supporting datum.

The mirror signs are also retained. On Q both sphere swaps induce $-R$; on Q^2 they induce $(q_+, q_-) \mapsto (Rq_-, Rq_+)$. Hence s_1 intertwines them. On $A(-1)$, the holomorphic map $z \mapsto -1/z$ induces $+R$, whereas the actual antiholomorphic map $z \mapsto -1/\bar{z}$ induces $-R$. Their supported degree-two maps are respectively $+R$ -swap and $-R$ -swap, so s_2 intertwines both. The annular actions are respectively $-R, +R$, exactly compatible with $\partial^1(a) = (-a, a)$. These are coefficient and orientation operations, not an exchanged label assigned to τ .

CW6. Which character ranges are actually separated

Under Θ , coefficient dilation is multiplication by a^s :

$$\Theta T_a = a^s \Theta, \quad LF = sF. \quad (\text{CW6.1})$$

Fix $a > 1$ when discussing logarithmic-modulus weights, and define

$$w_a(\lambda) = \frac{2 \log |\lambda|}{\log a} \quad (\lambda \neq 0). \quad (\text{CW6.2})$$

The actions themselves are defined for every $a > 0$, including the identity at $a = 1$.

The full jet at an actual zero ρ , in local coordinate $t = s - \rho$, has action

$$T_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_t^j, \quad M_t^{m_\rho} = 0. \quad (\text{CW6.3})$$

On the normal quotient $\mathcal{Q}(-1)$, the corresponding action is

$$a^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_t^j. \quad (\text{CW6.4})$$

The established open strip $0 < \Re \rho < 1$ gives the disjoint ranges

$$0 < w_a(a^\rho) < 2, \quad 2 < w_a(a^{\rho+1}) < 4. \quad (\text{CW6.5})$$

These preserve every nilpotent and do not impose $\Re \rho = 1/2$. They are ranges of the actual zero characters, not a claim that an infinite-dimensional complex representation is a mixed finite-dimensional étale module.

The endpoint lines retain weights $0, 2$ on E_+ , and $2, 0$ on E_- . Their action is explicitly $(c_0, c_1, d_0, d_1) \mapsto (c_0, ac_1, ad_0, d_1)$. The common-restriction image of $H^0(Y, \mathcal{F})$ is J , with action T_a . Its Fourier-graph parameter transforms as $h(v) \mapsto ah(av)$.

It is essential to compute the characters of the actual degree-one obstruction receiver $J(-1)$, rather than assign to it (CW6.5). For any $\lambda \in \mathbb{C}$, let $m_\lambda = \text{ord}_\lambda F_0$, equal to zero off $\mathcal{Z}$. For $F \in \mathcal{J}$, local division defines

$$\ell_\lambda(F) = \left(\frac{F}{F_0} \right) (\lambda) = \frac{F^{(m_\lambda)}(\lambda)}{F_0^{(m_\lambda)}(\lambda)}. \quad (\text{CW6.6})$$

The denominator is nonzero. All lower numerator and denominator derivatives vanish when $m_\lambda > 0$, giving the displayed ratio. Evaluation of a fixed derivative is continuous for the strip topology by the Cauchy integral formula on a small fixed circle. Thus ℓ_λ is continuous. Moreover

$$\ell_\lambda(F_0) = 1, \quad \ell_\lambda(a^s F) = a^\lambda \ell_\lambda(F). \quad (\text{CW6.7})$$

The first equality proves surjectivity to $\mathbb{C}$; the second follows either from local division or from Leibniz's formula with all lower vanishing derivatives retained.

Consequently $\mathcal{J}(-1)$, and hence $J(-1)$, has a continuous one-dimensional quotient character

$$a \mapsto a^{\lambda+1} \quad \text{for every } \lambda \in \mathbb{C}. \quad (\text{CW6.8})$$

At $0, 1, -2r$ the order m_λ is zero and the nonzero denominators are exactly (CW1.11), so there is no exceptional gap in this statement. For an actual original zero ρ , the choice $\lambda = \rho - 1$ gives the quotient character a^ρ , the same character that occurs in $\mathcal{Q}$. The point $\rho - 1$ is an evaluation point of the full original function; it has not been asserted to be a zero of ζ .

The same elementary evaluation on $\mathcal{B}$ gives all characters a^λ as quotient characters of A , and all $a^{\lambda+1}$ as quotient characters of $A(-1)$. Therefore the actual targets $J(-1)$ and $A(-1)$ in (CW4.2) and (CW5.3) do not satisfy a bound placing all their continuous character quotients above the original-zero range. The separated space is the *quotient* $Q(-1)$ of the full normal source, with its complete original-zero ideal removed.

This calculation concerns quotient characters. It does not claim that J or A has corresponding eigenvectors. In fact they have no nonzero arithmetic eigenvectors at all: an eigenfunction $F \in \mathcal{B}$ for one $a > 1$ would satisfy $(a^s - c)F(s) = 0$. The first factor is a nonzero entire function, so the product identity forces $F = 0$. The same argument, with a power of $a^s - c$, excludes nonzero generalized eigenvectors in these entire-function spaces.

CW7. A global vanishing theorem for the actual boundary maps

There is a complete global vanishing calculation even though the actual targets in CW6 do not have the proposed separated quotient-character bound. We prove it using the full quotient topology and a resolvent, without assuming that finite zero-jet subspaces are dense.

For $\mu \notin \mathcal{Z}$, define on $\mathcal{B}$

$$\mathcal{R}_\mu F(s) = \frac{F(s) - F_0(s)F(\mu)/F_0(\mu)}{\mu - s}. \quad (\text{CW7.1})$$

The numerator vanishes at $s = \mu$, so this is entire. It belongs to $\mathcal{B}$ continuously. To verify the topology directly, put $H = F - F_0 F(\mu)/F_0(\mu)$. Outside $|s - \mu| < 1$, division by $\mu - s$ is bounded by one. Inside that disk,

$$\frac{H(s)}{\mu - s} = - \int_0^1 H'(\mu + t(s - \mu)) dt.$$

Cauchy's estimate bounds $|H'|$ there by $\sup_{|z-\mu| \leq 2} |H(z)|$. For $M' = \max(M, |\Re \mu| + 2)$, this gives

$$b_{M,N}(\mathcal{R}_\mu F) \leq b_{M,N}(H) + (2 + |\Im \mu|)^N b_{M',0}(H). \quad (\text{CW7.2})$$

The scalar $F(\mu)/F_0(\mu)$ is bounded by $b_{M',0}(F)/|F_0(\mu)|$. Substituting it in (CW7.2) gives a finite sum of continuous seminorm bounds on F , with the complete F_0 retained.

If $F \in \mathcal{J}$, its numerator is in $\mathcal{J}$, and division by $\mu - s$ preserves all original-zero vanishing orders. Thus (CW7.1) descends continuously to $\mathcal{Q}$. Direct multiplication gives

$$(\mu - L)\mathcal{R}_\mu[F] = [F], \quad \mathcal{R}_\mu(\mu - L)[F] = [F]. \quad (\text{CW7.3})$$

The discarded term in the first equality is explicitly $F_0 F(\mu)/F_0(\mu) \in \mathcal{J}$; no factor of the original function has been suppressed. This rederives the needed RZ inverse on its exact full domain.

Let $H : \mathcal{Q} \rightarrow \mathcal{B}(-1)$ be continuous and complex-linear and commute with the recovered arithmetic actions. If equivariance is initially required only for positive integers, invertibility gives it for positive rationals; continuity and strong continuity of T_a give it for every $a > 0$. Differentiation at $a = e^t, t = 0$ is justified by

$$|e^{ts} - 1 - ts| \leq \frac{1}{2}t^2|s|^2e^{|t||\Re s|},$$

which proves convergence in every strip seminorm, and then in every quotient seminorm. The target generator is $L + 1$. Consequently

$$HL = (L + 1)H. \quad (\text{CW7.4})$$

For a fixed $\lambda \in \mathbb{C}$, evaluation gives

$$\text{ev}_\lambda H(L - (\lambda + 1)) = 0. \quad (\text{CW7.5})$$

When $\lambda + 1 \notin \mathcal{Z}$, (CW7.3) makes the operator in parentheses onto $\mathcal{Q}$. Hence $H(x)(\lambda) = 0$ for every $x \in \mathcal{Q}$. For each x , the entire function $H(x)$ therefore vanishes on the open complement of the discrete set $\mathcal{Z} - 1$, and is identically zero. We have proved

$$\boxed{\text{Hom}_{\text{cont}, T}(\mathcal{Q}, \mathcal{B}(-1)) = 0.} \quad (\text{CW7.6})$$

Since $\mathcal{J}(-1)$ is an invariant subspace, inclusion gives

$$\boxed{\text{Hom}_{\text{cont}, T}(\mathcal{Q}, \mathcal{J}(-1)) = 0.} \quad (\text{CW7.7})$$

Transport through Θ proves the corresponding statements for $Q \rightarrow A(-1)$ and $Q \rightarrow J(-1)$. Thus both the actual restriction $H^1(Y, \mathcal{F}) \rightarrow H^1(U, \underline{A})$ and the actual connecting map $\delta^1 : H^1(Y, \mathcal{G}) \rightarrow H^2(Y, \mathcal{F}_J)$ also vanish by this global operator calculation. Their explicit cochain proofs in CW4–CW5 identify which maps are being tested.

No assertion about a direct sum or unrestricted product of zero blocks entered the proof. Nor did overlapping quotient characters produce a nonzero map: the distinction between a quotient character and an entire-function eigenvector is retained.

For comparison, the separated original-zero targets also admit a global proof:

$$\text{Hom}_{\text{cont}, T}(\mathcal{Q}, \mathcal{Q}(-1)) = 0. \quad (\text{CW7.8})$$

Indeed, for a continuous complex-linear equivariant H , take the full m_ρ -jet of its target at an actual ρ . The target generator there is $(\rho + 1)I + M_t$, so that jet kills $(L_{\text{target}} - (\rho + 1))^{m_\rho}$. Equivariance and (CW7.3), with $\mu = \rho + 1 \notin \mathcal{Z}$, imply that every target jet of $H(x)$ is zero. The joint original-zero jets are faithful on $\mathcal{Q}$ by (CW1.15). Thus $H = 0$. This proves the global assertion behind the separated ranges in (CW6.5), including every multiplicity.

CW8. The normal quotient lifts classes but has no equivariant linear section

The actual degree-two row is

$$0 \rightarrow J(-1) \rightarrow A(-1) \xrightarrow{q} Q(-1) \rightarrow 0, \quad \text{or, under } \Theta, \quad 0 \rightarrow \mathcal{J}(-1) \rightarrow \mathcal{B}(-1) \xrightarrow{q} \mathcal{Q}(-1) \rightarrow 0. \quad (\text{CW8.1})$$

Every class lifts by its definition as a quotient; the whole fibre is (CW4.4). We now calculate the distinct question of a simultaneous arithmetic-equivariant linear choice of representatives.

For any actual zero ρ , put

$$e_\rho(s) = F_0(s)/(s - \rho). \quad (\text{CW8.2})$$

It is entire and in $\mathcal{B}$, by the same division estimate near ρ used above and polynomial division away from ρ . Its order at ρ is $m_\rho - 1$, so $[e_\rho] \neq 0$ in $\mathcal{Q}$. Moreover

$$(L - \rho)[e_\rho] = [F_0] = 0, \quad T_a[e_\rho] = a^\rho[e_\rho]. \quad (\text{CW8.3})$$

For the second equality, $(a^s - a^\rho)/(s - \rho)$ is entire and a multiplier with polynomial strip bounds away from a fixed disk; multiplied by F_0 , it lies in $\mathcal{J}$, with all required orders at every original zero.

The actual zero set is nonempty without an RH assumption. One way using the retained source proof is S2's order-at-most-one Hadamard factorization and $F_0(s) = F_0(1 - s)$. If it were zero-free, that factorization would give $F_0 = e^{a+bs}$; the symmetry forces $b = 0$. This contradicts $F_0(0) = 1/8$ and $F_0(2) = \pi/24$, since $\pi \neq 3$. This argument uses the original full function and does not assign a location to a fictional zero.

An arithmetic-equivariant linear section of (CW8.1) would send the nonzero class $[e_\rho]$, with its twisted eigencharacter $a^{\rho+1}$, to a nonzero entire function F . For one $a > 1$, equivariance would require

$$a^{s+1}F(s) = a^{\rho+1}F(s). \quad (\text{CW8.4})$$

The nonzero entire factor $a^{s+1} - a^{\rho+1}$ forces $F = 0$, a contradiction. Thus

$$\boxed{\text{The actual normal quotient has no arithmetic-equivariant linear section.}} \quad (\text{CW8.5})$$

Continuity is not needed for this no-section statement. In the continuous category, twisting both source and target by the same character and applying the proof of CW7 with target generator L also gives

$$\text{Hom}_{\text{cont}, T}(\mathcal{Q}(-1), \mathcal{B}(-1)) = 0. \quad (\text{CW8.6})$$

This is a statement about simultaneous compatible representatives, not about the existence of individual lifts. The latter are all present, with the exact affine fibres $a + J(-1)$. In particular (CW8.5) does not contradict the degree-one identity lift in CW4, the surjective degree-two map, or Deligne's surjectivity theorem. DC8 explicitly distinguished its canonical section into the intermediate B_i from an unspecified choice of lifts into the special-fibre group A_i .

CW9. The precise separated receiver obtained from the actual normal quotient

The geometric normal quotient is $\mathcal{Q}(-1)$, with action (CW6.4). It has the exact map into the GSL extension, rather than an identification with the kernel $J(-1)$ of (CW8.1).

Let

$$\mathcal{J}_+ = \{F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0 \quad (0 \leq j < m_\rho)\}, \quad \mathcal{E}_+ = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+). \quad (\text{CW9.1})$$

GSL4–GSL6 construct the entire multiplier E_+ , with jet one on the original divisor and jet zero on the divisor translated by $+1$, and prove its strip bounds

$$|E_+(s)| + |1 - E_+(s)| \leq C_M(1 + |\Im s|)^{d_M} \quad (|\Re s| \leq M). \quad (\text{CW9.2})$$

These give continuous multiplication on $\mathcal{B}$: $b_{M,N}(E_+F) \leq C_M b_{M,N+d_M}(F)$, and the analogous bound for $1 - E_+$. Those completed global bounds, not a formal blockwise interpolation, are the input from GSL.

With $VF(s) = F(s - 1)$, the complete maps are

$$\begin{aligned} j_{\text{high}} : \mathcal{Q}(-1) &\rightarrow \mathcal{E}_+, & [h] &\mapsto [(1 - E_+(s))h(s - 1)], \\ \pi_{\text{high}}[F] &= [F]_{\mathcal{J}}, & s_{\text{high}}[f] &= [E_+f]. \end{aligned} \quad (\text{CW9.3})$$

They give

$$0 \rightarrow \mathcal{Q}(-1) \xrightarrow{j_{\text{high}}} \mathcal{E}_+ \xrightarrow{\pi_{\text{high}}} \mathcal{Q} \rightarrow 0, \quad \pi_{\text{high}} s_{\text{high}} = 1. \quad (\text{CW9.4})$$

Independence of representatives follows from the two full jet identities, including all derivatives. The inverse comparison is

$$\mathcal{E}_+ \rightarrow \mathcal{Q} \oplus \mathcal{Q}(-1), \quad [F] \mapsto ([F]_{\mathcal{J}}, [F(s + 1)]_{\mathcal{J}}), \quad (\text{CW9.5})$$

whose inverse is

$$([f], [h]) \mapsto [E_+(s)f(s) + (1 - E_+(s))h(s - 1)]. \quad (\text{CW9.6})$$

The two inverse identities hold at every original and translated jet. Continuity follows from (CW9.2), translation $b_{M,N}(VF) \leq b_{M+1,N}(F)$, and the quotient seminorms. Equivariance retains the factor

$$a a^{s-1} = a^s. \quad (\text{CW9.7})$$

The complete original product defining the two divisors is

$$F_0(s)F_0(s-1) = \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right] \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right]. \quad (\text{CW9.8})$$

The translated factor has values $1/8$ at $s = 1, 2$, value (CW1.11) at $s = 1 - 2r$, and the same value at $s = 2 + 2r$. Its derivatives are exactly $F_0^{(j)}(s-1)$; derivatives of the product are the full sum

$$\sum_{j=0}^n \binom{n}{j} F_0^{(j)}(s) F_0^{(n-j)}(s-1). \quad (\text{CW9.9})$$

Original and translated endpoint and trivial-zero contributions therefore remain explicit.

The actual geometric normal map is

$$\mathbf{n} : A(-1) \rightarrow \mathcal{E}_+, \quad a \mapsto [(1 - E_+(s))(\Theta a)(s-1)]. \quad (\text{CW9.10})$$

Its factorization is exactly

$$A(-1) \xrightarrow{q} \mathcal{Q}(-1) \xrightarrow{j_{\text{high}}} \mathcal{E}_+. \quad (\text{CW9.11})$$

The second arrow is injective, so

$$\ker \mathbf{n} = J(-1), \quad \text{im } \mathbf{n} = \ker \pi_{\text{high}}. \quad (\text{CW9.12})$$

Thus the separated GSL kernel is the complete quotient of the actual normal source by its full original summation image. It is not the summation image itself.

The comparison on the actual connecting-map receiver can be computed without discarding that receiver. If $\iota : J(-1) \hookrightarrow A(-1)$ is the actual map in (CW4.2), then

$$q\iota = 0, \quad \mathbf{n}\iota = j_{\text{high}}q\iota = 0, \quad \mathbf{n}\iota\delta^1 = 0. \quad (\text{CW9.15})$$

The middle map is zero on the entire receiver, before considering the value of δ^1 . Therefore vanishing after this comparison is not an injective test for vanishing in $J(-1)$. The actual vanishing of δ^1 is instead proved by (CW4.3) and independently by (CW7.7). This identifies the exact comparison map and its kernel, rather than treating the two receiver presentations as unrelated.

The actual two-degree groups give the continuous equivariant isomorphism

$$H^1(Y, \mathcal{F}) \oplus H^2(Y, \mathcal{G}) \xrightarrow{\sim} \mathcal{E}_+ \quad (\text{CW9.13})$$

with formula (CW9.6). This map forgets the distinction between the two cohomological degrees only at its displayed direct-sum target. The degree-one connecting morphism of the actual sheaf sequence remains $\delta^1 : \mathcal{Q} \rightarrow J(-1)$, not a new morphism $\mathcal{Q} \rightarrow \mathcal{Q}(-1)$.

For precision, the split algebraic cross on this receiver is

$$0 \rightarrow \mathcal{Q}(-1) \rightarrow \mathcal{Q} \oplus \mathcal{Q}(-1) \xrightarrow{\text{pr}_1} \mathcal{Q} \rightarrow 0, \quad \partial_{\text{sum}} = \text{pr}_2. \quad (\text{CW9.14})$$

The actual degree-one group enters by $x \mapsto (x, 0)$, and its boundary is zero. The image of the kernel under ∂_{sum} is all $\mathcal{Q}(-1)$, so the DC-style obstruction quotient of this particular cross is zero. Transport through (CW9.13) gives exactly (CW9.4), whose continuous complex-linear equivariant section is unique by (CW7.8).

This cross is constructed from the computed cohomology groups and the exact normal quotient. It is not substituted for (CW4.2) or (CW5.3). In particular, the separated kernel in (CW9.14) is visible only after the stated quotient and assembly of two degrees.

CW10. Exact comparison with Deligne's invariant-cycle mechanism

DC3–DC9 derive, in Deligne's arithmetic reduction, the two linked rows

$$0 \rightarrow K_i = H^{i-1}(X_{\bar{\eta}}, E)_I(-1) \xrightarrow{j_i} B_i = H^i(X_{\eta}, E) \xrightarrow{\pi_i} C_i = H^i(X_{\bar{\eta}}, E)^I \rightarrow 0,$$

$$A_i = H^i(X_s, E) \xrightarrow{\alpha_i} B_i \xrightarrow{\partial_i} O_i = H^{2N-i-1}(X_s, E)^{\vee}(-N). \quad (\text{CW10.1})$$

Here N is the dimension of the smooth total model, not the generic fibre dimension. The invariant-cycle theorem's hypotheses are those of DC1–DC2, including the curve-trait setting and arithmetic reduction; they are not transferred to the sphere sheaf by its notation.

Its exact lifting obstruction is

$$\text{obs}_i(c) = [\partial_i b] \in O_i / \partial_i j_i(K_i), \quad \pi_i(b) = c. \quad (\text{CW10.2})$$

Changing b by $j_i(k)$ changes the boundary by $\partial_i j_i(k)$, proving well-definedness. Conversely, $\text{obs}_i(c) = 0$ permits subtracting such a $j_i(k)$ and then lifting through α_i by localization. Thus

$$\text{coker}(\pi_i \alpha_i) \simeq \text{im } \partial_i / \partial_i j_i(K_i). \quad (\text{CW10.3})$$

The geometric theorems prove

$$C_i, A_i \text{ have weights } \leq i, \quad K_i, O_i \text{ have weights } \geq i + 1. \quad (\text{CW10.4})$$

The bound on O_i is the full calculation

$$-(2N - i - 1) + 2N = i + 1;$$

the bound on K_i comes from dual invariant weights followed by the retained (-1) twist. Exactness of the weight cutoff in the finite-dimensional mixed category gives

$$W_i B_i \xrightarrow{\sim} C_i, \quad \partial_i(W_i B_i) = 0, \quad \text{im } \alpha_i = W_i B_i, \quad \partial_i j_i(K_i) = \text{im } \partial_i. \quad (\text{CW10.5})$$

The whole receiver need not vanish:

$$O_i / \partial_i j_i(K_i) \simeq \ker \text{sp}_{i+1}. \quad (\text{CW10.6})$$

The actual CSP sequence now has a fully determined comparison:

Actual calculation	Classes being lifted	Actual receiver or kernel	Calculated result
Quotient-sheaf degree one, (CW4.2)	$H^1 \mathcal{G} = Q$	$H^2 \mathcal{F}_J = J(-1)$	Boundary zero; unique cohomology lift, the identity
Quotient-sheaf degree two, (CW8.1)	$H^2 \mathcal{G} = Q(-1)$	Kernel $J(-1)$; next boundary target 0	Every class lifts; all lifts form $a + J(-1)$
Support to global degree one, (CW5.3)	$H^1 \mathcal{F} = Q$	Following receiver $A(-1)$	Following map zero; section $q \mapsto (q/2, -q/2)$
Global to annular degree zero	A	Q^2	Boundary diag q , image diag Q
Global to annular degree one	$A(-1)$	$A(-1)^2$	Injective boundary $a \mapsto (-a, a)$
Normal quotient assembled with degree one, (CW9.14)	Q	$Q(-1)$	Separated continuous split receiver; obstruction quotient zero

The first and third zero maps are proved both by their actual complexes and by the global continuous intertwiner calculation of CW7. Their full targets $J(-1)$ and $A(-1)$ have the character quotients of CW6, so Deligne’s finite-dimensional weight-cutoff proof (CW10.4) is not the proof of those target bounds here. This is an exact determination of the present objects and maps, not an assertion that no further geometric comparison can exist.

There is already a precise morphism to the separated normal quotient: (CW9.10)–(CW9.12), with the entire kernel $J(-1)$ and the entire image $\ker \pi_{\text{high}}$ calculated. It is this quotient and the two-degree comparison, rather than a renaming of $J(-1)$, that relates the actual CSP geometry to GSL’s separated row.

No map in this calculation forces the original characters a^ρ all to have weight one. The surviving original quotient still has its complete original zero jets, their nilpotent actions, and their actual strip positions. The results prove the stated actual lifts and their exact receivers; they do not assume a critical-line conclusion or construct a violation of it.

CW11. Prerequisites, source use, and retained scope

The operations used in this derivation have the following constructed prerequisites:

Operation	Exact prerequisite and proof
Vector sums, differences and $1/2$	Recovered complex coefficient spaces $A, S, V_\pm, Q$; CW1–CW2. None acts on τ .
Sheaf kernel and quotient	Stalkwise exact fibre-product and extension-by-zero maps; CSP2/CSP10 and CW2.1–CW2.3.
Long exact sequences and boundaries	Simultaneous finite complexes and their compatible contractions; CW2.3–CW2.4, CW3–CW5.
Ordinary equals Hausdorff cohomology	Actual closed-image/inverse theorem SSI1–SSI7 and ESI0–ESI8, imported through CSP1.7 with its full inverse CW1.17.
Normal degree character	Actual positive-degree covering $z \mapsto z^n$, annular period and oriented sphere integration; CSP8 and CW2.6.
Original-zero weights	Exact raw Mellin comparison, original zero ideal and full nilpotent action; CW1.6–CW1.18 and CW6.
Evaluation on the full kernel	Local division by the retained full F_0 , with all exceptional values; CW6.6–CW6.8.
Global vanishing of continuous intertwiners	Continuous full resolvent, strong differentiability and holomorphic uniqueness; CW7. No density of finite blocks is used.
Shifted normal comparison	GSL’s constructed multiplier with strip bounds, full translated factor and continuous CRT inverse; CW9.1–CW9.12.
Deligne comparison	The actual invariant/coinvariant, duality and weight maps of DC3–DC9; full total-dimension and Tate factors retained in CW10.

The new statements about absence of equivariant linear sections are restricted to the actual coefficient row (CW8.1). They preserve the whole original arithmetic and do not replace it by a different timing history. They do not deny individual lifting, and they do not identify that coefficient-row section problem with an unconstructed obstruction in the user’s complete construction.

The concrete mathematical conclusions are the actual cohomological identity lift, both zero connecting maps of (CW4.2), the full localization maps and their explicit sections, the global vanishing (CW7.6)–(CW7.8),

the exact normal lift fibres, and the quotient morphism to the separated receiver. Every comparison keeps the support, separate chart labels, full source kernel, original zeta factors, multiplicities, and cohomological degrees at the points where they are used.

Geometric trace and the full support-dual comparison

The ramified cover has an actual sheaf trace, including both pole multiplicities. Its transpose acts on the completed original-zeta residue receiver, retaining every full zero jet and the exact degree character. The continuous current construction realizes the complete signed support maps. Its inclusion into ordinary algebraic Verdier duality has a computed cone, with all coefficients, stalks and costalks retained.

The additional algebraic-dual sector admits inversion of every nonzero polynomial in the transposed Mellin multiplication operator. Consequently the inclusion of continuous into algebraic duals preserves all polynomial kernels and cokernels; every finite-dimensional invariant sector for that operator is already continuous. This compares the two categories on their exact finite spectral sectors. It does not differentiate discontinuous functionals or prove an unestablished weight bound.

The following five complete proofs retain the original zeta, endpoint terms, source image, winding signs and all operation domains. They extend RD/CDI and CSP/CW already included above. Deligne's required weight separation remains unproved.

Ramified trace, full residue completion, and the dual comparison

Exact coefficient maps after the complete arithmetic reconstruction; no coordinate is assigned to tau.

A. The actual cover and its trace: every recovered integer $n \geq 1$

$$\begin{array}{lll}
 0 \mapsto 0 & b_n(z) = z^n & \infty \mapsto \infty \\
 \text{Local multiplicity } n & \text{Interior fibre: } n \text{ sheets, each multiplicity } 1 & \text{Local multiplicity } n \\
 V_+ \rightarrow V_+ : v \mapsto nv & A^{\oplus n} \rightarrow A : (a_j) \mapsto \sum_{j=1}^n a_j & V_- \rightarrow V_- : v \mapsto nv \\
 \text{Tr}_n b_n^* = n \text{ id} & H^0 = Z, \quad H^1 = Q, \quad H^2 = A &
 \end{array}$$

B. The complete original-zeta receiver and the two different dual actions

$$\begin{array}{ccccccc}
 \mathcal{H}_{\text{res}} & \xrightarrow{\mathfrak{S} \cong} & Q'_\beta & \xrightarrow{\text{inclusion}} & Q^* & \xrightarrow{\text{quotient}} & Q^\dagger \\
 \mathfrak{S} \widehat{T}_n = U'_n \mathfrak{S}, \quad U'_n = n(T_{1/n}^Q)' & & & & \mathfrak{S} \widehat{V}_n = B'_n \mathfrak{S}, \quad B'_n = (T_n^Q)' & &
 \end{array}$$

The completion uses the exact residue seminorms, uniformly on bounded subsets of the original quotient.

C. The additional algebraic-dual sector has an exact operator structure

$$\begin{array}{ll}
 E^\dagger = E^* / E', \quad E = A, J, Q & L : F(s) \mapsto sF(s) \\
 L^\dagger - \mu : E^\dagger \xrightarrow{\cong} E^\dagger \quad (\mu \in \mathbb{C}) &
 \end{array}$$

Every finite generalized L^* -eigenvector in E^* is already continuous; the full quotient remains retained.

Proofs: GTR2-GTR6, GTR10; SDT4-SDT8; VSD7-VSD13. Maps shown are exact; no positivity or weight bound is asserted.

Figure 9: The ramified trace and its complete dual receiver. The middle row uses the explicitly specified strong residue topology. The right quotient is retained as an algebraic coefficient object. All identities apply globally in the indicated parameters. Proof locators GTR2-GTR6 and GTR10; SDT4-SDT8; VSD7-VSD13. Human-source comparisons: Connes and Consani, Schemes over F_1 and zeta functions, arXiv:0903.2024v3 section5; Deligne, La conjecture de Weil II, section3.6; Stacks Project, tag0GKE.

The derived direct image and its winding summand

Complete independent derivation, 24 September 2026. Proof locators CDW0–CDW10.

CDW0. The precise receiver and prerequisites

The supporting datum is still $\tau\langle Z_1; \text{no } Z_2 \rangle$. Every vector space, scalar, complex coordinate and differential form in this note belongs to the receiving geometry constructed after the complete arithmetic reconstruction. No coordinate, distance, midpoint or retracted addition is put on τ .

The complete connected user reconstruction and its correction table were read in the preceding CSP derivation; the current `READ_FIRST_USER_CONSTRUCTION.md`, PC01–PC05, and verbatim WU050–WU055 and WU061–WU063 were recalled for this continuation. In particular, the full source and both branch restrictions are retained. A selected arithmetic label does not initialize the construction, and a quotient of a kernel is not substituted for that kernel.

The existing proof `CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md`, CSP0–CSP13, constructs the space, coefficient sheaf, ordinary cohomology, costalks and all orientation conventions. Its source-reading record retains original author TeX from Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, and the field-valued charts and real structure in their later arXiv:2609.00299v1. The original-source coverage is exactly that recorded in CSP0, not a claim to have reread both entire papers during this continuation.

CS3 and CS11, CSL1 and CSL10, and SSI1–SSI7 with its independent ESI proof supply the actual continuous summation/Fourier maps and the equality of the original source image with the closed original-zero ideal under Mellin transformation. CSP2–CSP3 supply the fibre-product sheaf and the coefficient-valued Poincaré calculations. The present derivation constructs the entire derived direct image with its actions, rather than recovering it from its cohomology groups alone.

Let

$$\begin{aligned} X &= \{x_+, \eta, x_-\}, & U_+ &= \{x_+, \eta\}, & U_- &= \{x_-, \eta\}, & U &= \{\eta\}, \\ Y &= \mathbb{P}^1(\mathbb{C}), & Y_+ &= Y \setminus \{\infty\}, & Y_- &= Y \setminus \{0\}, & Y^\circ &= \mathbb{C}^\times. \end{aligned} \quad (\text{CDW0.1})$$

The continuous map $\pi : Y \rightarrow X$ sends 0 to x_+ , ∞ to x_- , and all of Y° to η . Thus $\pi^{-1}(\eta)$ is the entire punctured complex plane. The coordinate z used on that receiver is not a coordinate of τ .

CDW1. The original coefficient diagram

The exact spaces are

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, & V_\pm &= S \oplus E_\pm, & E_\pm &= \mathbb{C}^2, \\ A &= \left\{ a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \quad (N, j \geq 0) \right\}. \end{aligned} \quad (\text{CDW1.1})$$

Both S and A are Fréchet spaces. The two coordinates of each $E_\pm$ retain the source labels value at zero and real integral.

With the stated Fourier convention,

$$\begin{aligned} \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), & Ra(u) &= u^{-1}a(u^{-1}), & \hat{f}(t) &= \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv, \\ r_+(f, c_0, c_1) &= \Sigma f, & r_-(h, d_0, d_1) &= R\Sigma h = \hat{\Sigma} h. \end{aligned} \quad (\text{CDW1.2})$$

The last identity follows from the full Poisson formula

$$\Sigma \hat{h}(u) = u^{-1} \Sigma h(u^{-1}) + u^{-1} h(0) - \int_{\mathbb{R}} h(v) dv \quad (\text{CDW1.3})$$

and the two source conditions. No endpoint summand is identified with the other chart's summand.

The sheaf Ω on X is the diagram

$$V_+ \xrightarrow{r_+} A \xleftarrow{r_-} V_- \quad (\text{CDW1.4})$$

Its global sections are the fibre product. All morphisms of diagrams below consist of three continuous linear maps commuting with these two arrows. The pullback $\mathcal{F} = \pi^{-1}\Omega$ has the concrete presentation

$$\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-), \quad i : \{0, \infty\} \hookrightarrow Y, \quad (\text{CDW1.5})$$

with difference map $(a, v_+, v_-) \mapsto (a|_0 - r_+v_+, a|_\infty - r_-v_-)$. CSP2 proves this identification by an explicit adjunction map and identities on every stalk.

The actual image

$$J = \Sigma(S) = r_+(V_+) = r_-(V_-) \quad (\text{CDW1.6})$$

is closed, and $\Sigma : S \rightarrow J$ is a topological isomorphism by SSI1–SSI7. In particular $Q = A/J$ is Fréchet. Closedness is available here, but the derived splitting below does not use division by Σ or a choice of a continuous section $Q \rightarrow A$.

CDW2. A single resolution on the entire sphere

Let $\mathcal{E}_A^q$ be the sheaf of smooth A -valued differential q -forms on the underlying real two-dimensional manifold Y , with complex coefficients. Smoothness means that all coordinate derivatives exist in the Fréchet space A ; sections have the compact-set derivative seminorms induced by $p_{N,j}$.

The radial Poincaré operator on a coordinate ball,

$$(K\omega)_x = \int_0^1 t^{q-1} \iota_x \omega_{tx} dt, \quad (\text{CDW2.1})$$

exists because A is complete. For every continuous seminorm p ,

$$p \left(\int_a^b f(t) dt \right) \leq \int_a^b p(f(t)) dt. \quad (\text{CDW2.2})$$

Differentiation under the integral on compact sets gives the same bound for derivatives. The usual differentiation of the radial pullback, integrated from zero to one, proves $dK + Kd = 1$ in positive degree, and $Kd = 1 - \text{ev}_0$ in degree zero. Thus the coefficient-valued Poincaré lemma holds with continuous local primitives. Partitions of unity multiply these forms, making the sheaves fine. This is also the resolution used in CSP3.

Write $\text{ev} : \mathcal{E}_A^0 \rightarrow i_*(A \oplus A)$ for evaluation at the two poles. Define the single complex of sheaves on Y

$$\begin{aligned} \mathcal{K}^0 &= \mathcal{E}_A^0 \oplus i_*(V_+ \oplus V_-), \\ \mathcal{K}^1 &= \mathcal{E}_A^1 \oplus i_*(A \oplus A), \\ \mathcal{K}^2 &= \mathcal{E}_A^2, \end{aligned}$$

$$d^0(f, v) = (df, \text{ev}(f) - rv), \quad d^1(\omega, a) = d\omega. \quad (\text{CDW2.3})$$

The coordinates at a pole absent from an open are omitted. One has $d^1 d^0 = d^2 f = 0$. The augmentation $\mathcal{F} \rightarrow \mathcal{K}^0$ sends the locally constant function and its pole coordinates in (CDW1.5) to those same coordinates.

We verify exactness on every stalk. In degree zero, $df = 0$ says that f is locally constant; the second equation is exactly the fibre-product condition. Away from the poles the positive-degree assertion is the Poincaré lemma. At a pole p , a degree-one cocycle is a closed one-form ω and an arbitrary $a \in A$. Choose a local primitive g of ω , and replace it by $f = g + a - g(p)$. Then $df = \omega$ and $f(p) = a$, so

$$d^0(f, 0) = (\omega, a). \quad (\text{CDW2.4})$$

Every degree-two germ has a one-form primitive by (CDW2.1). This proves exactness of

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{K}^0 \rightarrow \mathcal{K}^1 \rightarrow \mathcal{K}^2 \rightarrow 0. \quad (\text{CDW2.5})$$

The zero-pole adjustment in (CDW2.4) is an operation on A , not on the supporting point τ .

Each $\mathcal{K}^q$ is acyclic on every open subset of Y . The form part is fine on this paracompact manifold and on its open subsets. The skyscraper summands are flabby: every restriction is a projection onto the pole coordinates still present and is surjective. For a direct image to the finite space X , a stalk of $R^q\pi_*\mathcal{K}^r$ is $H^q(\pi^{-1}U_x, \mathcal{K}^r)$, because x has its smallest open U_x . Those groups vanish for $q > 0$. Consequently

$$\boxed{\mathcal{C}^\bullet := \pi_*\mathcal{K}^\bullet \text{ represents } R\pi_*\mathcal{F}.} \quad (\text{CDW2.6})$$

This is one compatible resolution on all three preimage opens, not separate complexes whose gluing maps have been inferred from cohomology.

CDW3. The actual diagram of resolution sections

On either pole chart $Y_p \simeq \mathbb{C}$, the complex $\mathcal{C}(U_p)$ is

$$[C^\infty(Y_p, A) \oplus V_p] \rightarrow [\mathcal{E}_A^1(Y_p) \oplus A] \rightarrow \mathcal{E}_A^2(Y_p), \quad (\text{CDW3.1})$$

with exactly (CDW2.3). The inclusion

$$\iota_p : V_p[0] \rightarrow \mathcal{C}(U_p), \quad v \mapsto (\text{constant}(r_p v), v) \quad (\text{CDW3.2})$$

is a continuous quasi-isomorphism. Indeed global radial primitives on the plane chart repeat (CDW2.4); the projection of a degree-zero cocycle onto v is its continuous inverse on cohomology. Every positive-degree cocycle has the continuous primitive already written down.

On Y° , the skyscrapers vanish, so

$$\mathcal{C}(U) = \mathcal{E}_A^\bullet(\mathbb{C}^\times). \quad (\text{CDW3.3})$$

Keep the explicit real one-form

$$\vartheta = \frac{d \arg z}{2\pi} = \frac{1}{4\pi i} \left(\frac{dz}{z} - \frac{d\bar{z}}{\bar{z}} \right). \quad (\text{CDW3.4})$$

It is globally defined on $\mathbb{C}^\times$, closed, and has integral one on the positive unit circle. It is not defined at either pole.

For completeness, in $z = e^{t+i\theta}$, write a closed form as $\omega = P(t, \theta)dt + Q(t, \theta)d\theta$. The closure equation is $\partial_\theta P = \partial_t Q$. Its positive-circle period is

$$a = \int_0^{2\pi} Q(0, \theta)d\theta.$$

After subtracting $a\vartheta$, the explicit periodic primitive is

$$g(t, \theta) = \int_0^t P(x, 0)dx + \int_0^\theta (Q(t, \phi) - a/(2\pi)) d\phi. \quad (\text{CDW3.5})$$

Its two derivatives are P and $Q - a/(2\pi)$. The period is independent of t , so $g(t, 2\pi) = g(t, 0)$. Every two-form $G(t, \theta)dt \wedge d\theta$ has primitive $(\int_0^t G(x, \theta)dx)d\theta$. These formulas prove the continuous quasi-isomorphism

$$A[0] \oplus A[-1] \rightarrow \mathcal{C}(U), \quad a \text{ in degree } 0 \mapsto \text{constant}(a), \quad a \text{ in degree } 1 \mapsto a\vartheta. \quad (\text{CDW3.6})$$

In particular, the periods give a continuous inverse on H^1 , and exact forms are its closed kernel inside the closed one-forms. The two cohomology spaces have precisely the original Fréchet topology of A . There is no passage from a completion or an algebraic direct sum of spectral blocks in (CDW3.6).

The restrictions from (CDW3.1) to (CDW3.3) restrict the differential-form component and forget the pole coordinate. On the included constants (CDW3.2) these restrictions are exactly r_p , in degree zero, with no degree-one component.

CDW4. The explicit sheaf-complex splitting

Let $j_X : \{\eta\} \hookrightarrow X$. To retain the winding orientation, write $\mathbb{C}[\vartheta]$ for the one-dimensional vector space with basis the cohomology class of (CDW3.4), and define

$$\mathcal{W} = j_{X!}(A \otimes_{\mathbb{C}} \mathbb{C}[\vartheta]). \quad (\text{CDW4.1})$$

The tensor factor is one-dimensional, so this is just a labelled copy of the same Fréchet space A on $\{\eta\}$; no choice of an infinite tensor completion is involved. Its chart sections on U_+ and U_- are zero. The label records the precise period orientation and its transformation law.

Define the split complex

$$\mathcal{P}^\bullet = \Omega[0] \oplus \mathcal{W}[-1]. \quad (\text{CDW4.2})$$

Thus its degree-zero diagram is (CDW1.4), its degree-one diagram is $0 \rightarrow A[\vartheta] \leftarrow 0$, and its differential is zero. The restrictions out of the degree-zero charts are $r_\pm$.

Define

$$\Phi : \mathcal{P}^\bullet \longrightarrow \mathcal{C}^\bullet \quad (\text{CDW4.3})$$

as follows:

open	Φ^0	Φ^1
U_+	$v_+ \mapsto (\text{constant}(r_+ v_+), v_+)$	0
U_-	$v_- \mapsto (\text{constant}(r_- v_-), v_-)$	0
U	$a \mapsto \text{constant}(a)$	$a[\vartheta] \mapsto a\vartheta$.

(CDW4.4)

The degree-zero images are cocycles by (CDW2.3); the degree-one form is closed. Restrictions commute exactly because the chart constants restrict to the same $r_\pm v_\pm$. In degree one both chart domains are zero, so there is no condition requiring ϑ to extend through a pole.

This last point is part of the sheaf construction: a map out of $j_{X!}A[\vartheta]$ may have a nonzero value on the open $\{\eta\}$ and zero domain on the pole charts. It does not purport to define a global one-form on the sphere.

A complex of sheaves on X is a diagram of complexes on its three smallest opens. At $x_\pm$, (CDW4.4) is the chart quasi-isomorphism (CDW3.2); at η , it is (CDW3.6). Therefore Φ is a stalkwise quasi-isomorphism, and

$$\boxed{\Omega[0] \oplus \mathcal{W}[-1] = \mathcal{P}^\bullet \xrightarrow[\simeq]{\Phi} \mathcal{C}^\bullet \simeq R\pi_* \mathcal{F}.} \quad (\text{CDW4.5})$$

Here $\mathcal{C}^\bullet$ is the specified representative of $R\pi_* \mathcal{F}$, and the actual comparison arrow is $\Phi : \mathcal{P}^\bullet \rightarrow \mathcal{C}^\bullet$. This invertible comparison does not require a strict chain-level projection from arbitrary forms to both chart stalks.

Every map is continuous. For a compact $K \subset \mathbb{C}^\times$, a derivative order r , and a coefficient seminorm p ,

$$\|a\vartheta\|_{K,r,p} \leq C_{K,r,p}(a). \quad (\text{CDW4.6})$$

The constants-inclusion bound is immediate, and chart continuity uses the known continuous maps $r_\pm$. The inverses on chart and overlap cohomology were explicitly checked in CDW3. Thus this is a quasi-isomorphism with continuous maps and homeomorphisms on the stalk cohomology spaces. It is stronger than an abstract vector-space splitting inferred from hereditary properties of the underlying diagram.

CDW5. The geometric and arithmetic actions on the resolution

The coefficient-only action for $a > 0$ is

$$\begin{aligned} T_a b(u) &= b(u/a), \\ \rho_+(a)(f, c_0, c_1) &= (f(\cdot/a), c_0, ac_1), & \rho_-(a)(h, d_0, d_1) &= (ah(a \cdot), ad_0, d_1). \end{aligned} \quad (\text{CDW5.1})$$

CS8 gives $r_\pm \rho_\pm(a) = T_a r_\pm$. Apply T_a to every coefficient of a differential form and to the two skyscraper A -coordinates, and apply $\rho_\pm(a)$ to the pole $V_\pm$ -coordinates. Evaluation and differentiation commute with

these continuous linear maps, so this is a strict action on $\mathcal{K}^\bullet$ and $\mathcal{C}^\bullet$. It acts by T_a on the A -coefficient of $\mathcal{W}$. The map Φ strictly intertwines it.

For a recovered positive integer n , pullback by $b_n(z) = z^n$ is an additional, geometric chain map on $\mathcal{C}^\bullet$. It fixes the poles, acts identically on their $V_\pm$ and extra A coordinates, and pulls back the differential forms. Evaluation at a fixed pole commutes with pullback, so (CDW2.3) commutes with this operation even at a ramified pole. Directly from (CDW3.4),

$$b_n^* \vartheta = n \vartheta. \quad (\text{CDW5.2})$$

Thus the geometric action on $\mathcal{P}$ is the identity on Ω , and $n \text{ id}$ on $\mathcal{W}$. The two types of action commute because one changes the geometric argument and the other acts on the coefficient.

Their combined operation B_n has

$$\frac{\begin{array}{c|cc} & V_+ & V_- \\ \hline B_n & \rho_+(n) & \rho_-(n) \end{array} \quad \begin{array}{c} A \text{ in } \Omega \\ \hline T_n \end{array}, \quad B_n(a[\vartheta]) = n T_n a[\vartheta]. \quad (\text{CDW5.3})$$

The equality

$$T_n(a) b_n^* \vartheta = n T_n a \vartheta$$

proves strict equivariance of (CDW4.3), not merely equivariance of its cohomology.

The coefficient estimates are explicit. For $a > 0$,

$$p_{N,j}(T_a b) \leq \max(a^N, a^{-N}) p_{N,j}(b). \quad (\text{CDW5.4})$$

Pullback of smooth forms is continuous for compact derivative seminorms: on a fixed compact set the derivatives of b_n up to a prescribed order are bounded, and the chain rule bounds the result by finitely many seminorms on its compact image. The same statement holds in the $w = 1/z$ chart. No unbounded inverse-image set enters this estimate.

These are actual receiving-space covers for every $n \geq 1$. They are not asserted to extend the full signed CC structure to every even index. Composition $b_m b_n = b_{mn}$, together with (CDW5.1), gives $B_m B_n = B_{mn}$. The cohomological positive-rational and strongly continuous real extensions are those proved in CSP8; a nonintegral power is not being used as a globally defined holomorphic map.

CDW6. Both mirrors on the complete derived object

Retain the two different geometric maps

$$w(z) = -1/z, \quad \alpha(z) = -1/\bar{z}. \quad (\text{CDW6.1})$$

They have the same swap of closed points on X , but w is holomorphic and α is the antiholomorphic real structure from the later CC source.

Use the coefficient morphism already constructed in CS8: exchange the two pole spaces by identity on their correspondingly labelled source coordinates and act by R on all A -coordinates. Since $r_- = R r_+$, this gives a map over either base swap. On the resolution, pull back the forms by the specified geometric map and apply R to their coefficients; on the extra skyscraper coordinates use R and the pole swap. These operations commute with (CDW2.3).

The exact form identities are

$$w^* \vartheta = -\vartheta, \quad \alpha^* \vartheta = +\vartheta. \quad (\text{CDW6.2})$$

They follow either from (CDW3.4) or from the angle maps $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi + \theta$. Hence on the split object

$$\begin{array}{c|cc} & \Omega & \mathcal{W} \\ \hline w & \text{chart swap; overlap } R & -R \\ \alpha & \text{chart swap; overlap } R & +R. \end{array} \quad (\text{CDW6.3})$$

All these maps square to the identity and are strictly intertwined by Φ . The coefficient R is complex linear; if complex conjugation on coefficients is included, its separate conjugate-linear action must also be applied. The real form ϑ itself is unchanged by conjugating its scalar coefficients, so that additional convention has no hidden angular sign.

The maps w and α commute, with composite $z \mapsto \bar{z}$; with the two R -factors their composite on $\mathcal{W}$ is $-\text{id}$. This is the action of complex conjugation on the positive angular class. On Ω the two swaps and two R -factors compose to the identity.

For precision about actual geometric composition,

$$\begin{aligned} wb_n(z) &= -z^{-n}, & b_n w(z) &= (-1)^n z^{-n}, \\ \alpha b_n(z) &= -\bar{z}^{-n}, & b_n \alpha(z) &= (-1)^n \bar{z}^{-n}. \end{aligned} \quad (\text{CDW6.4})$$

Thus these maps commute with b_n for odd n , and for even n differ by $z \mapsto -z$. That rotation preserves constants and ϑ and therefore acts trivially on the split model. It is not asserted to be the identity on every smooth form in the resolution. The strict map Φ respects the actual transformations and these identities with their rotations retained.

CDW7. The Postnikov class and what has actually vanished

Write $\mathcal{R} = R\pi_* \mathcal{F}$. The cohomology sheaves of (CDW4.3) are identified as

$$\mathcal{H}^0(\mathcal{R}) = \Omega, \quad \mathcal{H}^1(\mathcal{R}) = \mathcal{W}, \quad \mathcal{H}^{q \neq 0,1}(\mathcal{R}) = 0. \quad (\text{CDW7.1})$$

The comparison on $\mathcal{H}^0$ is the original restriction diagram; on $\mathcal{H}^1$ it is the positive-period identification. Its Postnikov triangle is

$$\Omega \longrightarrow \mathcal{R} \longrightarrow \mathcal{W}[-1] \xrightarrow{\kappa} \Omega[1]. \quad (\text{CDW7.2})$$

The class κ is the particular extension datum that cohomology sheaves alone would leave unspecified.

Here it is calculated:

$$\boxed{\kappa = 0}. \quad (\text{CDW7.3})$$

Indeed (CDW4.3) is a continuous strict equivariant quasi-isomorphism from the direct sum, inducing the two specified cohomology identifications. The two inclusions followed by Φ give maps $j_0 : \Omega \rightarrow \mathcal{R}$ and $j_1 : \mathcal{W}[-1] \rightarrow \mathcal{R}$. The projections are represented by the explicit roofs

$$\mathcal{C}^\bullet \xleftarrow[\simeq]{\Phi} \mathcal{P}^\bullet \xrightarrow{\text{pr}_0} \Omega, \quad \mathcal{C}^\bullet \xleftarrow[\simeq]{\Phi} \mathcal{P}^\bullet \xrightarrow{\text{pr}_1} \mathcal{W}[-1]. \quad (\text{CDW7.4})$$

Every arrow in these roofs is a displayed continuous map. Inverting the specified quasi-isomorphism is the definition of these derived morphisms, not an assumed inverse of an analytic quotient. Their compositions satisfy

$$r_i j_i = 1, \quad r_0 j_1 = r_1 j_0 = 0, \quad j_0 r_0 + j_1 r_1 = 1. \quad (\text{CDW7.5})$$

This splits (CDW7.2), proving (CDW7.3) with all the arithmetic and mirror actions retained.

One may express this in the ordinary equivariant derived category of vector-space sheaves, or in the localization of the displayed continuous complexes at stalkwise quasi-isomorphisms inducing homeomorphisms on cohomology. The same actual comparison Φ exists before either localization, so neither assertion requires projectivity in a category of Fréchet spaces.

There is an additional uniqueness statement after forgetting topology and equivariance. For an arbitrary vector-space sheaf G on X ,

$$\text{Hom}_X(\mathcal{W}, G) = \text{Hom}_{\mathbb{C}}(A \otimes \mathbb{C}[\vartheta], G_\eta). \quad (\text{CDW7.6})$$

This follows directly from the diagram: the two chart maps from zero impose no restriction on its central map. Stalk evaluation is exact, and every complex vector space is projective, so the right-hand functor is exact. Therefore $\mathcal{W}$ is projective in that ordinary category and

$$\text{Ext}_X^k(\mathcal{W}, \Omega) = 0 \quad (k > 0). \quad (\text{CDW7.7})$$

The Postnikov class lies in its degree-two group; the difference between splittings inducing the fixed cohomology identifications lies in its degree-one group. Both vanish. This proves underlying derived uniqueness. It does not claim projectivity or uniqueness in a topological or equivariant exact category. Existence there has already been proved by the stronger explicit map, without using (CDW7.7).

The vanishing in (CDW7.3) concerns this actual derived direct image and its winding extension. It is not an identification of every lifting obstruction in Deligne's different geometric setting with this class.

CDW8. Naturality and the complete original source sequence

Let $(V_{\pm}, A, r_{\pm}) \rightarrow (V'_{\pm}, A', r'_{\pm})$ be any continuous coefficient-diagram morphism (f_+, f_-, g) , so $gr_{\pm} = r'_{\pm}f_{\pm}$. Its map on the resolution is g applied to each form coefficient and skyscraper A -coordinate, and $f_{\pm}$ on the pole coordinates. Differentiation and evaluation commute with these maps. Moreover

$$g(\text{constant}(a)) = \text{constant}(ga), \quad g(a\vartheta) = (ga)\vartheta. \quad (\text{CDW8.1})$$

Thus the square of sheaf complexes

$$\begin{array}{ccc} \Omega \oplus j_{X!}A[\vartheta][-1] & \xrightarrow{\Phi} & \pi_*\mathcal{K}_{\Omega} \\ \downarrow & & \downarrow \\ \Omega' \oplus j_{X!}A'[\vartheta][-1] & \xrightarrow{\Phi'} & \pi_*\mathcal{K}_{\Omega'} \end{array} \quad (\text{CDW8.2})$$

commutes strictly. This is naturality on the actual complexes and restrictions.

Apply it to the closed source $J \subset A$. Let Ω_J have the same pole spaces $V_{\pm}$, overlap J , and restrictions $r_{\pm}$ with that codomain. Let $\Omega_Q = j_{X!}Q$, with zero pole spaces and overlap $Q = A/J$. Then

$$0 \rightarrow \Omega_J \rightarrow \Omega \rightarrow \Omega_Q \rightarrow 0 \quad (\text{CDW8.3})$$

is an exact sequence of vector-space sheaves with the specified continuous maps. On the pole stalks its maps are $V_{\pm} \xrightarrow{1} V_{\pm} \rightarrow 0$; on the generic stalk it is the actual strict sequence $0 \rightarrow J \rightarrow A \rightarrow Q \rightarrow 0$. Inverse image is exact, as can be checked on stalks, and gives precisely

$$0 \rightarrow \mathcal{F}_J \rightarrow \mathcal{F} \rightarrow j_{Y!}Q_{Y^c} \rightarrow 0. \quad (\text{CDW8.4})$$

Its last sheaf identification follows from zero pole stalks and constant Q on the interior with zero pole generization; it is also the quotient constructed in CSP10.

All coefficient spaces J, A, Q are Fréchet, so the preceding resolution applies to each of these diagrams. Naturality represents the derived direct images in (CDW8.4) by the following degree-wise exact sequence of split complexes:

internal degree	$R\pi_*\mathcal{F}_J$	$R\pi_*\mathcal{F}$	$R\pi_*j_{Y!}Q$
0	Ω_J	Ω	$j_{X!}Q$
1	$j_{X!}J[\vartheta]$	$j_{X!}A[\vartheta]$	$j_{X!}Q[\vartheta]$

(CDW8.5)

The differentials in these split representatives are zero, and their horizontal maps are exactly the inclusions and quotient maps just specified. The degree-one arithmetic action is nT_n on all three terms; the degree-zero action is the original one. The two mirror actions on the winding row remain $-R, +R$.

This calculation retains the entire kernel J , including its topology. It does not replace it by the quotient Q , and it does not produce a continuous arithmetic-equivariant section of $A \rightarrow Q$ by declaring the vertical derived splitting. The split between two internal cohomological degrees and the horizontal source extension $J \rightarrow A \rightarrow Q$ are different specified maps.

The formula is also natural for an additional pole skyscraper summand. Such a summand has overlap coefficient zero, so contributes only degree zero and no winding term. Consequently the extra pole summands arising from the alternative moment-null generating-domain source in CS2A/SSI10 remain in degree zero under this calculation; they are not erased or turned into an additional zeta quotient.

CDW9. The exact comparison with positive sphere orientation

On the ordered cover (U_+, U_-) , use the Čech differential $r_+ - r_-$. Applying its derived global-section complex to (CDW4.2) gives the total model

$$V_+ \oplus V_- \xrightarrow{r_+ - r_-} A \quad \text{in total degrees } 0, 1, \quad A[\vartheta] \quad \text{in total degree } 2. \quad (\text{CDW9.1})$$

There is no map into or out of that last coefficient in this model. It is the group $H^1(X, \mathcal{W})$, with its positive angular generator.

Its comparison with the positive-integration $H^2(Y, \mathcal{F}) = A$ coordinate of CSP has a minus sign. To compute it, write $r = |z|$ and define smooth forms on the two respective charts

$$\beta_+ = \frac{r^2}{1+r^2} \vartheta, \quad \beta_- = -\frac{1}{1+r^2} \vartheta. \quad (\text{CDW9.2})$$

The first extends smoothly through $z = 0$; the second extends smoothly through $w = 1/z = 0$. Direct differentiation gives

$$d\beta_+ = d\beta_- = \omega_Y = \frac{1}{\pi} \frac{dx \wedge dy}{(1+|z|^2)^2}, \quad \beta_+ - \beta_- = \vartheta, \quad \int_Y \omega_Y = 1. \quad (\text{CDW9.3})$$

The Mayer–Vietoris fibre-cone differential is $d(v, b) = (dv, \text{res}(v) - db)$. Therefore

$$d((\beta_+, \beta_-), 0) = ((\omega_Y, \omega_Y), \vartheta). \quad (\text{CDW9.4})$$

The restriction representative of the positive global class $((\omega_Y, \omega_Y), 0)$ is consequently cohomologous to $(0, -\vartheta)$. Equivalently,

$$\boxed{H^1(X, \mathcal{W}) \ni a[\vartheta] \mapsto -a \in H^2(Y, \mathcal{F}) \text{ measured by positive complex integration.}} \quad (\text{CDW9.5})$$

The projection $\mathcal{F} \rightarrow \underline{A}_Y$ is an isomorphism on degree-two cohomology, so this scalar computation fixes the same sign for the original coefficient sheaf, not only for constant coefficients.

The two mirror actions now recover the CSP signs with no lost factor. On $H^1(X, \Omega)$ the chart swap gives the Čech sign, so both act as $-R$. On $H^1(X, \mathcal{W})$, the extra chart-order minus multiplies the angular action: w acts as $+R$, while α acts as $-R$. The common comparison sign in (CDW9.5) changes neither operator. Covers retain nT_n .

The same local cone convention sends the positive local angular class at a pole to minus its positive complex fundamental class. Since $d \arg(1/z) = -d \arg z$, the two-pole interior-loop boundary in positive integration coordinates is $a \mapsto (-a, a)$, exactly CSP6–CSP7. Thus the derived splitting, the costalk calculation and the global orientation comparison agree.

CDW10. Original-zeta comparison and the scope of the result

No zeta function or its source has changed in constructing this splitting. Keep the explicit raw transform and comparison

$$\begin{aligned} \mathcal{M}_0 a(s) &= \int_0^\infty a(u) u^s \frac{du}{u}, & \Theta a(s) &= \tfrac{1}{2} \mathcal{M}_0 a(s), \\ \mathcal{M}_0 \Sigma f(s) &= 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \end{aligned} \quad (\text{CDW10.1})$$

For the retained $f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$,

$$\Theta \Sigma f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{M}_0 \Sigma f_0 = 2F_0. \quad (\text{CDW10.2})$$

Its values $F_0(0) = F_0(1) = 1/8$, and

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \quad (r \geq 1) \quad (\text{CDW10.3})$$

retain the original endpoint and trivial-zero contributions. SSI supplies the complete inverse and every Taylor coefficient at those trivial-zero locations.

Let $\mathcal{B}$ be the entire functions rapidly decreasing on every bounded real strip, and let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ and } 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CDW10.4})$$

The exact source theorem gives $\Theta J = \mathcal{J}$ and the topological isomorphism $\mathcal{Q} = A/J \rightarrow \mathcal{Q}$. No multiplicity is suppressed.

In the positive-integration convention of CSP11, the geometric normal map to the GSL high-weight kernel is

$$a \mapsto [(1 - E_+(s))(\Theta a)(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{CDW10.5})$$

Here $\mathcal{J}_+$ is the full zero-jet ideal at every $\rho + 1$; E_+ is the globally controlled multiplier constructed in GSL, with jet one on the original divisor and zero on its $+1$ translate. Its target retains the full product

$$\left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \right] \left[\frac{(s-1)(s-2)}{8} \pi^{-(s-1)/2} \Gamma((s-1)/2) \zeta(s-1) \right]. \quad (\text{CDW10.6})$$

The second factor retains its values $1/8$ at $s = 1, 2$, and its values at $s = 1 - 2r$ are (CDW10.3). All derivative and nilpotent formulas are those of the full original and translated factors, as proved in CSP11 and GSL6.

In the winding coordinate of the actual derived splitting, (CDW9.5) must be applied first. The same map is therefore

$$\boxed{a[\vartheta] \mapsto -[(1 - E_+(s))(\Theta a)(s-1)]_{\mathcal{J} \cap \mathcal{J}_+}}. \quad (\text{CDW10.7})$$

Its kernel is exactly $J[\vartheta]$, because the inclusion of the GSL kernel is injective and $\Theta^{-1}J = J$. Its quotient is the entire $\mathcal{Q}(-1)$, with the GSL inclusion and the displayed orientation sign. For every recovered n ,

$$\Theta(nT_n a)(s-1) = n n^{s-1} \Theta a(s-1) = n^s \Theta a(s-1), \quad (\text{CDW10.8})$$

so the comparison remains equivariant, including both the geometric degree and Mellin exponent.

The new statement established here is the complete derived object

$$R\pi_* \pi^{-1} \Omega \simeq \Omega \oplus j_{X!}(A \otimes \mathbb{C}[\vartheta])[-1]$$

with a constructed continuous strict equivariant comparison, its naturality for the original source sequence, and its zero Postnikov class. The winding summand retains transforming information over the generic support. No finite-dimensional mixed étale realization is asserted, and no different lifting obstruction is identified with this Postnikov class without a map. The complete coefficient source J , its quotient $\mathcal{Q}$, their different roles in the horizontal source sequence, both geometric mirrors and all original factors remain specified.

Bounded lifts and strong continuous duals of the original-zeta coefficient

Independent derivation, 24 September 2026. Proof locators SDT0–SDT9.

SDT0. Actual input, source stage, and the assertion proved

This calculation starts on the coefficient spaces already constructed in CSP and CDI. The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All vector operations, seminorms, analytic coordinates, quotients and duals below belong to those reconstructed arithmetic coefficients. None is an operation or a newly assigned metric on that supporting datum.

The complete construction and its correction history were read for the preceding CS and CSB calculations. For this continuation the current construction guide, active-target and correction-precedence passages, and the full verbatim WU061–WU062 were reread. The mathematical input actually read here is CDI0–CDI3 and CDI8–CDI12, together with the exact coefficient definitions in CSP0–CSP1. The exact-image results SSI1–SSI7 and ESI0–ESI9 were read during the preceding calculation and are retained with their actual proof status.

The underlying human source is Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. Its original author TeX, retained at sources/CC_0903_2024_v3/author_source/announc3.tex, was previously read at lines 1376–1667. This note strengthens the topology of the already proved coefficient comparison; it is not a new reading of the entire human source or an identification with a Deligne dualizing complex.

Retain exactly

$$\begin{aligned} S &= \left\{ f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0 \right\}, \\ A &= \{ a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) < \infty \text{ for every } N, j \geq 0 \}, \quad p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)|, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), \quad J = \Sigma S, \quad Q = A/J, \quad q : A \longrightarrow Q. \end{aligned} \tag{SDT0.1}$$

Here N, j are nonnegative integers and the topology on A is the topology of all the displayed seminorms. SSI and ESI prove that J is closed and that $\Sigma : S \rightarrow J$ is a topological isomorphism.

We prove all of the following for these actual spaces:

1. Every bounded subset of A , and every bounded subset of Q , has compact closure.
2. Every compact subset of Q is the image of a compact subset of A ; consequently every bounded subset of Q has a bounded lift.
3. The continuous bijection in CDI3.3 is a topological isomorphism

$$q' : Q'_\beta \xrightarrow{\sim} J^\perp \subset A'_\beta. \tag{SDT0.2}$$

4. The evaluation maps identify the strong biduals of A and Q with their original topologies.
5. The span of all actual nontrivial-zero jet functionals is strongly dense in the entire continuous dual of the original-zeta quotient.

No nuclearity theorem or general bounded-lifting assertion is assumed. The compactness and lifting arguments are constructed below. In particular the statement is not deduced merely from the fact that a map of Fréchet spaces is onto.

SDT1. The original weighted topology and its quotient seminorms

Use the analytic change of variable $x = \log u$ only on A , writing

$$g(x) = a(e^x), \quad w_n(x) = e^{nx} + e^{-nx}, \quad P_n(a) = \max_{0 \leq j \leq n} \sup_{x \in \mathbb{R}} w_n(x) |g^{(j)}(x)|. \tag{SDT1.1}$$

Thus $P_n = \max_{j \leq n} p_{n,j}$ is expressed in the original seminorms, including both summands of the weight. No summand has been removed. The sequence P_n is increasing: $w_n(x) = 2 \cosh(nx)$ is increasing with $n \geq 0$, and the derivative range increases. For every N, j , choosing $n \geq \max(N, j)$ gives

$$p_{N,j} \leq P_n. \quad (\text{SDT1.2})$$

Conversely every P_n is a finite maximum of defining seminorms. Hence this is exactly the original topology. Each P_n is a norm on A .

For completeness, a sequence Cauchy in every P_n is Cauchy with every derivative uniformly on every compact interval in x . The corresponding uniform limits have the derivative compatibility obtained by integrating each first derivative on a compact interval. Inducting over derivatives gives a smooth limit g . For every n , the weighted derivatives are uniform Cauchy functions on the whole real axis, so their uniform limits are precisely $w_n g^{(j)}$. Set $a(u) = g(\log u)$. Then $P_n(a - a_k) \rightarrow 0$ and $P_n(a) < \infty$. This proves that A is complete. The metric

$$d_A(a, b) = \sum_{n=0}^{\infty} 2^{-n-1} \min\{1, P_n(a - b)\} \quad (\text{SDT1.3})$$

generates its topology. Thus A is a Hausdorff Fréchet space.

Define actual quotient seminorms

$$\overline{P}_n(q(a)) = \inf_{j \in J} P_n(a + j). \quad (\text{SDT1.4})$$

They are well defined, increasing, and subadditive, and they satisfy

$$q\{P_n < \varepsilon\} = \{\overline{P}_n < \varepsilon\}. \quad (\text{SDT1.5})$$

The equality uses the strict inequality: an infimum below ε supplies a representative with $P_n < \varepsilon$. These sets generate exactly the quotient topology. Images of open sets under q are open because their inverse images are the open sets $U + J$.

If all $\overline{P}_n(y)$ vanish, a representative a of y can be approximated by elements of J in every neighborhood defined by some P_n . Thus $a \in \overline{J} = J$ and $y = 0$. The quotient is therefore Hausdorff.

We also give its completeness argument. Let (y_ℓ) be Cauchy for every $\overline{P}_n$. Choose a subsequence $(y_{\ell_k})_{k \geq 1}$ so that

$$\overline{P}_k(y_{\ell_{k+1}} - y_{\ell_k}) < 2^{-k-1}.$$

Choose $b_k \in A$ with

$$q(b_k) = y_{\ell_{k+1}} - y_{\ell_k}, \quad P_k(b_k) < 2^{-k}. \quad (\text{SDT1.6})$$

For every fixed m , the tail from $k \geq m$ is absolutely summable in P_m , since $P_m \leq P_k$. The series $\sum_{k \geq 1} b_k$ therefore converges in A . Adding any representative of y_{ℓ_1} , then applying q , gives a limit of the chosen subsequence. The original Cauchy sequence has that same limit. The metric obtained from (SDT1.3) by replacing P_n by $\overline{P}_n$ is complete, so Q is Fréchet.

This proof uses the quotient's actual closed kernel, not a completion in which nonzero quotient classes are discarded.

SDT2. A finite-net estimate with the two original weights retained

Fix $n \geq 0$ and $M > 0$. Consider

$$\mathcal{U}_{n+1,M} = \{a \in A : P_{n+1}(a) \leq M\}. \quad (\text{SDT2.1})$$

This set need not be bounded in the full Fréchet topology. We prove the exact assertion needed: it is totally bounded for the single norm P_n .

For all $x \in \mathbb{R}$,

$$\frac{w_n(x)}{w_{n+1}(x)} = \frac{e^{nx} + e^{-nx}}{e^{(n+1)x} + e^{-(n+1)x}} \leq 2e^{-|x|}. \quad (\text{SDT2.2})$$

Indeed the numerator is at most $2e^{n|x|}$, while the denominator is at least $e^{(n+1)|x|}$. If $a, b \in \mathcal{U}_{n+1, M}$, then for $0 \leq j \leq n$,

$$\sup_{|x| > T} w_n(x) |g_a^{(j)}(x) - g_b^{(j)}(x)| \leq 4Me^{-T}. \quad (\text{SDT2.3})$$

For a prescribed $\varepsilon > 0$, choose $T > 0$ with $4Me^{-T} < \varepsilon/2$.

On $[-T, T]$, each $g_a^{(j)}$, $j \leq n+1$, has absolute value at most $M/2$, because $w_{n+1} \geq 2$. The functions $g_a^{(j)}$, $j \leq n$, are therefore Lipschitz with constant at most $M/2$. Put $W = w_n(T)$. Choose a finite mesh of $[-T, T]$ with every point at distance at most δ from a mesh point, where

$$0 < \delta < \frac{\varepsilon}{4MW}. \quad (\text{SDT2.4})$$

At every mesh point, quantize each of the finitely many complex numbers $g_a^{(j)}$, $j \leq n$, to a finite set with error at most η , where

$$0 < \eta < \frac{\varepsilon}{8W}. \quad (\text{SDT2.5})$$

The bound $M/2$ makes that quantization finite. There are finitely many resulting joint codes. For each occupied code choose one function b in the set with that code. If a has the same code, the two Lipschitz bounds and the two quantization errors give

$$\sup_{|x| \leq T} w_n(x) |g_a^{(j)}(x) - g_b^{(j)}(x)| \leq W(M\delta + 2\eta) < \varepsilon/2 \quad (0 \leq j \leq n). \quad (\text{SDT2.6})$$

Combining (SDT2.3) and (SDT2.6) proves $P_n(a - b) < \varepsilon$. The finitely many selected b 's form a finite P_n -net. For $M = 0$ the set is the singleton zero, so the same conclusion holds.

Every bounded $B \subset A$ has a finite bound for P_{n+1} , for each n . It is therefore totally bounded for every P_n . To see that this is total boundedness for d_A , first make the tail $\sum_{n > N} 2^{-n-1}$ small, and then use a finite sufficiently small P_N -net to control all terms with $n \leq N$. Since A is complete, $\overline{B}$ is compact.

The final metric fact can be verified directly: a totally bounded metric space permits extraction of a Cauchy subsequence from every sequence by successive finite covers of radii tending to zero; completeness supplies its limit. Thus its closed completion is compact. No additional locally convex compactness theorem is needed.

SDT3. Bounded quotient sets have compact closure

Let $B \subset Q$ be bounded in the actual quotient topology. For each n , put

$$M_n = \sup_{y \in B} \overline{P}_{n+1}(y) < \infty. \quad (\text{SDT3.1})$$

For each $y \in B$ choose a representative $a_y \in A$ with

$$q(a_y) = y, \quad P_{n+1}(a_y) < M_n + 1. \quad (\text{SDT3.2})$$

These choices are made separately for the present index n ; no assertion that they form a single Fréchet-bounded set is used here.

By SDT2, the set of these representatives admits a finite P_n -net. Applying q and using $\overline{P}_n(q(a)) \leq P_n(a)$ gives a finite $\overline{P}_n$ -net for B . Thus B is totally bounded in every quotient seminorm and hence in the complete quotient metric of SDT1. Its closure is compact:

$$\boxed{B \subset Q \text{ bounded} \implies \overline{B}^Q \text{ compact.}} \quad (\text{SDT3.3})$$

The same statement for A was proved in SDT2. In the usual terminology these Fréchet spaces are Montel: they are barrelled, and their closed bounded sets are compact. The barrelled assertion follows from the Baire argument supplied in SDT6. No quotient-stability assertion about general Montel spaces is required here.

Closed subspaces of A , including J , have the same bounded-compact property by intersection with their closed ambient subspace. Finite products of such spaces retain the property, because projections of a bounded set have compact closures and its closure is a closed subset of the product of those compact sets. In particular the actual graph coefficient $Z = J \oplus E_+ \oplus E_-$ also has this property.

SDT4. Constructive compact lifting through the actual quotient

We prove a compact lifting statement for this quotient directly. The argument works for a quotient of any Fréchet space by a closed subspace, but its hypotheses and construction will remain explicit.

For $n \geq 1$, define

$$\begin{aligned} U_n &= \{a \in A : P_n(a) < 2^{-n}\}, & W_n &= \{y \in Q : \bar{P}_n(y) < 2^{-n-1}\}, \\ C_n &= \{y \in Q : \bar{P}_n(y) \leq 2^{-n-1}\}. \end{aligned} \quad (\text{SDT4.1})$$

Then C_n is closed, W_n is an open neighborhood of zero,

$$\overline{W_n} \subset C_n \subset q(U_n). \quad (\text{SDT4.2})$$

The last inclusion follows from the strict gap $2^{-n-1} < 2^{-n}$ and (SDT1.5).

Let $K \subset Q$ be compact and nonempty. Choose $y_{0,1}, \dots, y_{0,r_0} \in K$ such that the translates $y_{0,i} + W_1$ cover K . Choose arbitrary lifts $a_{0,i} \in A$ with $q(a_{0,i}) = y_{0,i}$, and put

$$F_0 = \{a_{0,1}, \dots, a_{0,r_0}\}, \quad K_1 = \bigcup_{i=1}^{r_0} ((K - y_{0,i}) \cap C_1). \quad (\text{SDT4.3})$$

Each intersection is a closed subset of a compact translate of K . Hence K_1 is compact and lies in $C_1 \subset q(U_1)$. Every $y \in K$ has a representation $y = q(a_0) + r_1$ with $a_0 \in F_0$ and $r_1 \in K_1$.

Inductively suppose the compact set $K_n \subset C_n$ is constructed. Choose finitely many centers $y_{n,i} \in K_n$ whose translates $y_{n,i} + W_{n+1}$ cover K_n . Since $K_n \subset q(U_n)$, choose lifts

$$q(a_{n,i}) = y_{n,i}, \quad a_{n,i} \in U_n.$$

Define

$$F_n = \{a_{n,i}\}_i, \quad K_{n+1} = \bigcup_i ((K_n - y_{n,i}) \cap C_{n+1}). \quad (\text{SDT4.4})$$

Again K_{n+1} is compact and contained in C_{n+1} . The finite cover ensures that every residual $r_n \in K_n$ can be written

$$r_n = q(a_n) + r_{n+1}, \quad a_n \in F_n, \quad r_{n+1} \in K_{n+1}. \quad (\text{SDT4.5})$$

These finite sets are nonempty: each chosen center gives residual zero in the next residual set. Thus the displayed finite products are nonempty when K is nonempty.

Every choice from the product $\prod_{n=0}^{\infty} F_n$ has a convergent sum in A . For a fixed m and $N \geq \max(m, 1)$, its tails satisfy the uniform estimate

$$P_m \left(\sum_{n>N} a_n \right) \leq \sum_{n>N} P_m(a_n) \leq \sum_{n>N} 2^{-n} = 2^{-N}. \quad (\text{SDT4.6})$$

Completeness of A proves existence of each sum. The sum map

$$\Psi : \prod_{n=0}^{\infty} F_n \longrightarrow A, \quad \Psi((a_n)) = \sum_{n=0}^{\infty} a_n \quad (\text{SDT4.7})$$

is continuous: finite partial sums are continuous and (SDT4.6) controls the remainder uniformly in every defining seminorm.

The product of these finite discrete sets is compact. One can see this without an uncountable-product assertion: give it the metric recording the first coordinate at which two sequences differ; a diagonal subsequence argument gives a convergent subsequence from every sequence. Its image

$$B_K^* = \Psi \left(\prod_{n=0}^{\infty} F_n \right) \quad (\text{SDT4.8})$$

is therefore compact.

For any $y \in K$, choose a compatible residual path using (SDT4.3) and (SDT4.5). After stage N ,

$$y - q \left(\sum_{n=0}^N a_n \right) = r_{N+1} \in C_{N+1}.$$

For $N + 1 \geq m$,

$$\overline{P}_m(r_{N+1}) \leq \overline{P}_{N+1}(r_{N+1}) \leq 2^{-N-2}. \quad (\text{SDT4.9})$$

Thus the residual tends to zero in Q , and continuity of q gives $q(\Psi((a_n))) = y$. This proves $K \subset q(B_K^*)$. The compact set

$$B_K = B_K^* \cap q^{-1}(K) \quad (\text{SDT4.10})$$

then satisfies $q(B_K) = K$. The intersection is closed in B_K^* , because the compact set K is closed in the Hausdorff space Q . Empty K has the empty lift.

We have constructed, rather than assumed,

$$\boxed{K \subset Q \text{ compact} \implies \exists B_K \subset A \text{ compact with } q(B_K) = K.} \quad (\text{SDT4.11})$$

The construction supplies a compact set of lifts. It makes no assertion that q has a continuous linear section, and it does not choose a new topology on either coefficient.

SDT5. Bounded lifting and equality of the strong-dual topologies

Let $B \subset Q$ be bounded. SDT3 gives the compact closure $K = \overline{B}$. Lift K by SDT4 and define

$$\widetilde{B} = B_K \cap q^{-1}(B). \quad (\text{SDT5.1})$$

This set need not be closed or compact, but it is a subset of the compact set B_K , hence is bounded. Every $y \in B$ has a preimage in it. Therefore

$$\boxed{q(\widetilde{B}) = B, \quad \widetilde{B} \subset A \text{ bounded.}} \quad (\text{SDT5.2})$$

Compact sets are bounded because every continuous seminorm has a finite maximum on them; the seminorm characterization of boundedness then gives the assertion for their subsets.

For a locally convex space E , retain the strong-dual seminorms

$$p_D(\lambda) = \sup_{x \in D} |\lambda(x)|, \quad D \subset E \text{ bounded.} \quad (\text{SDT5.3})$$

CDI1 proves the algebraic identification $q'Q' = J^\perp$. If $C \subset A$ is bounded, then $q(C)$ is bounded and

$$p_C(q'\mu) = p_{q(C)}(\mu). \quad (\text{SDT5.4})$$

This proves continuity of $q' : Q'_\beta \rightarrow A'_\beta$. Conversely, for every bounded $B \subset Q$, choose its exact bounded lift from (SDT5.2). Then

$$p_B(\mu) = p_{\widetilde{B}}(q'\mu). \quad (\text{SDT5.5})$$

Every defining strong seminorm on Q' is thus the restriction of a defining strong seminorm from A' . Equations (SDT5.4) and (SDT5.5) prove

$$\boxed{Q'_\beta \xrightarrow[\mu \mapsto \mu \circ q]{\sim} (J^\perp, \text{subspace topology from } A'_\beta).} \quad (\text{SDT5.6})$$

The annihilator $J^\perp$ is closed, being the intersection of the continuous evaluation kernels at $j \in J$. This resolves the specific topology left open in CDI3.2–CDI3.3.

No openness conclusion for the separate restriction map $A'_\beta \rightarrow J'_\beta$ is needed for (SDT5.6), and no such conclusion is asserted here.

SDT6. The full strong bidual is the original coefficient

Let E be A or Q , with exactly the topology already proved. The argument also applies to the closed subspace J and the finite-product graph coefficient Z from SDT3.

First let $\Lambda : E'_\beta \rightarrow \mathbb{C}$ be continuous and linear. Continuity at zero gives finitely many bounded sets $B_1, \dots, B_r \subset E$ and a finite constant C such that

$$|\Lambda(\lambda)| \leq C \max_i p_{B_i}(\lambda) = p_{CB}(\lambda), \quad B = \bigcup_i B_i. \quad (\text{SDT6.1})$$

If $C = 0$, then $\Lambda = 0$. Otherwise let

$$D = \overline{\text{aconv}(CB)}^E, \quad (\text{SDT6.2})$$

where the absolutely convex hull consists of finite sums $\sum_i c_i b_i$ with $b_i \in CB$ and $\sum_i |c_i| \leq 1$. Every continuous seminorm is bounded on D by its supremum on CB . Thus D is bounded. By SDT2 or SDT3 it is compact, as well as closed and absolutely convex. Equation (SDT6.1) becomes

$$|\Lambda(\lambda)| \leq \sup_{x \in D} |\lambda(x)|. \quad (\text{SDT6.3})$$

For any finite family $\lambda_1, \dots, \lambda_r \in E'$, the vector $(\Lambda(\lambda_1), \dots, \Lambda(\lambda_r))$ belongs to the compact absolutely convex set

$$\{(\lambda_1(x), \dots, \lambda_r(x)) : x \in D\} \subset \mathbb{C}^r. \quad (\text{SDT6.4})$$

Indeed, otherwise finite-dimensional real separation would give a complex linear combination $\lambda = \sum_i c_i \lambda_i$ whose real value at that vector exceeds its supremum on the displayed set. Absolute convexity identifies that supremum with the supremum of the modulus. This contradicts (SDT6.3).

Hence the closed subsets $\{x \in D : \lambda(x) = \Lambda(\lambda)\}$, indexed by $\lambda \in E'$, have the finite intersection property. Compactness of D gives $x \in D$ with

$$\Lambda(\lambda) = \lambda(x) \quad \text{for every } \lambda \in E'. \quad (\text{SDT6.5})$$

Hahn–Banach separation of points in the Hausdorff locally convex space E makes x unique. Therefore

$$(E'_\beta)' = E \quad (\text{SDT6.6})$$

under the specified evaluation map, as vector spaces.

We prove equality of topologies as well. A bounded set $M \subset E'_\beta$ is pointwise bounded, since each singleton $\{x\} \subset E$ is bounded. Put

$$C_k = \{x \in E : \sup_{\lambda \in M} |\lambda(x)| \leq k\}, \quad k = 1, 2, \dots \quad (\text{SDT6.7})$$

Each C_k is closed and their union is E . The Baire theorem for the complete metric space E gives an interior point x_0 of one C_k . For a neighborhood U of zero with $x_0 + U \subset C_k$,

$$\sup_{\lambda \in M} |\lambda(u)| \leq 2k \quad (u \in U). \quad (\text{SDT6.8})$$

Thus M is equicontinuous, and the seminorm $x \mapsto \sup_{\lambda \in M} |\lambda(x)|$ is continuous on E . This proves continuity of evaluation into the strong bidual.

For the reverse topology, let P be any continuous seminorm on E , and set

$$M_P = \{\lambda \in E' : |\lambda(x)| \leq P(x) \text{ for all } x \in E\}. \quad (\text{SDT6.9})$$

For every bounded $B \subset E$,

$$\sup_{\lambda \in M_P} p_B(\lambda) \leq \sup_{x \in B} P(x) < \infty.$$

Hence M_P is bounded in E'_β . Complex Hahn–Banach, applied to each one-dimensional span, gives

$$P(x) = \sup_{\lambda \in M_P} |\lambda(x)|. \quad (\text{SDT6.10})$$

Every original continuous seminorm is therefore a strong-bidual seminorm. Together with (SDT6.5)–(SDT6.8), this proves the topological isomorphism

$$\boxed{E \xrightarrow{\sim} (E'_\beta)'_\beta, \quad x \mapsto (\lambda \mapsto \lambda(x)).} \quad (\text{SDT6.11})$$

For the terminology used in SDT3, the same Baire argument proves that E is barrelled. If $C \subset E$ is a closed, balanced, convex, absorbing set, then $E = \bigcup_{k \geq 1} kC$. Baire gives an interior point of some kC ; differences of two points in a small neighborhood of that point lie in $2kC$. Balanced convexity then makes C a neighborhood of zero. This proves the barrelled property from its definition.

SDT7. Actual zero jets are strongly dense

Retain the exact comparison coordinate, including its factor:

$$\mathcal{M}_0 a(s) = \int_0^\infty a(u) u^s \frac{du}{u}, \quad \kappa : Q \xrightarrow{\sim} \mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad \kappa([a]) = \left[\frac{1}{2} \mathcal{M}_0 a \right]. \quad (\text{SDT7.1})$$

Here $\mathcal{B}$ is the entire strip-Schwartz space with seminorms

$$b_{R,M}(F) = \sup_{\substack{|\sigma| \leq R \\ t \in \mathbb{R}}} (1 + |t|)^M |F(\sigma + it)|,$$

and

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of the original } \zeta, \ 0 \leq j < m_\rho\}. \quad (\text{SDT7.2})$$

These are the full actual multiplicities. The raw source identity remains

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{SDT7.3})$$

SSI's continuous inverse for a raw target $F \in \mathcal{J}$ is

$$f_F(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{F(2+it)}{2\zeta(2+it)} x^{-2-it} dt \quad (x > 0),$$

$$f_F(-x) = f_F(x), \quad f_F(0) = 0, \quad \int_{\mathbb{R}} f_F = 0, \quad \frac{f_F^{(2r)}(0)}{(2r)!} = \frac{F(-2r)}{2\zeta'(-2r)} \quad (r \geq 1). \quad (\text{SDT7.4})$$

The actual pole at 1 and trivial-zero residues are therefore not erased by the quotient-topology argument. In (SDT7.1)'s coordinate G , the raw target is $2G$.

Let

$$\delta_{\rho,j}([F]) = F^{(j)}(\rho), \quad \mathcal{D}_{\text{fin}} = \text{span}_{\mathbb{C}}\{\delta_{\rho,j} : \rho, 0 \leq j < m_\rho\} \subset \mathcal{Q}'. \quad (\text{SDT7.5})$$

Each jet is continuous: choose a circle about ρ contained in a fixed strip and use Cauchy's formula to bound its derivative by one $b_{R,0}$. It vanishes on $\mathcal{I}$, so factors continuously through the quotient. Most importantly, these functionals together separate every class:

$$\bigcap_{\rho,j} \ker \delta_{\rho,j} = \{0\} \subset \mathcal{Q}. \quad (\text{SDT7.6})$$

This is exactly the defining full-jet ideal (SDT7.2), not a finite truncation or a simplicity assumption.

The isomorphism κ transfers SDT3 and SDT6 to $\mathcal{Q}$. Suppose the strong closure of $\mathcal{D}_{\text{fin}}$ were a proper closed linear subspace of $\mathcal{Q}'_\beta$. Hahn–Banach separation in that Hausdorff locally convex space would give a nonzero continuous linear functional Λ vanishing on that subspace. By (SDT6.6), Λ is evaluation at a unique class $x \in \mathcal{Q}$. Since it vanishes on every $\delta_{\rho,j}$, (SDT7.6) gives $x = 0$, contrary to $\Lambda \neq 0$. Therefore

$$\overline{\mathcal{D}_{\text{fin}}}^{\beta(\mathcal{Q}', \mathcal{Q})} = \mathcal{Q}'_\beta. \quad (\text{SDT7.7})$$

Equivalently, for every $\lambda \in \mathcal{Q}'$, every finite family of bounded subsets $B_1, \dots, B_r \subset \mathcal{Q}$, and every $\varepsilon > 0$, there is a finite linear combination d of the actual full zero jets such that

$$\max_i \sup_{x \in B_i} |\lambda(x) - d(x)| < \varepsilon. \quad (\text{SDT7.8})$$

This is stronger than matching finitely many evaluations. It asserts density in the uniform-on-bounded-sets topology, with exactly the bounds that define that topology.

For precision about completion, E'_β is complete for each Fréchet E used here. Indeed a strong Cauchy net (λ_α) is pointwise Cauchy and defines a linear limit λ . On every bounded set B , the Cauchy estimate passes to the pointwise limit, so convergence is uniform on B . To show that λ is continuous, take any sequence $x_n \rightarrow 0$ in E . The compact set $\{0, x_1, x_2, \dots\}$ is bounded, and uniform convergence on it makes the restriction of λ a uniform limit of continuous functions. Thus $\lambda(x_n) \rightarrow 0$. Metrizability of E turns sequential continuity at zero into continuity. Hence $\lambda \in E'$, and the convergence already proved on every bounded set is strong convergence.

Consequently the Hausdorff completion of $\mathcal{D}_{\text{fin}}$, with the topology induced from $\mathcal{Q}'_\beta$, is canonically $\mathcal{Q}'_\beta$. This is a completion statement about jet functionals. It does not assert that the finite-support class space $\mathcal{Q}_{\text{fin}}$ is dense in $\mathcal{Q}$, nor that an unproved topology on that class space is its original quotient subspace topology.

SDT8. Strong-dual cohomology and the original residue map

Retain CDI's full endpoint and complex data:

$$\begin{aligned} E_+ &= \mathbb{C}e_{+,0} \oplus \mathbb{C}e_{+,1}, & E_- &= \mathbb{C}e_{-,0} \oplus \mathbb{C}e_{-,1}, & V_\pm &= S \oplus E_\pm, \\ r_+(f, e) &= \Sigma f, & r_-(h, e) &= R\Sigma h = \Sigma \hat{h}, & Ra(u) &= u^{-1}a(u^{-1}), \\ W &= V_+ \oplus V_-, & d(v_+, v_-) &= r_+v_+ - r_-v_-, & Z &= \ker d. \end{aligned} \quad (\text{SDT8.1})$$

Every original endpoint line remains present. The inverse $\Sigma^{-1} : J \rightarrow S$ gives the continuous splittings in CDI1:

$$V_\pm \simeq E_\pm \oplus J, \quad W \simeq Z \oplus J, \quad Z \simeq J \oplus E_+ \oplus E_-. \quad (\text{SDT8.2})$$

Finite topological direct sums transpose to finite topological direct sums for strong duals. To check this, a bounded set has bounded projections, and the image of a bounded set under either continuous coordinate inclusion is bounded. The strong seminorms in each direction are therefore bounded by finite sums or maxima of the coordinate strong seminorms.

For the actual global complex

$$D = [W \xrightarrow{d} A \xrightarrow{0} A] \quad \text{in degrees } 0, 1, 2,$$

the Hom convention $d_{D^\vee}^n \lambda = (-1)^{n+1} \lambda \circ d_D^{-n-1}$ gives

$$D^\vee = [A'_\beta \xrightarrow{0} A'_\beta \xrightarrow{\lambda \mapsto (r'_+ \lambda, -r'_- \lambda)} V'_{+\beta} \oplus V'_{-\beta}] \quad \text{in degrees } -2, -1, 0. \quad (\text{SDT8.3})$$

CDI1's exact transpose computation gives $\ker d' = J^\perp$ and $\text{im } d' = Z^\perp$. Equation (SDT5.6) identifies the kernel topology, while (SDT8.2) identifies the quotient topology in degree zero. Thus all three identifications are now strong topological isomorphisms:

$$\boxed{H^{-2}(D^\vee) = A'_\beta, \quad H^{-1}(D^\vee) \simeq Q'_\beta, \quad H^0(D^\vee) \simeq Z'_\beta.} \quad (\text{SDT8.4})$$

Here cohomology has its actual kernel/subspace and quotient topologies in the displayed coefficient complex.

For each actual pole-costalk complex $L_\pm = [V_\pm \rightarrow A \rightarrow A]$, the same argument gives

$$\boxed{H^{-2}(L_\pm^\vee) = A'_\beta, \quad H^{-1}(L_\pm^\vee) \simeq Q'_\beta, \quad H^0(L_\pm^\vee) \simeq E'_\pm \beta.} \quad (\text{SDT8.5})$$

The transposed two-support maps retain their original signs:

$$\begin{aligned} A'_\beta &\longrightarrow A'_\beta \oplus A'_\beta, & \lambda &\longmapsto (\lambda, \lambda) & \text{in degree } -2, \\ Q'_\beta &\longrightarrow Q'_\beta \oplus Q'_\beta, & \mu &\longmapsto (\mu, -\mu) & \text{in degree } -1. \end{aligned} \quad (\text{SDT8.6})$$

They are continuous for the proved topologies. No sign is changed by strengthening the identification in degree minus one.

Let $\mathcal{Q}_{\text{fin}}$ denote the classes whose full jet family vanishes outside a finite set of actual zeros; every order less than the actual multiplicity remains in each retained block. The original-zeta residue comparison of CDI9 and RD2–RD4 remains

$$\iota(G)(F) = \sum_\rho \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds, \quad \iota : \mathcal{Q}_{\text{fin}} \xrightarrow{\sim} \mathcal{D}_{\text{fin}}. \quad (\text{SDT8.7})$$

The sum is finite for $G \in \mathcal{Q}_{\text{fin}}$. Its denominator is the original ζ . At an actual zero ρ of multiplicity m , write

$$\zeta(\rho + t) = t^m u_\rho(t), \quad u_\rho(0) \neq 0.$$

In the monomial jet bases at ρ and $1 - \rho$, its full matrix and determinant are

$$P_{ij} = (-1)^j [t^{m-1-i-j}] \frac{1}{u_\rho(t)} \quad (0 \leq i, j < m), \quad \det P = u_\rho(0)^{-m} \neq 0. \quad (\text{SDT8.8})$$

Thus SDT7 proves that the actual image of (SDT8.7) is strongly dense. It does not equip its domain with a new unstated topology or extend the residue formula to an unproved infinite sum.

The unshifted arithmetic action remains $T_a[F(s)] = [a^s F(s)]$ for $a > 0$. Its source multiplier satisfies $b_{R,M}(a^s F) \leq \max(a^R, a^{-R}) b_{R,M}(F)$, so it is continuous on $\mathcal{B}$, preserves $\mathcal{J}$, and acts continuously on $\mathcal{Q}$. The exact covariance is

$$\iota(T_a G) = a(T_{a^{-1}})' \iota(G). \quad (\text{SDT8.9})$$

For the normal coefficient with action aT_a , the invariant-pairing contragredient is

$$\mathbf{N}_a^\vee = a^{-1}(T_{a^{-1}})', \quad \iota(T_a G) = a^2 \mathbf{N}_a^\vee \iota(G). \quad (\text{SDT8.10})$$

The strong topology does not remove either factor or the inverse parameter. Continuous linear maps take bounded sets to bounded sets, so all these individual transposed operators are strongly continuous. No positivity conclusion follows from reflexivity or strong density.

One can now construct an exact completed residue receiver. Give $\mathcal{Q}_{\text{fin}}$ the initial locally convex topology of ι , explicitly

$$p_B^{\text{res}}(G) = \sup_{F \in B} \left| \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds \right|, \quad B \subset \mathcal{Q} \text{ bounded.} \quad (\text{SDT8.11})$$

The local determinant (SDT8.8) makes this topology Hausdorff. By definition ι is a topological isomorphism onto $\mathcal{D}_{\text{fin}}$ with its induced strong topology. SDT7 proves that the latter is dense in the complete space $\mathcal{Q}'_{\beta}$. Therefore the actual Hausdorff completion is

$$\boxed{(\mathcal{Q}_{\text{fin}}, \widehat{(p_B^{\text{res}})_B}) \xrightarrow[\hat{\iota}]{} \mathcal{Q}'_{\beta}.} \quad (\text{SDT8.12})$$

This is a constructed residue completion. Its topology has been specified by (SDT8.11); it is not identified with the pre-existing subspace topology from $\mathcal{Q}$.

For completeness, the ordinary-transpose comparison also has a full formula. In this paragraph $R[F(s)] = [F(1-s)]$ is the coefficient reflection, so the ordered global degree-one mirror is $-R$, not R . Set

$$\mathbf{V}_a = aT_{a^{-1}} = RT_aR. \quad (\text{SDT8.13})$$

On $\mathcal{B}$, reflection satisfies $b_{R_0, M}(F(1-s)) \leq b_{R_0+1, M}(F)$; the different letter R_0 here is a strip width. The original functional equation preserves the full multiplicities at ρ and $1-\rho$, so reflection preserves $\mathcal{I}$. Both R and T_a therefore act continuously on $\mathcal{Q}$ and preserve the finite full-zero support, including the reflected multiplicities. Equation (SDT8.9) at the parameter a^{-1} , with the prefactor a retained, gives

$$\iota(\mathbf{V}_a G) = a a^{-1} (T_a)' \iota(G) = (T_a)' \iota(G). \quad (\text{SDT8.14})$$

The operator $\mathbf{V}_a$ is therefore continuous for (SDT8.11) and extends to the actual ordinary transpose T'_a on its completion. For the original action T_a , the extension is instead $a(T_{a^{-1}})'$, exactly as in (SDT8.9).

There is also an unreflected residue map

$$\phi = \iota R, \quad \phi(G)(F) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(s)}{\zeta(s)} ds, \quad \phi T_a = (T_a)' \phi. \quad (\text{SDT8.15})$$

The middle formula substitutes $RG(1-s) = G(s)$ before taking residues, so it introduces no change-of-variable sign. The last identity follows either directly by multiplying the integrand by a^s , or from $RT_a = \mathbf{V}_a R$ and (SDT8.14).

Its exact topological receiver can be specified without assuming continuity of R in an unexamined topology: equip $\mathcal{Q}_{\text{fin}}$ with

$$p_B^{\text{unref}}(G) = \sup_{F \in B} \left| \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(s)}{\zeta(s)} ds \right| = p_B^{\text{res}}(RG). \quad (\text{SDT8.16})$$

Since $R^2 = 1$ and $R\mathcal{Q}_{\text{fin}} = \mathcal{Q}_{\text{fin}}$, it is a homeomorphism from this unreflected topology to the reflected topology (SDT8.11). The map ϕ has the same range $\mathcal{D}_{\text{fin}}$, so its completion is also canonically $\mathcal{Q}'_{\beta}$, now intertwining T_a with the ordinary transpose T'_a . These are two explicitly related residue topologies; neither has been silently substituted for the inherited topology of $\mathcal{Q}_{\text{fin}} \subset \mathcal{Q}$.

SDT9. Exact implication and limits of this calculation

The proved chain is

$$\begin{array}{c}
\text{original weighted derivative estimates on } A \\
\Downarrow \text{SDT2-SDT3} \\
\text{every bounded subset of } Q \text{ has compact closure} \\
\Downarrow \text{SDT4-SDT5} \\
\text{compact lifts and exact bounded lifts through } q \\
\Downarrow \\
Q'_\beta \simeq J^\perp \subset A'_\beta,
\end{array} \tag{SDT9.1}$$

bounded-compact property and the actual Fréchet topology

$$\begin{array}{c}
\Downarrow \text{SDT6} \\
(Q'_\beta)' = Q \\
\Downarrow \text{SDT7} \\
\overline{\mathcal{D}_{\text{fin}}}^\beta = Q'_\beta.
\end{array}$$

The arrows name the exact proofs that justify the topology change. The compact lift is the finite-choice series (SDT4.7), not a claimed linear section. The dense space at the bottom right consists of functionals, not finite-support classes in the primal quotient.

This establishes the full strong-dual cohomology topology for the actual CDI coefficient complexes and strengthens their finite-jet detection to uniform approximation on bounded subsets. It neither constructs a Verdier dualizing sheaf nor identifies these infinite-dimensional coefficient duals with a finite-rank étale receiver. No weight-separation or RH conclusion is inferred from these topological statements.

The continuous support-dual sheaf on the actual sphere

Complete independent derivation, 24 September 2026. Proof locators CSD0–CSD12.

CSD0. Input stage and the category being constructed

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All coefficient vector spaces and all complex coordinates below belong to the receiving geometry after complete arithmetic reconstruction. No addition, numerical coordinate, midpoint or distance is assigned to τ .

For this continuation the current `READ_FIRST_USER_CONSTRUCTION.md`, the connected `USER_ARGUMENT_RECONSTRUCTION.md` and correction table, PC01–PC05 including its latest continuation, and verbatim WU050–WU055 and WU061–WU063 were recalled. The no-addition, no-metric, full-history and separate-branch corrections govern the same receiver as in CSP and CDW. The actual source maps are retained; the kernel J is not replaced by its quotient Q .

The original-source use is that recorded in CSP0 and CDW0: Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, and the field-valued charts and real structure of arXiv:2609.00299v1. Those source readings and the local reconstructions are different provenance claims; this continuation does not claim a fresh complete reading of either paper.

The complete current proof `CC_CONTINUOUS_DUAL_INDEPENDENT.md`, CDI0–CDI12, supplies the continuous duals of the already constructed global and support coefficient complexes. Its full maps are the comparison target here. CSP0–CSP13 and CDW0–CDW10 supply the sphere sheaf and one compatible fine resolution. SSI1–SSI7 and ESI give the closed image and continuous inverse of the full original summation source.

Before the final topology and transfer comparisons below, the complete accepted `CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md`, SDT0–SDT9, and `CC_RAMIFIED_TRACE_AND_RESIDUE_ACTI ON.md`, GTR0–GTR9, were read. Their bounded-lift theorem and separate weighted geometric trace are imported with their exact scope; neither is attributed to the preceding current construction.

An actual complex of sheaves with continuous maps is constructed below from continuous functionals on compactly supported tests. It is not defined by an unrestricted algebraic internal Hom. In particular A' always means the continuous complex-linear dual of the stated locally convex A , never its algebraic dual. No equivalence for arbitrary topological sheaves is assumed.

CSD1. Full primal source and single resolution

Let $Y = \mathbb{P}^1(\mathbb{C})$, $P = \{0, \infty\}$, and $i : P \hookrightarrow Y$. Retain

$$\begin{aligned}
 S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \quad V_{\pm} = S \oplus E_{\pm}, \quad E_{\pm} = \mathbb{C}^2, \\
 A &= \left\{ a \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j \geq 0 \right\}, \\
 \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \\
 \hat{f}(t) &= \int_{\mathbb{R}} f(v) e^{-2\pi i vt} dv, \\
 r_+(f, e) &= \Sigma f, \quad r_-(h, e) = R \Sigma h = \hat{\Sigma} h.
 \end{aligned} \tag{CSD1.1}$$

The last equality uses the full Poisson formula with terms $u^{-1}h(0) - \int h$, which vanish on this stated S . Both coordinates of each $E_{\pm}$, labelled value at zero and real integral, remain distinct.

The actual source image $J = \Sigma S$ is closed, $\Sigma : S \rightarrow J$ is a topological isomorphism, and $Q = A/J$ has its Fréchet quotient topology. The proved inverse-image sheaf is

$$\mathcal{F} = \pi^{-1}\Omega = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-). \tag{CSD1.2}$$

Here $\pi(0) = x_+$, $\pi(\infty) = x_-$, and $\pi(\mathbb{C}^\times) = \eta$. Its generic inverse image is the entire punctured plane, not a numerical coordinate assigned to τ .

Put $V = V_+ \oplus V_-$. The difference map is onto on stalks through its evaluation component and gives

$$0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_* V \xrightarrow{\text{ev}-r} i_* A^2 \rightarrow 0. \quad (\text{CSD1.3})$$

For smooth A -valued differential forms $\mathcal{E}_A^q$, the single resolution from CDW2 is

$$\begin{aligned} \mathcal{K}^0 &= \mathcal{E}_A^0 \oplus i_* V, & \mathcal{K}^1 &= \mathcal{E}_A^1 \oplus i_* A^2, & \mathcal{K}^2 &= \mathcal{E}_A^2, \\ d^0(f, v) &= (df, \text{ev}(f) - rv), & d^1(\omega, a) &= d\omega. \end{aligned} \quad (\text{CSD1.4})$$

Its degree-zero kernel is (CSD1.2). At a pole, a closed pair (ω, a) has primitive $f = g + a - g(p)$, $v = 0$, where $dg = \omega$. Away from the poles the coefficient-valued Poincaré lemma applies. This proves exactness in positive degrees with continuous local primitives. Forms and finite skyscrapers are fine.

This resolution also computes compactly supported cohomology. A section defined near a compact set extends after multiplication by a smooth cutoff equal to one on a smaller neighborhood of that set. The partition-of-unity contraction of the augmented Čech complex therefore works with compact supports: only finitely many members of a locally finite partition meet an input compact support, and all multiplications and sums are continuous. This proves the needed compact-support acyclicity for these form and skyscraper terms. It does not imply acyclicity for sections supported at a single point.

CSD2. Continuous compact-support contraction

Set $\mathcal{E}_c^q(U; A) = \Gamma_c(U, \mathcal{E}_A^q)$. Its topology is the locally convex inductive limit over compact supports, each given all smooth-derivative seminorms from A . Finite pole sums have their product topologies. Extension by zero of compactly supported forms is continuous.

An explicit compact-support Poincaré calculation is useful. In an oriented plane chart choose $\rho_x, \rho_y \in C_c^\infty(\mathbb{R})$ with integrals one. Define

$$I_x f(y) = \int_{\mathbb{R}} f(x, y) dx, \quad H_x f(x, y) = \int_{-\infty}^x f(t, y) dt - \left(\int_{-\infty}^x \rho_x(t) dt \right) I_x f(y), \quad (\text{CSD2.1})$$

and define H_y similarly. Completeness of A supplies the integrals. For every continuous seminorm,

$$p\left(\int f\right) \leq \int p(f), \quad (\text{CSD2.2})$$

which also proves continuity after any fixed number of derivatives.

For compactly supported forms put

$$\begin{aligned} K(P dx + Q dy) &= H_x P, \\ K(G dx \wedge dy) &= H_x G dy - \rho_x(x) H_y(I_x G)(y) dx, & K|_{\mathcal{E}_c^0} &= 0. \end{aligned} \quad (\text{CSD2.3})$$

Each output is supported in a fixed rectangle determined by the input support and the two fixed bumps. Thus these are continuous operators on the stated inductive limits. Direct differentiation gives

$$\begin{aligned} Kdf &= f, & dK\omega + Kd\omega &= \omega \quad (\deg \omega = 1), \\ dK\omega &= \omega - \rho_x \rho_y dx \wedge dy \int_{\mathbb{R}^2} \omega \quad (\deg \omega = 2). \end{aligned} \quad (\text{CSD2.4})$$

Indeed $\partial_x H_x f = f - \rho_x I_x f$ and $H_y \partial_y g = g$ for compactly supported g give every term in these identities. The bump two-form has positive integral one and is annihilated by K . This is a continuous contraction onto A in degree two. An orientation-preserving disk-to-plane diffeomorphism transfers it to every disk below.

On the cylinder $t = \log|z|$, $\theta = \arg z$, combine the one-dimensional compact contraction in t with the following circle homotopy: subtract the angular mean, integrate that mean-zero coefficient from zero to θ , then subtract the mean of the primitive. Its derivative is the original coefficient minus its mean and it is periodic. The surviving circle classes are constants and $d\theta/(2\pi)$. The compact cylinder classes are consequently

$$\rho(t) dt \quad (\deg 1), \quad \rho(t) dt \wedge \frac{d\theta}{2\pi} \quad (\deg 2), \quad \int \rho = 1. \quad (\text{CSD2.5})$$

All homotopies are continuous and use actual test forms, not an unproved homotopy-invariance assertion for topological coefficients.

CSD3. The continuous-current sheaf

For $-2 \leq n \leq 0$, define

$$\mathcal{T}_A^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad (dT)(\phi) = (-1)^{n+1}T(d\phi). \quad (\text{CSD3.1})$$

Other degrees vanish. Each dual may carry the weak topology of pointwise evaluation or the strong topology of uniform convergence on bounded test sets. Both use exactly the same continuous functionals.

Restriction is transpose to extension by zero. To prove the sheaf axiom, decompose a compactly supported test form by a locally finite subordinate partition of unity and evaluate compatible local functionals on its pieces. Only finitely many terms meet the compact support. Compatibility proves independence of the partition and uniqueness; continuity on every fixed-support space proves continuity on the specified inductive limit. Restriction and scalar multiplication of currents are continuous for both dual topologies: they transpose continuous operations on tests which take bounded sets to bounded sets.

The differential is the continuous Hom differential, so $d^2 = 0$. Smooth functions multiply currents on their test argument, making every term a fine sheaf. This gives ordinary-section acyclicity on all opens used below.

The positive complex orientation defines

$$\underline{A}'_Y[2] \longrightarrow \mathcal{T}_A^\bullet, \quad c_\lambda(\phi) = \int_U \lambda(\phi) \quad (\phi \text{ a compactly supported two-form}). \quad (\text{CSD3.2})$$

It is continuous and closed by Stokes. Transpose the explicit contraction (CSD2.4): this proves that (CSD3.2) is a local quasi-isomorphism with continuous inverse on cohomology. In particular its coefficient is the entire continuous A' . The weak-topology comparison is a homeomorphism. For this local coefficient the strong-topology comparison also follows from the two explicit bounded-set preserving test maps.

No tensor-product identification between scalar distributions and A' is used. Currents mean exactly (CSD3.1). At a pole define the positive Dirac current

$$\delta_p \lambda(f) = \lambda(f(p)), \quad i_* A' \longrightarrow \mathcal{T}_A^0. \quad (\text{CSD3.3})$$

This is the actual local Gysin chain map, with its orientation fixed below.

CSD4. The actual support-dual complex

Using the ordinary cone differential $d(b, a) = (d_B b + ga, -d_A a)$, put

$$\mathcal{D}_c \mathcal{F} = \text{Cone} \left(i_* A'^2 \xrightarrow{(\delta, -r')} \mathcal{T}_A^\bullet \oplus i_* V' \right). \quad (\text{CSD4.1})$$

The domain is in degree zero. Its complete terms and maps are

$$\begin{aligned} (\mathcal{D}_c \mathcal{F})^{-2} &= \mathcal{T}_A^{-2}, & (\mathcal{D}_c \mathcal{F})^{-1} &= \mathcal{T}_A^{-1} \oplus i_* A'^2, & (\mathcal{D}_c \mathcal{F})^0 &= \mathcal{T}_A^0 \oplus i_* V', \\ d^{-2} T &= (dT, 0), & d^{-1}(T, \lambda_+, \lambda_-) &= (dT + \delta_0 \lambda_+ + \delta_\infty \lambda_-, -r'_+ \lambda_+, -r'_- \lambda_-). \end{aligned} \quad (\text{CSD4.2})$$

Every map is continuous for both dual topologies.

The cone is the literal continuous transpose of the compact-support resolution: for every open U ,

$$\boxed{\Gamma(U, \mathcal{D}_c \mathcal{F}) = (\Gamma_c(U, \mathcal{K}^\bullet))^\vee, \quad (C^\vee)^n = (C^{-n})', \quad d_n^\vee = (-1)^{n+1} (d_C^{-n-1})'.} \quad (\text{CSD4.3})$$

For degree -1 , transposing $d^0(f, v)$ gives $T(df) + \sum_p \lambda_p(f(p) - r_p v_p)$, exactly (CSD4.2). Degree -2 has the minus sign in (CSD3.1). Restriction in (CSD4.3) is adjoint to extension by zero.

Evaluation is a cochain pairing: its two differential contributions are $(-1)^{n+1} \lambda(dc)$ and $(-1)^n \lambda(dc)$, whose sum is zero. Formula (CSD4.3), together with the proved compact-support resolution, is the precise continuous support-duality meaning used here. It is an actual sheaf complex, not a declaration that general continuous dualization is exact.

Using (CSD3.2), it represents the candidate

$$\text{Cone}(i_* A'^2 \xrightarrow{(\text{Gys}, -r')} \underline{A}'_Y[2] \oplus i_* V'), \quad (\text{CSD4.4})$$

whose derived Gysin arrow is represented exactly by (CSD3.3).

CSD5. Pole stalk and original compact-support pairing

Use the positive local coordinate $z_p = z$ at zero and $z_p = 1/z$ at infinity. The form

$$\vartheta_p = \frac{d \arg z_p}{2\pi} \quad (\text{CSD5.1})$$

is locally integrable. Its degree-minus-one current is $h_p \lambda(\phi) = \int \vartheta_p \wedge \lambda(\phi)$. The local $1/r$ coefficient is integrable against the area factor $r dr d\theta$. Stokes on a punctured disk with compact test f gives

$$d(h_p \lambda)(f) = h_p \lambda(df) = \lambda(f(p)), \quad dh_p \lambda = \delta_p \lambda. \quad (\text{CSD5.2})$$

The positive small-circle limiting integral fixes the sign.

The primal pole costalk is $L_p = [V_p \xrightarrow{r_p} A \xrightarrow{0} A]$ in degrees $0, 1, 2$. Its dual differential is r'_p in degrees $-1, 0$. The actual stalk comparison is

degree	$L_p^\vee \rightarrow (\mathcal{D}_c \mathcal{F})_p$
-2	$\lambda \mapsto c_\lambda$
-1	$\mu \mapsto (h_p \mu, -\mu)$
0	$\nu \mapsto (0, \nu)$.

(CSD5.3)

Indeed $d(h_p \mu, -\mu) = (0, r'_p \mu)$. The local current contraction leaves only A' in degree -2 ; the Dirac arrow is locally null-homotopic through h_p , leaving exactly r'_p . This proves the quasi-isomorphism with continuous maps. The same construction on shrinking disks gives the actual stalk.

For a direct pairing check choose $\chi \in C_c^\infty$ equal to one near p , and a compact positive two-form ω_c of integral one. The primal compact-source chain map is

$$L_p \rightarrow \Gamma_c(U, \mathcal{K}), \quad v \mapsto (\chi r_p v, v), \quad a \text{ in degree } 1 \mapsto (d\chi a, 0), \quad a \text{ in degree } 2 \mapsto \omega_c a. \quad (\text{CSD5.4})$$

Its differential is r_p . The compact contraction (CSD2.4) followed by elimination of the evaluation coordinate proves it is a quasi-isomorphism. Pairing it with (CSD5.3) gives

$$\lambda(a), \quad \int \vartheta_p \wedge d\chi \mu(a) = \mu(a), \quad \nu(v). \quad (\text{CSD5.5})$$

The middle identity is (CSD5.2) on test χ . This is precisely CDI's positive-orientation costalk pairing.

For clarity about the cohomology topology, on each fixed disk U , transposing the compact-test map (CSD5.4) gives a continuous reverse chain map from sections of the current complex on that disk. It is not an evaluation of an arbitrary current germ on a fixed test support. In degrees $-2, -1, 0$ it is

$$\begin{aligned} T &\mapsto (a \mapsto T(\omega_c a)), & (T, \lambda) &\mapsto (a \mapsto T(d\chi a)), \\ (T, \nu) &\mapsto \nu + r'_p(a \mapsto T(\chi a)). \end{aligned} \quad (\text{CSD5.7})$$

It composes with the disk version of (CSD5.3) to the identity. Because that map is already a quasi-isomorphism, the induced cohomology maps are inverse continuous maps. Their cohomology identifications agree under shrinking disks: degree-minus-two cocycles are constant currents; for a degree-minus-one cocycle (T, λ) , the equation $dT + \delta_p \lambda = 0$ gives $T(d\chi a) = -\lambda(a)$, independent of the cutoff; and degree-zero cutoff changes lie in $\text{im } r'_p$. The underlying sheaf stalk therefore has these groups. Its cohomology topology in this proof means the stable small-disk cohomology topology, identified by these identity transition maps. It is not a claim that arbitrary locally convex inductive limits of germs commute with forming cohomology topologies. This gives the precise topological comparison for both stated dual topologies before identifying the quotient coefficient through SDT below.

Consequently

$$H^{-2}(i_p^* \mathcal{D}_c \mathcal{F}) = A', \quad H^{-1}(i_p^* \mathcal{D}_c \mathcal{F}) = Q', \quad H^0(i_p^* \mathcal{D}_c \mathcal{F}) = E'_p. \quad (\text{CSD5.6})$$

The middle identification is $\ker r'_p = J^\perp = q'Q'$. The last uses the actual splitting $V_p = E_p \oplus J$ from Σ^{-1} , and Hahn–Banach extends every continuous functional on closed J to A . Thus these are the entire continuous duals.

CSD6. Derived point support, computed separately

Fineness does not make a current sheaf automatically acyclic for support at a point. A current on a punctured disk need not extend across its missing point. Compute support by the actual restriction fibre cone

$$R\Gamma_{\{p\}}(U, \mathcal{D}_c \mathcal{F}) = \text{Cone}(\Gamma(U, \mathcal{D}_c \mathcal{F}) \rightarrow \Gamma(U \setminus \{p\}, \mathcal{D}_c \mathcal{F}))[-1]. \quad (\text{CSD6.1})$$

The ordinary sections here compute hypercohomology because the displayed terms are fine. They are not replaced by termwise point-supported currents.

On a disk the current complex has A' in degree -2 . On its puncture it has A' in degrees $-2, -1$, represented by $c_\lambda, h_p \lambda$, by the cylinder contraction (CSD2.5). In particular $h_p \lambda$ pairs as $-\lambda(a)$ with $\rho(t)dt a$, since $d\theta \wedge dt = -dt \wedge d\theta$. This radial sign is retained.

Restriction on (CSD5.3) is identity in degrees $-2, -1$ and zero in degree zero. In the fibre-cone convention $d(x, y) = (dx, \text{res}(x) - dy)$, cancel its contractible degree-minus-two pair. The remaining complex is

$$A' \longrightarrow V'_p \oplus A', \quad \mu \longmapsto (r'_p \mu, \mu). \quad (\text{CSD6.2})$$

The last coordinate records the punctured angular current. The positive supported coordinate is its negative: in the same cone, $d(h_p \lambda, 0) = (\delta_p \lambda, h_p \lambda)$, so $(\delta_p \lambda, 0)$ is cohomologous to $(0, -h_p \lambda)$. Changing to that positive supported coordinate and negating the source rewrites (CSD6.2) as

$$A' \xrightarrow{\lambda \mapsto (\lambda, -r'_p \lambda)} A' \oplus V'_p \quad \text{in degrees } -1, 0. \quad (\text{CSD6.3})$$

The first target is now measured by the positive Dirac orientation.

The continuous deformation retraction onto $V'_p[0]$ is

$$P(a, \nu) = \nu + r'_p a, \quad S(\nu) = (0, \nu), \quad h(a, \nu) = a. \quad (\text{CSD6.4})$$

One has $PS = 1$, $hd = 1$ on degree -1 , and $dh(a, \nu) = (a, -r'_p a) = (a, \nu) - SP(a, \nu)$. Therefore

$$\boxed{i_p^! \mathcal{D}_c \mathcal{F} \simeq V'_p[0] = (i_p^* \mathcal{F})^\vee.} \quad (\text{CSD6.5})$$

Costalk-to-stalk is $\nu \mapsto (0, \nu)$ in (CSD5.3); on degree-zero cohomology it is restriction $V'_p \rightarrow E'_p$. Together with CSD5 this proves both support/stalk comparisons from actual local complexes.

CSD7. Global angular current and the CDI chain comparison

The angular form $\vartheta = d \arg z / (2\pi)$ is locally integrable on the sphere. At infinity it is $-\vartheta_-$; hence

$$d(\vartheta\lambda) = \delta_0\lambda - \delta_\infty\lambda. \quad (\text{CSD7.1})$$

The original global complex and its continuous dual are

$$\begin{aligned} D &= [V_+ \oplus V_- \xrightarrow{r_+ - r_-} A \xrightarrow{0} A], \\ D^\vee &= [A' \xrightarrow{0} A' \xrightarrow{\mu \mapsto (r'_+\mu, -r'_-\mu)} V'_+ \oplus V'_-] \end{aligned} \quad (\text{CSD7.2})$$

in degrees 0, 1, 2 and $-2, -1, 0$, respectively. There is an actual continuous chain map

$$\Psi : D^\vee \longrightarrow \Gamma(Y, \mathcal{D}_c\mathcal{F}),$$

$$\lambda \mapsto c_\lambda \quad (\deg -2), \quad \mu \mapsto (\vartheta\mu; -\mu, +\mu) \quad (\deg -1), \quad (\nu_+, \nu_-) \mapsto (0; \nu_+, \nu_-) \quad (\deg 0). \quad (\text{CSD7.3})$$

The middle differential's current terms cancel by (CSD7.1), leaving precisely $(r'_+\mu, -r'_-\mu)$.

To prove this is a quasi-isomorphism, compute the global current complex using the local and cylinder contractions (CSD2.4)–(CSD2.5) and two-chart Mayer–Vietoris. Its cohomology is A' in degrees $-2, 0$ and zero in degree -1 . The first class is c_λ ; the last is evaluated on constant test functions. Both positive Dirac currents have that last value λ . All comparisons are finite sums, the written integral operators and partition-of-unity maps, so they are continuous. This computes the current cohomology without assuming arbitrary exactness of continuous dualization.

The cone then has the finite model

$$\begin{aligned} P^{-2} &= A', \quad P^{-1} = A'^2, \quad P^0 = A' \oplus V'_+ \oplus V'_-, \\ d^{-1}(\lambda_+, \lambda_-) &= (\lambda_+ + \lambda_-, -r'_+\lambda_+, -r'_-\lambda_-). \end{aligned} \quad (\text{CSD7.4})$$

Its explicit deformation retraction is

degree	$P \rightarrow D^\vee$	$D^\vee \rightarrow P$
-2	1	1
-1	$(\lambda_+, \lambda_-) \mapsto -\lambda_+$	$\mu \mapsto (-\mu, \mu)$
0	$(a, \nu_+, \nu_-) \mapsto (\nu_+, \nu_- + r'_-a)$	$(\nu_+, \nu_-) \mapsto (0, \nu_+, \nu_-)$.

(CSD7.5)

The homotopy $h^0(a, \nu_+, \nu_-) = (0, a)$ satisfies $dh + hd = 1 - SP$; substitute into (CSD7.4) to check both degrees. The current map (CSD7.3) induces the section in (CSD7.5), proving its quasi-isomorphism.

There is also a direct pairing check. Set

$$\omega_Y = \frac{1}{\pi} \frac{dx \wedge dy}{(1 + |z|^2)^2}, \quad \int_Y \omega_Y = 1.$$

The primal finite complex maps into global sections of $\mathcal{K}$ by

$$(v_+, v_-) \mapsto (\text{constant}(r_-v_-), v_+, v_-), \quad b \text{ in degree } 1 \mapsto (0; -b, 0), \quad a \text{ in degree } 2 \mapsto \omega_Y a. \quad (\text{CSD7.6})$$

Its differential is $r_+ - r_-$, including the displayed degree-one minus. Pairing (CSD7.3) and (CSD7.6) gives the original evaluation in every degree. Thus this realizes CDI's pairings and signs, not only its cohomology groups.

The continuous transpose of (CSD7.6) is an explicit cohomological inverse to (CSD7.3). Put $\ell_T(a) = T(\text{constant}(a))$ for a degree-zero current. In degrees $-2, -1, 0$, this reverse chain map is

$$T \mapsto (a \mapsto T(\omega_Y a)), \quad (T, \lambda_+, \lambda_-) \mapsto -\lambda_+,$$

$$(T, \nu_+, \nu_-) \mapsto (\nu_+, \nu_- + r'_- \ell_T). \quad (\text{CSD7.8})$$

For degree -1 , its chain identity is $(-r'_+ \lambda_+, r'_- \lambda_+) = d_{D^\vee}(-\lambda_+)$. It composes with Ψ to the identity in every degree. All test embeddings in this formula are continuous and take bounded sets to bounded sets. Therefore the induced inverse cohomology maps are continuous for both weak and strong dual topologies. No theorem that continuous dualization preserves arbitrary quasi-isomorphisms is used to prove this assertion.

In particular

$$\begin{aligned} H^{-2}(Y, \mathcal{D}_c \mathcal{F}) &= A', & H^{-1}(Y, \mathcal{D}_c \mathcal{F}) &= Q', & H^0(Y, \mathcal{D}_c \mathcal{F}) &= Z', \\ Z &= \{(v_+, v_-) : r_+ v_+ = r_- v_-\} \simeq J \oplus E_+ \oplus E_-. \end{aligned} \quad (\text{CSD7.7})$$

The last comparison uses the original Fourier-source inverse and retains all four endpoint lines.

CSD8. Restriction realizes the transposed support map

Restriction of the global angular current is h_+ at zero and $-h_-$ at infinity. Thus (CSD7.3) and (CSD5.3) give exactly

$$\begin{aligned} D^\vee \rightarrow L_+^\vee : & \quad (\deg 0, \deg -1, \deg -2) = (\text{pr}_{V'_+}, +1, +1), \\ D^\vee \rightarrow L_-^\vee : & \quad (\deg 0, \deg -1, \deg -2) = (\text{pr}_{V'_-}, -1, +1). \end{aligned} \quad (\text{CSD8.1})$$

These are the transposes of the original support-to-global map

$$L_+ \oplus L_- \rightarrow D, \quad k^0 = 1, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{CSD8.2})$$

Consequently degree-minus-one is anti-diagonal and degree-minus-two diagonal:

$$Q' \rightarrow Q'^2 : \mu \mapsto (\mu, -\mu), \quad A' \rightarrow A'^2 : \lambda \mapsto (\lambda, \lambda). \quad (\text{CSD8.3})$$

Degree zero is endpoint restriction $Z' \rightarrow E'_+ \oplus E'_-$.

The complete cone comparison is

$$\text{Cone}(k)^\vee \rightarrow \text{Cone}(k^\vee)[-1], \quad (\phi, \psi) \mapsto ((-1)^{n+1} \psi, \phi) \text{ in degree } n. \quad (\text{CSD8.4})$$

Substitution into the Hom and shifted-cone differentials proves it is a chain isomorphism. The dual long row is therefore the full CDI5 row

$$0 \rightarrow A' \xrightarrow{\Delta} A'^2 \xrightarrow{-\lambda_+ + \lambda_-} A' \xrightarrow{0} Q' \xrightarrow{(\mu, -\mu)} Q'^2 \xrightarrow{q'(\mu_+ + \mu_-)} A' \xrightarrow{r'} Z' \rightarrow E'_+ \oplus E'_- \rightarrow 0. \quad (\text{CSD8.5})$$

Here $r : Z \rightarrow A$ is common restriction and $r' \lambda = (\lambda|_J, 0, 0)$ in (CSD7.7). Diagonal/difference identities and Hahn–Banach on $J \subset A$ prove exactness. Thus the earlier continuous coefficient duality has an actual sheaf support construction realizing its maps.

The ordinary annular dual in (CSD8.4) has degrees $-1, 0$. It is not confused with $R\Gamma(\mathbb{C}^\times, \mathcal{T}_A)$, whose degrees are $-2, -1$ because it is dual to compactly supported annular cohomology. Formula (CSD4.3) specifies the support convention.

CSD9. Proper geometric transposes and both mirrors

A continuous coefficient-diagram map acts on compact tests and, contravariantly, on this current complex. All evaluation comparisons are natural for these actual transposes.

For a recovered positive integer n , $b_n(z) = z^n$ is proper. Hence test pullback preserves compact supports and its transpose defines

$$(b_{n*} T)(\phi) = T(b_n^* \phi). \quad (\text{CSD9.1})$$

After including the transposed coefficient action, this is the proper current pushforward $b_{n*} \mathcal{D}_c \mathcal{F} \rightarrow \mathcal{D}_c \mathcal{F}$, adjoint to primal pullback. It is not pullback on the dual object and is not an inverse root map.

The primal coefficient action is

$$T_a b(u) = b(u/a), \quad \rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \quad \rho_-(a)(h, d_0, d_1) = (ah(a \cdot), ad_0, d_1). \quad (\text{CSD9.2})$$

The current transpose uses T'_n and $\rho_\pm(n)'$. Under (CSD7.3) its complete finite-model action is

$$(nT'_n, T'_n, \rho_+(n)' \oplus \rho_-(n)') \quad \text{in degrees } -2, -1, 0. \quad (\text{CSD9.3})$$

For a direct check, $b_{n*}c_\lambda = nc_\lambda$, $b_{n*}\vartheta = \vartheta$, and $b_{n*}\delta_p = \delta_p$. On the punctured receiver the n inverse branches each contribute $1/n$ of the angular form to its trace. The locally integrable current equality extends across the poles. These calculations verify the precise current/global comparison. The broader geometric trace with other coefficient conventions is a separate receiver map, not identified with this one by terminology.

The specified mirrors and their sphere orientation degrees are

$$w(z) = -1/z, \quad \varepsilon_w = +1; \quad \alpha(z) = -1/\bar{z}, \quad \varepsilon_\alpha = -1. \quad (\text{CSD9.4})$$

Both transpose the pole swap and use R' . For a current represented by a form, proper pushforward acts as $\varepsilon_g(g^{-1})^*$ because the integration orientation is fixed. Thus

$$g_*c_\lambda = \varepsilon_g c_\lambda, \quad g_*\vartheta = -\vartheta \quad (g = w, \alpha), \quad (\text{CSD9.5})$$

using $w^*\vartheta = -\vartheta$, $\alpha^*\vartheta = +\vartheta$. The global dual actions are exactly

$$(\varepsilon_g R', -R', \text{dual pole swap}) \quad \text{in degrees } -2, -1, 0. \quad (\text{CSD9.6})$$

At pole stalks each angular current uses its own positive coordinate. Degree minus one is $+R'$ with pole swap, while degree minus two is $\varepsilon_g R'$ with swap. Hence (CSD8.1) intertwines the mirrors with all CDI7 signs.

The coefficient contragredient giving an invariant pairing is different. On the already constructed cohomological representation extended to $a > 0$, it is

$$(a^{-1}(T_{a^{-1}})', (T_{a^{-1}})', \rho_+(a^{-1})' \oplus \rho_-(a^{-1})'). \quad (\text{CSD9.7})$$

It satisfies $\langle B_a^\vee \lambda, B_a x \rangle = \langle \lambda, x \rangle$. This is CDI6's inverse-parameter coefficient operation. It is not asserted to arise from a holomorphic $z^{1/a}$ and is not substituted for (CSD9.1).

There is now also a separately constructed weighted trace on the primal sheaf. GTR2 uses the sum over all n interior sheets and n times the identity on each ramified pole space, including both endpoint coordinates. It proves $\text{Tr}_n b_n^* = n \cdot 1$ on that sheaf. GTR3–GTR4 prove its global degree factors $(n, n, 1)$, and, after composing with the inverse coefficient action, the representative

$$\mathbf{U}_n = (n\rho(n^{-1}), nT_{n^{-1}}, T_{n^{-1}}), \quad \mathbf{U}_n \mathbf{B}_n = \mathbf{B}_n \mathbf{U}_n = n \cdot 1 \quad (\text{CSD9.8})$$

in primal degrees $0, 1, 2$, where $\rho = \rho_+ \oplus \rho_-$. Its transpose on the full coefficient dual consequently is

$$\mathbf{U}'_n = (T'_{n^{-1}}, nT'_{n^{-1}}, n\rho(n^{-1})') = n(\mathbf{B}_n^{-1})' \quad (\text{CSD9.9})$$

in degrees $-2, -1, 0$. These are the induced maps and the explicitly constructed coefficient representative from GTR, not a claim that a chain lift to currents is uniquely determined by its cohomology. In particular the middle operator $nT'_{n^{-1}}$ in (CSD9.9) is different from T'_n in the current pushforward (CSD9.3). GTR5 proves that the original reflected residue map intertwines the former operator. The current support-duality comparison keeps these maps distinct and retains the ramification factor.

CSD10. Topologies and the proved extent of duality

Every current and functional is continuous for the actual Fréchet and compact-test topologies. The comparisons use explicit integrations, cutoffs and continuous finite operations. Exactness of continuous dualization for arbitrary locally convex complexes has not been assumed.

The weak-dual cohomology topologies are precisely CDI3's. The annihilator $J^\perp \subset A'$ has the weak topology of Q' : evaluation at a is evaluation at $q(a)$, and every quotient vector has a representative. The actual continuous decompositions $V_p = E_p \oplus J$ and $V_+ \oplus V_- = Z \oplus J$ transpose to finite decompositions, identifying the degree-zero quotient topologies. The current homotopies and comparison maps preserve those identifications. All these pairings are separated and involve the entire continuous dual.

For strong duals every displayed map remains continuous. The induced topology on $Q' \subset A'_\beta$ initially has the seminorm family

$$\mu \mapsto \sup_{a \in B} |\mu(q(a))|, \quad B \subset A \text{ bounded.} \quad (\text{CSD10.1})$$

The accepted SDT2–SDT5 proof now identifies this family with the full strong topology of Q' . It proves compact closure of every bounded subset of Q using the original two-sided weighted derivative bounds, then constructs a compact lift through q as a uniformly convergent series of finite choices. Intersecting that compact lift with the inverse image of a bounded subset $B \subset Q$ gives a bounded set $\widetilde{B} \subset A$ with $q(\widetilde{B}) = B$. Hence

$$\sup_{y \in B} |\mu(y)| = \sup_{a \in \widetilde{B}} |q' \mu(a)|, \quad Q'_\beta \xrightarrow[q']{\sim} J^\perp \subset A'_\beta. \quad (\text{CSD10.2})$$

Conversely, continuous q takes bounded subsets of A to bounded subsets of Q , giving the other topology inclusion.

The finite source splittings used above give the strong topology on the degree-zero groups, and the fixed-test maps (CSD5.7) and (CSD7.8) identify the current cohomology topologies with those of the finite coefficient complexes. Thus the global identifications are strong topological isomorphisms with $A'_\beta, Q'_\beta, Z'_\beta$. The pole stalk groups, equipped with the stable small-disk cohomology topology specified in CSD5, are strongly topologically isomorphic to $A'_\beta, Q'_\beta, E'_{p\beta}$. The supported comparison (CSD6.4) uses continuous finite operations on the local restriction cone and gives $V'_{p\beta}$ with the analogous stable local topology. This imports exactly SDT's proved bounded-lift result; it does not assume or assert strong openness of the separate restriction $A'_\beta \rightarrow J'_\beta$.

Ordinary algebraic sheaf Hom is not invoked to turn these continuous duals into algebraic duals. The exact support-dual object established here is (CSD4.1), with test-functional characterization (CSD4.3) and its proved local/global comparisons.

CSD11. Original zeta and every coefficient contribution

Neither A nor its original source J has changed. Retain

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{CSD11.1})$$

For $f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$,

$$\Theta \Sigma f_0(s) = F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{CSD11.2})$$

The endpoint values $F_0(0) = F_0(1) = 1/8$ and all trivial-zero comparisons remain

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{CSD11.3})$$

The complete inverse and these contributions are proved in SSI/ESI and retained in CSP11/CDW10.

For $\mathcal{B}$, the entire functions rapidly decreasing on every bounded real strip, let

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho, 0 \leq j < m_\rho\}.$$

The proved map is the topological isomorphism $\kappa : Q \rightarrow Q = \mathcal{B}/\mathcal{J}$, $\kappa([a]) = [\Theta a]$, with $\Theta J = \mathcal{J}$. Its actual continuous transpose is

$$\kappa' : Q' \rightarrow Q', \quad (\kappa' \lambda)([a]) = \lambda([\Theta a]). \quad (\text{CSD11.4})$$

Thus the factor $1/2$ has not disappeared in duality. Every jet $\delta_{\rho,j}([a]) = (\Theta a)^{(j)}(\rho)$ is a continuous element of the Q' in CSD5 and CSD7. Its entire transpose matrix is

$$T'_a \delta_{\rho,j} = a^\rho \sum_{\ell=0}^j \binom{j}{\ell} (\log a)^{j-\ell} \delta_{\rho,\ell}. \quad (\text{CSD11.5})$$

For the normal contragredient the scalar is $a^{-\rho-1}$ and each logarithm is $-\log a$, as in CDI6.8. Every multiplicity is retained without presuming a location inside the known open critical strip.

The notation in (CSD11.5) uses raw derivatives. GTR and RD also use Taylor-coefficient functionals. Their exact comparison is

$$\delta_{\rho,j}^{\text{raw}} = j! \delta_{\rho,j}^{\text{Taylor}}, \quad \delta_{\rho,j}^{\text{Taylor}}([a]) = \frac{(\Theta a)^{(j)}(\rho)}{j!}. \quad (\text{CSD11.6})$$

In that latter basis the actual transfer transpose from (CSD9.9) is

$$\mathcal{U}'_n \delta_{\rho,j}^{\text{Taylor}} = n^{1-\rho} \sum_{h=0}^j \frac{(-\log n)^{j-h}}{(j-h)!} \delta_{\rho,h}^{\text{Taylor}}. \quad (\text{CSD11.7})$$

Multiplying by $j!$ gives the raw-derivative formula with $\binom{j}{h}$. Thus the factorials, inverse parameter and ramification factor are all explicit in the comparison to GTR.

This construction does not promote the finite-residue identification to a topological self-duality of all Q , replace J by Q , or infer positivity of a Hermitian form from separated evaluation.

CSD12. Constructed support-duality result

The actual sheaf complex (CSD4.1) is the continuous current dual of the compact-support resolution. Its Gysin and endpoint arrow is $(\delta, -r')$. The local angular current and restriction fibre cone prove

$$i_p^* \mathcal{D}_c \mathcal{F} \simeq (i_p^! \mathcal{F})^\vee, \quad i_p^! \mathcal{D}_c \mathcal{F} \simeq (i_p^* \mathcal{F})^\vee.$$

The global chain map (CSD7.3) realizes CDI's entire continuous dual of the original sphere complex. Its restrictions give the full diagonal and anti-diagonal support maps. Proper cover transposes and both actual mirrors retain their degree and orientation factors.

This proves continuous support duality for the specified receiver. It assigns no operation to τ , introduces no finite-dimensional coefficient replacement, and asserts no RH or Deligne-purity conclusion.

The actual algebraic Verdier dual and its continuous coefficient comparison

Independent mathematical derivation, 24 September 2026. Proof locators VSD0–VSD14.

VSD0. Input, categories, and prerequisites

This calculation takes place after the complete arithmetic reconstruction. The user's notation $Z_0, Z_1, \dots, \tau$ is unchanged. There is no addition, numerical coordinate, distance, or parity operation on τ here. The coefficient spaces and their scalar operations are the already constructed receiving objects of CSP and CDI. Dualization is an operation on those coefficients, not a new construction of the counting unit or of selected primes.

The corpus rule argument_reconstruction/CORPUS_AND_OPERATION_RULE.md was read. The complete coefficient-dual proof CDI0–CDI12 and complete original-zeta residue proof RD0–RD9 were read. CSP0–CSP7, including the inverse-image sheaf, finite complexes, local orientations and support maps, were read in full. CDW9–CDW10's positive-integration and original-zeta comparisons were also read. This is an independent calculation of their sheaf-duality receiver, not a claimed new reading of the entire original Connes–Consani or Deligne papers.

Let $Y = \mathbb{P}^1(\mathbb{C})$ have its usual topology and positive complex orientation. Write $D = \{p_+, p_-\} = \{0, \infty\}$, $i : D \hookrightarrow Y$, and $j : U = Y \setminus D \hookrightarrow Y$. Retain

$$A, \quad J = \Sigma S \subset A, \quad Q = A/J, \quad V_p = S \oplus E_p, \quad E_p = \mathbb{C}^2, \quad r_p : V_p \rightarrow A. \quad (\text{VSD0.1})$$

Here $r_+ = \Sigma$, $r_- = R\Sigma = \Sigma^\wedge$ on the source summands, and both vanish on the stated endpoint summands. SSI and CDI1 prove that J is closed, $\Sigma : S \rightarrow J$ is a topological isomorphism, and continuous sections $s_p : J \rightarrow V_p$ satisfy

$$r_p s_p = 1_J, \quad V_p \simeq J \oplus E_p. \quad (\text{VSD0.2})$$

These coefficient spaces are Hausdorff Fréchet spaces. Put

$$W = V_+ \oplus V_-, \quad d(v_+, v_-) = r_+ v_+ - r_- v_-, \quad Z = \ker d \simeq J \oplus E_+ \oplus E_-. \quad (\text{VSD0.3})$$

The actual sheaf is

$$\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_* W, \quad (\text{VSD0.4})$$

with exact presentation

$$0 \rightarrow \mathcal{F} \rightarrow \underline{A}_Y \oplus i_* W \xrightarrow{\epsilon - r} i_*(A \oplus A) \rightarrow 0. \quad (\text{VSD0.5})$$

The two output components are $a|_{p_+} - r_+ v_+$ and $a|_{p_-} - r_- v_-$. Surjectivity is stalkwise using the constant-sheaf coordinate at the chosen pole.

Our ordinary sheaf category is the abelian category of sheaves of complex vector spaces. It imposes no continuity on coefficient linear maps. Distinguish

$$E^* = \text{Hom}_{\mathbb{C}}(E, \mathbb{C}), \quad E' = \text{Hom}_{\text{cont}, \mathbb{C}}(E, \mathbb{C}), \quad E^\dagger = E^*/E'. \quad (\text{VSD0.6})$$

The first is the full algebraic dual, the second the continuous dual with weak topology when needed, and the third initially only a vector space. No finite-dimensional or reflexive hypothesis is inserted.

Complexes are cohomological: $(C[k])^n = C^{n+k}$, so $\underline{E}[2]$ has its coefficient in degree -2 . Our Hom differential and cone differential are

$$d_{C^*}^n \lambda = (-1)^{n+1} \lambda d_C^{-n-1}, \quad d_{\text{Cone}(f)}(b, a) = (d_B b + f(a), -d_A a). \quad (\text{VSD0.7})$$

VSD1. Constant coefficients of arbitrary dimension

The dualizing complex on the oriented real two-manifold Y is

$$\omega_Y = \mathbb{C}_Y[2]. \quad (\text{VSD1.1})$$

Its orientation is fixed by integration of a compactly supported two-form with positive integral one in each positive complex coordinate. The local compact-support de Rham calculation on an oriented disk gives one coefficient in degree two and zero in the other degrees; extension by zero and integration give the trace map. These local orientation identifications glue to the displayed dualizing complex. Neither pole orientation is reversed.

Define ordinary Verdier duality by

$$\mathbb{D}_Y K = R\mathcal{H}om_Y(K, \omega_Y). \quad (\text{VSD1.2})$$

For every vector space E , including arbitrary infinite dimension,

$$\boxed{\mathbb{D}_Y E_Y \simeq E_Y^*[2]}. \quad (\text{VSD1.3})$$

To verify the coefficient claim without finite rank, on any open disk B constant-sheaf adjunction gives

$$\text{Hom}_{\text{Sh}(B)}(\underline{E}_B, K) = \text{Hom}_{\mathbb{C}}(E, \Gamma(B, K)). \quad (\text{VSD1.4})$$

A sheaf map is determined by the images of constant sections; conversely a linear map into sections extends to locally constant sections by gluing on their regions of constancy. Resolve K by injective sheaves. Vector-space Hom from E is exact: choose a basis and lift each basis image through any surjection. The Hom complex therefore has cohomology

$$\text{Ext}_{\text{Sh}(B)}^q(\underline{E}_B, K) = \text{Hom}_{\mathbb{C}}(E, H^q(B, K)). \quad (\text{VSD1.5})$$

For constant $K = \mathbb{C}_B$, the disk contraction gives $H^0 = \mathbb{C}$, $H^{q>0} = 0$. Internal derived-Hom stalks are the filtered colimit of these local Ext groups; higher internal Ext stalks vanish. In degree zero the internal Hom is constant with coefficient E^* : on a connected disk each scalar image of a constant section is constant, defining a single linear functional on all of E . Thus (VSD1.3) follows.

For a point p and every vector space B ,

$$i_p^! \underline{B}_Y = B[-2], \quad i_p^! \underline{B}_Y[2] = B. \quad (\text{VSD1.6})$$

Indeed the disk/punctured-disk relative complex has identity restriction in degree zero, while the punctured disk has its independent degree-one angular coefficient B . Its fibre is the local fundamental class in degree two. This is the coefficient version of CSP6 and uses no dimension bound.

VSD2. Pole duals and the signed Gysin map

For a skyscraper coefficient,

$$\boxed{\mathbb{D}_Y i_{p*} E = i_{p*} E^*}. \quad (\text{VSD2.1})$$

Closed-inclusion adjunction identifies derived Hom from $i_{p*} E$ into ω_Y with the pushforward of derived Hom from E into $i_p^! \omega_Y = \mathbb{C}$. At the point vector-space Hom is exact, so no higher Ext remains. This proves (VSD2.1) in arbitrary dimension.

For any vector space B , define

$$g_{p,B} : i_{p*} B \longrightarrow \underline{B}_Y[2] \quad (\text{VSD2.2})$$

as the adjunction counit corresponding to the identity of $B = i_p^! \underline{B}_Y[2]$, where the identification is positive local complex orientation. Derived adjunction specifies this fully:

$$\text{Hom}_{D(Y)}(i_{p*} B, \underline{B}_Y[2]) = \text{Hom}_{\mathbb{C}}(B, B). \quad (\text{VSD2.3})$$

The dual of restriction $\underline{E}_Y \rightarrow i_{p*}E$ is g_{p,E^*} with positive sign. To check it, pair e with λ ; the corresponding local compactly supported two-form has integral $\lambda(e)$, using a positive fundamental form. Extension by zero to the sphere preserves that integral, so the global scalar is $+\lambda(e)$.

The angular connecting sign is separate. For $\chi = 1$ near the pole and zero outside a disk,

$$-d\chi \wedge \frac{d\theta_p}{2\pi}, \quad \int -d\chi \wedge \frac{d\theta_p}{2\pi} = 1 \quad (\text{VSD2.4})$$

is positive local orientation. In the CSP relative cone the boundary of the positive angular class is the negative of this positive-integral class. Gysin then carries the positive fundamental class positively to the global one. These signs belong to different maps.

For the two poles,

$$g_B = g_{p_+,B} + g_{p_-,B} : i_*(B \oplus B) \rightarrow \underline{B}_Y[2] \quad (\text{VSD2.5})$$

induces

$$B \oplus B \rightarrow H^0(Y, \underline{B}[2]) = B, \quad (b_+, b_-) \mapsto b_+ + b_-. \quad (\text{VSD2.6})$$

Both local coordinates $z, w = 1/z$ are complex oriented. The identity $d \arg w = -d \arg z$ belongs to the annular comparison, not to the Gysin sum.

VSD3. The actual algebraic Verdier triangle

Dualizing (VSD0.5) with those explicitly computed maps gives

$$i_*((A^*)^2) \xrightarrow{(g_{A^*}, -r^*)} \underline{A}_Y^*[2] \oplus i_*W^* \rightarrow \mathbb{D}_Y\mathcal{F} \rightarrow i_*((A^*)^2)[1]. \quad (\text{VSD3.1})$$

The restriction component is

$$(\lambda_+, \lambda_-) \mapsto (-r_+^* \lambda_+, -r_-^* \lambda_-). \quad (\text{VSD3.2})$$

It retains the minus in $\epsilon - r$; Gysin retains the positive sign of ϵ . Thus, representing the Gysin maps on a resolution when forming a cone,

$$\boxed{\mathbb{D}_Y\mathcal{F} \simeq \text{Cone} \left(i_*((A^*)^2) \xrightarrow{(g_{A^*}, -r^*)} \underline{A}_Y^*[2] \oplus i_*W^* \right)}. \quad (\text{VSD3.3})$$

This constructs the sheaf dual. It does not declare that a continuous-dual complex is already the ordinary Verdier dual.

VSD4. Every stalk and costalk

On the interior,

$$j^*\mathbb{D}_Y\mathcal{F} = \underline{A}_U^*[2]. \quad (\text{VSD4.1})$$

At a pole the derived stalk of Gysin is zero as a derived vector-space map from degree zero to degree minus two: vector-space Ext in degree two is zero. In the local cone, multiplying the degree-minus-one source coordinate by minus one turns $-r_p^*$ into $+r_p^*$. The stalk model is

$$A^* \text{ in degree } -2, \quad A^* \xrightarrow{r_p^*} V_p^* \text{ in degrees } -1, 0. \quad (\text{VSD4.2})$$

It is the transpose of CSP's positive-integration costalk $L_p = [V_p \xrightarrow{r_p} A \rightarrow^0 A]$. Vanishing of the stalk Gysin map does not make the global Gysin morphism vanish.

The algebraic dual restriction sequence is exact:

$$0 \rightarrow Q^* \xrightarrow{q^*} A^* \rightarrow J^* \rightarrow 0. \quad (\text{VSD4.3})$$

Surjectivity follows by extending a basis of J to one of A . Since $V_p = J \oplus E_p$, the image of r_p^* is precisely the annihilator of E_p . Thus

$$\mathcal{H}^{-2}(\mathbb{D}\mathcal{F})_p = A^*, \quad \mathcal{H}^{-1}(\mathbb{D}\mathcal{F})_p = Q^*, \quad \mathcal{H}^0(\mathbb{D}\mathcal{F})_p = E_p^*. \quad (\text{VSD4.4})$$

All other pole cohomology stalks vanish. The full cohomology sheaves are

$$\mathcal{H}^{-2}(\mathbb{D}\mathcal{F}) = \underline{A}_Y^*, \quad \mathcal{H}^{-1}(\mathbb{D}\mathcal{F}) = i_*(Q^* \oplus Q^*), \quad \mathcal{H}^0(\mathbb{D}\mathcal{F}) = i_*(E_+^* \oplus E_-^*). \quad (\text{VSD4.5})$$

Degree-minus-two generization is identity. This list does not replace the cone and its Gysin extension class.

Applying $i_p^!$ to (VSD3.1), the map becomes

$$A^* \rightarrow A^* \oplus V_p^*, \quad \lambda \mapsto (\lambda, -r_p^* \lambda). \quad (\text{VSD4.6})$$

Its cone contracts to $V_p^*[0]$. Explicitly the projection

$$(a, \phi) \mapsto \phi + r_p^* a \quad (\text{VSD4.7})$$

is onto with kernel precisely (VSD4.6). Therefore

$$\boxed{i_p^! \mathbb{D}_Y \mathcal{F} \simeq V_p^*[0]}. \quad (\text{VSD4.8})$$

At an interior point x , $i_x^! \mathbb{D}\mathcal{F} = A^*[0]$. Both stalk/costalk interchange statements are established by these actual local complexes.

VSD5. Global cohomology and explicit contraction

Derived global sections of (VSD3.3) have finite model

$$\begin{aligned} K_*^{-2} &= A^*, \quad K_*^{-1} = (A^*)^2, \quad K_*^0 = A^* \oplus V_+^* \oplus V_-^*, \\ d^{-2} &= 0, \quad d^{-1}(\lambda_+, \lambda_-) = (\lambda_+ + \lambda_-, -r_+^* \lambda_+, -r_-^* \lambda_-). \end{aligned} \quad (\text{VSD5.1})$$

The degree-minus-two coefficient comes from sphere degree zero of the unshifted constant sheaf. The degree-zero A^* comes from its positive sphere degree-two class; its two Gysin contributions are the sum in (VSD2.6). This is precisely the algebraic Hom complex of CSP4.1.

It contracts onto

$$D_*^{-2} = A^*, \quad D_*^{-1} = A^*, \quad D_*^0 = V_+^* \oplus V_-^*, \quad d^{-2} = 0, \quad d^{-1} \lambda = (r_+^* \lambda, -r_-^* \lambda). \quad (\text{VSD5.2})$$

The inclusion $u : D_* \rightarrow K_*$ and retraction $v : K_* \rightarrow D_*$ are

$$u^{-2} = 1, \quad u^{-1} \lambda = (-\lambda, \lambda), \quad u^0(\phi_+, \phi_-) = (0, \phi_+, \phi_-), \quad (\text{VSD5.3})$$

$$v^{-2} = 1, \quad v^{-1}(\lambda_+, \lambda_-) = -\lambda_+, \quad v^0(a, \phi_+, \phi_-) = (\phi_+, \phi_- + r_-^* a). \quad (\text{VSD5.4})$$

Substitution shows they are chain maps and $vu = 1$. A homotopy for $1 - uv$ is

$$h^0(a, \phi_+, \phi_-) = (0, a) \quad (\text{VSD5.5})$$

with all other components zero: dh in degree zero is $(a, 0, -r_-^* a)$, and hd in degree minus one is $(0, \lambda_+ + \lambda_-)$, exactly $1 - uv$.

Since $W \simeq Z \oplus J$ by CDI1's explicit section, restriction and extension of algebraic functionals give

$$\boxed{H^{-2}(Y, \mathbb{D}\mathcal{F}) = A^*, \quad H^{-1}(Y, \mathbb{D}\mathcal{F}) = Q^*, \quad H^0(Y, \mathbb{D}\mathcal{F}) = Z^*}. \quad (\text{VSD5.6})$$

Their pairings with CSP's A, Q, Z are ordinary evaluation on each full algebraic dual. No bidual surjectivity is asserted.

VSD6. The complete transposed support maps

The source support map has components

$$k^0 = 1_W, \quad k^1(b_+, b_-) = b_+ - b_-, \quad k^2(c_+, c_-) = c_+ + c_-. \quad (\text{VSD6.1})$$

Under the exact local and global dual models, restriction of the dual global complex to the two dual stalk models is

$$(k^*)^0 = 1_{W^*}, \quad (k^*)^{-1}\lambda = (\lambda, -\lambda), \quad (k^*)^{-2}\lambda = (\lambda, \lambda). \quad (\text{VSD6.2})$$

The pole differential sends (λ_+, λ_-) to $(r_+^*\lambda_+, r_-^*\lambda_-)$, so chain compatibility follows directly. The cohomology maps are diagonal on A^* , anti-diagonal on Q^* , and endpoint restriction on Z^* .

The complete nonzero part of the algebraically dual localization row is

$$\begin{aligned} 0 \rightarrow A^* &\xrightarrow{\lambda \mapsto (\lambda, \lambda)} (A^*)^2 \xrightarrow{(\lambda_+, \lambda_-) \mapsto -\lambda_+ + \lambda_-} A^* \xrightarrow{0} Q^* \\ &\xrightarrow{\mu \mapsto (\mu, -\mu)} (Q^*)^2 \xrightarrow{(\mu_+, \mu_-) \mapsto q^*(\mu_+ + \mu_-)} A^* \xrightarrow{r^*} Z^* \xrightarrow{\text{res}} E_+^* \oplus E_-^* \rightarrow 0. \end{aligned} \quad (\text{VSD6.3})$$

Here $r : Z \rightarrow A$ is common restriction and $r^*\lambda = (\lambda|_J, 0, 0)$. Exactness follows from the displayed sum/difference kernels, $\ker(A^* \rightarrow J^*) = q^*Q^*$, extension onto J^* , and the split endpoint coordinates. The annular arrow is the transpose of $a \mapsto (-a, a)$; it is not the transpose of the Gysin sum.

There is also the support-to-global map for the dual sheaf itself. Its two pole costalks, computed in (VSD4.8), are $V_+^*[0] \oplus V_-^*[0]$. On the global model (VSD5.2) their inclusion is the identity into degree zero W^* . On cohomology this is restriction of a functional on W to its subspace Z . Thus

$$(\phi_+, \phi_-) \mapsto (s_+^*\phi_+ + s_-^*\phi_-, \phi_+|_{E_+}, \phi_-|_{E_-}) \quad \text{in } Z^* \simeq J^* \oplus E_+^* \oplus E_-^*. \quad (\text{VSD6.4})$$

Both pole contributions have a positive sign. Its kernel is precisely $\{(r_+^*\lambda, -r_-^*\lambda) : \lambda \in A^*\}$: it consists of the functionals annihilating Z , which factor through $d : W \rightarrow J$, and every such functional on J extends to A . Surjectivity follows from the explicit splitting $W \simeq Z \oplus J$. Hence (VSD6.4) is also completely calculated; it is a different map from the dual-global-to-dual-stalk restriction in (VSD6.2).

VSD7. A continuous subcomparison as an actual sheaf object

The continuous candidate is constructed in the same ordinary derived sheaf category, with the specified continuous-dual coefficient subspaces:

$$\mathcal{D}_c\mathcal{F} = \text{Cone} \left(i_*((A')^2) \xrightarrow{(g_{A'}, -r')} \underline{A}'_Y[2] \oplus i_*W' \right). \quad (\text{VSD7.1})$$

This defines actual constant and skyscraper sheaves, using the constructed Gysin map and continuous transposes. It is not asserted to be a general dualizing functor on all topological sheaves. It supplies an actual receiving object for this particular $\mathcal{F}$.

The inclusions $E' \hookrightarrow E^*$ commute with all transposes. Gysin is coefficient-natural by its identity characterization (VSD2.3). They therefore produce

$$\boxed{c : \mathcal{D}_c\mathcal{F} \longrightarrow \mathbb{D}_Y\mathcal{F}.} \quad (\text{VSD7.2})$$

The functorial resolution of the coefficient-natural Gysin diagrams produces a map of ordinary cone complexes, so (VSD7.2) is not merely a comparison between cohomology lists.

The preceding finite calculations hold with primes instead of stars. Continuous Hahn–Banach on the closed subspace $J \subset A$ supplies the required surjectivity of restriction; CDI1 proves that step with these actual topologies. Consequently

$$\begin{array}{c|ccc} & -2 & -1 & 0 \\ \hline H^\bullet(Y, \mathcal{D}_c\mathcal{F}) & A' & Q' & Z' \\ \mathcal{H}^\bullet(\mathcal{D}_c\mathcal{F})_p & A' & Q' & E'_p \end{array} \quad (\text{VSD7.3})$$

and $i_p^! \mathcal{D}_c \mathcal{F} = V_p'[0]$; the interior restriction is $\underline{A}'[2]$. The global finite complex is exactly CDI2.2, the pole complex CDI2.6, and support maps are CDI4–CDI5. Each cohomology map of c is the inclusion of continuous into full algebraic dual of the corresponding original group. With weak coefficient topologies the finite-model cohomology topologies are the exact weak topologies proved in CDI3. The ordinary sheaf category records the underlying vector spaces.

Thus the continuous coefficient model now has an actual sheaf receiver, with a proved map to ordinary Verdier duality. The two duals have not been identified without a map.

VSD8. The comparison cone and every one of its groups

Write

$$\mathcal{E} = \text{Cone}(c). \quad (\text{VSD8.1})$$

The exact comparison object is

$$\mathcal{E} \simeq \text{Cone} \left(i_*((A^\dagger)^2) \xrightarrow{(g_{A^\dagger}, -r^\dagger)} \underline{A}_Y^\dagger[2] \oplus i_* W^\dagger \right). \quad (\text{VSD8.2})$$

For each coefficient, $0 \rightarrow E' \rightarrow E^* \rightarrow E^\dagger \rightarrow 0$ induces exact sequences of constant sheaves and skyscrapers, since exactness is stalkwise. The coefficient-natural Gysin square gives a morphism of triangles. Its cone is (VSD8.2): one may compute it by functorial resolutions and the two cone differentials, or by the resulting exact quotient diagram. No topology on $E^\dagger$ is used here.

The continuous and algebraic restriction rows form the exact diagram

$$\begin{array}{ccccccccc} 0 & \rightarrow & Q' & \rightarrow & A' & \rightarrow & J' & \rightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \rightarrow & Q^* & \rightarrow & A^* & \rightarrow & J^* & \rightarrow & 0. \end{array} \quad (\text{VSD8.3})$$

All vertical maps are injections. The quotient row is therefore

$$0 \rightarrow Q^\dagger \rightarrow A^\dagger \rightarrow J^\dagger \rightarrow 0. \quad (\text{VSD8.4})$$

To verify its middle kernel directly, an algebraic functional on A whose restriction to J is continuous differs, by a continuous Hahn–Banach extension of that restriction, from a functional factoring algebraically through Q . Surjectivity on the right follows by algebraic extension from J .

The endpoint spaces are finite dimensional, so all their functionals are continuous. The actual splittings in VSD0 give

$$V_p^\dagger \simeq J^\dagger, \quad Z^\dagger \simeq J^\dagger, \quad E_p^\dagger = 0. \quad (\text{VSD8.5})$$

The induced $r_p^\dagger : A^\dagger \rightarrow V_p^\dagger$ is onto and has kernel $Q^\dagger$. The finite complexes now give

$$\begin{array}{lll} \mathcal{H}^{-2}(\mathcal{E}) = \underline{A}_Y^\dagger, & \mathcal{H}^{-1}(\mathcal{E}) = i_*(Q^\dagger \oplus Q^\dagger), & \mathcal{H}^0(\mathcal{E}) = 0, \\ i_p^! \mathcal{E} = V_p^\dagger[0] \simeq J^\dagger[0], & & \\ H^{-2}(Y, \mathcal{E}) = A^\dagger, & H^{-1}(Y, \mathcal{E}) = Q^\dagger, & H^0(Y, \mathcal{E}) = Z^\dagger \simeq J^\dagger. \end{array} \quad (\text{VSD8.6})$$

In particular a zero degree-zero cohomology sheaf does not force zero degree-zero global cohomology or zero pole costalk. The Gysin extension in (VSD8.2) retains this information.

All support maps are (VSD6.2)–(VSD6.3) on the dagger coefficients, with finite endpoint terms zero. This comparison defect has specified support, maps and computed cohomology; it is not an unnamed failure of an analogy.

VSD9. Nonzero defect for the actual infinite coefficients

Every infinite-dimensional Hausdorff Fréchet space E has discontinuous algebraic functionals. For a defining seminorm sequence $p_1, p_2, \dots$, choose independent x_n and scale them by nonzero constants to obtain independent v_n with

$$\max_{1 \leq j \leq n} p_j(v_n) \leq 2^{-n}. \quad (\text{VSD9.1})$$

Then $v_n \rightarrow 0$. Assign $\lambda(v_n) = 1$ on their span and extend the independent family to an algebraic basis of E , assigning zero to the additional basis vectors. This defines an algebraic functional which is not continuous. Hence $E^\dagger \neq 0$.

The actual A is infinite dimensional: smooth functions with pairwise disjoint compact supports in $\mathbb{R}_{>0}$ lie in A and are independent. The actual S is infinite dimensional too. In pairwise disjoint positive intervals choose nonzero compactly supported smooth functions of integral zero, for example derivatives of interior bump functions, and extend them evenly. They vanish at zero, retain integral zero on the real line, and are independent members of S . The proved topological isomorphism $\Sigma : S \rightarrow J$ gives

$$A^\dagger \neq 0, \quad J^\dagger \neq 0. \quad (\text{VSD9.2})$$

Consequently c is not a quasi-isomorphism: its open-stratum degree-minus-two cokernel is the nonzero $A^\dagger$. Its pole costalk comparison also has nonzero quotient $J^\dagger$. This proves the precise difference between the two defined dual constructions; it says nothing against the original support τ or its lifting programme.

VSD10. Weak completion and the original residue family

For every Hausdorff locally convex E ,

$$\overline{E'}^{\sigma(E^*, E)} = E^*. \quad (\text{VSD10.1})$$

Indeed fix $\lambda \in E^*$ and finitely many $e_1, \dots, e_m$. On their finite-dimensional span L , the restriction of λ is continuous, because the Hausdorff induced topology on L is its ordinary finite-dimensional topology. It is bounded by a continuous seminorm induced from E . Hahn–Banach extends it continuously to all of E , agreeing exactly on the prescribed vectors. Every pointwise neighborhood of λ therefore meets E' .

The full algebraic dual is complete in pointwise convergence. It is the closed subspace of the complete product $\mathbb{C}^E$ cut out by every linearity equation

$$f(x + y) = f(x) + f(y), \quad f(cx) = cf(x). \quad (\text{VSD10.2})$$

Its topology on E' is exactly the weak dual topology. Thus

the Hausdorff completion of E'_σ is E^* with its pointwise topology.

(\text{VSD10.3})

Weak completion and weak closure inside a specified continuous dual are different operations; neither is substituted for the other.

The quotient topology on E^*/E' from $\sigma(E^*, E)$ is indiscrete. To see it, let a nonempty open saturated set contain $x + U$, with U an open neighborhood of zero. For any y , density makes $y - x - U$ meet E' , so $y \in x + U + E'$, and saturation places every y in the set. Its only open subsets are therefore empty and whole. The nonzero algebraic quotients in VSD9 are not Hausdorff quotients for this particular topology. They remain the exact ordinary-sheaf coefficients in VSD8.

RD's finite-support original-zeta residue family $\mathcal{D}$ satisfies both

$$\overline{\mathcal{D}}^{\sigma(Q', Q)} = Q', \quad \overline{\mathcal{D}}^{\sigma(Q^*, Q)} = Q^*. \quad (\text{VSD10.4})$$

The first is RD4. For the second, take any algebraic λ and finitely many $x_1, \dots, x_m \in Q$. If their λ -values were outside the finite-dimensional image $\{(d(x_i))_i : d \in \mathcal{D}\}$, a separating linear relation would annihilate every full zero jet of $\sum c_i x_i$, hence that class would be zero by the definition of Q , while its λ -value would be nonzero. This contradiction proves exact interpolation on every finite set, hence the second density. These equations have different stated ambient spaces. Neither makes a discontinuous functional continuous or changes the residue formula.

VSD11. Original-zeta data and receiving morphisms

The actual coefficient comparison remains

$$\kappa : Q \longrightarrow \mathcal{Q}, \quad [a] \mapsto \left[\frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u} \right], \quad (\text{VSD11.1})$$

with $\Sigma f = 2 \sum_{n \geq 1} f(nu)$. Its exact dual square is

$$\begin{array}{ccc} \mathcal{Q}' & \hookrightarrow & \mathcal{Q}^* \\ \downarrow \kappa' & & \downarrow \kappa^* \\ \mathcal{Q}' & \hookrightarrow & \mathcal{Q}^*. \end{array} \quad (\text{VSD11.2})$$

Both vertical maps are isomorphisms, and so is the induced map on their algebraic quotients. RD's finite-support residue map therefore enters the two sheaf-dual receivers through the actual inclusion:

$$\mathcal{Q}_{\text{fin}} \xrightarrow{\kappa' \iota} H^{-1}(Y, \mathcal{D}_c \mathcal{F}) \xrightarrow{H^{-1}(c)} H^{-1}(Y, \mathbb{D} \mathcal{F}). \quad (\text{VSD11.3})$$

Its first arrow is

$$G \mapsto \left([a] \mapsto \sum_{\rho} \text{Res}_{s=\rho} \frac{\frac{1}{2} \mathcal{M}_0 a(s) G(1-s)}{\zeta(s)} ds \right). \quad (\text{VSD11.4})$$

The denominator is the original zeta, and the residues and ideal retain all zero multiplicities. The comparison does not replace it by the source transform

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{VSD11.5})$$

All of that transform's factors, endpoint values, trivial-zero derivatives and local-unit derivatives remain those proved in RD8 and CDI12. The four endpoint lines are explicitly retained in $E_+^* \oplus E_-^*$ and $E_+^* \oplus E_-^*$. Their dagger quotient vanishes because they are finite dimensional, not because their original contributions were deleted.

Every continuous coefficient operator preserves $E' \subset E^*$ under ordinary transposition and thus induces an operator on $E^\dagger$. Gysin is coefficient-natural. Thus the entire comparison triangle intertwines the already proved coefficient, cover and mirror maps, retaining their signs and factors. This does not choose a new arithmetic degree convention; the cover/trace derivation is a separate calculation. Ordinary transpose, inverse-parameter contragredient, and residue degree-character comparison remain the distinct maps proved in CDI6–CDI10.

VSD12. Actual output for the support-dual calculation

The actual $\mathcal{F}$ now has an explicit ordinary algebraic Verdier dual, including both pole Gysin maps, positive orientation, the minus on dualized restrictions, every stalk and costalk, and full global cohomology. These proofs apply to the actual infinite coefficient spaces without a finite-rank duality assumption.

The continuous-dual coefficient complexes have an actual sheaf receiver $\mathcal{D}_c \mathcal{F}$ and a proved map to ordinary Verdier duality. Its cone is the specific object in VSD8, with every coefficient group and support map computed. Its nonzero difference is retained and studied in VSD9–VSD10. This supplies the exact comparison required before using either category in the support-duality part of a Deligne-style argument.

No finite-rank étale realization, Deligne weight bound, arithmetic purity, or RH conclusion is an assumption or conclusion here. The subsequent weight calculation can use the proved cone and coefficient-category maps instead of a presumed equality of the two duals.

VSD13. The exact comparison defect admits every rational operation in the Mellin multiplication operator

The nonzero comparison object has additional algebraic structure which can be calculated globally. Let

$$\mathcal{B} = \left\{ F \text{ entire} : \sup_{|\Re s| \leq R} (1 + |\Im s|)^M |F(s)| < \infty \text{ for all } R, M \in \mathbb{N} \right\}, \quad \mathcal{J} = \Theta \mathcal{J}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J},$$

$$L : \mathcal{B} \rightarrow \mathcal{B}, \quad (LF)(s) = sF(s). \quad (\text{VSD13.1})$$

The topological isomorphism $\Theta : A \rightarrow \mathcal{B}$ and its restriction $J \rightarrow \mathcal{J}$ are the actual Mellin comparison of CDI6 and SSI, retaining $\Theta = \frac{1}{2}\mathcal{M}_0$. The operator L is continuous, preserves $\mathcal{J}$, and induces an operator on $\mathcal{Q}$. These are the domains and maps being transposed below.

For every $\mu \in \mathbb{C}$, multiplication

$$M_\mu = L - \mu : \mathcal{B} \longrightarrow \mathcal{B} \quad (\text{VSD13.2})$$

is injective and has image exactly the closed hyperplane $F(\mu) = 0$. Its inverse on that image is division by $s - \mu$. That inverse is continuous: outside the disk $|s - \mu| < 1$, the quotient is bounded by the original function; inside, the maximum principle applied to the holomorphic quotient bounds it using a circle of radius two centered at μ . Every coefficient seminorm on that circle is bounded by one of the larger-strip seminorms on $\mathcal{B}$, and the polynomial imaginary-coordinate weight is bounded on the disk. This gives a finite seminorm bound for each target seminorm. Evaluation onto $\mathbb{C}$ is surjective, since $e^{(s-\mu)^2} \in \mathcal{B}$ has value one at μ . Thus the image really has codimension one.

The same multiplication on $\mathcal{J}$ is injective with closed codimension-one image. To prove this at every μ , retain the complete function

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) \in \mathcal{J}. \quad (\text{VSD13.3})$$

Its zero divisor is the complete actual nontrivial-zero divisor, with all multiplicities, as retained in RD1 and RD8. Every $F \in \mathcal{J}$ therefore has an entire quotient F/F_0 ; no claim that this quotient belongs to $\mathcal{B}$ is needed. Define

$$\ell_\mu(F) = \begin{cases} F(\mu)/F_0(\mu), & F_0(\mu) \neq 0, \\ F^{(m)}(\mu)/F_0^{(m)}(\mu), & F_0 \text{ has a zero of order } m \text{ at } \mu. \end{cases} \quad (\text{VSD13.4})$$

This is the precise removable value $(F/F_0)(\mu)$. Cauchy's estimate makes it continuous; the nonzero denominator derivative is retained. Since $\ell_\mu(F_0) = 1$, it is onto. The image of $M_\mu : \mathcal{J} \rightarrow \mathcal{J}$ is exactly $\ker \ell_\mu$. Necessity follows by taking the quotient by F_0 . Conversely, when $\ell_\mu(F) = 0$, division by $s - \mu$ leaves the full required order of vanishing at μ and preserves every other required zero jet, so the quotient belongs to $\mathcal{J}$. Its continuity follows from the division estimate just proved in $\mathcal{B}$, restricted to these closed subspaces. This proves the entire statement, including at every multiple zero and at each endpoint or trivial-zero location where F_0 is nonzero.

Here is the general transpose calculation applied to these two actual operators. Suppose $M : E \rightarrow E$ is a continuous injective map onto a closed subspace of finite codimension, with continuous inverse on that image. Its algebraic transpose $M^* : E^* \rightarrow E^*$ is onto, by algebraically extending the functional $Mx \mapsto \lambda(x)$ from the image. Its continuous transpose $M' : E' \rightarrow E'$ is onto as well: the same functional is continuous on the image when λ is continuous, and Hahn–Banach extends it to E . Moreover

$$\ker M^* = \ker M' \quad (\text{VSD13.5})$$

as subspaces of E^* . Both are all functionals on the finite-dimensional Hausdorff quotient $E/M(E)$, and every such functional is continuous. Consequently

$$M^\dagger : E^\dagger \longrightarrow E^\dagger \quad (\text{VSD13.6})$$

is bijective. Surjectivity follows from surjectivity of M^* . For injectivity, if $M^*\lambda$ is continuous, choose a continuous ν with $M'\nu = M^*\lambda$; then $\lambda - \nu \in \ker M^* = \ker M'$, so λ itself is continuous.

Applying this proved calculation to (VSD13.2) and to its restriction to $\mathcal{I}$ yields

$$L^\dagger - \mu : \mathcal{B}^\dagger \xrightarrow{\sim} \mathcal{B}^\dagger, \quad L^\dagger - \mu : \mathcal{J}^\dagger \xrightarrow{\sim} \mathcal{J}^\dagger \quad (\mu \in \mathbb{C}). \quad (\text{VSD13.7})$$

Here $L^\dagger$ means the operator induced by the ordinary transpose L^* on the indicated dual quotient. It is not multiplication on the original primal space.

The exact dual-quotient row

$$0 \rightarrow \mathcal{Q}^\dagger \rightarrow \mathcal{B}^\dagger \rightarrow \mathcal{J}^\dagger \rightarrow 0 \quad (\text{VSD13.8})$$

commutes with these operators. It follows, by a direct diagram chase, that

$$\boxed{L^\dagger - \mu : \mathcal{Q}^\dagger \xrightarrow{\sim} \mathcal{Q}^\dagger \quad \text{for every } \mu \in \mathbb{C}.} \quad (\text{VSD13.9})$$

Injectivity follows from the middle bijection. For surjectivity, solve the equation in $\mathcal{B}^\dagger$; the image of its solution in $\mathcal{J}^\dagger$ is killed by $L^\dagger - \mu$, so it is zero by (VSD13.7). The solution therefore lies in the injected $\mathcal{Q}^\dagger$. Transport through Θ gives the same statements on $A^\dagger, J^\dagger, Q^\dagger$, and through the explicit endpoint splittings on $V_p^\dagger, Z^\dagger$.

Every nonzero polynomial P over $\mathbb{C}$ factors into a nonzero scalar times linear factors. Thus $P(L^\dagger)$ is bijective on all these defect coefficients. The usual commuting inverses define a unique extension of their $\mathbb{C}[t]$ -module structure to a vector-space structure over the rational-function field $\mathbb{C}(t)$:

$$\frac{P(t)}{Q(t)} \cdot x = P(L^\dagger) Q(L^\dagger)^{-1} x, \quad Q \neq 0. \quad (\text{VSD13.10})$$

This is independent of the fractional presentation because cross multiplication gives identical commuting operators. Every map in the dagger coefficient diagram commutes with $L^\dagger$ and hence with these inverses. Consequently the entire comparison object (VSD8.2), including its Gysin maps, can be formed on these $\mathbb{C}(t)$ -coefficient spaces.

In particular none of these coefficients has a nonzero generalized $L^\dagger$ -eigenvector of any finite order: $(L^\dagger - \mu)^m x = 0$ implies $x = 0$. The actual finite zero-jet residue vectors are in the continuous subobject, and their image in the dagger quotient is zero. This proves a global property of the precise extra coefficient sector rather than stopping at its nonzero existence.

One distinction is essential. On a discontinuous algebraic functional, one cannot move that functional through a limit merely because the original coefficient action is differentiable. The $L^\dagger$ above is the induced transpose of the explicitly defined operator (VSD13.1). This proof does not identify it with differentiation of the algebraic action T_a^* on arbitrary discontinuous functionals. Therefore the absence of generalized $L^\dagger$ -eigenvectors by itself does not classify all characters of that algebraic dilation action, does not prove a weight bound, and does not erase the comparison sector. It gives the exact rational-operator structure with the domains, maps and limitation of the inference specified.

There is an exact finite-sector comparison in both kernel and cokernel. For $E = A, J, Q$ with these transported operators, and every nonzero polynomial P , inclusion $E' \hookrightarrow E^*$ induces

$$\boxed{\ker P(L') \xrightarrow{\sim} \ker P(L^*), \quad \text{coker } P(L') \xrightarrow{\sim} \text{coker } P(L^*).} \quad (\text{VSD13.11})$$

These are algebraic kernel and cokernel identifications with their actual maps; no topological quotient assertion is inserted. For the kernel, a functional killed by $P(L^*)$ has dagger class killed by the bijection $P(L^\dagger)$, so it is continuous. For surjectivity of the cokernel map, given $\lambda \in E^*$, solve $P(L^\dagger)\bar{\mu} = \bar{\lambda}$, lift $\bar{\mu}$ to $\mu \in E^*$, and observe $\lambda - P(L^*)\mu \in E'$. For injectivity, if continuous η equals $P(L^*)\lambda$, bijectivity on the dagger quotient forces $\lambda \in E'$.

Every finite-dimensional L^* -invariant subspace $M \subset E^*$ is consequently contained in E' . Its finite-dimensional operator has a nonzero characteristic polynomial P annihilating M ; the adjugate identity for $tI - L^*|_M$, evaluated at that operator, proves the assertion. Therefore

$$\boxed{M \subset \ker P(L^*) = \ker P(L') \subset E'.} \quad (\text{VSD13.12})$$

For every μ and $m \geq 1$, the entire kernels of $(L^* - \mu)^m$ and $(L' - \mu)^m$ agree, with no dimension bound needed for that kernel equality. Thus the two categories have identical full polynomial kernels and cokernels through their stated inclusion.

Replacing L by $L + k$, for any stated scalar k , preserves this result because $P(t + k)$ is a nonzero polynomial. The retained normal-degree shift therefore leaves the finite-sector comparison valid. On Z and V_p , retain their actual continuous endpoint blocks separately; all functionals on those finite blocks are continuous, and the explicit splitting reduces the other block to the J calculation. This extends (VSD13.11)–(VSD13.12) to those specified operators without dropping endpoints.

These consequences concern the explicit multiplication operator. They do not assume differentiation of discontinuous dual functionals or classify arbitrary dilation characters. They supply the exact finite-sector comparison from the calculated defect.

VSD14. Independent check of the completed trace receiver

After VSD0–VSD13 were completed, the entire SDT0–SDT9 proof was read, together with GTR10 in full and the exact GTR4.5–GTR5.4 action formulas used there. The checked SDT file is `../tau_weight_cohomology_20260924/CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md`, SHA256 `7f1a03a6225f1dfc22b167713c0ddbb154635b06f9479f6137b229845b67aefc`. The checked GTR file is `CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md`, SHA256 `8ede942ee574e1f96fdc2b6247c8786491abe3fc6401c24f157625f8f2fb9811`. This check found no mathematical correction to GTR10.

SDT's weighted compactness estimate proves total boundedness one seminorm at a time with both original exponential weights retained. Its quotient argument makes only the needed per-seminorm lift choices; the independent compact-lift construction subsequently supplies a single compact set of lifts by its finite-choice series and uniform tails. Thus the bounded-lift conclusion used in the strong-dual embedding does not assume a continuous linear section. The stated strong topology on Q' is exactly the subspace topology under q' , because each bounded quotient set has such an exact bounded lift.

The compact absolutely convex hull argument in SDT6 identifies every continuous functional on the strong dual with evaluation at an actual coefficient vector. Its Baire and polar estimates recover the original bidual topology. SDT7 then obtains strong density of full zero jets by separation, and proves strong-dual completeness by uniform convergence on each bounded set followed by sequential continuity on the Fréchet primal space. These steps establish the precise strong completion imported by GTR10. They are compatible with VSD10: the strong and weak topologies have different completions, and neither proof interchanges them.

Here is a direct verification of the new action on that completion. Put

$$p_B(G) = \sup_{F \in B} \left| \sum_{\rho} \operatorname{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds \right|, \quad \mathbf{V}_a = aT_{a^{-1}}, \quad (\text{VSD14.1})$$

where $B \subset \mathcal{Q}$ is bounded, $G \in \mathcal{Q}_{\text{fin}}$, and the original denominator and multiplicities are retained. In the reflected second argument, T_a contributes a^{1-s} , whereas $\mathbf{V}_a$ contributes $a a^{s-1} = a^s$. Since $a > 0$,

$$p_B(T_a G) = a p_{T_{a^{-1}}B}(G), \quad p_B(\mathbf{V}_a G) = p_{T_a B}(G). \quad (\text{VSD14.2})$$

Both right-hand sets are bounded. These are exact seminorm equalities, so they prove continuity of both maps and their inverses for the actual residue topology. They extend uniquely to its Hausdorff completion, and their products are obtained by density from $T_a T_b = T_{ab}$, $\mathbf{V}_a \mathbf{V}_b = \mathbf{V}_{ab}$, and $T_a \mathbf{V}_a = a \cdot 1$.

The complete residue isomorphism $\hat{\iota} : \mathcal{H}_{\text{res}} \rightarrow \mathcal{Q}'_{\beta}$ is the completion of the already injective residue map. Transposing the topological isomorphism κ is legitimate for strong duals: bounded sets have bounded images in each direction. Therefore $\mathfrak{J} = \kappa' \hat{\iota}$ identifies $\mathcal{H}_{\text{res}}$ with the entire actual degree-minus-one dual cohomology. The finite-domain identities and (VSD14.2) give

$$\mathfrak{J} \widehat{T}_n = n T'_{n^{-1}} \mathfrak{J} = \mathbf{U}'_n \mathfrak{J}, \quad \mathfrak{J} \widehat{\mathbf{V}}_n = T'_n \mathfrak{J} = \mathbf{B}'_n \mathfrak{J}. \quad (\text{VSD14.3})$$

Continuity extends these identities from the dense finite-domain image to the whole completion. The first action retains the transpose of inverse-coefficient transfer; the second retains the ordinary transpose of pullback. No factor n or inverse parameter is lost.

Evaluation then defines the full pairing without an unproved infinite residue sum. A convergent net in $\mathcal{H}_{\text{res}}$ gives uniform convergence of its finite residue pairings on every bounded primal set. This proves independence of representatives and both identities

$$\widehat{\mathcal{R}}(T_n F, \widehat{T}_n h) = n \widehat{\mathcal{R}}(F, h), \quad \widehat{\mathcal{R}}(F, \widehat{V}_n h) = \widehat{\mathcal{R}}(T_n F, h). \quad (\text{VSD14.4})$$

The anti-diagonal support map is continuous in these strong topologies, by the sum of seminorms on the two bounded projections. It is exactly $h \mapsto (h, -h)$ in the completed residue coordinates. Its map into the overlap dual remains

$$h \mapsto q' \mathcal{J}(h), \quad (q' \mathcal{J}(h))(a) = \widehat{\mathcal{R}}([\Theta a], h), \quad \Theta = \tfrac{1}{2} \mathcal{M}_0. \quad (\text{VSD14.5})$$

The kernel condition follows from $\Theta J = \mathcal{J}$; SDT's bounded-lift argument proves the stated strong topological embedding. These verifications establish GTR10 exactly as written, with the complete original zeta and retained support signs. Neither a primal self-duality nor a positive form is inferred.

Ramified geometric trace and the original-zeta dual action

Complete derivation, 24 September 2026. Proof locators GTR0–GTR11.

GTR0. Construction stage and sources

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The user's complete arithmetic reconstruction precedes every coefficient space, scalar, positive integer, and analytic coordinate below. No addition or numerical coordinate is assigned to τ . The governing corpus and prerequisite rule and correction chains were read; the full passages rejecting an added distance/vector and the pooling of separate branch quotients were reread. The geometric sheets of an already constructed cover are not pooled arithmetic histories. The original two labelled support charts remain separate throughout.

The inputs are the complete CSP0–CSP13 sphere construction and CDW0–CDW10 explicit derived direct image in the receiving task. Their exact coefficients, restriction maps, periods and positive integration conventions are retained. The continuous transpose comparison is CDI0–CDI12, and the original-zeta residue comparison is RD0–RD9. These proofs were read in full. No continuous-dual complex is identified here with an ordinary Verdier dual without a separate constructed comparison.

The literature comparison is the Stacks Project, tag 0GKE, especially the trace existence, composition and flat quasi-finite weighting statements. Original author TeX `more-etale.tex`, lines1761–2127, was read at pinned commit a04446e57ec1fbc252a871afcec7752fb2807b14; its source receipt is retained locally. That source concerns étale sheaves. The topological sheaf trace used here is constructed directly below, so its existence is not inferred by changing categories in the cited theorem. The original Connes–Consani source identity and reading coverage remain CSP0's arXiv:0903.2024v3, §5, and the later signed chart construction. Deligne's comparison remains Weil II, §3.6, printed213–214.

GTR1. The actual sheaf and positive-degree covers

Use $Y = \mathbb{P}^1(\mathbb{C})$, $D = \{0, \infty\}$, $Y^\circ = \mathbb{C}^\times$, and the CSP coefficient spaces

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-x) = f(x), f(0) = 0, \int_{\mathbb{R}} f = 0\}, \quad V_p = S \oplus E_p, \quad E_p = \mathbb{C}^2 \quad (p = +, -),$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u d_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\}. \quad (\text{GTR1.1})$$

The full restrictions are

$$\Sigma f(u) = 2 \sum_{k \geq 1} f(ku), \quad Ra(u) = u^{-1}a(u^{-1}), \quad r_+(f, e) = \Sigma f, \quad r_-(h, e) = R\Sigma h. \quad (\text{GTR1.2})$$

Write $J = \Sigma S$, $Q = A/J$. The actual source-image theorem proves closedness of J and the continuous inverse of $\Sigma : S \rightarrow J$. With $i : D \hookrightarrow Y$,

$$\mathcal{F} = \underline{A}_Y \times_{i_*(A \oplus A)} i_*(V_+ \oplus V_-). \quad (\text{GTR1.3})$$

On a connected open meeting one pole, a section consists of a constant $a \in A$ and $v_p \in V_p$ with $a = r_p v_p$; on an interior connected open it is a constant a . This follows directly from the fibre product, and specifies the local maps used below.

For every recovered integer $n \geq 1$, let $b_n(z) = z^n$, with both poles fixed. The equality $\pi b_n = \pi$ supplies the canonical identification $b_n^{-1}\mathcal{F} = \mathcal{F}$. This is an identification of sheaves pulled from the same three-point diagram, not an inverse geometric map.

The local multiplicities are

$$e_n(x) = \begin{cases} n, & x = 0 \text{ or } \infty, \\ 1, & x \in \mathbb{C}^\times, \end{cases} \quad \sum_{b_n(x)=y} e_n(x) = n. \quad (\text{GTR1.4})$$

At zero the local ring quotient is $\mathbb{C}[z]_{(z)}/(z^n)$, of length n with basis $1, z, \dots, z^{n-1}$; the coordinate $w = 1/z$ gives the same calculation at infinity. Away from the poles the derivative nz^{n-1} is nonzero, and there are exactly n distinct roots. Thus (GTR1.4) includes both ramified points and every interior sheet. It agrees with the flat quasi-finite weighting in the cited source, without removing the ramification.

GTR2. The weighted trace exists on the actual topological sheaf

Define

$$\mathrm{Tr}_n : b_{n*} b_n^{-1} \mathcal{F} = b_{n*} \mathcal{F} \longrightarrow \mathcal{F} \quad (\text{GTR2.1})$$

as follows. On a sufficiently small interior disk the inverse image is a disjoint union of n disks, and the map on sections is

$$(a_1, \dots, a_n) \mapsto \sum_{j=1}^n a_j. \quad (\text{GTR2.2})$$

On a sufficiently small disk about a pole, the inverse image is one pole disk and a section is $(r_p v_p, v_p)$. Its image is

$$(r_p v_p, v_p) \mapsto (nr_p v_p, nv_p). \quad (\text{GTR2.3})$$

The restriction of (GTR2.3) to an interior subdisk equals (GTR2.2): all n sheet values are the restriction $r_p v_p$. Reordering the sheets does not change the sum. These formulas therefore agree on a basis of neighborhoods and every overlap, and define a sheaf morphism. Each local map is a finite continuous linear operation on the specified Fréchet spaces. At each pole both endpoint coordinates are multiplied by n ; they are not discarded because their generic restriction vanishes.

These local descriptions determine the map uniquely once its full weighted stalk values are specified. Generic-sheet agreement alone would not determine a map on the endpoint kernels E_p . Formula (GTR2.3) supplies that missing value explicitly from the local multiplicity.

The pullback unit $u_n : \mathcal{F} \rightarrow b_{n*} b_n^{-1} \mathcal{F}$ is diagonal on an interior disk and is the identity on a pole stalk. Hence

$$\boxed{\mathrm{Tr}_n u_n = n \, 1_{\mathcal{F}}}. \quad (\text{GTR2.4})$$

This is an equality on every stalk and therefore on sheaves. No division by n is used in constructing the trace.

The finite proper map b_n has exact direct image on vector-space sheaves. To see this without a finite-rank hypothesis, its direct-image stalk at y is the finite direct sum of the stalks at $b_n^{-1}(y)$. Indeed small disjoint neighborhoods of the finite fibre contain the inverse image of some neighborhood of y : otherwise properness supplies an accumulation point outside those neighborhoods mapping to y , a contradiction. Taking the corresponding neighborhood limits proves the stalk formula. Finite direct sums and stalk functors are exact. This also proves $Rb_{n*} = b_{n*}$; derived global sections of the pushforward equal those on the source. Thus (GTR2.1) gives an actual transfer on all cohomology groups.

The formula is natural in every continuous morphism of coefficient diagrams, since such maps commute with finite sums and multiplication by n . In particular it is natural for the actual sequence with overlap coefficients $J \rightarrow A \rightarrow Q$, retaining the entire J . Composition is exact: grouping roots in the composite cover multiplies their local multiplicities,

$$e_{mn}(x) = e_n(x) e_m(b_n(x)),$$

and the two successive finite sums equal the single weighted sum. Consequently the transfer for b_{mn} equals the composite transfer for b_n and b_m , with the direct-image identifications retained.

GTR3. Transfer on the full cohomology and support diagrams

Let P_n denote geometric pullback alone. The positive angular form is

$$\vartheta = \frac{d \arg z}{2\pi}, \quad b_n^* \vartheta = n \vartheta. \quad (\text{GTR3.1})$$

The positive sphere form and its full integral are

$$\omega_Y = \frac{dx \wedge dy}{\pi(1+|z|^2)^2}, \quad \int_Y b_n^* \omega_Y = 2n^2 \int_0^\infty \frac{r^{2n-1}}{(1+r^{2n})^2} dr = n. \quad (\text{GTR3.2})$$

CSP and CDW give the explicit global coefficient complex

$$D^0 = V_+ \oplus V_- \xrightarrow{r_+ - r_-} D^1 = A \xrightarrow{0} D^2 = A. \quad (\text{GTR3.3})$$

Its groups are $H^0 = Z = \ker(r_+ - r_-)$, $H^1 = Q$, $H^2 = A$. On this model geometric pullback is $(1, 1, n)$. In degree one it is the identity because the original degree-zero overlap constants are unchanged; in degree two (GTR3.2) gives the factor. These are invertible operators on these already constructed complex coefficient groups.

Taking cohomology in (GTR2.4) and using the proved pullback values determines the actual transfer:

	$H^0(Y, \mathcal{F}) = Z$	$H^1(Y, \mathcal{F}) = Q$	$H^2(Y, \mathcal{F}) = A$	
P_n	1	1	n	
Tr_n	n	n	1.	(\text{GTR3.4})

The map $(n, n, 1)$ on (GTR3.3) is a continuous cochain representative of these transfer values: its only nonzero differential commutes with n . No assertion that the map $z \mapsto z^n$ has a single-valued inverse is involved. This statement identifies the induced cohomology maps and an explicit compatible coefficient representative; it does not infer uniqueness of chain representatives from cohomology alone.

At either pole the costalk groups are E_p, Q, A in degrees 0, 1, 2; their geometric pullback and transfer factors are again $(1, 1, n)$ and $(n, n, 1)$. This follows from the actual local support complex $[V_p \rightarrow A \rightarrow 0A]$, the local angular degree n , and (GTR2.4) with supports. On the punctured sphere the factors in degrees 0, 1 are $(1, n)$ and $(n, 1)$. These maps commute with the entire CSP localization sequence: the maps $Q^2 \rightarrow Q$, $(x_+, x_-) \mapsto x_+ - x_-$, have transfer factor n on both sides; the maps $A \rightarrow A^2 \rightarrow A$, $a \mapsto (-a, a) \mapsto c_+ + c_-$, have factor one throughout their degree-two terms. The endpoint map has factor n . Thus every support sign and both endpoint kernels survive the transfer.

GTR4. The inverse coefficient map and the two different adjoints

The already constructed coefficient action is

$$T_a b(u) = b(u/a),$$

$$\rho_+(a)(f, c_0, c_1) = (f(\cdot/a), c_0, ac_1), \quad \rho_-(a)(h, d_0, d_1) = (ah(a \cdot), ad_0, d_1), \quad a > 0. \quad (\text{GTR4.1})$$

Direct substitution proves that their inverses are the same displayed formulas at a^{-1} ; they commute with the restrictions as $r_p \rho_p(a) = T_a r_p$. This inverse is on coefficient spaces, not on the geometric cover. It is also the required coefficient map on the dual: evaluation satisfies $\lambda(\rho(a)^{-1} \rho(a)x) = \lambda(x)$. Separation of evaluation proves uniqueness of that coefficient dual map if evaluation is to be preserved.

Set $\rho(a) = \rho_+(a) \oplus \rho_-(a)$. The full pullback from CSP is

$$\mathbf{B}_n = (\rho(n), T_n, nT_n) \quad (\text{GTR4.2})$$

on degrees 0, 1, 2. Combine the actual geometric transfer with the proved inverse coefficient action. Its compatible coefficient representative is

$$\mathbf{U}_n = (n\rho(n^{-1}), nT_{n^{-1}}, T_{n^{-1}}). \quad (\text{GTR4.3})$$

The order of these two operations is immaterial, since the trace is natural in coefficient maps. On each degree of the displayed complex,

$$\boxed{\mathbf{U}_n \mathbf{B}_n = \mathbf{B}_n \mathbf{U}_n = n \mathbf{1}.} \quad (\text{GTR4.4})$$

The scalar n therefore comes from the full ramified transfer, rather than being chosen to obtain a desired spectral formula. Composition of covers and coefficient actions gives $U_m U_n = U_{mn}$.

For the continuous dual complex use CDI's actual Hom differential. Ordinary transposition of a degree-zero cochain map adds no new sign. There are three specified maps:

$$\begin{array}{c|ccc}
\text{dual degree} & -2 & -1 & 0 \\
\hline
B'_n & nT'_n & T'_n & \rho(n)' \\
(B_n^{-1})' & n^{-1}T'_{n^{-1}} & T'_{n^{-1}} & \rho(n^{-1})' \\
U'_n & T'_{n^{-1}} & nT'_{n^{-1}} & n\rho(n^{-1})'.
\end{array} \tag{GTR4.5}$$

The first is adjoint to original pullback; the second is the contragredient; the third is adjoint to the constructed transfer with inverse coefficients. Thus

$$\begin{aligned}
U'_n &= n(B_n^{-1})', \quad B'_n U'_n = U'_n B'_n = n1, \\
\langle U'_n \lambda, B_n x \rangle &= n \langle \lambda, x \rangle.
\end{aligned} \tag{GTR4.6}$$

All identities hold on the complete coefficient complex and hence on its actual cohomology. They do not require a spectral decomposition or a density assertion. The induced degree-zero dual endpoint characters of U'_n are $n, 1, 1, n$, in the retained order c_0, c_1, d_0, d_1 .

GTR5. The original-zeta residue map receives this exact transfer action

Retain the original Mellin comparison

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta T_n b(s) = n^s \Theta b(s), \quad \Theta J = \mathcal{J}. \tag{GTR5.1}$$

Here $\mathcal{B}$ is the entire functions rapidly decreasing on every bounded real strip, $\mathcal{J}$ consists of all functions vanishing to the actual full multiplicity at every nontrivial zero of original ζ , and $\mathcal{Q} = \mathcal{B}/\mathcal{J}$. Write $\kappa : Q \rightarrow Q$, $[b] \mapsto [\Theta b]$. For $g = [G]$ with finitely supported full zero jets, RD proves

$$\iota(g)([F]) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds. \tag{GTR5.2}$$

Each member of this finite-jet subspace has a representative satisfying the required vanishing at every other zero; it is not a cutoff of the underlying arithmetic. RD constructs those representatives, proves that the sum is finite, and proves invertibility of every local multiplicity matrix. Its map ι is a linear isomorphism onto the finite-jet dual inside the entire continuous dual.

The constructed transfer on degree one is $nT_{n^{-1}}$; hence its transpose on $H^{-1}(D^\vee) = Q'$ is $nT'_{n^{-1}}$. Substituting in the complete residue integrand gives

$$\boxed{\kappa' \iota T_n = U'_n \kappa' \iota}. \tag{GTR5.3}$$

Indeed replacing $G(1-s)$ by $n^{1-s}G(1-s)$ is exactly multiplying the first argument by n^{-s} and the result by n . This proves the diagram with the actual geometric transfer. The factor in RD5 is now received by a constructed sheaf operation, rather than merely compared as an identical scalar.

For Taylor-coefficient functionals $\delta_{\rho,j}([F]) = F^{(j)}(\rho)/j!$, the whole formula is

$$U'_n \kappa' \delta_{\rho,j} = n^{1-\rho} \sum_{h=0}^j \frac{(-\log n)^{j-h}}{(j-h)!} \kappa' \delta_{\rho,h}. \tag{GTR5.4}$$

It follows by Taylor multiplication, retaining every nilpotent entry and factorial. No zero simplicity is assumed. The finite-jet image is invariant, and U'_n is continuous for the weak topology: evaluation at x after transposition is evaluation at the already constructed $U_n x$. Consequently RD's weakly dense receiver and

its actual ambient continuous dual carry compatible transfer actions. This does not replace that ambient dual by its completion.

The geometric construction applies to every recovered $n \geq 1$. The corresponding formulas for arbitrary $a > 0$ define the already available continuous coefficient representation; no nonintegral holomorphic cover is asserted.

GTR6. Normal direction, winding orientation and the retained extra factors

The primal normal action on H^2 is nT_n on A . Write T_a^A and T_a^Q when distinguishing the coefficient space from its quotient, and retain $q : A \rightarrow Q$. Since $qT_a^A = T_a^Qq$, transposition gives $(T_a^A)'q' = q'(T_a^Q)'$. The specified three maps are

$$S_{\text{res}} = n(T_{n-1}^Q)' \quad \text{on } Q', \quad N_{\text{contra}} = n^{-1}(T_{n-1}^A)' \quad \text{on } A', \quad N_{\text{transfer}} = (T_{n-1}^A)' \quad \text{on } A'.$$

Their exact comparison, with its domain and receiving inclusion retained, is

$$q'S_{\text{res}} = n^2N_{\text{contra}}q' = nN_{\text{transfer}}q' : Q' \rightarrow A'. \quad (\text{GTR6.0})$$

Here q' identifies Q' with $J^\perp$; SDT5 proves this is also a closed strong topological embedding. Thus the residue action has the displayed extra factors on this specific subspace of the full normal dual, not on every functional in A' . None of the factors or the inclusion may be removed.

CDW's winding coordinate $a[\vartheta]$ maps to $-a$ in positive sphere integration. Under that comparison, the GSL normal map is exactly

$$a[\vartheta] \mapsto -[(1 - E_+(s))(\Theta a)(s - 1)]_{\mathcal{J} \cap \mathcal{J}_+}, \quad (\text{GTR6.1})$$

with kernel $J[\vartheta]$. The sign commutes with both nT_n and T_{n-1} , so the trace and pullback tables are unchanged by this explicit coordinate comparison. It is not permission to omit the sign from the normal map itself.

The two covers' pole trace maps remain separate. Likewise the holomorphic swap $-1/z$ and antiholomorphic swap $-1/\bar{z}$ retain their original CDW angular signs $-1, +1$ and their CSP degree-two signs $+1, -1$. Their current pushforwards are adjoints to their specified pullbacks. No positive degree is assigned to the antiholomorphic orientation reversal by (GTR3.2), which concerns only the holomorphic covers b_n .

GTR7. Naturality with the actual lifting boundary

The exact source sequence is

$$0 \rightarrow \mathcal{F}_J \rightarrow \mathcal{F} \rightarrow j_!Q \rightarrow 0. \quad (\text{GTR7.1})$$

The weighted trace is defined on all three terms by the same fibre multiplicities and is natural in their coefficient maps by GTR2. Therefore its induced long exact sequences commute with transfer, as they do with pullback. For the actual connecting map

$$\delta : Q \rightarrow J(-1), \quad (\text{GTR7.2})$$

this gives both

$$\delta T_n = nT_n \delta, \quad \delta(nT_{n-1}) = T_{n-1} \delta. \quad (\text{GTR7.3})$$

The second relation also follows algebraically from the first because the specified coefficient actions are invertible. Thus the new trace construction does not falsely add an independent restriction to the lifting problem. CW4 and CW7 prove $\delta = 0$ on the entire source; the present construction proves how this already established lift is carried through the actual geometric trace and its continuous dual.

The identity (GTR5.4) pairs actual exponents ρ and $1 - \rho$ through the full functional equation. It retains their sum and every multiplicity coordinate. It does not supply an absolute-value estimate for either eigenvalue. Deriving the support-dual weight bounds needed in Deligne §3.6 remains a mathematical task; these constructed maps are the receiving operations for that task.

GTR8. Original arithmetic contributions retained

The source arithmetic remains

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{m \geq 1} (\log p) p^{-ms} \quad (\Re s > 1). \quad (\text{GTR8.1})$$

The original Σ still has both rational signs and the complete Mellin identity

$$\mathcal{M}_0 \Sigma f(s) = 2\zeta(s) \int_0^\infty f(x) x^s \frac{dx}{x}, \quad \mathcal{M}_0 = 2\Theta. \quad (\text{GTR8.2})$$

For the source used to construct the global residue isolators, retain

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{GTR8.3})$$

RD and SSI give their full unit derivatives, inverse Mellin formula and trivial-zero Taylor observations. The residue denominator in (GTR5.2) is original ζ , not F_0 . Its reflection retains

$$\chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \quad \zeta(s) = \chi(s) \zeta(1-s), \quad (\text{GTR8.4})$$

and the residue reversal has the sign from $ds = -d(1-s)$, with every local derivative of χ retained as in RD6. The new transfer does not modify any of these functions or their exceptional contributions.

GTR9. Exact result received by the support-duality programme

The factor n in the original-zeta residue covariance now has the specific geometric source (GTR2.1): the weighted trace for the actual ramified cover of the coefficient sheaf. Its inverse coefficient map is explicit, its two ramified stalks have their full multiplicities, and its action is calculated on every global, supported and annular group. The original-zeta finite-jet receiver intertwines its transpose exactly by (GTR5.3). The adjoint of pullback, the contragredient, and the adjoint of transfer are linked by (GTR4.6), rather than identified without their degree factors. This closes that action-comparison step. It does not identify the ordinary algebraic Verdier dual with the continuous dual or establish the missing Deligne weight bounds.

GTR10. The completed residue receiver and the full trace action

The completed strong-topology proof SDT0–SDT9 was read in full. Its accepted source is `../tau_weight_cohomology_20260924/CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md`, SHA256 `7f1a03a6225f1d fc22b167713c0ddb154635b06f9479f6137b229845b67aefc`. SDT4 constructs compact lifts through $A \rightarrow Q$; SDT5 proves the strong topological inclusion $Q'_\beta \simeq J^\perp \subset A'_\beta$; SDT6–SDT8 prove strong reflexivity, strong density of the full finite-jet functionals, and their exact Hausdorff completion. These results concern the actual coefficient spaces, not bounded samples of zeros.

Define the residue seminorms on the already constructed finite-support class space by

$$p_B^{\text{res}}(G) = \sup_{F \in B} \left| \sum_\rho \text{Res}_{s=\rho} \frac{F(s)G(1-s)}{\zeta(s)} ds \right|, \quad B \subset \mathcal{Q} \text{ bounded}. \quad (\text{GTR10.1})$$

The sum for each such G is finite and contains every multiplicity coordinate in its support. Write

$$\mathcal{H}_{\text{res}} = (\mathcal{Q}_{\text{fin}}, \widehat{(p_B^{\text{res}})_B}). \quad (\text{GTR10.2})$$

This names the specified Hausdorff completion, not a new topology silently put on the entire original quotient. SDT8.12 supplies the topological isomorphism

$$\hat{\iota} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} Q'_\beta, \quad \mathfrak{I} = \kappa' \hat{\iota} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} Q'_\beta = H^{-1}(D^\vee). \quad (\text{GTR10.3})$$

The factor $1/2$ is still in $\kappa[a] = [\Theta a]$. To verify that its transpose is a strong topological isomorphism, the continuous maps κ and κ^{-1} carry bounded sets to bounded sets. Each defining strong seminorm therefore pulls back to the seminorm on the corresponding bounded image. This proves both directions of the topological assertion in (GTR10.3).

For every $a > 0$, put $V_a = aT_{a^{-1}}$ on $\mathcal{Q}_{\text{fin}}$. Direct substitution into the full residue, before any completion, proves the exact seminorm identities

$$p_B^{\text{res}}(T_a G) = a p_{T_{a^{-1}}B}^{\text{res}}(G), \quad p_B^{\text{res}}(V_a G) = p_{T_a B}^{\text{res}}(G). \quad (\text{GTR10.4})$$

Indeed the respective multipliers on the second argument at $1-s$ are a^{1-s} and $a a^{-(1-s)} = a^s$. The sets on the right are bounded because T_a and its inverse are continuous on $\mathcal{Q}$. Thus both maps and their inverses extend uniquely to continuous automorphisms $\hat{T}_a$ and $\hat{V}_a$ of (GTR10.2). Their products extend those on the dense domain, so

$$\hat{T}_a \hat{T}_b = \hat{T}_{ab}, \quad \hat{V}_a \hat{V}_b = \hat{V}_{ab}, \quad \hat{T}_a \hat{V}_a = a \cdot 1. \quad (\text{GTR10.5})$$

For the recovered integer covers this gives, on the entire degree-minus-one strong cohomology group,

$$\boxed{\mathfrak{I} \hat{T}_n = U'_n \mathfrak{I}, \quad \mathfrak{I} \hat{V}_n = B'_n \mathfrak{I}.} \quad (\text{GTR10.6})$$

Here $U'_n = nT'_{n^{-1}}$ and $B'_n = T'_n$ in degree minus one, exactly as in (GTR4.5). The equalities follow on the dense subspace from (GTR5.3) and (GTR10.4). All maps are continuous into the Hausdorff space Q'_β , so their equalizers are closed; the equalities hold on the whole completion. This argument proves the full action without assuming a convergent infinite sum of residues.

The complete pairing is the actual evaluation

$$\widehat{\mathcal{R}}(F, h) = \hat{\iota}(h)(F), \quad F \in \mathcal{Q}, \quad h \in \mathcal{H}_{\text{res}}. \quad (\text{GTR10.7})$$

For a net $G_\alpha \rightarrow h$ in the specified completion, its original finite residue pairings converge to (GTR10.7) uniformly on every bounded $B \subset \mathcal{Q}$. This is precisely convergence in the strong dual under (GTR10.3), so the value is independent of the representing net. Nondegeneracy in h follows from injectivity of $\hat{\iota}$. Nondegeneracy in F follows from the full zero-jet functionals, which separate $\mathcal{Q}$ by the definition of $\mathcal{I}$. No positive Hermitian form is inferred from this bilinear nondegeneracy.

The full transfer identities become

$$\widehat{\mathcal{R}}(T_n F, \hat{T}_n h) = n \widehat{\mathcal{R}}(F, h), \quad \widehat{\mathcal{R}}(F, \hat{V}_n h) = \widehat{\mathcal{R}}(T_n F, h). \quad (\text{GTR10.8})$$

They follow by evaluation of (GTR10.6), or by passing the exact finite residue identities through the above uniform convergence. The first retains the geometric degree; the second identifies the transpose of pullback. All original nilpotent entries of (GTR5.4) persist in this extension.

Finally, the actual two-support restriction in degree minus one is

$$Q'_\beta \longrightarrow Q'_\beta \oplus Q'_\beta, \quad \lambda \longmapsto (\lambda, -\lambda). \quad (\text{GTR10.9})$$

It is continuous: the seminorm of its value on a bounded pair is bounded by the sum of the seminorms on its two bounded projections. In the completed residue coordinates it is exactly

$$h \longmapsto (h, -h). \quad (\text{GTR10.10})$$

This follows from (GTR10.3) and linearity, retains both separate poles, and commutes with both actions in (GTR10.6). The embedding into the overlap dual is explicitly

$$h \longmapsto q' \mathfrak{I}(h) \in J^\perp \subset A'_\beta, \quad (q' \mathfrak{I}(h))(a) = \widehat{\mathcal{R}}([\Theta a], h). \quad (\text{GTR10.11})$$

Its vanishing on J follows because $\Theta J = \mathcal{J}$; SDT5 proves it is a strong topological isomorphism onto that annihilator. Thus the completed residue receiver maps into the entire signed support-dual diagram by the original quotient and Mellin maps. Neither the endpoint groups in degrees zero nor the full normal group in degree minus two is deleted. This proves the completed trace comparison; it does not assert that $\mathcal{H}_{\text{res}}$ is the original primal quotient with its original topology, or that its evaluation pairing is positive.

GTR11. Identification of the actual trace on the derived direct image

There is a stronger receiving statement than the cohomology calculation of GTR3. Work in the ordinary derived category of complex vector-space sheaves on the actual three-point space X . CDW4 constructs the specific quasi-isomorphism

$$\Phi : \Omega \oplus \mathcal{W}[-1] \xrightarrow{\sim} R\pi_* \mathcal{F}, \quad \mathcal{W} = j_{\eta!} \underline{A}_{\{\eta\}}, \quad (\text{GTR11.1})$$

represented by constants and the positive angular form. Here $\mathcal{W}$ is a sheaf on X , not the previously used vector space $W = V_+ \oplus V_-$. Its only nonzero stalk is A at η . Every map in the stated representative is continuous on the original coefficient spaces. CDW5 proves, before taking global cohomology, that geometric pullback P_n obeys

$$P_n \Phi = \Phi \begin{pmatrix} 1_{\Omega} & 0 \\ 0 & n1_{\mathcal{W}[-1]} \end{pmatrix}. \quad (\text{GTR11.2})$$

Consequently P_n is an automorphism of this actual derived direct image. Its inverse is represented through Φ by $\text{diag}(1, n^{-1})$; no inverse geometric root map is used.

Let $\mathfrak{t}_n$ be the morphism on $R\pi_* \mathcal{F}$ induced by the actual sheaf trace of GTR2. Since $\pi b_n = \pi$, the intermediate object $R\pi_* b_{n*} \mathcal{F}$ is canonically $R\pi_* \mathcal{F}$. To justify this derived identification, b_{n*} is exact by GTR2 and preserves injectives because it is right adjoint to the exact inverse-image functor. An injective resolution therefore computes both sides of the composite direct image. The sheaf equality (GTR2.4) yields the equality of derived morphisms

$$\mathfrak{t}_n P_n = n1_{R\pi_* \mathcal{F}}. \quad (\text{GTR11.3})$$

Explicitly, for the preceding identification $\alpha : R\pi_* b_{n*} \mathcal{F} \xrightarrow{\sim} R\pi_* \mathcal{F}$, the two maps are $P_n = \alpha R\pi_*(u_n)$ and $\mathfrak{t}_n = R\pi_*(\text{Tr}_n)\alpha^{-1}$. CDW's chain pullback extends the sheaf unit through its augmentation; it therefore represents this same P_n . Compose on the right with the already constructed inverse of P_n . This proves

$$\boxed{\mathfrak{t}_n = nP_n^{-1}, \quad \mathfrak{t}_n \Phi = \Phi \begin{pmatrix} n1_{\Omega} & 0 \\ 0 & 1_{\mathcal{W}[-1]} \end{pmatrix}.} \quad (\text{GTR11.4})$$

This identifies the actual trace morphism in the derived category, including its possible extension information. It does not infer equality of morphisms merely from equal induced cohomology maps.

The coefficient action of (GTR4.1) commutes with Φ , preserves both summands, and has an explicit inverse. Thus the combined pullback and inverse-coefficient transfer are represented on (GTR11.1) by

$$\mathbf{B}_n = \begin{pmatrix} \rho_{\Omega}(n) & 0 \\ 0 & nT_n \end{pmatrix}, \quad \mathbf{U}_n = \begin{pmatrix} n\rho_{\Omega}(n^{-1}) & 0 \\ 0 & T_{n^{-1}} \end{pmatrix}. \quad (\text{GTR11.5})$$

The sheaf morphism $\rho_{\Omega}(a)$ means $\rho_+(a), \rho_-(a), T_a$ on the two chart stalks and the generic stalk, with the original restrictions retained. Taking derived global sections and the simultaneous contractions of CW2 gives exactly the full representatives (GTR4.2)–(GTR4.3). In particular the degree-one factor n comes from the Ω summand's trace, while the degree-two factor one comes from its separate angular summand. CDW9's minus between the angular coordinate and positive sphere integration remains in that comparison.

Every step is natural in the diagrams for $J \rightarrow A \rightarrow Q$, since CDW8's Φ is natural and the trace was constructed naturally in GTR2. Thus the same derived identification applies simultaneously to the actual source sequence (GTR7.1), with its boundary maps, rather than to separately chosen cohomology lists. This strengthens the precise scope of GTR3 to an actual ordinary derived-sheaf comparison with an explicit continuous coefficient representative. It makes no assertion of a strict current pullback through a ramified root map or of uniqueness in an unspecified category of topological complexes.

Original divisor, residue trace, and the positive-transfer attempt

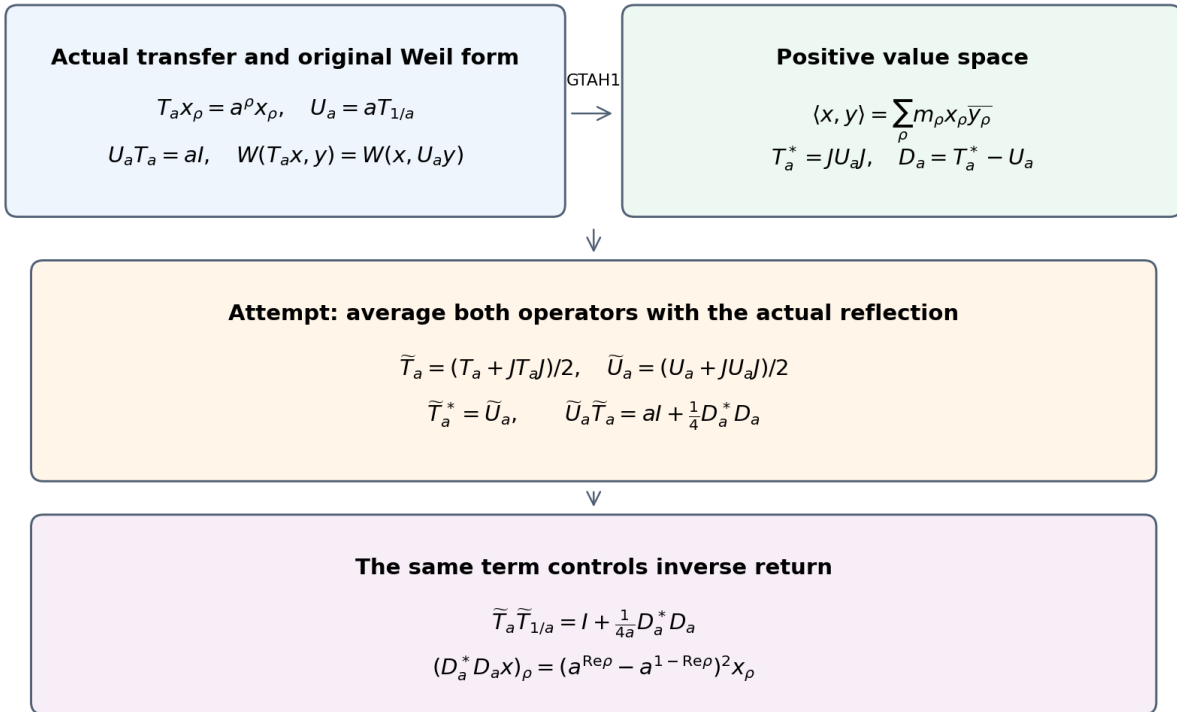
The full generator comparison has an actual holomorphic jet matrix and a meromorphic line-module inverse recovering the original zeta function. The retained pole and trivial-zero frames carry every infinitesimal order. These are the complete GCF and PGF proofs, previously received by the linked programme calculation.

The next attempted weight argument uses the original residue receiver and the actual geometric transfer. It calculates the source map to the full Weil trace and its kernel. The positive value completion has an exact adjoint defect. Averaging with the actual reflection repairs the adjoint identity, but the full degree and return composition acquire the same positive defect. Every formula is proved for the complete original zero set and all multiplicities. No numerical zero range or assumed purity is used.

The complete GMS source proof and the new GSR receiving calculation follow these attempts. GMS constructs full derived separation using the entire multiplier algebra. GSR proves its exact original-residue covariance, retains the complete kernel of the representative lift, and calculates that the separator acts as the identity on every original-zero jet. Its action therefore preserves the already measured positive-adjoint defect.

The positive-adjoint repair keeps an exact degree defect

All actual nontrivial zeros and their multiplicities; no finite zero sample



Proofs: GTAH0-GTAH5. J exchanges ρ with $1-\text{conjugate}(\rho)$; $a > 0$. Original factors and full jets remain in the source.

Figure 10: The actual transfer, the positive adjoint, and the attempted reflection repair. The extra term is calculated in the entire value completion; its source retains every higher jet and original multiplier. Proofs GTAH0-GTAH5; original source and geometric maps RD/GTR and GIQ.

The exact definitions and complete proofs follow. The Deligne weight-separation goal remains active. The displayed positive defect is a calculated object to study through the source map, not an assertion of its vanishing.

The full generator cokernel map and the original zeta divisor

24 September 2026. Proof locators GCF0–GCF8. This derives an actual map in the existing coefficient sequence and its global comparison with the original meromorphic zeta function. It does not establish a Deligne weight bound.

GCF0. Prerequisites and the precise receiving object

The supporting datum remains the user's τ , with the information-layer notation $Z_0, Z_1, Z_2, \dots$. All operations in this note occur after the complete arithmetic reconstruction. No addition, arithmetic value, distance, or parity is assigned to τ . Distinct reconstructed branches are not pooled.

The actual source comparison is SSI0–SSI8 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md, together with S1–S7 in GLOBAL_MELLIN_SYNTHESIS.md. It gives the topological isomorphism

$$\Theta : A \longrightarrow \mathcal{B}, \quad \Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad J = \Sigma \mathcal{S}_0^{\text{even}}, \quad \Theta J = \mathcal{J}, \quad \mathcal{Q} = A/J \simeq \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{GCF0.1})$$

Here A has the seminorms $\sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)|$, and $\Sigma f(u) = 2 \sum_{k \geq 1} f(ku)$. The coefficient space and its closed subspace are

$$\mathcal{B} = \left\{ F \text{ entire} : b_{R,M}(F) := \sup_{|\Re s| \leq R} (1 + |\Im s|)^M |F(s)| < \infty \text{ for every } R, M \in \mathbb{N} \right\},$$

$$\mathcal{J} = \{ F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of } \zeta, \ 0 \leq j < m_\rho \}. \quad (\text{GCF0.2})$$

All multiplicities m_ρ are retained. The entire source element, obtained from the original Gaussian-polynomial source, is

$$F_0(s) = c(s)\zeta(s), \quad c(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2). \quad (\text{GCF0.3})$$

It belongs to $\mathcal{J}$, has exactly its prescribed zero divisor, and satisfies $F_0(0) = F_0(1) = 1/8$. These are proved source properties, not a definition of the original ζ . In particular the recovered original series still has unit term 1 and every integer term, and its Euler expansion still contains every prime-power repetition. The multiplier in (GCF0.3) is never suppressed below.

The operator is the continuous map $LF(s) = sF(s)$ on $\mathcal{B}$, restricted to $\mathcal{J}$ and induced on $\mathcal{Q}$. Under (GCF0.1) it corresponds to $-u\partial_u$. Thus neither a numerical coordinate for τ nor an unconstructed arithmetic action is being introduced. CW6–CW8 already calculate its quotient characters and resolvent; VSD13 proves its closed codimension-one images on $\mathcal{B}$ and $\mathcal{J}$. RZ1–RZ6 retain the full quotient resolvent and multiplicity projectors. The present new receiver is the map between those two cokernels as the parameter varies, including its original meromorphic divisor.

GCF1. Both cokernel lines, with their actual quotient maps

For every $\lambda \in \mathbb{C}$, define

$$\epsilon_\lambda(F) = F(\lambda) \quad (F \in \mathcal{B}), \quad \ell_\lambda(F) = (F/F_0)(\lambda) \quad (F \in \mathcal{J}). \quad (\text{GCF1.1})$$

The second quotient is entire because the complete zero divisor of F_0 is imposed on every $F \in \mathcal{J}$. It need not belong to $\mathcal{B}$. If F_0 has order m at λ , its removable value is $F^{(m)}(\lambda)/F_0^{(m)}(\lambda)$; the nonzero denominator is retained.

Both maps are continuous and onto. For ϵ_λ , a section is $z \mapsto ze^{(s-\lambda)^2}$. For ℓ_λ , a section is $z \mapsto zF_0$. Continuity of the latter follows either by the displayed derivative formula or by a contour estimate: on a circle avoiding the zeros of F_0 , division by F_0 is bounded, and the maximum principle bounds the entire quotient on the enclosed compact set by this boundary bound. A larger-strip seminorm controls the numerator on that circle.

Division by $s - \lambda$ is continuous on the closed subspace of $\mathcal{B}$ vanishing at λ . Outside a fixed disk the denominator is bounded away from zero. Inside the disk, the maximum principle for the removable quotient bounds it on a larger circle. This proves a finite larger-strip seminorm bound for every target seminorm. On $\mathcal{J}$, division preserves its prescribed orders precisely when $\ell_\lambda(F) = 0$. Consequently the two exact rows are

$$0 \rightarrow \mathcal{B} \xrightarrow{L-\lambda} \mathcal{B} \xrightarrow{\epsilon_\lambda} \mathbb{C} \rightarrow 0, \quad 0 \rightarrow \mathcal{J} \xrightarrow{L-\lambda} \mathcal{J} \xrightarrow{\ell_\lambda} \mathbb{C} \rightarrow 0. \quad (\text{GCF1.2})$$

Every image is closed, and the inverse on the image is continuous. The inclusion $\mathcal{J} \hookrightarrow \mathcal{B}$ induces the exact scalar map

$$\begin{array}{ccc} \mathcal{J}/(L-\lambda)\mathcal{J} & \longrightarrow & \mathcal{B}/(L-\lambda)\mathcal{B} \\ \ell_\lambda \downarrow \cong & & \cong \downarrow \epsilon_\lambda \\ \mathbb{C} & \xrightarrow{\cdot F_0(\lambda)} & \mathbb{C}. \end{array} \quad (\text{GCF1.3})$$

Indeed $F(\lambda) = F_0(\lambda)(F/F_0)(\lambda)$, including removable values at every zero. Thus the full source expression in (GCF0.3) is an actual cokernel comparison, not a chosen positivity factor.

GCF2. Holomorphic parameter dependence

The quotient-line comparison is a holomorphic family over the entire parameter plane, including the zero set. Here is an explicit verification avoiding any assumption about holomorphic lifts through $\mathcal{B} \rightarrow \mathcal{Q}$.

Put

$$v_{\mathcal{B},\lambda}(s) = e^{(s-\lambda)^2}, \quad v_{\mathcal{J},\lambda}(s) = F_0(s), \quad \eta_{\mathcal{B},\lambda} = \epsilon_\lambda, \quad \eta_{\mathcal{J},\lambda} = \ell_\lambda. \quad (\text{GCF2.1})$$

For $E = \mathcal{B}, \mathcal{J}$, the maps

$$S_{E,\lambda}F(s) = \frac{F(s) - v_{E,\lambda}(s)\eta_{E,\lambda}(F)}{s - \lambda} \quad (\text{GCF2.2})$$

take values in E , and direct multiplication gives

$$(L - \lambda)S_{E,\lambda}F = F - v_{E,\lambda}\eta_{E,\lambda}(F), \quad S_{E,\lambda}(L - \lambda)F = F. \quad (\text{GCF2.3})$$

For $E = \mathcal{J}$, the numerator has one additional order at $s = \lambda$ relative to the required divisor, since its quotient by F_0 vanishes there. This proves the asserted target of (GCF2.2), also at a multiple zero.

On any compact set of parameters λ , the division bounds in GCF1 can use one larger disk and strip. The Gaussian and all its parameter derivatives have uniform bounds in every $\mathcal{B}$ seminorm on that parameter set. The map $F \mapsto F/F_0$ is continuous from $\mathcal{J}$ to entire functions with compact convergence, by the same boundary-circle argument. Evaluations and all their parameter derivatives are therefore bounded by finitely many source seminorms uniformly on a compact parameter set. Formula (GCF2.2) has a removable singularity along $s = \lambda$; Cauchy's integral formula and these bounds prove holomorphic dependence with values in every coefficient seminorm. They also prove that (GCF2.2) carries a holomorphic E -valued family to a holomorphic E -valued family.

Let $\mathcal{H}_E(U)$ denote holomorphic E -valued maps on an open parameter set $U \subset \mathbb{C}$. Equations (GCF2.1)–(GCF2.3) prove, on every such set,

$$\mathcal{H}_E(U)/(L - \lambda)\mathcal{H}_E(U) \xrightarrow{\eta_{E,\lambda}} \mathcal{O}(U). \quad (\text{GCF2.4})$$

This is an isomorphism: the stated section proves surjectivity and $S_{E,\lambda}$ proves its exact kernel. Inclusion of the coefficient spaces is consequently represented by the global map of these two quotient line sheaves

$$\mathcal{O}_{\mathbb{C}} \xrightarrow{\cdot F_0} \mathcal{O}_{\mathbb{C}}. \quad (\text{GCF2.5})$$

Its determinant is the displayed one-by-one entry F_0 , with its exact source factor $1/8$. No determinant or trace of an infinite-dimensional operator has been assumed. The frame of the first line uses the actual source element F_0 ; changing that frame would change the written entry and its inverse frame map together.

GCF3. Exact relation with the actual quotient obstruction

Apply the kernel/cokernel sequence to multiplication by $L - \lambda$ in the actual row $0 \rightarrow \mathcal{I} \rightarrow \mathcal{B} \rightarrow \mathcal{Q} \rightarrow 0$. Since the first two multiplications are injective, it gives

$$0 \rightarrow \ker(L - \lambda \mid \mathcal{Q}) \xrightarrow{\partial_\lambda} \mathbb{C} \xrightarrow{\cdot F_0(\lambda)} \mathbb{C} \rightarrow \text{coker}(L - \lambda \mid \mathcal{Q}) \rightarrow 0. \quad (\text{GCF3.1})$$

The boundary has a literal formula. Represent a kernel class by $G \in \mathcal{B}$, so $(s - \lambda)G \in \mathcal{I}$. Then

$$\partial_\lambda[G] = \left(\frac{(s - \lambda)G(s)}{F_0(s)} \right)_{s=\lambda}. \quad (\text{GCF3.2})$$

Changing G by an element of $\mathcal{I}$ changes the numerator quotient by a multiple of $s - \lambda$, so this is well defined. These computations prove (GCF3.1), including its sign, directly.

When $F_0(\lambda) \neq 0$, both outside terms vanish. When $\lambda = \rho$ is an actual nontrivial zero, both outside terms are one-dimensional and

$$G_\rho(s) = \frac{F_0(s)}{s - \rho} \in \mathcal{B}, \quad \partial_\rho[G_\rho] = 1. \quad (\text{GCF3.3})$$

This remains true for a multiple zero: the class has its last retained nonzero jet of order $m_\rho - 1$. All statements quantify over the full actual zero set.

This boundary is the obstruction to lifting an eigenclass of $\mathcal{Q}$ to a class in $\mathcal{B}$ satisfying that same generator equation. It is not CW4's map $\delta^1 : \mathcal{Q} \rightarrow J(-1)$. CW4 constructs a different, geometric cohomology lift and proves its boundary zero. Here the explicit nonzero boundary (GCF3.3) specifies its own different lifting problem; it does not undo the geometric lift or imply an obstruction at τ .

For the shifted normal action $L + 1$, exactly the same comparison has entry $F_0(\lambda - 1)$. On $\Re \lambda \in (0, 1)$ this entry is nonzero, since all zeros of F_0 have real part in $(0, 1)$. This recovers the already proved separation of the $\mathcal{Q}$ and $\mathcal{Q}(-1)$ spectra. It does not replace the full target $J(-1)$ by its quotient.

GCF4. The complete divisor module of the original function

Let $\mathcal{M}$ be the sheaf of meromorphic functions on the parameter plane. Write $[a]$ for the divisor at the scalar point a , and set

$$T = \sum_{n \geq 1} [-2n], \quad D = [1] - T. \quad (\text{GCF4.1})$$

The sum is locally finite, so its orders define a divisor on the whole plane. The sheaf $\mathcal{O}(D) \subset \mathcal{M}$ consists of meromorphic functions g for which $\text{ord}_a(g) + D(a) \geq 0$ at every point. Locally it is a free rank-one $\mathcal{O}$ -module; at 1 its frame is $(s - 1)^{-1}$, and at $-2n$ its frame is $s + 2n$.

The full factor c in (GCF0.3) has divisor exactly D . Indeed Gamma has no zeros, its simple pole at zero cancels the displayed factor s , the factor $s - 1$ gives a simple zero at 1, and its remaining simple poles occur at every $-2n$. The recurrence for Gamma retains their residues in GCF5 below. The power of π is an everywhere nonzero entire factor. There are no further zeros or poles. Hence multiplication gives a global module isomorphism

$$\mu_c : \mathcal{O}(D) \xrightarrow{\sim} \mathcal{O}, \quad g \mapsto cg, \quad f \mapsto f/c. \quad (\text{GCF4.2})$$

The order inequalities prove both directions at every exceptional point, so (GCF4.2) is not division by a supposed holomorphic unit there.

The original meromorphic ζ belongs to $\mathcal{O}(D)$: it has its simple pole at 1, its simple zeros at every $-2n$, and no other poles. The complete comparison is

$$\begin{array}{ccc} \mathcal{O} & \xrightarrow{\cdot \zeta} & \mathcal{O}(D) \subset \mathcal{M} \\ \text{id} \downarrow & & \downarrow \mu_c \cong \\ \mathcal{O} & \xrightarrow{\cdot F_0} & \mathcal{O}. \end{array} \quad (\text{GCF4.3})$$

It is a commutative diagram of actual holomorphic line modules with their specified meromorphic embeddings. Its inverse recovers $\zeta = F_0/c$ at every point. Retaining D , μ_c , and the inclusion into $\mathcal{M}$ retains the original zero and pole orders. Keeping only the bottom scalar function would not specify those embeddings.

The cokernel of either horizontal module map is supported precisely on the nontrivial zero divisor, with length m_ρ at ρ . To prove this, at a nontrivial zero the multiplier is h^{m_ρ} times its retained holomorphic unit, where $h = s - \rho$; the quotient has basis $1, h, \dots, h^{m_\rho-1}$. At every other point the horizontal map is multiplication by a unit in the displayed local frames. GCF5 computes those frames and units at all cancellation points.

GCF5. Every exceptional fibre and every infinitesimal order

At $s_0 = -2n$, $n \geq 1$, put $h = s + 2n$. The exact local factorizations are

$$\zeta(-2n + h) = h a_n(h), \quad c(-2n + h) = h^{-1} b_n(h), \quad F_0(-2n + h) = a_n(h) b_n(h), \quad (\text{GCF5.1})$$

where the holomorphic units and their constants are

$$a_n(0) = \zeta'(-2n), \quad b_n(0) = \frac{(-2n)(-2n-1)}{8} \pi^n \frac{2(-1)^n}{n!} = \frac{n(2n+1)(-1)^n \pi^n}{2n!}. \quad (\text{GCF5.2})$$

Gamma's residue follows by applying $\Gamma(z+1) = z\Gamma(z)$ repeatedly to $\Gamma(h/2)$. Nonvanishing of $a_n(0)$ follows from the original functional equation and the positive Dirichlet value at $1+2n$; explicitly

$$a_n(0) b_n(0) = \frac{(1+2n)(2n)}{8} \pi^{-(1+2n)/2} \Gamma((1+2n)/2) \zeta(1+2n) \neq 0. \quad (\text{GCF5.3})$$

The fibre of the original target line is $h\mathcal{O}/h^2\mathcal{O}$, and the map μ_c sends ha_n to b_na_n . It therefore retains $\zeta'(-2n)$ and its full multiplier, even though the bottom function is nonzero there.

At $s = 1$, put $h = s - 1$. Then

$$\zeta(1+h) = h^{-1} a_1^{\text{pole}}(h), \quad c(1+h) = h b_1^{\text{pole}}(h), \quad a_1^{\text{pole}}(0) = 1, \quad b_1^{\text{pole}}(0) = 1/8. \quad (\text{GCF5.4})$$

The original target fibre is $h^{-1}\mathcal{O}/\mathcal{O}$. Its residue coefficient 1 is carried to $1/8$. At $s = 0$, the target is $\mathcal{O}$, $c(0) = -1/4$, and $\zeta(0) = -1/2$, again giving $F_0(0) = 1/8$. Neither zero nor pole data has been silently removed.

All higher orders are equally explicit. At any point, write $\mathcal{O}(D) = h^{-d}\mathcal{O}$ and $c = h^d b(h)$, with $d = D(s_0)$ and $b(0) \neq 0$. On the full length- r jet module, (GCF4.2) is

$$h^{-d}\mathcal{O}/h^{r-d}\mathcal{O} \longrightarrow \mathcal{O}/h^r\mathcal{O}, \quad [h^{-d}a(h)] \longmapsto [b(h)a(h)], \quad (\text{GCF5.5})$$

for every $r \geq 1$. In ordered bases $h^{-d}, \dots, h^{r-1-d}$ and $1, \dots, h^{r-1}$, its matrix has entries

$$B_{ij} = \begin{cases} b^{(i-j)}(0)/(i-j)!, & i \geq j, \\ 0, & i < j, \end{cases} \quad 0 \leq i, j < r, \quad \det B = b(0)^r. \quad (\text{GCF5.6})$$

Its inverse is the same matrix rule with every coefficient of $1/b$. Thus every infinitesimal coefficient is retained; the comparison is not only a statement about values or a first-order approximation.

GCF6. The transported involution on the original target

The source functional equation gives $F_0(s) = F_0(1-s)$. On the original target module, the exact transported reflection is

$$(\mathcal{R}_D g)(s) = \frac{c(1-s)}{c(s)} g(1-s), \quad \mu_c \mathcal{R}_D = \mathcal{R} \mu_c, \quad (\mathcal{R} f)(s) = f(1-s). \quad (\text{GCF6.1})$$

It is defined globally as a map of line modules over the reflection, including exceptional points: by (GCF4.2), it is $\mu_c^{-1} \mathcal{R} \mu_c$. Thus its square is the identity, and $\mathcal{R}_D \zeta = \zeta$. The ratio is explicitly

$$\frac{c(1-s)}{c(s)} = \frac{\frac{(1-s)((1-s)-1)}{8} \pi^{-(1-s)/2} \Gamma((1-s)/2)}{\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2)}. \quad (\text{GCF6.2})$$

This is an isomorphism of the indicated line modules over reflection, not an assertion that the ratio is an everywhere holomorphic unit on scalar functions. In particular its zeros and poles implement the frames in GCF5. On coefficient functions, $\mathcal{R}(L-\lambda) = -(L-(1-\lambda))\mathcal{R}$. On length- r jets, reflection sends h^j to $(-1)^j h^j$ between the two centres. Consequently the full generator comparison respects the reflected parameter and every derivative sign. No radius or distance from τ was used to construct this action.

GCF7. The original logarithmic derivative remains complete

Off the zeros and poles, logarithmic differentiation of the actual module comparison gives

$$\frac{F'_0}{F_0} = \frac{\zeta'}{\zeta} + \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}. \quad (\text{GCF7.1})$$

It then holds meromorphically. At a nontrivial zero its residue is the complete multiplicity. At 1, the residues -1 from ζ'/ζ and $+1$ from $1/(s-1)$ cancel. At each $-2n$, the residues $+1$ from the trivial zero and -1 from Gamma cancel. At zero the residues $+1$ from $1/s$ and -1 from Gamma cancel; ζ is nonzero there. These are the same contributions encoded by the frames in GCF5, with the power of π and every regular derivative term still present in (GCF7.1).

GCF8. Consequence for the requested weight calculation

The actual coefficient inclusion has now produced a holomorphic one-by-one determinant map, and (GCF4.3) returns it to the original ζ as an isomorphism of explicitly embedded divisor modules. The precise spectral lifting boundary is (GCF3.2). It is nonzero at every actual nontrivial zero, on or off the critical line, and retains multiplicity through the complete jet modules. Thus it cannot by itself distinguish the two locations. This is a calculation of the actual boundary, rather than a hypothetical counterexample constructed outside the programme.

CW4's geometric lifting boundary remains zero. Deligne's argument additionally separates the weights of its entire source and target; no such bound follows merely from the determinant map just derived. The original full target $J(-1)$, the degree factor, and the CW quotient maps remain present. The new comparison supplies the exact original-zeta object to which any further weight or trace argument must apply. It does not replace that argument with a desired assumption.

Human source comparisons: Connes and Consani, *Schemes over $F1$ and zeta functions*, arXiv:0903.2024v3, §5, for the source summation/sheaf construction; Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, arXiv:2602.04022v1, `rhready.tex` lines 526–545, for the retained functional equation; Deligne, *La conjecture de Weil. II*, IHÉS52 (1980), 137–252, §3.6, for the distinct weight-separation lifting mechanism. Source-reading coverage and hashes are recorded privately; these citations do not assert a new reading of all external dependencies.

The full parameter and jet family of the original-zeta generator sequence

Independent derivation, 24 September 2026. Complete proof locators PGF0–PGF9.

PGF0. Scope, prerequisites, and actual sources read

This calculation is on the arithmetic coefficient spaces after their complete global reconstruction. It does not construct arithmetic by selecting a few primes. The user’s supporting notation $Z_0, Z_1, Z_2, \dots, \tau$ remains unchanged. Addition at τ is retracted. None of the additions, derivatives, scalar operations, or coordinates below are operations on τ .

The governing argument `_reconstruction/CORPUS_AND_OPERATION_RULE.md` was read, including the recorded corrections concerning prerequisites and the full obstruction receiver. This derivation uses the already constructed exact row

$$0 \longrightarrow \mathcal{J} \xrightarrow{\iota} \mathcal{B} \xrightarrow{q} \mathcal{Q} \longrightarrow 0. \quad (\text{PGF0.1})$$

It does not assert a new weight bound or an RH conclusion. The map calculated here is the parameter-dependent inclusion between the two generator cokernels and its actual snake boundary. It is not the sheaf connecting map $\mathcal{Q} \rightarrow \mathcal{J}(-1)$, whose vanishing is already proved in CW4 and CW7.

Programme sources read for this derivation were:

- `GLOBAL_MELLIN_SYNTHESIS.md`, S1–S3, including the complete source multiplier and every exceptional value.
- `EXACT_SCHWARTZ_SUMMATION_IMAGE.md`, SSI0–SSI3, for the exact statement, source conventions, and original-zeta receiver. The completed SSI exact-image theorem is used as an established input, not reproved here.
- `../tau_weight_cohomology_20260924/CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md`, CW0–CW9, for the actual source row, its full kernel, and the distinction between quotient characters and eigenvectors.
- `../tau_weight_cohomology_20260924/ORIGINAL_ZETA_RESOLVENT_AND_PROJECTORS.md`, RZ1–RZ8, for whole-space division estimates, holomorphic resolvents, and the already constructed generalized eigenspaces. The present proof does not claim those prior constructions as new.
- `CC_VERDIER_SUPPORT_INDEPENDENT.md`, VSD13, for the first-order closed-image statement and its separate algebraic-dual consequence. The present calculation concerns the primal sequence; it does not duplicate the dagger-dual lifting calculation.

The upstream source construction is Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. SSI and CW record the original author-file identity, exact reading coverage, and receiving maps. This independent note makes no additional claim of having reread that complete author source. All new calculations used here are proved below.

PGF1. Objects, operators, and the retained entire source function

Let $\mathcal{B}$ be the complex Fréchet space of entire functions with

$$b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \quad (A, M \in \mathbb{N}). \quad (\text{PGF1.1})$$

Let $\mathcal{Z}$ be the actual nontrivial-zero set of the original Riemann zeta function, and let m_ρ be each full multiplicity. Put

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for all } \rho \in \mathcal{Z}, 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{PGF1.2})$$

The ideal is closed because each indicated jet functional is continuous by Cauchy’s formula. The quotient uses its actual Fréchet quotient topology.

Multiplication by s , written L on each space where it is defined, is continuous and preserves $\mathcal{J}$. For example,

$$b_{A,M}(LF) \leq (A+1)b_{A,M+1}(F). \quad (\text{PGF1.3})$$

For every $\lambda \in \mathbb{C}$ and positive integer r , define

$$M_{\lambda,r} = (L - \lambda)^r. \quad (\text{PGF1.4})$$

This is an everywhere-defined continuous linear map on $\mathcal{B}, \mathcal{J}, \mathcal{Q}$. The arithmetic coefficient field, polynomial, and integer r here are already reconstructed receivers.

Retain the entire source function with all its original factors:

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{PGF1.5})$$

It belongs to $\mathcal{J}$, and its divisor consists exactly of (ρ, m_ρ) . At the exceptional points its values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad (\text{PGF1.6})$$

$$F_0(-2n) = \frac{n(2n+1)(-1)^n \pi^n}{2n!} \zeta'(-2n) = \frac{(1+2n)(2n)}{8} \pi^{-(1+2n)/2} \Gamma((1+2n)/2) \zeta(1+2n) \neq 0 \quad (n \geq 1). \quad (\text{PGF1.7})$$

These are the retained S2 and CW1 identities. In particular the original zeta pole at 1 and the trivial zeros at $-2n$ are not additional zeros of this specified source function.

Set

$$m_\lambda = \text{ord}_\lambda F_0, \quad (\text{PGF1.8})$$

with order zero where $F_0(\lambda) \neq 0$. Every $F \in \mathcal{J}$ has an entire quotient F/F_0 , by local Taylor division at each zero. This quotient is not asserted to belong to $\mathcal{B}$.

For later continuity, the division map

$$\mathcal{D} : \mathcal{J} \longrightarrow \mathcal{O}(\mathbb{C}), \quad F \mapsto F/F_0 \quad (\text{PGF1.9})$$

is continuous for the compact-open topology on the target. To prove it, put any given compact set inside a circle $|s| = R$ which contains no zero of F_0 on its boundary. Such an R exists because the zeros are discrete. The maximum principle applied to the entire quotient gives

$$\sup_{|s| \leq R} |F(s)/F_0(s)| \leq \frac{b_{A,0}(F)}{\min_{|s|=R} |F_0(s)|}, \quad A \geq R. \quad (\text{PGF1.10})$$

The denominator is positive. Cauchy's formula on a slightly larger circle gives the same conclusion for every fixed derivative and every compact set.

PGF2. Both complete jet quotients, with their exact topology

For a fixed $r \geq 1$, write

$$R_r = \mathbb{C}[h]/(h^r), \quad j_\lambda^r H = \sum_{j=0}^{r-1} \frac{H^{(j)}(\lambda)}{j!} h^j \quad \text{for an entire } H. \quad (\text{PGF2.1})$$

Here h is the jet coordinate $s - \lambda$ in an arithmetic coefficient receiver; it is not a coordinate of τ . Define

$$\beta_{\lambda,r} : \mathcal{B} \rightarrow R_r, \quad F \mapsto j_\lambda^r F, \quad \alpha_{\lambda,r} : \mathcal{J} \rightarrow R_r, \quad F \mapsto j_\lambda^r (F/F_0). \quad (\text{PGF2.2})$$

Both are continuous. This follows for β from Cauchy estimates and for α from (PGF1.9)–(PGF1.10).

They are both onto, with explicit continuous linear sections. If P is the unique polynomial of degree below r representing its class in R_r , put

$$\sigma_{\lambda,r}^I(P)(s) = F_0(s)P(s-\lambda). \quad (\text{PGF2.3})$$

The product belongs to $\mathcal{J}$, since polynomial multiplication preserves $\mathcal{B}$ and all required zero orders. Its quotient by F_0 is exactly $P(s-\lambda)$, so $\alpha\sigma^I = 1$.

For the other section, retain the Gaussian and every truncated coefficient:

$$\sigma_{\lambda,r}^B(P)(s) = e^{(s-\lambda)^2} \text{Tay}_{h,<r}(e^{-h^2}P(h))\Big|_{h=s-\lambda}. \quad (\text{PGF2.4})$$

The Gaussian times this finite polynomial belongs to $\mathcal{B}$: on every bounded real strip the factor has modulus $e^{(\Re s - \Re \lambda)^2 - (\Im s - \Im \lambda)^2}$, which dominates every polynomial decay requirement. Its Taylor polynomial at $s = \lambda$ equals P modulo h^r , so $\beta\sigma^B = 1$. These are sections of the specified quotient maps, not a section of $\mathcal{B} \rightarrow \mathcal{Q}$.

The kernels are exactly

$$\ker \beta_{\lambda,r} = (s-\lambda)^r \mathcal{B}, \quad \ker \alpha_{\lambda,r} = (s-\lambda)^r \mathcal{J}. \quad (\text{PGF2.5})$$

For the first identity, vanishing of the jet is precisely divisibility to order r at λ . The quotient is entire and remains in $\mathcal{B}$. An explicit division estimate is given below. For the second, $\alpha(F) = 0$ means that F/F_0 has a zero of order at least r . Thus $F/(s-\lambda)^r$ retains the full zero order m_λ at λ , as well as the full orders at all other original zeros, and belongs to $\mathcal{J}$. The converse implications follow by direct multiplication.

For precision, if $H \in \mathcal{B}$ vanishes to order at least r at λ , its entire quotient obeys

$$b_{A,M}\left(\frac{H}{(s-\lambda)^r}\right) \leq b_{A,M}(H) + 2^{-r}(2 + |\Im \lambda|)^M b_{A',0}(H), \quad A' \geq \max(A, |\Re \lambda| + 2). \quad (\text{PGF2.6})$$

Outside the radius-one disk, the denominator has modulus at least one. Inside that disk, apply the maximum principle to the quotient on the radius-two disk; its boundary value is bounded by $2^{-r} \sup |H|$. The polynomial imaginary-coordinate weight is bounded by $(2 + |\Im \lambda|)^M$ on the smaller disk. This proves the estimate, which is uniform for λ in a compact set after fixing A' and the displayed maximum of $|\Im \lambda|$.

Consequently the maps

$$\bar{\alpha}_{\lambda,r} : \mathcal{J}/(L-\lambda)^r \mathcal{J} \xrightarrow{\sim} R_r, \quad \bar{\beta}_{\lambda,r} : \mathcal{B}/(L-\lambda)^r \mathcal{B} \xrightarrow{\sim} R_r \quad (\text{PGF2.7})$$

are topological linear isomorphisms. Their continuous inverses are the quotient classes of (PGF2.3) and (PGF2.4). On both sides L acts as multiplication by $\lambda + h$, so these are also exact $\mathbb{C}[L]$ -module isomorphisms.

Their multiplicative meanings differ and must be retained. The map $\bar{\beta}$ is an algebra isomorphism to the usual truncated polynomial algebra; its identity is the class represented by (PGF2.4) with $P = 1$. On the $\mathcal{J}$ -quotient, (PGF2.7) transports the inherited product to

$$P \star Q = f_{\lambda,r}(h)P(h)Q(h) \pmod{h^r}, \quad f_{\lambda,r}(h) := j_\lambda^r F_0. \quad (\text{PGF2.8})$$

Indeed $(FG)/F_0 = F_0(F/F_0)(G/F_0)$. Thus $\bar{\alpha}$ is a linear and module isomorphism; it is not an identification with the usual multiplication on R_r . This retains the full source coefficient rather than silently replacing its product.

PGF3. The inclusion matrix and every retained factor

The inclusion $\iota : \mathcal{J} \rightarrow \mathcal{B}$ induces, in (PGF2.7), exactly

$$\mathbb{T}_r(\lambda) : R_r \longrightarrow R_r, \quad P \longmapsto f_{\lambda,r}P \pmod{h^r}. \quad (\text{PGF3.1})$$

Proof: $F = F_0(F/F_0)$ as an identity of entire functions, so taking its full r -jet gives (PGF3.1). No local unit is removed in this map.

In the ordered basis $1, h, \dots, h^{r-1}$, write

$$c_j(\lambda) = \frac{F_0^{(j)}(\lambda)}{j!}. \quad (\text{PGF3.2})$$

Then the complete matrix is

$$\mathsf{T}_r(\lambda)_{ij} = \begin{cases} c_{i-j}(\lambda), & 0 \leq j \leq i < r, \\ 0, & 0 \leq i < j < r. \end{cases} \quad (\text{PGF3.3})$$

It follows by the triangular determinant formula that

$$\det \mathsf{T}_r(\lambda) = F_0(\lambda)^r = \left[\frac{\lambda(\lambda-1)}{8} \pi^{-\lambda/2} \Gamma(\lambda/2) \zeta(\lambda) \right]^r, \quad (\text{PGF3.4})$$

where the expression on the right has its full entire continuation and the values (PGF1.6)–(PGF1.7). The exponent r is the matrix rank of this r -jet comparison. It is not a pooling of two arithmetic branches or a replacement of the original return measure by a power of zeta.

At a point where the individual factors are holomorphic, all coefficients in the matrix retain the full Leibniz sum. With $P(s) = s(s-1)$, this is

$$c_j(\lambda) = \frac{\pi^{-\lambda/2}}{8} \sum_{\substack{a+b+c+d=j \\ 0 \leq a \leq 2}} \frac{P^{(a)}(\lambda)}{a!} \frac{(-\log \pi/2)^b}{b!} \frac{2^{-c} \Gamma^{(c)}(\lambda/2)}{c!} \frac{\zeta^{(d)}(\lambda)}{d!}. \quad (\text{PGF3.5})$$

At $0, 1, -2, -4, \dots$, the definition (PGF3.2) takes the derivatives of the full analytic product. Equivalently, its coefficient is the coefficient of h^j in the product of the full convergent Laurent expansions of

$$\frac{(\lambda+h)(\lambda+h-1)}{8}, \quad \pi^{-(\lambda+h)/2}, \quad \Gamma((\lambda+h)/2), \quad \zeta(\lambda+h). \quad (\text{PGF3.6})$$

Each has only the stated finite principal part at that point, so the coefficient of each fixed nonnegative power is a finite sum over the allowed Laurent exponents. This prescription keeps the cancellation derivatives, including the zeta residue and the trivial-zero derivatives. In particular, the diagonal is $1/8$ at both endpoints, and equals (PGF1.7) at each negative even integer; the determinant there is the corresponding nonzero r -th power. The original zeta's pole and trivial zeros remain in (PGF3.6), rather than being asserted absent from the original function.

PGF4. The full snake sequence and its actual boundary

Apply $M_{\lambda,r}$ to all three terms of (PGF0.1). The first two vertical maps are injective, since an entire function multiplied by the nonzero polynomial $(s-\lambda)^r$ can vanish identically only when that entire function is zero. The full resulting snake sequence is

$$0 \longrightarrow K_{\lambda,r} \xrightarrow{\partial_{\lambda,r}} R_r \xrightarrow{\mathsf{T}_r(\lambda)} R_r \xrightarrow{\pi_{\lambda,r}} C_{\lambda,r} \longrightarrow 0, \quad (\text{PGF4.1})$$

where

$$K_{\lambda,r} = \ker((L-\lambda)^r : \mathcal{Q} \rightarrow \mathcal{Q}), \quad C_{\lambda,r} = \mathcal{Q}/(L-\lambda)^r \mathcal{Q}. \quad (\text{PGF4.2})$$

Here is the explicit boundary, with its sign fixed:

$$\partial_{\lambda,r}[H] = j_\lambda^r \left(\frac{(s-\lambda)^r H(s)}{F_0(s)} \right). \quad (\text{PGF4.3})$$

The hypothesis defining the kernel means that the numerator is in $\mathcal{J}$, so its quotient is entire. If the representative H is changed by $G \in \mathcal{J}$, the displayed quotient changes by $(s-\lambda)^r G/F_0$, whose first r Taylor coefficients vanish. Thus the boundary is well defined. Its positive sign is the diagram chase: lift $[H]$ to $H \in \mathcal{B}$, apply $(s-\lambda)^r$, and take the resulting element of $\mathcal{J}$, without a reversal of the subtraction order.

The last map sends a polynomial class $P \in R_r$ to the class of any lift of P under $\beta_{\lambda,r}$, followed by q and then the quotient in (PGF4.2). Different lifts differ by $(s - \lambda)^r \mathcal{B}$, whose image vanishes there. It is onto because every $\mathcal{Q}$ -class has a representative in $\mathcal{B}$.

For a direct proof of every exactness claim, first suppose $\partial[H] = 0$. Then $(s - \lambda)^r H = (s - \lambda)^r G$ for some $G \in \mathcal{J}$, by (PGF2.5). Injectivity of polynomial multiplication on $\mathcal{B}$ gives $H = G$, so $[H] = 0$. Next, if $\partial[H] = P$, the jet of $(s - \lambda)^r H$ is zero, so $\mathsf{T}_r P = 0$. Conversely, if $\mathsf{T}_r P = 0$, take its unique polynomial representative of degree below r . The function

$$H_P(s) = \frac{F_0(s)P(s - \lambda)}{(s - \lambda)^r} \quad (\text{PGF4.4})$$

is entire by the vanishing condition $\mathsf{T}_r P = 0$, belongs to $\mathcal{B}$ by (PGF2.6), satisfies $(s - \lambda)^r H_P \in \mathcal{J}$, and has boundary exactly P . Finally a jet maps to zero in $C_{\lambda,r}$ exactly when a representative is the sum of an element of $\mathcal{J}$ and an element of $(s - \lambda)^r \mathcal{B}$; its jet is therefore precisely in the image of T_r . This proves (PGF4.1) without replacing the full quotient by a product of jets.

Put $m = m_\lambda$ and $d = \min(r, m)$. At the fixed parameter, write

$$F_0(\lambda + h) = h^m u_\lambda(h), \quad u_\lambda(0) = F_0^{(m)}(\lambda)/m! \neq 0. \quad (\text{PGF4.5})$$

This is a local factorization with the unit retained. Multiplication by its full truncated jet is an invertible map of R_r , with inverse the truncated Taylor series of $1/u_\lambda$. It follows that

$$\ker \mathsf{T}_r(\lambda) = h^{r-d} R_r, \quad \text{im } \mathsf{T}_r(\lambda) = h^d R_r, \quad (\text{PGF4.6})$$

where $h^r R_r = 0$. Consequently

$$\dim K_{\lambda,r} = \dim C_{\lambda,r} = d. \quad (\text{PGF4.7})$$

The actual kernel basis and its boundary are

$$q_{\lambda,k} = \left[\frac{F_0(s)}{(s - \lambda)^k} \right], \quad \partial_{\lambda,r} q_{\lambda,k} = h^{r-k}, \quad 1 \leq k \leq d. \quad (\text{PGF4.8})$$

Each representative is entire because $k \leq m$, and its membership in $\mathcal{B}$ follows from (PGF2.6). The distinct monomials prove independence. They span because those monomials span the kernel in (PGF4.6) and (PGF4.3) is an isomorphism onto that kernel. This gives the previously known generalized eigenspaces a full source-boundary comparison, including every r , rather than asserting a new spectral location theorem.

The cokernel has an equally explicit map:

$$C_{\lambda,r} \xrightarrow{\sim} R_d, \quad [[H]] \mapsto j_\lambda^d H, \quad (\text{PGF4.9})$$

where $R_0 = 0$. It is well defined since $d \leq m$ and $d \leq r$. Its kernel can be checked constructively. If $m \geq r$, a representative whose r -jet vanishes is directly divisible by $(s - \lambda)^r$ in $\mathcal{B}$. If $m < r$ and H vanishes to order at least m , its local quotient H/F_0 is holomorphic at λ . Let

$$P(h) = \text{Tay}_{h, < r-m}(H(\lambda + h)/F_0(\lambda + h)). \quad (\text{PGF4.10})$$

Then $H(s) - F_0(s)P(s - \lambda)$ vanishes to order at least r , belongs to $\mathcal{B}$, and is divisible there by $(s - \lambda)^r$. Since the subtracted term belongs to $\mathcal{J}$, the quotient class of H lies in $(L - \lambda)^r \mathcal{Q}$. The same argument with $m = 0$ shows that this map is onto all of $\mathcal{Q}$ when $\lambda \notin \mathcal{Z}$.

The map $\mathcal{Q} \rightarrow R_d$ in (PGF4.9) is continuous, since it descends from continuous jets on $\mathcal{B}$. Its kernel is thus closed; this proves closedness of $(L - \lambda)^r \mathcal{Q}$. All finite-dimensional identifications in (PGF4.1), (PGF4.8), and (PGF4.9) are topological. No density of the sum of the primal generalized eigenspaces is used or concluded.

PGF5. The boundary in the original Schwartz and Mellin sources

The exact source spaces and map from SSI and CW are

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \\ A &= \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j\}, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), \quad J = \Sigma S, \quad Q = A/J, \end{aligned} \tag{PGF5.1}$$

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \tag{PGF5.2}$$

The established exact-image theorem says that $\Sigma : S \rightarrow J$ and $\Theta : A \rightarrow \mathcal{B}$ are topological isomorphisms onto their stated targets, with $\Theta J = \mathcal{J}$. The quotient map $\kappa : Q \rightarrow \mathcal{Q}$ is $[a] \mapsto [\Theta a]$. Every factor $2, 1/2, 1/\pi$ in these maps remains explicit.

Define the actual source generators

$$D_S f(v) = -v f'(v), \quad D_A a(u) = -u a'(u). \tag{PGF5.3}$$

They preserve their indicated spaces continuously. Evenness and Schwartz estimates show this for S ; its value condition is preserved by the factor v , and integration by parts gives $\int D_S f = \int f = 0$. The weighted logarithmic-derivative seminorms prove the statement on A . Termwise differentiation of the convergent Schwartz sum, with the same endpoint estimates as SSI2, and integration by parts in the Mellin integral give

$$\Sigma D_S = D_A \Sigma, \quad \Theta D_A = L \Theta. \tag{PGF5.4}$$

All boundary terms vanish by the stated seminorms. These identities transport (PGF4.1) to the actual row $0 \rightarrow S \xrightarrow{\Sigma} A \rightarrow Q \rightarrow 0$.

For the original Gaussian source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad \Theta \Sigma f_0 = F_0, \tag{PGF5.5}$$

the kernel class in (PGF4.8) has the exact A -representative

$$a_{\lambda,k}(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} \frac{F_0(1/2 + it)}{(1/2 + it - \lambda)^k} u^{-it} dt, \quad 1 \leq k \leq \min(r, m_\lambda). \tag{PGF5.6}$$

The integrand uses its removable entire value if $1/2 + it = \lambda$; no singular integral prescription is imposed. The numerator divided by that power belongs to $\mathcal{B}$, so the established Mellin inverse estimates prove $a_{\lambda,k} \in A$.

Using (PGF5.4) gives the equality in the actual prequotient source:

$$(D_A - \lambda)^r a_{\lambda,k} = \Sigma((D_S - \lambda)^{r-k} f_0). \tag{PGF5.7}$$

Indeed the Mellin transforms of the two sides both equal $(s - \lambda)^{r-k} F_0(s)$, and Θ is injective. Thus the original-source snake boundary is

$$[a_{\lambda,k}] \mapsto [(D_S - \lambda)^{r-k} f_0] \quad \text{in } S/(D_S - \lambda)^r S. \tag{PGF5.8}$$

This proves the full map between the kernel class and its genuine Schwartz-source boundary. The representation does not assign a scalar or arithmetic operation to the support.

PGF6. Holomorphic parameter dependence and the exact family cokernel

All entries of (PGF3.3) are entire in λ . The quotient coordinates themselves form compatible holomorphic families; this can be proved directly, without assuming a holomorphic section of $\mathcal{B} \rightarrow \mathcal{Q}$.

First $\lambda \mapsto (L - \lambda)^r$ is a polynomial family of continuous operators on $\mathcal{B}$ and $\mathcal{I}$, by expansion into the finitely many continuous powers of L . The section $\sigma_{\lambda,r}^I$ is polynomial in λ with values in $\mathcal{I}$. The Gaussian section $\sigma_{\lambda,r}^B$ is entire with values in $\mathcal{B}$: for λ in any compact set, each strip seminorm of the function and each parameter derivative is bounded by a constant times a Gaussian in $\Im s$ times a fixed polynomial. Pointwise parameter Cauchy formulas can therefore be integrated in each seminorm; the corresponding Cauchy remainder estimates prove convergence of difference quotients in the Fréchet topology. Since the input space R_r is finite dimensional, this proves the assertion for the section as an operator family.

The jet maps $\alpha_{\lambda,r}$ and $\beta_{\lambda,r}$ are likewise holomorphic in λ , uniformly on bounded input sets. For β , apply Cauchy's formula to derivatives of F on a fixed larger compact set. For α , apply the same argument to the continuous map (PGF1.9). The estimate (PGF1.10) bounds the result by one input seminorm for every compact parameter set and derivative order. It also proves holomorphy on a parameter-dependent holomorphic $\mathcal{I}$ -valued family by joint evaluation and the parameter Cauchy formula.

For completeness, these give exact sheaf rows of holomorphic families. Write $\mathcal{O}_U(E)$ for holomorphic functions from an open parameter set $U \subset \mathbb{C}$ into the indicated Fréchet space E . For $E = \mathcal{B}$ use β , and for $E = \mathcal{I}$ use α . Then

$$0 \longrightarrow \mathcal{O}_U(E) \xrightarrow{(s-\lambda)^r} \mathcal{O}_U(E) \xrightarrow{j_E} \mathcal{O}_U^r \longrightarrow 0 \quad (\text{PGF6.1})$$

is exact, with the sections just constructed. To check the sole analytic divisibility issue, take a holomorphic family H_λ in the kernel. Pointwise division $H_\lambda(s)/(s-\lambda)^r$ is entire in s . In local coordinates $(\lambda, h = s - \lambda)$, the first r coefficients of $H_\lambda(\lambda + h)$ vanish identically, so Taylor division shows joint holomorphy of the quotient, including the diagonal. Estimate (PGF2.6), uniformly on compact parameter sets, bounds each output seminorm by input seminorms. Cauchy's parameter formula and remainder estimates therefore prove Fréchet holomorphy. For $E = \mathcal{I}$, (PGF2.5) additionally proves that every divided value lies in $\mathcal{I}$; since it is closed, the holomorphic family lies in that space. This proves exactness of both rows with their actual topologies.

The inclusion between the rows of (PGF6.1) induces the entire matrix map (PGF3.3). Define its coherent cokernel sheaf on the parameter plane by

$$\mathcal{C}_r = \text{coker}(\mathcal{O}_{\mathbb{C}}^r \xrightarrow{\tau_r} \mathcal{O}_{\mathbb{C}}^r). \quad (\text{PGF6.2})$$

The displayed map of sheaves is injective, although some of its fixed-parameter matrices have nonzero kernels. To prove injectivity, multiply a germ in its kernel by the adjugate matrix. Each component is then killed by the nonzero holomorphic germ $F_0(\lambda)^r$. The local holomorphic ring is an integral domain, so each component is zero. Hence

$$0 \rightarrow \mathcal{O}_{\mathbb{C}}^r \xrightarrow{\tau_r} \mathcal{O}_{\mathbb{C}}^r \rightarrow \mathcal{C}_r \rightarrow 0 \quad (\text{PGF6.3})$$

is a free resolution of the exact family cokernel. No exactness assertion for the unconstructed sheaf of holomorphic $\mathcal{Q}$ -valued lifts is needed for this statement.

Fix an actual zero ρ , put $x = \lambda - \rho$ and $m = m_\rho$, and write

$$F_0(\rho + y) = y^m u_\rho(y), \quad u_\rho(0) \neq 0. \quad (\text{PGF6.4})$$

In the local free $\mathbb{C}\{x\}$ -module $\mathbb{C}\{x\}[h]/(h^r)$, the original map is multiplication by

$$(x + h)^m u_\rho(x + h) \pmod{h^r}. \quad (\text{PGF6.5})$$

Its unit comparison is the explicit domain automorphism given by multiplication by the full jet of $u_\rho(x+h)$; its inverse uses the full jet of $1/u_\rho(x+h)$. Multiplication by $(x+h)^m$ composed with that domain automorphism is exactly the original map (PGF6.5). Thus the original unit has a stated map and inverse throughout the comparison.

The resulting cokernel is consequently

$$(\mathcal{C}_r)_\rho \cong \mathbb{C}\{x, h\}/(h^r, (x+h)^m) \cong \mathbb{C}\{y, h\}/(h^r, y^m), \quad y = x + h. \quad (\text{PGF6.6})$$

The two maps in the second isomorphism are exactly $y \mapsto x+h$ and $x \mapsto y-h$, with h fixed. Every convergent power series reduces uniquely to $\sum_{0 \leq i < m, 0 \leq j < r} a_{ij} y^i h^j$, so the length of this local parameter module is

$$\text{length}_{\mathbb{C}\{x\}}(\mathcal{C}_r)_\rho = mr. \quad (\text{PGF6.7})$$

It is supported at $x = 0$: the action of x is $y - h$, and $(y - h)^{m+r-1} = 0$. Its power $(y - h)^{m+r-2}$ is nonzero, because its only surviving monomial is

$$\binom{m+r-2}{m-1} (-1)^{r-1} y^{m-1} h^{r-1}, \quad (\text{PGF6.8})$$

with nonzero coefficient in the already recovered complex coefficient field. The length in (PGF6.7) includes these parameter-direction extensions; it must not be replaced by the dimension of a fixed fibre.

Indeed the fibre at ρ is

$$\mathcal{C}_r \otimes_{\mathcal{O}} \mathbb{C}_\rho \cong \mathbb{C}[h]/(h^r, h^m) \cong R_{\min(r,m)}. \quad (\text{PGF6.9})$$

Applying the fibre functor to the proved free resolution (PGF6.3) also gives

$$0 \rightarrow \text{Tor}_1^{\mathcal{O}}(\mathcal{C}_r, \mathbb{C}_\lambda) \rightarrow R_r \xrightarrow{\text{T}_r(\lambda)} R_r \rightarrow \mathcal{C}_r \otimes \mathbb{C}_\lambda \rightarrow 0. \quad (\text{PGF6.10})$$

This is the same finite sequence as (PGF4.1), with the explicit isomorphism from the actual primal kernel supplied by its boundary (PGF4.3). Therefore

$$K_{\lambda,r} \xrightarrow[\partial_{\lambda,r}]{} \text{Tor}_1^{\mathcal{O}}(\mathcal{C}_r, \mathbb{C}_\lambda), \quad C_{\lambda,r} \xrightarrow{\sim} \mathcal{C}_r \otimes \mathbb{C}_\lambda. \quad (\text{PGF6.11})$$

The first arrow names the kernel inclusion in (PGF6.10) and uses its fixed positive sign. These are exact maps explaining why a sheaf injection can have a noninjective specialization; the missing fibre kernel is retained as the displayed Tor group.

PGF7. Arithmetic actions and the full coefficient reflection

For every recovered positive real a , the coefficient action is

$$T_a F(s) = a^s F(s). \quad (\text{PGF7.1})$$

On each of the two jet quotients in (PGF2.7), it acts as

$$a^\lambda \sum_{j=0}^{r-1} \frac{(\log a)^j}{j!} M_h^j. \quad (\text{PGF7.2})$$

This follows by taking the full jet of $a^{\lambda+h}$; in the $\mathcal{J}$ -quotient the division by F_0 leaves exactly the same multiplier. The matrix T_r commutes with that action because both are multiplication operators in R_r . The boundary is equivariant by (PGF4.3), since T_a commutes with $(L - \lambda)^r$. On its explicit kernel basis the whole action is

$$T_a q_{\lambda,k} = a^\lambda \sum_{j=0}^{k-1} \frac{(\log a)^j}{j!} q_{\lambda,k-j}. \quad (\text{PGF7.3})$$

To verify the equality on the full quotient, subtract its right side from its left. The numerator of the difference is F_0 times the remainder after subtracting the first k Taylor coefficients of a^s at λ . That remainder is divisible by $(s - \lambda)^k$; after division the result belongs to $\mathcal{J}$, with the strip estimates following from boundedness of a^s on each fixed strip and (PGF2.6). Thus every nilpotent coefficient is retained, not only the leading character.

If the same coefficient representation is given the actual normal-degree twist, its action is aT_a ; equations (PGF7.2)–(PGF7.3) acquire exactly the factor a , equivalently $a^{\lambda+1}$ instead of a^λ . This is a receiving comparison; it does not identify $\mathcal{J}(-1)$ with $\mathcal{Q}(-1)$.

The coefficient reflection is

$$\mathcal{R}F(s) = F(1-s), \quad F_0(1-s) = F_0(s). \quad (\text{PGF7.4})$$

The identity for F_0 is the original source Fourier/Mellin identity retained in S2. Define $C_r P(h) = P(-h)$. Then

$$\beta_{1-\lambda,r}(\mathcal{R}F) = C_r \beta_{\lambda,r}(F), \quad \alpha_{1-\lambda,r}(\mathcal{R}F) = C_r \alpha_{\lambda,r}(F), \quad (\text{PGF7.5})$$

and

$$\mathbb{T}_r(1-\lambda)C_r = C_r \mathbb{T}_r(\lambda). \quad (\text{PGF7.6})$$

Each equality follows by substituting $s = 1 - \lambda + h$, so that $1 - s = \lambda - h$, retaining every power of -1 .

There is also a required sign in the snake boundary:

$$\boxed{\partial_{1-\lambda,r}(\mathcal{R}x) = (-1)^r C_r \partial_{\lambda,r}(x)}. \quad (\text{PGF7.7})$$

Indeed $(L - (1 - \lambda))^r \mathcal{R} = (-1)^r \mathcal{R}(L - \lambda)^r$, as multiplication of entire functions verifies. Substituting this identity into (PGF4.3) proves (PGF7.7). On the basis it reads

$$\mathcal{R}q_{\lambda,k} = (-1)^k q_{1-\lambda,k}, \quad (\text{PGF7.8})$$

and the two signs in (PGF7.7) give the same $(-1)^k$. These are coefficient-reflection statements. Any additional sphere-orientation or cohomological-degree sign belongs to its separately specified geometric map and has not been silently added or removed here.

PGF8. Exact information supplied to the lifting calculation

The parameter-dependent inclusion of the full primal coefficient ideals is now an explicit holomorphic matrix family, not merely a list of quotient characters. For every complex parameter and every jet order, its full entries are (PGF3.3), its actual source boundary is (PGF5.8), and its specialization kernel and cokernel are (PGF4.8)–(PGF4.9). The coherent family retains the longer parameter extension (PGF6.6), whose length is rm_ρ , together with both the Tor kernel and ordinary fibre of dimension $\min(r, m_\rho)$.

Off the actual nontrivial-zero divisor, the inclusion between the two jet quotients is an isomorphism with inverse multiplication by the complete truncated Taylor series of $1/F_0$. At an actual zero, it is the full nilpotent multiplication by $h^{m_\lambda} u_\lambda(h)$, with the unit and every derivative retained in the matrix. At $0, 1, -2, -4, \dots$, the nonzero exceptional values (PGF1.6)–(PGF1.7) remain in this map. This describes precisely the divisor detected by this source construction; the original zeta's endpoint pole and trivial zeros still require their full comparison factors, already displayed in (PGF1.5) and (PGF3.6).

The family does not place every actual zero on a vertical line and does not turn a quotient character into an eigenvector of $\mathcal{J}$. The calculation also does not claim the requested Deligne weight separation for the whole obstruction target $\mathcal{J}(-1)$. It supplies exact primal maps which any such calculation must use, while keeping the already established zero boundary of the actual sheaf lift distinct from the generally nonzero boundary (PGF4.3).

PGF9. Independent internal checks of the formulas

These checks are symbolic identities for arbitrary r, λ, m_λ , not bounded numerical tests.

1. Both maps in (PGF2.7) are proved with explicit continuous sections and an exact kernel calculation; there is no appeal to an unspecified quotient topology.
2. The matrix (PGF3.3) is the full product $\text{jet } j^r F_0 j^r(F/F_0)$. Its determinant, local unit, and all exceptional-point contributions are retained; the transported product on $\mathcal{J}$ is separately recorded in (PGF2.8).

3. For every boundary basis vector, multiplication by $(s - \lambda)^r$ gives exactly $F_0(s)(s - \lambda)^{r-k}$. Division by F_0 therefore gives h^{r-k} , fixing the boundary sign and verifying its actual Schwartz representative.
4. At $m_\lambda = 0$, the matrix is invertible and both primal kernel and cokernel vanish. At $0 < m_\lambda < r$, they each have dimension m_λ . At $m_\lambda \geq r$, the matrix specialization is zero and both have dimension r . These exhaust all cases without assuming zero simplicity.
5. The sheaf map is injective because its determinant is a nonzero holomorphic germ, while its fibre kernel is exactly the Tor group in (PGF6.10). Thus no interchange of a sheaf kernel and its fibre has occurred.
6. The local parameter length rm_ρ , fibre dimension $\min(r, m_\rho)$, and nilpotence exponent $m_\rho + r - 1$ follow from the same explicit module (PGF6.6); they have not been identified with one another.
7. Reflection changes the parameter by $\lambda \mapsto 1 - \lambda$, the jet coordinate by $h \mapsto -h$, and the boundary by the additional factor $(-1)^r$. This is checked independently on every kernel basis vector in (PGF7.8).

No new purity statement, positivity statement, or conclusion about RH is asserted by these identities.

The exact original-zeta map from the Weil trace to the completed residue receiver

24 September 2026. Independent derivation, RTT0–RTT11.

RTT0. Source objects, prerequisites, and conventions

The support remains the user's $\tau\langle Z_1; \text{no } Z_2 \rangle$. The complete arithmetic reconstruction precedes the coefficient spaces and operations below. Addition at τ remains retracted. No number, parity, metric, midpoint, or additional source operation is assigned to that support. The corpus-and-operation rule was read before this derivation. The new workflow rule is implemented here by constructing the actual comparison exposed by the previous distinction between residue duality and the Weil trace.

The complete GTR0–GTR11 proof, RD0–RD9, and the relevant GIQ2, GIQ3, GIQ8 and GIQ9 calculations were read. SDT5–SDT7 was reread for the strong topology and bidual step. GTR10 supplies a proved topological isomorphism, rather than a proposed self-duality:

$$\hat{\iota} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} \mathcal{Q}'_{\beta}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad (\text{RTT0.1})$$

where

$$\mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for every } A, M\},$$

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of } \zeta, \ 0 \leq j < m_{\rho}\}. \quad (\text{RTT0.2})$$

Here m_{ρ} is the full multiplicity. The completion in (RTT0.1) is the completion of $\mathcal{Q}_{\text{fin}}$ for the residue seminorms, not the original topology of $\mathcal{Q}$. On its dense finite-support subspace,

$$\iota h(F) = \mathcal{R}(F, h) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)h(1-s)}{\zeta(s)} ds. \quad (\text{RTT0.3})$$

The denominator is original ζ . RD1 constructs the global isolators $e_{\rho,j}$, whose full jet at ρ is $(s - \rho)^j$ and whose required jets at every other zero vanish. No density of their span in the original $\mathcal{Q}$ is asserted or used here.

Use the following explicit convention for the Weil form: linear in its first argument and conjugate-linear in its second,

$$W(F, G) = \sum_{\rho} m_{\rho} F(\rho) \overline{G(1 - \bar{\rho})}. \quad (\text{RTT0.4})$$

This is GIQ's form with the two arguments exchanged. Since that form is Hermitian, the change of convention does not change any diagonal value. The exact maps proving its continuity and identifying its radical are given again below in the present $\mathcal{B}$ coordinate.

RTT1. The global value map and the full strong-dual functional

Let

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_{\rho} m_{\rho} x_{\rho} \overline{y_{\rho}}, \quad \rho^{\#} = 1 - \bar{\rho}, \quad (Jx)_{\rho} = x_{\rho^{\#}}. \quad (\text{RTT1.1})$$

The functional equation and complex conjugation preserve multiplicities, so J is a self-adjoint unitary involution. Every zero here is in the entire actual nontrivial divisor, not a numerical selection.

The map

$$E : \mathcal{Q} \longrightarrow H, \quad E[F] = (F(\rho))_{\rho} \quad (\text{RTT1.2})$$

is well defined and continuous. Indeed, the unconditional zero count $\sum_{|\Im \rho| \leq T} m_{\rho} = O(T \log(T + 2))$, already used in GIQ2, implies

$$\|E[F]\|_+^2 \leq b_{1,2}(F)^2 \sum_{\rho} m_{\rho} (1 + |\Im \rho|)^{-4} < \infty. \quad (\text{RTT1.3})$$

The corresponding map on $\mathcal{B}$ kills $\mathcal{I}$, so the quotient topology makes (RTT1.2) continuous. RD1's isolator $e_{\rho,0}$ maps to the coordinate vector at ρ . Therefore $E(\mathcal{Q})$ contains every finitely supported value sequence and is dense in H . This proves density of the value image only; it does not assert density of $\mathcal{Q}_{\text{fin}}$ in $\mathcal{Q}$.

Now

$$W(F, G) = \langle EF, JEG \rangle_+. \quad (\text{RTT1.4})$$

In particular its full sum converges absolutely by Cauchy–Schwarz. For a bounded set $B \subset \mathcal{Q}$, put $c_B = \sup_{F \in B} \|EF\|_+ < \infty$. Then

$$\sup_{F \in B} |W(F, G)| \leq c_B \|EG\|_+. \quad (\text{RTT1.5})$$

Consequently

$$W : \mathcal{Q} \longrightarrow \mathcal{Q}'_\beta, \quad W(G)(F) = W(F, G) \quad (\text{RTT1.6})$$

is a continuous anti-linear map. Its exact kernel is

$$\mathcal{N}_0 = \{[G] \in \mathcal{Q} : G(\rho) = 0 \text{ for every } \rho\}. \quad (\text{RTT1.7})$$

One containment follows from the formula; conversely testing against $e_{\rho,0}$ detects $m_\rho \overline{G(\rho^\#)}$. Thus this is exactly the higher-jet radical already present in GIQ2, not a discarded multiplicity block.

RTT2. A continuous original-zeta derivative multiplier on the full quotient

Set

$$(CG)(s) = \overline{G(\bar{s})}, \quad H_\zeta(s) = s^2 \zeta'(1-s). \quad (\text{RTT2.1})$$

The first map is anti-linear, isometric for the displayed $b_{A,M}$, and preserves $\mathcal{I}$ because complex conjugation preserves the full original zero divisor. The second function is entire: the only pole of $\zeta'(1-s)$ is the double pole at $s = 0$. Since the residue of original ζ at 1 is 1,

$$\zeta'(1-s) = -s^{-2} + \text{a holomorphic germ at } 0, \quad H_\zeta(0) = -1. \quad (\text{RTT2.2})$$

This original pole and its exact coefficient are retained.

For completeness, H_ζ has at most polynomial growth on every bounded real strip. An elementary exact estimate follows from Euler–Maclaurin, with all its terms retained:

$$\zeta(z) = \frac{1}{z-1} + \frac{1}{2} + \sum_{k=1}^M \frac{B_{2k}}{(2k)!} (z)_{2k-1} - \frac{(z)_{2M}}{(2M)!} \int_1^\infty \widetilde{B}_{2M}(x) x^{-z-2M} dx, \quad \Re z > 1 - 2M. \quad (\text{RTT2.3})$$

Here $(z)_r = z(z+1)\cdots(z+r-1)$, with $(z)_0 = 1$, and $\widetilde{B}_{2M}(x) = B_{2M}(x - [x])$. The formula follows by applying integration by parts successively on integer intervals to x^{-z} , using $B'_j = jB_{j-1}$, summing the boundary terms, and taking the upper endpoint to infinity first on $\Re z > 1$. The displayed integral converges on the larger stated half-plane and hence gives the analytic continuation there. At the lower endpoint the even Bernoulli boundary values give precisely the displayed rising-factorial terms; the remainder retains its minus sign.

Choose M so that a prescribed strip satisfies $\Re z + 2M - 1 \geq \epsilon > 0$. Boundedness of the periodic polynomial gives bounds for the integral and its first derivative by a constant times ϵ^{-1} and ϵ^{-2} , respectively; the derivative contributes $-\log x$ under the integral. Differentiating (RTT2.3) therefore proves a polynomial bound for $\zeta'(z) + (z-1)^{-2}$ on that strip. Substitution $z = 1-s$, multiplication by s^2 , and (RTT2.2) prove the assertion for H_ζ . Thus for each A there are finite d_A, C_A with

$$b_{A,M}(H_\zeta F) \leq C_A b_{A,M+d_A}(F). \quad (\text{RTT2.4})$$

Multiplication by H_ζ is consequently a continuous endomorphism of $\mathcal{B}$ preserving $\mathcal{I}$.

Retain the actual isolator source

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_0(0) = F_0(1) = \frac{1}{8}. \quad (\text{RTT2.5})$$

Its complete factors are used here to construct an inverse to $L[F] = [sF]$, not to replace original ζ . Define

$$(D_0 F)(s) = \frac{F(s) - F_0(s)F(0)/F_0(0)}{s}. \quad (\text{RTT2.6})$$

The numerator vanishes at zero. Outside a fixed disk, division by s has the required polynomial bound; inside the disk Cauchy's formula bounds the removable quotient on a smaller circle by the numerator on a larger circle. This proves $D_0 : \mathcal{B} \rightarrow \mathcal{B}$ continuous. It preserves $\mathcal{I}$, because no nontrivial zero is zero and the numerator retains all required vanishing orders. Moreover

$$L[D_0 F] = [F], \quad [D_0(sF)] = [F]. \quad (\text{RTT2.7})$$

The first identity uses $F_0 \in \mathcal{I}$; the second uses $(sF)(0) = 0$. Thus the induced operator is the unique continuous inverse L^{-1} . No choice of isolator changes this quotient inverse.

The resulting actual primal map is

$$\boxed{\mathbf{P} = L^{-2} M_{H_\zeta} \mathbf{C} : \mathcal{Q} \longrightarrow \mathcal{Q}.} \quad (\text{RTT2.8})$$

It is continuous and anti-linear. At every nontrivial zero its full jet is exactly the jet of

$$\zeta'(1-s)\overline{G(\bar{s})}. \quad (\text{RTT2.9})$$

Indeed the local inverse of L^2 is multiplication by s^{-2} , a holomorphic unit on each such germ. Formula (RTT2.8) proves existence on the entire quotient even though the uncorrected global expression (RTT2.9) has a pole at zero. The double-pole correction and the explicit inverse are not omitted.

RTT3. The exact local rank and every nilpotent coordinate

Fix an output zero σ , put $\rho = 1 - \sigma$, $\eta = \bar{\sigma}$, and $m = m_\sigma = m_\rho = m_\eta$. In the retained local parameter $t = s - \sigma$, write

$$\zeta(\rho + v) = v^m u_\rho(v), \quad u_\rho(0) = \frac{\zeta^{(m)}(\rho)}{m!} \neq 0. \quad (\text{RTT3.1})$$

Direct differentiation before taking a jet gives

$$\zeta'(\rho - t) = m(-t)^{m-1} u_\rho(-t) + (-t)^m u'_\rho(-t). \quad (\text{RTT3.2})$$

Multiplying the entire Taylor series of CG , and reducing only modulo the actual defining ideal (t^m) , gives

$$j_\sigma(\text{PG}) = m(-1)^{m-1} u_\rho(0) \overline{G(\eta)} t^{m-1}. \quad (\text{RTT3.3})$$

Every omitted term in this displayed quotient calculation has degree at least m ; the full original expression is (RTT3.2). Thus the anti-linear block map from the input block at η to the output block at $\bar{\eta}$ has rank exactly one. Its kernel is $t_\eta \mathbb{C}[t_\eta]/(t_\eta^m)$, of complex dimension $m - 1$, and its image is the highest-jet line $\mathbb{C} t_\sigma^{m-1}$. These statements include $m = 1$: then the rank is one, the kernel is zero, and the coefficient is the original $\zeta'(\rho)$.

All coefficients are nonzero on their indicated one-dimensional maps. Hence

$$\ker \mathbf{P} = \mathcal{N}_0. \quad (\text{RTT3.4})$$

This follows globally because membership in $\mathcal{I}$ is exactly vanishing of every full required jet. For $m > 1$ the image is in the local socle, the annihilator of the local maximal ideal; this is a statement about the target line, not the deletion of that maximal ideal from the original quotient. The complete residue matrix in RD3 remains invertible on all m jets. The rank loss belongs to $\mathbf{P}$, the specific map from the Weil trace to that full residue receiver.

RTT4. The finite-support identity retains the original logarithmic derivative

For $G \in \mathcal{Q}_{\text{fin}}$, (RTT3.3) shows $\mathbf{P}G \in \mathcal{Q}_{\text{fin}}$. For arbitrary $F \in \mathcal{Q}$, use the exact germs (RTT2.9) in the original residue:

$$\begin{aligned}\mathcal{R}(F, \mathbf{P}G) &= \sum_{\rho} \text{Res}_{s=\rho} F(s) \overline{G(1-\bar{s})} \frac{\zeta'(s)}{\zeta(s)} ds \\ &= \sum_{\rho} m_{\rho} F(\rho) \overline{G(1-\bar{\rho})} = W(F, G).\end{aligned}\tag{RTT4.1}$$

The sum is finite in this section. At a multiplicity- m zero,

$$\frac{\zeta'(\rho+t)}{\zeta(\rho+t)} = \frac{m}{t} + \frac{u'_{\rho}(t)}{u_{\rho}(t)}.\tag{RTT4.2}$$

The second term is holomorphic and therefore contributes zero residue. This proves why the trace keeps precisely the constant observations and the multiplicity m , while the residue pairing alone sees the whole jet algebra. No derivative factor was set equal to one.

RTT5. The full completed map, without a primal-density assumption

Define

$$\boxed{\mathbf{S} = \hat{\iota}^{-1}\mathbf{W} : \mathcal{Q} \longrightarrow \mathcal{H}_{\text{res}}}.\tag{RTT5.1}$$

It is continuous and anti-linear by (RTT0.1) and (RTT1.5). It satisfies, on its entire domain,

$$\widehat{\mathcal{R}}(F, \mathbf{S}G) = W(F, G), \quad \ker \mathbf{S} = \mathcal{N}_0.\tag{RTT5.2}$$

Here the completed residue is evaluation as defined in GTR10. Formula (RTT4.1) proves that $\mathbf{S}G$ is the canonical image of $\mathbf{P}G$ in the completion whenever G is in $\mathcal{Q}_{\text{fin}}$. The assertion on all of $\mathcal{Q}$ follows from the direct construction (RTT5.1), not an extension by unproved density.

There is also an explicit convergent residue construction for every G . Direct the finite subsets $E \subset \mathcal{Z}$ by inclusion and put

$$h_E(G) = \sum_{\rho \in E} m_{\rho} (-1)^{m_{\rho}-1} u_{\rho}(0) \overline{G(\rho^{\#})} e_{1-\rho, m_{\rho}-1}.\tag{RTT5.3}$$

By RD3's exact matrix, or by substituting $(s - \rho) \mapsto -(s - \rho)$,

$$\iota h_E(G)(F) = \sum_{\rho \in E} m_{\rho} F(\rho) \overline{G(\rho^{\#})}.\tag{RTT5.4}$$

For every bounded $B \subset \mathcal{Q}$, Cauchy–Schwarz gives

$$\sup_{F \in B} |W(F, G) - \iota h_E(G)(F)| \leq c_B \left(\sum_{\rho \notin E} m_{\rho} |G(\rho^{\#})|^2 \right)^{1/2} \longrightarrow 0.\tag{RTT5.5}$$

This is convergence in the full strong dual. By (RTT0.1), $h_E(G)$ converges in the specified residue completion to $\mathbf{S}G$. It treats every actual zero and all multiplicities; the finite sets are an exact convergent net, not a bounded numerical argument.

Equations (RTT3.3) and (RTT5.3) also give a precise relationship to the primal map. If $\mathbf{P}G = \mathbf{P}K$, then $G - K \in \mathcal{N}_0$, so $\mathbf{S}G = \mathbf{S}K$. There is consequently a well-defined injective linear map

$$\mathbf{j}_{\text{tr}} : \mathbf{P}(\mathcal{Q}) \longrightarrow \mathcal{H}_{\text{res}}, \quad \mathbf{j}_{\text{tr}}(\mathbf{P}G) = \mathbf{S}G.\tag{RTT5.6}$$

Linearity follows because both $\mathbf{P}$ and $\mathbf{S}$ are anti-linear. Give $\mathbf{P}(\mathcal{Q})$ the quotient topology transported from $\mathcal{Q}/\mathcal{N}_0$ by $\mathbf{P}$; then $\mathbf{j}_{\text{tr}}$ is continuous by (RTT5.1). This specifies its topology. No assertion that this quotient topology is the subspace topology from the original $\mathcal{Q}$, or that every primal class embeds in the residue completion, is needed.

RTT6. The exact closed receiver of the trace

The closure of the trace image in the full strong dual is

$$\overline{W(\mathcal{Q})}^\beta = \mathcal{N}_0^\perp \subset \mathcal{Q}'_\beta. \quad (\text{RTT6.1})$$

To prove this, each $W(G)$ annihilates $\mathcal{N}_0$, so the closure is contained in the closed annihilator. Conversely, $W(\mathcal{Q})$ contains each value functional $\delta_{\rho,0}$ up to its nonzero multiplicity scalar, by choosing $G = e_{\rho^\#,0}$. Let D_0 be their linear span. SDT6 proves that every continuous linear functional on $\mathcal{Q}'_\beta$ is evaluation at an actual $F \in \mathcal{Q}$. If a member $\lambda \in \mathcal{N}_0^\perp$ were outside the strong closure of D_0 , locally convex separation would give such an F with every $\delta_{\rho,0}(F) = 0$ but $\lambda(F) \neq 0$. The former says $F \in \mathcal{N}_0$, contradicting the latter. This proves (RTT6.1).

Accordingly the exact closed subreceiver inside $\mathcal{H}_{\text{res}}$ is $\widehat{\iota}^{-1}(\mathcal{N}_0^\perp)$. It retains all zero values. If any actual zero has multiplicity at least two, then $\mathcal{N}_0 \neq 0$, as witnessed by $e_{\rho,1}$, and this closed subreceiver is proper because $\delta_{\rho,1}$ does not annihilate that witness. This statement does not postulate the existence of a multiple zero. It identifies exactly what the two different receivers record for every multiplicity allowed by the original divisor.

The full positive value completion also has an actual map into the residue receiver:

$$A : H \longrightarrow \mathcal{H}_{\text{res}}, \quad (\widehat{\iota}Ay)(F) = \langle EF, Jy \rangle_+, \quad S = AE. \quad (\text{RTT6.2})$$

This map is anti-linear and continuous from the Hilbert norm to the residue strong topology by the bound $c_B \|y\|_+$. It is injective because $E(\mathcal{Q})$ is dense: if all pairings with Jy vanish, then $Jy = 0$, hence $y = 0$. In particular the trace-to-residue map does not collapse a nonzero value direction, even though it kills the separate higher-jet radical in its original source. For a bounded operator $D : H \rightarrow H$, the map AD is therefore zero exactly when $D = 0$. This proves an exact detection statement for the positive-adjoint defect calculated in GTAH, without asserting a lift of that possibly nonholomorphic multiplier to $\mathcal{Q}$, or giving $A(H)$ an unproved subspace-topology inverse.

RTT7. Dilation, generator, and the actual transfer

On the original quotient retain

$$T_a[F] = [a^s F(s)], \quad L[F] = [sF(s)], \quad a > 0. \quad (\text{RTT7.1})$$

The quotient map P commutes with T_a : C commutes with this multiplier because a is positive real, and the other operators in (RTT2.8) are commuting multipliers and the inverse of L . It also commutes with L . These are anti-linear commutation statements with the indicated real parameter; no complex scalar is moved through P without conjugation.

The continuous dual map has the different, exact action

$$\boxed{WT_a = a(T_{a^{-1}})'W, \quad WL = (1 - L')W.} \quad (\text{RTT7.2})$$

For the first identity, the multiplier in $W(F, T_a G)$ is $\overline{a^{1-\bar{\rho}}} = a^{1-\rho}$, which is exactly the multiplier obtained by applying $aT_{a^{-1}}$ to F . For the second it is $1 - \bar{\rho} = 1 - \rho$. Absolute convergence follows as before; the extra factor ρ is controlled by one additional defining strip seminorm. Thus no differentiation of a merely weakly convergent sum is used.

GTR10 proves $\widehat{\iota}T_a = a(T_{a^{-1}})'\widehat{\iota}$. Therefore (RTT7.2) is equivalent to

$$\boxed{ST_a = \widehat{T}_a S.} \quad (\text{RTT7.3})$$

The generator on the residue completion is explicitly $\widehat{L} = \widehat{\iota}^{-1}(1 - L')\widehat{\iota}$, a continuous operator extending the finite-support residue generator. Hence $SL = \widehat{L}S$. Transposition of a continuous operator is strong-continuous because it maps each bounded set to a bounded set; this verifies continuity of the displayed operator.

For recovered integer covers, set $U_n = nT_{n-1}$ on actual degree-one cohomology. GTR2–GTR4 construct it from the weighted ramified sheaf trace and inverse coefficient action. Retaining $\kappa : Q_A = A/J \rightarrow Q$, $\kappa[a] = [\frac{1}{2} \int_0^\infty a(u)u^s du/u]$, the actual receiving map is

$$S_A = S\kappa : Q_A \longrightarrow \mathcal{H}_{\text{res}}, \quad \mathfrak{J} = \kappa' \hat{t} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} Q'_{A,\beta}. \quad (\text{RTT7.4})$$

It obeys

$$\mathfrak{J} S_A T_n = U'_n \mathfrak{J} S_A. \quad (\text{RTT7.5})$$

All domains and the Mellin factor $1/2$ are retained. The two pole restrictions are still $h \mapsto (h, -h)$, so composing (RTT7.4) gives $G \mapsto (S_A G, -S_A G)$. No supported pole is identified with the other one.

RTT8. The geometric transfer and the Hilbert adjoint have a proved comparison

The Weil form has the exact adjoint identity

$$W(T_a F, G) = W(F, aT_{a^{-1}} G). \quad (\text{RTT8.1})$$

Indeed the multiplier on the right is $\overline{a^{1-\rho^\#}} = a^\rho$, the multiplier on the left. This is an adjoint with respect to the specified possibly indefinite and possibly degenerate Weil form. On $Q/\mathcal{N}_0$ it is nondegenerate; on the full Q , an adjoint identity alone cannot determine changes whose image is in $\mathcal{N}_0$. The chosen full operator in (RTT8.1) is nevertheless already fixed by the geometric construction and has all its higher jets.

On the positive auxiliary space H , write

$$(D_a x)_\rho = a^\rho x_\rho, \quad (U_a x)_\rho = a^{1-\rho} x_\rho. \quad (\text{RTT8.2})$$

These are bounded operators, since $0 < \Re \rho < 1$. Direct evaluation in $\langle \cdot, \cdot \rangle_+$ gives

$$(D_a^* x)_\rho = a^{\bar{\rho}} x_\rho, \quad \boxed{U_a = J D_a^* J}. \quad (\text{RTT8.3})$$

The second identity follows from $1 - \rho^\# = \bar{\rho}$. Thus the constructed geometric transfer receives the adjoint of pullback for the actual Weil form, and its relation to the positive Hilbert adjoint is the explicit conjugation (RTT8.3). The ζ' multiplier in the trace-to-residue map does not remove that involution; (RTT5.2) proves that it recovers exactly the old Weil values.

This answers what additional comparison the residue bridge makes available: any proposed change of the positive adjoint can now be transported through S and tested by (RTT7.2)–(RTT8.3) on the same original quotient. One cannot replace U_a by D_a^* merely because a residue pairing is nondegenerate. Their exact relation, including J , has been calculated here; further positivity work must use it rather than assume that it is the identity relation.

RTT9. Original functional-equation factors and the full arithmetic form

No completed zeta replaces original ζ in this construction. The complete original functional equation is

$$\zeta(s) = \chi(s) \zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{RTT9.1})$$

Its exact differentiated identity is

$$\zeta'(1-s) = \frac{\chi'(s)}{\chi(s)^2} \zeta(s) - \frac{1}{\chi(s)} \zeta'(s). \quad (\text{RTT9.2})$$

At a nontrivial zero, χ is a holomorphic unit. The first term vanishes to the full original multiplicity; hence the full jet of (RTT2.9) can also be computed from the second term, retaining its minus sign and unit. Equation (RTT9.2), rather than a removal of those factors, proves that local quotient comparison. Globally

the definition remains (RTT2.8), and it does not require a meromorphic multiplier involving χ^{-1} to preserve $\mathcal{B}$.

The use of F_0 in (RTT2.5) retains the endpoint values and all its original trivial-zero values

$$F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{RTT9.3})$$

They are not nontrivial-zero coordinates in $\mathcal{Q}$; their role in the full source reconstruction and the four endpoint lines remains RD8 and SSI.

More directly, (RTT5.2) is an equality with the existing complete original arithmetic form, not a newly selected sign form. For $A_{F,G}(s) = F(s)G(1-\bar{s})$, choose the logarithmic test h with $A_{F,G}(s) = \int_{\mathbb{R}} h(v) e^{-(s-1/2)v} dv$, whose existence in the full test space follows by the Mellin comparison and convolution proof of GIQ9. Retain $\hat{h}(t) = \int h(v) e^{-itv} dv$. Then

$$\widehat{\mathcal{R}}(F, SG) = A_{F,G}(0) + A_{F,G}(1) + A_{\infty}(h) - P_{\text{hist}}(h), \quad (\text{RTT9.4})$$

where every original term is

$$P_{\text{hist}}(h) = \sum_{n \geq 2} \frac{\log L_n - \log L_{n-1}}{\sqrt{n}} (h(\log n) + h(-\log n)), \quad L_n = \text{lcm}(1, \dots, n),$$

$$A_{\infty}(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt. \quad (\text{RTT9.5})$$

The coefficient $\log L_n - \log L_{n-1} = \Lambda(n)$ retains all prime-power repetitions from the complete history. For each finite trivial-zero cutoff B , the raw divisor identity is also retained:

$$V_{\zeta,B} = W(F, G) + \sum_{r=1}^B A_{F,G}(-2r) - A_{F,G}(1),$$

$$G_B = A_{\infty}(h) + A_{F,G}(0) + \sum_{r=1}^B A_{F,G}(-2r), \quad V_{\zeta,B} = G_B - P_{\text{hist}}(h). \quad (\text{RTT9.6})$$

No divergent infinite sum over trivial zeros is introduced. These equalities follow from the fully proved GIQ9 identity and (RTT5.2). The same support coefficients therefore carry exactly the same arithmetic compensation before and after applying the new map.

RTT10. Labels and the coefficient extension

Let K be the complex vector space of the retained formal coefficient or support records, or their already constructed algebraic tensor product. Because $\mathbf{P}$ and $\mathbf{S}$ are anti-linear, an ordinary tensor with 1_K must first have its domain typed correctly. Write $\overline{\mathcal{Q}}$ for the conjugate vector space, with

$$\lambda \cdot \bar{q} = \overline{\lambda q}.$$

The maps

$$\tilde{\mathbf{P}} : \overline{\mathcal{Q}} \rightarrow \mathcal{Q}, \quad \bar{q} \mapsto \mathbf{P}q, \quad \tilde{\mathbf{S}} : \overline{\mathcal{Q}} \rightarrow \mathcal{H}_{\text{res}}, \quad \bar{q} \mapsto \mathbf{S}q \quad (\text{RTT10.1})$$

are complex-linear: for example $\mathbf{S}(\bar{\lambda}q) = \lambda \mathbf{S}q$. Thus their tensor extensions with 1_K have domain $\overline{\mathcal{Q}} \otimes_{\mathbb{C}} K$. Balancing holds explicitly:

$$(\tilde{\mathbf{S}} \otimes 1_K)((\lambda \bar{q}) \otimes k) = \lambda \mathbf{S}q \otimes k = \mathbf{S}q \otimes \lambda k = (\tilde{\mathbf{S}} \otimes 1_K)(\bar{q} \otimes \lambda k). \quad (\text{RTT10.2})$$

The identical argument applies to $\tilde{\mathbf{P}}$. A finite expansion in a basis of K proves that both kernels are exactly $\overline{\mathcal{N}_0} \otimes_{\mathbb{C}} K$. Every record is unchanged.

There is also an explicitly anti-linear presentation on $\mathcal{Q} \otimes_{\mathbb{C}} K$ if the specified formal/support basis $(k_b)_b$ is retained. Define $C_K(\sum_b c_b k_b) = \sum_b \bar{c}_b k_b$; this fixes each label and conjugates only its complex coefficient. Then

$$q \otimes k \mapsto Sq \otimes C_K k \quad (\text{RTT10.3})$$

is well defined and anti-linear. Indeed the two representatives $(\lambda q) \otimes k$ and $q \otimes \lambda k$ both map to $\bar{\lambda}Sq \otimes C_K k$. Its kernel is $\mathcal{N}_0 \otimes_{\mathbb{C}} K$ by the same basis argument. The formula with P has the same properties. This second presentation specifies the needed coefficient conjugation instead of implicitly identifying $\bar{\mathcal{Q}}$ with $\mathcal{Q}$.

Neither presentation identifies $[\tau]$ with $[1]$ or either support branch with the other. No multiplication or addition of the source τ is inferred from operations on its already constructed coefficient receiver.

RTT11. Exact outcome and receiving calculation

There are now two proved maps with different codomains:

$$P = L^{-2} M_{s^2 \zeta'(1-s)} C : \mathcal{Q} \rightarrow \mathcal{Q}, \quad S = \hat{\iota}^{-1} W : \mathcal{Q} \rightarrow \mathcal{H}_{\text{res}}. \quad (\text{RTT11.1})$$

Both are continuous and anti-linear, and both have exactly the full value-zero kernel $\mathcal{N}_0$. Their finite-support comparison and full strong limit are (RTT4.1) and (RTT5.3)–(RTT5.6). Locally the exact original ζ' coefficient maps each multiplicity block onto its conjugate block's highest-jet line. Globally the completed residue pairing of SG is exactly the original full Weil pairing of G , including every arithmetic, Gamma, endpoint and finite trivial-divisor contribution.

The receiving action is the actually constructed geometric transfer transpose, (RTT7.5); the relation to the positive Hilbert adjoint is (RTT8.3). Those maps make a further positivity calculation concrete without changing the original Weil values or assuming purity. The comparison itself establishes neither the absence of off-line zeros nor a new positivity theorem.

Proof-source paths actually used

- ORIGINAL_ZETA_RESIDUE_DUAL_AND_SUPPORT_MAPS.md, RD0–RD9: entire quotient, global isolators, original residue matrices, full functional-equation units, sheaf comparison and original exceptional terms.
- CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md, GTR0–GTR11: actual weighted trace, inverse coefficient action, strong completion, signed support maps and derived transfer.
- GLOBAL_INFINITESIMAL_QUOTIENT.md, GIQ2–GIQ3 and GIQ8–GIQ9: value form, nilpotent radical, actual dilation and complete original arithmetic identity.
- ../tau_weight_cohomology_20260924/CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md, SDT5–SDT7: exact strong topology, strong bidual and separation argument. The completed residue is used through GTR10's already proved topological isomorphism.

No numerical test, simplicity assumption, finite-support primal-density claim, or new publication receipt occurs in this derivation.

Geometric transfer, the positive adjoint, and the exact reflection-average defect

24 September 2026. Proof locators GTA0–GTA7. This carries out the next attempt requested by the user’s rule: after identifying what a result has not established, ask what would establish it using the entire known programme, and try that construction.

GTA0. The actual objects already constructed

All arithmetic and coefficient operations below follow the complete arithmetic reconstruction. The supporting datum remains $\tau\langle Z_1; \text{no intrinsic } Z_2 \rangle$; none of these operations defines addition, coordinates, or a distance on that datum. The complete source quotient $\mathcal{Q} = \mathcal{B}/\mathcal{I}$, its full multiplicity jets and all its original-zeta factors remain as in SSI, RD, GCF and PGF. This calculation does not replace that quotient by its value observation.

GIQ2 proves the continuous value map with dense image

$$E_0 : \mathcal{Q} \longrightarrow \mathcal{H}, \quad E_0[F] = (F(\rho))_{\rho \in \mathcal{Z}}, \quad \mathcal{H} = \ell^2(\mathcal{Z}, m_\rho), \quad (\text{GTA0.1})$$

where $\mathcal{Z}$ is the entire set of distinct nontrivial zeros of the original ζ , and m_ρ is its actual multiplicity. Its kernel is exactly the ideal of classes with zero values at every zero; higher jets remain in this kernel and in the source quotient. The topology here is the stated positive value norm, not a new topology silently assigned to $\mathcal{Q}$. RD1 supplies explicit representatives with a prescribed full jet at one zero and every required vanishing jet at all other zeros. Thus finite value vectors actually lie in the image, which also proves its density after the SSI identification.

Use the inner product linear in its first variable,

$$\langle x, y \rangle = \sum_{\rho} m_{\rho} x_{\rho} \overline{y_{\rho}}. \quad (\text{GTA0.2})$$

The original functional equation gives the involution $\rho^{\#} = 1 - \bar{\rho}$, with $m_{\rho^{\#}} = m_{\rho}$. The operator

$$(\mathbf{J}x)_{\rho} = x_{\rho^{\#}}, \quad \mathbf{J}^2 = I, \quad \mathbf{J}^* = \mathbf{J} \quad (\text{GTA0.3})$$

is a unitary involution on this complete value space. The full original Weil form, including its original arithmetic expression in GIQ9, is

$$W(x, y) = \langle x, \mathbf{J}y \rangle = \sum_{\rho} m_{\rho} x_{\rho} \overline{y_{\rho^{\#}}}. \quad (\text{GTA0.4})$$

GIQ9 proves the receiving map from the original tests and the complete prime, Gamma, endpoint and finite trivial-zero formulas. Only the inner-product convention changes here: GIQ uses conjugate linearity in the first variable; (GTA0.4) is its complex conjugate. No sign changes in quadratic values result.

The arithmetic dilation and the actual degree-weighted transfer on this receiver are

$$(\mathbf{T}_a x)_{\rho} = a^{\rho} x_{\rho}, \quad \mathbf{U}_a = a \mathbf{T}_{1/a}, \quad a > 0. \quad (\text{GTA0.5})$$

For recovered positive integers $a = n$, the factor n is the actual cover degree proved in GTR0–GTR11, including the multiplicities at both poles. For general $a > 0$ these are coefficient operators; no nonintegral sphere cover is asserted. The known strip $0 < \Re \rho < 1$ makes both operators bounded, with bounded inverses. Direct multiplication gives

$$\mathbf{U}_a \mathbf{T}_a = \mathbf{T}_a \mathbf{U}_a = aI, \quad \mathbf{T}_a \mathbf{T}_b = \mathbf{T}_{ab}, \quad \mathbf{U}_a \mathbf{U}_b = \mathbf{U}_{ab}. \quad (\text{GTA0.6})$$

The original coefficient maps commute with E_0 . These statements hold on the full actual zero set, without a zero-height cutoff or simplicity assumption.

GTAH1. What would give the desired modulus, and the direct calculation

The existing geometric degree identity would give the common modulus if its transfer were the adjoint for the actual positive value norm. The direct calculation of that adjoint is

$$(\mathsf{T}_a^* x)_\rho = a^{\bar{\rho}} x_\rho, \quad \mathsf{J} \mathsf{U}_a \mathsf{J} = \mathsf{T}_a^*. \quad (\text{GTAH1.1})$$

Indeed substitution in (GTAH0.2) proves the first equality. For the second, the multiplier on the left is $a^{1-\rho^\#} = a^{\bar{\rho}}$. No use of a conjectural real part is made.

For the actual Weil form the transfer already is the adjoint:

$$W(\mathsf{T}_a x, y) = W(x, \mathsf{U}_a y), \quad W(\mathsf{T}_a x, \mathsf{T}_a y) = a W(x, y). \quad (\text{GTAH1.2})$$

Both follow directly from $\overline{\rho^\#} = 1 - \rho$. Thus the precise discrepancy between the two adjoints is the bounded operator

$$\mathsf{D}_a := \mathsf{T}_a^* - \mathsf{U}_a, \quad (\mathsf{D}_a x)_\rho = (a^{\bar{\rho}} - a^{1-\rho}) x_\rho. \quad (\text{GTAH1.3})$$

For $\rho = \sigma + i\gamma$, all scalar factors remain explicit:

$$a^{\bar{\rho}} - a^{1-\rho} = e^{-i\gamma \log a} (a^\sigma - a^{1-\sigma}). \quad (\text{GTAH1.4})$$

For any fixed $a > 1$, its kernel is exactly the closed subspace supported on the actual critical-line zeros. This is proved by strict monotonicity of the real function a^x , applied coordinatewise. In particular, every possible off-line value is retained by this defect, whereas every higher jet lies in the separately retained kernel of E_0 . This is a computation of the candidate positive adjoint, not a conclusion about which off-line zeros exist.

GTAH2. Reflection averaging is a concrete second attempt

The positive form is obtained from the existing Weil form by $\langle x, y \rangle = W(x, \mathsf{J}y)$. Rather than declare this change harmless, retain J and test the reflected average of the actual operators:

$$\tilde{\mathsf{T}}_a = \frac{1}{2}(\mathsf{T}_a + \mathsf{J} \mathsf{T}_a \mathsf{J}), \quad \tilde{\mathsf{U}}_a = \frac{1}{2}(\mathsf{U}_a + \mathsf{J} \mathsf{U}_a \mathsf{J}). \quad (\text{GTAH2.1})$$

These are bounded operators on the entire value space, and

$$\tilde{\mathsf{T}}_a^* = \tilde{\mathsf{U}}_a. \quad (\text{GTAH2.2})$$

Proof: transpose (GTAH2.1), use (GTAH1.1), and use $\mathsf{J}^* = \mathsf{J}$. Thus this attempt does repair the positive-adjoint identity. It is not asserted that J or these averages lift to the original test quotient; their domain is the proved value completion.

The complete scalar multiplier of the first average is

$$\tilde{t}_a(\rho) = \frac{a^\rho + a^{\rho^\#}}{2} = e^{i\gamma \log a} \frac{a^\sigma + a^{1-\sigma}}{2}. \quad (\text{GTAH2.3})$$

Its degree relation is consequently

$$\boxed{\tilde{\mathsf{U}}_a \tilde{\mathsf{T}}_a = aI + \frac{1}{4} \mathsf{D}_a^* \mathsf{D}_a.} \quad (\text{GTAH2.4})$$

To prove this identity, square the absolute value in (GTAH2.3), retaining the cross term $2a$, and subtract a :

$$\left(\frac{a^\sigma + a^{1-\sigma}}{2} \right)^2 - a = \frac{(a^\sigma - a^{1-\sigma})^2}{4}. \quad (\text{GTAH2.5})$$

This is exactly the diagonal multiplier of the extra operator in (GTAH2.4). Hence the attempted positive repair produces an explicitly positive degree defect rather than making the original degree identity automatic.

GTAH3. All arithmetic compositions and the return-time defect

The same defect also measures failure of the averaged operators to preserve the full multiplicative arithmetic action. Put

$$E_a = T_a - JT_aJ. \quad (\text{GTAH3.1})$$

All these operators are diagonal, so they commute. Expanding the four terms gives, for every $a, b > 0$,

$$\boxed{\widetilde{T}_a \widetilde{T}_b - \widetilde{T}_{ab} = -\frac{1}{4}E_a E_b.} \quad (\text{GTAH3.2})$$

The proof uses both original group laws in (GTAH0.6) and the same laws after conjugation by J ; it makes no selection of two named primes. The right side is a global formula for every recovered arithmetic parameter and all its repetitions.

Taking $b = 1/a$ in the exact scalar multipliers yields

$$\boxed{\widetilde{T}_a \widetilde{T}_{1/a} = I + \frac{1}{4a}D_a^* D_a.} \quad (\text{GTAH3.3})$$

Thus averaging is compatible with inverse return only on the kernel already calculated in GTAH1. The calculation connects the positive repair to the complete return action, rather than treating a changed quadratic form as if all earlier maps remained fixed.

The scalar formula also gives $\widetilde{U}_a = a\widetilde{T}_{1/a}$, so (GTAH3.3) recovers (GTAH2.4) with the same degree. Both defects are the same operator with the explicitly retained factor a .

GTAH4. A single bounded self-adjoint operator controls the whole defect

On the value space let the full generator L have its maximal diagonal domain

$$\text{Dom } L = \{x : \sum_{\rho} m_{\rho} |\rho|^2 |x_{\rho}|^2 < \infty\}, \quad (Lx)_{\rho} = \rho x_{\rho}. \quad (\text{GTAH4.1})$$

Finite value vectors form a dense domain. Direct use of (GTAH0.2) shows that its adjoint has the same domain and multiplier $\bar{\rho}$. The operator $L + L^* - I$ on that domain has the bounded extension with multiplier $2\Re \rho - 1$. Define its half by the actual comparison

$$C = \frac{1}{2}(L + L^* - I), \quad (Cx)_{\rho} = (\sigma - \frac{1}{2})x_{\rho}, \quad \|C\| \leq \frac{1}{2}, \quad JCJ = -C. \quad (\text{GTAH4.2})$$

This notation records all terms of the generator comparison; the scalar $1/2$ is not a coordinate assigned to τ . It does not replace L , ζ , or their original factors.

The exact identity for every $a > 0$ is

$$D_a^* D_a = 4a \sinh^2((\log a)C). \quad (\text{GTAH4.3})$$

It follows by writing $a^{\sigma} - a^{1-\sigma} = 2a^{1/2} \sinh((\sigma - 1/2) \log a)$ in (GTAH1.4), keeping both powers in the preceding expression. Bounded functional calculus here means simply applying the indicated continuous function to the diagonal real coordinates, so no spectral-domain assertion is concealed.

For $a > 1$, the complete operator estimates are

$$4a(\log a)^2 C^2 \leq D_a^* D_a \leq 4(a-1)^2 C^2. \quad (\text{GTAH4.4})$$

Indeed for $0 \leq x \leq 1/2$, $\sinh((\log a)x) \geq (\log a)x$ by its power series, while convexity and its zero value at zero give $\sinh((\log a)x) \leq 2x \sinh((\log a)/2)$. Squaring, multiplying by $4a$, and using $4a \sinh^2((\log a)/2) = (a-1)^2$ proves the bounds. Evenness handles negative x . These are global analytic estimates, not numerical bounds on a finite list of zeros.

GTAH5. The exact positive trace with all multiplicities

The ordinary operator trace on $\ell^2(\mathcal{Z}, m_\rho)$ does not itself insert multiplicity m_ρ : rescaling an orthonormal basis would remove the weight. To preserve the original zeta multiplicities, use the actual diagonal trace

$$\mathrm{tr}_\zeta(M_f) = \sum_\rho m_\rho f(\rho) \quad (\text{GTAH5.1})$$

on the diagonal algebra, first for nonnegative f , and then when the absolute weighted sum converges. This is the trace received from the full local multiplication algebra: multiplication by a germ $f(\rho+h)$ on $\mathbb{C}[h]/h^{m_\rho}$ has triangular matrix with all m_ρ diagonal entries $f(\rho)$. Its trace is therefore $m_\rho f(\rho)$, also equal to $\mathrm{Res}_\rho f \zeta' / \zeta$. The nilpotent terms remain in that matrix, although their trace is zero. This is the established PCJ10/GIQ trace, with its original-zeta receiver.

For every $t > 0$, set $k_t(\rho) = e^{-t(\Im \rho)^2}$. The unconditional full zero-count estimate $\sum_{|\rho| \leq R} m_\rho = O(R \log(R+2))$ proves $\sum m_\rho k_t(\rho) < \infty$: bound each strip $j \leq |\Im \rho| < j+1$ by its containing disk count and sum the convergent series $O((j+2) \log(j+3) e^{-tj^2})$. Since $|a^\sigma - a^{1-\sigma}| \leq a - 1$ for $a > 1$, the entire positive defect trace is finite and exactly

$$\mathcal{E}(a, t) := \mathrm{tr}_\zeta(M_{k_t} D_a^* D_a) = \sum_\rho m_\rho e^{-t(\Im \rho)^2} (a^{\Re \rho} - a^{1-\Re \rho})^2. \quad (\text{GTAH5.2})$$

No zero term is cut off. Reflection preserves every summand, since it negates the difference before squaring and leaves $\Im \rho$ unchanged. Thus the two reflected directions add to this positive quantity; their signed first differences cancel, but their exact degree defect does not.

Taking this faithful positive diagonal trace in (GTAH4.4) gives

$$4a(\log a)^2 \sum_\rho m_\rho k_t(\rho) (\Re \rho - \tfrac{1}{2})^2 \leq \mathcal{E}(a, t) \leq 4(a-1)^2 \sum_\rho m_\rho k_t(\rho) (\Re \rho - \tfrac{1}{2})^2. \quad (\text{GTAH5.3})$$

The trace is a receiver on actual spectral values; no interchange of a nonholomorphic test with the holomorphic prime explicit formula has been asserted. Its vanishing has not been presumed. The purpose of (GTAH5.2) is to retain and measure the exact error in the attempted construction, not to rename a vanishing conjecture as a theorem.

GTAH6. The result of the attempted repair and its next mathematical receiver

The attempt began with the actual geometric transfer and the positive norm already constructed in GIQ. Its adjoint was calculated, not assigned. Reflection averaging repairs that adjoint identity, but the original degree and arithmetic composition laws acquire the same positive defect (GTAH2.4) and (GTAH3.3). The full scalar, operator and multiplicity-trace descriptions of that defect are (GTAH1.4), (GTAH4.3) and (GTAH5.2).

The next useful target from this finding and the complete programme is the original source-to-dual map, before its value quotient: can the original-zeta residue duality provide an adjoint comparison whose image forces this defect to vanish? That attempt is now carried out in `RESIDUE_TRACE_TRANSFER_INDEPENDENT.md`, `RTT0-RTT11`, read in full for this receiving calculation. It constructs the actual continuous anti-linear map

$$P = L^{-2} M_{s^2 \zeta'(1-s)} K : \mathcal{Q} \rightarrow \mathcal{Q}, \quad (KG)(s) = \overline{G(\bar{s})}, \quad (\text{GTAH6.1})$$

and the full strong residue map $S : \mathcal{Q} \rightarrow \mathcal{H}_{\mathrm{res}}$ with

$$\widehat{\mathcal{R}}(F, SG) = W(E_0 F, E_0 G), \quad \ker S = \ker E_0 = \mathcal{N}_0. \quad (\text{GTAH6.2})$$

The notation K here is RTT's conjugation, distinct from the self-adjoint operator C in GTAH4. The multiplier has the exact endpoint value $s^2 \zeta'(1-s)|_{s=0} = -1$; the inverse of L is proved on $\mathcal{Q}$, rather than division through an unremoved source pole. RTT3 calculates rank one on every full multiplicity block, with its complete nonzero ζ' coefficient. RTT5 proves the strong limit on the entire quotient without a finite-support density assumption.

This map permits a further exact test on the original source. Define the positive continuous form

$$\mathfrak{d}_a(F, G) = \langle D_a E_0 F, D_a E_0 G \rangle. \quad (\text{GTAH6.3})$$

Continuity follows from continuity of E_0 and boundedness of D_a . Its kernel for $a > 1$ consists exactly of classes whose value is zero at every off-line zero; the higher-jet kernel $\mathcal{N}_0$ is contained in it. Since (GTAH6.2) identifies exactly that higher-jet kernel, (GTAH6.3) descends without further loss to the trace image: give $S(Q)$ the quotient topology from $\mathcal{Q}/\mathcal{N}_0$, and define

$$\mathfrak{d}_{a,\text{res}}(SF, SG) = \overline{\mathfrak{d}_a(F, G)}. \quad (\text{GTAH6.4})$$

The conjugation is required because S is anti-linear; it makes the displayed form linear in its first argument on the target. Changing representatives by $\mathcal{N}_0$ leaves it unchanged. This proves a positive continuous target form for the stated quotient topology, not for an unverified subspace topology of $\mathcal{H}_{\text{res}}$. Its quadratic value is exactly (GTAH6.3), so no off-line value is erased by this source-to-residue operation.

The signed test is equally explicit. On any two-element orbit $\{\rho, \rho^\#\}$ of the actual zero set, use the actual global isolators and put $F = e_{\rho,0} - e_{\rho^\#,0}$. Then

$$W(E_0 F, E_0 F) = -2m_\rho, \quad \widehat{\mathcal{R}}(F, SF) = -2m_\rho, \quad (\text{GTAH6.5})$$

by its two nonzero values and (GTAH6.2). This quantifies over the actual orbits; it does not assert that any off-line orbit exists. It proves that the infinitesimal kernel in this particular map cannot secretly remove such a direction while retaining the original Weil pairing.

There is also a direct comparison for the entire value Hilbert space, rather than only the quotient image just used. RTT6.2 constructs

$$A : H \longrightarrow \mathcal{H}_{\text{res}}, \quad (\widehat{A}y)(F) = \langle E_0 F, Jy \rangle, \quad S = AE_0. \quad (\text{GTAH6.6})$$

For a bounded subset $B \subset \mathcal{Q}$, its defining strong seminorm is at most $\sup_{F \in B} \|E_0 F\| \|y\|$, so A is continuous and anti-linear. If $Ay = 0$, then Jy is orthogonal to the dense subspace $E_0(Q)$; hence $Jy = 0$ and $y = 0$. Applying this injectivity to each vector gives the exact equivalence

$$AD_a^* D_a = 0 \iff D_a^* D_a = 0 \iff D_a = 0. \quad (\text{GTAH6.7})$$

The last equivalence follows from $\langle D_a^* D_a x, x \rangle = \|D_a x\|^2$ for every $x \in H$. Thus the complete residue receiver detects the positive degree defect faithfully. This does not assert that $A(H)$ has a continuous inverse with its subspace topology, or that the nonholomorphic value multiplier lifts to $\mathcal{Q}$.

Thus both attempts are calculated: reflection averaging and the full source-to-residue trace map. The first yields the exact degree defect; the second carries that defect, with all value information, into the specified residue receiver. No conclusion about absence of off-line zeros or the full target's Deligne weights has been assumed. The next source-level construction must now control this explicit surviving form while preserving the original transfer and complete return action; the linked full strip-multiplier calculation is addressing the source modules rather than substituting another scalar criterion.

GTAH7. Exact source and proof provenance

Programme proofs used: GIQ2,3,8,9 in GLOBAL_INFINITESIMAL_QUOTIENT.md; RD1–RD5 in ORIGINAL_ZETA_RESIDUE_DUAL_AND_SUPPORT_MAPS.md; GTR0–GTR11 in CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md; SSI0–SSI8 in EXACT_SCHWARTZ_SUMMATION_IMAGE.md; PCJ10 in POSITIVE_COMPLETION_AND_ZERO_JETS.md. These inputs already prove the underlying zero/value completion, trace radical, actual degree transfer, full source ideal and all-zero pairing. The new calculation is the exact adjoint-repair attempt with the full transfer composition and its positive defect; it does not claim the earlier constructions as new.

Human-source route: Connes and Consani, *Schemes over F_1 and zeta functions*, arXiv:0903.2024v3, §5, for the original summation/cohomology construction; Deligne, *La conjecture de Weil. II*, IHÉS52(1980),137–252, §§1.5 and 3.6, for the distinct use of positivity and weight separation. No finite-rank étale realization or Deligne weight theorem is assumed for the spaces in this note.

Global multiplier separation and the full normal source

Complete independent derivation. Stable proof locators GMS0–GMS11.

GMS0. Stage, sources, and the operation being calculated

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Every scalar, function, module, and arithmetic action below belongs to the constructed coefficient receiver after complete-history arithmetic reconstruction. No operation below adds, subtracts, measures, or assigns coordinates to τ . The two branches retain their separately recovered counters and return measures.

The governing construction and correction chain are READ_FIRST_USER_CONSTRUCTION.md, USER_ARGUMENT_RECONSTRUCTION.md, and the retained WU050–WU055, WU061–WU065 passages in USER_CONSTRUCTION_FULL_LOGBOOK.md. In particular, a finite pair of named primes does not initialize this calculation, and an unrelated timing history does not replace the original arithmetic. WU065 asks that a limited result lead to a concrete next calculation using the accumulated findings. Here that calculation is to combine CLP13’s finite coherent separation with GSL’s already constructed global interpolation multiplier, and then apply it to the complete actual localization row.

The local source proofs used are:

- GLOBAL_SHIFTED_ZETA_LIFT.md, GSL1–GSL6: the actual entire-function space, original source factor, corridor integral, strip estimates, all-multiplicity interpolation, and translated quotient.
- CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md, CW1–CW8: the original half-Mellin transform, exact source image, actual localization maps and generators, full resolvent, and actual normal extension.
- CC_LOCALIZATION_POLYNOMIAL_WEIGHTS.md, CLP1–CLP3 and CLP12–CLP13: the simultaneous source complexes, symmetric continuous section, retained nonzero normal extension, and finite coherent comparison.

GSL4–GSL6 and CLP12–CLP13 were read in full for this continuation; the underlying CW and CLP source calculations were previously read in full and the formulas used below were recalled directly. The independent full-source theorem $\Sigma : S \rightarrow J$ and its continuous inverse are the SSI/ESI results used in CW1 and CLP1. These are named proved prerequisites, not new assumptions about a hypothetical source.

The original human-source constructions remain Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, §5, arXiv:0903.2024v3, and the separately recorded signed-chart source arXiv:2609.00299v1. The lifting comparison is with Deligne, *La conjecture de Weil. II*, §3.6, reconstructed in DC with its geometric hypotheses. This note proves the new module and chain maps below. It does not assert a fresh reading of those whole human sources or identify the coefficient category with Deligne’s étale category.

GMS1. The actual multiplier algebra and its complete modules

Let $\mathcal{Z}$ denote the actual nontrivial-zero divisor of the original ζ , and retain each multiplicity m_ρ . The established zero-free boundary lines give

$$0 < \Re \rho < 1 \quad (\rho \in \mathcal{Z}). \quad (\text{GMS1.1})$$

Define the unital commutative complex algebra

$$M = \left\{ h \in \mathcal{O}(\mathbb{C}) : \begin{array}{l} \text{for every } A \geq 0 \text{ there are } C_A > 0, N_A \in \mathbb{N} \\ \text{with } |h(x + it)| \leq C_A(1 + |t|)^{N_A} \\ \text{whenever } |x| \leq A \end{array} \right\}. \quad (\text{GMS1.2})$$

The operations are pointwise. Sums satisfy the estimate with the maximum of the two exponents and the sum of the constants; products satisfy it with the sum of the exponents and product of the constants. The functions 1, s , and $a^s = \exp(s \log a)$ for every recovered real $a > 0$ belong to M . Real translation by b

preserves M , using the bound on the strip of width $A + |b|$. No topology on the algebra M is needed for algebraic derived Hom below.

The actual coefficient space and its ideals are

$$\begin{aligned} \mathcal{B} &= \left\{ F \in \mathcal{O}(\mathbb{C}) : b_{A,N}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^N |F(s)| < \infty \text{ for all } A, N \geq 0 \right\}, \\ \mathcal{J} &= \{ F \in \mathcal{B} : F^{(j)}(\rho) = 0, \rho \in \mathcal{Z}, 0 \leq j < m_\rho \}, \\ \mathcal{J}_+ &= \{ F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0, \rho \in \mathcal{Z}, 0 \leq j < m_\rho \}, \\ \mathcal{Q} &= \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+. \end{aligned} \tag{GMS1.3}$$

These are the full modules, without a finite-support restriction or a replacement by a direct sum of jets. The countable seminorms at integer A, N make $\mathcal{B}$ Fréchet. To check completeness, a Cauchy sequence converges uniformly on every compact set to an entire function; its Cauchy property in each weighted strip supremum then gives convergence in that seminorm to the same function. Evaluations of every fixed derivative are continuous by Cauchy's estimate on a larger strip. Thus both ideals are closed and both quotient spaces are Fréchet with their quotient topology.

For fixed $h \in M$, multiplication is continuous, with

$$b_{A,N}(hF) \leq C_A b_{A,N+N_A}(F). \tag{GMS1.4}$$

The full Leibniz formula

$$(hF)^{(j)}(z) = \sum_{k=0}^j \binom{j}{k} h^{(k)}(z) F^{(j-k)}(z) \tag{GMS1.5}$$

shows that it preserves both ideals. Hence $\mathcal{B}, \mathcal{J}, \mathcal{J}_+, \mathcal{Q}, \mathcal{Q}_+$ are actual M -modules, and each fixed multiplier acts continuously. We use this last property, together with strict exact sequences of Fréchet spaces, when discussing continuous extensions. We do not infer this structure for an arbitrary continuous module carrying only a generator L .

GMS2. The normal coordinate and every original source factor

The full original arithmetic is retained:

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \tag{GMS2.1}$$

The half-Mellin source from CW1 and CLP1 is

$$\Theta b(s) = \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \tag{GMS2.2}$$

It is a topological isomorphism $A \rightarrow \mathcal{B}$ and takes the exact original source $J = \Sigma S$ onto $\mathcal{J}$, where

$$\Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad S = \{ f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f = 0 \}. \tag{GMS2.3}$$

For $\Re s > 1$, absolute summation proves

$$\Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v}. \tag{GMS2.4}$$

The particular source

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

has the entire transform

$$F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = C_0(s) \zeta(s). \quad (\text{GMS2.5})$$

It belongs to $\mathcal{B}$ and has precisely $\mathcal{Z}$ as its zero divisor. Formula (GMS2.5) is an exact source comparison; the original ζ and the full multiplier C_0 remain in every local comparison.

Define $VF(\lambda) = F(\lambda - 1)$ and $UF(s) = F(s + 1)$. Then

$$b_{A,N}(VF), \quad b_{A,N}(UF) \leq b_{A+1,N}(F), \quad V\mathcal{J} = \mathcal{J}_+. \quad (\text{GMS2.6})$$

The actual normal coefficient module before translation has the action

$$h \cdot_{\text{normal}} F(s) = h(s + 1)F(s). \quad (\text{GMS2.7})$$

This is the M -action on $\mathcal{B}(-1), \mathcal{J}(-1), \mathcal{Q}(-1)$. In particular its generator is $s + 1$, and its dilation is a^{s+1} , including the factor a . Direct substitution gives

$$V(h(s + 1)F(s))(\lambda) = h(\lambda)VF(\lambda). \quad (\text{GMS2.8})$$

Thus V gives the actual continuous M -linear isomorphism from the full normal row to

$$0 \longrightarrow \mathcal{J}_+ \xrightarrow{\iota} \mathcal{B} \xrightarrow{q_+} \mathcal{Q}_+ \longrightarrow 0. \quad (\text{GMS2.9})$$

Its kernel is the entire $\mathcal{J}_+$. In the original source coordinates it remains the entire $\mathcal{J}(-1)$.

The translated original factor is

$$F_+(\lambda) = VF_0(\lambda) = \frac{(\lambda - 1)(\lambda - 2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda - 1)/2) \zeta(\lambda - 1). \quad (\text{GMS2.10})$$

It has every zero $\rho + 1$ to the original multiplicity. Its exceptional values, including the retained cancellations, are

$$\begin{aligned} F_+(1) = F_+(2) &= \frac{1}{8}, \quad F_+(0) = F_0(-1) = \frac{\pi}{24}, \\ F_+(1 - 2r) &= \frac{r(2r + 1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= F_+(2 + 2r) = \frac{(1 + 2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1 + 2r)/2) \zeta(1 + 2r) \neq 0 \quad (r \geq 1). \end{aligned} \quad (\text{GMS2.11})$$

The corresponding unshifted values are $F_0(0) = F_0(1) = 1/8$ and the second displayed formula with arguments $-2r, 1 + 2r$. All derivatives of the translated entire function satisfy $F_+^{(j)}(\lambda) = F_0^{(j)}(\lambda - 1)$. In particular, at a nontrivial zero the full local unit is

$$F_+(\rho + 1 + z) = z^{m_\rho} C_0(\rho + z) \frac{\zeta(\rho + z)}{z^{m_\rho}}, \quad C_0(\rho) \frac{\zeta^{(m_\rho)}(\rho)}{m_\rho!} \neq 0. \quad (\text{GMS2.12})$$

GMS3. The already constructed global separator, with its exact scope

GSL4 constructs an entire multiplier E with polynomial bounds on each vertical strip and with the full jets

$$E - 1 \in (s - \rho)^{m_\rho} \mathcal{O}_\rho, \quad E \in (s - (\rho - 1))^{m_\rho} \mathcal{O}_{\rho-1}. \quad (\text{GMS3.1})$$

For specificity, the actual construction is $G(s) = F_0(s)F_0(s + 1)$, $r(t) = \varepsilon(1 + t^2)^{-3}$, $\chi(x + it) = \eta(x/r(t))$, with the smooth cutoff η equal to zero on $(-\infty, -1]$ and one on $[1, \infty)$. The original-zeta zero-free corridor chooses $\varepsilon > 0$ so G has no zeros on the cutoff-transition corridor. Put

$$v(w) = \frac{\bar{\partial}\chi(w)}{G(w)} \quad \text{on that corridor, and } v = 0 \text{ off it,} \quad u(s) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{e^{(s-w)^2}}{s - w} v(w) dA(w),$$

$$E(s) = \chi(s) - G(s)u(s). \quad (\text{GMS3.2})$$

GSL4.4–GSL4.14 prove absolute convergence, $\bar{\partial}u = v$, entire continuation of E , and

$$|E(x + it)| + |1 - E(x + it)| \leq C_A(1 + |t|)^{d_A} \quad (|x| \leq A). \quad (\text{GMS3.3})$$

Those estimates use the full two source factors in G ; they are the proved global input here. The full vanishing orders in (GMS3.1) follow from (GMS3.2) at the two divisors. No convergence of a sum of finite spectral projectors supplies this multiplier.

Set

$$c(\lambda) = E_+(\lambda) = 1 - E(\lambda - 1) \in M. \quad (\text{GMS3.4})$$

Translating (GMS3.1) gives

$$c - 1 \in (\lambda - \rho)^{m_\rho} \mathcal{O}_\rho, \quad c \in (\lambda - (\rho + 1))^{m_\rho} \mathcal{O}_{\rho+1}. \quad (\text{GMS3.5})$$

Consequently, with every multiplicity retained,

$$\begin{aligned} c\mathcal{B} &\subset \mathcal{J}_+, & (1 - c)\mathcal{B} &\subset \mathcal{J}, \\ c_{\mathcal{Q}} &= 1_{\mathcal{Q}}, & c_{\mathcal{Q}_+} &= 0_{\mathcal{Q}_+}. \end{aligned} \quad (\text{GMS3.6})$$

All inclusions follow by (GMS1.5). These formulas establish both identities on the complete quotient modules, including classes with infinitely many nonzero jets.

The function c is not asserted to satisfy $c^2 = c$ in M . Indeed it takes both values zero and one and is entire, so it is nonconstant, whereas an entire solution of $c^2 = c$ on $\mathbb{C}$ is constant. Thus it is not an idempotent on $\mathcal{B}$ or on its nonzero ideal $\mathcal{J}_+$. It is an idempotent on $\mathcal{Q} \oplus \mathcal{Q}_+$. No value zero or one at the separate endpoint parameters 0, 1 is inferred from (GMS3.5).

GMS4. Full derived separation, with an explicit algebraic contraction

We prove the required derived statements over the actual ring M . The proof uses arbitrary algebraic projective resolutions; it requires no finite generation of the modules and makes no topological projectivity claim.

Choose a projective resolution $P^\bullet \rightarrow X$ in degrees at most zero. Let $a \in M$ act as zero on X . There is a degree-minus-one M -linear map $h : P^\bullet \rightarrow P^{\bullet-1}$ with

$$d_P h + h d_P = a_P. \quad (\text{GMS4.1})$$

Here is the full lifting induction. If $\epsilon : P^0 \rightarrow X$ is the augmentation, aP^0 lands in $\ker \epsilon = \text{im}(d_P : P^{-1} \rightarrow P^0)$, so projectivity of P^0 lifts a_{P^0} to $h^0 : P^0 \rightarrow P^{-1}$. Suppose h^{j+1} and all higher components have been chosen. On P^j the map $a_{P^j} - h^{j+1} d_P^j$ lands in $\ker d_P^j$: applying d_P^j and the already established equation in degree $j+1$ gives zero. Exactness gives $\ker d_P^j = \text{im } d_P^{j-1}$; projectivity of P^j lifts this map to $h^j : P^j \rightarrow P^{j-1}$. Induction proves (GMS4.1) in every degree. An arbitrary module has a free resolution by successively choosing a free module surjecting onto each kernel, so the construction applies to the actual full modules.

For a module or bounded complex $N^\bullet$, the Hom differential and its degree-minus-one homotopy are

$$Df = d_N f - (-1)^n f d_P, \quad (Hf) = (-1)^n f h \quad (f \in \text{Hom}_M(P, N)^n). \quad (\text{GMS4.2})$$

Multiplication gives the exact identity

$$D(Hf) + H(Df) = f(hd_P + d_P h) = f a_P = a_N f. \quad (\text{GMS4.3})$$

The two $d_N f h$ terms cancel. The last equality uses M -linearity and centrality. The bounded-above projective resolution computes derived Hom into modules and into the finite coefficient complexes used here. All operations in (GMS4.2) are algebraic compositions; no infinite series or density assertion occurs.

Apply this with $X = \mathcal{Q}$, $a = 1 - c$, and $N = \mathcal{Q}_+$. Equation (GMS3.6) says $a_X = 0$, $a_N = 1$, so (GMS4.3) is a contraction of the full Hom complex. For the reverse direction resolve $X = \mathcal{Q}_+$, use $a = c$, and take $N = \mathcal{Q}$. Again $a_X = 0$, $a_N = 1$. Therefore

$$\boxed{\mathrm{RHom}_M(\mathcal{Q}, \mathcal{Q}_+) \simeq 0, \quad \mathrm{RHom}_M(\mathcal{Q}_+, \mathcal{Q}) \simeq 0.} \quad (\text{GMS4.4})$$

In particular every algebraic Ext_M^k in either displayed direction is zero for $k \geq 0$. The proof applies to the whole actual modules in (GMS1.3), without presenting either as a limit of the finite coherent families from CLP13.

There is also a fully specified resolution and homotopy with no choices of projective lifts. For an M -module X , let

$$P_n(X) = M \otimes_{\mathbb{C}} M^{\otimes_{\mathbb{C}} n} \otimes_{\mathbb{C}} X \quad (n \geq 0), \quad \epsilon(a_0 \otimes x) = a_0 x. \quad (\text{GMS4.5})$$

The left M -action is on the first factor. Each term is a free M -module: its remaining tensor factors are a complex vector space, and a vector-space basis identifies the tensor product with a direct sum of copies of M . The differential is

$$\begin{aligned} d_n(a_0 \otimes \cdots \otimes a_n \otimes x) &= \sum_{i=0}^{n-1} (-1)^i a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_n \otimes x \\ &\quad + (-1)^n a_0 \otimes \cdots \otimes a_{n-1} \otimes a_n x. \end{aligned} \quad (\text{GMS4.6})$$

In $d_{n-1}d_n$, the terms performing two disjoint multiplications cancel in opposite orders; the terms multiplying three adjacent factors cancel by associativity; the two final terms cancel by the module law. Hence $d^2 = 0$. The complex-linear augmented homotopy inserting 1 in front, $s_n(a_0 \otimes \cdots \otimes a_n \otimes x) = 1 \otimes a_0 \otimes \cdots \otimes a_n \otimes x$, and $s_{-1}(x) = 1 \otimes x$, satisfies $ds + sd = 1$: the first multiplication in ds is the identity term, and its other terms cancel term by term with sd . Therefore the augmented complex is exact. This proves it is an actual free resolution, although this augmentation contraction is not asserted M -linear.

Its M -linear insertion of the specified central multiplier is

$$H_n(a_0 \otimes \cdots \otimes a_n \otimes x) = \sum_{i=0}^n (-1)^i a_0 \otimes \cdots \otimes a_i \otimes c \otimes a_{i+1} \otimes \cdots \otimes a_n \otimes x. \quad (\text{GMS4.7})$$

In $dH + Hd$, multiplications not involving the inserted c cancel with the identical insertion after that multiplication, with opposite sign. Multiplication of c with an interior factor occurs in two adjacent insertion terms, whose signs are opposite; centrality makes their values equal. The two terms left are the first-factor multiplication by c and minus the last-factor action by c . Thus in every degree

$$d_{n+1}H_n + H_{n-1}d_n = c_{P_n(X)} - P_n(c_X), \quad H_{-1} = 0. \quad (\text{GMS4.8})$$

Here $P_n(c_X)$ applies c_X only to the last factor. In the Hom cochain convention $C^n = \mathrm{Hom}_M(P_n(X), N)$, $\delta f = f d_{n+1}$, put $k^n(f) = f H_{n-1}$, with $k^0 = 0$. Equations (GMS4.7)–(GMS4.8) give

$$\delta k + k\delta = c_N - (\text{precomposition by } P_{\bullet}(c_X)). \quad (\text{GMS4.9})$$

For $X = \mathcal{Q}$, this is $c_N - 1$: consequently $-k$ contracts the complex when $N = \mathcal{Q}_+$, and gives the identity-minus-multiplier homotopies for $N = \mathcal{B}, \mathcal{J}_+$. For $X = \mathcal{Q}_+$, $N = \mathcal{Q}$, it is the identity, so k contracts the reverse Hom. This is a concrete alternative realization of every module homotopy used in (GMS4.4) and (GMS5.3). The convention $\delta f = f d$ is isomorphic to the ordinary cohomological Hom convention (GMS4.2) after multiplication of degree n by $(-1)^{n(n+1)/2}$. Each insertion sum is finite in its degree; these are algebraic tensor products, with no assertion about completed tensor products or continuous free resolutions.

GMS5. The exact derived inverse for the entire normal ideal

The separator gives an actual continuous M -linear map

$$\beta : \mathcal{B} \longrightarrow \mathcal{J}_+, \quad \beta(F) = cF. \quad (\text{GMS5.1})$$

Its codomain follows from (GMS3.6), its continuity from (GMS1.4), and its M -linearity from pointwise commutativity. With the actual inclusion ι from (GMS2.9),

$$\iota\beta = c_{\mathcal{B}}, \quad \beta\iota = c_{\mathcal{J}_+}. \quad (\text{GMS5.2})$$

These are multiplier identities, not identities of the original modules.

Resolve $\mathcal{Q}$ by P , and retain the homotopy (GMS4.1) for $a = 1 - c$. For either $N = \mathcal{B}$ or $N = \mathcal{J}_+$, (GMS4.3) gives

$$DH_N + H_N D = 1 - c_N \quad \text{on } \text{Hom}_M(P, N). \quad (\text{GMS5.3})$$

Hence postcomposition by β and by ι are explicit chain-homotopy inverses. We have proved

$$\boxed{\text{RHom}_M(\mathcal{Q}, \mathcal{J}_+) \xrightarrow{\iota_*} \text{RHom}_M(\mathcal{Q}, \mathcal{B}) \text{ is a quasi-isomorphism, with inverse } \beta_*}. \quad (\text{GMS5.4})$$

The homotopies are (GMS5.3); in particular $1 - \iota_*\beta_*$ and $1 - \beta_*\iota_*$ are the stated Hom differentials of those homotopies. Nothing has quotiented $\mathcal{J}_+$ further.

Equivalently, for every integer k , postcomposition with $\iota[k]$ is a bijection of derived morphism groups

$$\text{Hom}_{D(M)}(\mathcal{Q}, \mathcal{J}_+[k]) \xrightarrow{\sim} \text{Hom}_{D(M)}(\mathcal{Q}, \mathcal{B}[k]); \quad [f] \mapsto [\iota f], \quad [g] \mapsto [\beta g]. \quad (\text{GMS5.5})$$

Uniqueness here is uniqueness of the derived morphism class; it is not uniqueness of each cochain representative. In degree zero both groups are in fact zero. For an M -map $g : \mathcal{Q} \rightarrow \mathcal{B}$, $(1 - c)g(q) = g((1 - c)q) = 0$. Since $1 - c$ is a nonzero entire function, multiplication by it is injective on $\mathcal{B}$: the product identity forces the entire second factor to vanish on its nonempty open nonzero set and hence everywhere. Thus $g = 0$; inclusion proves the same for target $\mathcal{J}_+$. This does not assert that the higher groups in (GMS5.5) separately vanish.

The same normal triangle and the reverse separation in (GMS4.4) also give

$$\text{RHom}_M(\mathcal{B}, \mathcal{Q}) \longrightarrow \text{RHom}_M(\mathcal{J}_+, \mathcal{Q}) \quad \text{a quasi-isomorphism.} \quad (\text{GMS5.6})$$

Indeed applying derived Hom into $\mathcal{Q}$ to the exact triangle of (GMS2.9) puts its cone next to $\text{RHom}_M(\mathcal{Q}_+, \mathcal{Q}) \simeq 0$. This is a statement about the same full kernel and quotient.

If another GSL cutoff yields c' satisfying (GMS3.5), its map β' is another chain-homotopy inverse of ι_* by the same proof. Therefore β_* and β'_* coincide in the derived category. For example, $\beta_* - \beta'_* = \beta_*(1 - \iota_*\beta'_*) + (\beta_*\iota_* - 1)\beta'_*$, and (GMS5.3) explicitly makes both terms null-homotopic. The derived inverse does not depend on the cutoff choices.

GMS6. Exact meaning for pullbacks and the surviving normal extension

The normal extension (GMS2.9) has its Yoneda class

$$e_+ \in \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+), \quad e_+ : \mathcal{Q}_+ \longrightarrow \mathcal{J}_+[1]. \quad (\text{GMS6.1})$$

Its domain is $\mathcal{Q}_+$. It is not one of the two cross groups in (GMS4.4).

For an ordinary M -map $f : \mathcal{Q} \rightarrow \mathcal{Q}_+$, (GMS4.4) gives $f = 0$. Its actual pullback middle module is therefore

$$\{(b, q) \in \mathcal{B} \oplus \mathcal{Q} : q_+(b) = f(q) = 0\} = \mathcal{J}_+ \oplus \mathcal{Q}. \quad (\text{GMS6.2})$$

The section is $q \mapsto (0, q)$; it is M -linear and continuous in the product topology. For any derived morphism $f : \mathcal{Q} \rightarrow \mathcal{Q}_+[k]$, all such classes also vanish by (GMS4.4), so the composition with the corresponding shift of e_+ is zero. This is the exact pullback statement supplied by the separation. It does not produce a section $\mathcal{Q}_+ \rightarrow \mathcal{B}$ of the normal row.

That normal row remains non-split even algebraically over M . Choose an actual $\rho \in \mathcal{Z}$, whose existence is the proved original-zeta fact used in CW8. Put $\lambda_0 = \rho + 1$. Entire division and the strip estimates give

$$v_\rho(\lambda) = \frac{F_+(\lambda)}{\lambda - \lambda_0} \in \mathcal{B}, \quad [v_\rho] \neq 0 \text{ in } \mathcal{Q}_+, \quad (\lambda - \lambda_0)[v_\rho] = [F_+] = 0. \quad (\text{GMS6.3})$$

The nonzero quotient class follows from its exact order $m_\rho - 1$ at λ_0 ; (GMS2.12) retains the local unit. An M -linear section would send it to an entire function F satisfying $(\lambda - \lambda_0)F = 0$, hence to zero, contradicting its quotient. Thus $e_+ \neq 0$. The nonzero class in the polynomial source test is exactly

$$[F_+] \in \mathcal{J}_+ / (\lambda - \lambda_0)\mathcal{J}_+. \quad (\text{GMS6.4})$$

It cannot vanish, since an identity $F_+ = (\lambda - \lambda_0)H$ with $H \in \mathcal{J}_+$ would force $H = v_\rho$, which has insufficient vanishing order at λ_0 . This is CLP12's positive normal boundary with its whole ideal and full original factor. It coexists with (GMS5.4) because the domains of the two statements are different specified modules.

One can also calculate the actual pushout that the multiplier makes split. Push (GMS2.9) out along $c : \mathcal{J}_+ \rightarrow \mathcal{J}_+$. Its middle module is

$$\mathcal{P}_c = (\mathcal{J}_+ \oplus \mathcal{B}) / \{(cj, -j) : j \in \mathcal{J}_+\}, \quad i_c(i) = [(i, 0)], \quad p_c([(i, b)]) = q_+(b). \quad (\text{GMS6.5})$$

The relation subspace is closed: convergence of $(cj_\nu, -j_\nu)$ implies that j_ν converges in the closed subspace $\mathcal{J}_+$ to $-b$, and continuity of multiplication gives the first coordinate $-cb$. Thus $\mathcal{P}_c$ has its actual Fréchet quotient topology. Define

$$\Phi_c : \mathcal{P}_c \longrightarrow \mathcal{J}_+ \oplus \mathcal{Q}_+, \quad [(i, b)] \mapsto (i + cb, q_+(b)). \quad (\text{GMS6.6})$$

It kills precisely the displayed relations: if both coordinates are zero, then $b = j \in \mathcal{J}_+$ and $i = -cj$, which is the negative of one relation. It is onto, since for (i_0, q) any lift b of q gives the preimage $[(i_0 - cb, b)]$. The latter is independent of the lift: changing b by j changes the pair by $(-cj, j)$, a relation. Continuity of the inverse follows because the continuous map $(i_0, b) \mapsto [(i_0 - cb, b)]$ kills $\{0\} \oplus \mathcal{J}_+$ and descends through the product quotient $\mathcal{J}_+ \oplus \mathcal{B} \rightarrow \mathcal{J}_+ \oplus \mathcal{Q}_+$. Thus Φ_c is an explicit continuous M -linear isomorphism of extensions.

Its section and its Yoneda interpretation are

$$q \mapsto [(-cb, b)] \quad (q_+(b) = q), \quad c_*e_+ = 0 \text{ in } \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+). \quad (\text{GMS6.7})$$

This constructs the annihilation of the extension class by the central multiplier, while (GMS6.3)–(GMS6.4) retain $e_+ \neq 0$ before that pushout. Its kernel is still a copy of the full $\mathcal{J}_+$, with the specified changed map into the middle module.

GMS7. Continuous splitting for actual strict multiplier extensions

There is also a continuous statement that does not use an unconstructed Fréchet projective resolution. Suppose

$$0 \longrightarrow \mathcal{Q}_+ \xrightarrow{j} X \xrightarrow{\pi} \mathcal{Q} \longrightarrow 0 \quad (\text{GMS7.1})$$

is a strict exact sequence of Fréchet M -modules: j is a topological embedding with closed image, π induces the quotient topology, and every fixed multiplier acts continuously on X . Such an extension is an explicitly specified class of objects; a continuous L -equivariant extension is not declared to satisfy these additional hypotheses.

For $q \in \mathcal{Q}$, take any lift $x \in X$, and set

$$s(q) = c_X x. \quad (\text{GMS7.2})$$

If the lift changes by $j(y)$, the result changes by $j(cy) = 0$, proving well-definedness. Moreover $\pi s(q) = cq = q$ and $s\pi = c_X$. It is M -linear because c is central. Its continuity follows exactly from the quotient topology: $s\pi = c_X$ is continuous. It is the unique M -linear section, since the difference of two sections factors through j as an M -map $\mathcal{Q} \rightarrow \mathcal{Q}_+$, which is zero by (GMS4.4).

There is an actual idempotent on this extension:

$$c_X^2 = c_X. \quad (\text{GMS7.3})$$

Indeed $(c_X - 1)x \in j\mathcal{Q}_+$, and applying c_X kills that element. The complete continuous splitting and inverse are

$$\begin{aligned} \mathcal{Q}_+ \oplus \mathcal{Q} &\longrightarrow X, & (y, q) &\longmapsto j(y) + s(q), \\ X &\longrightarrow \mathcal{Q}_+ \oplus \mathcal{Q}, & x &\longmapsto (j^{-1}((1 - c_X)x), \pi x). \end{aligned} \quad (\text{GMS7.4})$$

The first coordinate of the inverse is well-defined since $\pi(1 - c_X) = 0$, and is continuous because j^{-1} is continuous on its image. Direct substitution and (GMS7.3) prove that the maps are inverse.

For the reversed strict extension $0 \rightarrow \mathcal{Q} \rightarrow X \rightarrow \mathcal{Q}_+ \rightarrow 0$, the identical proof uses $1 - c$ on a lift to construct its unique continuous M -linear section. If a second permissible separator c' is used in (GMS7.2), both resulting maps are sections, so uniqueness makes them equal. More directly, $c_X - c'_X$ kills the kernel and has image in that kernel, hence factors as an M -map $\mathcal{Q} \rightarrow \mathcal{Q}_+$, which vanishes. This proves cutoff independence on the actual extension.

For example, the existing GSL high row has

$$X = \mathcal{B}/(\mathcal{J} \cap \mathcal{J}_+), \quad j([g]_{\mathcal{J}_+}) = [(1 - c)g]_{\mathcal{J} \cap \mathcal{J}_+}, \quad \pi([F]) = [F]_{\mathcal{J}}. \quad (\text{GMS7.5})$$

The section from (GMS7.2) is exactly $[f]_{\mathcal{J}} \mapsto [cf]$. Independence of representatives follows from the two full jet conditions. The full inverse of the quotient map to $\mathcal{Q} \oplus \mathcal{Q}_+$ is

$$([f]_{\mathcal{J}}, [g]_{\mathcal{J}_+}) \mapsto [cf + (1 - c)g]_{\mathcal{J} \cap \mathcal{J}_+}. \quad (\text{GMS7.6})$$

Products $c\mathcal{J}$ and $(1 - c)\mathcal{J}_+$ lie in the intersection, so this is well-defined. Its two quotient images are $[f]$ and $[g]$; the reverse composition uses $cF + (1 - c)F = F$. The estimates (GMS1.4), with independent infima over both representatives, prove continuity in both directions. Thus the example is strict and realizes the exact hypotheses of (GMS7.1).

GMS8. The actual original-source map behind the derived inverse

Returning (GMS5.1) through V gives

$$V^{-1}\beta V(F)(s) = c(s + 1)F(s) = (1 - E(s))F(s). \quad (\text{GMS8.1})$$

The actual map on the full unshifted source spaces with their normal actions is therefore

$$\mathfrak{b} : A(-1) \longrightarrow J(-1), \quad \mathfrak{b}(a) = \Theta^{-1}[(1 - E(s))\Theta a(s)]. \quad (\text{GMS8.2})$$

Its image is in J because $(1 - E)\mathcal{B} \subset \mathcal{J} = \Theta J$. Its exact integral representative is

$$\mathfrak{b}(a)(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} (1 - E(1/2 + it))\Theta a(1/2 + it)u^{-it} dt. \quad (\text{GMS8.3})$$

The bound (GMS3.3) multiplied by arbitrarily rapid strip decay of Θa proves absolute convergence after any fixed number of $u\partial_u$ derivatives. The original inverse Mellin estimates in (GMS2.2), applied to the strip seminorm bound (GMS1.4), prove continuity into every seminorm of A . Its image is already in the closed source J , and $H = \Sigma^{-1} : J \rightarrow S$ is the actual continuous source inverse. Consequently

$$a \mapsto f_a = H\mathfrak{b}(a) \in S, \quad \Sigma f_a = \mathfrak{b}(a), \quad (\text{GMS8.4})$$

is a continuous original Schwartz-source representative with its value and integral conditions retained.

The two composites with the original inclusion $J(-1) \hookrightarrow A(-1)$ are multiplication by $1 - E(s)$ under Θ . The derived identity in (GMS5.4) follows from the action of the same ring element $c(\lambda)$ on the unshifted source $\mathcal{Q}$, where it is the identity. The shift in (GMS8.1) is essential; it is not permission to replace the normal action $h(s + 1)$ by $h(s)$.

GMS9. The multiplier action on the complete simultaneous complexes

We construct the action, rather than presuming a general functional calculus for an arbitrary operator. Let

$$\begin{aligned} V_{\pm} &= S \oplus \mathbb{C}^2, \quad W = V_+ \oplus V_-, \quad r_+(f, c_0, c_1) = \Sigma f, \quad r_-(g, d_0, d_1) = R\Sigma g = \Sigma \hat{g}, \\ Rb(u) &= u^{-1}b(u^{-1}), \quad \hat{g}(v) = \int_{\mathbb{R}} g(x)e^{-2\pi i x v} dx. \end{aligned} \quad (\text{GMS9.1})$$

The full preceding Poisson identity is $\Sigma \hat{g}(u) = u^{-1}\Sigma g(u^{-1}) + u^{-1}g(0) - \int g$. Its last two terms vanish on S ; the four separate endpoint coordinates are retained. The source sections are $s_+(b) = (Hb, 0, 0)$ and $s_-(b) = (\widehat{Hb}, 0, 0)$, with $r_{\pm}s_{\pm} = 1_J$. Put $h_J = \Theta^{-1}m_h\Theta$ on J . Define

$$\begin{aligned} h_{V_+}(f, c_0, c_1) &= s_+(h_J \Sigma f) + (0, h(0)c_0, h(1)c_1), \\ h_{V_-}(g, d_0, d_1) &= s_-(h_J R\Sigma g) + (0, h(1)d_0, h(0)d_1). \end{aligned} \quad (\text{GMS9.2})$$

All displayed maps are continuous. On each chart the decomposition into its source summand and endpoint summand is direct. Since $r_{\pm}s_{\pm} = 1$, composition of (GMS9.2) for multipliers h, k is the action for hk , addition is the action for $h+k$, and 1 acts as the identity. This proves an actual M -module structure on W , with

$$r_{\pm}h_{V_{\pm}} = h_J r_{\pm}. \quad (\text{GMS9.3})$$

For $h(s) = s$ this is the previously computed generator G on W , with endpoint eigenvalues $(0, 1, 1, 0)$. For $h(s) = a^s$, injectivity of the source restrictions and (GMS9.3) identify its source components with $f(v/a)$ and $ag(av)$, respectively. Its endpoint components are (c_0, ac_1, ad_0, d_1) . Thus it extends the actual coefficient actions, with every factor of a .

The actual simultaneous complexes, in degrees 0, 1, 2, are

$$D_J = (W \xrightarrow{r_+ - r_-} J \xrightarrow{0} J(-1)), \quad D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)), \quad D_G = (0 \rightarrow Q \xrightarrow{0} Q(-1)). \quad (\text{GMS9.4})$$

Use (GMS9.2) in degree zero, multiplication by $h(s)$ under Θ in degree one, and multiplication by $h(s+1)$ in degree two. Equation (GMS9.3) proves compatibility with the first differential; the second is zero. Inclusions, quotients, and the degree-zero identity in the row $0 \rightarrow D_J \rightarrow D_F \rightarrow D_G \rightarrow 0$ are M -linear.

The symmetric section, projection, and exact source decomposition are

$$\begin{aligned} \sigma b &= \tfrac{1}{2}(s_+b, -s_-b), \quad d\sigma = 1_J, \quad p_Z = 1_W - \sigma d, \quad Z = \ker d, \\ W &\xrightarrow{\sim} Z \oplus J, \quad w \mapsto (p_Z w, dw), \quad (z, b) \mapsto z + \sigma b. \end{aligned} \quad (\text{GMS9.5})$$

They are M -linear by (GMS9.2)–(GMS9.3). Moreover

$$Z \simeq J \oplus \mathbb{C}_+^2 \oplus \mathbb{C}_-^2, \quad (b, e_+, e_-) \mapsto (s_+b + e_+, s_-b + e_-), \quad (\text{GMS9.6})$$

where the endpoint multipliers are exactly $h(0), h(1), h(1), h(0)$. Thus c retains the values $c(0), c(1)$ there; they are not replaced by spectral zero or one values.

Applying this decomposition contracts the $[J \xrightarrow{1} J]$ summands and gives the simultaneous quasi-isomorphisms of CLP12, now proved M -linear:

$$\begin{aligned} D_J &\simeq Z[0] \oplus J(-1)[-2], \\ D_F &\simeq Z[0] \oplus Q[-1] \oplus A(-1)[-2], \\ D_G &= Q[-1] \oplus Q(-1)[-2]. \end{aligned} \quad (\text{GMS9.7})$$

The vertical maps are p_Z in degree zero, respectively zero or the quotient q in degree one, and the identity in degree two. Their first two kernels are $[\sigma J \xrightarrow{d} J]$, contracted by σ . This proves each claim in (GMS9.7), including the retained normal row (GMS2.9). No M -linear section of $A \rightarrow Q$ has been inserted.

The ring element c acts as an actual projection on D_G : it is one on $Q[-1]$ and zero on $Q(-1)[-2]$, by (GMS2.8) and (GMS3.6). It is not an idempotent on the full D_F, D_J . In particular applying it does not erase their endpoints or replace their normal source ideal.

GMS10. Full derived localization and its exact geometric action scope

Transport by Θ , and by V on the normal terms, identifies Q with $\mathcal{Q}$ and $Q(-1)$ with $\mathcal{Q}_+$. Derived Hom from the complete $\mathcal{Q}$ applied to (GMS9.7) gives

$$\mathrm{RHom}_M(\mathcal{Q}, D_G) \simeq \mathrm{RHom}_M(\mathcal{Q}, \mathcal{Q})[-1]. \quad (\text{GMS10.1})$$

The projection and inclusion of the degree-one summand are inverse up to the explicit contraction of its normal Hom summand from (GMS4.2)–(GMS4.4). This is a full-module statement, not an inference from finite spectral vectors.

The actual connecting morphism in the localization triangle is zero on $Q[-1]$ and is $e_+[-2]$ on $\mathcal{Q}_+[-2]$, by the simultaneous maps (GMS9.7). Consequently

$$\mathrm{RHom}_M(\mathcal{Q}, D_G) \longrightarrow \mathrm{RHom}_M(\mathcal{Q}, D_J[1]) \quad \text{is zero in the derived category.} \quad (\text{GMS10.2})$$

Indeed the only nonzero candidate component factors through the contractible complex $\mathrm{Hom}_M(P, \mathcal{Q}_+)$. Its contraction was constructed in (GMS4.2). Equivalently, the normal comparison inside the source triangle is precisely the chain-homotopy equivalence (GMS5.4). The Z term remains; the degree-one component lifts through the identity on $Q[-1]$ in (GMS9.7). Thus (GMS10.2) describes the actual full localization maps, with the full normal ideal retained.

The underlying source geometry gives the cover action for each integer $n \geq 1$:

$$(\rho_+(n) \oplus \rho_-(n), T_n, nT_n) \quad (\text{GMS10.3})$$

on degrees zero, one, and two. Its last factor is the positive degree of the actual holomorphic cover $z \mapsto z^n$. These are exactly the actions of $h(s) = n^s$ from (GMS9.4), since the normal component uses n^{s+1} . The same formulas for real $a > 0$ are coefficient actions; there is no asserted nonintegral holomorphic covering. All maps in (GMS5.1), (GMS8.2), and (GMS9.7) commute with the specified M -action and hence with these covers' coefficient actions.

Mirror actions retain their degree-dependent semilinearity. Under the actual half-Mellin transform, R is $F(s) \mapsto F(1-s)$, since substituting $u \mapsto u^{-1}$ in (GMS2.2) gives this identity. Thus the low-degree mirror changes $h(s)$ to $h(1-s)$. On the normal coordinate $\lambda = s+1$, the same source reflection changes λ to $3-\lambda$, so its multiplier is $h(3-\lambda)$. Both substitutions preserve M by its strip bounds. The geometric sign is retained separately: the antipodal map $-1/\bar{z}$ has degree minus one on the complex-oriented sphere and degree plus one on the equatorial loop; the holomorphic map $-1/z$ has degree plus one on the sphere and degree minus one on that loop. CSP/CDW compute their corresponding actions (chart swap, $-R, \varepsilon R$) on the finite model, with $\varepsilon = -1$ and $+1$, respectively. Neither geometric sign changes the stated conjugation of the generator.

In particular, a single substitution $\lambda \mapsto 1-\lambda$ is not used in both degrees, and no symmetry of the particular cutoff function E is presumed. Conjugating a multiplier map by a specified mirror gives the map with its conjugated multiplier and the retained geometric signs. The vanishing in (GMS4.4) and the derived inverse in (GMS5.4) remain statements for the displayed domains and actions.

GMS11. What the calculation establishes and retains

This continuation constructs the M -action on the entire actual coefficient row, and proves the two full derived separations (GMS4.4). It then supplies the explicit actual map $\beta : \mathcal{B} \rightarrow \mathcal{I}_+$, the full-source representative (GMS8.3)–(GMS8.4), and their chain-homotopy inverse property after derived Hom from the complete original quotient. The same separator constructs the unique continuous splitting of every strict Fréchet M -extension with these two separated quotient terms. Each operation has its exact domain and topology above.

The normal extension e_+ itself remains nonzero, with the original-factor boundary (GMS6.4). The global conclusion is its zero pullback against the separated original quotient and the full derived comparison (GMS5.4), not deletion of its kernel or a splitting of $\mathcal{B} \rightarrow \mathcal{Q}_+$. Likewise, the logarithmic-modulus ranges of the two

actual quotient spectra are $(0, 2)$ and $(2, 4)$, with scalars a^ρ and $a^{\rho+1}$ and all nilpotent coefficients. The multiplier argument controls the connecting maps between these specified full modules; it does not alter the original zero positions or assert a common pure weight.

The algebraic derived Hom statements use M -projective resolutions with algebraic Hom. The continuous splitting and source maps use explicit continuous formulas and quotient topologies. No claim about a general continuous derived category, an unconstructed extension of an arbitrary L -module to M , or a replacement of Deligne's geometric hypotheses is needed. The new global chain contraction is the completed attempted step beyond CLP13's finite coherent separation.

Two additional comparison boundaries are exact. First, the complete jet identities make c one on each actual original block of length m_ρ , zero on its translated block, and hence give those identities on the actual kernels $Q[(L - \rho)^r]$. In CLP13's family stalk the arithmetic variable is $y = s - \rho$ with $y^{m_\rho} = 0$, so the same identity holds there even though the parameter variable has longer nilpotence. The identities do not make c one on an arbitrary test module $\mathbb{C}[t]/(t - \rho)^r$ when $r > m_\rho$; its higher Taylor coefficients can act there. Nor does restriction of scalars from M replace a $\mathbb{C}[t]$ -derived Hom calculation by an M -derived Hom calculation. Second, no cyclicity or algebra unit of $\mathcal{Q} = \mathcal{B}/\mathcal{I}$ has been used. The entire quotient c/F_+ exists by the vanishing orders, but it has not been declared to belong to M . The proof uses the actual map $c\mathcal{B} \subset \mathcal{I}_+$, not an unproved multiplier Bézout identity or a claim that $\mathcal{I}_+ = F_+M$.

The full source separator and the original residue receiver

24 September 2026. Complete receiving proof, GSR0–GSR4. This calculation applies the full-module separation just constructed in GMS to the actual pairing and transfer in RTT and GTAH. Every coefficient operation occurs after complete-history arithmetic reconstruction. No addition, numerical coordinate, or parity is assigned to the supporting datum τ .

GSR0. Actual source and the comparison being tested

The receiving source is `../tau_weight_cohomology_20260924/CC_GLOBAL_MULTIPLIER_SEPARATION.md`, GMS0–GMS11, read completely at SHA256 3164c3ae00fbfbaf17bce809d48b7d0e56ff8a4c0991fdb374a4b11dd18bec59. Its underlying separator is the already proved GSL corridor construction. Here M is its entire multiplier algebra, $\mathcal{B}$ is the entire rapid-strip space, $\mathcal{J}$ is the full original nontrivial-zero jet ideal, and $\mathcal{J}_+$ is the full translated ideal. Put

$$\mathcal{Q} = \mathcal{B}/\mathcal{J}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+, \quad q : \mathcal{B} \longrightarrow \mathcal{Q}. \quad (\text{GSR0.1})$$

The actual global multiplier $c = E_+ \in M$ satisfies, with every original multiplicity,

$$c\mathcal{B} \subset \mathcal{J}_+, \quad (1 - c)\mathcal{B} \subset \mathcal{J}, \quad c_{\mathcal{Q}} = 1, \quad c_{\mathcal{Q}_+} = 0. \quad (\text{GSR0.2})$$

These are GMS3.5–GMS3.6. They hold on the complete modules, not only on finite jets. The map $\beta : \mathcal{B} \rightarrow \mathcal{J}_+$ is $F \mapsto cF$. GMS5 constructs its chain-homotopy inverse property after derived Hom from $\mathcal{Q}$; GMS6 retains the nonzero normal extension. We now calculate the effect of this particular map on the complete original Weil pairing and its residue receiver.

GSR1. The exact multiplier involution in the residue pairing

For $h \in M$, define the multiplier

$$h^\#(s) = \overline{h(1 - \overline{s})}. \quad (\text{GSR1.1})$$

This is an entire function: conjugating a Taylor expansion of h at $1 - \overline{s_0}$ gives its Taylor expansion at s_0 , with coefficient $(-1)^j \overline{h^{(j)}(1 - \overline{s_0})}/j!$. Its polynomial bound on a strip of width A follows from that of h on width $A + 1$. Thus $h^\# \in M$. Direct substitution proves $(h^\#)^\# = h$, $(hk)^\# = h^\#k^\#$, and $(\alpha h)^\# = \overline{\alpha}h^\#$. These are statements about reconstructed coefficient functions.

Let $\rho^\# = 1 - \overline{\rho}$. The exact functional equation of the original zeta function gives this involution on its nontrivial-zero divisor, preserving multiplicities. RTT9 retains its complete multiplier and exceptional terms. For $F, G \in \mathcal{Q}$, retain

$$W(F, G) = \sum_{\rho} m_{\rho} F(\rho) \overline{G(\rho^\#)}. \quad (\text{GSR1.2})$$

Each fixed multiplier has polynomial growth on the zero strip, while representatives of F, G have arbitrary rapid strip decay. The zero-count bound used in RTT1 therefore proves absolute convergence after multiplication by h or $h^\#$. Since $\overline{h^\#(\rho^\#)} = h(\rho)$, termwise substitution proves

$$W(m_h F, G) = W(F, m_{h^\#} G). \quad (\text{GSR1.3})$$

Let $S : \mathcal{Q} \rightarrow \mathcal{H}_{\text{res}}$ and $\hat{S} : \mathcal{H}_{\text{res}} \rightarrow \mathcal{Q}'_{\beta}$ be the actual RTT5 and GTR10 maps. Equation (GSR1.3), with h replaced by $h^\#$, gives the complete typed covariance

$$\hat{S} m_h = m'_{h^\#} \hat{S}. \quad (\text{GSR1.4})$$

Indeed both sides evaluated at G and then at F are $W(F, m_h G)$. The transpose is strong-continuous because the continuous multiplier sends bounded subsets of $\mathcal{Q}$ to bounded subsets. For $h(s) = a^s$, its reflected multiplier is exactly a^{1-s} , so (GSR1.4) recovers the full degree-weighted transfer $a(T_{1/a})'$. No degree factor is removed.

GSR2. Every full jet survives the separator on the original quotient

At each original zero ρ , the germ $c - 1$ vanishes to order m_ρ . Applying the Taylor formula from GSR1 at $\rho^\#$, and using $m_{\rho^\#} = m_\rho$, proves that $c^\# - 1$ has the same vanishing order at ρ . The full Leibniz formula then gives

$$m_c = 1_Q, \quad m_{c^\#} = 1_Q, \quad Sm_c = S, \quad m'_{c^\#} = 1_{Q'_\beta}. \quad (\text{GSR2.1})$$

This includes every higher jet in $\mathcal{Q}$. It is stronger than merely saying that $c(\rho) = 1$. In particular the exact residue identity is

$$\widehat{\mathcal{R}}(F, S(cG)) = \widehat{\mathcal{R}}(F, SG) = W(F, G). \quad (\text{GSR2.2})$$

On the value Hilbert space H , the multiplier induced by c is the identity as well. With GTAH's actual positive-adjoint defect $D_a = T_a^* - aT_{1/a}$, we therefore obtain

$$m_c D_a^* D_a = D_a^* D_a m_c = D_a^* D_a. \quad (\text{GSR2.3})$$

The injective anti-linear receiver $A : H \rightarrow \mathcal{H}_{\text{res}}$ from RTT6.2 retains this entire operator by GTAH6.7. Thus the full separator has an exact compatible action on the residue pairing, but that action does not change its value or its positive-adjoint defect. This is a calculation of the proposed use of c , not a conclusion about other maps supplied by the full source geometry.

GSR3. The representative lift and its complete kernel

The actual inclusion $\mathcal{J}_+ \hookrightarrow \mathcal{B}$ followed by q gives a continuous surjection

$$q_{\text{res}} : \mathcal{J}_+ \longrightarrow \mathcal{Q}, \quad F \longmapsto [F]_{\mathcal{J}}. \quad (\text{GSR3.1})$$

This map restricts the original quotient map to the full translated ideal. To prove surjectivity, take any representative $F \in \mathcal{B}$. Then $cF \in \mathcal{J}_+$ and $q(cF) = q(F)$ by (GSR0.2). Its kernel is exactly $\mathcal{J} \cap \mathcal{J}_+$. Consequently

$$0 \longrightarrow \mathcal{J} \cap \mathcal{J}_+ \longrightarrow \mathcal{J}_+ \xrightarrow{q_{\text{res}}} \mathcal{Q} \longrightarrow 0, \quad q_{\text{res}}\beta = q \quad (\text{GSR3.2})$$

is a strict exact sequence of Fréchet spaces and continuous M -maps. The ideals are closed; the open mapping theorem applies to the displayed continuous surjection between Fréchet spaces. This proves strictness, retaining its entire kernel.

There is no M -linear section of this displayed surjection. Fix an actual original zero ρ and use the original entire source factor

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad v_\rho(s) = \frac{F_0(s)}{s-\rho} \in \mathcal{B}. \quad (\text{GSR3.3})$$

The original quotient class $[v_\rho]$ is nonzero because its vanishing order at ρ is exactly $m_\rho - 1$, whereas $(s-\rho)[v_\rho] = [F_0] = 0$. An M -linear section would send this nonzero class to an entire function killed by $s-\rho$. That function must vanish on $\mathbb{C} \setminus \{\rho\}$ and hence everywhere, a contradiction. The existence of an actual nontrivial zeta zero here is the established original-zeta fact already used in CW8 and GMS6; no off-line zero is presumed.

This identifies the precise kind of lift already constructed: β lifts representatives from $\mathcal{B}$, and its derived comparison is GMS5. It does not supply a section from $\mathcal{Q}$. The dual restriction has the injective continuous map

$$q'_{\text{res}} : \mathcal{Q}'_\beta \longrightarrow (\mathcal{J}_+)'_\beta, \quad \lambda \longmapsto \lambda \circ q_{\text{res}}, \quad \beta' q'_{\text{res}} = q'. \quad (\text{GSR3.4})$$

Injectivity follows from surjectivity of q_{res} ; continuity follows from bounded-set preservation. The last identity is the transpose of (GSR3.2). In particular it applies to $\lambda = \widehat{\mathcal{R}}SG$, giving an exact comparison of the original trace through the full normal source. No assertion about an inverse for the strong subspace topology is needed.

GSR4. What this test supplies for the next attempt

GMS's full-module separation now has its exact receiving map to the original-zeta residue trace: (GSR1.4), (GSR2.2), and (GSR3.4). The local and global source factors, all multiplicities, all nilpotent coordinates, and the entire normal kernel remain. The derived lifting construction and the positive-adjoint calculation concern these actual connected objects.

For the requested positivity step, this test rules out treating the separator's action itself as a sign-changing or defect-annihilating operation on $\mathcal{Q}$: that action is exactly the identity. What would advance the argument is a source-derived adjoint identity for the actual transfer, rather than its central multiplier action. The concrete target is GTAH's $T_a^* = aT_{1/a}$ on the established value completion, with its actual maps back to the full source and residue dual. Its first attempted construction, reflection averaging, has already been calculated in GTAH2–GTAH3 and has the retained positive degree defect; its second attempted source correction, the full separator, is calculated here. Further work must use additional source structure and these measured defects, without declaring the target identity or positivity as an assumption.

Human-source context remains Connes–Consani's actual source sheaf, arXiv:0903.2024v3, §5, and Deligne's weight-controlled lifting argument, Weil II, §3.6. This note does not claim a new reading of either entire paper or an established purity theorem.

Actual extension classes, full source maps, and residue involutions

The complete received FOD and NEA proofs below decompose the actual source extension and compute the exact annihilators of its two cyclic class modules. The normal boundary is annihilated by the constructed separator. Its original component is retained by that same separator, through the exact pushout and scalar maps. Every full kernel and multiplicity is retained.

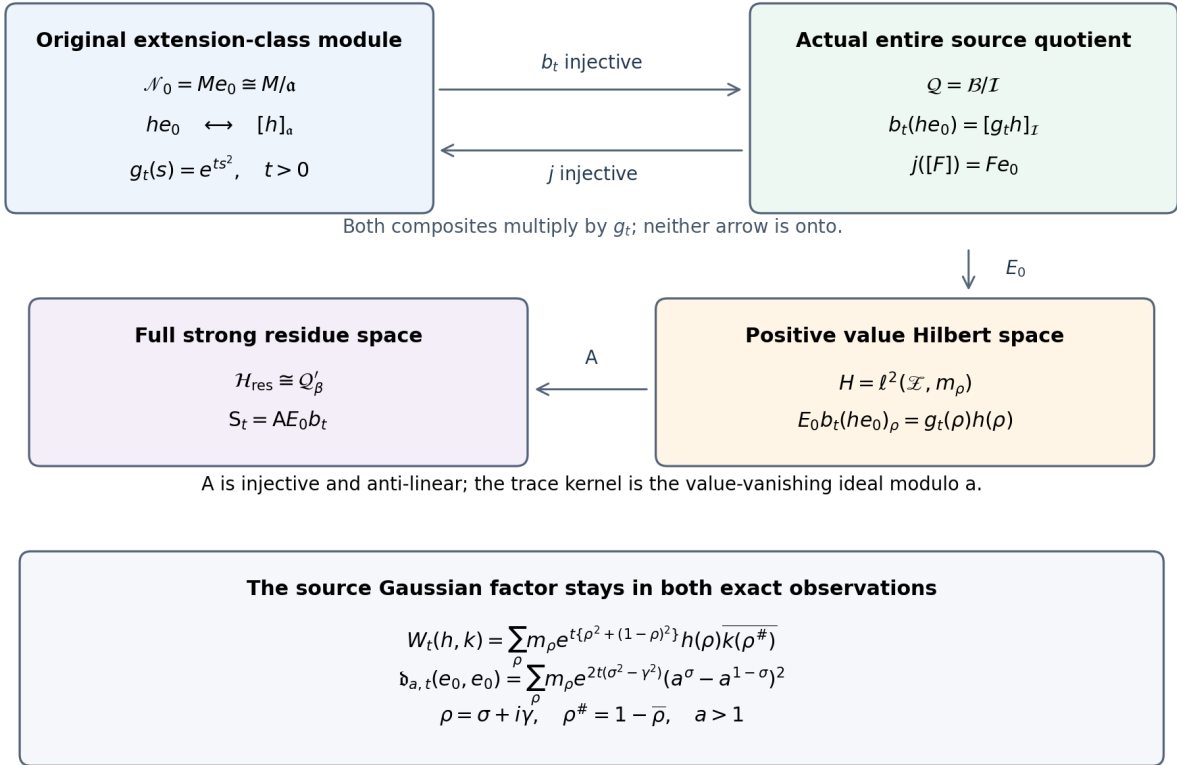
ECR constructs proper injective maps in both directions between the original cyclic extension-class module and the actual entire source quotient. Both composites multiply by the specified Gaussian source function; neither composite is declared an identity. Every local jet is preserved. The actual extension generator then tests the entire positive-adjoint defect, with its original Gaussian coefficient, through a proved map to the value Hilbert space and strong residue space. The independent ECRA review supplies direct source seminorm estimates and the exact comparison with the earlier global defect sum.

ECI constructs the actual original and normal involutions, their exact coordinate translation, and their maps into the entire strong residue receiver. The residue-dual involution contains the original functional-equation unit and reflection sign. Every polynomial-growth cyclic class has a canonical trace functional, whose strong convergence is proved. Its comparison with ECR retains the Gaussian factor and its involution factor. The normal pairing has degree factor a cubed, whereas the normal coefficient transfer has the factor a ; the resulting factor a squared is retained in their comparison. For positive integer $a=n$ these coefficient maps are realized by the actual geometric degree- n cover. These statements are on the specified domains and do not change the original degree-one pairing.

The complete proofs follow. The restored research goal remains active. These computations receive the source-extension separation and test its involution and duality. They do not assert that the surviving original positive-adjoint defect vanishes or that the Riemann hypothesis has been proved. The next calculation must use these actual maps, including their degree factors and kernels, rather than erase the measured difference.

The actual extension class reaches the full original residue receiver

Every actual zero and multiplicity; the two embeddings retain all jets



Proofs: ECR1-ECR6, RTT5-RTT6 and FOD4-FOD6. Original prime, Gamma, endpoint and finite trivial-divisor terms remain in ECR4.

Figure 11: The actual extension class and its exact source, Hilbert and residue maps. ECR1–ECR6 prove the two embeddings, their composites, their full jets, and both displayed complete sums. RTT5–RTT6 prove the residue receiver. FOD4–FOD6 identify the source-extension components. The full arithmetic terms remain in ECR4.

The complete source extension and its two actual obstruction components

Complete derivation. Stable proof locators FOD0–FOD8.

FOD0. The constructed source and the question being answered

The supporting datum is $\tau\langle Z_1; \text{no } Z_2 \rangle$. All operations below take place on the coefficient spaces after the complete-history arithmetic reconstruction. The separate branch counters and their complete original arithmetic remain. No addition, midpoint, metric, vector coordinate or numerical origin is assigned to that support.

The connected argument, its correction table, the verbatim WU050–WU055 and WU061–WU065 passages, and the task-local prerequisite instructions were reread for this calculation. PC12 records the check. The actual question after GMS is what its separator does to the full source extension, retaining the intersection ideal rather than keeping only the already split quotient. This question follows WU065: use the new result together with the full previous work, construct the next maps, and calculate them.

Inputs, retained with complete proofs in the cumulative repository, are GSL4–GSL6 for the actual global entire separator; CW1–CW8 for the full original source, topology and boundary maps; CLP12 for the actual derived localization boundary; and GMS0–GMS11 for the full multiplier action and complete-module homotopies. Root read those operative sections and the complete current GMS proof. The calculation below derives its source extension directly; it does not depend on acceptance of a separate peer draft.

The original source geometry is Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, with the signed-chart comparison in arXiv:2609.00299v1. The distinct lifting mechanism in Deligne, *La conjecture de Weil. II*, §3.6, is reconstructed in DC5–DC9 with its exact obstruction quotient. These are the previously recorded source readings, not a claim to have read new original sources in this continuation.

FOD1. Entire coefficients, both full ideals, and the original factors

Let M be the unital commutative algebra of entire h such that for every $A \geq 0$ there are C_A, N_A with

$$|h(x + it)| \leq C_A(1 + |t|)^{N_A} \quad (|x| \leq A). \quad (\text{FOD1.1})$$

Let $\mathcal{B}$ be the Fréchet space of entire F with

$$b_{A,N}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^N |F(s)| < \infty \quad \text{for every } A, N \geq 0. \quad (\text{FOD1.2})$$

Pointwise multiplication makes $\mathcal{B}$ an ideal in M ; a fixed h acts continuously by $b_{A,N}(hF) \leq C_A b_{A,N+N_A}(F)$.

Write $\mathcal{Z}$ for the distinct actual nontrivial zeros of the original ζ , each with multiplicity m_ρ . Their established open strip is $0 < \Re \rho < 1$. Put

$$\begin{aligned} \mathcal{I} &= \{F \in \mathcal{B} : \text{ord}_\rho F \geq m_\rho \ \forall \rho\}, \\ \mathcal{I}_+ &= \{F \in \mathcal{B} : \text{ord}_{\rho+1} F \geq m_\rho \ \forall \rho\}, \\ \mathcal{K} &= \mathcal{I} \cap \mathcal{I}_+, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{I}_+. \end{aligned} \quad (\text{FOD1.3})$$

The derivative evaluations are continuous by Cauchy’s formula on a larger strip, so the three ideals are closed. All their displayed quotient spaces carry their actual Fréchet quotient topologies.

The original source comparison is

$$\begin{aligned} \Theta a(s) &= \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, & \Theta^{-1} F(u) &= \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt, \\ \Sigma f(u) &= 2 \sum_{n \geq 1} f(nu), & \Theta \Sigma f(s) &= \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \end{aligned} \quad (\text{FOD1.4})$$

Here S consists of even Schwartz functions satisfying $f(0) = 0$ and $\int_{\mathbb{R}} f = 0$; the proved exact image is $J = \Sigma S = \Theta^{-1}\mathcal{J}$, with the continuous inverse Σ^{-1} . The coefficient space $A = \Theta^{-1}\mathcal{B}$ has all seminorms $\sup_{u>0}(u^N + u^{-N})|(u\partial_u)^j a(u)|$.

Retain the full source functions

$$f_0(v) = \frac{\pi}{2}v^2(2\pi v^2 - 3)e^{-\pi v^2}, \quad F_0(s) = \Theta\Sigma f_0(s) = \frac{s(s-1)}{8}\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$F_+(s) = F_0(s-1) = \frac{(s-1)(s-2)}{8}\pi^{-(s-1)/2}\Gamma((s-1)/2)\zeta(s-1). \quad (\text{FOD1.5})$$

Both belong to $\mathcal{B} \subset M$. Their divisors are exactly $\mathcal{Z}$ and $\mathcal{Z} + 1$, respectively, with the complete multiplicities. They are not substituted for the original function:

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\zeta'(s)/\zeta(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{FOD1.6})$$

The endpoint and trivial-zero comparisons retain

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = \frac{\pi}{24}, \quad F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2^r r!} \zeta'(-2r) \neq 0 \quad (r \geq 1). \quad (\text{FOD1.7})$$

Every value and derivative of F_+ is the corresponding value and derivative of F_0 at $s-1$. At an original zero,

$$F_0(\rho+z) = z^{m_\rho} U_\rho(z), \quad U_\rho(z) = \frac{(\rho+z)(\rho+z-1)}{8} \pi^{-(\rho+z)/2} \Gamma((\rho+z)/2) \frac{\zeta(\rho+z)}{z^{m_\rho}}, \quad (\text{FOD1.8})$$

and $U_\rho(0) \neq 0$. All products and divisions below use this complete local unit, not a replacement of it by one.

The proved separator is $c(s) = E_+(s) = 1 - E(s-1) \in M$, with

$$c-1 \in (s-\rho)^{m_\rho} \mathcal{O}_\rho, \quad c \in (s-\rho-1)^{m_\rho} \mathcal{O}_{\rho+1}. \quad (\text{FOD1.9})$$

GSL constructs E by its full Gaussian Cauchy integral and proves the strip estimates; GMS3 recalls that construction. Hence

$$c\mathcal{B} \subset \mathcal{J}_+, \quad (1-c)\mathcal{B} \subset \mathcal{J}, \quad c_{\mathcal{Q}} = 1, \quad c_{\mathcal{Q}_+} = 0. \quad (\text{FOD1.10})$$

FOD2. The complete source extension before discarding its kernel

Let q, q_+ be the two quotient maps. The actual joint map is

$$q_{\text{joint}} : \mathcal{B} \longrightarrow \mathcal{Q} \oplus \mathcal{Q}_+, \quad F \longmapsto (qF, q_+F). \quad (\text{FOD2.1})$$

It is onto: for representatives $f, g \in \mathcal{B}$, the function $cf + (1-c)g$ has the requested two quotient images by (FOD1.10). Its kernel is exactly $\mathcal{K}$, and

$$\bar{q}_{\text{joint}} : \mathcal{B}/\mathcal{K} \xrightarrow{\sim} \mathcal{Q} \oplus \mathcal{Q}_+, \quad [F]_{\mathcal{K}} \longmapsto (qF, q_+F),$$

$$\bar{q}_{\text{joint}}^{-1} : \mathcal{Q} \oplus \mathcal{Q}_+ \longrightarrow \mathcal{B}/\mathcal{K}, \quad ([f]_{\mathcal{J}}, [g]_{\mathcal{J}_+}) \longmapsto [cf + (1-c)g]_{\mathcal{K}} \quad (\text{FOD2.2})$$

is a continuous inverse. Independence of representatives uses $c\mathcal{J} \subset \mathcal{K}$ and $(1-c)\mathcal{J}_+ \subset \mathcal{K}$. To check continuity of the inverse, the continuous map on $\mathcal{B} \oplus \mathcal{B}$ kills $\mathcal{J} \oplus \mathcal{J}_+$ and therefore descends through that product quotient. Thus the full exact sequence is strict:

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{B} \xrightarrow{q_{\text{joint}}} \mathcal{Q} \oplus \mathcal{Q}_+ \longrightarrow 0. \quad (\text{FOD2.3})$$

Equation (FOD2.2) splits the quotient into two summands. It does not split (FOD2.3).

Pulling (FOD2.3) back along the two actual summand inclusions gives

$$\begin{aligned} e_{\text{low}} : \quad 0 \rightarrow \mathcal{K} \rightarrow \mathcal{J}_+ \xrightarrow{q|_{\mathcal{J}_+}} \mathcal{Q} \rightarrow 0, \\ e_{\text{high}} : \quad 0 \rightarrow \mathcal{K} \rightarrow \mathcal{J} \xrightarrow{q_+|_{\mathcal{J}}} \mathcal{Q}_+ \rightarrow 0. \end{aligned} \quad (\text{FOD2.4})$$

These are actual pullbacks: the zero normal quotient condition is precisely membership in $\mathcal{J}_+$, and the zero original quotient condition is precisely membership in $\mathcal{J}$.

The first row is strict with explicit quotient inverse $[f]_{\mathcal{J}} \mapsto [cf]_{\mathcal{K}}$. For $b \in \mathcal{J}_+$, $cb - b = -(1-c)b \in \mathcal{K}$, which proves the reverse composition. The second row has inverse $[g]_{\mathcal{J}_+} \mapsto [(1-c)g]_{\mathcal{K}}$, by the same two full jet conditions. All these maps are continuous by the actual multiplier estimates.

FOD3. Pushouts recover the original and normal rows exactly

Push the first row of (FOD2.4) out along $\mathcal{K} \hookrightarrow \mathcal{J}$. Its middle module and comparison map are

$$\mathcal{P}_0 = (\mathcal{J} \oplus \mathcal{J}_+)/\{(k, -k) : k \in \mathcal{K}\}, \quad [(i, b)] \mapsto i + b \in \mathcal{B}. \quad (\text{FOD3.1})$$

This map is onto by $F = (1-c)F + cF$. Its kernel is exactly the relation subspace, since $i + b = 0$ forces $i = -b \in \mathcal{K}$. Its continuous inverse is

$$F \mapsto [((1-c)F, cF)]. \quad (\text{FOD3.2})$$

To verify the inverse on a pair (i, b) , subtract (i, b) from the pair in (FOD3.2) at $F = i + b$. The result is $(k, -k)$, where $k = -ci + (1-c)b \in \mathcal{K}$. This also proves the inverse without selecting representatives of either quotient. The relation subspace is closed because $\mathcal{K}$ is closed, so all middle spaces have the stated Fréchet quotient topology.

Consequently the pushout is exactly

$$e_0 : \quad 0 \rightarrow \mathcal{J} \rightarrow \mathcal{B} \xrightarrow{q} \mathcal{Q} \rightarrow 0. \quad (\text{FOD3.3})$$

Interchanging the ideals and replacing c by $1-c$ gives, with all signs the same,

$$\begin{aligned} (\mathcal{J}_+ \oplus \mathcal{J})/\{(k, -k) : k \in \mathcal{K}\} &\xrightarrow{\sim} \mathcal{B}, \quad [(i, b)] \mapsto i + b, \\ e_+ : \quad 0 \rightarrow \mathcal{J}_+ \rightarrow \mathcal{B} &\xrightarrow{q_+} \mathcal{Q}_+ \rightarrow 0. \end{aligned} \quad (\text{FOD3.4})$$

Its inverse is $F \mapsto [(cF, (1-c)F)]$. Thus e_0 and e_+ are the specified pushouts of the same actual full-source extension.

FOD4. Every multiplier that kills each component

Use algebraic M -module Ext , with the strict continuous conclusions proved separately below. Define two ideals of the multiplier ring itself:

$$\mathfrak{a} = \{h \in M : \text{ord}_{\rho} h \geq m_{\rho} \ \forall \rho\}, \quad \mathfrak{a}_+ = \{h \in M : \text{ord}_{\rho+1} h \geq m_{\rho} \ \forall \rho\}. \quad (\text{FOD4.1})$$

These are not renamed copies of $\mathcal{J}, \mathcal{J}_+$: their functions obey polynomial strip bounds, not necessarily rapid strip decay. Precisely, $\mathcal{J} = \mathcal{B} \cap \mathfrak{a}$ and $\mathcal{J}_+ = \mathcal{B} \cap \mathfrak{a}_+$.

For an actual row $0 \rightarrow N \rightarrow X \rightarrow Y \rightarrow 0$, the pushout along multiplication by h has middle module

$$P_h = (N \oplus X)/\{(hn, -n) : n \in N\}. \quad (\text{FOD4.2})$$

It splits as an M -extension exactly when there is an M -linear $r : X \rightarrow N$ with $r|_N = h$. In one direction, such r gives the retraction $[(n, x)] \mapsto n + r(x)$, and hence a splitting. Conversely, a split extension has

an M -linear retraction onto its kernel, obtained by subtracting its section; composing that retraction with $x \mapsto [(0, x)]$ gives r , because $[(0, n)] = [(hn, 0)]$.

Apply this first to e_{low} , with $N = \mathcal{K}$ and $X = \mathcal{J}_+$. Put

$$G(s) = F_0(s)F_+(s) \in \mathcal{K} \subset M. \quad (\text{FOD4.3})$$

This is a nonzero entire function, with all factors from (FOD1.5). For $b \in \mathcal{J}_+$, $Gb \in \mathcal{K}$. Therefore any r supplied by a splitting satisfies

$$Gr(b) = r(Gb) = hGb. \quad (\text{FOD4.4})$$

Cancellation is valid as an identity of entire functions: off the discrete zeros of G the two functions are equal, and the identity theorem extends that equality everywhere. Hence $r(b) = hb$.

In particular $hF_+ \in \mathcal{K}$. At each original ρ , the function F_+ is a unit, since $-1 < \Re(\rho - 1) < 0$ and the divisor of F_0 is $\mathcal{Z}$. Thus h vanishes to order at least m_ρ at every ρ , proving $h \in \mathfrak{a}$. Conversely, $h \in \mathfrak{a}$ gives $h\mathcal{J}_+ \subset \mathcal{K}$ by the full Leibniz formula, so $r(b) = hb$ constructs the splitting. We have proved

$$\text{Ann}_M(e_{\text{low}}) = \mathfrak{a}. \quad (\text{FOD4.5})$$

The identical argument for e_{high} uses F_0 , which is a unit at every $\rho + 1$, and proves

$$\text{Ann}_M(e_{\text{high}}) = \mathfrak{a}_+. \quad (\text{FOD4.6})$$

For the original row e_0 , use $G = F_0$ in the cancellation argument with $N = \mathcal{J}$, $X = \mathcal{B}$. It forces $r(b) = hb$. The function $b(s) = e^{s^2}$ belongs to $\mathcal{B}$, since its modulus is $e^{x^2 - t^2}$ on $s = x + it$, and it is nowhere zero. Thus $hb \in \mathcal{J}$ forces all the original zero jets of h . The converse is again multiplication. Replacing $F_0, \mathcal{J}$ by $F_+, \mathcal{J}_+$ proves the normal result. Therefore

$$\text{Ann}_M(e_0) = \mathfrak{a}, \quad \text{Ann}_M(e_+) = \mathfrak{a}_+. \quad (\text{FOD4.7})$$

Every splitting in the sufficient direction is continuous and explicit:

$$[(n, x)] \mapsto (n + hx, [x]), \quad (n_0, [x]) \mapsto [(n_0 - hx, x)]. \quad (\text{FOD4.8})$$

The second formula is independent of the lift x , since changing it by n changes the pair by $(-hn, n)$, a relation. Its continuity follows by descending the continuous map on $N \oplus X$ through the quotient $N \oplus X \rightarrow N \oplus Y$. The relation subspace in (FOD4.2) is closed: convergence of $(hn_\nu, -n_\nu)$ forces convergence of n_ν in the closed space N , and continuity of multiplication gives the first coordinate. Thus (FOD4.8) is an actual Fréchet isomorphism. Necessity was proved even for algebraic splittings, so these annihilator criteria are also exact for continuous M -splittings.

FOD5. The full obstruction decomposition and the separator

Let e_{joint} denote the Yoneda class of (FOD2.3). Resolving the two summands separately and taking their direct sum gives the exact identification

$$\begin{aligned} \text{Ext}_M^1(\mathcal{Q} \oplus \mathcal{Q}_+, \mathcal{K}) &\simeq \text{Ext}_M^1(\mathcal{Q}, \mathcal{K}) \oplus \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{K}), \\ e_{\text{joint}} &= (e_{\text{low}}, e_{\text{high}}). \end{aligned} \quad (\text{FOD5.1})$$

The identification is pullback along the two summand inclusions, which were calculated in (FOD2.4). Thus

$$\text{Ann}_M(e_{\text{joint}}) = \mathfrak{a} \cap \mathfrak{a}_+. \quad (\text{FOD5.2})$$

By (FOD1.9), $c \in \mathfrak{a}_+$, $1 - c \in \mathfrak{a}$, so these two multiplier ideals are comaximal. Their intersection equals their product: if z belongs to both, $z = zc + z(1 - c)$ is a sum of two products, and the reverse inclusion follows from their being ideals. Consequently the actual cyclic image of the full extension has the exact description

$$Me_{\text{joint}} \simeq M/(\mathfrak{a} \cap \mathfrak{a}_+) \xrightarrow{\sim} M/\mathfrak{a} \oplus M/\mathfrak{a}_+. \quad (\text{FOD5.3})$$

The first isomorphism sends the class of h to he_{joint} , with kernel just proved. The second has inverse $([h], [k]) \mapsto [ch + (1-c)k]$; independence and both compositions follow from the same full jet identities as in (FOD2.2).

The complete resulting actions are

$$\begin{aligned} ce_{\text{joint}} &= (e_{\text{low}}, 0), & (1-c)e_{\text{joint}} &= (0, e_{\text{high}}), \\ ce_{\text{low}} &= e_{\text{low}}, & ce_{\text{high}} &= 0, \\ ce_0 &= e_0, & ce_+ &= 0. \end{aligned} \tag{FOD5.4}$$

None of $e_{\text{low}}, e_{\text{high}}, e_0, e_+$ is zero: the original zeta has a nontrivial zero, so each ideal in (FOD4.1) is proper, and its annihilator does not contain one. A direct witness for e_{low} is

$$v_\rho(s) = F_0(s)/(s-\rho), \quad qv_\rho \neq 0, \quad (L-\rho)qv_\rho = 0. \tag{FOD5.5}$$

The function is entire, rapidly decreasing on closed vertical strips, and has exact order $m_\rho - 1$ at ρ . Any M -section into $\mathcal{J}_+$ would send this nonzero quotient class to an entire function killed by $s-\rho$, which must be zero. The normal witness is $F_+(s)/(s-\rho-1)$. These use actual zeros and are not hypothetical alternative arithmetic.

The pushouts in FOD3 induce isomorphisms on the respective cyclic class modules $Me_{\text{low}} \rightarrow Me_0$ and $Me_{\text{high}} \rightarrow Me_+$: they are onto by their displayed generators, and injective by the equal annihilators (FOD4.5)–(FOD4.7). No claim of an isomorphism of their entire Ext groups follows.

FOD6. The exact original quotient retained in the normal ideal

The strict map $q_{\text{res}} = q|_{\mathcal{J}_+} : \mathcal{J}_+ \rightarrow \mathcal{Q}$ satisfies

$$q_{\text{res}}\beta = q, \quad \beta : \mathcal{B} \rightarrow \mathcal{J}_+, \quad \beta(F) = cF. \tag{FOD6.1}$$

Both equalities are for the actual source maps, not an inverse of q_{res} . Their continuous strong transposes satisfy

$$\beta' q'_{\text{res}} = q'. \tag{FOD6.2}$$

Indeed composition with any continuous linear functional proves the identity; continuity for strong duals follows because a continuous linear map sends bounded sets to bounded sets. The kernel of q_{res} remains $\mathcal{K}$, and FOD4–FOD5 show exactly why c does not split this row.

For any continuous pairing P already defined on $\mathcal{Q}$ and any of its established dual receivers, composing its first argument with $q_{\text{res}}\beta$ gives the original value $P(qF, \cdot)$ by (FOD6.1). This is an identity for existing pairings; it introduces neither a positive form nor a replacement of the complete quotient by values alone.

Define $V : \mathcal{B}_s \rightarrow \mathcal{B}_\lambda$ by $(VF)(\lambda) = F(\lambda - 1)$. This is the actual coordinate transport from the original normal variable $s = \lambda - 1$; it sends $\mathcal{J}$ to $\mathcal{J}_+$. In that variable the representative $\Theta^{-1}V^{-1}\beta V\Theta : A \rightarrow J$ is exactly

$$a \mapsto \Theta^{-1}[(1 - E(s))\Theta a(s)](u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} (1 - E(1/2 + it))\Theta a(1/2 + it)u^{-it} dt, \tag{FOD6.3}$$

followed by the proved inverse $\Sigma^{-1} : J \rightarrow S$ when a Schwartz representative is wanted. The original normal multiplier is $h(s+1)$, and the dilation is a^{s+1} , retaining its factor a . Translating to λ changes that action to $h(\lambda)$. The map q_{res} in (FOD6.1) is stated in this translated coordinate, as are the two ideals in FOD1.3. Its source-coordinate identity is $q_{\text{res}}\beta V = qV$; thus (FOD6.3) does not silently replace the translated quotient by the original source-variable quotient.

FOD7. The actual localization boundary and the two actions on it

The proved actual coefficient complexes are

$$D_J = (W \xrightarrow{r_+ - r_-} J \xrightarrow{0} J(-1)), \quad D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)), \quad D_G = (0 \rightarrow Q \xrightarrow{0} Q(-1)), \quad (\text{FOD7.1})$$

in degrees 0, 1, 2, with zero second differential. All four endpoint characters in W remain $h(0), h(1), h(1), h(0)$. GMS9 proves simultaneous M -linear quasi-isomorphisms

$$D_J \simeq Z[0] \oplus \mathcal{I}_+[-2], \quad D_F \simeq Z[0] \oplus \mathcal{Q}[-1] \oplus \mathcal{B}[-2], \quad D_G = \mathcal{Q}[-1] \oplus \mathcal{Q}_+[-2]. \quad (\text{FOD7.2})$$

Here the normal terms have been transported by Θ and V , and $Z = \ker(r_+ - r_-)$. These are the already constructed quasi-isomorphisms with explicitly contractible kernels, not inserted sections of $\mathcal{B} \rightarrow \mathcal{Q}$.

The actual connecting morphism $\delta : D_G \rightarrow D_J[1]$ is zero on the low summand and is $e_+[-2]$ on the normal summand. Let

$$\mathcal{O}_{\text{loc}} = \{h\delta : h \in M\} \subset \text{Hom}_{D(M)}(D_G, D_J[1]), \quad (\text{FOD7.3})$$

where $h\delta$ is central postcomposition on the target, equivalently precomposition on the source. Taking the normal inclusion and projection in (FOD7.2) gives

$$h\delta = 0 \iff he_+ = 0, \quad \mathcal{O}_{\text{loc}} \simeq M/\mathfrak{a}_+, \quad c\delta = 0. \quad (\text{FOD7.4})$$

The forward implication is obtained by that inclusion and projection; the reverse follows from the zero low component. Thus this is the cyclic image of the actual localization boundary, with its full normal ideal retained.

For every derived M -morphism $f : \mathcal{Q}[k] \rightarrow D_G$, centrality and $c_{\mathcal{Q}} = 1$ give

$$\delta f = \delta f c_{\mathcal{Q}[k]} = \delta c_{D_G} f = c_{D_J[1]} \delta f = 0. \quad (\text{FOD7.5})$$

This proves the actual lifting obstruction for this specified derived source test is zero. The exact Hom sequence of the actual localization triangle therefore gives a lift $\tilde{f} : \mathcal{Q}[k] \rightarrow D_F$. More concretely, (FOD7.2) and the complete contraction of $\text{RHom}_M(\mathcal{Q}, \mathcal{Q}_+)$ in GMS4 show that f has zero normal component in the derived category; retaining its low component as the $\mathcal{Q}[-1]$ component of D_F constructs a lift. The complete set of lifts is a coset of the image of $\text{Hom}_{D(M)}(\mathcal{Q}[k], D_J) \rightarrow \text{Hom}_{D(M)}(\mathcal{Q}[k], D_F)$. No unique cochain representative is asserted.

The action used in (FOD7.3) must be distinguished from conjugation. For the actual arithmetic operator T_a , equivariance is

$$(T_a)_{D_J[1]} \delta = \delta (T_a)_{D_G}.$$

Consequently

$$(T_a)_{D_J[1]} \delta (T_a)_{D_G}^{-1} = \delta. \quad (\text{FOD7.6})$$

The boundary is fixed under this conjugation. Its cyclic module under postcomposition in (FOD7.3) is different; a weight calculated for that module is not a conjugation weight of the boundary map.

FOD8. Exact comparison with the lifting target

The construction now retains three different facts simultaneously. The quotient $\mathcal{B}/\mathcal{K}$ splits into its original and normal pieces by (FOD2.2). The source extension before that quotient has two nonzero components, and the separator retains one while annihilating the other, exactly as in (FOD5.4). The actual derived localization boundary uses the normal component, so its composition with every displayed complete-original-quotient source class vanishes by (FOD7.5).

The two full cyclic class modules are not declared to be the full coefficient ideals or the full quotients $\mathcal{Q}, \mathcal{Q}_+$. Their exact descriptions are $M/\mathfrak{a}$ and $M/\mathfrak{a}_+$, with every jet retained. The calculation is an actual boundary calculation, not the splitting of an independently chosen extension. It also proves that moving an original class into the full normal ideal through β does not erase the original class or its existing pairing.

Deligne's precise obstruction receiver is $\text{im } \partial/\partial j(K)$, with its specialization and inertia maps, DC5.3–DC5.4. The module $\mathcal{O}_{\text{loc}}$ in (FOD7.3) is instead the cyclic image of a specified derived boundary under central coefficients. Equations (FOD7.4)–(FOD7.5) prove its actual effect without identifying those two receivers by their names. No new mixed étale realization, common modulus of original zeros, or RH conclusion is inferred. The source and full kernel comparisons proved above remain available for constructing that further geometric identification.

The exact annihilator of the actual normal extension

Complete independent derivation. Stable proof locators NEA0–NEA10.

NEA0. Construction stage and source coverage

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The present operations act on the complete arithmetic coefficient receiver already constructed after whole-history reconstruction. The two branches retain separate counters. No addition, subtraction, midpoint, metric, or numerical coordinate is assigned to τ .

The current READ_FIRST_USER_CONSTRUCTION.md, correction-precedence table and latest stages of USER_ARGUMENT_RECONSTRUCTION.md, and PC10–PC11 prerequisite records were recalled for this continuation. The complete correction corpus remains the governing record. WU061 concerns the actual lifting map; WU062 requires checking every operation against the construction; WU065 requires a concrete attempted next step using the accumulated findings. Here the input is the actual nonzero normal extension retained by GMS, and the step is to identify exactly its multiplier annihilator and then the module generated by the actual connecting morphism.

The direct mathematical prerequisites are CC_GLOBAL_MULTIPLIER_SEPARATION.md, GMS0–GMS11, completely derived and independently reviewed at SHA256 3164C3AE00FBFBAF17BCE809D48B7D0E56FF8A4C0991FDB374A4B11DD18BEC59; CC_LOCALIZATION_POLYNOMIAL_WEIGHTS .md, CLP12–CLP13; and CC_ACTUAL_WEIGHT_LIFT_COMPARISON.md, CW1–CW8. The full current GMS proof and its actual normal row are the source for the operations below. The exact source theorem and continuous Schwartz inverse are the SSI/ESI results received through CW1 and GMS2/GMS8. The human-source relation is the previously read Connes–Consani §5 construction and the Deligne §3.6 comparison recorded in those notes. No new human-source reading or new geometric specialization map is claimed here.

NEA1. The actual ring, full ideal, and original source

Use the unital commutative ring

$$M = \{h \in \mathcal{O}(\mathbb{C}) : \forall A \geq 0 \exists C_A > 0, N_A \in \mathbb{N}, |h(x + it)| \leq C_A(1 + |t|)^{N_A} \text{ for } |x| \leq A\}, \quad (\text{NEA1.1})$$

and the actual Fréchet module

$$\mathcal{B} = \{F \in \mathcal{O}(\mathbb{C}) : b_{A,N}(F) = \sup_{|\Re \lambda| \leq A} (1 + |\Im \lambda|)^N |F(\lambda)| < \infty \text{ for all } A, N \geq 0\}. \quad (\text{NEA1.2})$$

Let $\mathcal{Z}$ be the original nontrivial-zero divisor of ζ , with multiplicities m_ρ . Its translated divisor is

$$\mathcal{D}_+ = \sum_{\rho \in \mathcal{Z}} m_\rho [\rho + 1], \quad 1 < \Re(\rho + 1) < 2. \quad (\text{NEA1.3})$$

The support notation counts each distinct zero once, with its separately retained multiplicity. Define

$$\begin{aligned} \mathcal{J}_+ &= \{F \in \mathcal{B} : F^{(j)}(\rho + 1) = 0 \text{ for } 0 \leq j < m_\rho, \rho \in \mathcal{Z}\}, \\ \mathcal{Q}_+ &= \mathcal{B}/\mathcal{J}_+, \\ K_+ &= \{h \in M : h^{(j)}(\rho + 1) = 0 \text{ for } 0 \leq j < m_\rho, \rho \in \mathcal{Z}\}. \end{aligned} \quad (\text{NEA1.4})$$

Thus $\mathcal{J}_+ = \mathcal{B} \cap K_+$; K_+ is an ideal of M , whereas $\mathcal{J}_+$ is the full closed ideal within the actual source module. They are not identified with each other. Every multiplier acts continuously on $\mathcal{B}$ and its closed invariant subspaces and quotients, since

$$b_{A,N}(hF) \leq C_A b_{A,N+N_A}(F). \quad (\text{NEA1.5})$$

The derivative product formula preserves all the vanishing orders.

The complete original transform is

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = C_0(s) \zeta(s),$$

$$F_+(\lambda) = F_0(\lambda-1) = \frac{(\lambda-1)(\lambda-2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda-1)/2) \zeta(\lambda-1). \quad (\text{NEA1.6})$$

Both are the actual entire source functions from GMS2. In particular $F_+ \in \mathcal{J}_+ \subset \mathcal{B} \subset M$, it is nonzero, and its zero divisor is exactly $\mathcal{D}_+$. The original arithmetic is still

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{NEA1.7})$$

Thus the unit and every prime-power repetition remain. The exceptional entire values of the full source factor are

$$F_+(1) = F_+(2) = \frac{1}{8}, \quad F_+(0) = \frac{\pi}{24},$$

$$F_+(1-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r)$$

$$= F_+(2+2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1). \quad (\text{NEA1.8})$$

They retain the pole and Gamma cancellations and the trivial-zero derivatives; those contributions are not replaced by extra zeros of F_+ .

The object under study is exactly

$$e_+ : 0 \longrightarrow \mathcal{J}_+ \xrightarrow{\iota} \mathcal{B} \xrightarrow{q_+} \mathcal{Q}_+ \longrightarrow 0, \quad [e_+] \in \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+). \quad (\text{NEA1.9})$$

We write e_+ for its class when no ambiguity occurs. It is the translated normal row of CW/CLP, not a separately chosen extension. The M -action on the unshifted normal source is $h(s+1)$; translation $VF(\lambda) = F(\lambda-1)$ gives the displayed $h(\lambda)$ -action and retains the dilation factor a^{s+1} .

NEA2. Every multiplier pushout, with its actual topology

For $h \in M$, pushing (NEA1.9) out along multiplication by h on $\mathcal{J}_+$ gives

$$\mathcal{P}_h = (\mathcal{J}_+ \oplus \mathcal{B}) / \{(hj, -j) : j \in \mathcal{J}_+\}, \quad i_h(i) = [(i, 0)], \quad p_h([(i, b)]) = q_+(b). \quad (\text{NEA2.1})$$

The relation subspace is closed: convergence of $(hj_\nu, -j_\nu)$ in the product implies convergence of j_ν in the closed subspace $\mathcal{J}_+ \subset \mathcal{B}$, and continuity of multiplication identifies the first limit as multiplication by h of that limit. Thus $\mathcal{P}_h$ is Fréchet. Each fixed multiplier acts continuously on it, because it preserves the relation subspace.

The maps form the exact row

$$0 \longrightarrow \mathcal{J}_+ \xrightarrow{i_h} \mathcal{P}_h \xrightarrow{p_h} \mathcal{Q}_+ \longrightarrow 0. \quad (\text{NEA2.2})$$

Indeed i_h is injective since a relation with second coordinate zero has $j = 0$; p_h is onto; and when $b \in \mathcal{J}_+$, the exact identity

$$[(i, b)] = [(i + hb, 0)] \quad (\text{NEA2.3})$$

identifies its kernel with the image of i_h . These are continuous maps. The Fréchet open mapping theorem applied to the continuous bijection $\mathcal{J}_+ \rightarrow \ker p_h$ makes it a topological isomorphism; applied to the continuous surjection p_h , it gives the quotient topology. The hypotheses hold because $\ker p_h$ is closed and all the spaces are Fréchet. Thus this is a strict extension, not merely an algebraic presentation endowed with an unspecified topology. Its Yoneda class is he_+ .

We now prove the splitting criterion directly. If (NEA2.2) has any algebraic M -linear section s , define

$$R = i_h^{-1}(1 - sp_h) : \mathcal{P}_h \rightarrow \mathcal{J}_+, \quad r(b) = R([(0, b)]). \quad (\text{NEA2.4})$$

Since $(1 - sp_h)x$ lies in the kernel, the definition is valid, and $Ri_h = 1$. Formula (NEA2.3) implies

$$r|_{\mathcal{J}_+} = h|_{\mathcal{J}_+}. \quad (\text{NEA2.5})$$

Conversely, if an M -linear map $r : \mathcal{B} \rightarrow \mathcal{J}_+$ satisfies (NEA2.5), the map $[(i, b)] \mapsto i + r(b)$ is well-defined, and a section is

$$q \mapsto [(-r(b), b)] \quad (q_+(b) = q). \quad (\text{NEA2.6})$$

Changing b by $j \in \mathcal{J}_+$ changes this pair by $(-hj, j)$, a relation. Thus

$$he_+ = 0 \iff \text{there is an } M\text{-map } r : \mathcal{B} \rightarrow \mathcal{J}_+ \text{ extending } h|_{\mathcal{J}_+}. \quad (\text{NEA2.7})$$

This proves the specific endomorphism-lifting assertion usually expressed by the connecting homomorphism in Hom-Ext, without assuming it as an uncomputed criterion.

NEA3. Necessity, sufficiency, and uniqueness

Suppose r exists in (NEA2.7). For every $b \in \mathcal{B}$, multiplication by the actual original source $F_+ \in M$ gives

$$F_+r(b) = r(F_+b) = hF_+b. \quad (\text{NEA3.1})$$

Here $F_+b \in \mathcal{J}_+$: F_+ has all the required zeros, and it is an M -multiplier of $\mathcal{B}$. The middle equality consequently uses the prescribed restriction (NEA2.5). The nonzero entire function F_+ can be cancelled from this identity: outside its discrete zero set the other two factors agree, hence their entire continuations agree. Thus necessarily

$$r(b) = hb \quad \text{for every } b \in \mathcal{B}, \quad h\mathcal{B} \subset \mathcal{J}_+. \quad (\text{NEA3.2})$$

To read the precise multiplicities from this inclusion, take the single zero-free source-space test function

$$g(\lambda) = e^{\lambda^2} \in \mathcal{B}. \quad (\text{NEA3.3})$$

It belongs to the existing coefficient space because $b_{A,N}(g) \leq e^{A^2} \sup_t (1 + |t|)^N e^{-t^2} < \infty$. It is a unit in each local holomorphic ring. Thus $hg \in \mathcal{J}_+$ forces $\text{ord}_{\rho+1} h \geq m_\rho$ for every actual zero. This test is a function in the already constructed source space, not an alternative arithmetic or an alternative zeta function.

Conversely, these full vanishing orders make $hb \in \mathcal{J}_+$ for every $b \in \mathcal{B}$ by the derivative product formula. The map $r_h(b) = hb$ is continuous by (NEA1.5), so (NEA2.6) gives a continuous splitting. We have proved

$$\boxed{\text{Ann}_M(e_+) = K_+ = \{h \in M : \text{ord}_{\rho+1} h \geq m_\rho \text{ for every } \rho \in \mathcal{Z}\}.} \quad (\text{NEA3.4})$$

The same source-space test shows

$$K_+ = \text{Ann}_M(\mathcal{Q}_+), \quad (\text{NEA3.5})$$

since annihilation of the quotient is exactly the inclusion $h\mathcal{B} \subset \mathcal{J}_+$.

For every $h \in K_+$, the full continuous isomorphism of its pushout with the split row is

$$\Phi_h([(i, b)]) = (i + hb, q_+(b)), \quad \Phi_h^{-1}(i_0, q) = [(i_0 - hb, b)] \quad (q_+(b) = q). \quad (\text{NEA3.6})$$

The first coordinate is in the full $\mathcal{J}_+$. Both formulas are independent of representatives by (NEA2.1). Their composites are identities by substitution. For continuity of the inverse, the map $(i_0, b) \mapsto [(i_0 - hb, b)]$ is continuous and kills $\{0\} \oplus \mathcal{J}_+$; it therefore descends through the product quotient. The resulting section is $q \mapsto [(-hb, b)]$.

Every such section is unique, even among algebraic M -linear sections. Indeed

$$\text{Hom}_M(\mathcal{Q}_+, \mathcal{J}_+) = 0, \quad \text{Hom}_M(\mathcal{Q}_+, \mathcal{B}) = 0. \quad (\text{NEA3.7})$$

To prove both statements, observe that F_+ annihilates $\mathcal{Q}_+$ and acts injectively by multiplication on either entire-function target. For an M -map v , the identity $F_+v(q) = v(F_+q) = 0$ therefore forces $v = 0$. The difference of two sections of (NEA2.2) is a map into its kernel, so it is zero. Consequently any algebraic splitting that exists is automatically the explicit continuous one in (NEA3.6); no continuity of a hypothetical algebraic section was assumed in the necessity proof.

The actual extension is nonzero: $1 \notin K_+$, since the actual divisor is nonempty, as established in CW8. Thus (NEA3.4) includes $e_+ \neq 0$. It also proves that no classification of arbitrary endomorphisms of $\mathcal{B}$, no principal-ideal assertion, and no unproved growth bound for h/F_+ is needed.

NEA4. Pullback of every multiplier on the quotient

For clarity, lifting the quotient endomorphism $h_{\mathcal{Q}_+}$ is a different presentation of the same killed class. Its actual pullback is

$$\mathcal{E}_h = \{(b, q) \in \mathcal{B} \oplus \mathcal{Q}_+ : q_+(b) = hq\}, \quad i'_h(j) = (j, 0), \quad p'_h(b, q) = q. \quad (\text{NEA4.1})$$

It is a closed subspace of the product, with its Fréchet topology, and is a strict exact extension by the same open-mapping argument as in NEA2. There is an explicit continuous M -linear isomorphism from the pushout to this pullback:

$$\mathcal{P}_h \longrightarrow \mathcal{E}_h, \quad [(i, b)] \mapsto (i + hb, q_+(b)). \quad (\text{NEA4.2})$$

For arbitrary h , the first coordinate here is in $\mathcal{B}$, not asserted to be in $\mathcal{J}_+$. It has the required quotient $hq_+(b)$. The relations are killed. The map is onto: if $(b_0, q) \in \mathcal{E}_h$, choose a lift b of q ; then $i = b_0 - hb \in \mathcal{J}_+$, and $[(i, b)]$ maps to (b_0, q) . Its kernel is exactly the relation subspace, by (NEA2.3). Thus it is a continuous bijection of Fréchet spaces and its inverse is continuous by the open mapping theorem. It is the identity on the displayed kernel and quotient. This proves concretely that pushout and pullback define the same scalar action he_+ .

A section of (NEA4.1) is equivalent to an M -map $v : \mathcal{Q}_+ \rightarrow \mathcal{B}$ with $q_+v = h_{\mathcal{Q}_+}$. By (NEA3.7) the only possible v is zero. Hence this lift exists precisely when $h \in K_+$, and then the section is $q \mapsto (0, q)$. Under (NEA4.2) it is exactly $q \mapsto [(-hb, b)]$. This does not identify the nonzero source-return map $b \mapsto hb : \mathcal{B} \rightarrow \mathcal{J}_+$ with the zero quotient-to-source lift; their domains and purposes are different and their exact relation is (NEA4.2).

NEA5. Full multiplicity probes and exact residual divisors

Fix an actual $\lambda_0 = \rho + 1$, put $z = \lambda - \lambda_0$, and retain $m = m_\rho$. The full local source factor is

$$F_+(\lambda_0 + z) = z^m b_\rho(z), \quad b_\rho(z) = C_0(\rho + z) \frac{\zeta(\rho + z)}{z^m}, \quad b_\rho(0) = C_0(\rho) \frac{\zeta^{(m)}(\rho)}{m!} \neq 0. \quad (\text{NEA5.1})$$

For example all its Taylor coefficients, without dropping the completion derivatives, are

$$[z^j]b_\rho(z) = \sum_{\ell=0}^j \frac{C_0^{(\ell)}(\rho)}{\ell!} \frac{\zeta^{(m+j-\ell)}(\rho)}{(m+j-\ell)!}. \quad (\text{NEA5.2})$$

Here $C_0(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$ is the full function in (NEA1.6).

Division by a fixed polynomial preserves M whenever the numerator vanishes at every root to the needed order. To verify this, remove fixed small disks around the finitely many roots; on the complement the reciprocal polynomial has at most a fixed polynomial bound, and in each disk the entire quotient is bounded on a compact set. The same assertion holds for $\mathcal{B}$, retaining all its rapid strip seminorms. In particular $M/(z^k)$ is exactly $\mathbb{C}[z]/z^k$: the Taylor polynomial supplies every jet, and a zero jet is divisible by z^k with quotient in M .

For $1 \leq k \leq m$, the actual full source gives the M -linear injection

$$\iota_{\rho,k} : M/(z^k) \longrightarrow \mathcal{Q}_+, \quad [a] \mapsto \left[a \frac{F_+}{z^k} \right]. \quad (\text{NEA5.3})$$

The source function F_+/z^k is entire and in $\mathcal{B}$; at the chosen zero it has exact order $m-k$, and at every other zero it retains the required order. Its product with a lies in $\mathcal{J}_+$ exactly when a has a zero of order at least k at λ_0 . This proves both well-definedness and injectivity, with every other zero still retained.

The two free M -modules in $0 \rightarrow M \xrightarrow{z^k} M \rightarrow M/(z^k) \rightarrow 0$ form a resolution, since M is a ring of entire functions and multiplication by z^k is injective. Pulling the class he_+ back along (NEA5.3) gives exactly

$$\iota_{\rho,k}^*(he_+) = [hF_+] \in \text{Ext}_M^1(M/(z^k), \mathcal{J}_+) = \mathcal{J}_+/z^k \mathcal{J}_+. \quad (\text{NEA5.4})$$

To check the positive sign, use the lift F_+/z^k of the generator in the original extension. Its relation is $z^k(F_+/z^k) = F_+$; pushing its kernel by h changes that relation to hF_+ . This is the positive boundary convention used in CLP12.

There is an exact full-factor coordinate and inverse

$$\mathcal{J}_+/z^k \mathcal{J}_+ \xrightarrow{\sim} \mathbb{C}[z]/z^k, \quad [j] \mapsto \text{Tay}_{\lambda_0, < k}(j/F_+), \quad [p] \mapsto [F_+p]. \quad (\text{NEA5.5})$$

The quotient j/F_+ in this formula is taken locally at λ_0 , where (NEA5.1) is the retained unit comparison. If its first k coefficients vanish, then j/z^k is entire, lies in $\mathcal{B}$, and lies in $\mathcal{J}_+$: at the chosen point it still vanishes to order at least m , and at the other points division is by a unit. This proves the kernel claim. Multiplication by F_+ times a polynomial proves surjectivity and the two inverse identities. Derivative evaluations are continuous on the actual Fréchet ideal. Thus this finite-dimensional coordinate is continuous as well; no global claim that $j/F_+ \in M$ is made.

Under (NEA5.5), the boundary (NEA5.4) is the complete k -jet of h . In particular the $k = m$ probe vanishes precisely when $\text{ord}_{\lambda_0} h \geq m$. The eigenvector probe $k = 1$ detects only $h(\lambda_0)$ and would miss higher multiplicities. If $v = \text{ord}_{\lambda_0} h$, the local class has exact z -nilpotence exponent $\max(k-v, 0)$, with exponent zero meaning the class is zero. Indeed its first nonzero Taylor term is a nonzero coefficient times z^v , and multiplication by z raises this order by exactly one until it reaches k .

For the complete global class the exact residual annihilator is consequently

$$\text{Ann}_M(he_+) = (K_+ : h) = \{a \in M : \text{ord}_{\rho+1} a \geq \max(m_\rho - \text{ord}_{\rho+1} h, 0) \text{ for every } \rho\}. \quad (\text{NEA5.6})$$

This follows also directly by applying (NEA3.4) to ah . When $h = 0$, all residual multiplicities are zero and the annihilator is M . This classifies the effect of every multiplier on the actual extension, retaining its entire divisor rather than only finitely many tests.

NEA6. The actual cyclic module of extension classes

Define the actual image

$$\mathcal{N}_+ = Me_+ \subset \text{Ext}_M^1(\mathcal{Q}_+, \mathcal{J}_+). \quad (\text{NEA6.1})$$

It is the image of the connecting homomorphism on the specified multiplier endomorphisms of $\mathcal{J}_+$. No claim that these are all endomorphisms is needed. The exact map and its kernel give

$$M/K_+ \xrightarrow{\sim} \mathcal{N}_+, \quad [h] \mapsto he_+. \quad (\text{NEA6.2})$$

The isomorphism is canonical for the actual displayed extension, with $[1]$ mapped to e_+ . It is an algebraic M -module statement. We assign no unproved locally convex quotient topology to M/K_+ or to the ambient Ext group.

The full jet map gives a faithful concrete description of this module:

$$M/K_+ \rightarrow \prod_{\rho \in \mathcal{Z}} \mathbb{C}[z]/z^{m_\rho}, \quad [h] \mapsto \left(\sum_{j=0}^{m_\rho-1} \frac{h^{(j)}(\rho+1)}{j!} z^j \right)_\rho. \quad (\text{NEA6.3})$$

It is injective by definition of K_+ . Its image is the image of the actual strip-polynomial multiplier functions. It is not declared to be the unrestricted product. The projection onto any specified finite set of these jet factors is surjective: the usual finite polynomial remainder construction follows from Bézout for the pairwise coprime polynomials $(\lambda - \rho - 1)^{m_\rho}$, and the resulting polynomial belongs to M . This finite assertion neither supplies arbitrary infinite interpolation nor assumes convergence of finite spectral expansions.

There is also the exact injective M -map from the original normal quotient

$$\mathcal{Q}_+ = \mathcal{B}/\mathcal{J}_+ \longrightarrow M/K_+, \quad [F]_{\mathcal{J}_+} \longmapsto [F]_{K_+}. \quad (\text{NEA6.4})$$

It is induced by $\mathcal{B} \subset M$, and injectivity is precisely $\mathcal{J}_+ = \mathcal{B} \cap K_+$. In terms of (NEA6.2), it sends $[F]$ to Fe_+ . We do not identify the two full quotient modules or assert surjectivity of (NEA6.4).

The ideal K_+ , hence the annihilator of the single actual extension class, recovers the complete translated zero divisor: its common zero set is exactly the zero set of F_+ , since $F_+ \in K_+$, and at $\lambda_0 = \rho + 1$,

$$\min_{0 \neq h \in K_+} \text{ord}_{\lambda_0} h = m_\rho. \quad (\text{NEA6.5})$$

Every member has at least that order, and the original full factor F_+ attains it. This is a precise recovery of the divisor and its multiplicities from the annihilator, not a claim about where the original zeros lie within their strip.

NEA7. The exact full-module generator resolvent

Write $T[h] = [\lambda h]$ on M/K_+ , equivalently postcomposition by the normal generator on $\mathcal{N}_+$. For a scalar $\mu \notin \text{Supp } \mathcal{D}_+$, define

$$\mathcal{R}_\mu h(\lambda) = \frac{h(\lambda) - F_+(\lambda)h(\mu)/F_+(\mu)}{\lambda - \mu}. \quad (\text{NEA7.1})$$

The denominator of the retained full scalar is nonzero. The numerator has a zero at μ , so the quotient is entire. It belongs to M : outside a fixed disk around μ this follows from the strip bounds for h and F_+ , while on the disk the entire quotient is bounded on a compact set. If $h \in K_+$, both numerator terms vanish to every required translated order, and the denominator is a unit at each translated zero. Thus $\mathcal{R}_\mu$ preserves K_+ and induces an operator on the whole quotient.

Multiplication gives the two exact identities

$$(T - \mu)\mathcal{R}_\mu[h] = [h], \quad \mathcal{R}_\mu(T - \mu)[h] = [h]. \quad (\text{NEA7.2})$$

For the first, the retained difference from h is $F_+h(\mu)/F_+(\mu) \in K_+$; for the second, the evaluated numerator $((\lambda - \mu)h)(\mu)$ is zero. The inverse is M -linear on the quotient: $T - \mu$ is M -linear and a bijection, so its inverse is M -linear. Formula (NEA7.1) already specifies that inverse on each representative, without a finite-block-density argument.

At an actual $\lambda_0 = \rho + 1$, the nonzero class

$$\left[\frac{F_+}{\lambda - \lambda_0} \right]_{K_+} \quad (\text{NEA7.3})$$

is an eigenvector of T with eigenvalue λ_0 : it has order $m_\rho - 1$ there and is killed by $T - \lambda_0$. Hence $T - \lambda_0$ is not injective. We have therefore computed the entire algebraic resolvent set and its complement:

$$\{\mu \in \mathbb{C} : T - \mu \text{ is not bijective on } \mathcal{N}_+\} = \text{Supp } \mathcal{D}_+ = \{\rho + 1 : \rho \in \mathcal{Z}\}. \quad (\text{NEA7.4})$$

This statement specifies algebraic bijectivity; it does not invoke the spectrum of an operator on an unassigned Banach or Fréchet topology.

The corresponding complete local chains are retained as actual submodules. At λ_0 , the map

$$M/((\lambda - \lambda_0)^m) \longrightarrow M/K_+, \quad [a] \longmapsto \left[a \frac{F_+}{(\lambda - \lambda_0)^m} \right] \quad (\text{NEA7.5})$$

is injective by the same full-order argument as (NEA5.3). On this block, multiplication by a_0^λ , for recovered $a_0 > 0$, is exactly

$$a_0^{\rho+1} \sum_{j=0}^{m-1} \frac{(\log a_0)^j}{j!} M_z^j. \quad (\text{NEA7.6})$$

The map includes the entire factor in (NEA5.1); no derivative coefficients are removed by this expression for the action.

Every unital complex-algebra character of M/K_+ is evaluation at exactly one translated zero. To prove this, let χ be such a character and put $\mu = \chi([\lambda])$. For every $h \in M$, the function $(h(\lambda) - h(\mu))/(\lambda - \mu)$ is entire and belongs to M , by the division estimate used above. Applying χ gives $\chi([h]) = h(\mu)$. Since $F_+ \in K_+$, it follows that $F_+(\mu) = 0$, so $\mu \in \text{Supp } \mathcal{D}_+$. Conversely evaluation at any such point annihilates K_+ and defines a character. Thus their actual dilation characters are $a_0^{\rho+1}$. For $a_0 > 1$, their logarithmic-modulus values are

$$\frac{2 \log |a_0^{\rho+1}|}{\log a_0} = 2\Re \rho + 2 \in (2, 4). \quad (\text{NEA7.7})$$

These are the scalar values for the postcomposition module action just defined. They are not weights of a conjugation action on equivariant Hom; NEA10 retains that distinction.

NEA8. Separation of the actual class module from the original quotient

The existing global separator is $c(\lambda) = E_+(\lambda) = 1 - E(\lambda - 1)$. GMS3 proves that $c \in K_+$, with full orders, and that it acts as the identity on the complete original quotient $\mathcal{Q} = \mathcal{B}/\mathcal{I}$. Consequently

$$c\mathcal{N}_+ = 0, \quad c\mathcal{Q} = \mathcal{Q} \text{ with } c_{\mathcal{Q}} = 1. \quad (\text{NEA8.1})$$

The second equality alone is not the reason; its stated identity action is the property used.

Apply the explicit bar-resolution homotopy of GMS4 to these actual modules. If $P \rightarrow \mathcal{Q}$ is a projective resolution, multiplication by $1 - c$ lifts zero on its augmentation, and there is $h : P \rightarrow P[-1]$ with $dh + hd = 1 - c$. The Hom homotopy $H^n(f) = (-1)^n fh$ satisfies

$$DH + HD = 1 - c_{\mathcal{N}_+} = 1 \quad \text{on } \text{Hom}_M(P, \mathcal{N}_+). \quad (\text{NEA8.2})$$

For the reverse direction, resolve $\mathcal{N}_+$, lift its zero action of c , and use the same Hom formula; the target action of c on $\mathcal{Q}$ is one, giving the reverse contraction. Hence

$$\boxed{\text{RHom}_M(\mathcal{Q}, \mathcal{N}_+) \simeq 0, \quad \text{RHom}_M(\mathcal{N}_+, \mathcal{Q}) \simeq 0.} \quad (\text{NEA8.3})$$

The complete bar insertion formulas in GMS4.5–GMS4.9 apply without any finiteness condition on $\mathcal{N}_+$. This is separation of a module generated by the actual extension class. It does not assign the translated-zero spectrum to the entire kernel $\mathcal{I}_+$, whose other quotient characters remain as calculated in CW6.

NEA9. Original-source maps for every annihilating multiplier

For every $h \in K_+$, the unique source-return map in (NEA3.2) becomes, in the original normal variable,

$$V^{-1}r_h V(F)(s) = h(s+1)F(s). \quad (\text{NEA9.1})$$

Its exact source representative is

$$\mathfrak{r}_h(a)(u) = \Theta^{-1}[h(s+1)\Theta a(s)](u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} h(3/2 + it)\Theta a(1/2 + it)u^{-it} dt. \quad (\text{NEA9.2})$$

The shift $3/2 + it$ is required by the actual normal action. The multiplier bound, rapid strip decay, and the full inverse Mellin estimates from GMS2 prove absolute differentiated convergence and continuity $A(-1) \rightarrow$

$J(-1)$. It lands in J because $h(s+1)$ has the required original-zero orders and $\Theta J = \mathcal{J}$. Composing with the exact continuous inverse $H = \Sigma^{-1} : J \rightarrow S$ gives

$$f_{h,a} = H\mathfrak{r}_h(a), \quad \Sigma f_{h,a} = \mathfrak{r}_h(a), \quad f_{h,a}(0) = 0, \quad \int_{\mathbb{R}} f_{h,a} = 0. \quad (\text{NEA9.3})$$

All sums, factors 2, $1/2$, $u^{-1/2}/\pi$, and endpoint conditions are the original ones from GMS2/GMS8. For $h = c$, formula (NEA9.2) reduces exactly to multiplication by $1 - E(s)$, because $c(s+1) = 1 - E(s)$. The classification (NEA3.4) says precisely that these, and only these multipliers, admit the displayed lift into the full source kernel.

NEA10. The actual geometric connecting morphism and its class module

The simultaneous complexes of the actual sheaf row, with their constructed M -actions, are

$$D_J = (W \xrightarrow{r_+ - r_-} J \xrightarrow{0} J(-1)), \quad D_F = (W \xrightarrow{r_+ - r_-} A \xrightarrow{0} A(-1)), \quad D_G = \mathcal{Q}[-1] \oplus \mathcal{Q}(-1)[-2]. \quad (\text{NEA10.1})$$

The low generator is L , the normal generator is $L + 1$, and GMS9 retains the endpoint characters $(h(0), h(1), h(1), h(0))$. All identifications below use Θ on low coefficients and $V\Theta$ on normal coefficients. The explicit symmetric source contraction in CLP12/GMS9 gives

$$D_J \simeq Z[0] \oplus \mathcal{J}_+[-2], \quad D_F \simeq Z[0] \oplus \mathcal{Q}[-1] \oplus \mathcal{B}[-2], \quad D_G \simeq \mathcal{Q}[-1] \oplus \mathcal{Q}_+[-2]. \quad (\text{NEA10.2})$$

These are the compatible M -linear quasi-isomorphisms of the actual row, and retain the entire ideal. Its connecting morphism is

$$\delta : D_G \longrightarrow D_J[1], \quad \delta|_{\mathcal{Q}[-1]} = 0, \quad \delta|_{\mathcal{Q}_+[-2]} = e_+[-2] \text{ into } \mathcal{J}_+[-1]. \quad (\text{NEA10.3})$$

Let M act on this derived Hom group by postcomposition on $D_J[1]$; since δ is M -linear, this equals precomposition on its source. Inclusion and projection of the normal summands in (NEA10.2) prove both directions of

$$h\delta = 0 \iff he_+ = 0. \quad (\text{NEA10.4})$$

Therefore the actual cyclic boundary-image module is

$$M\delta \subset \text{Hom}_{D(M)}(D_G, D_J[1]), \quad M/K_+ \xrightarrow{\sim} M\delta, \quad [h] \mapsto h\delta, \\ M\delta \xrightarrow{\sim} \mathcal{N}_+, \quad h\delta \mapsto he_+. \quad (\text{NEA10.5})$$

This identifies the class receiver directly inside the actual localization triangle; it does not replace its full source or kernel. Its exact annihilator, resolvent, and full translated divisor are (NEA3.4), (NEA7.1)–(NEA7.4), and (NEA6.5).

In particular $c\delta = 0$, whereas c acts as the identity on $\mathcal{Q}[k]$ for every integer k . For every derived M -morphism $f : \mathcal{Q}[k] \rightarrow D_G$, naturality of the central action gives the actual lifting equation

$$\delta f = \delta f c_{\mathcal{Q}[k]} = \delta c_{D_G} f = c_{D_J[1]} \delta f = 0. \quad (\text{NEA10.6})$$

It follows that f lifts through $D_F \rightarrow D_G$. To display the lift with the actual decompositions, the normal component of f is zero in the derived category by GMS4.4. Write its remaining component as $f_1 : \mathcal{Q}[k] \rightarrow \mathcal{Q}[-1]$. Then $(0, f_1, 0)$ in the middle decomposition of (NEA10.2) is a lift, transported back through its specified quasi-isomorphism. This proves existence; it does not claim uniqueness of all lifts through D_F , whose differences can come from D_J .

The action used in (NEA10.5)–(NEA10.6) must not be confused with conjugation by dilation. Postcomposition by the multiplier a^λ produces the module action whose scalar values are (NEA7.6)–(NEA7.7). The conjugation action on a derived morphism is instead

$$\delta \mapsto T_{a, D_J[1]} \delta T_{a, D_G}^{-1}. \quad (\text{NEA10.7})$$

It fixes δ , because δ is equivariant. No translated-zero weight is assigned to this conjugation action. Likewise no identification of $M\delta$ with Deligne's geometric obstruction quotient is asserted without its comparison map. The proved conclusion is the explicit annihilator, spectrum and separation of the module generated by the actual localization class, and the resulting actual lift (NEA10.6), with all original kernels, factors, multiplicities and degrees retained.

The actual extension class in the original source and residue spaces

24 September 2026. Complete receiving derivation, ECR0–ECR7.

ECR0. The constructed prerequisites and the attempted step

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The functions, complex coefficients and positive parameters in this calculation belong to the already reconstructed complete arithmetic receiver. None is an arithmetic value, parity, coordinate or addition assigned to τ . Separate branch histories and their coefficient records remain.

The complete received proofs are FOD0–FOD8 in `../tau_weight_cohomology_20260924/CC_FULL_SO URCE_OBSTRUCTION_DECOMPOSITION.md`, SHA256 0f16829aa955a3af1bd8b16d70851df5467040b9685d607a7a1b8aa3a0cae0a3, and NEA0–NEA10 in `../tau_weight_cohomology_20260924/CC_NORMAL_EXTENSION_ANNIHILATOR.md`, SHA256 817455b79efe7a3cddb5310f760cfef86f5a9bc02a2498a116df0253c8f3172b. Both were read completely for this receiving calculation. RTT0–RTT11, GTAH0–GTAH7 and GSR0–GSR4 supply the existing residue, adjoint and multiplier comparisons.

FOD proves the actual original extension

$$e_0 : 0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{B} \longrightarrow \mathcal{Q} \longrightarrow 0, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad (\text{ECR0.1})$$

and its exact annihilator $\mathfrak{a} \subset M$, the ideal of entire strip-polynomial multipliers vanishing at each actual nontrivial zeta zero ρ to multiplicity m_ρ . The source ideal is $\mathcal{I} = \mathcal{B} \cap \mathfrak{a}$, where $\mathcal{B}$ is the entire rapid-strip space. Consequently

$$\mathcal{N}_0 = Me_0 \simeq M/\mathfrak{a}, \quad [h] \mapsto he_0. \quad (\text{ECR0.2})$$

The symbol $\mathcal{N}_0$ in this note is the cyclic extension-class module. RTT's value-zero kernel is written $\mathcal{N}_{\text{jet}}$ here to avoid a collision of names.

What would connect this new class module to the existing positivity calculation is a proved map into the actual source quotient and residue space, rather than assigning a norm to an unspecified Ext topology. We now construct a full family of such maps and calculate its effect on the original trace and on the measured adjoint defect.

ECR1. Injective maps in both directions, with their full composites

For every $t > 0$, use the specified entire test function

$$g_t(s) = \exp(ts^2). \quad (\text{ECR1.1})$$

It is nowhere zero and belongs to $\mathcal{B} \subset M$, because for $|x| \leq A$,

$$(1 + |y|)^N |g_t(x + iy)| \leq e^{tA^2} (1 + |y|)^N e^{-ty^2}, \quad (\text{ECR1.2})$$

whose supremum is finite for every A, N . If $h \in M$ has growth exponent N_A on that strip, the same estimate with exponent $N + N_A$ proves $g_t h \in \mathcal{B}$.

Define actual M -linear maps

$$j : \mathcal{Q} \longrightarrow \mathcal{N}_0, \quad [F]_{\mathcal{I}} \mapsto Fe_0, \quad b_t : \mathcal{N}_0 \longrightarrow \mathcal{Q}, \quad he_0 \mapsto [g_t h]_{\mathcal{I}}. \quad (\text{ECR1.3})$$

For j , the kernel condition is exactly $\mathcal{B} \cap \mathfrak{a} = \mathcal{I}$. For b_t , a change of representative by $\mathfrak{a}$ changes the product by $\mathcal{I}$, so it is well defined. If $g_t h \in \mathcal{I}$, the fact that g_t is a unit in every local holomorphic ring forces the full vanishing order of h at every original zero. Hence $h \in \mathfrak{a}$. Both maps are therefore injective, with every multiplicity coordinate retained. Their composites are exactly

$$jb_t = m_{g_t} \quad \text{on } \mathcal{N}_0, \quad b_t j = m_{g_t} \quad \text{on } \mathcal{Q}. \quad (\text{ECR1.4})$$

These are not identity maps. No topology is imposed on the ambient Ext group or on $M/\mathfrak{a}$.

Neither embedding in (ECR1.3) is surjective. If e_0 were in the image of j , an $F \in \mathcal{B}$ would have $F(\rho) = 1$ at every nontrivial zero. The unbounded imaginary parts of the actual zeta zeros and rapid strip decay contradict this. Here unboundedness follows from the established zero-count asymptotic for the original zeta, not an off-line-zero assertion. If $[g_{t/2}]$ were in the image of b_t , some $h \in M$ would satisfy

$$h(\rho) = g_{t/2}(\rho)/g_t(\rho) = e^{-t\rho^2/2}. \quad (\text{ECR1.5})$$

Writing $\rho = \sigma + i\gamma$, its modulus is $e^{t(\gamma^2 - \sigma^2)/2}$. Since $0 < \sigma < 1$ and $|\gamma|$ is unbounded, it cannot obey a polynomial bound on the fixed zero strip. This proves the second nonsurjectivity. The same argument applied to $g_t h = 1$ modulo $\mathfrak{a}$ proves that $[g_t]$ is not a unit in $M/\mathfrak{a}$. Its absence of zeros does not license division in this multiplier quotient.

ECR2. Every local jet and the exact relation between test parameters

At an actual zero ρ of multiplicity m , put $z = s - \rho$. The complete multiplier is

$$g_t(\rho + z) = e^{t\rho^2} e^{2t\rho z} e^{tz^2}, \quad [z^r]g_t(\rho + z) = e^{t\rho^2} \sum_{v=0}^{\lfloor r/2 \rfloor} \frac{(2t\rho)^{r-2v} t^v}{(r-2v)!v!}. \quad (\text{ECR2.1})$$

Thus on the entire multiplicity block, the matrix of b_t in ordinary Taylor coordinates is lower triangular with entry $[z^{i-j}]g_t(\rho + z)$ for $i \geq j$, and zero otherwise. Its determinant is $e^{mt\rho^2} \neq 0$. Its local inverse retains all coefficients of $e^{-t\rho^2} e^{-2t\rho z} e^{-tz^2}$, truncated only at the stated length m . This is a local finite-length inverse, not an inverse in M or of either global embedding.

For $u > t > 0$, the full global identity is

$$g_u = g_{u-t} g_t, \quad b_u = m_{g_{u-t}} b_t. \quad (\text{ECR2.2})$$

All multipliers in this formula belong to $\mathcal{B} \subset M$. It compares the entire family without identifying its different test functions or cancelling any factor.

ECR3. The actual source representative

The original half-Mellin map is the already proved isomorphism

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + iy) u^{-iy} dy. \quad (\text{ECR3.1})$$

For the extension-class representative h , the actual source of $b_t(h e_0)$ is

$$a_{t,h}(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} e^{t(1/2+iy)^2} h(1/2 + iy) u^{-iy} dy. \quad (\text{ECR3.2})$$

Gaussian decay and polynomial strip growth prove absolute convergence after every fixed number of logarithmic derivatives; (ECR1.2) and the full inverse-Mellin estimates give every source seminorm. This is the entire actual source space A , not necessarily its subspace J .

For the actual generator e_0 , retain the elementary Gaussian integral explicitly:

$$a_{t,1}(u) = \frac{e^{t/4}}{\sqrt{\pi t}} u^{-1/2} \exp\left(-\frac{(\log u - t)^2}{4t}\right). \quad (\text{ECR3.3})$$

Indeed $e^{t(1/2+iy)^2} = e^{t/4} e^{-ty^2} e^{ity}$, and integrating $e^{-ty^2} e^{-iy(\log u - t)}$ gives $\sqrt{\pi/t} \exp(-(\log u - t)^2/(4t))$. Every original factor in (ECR3.1) remains. This is a chosen source test, not a replacement of the original zeta or its theta transform.

ECR4. Exact residue and full original Weil pairing

Let $S : \mathcal{Q} \rightarrow \mathcal{H}_{\text{res}}$ be RTT5's continuous anti-linear residue receiver. The actual map on extension classes is

$$S_t = Sb_t : \mathcal{N}_0 \longrightarrow \mathcal{H}_{\text{res}}. \quad (\text{ECR4.1})$$

No continuity relative to an unspecified topology on $\mathcal{N}_0$ is claimed. Introduce the value-vanishing ideal

$$\mathfrak{a}_{\text{val}} = \{h \in M : h(\rho) = 0 \text{ for all actual } \rho\}. \quad (\text{ECR4.2})$$

RTT5 gives precisely

$$\ker S_t = \mathfrak{a}_{\text{val}}/\mathfrak{a} \quad \text{under (ECR0.2)}. \quad (\text{ECR4.3})$$

The reason is $g_t(\rho) \neq 0$ for every zero. This kernel has zero constant coordinate in every multiplicity block, but we do not identify it with the global nilradical without a proof controlling all multiplicities simultaneously.

Put $\rho^\# = 1 - \bar{\rho}$. RTT5 now gives the full exact formula

$$\begin{aligned} W_t(h, k) &= \widehat{\mathcal{R}}(b_t(he_0), S_t(ke_0)) \\ &= \sum_{\rho} m_{\rho} e^{t\{\rho^2 + (1-\rho)^2\}} h(\rho) \overline{k(\rho^\#)}. \end{aligned} \quad (\text{ECR4.4})$$

The exponential factor follows from $g_t(\rho) \overline{g_t(\rho^\#)} = e^{t\rho^2} e^{t(1-\rho)^2}$; it has not been replaced by one or by a real modulus. Absolute convergence follows from the Gaussian decay, polynomial multiplier bounds and the original zero-count estimate. This proves a Hermitian form by transport of the established form, or directly by exchanging ρ and $\rho^\#$.

For completeness its complete arithmetic receiver has the original entire test

$$A_{t,h,k}(s) = e^{t\{s^2 + (1-s)^2\}} h(s) \overline{k(1 - \bar{s})}. \quad (\text{ECR4.5})$$

Let $v_{t,h,k}$ be its actual logarithmic inverse transform, with $A(s) = \int_{\mathbb{R}} v(v_0) e^{-(s-1/2)v_0} dv_0$ and $\hat{v}(y) = \int v(v_0) e^{-iyv_0} dv_0$. It belongs to the complete source test space by (ECR3.2), reflection, and the convolution proof in GIQ9, reread for this calculation. Then

$$W_t(h, k) = A(0) + A(1) + \mathcal{A}_{\infty}(v) - P_{\text{hist}}(v), \quad (\text{ECR4.6})$$

where the full terms are

$$\begin{aligned} \mathcal{A}_{\infty}(v) &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{v}(y) \left(\Re \frac{\Gamma'(1/4 + iy/2)}{\Gamma(1/4 + iy/2)} - \log \pi \right) dy, \\ P_{\text{hist}}(v) &= \sum_{n \geq 2} \frac{\log L_n - \log L_{n-1}}{\sqrt{n}} (v(\log n) + v(-\log n)), \quad L_n = \text{lcm}(1, \dots, n). \end{aligned} \quad (\text{ECR4.7})$$

In particular $A(0) = e^t h(0) \overline{k(1)}$ and $A(1) = e^t h(1) \overline{k(0)}$. The original pole and trivial divisor remain at every finite cutoff B :

$$\begin{aligned} V_{\zeta,B} &= W_t(h, k) + \sum_{r=1}^B A(-2r) - A(1), \quad G_B = \mathcal{A}_{\infty}(v) + A(0) + \sum_{r=1}^B A(-2r), \\ V_{\zeta,B} &= G_B - P_{\text{hist}}(v), \quad A(-2r) = e^{t\{4r^2 + (1+2r)^2\}} h(-2r) \overline{k(1 + 2r)}. \end{aligned} \quad (\text{ECR4.8})$$

No divergent infinite trivial-zero sum is asserted. GIQ9 proves convergence of all the other displayed complete sums and the full source-domain extension of the explicit formula.

ECR5. The positive value form and exact defect on the actual class module

The positive auxiliary form pulled back by the actual embedding is

$$\langle he_0, ke_0 \rangle_t = \sum_{\rho} m_{\rho} e^{2t\Re(\rho^2)} h(\rho) \overline{k(\rho)}. \quad (\text{ECR5.1})$$

It converges absolutely by the same estimates, and has exactly the kernel in (ECR4.3). It is positive semidefinite, since its diagonal is a sum of nonnegative terms with strictly positive coefficients. All these value-zero classes remain in the original extension-class module and in the full jet injection b_t .

For every recovered $a > 0$, retain the coefficient actions

$$T_a(he_0) = a^s he_0, \quad U_a(he_0) = a^{1-s} he_0. \quad (\text{ECR5.2})$$

Both multipliers are in M , and b_t commutes with them because all these are pointwise multipliers. Thus the coefficient transfer is preserved for every $a > 0$. For $a = n$ a positive integer, GTR's actual degree- n covering realizes this as its geometric transfer. No covering with nonintegral degree is asserted. The full coefficient identities are

$$W_t(T_a x, y) = W_t(x, U_a y), \quad U_a T_a = aI. \quad (\text{ECR5.3})$$

These follow either by transport through b_t or directly from the full sum (ECR4.4).

On the established value Hilbert space, $D_a = T_a^* - U_a$ has multiplier $e^{-i\gamma \log a} (a^{\sigma} - a^{1-\sigma})$, where $\rho = \sigma + i\gamma$. Its exact pullback to the cyclic class module is

$$\mathfrak{d}_{a,t}(he_0, ke_0) = \sum_{\rho} m_{\rho} e^{2t(\sigma^2 - \gamma^2)} (a^{\sigma} - a^{1-\sigma})^2 h(\rho) \overline{k(\rho)}. \quad (\text{ECR5.4})$$

This is $\langle D_a E_0 b_t(he_0), D_a E_0 b_t(ke_0) \rangle$, so it is positive semidefinite with its actual source map exhibited. For fixed $a > 1$, the squared factor is at most $(a-1)^2$, while $e^{2t\sigma^2} \leq e^{2t}$. These inequalities and polynomial multiplier growth prove convergence over all actual zeros. No nonholomorphic multiplier has been substituted into (ECR4.6).

For the particular actual extension generator the formula is

$$\mathfrak{d}_{a,t}(e_0, e_0) = \sum_{\rho} m_{\rho} e^{2t(\sigma^2 - \gamma^2)} (a^{\sigma} - a^{1-\sigma})^2. \quad (\text{ECR5.5})$$

Every term is nonnegative, and for $a > 1$ its vanishing is equivalent to $\sigma = 1/2$. Thus this single actual class tests the entire measured defect; it is not a finite zero sample. This computes an exact detection property, not the vanishing of the sum. The coefficient is the full original-source Gaussian weight, including $e^{2t\sigma^2}$.

ECR6. What the separator and the full source extension do to this test

FOD proves $ce_0 = e_0$, $ce_+ = 0$, and the decomposition of the full joint class into the low and high components, retaining its intersection kernel. By M -linearity, (ECR1.3) therefore gives

$$b_t(cx) = b_t(x), \quad S_t(cx) = S_t(x), \quad \mathfrak{d}_{a,t}(cx, cy) = \mathfrak{d}_{a,t}(x, y) \quad (x, y \in \mathcal{N}_0). \quad (\text{ECR6.1})$$

The annihilation of the high component is consequently compatible with, and does not cancel, the surviving defect of the original component. The connecting map is the one actually proved in FOD3: pushout takes the low component to e_0 and induces an isomorphism on their cyclic modules because both annihilators are $\mathfrak{a}$. Composing that isomorphism with b_t and S_t supplies the same formulas directly on the actual low source-extension component. Its full kernel $\mathcal{J} \cap \mathcal{J}_+$ remains in that extension.

The two normal and original components have separate degree actions. Transport of the normal source by $\lambda = s+1$ has character a^{s+1} , not a^s . No cancellation between their norms or trace terms follows by erasing that extra factor. The complete involution and degree comparison is calculated separately in the receiving extension-class involution note.

ECR7. Scope and the next receiving step

This calculation places the actual extension-class module into the entire original source by explicit proper embeddings, preserves every local jet, and maps its value trace into the complete residue space. It exhibits the actual extension generator as a global test of the previously measured positive-adjoint defect. The full original-zeta arithmetic formula, pole, Gamma, endpoints and finite trivial-divisor terms remain (ECR4.6)–(ECR4.8).

The multiplier that annihilates the normal localization component leaves this original class and its defect fixed. What would advance the purity argument is therefore additional control on the original component through the actual involution and duality, with both degree actions retained. The separate extension-class involution calculation is being carried out on these modules; its result must be compared using the maps (ECR1.3) and (ECR4.1), rather than by assigning a positive form to an unconstructed Ext topology. No claim of Deligne weight purity or RH is made by the detection identity (ECR5.5).

Independent audit of the extension-class source and residue maps

24 September 2026. Mathematical review only. No audited proof file was edited.

The complete ECR0–ECR7 proof was checked at SHA256 0da1815f474c5279012a94be8f65fdb8dbf3a178f56e97ff7b543a2d04c82bbd. No required correction was found. The conclusions concern the actual complete arithmetic receiver. They assign no arithmetic, coordinate, metric, midpoint, or parity to the support, and make no cyclic-group identification of the user’s information layers.

Reading coverage and exact versions

The following programme proofs were read over their entire indicated ranges. Their more remote GMS, SSI, GTR and SDT dependencies were received through these complete proofs; no new independent audit of every remote theorem or new complete human-source reading is claimed.

File	Coverage	SHA256
../tau_weight_cohomology_20260924/CC_FULL_SOURCE_OBSTRUCTION_DECOMPOSITION.md	FOD0–FOD8	0f16829aa955a3af1bd8b16d70851df5467040b9685d607a7a1b8aa3a0cae0a3
../tau_weight_cohomology_20260924/CC_NORMAL_EXTENSION_ANNIHILATOR.md	NEA0–NEA10	817455b79efe7a3cdbb5310f760cfe86f5a9bc02a2498a116df0253c8f3172b
RESIDUE_TRACE_TRANSFER_INDEPENDENT.md	RTT0–RTT11	6a83d81039b4eb3e49c4054b4ff157896709647d6a5546751362b05a21de25b2
GEOMETRIC_TRANSFER_POSITIVE_ADJOINT_DEFECT.md	GTAH0–GTAH7	dd27d0022c6cc0b82fc0cabcc5d43aed84d18fe762f382ba3be16e2fa50dc739
GLOBAL_SEPARATOR_TO_RESIDUE_COMPARISON.md	GSR0–GSR4	8abc2cbe84006219c2a49a38c625a9462ba3afd93d87fae65fd39e4b6be1e8a5
GLOBAL_INFINITESIMAL_QUOTIENT.md	GIQ9, including the full-source extension and convolution estimates	7c6dc699f7b028767d468bbe8e40bc4b83629b5da4d5cb0be685aad05371050e

The FOD and NEA hashes agree with ECR0. Their annihilator proofs use entire-function cancellation and a nowhere-zero Gaussian test; they require neither a principal-ideal assertion nor arbitrary infinite interpolation. FOD3–FOD5 prove the low-component pushout and cyclic-module isomorphism used in ECR6.

Embeddings, local jets, and the actual source

For $t > 0$, $g_t(s) = e^{ts^2}$ belongs to $\mathcal{B}$, and $g_t M \subset \mathcal{B}$. The maps in ECR1.3 are well defined and M -linear. The first has kernel $\mathcal{B} \cap \mathfrak{a} = \mathcal{I}$. The second is injective because g_t is a unit in each local holomorphic ring at an actual zero. Their composites are multiplication by g_t , exactly as stated.

Both properness proofs hold on the complete actual divisor. An image of e_0 under j would give a rapid-strip function with value one at every zero. An image of $[g_t/2]$ under b_t would give a function in M with values $e^{-t\rho^2/2}$, of modulus $e^{t(\gamma^2-\sigma^2)/2}$. The former contradicts rapid decay, and the latter contradicts polynomial growth on the fixed zero strip along the unbounded actual zero heights. These necessary value conditions use no interpolation assertion. The same argument proves that $[g_t]$ is not a unit of $M/\mathfrak{a}$.

Expanding $e^{t\rho^2}e^{2t\rho z}e^{tz^2}$ gives ECR2.1, including all factorials and summands. On a length- m block the matrix is lower triangular with diagonal $e^{t\rho^2}$, hence determinant $e^{mt\rho^2}$. Its inverse retains the complete

Taylor jet of $e^{-t\rho^2}e^{-2t\rho z}e^{-tz^2}$. This finite local inverse asserts no global division in M . The parameter identity $b_u = m_{g_{u-t}}b_t$ has the stated domain $u > t > 0$.

The source estimate in ECR3.2 can also be checked directly. For $F(s) = e^{ts^2}h(s)$, shifting from the line $1/2$ to any fixed real c gives

$$a_{t,h}(u) = \frac{u^{-c}}{\pi} \int_{\mathbb{R}} F(c + iy) u^{-iy} dy. \quad (\text{ECRA1})$$

The horizontal integrals tend to zero because the integrand on the intervening strip is bounded by a polynomial in the height times e^{-ty^2} , for fixed $u > 0$. Differentiation under the absolutely convergent integral gives, for $D = u\partial_u$,

$$|D^j a_{t,h}(u)| \leq \frac{u^{-c}}{\pi} \int_{\mathbb{R}} |c + iy|^j |F(c + iy)| dy. \quad (\text{ECRA2})$$

For a source seminorm of exponent N , choose $c = N + 1$ on $u \geq 1$ and $c = -N - 1$ on $0 < u \leq 1$. The integrals are finite, and $(u^N + u^{-N})|D^j a_{t,h}(u)|$ is bounded. This proves every actual source seminorm. Fourier inversion retains the coefficient $1/\pi$ and the original half-Mellin factor $1/2$.

For $h = 1$, the integral is exactly $\sqrt{\pi/t} \exp(-(\log u - t)^2/(4t))$. Multiplying by the original $u^{-1/2}e^{t/4}/\pi$ gives ECR3.3 with all factors and signs retained. This source need not lie in J .

Residue, arithmetic, and the value kernel

RTT5 applies to $g_t h, g_t k \in \mathcal{B}$. The product is exactly

$$g_t(\rho) \overline{g_t(\rho^\#)} = e^{t\rho^2} e^{t(1-\rho)^2}, \quad \rho^\# = 1 - \bar{\rho}. \quad (\text{ECRA3})$$

Its modulus is $e^{t\{\sigma^2 + (1-\sigma)^2 - 2\gamma^2\}}$. This Gaussian bound, polynomial multiplier growth, and the full zero-count estimate prove absolute convergence. Exchanging ρ and $\rho^\#$, with equal multiplicities, proves Hermitian symmetry. Replacing the product by $|g_t(\rho)|^2$ would change the original Weil form; ECR does not do that.

The kernel of S_t and of the positive value form is exactly $\mathfrak{a}_{\text{val}}/\mathfrak{a}$, because all Gaussian values are nonzero. ECR correctly avoids identifying this with the global nilradical. Its exact criterion is

$$\sqrt{\mathfrak{a}}/\mathfrak{a} = \{[h] : \exists n \geq 1 \ \forall \rho, \ n \text{ ord}_\rho h \geq m_\rho\}. \quad (\text{ECRA4})$$

Indeed $[h]^n = 0$ is equivalent to $h^n \in \mathfrak{a}$, and then to the displayed inequalities. Value vanishing requires only $\text{ord}_\rho h \geq 1$; it does not furnish a single exponent valid at all zeros. No uniform multiplicity bound is assumed.

The entire $A_{t,h,k}$ is a product of the specified rapid-strip transforms. GIQ9's reflection and convolution estimates therefore place its inverse in the complete logarithmic test space. ECR4.6–ECR4.7 retain the endpoints, $1/(2\pi)$, digamma argument $1/4 + iy/2$, $-\log \pi$, and every prime-power coefficient. Substitution at $0, 1, -2r$ checks every term of ECR4.8. The trivial-divisor sum is finite as stated.

An additional constant check, for the actual generator in both slots, is

$$v_{t,1,1}(v_0) = \frac{e^{t/2}}{\sqrt{8\pi t}} \exp\left(-\frac{v_0^2}{8t}\right). \quad (\text{ECRA5})$$

Its Fourier transform is $e^{t/2}e^{-2ty^2}$, exactly $A_{t,1,1}(1/2 + iy)$. Its two endpoint Laplace values are e^t . This checks the constants against the original arithmetic receiver.

Positive defect and the source separator

The auxiliary positive form has weight $|g_t(\rho)|^2 = e^{2t(\sigma^2 - \gamma^2)}$. Squaring the modulus of GTAH's exact multiplier $e^{-i\gamma \log a}(a^\sigma - a^{1-\sigma})$ gives ECR5.4. The Weil transfer identity follows from $\overline{a^{1-\rho^\#}} = a^\rho$; $U_a T_a = aI$ holds on the complete class module, including its jets.

For $a > 1, t > 0$, every term in ECR5.5 is nonnegative and has a strictly positive Gaussian weight. Its vanishing consequently tests every actual zero at once. This is detection, not a proof of vanishing. What further calculation is available from this result and GTAH5? Retaining the full original-source weight gives

$$\mathcal{E}(a, 2t) \leq \mathfrak{d}_{a,t}(e_0, e_0) \leq e^{2t} \mathcal{E}(a, 2t). \quad (\text{ECRA6})$$

Proof: multiply the nonnegative summands of GTAH5.2 by the retained factor $e^{2t\sigma^2}$, and use $1 \leq e^{2t\sigma^2} \leq e^{2t}$ on the established strip. Every sum converges, so the inequalities hold for the complete sums. This calculates the comparison between the earlier global weighted trace and the actual extension generator.

Finally, FOD's identity $ce_0 = e_0$ gives $cx = x$ on the full module Me_0 , proving ECR6.1. FOD3's pushout and FOD4's equal annihilators give the exact connection to the low source-extension component, retaining its intersection kernel. The normal action still has a^{s+1} ; no extra factor is erased and no postcomposition action is replaced by conjugation on a derived morphism. The separate involution derivation is outside this audit.

The audit approves the stated maps and identities. It does not assert purity, place an unspecified topology on Ext, replace a source quotient by an unrestricted infinite jet product, or replace original zeta and its complete factors.

Involutions of the actual extension-class modules and their original-zeta residue maps

24 September 2026. Complete independent derivation ECI0–ECI13.

ECI0. The constructed objects and the next calculation

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All scalars, functions, modules and dual operations below occur after complete-history arithmetic reconstruction. The two branch counters remain separate. Addition at τ remains retracted, and no numerical coordinate, midpoint, metric or parity is assigned to it. The current corpus-and-operation rule and `READ_FIRST_USER_CONSTRUCTION.md` were read again before selecting these operations. The amended forward-step rule asks for an actual next calculation using all established findings. Here that calculation is the involution and duality of the actual cyclic extension modules just constructed, followed by their map to the existing residue receiver.

The complete FOD0–FOD8 and NEA0–NEA10 proofs were read, together with the already checked GMS, GSR, RTT and GTAH comparisons. Their exact annihilator statements are used, not an assumption that the full normal ideal has a quotient’s spectrum. Write

$$M = \{h \text{ entire} : \text{on each bounded real strip, } h \text{ has at most polynomial vertical growth}\},$$

$$\mathcal{B} = \{F \text{ entire} : b_{A,N}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^N |F(s)| < \infty \text{ for every } A, N\}. \quad (\text{ECI0.1})$$

For every actual original nontrivial zero ρ of ζ , retain its full multiplicity m_ρ . The ideals are

$$\mathfrak{a} = \{h \in M : \text{ord}_\rho h \geq m_\rho \ \forall \rho\}, \quad \mathfrak{a}_+ = \{h \in M : \text{ord}_{\rho+1} h \geq m_\rho \ \forall \rho\},$$

$$\mathcal{I} = \mathcal{B} \cap \mathfrak{a}, \quad \mathcal{I}_+ = \mathcal{B} \cap \mathfrak{a}_+, \quad \mathcal{Q} = \mathcal{B}/\mathcal{I}, \quad \mathcal{Q}_+ = \mathcal{B}/\mathcal{I}_+,$$

$$\mathcal{C} = M/\mathfrak{a}, \quad \mathcal{C}_+ = M/\mathfrak{a}_+. \quad (\text{ECI0.2})$$

FOD and NEA prove the identifications

$$\mathcal{C} \simeq M e_0 \simeq M e_{\text{low}}, \quad \mathcal{C}_+ \simeq M e_+ \simeq M e_{\text{high}} \simeq M \delta. \quad (\text{ECI0.3})$$

Each sends $[h]$ to the displayed actual class multiplied by h . The generator e_0 is the original source row, e_+ is the actual normal source row, and δ is the actual localization boundary with the shifts retained in FOD7 and NEA10. These are isomorphisms of the cyclic images only, not of full Ext groups or full source ideals. No unproved locally convex topology is assigned to M , $\mathcal{C}$, or $\mathcal{C}_+$. Their embeddings of $\mathcal{Q}$ and $\mathcal{Q}_+$ remain the injective maps induced by $\mathcal{B} \subset M$.

ECI1. Conjugation, reflection, and the two distinct Weil involutions

On the original coordinate define

$$Ch(s) = \overline{h(\bar{s})}, \quad R_1 h(s) = h(1-s), \quad K_1 h(s) = h^{\#1}(s) = \overline{h(1-\bar{s})}. \quad (\text{ECI1.1})$$

The first and third maps are anti-linear, the second is linear, and $K_1 = CR_1 = R_1C$. Taylor expansion proves that their outputs are entire. A strip of width A is carried into one of width at most $A+1$; imaginary parts only change sign. Thus each preserves M and is continuous on $\mathcal{B}$. Each is multiplicative and involutive, with the appropriate conjugation of complex scalars.

The actual original functional equation and conjugation give the multiplicity-preserving permutations $\rho \mapsto \bar{\rho}$, $\rho \mapsto 1-\rho$, and $\rho \mapsto \rho^\# = 1-\bar{\rho}$. Hence all three maps preserve $\mathfrak{a}$ and $\mathcal{I}$, and induce the corresponding maps on $\mathcal{C}$ and $\mathcal{Q}$. In full local jets,

$$j_\rho(K_1 h) = \sum_{j=0}^{m_\rho-1} \frac{(-1)^j \overline{h^{(j)}(\rho^\#)}}{j!} (s-\rho)^j. \quad (\text{ECI1.2})$$

This is the complete jet map, not only its value.

In the translated normal coordinate λ , the corresponding maps are

$$R_3 h(\lambda) = h(3 - \lambda), \quad K_3 h(\lambda) = h^{\#3}(\lambda) = \overline{h(3 - \bar{\lambda})}. \quad (\text{ECI1.3})$$

Conjugation remains $Ch(\lambda) = \overline{h(\bar{\lambda})}$. These preserve $\mathfrak{a}_+$ and J_+ , since

$$3 - \overline{(\rho + 1)} = \rho^\# + 1. \quad (\text{ECI1.4})$$

Formula (ECI1.2) holds with $\rho, \rho^\#$ replaced by $\rho + 1, \rho^\# + 1$, with exactly the same factorials and signs. In particular $\#_1$ must not be used as the normal involution without coordinate transport. It carries the translated divisor to $\mathcal{Z} - 1$, since $1 - \overline{(\rho + 1)} = \rho^\# - 1$. Likewise $\#_3$ carries the unshifted divisor to $\mathcal{Z} + 2$. These are exact maps to other translated ideals, not identifications with the original ideal.

The induced operations on the cyclic extension modules are, for example,

$$K_{e_0}(he_0) = h^{\#1}e_0, \quad K_{e_+}(he_+) = h^{\#3}e_+. \quad (\text{ECI1.5})$$

Well-definedness is precisely invariance of the proved annihilator ideals. On the actual original row, the anti-linear semilinear maps K_1 on its kernel, middle module and quotient commute with the inclusion and quotient maps. On the normal row the analogous statement uses K_3 . Consequently these are also the maps on the cyclic classes obtained by applying those explicit semilinear exact equivalences to the rows, fixing the respective generator. No homological sign is introduced by this covariant exact operation. Transport to the low and high class modules uses the exact cyclic pushout isomorphisms (ECI0.3); it does not claim that K_1 alone preserves the full intersection ideal of the joint row.

ECI2. Translation and every scalar degree

Retain the actual maps

$$(Vh)(\lambda) = h(\lambda - 1), \quad (Uk)(s) = k(s + 1), \quad UV = VU = 1. \quad (\text{ECI2.1})$$

They preserve polynomial strip bounds and rapid strip bounds, using strips enlarged by one. They give inverse ring isomorphisms $M_s \leftrightarrow M_\lambda$, take $\mathfrak{a} \leftrightarrow \mathfrak{a}_+$, and induce inverse maps $\mathcal{C} \leftrightarrow \mathcal{C}_+$, as well as the established topological maps $\mathcal{Q} \leftrightarrow \mathcal{Q}_+$. They are semilinear for this translation of scalars:

$$V(hx) = (Vh)(Vx). \quad (\text{ECI2.2})$$

Equivalently, equip the unshifted normal module with the actual action $h \cdot x = h(s + 1)x(s)$; then V is M -linear to the normal module in the λ -coordinate.

Direct substitution proves

$$CV = VC, \quad R_3V = VR_1, \quad K_3V = VK_1. \quad (\text{ECI2.3})$$

Under (ECI0.3), $he_0 \mapsto (Vh)e_+$ is the corresponding semilinear isomorphism of cyclic extension modules. At the level of the actual rows it is induced by translating the entire original source sequence, including its kernel. It is not the identity on the scalar ring's coordinate.

Write $L[h] = [sh]$, $L_+[k] = [\lambda k]$, $D_a[h] = [a^s h]$, and $D_a^+[k] = [a^\lambda k]$, for every already recovered $a > 0$. Then

$$L_+V = V(L + 1), \quad D_a^+V = aVD_a, \quad UD_a^+ = aD_aU. \quad (\text{ECI2.4})$$

Their involution identities are

$$\begin{aligned} K_1L &= (1 - L)K_1, & K_3L_+ &= (3 - L_+)K_3, \\ \boxed{K_1D_a &= aD_{1/a}K_1, & K_3D_a^+ &= a^3D_{1/a}^+K_3.} \end{aligned} \quad (\text{ECI2.5})$$

Both scalar degree factors follow by substituting the full coordinate in the exponential; neither is selected from a desired weight. On a length- m_ρ block they retain

$$D_a = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_t^j, \quad D_a^+ = a^{\rho+1} \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} M_t^j. \quad (\text{ECI2.6})$$

Translation leaves the local parameter t unchanged; the reflected anti-linear jet map changes it to $-t$ while conjugating coefficients as in (ECI1.2). These formulas retain every nilpotent term.

ECI3. The joint class module has a componentwise involution with an explicit source representative

FOD proves $c \in \mathfrak{a}_+$, $1 - c \in \mathfrak{a}$, and

$$M/(\mathfrak{a} \cap \mathfrak{a}_+) \xrightarrow{\sim} \mathcal{C} \oplus \mathcal{C}_+, \quad [h] \mapsto ([h]_{\mathfrak{a}}, [h]_{\mathfrak{a}_+}), \quad (\text{ECI3.1})$$

with inverse $([f], [g]) \mapsto [cf + (1 - c)g]$. The componentwise operation $(x, y) \mapsto (\mathbf{K}_1 x, \mathbf{K}_3 y)$ therefore induces an exact anti-linear multiplicative involution on the actual joint cyclic module. In the single quotient presentation it is

$$[h] \mapsto [ch^{\#_1} + (1 - c)h^{\#_3}] \mod (\mathfrak{a} \cap \mathfrak{a}_+). \quad (\text{ECI3.2})$$

If a representative changes by the intersection ideal, the first component of its $\#_1$ change vanishes in $\mathcal{C}$, and the second component of its $\#_3$ change vanishes in $\mathcal{C}_+$. The separator kills the opposite components. This proves well-definedness directly. Multiplicativity and the square identity follow from (ECI3.1) on both components. In particular the two actual component idempotents represented by c and $1 - c$ are fixed.

There is an actual anti-linear continuous source representative on $\mathcal{B}$:

$$\mathbf{A}_c F = c\mathbf{K}_1 F + (1 - c)\mathbf{K}_3 F. \quad (\text{ECI3.3})$$

It preserves $\mathcal{K} = \mathcal{J} \cap \mathcal{J}_+$, by exactly the preceding full-jet argument, and induces the componentwise involution on $\mathcal{B}/\mathcal{K}$. Its complete square before quotient is

$$\begin{aligned} \mathbf{A}_c^2 F &= [cc^{\#_1} + (1 - c)(1 - c)^{\#_3}] F \\ &\quad + c(1 - c)^{\#_1} F(s + 2) + (1 - c)c^{\#_3} F(s - 2). \end{aligned} \quad (\text{ECI3.4})$$

This follows by expanding the four terms and using $\mathbf{K}_1 \mathbf{K}_3 F = F(s + 2)$, $\mathbf{K}_3 \mathbf{K}_1 F = F(s - 2)$. Each cross-term multiplier vanishes to the full orders at both required divisors. The coefficient of F , minus one, does as well. Thus $\mathbf{A}_c^2 F - F \in \mathcal{K}$, with the entire difference displayed. No equality $\mathbf{A}_c^2 = 1$ on $\mathcal{B}$, and no single scalar-ring involution preserving the whole original and normal row at once, has been assumed. The actual full-source extension and this explicit error remain available.

ECI4. The dual involution on the strong residue receiver contains the functional-equation unit

Retain RTT's $\hat{\iota} : \mathcal{H}_{\text{res}} \xrightarrow{\sim} \mathcal{Q}'_\beta$. The continuous anti-linear involution $\mathbf{K}_1$ on $\mathcal{Q}$ induces

$$(\mathbf{K}_1^\vee \lambda)(F) = \overline{\lambda(\mathbf{K}_1 F)}, \quad \mathbf{K}_1^\vee : \mathcal{Q}'_\beta \rightarrow \mathcal{Q}'_\beta. \quad (\text{ECI4.1})$$

The value is linear in F , anti-linear in λ , and the square is the identity. For a bounded $B \subset \mathcal{Q}$, its strong seminorm is $p_{\mathbf{K}_1 B}(\lambda)$. Thus it is strong-continuous. Define the actual continuous involution

$$\mathbf{K}_{\text{res}} = \hat{\iota}^{-1} \mathbf{K}_1^\vee \hat{\iota}. \quad (\text{ECI4.2})$$

This definition is on the full completion and requires no global meromorphic multiplier theorem.

Its finite-jet formula retains a nontrivial unit and sign. Original zeta has

$$\zeta(s) = \chi(s)\zeta(1-s), \quad \chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}. \quad (\text{ECI4.3})$$

For finite-support g , conjugating its residue pairing against $K_1 F$, using $\overline{\zeta(\bar{s})} = \zeta(s)$, and then substituting $s = 1 - z$, gives

$$\begin{aligned} (K_1^\vee \iota g)(F) &= \sum_{\rho} \text{Res}_{s=\rho} \frac{F(1-s)(Cg)(1-s)}{\zeta(s)} ds \\ &= - \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)(Cg)(s)\chi(s)}{\zeta(s)} ds. \end{aligned} \quad (\text{ECI4.4})$$

The minus sign is the derivative of the reflection. Accordingly $K_{\text{res}} g$ is the unique finite-support class with local jets

$$\boxed{j_{\sigma}(K_{\text{res}} g) = j_{\sigma}[-\chi(1-s)g^{\#1}(s)]}. \quad (\text{ECI4.5})$$

Every required germ of $\chi(1-s)$ is a holomorphic unit; RD's global finite isolators realize those jets. Their derivatives are included in the product in (ECI4.5), not replaced by the unit's value. The formula does not assert that its meromorphic expression on the entire plane preserves $\mathcal{B}$. The extension to the full residue completion is instead already proved by (ECI4.1)–(ECI4.2). Locally its square is also directly one, since $\chi(1-s)\chi(s) = 1$ and the two minus signs cancel.

For reference, conjugation alone on the dual, $\lambda \mapsto (F \mapsto \overline{\lambda(CF)})$, corresponds on finite residue classes to $g \mapsto Cg$. Linear reflection on the dual corresponds to local jets $-\chi(1-s)g(1-s)$. These follow by the same computation without one of its conjugations. Thus the source involution and its actual residue-dual representative have a proved morphism retaining their functional-equation defect; they are not silently identified.

ECI5. Every cyclic class has a canonical original-zeta trace functional

An element $[h] \in \mathcal{C}$ need not be in the rapid-decay quotient $\mathcal{Q}$. Nevertheless the exact expression

$$W_M([h])(F) = \sum_{\rho} m_{\rho} F(\rho) \overline{h(\rho^{\#})}, \quad F \in \mathcal{Q}, \quad (\text{ECI5.1})$$

defines a continuous complex-linear functional on $\mathcal{Q}$. It is anti-linear in $[h]$. To prove the assertion with its topology, let $w_{\rho} = 1 + |\mathcal{J}\rho|$. The strip bound for this particular h gives $|h(\rho^{\#})| \leq Cw_{\rho}^d$. For every integer $k \geq 0$,

$$E_k : \mathcal{Q} \rightarrow H, \quad (E_k F)_{\rho} = w_{\rho}^k F(\rho) \quad (\text{ECI5.2})$$

is continuous: its squared norm is bounded on $\mathcal{B}$ by $b_{1,k+2}(F)^2 \sum_{\rho} m_{\rho} w_{\rho}^{-4}$, and it kills $\mathcal{J}$. Choose $k \geq d + 2$. Cauchy–Schwarz gives

$$|W_M([h])(F)| \leq \|E_k F\|_+ \left(\sum_{\rho} m_{\rho} w_{\rho}^{-2k} |h(\rho^{\#})|^2 \right)^{1/2} < \infty. \quad (\text{ECI5.3})$$

The unconditional full zero count makes the last sum finite. This proves both absolute convergence and continuity on the actual quotient. Changing h by $\mathfrak{a}$, or F by $\mathcal{J}$, changes no term.

Define

$$\boxed{S_M = \hat{\iota}^{-1} W_M : \mathcal{C} \longrightarrow \mathcal{H}_{\text{res}}}. \quad (\text{ECI5.4})$$

This is an anti-linear map from the specified algebraic module; no continuity claim is made for an unassigned topology on $\mathcal{C}$. Its restriction along $\mathcal{Q} \hookrightarrow \mathcal{C}$ is exactly the already proved continuous map S from RTT.

There is an exact strong residue realization of every value in (ECI5.4). For finite $E \subset \mathcal{Z}$, put

$$r_E(h) = \sum_{\rho \in E} m_\rho (-1)^{m_\rho - 1} u_\rho(0) \overline{h(\rho^\#)} e_{1-\rho, m_\rho - 1}, \quad \zeta(\rho + t) = t^{m_\rho} u_\rho(t). \quad (\text{ECI5.5})$$

Its residue functional is the finite partial sum in (ECI5.1), by RTT's exact local matrix. For a bounded $B \subset \mathcal{Q}$, the tail seminorm is at most

$$\sup_{F \in B} \|E_k F\|_+ \left(\sum_{\rho \notin E} m_\rho w_\rho^{-2k} |h(\rho^\#)|^2 \right)^{1/2} \rightarrow 0. \quad (\text{ECI5.6})$$

This proves strong convergence in the entire residue receiver. No density of finite-support primal classes, chosen Gaussian window, or unproved infinite residue series is needed.

Let $\mathfrak{r} = \{h \in M : h(\rho) = 0 \ \forall \rho\}$. Global isolators tested in (ECI5.1) give the exact kernel

$$\ker S_M = \mathfrak{r}/\mathfrak{a}. \quad (\text{ECI5.7})$$

Locally this is the higher-jet ideal in every multiplicity block. No uniform nilpotence exponent, or equality with an algebraic nilradical when multiplicities are unbounded, is asserted globally. Each original zero value survives.

ECI6. A full-jet primal multiplier on the cyclic module, and involution covariance

The multiplier $H_\zeta = s^2 \zeta'(1-s)$ belongs to M by RTT2's full Euler–Maclaurin estimate, and $H_\zeta(0) = -1$. The original $F_0 \in \mathfrak{a}$ has $F_0(0) = 1/8$. Thus

$$[h] \mapsto \left[\frac{h(s) - F_0(s)h(0)/F_0(0)}{s} \right] \quad (\text{ECI6.1})$$

is the inverse of L on the entire $\mathcal{C}$. Entire division at the one removable point preserves the polynomial strip bounds, and the same local argument preserves $\mathfrak{a}$. The two inverse identities follow by multiplication, exactly as in NEA7 and RTT2. This constructs

$$\mathbf{P}_M = L^{-2} M_{s^2 \zeta'(1-s)} \mathbf{C} : \mathcal{C} \rightarrow \mathcal{C}. \quad (\text{ECI6.2})$$

Its full local germ at a zero is $\zeta'(1-s)\overline{h(\bar{s})}$. If the output zero is σ , put $\rho = 1 - \sigma$, $m = m_\rho$. Then

$$j_\sigma(\mathbf{P}_M h) = m(-1)^{m-1} u_\rho(0) \overline{h(\bar{\sigma})} (s - \sigma)^{m-1}. \quad (\text{ECI6.3})$$

The proof differentiates $\zeta(\rho - t) = (-t)^m u_\rho(-t)$ with respect to its own argument before taking the full quotient jet, as in RTT3. The coefficient is nonzero, so the block map has rank one, with the exact kernel of all positive-order input jets. Its global kernel is again $\mathfrak{r}/\mathfrak{a}$. The map restricts to RTT's continuous $\mathbf{P}$ on $\mathcal{Q}$; the target is otherwise the explicitly algebraic module $\mathcal{C}$.

For any $b, h \in M$, the full convergent trace formula proves

$$\widehat{S}_M(bh) = m'_{b\#1} \widehat{S}_M(h). \quad (\text{ECI6.4})$$

The transpose acts on the original continuous dual; its strong continuity follows because the multiplier on $\mathcal{Q}$ preserves bounded sets. This is the extended exact GSR covariance, with its scalar conjugation intact.

The involution comparison is

$$\boxed{S_M K_1 = K_{\text{res}} S_M}. \quad (\text{ECI6.5})$$

Indeed evaluating the right side at F gives $\overline{\sum m_\rho F(\rho^\#) \overline{h(\rho^\#)}} = \sum m_\rho F(\rho) h(\rho)$, after reindexing by $\#$. Evaluating the left side gives the same expression. Its absolute convergence follows from (ECI5.3). Thus this is a whole-module equality, not an extension inferred from finite-support density.

The local formula (ECI4.5) agrees with (ECI6.5) through the exact differentiated functional equation

$$\zeta'(1-s) = \frac{\chi'(s)}{\chi(s)^2} \zeta(s) - \frac{1}{\chi(s)} \zeta'(s). \quad (\text{ECI6.6})$$

The first term retains the full original multiplicity and is zero only after passage to that actual jet quotient; the second term retains the minus sign and all unit derivatives. This proves the relation between the source involution, the ζ' trace map and the residue-dual involution, without replacing the original function by its completion or treating χ as a global entire multiplier.

ECI7. The normal class module maps to the same existing residue receiver

Use the actual inverse translation $U : \mathcal{C}_+ \rightarrow \mathcal{C}$. Define

$$\mathbf{S}_+ = \mathbf{S}_M U : \mathcal{C}_+ \rightarrow \mathcal{H}_{\text{res}}. \quad (\text{ECI7.1})$$

For $X \in \mathcal{Q}_+$, its precise evaluation is

$$(\hat{\mathbf{S}}_+ k)(UX) = \sum_{\rho} m_{\rho} X(\rho+1) \overline{k(\rho^{\#}+1)}. \quad (\text{ECI7.2})$$

Thus it is a map from the actual cyclic normal-extension or boundary module, by (ECI0.3), to the original strong residue receiver, tested through the exact coordinate map U . Its kernel is $\mathfrak{r}_+/\mathfrak{a}_+$, where $\mathfrak{r}_+ = V\mathfrak{r}$. All zero values and all original multiplicities in the normal module are retained.

For the normal involution and arbitrary multipliers,

$$\mathbf{S}_+ \mathbf{K}_3 = \mathbf{K}_{\text{res}} \mathbf{S}_+, \quad \hat{\mathbf{S}}_+(bk) = m'_{U(b\#_3)} \hat{\mathbf{S}}_+(k). \quad (\text{ECI7.3})$$

These follow from $UK_3 = K_1 U$, $U(bk) = (Ub)(Uk)$, and (ECI6.4). Specializing to $b(\lambda) = a^{\lambda}$ retains

$$\boxed{\mathbf{S}_+ D_a^+ = a \hat{T}_a \mathbf{S}_+, \quad \hat{\mathbf{S}}_+ D_a^+ = a^2 (T_{1/a})' \hat{\mathbf{S}}_+.} \quad (\text{ECI7.4})$$

Here $\hat{T}_a$ is the existing residue action from GTR10, satisfying $\hat{\mathbf{S}}_+ \hat{T}_a = a (T_{1/a})' \hat{\mathbf{S}}_+$. The extra factor in (ECI7.4) is the actual normal translation factor in (ECI2.4).

There is an exact translated primal map as well:

$$\mathbf{P}_+ = V \mathbf{P}_M U = (L_+ - 1)^{-2} M_{(\lambda-1)^2 \zeta'(2-\lambda)} \mathbf{C}. \quad (\text{ECI7.5})$$

The only derivative pole being removed is at $\lambda = 1$, and the multiplier's value there is -1 . This point is outside the actual normal divisor, so the inverse of $L_+ - 1$ exists on $\mathcal{C}_+$ by NEA7. Every full block formula is (ECI6.3) translated by one, with the local coordinate unchanged. No additional spectrum is inserted.

ECI8. Normal residue duality and the distinction from geometric normal transfer

The translated residue pairing on actual finite-support normal classes is

$$\mathcal{R}_+(X, Y) = \sum_{\rho} \text{Res}_{\lambda=\rho+1} \frac{X(\lambda)Y(3-\lambda)}{\zeta(\lambda-1)} d\lambda. \quad (\text{ECI8.1})$$

It is a coordinate transport of the original residue, not a replacement arithmetic function:

$$\mathcal{R}_+(VF, VG) = \mathcal{R}(F, G). \quad (\text{ECI8.2})$$

Proof: put $\lambda = s + 1$. The denominator becomes original $\zeta(s)$, the second numerator becomes $G(1-s)$, and $d\lambda = ds$. Both multiplicities and positive local residue orientations are unchanged. The completion for its corresponding bounded-set seminorms is therefore topologically isomorphic to the existing $\mathcal{H}_{\text{res}}$, via the

extension of V ; this follows because V, U on the test quotients carry bounded sets to bounded sets. No extra completion is needed in (ECI7.1).

The full normal functional equation is

$$\zeta(\lambda - 1) = \chi(\lambda - 1)\zeta(2 - \lambda). \quad (\text{ECI8.3})$$

Consequently the normal residue-dual involution has local jets

$$Y(\lambda) \mapsto -\chi(2 - \lambda)Y^{\#_s}(\lambda). \quad (\text{ECI8.4})$$

This is exactly (ECI4.5) translated, with its full Gamma, sine, power and derivative data from (ECI4.3). Its completed action is the same transported continuous dual involution, not an assumed entire multiplier.

The normal pairing's full dilation factor is

$$\mathcal{R}_+(D_a^+X, D_a^+Y) = a^3\mathcal{R}_+(X, Y), \quad W_+(D_a^+X, Y) = W_+(X, a^3D_{1/a}^+Y), \quad (\text{ECI8.5})$$

where W_+ is the exact value sum in (ECI7.2). In the residue integrand the multiplier is $a^\lambda a^{3-\lambda} = a^3$. In the value formula the equality follows from $\overline{3 - \lambda} = 3 - \lambda$. These are whole-divisor identities.

For $a = n$ a positive integer, GTR's actual degree- n covering has normal pullback aT_a in the original source coordinate and transfer with inverse coefficients $T_{1/a}$. For general $a > 0$, these same expressions remain the specified coefficient operators; no covering with nonintegral degree is asserted. Transporting both coefficient operators gives

$$V(aT_a)U = D_a^+, \quad VT_{1/a}U = aD_{1/a}^+. \quad (\text{ECI8.6})$$

Their product remains aI . Therefore the normal Weil/residue adjoint operator in (ECI8.5) is precisely a^2 times the displayed normal coefficient transfer, and, for positive integer $a = n$, precisely n^2 times the actual geometric normal transfer. The three identities (ECI7.4), (ECI8.5) and (ECI8.6) agree with each other; none permits deletion of that factor. This calculation concerns the degree-shifted normal class module. It does not assign its a^3 factor to the original degree-one Weil pairing, whose factor remains a .

ECI9. The separator's zero and identity actions remain different after translation

FOD and NEA prove

$$c = 0 \text{ on } \mathcal{C}_+, \quad c = 1 \text{ on } \mathcal{C}, \quad c\delta = 0. \quad (\text{ECI9.1})$$

The two actions are compatible with the actual translation because

$$(Uc)(s) = c(s + 1) = 1 - E(s) \in \mathfrak{a}, \quad (\text{ECI9.2})$$

whereas $c(s) - 1 \in \mathfrak{a}$. Thus translating the annihilation equation on the normal class module gives multiplication by Uc , not multiplication by c , on the original module. Explicitly

$$S_+(ck) = S_M((Uc)(Uk)) = 0, \quad S_M(ch) = S_M(h). \quad (\text{ECI9.3})$$

The first vanishes because its argument is zero in $\mathcal{C}$; the second retains the entire original trace. These equations are the exact connecting map between the two assertions. They do not identify their source classes or their scalar actions. In particular the normal annihilation $c\delta = 0$, which proves the actual lift in FOD7/NEA10, does not become an annihilation of original Weil values by silently removing the coordinate translation.

ECI10. What the extension-class residue maps do to the positive-adjoint defect

On the established value Hilbert space retain GTAH's bounded operator

$$(D_a x)_\rho = (a^{\bar{\rho}} - a^{1-\rho})x_\rho, \quad a > 1. \quad (\text{ECI10.1})$$

It acts on this explicitly constructed value space, not as an asserted entire multiplier on all of $\mathcal{C}$. RTT6.2 proves an injective continuous anti-linear map

$$A : H \rightarrow \mathcal{H}_{\text{res}}, \quad (\hat{A}y)(F) = \langle EF, Jy \rangle, \quad S = AE. \quad (\text{ECI10.2})$$

The new extension-class map restricts to exactly this map on $\mathcal{Q} \subset \mathcal{C}$, by (ECI5.1). Every finite value vector is attained there by the actual global isolators. Consequently this extension of the domain loses no original value direction, and

$$AD_a^* D_a = 0 \iff D_a^* D_a = 0. \quad (\text{ECI10.3})$$

The reverse implication is immediate; the forward implication is injectivity applied to each vector. Positivity of the Hilbert norm then gives equivalence to $D_a = 0$, exactly as in GTAH6.7.

There is a specific full-source test for every actual zero, avoiding an unproved nonholomorphic multiplier lift. Fix ρ and let $e_{\rho,0}$ be its global full-jet isolator, viewed in $\mathcal{Q} \subset \mathcal{C}$. Put

$$d_\rho^2 = |a^{\bar{\rho}} - a^{1-\rho}|^2, \quad h_\rho = d_\rho^2 e_{\rho,0} \in \mathcal{B} \subset M. \quad (\text{ECI10.4})$$

Then $E[h_\rho] = D_a^* D_a E[e_{\rho,0}]$, and evaluation on the other actual isolator gives

$$(\hat{S}_M[h_\rho])(e_{\rho^\#,0}) = m_\rho d_\rho^2. \quad (\text{ECI10.5})$$

This is nonzero precisely when the displayed scalar defect is nonzero. It quantifies over the actual divisor and does not postulate an off-line zero. Translating h_ρ to $Vh_\rho \in \mathcal{C}_+$ gives the same result through $S_+ V = S_M$. Thus neither the extension-class involution nor this residue comparison kills a nonzero positive-adjoint defect. The statement is established by the actual maps and witnesses, not by treating an RH-equivalent criterion as an achievement.

For every actual two-point orbit, the signed test also remains $e_{\rho,0} - e_{\rho^\#,0}$, with original Weil value $-2m_\rho$ as calculated in GTAH6.5. Both new class-module maps restrict to that same residue value. The higher-jet kernel (ECI5.7) is the exact information they remove; it contains no nonzero isolated value vector.

ECI11. Original factors, class actions and category scope

Every source factor used above remains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_+(\lambda) = \frac{(\lambda-1)(\lambda-2)}{8} \pi^{-(\lambda-1)/2} \Gamma((\lambda-1)/2) \zeta(\lambda-1). \quad (\text{ECI11.1})$$

Their endpoints are $F_0(0) = F_0(1) = F_+(1) = F_+(2) = 1/8$, and the original trivial-zero comparison is

$$F_0(-2r) = F_+(1-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) = \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r), \quad r \geq 1. \quad (\text{ECI11.2})$$

The function in every residue denominator remains ζ in its explicitly related coordinate. Its arithmetic is still

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'}{\zeta}(s) = \sum_p \sum_{k \geq 1} (\log p) p^{-ks}, \quad \Re s > 1. \quad (\text{ECI11.3})$$

On the original test subspace the new trace maps are exactly RTT's map, so RTT9's complete prime, Gamma, endpoint and finite trivial-divisor identity remains unchanged. Formula (ECI5.1) on a general polynomial-growth multiplier is proved directly as a continuous spectral functional; no extension of the prime explicit formula to an unspecified larger test class is inferred.

The class-module action above is multiplication, or central postcomposition on the actual cyclic extension classes. As FOD7 and NEA10 prove, conjugation of the equivariant boundary by dilation fixes that boundary. The scalar factors in ECI2 and ECI8 concern the specified multiplier and pairing actions; they are not reassigned as conjugation weights of δ . Likewise the semilinear involutions of ECI1 are explicit coefficient and row operations, not an unproved identification with ordinary or étale Verdier duality. The residue-dual comparison they do supply is (ECI4.1)–(ECI4.5), and its relation to the actual extension modules is (ECI6.5) and (ECI7.3).

ECI12. Exact comparison with the Gaussian source embedding

The complete ECR0–ECR7 derivation in `EXTENSION_CLASS_TO_ORIGINAL_RESIDUE.md` was read for this comparison. It constructs, for each $t > 0$, the entire source multiplier $g_t(s) = e^{ts^2}$ and the injective M -map

$$b_t : \mathcal{C} \rightarrow \mathcal{Q}, \quad [h] \mapsto [g_t h], \quad j : \mathcal{Q} \hookrightarrow \mathcal{C}, \quad j b_t = m_{g_t}. \quad (\text{ECI12.1})$$

Its coefficient-to-source estimate retains $|g_t(\sigma + i\gamma)| = e^{t(\sigma^2 - \gamma^2)}$, so multiplying a polynomial-growth representative by g_t really lands in $\mathcal{B}$. The target and purpose differ from the direct trace functional of ECI5: b_t reaches the actual source quotient and hence the positive value Hilbert space, while (ECI5.4) reaches its strong dual receiver. For example the value sequence of $[1] \in \mathcal{C}$ is not in H , since there are infinitely many actual zeros, but (ECI5.1) at $h = 1$ is a continuous functional on every original rapid test. There is no identification of those two receiving spaces.

The exact comparison is

$$\boxed{S b_t = S_M m_{g_t} : \mathcal{C} \rightarrow \mathcal{H}_{\text{res}}.} \quad (\text{ECI12.2})$$

Indeed $S_M j = S$ by (ECI5.1); compose this equality with b_t and use (ECI12.1). Explicitly both sides evaluated against F give $\sum_{\rho} m_{\rho} F(\rho) \overline{g_t(\rho^{\#}) h(\rho^{\#})}$, with absolute and strong convergence already proved. Because g_t is a unit in each actual local ring, both maps have the same value-zero kernel $\mathfrak{r}/\mathfrak{a}$. This is not a claim that $[g_t]$ is invertible in the global multiplier quotient; ECR1 proves it is not.

The embedding also has an exact involution factor which should be retained in any dual comparison:

$$g_t^{\#1}(s) = e^{t(1-s)^2} = d_t(s) g_t(s), \quad d_t(s) = e^{t(1-2s)}, \quad d_t^{\#1} = d_t^{-1}. \quad (\text{ECI12.3})$$

Both d_t and its inverse are in M . Consequently

$$K_1 b_t = m_{d_t} b_t K_1. \quad (\text{ECI12.4})$$

This is equality on full source jets: it follows from multiplication of the entire functions before quotient. No coordinate or coefficient has been normalized. Conjugation alone commutes with b_t , since g_t has real Taylor coefficients; reflection alone satisfies the same d_t factor as (ECI12.4).

Write $S_t = S b_t$, as in ECR4. Combining (ECI6.5), (ECI12.4), and the multiplier covariance gives the complete dual version

$$\hat{I} K_{\text{res}} S_t = m'_{d_t^{-1}} \hat{I} S_t K_1. \quad (\text{ECI12.5})$$

The inverse factor arises from the involution on the multiplier, not an inverse of g_t . For the normal class module the transported source Gaussian is $(V g_t)(\lambda) = e^{t(\lambda-1)^2}$, and its corresponding factor is $(V d_t)(\lambda) = e^{t(3-2\lambda)}$. These follow directly from (ECI2.3), retaining the actual shift instead of choosing a different normal Gaussian.

For two class-module inputs, (ECI12.2) gives precisely ECR's full trace

$$\widehat{\mathcal{R}}(b_t h, S_t k) = \sum_{\rho} m_{\rho} e^{t\{\rho^2 + (1-\rho)^2\}} h(\rho) \overline{k(\rho^{\#})}. \quad (\text{ECI12.6})$$

The positive value receiver instead has the actual modulus weight $e^{2t(\sigma^2 - \gamma^2)}$. The two weights are not substituted for each other. ECR4 proves the former's complete original prime/Gamma/endpoint formula; ECR5 proves the latter's positive defect formula through its stated Hilbert-space map. Thus the canonical residue map here and ECR's Gaussian source embedding agree by an exact morphism while preserving their different domains and purposes.

ECI13. Exact receiving result

The new cyclic modules now have complete involution maps, including their full jets, translation, and actual row origins. Their original and normal involutions use $1 - s$ and $3 - \lambda$, respectively, with the exact

translation diagram (ECI2.3). Their residue-dual involutions retain the functional-equation units and minus signs (ECI4.5), (ECI8.4). Every polynomial-growth extension class has a canonical, explicitly convergent map to the original strong residue receiver, extending RTT without changing its kernel on original tests. Its covariance retains every factor a , a^2 , and a^3 in the different specified domains.

This calculates the proposed use of the new annihilator modules rather than merely naming their resemblance to duality. The normal boundary's annihilation and the unchanged original positive-adjoint defect coexist through the proved translation (ECI9.2). The remaining attempt toward the active goal must supply additional source information controlling that retained defect; the independent positive source-pairing calculation can now use the exact involution and degree dictionary here. No pure weight, positive original Weil form, or RH conclusion is assumed.

Proof sources actually read for this continuation: ../tau_weight_cohomology_20260924/CC_FULL_SOURCE_OBSTRUCTION_DECOMPOSITION.md, FOD0–FOD8; ../tau_weight_cohomology_20260924/CC_NORMAL_EXTENSION_ANNIHILATOR.md, NEA0–NEA10; EXTENSION_CLASS_TO_ORIGINAL_RESIDUE.md, ECR0–ECR7; current construction/prerequisite instructions; and the existing checked RTT, GSR, GTAH and GMS maps cited at their equation locators above. Human-source context remains the original Connes–Consani §5 construction and Deligne's Weil II §3.6, with their source-reading coverage recorded in those receiving proofs. This continuation makes no new complete-reading claim for those human papers and no public release claim.

Independent audit of the extension-class involution calculation

24 September 2026. Complete ECI0–ECI13 reviewed initially at SHA256 4a69cc5823e6827f9a464cfc98f835a00e01e700b47efe33cd94845be45a0e16, followed by review of the corrected ECI8 and ECR5 passages and verification of the final hashes recorded below.

All displayed identities checked in this review are correct. One scope qualification was required in ECI8 and in ECR5’s inherited geometric-transfer wording: GTR constructs actual geometric covers for recovered positive integers n , whereas the coefficient identities hold for every specified $a > 0$. Root applied this correction, and the revised passages have been checked. No proof source was edited by this independent reviewer.

Coverage

Read ECI0–ECI13 completely. The complete FOD0–FOD8, NEA0–NEA10, ECR0–ECR7, RTT0–RTT11, GTAH0–GTAH7, GSR0–GSR4 and GIQ9 had been read for the preceding source audit in this same independent review session. Their exact hashes and coverage are recorded in `EXTENSION_CLASS_SOURCE_INDEPENDENT_AUDIT.md`. The present calculation reuses those checked annihilators, source inclusions, residue convention, strong-dual receiver and value-map estimates.

Additionally read GTR1, GTR4, GTR6, and GTR10 through equation GTR10.8 and its proof in `CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md`, SHA256 6fb23be5d0f869af5c9bf4a5ba8f00584bcf5718079c4c3139b4b88fc3b15f04. These passages specify the integer covers, normal pullback and transfer, full source-to-normal-dual comparison, and the completed residue action. This is bounded programme-proof coverage, not a new complete reading of the remote GMS/SSI/SDT sources or of the human bibliography.

The support and its information layers retain the user’s definitions and retractions. None of the following operations is assigned to the support itself.

Full jets, translation, and the joint source map

The strip estimates and multiplicity-preserving zero permutations make C, K_1, K_3 well defined on their stated rings, ideals and quotients. The factor $(-1)^j$ in ECI1.2 follows by differentiating the reflected Taylor series before truncating at the actual multiplicity. ECI1.5 is the induced semilinear exact operation on the respective rows. It does not require the original involution to preserve the full intersection ideal.

The relations $L_+ V = V(L+1)$, $D_a^+ V = a V D_a$ and $K_3 V = V K_1$ follow by substitution at $\lambda-1$. Consequently ECI2.5 retains the distinct factors a and a^3 , with every Taylor term in ECI2.6 present.

The Chinese-remainder description in ECI3 is exactly the one proved by FOD’s full separator. Direct expansion gives

$$K_1 K_3 F(s) = F(s+2), \quad K_3 K_1 F(s) = F(s-2).$$

This verifies both shifted terms and their coefficients in ECI3.4. On the original divisor, $c = 1$ and $c^{\#1} = 1$ to every required order; on the normal divisor, $c = 0$ and $c^{\#3} = 0$ to every required order. Thus each cross coefficient and the diagonal coefficient minus one belong to both required vanishing ideals. The conclusion is precisely $A_c^2 F - F \in \mathcal{K}$, not equality before quotient.

Residue-dual sign and functional-equation unit

The bilinear residue convention is

$$\mathcal{R}(F, g) = \sum_{\rho} \operatorname{Res}_{s=\rho} \frac{F(s)g(1-s)}{\zeta(s)} ds.$$

Conjugating $\mathcal{R}(K_1 F, g)$, reindexing by conjugation, and then substituting $s = 1 - z$ gives

$$(\mathcal{K}_1^{\vee} \iota g)(F) = - \sum_{\rho} \operatorname{Res}_{z=\rho} \frac{F(z)(Cg)(z)}{\zeta(1-z)} dz = - \sum_{\rho} \operatorname{Res}_{z=\rho} \frac{F(z)(Cg)(z)\chi(z)}{\zeta(z)} dz.$$

The minus sign comes from $ds = -dz$. To express this again as $\mathcal{R}(F, g_{\text{new}})$, one must have

$$g_{\text{new}}(1-z) = -\chi(z)(\mathbb{C}g)(z), \quad j_{\sigma}g_{\text{new}} = j_{\sigma}[-\chi(1-s)g^{\#_1}(s)].$$

This proves ECI4.4–ECI4.5 with the asserted sign, reflected argument and full unit derivatives. The unit is only required locally at nontrivial zeros. The completed operator is instead defined by the proved strong dual involution, so no invalid entire multiplier assertion is used.

Its local square multiplies by $\chi(1-s)\chi(s) = 1$, with two minus signs. Differentiating the full original functional equation gives ECI6.6; the term involving $\chi'\zeta/\chi^2$ vanishes only after passage to the actual full-multiplicity quotient. Translation gives $-\chi(2-\lambda)Y^{\#_3}$ in ECI8.4.

Canonical class functionals and their exact kernels

For each fixed $h \in M$, choose $k \geq d+2$ from its strip growth exponent d . The weighted Cauchy–Schwarz estimate ECI5.3 then bounds the tail by the convergent full sum of $m_{\rho}(1+|\Im\rho|)^{-4}$. The same estimate is uniform over any bounded subset of the actual quotient because E_k is continuous. This verifies both the original functional and strong convergence of ECI5.5. No arbitrary infinite interpolation or finite-support density in the primal quotient is used.

Global isolators prove exactly the kernel $\mathfrak{r}/\mathfrak{a}$. This is the value-zero ideal; the note correctly refrains from identifying it with a global nilradical. The inverse of L in ECI6.1 removes its single apparent singularity by the full $F_0(s)h(0)/F_0(0)$ term. Its two quotient inverse identities and polynomial strip bounds hold, giving the full class-module map $\mathbf{P}_M$. ECI6.3 retains rank one and its complete nonzero original-zeta coefficient on each actual multiplicity block.

Evaluation of both sides verifies ECI6.4–ECI6.5. Their translation gives ECI7.1–ECI7.5, including the removed derivative pole at $\lambda = 1$ and its value -1 .

Every degree factor and the required scope repair

For $b(\lambda) = a^{\lambda}$,

$$U(b^{\#_3})(s) = a^{2-s}.$$

Thus ECI7.4 is exactly

$$\hat{t}S_+D_a^+ = a^2(T_{1/a})'\hat{t}S_+ = \hat{t}a\hat{T}_a S_+.$$

The residue product in the normal coordinate is $a^{\lambda}a^{3-\lambda} = a^3$, giving ECI8.5. Separately,

$$V(aT_a)U = D_a^+, \quad VT_{1/a}U = aD_{1/a}^+,$$

whose product is aI . Therefore the normal Weil adjoint $a^3D_{1/a}^+$ is exactly a^2 times the displayed normal transfer operator. No factor is missing.

The initial ECI8 wording called these actual geometric maps for the general positive parameter used in ECI2. GTR1 constructs the covers only for recovered integers $n \geq 1$, and GTR4.2–GTR4.3 then give nT_n and $T_{1/n}$ in normal degree. The required qualification was:

For each recovered positive integer $a = n$, GTR constructs the actual geometric normal pullback nT_n and inverse-coefficient transfer $T_{1/n}$. For general $a > 0$, aT_a and $T_{1/a}$ are the corresponding coefficient operators. The following transport identities hold for every $a > 0$; their interpretation as these geometric cover maps is restricted to the recovered positive integers.

Root’s revised ECI8 now states this qualification, and also distinguishes the factor a^2 for coefficient operators from n^2 for the actual integer-cover transfer. All displayed formulas remain unchanged and correct. The corresponding old ECR5 phrase, “the actual geometric transfer is preserved,” needed the same qualification. Revised ECR5 now retains the coefficient identities for all $a > 0$ and identifies their geometric realization only at $a = n$ a positive integer. Both corrected passages were read directly after the update.

Separator and Gaussian covariance

The exact scalar transport is $Uc(s) = c(s+1) = 1 - E(s) \in \mathfrak{a}$. In contrast, $c(s) - 1 \in \mathfrak{a}$. ECI9.3 therefore compares two different scalar actions by their actual map; normal annihilation does not turn into annihilation of the original trace.

For the Gaussian,

$$g_t^{\#1} = d_t g_t, \quad d_t = e^{t(1-2s)}, \quad d_t^{\#1} = d_t^{-1}.$$

Both d_t and its inverse are strip-polynomial multipliers. Using the already proved covariance gives the full derivation

$$\begin{aligned} \hat{\iota} K_{\text{res}} S_t &= \hat{\iota} S K_1 b_t \\ &= \hat{\iota} S m_{d_t} b_t K_1 \\ &= m'_{d_t^{\#1}} \hat{\iota} S b_t K_1 \\ &= m'_{d_t^{-1}} \hat{\iota} S_t K_1. \end{aligned}$$

This proves ECI12.5, including its inverse factor and both anti-linear maps. It uses no inverse of g_t . Translation gives $e^{t(\lambda-1)^2}$ and $e^{t(3-2\lambda)}$, as stated.

ECI10's isolator test evaluates to $m_\rho |a^{\bar{\rho}} - a^{1-\rho}|^2$ with the specified reflected test point; its vanishing is not assumed. ECI11 retains the full original factors and restricts the prime explicit formula to its established test domain. ECI12.6 agrees with ECR's exact product weight rather than its different positive modulus weight. ECI13 accurately describes the result with the geometric parameter qualification now included.

Final accepted versions and relation to the preceding audit

The corrected proof hashes were read from disk and verified:

Proof	Accepted SHA256
EXTENSION_CLASS_TO_ORIGINAL_RESIDU E.md	0a4e004a008422f622beeca7af8bab9a9f6527c0d4eb4 3571ea6443c0975250
EXTENSION_CLASS_INVOLUTION_INDEPEN DENT.md	ba951baf03f04edfeed66f98708e3c3ae5c6b2657b23b5 69cb108af4316aec31

EXTENSION_CLASS_SOURCE_INDEPENDENT_AUDIT.md remains unchanged at SHA256 1deb16eb34459051ba337c1a6c801a19bab506c605353f6d3c749533047c4fd0. It records full coverage of the original ECR proof at SHA256 0da1815f474c5279012a94be8f65fdb8dbf3a178f56e97ff7b543a2d04c82bbd. This present audit supplements its acceptance with the small geometric-parameter scope correction just described. No original factor, map, sum, jet or displayed identity was changed by that correction. Root preserved the previous proof versions under history/before_cover_parameter_scope_20260924.

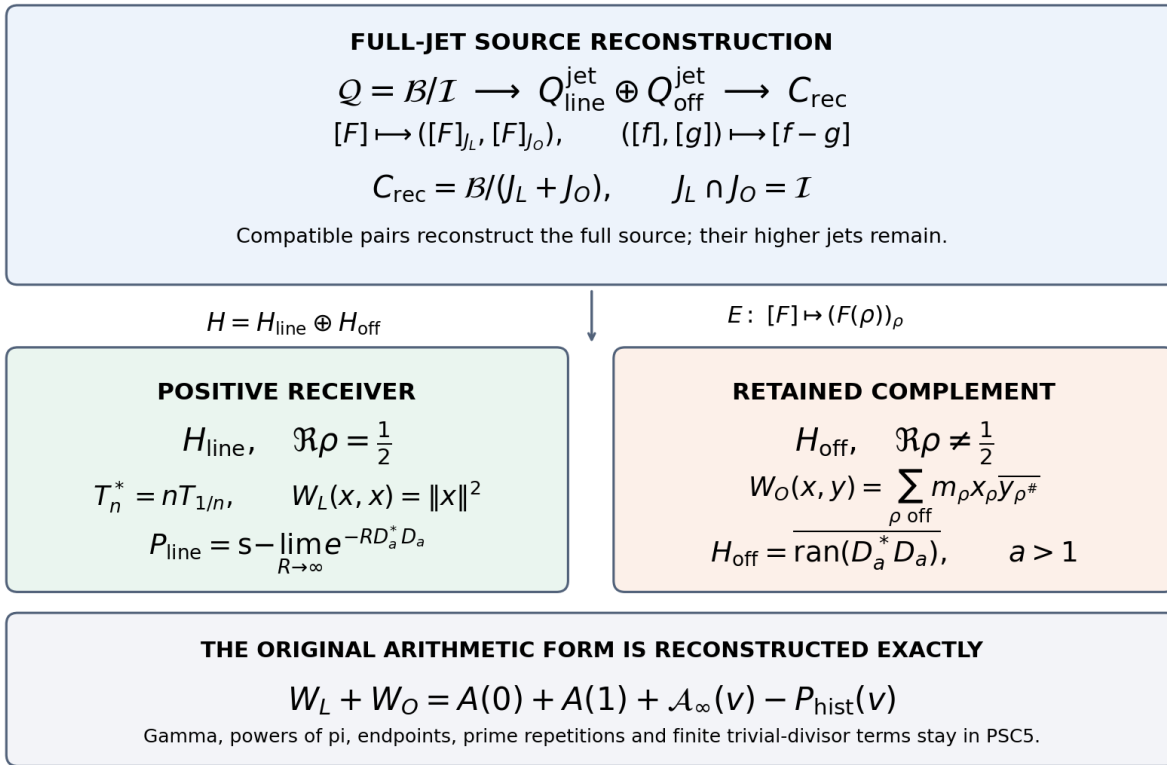
Positive transfer quotient and reconstruction of the complete source

The next attempt constructs the maximal positive Hilbert receiver preserving the actual transfer as adjoint, on the complete original value space. Its quotient, universal mapping property and all bounded positive invariant forms are proved. The original indefinite Weil form and the precise data removed by the projection remain explicit.

The source calculation then constructs the corresponding full-jet and value quotients, with every kernel. The complete source is recovered as a fibre product over a calculated compatibility quotient. The topology of that reconstruction, the exact original-zeta arithmetic difference, the existing extension-class maps and the full-jet multiplier problem are all proved. A global heat-operator limit constructs the Hilbert projector without assuming a source interpolation theorem or a zero-height cutoff.

The positive quotient and the complete source it comes from

All actual zeros and multiplicities; no numerical cutoff or assumed purity



Proofs: PSC1-PSC8 and PTQ. Integer n is the geometric cover degree; a is a coefficient parameter.

Figure 12: The positive transfer quotient and the complete source reconstruction. Every actual zero and its multiplicity remains in the two projections and their compatibility data. The original arithmetic form is the sum of both contributions. Proofs PSC1–PSC8 and the independent universal-quotient derivation; full Gamma, endpoint, prime-history and finite trivial-divisor formulas are PSC5.

The complete proofs and mathematical review follow. The construction preserves the original function and its discarded contribution as separate specified data. It does not infer the vanishing of that contribution from positivity of the compressed receiver. The actual Deligne lifting and original-zeta positivity goal remains active.

The universal positive receiver of the actual geometric transfer

24 September 2026. Complete independent derivation PTQ0–PTQ11.

PTQ0. Actual prerequisites and the quotient being constructed

The classical location and functional-equation facts used here are recorded in NIST DLMF §25.10(i) and §25.4; neither simplicity of the zeros nor RH is assumed.

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The coefficient spaces, integers, primes, dilation operators and Hilbert norms in this proof are used only after the complete-history arithmetic reconstruction. No addition, numerical value, parity, midpoint or coordinate is assigned to τ . The source kernels and branch records remain those of the existing construction. This calculation implements the next requested step: construct the positive receiver forced by the measured adjoint defect, and prove its exact universal property, rather than assume that the original defect vanishes.

The inputs are the actual complete-zero value receiver in GIQ/RTT, the geometric trace and its degree in GTR, the computed positive-adjoint defect in GTAH, and the full source and class-module maps in ECR/ECI. Their relevant full proofs were read and derived in the preceding comparisons; the current ECR source/Hilbert comparison and ECI degree and involution formulas were recalled directly. In particular the value observation is a proved map from the complete original quotient, and is not an arbitrary alternative spectrum.

Let $\mathcal{Z}$ be the set of all distinct actual nontrivial zeros of original ζ , with the full multiplicities m_ρ . Write $\rho = \sigma + i\gamma$. The established zero-free boundary lines give $0 < \sigma < 1$. Set

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_{\rho} m_{\rho} x_{\rho} \overline{y_{\rho}},$$

$$\rho^{\#} = 1 - \bar{\rho}, \quad (Jx)_{\rho} = x_{\rho^{\#}}, \quad W(x, y) = \langle x, Jy \rangle_+. \quad (\text{PTQ0.1})$$

The inner product is linear in its first argument. The original functional equation preserves the multiplicities and gives $J^* = J = J^{-1}$. For every recovered positive integer n , retain the original operators

$$(T_n x)_{\rho} = n^{\rho} x_{\rho}, \quad (U_n x)_{\rho} = n^{1-\rho} x_{\rho}, \quad U_n = nT_{1/n}. \quad (\text{PTQ0.2})$$

The positive integer n is the actual ramified cover degree from GTR. The inverse parameter in $T_{1/n}$ is the coefficient inverse, not a cover of fractional degree. All these operators are bounded and invertible on H , and

$$T_m T_n = T_{mn}, \quad U_m U_n = U_{mn}, \quad T_m U_n = U_n T_m, \quad T_n U_m = U_m T_n = nI. \quad (\text{PTQ0.3})$$

Their exact positive-adjoint comparison, already calculated in GTAH, is

$$T_n^* = JU_n J, \quad D_n = T_n^* - U_n, \quad (D_n x)_{\rho} = e^{-i\gamma \log n} (n^{\sigma} - n^{1-\sigma}) x_{\rho}. \quad (\text{PTQ0.4})$$

All subsequent uses of a fixed recovered integer concern this already constructed whole-spectrum representation. No finite collection of arithmetic labels is being used to initialize the programme's arithmetic.

PTQ1. The closed space generated by the positive-adjoint defects

Define the actual subsets

$$\mathcal{Z}_{\text{line}} = \{\rho \in \mathcal{Z} : \Re \rho = 1/2\}, \quad \mathcal{Z}_{\text{off}} = \mathcal{Z} \setminus \mathcal{Z}_{\text{line}},$$

$$H_{\text{line}} = \{x \in H : x_{\rho} = 0 \text{ for } \rho \in \mathcal{Z}_{\text{off}}\}, \quad H_{\text{off}} = H_{\text{line}}^{\perp}. \quad (\text{PTQ1.1})$$

These definitions are applied to the actual full zero set. They assume neither an off-line zero nor its absence. Denote the respective orthogonal projections by P_{line} and P_{off} .

For each recovered $n > 1$, monotonicity of $x \mapsto n^x$ proves

$$\ker D_n = \ker(D_n^* D_n) = H_{\text{line}}, \quad \overline{\text{Ran } D_n} = \overline{\text{Ran}(D_n^* D_n)} = H_{\text{off}}. \quad (\text{PTQ1.2})$$

Indeed the multiplier in (PTQ0.4) is zero exactly at $\sigma = 1/2$. Both ranges are supported in $\mathcal{Z}_{\text{off}}$. Conversely each coordinate vector at an off-line zero is in either range, by dividing that single vector by its specified nonzero scalar multiplier. Finite coordinate vectors are dense in the stated closed coordinate subspace; this proves the closure assertions. No uniformly bounded reciprocal multiplier is presumed, so the ranges themselves are not declared closed.

In particular the closed span of the ranges of all the actual defects, or of all their positive squares, is exactly H_{off} . The entire-family quotient is therefore independent of which recovered $n > 1$ is used to express this same closed space.

The projections can also be obtained from the positive defect itself by the exact strong limits

$$P_{\text{line}} = \text{s-lim}_{r \rightarrow \infty} \exp(-r D_n^* D_n), \quad P_{\text{off}} = \text{s-lim}_{r \rightarrow \infty} (I - \exp(-r D_n^* D_n)). \quad (\text{PTQ1.3})$$

For every coordinate, the exponential tends to one if its defect is zero, and to zero otherwise. Its modulus is at most one. Applying dominated convergence to the complete weighted square sum of any fixed $x \in H$ proves strong convergence. This constructs the closed quotient from the actual bounded positive operators, without selecting a numerical zero cutoff or assuming an entire-function projector.

PTQ2. The positive quotient retains the original degree and every arithmetic composition

Form the Hilbert quotient

$$H_{\text{pos}} = H/H_{\text{off}}, \quad q_{\text{pos}} : H \rightarrow H_{\text{pos}}. \quad (\text{PTQ2.1})$$

Its quotient norm is exactly $\|q_{\text{pos}}x\| = \|P_{\text{line}}x\|_+$. The map $q_{\text{pos}}x \mapsto P_{\text{line}}x$ is therefore a canonical unitary isomorphism from the quotient onto H_{line} . We use this specified identification below; it is not a new norm silently assigned to the original source quotient.

Both coordinate subspaces reduce every T_n , U_n , and J . The induced operators T_n^{pos} , U_n^{pos} consequently have the unchanged multipliers on the retained coordinates and all the original relations (PTQ0.3). On each retained coordinate,

$$n^{\bar{\rho}} = n^{1-\rho}.$$

Thus

$$\boxed{(T_n^{\text{pos}})^* = U_n^{\text{pos}}, \quad (T_n^{\text{pos}})^* T_n^{\text{pos}} = T_n^{\text{pos}} (T_n^{\text{pos}})^* = nI.} \quad (\text{PTQ2.2})$$

The degree is still n . No scalar factor has been removed by rescaling the representation. These are the actual original pullback and transfer maps after the specified quotient, not the reflected averages of GTAH.

PTQ3. Universal property among bounded intertwiners to positive adjoint representations

Here is the category in which maximality is proved. A receiving Hilbert representation consists of a Hilbert space K and bounded operators A_n, B_n , indexed by the recovered positive integers, with the full relations (PTQ0.3) after replacing T, U by A, B , with $A_1 = B_1 = I$, and with $B_n = A_n^*$. A map from the actual representation is a bounded linear operator $X : H \rightarrow K$ satisfying

$$XT_n = A_n X, \quad XU_n = B_n X \quad \text{for every recovered } n \geq 1. \quad (\text{PTQ3.1})$$

No surjectivity, isometry, simplicity of zeros, or finite dimension of the target is required.

Every such map has a unique bounded factorization

$$\boxed{X = Y q_{\text{pos}}, \quad Y : H_{\text{pos}} \rightarrow K,} \quad (\text{PTQ3.2})$$

and Y intertwines every original retained pullback and transfer. Under the identification with H_{line} , it is simply $X|_{H_{\text{line}}}$, and $\|Y\| = \|X\|$.

To prove the assertion, let ε_ρ be the coordinate vector with value one at ρ . Its squared norm is m_ρ ; no multiplicity is suppressed. Put $v = X\varepsilon_\rho$. Then $A_nv = n^\rho v$, while $A_n^*A_n = nI$ gives

$$n\|v\|_K^2 = \|A_nv\|_K^2 = n^{2\Re\rho}\|v\|_K^2. \quad (\text{PTQ3.3})$$

For $\rho \in \mathcal{Z}_{\text{off}}$ and any recovered $n > 1$, the two real scalars differ, so $v = 0$. Boundedness of X then kills the closure of all off-line coordinate vectors, namely H_{off} . This proves existence of the bounded factorization through the Hilbert quotient. The quotient is onto, so its factor is unique. Restriction to the reducing subspace proves the intertwining assertions, and $X = XP_{\text{line}}$ proves equality of norms.

This universal property proves maximality rather than merely exhibiting one positive receiver: every bounded positive-adjoint receiver of the unchanged geometric transfer must discard at least this entire closed space, and the quotient constructed in PTQ2 discards exactly that space. In fact the proof needs only (PTQ3.1) and $B_n = A_n^*$ on the closed span of $X(H)$; the original degree relation then implies $A_n^*A_n = nI$ there by density. The category stated above retains all original relations on the whole receiving space.

PTQ4. All bounded positive transfer-adjoint forms on the original value space

The universal quotient has a complete form-level classification. Let B be a bounded positive semidefinite Hermitian form on H , linear in its first variable. The condition

$$B(T_n x, y) = B(x, U_n y) \quad \text{for every } n \geq 1 \quad (\text{PTQ4.1})$$

is equivalent to

$$B(T_n x, T_n y) = nB(x, y) \quad \text{for every } n \geq 1. \quad (\text{PTQ4.2})$$

One implication uses $U_n T_n = nI$. Conversely replace y in (PTQ4.2) by $T_n^{-1}y$, retaining the real scalar n , and use $U_n = nT_n^{-1}$.

All such bounded forms are precisely

$$B(x, y) = \sum_{\rho \in \mathcal{Z}_{\text{line}}} m_\rho c_\rho x_\rho \overline{y_\rho}, \quad 0 \leq c_\rho \leq C \text{ for a finite } C. \quad (\text{PTQ4.3})$$

Here is a full proof. Substituting $x = y = \varepsilon_\rho$ in (PTQ4.2) forces the diagonal value to be zero at every off-line zero, by the same scalar argument as (PTQ3.3). Positivity implies Cauchy–Schwarz: expanding $B(x + zy, x + zy) \geq 0$ and minimizing its quadratic in the complex scalar z proves $|B(x, y)|^2 \leq B(x, x)B(y, y)$. Thus each off-line coordinate pairs to zero with every vector.

For two distinct critical-line zeros $\rho = 1/2 + i\gamma$, $\eta = 1/2 + i\gamma'$, (PTQ4.2) says

$$(e^{i(\gamma-\gamma')\log n} - 1)B(\varepsilon_\rho, \varepsilon_\eta) = 0 \quad \text{for every recovered } n \geq 1. \quad (\text{PTQ4.4})$$

If this cross value were nonzero, every displayed phase would be one. Ratios at consecutive integers would then give $e^{i(\gamma-\gamma')\log((n+1)/n)} = 1$ for every n . If $\gamma - \gamma' \neq 0$, the absolute value of its real exponent is strictly between zero and 2π for all sufficiently large n , a contradiction. Hence the frequencies would be equal, and two critical-line zeros with equal frequency are the same point. Thus every distinct cross value vanishes. This argument uses the entire recovered integer family, rather than taking a finite pair of named primes as the source of arithmetic.

Set $c_\rho = B(\varepsilon_\rho, \varepsilon_\rho)/m_\rho$ on the line. Positivity gives $c_\rho \geq 0$, and boundedness of the form gives a common upper bound. Its values on finite coordinate sums are exactly (PTQ4.3). Density of those vectors in H , and boundedness of the form and of the displayed weighted sum, prove the identity on all of H . Conversely every family in (PTQ4.3) defines a bounded positive semidefinite form and satisfies (PTQ4.1) directly.

The maximal form within the bound $B(x, x) \leq \|x\|_+^2$ is therefore $\langle P_{\text{line}}x, P_{\text{line}}y \rangle_+$. More generally any form in (PTQ4.3) is bounded above by a constant times this form. This is a classification for bounded forms on the specified actual Hilbert space; no unrestricted form on an untopologized Ext group is being classified. For every intertwiner in PTQ3, its pulled-back form $\langle Xx, Xy \rangle_K$ belongs to this list.

PTQ5. The full Weil form does not automatically descend through this quotient

The original form remains

$$W(x, y) = \langle P_{\text{line}}x, P_{\text{line}}y \rangle_+ + \langle P_{\text{off}}x, JP_{\text{off}}y \rangle_+. \quad (\text{PTQ5.1})$$

Both cross terms vanish because the two subspaces are orthogonal and J preserves them; on H_{line} , $J = I$. Thus the positive quotient form is the first displayed compression. The second term is the complete discarded indefinite form, with its domain H_{off} and involution retained.

A form descends through a vector-space quotient exactly when its kernel space pairs to zero with every vector: this is necessary by replacing either argument by an equivalent representative, and sufficient by the same calculation. In this case W is nondegenerate on H , since J is invertible. Therefore

$$W \text{ descends through } q_{\text{pos}} \iff H_{\text{off}} = 0. \quad (\text{PTQ5.2})$$

One can check the failure on every applicable actual pair without invoking general nondegeneracy. If $\rho \in \mathcal{Z}_{\text{off}}$, then $\rho^\# \neq \rho$, both coordinate vectors are killed by q_{pos} , but

$$W(\varepsilon_\rho, \varepsilon_{\rho^\#}) = m_\rho \neq 0. \quad (\text{PTQ5.3})$$

Their two combinations have

$$\begin{aligned} W(\varepsilon_\rho + \varepsilon_{\rho^\#}, \varepsilon_\rho + \varepsilon_{\rho^\#}) &= 2m_\rho, \\ W(\varepsilon_\rho - \varepsilon_{\rho^\#}, \varepsilon_\rho - \varepsilon_{\rho^\#}) &= -2m_\rho. \end{aligned} \quad (\text{PTQ5.4})$$

The universal positive-transfer quotient discards both directions of each such orbit, including its positive direction. It is not a quotient of the original Weil form obtained by removing its radical; that radical on H is zero. Formula (PTQ5.1) states exactly the compression and the removed contribution. This determines its mathematical scope without assuming whether the discarded sector is empty.

PTQ6. Every original source kernel and multiplicity block

Retain the actual original quotient $\mathcal{Q} = \mathcal{B}/\mathcal{J}$ with its Fréchet quotient topology, and the proved continuous value observation

$$E : \mathcal{Q} \rightarrow H, \quad E[F] = (F(\rho))_\rho. \quad (\text{PTQ6.1})$$

Define the actual composite

$$E_{\text{pos}} = q_{\text{pos}}E : \mathcal{Q} \rightarrow H_{\text{pos}}, \quad \mathcal{K}_{\text{pos}} = \ker E_{\text{pos}} = \{[F] : F(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}_{\text{line}}\}. \quad (\text{PTQ6.2})$$

It is continuous, and its kernel is a closed subspace of $\mathcal{Q}$. It contains the original full value-zero kernel $\mathcal{N}_{\text{jet}} = \ker E$, but no equality is presumed. The global isolators from RD give every finite retained value vector. Hence $E_{\text{pos}}(\mathcal{Q})$ is dense in H_{pos} .

The induced map $\mathcal{Q}/\mathcal{K}_{\text{pos}} \rightarrow H_{\text{pos}}$ is injective with dense image and becomes an isometry when its source is equipped with the explicit positive norm $\|E_{\text{pos}}F\|$. Its completion for that norm is exactly H_{pos} . This specifies the norm and its completion, not an assertion that the original Fréchet quotient topology is that norm topology.

At the level of each original full multiplicity block, the map is exactly

$$\mathbb{C}[t]/(t^{m_\rho}) \longrightarrow H_{\text{pos}}, \quad \sum_{j=0}^{m_\rho-1} a_j t^j \mapsto \begin{cases} a_0 \varepsilon_\rho, & \rho \in \mathcal{Z}_{\text{line}}, \\ 0, & \rho \in \mathcal{Z}_{\text{off}}. \end{cases} \quad (\text{PTQ6.3})$$

This uses the actual globally realized finite block from RD, not an assertion that every unrestricted infinite jet pattern is in $\mathcal{Q}$. Thus its local kernel has dimension $m_\rho - 1$ on a retained block and m_ρ on a discarded block. All those classes remain in the original quotient; (PTQ6.3) states exactly what this receiver observes. In the trace below, a retained value still has weight m_ρ ; its one-dimensional observed value does not replace that original multiplicity by one.

Every T_n, U_n on the full source commutes with (PTQ6.1) and its composite, so the quotient preserves their exact degree factors. No operator lifting P_{line} back to $\mathcal{Q}$, and no arbitrary subset interpolation in its entire source, is asserted. The maps here are the actual defined composites into the Hilbert receiver.

PTQ7. Extension-class source maps and the original residue receiver

ECR constructs $\mathcal{C} = M/\mathfrak{a}$, the actual cyclic original extension-class module, and the injective maps

$$b_t : \mathcal{C} \rightarrow \mathcal{Q}, \quad [h] \mapsto [e^{ts^2}h], \quad t > 0. \quad (\text{PTQ7.1})$$

Their composites with this universal receiver are

$$E_{\text{pos},t} = q_{\text{pos}} E b_t, \quad \ker E_{\text{pos},t} = \{h \in M : h(\rho) = 0 \ \forall \rho \in \mathcal{Z}_{\text{line}}\} / \mathfrak{a}. \quad (\text{PTQ7.2})$$

The kernel follows because $e^{t\rho^2} \neq 0$ at every actual zero. This is an algebraic domain with an explicit map to a Hilbert space, not an unspecified Hilbert topology on the ambient Ext group. Its full multiplicity maps still include ECR's invertible triangular Gaussian matrices before the value observation (PTQ6.3). Their original class and higher jets are not altered by the construction of this new receiver.

RTT supplies the injective continuous anti-linear map

$$\mathbf{A} : H \rightarrow \mathcal{H}_{\text{res}}, \quad (\hat{\mathbf{A}}y)(F) = \langle EF, Jy \rangle_+, \quad \mathbf{S} = \mathbf{A}E. \quad (\text{PTQ7.3})$$

Define on the complete original source

$$\mathbf{S}_{\text{line}} = \mathbf{A}P_{\text{line}}E, \quad \mathbf{S}_{\text{off}} = \mathbf{A}P_{\text{off}}E. \quad (\text{PTQ7.4})$$

Both are continuous anti-linear maps to the already constructed strong residue completion, and

$$\begin{aligned} \mathbf{S} &= \mathbf{S}_{\text{line}} + \mathbf{S}_{\text{off}}, \\ \widehat{\mathcal{R}}(F, \mathbf{S}_{\text{line}}G) &= \sum_{\rho \in \mathcal{Z}_{\text{line}}} m_\rho F(\rho) \overline{G(\rho)}, \\ \widehat{\mathcal{R}}(F, \mathbf{S}_{\text{off}}G) &= \sum_{\rho \in \mathcal{Z}_{\text{off}}} m_\rho F(\rho) \overline{G(\rho^\#)}. \end{aligned} \quad (\text{PTQ7.5})$$

Absolute convergence follows from the original H estimates. Injectivity of $\mathbf{A}$ proves $\ker \mathbf{S}_{\text{line}} = \mathcal{K}_{\text{pos}}$, and proves the corresponding off-line value-zero kernel for $\mathbf{S}_{\text{off}}$. Their intersection is exactly $\mathcal{N}_{\text{jet}}$. These formulas construct actual residue maps for both parts of (PTQ5.1); they do not assert a continuous projection on the entirety of $\mathcal{H}_{\text{res}}$, or an inverse to $\mathbf{A}$ with an unproved subspace topology.

The original source Weil form descends to $\mathcal{Q}/\mathcal{K}_{\text{pos}}$ exactly when $\mathcal{K}_{\text{pos}} \subset \mathcal{N}_{\text{jet}}$, since RTT proves that $\mathcal{N}_{\text{jet}}$ is its full radical. The actual off-line isolator, when there is one, violates this containment. Thus the source version has the same exact distinction between a compression and descent as PTQ5, with every higher-jet class accounted for.

PTQ8. Multiplicity trace of the retained and discarded sectors

Human-source attribution: the unconditional zero-counting estimate is the classical Riemann–von Mangoldt theorem. The inspected native-TeX witness is Alain Connes, The Riemann Hypothesis: Past, Present and a Letter Through Time, the Hardy–Littlewood discussion immediately before “Zero-free regions and zero-density estimates” (author TeX line499). It states the asymptotic that implies the bound used here; this is not a new zero-counting theorem.

The required trace is the original multiplicity-weighted diagonal trace, not the ordinary Hilbert operator trace after rescaling an orthonormal basis:

$$\text{tr}_\zeta(M_f) = \sum_\rho m_\rho f(\rho). \quad (\text{PTQ8.1})$$

For nonnegative diagonal functions it is allowed to take the value $+\infty$; for complex functions it is used when the absolute weighted sum converges. Its multiplicity is the trace of multiplication on the full local algebra

$\mathbb{C}[t]/t^{m_\rho}$, whose diagonal entries are all $f(\rho)$, and is also the residue of $f\zeta'/\zeta$ for a holomorphic local germ. This explains the factor m_ρ without replacing the entire jet algebra by a single trace coordinate.

For every nonnegative diagonal observable,

$$\begin{aligned}\mathrm{tr}_\zeta(M_f) &= \mathrm{tr}_\zeta(P_{\mathrm{line}}M_f) + \mathrm{tr}_\zeta(P_{\mathrm{off}}M_f), \\ \mathrm{tr}_\zeta(P_{\mathrm{line}}M_f) &= \sum_{\rho \in \mathcal{Z}_{\mathrm{line}}} m_\rho f(\rho), \quad \mathrm{tr}_\zeta(P_{\mathrm{off}}M_f) = \sum_{\rho \in \mathcal{Z}_{\mathrm{off}}} m_\rho f(\rho).\end{aligned}\tag{PTQ8.2}$$

The same identity holds for absolutely summable complex observables. In particular for every $t > 0$, the strictly positive weight $k_t(\rho) = e^{-t(\Im \rho)^2}$ gives finite quantities on both sectors. The unconditional count $\sum_{|\Im \rho| \leq R} m_\rho = O(R \log(R+2))$ bounds the complete sum by a convergent sum of a polynomial-logarithmic factor times e^{-tj^2} . Thus the discarded trace is the exact nonnegative quantity

$$\mathrm{tr}_\zeta(P_{\mathrm{off}}M_{k_t}) = \sum_{\rho \in \mathcal{Z}_{\mathrm{off}}} m_\rho e^{-t\gamma^2} = \sum_{\{\rho, \rho^\#\} \subset \mathcal{Z}_{\mathrm{off}}} 2m_\rho e^{-t\gamma^2}.\tag{PTQ8.3}$$

The second sum is over the actual two-element reflection orbits, each once. It is zero exactly when that actual sector is empty, because every weight is strictly positive. Its value has not been assumed zero.

Every positive-adjoint defect is entirely in the discarded sector:

$$\begin{aligned}P_{\mathrm{line}}D_n^*D_n &= 0, \quad P_{\mathrm{off}}D_n^*D_n = D_n^*D_n. \\ \mathrm{tr}_\zeta(M_{k_t}D_n^*D_n) &= \sum_{\rho \in \mathcal{Z}_{\mathrm{off}}} m_\rho e^{-t\gamma^2} (n^\sigma - n^{1-\sigma})^2.\end{aligned}\tag{PTQ8.4}$$

The sum converges because the squared scalar is bounded by $(n-1)^2$. Within each two-point orbit the two terms are equal and add. The zero weighted trace of the off-diagonal exchange matrix J on such a two-dimensional orbit does not remove this positive trace or its two value directions: its matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, while the positive trace of the identity on that orbit, with the original multiplicity, is $2m_\rho$. The two signs in (PTQ5.4) therefore cannot be discarded merely because the exchange matrix has zero trace.

The functions k_t and the sector indicators are value observables. No assertion here inserts these nonholomorphic functions into the original holomorphic prime explicit formula. The complete original arithmetic of the removed form is a separate source comparison, for which the exact removed term is already (PTQ7.5).

PTQ9. The full original-zeta formula retains the discarded term

Human-source attribution: the explicit formula is due to A. P. Guinand, *A summation formula in the theory of prime numbers*, Proc. London Math. Soc. (2)50 (1949), 107–119, and André Weil, *Sur les formules explicites de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund (1952), 252–265. The inspected native-TeX witness is Alain Connes, arXiv:2602.04022v1, “Riemann’s formula, von Mangoldt paper”, equations `mellin`, `bombieriexplicit`, `bombieriexplicit1`, `bombieriexplicit2` (lines 463–482). The historical originals are credited through that witness; no new reading of those originals is claimed. The full logarithmic-coordinate transport and admissible-test estimates used in this programme remain in GIQ9 and the earlier complete trace derivation.

For original source classes F, G , the sum of the two residue values in (PTQ7.5) is exactly the original Weil value, not a replacement of it. With the complete test

$$\begin{aligned}A_{F,G}(s) &= F(s)\overline{G(1-\bar{s})}, \quad A(s) = \int_{\mathbb{R}} h(v)e^{-(s-1/2)v} dv, \\ \hat{h}(t) &= \int_{\mathbb{R}} h(v)e^{-itv} dv,\end{aligned}\tag{PTQ9.1}$$

RTT9/GIQ9 prove

$$\widehat{\mathcal{R}}(F, S_{\text{line}}G) + \widehat{\mathcal{R}}(F, S_{\text{off}}G) = A(0) + A(1) + A_{\infty}(h) - P_{\text{hist}}(h), \quad (\text{PTQ9.2})$$

where the retained complete terms are

$$A_{\infty}(h) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt,$$

$$P_{\text{hist}}(h) = \sum_{n \geq 2} \frac{\log L_n - \log L_{n-1}}{\sqrt{n}} (h(\log n) + h(-\log n)), \quad L_n = \text{lcm}(1, \dots, n). \quad (\text{PTQ9.3})$$

The exact identity is merely received through the proved splitting of its left side; no separate prime formula for a projected test is asserted. At every finite trivial-zero cutoff B , its raw divisor still satisfies

$$V_{\zeta, B} = W(EF, EG) + \sum_{r=1}^B A(-2r) - A(1), \quad G_B = A_{\infty}(h) + A(0) + \sum_{r=1}^B A(-2r), \quad V_{\zeta, B} = G_B - P_{\text{hist}}(h). \quad (\text{PTQ9.4})$$

The original unit, full prime-power repetitions, Gamma term, endpoints, original pole and finite trivial-divisor contribution are consequently unchanged. The new positive compression alone equals the right side of (PTQ9.2) only after subtracting the displayed full off-line term; omitting that term would be a different identity.

PTQ10. Relation to the full-jet and extension-class representations

The universal theorem is stated for bounded intertwiners from the actual value Hilbert representation. It does not assert that every algebraic map from the entire rapid-strip quotient, or every map from the untologized extension-class module, is bounded for that value norm. The full source maps used here are the constructed E , b_t , S and A , with their exact kernels given in PTQ6–PTQ7. In particular a nilpotent jet is already invisible to the value Hilbert representation; it remains an actual nonzero source class when its multiplicity permits it. The kernel of the new positive receiver adds every off-line value block to that known kernel.

On the normal extension-class module, ECI's exact translation V and inverse U carry a zero ρ to $\rho + 1$, with the full scalar n in $D_n^+ V = nV D_n$. Its normal involution is around 3, and its pairing degree is n^3 , while its actual geometric normal transfer still has product degree n as ECI8 proves. The present quotient concerns the original degree-one representation (PTQ0.2). It must not be transported to a different normal-adjoint assertion by deleting those degree factors. Nothing in the proof identifies the original coefficient quotient with the normal extension's full ideal or with its cyclic class module.

PTQ11. Established outcome and exact remaining source question

The quotient of the actual positive-adjoint defect has now been constructed and proved universal in the stated bounded Hilbert category. Its kernel is exactly the closed off-line coordinate space; its retained operators satisfy the original geometric degree and every integer composition law, and every bounded positive-adjoint receiver factors through it uniquely. Its complete bounded positive forms are (PTQ4.3), with the original multiplicities retained.

The original Weil form is carried by the sum of its positive compression and the explicit discarded indefinite form. This quotient therefore does not prove the original form positive by deleting its negative directions: it discards both signs in every off-line pair, and the original arithmetic identity retains their whole contribution. The actual source question is now precise: construct the relevant source map to this universal receiver while accounting for the full kernel and the removed original-zeta term. This note supplies all receiving maps and kernels without presuming an entire source projector; that source-level construction is the complementary calculation being pursued.

Proof sources: GTR0–GTR11 for the actual geometric transfer and degree; GTAH0–GTAH6 for its positive-adjoint defect; RTT0–RTT11 for the complete quotient, full-jet isolators, value/radical map and strong

residue receiver; ECR0–ECR7 for the Gaussian source maps of the actual cyclic extension module; ECI0–ECI13 for its exact involutions and normal degree comparisons. These are the existing original-zeta programme objects and their completed proofs. This derivation neither asserts an off-line zero nor assumes their absence, and supplies no numerical interval evidence or new publication claim.

The positive transfer quotient, its full source, and the retained arithmetic defect

24 September 2026. Complete derivation PSC0–PSC9.

PSC0. Actual prerequisites and the attempted construction

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Arithmetic and coefficient operations here follow the complete-history reconstruction; no addition, numerical value, parity, midpoint or coordinate is assigned to the support. The two branch histories remain separate. The corpus corrections concerning source maps, quotienting, retained counting data and admissibility were reread before this construction, including the complete direct global-quotient argument. The current workflow expressly requests the canonical quotient of a calculated pairing defect and its induced trace and transfer. This note carries out that step and computes the data needed to reconstruct the original source.

Use the proved entire strip-polynomial multiplier ring M , its rapid-strip ideal $\mathcal{B}$, full original-zero ideal $\mathcal{J}$, and quotient $\mathcal{Q} = \mathcal{B}/\mathcal{J}$. GTAH0–GTAH1 and RTT5–RTT6 give

$$E : \mathcal{Q} \longrightarrow H = \ell^2(\mathcal{Z}, m_\rho), \quad E[F]_\rho = F(\rho), \quad W(x, y) = \langle x, Jy \rangle, \quad (Jx)_\rho = x_{\rho^\#}, \quad \rho^\# = 1 - \bar{\rho}. \quad (\text{PSC0.1})$$

The inner product is linear in its first argument. Every multiplicity m_ρ is retained. The actual global full-jet isolators $e_{\rho, j} \in \mathcal{Q}$ are given by RD1. No density of finite-support classes in the original Fréchet topology is assumed. Finite value vectors are in $E\mathcal{Q}$, which is dense in H .

For $a > 0$, the coefficient operators are $T_a x_\rho = a^\rho x_\rho$ and $U_a = aT_{1/a}$. For a recovered positive integer $a = n$, GTR constructs the actual degree- n cover and its transfer; no nonintegral cover is asserted. GTAH proves

$$D_a = T_a^* - U_a, \quad (D_a x)_\rho = e^{-i\Im \rho \log a} (a^{\Re \rho} - a^{1-\Re \rho}) x_\rho. \quad (\text{PSC0.2})$$

RTT supplies the continuous injective anti-linear map $A : H \rightarrow \mathcal{H}_{\text{res}}$, with

$$(\hat{A}y)(F) = \langle EF, Jy \rangle, \quad S = AE. \quad (\text{PSC0.3})$$

ECR and ECI construct the actual cyclic extension-class maps to these spaces. Their full original factors and degree actions are retained below.

PSC1. The canonical projector is constructed from the measured defect

Partition the actual divisor, without asserting either part empty, as

$$\mathcal{Z}_{\text{line}} = \{\rho \in \mathcal{Z} : \Re \rho = 1/2\}, \quad \mathcal{Z}_{\text{off}} = \{\rho \in \mathcal{Z} : \Re \rho \neq 1/2\}. \quad (\text{PSC1.1})$$

Both subsets are preserved by conjugation and $\#$, with their original multiplicities. Let $H_{\text{line}}, H_{\text{off}}$ be the corresponding closed coordinate subspaces and $P_{\text{line}}, P_{\text{off}}$ their orthogonal projections. These are observations on the complete arithmetic receiver, not alternative definitions of its original divisor.

For any fixed recovered $a > 1$, put $B_a = D_a^* D_a$. This is bounded and positive, with exact diagonal coefficient

$$b_a(\rho) = (a^{\Re \rho} - a^{1-\Re \rho})^2. \quad (\text{PSC1.2})$$

Its zero coefficients are exactly the line subset. Thus

$$\ker B_a = H_{\text{line}}, \quad \overline{\text{ran } B_a} = H_{\text{off}}, \quad P_{\text{line}} = \text{s-lim}_{R \rightarrow +\infty} e^{-RB_a}. \quad (\text{PSC1.3})$$

For the range assertion, every finite vector supported off the line is an image: divide its finitely many nonzero coordinates by their strictly positive $b_a(\rho)$. Such vectors are dense in H_{off} . Conversely every image

is supported there. For the strong limit, the exponential is the norm-convergent power series of the bounded operator, hence has coefficient $e^{-Rb_a(\rho)}$. For each $x \in H$,

$$\|(e^{-RB_a} - P_{\text{line}})x\|^2 = \sum_{\rho \in \mathcal{Z}_{\text{off}}} m_\rho e^{-2Rb_a(\rho)} |x_\rho|^2 \longrightarrow 0. \quad (\text{PSC1.4})$$

Dominated convergence applies with the actual summable majorant $m_\rho |x_\rho|^2$. This constructs the projector globally, without a numerical zero-height cutoff, a spectral gap, or simplicity. The outcome is independent of the chosen recovered $a > 1$, since its diagonal zero set is the same.

All T_a, U_a, J commute with these two projections. On H_{line} , direct substitution gives

$$(T_a|_{H_{\text{line}}})^* = U_a|_{H_{\text{line}}}, \quad U_a T_a = aI, \quad W|_{H_{\text{line}}} = \langle \cdot, \cdot \rangle. \quad (\text{PSC1.5})$$

The complete degree factor remains a . We have constructed the positive receiver on the quotient $H/H_{\text{off}} \simeq H_{\text{line}}$, together with its section as the orthogonal subspace of H . The quotient does not erase the retained complementary space.

PSC2. The full source quotient and its higher jets

Human-source attribution: the topological inverse used here is the classical open mapping theorem for Fréchet spaces. See Terence Tao, 245B, Notes11: The strong and weak topologies, §1, Remark1 (21 February2009), explicitly recording the Fréchet extension, and Notes9, Theorem3 for the Banach proof. The present proof checks the closedness and completeness required for its application; the cited Remark records the extension rather than supplying a full Fréchet proof.

For either subset $T = \mathcal{Z}_{\text{line}}$ or $T = \mathcal{Z}_{\text{off}}$, define actual closed M -submodules

$$\mathcal{J}_T^{\text{jet}} = \{F \in \mathcal{B} : \text{ord}_\rho F \geq m_\rho \ (\rho \in T)\}, \quad \mathcal{J}_T^{\text{val}} = \{F \in \mathcal{B} : F(\rho) = 0 \ (\rho \in T)\}. \quad (\text{PSC2.1})$$

Closedness follows by intersecting the kernels of the continuous derivative evaluations. Their continuity follows from Cauchy's estimate on a fixed circle around each specified point, inside a bounded real strip. Multiplication by each $h \in M$ preserves these kernels and is continuous on $\mathcal{B}$, by its polynomial strip bounds and the defining seminorms of $\mathcal{B}$. Put

$$Q_T^{\text{jet}} = \mathcal{B}/\mathcal{J}_T^{\text{jet}}, \quad Q_T^{\text{val}} = \mathcal{B}/\mathcal{J}_T^{\text{val}}. \quad (\text{PSC2.2})$$

These are Fréchet quotient spaces, and the exact sequence

$$0 \longrightarrow \mathcal{J}_T^{\text{val}}/\mathcal{J}_T^{\text{jet}} \longrightarrow Q_T^{\text{jet}} \longrightarrow Q_T^{\text{val}} \longrightarrow 0 \quad (\text{PSC2.3})$$

retains every higher jet as its stated kernel. It is a strict sequence by the open mapping theorem applied to the displayed Fréchet quotients.

The original $\mathcal{J}$ is contained in both ideals of (PSC2.1). Hence the original $\mathcal{Q}$ maps onto both spaces. The precise kernel of the map to the positive value receiver is

$$\ker(P_{\text{line}}E) = \mathcal{J}_{\mathcal{Z}_{\text{line}}}^{\text{val}}/\mathcal{J}. \quad (\text{PSC2.4})$$

The induced map $Q_{\mathcal{Z}_{\text{line}}}^{\text{val}} \rightarrow H_{\text{line}}$ is continuous, injective and has dense image, because every finite line-value vector has an original full-jet isolator representative. Its norm completion is exactly H_{line} . No assertion that the Fréchet topology equals the value norm topology is made. The analogous statements hold for the off-line subset.

The induced source operations T_a, U_a, L , conjugation and $K_1 F(s) = \overline{F(1-\bar{s})}$ are well-defined on each of the jet and value quotients; their defining ideals are invariant. Every original multiplicity is retained in the value norm, even after the nonconstant jet coordinates pass into (PSC2.3).

PSC3. Exact reconstruction of the full source from the two observations

Write $J_L = \mathcal{J}_{\mathcal{Z}_{\text{line}}}^{\text{jet}}$, $J_O = \mathcal{J}_{\mathcal{Z}_{\text{off}}}^{\text{jet}}$. Then

$$J_L \cap J_O = \mathcal{J}. \quad (\text{PSC3.1})$$

Retain the actual reconstruction defect module

$$C_{\text{rec}} = \mathcal{B}/(J_L + J_O). \quad (\text{PSC3.2})$$

This is an algebraic quotient equipped, when used topologically, with its quotient locally convex topology; no closedness of the sum or Hausdorff property is asserted. There is a completely explicit exact sequence of M -modules

$$0 \longrightarrow \mathcal{Q} \xrightarrow{\Delta} Q_{\mathcal{Z}_{\text{line}}}^{\text{jet}} \oplus Q_{\mathcal{Z}_{\text{off}}}^{\text{jet}} \xrightarrow{d} C_{\text{rec}} \longrightarrow 0, \quad (\text{PSC3.3})$$

where $\Delta[F] = ([F]_{J_L}, [F]_{J_O})$ and $d([f], [g]) = [f - g]$. Independence of representatives follows from the denominator of (PSC3.2); surjectivity follows by using $([f], 0)$. Injectivity of Δ is (PSC3.1). If $f - g = l + o$, with $l \in J_L, o \in J_O$, then $F = f - l = g + o$ is an original representative mapping to the given pair. This proves exactness at the middle and the complete inverse on compatible pairs.

Thus the original full-jet source is exactly the algebraic fibre product over C_{rec} , with explicit inverse. The maps in (PSC3.3) are continuous. The inverse onto the image of Δ is continuous exactly when $J_L + J_O$ is closed. Indeed closedness makes C_{rec} Fréchet and $\ker d$ a closed Fréchet subspace of the product, so the open mapping theorem applies to the bijection $\Delta : \mathcal{Q} \rightarrow \ker d$. Conversely, a continuous inverse makes its image complete in the induced Hausdorff locally convex topology, hence closed. The preimage of that closed image under $f \mapsto ([f]_{J_L}, 0)$ is precisely $J_L + J_O$, proving closedness.

The exact closure of the image is also calculable:

$$\overline{\Delta(\mathcal{Q})} = \ker(Q_{\mathcal{Z}_{\text{line}}}^{\text{jet}} \oplus Q_{\mathcal{Z}_{\text{off}}}^{\text{jet}} \longrightarrow \mathcal{B}/\overline{J_L + J_O}), \quad (\text{PSC3.3a})$$

where the map is again the difference. Containment follows by continuity. For the reverse, write $f - g = \lim(l_j + o_j)$ in the Fréchet space, with $l_j \in J_L, o_j \in J_O$. Then $F_j = f - l_j$ has first observation exactly $[f]_{J_L}$, and its second tends to $[g]_{J_O}$, since $F_j - g - o_j \rightarrow 0$. This proves the formula. No arbitrary-subset interpolation or closedness assumption is made. The kernel discarded by the positive jet quotient has its further exact description

$$0 \longrightarrow J_L/\mathcal{J} \xrightarrow{[F] \mapsto [F]_{J_O}} Q_{\mathcal{Z}_{\text{off}}}^{\text{jet}} \longrightarrow C_{\text{rec}} \longrightarrow 0. \quad (\text{PSC3.4})$$

Its image is $(J_L + J_O)/J_O$, which proves the assertion. The same construction with value ideals has intersection $\mathcal{J}_{\mathcal{Z}}^{\text{val}}$, retaining the separate passage from the original jets to the value observation.

These formulas preserve both source observations and their compatibility data. They do not replace the full source by a freely chosen direct sum. If either divisor subset is empty, its ideal is $\mathcal{B}$, and the formulas correctly give the corresponding zero quotient and zero reconstruction defect.

PSC4. Original and compressed residue maps

On the original source define actual maps to the already constructed strong residue space by

$$S_L = AP_{\text{line}}E, \quad S_O = AP_{\text{off}}E, \quad S = S_L + S_O. \quad (\text{PSC4.1})$$

They are continuous and anti-linear, and their kernels are exactly the respective value ideals modulo $\mathcal{J}$, by injectivity of A . Evaluation against $F \in \mathcal{Q}$ gives

$$\widehat{\mathcal{R}}(F, S_T G) = W_T(F, G) = \sum_{\rho \in T} m_{\rho} F(\rho) \overline{G(\rho^{\#})}. \quad (\text{PSC4.2})$$

The sum converges absolutely by Cauchy–Schwarz and is bounded on each bounded test set by the corresponding full E -norm bound. Both subsets are $\#$ -stable. Consequently

$$W = W_L + W_O, \quad W_L(F, F) = \sum_{\rho \in \mathcal{Z}_{\text{line}}} m_{\rho} |F(\rho)|^2 \geq 0, \quad (\text{PSC4.3})$$

and all cross pairings between their Hilbert subspaces vanish.

For every actual off-line orbit $\{\rho, \rho^\#\}$, the original isolators give

$$W_O(e_{\rho,0} - e_{\rho^\#,0}, e_{\rho,0} - e_{\rho^\#,0}) = -2m_\rho, \quad W_O(e_{\rho,0}, e_{\rho^\#,0}) = m_\rho. \quad (\text{PSC4.4})$$

These are statements about each orbit in the actual set, not a claim that such an orbit exists. They prove that quotienting by H_{off} is a compression of the original Weil form, not its descent unless the discarded subspace is zero. More generally, the radical of W on H is zero: $W(x, Jx) = \|x\|^2$. A sesquilinear form descends through a quotient exactly when its kernel pairs to zero with every vector, as follows by changing either representative. This gives the exact descent test without identifying the quotient's new positive form with the old form.

The projector construction in PSC1 yields an actual residue-space limit

$$Ae^{-RB_a}EG \longrightarrow S_L G \quad \text{in the strong residue topology for every } G \in \mathcal{Q}. \quad (\text{PSC4.5})$$

Indeed (PSC1.4) is norm convergence in H , and A is continuous into that topology. Finite original isolators also give explicit source representatives for each value approximation:

$$G_{R,E_0} = \sum_{\rho \in E_0} e^{-Rb_a(\rho)} G(\rho) e_{\rho,0}, \quad E_0 \subset \mathcal{Z} \text{ finite}. \quad (\text{PSC4.6})$$

Their value vectors converge to $e^{-RB_a}EG$ in H , so their actual residue images converge to the left side of (PSC4.5). No convergence in the original full-jet source topology is asserted, and the omitted higher jets remain exactly (PSC2.3). This is a convergent global construction, not a bounded numerical test.

PSC5. The original arithmetic formula and the exact discarded contribution

Human-source attribution: the explicit formula is due to A. P. Guinand, *A summation formula in the theory of prime numbers*, Proc. London Math. Soc. (2)50 (1949), 107–119, and André Weil, *Sur les formules explicites de la théorie des nombres premiers*, Comm. Sémin. Math. Univ. Lund (1952), 252–265. The inspected native-TeX witness is Alain Connes, arXiv:2602.04022v1, “Riemann’s formula, von Mangoldt paper”, equations `mellin`, `bombieriexplicit`, `bombieriexplicit1`, `bombieriexplicit2` (lines 463–482). The historical originals are credited through that witness; no new reading of those originals is claimed. The full logarithmic-coordinate transport and admissible-test estimates used in this programme remain in GIQ9 and the earlier complete trace derivation.

Take actual $F, G \in \mathcal{B}$ and retain

$$A_{F,G}(s) = F(s) \overline{G(1-\bar{s})}, \quad A(s) = \int_{\mathbb{R}} v(u) e^{-(s-1/2)u} du, \quad \hat{v}(y) = A(1/2 + iy). \quad (\text{PSC5.1})$$

GIQ9 proves that the inverse test v is in the complete admissible logarithmic test space, by the original inverse-Mellin, reflection and convolution estimates. The full original formula is

$$W_L(F, G) + W_O(F, G) = A(0) + A(1) + \mathcal{A}_\infty(v) - P_{\text{hist}}(v), \quad (\text{PSC5.2})$$

where no term is changed:

$$\begin{aligned} \mathcal{A}_\infty(v) &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{v}(y) \left(\Re \frac{\Gamma'(1/4 + iy/2)}{\Gamma(1/4 + iy/2)} - \log \pi \right) dy, \\ P_{\text{hist}}(v) &= \sum_{n \geq 2} \frac{\log L_n - \log L_{n-1}}{\sqrt{n}} (v(\log n) + v(-\log n)), \quad L_n = \text{lcm}(1, \dots, n). \end{aligned} \quad (\text{PSC5.3})$$

Thus the positive compression has the exact arithmetic expression

$$W_L(F, G) = A(0) + A(1) + \mathcal{A}_\infty(v) - P_{\text{hist}}(v) - W_O(F, G). \quad (\text{PSC5.4})$$

The discarded term is explicitly the complete off-line sum, not an endpoint, completion multiplier, or adjustment to the prime counter. In particular both endpoints remain $A(0) = F(0)\overline{G(1)}$ and $A(1) = F(1)\overline{G(0)}$.

At every finite trivial-divisor cutoff $b \geq 1$, preserve

$$V_{\zeta,b} = W_L + W_O + \sum_{r=1}^b A(-2r) - A(1), \quad G_b = \mathcal{A}_\infty(v) + A(0) + \sum_{r=1}^b A(-2r),$$

$$V_{\zeta,b} = G_b - P_{\text{hist}}(v), \quad V_{L,b} = G_b - P_{\text{hist}}(v) - W_O, \quad (\text{PSC5.5})$$

where $V_{L,b} = W_L + \sum_{r=1}^b A(-2r) - A(1)$. Here $A(-2r) = F(-2r)\overline{G(1+2r)}$. No divergent infinite trivial-zero sum or new zeta function is introduced.

The exact morphism back to the original Weil functional is addition of the retained functional W_O , as proved term by term in (PSC5.2). Formula (PSC4.4) proves that this functional is zero on all original pairs exactly when the actual off-line divisor is empty. This calculates the precise information needed to reconstruct the original trace after positive compression.

PSC6. Exact transport from the original extension-class generator

ECR1 and ECI12 give $\mathcal{C} = M/\mathfrak{a} \simeq Me_0$, $g_t(s) = e^{ts^2}$, and $b_t[h] = [g_th]$. Define

$$S_{L,t} = AP_{\text{line}}Eb_t, \quad S_{O,t} = AP_{\text{off}}Eb_t, \quad Sb_t = S_{L,t} + S_{O,t}. \quad (\text{PSC6.1})$$

These are algebraic-domain anti-linear maps to the strong residue space; no topology on the ambient Ext group is assumed. Every required source estimate is already proved in ECR1–ECR3. For T equal to either subset, the resulting exact trace is

$$W_{T,t}(h, k) = \sum_{\rho \in T} m_\rho e^{t\{\rho^2 + (1-\rho)^2\}} h(\rho) \overline{k(\rho^\#)}. \quad (\text{PSC6.2})$$

Its original source test is the entire function $e^{t\{s^2 + (1-s)^2\}} h(s) \overline{k(1-\bar{s})}$; substituting it into PSC5 retains every original arithmetic term. On the line its coefficient is $e^{2t(1/4 - (\Im \rho)^2)}$, which follows by direct substitution without dropping the factor $e^{t/2}$.

The measured defect is entirely retained on the complementary value receiver:

$$\|D_a Eb_t(h)\|^2 = \sum_{\rho \in \mathcal{Z}_{\text{off}}} m_\rho e^{2t\{(\Re \rho)^2 - (\Im \rho)^2\}} b_a(\rho) |h(\rho)|^2. \quad (\text{PSC6.3})$$

Every coefficient is the one proved in ECR5, and the line terms are zero. For $h = 1$, this is the original actual extension generator. The separator c of FOD satisfies $[c] = 1$ in $\mathcal{C}$, so it fixes both terms of (PSC6.1) and (PSC6.3). Its annihilation of the distinct normal class is transported by $Uc = c(s+1)$, as proved in ECI9. On the original value space it acts as the identity on both pieces, supplying no separation between them. Its induced value action equals P_{line} exactly when $H_{\text{off}} = 0$, since $I - P_{\text{line}} = P_{\text{off}}$. This proves the precise comparison without presupposing an off-line zero.

PSC7. The exact space of full-jet holomorphic projectors

Define multiplier ideals $\mathfrak{a}_T \subset M$ by full order m_ρ at every $\rho \in T$. Then $\mathfrak{a}_{\text{line}} \cap \mathfrak{a}_{\text{off}} = \mathfrak{a}$. The space of multipliers realizing the full-jet line projector is precisely

$$\mathcal{P} = \{p \in M : p \in \mathfrak{a}_{\text{off}}, 1-p \in \mathfrak{a}_{\text{line}}\}. \quad (\text{PSC7.1})$$

This is an explicitly defined space of coefficient maps to investigate, not an assumed source splitting. Its elements are exactly the solutions to $p + (1-p) = 1$ with the two indicated ideal memberships. Therefore it is nonempty exactly when $\mathfrak{a}_{\text{line}} + \mathfrak{a}_{\text{off}} = M$, and any two of its elements differ by $\mathfrak{a}$. Each represents the same

idempotent in $M/\mathfrak{a}$, since $p^2 - p$ vanishes to full order on both subsets. These statements follow directly from the displayed ideal equalities.

Every $p \in \mathcal{P}$ constructs the full source splitting, with inverse

$$([f]_{J_L}, [g]_{J_O}) \mapsto [pf + (1-p)g]_{\mathcal{J}}. \quad (\text{PSC7.2})$$

It is independent of representatives because $pJ_L \subset \mathcal{J}$ and $(1-p)J_O \subset \mathcal{J}$. Its two compositions are identities by the full-jet conditions, and it is continuous because multiplication is continuous. This proves that its existence forces $C_{\text{rec}} = 0$, with the actual splitting exhibited. Neither $\mathcal{P} \neq \emptyset$ nor $C_{\text{rec}} = 0$ is assumed here.

For each T , $\mathcal{J}_T^{\text{jet}} = \mathcal{B} \cap \mathfrak{a}_T$. Thus the maps $Q_T^{\text{jet}} \rightarrow M/\mathfrak{a}_T$ induced by inclusion and $M/\mathfrak{a}_T \rightarrow Q_T^{\text{jet}}$ induced by multiplication by g_t are injective, with both composites multiplying by g_t . The proof is ECR1's strip bound and local-unit argument at the specified subset. This proves the relation between the source and multiplier versions of the reconstruction problem, without declaring their defect quotients equal.

PSC8. A global estimate forced by a holomorphic projector

Every actual $p \in \mathcal{P}$ has constants $C > 0$ and an integer $N \geq 0$, depending on that entire multiplier, such that for every $\rho \in \mathcal{Z}_{\text{line}}$ and $\eta \in \mathcal{Z}_{\text{off}}$ with $|\rho - \eta| \leq 1$,

$$1 \leq C(1 + |\mathfrak{J}\rho|)^N |\rho - \eta|. \quad (\text{PSC8.1})$$

Proof: on $|\Re s| \leq 3$, polynomial strip growth bounds $|p(s)|$ by $C_0(1 + |\mathfrak{J}s|)^N$. Every unit-radius circle centered on the segment from ρ to η stays in this strip, and its imaginary height differs from $\mathfrak{J}\rho$ by at most two. Cauchy's derivative formula therefore bounds $|p'|$ on the segment by $C(1 + |\mathfrak{J}\rho|)^N$, with C increased by the explicit bounded-height comparison. Integrating p' along the segment and using $p(\rho) = 1, p(\eta) = 0$ proves (PSC8.1). If the two points are farther than one, their separation already exceeds one. The result treats every actual pair; it assumes no existence of a near pair or off-line zero.

Consequently a bounded Hilbert projector alone supplies neither this necessary entire-function estimate nor the full-jet interpolation of PSC7. The exact morphisms relating these constructions are PSC2, PSC3 and PSC7. The heat projector in PSC1 is constructed without this assumption, and its actual residue realization is PSC4.5–PSC4.6. This records the additional structure precisely and retains the reconstruction defect rather than discarding it.

PSC9. Relation to the lifting goal and completed outcome of the attempt

FOD7 and NEA10 prove the specified derived original-quotient-source lifting vanishing while retaining the normal extension. Deligne's actual invariant-cycle cross instead has obstruction image $\text{im } \partial/\partial j(K)$, as established in DC5.3–DC5.4 from the full localization and inertia sequences. Neither this image quotient nor its duality weights is replaced here by a Hilbert quotient.

The present calculation implements the requested next attempt on the exact residue/positivity receiver. It constructs its canonical positive transfer quotient by a global operator limit; supplies its actual source quotient and all higher-jet kernels; reconstructs the full source as an explicit fibre product; and calculates the exact original-zeta arithmetic contribution that the compression discards. The additional full-jet multiplier problem is the concrete space $\mathcal{P}$, with its source map and proved estimate. No arbitrary splitting or lossless normalization is presumed.

The original source and off-line contribution are retained for the next Deligne specialization/duality comparison. Their formulas quantify over the complete actual divisor. They neither posit an alternative arithmetic nor use a numerical sample to prove a global assertion. The measured defect's vanishing remains the active target, and is not inferred from the positive quotient obtained by removing its range.

Proof sources actually used: GTAH0–GTAH7, RTT5–RTT7 and RD1; ECR0–ECR7 and ECI0–ECI13 at their accepted versions; GIQ9's full original explicit formula and source-domain proof; FOD7, NEA10 and DC3.7–DC7 for the exact lifting comparison. Human-source identities and prior reading coverage are retained in the CC and original Deligne ledgers; no new whole-paper reading is claimed by this derivation.

Independent audit of the positive quotient, full source, and arithmetic comparison

24 September 2026. Complete mathematical review of PSC0–PSC9 in POSITIVE_QUOTIENT_SOURCE_AND_ARITHMETIC.md.

Accepted final proof SHA256: 2d188377c6b694e363e5b32a331a8af2ef4562d2cb8aab272d0ce1149b6f1171.

No mathematical repair remains outstanding in this version. The review identified the need to state the precise topological scope of PSC3 and to correct the unconditional separator comparison at the end of PSC6. Root incorporated both corrections. The resulting passages and final hash were read and verified. Missing TeX command escapes in an earlier draft were also repaired before acceptance.

Exact coverage

The complete PSC0–PSC9 text was read initially and then completely reread after the command-escape repair, at SHA256 3dbddd956da6637402626a816fb85989c94e9572fe8d6bf919089cc037d51d08. The final added topology criterion, closure formula PSC3.3a, and corrected PSC6 comparison were subsequently read directly.

The complete FOD0–FOD8, NEA0–NEA10, GTAH0–GTAH7, RTT0–RTT11, GSR0–GSR4, ECR0–ECR7, ECI0–ECI13, and GIQ9 were read in this same independent-review session. Their reading and version records are in EXTENSION_CLASS_SOURCE_INDEPENDENT_AUDIT.md and EXTENSION_CLASS_INVOLUTION_AUDIT.md. The accepted ECR and ECI versions used here are respectively 0a4e004a008422f622beeca7af8babc9a9f6527c0d4eb43571ea6443c0975250 and ba951baf03f04edfeed66f98708e3c3ae5c6b2657b23b569cb108af4316aec31.

Additional direct source coverage for this audit:

- ORIGINAL_ZETA_RESIDUE_DUAL_AND_SUPPORT_MAPS.md, complete RD1, including the full-factor recursive construction of every global isolator; SHA256 4dbde6fb4a959d74d45c203dcc5ae154d1ee74a901f9f130f42cc7b49caea52c.
- ../tau_weight_cohomology_20260924/DELIGNE_INVARIANT_CYCLE_QUOTIENT.md, DC0–DC1, equations DC3.4–DC3.7 with their surrounding argument, DC4.1–DC4.8, and DC5 through the exact quotient calculation DC5.4 and its proof; SHA256 3a9afdaeacd7db0f154348f6487a204852459b08ae8e0f90017e7755fd627521.

This is programme-proof coverage. No new whole-paper reading of the human bibliography or complete rereading of all remote source-image and topology proofs is claimed. The support, separate branch histories and information layers retain the user’s definitions; no arithmetic operation, coordinate or group structure is assigned to that support.

PSC1–PSC2: complete projector and exact source domains

For fixed $a > 1$, $b_a(\rho) = (a^{\Re \rho} - a^{1-\Re \rho})^2$ is bounded and nonnegative, and vanishes exactly at $\Re \rho = 1/2$. Thus the kernel is H_{line} . Every finitely supported vector in H_{off} is in the range of B_a , by division by finitely many strictly positive coefficients; this proves the range closure without a lower bound on all off-line coefficients.

The norm-convergent exponential series has the stated diagonal coefficients. For every $x \in H$, the squared difference in PSC1.4 is dominated by the original summable sequence $m_\rho |x_\rho|^2$, giving strong convergence on the complete divisor. This proves neither operator-norm convergence nor a decay gap, and neither is used. The zero set of b_a , and hence the projector, is independent of the chosen $a > 1$.

The two coordinate subsets are invariant under conjugation and $\#$; consequently J preserves both. On the line $\rho^\# = \rho$, so the original Weil form restricts to the positive value inner product. The coefficients $a^{\bar{\rho}}$ and $a^{1-\rho}$ agree there, proving the adjoint relation with the original degree factor a . Geometric covering interpretations are correctly restricted to recovered positive integers.

The jet and value ideals in PSC2 are closed by the specified continuous evaluations. Their quotients and the kernel in PSC2.3 are Fréchet, so that exact sequence is strict. Factoring the continuous map $P_T E : \mathcal{B} \rightarrow H_T$ through its exact kernel gives the injective continuous map from Q_T^{val} ; the global isolators put every finite value vector in its image. Its induced norm completion is therefore H_T , without equating the original Fréchet and norm topologies. Multipliers, the generator, conjugation and the reflected involution preserve all stated defining ideals.

PSC3: algebraic reconstruction and its exact topology criterion

The intersection $J_L \cap J_O = \mathcal{J}$ proves injectivity of Δ . The difference map is well defined and onto. For a compatible pair with $f - g = l + o$, the formula

$$F = f - l = g + o$$

gives both requested observations. Changing representatives or the decomposition changes F only by the intersection ideal, by the already proved injectivity. This proves the algebraic fibre-product description and its actual inverse. The complementary exact sequence PSC3.4 follows because the kernel of $Q_O^{\text{jet}} \rightarrow C_{\text{rec}}$ is exactly $(J_L + J_O)/J_O$. Its preimage in J_L has kernel $\mathcal{J}$. The empty-subset cases are consistent with all these formulas.

The topology is governed precisely by

$$\Delta^{-1} : \Delta(\mathcal{Q}) \rightarrow \mathcal{Q} \text{ is continuous} \iff J_L + J_O \text{ is closed in } \mathcal{B}. \quad (\text{PSCA1})$$

For the forward direction, the image becomes topologically isomorphic to the complete Fréchet space $\mathcal{Q}$, hence is a complete subspace of a Hausdorff locally convex space and therefore closed. Its preimage under the continuous map $f \mapsto ([f]_{J_L}, 0)$ is exactly $J_L + J_O$. Conversely, closedness of that sum makes C_{rec} Hausdorff Fréchet; $\ker d$ is then a closed Fréchet subspace of the product, and the continuous bijection $\Delta : \mathcal{Q} \rightarrow \ker d$ has continuous inverse by the open mapping theorem.

The map d is itself open for the specified quotient topology, even before closedness of the sum is established. If $\pi = q_L \times q_O$, then

$$d\pi = q_{\text{sum}} \circ (f, g \mapsto f - g).$$

For an open subset U of the product quotient, $\pi^{-1}U$ is open; its image under the open difference map and then the quotient map is $d(U)$, hence open.

The more precise closure calculation in PSC3.3a is also correct. Let d_{sep} be the difference map into $\mathcal{B}/\overline{J_L + J_O}$. Continuity gives $\overline{\Delta(\mathcal{Q})} \subset \ker d_{\text{sep}}$. For the reverse inclusion choose $l_j + o_j \rightarrow f - g$, possible with a sequence because $\mathcal{B}$ is Fréchet. Then $F_j = f - l_j$ has first observation exactly $[f]_{J_L}$ and second observation tending to $[g]_{J_O}$, because

$$F_j - g - o_j = (f - g) - (l_j + o_j) \rightarrow 0.$$

This proves the reverse inclusion and retains the exact topological defect. No closedness or subset interpolation is assumed. PSC's C_{rec} denotes this reconstruction quotient; it is not identified with any earlier coefficient-record algebra.

PSC4–PSC5: residue compression and every arithmetic contribution

Continuity and anti-linearity of $S_T = AP_T E$ follow from the actual maps. Injectivity of A gives the exact value-ideal kernel. The $\#$ -stability of each subset proves PSC4.2 and removes cross terms. For an actual two-point orbit, direct substitution of the isolator values gives the stated $-2m_\rho$ diagonal value and m_ρ cross value. These statements do not assume the existence of such an orbit.

The original form has zero radical on H , because

$$W(x, Jx) = \langle x, J^2 x \rangle = \|x\|^2.$$

It therefore descends to H/H_{off} only when the quotient kernel is zero. The positive form obtained using P_{line} is its specified compression. The strong residue limit in PSC4.5 follows from norm convergence in H and continuity of A . For every fixed R , the finite-isolator value vectors in PSC4.6 converge in H to $e^{-RB_a}EG$; their residue images consequently converge in the stated strong topology. No primal Fréchet convergence or higher-jet reconstruction is inferred from this norm approximation.

GIQ9 applies to the actual product

$$A_{F,G}(s) = F(s)\overline{G(1-\bar{s})}.$$

Thus PSC5.2 retains the original complete zero sum, both endpoints, the factor $1/(2\pi)$, the digamma argument $1/4 + iy/2$, $-\log \pi$, the factors $1/\sqrt{n}$, and every increment $\log L_n - \log L_{n-1}$. Splitting the absolutely convergent zero sum and moving W_O to the other side gives the minus sign in PSC5.4. The test has not been replaced by a subset interpolation.

The finite-cutoff formulas retain the original pole term $-A(1)$ and each $A(-2r) = F(-2r)\overline{G(1+2r)}$. Subtracting W_O gives exactly the stated $V_{L,b}$, with no limiting infinite trivial-zero sum. Adding the retained W_O reconstructs the original trace. Its being identically zero on original pairs is equivalent to an empty off-line subset: necessity is the actual cross-isolator test, and sufficiency is the empty sum.

PSC6: original class maps and the corrected separator comparison

The Gaussian embeddings give actual source functions before either projection is applied. Their full Weil product is $e^{t\{\rho^2+(1-\rho)^2\}}$; on the line it is $e^{2t(1/4-(\Im\rho)^2)}$, including $e^{t/2}$. The positive defect instead has the distinct original weight $e^{2t\{\Re\rho^2-(\Im\rho)^2\}}$. Its line terms vanish because $b_a = 0$ there, giving PSC6.3 without replacing either weight by the other.

The earlier unconditional sentence that c is not the line projector was incorrect when read without an off-line existence assumption. FOD gives $m_c = I$ on the original quotient and hence on its dense value image; the induced value action is I_H . Therefore

$$m_c^H - P_{\text{line}} = P_{\text{off}}, \quad m_c^H = P_{\text{line}} \iff H_{\text{off}} = 0. \quad (\text{PSCA2})$$

The final PSC6 now states this exact comparison and is correct. The analogous full-jet assertion is that the class represented by c realizes PSC7's line projector exactly when the off-line subset is empty: otherwise its value is one at an actual off-line zero where that projector would have value zero. No such zero is assumed. The normal annihilation is still transported by $Uc = c(s+1)$, retaining its distinct scalar action.

PSC7–PSC8: exact multiplier space and separation estimate

The classification of $\mathcal{P}$ is exact. If $1 = l + o$ with $l \in \mathfrak{a}_{\text{line}}$ and $o \in \mathfrak{a}_{\text{off}}$, then $p = o$ has the required full jets; the reverse implication uses $1 = (1-p) + p$. Two solutions differ by the intersection $\mathfrak{a}$, and $p(p-1)$ belongs to that intersection. These are statements about the explicitly defined solution space, not assertions of its nonemptiness.

For such a p , multiplication gives the actual source decomposition

$$F = (1-p)F + pF, \quad (1-p)F \in J_L, \quad pF \in J_O.$$

Thus $J_L + J_O = \mathcal{B}$, and PSC7.2 is the correct continuous inverse to the resulting product map. Its representative independence follows from $pJ_L, (1-p)J_O \subset \mathcal{J}$. The injections between each subset source quotient and $M/\mathfrak{a}_T$ use exactly the Gaussian strip estimate and the local unit at every selected zero; neither requires arbitrary interpolation.

The constant in PSC8 can be made explicit from its stated strip bound. On a unit circle centered at a point of the segment $[\rho, \eta]$, with $|\rho - \eta| \leq 1$,

$$1 + |\Im s| \leq 1 + |\Im \rho| + 2 \leq 3(1 + |\Im \rho|).$$

Cauchy's derivative estimate therefore gives

$$|p'(z)| \leq C_0 3^N (1 + |\Im \rho|)^N$$

on the entire segment. Since $p(\rho) - p(\eta) = 1$, integration yields PSC8.1 with $C = C_0 3^N$. This necessary estimate controls all actual pairs and does not assume a near pair, a spectral gap, or an off-line zero.

PSC9 and final scope

The cited original-source derived lifting vanishing is the one in FOD7/NEA10. DC5.3–DC5.4 give the distinct specialization obstruction image $\text{im } \partial/\partial j(K)$ by their exact cross. The positive Hilbert quotient is not identified with that image or its weight data.

The final note constructs the specified complete positive receiver, its source observations, the exact reconstruction fibre product and topology criterion, and the original arithmetic correction. It preserves the complementary source and value data. Acceptance does not assert absence or existence of off-line zeros, simplicity, an arbitrary-subset interpolation theorem, or vanishing of the original defect.

Independent review of the universal receiver, positive forms, and multiplicity trace

24 September 2026. Scoped review of POSITIVE_TRANSFER_UNIVERSAL_QUOTIENT.md at SHA256 b70a8f0edf765dcfb3259735eb4a8f90a31c649ad308f55d821dcf5cccdade930.

No repair was found necessary for the claims reviewed here.

Coverage and scope

The independent proof verification in this review covers the definitions and constructions PTQ0–PTQ2, the complete universal property PTQ3, the complete bounded-positive-form classification PTQ4, and the complete multiplicity-weighted trace calculation PTQ8. The finite two-point orbit and sign convention PTQ5.3–PTQ5.4 were checked where used in PTQ8. The rest of PTQ5–PTQ11 is not certified as a separate complete proof audit by this review.

The source spaces, full divisor, multiplicities and operator formulas are the ones already read in GTAH0–GTAH7 and RTT0–RTT11 during this independent-review session. The corresponding source, residue and class-map checks are recorded separately in EXTENSION_CLASS_SOURCE_INDEPENDENT_AUDIT.md, EXTENSION_CLASS_INVOLUTION_AUDIT.md and POSITIVE_QUOTIENT_SOURCE_AUDIT.md. No new complete reading of their human-source bibliography is claimed. All coefficient operations follow the established arithmetic reconstruction; no operation or numerical coordinate is assigned to the support or its information layers.

Universal receiver

The ranges in PTQ1 are supported off the line. Every finite off-line coordinate vector is the image of a finite vector after division by its finitely many nonzero defect coefficients. Their closure is therefore exactly H_{off} , without asserting a bounded reciprocal or a closed range. Dominated convergence in the original multiplicity-weighted square norm proves the strong exponential limits. Both coordinate subspaces reduce the specified original operators. On the retained coordinates the identity $n^{\bar{\rho}} = n^{1-\rho}$ gives PTQ2.2, preserving the degree n .

In the stated receiving category,

$$B_n = A_n^*, \quad B_n A_n = A_n B_n = nI$$

imply $A_n^* A_n = nI$. For the original coordinate vector ε_ρ , the intertwiner gives

$$A_n X \varepsilon_\rho = n^\rho X \varepsilon_\rho.$$

Taking squared norms produces exactly

$$n \|X \varepsilon_\rho\|^2 = n^{2\Re \rho} \|X \varepsilon_\rho\|^2.$$

For any recovered $n > 1$, the two scalars differ off the line. Hence X kills each such coordinate and, by boundedness, all of H_{off} . The quotient factor is well defined by $Y q_{\text{pos}} x = Xx$, is unique because the quotient is onto, and is bounded by testing its unique line representative. Moreover $X = X P_{\text{line}}$ gives

$$\|Y\| = \|X|_{H_{\text{line}}}\| = \|X\|.$$

The quotient intertwining identities follow by composing with the surjective quotient map. This is the required universal property in the explicitly stated bounded Hilbert category.

PTQ3's smaller-image observation is valid as well. The closed span of $X(H)$ is invariant under A_n, B_n , by the intertwining relations and boundedness. On its dense image the original degree identity gives $B_n A_n = nI$, which extends continuously to that closed span. If $B_n = A_n^*$ there, the same norm argument applies. No claim about arbitrary algebraic maps from the entire source or an untopologized Ext group follows.

Classification of bounded positive forms

The equivalence of PTQ4.1 and PTQ4.2 uses $U_n T_n = nI$ in one direction and $U_n = nT_n^{-1}$ in the other. The coefficient n is real, so its location in the conjugate-linear argument causes no change.

For a positive semidefinite Hermitian form, Cauchy–Schwarz shows that a vector with zero diagonal value pairs to zero with every vector. Applying the degree identity to ε_ρ therefore kills every off-line row and column of the form.

For distinct retained zeros, put $\delta = \Im \rho - \Im \eta$. A nonzero cross coefficient would force $e^{i\delta \log n} = 1$ for every recovered positive integer. Dividing the phases at consecutive integers gives

$$e^{i\delta \log(1+1/n)} = 1.$$

If $\delta \neq 0$, then for sufficiently large n ,

$$0 < |\delta| \log(1 + 1/n) < 2\pi,$$

which is impossible for a real exponent whose complex exponential is one. Thus $\delta = 0$, and two retained zeros with that same frequency are the same complex point. Every distinct cross coefficient consequently vanishes. This uses the already reconstructed complete integer family.

The coordinate norm is $\|\varepsilon_\rho\|^2 = m_\rho$. Therefore

$$c_\rho = \frac{B(\varepsilon_\rho, \varepsilon_\rho)}{m_\rho}$$

is nonnegative and bounded uniformly by the bound of B . Finite-coordinate expansion gives PTQ4.3, and the estimate

$$\left| \sum_{\rho \in \mathcal{Z}_{\text{line}}} m_\rho c_\rho x_\rho \overline{y_\rho} \right| \leq C \|x\|_+ \|y\|_+$$

extends the identity to the complete Hilbert space. Conversely every displayed nonnegative bounded family defines such a form and satisfies the transfer-adjoint relation by direct substitution. The maximality under the bound $B(x, x) \leq \|x\|_+^2$ is precisely the coordinate inequality $c_\rho \leq 1$. Pullback along any PTQ3 intertwiner satisfies the same relation, so it belongs to this classification.

This classification is for the stated value Hilbert space with one coordinate per distinct zero and its original weight m_ρ . It is not a classification of unrestricted forms on full local jet algebras.

Multiplicity trace

For a diagonal value observable, PTQ8 uses the stipulated weight

$$\text{tr}_\zeta(M_f) = \sum_\rho m_\rho f(\rho).$$

This differs from the ordinary Hilbert operator trace on H : the orthonormal vector at a distinct zero is $\varepsilon_\rho / \sqrt{m_\rho}$, and its operator eigenvalue is only $f(\rho)$. The factor m_ρ is instead exactly the trace of multiplication by a local germ on the original length- m_ρ algebra, where every diagonal entry is its value $f(\rho)$. For a holomorphic germ it is also the coefficient of $(s - \rho)^{-1}$ in $f(s)\zeta'(s)/\zeta(s)$. Arbitrary value observables here do not assert a global entire interpolation.

For nonnegative sequences the trace identities are extended nonnegative sums; for complex sequences they are used under absolute summability. No unbounded-operator product is needed to establish these coefficientwise identities. The projection partition proves PTQ8.2 in both stated domains.

The full zero count and the Gaussian $e^{-t\gamma^2}$ make both sector sums finite. Reflection $\rho \mapsto \rho^\#$ preserves γ and multiplicity, and every off-line orbit has exactly two distinct elements. This verifies the factor $2m_\rho$ in PTQ8.3. Strict positivity of every weight proves its exact zero criterion without assuming either outcome.

The defect vanishes on line coordinates, and

$$|n^\sigma - n^{1-\sigma}| \leq n - 1 \quad (0 < \sigma < 1)$$

proves absolute convergence in PTQ8.4. Reflection negates the difference and hence preserves its square, so the two orbit terms add. On a two-coordinate orbit the finite-dimensional multiplicity trace is $m_\rho \text{Tr}$; it gives zero on the exchange matrix and $2m_\rho$ on the identity. Thus the comparison with PTQ5.4 is valid and does not remove either value direction.

The last paragraph of PTQ8 correctly keeps these diagonal observables outside the original holomorphic prime explicit formula. The reviewed results preserve the complete integer cover degrees, original multiplicities, and bounded-map domains; they assert no arbitrary source projector or vanishing of the original defect.

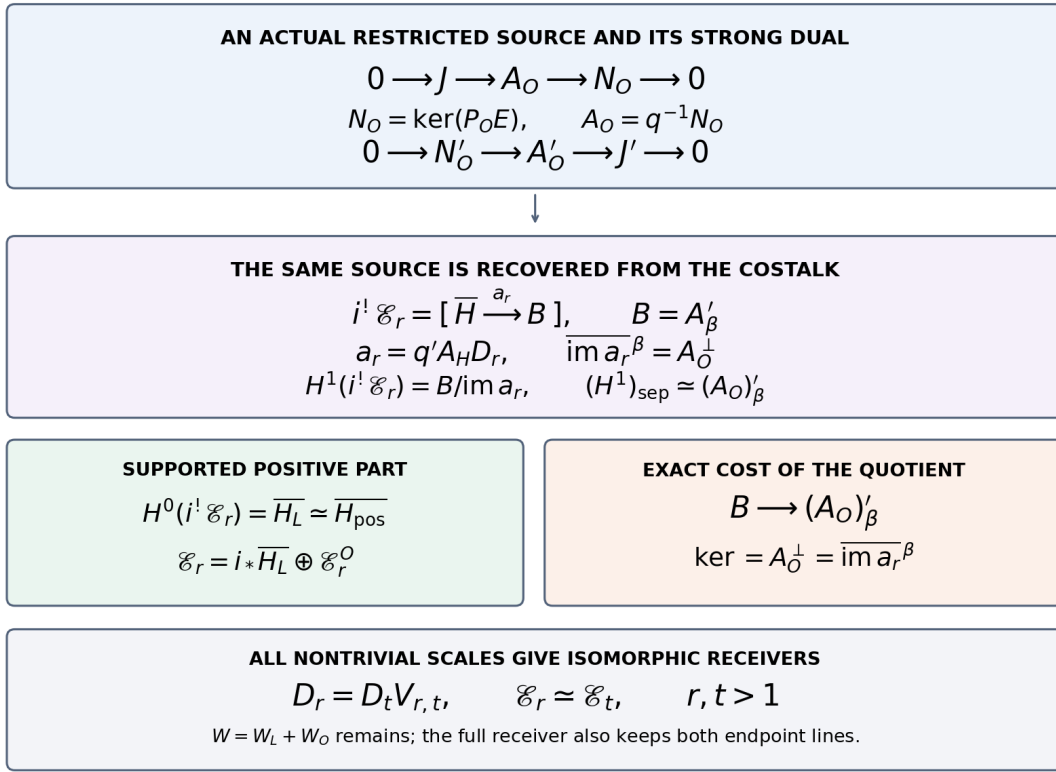
The positive quotient and the actual supported Gysin receiver

The complete received RGR proof constructs the actual continuous-current Gysin receiver from the original source, its signed differential, both endpoint lines, its dual original extension and the separate positive-adjoint obstruction. The next GDC proof calculates its complete strong-dual image, raw and separated costalk quotients, and the canonical maps to the already constructed positive receiver.

The original source appears again as an explicit restricted Fréchet extension. Its transpose is the actual separated costalk row, with topological inverses proved through compactness and Hahn–Banach. A continuous-current quotient kills the exact closed defect image; its full coefficient kernel remains in a proved triangle. All nontrivial positive coefficient scales have the same unclosed image and isomorphic receivers, with explicit bounded inverses and a cocycle law.

The original source inside the supported defect receiver

Exact maps on the complete zero divisor, with every multiplicity retained



Proofs: GDC1–GDC12, with RGR0–RGR12 and PTQ. r is a coefficient parameter; integer n is a cover degree.

Figure 13: Exact source and supported-cohomology comparison. Here N_O is the source whose off-line values vanish, A_O is its inverse image in the original A , and B is the strong dual of A . The reduced receiver retains its raw degree-one quotient and its separated comparison; the full receiver additionally retains both labelled endpoint lines by GDC4.7. The lower-left summand is the conjugate universal positive quotient. The original Weil form retains both terms. Complete proofs: GDC1–GDC12 and received RGR0–RGR12; the positive universal property is PTQ3–PTQ4.

The full proofs follow. The final GDC text has complete root verification, with independent constituent derivations of the scale and topology comparisons; a complete independent final-text audit is not claimed. The complete original lifting and arithmetic weight-separation target remains active. The receiving quotient has a proved kernel; its existence does not itself make the original defect vanish.

The supported residue map, its signed Gysin obstruction, and its actual receiver

Independent derivation, 24 September 2026. Proof locators RGR0–RGR12.

RGR0. Construction stage and the precise problem

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. The operations in this note are on the already constructed arithmetic coefficient sheaf. They give no addition, subtraction, midpoint, metric, vector coordinate, or extra parity to τ . The complete-history reconstruction and separate branch counters precede these coefficient operations. The connected user argument, its correction table, and the full passages U126–U128, U135–U141, WU050–WU055 and WU061–WU062 were consulted before this derivation. In particular, a local analytic lifting problem is not identified with the user’s entire construction or with Deligne’s arithmetic lifting problem merely by using the word obstruction.

The actual inputs are CSP’s inverse-image sheaf, CSD’s continuous compact-test current dual, and RTT’s original-zeta residue map. The source extension is the retained original component e_0 of FOD3. The calculation asks whether the residue map on the supported quotient lifts to the existing local dual sheaf. It computes the exact obstruction and then constructs its homotopy-pullback receiver. RGR9 derives its relation to the original coefficient extension. RGR11 also carries the already computed positive-adjoint defect into the same local receiver; it does not substitute the nonzero total residue form for that defect.

All derived sheaf calculations below take place in sheaves of complex vector spaces. The specified maps between coefficient spaces and currents remain continuous. No derived category of arbitrary locally convex spaces, finite-dimensionality of the coefficient spaces, Artinian perverse category, or equality of continuous and algebraic duals is assumed.

The primary geometric provenance is Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, transported by the explicit map in CSP0.2. The independent calculations used here are complete local proof files retained beside this note: CC_SPHERE_PULLBACK_AND_NORMAL_DIRECTION.md, CSP0–CSP7; CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md, CSD1–CSD6 and CSD10–CSD11; CC_FULL_SOURCE_OBSTRUCTION_DECOMPOSITION.md, FOD1–FOD5; and the peer’s RESIDUE_TRACE_TRANSFER_INDEPENDENT.md, RTT0–RTT6. The positive-adjoint receiver in RGR11 uses GEOMETRIC_TRANSFER_POSITIVE_ADJOINT_DEFECT.md, GTAH0–GTAH2. The exact proof dependence is given rather than attributing these programme calculations to the primary paper.

RGR1. Full coefficient source and original-zeta factors

Use

$$\begin{aligned} A &= \left\{ a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \quad (N, j \geq 0) \right\}, \\ S &= \{ f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), \ f(0) = 0, \ \int_{\mathbb{R}} f(v) \, dv = 0 \}, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \\ J &= \Sigma S \subset A, \quad Q = A/J, \quad q : A \longrightarrow Q. \end{aligned} \tag{RGR1.1}$$

The exact summation-image theorem used by CSP proves that J is closed and that $\Sigma : S \rightarrow J$ is a topological isomorphism. Both A and Q carry their proved Fréchet topologies.

The full Mellin comparison is

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} \, dt. \tag{RGR1.2}$$

It identifies A with the entire rapid-strip space

$$\mathcal{B} = \{ F \text{ entire} : b_{R,N}(F) = \sup_{|\Re s| \leq R} (1 + |\Im s|)^N |F(s)| < \infty \text{ for every } R, N \geq 0 \}$$

and J with

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every actual nontrivial zero } \rho \text{ of } \zeta, 0 \leq j < m_\rho\}. \quad (\text{RGR1.3})$$

Thus $\Theta_Q : Q \rightarrow \mathcal{Q} = \mathcal{B}/J$ is the exact quotient isomorphism. The summation formula before continuation is

$$\Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \quad (\text{RGR1.4})$$

In particular the original Schwartz function and its entire image remain

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{RGR1.5})$$

The exceptional values are

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = \frac{\pi}{24}, \quad F_0(-2r) = \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \quad (r \geq 1). \quad (\text{RGR1.6})$$

At a nontrivial zero ρ ,

$$F_0(\rho+t) = t^{m_\rho} \frac{(\rho+t)(\rho+t-1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho+t)/2) \frac{\zeta(\rho+t)}{t^{m_\rho}}. \quad (\text{RGR1.7})$$

The full local holomorphic unit near $t = 0$ on the right is retained, including every derivative. It is not asserted to be an entire zero-free function. The working residue denominator below is original ζ , not F_0 . The original arithmetic is still

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{RGR1.8})$$

Fix one positively oriented small disk D about a pole, with inclusion $i : \{p\} \hookrightarrow D$. The local coordinate is z at the positive pole and $1/z$ at the other pole. CSP and CSD give the full local source $V_p = S \oplus E_p$, where the two lines of E_p record value at zero and real integral. Its restriction is Σ , or $R\Sigma = \hat{\Sigma}$, respectively, with

$$Ra(u) = u^{-1} a(u^{-1}), \quad \hat{f}(t) = \int_{\mathbb{R}} f(v) e^{-2\pi i v t} dv.$$

The full Poisson correction is $u^{-1} f(0) - \int f$; it is zero on this stated S , and its two coordinates are retained in E_p .

Because the endpoint coordinates are explicit direct summands, the local sheaf has the actual splitting

$$\mathcal{F}|_D = \mathcal{F}_{\text{red}} \oplus i_* E_p, \quad \mathcal{F}_{\text{red}} = \ker(\underline{A}_D \xrightarrow{q \text{ ev}_p} i_* Q). \quad (\text{RGR1.9})$$

Set $P = \mathcal{F}_{\text{red}}[1]$. The endpoint summand is restored in RGR10. This splitting is proved by sending a source vector (f, e) to $(\Sigma f, e)$, or $(R\Sigma f, e)$, using the proved inverse on J . It does not remove a source contribution without its comparison map.

RGR2. The actual residue map, with its complete radical

Let $\mathcal{Z}$ denote the set of distinct actual nontrivial zeros, retaining multiplicities m_ρ , and put $\rho^\# = 1 - \bar{\rho}$. Define

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_\rho m_\rho x_\rho \overline{y_\rho}, \quad (\mathcal{J}x)_\rho = x_{\rho^\#},$$

$$E : Q \longrightarrow H, \quad E(F)_\rho = (\Theta_Q F)(\rho). \quad (\text{RGR2.1})$$

Evaluation is independent of representatives. For $f \in A$ representing a class $[f] \in Q$, the unconditional zero count and the rapid-strip seminorm defined in RGR1 give the convergent estimate

$$\|E[f]\|_+^2 \leq b_{1,2}(\Theta f)^2 \sum_{\rho} m_{\rho}(1 + |\Im \rho|)^{-4}. \quad (\text{RGR2.2})$$

The map on A kills J , so it is continuous on the quotient. RTT's full isolators give every finite value sequence, hence $E(Q)$ is dense in H . This asserts density of values, not density of finite full-jet classes in the primal topology.

Use the conjugate space $C = \overline{Q}$, with scalar action $\lambda \overline{G} = \overline{\lambda G}$. Define the linear continuous map

$$\sigma : C \rightarrow Q'_{\beta}, \quad \sigma(\overline{G})(F) = \langle EF, JEG \rangle_+ = \sum_{\rho} m_{\rho}(\Theta_Q F)(\rho) \overline{(\Theta_Q G)(\rho^{\#})}. \quad (\text{RGR2.3})$$

The prime on a coefficient space always means its continuous complex-linear dual; the subscript β means uniform convergence on bounded sets. For bounded $B_0 \subset Q$,

$$\sup_{F \in B_0} |\sigma(\overline{G})(F)| \leq \left(\sup_{F \in B_0} \|EF\|_+ \right) \|EG\|_+. \quad (\text{RGR2.4})$$

This proves absolute convergence and strong continuity. Testing against each value isolator proves

$$\ker \sigma = \overline{\mathcal{N}_0}, \quad \mathcal{N}_0 = \{G \in Q : (\Theta_Q G)(\rho) = 0 \text{ for every } \rho\}. \quad (\text{RGR2.5})$$

Higher jets remain in Q ; (RGR2.5) records exactly which of them this map does not detect. No simplicity assumption is used.

The comparison with the full residue completion is RTT's proved isomorphism $\hat{\iota} : \mathcal{H}_{\text{res}} \rightarrow Q'_{\beta}$. In the Q coordinate its finite expression is

$$\mathcal{R}(F, h) = \sum_{\rho} \text{Res}_{s=\rho} \frac{F(s)h(1-s)}{\zeta(s)} ds. \quad (\text{RGR2.6})$$

For $\zeta(\rho + t) = t^{m_{\rho}} u_{\rho}(t)$, the exact finite approximants are

$$h_B(G) = \sum_{\rho \in B} m_{\rho} (-1)^{m_{\rho}-1} u_{\rho}(0) \overline{G(\rho^{\#})} e_{1-\rho, m_{\rho}-1}. \quad (\text{RGR2.7})$$

Their residue functionals converge to the full sum (RGR2.3), uniformly on each bounded test set, by the Cauchy–Schwarz tail estimate in (RGR2.4). Thus σ is the continuous-dual version of RTT's residue map, with the stated Θ_Q and $\hat{\iota}$ comparisons. It is not an identification of the entire primal quotient with its dual.

Write

$$B = A'_{\beta}, \quad j = q' : Q'_{\beta} \rightarrow B, \quad a = j\sigma : C \rightarrow B. \quad (\text{RGR2.8})$$

Here j is the actual transpose, $j(\lambda)(f) = \lambda(qf)$. CSD10 and SDT's bounded-lift theorem give its strong topological identification with $J^{\perp} \subset B$. In particular it is injective and

$$\ker a = \overline{\mathcal{N}_0}. \quad (\text{RGR2.9})$$

RGR3. The cone sign and the positive angular quotient

For compactly supported smooth A -valued forms, use CSD's current complex

$$\mathcal{T}_A^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad dT(\phi) = (-1)^{n+1} T(d\phi), \quad -2 \leq n \leq 0. \quad (\text{RGR3.1})$$

It represents $\underline{B}_D[2]$, by the explicit compact-support integration contraction in CSD2. Its positive Dirac current and positive angular current satisfy

$$\delta_p \lambda(f) = \lambda(f(p)), \quad h_p \lambda(\omega) = \int_D \frac{d \arg z_p}{2\pi} \wedge \lambda(\omega), \quad dh_p \lambda = \delta_p \lambda. \quad (\text{RGR3.2})$$

The last equality follows by Stokes on a punctured disk; the small positive circle has integral one. These currents are continuous on the specified compact-test spaces. No tensor-product description of arbitrary vector-valued distributions is needed.

The reduced sheaf is resolved by

$$\mathcal{K}^0 = \mathcal{E}_A^0, \quad \mathcal{K}^1 = \mathcal{E}_A^1 \oplus i_* Q, \quad \mathcal{K}^2 = \mathcal{E}_A^2, \quad d^0 f = (df, qf(p)), \quad d^1(\omega, x) = d\omega. \quad (\text{RGR3.3})$$

At the pole a primitive of a closed one-form can be altered by a constant representative of its required quotient value; q is onto. This proves algebraic stalkwise exactness, and the written differentials are continuous. It does not provide a continuous linear choice of representatives $Q \rightarrow A$. On the puncture it is the coefficient de Rham resolution. Its continuous transpose has the cone written below, but identifying that cone with the actual CSD current dual requires the separate comparison that follows; no general exactness of continuous dualization is being used.

$$D_c \mathcal{F}_{\text{red}} \simeq \text{Cone}(g), \quad g = \delta_p j : i_* Q'_\beta \longrightarrow \mathcal{T}_A, \quad D_c P \simeq \text{Cone}(g)[-1]. \quad (\text{RGR3.4})$$

Here is the comparison with the already constructed current dual. Use the reduced version of CSD4, whose literal complex is

$$D_{\text{old}} = \text{Cone}(i_* A' \xrightarrow{(\delta_p, -k')} \mathcal{T}_A \oplus i_* J'). \quad (\text{RGR3.4a})$$

There is a continuous cochain inclusion $\text{Cone}(g) \rightarrow D_{\text{old}}$: identity on the current terms, $(T, \lambda) \mapsto (T, q'\lambda)$ in degree minus one, and $T \mapsto (T, 0)$ in degree zero. The differential commutes because $k'q' = 0$. The quotient complex is

$$i_*(A'/q'Q') \xrightarrow{-k'} i_* J' \quad \text{in degrees } -1, 0. \quad (\text{RGR3.4b})$$

This is algebraically acyclic: the induced map is bijective by the quotient definition and continuous Hahn-Banach extension from closed J to A . Hence the inclusion is a sheaf quasi-isomorphism. No continuous splitting $Q \rightarrow A$, no strong openness of $A' \rightarrow J'$, and no termwise exactness assertion about arbitrary dual complexes was needed. The local cohomology topologies of the small cone are computed directly by the same angular-current and fixed-test maps as CSD5. In particular the middle coefficient is Q'_β by the proved strong embedding q' . We use the small cone as this verified representative of $D_c P$ below.

For the standard cone $d(b, x) = (d_B b + fx, -d_X x)$, the latter representative has

$$(D_c P)^{-1} = \mathcal{T}_A^{-2}, \quad (D_c P)^0 = \mathcal{T}_A^{-1} \oplus i_* Q', \quad (D_c P)^1 = \mathcal{T}_A^0, \\ d^{-1}b = (-d_T b, 0), \quad d^0(b, \lambda) = -d_T b - g\lambda. \quad (\text{RGR3.5})$$

With literal projection $\pi(b, \lambda) = \lambda$, the connecting arrow of the short exact sequence of complexes is $-g$: lift λ as $(0, \lambda)$, whose differential is $-g\lambda$. This is the dual counterpart of the negative connecting arrow in the coordinate triangle $i_* Q \rightarrow \text{Cone}(q) \rightarrow \underline{A}[1] \rightarrow i_* Q[1]$.

The positive angular stalk class labelled $\mu \in Q'$ is

$$(h_p j \mu, -\mu). \quad (\text{RGR3.6})$$

Its differential is zero by (RGR3.2). It restricts to the positive angular class $h_p j \mu$. Consequently the quotient map recovering that label is

$$\gamma : D_c P \longrightarrow i_* Q', \quad \gamma(b, \lambda) = -\lambda. \quad (\text{RGR3.7})$$

This is an explicit change of the quotient coordinate, not a suppressed sign. With this convention the actual triangle is

$$\mathcal{T}_A[-1] \longrightarrow D_c P \xrightarrow{\gamma} i_* Q' \xrightarrow{g} \mathcal{T}_A. \quad (\text{RGR3.8})$$

Indeed the lift of μ in this quotient coordinate is $(0, -\mu)$, whose differential is $+g\mu$. The first arrow is the unchanged inclusion of currents.

RGR4. Gysin adjunction and the exact lifting obstruction

We ask for a morphism $\ell : i_*C \rightarrow D_cP$ with $\gamma\ell = i_*\sigma$. The exact Hom sequence of (RGR3.8) identifies its obstruction as

$$\boxed{\mathfrak{o}_\sigma = g i_*\sigma = \text{Gys}_B \circ i_*a : i_*C \longrightarrow \underline{B}_D[2]}. \quad (\text{RGR4.1})$$

At the current level this is exactly $x \mapsto \delta_p(ax)$, with positive sign.

Here the Gysin adjunction is valid for these full coefficient spaces. To see it without a finite-rank hypothesis, compute supported cohomology of $\mathcal{T}_A$ by the restriction fibre from a disk to its puncture. The disk current cohomology is B in degree -2 . The puncture has B in degrees $-2, -1$, represented by constant and angular currents. Cancelling the identity constant restriction leaves B in degree zero. The positive generator is the class of the supported positive Dirac current. Equivalently, in the restriction fibre, $(\delta_p\lambda, 0)$ is cohomologous to $(0, -h_p\lambda)$. Thus

$$i^! \underline{B}_D[2] \simeq B[0], \quad \text{Hom}_{D(D)}(i_*C, \underline{B}_D[2]) \simeq \text{Hom}_{\mathbb{C}}(C, B), \quad (\text{RGR4.2})$$

and (RGR4.1) corresponds to $+a$. The derived adjunction uses the exact point functor and the category of vector spaces; no dimension bound is needed. Every displayed continuous coefficient map gives a continuous current representative. The Hom group in (RGR4.2) is not claimed to contain only continuous coefficient maps.

Since j is injective,

$$\mathfrak{o}_\sigma = 0 \iff a = 0 \iff \sigma = 0. \quad (\text{RGR4.3})$$

For the actual residue map it is nonzero. Choose any actual zero ρ , its full value isolator F , and the isolator at $\rho^\#$ as G . Equation (RGR2.3) gives $\sigma(\overline{G})(F) = m_\rho \neq 0$. All arithmetic has already been reconstructed before this use of its zero divisor. In particular the same witness works at a zero on the critical line: this total-residue obstruction does not distinguish on-line from off-line zeros.

The identity $dh_pax = \delta_pax$ does not supply a supported sheaf null-homotopy. The current h_pax restricts to a nonzero angular current on the puncture whenever $ax \neq 0$, while a section of i_*C restricts to zero there. Thus $x \mapsto h_pax$ is a primitive on a disk or at its stalk, but it cannot be a morphism $i_*C \rightarrow \mathcal{T}_A^{-1}$ for nonzero a . This is the precise reason a zero stalkwise Gysin map can represent the nonzero supported derived morphism (RGR4.1).

For an arbitrary vector subspace $C_0 \subset C$, the restricted map lifts exactly when $C_0 \subset \overline{\mathcal{N}_0}$. This follows by applying the same adjunction to the restriction of a . Thus the exact largest liftable part of the supported source is its retained value-zero radical. No absence of higher jets is presumed.

RGR5. The actual receiver constructed from the obstruction

Use the sign in (RGR3.8) and define

$$\mathcal{E}_\sigma = \text{Cone}(-\delta_p a)[-1]. \quad (\text{RGR5.1})$$

The entire complex is

$$\begin{aligned} \mathcal{E}_\sigma^{-1} &= \mathcal{T}_A^{-2}, & \mathcal{E}_\sigma^0 &= \mathcal{T}_A^{-1} \oplus i_*C, & \mathcal{E}_\sigma^1 &= \mathcal{T}_A^0, \\ d^{-1}b &= (-d_T b, 0), & d^0(b, x) &= -d_T b + \delta_p(ax). \end{aligned} \quad (\text{RGR5.2})$$

It has the explicit morphisms

$$p_\sigma : \mathcal{E}_\sigma \longrightarrow i_*C, \quad (b, x) \mapsto x, \quad v_\sigma : \mathcal{E}_\sigma \longrightarrow D_cP, \quad (b, x) \mapsto (b, -\sigma x), \quad (\text{RGR5.3})$$

and identity on the other current terms. Substitution into (RGR3.5) gives $-d_T b - g(-\sigma x) = -d_T b + \delta_p ax$, so this is a cochain map, and

$$\gamma v_\sigma = i_*\sigma p_\sigma. \quad (\text{RGR5.4})$$

Because γ is a degreewise split surjection of the written complexes, their strict pullback is the homotopy pullback. Its equation is $-\lambda = \sigma x$, yielding exactly (RGR5.2). The associated triangle is

$$\underline{B}_D[1] \longrightarrow \mathcal{E}_\sigma \xrightarrow{p_\sigma} i_*C \xrightarrow{\mathfrak{o}_\sigma} \underline{B}_D[2]. \quad (\text{RGR5.5})$$

This receiver retains the obstruction as its attaching map. It does not declare that the original unsupported lift exists.

RGR6. Stalk, costalk, nearby cycles, and variation

On the puncture, $\mathcal{E}_\sigma$ is the current realization of $\underline{B}[1]$, so its local system has monodromy identity. On a small disk its two cohomology groups have explicit representatives

$$B \ni \lambda \mapsto c_\lambda \quad (\deg -1), \quad C \ni x \mapsto (h_p ax, x) \quad (\deg 0). \quad (\text{RGR6.1})$$

The second is closed because $dh_p ax = \delta_p ax$. CSD's compact-test contraction kills the remaining current terms and provides inverse continuous maps on the cohomology of each fixed disk. Consequently

$$i^* \mathcal{E}_\sigma \simeq B[1] \oplus C[0]. \quad (\text{RGR6.2})$$

Restriction to the punctured disk is identity on the degree-minus-one constant class and is $x \mapsto ax$ on the degree-zero angular class. The supported restriction fibre cancels the constant identity. With the positive Dirac convention of (RGR4.2), the remaining costalk complex is

$$\boxed{i^! \mathcal{E}_\sigma \simeq [C \xrightarrow{a} B] \quad \text{in degrees } 0, 1.} \quad (\text{RGR6.3})$$

For an explicit cancellation, the restriction fibre in the convention $d(s, t) = (ds, \text{res}(s) - dt)$ has coefficient complex

$$B \xrightarrow{b \mapsto (0, b)} C \oplus B \xrightarrow{(x, b) \mapsto ax} B \quad \text{in degrees } -1, 0, 1. \quad (\text{RGR6.3a})$$

Eliminating the identity pair gives (RGR6.3). In the shifted current complex, $dh_p b = -\delta_p b$; hence in this fibre $d(h_p b, 0) = (-\delta_p b, h_p b)$, so $(\delta_p b, 0)$ and $(0, h_p b)$ give the same positive supported class. Thus the target B in (RGR6.3) uses positive Dirac orientation.

The literal standard expression $\text{Cone}(\text{var})[-1]$, for $\text{var} = a : C \rightarrow B$, instead has differential $-a$. Its exact comparison with (RGR6.3) is identity on C and multiplication by -1 on the target B . Thus its literal target coordinate is the negative of the positive supported coordinate just specified. This sign change does not change $\text{var} = +a$ in (RGR6.5). For the comparison to $[Q' \xrightarrow{j} B]$, change the target coordinate on both complexes together: the costalk arrow remains $(\sigma, 1_B)$. For $[C \rightarrow Q'] \rightarrow [C \rightarrow B]$ in RGR9, change both target coordinates together: its arrow remains $(1_C, j)$. This explicitly reconciles the standard shifted-cone convention with the positive-Dirac restriction-fibre convention.

The sign can also be read without cancellation: applying $i^!$ to (RGR5.5), its connecting morphism is $+a : C \rightarrow B$ by (RGR4.2). The costalk cohomology is therefore

$$H^0(i^! \mathcal{E}_\sigma) = \overline{\mathcal{N}_0}, \quad H^1(i^! \mathcal{E}_\sigma) = B/a(C). \quad (\text{RGR6.4})$$

These are the algebraic cohomology groups with their indicated kernel and quotient topologies. No closedness of $a(C)$, Hausdorffness of its quotient, or strong-open mapping theorem for a dual restriction map is asserted.

For completeness one can read the nearby and vanishing maps directly. Pull back the constant punctured local system to the universal cover and contract there; its shifted nearby complex is $B[0]$. The ordinary restriction model on the puncture is $B[1] \oplus B[0]$. In the stalk model (RGR6.2), its first component is projection onto $B[1]$, and its angular component is $a : C \rightarrow B$. The shifted vanishing cone of the first component cancels this identity pair and leaves $C[0]$. Thus, in the positive angular convention,

$$\Psi(\mathcal{E}_\sigma) = B, \quad \Phi(\mathcal{E}_\sigma) = C, \quad \text{can} = 0, \quad \text{var} = a, \quad T = 1. \quad (\text{RGR6.5})$$

Here $\Psi = R\psi[-1]$ and $\Phi = R\phi[-1]$. Both composites of can and var are zero, agreeing with $1 - T = 0$. This is a direct calculation on the specified complexes, rather than an invocation of finite-constructible gluing theory.

The full morphism to $D_c P$ is

$$(\Psi(v_\sigma), \Phi(v_\sigma)) = (1_B, \sigma). \quad (\text{RGR6.6})$$

On costalks it is the map $[C \xrightarrow{a} B] \rightarrow [Q' \xrightarrow{j} B]$ with components $(\sigma, 1_B)$. In particular the radical appears as an actual supported cohomology group; it has not been erased by passing to values.

RGR7. Exact supported kernel and cokernel of the receiver map

The current term is common to the two triangles (RGR3.8) and (RGR5.5), and the map there is identity. Taking the cone of the displayed morphism of triangles gives

$$\text{Cone}(v_\sigma) \simeq i_* \text{Cone}(\sigma), \quad \text{Cone}(\sigma) = [C \xrightarrow{\sigma} Q'] \text{ in degrees } -1, 0. \quad (\text{RGR7.1})$$

Here is an explicit cone comparison. Put $L = \mathcal{T}_A[-1]$ and write a cone element as $(b, \lambda; u, x) \in (D_c P)^n \oplus \mathcal{E}_\sigma^{n+1}$. The map

$$(b, \lambda; u, x) \mapsto (-\lambda, x) \quad (\text{RGR7.1a})$$

to $i_* \text{Cone}(\sigma)$ is a surjective cochain map: the target differential is $(\eta, x) \mapsto (\sigma x, 0)$. Its kernel is $\text{Cone}(1_L)$, with contraction $(b; u) \mapsto (0; b)$. Indeed the sum of the two compositions with the cone differential is $(b; u)$. This proves (RGR7.1) and its sign. It follows that

$$\mathcal{H}^{-1} \text{Cone}(v_\sigma) = i_* \overline{\mathcal{N}_0}, \quad \mathcal{H}^0 \text{Cone}(v_\sigma) = i_*(Q'/\sigma C). \quad (\text{RGR7.2})$$

The inclusion of $i_* \overline{\mathcal{N}_0}$ into $\mathcal{E}_\sigma$ is literal: send x to $(0, x)$ in degree zero. Its differential vanishes because $ax = 0$. This supported map records the maximal liftable subspace from RGR4.

Applying (RGR6.3) with $C = Q'$, $a = j$, gives

$$i^! D_c P \simeq [Q' \xrightarrow{j} B] \simeq J'[-1] \quad (\text{RGR7.3})$$

as complexes of vector spaces. The last assertion uses Hahn–Banach to identify the algebraic cokernel with J' . It asserts no strong-topology inverse. Since this has cohomology only in degree one, $\text{Hom}_{D(D)}(i_* C, D_c P) = 0$. Thus a lift of a zero supported residue map is the zero morphism in this derived category; the failure to lift the actual nonzero residue map is not an ambiguity among several lifts.

RGR8. The image topology and the retained multiplicity information

RTT6 proves in the strong dual

$$\overline{\sigma(C)} = \mathcal{N}_0^\perp \subset Q'_\beta. \quad (\text{RGR8.1})$$

One proof is to note that the image contains every value functional. A continuous functional on the strong dual separating its closure from an element of $\mathcal{N}_0^\perp$ is, by SDT's proved bidual result, evaluation at an actual class in Q . Vanishing on every value functional places that class in $\mathcal{N}_0$, a contradiction. Conversely every image functional annihilates $\mathcal{N}_0$.

Since j is a strong topological embedding with closed image, this gives

$$\overline{a(C)} = j(\mathcal{N}_0^\perp) \subset J^\perp \subset B. \quad (\text{RGR8.2})$$

The separated quotient of the second group in (RGR6.4) is therefore $B/j(\mathcal{N}_0^\perp)$. The unseparated quotient $B/a(C)$ remains the actual cohomology with its original quotient topology. The two are not silently identified.

At a zero of multiplicity m , the full primal local block is $\mathbb{C}[t]/(t^m)$. Its value-zero part $t\mathbb{C}[t]/(t^m)$ has dimension $m - 1$. The actual residue numerator in (RGR2.7) uses $m(-1)^{m-1}u_\rho(0)$ and the highest-jet line of the reflected block. Thus the receiver retains the full input block and records the precise rank-one value observation; it does not replace multiplicity by one or identify the value radical with a removed part of the sheaf.

RGR9. Exact relation to the retained original extension

Under (RGR1.2), FOD's actual original source row is

$$e_0 : 0 \longrightarrow J \xrightarrow{k} A \xrightarrow{q} Q \longrightarrow 0. \quad (\text{RGR9.1})$$

Its continuous transpose is the vector-space exact sequence

$$0 \longrightarrow Q' \xrightarrow{j=q'} B = A' \xrightarrow{k'} J' \longrightarrow 0. \quad (\text{RGR9.2})$$

Injectivity and the kernel follow from the quotient definition. Surjectivity is precisely Hahn–Banach for continuous functionals on the closed subspace $J \subset A$. All three maps are continuous for strong duals because the primal maps carry bounded sets to bounded sets. We do not infer that k' is strongly open or that a continuous section exists.

There is a further actual receiver

$$\mathcal{K}_\sigma = \text{Fib}(i_* C \xrightarrow{\text{Gys}_{Q'} \sigma} \underline{Q}'_D[2]), \quad (\text{RGR9.3})$$

where Fib denotes the positive-quotient convention used in (RGR5.1)–(RGR5.5). It has a continuous-current representative. Form $\mathcal{T}_Q$ from compactly supported smooth Q -valued tests by exactly (RGR3.1). The compact-support integrations of CSD2 apply because Q is Fréchet and complete. They identify $\mathcal{T}_Q \simeq \underline{Q}'[2]$, with its actual strong coefficient topology on local cohomology. Then

$$\mathcal{K}_\sigma^n = \mathcal{T}_Q^{n-1} \oplus i_* C^n, \quad d(v, x) = (-dv + \delta_Q \sigma x, 0). \quad (\text{RGR9.3a})$$

Transposing the continuous maps on compactly supported tests induced by $q : A \rightarrow Q$ and $k : J \rightarrow A$ gives

$$q^\vee : \mathcal{T}_Q \rightarrow \mathcal{T}_A, \quad k^\vee : \mathcal{T}_A \rightarrow \mathcal{T}_J, \quad k^\vee q^\vee = 0. \quad (\text{RGR9.3b})$$

Their maps on constant-current coefficients are precisely $j = q'$ and k' . The resulting receiver maps are $(v, x) \mapsto (q^\vee v, x)$ and $(b, x) \mapsto k^\vee b$. These are continuous cochain maps. No assertion that the three current complexes form a termwise exact strong-dual sequence is required. The same object can alternatively be defined by the positively oriented scalar Gysin map tensored with the coefficient vector space in the stated vector-sheaf category.

Naturality of the oriented Gysin map gives

$$\text{Gys}_B j\sigma = \underline{j}[2] \text{Gys}_{Q'} \sigma. \quad (\text{RGR9.4})$$

Apply the fibre triangle for this composite. The fibre of $\underline{j}[2]$ is $\underline{J}'[1]$ by the exact row (RGR9.2). For the written current complexes this identification also follows from the continuous constant-current comparisons: after cancelling the common supported C coordinate, the cone is $\text{Cone}(\mathcal{T}_Q[-1] \rightarrow \mathcal{T}_A[-1])$, and its comparison to $\mathcal{T}_J[-1]$ is a quasi-isomorphism by the coefficient row (RGR9.2). Consequently there is a precise triangle

$$\boxed{\mathcal{K}_\sigma \longrightarrow \mathcal{E}_\sigma \longrightarrow \underline{J}'_D[1] \longrightarrow \mathcal{K}_\sigma[1].} \quad (\text{RGR9.5})$$

The map on nearby coefficients is $j : Q' \rightarrow B$, and on vanishing coefficients it is 1_C . The second map on nearby coefficients is $k' : B \rightarrow J'$, and on vanishing coefficients it is zero. These formulas prove compatibility since $k'j\sigma = 0$. In an explicit cone proof, eliminate the common supported C coordinates; what remains is the coefficient cone of j , giving $\underline{J}'[1]$. This also fixes the arrows in (RGR9.5), rather than asserting a decomposition of its middle object.

The complete local data of $\mathcal{K}_\sigma$ are

$$\begin{aligned} \Psi \mathcal{K}_\sigma &= Q', \quad \Phi \mathcal{K}_\sigma = C, \quad \text{can} = 0, \quad \text{var} = \sigma, \\ i^* \mathcal{K}_\sigma &\simeq Q'[1] \oplus C, \quad i^! \mathcal{K}_\sigma \simeq [C \xrightarrow{\sigma} Q'] \quad (\deg 0, 1). \end{aligned} \quad (\text{RGR9.6})$$

Its costalk groups are $\overline{\mathcal{N}}_0$ and $Q'/\sigma C$. The latter maps injectively into $B/a(C)$; explicitly,

$$0 \longrightarrow Q'/\sigma C \xrightarrow{[\lambda] \mapsto [j\lambda]} B/a(C) \xrightarrow{[b] \mapsto k'b} J' \longrightarrow 0. \quad (\text{RGR9.7})$$

Exactness follows directly from (RGR9.2). This row is the pushout of the actual dual row (RGR9.2) by $Q' \rightarrow Q'/\sigma C$: the pushout map sends $[(b, [\lambda])]$ to $[b + j\lambda]$, and the relation $(j\mu, -[\mu])$ maps to zero. Its inverse is $[b] \mapsto [(b, 0)]$. These are continuous maps for the stated quotient topologies, with no assertion that these quotient spaces are separated.

Thus the connection to e_0 is exact: its transpose inclusion is the coefficient of the Gysin obstruction, and its transpose extension produces both the triangle (RGR9.5) and the pushout row (RGR9.7). The class $\mathfrak{o}_\sigma \in \text{Hom}(i_*C, \underline{B}[2])$ is not declared equal to an M -module Ext^1 class in a different category. No M -linearity of the untwisted anti-linear Weil map is inferred; its reflected multiplier covariance is the one proved in RTT and ECI.

The concrete next calculation after the nonzero obstruction is therefore already performed: isolate the arithmetic dual coefficient Q' inside the full nearby coefficient A' , retain the dual original extension through (RGR9.5), and calculate the exact remaining quotient. Pushing the obstruction along k' gives zero because $k'j = 0$, but this kills the entire residue coefficient. It cannot be described as a proof that the original residue map lifted unchanged.

RGR10. Both endpoint lines remain in the complete local receiver

From the explicit source splitting (RGR1.9),

$$P_{\text{full}} = \mathcal{F}|_D[1] = P \oplus i_*E_p[1], \quad D_cP_{\text{full}} = D_cP \oplus i_*E'_p[-1]. \quad (\text{RGR10.1})$$

The two endpoint lines remain separate, with their original source labels. The quotient to i_*Q' is γ on the first factor and zero on the endpoint factor. The complete homotopy pullback is consequently

$$\mathcal{E}_{\sigma, \text{full}} = \mathcal{E}_\sigma \oplus i_*E'_p[-1]. \quad (\text{RGR10.2})$$

The endpoint summand has no nearby part; its shifted vanishing part, stalk, and costalk are all $E'_p[-1]$. The additional summand cannot cancel (RGR4.1), because the connecting map factors through the first summand's quotient and is exactly the same map. At infinity the positive coordinate is $1/z$; hence the same local Gysin formulas hold with its own positive angular orientation. No assertion about cover-degree transport is added here; those factors remain in CSD9 and the separate full-cover calculation.

RGR11. The already computed positive-adjoint defect has its own exact Gysin receiver

The total residue form is nonzero even on the critical line. To test the desired adjoint identity one must instead use the actual discrepancy already calculated on the full value completion. For every recovered parameter $r > 1$, retain

$$(T_r y)_\rho = r^\rho y_\rho, \quad U_r = rT_{1/r}, \quad D_r = T_r^* - U_r, \\ (D_r y)_\rho = e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}) y_\rho. \quad (\text{RGR11.1})$$

The known open strip makes these bounded operators. Integer $r = n$ is the actual geometric cover degree; arbitrary positive r is only a coefficient parameter. No nonintegral geometric cover is asserted.

RTT6 supplies the continuous injective anti-linear map

$$A_H : H \longrightarrow Q'_\beta, \quad A_H(y)(F) = \langle EF, Jy \rangle_+. \quad (\text{RGR11.2})$$

Its continuity follows from (RGR2.4) with $\|y\|_+$, and injectivity follows from density of $E(Q)$. Put

$$\sigma_r : \overline{H} \rightarrow Q'_\beta, \quad \sigma_r(\overline{y}) = A_H(D_r y), \quad a_r = j\sigma_r. \quad (\text{RGR11.3})$$

The general current-cone, adjunction, homotopy-pullback, stalk/costalk, and dual-row pushout constructions of RGR3–RGR7 and RGR9 apply to this actual map with source $\overline{H}$, using its own $\ker \sigma_r$ and $\operatorname{im} \sigma_r$. Those constructions use a continuous coefficient map, not finite dimension or a primal holomorphic lift of D_r . The value-source radical and closure formulas of RGR2 and RGR8 are not transferred to this different source. The resulting objects are

$$\mathfrak{o}_r = \operatorname{Gys}_B i_* a_r, \quad \mathcal{E}_r = \operatorname{Cone}(-\delta_p a_r)[-1], \quad i^! \mathcal{E}_r \simeq [\overline{H} \xrightarrow{a_r} B]. \quad (\text{RGR11.4})$$

Both j and A_H are injective. The oriented adjunction therefore proves

$$\mathfrak{o}_r = 0 \iff D_r = 0, \quad H^0(i^! \mathcal{E}_r) = \overline{\ker D_r}. \quad (\text{RGR11.5})$$

For the actual diagonal operator in (RGR11.1), $\ker D_r$ is exactly the value subspace supported on the critical-line zeros: the real function r^x is strictly increasing, so its displayed difference vanishes exactly at $x = 1/2$. This proves the precise detection map on the entire zero set. It neither postulates nor numerically selects an off-line zero.

The adjoint defect and the total residue map have consequently been given different, fully constructed local receivers. Their costalk kernels are different: $\overline{\mathcal{N}}_0$ for the full primal residue source, and the critical-line value subspace for the defect source. The coefficient extension still enters through the same q' , and both retain the exact Gysin sign and degree. No weight separation proving $\mathfrak{o}_r = 0$ is asserted by this construction; its contribution is the actual sheaf-level map and receiver on which such a calculation would have to act.

RGR12. Proved scope and continuation data

The lift of the total supported residue map fails by the explicitly nonzero class (RGR4.1). Its meaning is exactly local sheaf support: it is present at an ordinary critical-line zero as well. The full homotopy-pullback receiver, its current complex, all four local functors, its radical, its image topology, and its connection to the actual original source extension are computed in RGR5–RGR10. The separate arithmetic positive-adjoint defect is carried to the same exact geometry in RGR11, without inserting a conclusion about its vanishing.

For a Deligne comparison, retain the actual image quotient in `DELIGNE_INVARIANT_CYCLE_QUOTIENT.md`, DC5.3–DC5.4. The present Gysin class is not that quotient merely because it is a connecting morphism. The source P is the explicit singular coefficient sheaf (RGR1.9); Deligne’s proper smooth-total family, inertia, and arithmetic weight hypotheses are additional geometric data. This note constructs the receiver needed to compare them instead of assuming those data. It also does not alter FOD7’s previously proved vanishing of the specified global derived original-quotient lift: that global map and (RGR4.1) have different sources and targets, both retained here.

The original ζ , its unit term, all prime-power repetitions, the Gamma and endpoint factors, the full non-trivial multiplicities, and the trivial-zero derivative values remain in RGR1–RGR2 and their exact source comparisons. The note supplies no arithmetic operation on τ , no arbitrary positive form on the source, and no claim of RH or completion of WU061.

The Gysin defect closure, the positive receiver, and the dual original extension

Independent derivation with root topology strengthening, 24 September 2026. Proof locators GDC0–GDC14. The independent reviewer derived the restriction-topology argument separately; the root verified and integrated its full proof below. The final written GDC0–GDC14 received root mathematical verification, not a completed independent audit of every final line.

GDC0. Construction order, read sources, and notation

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Every coefficient operation below occurs after the complete-history arithmetic reconstruction. No addition, arithmetic value, parity, coordinate, distance or midpoint is assigned to τ . The notation Z_n continues to denote the user's information layers. Both support branches and their separate records remain present. This calculation constructs the next map from the already measured defect; it does not insert weight purity into the hypotheses.

The complete proofs RGR0–RGR12, PTQ0–PTQ11 and PSC0–PSC9 were read for this derivation, as were SDT0–SDT9, including its full bidual proof. RTT is the existing independently derived residue comparison. The current definition-source instruction and amended what-would workflow were reread. The exact versions and coverage are recorded in GDC14.

Use RGR's original coefficient extension, with its actual Fréchet topologies,

$$0 \longrightarrow J \xrightarrow{k} A \xrightarrow{q} Q \longrightarrow 0, \quad B = A'_\beta, \quad j = q' : Q'_\beta \longrightarrow B. \quad (\text{GDC0.1})$$

The prime means the continuous complex-linear dual; β means uniform convergence on bounded sets. SDT proves that j is a strong topological embedding with closed image $J^\perp$, and that $(Q'_\beta)' = Q$ by evaluation. In the original Mellin coordinate

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad Q \simeq \mathcal{B}/\mathcal{I}, \quad (\text{GDC0.2})$$

$\mathcal{B}$ is the entire rapid-strip space and $\mathcal{I}$ is the ideal vanishing to order m_ρ at every actual nontrivial zero of original ζ . We write $F(\rho)$ for the value of the class after this specified comparison, not an assigned value of the support.

Retain the entire actual divisor $\mathcal{Z}$, its multiplicities, and

$$H = \ell^2(\mathcal{Z}, m), \quad \rho^\# = 1 - \bar{\rho}, \quad (\text{J}y)_\rho = y_{\rho^\#}, \quad E : Q \rightarrow H, \\ A_H(y)(F) = \langle EF, \text{J}y \rangle_+ = \sum_\rho m_\rho F(\rho) \overline{y_{\rho^\#}}. \quad (\text{GDC0.3})$$

The inner product is linear in its first variable. The map E is continuous and has dense image; A_H is continuous, injective and anti-linear. Density follows from the actual full-jet isolators realizing finite value sequences, not from a claim about finite-support density in the primal Fréchet topology.

For $r > 1$ set

$$D_r = T_r^* - U_r, \quad U_r = rT_{1/r}, \quad d_r(\rho) = e^{-i\gamma \log r} (r^\sigma - r^{1-\sigma}), \quad (D_r y)_\rho = d_r(\rho) y_\rho, \quad (\text{GDC0.4})$$

where $\rho = \sigma + i\gamma$ and $0 < \sigma < 1$. For the conjugate Hilbert space $\overline{H}$,

$$\sigma_r(\overline{y}) = A_H(D_r y), \quad a_r = j\sigma_r. \quad (\text{GDC0.5})$$

These are continuous complex-linear maps on $\overline{H}$. An arbitrary $r > 1$ here is a coefficient parameter; only a recovered positive integer n is asserted to be an actual geometric cover degree.

The classical functional-analysis ingredients used below are complex Hahn–Banach extension and separation, and the Baire-category argument for equicontinuity. For accessible proofs of those classical tools see Tao, duality and Hahn–Banach, Theorem 1 and its complex extension and Tao, Baire-category consequences. GDC7 supplies the particular compact-polar restriction proof for the coefficient spaces used here; that programme-specific conclusion is not attributed to those sources.

GDC1. Exact signs and the annihilator on the original source

Let H_L, H_O be the closed subspaces supported respectively on the actual line and off-line parts of the divisor, with projections P_L, P_O . Both subsets are preserved by $\#$. The sign under reflection is

$$d_r(\rho^\#) = -d_r(\rho). \quad (\text{GDC1.1})$$

Consequently the complete functional is

$$\sigma_r(\bar{y})(F) = \sum_{\rho} m_{\rho} F(\rho) \overline{d_r(\rho^\#) y_{\rho^\#}} = - \sum_{\rho} m_{\rho} F(\rho) e^{i\gamma \log r} (r^{\sigma} - r^{1-\sigma}) \overline{y_{\rho^\#}}. \quad (\text{GDC1.2})$$

The minus sign in the second expression comes from (GDC1.1); it is independent of the positive Dirac orientation of the Gysin map. Both sums converge absolutely by Cauchy–Schwarz and boundedness of D_r .

Define closed source subspaces

$$\begin{aligned} N_O &= \ker(P_O E) = \{F \in Q : F(\rho) = 0 \ \forall \rho \in \mathcal{Z}_O\}, \\ N_L &= \ker(P_L E), \quad N_0 = N_L \cap N_O = \ker E, \quad A_O = q^{-1}(N_O) \subset A. \end{aligned} \quad (\text{GDC1.3})$$

The subscript records the values required to vanish. In particular N_O retains all higher jets and the line values. No direct-sum source interpolation is included in its definition.

The exact preannihilator of the defect image is

$$\boxed{\{F \in Q : \sigma_r(\bar{y})(F) = 0 \ \forall y \in H\} = N_O.} \quad (\text{GDC1.4})$$

One proof reads (GDC1.2). All its coefficients vanish on line points. At any actual off-line ρ , choose y with the sole nonzero coordinate $y_{\rho^\#} = 1/(m_{\rho} d_r(\rho^\#))$. Its functional is exactly $F \mapsto F(\rho)$. Thus annihilating every image functional forces precisely the off-line value conditions, and those conditions conversely kill the entire sum. This uses actual individual isolating coordinates in H , not an unrestricted holomorphic interpolation claim.

In particular $\ker \sigma_r = \overline{H_L}$. The overline here denotes the conjugate vector space, not a closure; closures below carry a topology label.

GDC2. The actual strong-dual closure

Write $I_r = \sigma_r(\overline{H}) \subset Q'_{\beta}$. Then

$$\boxed{\overline{I_r}^{\beta} = S_O := N_O^{\perp}, \quad \overline{a_r(\overline{H})}^{\beta} = jS_O = A_O^{\perp} \subset B.} \quad (\text{GDC2.1})$$

The first inclusion into $N_O^{\perp}$ follows from (GDC1.4), and this annihilator is strongly closed. If a functional in $N_O^{\perp}$ lay outside the strong closure of I_r , Hahn–Banach separation of that closed linear subspace would give a continuous linear functional on Q'_{β} vanishing on I_r but not on the stated functional. SDT’s full bidual result identifies the separator as evaluation at some $F \in Q$. Equation (GDC1.4) then forces $F \in N_O$, a contradiction. This proves equality in the strong, not only weak, topology.

For the second assertion, the closed strong embedding j takes the first closure to the closure inside B . A functional $j\lambda$ annihilates A_O exactly when λ annihilates N_O . Conversely any continuous functional on A annihilating A_O annihilates $J \subset A_O$, hence is a unique $j\lambda$. This proves $jS_O = A_O^{\perp}$.

There is also a useful intermediate equality:

$$\overline{A_H(H_O)}^\beta = S_O. \quad (\text{GDC2.2})$$

Indeed $I_r \subset A_H(H_O) \subset S_O$, while (GDC2.1) supplies the closure. These are closure identities. They do not declare I_r or $A_H(H_O)$ closed, equal, or topologically isomorphic to a Hilbert space with the strong subspace topology.

GDC3. Every nontrivial coefficient parameter has the same unclosed image

A stronger comparison than (GDC2.1) is available without changing the source. For $r, t > 1$, define a diagonal operator $V_{r,t}$ on H by

$$v_{r,t}(\rho) = \begin{cases} e^{-i\gamma \log(r/t)} \frac{r^\sigma - r^{1-\sigma}}{t^\sigma - t^{1-\sigma}}, & \sigma \neq 1/2, \\ 1, & \sigma = 1/2. \end{cases} \quad (\text{GDC3.1})$$

It is bounded and has bounded inverse $V_{t,r}$. To prove this, put $c = \sigma - 1/2$. The absolute value of its off-line coefficient is

$$\sqrt{r/t} \frac{\sinh(c \log r)}{\sinh(c \log t)}. \quad (\text{GDC3.2})$$

The ratio has a positive continuous extension at $c = 0$, of value $\sqrt{r/t} \log r / \log t$, and is positive on the compact interval $[-1/2, 1/2]$. Its minimum there is positive and its maximum finite. The exponential in (GDC3.1) has modulus one. Assigning the actual line value one therefore keeps both bounds. No lower bound on an off-line zero's distance from the line is needed.

The same comparison has explicit bounds without taking an unspecified compact minimum. Retain the original difference as

$$r^\sigma - r^{1-\sigma} = c L_r(c), \quad L_r(c) = \int_0^1 (\log r) (r^{1/2+uc} + r^{1/2-uc}) du.$$

This follows by integrating the derivative of $u \mapsto r^{1/2+uc} - r^{1/2-uc}$; both endpoint terms remain. For $|c| \leq 1/2$, arithmetic-geometric mean and monotonicity in $|uc|$ give

$$2\sqrt{r} \log r \leq L_r(c) \leq (r+1) \log r.$$

Consequently every off-line diagonal coefficient in (GDC3.1) satisfies

$$\frac{2\sqrt{r} \log r}{(t+1) \log t} \leq |v_{r,t}(\rho)| \leq \frac{(r+1) \log r}{2\sqrt{t} \log t}. \quad (\text{GDC3.2a})$$

The line value one is retained separately. These are bounds on the actual parameter-change map, not a change of the original operators.

Direct coordinate substitution gives

$$D_r = D_t V_{r,t}, \quad V_{r,t} V_{t,u} = V_{r,u}, \quad V_{r,t}|_{H_L} = I. \quad (\text{GDC3.3})$$

All $V_{r,t}$ commute with every coefficient dilation and transfer. In particular

$$\boxed{I_r = I_t \quad (r, t > 1).} \quad (\text{GDC3.4})$$

This is equality of the actual images before closure. It does not say that the operator norms, pointwise defect sizes, or heat-weighted traces of the defects agree. At $r = 1$, $D_1 = 0$, so that parameter is deliberately excluded from (GDC3.4).

GDC4. Gysin complexes, the unseparated costalk, and parameter isomorphisms

Use the positively oriented coordinate and currents of RGR, with $dh_p b = \delta_p b$. Its verified defect receiver is

$$\begin{aligned} \mathcal{E}_r &= \text{Cone}(-\delta_p a_r)[-1], & \mathcal{E}_r^{-1} &= \mathcal{T}_A^{-2}, & \mathcal{E}_r^0 &= \mathcal{T}_A^{-1} \oplus i_* \overline{H}, & \mathcal{E}_r^1 &= \mathcal{T}_A^0, \\ d^{-1}b &= (-d_T b, 0), & d^0(b, \overline{y}) &= -d_T b + \delta_p a_r(\overline{y}). \end{aligned} \quad (\text{GDC4.1})$$

The positive angular quotient gives the triangle

$$\underline{B}_D[1] \rightarrow \mathcal{E}_r \rightarrow i_* \overline{H} \xrightarrow{+\text{Gys}_B a_r} \underline{B}_D[2]. \quad (\text{GDC4.2})$$

The costalk in positive Dirac coordinates is exactly

$$i^! \mathcal{E}_r = [\overline{H} \xrightarrow{+a_r} B] \quad (\deg 0, 1). \quad (\text{GDC4.3})$$

RGR proves this by its restriction-fibre cancellation; the literal shifted cone has differential $-a_r$, compared with (GDC4.3) by minus identity on the target. The positive sign of (GDC4.2) and (GDC4.3) is not changed by the coefficient minus sign in (GDC1.2).

Its actual cohomology with the kernel and quotient topologies is

$$H^0(i^! \mathcal{E}_r) = \overline{H}_L, \quad H^1(i^! \mathcal{E}_r) = B/jI_r. \quad (\text{GDC4.4})$$

The second space is retained even if unseparated. Its separated quotient is exactly

$$\boxed{(H^1(i^! \mathcal{E}_r))_{\text{sep}} = B/A_O^\perp}. \quad (\text{GDC4.5})$$

In fact the closure of zero in (GDC4.4) is $A_O^\perp/jI_r$. This follows by applying the open quotient map to (GDC2.1): a class belongs to the closure of zero precisely when every neighborhood of a representative intersects jI_r . No completion or replacement of the unseparated cohomology has been made.

Let $\overline{V}_{r,t}(\overline{y}) = \overline{V}_{r,t}y$. The continuous cochain isomorphism

$$\mathcal{E}_r \longrightarrow \mathcal{E}_t, \quad (b, \overline{y}) \mapsto (b, \overline{V}_{r,t}\overline{y}) \quad (\text{GDC4.6})$$

is identity on the other current terms. The differential commutes because $a_t \overline{V}_{r,t} = a_r$; its inverse uses $V_{t,r}$. These isomorphisms satisfy the cocycle law, fix the entire nearby coefficient B , and fix the supported kernel $\overline{H}_L$. Thus the parameter-independent costalk quotient has an explicit receiver-level comparison, not merely matching dimensions.

These are the reduced local summand's formulas, in the precise sense of RGR1.9. Both original endpoint lines are restored by the same explicit direct summand as RGR10:

$$\mathcal{E}_{r,\text{full}} = \mathcal{E}_r \oplus i_* E'_p[-1], \quad i^! \mathcal{E}_{r,\text{full}} = [\overline{H} \xrightarrow{a_r} B] \oplus E'_p[-1]. \quad (\text{GDC4.7})$$

Thus degree zero is still $\overline{H}_L$, while degree one is $(B/jI_r) \oplus E'_p$, and its separated version is $(B/A_O^\perp) \oplus E'_p$. Here E'_p keeps its two labelled source coordinates, value at zero and real integral. All parameter isomorphisms and receiving maps below extend by the identity on this endpoint summand. At the other pole the coordinate is $1/z$, with its own positive angular orientation as in RGR10. Neither endpoint line is absorbed into the defect quotient.

GDC5. The universal positive quotient occurs as a supported summand

PTQ proves that the universal bounded positive-adjoint receiver for all actual integer transfers is

$$q_+ : H \rightarrow H/H_O =: H_{\text{pos}}, \quad u : H_{\text{pos}} \xrightarrow{\sim} H_L, \quad u(q_+ y) = P_L y. \quad (\text{GDC5.1})$$

Its original degree remains n : $(T_n^{\text{pos}})^* = U_n^{\text{pos}}$ and $U_n^{\text{pos}} T_n^{\text{pos}} = nI$. Since a_r vanishes on $\overline{H_L}$, the orthogonal decomposition of its domain gives the literal complex decomposition

$$\boxed{\mathcal{E}_r = i_* \overline{H_L} \oplus \mathcal{E}_r^O, \quad \mathcal{E}_r^O = \text{Cone}(-\delta_p a_r|_{\overline{H_O}})[-1].} \quad (\text{GDC5.2})$$

The first summand is in degree zero. Its inclusion sends $\overline{q_+ y}$ to $(0, \overline{P_L y})$, using u . Its projection sends $(b, \overline{y})$ to $\overline{q_+ y}$. Their composition is identity; substitution into (GDC4.1) verifies both cochain maps. The complement has zero degree-zero costalk kernel, since A_H , j , and $D_r|_{H_O}$ are injective. Its degree-one costalk is still B/jI_r .

This identifies the maximal supported source on which this particular Gysin obstruction vanishes. Any vector subspace of $\overline{H}$ has zero restricted obstruction exactly when it is contained in $\overline{H_L}$, by oriented Gysin adjunction and injectivity of jA_H . It is the conjugate of the PTQ receiver through (GDC5.1), not the kernel of the total residue map. The total residue map is nonzero on actual line values, as RGR4 proves.

GDC6. The full dual-extension rows, before and after separation

The arithmetic receiver $\mathcal{K}_r$ is RGR9's fibre with nearby coefficient Q' , supported coefficient $\overline{H}$, and variation $+\sigma_r$. Its costalk is $[\overline{H} \xrightarrow{\sigma_r} Q']$. The existing original dual row produces

$$0 \rightarrow Q'/I_r \xrightarrow{[\lambda] \mapsto [j\lambda]} B/jI_r \xrightarrow{[b] \mapsto k'b} J' \rightarrow 0. \quad (\text{GDC6.1})$$

This is an exact row of vector spaces with continuous maps for the indicated quotient topologies. To check its middle kernel, $k'b = 0$ gives $b = j\lambda$; its left kernel is zero by injectivity of j . Surjectivity is continuous Hahn–Banach extension from the closed subspace $J \subset A$. The row is the actual pushout of the transpose of e_0 , as witnessed by $[(b, [\lambda])] \mapsto [b + j\lambda]$.

Passing to the explicitly calculated separated quotients gives another exact row,

$$\boxed{0 \rightarrow Q'_\beta/S_O \xrightarrow{\bar{j}} B/A_O^\perp \xrightarrow{\bar{k}'} J'_\beta \rightarrow 0.} \quad (\text{GDC6.2})$$

The left and middle spaces here have their quotient locally convex topologies. Exactness follows from precisely the same calculation, since $A_O^\perp = jS_O \subset jQ'$. The left map is a closed topological embedding: quotienting the closed embedded subspace $jQ' \subset B$ by its subspace jS_O gives the induced subspace topology in B/jS_O . Indeed an open set $U \subset B$ satisfies $q(U) \cap q(jQ') = q(U \cap jQ')$, because the quotient kernel lies in jQ' . Its image is closed because jQ' is closed.

The natural vertical maps from (GDC6.1) to (GDC6.2) have kernels

$$S_O/I_r \quad \text{and} \quad A_O^\perp/jI_r, \quad (\text{GDC6.3})$$

which are identified by j . They are the respective closures of zero and have been retained, not treated as nonexistent cohomology. The strong openness needed for the separated extension is proved directly in GDC7.

GDC7. The separated row is the transpose of an actual restricted original extension

The inclusion $A_O = q^{-1}N_O$ constructs the strict Fréchet row

$$e_O : \quad 0 \rightarrow J \xrightarrow{k_O} A_O \xrightarrow{q_O} N_O \rightarrow 0. \quad (\text{GDC7.1})$$

Both subspaces are closed; q_O is onto by its preimage definition. Its quotient topology is the subspace topology of N_O : if $U \subset A$ is open, then $q(U \cap A_O) = q(U) \cap N_O$, since any representative of a class in N_O belongs to A_O . Thus it is the pullback of the actual original extension along $N_O \hookrightarrow Q$, with no chosen section.

Continuous restriction gives the following bijections; their strong topological inverses are proved below:

$$\begin{aligned}\alpha : Q'_\beta/S_O &\longrightarrow (N_O)'_\beta, & \alpha[\lambda] &= \lambda|_{N_O}, \\ \eta : B/A_O^\perp &\longrightarrow (A_O)'_\beta, & \eta[b] &= b|_{A_O}.\end{aligned}\tag{GDC7.2}$$

They are surjective by Hahn–Banach and injective by their displayed kernels. They are continuous because a bounded subset of a closed subspace is bounded in the ambient space. Their exact commutative diagram is

$$\begin{array}{ccccccccc} 0 & \rightarrow & Q'_\beta/S_O & \xrightarrow{\bar{j}} & B/A_O^\perp & \xrightarrow{\bar{k}'} & J' & \rightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \eta & & \parallel & & \\ 0 & \rightarrow & (N_O)' & \xrightarrow{q'_O} & (A_O)' & \xrightarrow{k'_O} & J' & \rightarrow & 0.\end{array}\tag{GDC7.3}$$

Every displayed map is continuous for the topologies just specified. Both rows are algebraically exact. In particular the middle separated costalk is explicitly related to the dual of a source that still contains the complete original J . The following argument proves both rows strict and both vertical arrows topological isomorphisms.

The map q'_O is also a closed strong embedding. To verify this rather than assuming it from surjectivity, a bounded set of N_O has compact closure: it is a bounded subset of the Montel space Q , and N_O is closed. SDT4's finite-choice compact-lift construction applies to the strict quotient (GDC7.1) of Fréchet spaces, using its increasing quotient seminorms. It lifts that compact closure to a compact subset of A_O , and hence gives bounded lifts of the original bounded set. For each bounded $C \subset N_O$ with bounded lift $\tilde{C} \subset A_O$,

$$\sup_{x \in C} |\lambda(x)| = \sup_{a \in \tilde{C}} |(q'_O \lambda)(a)|.\tag{GDC7.4}$$

Together with the continuous-forward seminorm estimate this proves the strong embedding. Its image is $J^\perp$ inside $(A_O)'$, hence closed.

Here is the required restriction-topology argument for the actual spaces. Let X be any one of A, Q, A_O, N_O , with its established Fréchet topology, and let $L \subset X$ be one of its closed subspaces used below. Every bounded subset of X has compact closure: SDT proves this for A, Q , and it passes directly to their closed subspaces. The closed absolutely convex hull of a bounded subset is still bounded, because every continuous seminorm is bounded on that hull. Its closure is therefore compact as well.

The restriction

$$R_L : X'_\beta \longrightarrow L'_\beta\tag{GDC7.5}$$

is continuous since bounded sets in L remain bounded in X , and surjective by continuous complex Hahn–Banach extension. We prove it is open. For a compact absolutely convex $K \subset X$, put

$$\mathcal{U}_K = \{\mu \in X' : \sup_{x \in K} |\mu(x)| < 1\}, \quad \mathcal{V}_K = \{\lambda \in L' : \sup_{x \in K \cap L} |\lambda(x)| < 1/2\}.$$

Such $\mathcal{U}_K$ form a base of strong neighborhoods after scaling and finite unions of bounded sets followed by closed absolutely convex hull. The set $\mathcal{V}_K$ is a strong neighborhood since $K \cap L$ is bounded in L .

Fix $\lambda \in \mathcal{V}_K$. There is an open absolutely convex neighborhood U of zero in X for which

$$|\lambda(x)| < 1 \quad (x \in (K + U) \cap L).\tag{GDC7.6}$$

If not, choose a decreasing neighborhood base U_j , and $x_j = k_j + u_j \in L$ with $k_j \in K$, $u_j \in U_j$ and $|\lambda(x_j)| \geq 1$. Compactness supplies a subsequence $k_j \rightarrow k$; then $u_j \rightarrow 0$, so $x_j \rightarrow k$. Closedness gives $k \in K \cap L$, and continuity on L contradicts $|\lambda(k)| < 1/2$. This proves (GDC7.6).

The Minkowski functional p of the open absolutely convex absorbing set $K + U$ is a continuous seminorm. Scaling (GDC7.6) shows $|\lambda(x)| \leq p(x)$ for every $x \in L$: test each positive $a > p(x)$, for which $x/a \in K + U$,

and let a decrease to $p(x)$. Complex Hahn–Banach now gives a continuous extension μ with $|\mu| \leq p$ on X . Since K is compact and lies in the open unit ball of p ,

$$\sup_{x \in K} |\mu(x)| \leq \max_{x \in K} p(x) < 1.$$

Thus $\mathcal{V}_K \subset R_L(\mathcal{U}_K)$, proving strong openness. This proves

$$X'_\beta / L^\perp \xrightarrow[\text{restriction}]{\sim} L'_\beta \quad (\text{GDC7.7})$$

with the actual quotient and strong dual topologies.

Restriction also admits bounded lifts of bounded sets, without a linear section. A strongly bounded family $\mathcal{C} \subset L'$ is pointwise bounded. The Baire argument on the Fréchet space L makes it equicontinuous: the closed sets where $\sup_{\lambda \in \mathcal{C}} |\lambda(x)| \leq j$ cover L , one has interior, and taking differences gives a neighborhood of zero with a uniform bound. Consequently there are a continuous seminorm p_X on X and $C > 0$ such that $|\lambda(x)| \leq Cp_X(x)$ for all $\lambda \in \mathcal{C}, x \in L$. Choose each Hahn–Banach extension with this same bound. On every bounded $B_0 \subset X$, their values are bounded by $C \sup_{B_0} p_X$, so the chosen family is strongly bounded in X' .

Apply (GDC7.7) to $(X, L) = (Q, N_O)$ and (A, A_O) . This proves that α, η in (GDC7.3) are strong topological isomorphisms. Apply it also to (A_O, J) ; then k'_O is strongly open. Together with the closed embedding q'_O , both rows in (GDC7.3) are strict exact sequences of the specified locally convex spaces. In particular,

$$\boxed{(H^1(i^! \mathcal{E}_r))_{\text{sep}} \simeq (A_O)'_\beta.} \quad (\text{GDC7.8})$$

The unseparated cohomology B/jI_r , its closure of zero, and every map in (GDC6.3) remain part of the construction. Equation (GDC7.8) does not identify that original unseparated quotient with its separated quotient.

GDC8. The induced original residue map is exactly the positive compression

Let $p_S : Q' \rightarrow Q'/S_O$ be the quotient. Since $A_H(H_O) \subset S_O$,

$$p_S A_H y = p_S A_H P_L y. \quad (\text{GDC8.1})$$

The right-hand map is injective on H_L : an element $A_H x$, $x \in H_L$, annihilating N_O pairs to zero with every line-value isolator, so all coordinates of x vanish. Thus

$$\overline{H_{\text{pos}}} \longrightarrow Q'/S_O, \quad \overline{q_+ y} \mapsto p_S A_H y \quad (\text{GDC8.2})$$

is a well-defined continuous injective linear map. It is not asserted to be a strong topological embedding or onto the entire coefficient quotient.

For $F \in N_O$, the exact evaluation after restriction is

$$(\alpha p_S A_H y)(F) = \sum_{\rho \in \mathcal{Z}_L} m_\rho F(\rho) \overline{y_\rho} = \langle EF, P_L y \rangle_+. \quad (\text{GDC8.3})$$

This is a direct original-residue realization of the positive receiver, with no nonholomorphic source projector. The source N_O maps densely into H_L , because every finite line-value isolator lies in N_O . Its kernel is exactly N_0 . Thus N_O/N_0 , equipped with its explicit value norm, completes to H_L ; its Fréchet quotient topology is not equated with that norm.

For the original total map $\sigma(\overline{G}) = A_H EG$, the composite $p_S \sigma$ has exactly the kernel $\overline{N_L}$, by (GDC8.3) and those isolators. Its strong closure in the quotient is

$$\boxed{\overline{p_S \sigma(\overline{Q})} = N_0^\perp / S_O.} \quad (\text{GDC8.4})$$

To prove this, RTT/RGR8 gives $\overline{\sigma(Q)}^\beta = N_0^\perp$. This closed subspace contains S_O , since $N_0 \subset N_O$. Its quotient is a closed subspace of Q'/S_O ; continuity of the quotient and density give exactly (GDC8.4).

The remaining quotient in the exact row

$$0 \rightarrow N_0^\perp/S_O \rightarrow Q'/S_O \rightarrow Q'/N_0^\perp \rightarrow 0 \quad (\text{GDC8.5})$$

retains the functionals on the original value-zero source, through the strong topological restriction isomorphism $Q'/N_0^\perp \rightarrow (N_0)_\beta'$ supplied by (GDC7.7) with $X = Q, L = N_0$. Thus the coefficient quotient has not erased the higher jets simply because its residue part observes only values.

Evaluation of an arbitrary class of Q'/S_O is defined on N_O , not on all of Q . This domain is essential. On the image of the original residue map the canonical representative $A_H P_L E G \in Q'$ does produce the full positive compression on all original tests; that representative is supplied by the proved Hilbert map. We do not extend it to an unproved projection on all Q'_β .

GDC9. The remaining source compatibility is preserved exactly

The restricted source and the positive source quotient are related by

$$0 \rightarrow N_O/N_0 \rightarrow Q/N_L \rightarrow Q/(N_L + N_O) \rightarrow 0. \quad (\text{GDC9.1})$$

The first map sends the same class to its class modulo N_L ; its kernel is $N_L \cap N_O = N_0$. Its image is $(N_L + N_O)/N_L$, proving the last cokernel. All maps are continuous for their subspace/quotient topologies, and the last quotient is retained even if unseparated.

This is the value version of PSC's reconstruction defect. It proves the strongest immediate source map supplied by (GDC8.3) while keeping its cokernel, rather than assuming that every positive value quotient has a representative vanishing at all off-line zeros. Its image is dense after the explicitly specified Hilbert completion, since it contains every finite retained value vector. Neither surjectivity in (GDC9.1) at the first arrow nor primal topological density is inferred from that Hilbert statement.

The whole Weil form remains the exact sum

$$W(F, G) = \langle P_L E F, P_L E G \rangle_+ + \langle P_O E F, J P_O E G \rangle_+. \quad (\text{GDC9.2})$$

The second term need not descend to Q/N_L : on each actual off-line reflected pair it has the two signed values $+2m_\rho$ and $-2m_\rho$ of PTQ5. Their existence as formulas does not assert that such a pair exists. The restriction $F \in N_O$ kills this term by an exact source condition. It does not prove that it is zero on the entire original source.

GDC10. A constructed coefficient map kills the defect while retaining the positive residue

The continuous restriction

$$R_O : B \rightarrow (A_O)_\beta', \quad R_O b = b|_{A_O} \quad (\text{GDC10.1})$$

has kernel exactly $A_O^\perp = \overline{a_r(\overline{H})}^\beta$. Therefore it kills the actual Gysin coefficient a_r , and kills exactly its closed linear envelope. Unlike restriction to J alone, it retains the line-residue part in (GDC8.3), transported by the injective q'_O .

There is a literal continuous-current version. Inclusion $\iota_O : A_O \hookrightarrow A$ induces continuous maps on compactly supported smooth test forms. Their transposes give a cochain map

$$\iota_O^\vee : \mathcal{T}_A \rightarrow \mathcal{T}_{A_O}. \quad (\text{GDC10.2})$$

It commutes with the current differential and sends $\delta_p b$ to $\delta_p R_O b$. Hence

$$\mathcal{E}_r \rightarrow \mathcal{T}_{A_O}[-1] \oplus i_* \overline{H}, \quad (b, \overline{y}) \mapsto (\iota_O^\vee b, \overline{y}) \quad (\text{GDC10.3})$$

is a cochain map: its target has zero attaching map, because $R_O a_r = 0$. The coefficient space A_O is closed Fréchet and complete, so RGR's compact-test current contraction applies to it, giving the actual target $(A_O)'[1] \oplus i_* \overline{H}$. This is an exhibited receiving map, not merely a proposed future nullification.

For precision about universality, set $B_{\text{sep}} = B/A_O^\perp$. Every continuous linear map $f : B \rightarrow K$ into a Hausdorff locally convex space that kills a_r kills its closure and factors uniquely and continuously through B_{sep} . The quotient topology proves continuity of the factor. Thus the separated coefficient quotient is the universal such coefficient target. In vector-space sheaves its zero attaching map gives the split receiver

$$\mathcal{E}_{r,\text{sep}} = \underline{B_{\text{sep}}}[1] \oplus i_* \overline{H}.$$

The quotient map induces an exact triangle

$$\underline{A_O^\perp}[1] \longrightarrow \mathcal{E}_r \longrightarrow \mathcal{E}_{r,\text{sep}} \longrightarrow \underline{A_O^\perp}[2]. \quad (\text{GDC10.4})$$

To verify it, use the two Gysin triangles with common supported term and the coefficient quotient $B \rightarrow B_{\text{sep}}$. Its fibre is $A_O^\perp$ in degree zero. Cancelling the common supported term leaves that coefficient fibre shifted by one. This calculation takes place in vector-space sheaves; it does not assert exactness of an unspecified derived category of locally convex spaces. The separately constructed map (GDC10.3) gives the actual continuous-current comparison with the strong dual $(A_O)'$.

The original nonzero obstruction, if present, is thereby killed by a specified coefficient quotient with its entire kernel retained in (GDC10.4). This is not a null-homotopy of the original attaching map in its unchanged coefficient sheaf. It is the exact next construction yielded by its measured closure, together with a retained positive residue and the dual source extension (GDC7.3).

GDC11. Every full multiplicity block and original residue factor

Let an actual zero have multiplicity m . On its globally realized finite local block $\mathbb{C}[t]/t^m$, the condition defining N_O is

$$N_O|_\rho = \begin{cases} \mathbb{C}[t]/t^m, & \rho \in \mathcal{Z}_L, \\ t\mathbb{C}[t]/t^m, & \rho \in \mathcal{Z}_O. \end{cases} \quad (\text{GDC11.1})$$

Thus the closed defect image removes no dual jet at a line point and removes exactly the value-functional line at an off-line point. In Q'/S_O , all m finite dual-jet directions at a line point and the $m - 1$ derivative directions at an off-line point remain. Independence can be checked against their actual primal full-jet isolators, which lie in N_O for exactly these stated directions. These finite-block assertions do not decompose the infinite source as an unrestricted product.

The actual residue map continues to use original ζ :

$$\mathcal{R}(F, h) = \sum_\rho \text{Res}_{s=\rho} \frac{F(s)h(1-s)}{\zeta(s)} ds. \quad (\text{GDC11.2})$$

Writing $\zeta(\rho + t) = t^{m_\rho} u_\rho(t)$, the finite residue numerator for (GDC1.2) is exactly

$$h_{r,C}(y) = \sum_{\rho \in C} m_\rho (-1)^{m_\rho - 1} u_\rho(0) \overline{d_r(\rho^\#) y_{\rho^\#}} e_{1-\rho, m_\rho - 1}, \quad C \subset \mathcal{Z} \text{ finite.} \quad (\text{GDC11.3})$$

Every factor is retained. The sign from reflection of the residue jet is $(-1)^{m_\rho - 1}$; the additional sign of (GDC1.1) is separate. At the numerator's complementary point $1 - \rho$, the highest-jet line pairs to the value functional with exactly this full local unit factor. All derivatives of the unit remain in the original full residue pairing; (GDC11.3) records the particular rank-one numerator, not a replacement of that pairing matrix.

For a bounded test set $K \subset Q$, the tail of the residue functionals is bounded by

$$\sup_{F \in K} \|EF\|_+ \left(\sum_{\rho \notin C} m_\rho |d_r(\rho^\#) y_{\rho^\#}|^2 \right)^{1/2}. \quad (\text{GDC11.4})$$

It tends to zero along all finite subsets, since $D_r y \in H$. This proves the exact strong residue realization, without assuming that the finite primal numerators converge in Q .

The coefficient comparison still retains

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_0(0) = F_0(1) = \frac{1}{8},$$

$$F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) \quad (k \geq 1). \quad (\text{GDC11.5})$$

Nothing here changes the original unit term, Euler prime-power weights, Gamma terms, endpoint terms or finite trivial-divisor sums of PSC5. The coefficient quotient's positive residue corresponds to that full arithmetic identity with its explicit off-line term retained; it supplies no replacement prime formula with that term silently omitted.

GDC12. Exact coefficient actions and degree scope

For every recovered positive integer n , the original source operators act by $T_n F(s) = n^s F(s)$ and $U_n = nT_{1/n}$. They preserve N_O, N_L, N_0, A_O and every full-jet ideal. Let the same notation on A denote the actual conjugate by Θ . Define on the dual coefficients

$$\mathcal{T}_n^Q = n(T_{1/n}^Q)', \quad \mathcal{U}_n^Q = (T_n^Q)', \quad \mathcal{T}_n^B = n(T_{1/n}^A)', \quad \mathcal{U}_n^B = (T_n^A)'. \quad (\text{GDC12.1})$$

All are strongly continuous, because their primal operators take bounded sets to bounded sets. The original degree remains $\mathcal{U}_n \mathcal{T}_n = \mathcal{T}_n \mathcal{U}_n = nI$.

The exact covariance is

$$\sigma_r \bar{T}_n = \mathcal{T}_n^Q \sigma_r, \quad \sigma_r \bar{U}_n = \mathcal{U}_n^Q \sigma_r, \quad j \mathcal{T}_n^Q = \mathcal{T}_n^B j, \quad j \mathcal{U}_n^Q = \mathcal{U}_n^B j. \quad (\text{GDC12.2})$$

For example direct substitution into (GDC0.3) gives $A_H T_n = n(T_{1/n}^Q)' A_H$: the conjugated multiplier at $\rho^\#$ is $n^{1-\rho}$. Then $D_r T_n = T_n D_r$ gives the first identity. Applying the same identity at $1/n$ and retaining the leading scalar n proves the second. Naturality of q proves the last two. This proves covariance without confusing an anti-linear map with a complex-linear transpose; $\bar{T}_n$ and $\bar{U}_n$ act on $\bar{H}$.

The images, their closures, and both rows (GDC6.1)–(GDC7.3) inherit these exact actions. The parameter comparisons $V_{r,t}$ commute with the source actions and hence with their receiver isomorphisms. Coefficient naturality commutes with the oriented Gysin map, so all asserted diagrams hold on the fixed disk with the positive sign already proved. These coefficient comparisons do not assert an additional nonintegral cover, or drop the separate angular degree of an actual disk cover. In particular the arbitrary parameter r in the measured defect and the integer cover degree n have not been identified.

GDC13. What this construction contributes to the lifting target

The exact common receiver is now calculated beyond its kernel: its strong closure is $N_O^\perp$, its entire unseparated image is independent of $r > 1$, and the parameter change is a specified cochain isomorphism. Its supported kernel is canonically the conjugate of the universal positive quotient. Its separated costalk is the quotient by $A_O^\perp$, with the original dual extension and all higher jets retained.

The next construction has also been carried out: restricting to the actual source A_O kills the closed defect coefficient, produces the concrete split receiving complex (GDC10.3), and retains the positive residue through (GDC8.3). Its exact cost is the constant coefficient kernel in (GDC10.4), together with the source compatibility quotient (GDC9.1). These are existing-object maps with proved kernels, not assumed interpolation or comaximality. They show precisely which source and quotient enter any subsequent specialization comparison.

Deligne's invariant-cycle obstruction quotient is still the separately derived $\text{im } \partial/\partial j(K)$. The present current map has its own stated source, target and attaching class; it has not been declared that arithmetic quotient.

A vanishing achieved after (GDC10.1) is a proved coefficient-quotient vanishing, with its retained kernel, and does not assert weight-separated vanishing in the original coefficient. The calculation supplies an actual dual-extension and Gysin diagram to compare with that target while preserving the original positive and discarded forms. No existence of an off-line zero, absence of one, simplicity, arbitrary subset interpolation or vanishing of the original defect has been assumed.

GDC14. Exact source coverage and version record

Complete current reads for this derivation:

- CC_RESIDUE_GYSIN_RECEIVER_INDEPENDENT.md, RGR0–RGR12; SHA256 dd0c5fe172b2d4e2c18a337a1a92d06d9a523196f349ffa34bdb6cef1969e814.
- POSITIVE_TRANSFER_UNIVERSAL_QUOTIENT.md, PTQ0–PTQ11; SHA256 b70a8f0edf765dcfb3259735eb4a8f90a31c649ad308f55d821dcf5cccdac930.
- POSITIVE_QUOTIENT_SOURCE_AND_ARITHMETIC.md, PSC0–PSC9; SHA256 2d188377c6b694e363e5b32a331a8af2ef4562d2cb8aab272d0ce1149b6f1171.
- CC_STRONG_DUAL_TOPOLOGY_AND_JETS.md, SDT0–SDT9, with the complete SDT6 proof reread separately to recover the tool-truncated passage; SHA256 7f1a03a6225f1dfc22b167713c0ddbb154635b06f9479f6137b229845b67aefc.

Retained complete independent derivation: RESIDUE_TRACE_TRANSFER_INDEPENDENT.md, RTT0–RTT11; SHA256 6a83d81039b4eb3e49c4054b4ff157896709647d6a5546751362b05a21de25b2. The exact original-zeta residue, strong-dual topology, and Hilbert receiver formulas used from it are restated and proved in the receiving calculations above. RGR records the original Connes–Consani source provenance and its actual geometric transport; no new whole-paper reading of Connes–Consani or Deligne is claimed here.

The independent derivation initially created only this proof file. The root then verified GDC0–GDC14, corrected the displayed costalk spacing command, added the independently derived explicit parameter bounds and proved the full restriction-topology strengthening in GDC7. The pre-strengthening version is preserved under history/before_gysin_topology_strengthening_20260924. The independent review session supplied the separate mathematical derivations of the scale ratio and strong restriction theorem but ended before completing an audit of this final written version. Final-version review coverage must therefore be reported as complete root verification, with those independent constituent checks, rather than a complete independent final-version audit.

Mathematical verification of the complete Gysin comparison

24 September 2026. Complete root verification of GDC0–GDC14 in GYSIN_DEFECT_CLOSURE_AND_POSITIVE_RECEIVER.md, SHA256 5d0f471b3250e4ee894caca68232c556d64a137a49a9fe25a641db61b94f79ab. This includes the typesetting repair placing parentheses around N_0 before its strong-dual subscript.

Coverage and review provenance

The root read every section and formula of the independent first derivation, then checked and integrated the complete restriction-topology proof, explicit bounds, and restoration of both endpoint lines. A separate mathematical reviewer read complete RGR0–RGR12 and independently derived the scale ratio, strong annihilator argument and strong-open restriction theorem. That session ended before a complete audit of the final written GDC version. This note records complete root verification and those independent constituent derivations; it does not claim a complete independent final-version audit.

The received RGR source has SHA256 dd0c5fe172b2d4e2c18a337a1a92d06d9a523196f349ffa34bdb6cef1969e814. Its complete text and originating task’s review receipt are retained under sources/programme/cc_sheaf. Root read RGR0–RGR12 completely. PTQ0–PTQ11 and PSC0–PSC9 are already present with their precise review scopes. SDT supplies the proved topology of the actual coefficient spaces; no new complete reading of a human paper is claimed here.

Coefficients, conjugation and closure

GDC0 uses the original source row $0 \rightarrow J \rightarrow A \rightarrow Q \rightarrow 0$ after complete arithmetic reconstruction. It assigns no coefficient value or operation to tau. In

$$A_H(y)(F) = \sum_{\rho} m_{\rho} F(\rho) \overline{y_{\rho\#}},$$

the Hilbert inner product is linear in the first variable. Thus A_H is anti-linear and $\sigma_r : \overline{H} \rightarrow Q'$ is linear. Reflection fixes γ , changes σ to $1 - \sigma$, and negates d_r , producing exactly the minus sign in GDC1.2.

The coordinate choice $y_{\rho\#} = 1/(m_{\rho} d_r(\rho\#))$ gives the value functional at each actual off-line point, because conjugating the resulting product still gives 1. This proves the preannihilator N_O , without unrestricted source interpolation. SDT’s strong-bidual theorem makes separation of the strong closure an evaluation at an actual $F \in Q$; that proves $\overline{I}_r^{\beta} = N_O^{\perp}$. The closed strong embedding q' then proves the ambient closure $A_O^{\perp}$. Neither image is presumed closed.

Scale maps and supported complexes

The unscaled differences factor as

$$r^{\sigma} - r^{1-\sigma} = (\sigma - \tfrac{1}{2}) \int_0^1 (\log r) \left(r^{1/2+u(\sigma-1/2)} + r^{1/2-u(\sigma-1/2)} \right) du.$$

The two positive bounds in GDC3.2a control the actual diagonal ratio and its reciprocal on the entire open strip. Setting its line value to 1 preserves bounded invertibility; no analytic dependence on a hypothetical moving zero is required. Direct coordinate multiplication proves $D_r = D_t V_{r,t}$, the cocycle identity, and equality of the images before closure. The parameter 1 is explicitly excluded.

The current differential in GDC4.1 uses $+\delta_p a_r$. Therefore the map to scale t is identity on currents and $\overline{V}_{r,t}$ on the supported coefficient. Substitution proves it is a cochain isomorphism with the specified inverse. The positive-Dirac costalk is $[\overline{H} \xrightarrow{+a_r} B]$; the literal shifted-cone target coordinate has the opposite sign, exactly as in RGR6. The coefficient reflection sign is unrelated to this cone-coordinate comparison.

The domain decomposition supplies the literal direct summand $i_* \overline{H_L}$, and the complement has injective costalk differential. Thus $H^0(i^! \mathcal{E}_r)$ is exactly the conjugate of the universal positive receiver. The raw

degree-one quotient is retained. Restoring $i_*E'_p[-1]$ adds both original endpoint lines in degree one, with all receiving maps identity on them.

Topology of the original dual extension

The source row $0 \rightarrow J \rightarrow A_O \rightarrow N_O \rightarrow 0$ is strict by the preimage definition and openness of the original quotient. The transpose's left strong embedding follows from SDT's constructive compact lifting, applied to this actual quotient. The unseparated pushout row GDC6.1 follows by direct kernel and image calculation. Its closure-of-zero row is retained explicitly.

For the stronger right-hand topology, the proof applies to each actual Fréchet space X whose bounded sets have compact closure and each stated closed subspace L . For every compact absolutely convex $K \subset X$, an element $\lambda \in L'$ bounded by $1/2$ on $K \cap L$ extends to an element of X' bounded strictly by 1 on K . The proof of GDC7.6 is valid: a contrary sequence approaches a point of $K \cap L$, and continuity of λ there contradicts its bound. The gauge of $K + U$ is a continuous seminorm. Complex Hahn–Banach extends the functional with this same gauge bound; compactness makes the bound on K strictly below 1 . These polars form a base of strong neighborhoods, proving openness.

For bounded lifts of dual families, strong boundedness implies pointwise boundedness. The given Baire argument makes the family equicontinuous on the closed Fréchet subspace. Its inherited topology supplies one ambient seminorm controlling the family. Extending every member with this same seminorm gives a strongly bounded family in the ambient dual. This proves the asserted bounded lifting without constructing a linear section.

It follows that the maps α, η in GDC7.3 are actual strong topological isomorphisms and both rows are strict. This is a proved strengthening of the initial draft's continuous bijections. It does not replace the unseparated quotient by the separated one.

Residue, jets and receiving triangle

On N_O , the quotient residue is exactly the positive line sum. Line isolators prove injectivity of the map from H_{pos} ; they do not prove surjectivity onto the full source dual. The closure of the original residue image is $N_0^\perp/S_O$, whose complementary quotient retains all functionals on N_0 . The source compatibility quotient $Q/(N_L + N_O)$ remains in the exact row GDC9.1.

The compact-test inclusion $A_O \hookrightarrow A$ transposes to the displayed continuous cochain map of current complexes. Restriction kills exactly $A_O^\perp$, the closed defect image, so the target attaching map is zero. The source complex is not declared split. In the vector-sheaf category, the comparison of the two fibre triangles with identical supported term gives the retained coefficient fibre $A_O^\perp[1]$ and the exact triangle GDC10.4. This is compatible with the independently exhibited current map and the proved strong topological identification of its coefficient quotient.

At every actual multiplicity block, the off-line condition kills its value and retains its $m_\rho - 1$ higher-jet coordinates; at line points it retains all m_ρ coordinates. Original residue unit, complementary point, signs and multiplicity factors in GDC11.3 agree with RGR2.7. The Cauchy–Schwarz tail estimate proves strong convergence of the residue functionals, not convergence of primal finite numerators.

Finally $A_H T_n = n(T_{1/n})' A_H$ follows directly by conjugating the multiplier at $\rho^\#$, yielding $n^{1-\rho}$. This gives GDC12's source-conjugate covariance and full degree n . No noninteger geometric cover or extra angular degree is inferred. Original ζ , full F_0 comparison, endpoint and trivial-zero values and the two Weil contributions remain as written.

The outcome is the actual closure, dual source, scale isomorphism and quotient receiving map. It does not establish vanishing of the original Gysin defect or identify this analytic coefficient construction with Deligne's complete arithmetic weight argument.

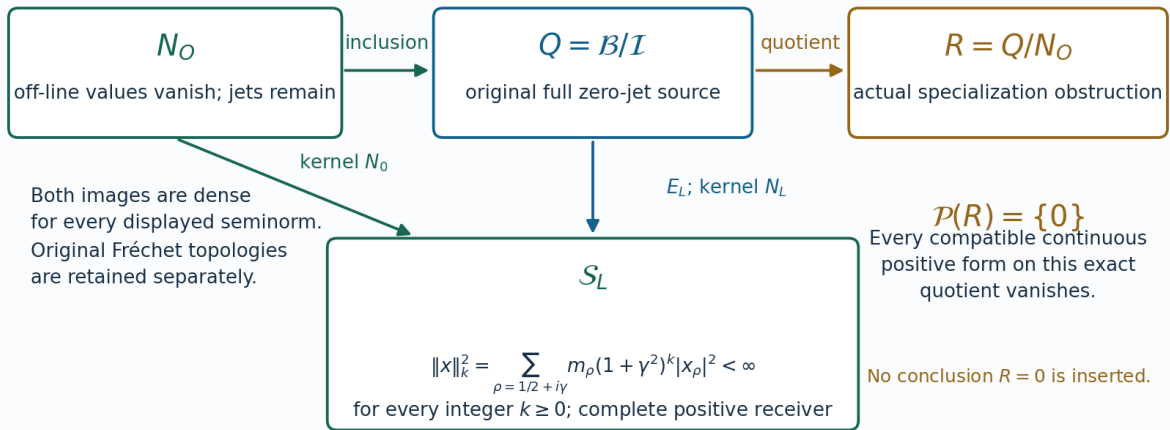
Continuous positivity on the full source and the specialization obstruction

The complete receiving sources follow first: ACD identifies the actual continuous dual, ADC its two invariant-cycle crosses, DCA every genuine finite-sheet cover and trace map, and AST the exact specialization quotient with its strong topology. Their full source, endpoints, signs, multiplicities and comparison maps are retained.

The new CFP proof classifies every continuous positive transfer-adjoint form on the actual Fréchet original-zeta source. It proves the required unitary spectral decomposition through an explicit Gaussian-measure argument; no density of finite primal jets or boundedness in the earlier value norm is assumed. The complete original multiplier and all exceptional terms are retained. CPS then constructs the common positive completion and proves that the actual specialization obstruction quotient has only the zero compatible positive form. Its original coefficient and strong dual remain present.

All continuous positive transfer-adjoint forms on the original source

$B(T_n F, G) = B(F, nT_{1/n} G)$ for every recovered integer n ; all multiplicities retained



$$B_c(F, G) = \sum_{\rho \in \mathbb{Z}_L} m_\rho c_\rho F(\rho) \overline{G(\rho)}, \quad c_\rho \geq 0 \text{ with polynomial growth}$$

$W = W_L + W_O$ remains the original Weil form. The positive receiver returns W_L ; the full W_O is retained.

Exact comparison diagram, not a zero-location plot. Full proofs: CFP and CPS1–CPS6; source quotient: AST1–AST5.

Coefficient source: Connes–Consani, arXiv:0903.2024v3, §5. No operation or coordinate is assigned to τ .

Figure 14: Exact source and positive-receiver maps. All positive continuous forms are classified in CFP; CPS1–CPS5 give their common radical, completion and the zero positive receiver of the actual specialization obstruction. Both maps into the lower space have dense image for the displayed seminorms. The original Fréchet topologies, source kernels and full original Weil complement remain distinct and explicit. Source quotient and Gysin comparison: AST1–AST8.

The complete proofs and the precise independent/root review scopes are included. These results close the restriction of the earlier PTQ classification to bounded Hilbert forms. The original Deligne weight-separation target remains active and unfinished; none of the positive receiving maps has been declared faithful on an unproved part of the original source.

The adjoint-defect cone: continuous duality and the full original-source comparison

Complete derivation, with proof locators **ACD0**–**ACD8**. The objects and all maps below use the already reconstructed original arithmetic. The calculation constructs the duality comparison, then follows the same boundary through both poles and the genuine geometric covers.

ACD0. Input stage and conventions

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. No addition, numerical origin, metric, midpoint, or vector coordinate is assigned to it. Complete-history reconstruction precedes the coefficient spaces below. Winding information and the carried grading remain in the original source; the two branches retain their own recovered counters. The coordinate of a pole disk is a coordinate in the receiving sphere of CSP, whose punctured part lies over the generic point, not a coordinate on this supporting datum.

We use the source and quotient from ADC1 and CSD1, with their proved topologies:

$$S = \{f \in \mathcal{S}(\mathbb{R}) : f(-v) = f(v), f(0) = 0, \int f = 0\}, \quad \Sigma f(u) = 2 \sum_{k \geq 1} f(ku),$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for all } N, j\}, \quad J = \Sigma S, \quad Q = A/J. \quad (\text{ACD0.1})$$

The quotient map is q . The subspace J is closed; the inverse of $\Sigma : S \rightarrow J$ is continuous. The original transform and its full source divisor are

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}. \quad (\text{ACD0.2})$$

These factors are not replaced by ζ or by a different completed function. At the exceptional arguments,

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = F_0(2) = \frac{\pi}{24}, \quad F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k). \quad (\text{ACD0.3})$$

The full jets at an actual nontrivial zero ρ are those of

$$F_0(\rho + t) = t^{m_\rho} \frac{(\rho + t)(\rho + t - 1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho + t)/2) \frac{\zeta(\rho + t)}{t^{m_\rho}}. \quad (\text{ACD0.4})$$

In particular the quotient Q retains all multiplicities and higher jets. The unit and prime-power repetitions remain in the original source identities

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{ACD0.5})$$

Here these are properties of the already recovered arithmetic, not initial labels supplied before its reconstruction.

Let $H = \ell^2(\mathcal{Z}, m)$ for the actual nontrivial zeros, with inner product $\langle x, y \rangle_+ = \sum_\rho m_\rho x_\rho \overline{y_\rho}$, linear in its first variable, and put

$$\rho^\# = 1 - \bar{\rho}, \quad (\mathcal{J}y)_\rho = y_{\rho^\#}, \quad E : Q \rightarrow H, \quad (EF)_\rho = (\Theta_Q F)(\rho). \quad (\text{ACD0.6})$$

The map E is continuous, has dense image containing all finite value vectors, and retains its higher-jet kernel. These are the actual source results RTT/GTAH/ADC1. Fix a recovered coefficient parameter $r > 1$. Define the bounded operators

$$(T_r y)_\rho = r^\rho y_\rho, \quad U_r = r T_{1/r}, \quad D_r = T_r^* - U_r, \quad d_r(\rho) = e^{-i\mathcal{J}\rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}). \quad (\text{ACD0.7})$$

We write $B = A'_\beta$, $C = \overline{H}$, with the strong continuous dual on B . RGR11's actual map is

$$\sigma_r(\overline{y})(F) = \langle EF, JD_r y \rangle_+, \quad a_r = q' \sigma_r : C \rightarrow B. \quad (\text{ACD0.8})$$

No off-line zero is assumed to exist. Images below retain their quotient or ambient subspace topology as explicitly specified; closedness is not inferred from the notation for an image.

ACD1. The two continuous transposes

Classical human-source attribution: the complex Hilbert-space Riesz representation theorem, with the conjugate-space convention, is recorded in Terence Tao, 245B Notes 5, Theorem 1, Remark 2 and Exercise 15. The explicit maps and signs used here remain displayed below.

The linear isomorphism

$$R : C \longrightarrow H'_\beta, \quad R(\overline{y})(h) = \langle h, y \rangle_+ \quad (\text{ACD1.1})$$

is the Riesz isomorphism with the conjugate complex structure explicitly retained. It is isometric for the dual norm. Similarly $C'_\beta \simeq H$ sends h to $\overline{y} \mapsto \langle h, y \rangle_+$. SDT6 proves the strong topological evaluation isomorphism $A \simeq (A'_\beta)'_\beta$. Thus taking the continuous transpose of (ACD0.8) on the entire bidual gives exactly

$$b_r : A \rightarrow H, \quad b_r = D_r^* J E q, \quad \beta_r : Q \rightarrow H, \quad \beta_r = D_r^* J E, \quad b_r = \beta_r q. \quad (\text{ACD1.2})$$

Indeed, for every $f \in A, y \in H$,

$$a_r(\overline{y})(f) = \langle E q f, J D_r y \rangle_+ = \langle D_r^* J E q f, y \rangle_+. \quad (\text{ACD1.3})$$

This proves both $a'_r \text{ev}_A = b_r$ under $C' = H$ and

$$b'_r R = a_r. \quad (\text{ACD1.4})$$

No section of q is used. For a bounded subset $K \subset Q$,

$$\sup_{F \in K} |\sigma_r(\overline{y})(F)| \leq \sup_{F \in K} \|EF\|_+ \|D_r\| \|y\|_+. \quad (\text{ACD1.5})$$

This proves the needed strong continuity; continuity of b_r follows from its three continuous factors.

Write

$$Z_r = \ker \beta_r = \{F \in Q : F(\rho) = 0 \text{ at every actual off-critical zero}\}, \quad C_{\text{crit}} = \overline{\ker D_r}. \quad (\text{ACD1.6})$$

The equality uses (ACD0.7), reflection of the zero set, and the coordinate formula for β_r . All higher-jet classes with zero values belong to Z_r . The exact relations are

$$\ker a_r = C_{\text{crit}}, \quad \ker b_r = q^{-1} Z_r, \quad b_r A = \beta_r Q, \quad b_r = 0 \iff D_r = 0. \quad (\text{ACD1.7})$$

Injectivity of the anti-linear map $y \mapsto \langle E(-), Jy \rangle$ follows from density of $E(Q)$, proving the first equality. For the last equivalence, vanishing of b_r makes the bounded operator $D_r^* J$ zero on that same dense image, and therefore zero on H . These arguments do not discard its kernel or any source jet.

ACD2. A fine resolution of the actual transpose cone

On a positively oriented pole disk D , let $i : \{p\} \hookrightarrow D$. Form the specified sheaf complex

$$\mathcal{P}_r = [A^{-1} \xrightarrow{+b_r \text{ev}_p} i_* H^0]. \quad (\text{ACD2.1})$$

This is the cone presentation of ADC8; the plus sign is part of this representative. Its fine resolution has

$$\mathcal{L}_r^{-1} = \mathcal{E}_A^0, \quad \mathcal{L}_r^0 = \mathcal{E}_A^1 \oplus i_* H, \quad \mathcal{L}_r^1 = \mathcal{E}_A^2,$$

$$d^{-1}f = (-df, b_r f(p)), \quad d^0(\omega, h) = -d\omega. \quad (\text{ACD2.2})$$

The map from (ACD2.1) inserts locally constant functions and the unchanged H coordinate. The coefficient-valued Poincaré contraction proves it is a quasi-isomorphism, including at the pole: the de Rham resolution of $\underline{A}$ and the identity on the skyscraper give the map of the two cones. It neither requires b_r to be onto nor its image to be closed. Smooth partitions of unity act on the forms and by their value at p on the skyscraper. CSD1–2’s cutoff and contraction proof therefore establishes compact-support acyclicity and continuity for these exact terms too.

Define the literal continuous compact-test Hom complex

$$\mathcal{D}_r^k(U) = \text{Hom}_{\text{cont}}(\Gamma_c(U, \mathcal{L}_r^{-k}), \mathbb{C}), \quad d_{\mathcal{D}}^k \ell = (-1)^{k+1} \ell d_{\mathcal{L}}. \quad (\text{ACD2.3})$$

Tests have the fixed-support smooth seminorms and their locally convex inductive limit. Currents restrict by extension-by-zero transpose. A locally finite partition of unity proves the sheaf property exactly as in CSD3. This constructs a sheaf complex; it does not declare continuous dualization exact on all locally convex complexes.

Put $\mathcal{T}_A^j = (\mathcal{E}_c^{-j}(A))'$, $d_T v(\phi) = (-1)^{j+1} v(d\phi)$, and $\delta_p \lambda(f) = \lambda(f(p))$. In degrees $(-1, 0, 1)$, (ACD2.3) is

$$\begin{aligned} \mathcal{T}_A^{-2}, \quad \mathcal{T}_A^{-1} \oplus i_* H', \quad \mathcal{T}_A^0, \\ d_{\mathcal{D}}^{-1} v = (d_T v, 0), \quad d_{\mathcal{D}}^0(u, \lambda) = d_T u - \delta_p b'_r \lambda. \end{aligned} \quad (\text{ACD2.4})$$

For example, on a degree-minus-one test f , the degree-zero Hom differential is $-u(-df) - \lambda(b_r f(p)) = u(df) - \lambda(b_r f(p))$, which is the second formula. The first follows by applying a degree-minus-one functional to $-d\omega$. These two computations fix every sign.

ACD3. The exact current-cone identification and pairing

The previously constructed receiver is

$$\begin{aligned} \mathcal{E}_r = \text{Cone}(-\delta_p a_r)[-1], \\ d_{\mathcal{E}}^{-1} v = (-d_T v, 0), \quad d_{\mathcal{E}}^0(u, x) = -d_T u + \delta_p a_r x. \end{aligned} \quad (\text{ACD3.1})$$

Using (ACD1.1), define the degree-wise map $\mathcal{D}_r \rightarrow \mathcal{E}_r$ by

$$v \mapsto -v \quad (\deg -1), \quad (u, Rx) \mapsto (u, x) \quad (\deg 0), \quad w \mapsto -w \quad (\deg 1). \quad (\text{ACD3.2})$$

It is a continuous chain isomorphism. In degree minus one its two differential composites are $(d_T v, 0)$; in degree zero both are $-d_T u + \delta_p a_r x$, by (ACD1.4). The inverse uses the same signs and R . Thus this is the exact continuous support-dual comparison for the constructed resolution, not a general assertion about dualizing an unseparated image quotient.

The evaluation pairing of $\mathcal{E}_r$ against compact tests of $\mathcal{L}_r$ is accordingly

degree of current	test	pairing
-1	$\eta \in \mathcal{E}_c^2(A)$	$-v(\eta)$
0	$(\omega, h) \in \mathcal{E}_c^1(A) \oplus H$	$u(\omega) + \langle h, y \rangle_+, \quad x = \bar{y}$
1	$f \in \mathcal{E}_c^0(A)$	$-w(f).$

(ACD3.3)

For a current of degree k , substitution into (ACD2.2) and (ACD3.1) gives $\langle dv, c \rangle + (-1)^k \langle v, dc \rangle = 0$. For degree zero, the two terms containing $b_r f(p)$ cancel precisely because of (ACD1.3). Hence (ACD3.3) descends to the supported cohomology pairings.

The positive angular current is $h_p \lambda$, with $d_T h_p \lambda = \delta_p \lambda$. The stalk representatives of $\mathcal{E}_r$ are the constant current c_λ in degree -1 and $(h_p a_r x, x)$ in degree zero. Their restriction is $(1_B, a_r)$. These are exactly RGR/ADC’s representatives: no sign of the point coordinate was changed in (ACD3.2). A compact positive two-form ω_c with integral one represents the compact A class in degree one of $\mathcal{L}_r$; a point vector h represents

its degree-zero H class. Pairing gives $-\lambda(a)$ and $\langle h, y \rangle_+$, respectively. The first minus is retained rather than hidden in an orientation change. CSD2's compact contraction proves these classes compute the entire compact model $H[0] \oplus A[-1]$.

The full endpoint term in the original source is $i_*E_p[1]$, with $E_p = \mathbb{C}^2$ carrying the separate value-at-zero and integral labels. It pairs with $i_*E'_p[-1]$ by evaluation. This summand has no nearby group; its special and supported degrees are -1 in the primal and $+1$ in the dual. Both labels and both poles are retained.

ACD4. Ordinary algebraic duality and the original source map

For the unrestricted algebraic dual, replace $\mathcal{T}_A$ by

$$\mathcal{T}_{A,\text{alg}}^j(U) = \text{Hom}_{\mathbb{C}}(\mathcal{E}_c^{-j}(U; A), \mathbb{C}) \quad (\text{ACD4.1})$$

and H' by H^* . The same Hom differential and degree signs prove the analogous cone formula with attaching map $b_r^* : H^* \rightarrow A^*$. CV2–CV4's compact contraction gives $\mathcal{T}_{A,\text{alg}} \simeq \underline{A}^*[2]$ and identifies its positive Dirac map with Gysin. Together with the exact skyscraper comparison this is the ordinary sheaf Verdier dual of (ACD2.1), through these explicit resolutions. No finite-dimensional bidual identity is used.

There is a literal chain injection from (ACD2.3) to this algebraic Hom model: continuous test functionals are algebraic test functionals, $H' \hookrightarrow H^*$, and $b_r^*|_{H'} = b_r'$. In cone coordinates it is

$$(v, (u, x), w) \mapsto (v, (u, Rx), w). \quad (\text{ACD4.2})$$

Its target has the same degree-sign convention (ACD3.2). The inclusion need not be an isomorphism. This supplies the comparison instead of conflating the two dual categories.

The original reduced local source is $P = \mathcal{F}_{\text{red}}[1] \simeq [\underline{A}^{-1} \xrightarrow{+q} i_*Q^0]$. Its actual map into the new cone is

$$f_r : P \rightarrow \mathcal{P}_r, \quad f_r^{-1} = 1_A, \quad f_r^0 = \beta_r. \quad (\text{ACD4.3})$$

It commutes with the differential by $\beta_r q = b_r$. Add the identity on each original $E_p[1]$ for the full source. On fine resolutions this is 1 on forms and β_r on the point coefficient. Its continuous transpose is therefore 1 on currents and $\beta_r' R = \sigma_r$ on the point functional.

There is a necessary representative sign here. The accepted RGR model of $D_c P$ is $\text{Cone}(+\delta_p q')[-1]$, whereas the literal Hom dual of the plus- q cone just used, after (ACD3.2), is $\text{Cone}(-\delta_p q')[-1]$. Negating its point coordinate and leaving its currents unchanged is a chain isomorphism between these representatives. Hence in the accepted RGR representative the transpose of (ACD4.3) is exactly

$$\mathcal{E}_r \rightarrow D_c P, \quad (u, x) \mapsto (u, -\sigma_r x) \quad \text{in degree zero}, \quad (\text{ACD4.4})$$

with identity on the other current degrees. Substitution gives $-d_T u - \delta_p q'(-\sigma_r x) = -d_T u + \delta_p a_r x$, proving the differential square. Thus this is the previously constructed RGR map, with its actual source and signs, now proved to be the continuous transpose of the original-source map. A sign-free formal slogan about the dual of a cone would not supply this comparison.

ACD5. The specialization image and its full local pairing

The local primal stalk is $[A^{-1} \xrightarrow{b_r} H^0]$, its punctured model is $A[1] \oplus A[\vartheta_p]$, and restriction is identity on the first A and zero on H . The restriction fibre is

$$A \xrightarrow{a \mapsto (b_r a, a)} H \oplus A \xrightarrow{0} A \quad \text{in degrees } -1, 0, 1. \quad (\text{ACD5.1})$$

The quotient coordinate $(h, a) \mapsto h - b_r a$ cancels the first identity pair. Therefore the two positive-orientation localization boundaries are

$$\partial_{-1} = -b_r : A \rightarrow H, \quad \partial_0 = +1 : A[\vartheta_p] \rightarrow A. \quad (\text{ACD5.2})$$

In degree -1 the angular coinvariant term of DC's cross is zero. Its exact obstruction image is consequently $b_r A$, and

$$\text{coker } \text{sp}_{-1} = A / \ker b_r \simeq Q / Z_r, \quad [a] \mapsto -b_r a \quad (\text{ACD5.3})$$

is the actual boundary comparison to the image. The first isomorphism is a topological quotient isomorphism, induced by the quotient map q . Its bijection to $b_r A$ is continuous with the ambient Hilbert subspace topology, but inverse continuity is not asserted. In degree zero the entire punctured term is coinvariant, so the quotient of boundary images is zero. The endpoint degrees are those of ACD3 and the full table ADC8.

The original boundary is $-q : A \rightarrow Q$. The map (ACD4.3) gives its exact commuting square with $-b_r$ by β_r . Thus the map on original local obstruction images is

$$Q \xrightarrow{\beta_r} \beta_r Q, \quad \ker \beta_r = Z_r. \quad (\text{ACD5.4})$$

The complementary current-side source is paired with it by

$$(Q / Z_r) \times (C / C_{\text{crit}}) \rightarrow \mathbb{C}, \quad ([F], [\bar{y}]) \mapsto \langle \beta_r F, y \rangle_+ = \sigma_r(\bar{y})(F). \quad (\text{ACD5.5})$$

Both radicals vanish: the left by the definition of Z_r , the right by injectivity of the map from H to value functionals and (ACD1.7). This is the exact pairing, not a claim that these images exhaust each other's topological duals. The current-side invariant-cycle quotient is zero for every a_r , while its coinvariant-boundary kernel is $a_r C$ (ADC3). Equations (ACD3.2), (ACD4.4) and (ACD5.5) identify the precise duality relation between these different tests.

ACD6. Both poles and the global original-source map

On $Y = \mathbb{P}^1(\mathbb{C})$, use the existing common coefficient trivialization of CSD, with restriction maps $r_+ = \Sigma$ and $r_- = R\Sigma$ into A . Their common image is J , so after the proved source inverses the reduced source is the kernel of $\underline{A}_Y \rightarrow i_+ Q \oplus i_- Q$, with q at both poles. Define the new global cone by the same actual quotient map followed by β_r at each pole:

$$\mathcal{P}_r^Y = [\underline{A}_Y^{-1} \xrightarrow{(b_r, b_r)} i_+ H \oplus i_- H], \quad \mathcal{E}_r^Y = \text{Cone}(-\delta_+ a_r - \delta_- a_r)[-1]. \quad (\text{ACD6.1})$$

The local signs (ACD2–3) glue, giving the same continuous dual comparison globally. Dirac currents at both poles are positive for the complex orientation; the expression in (ACD6.1) is a sum, not an unrecorded cancellation. Add $(i_+ E_+ \oplus i_- E_-)[1]$ in the primal and its full continuous dual shifted by -1 .

Here is a full global cochain calculation. Cover the sphere by the two pole disks with annular overlap. The disk complexes and overlap contraction from ACD5 give the Mayer–Vietoris model

$$A \oplus A \xrightarrow{d^{-1}} H \oplus H \oplus A \xrightarrow{0} A, \quad d^{-1}(x_+, x_-) = (b_r x_+, b_r x_-, x_+ - x_-) \quad (\text{ACD6.2})$$

in degrees $(-1, 0, 1)$. The last A is the positively oriented angular overlap class. It is the sphere's degree-two constant-coefficient class after the shift. Its orientation comparison includes the shift sign: take a partition function χ_+ equal to one near zero and zero near infinity, and $\vartheta = d \arg z / (2\pi)$ on the overlap. In the resolution with differential $-d$, the overlap class ϑa maps to the global two-form $-d\chi_+ \wedge \vartheta a$. Its integral is $-\int_0^\infty \chi'_+(r) dr a = a$. The corresponding unshifted $+d$ comparison is the negative sign recorded in CDW9. Thus no positive angular coordinate has been silently identified with the unshifted positive integration coordinate. The projection to

$$K_r^Y = [A^{-1} \xrightarrow{(b_r, b_r)} H^0 \oplus H^0 \xrightarrow{0} A^1] \quad (\text{ACD6.3})$$

is $p^{-1}(x_+, x_-) = x_-$, $p^0(h_+, h_-, c) = (h_+ - b_r c, h_-)$, and $p^1 = 1$. Its section is $a \mapsto (a, a)$ and $(h_+, h_-) \mapsto (h_+, h_-, 0)$. The homotopy $h^0(h_+, h_-, c) = (c, 0)$, zero in other degrees, satisfies $dh + hd = 1 - ip$. Every identity is verified by inserting (ACD6.2). Thus (ACD6.3) is a continuous cochain equivalence, not only a dimension count.

Its groups, with all endpoint coordinates reattached, are

$$\begin{aligned}\mathbb{H}^{-1}(Y, \mathcal{P}_{r,\text{full}}^Y) &= \ker b_r \oplus E_+ \oplus E_-, \\ \mathbb{H}^0(Y, \mathcal{P}_{r,\text{full}}^Y) &\simeq H \oplus (H/b_r A), \quad \mathbb{H}^1(Y, \mathcal{P}_{r,\text{full}}^Y) = A.\end{aligned}\tag{ACD6.4}$$

The degree-zero isomorphism is $[(h_+, h_-)] \mapsto (h_+ - h_-, [h_-])$, with inverse $(h, [k]) \mapsto [(h+k, k)]$. Changing k by $b_r a$ changes the pair by $(b_r a, b_r a)$. Both maps are continuous for the stated quotient topologies, including when the second factor is not separated.

Replace H, b_r by Q, q to compute the original source. Since q is onto, its degree-zero group is exactly Q , by difference of the two quotient coordinates; its other groups are $J \oplus E_+ \oplus E_-$ and A . The map from the original global source into (ACD6.4) is explicitly

$$J \oplus E_+ \oplus E_- \hookrightarrow \ker b_r \oplus E_+ \oplus E_-, \quad Q \rightarrow H \oplus (H/b_r A), \quad F \mapsto (\beta_r F, 0), \quad A \xrightarrow{1} A.\tag{ACD6.5}$$

The zero second coordinate follows because $\beta_r Q = b_r A$, not because the second quotient is discarded. In degree zero the exact kernel is Z_r , just as for the local specialization image.

For the current dual, the corresponding global finite cochain model is

$$B^{-1} \xrightarrow{0} C^0 \oplus C^0 \xrightarrow{(x_+, x_-) \mapsto a_r x_+ + a_r x_-} B^1,\tag{ACD6.6}$$

with endpoint $E'_+ \oplus E'_-$ in degree one. It follows either by the same global current contraction or by applying (ACD3.2) to the continuous Hom of the explicit finite cochain model (ACD6.3). All homotopies used to get that model are continuous and can be transposed. Its groups are

$$\mathbb{H}^{-1} = B, \quad \mathbb{H}^0 = \{(x_+, x_-) : a_r(x_+ + x_-) = 0\}, \quad \mathbb{H}^1 = (B/a_r C) \oplus E'_+ \oplus E'_-.\tag{ACD6.7}$$

The middle kernel is continuously isomorphic to $C \oplus C_{\text{crit}}$ by $(x_+, x_-) \mapsto (x_+, x_+ + x_-)$; its inverse is $(x, k) \mapsto (x, k - x)$. In particular the anti-diagonal classes are always present. They express cancellation between the two retained pole contributions; they do not imply that either map a_r vanishes. The original global image (ACD6.5) and the local obstruction (ACD5.4) retain precisely the same defect despite this global cancellation.

ACD7. Genuine covers and their transpose actions

For a recovered integer $n \geq 1$, let $f_n(z) = z^n$. On constant coefficients the sheaf unit is the diagonal into the n branches away from a pole. The sheaf trace is branch sum there and multiplication by n on the single constant pole stalk. The latter follows directly from restriction: a constant pole section restricts to the same value on all branches. These formulas define sheaf maps on disks, and agree on overlaps. Direct image for this finite proper map is exact: its stalks are the finite sums over the fibre, and a disk about a branch value has one connected inverse disk. There is no replacement of all nearby branches by their invariant diagonal.

The actual identities proved in ADC8 are

$$b_r T_n^A = U_n^* b_r, \quad b_r U_n^A = T_n^* b_r.\tag{ACD7.1}$$

They also follow immediately from (ACD1.2), $ET_n^Q = T_n E$, $JT_n = U_n^* J$, and commutation of diagonal operators. Consequently the literal cones have two sheaf maps

$$U_n : \mathcal{P}_r \rightarrow f_{n*} \mathcal{P}_r, \quad V_n : f_{n*} \mathcal{P}_r \rightarrow \mathcal{P}_r.\tag{ACD7.2}$$

Their complete nearby maps, constant pole maps, and point H maps are

	nearby	constant pole A	point H
U_n	$a \mapsto (T_n a, \dots, T_n a)$	T_n	U_n^*
V_n	$(a_j) \mapsto T_{1/n} \sum_j a_j$	$n T_{1/n} = U_n$	T_n^*

(ACD7.3)

The two differential squares are exactly (ACD7.1). Thus these maps are constructed before dualization and before a weight claim. The trace here is a sheaf trace on locally constant branches; no smooth pullback of a ramified arbitrary distribution is presumed.

The full vanishing coefficient can also be computed, without replacing the cover by its invariant part. Put $\Delta a = (a, \dots, a)$, $\Sigma v = \sum_j v_j$, and let P_n be the cyclic deck permutation of A^n . The shifted specialization cone has injective first differential $a \mapsto (-\Delta a, b_r a)$. Its sole remaining vanishing coefficient is therefore

$$V_n = (A^n \oplus H) / \{(-\Delta a, b_r a) : a \in A\}. \quad (\text{ACD7.3a})$$

This quotient is separated and has the explicit continuous decomposition

$$[(v, h)] \mapsto (v - \Delta m(v), h + b_r m(v)) \in \ker \Sigma \oplus H, \quad m(v) = \Sigma v / n. \quad (\text{ACD7.3b})$$

The inverse sends (w, k) to $[(w, k)]$. Replacing (v, h) by $(v - \Delta a, h + b_r a)$ leaves both displayed coordinates unchanged. Conversely their difference from (v, h) is the defining relation with $a = m(v)$. This proves the isomorphism and closedness of the relation subspace, without requiring $b_r A$ closed.

With the same $1 - P_n$ variation convention as the positive-support cut model, the full arrows are

$$\text{can}(v) = [(v, 0)], \quad \text{var}[(v, h)] = (1 - P_n)v. \quad (\text{ACD7.3c})$$

Variation is well-defined since $(1 - P_n)\Delta = 0$; the vanishing monodromy is $[(v, h)] \mapsto [(P_n v, h)]$. The two composites of canonical and variation maps are the corresponding $1 - T$ operators. For $n = 1$, (ACD7.3b) identifies V_1 with H , and canonical becomes b_r , variation zero, as required. To identify the actual variation rather than only its composites, the deck-cut calculation on $[(v, 0)]$ is $(1 - P_n)v$. Naturality for the supported inclusion $i_* H \rightarrow f_{n*} \mathcal{P}_r$ makes variation zero on $[(0, h)]$, because that source has zero nearby cycles. These two types span V_n , proving exactly (ACD7.3c).

The two maps (ACD7.2) induce respectively

$$\begin{aligned} H &\longrightarrow V_n, & h &\longmapsto [(0, U_n^* h)], \\ V_n &\longrightarrow H, & [(v, h)] &\longmapsto b_r T_{1/n} \Sigma v + T_n^* h. \end{aligned} \quad (\text{ACD7.3d})$$

For the second map a defining relation changes its value by $-b_r U_n a + T_n^* b_r a = 0$. For the first canonical square, $[(\Delta T_n a, 0)] = [(0, b_r T_n a)] = [(0, U_n^* b_r a)]$. For the second square apply its displayed formula to $[(v, 0)]$. Both variation squares follow from $(1 - P_n)\Delta = 0$ and $\Sigma(1 - P_n) = 0$. Thus all augmentation coordinates, canonical maps and variation maps have been constructed, not only their invariant or coinvariant reductions.

The composite $V_n U_n$ is n times identity: on A , $U_n T_n = n$; on H , $T_n^* U_n^* = (U_n T_n)^* = n$. The reverse composite on the full nearby object is $\Delta \Sigma$, the deck norm. It is not n times identity on the augmentation part. On the pole H term it is n times identity. These are the complete constant/skyscraper sheaf maps, so the equalities hold as cone maps. A raw-current cut presentation can have the additional homotopies calculated in DCA without changing these derived maps.

Under the pairing (ACD3.3), the point transpose of U_n^* is $\overline{U}_n$, since $\langle U_n^* h, y \rangle_+ = \langle h, U_n y \rangle_+$. Similarly the transpose of T_n^* is $\overline{T}_n$. On the open coefficient, unit transposes to sum with T_n' and trace transposes to diagonal with $T_{1/n}'$. Thus the resulting current maps are precisely the DCA directions

$$\begin{aligned} f_{n*} \mathcal{E}_r &\rightarrow \mathcal{E}_r : & \Psi &= T_n' \Sigma, & i^* &= (n T_n', \overline{U}_n), & i^! &= (\overline{U}_n, T_n'), \\ \mathcal{E}_r &\rightarrow f_{n*} \mathcal{E}_r : & \Psi &= \Delta T_{1/n}', & i^* &= (T_{1/n}', \overline{T}_n), & i^! &= (\overline{T}_n, n T_{1/n}'). \end{aligned} \quad (\text{ACD7.4})$$

The angular factor in the second line is n , because the positive circle has covering degree n . The first line's pole-constant factor n comes from summing the n nearby branches; its Dirac factor is one before T_n' . DCA gives the explicit raw cut homotopy realizing these arrows. All factors are retained in (ACD7.4).

On the original endpoint labels, ordered as zero-pole value, zero-pole integral, infinity-pole value, infinity-pole integral, the primal unit actions are $(1, n, n, 1)$ and the primal weighted trace actions are $(n, 1, 1, n)$, as in GTR. Their continuous transposes give the same scalar lists on the dual endpoint coordinates. The endpoint map from the original source in (ACD4.3) is identity and respects both lists.

ACD8. The actual weight comparison and its scope

For the unit direction in (ACD7.3), the local degree-minus-one invariant-cycle source is A with T_n ; its supported target is H with U_n^* . On a value class in Q/Z_r concentrated at an actual off-critical ρ , its image under β_r is at $\rho^\#$. The two eigenvalues are

$$n^\rho \quad \text{and} \quad n^{1-\rho^\#} = n^\rho. \quad (\text{ACD8.1})$$

For weighted trace they are $n^{1-\rho}$ and $n^{\rho^\#} = n^{1-\rho}$. The eigenvalue statements apply to the indicated quotient value class. A representative in full Q may also carry higher-jet terms, retained in the kernel Z_r . The entire continuous intertwining identities are (ACD7.1), not a conclusion drawn only from these single lines.

For the global cone, pullback acts on the degree-one A in (ACD6.3) by nT_n , from the oriented sphere degree. Weighted trace acts there by $T_{1/n}$; their composite is n as required. On the degree-minus-one A the actions are T_n and U_n . On $H \oplus H$ in cochain degree zero they are U_n^* and T_n^* on each copy. The diagonal boundary commutes with these actions by (ACD7.1). Thus the global degree factor has not been suppressed; it lies on the other cohomological component and does not alter (ACD8.1).

Deligne's §3.6 argument, reconstructed with its exact hypotheses in DC, separates special-source weights from the supported target by a genuine dual geometric estimate. Here the constructed continuous dual, the source map, both actual cover directions, and the full two-pole cohomology have been computed. They give the same eigenvalue on every nonzero image of the particular specialization boundary $-b_r$. The normal multiplier separator of GMS is identity on the original quotient Q , hence identity on its image $\beta_r Q$; its annihilation of the different translated normal extension does not annihilate this image.

The calculation therefore identifies what the current geometry actually supplies: (ACD5.4) is the original-source obstruction quotient, (ACD3.2) its continuous duality comparison, (ACD6.5) its global image, and (ACD7.3) its genuine geometric actions. No vanishing of D_r , no pure Frobenius realization, and no RH resolution is asserted. This is a completed attempt to derive the missing weight distinction from the accumulated duality and cover data, rather than assigning a new twist or replacing the original arithmetic.

Source provenance is inherited with its exact reading scope from CSP/CSD and DC: Connes–Consani, arXiv:0903.2024v3, §5, and arXiv:2609.00299v1, signed charts; Deligne, *La conjecture de Weil II*, §3.6, read here through the separately identified French transcription, not claimed to be author TeX. The definitions of nearby/vanishing cycles are compared with Reich's retained author source, arXiv:1002.1686v4; no finite-rank theorem is imported for these infinite coefficients. Complete local proofs used here are retained in this repository as SDT, CSD, CV, LNC, LVD, RGR, ADC, DCA, GTAH and RTT, with their individual proof locators and acceptance scopes.

The adjoint-defect receiver and its actual invariant-cycle cross

Complete local derivation. Proof locators **ADC0–ADC9**. This note calculates the invariant-cycle cross of the existing receiver of the positive-adjoint discrepancy. Its attaching map is retained; neither a new coefficient system nor a purity hypothesis is inserted.

ADC0. Construction stage, correction audit, and inputs

The supporting datum is $\tau\langle Z_1; \text{no } Z_2 \rangle$. All operations here take place on the coefficient spaces after complete-history arithmetic reconstruction. No addition, coordinate, metric, midpoint, or extra parity is assigned to the support. The two arithmetic branches retain their own recovered counters.

Before this calculation the connected `USER_ARGUMENT_RECONSTRUCTION.md`, its correction-precedence table, `READ_FIRST_USER_CONSTRUCTION.md`, the verbatim WU062 passage, and the WU064–WU065 amendment were read. In particular, an obstruction in a specified coefficient receiver must be connected by actual maps to the other programme objects; an independently selected extension cannot replace the desired lifting problem. WU065 also requires attempting the next calculation using all established results when a preceding result does not give the sought conclusion.

The actual inputs are RGR0–RGR12 in `CC_RESIDUE_GYSIN_RECEIVER_INDEPENDENT.md`; LNC0–LNC11 in `CC_LOCAL_NEARBY_CYCLES_AND_SPECIALIZATION.md`; LVD0–LVD9 in `CC_LOCAL_VANISHING_DUAL_ARROWS.md`; and DC0–DC12 in `DELIGNE_INVARIANT_CYCL E_QUOTIENT.md`. RGR constructs the source map and current cone. LNC constructs nearby cycles, the exact local cross, and the proper-current/cover signs. DC proves the two weight directions and the exact quotient for Deligne’s distinct geometric objects. The positive-adjoint operator is the actual GTAH0–GTAH1 operator in `GEOMETRIC_TRANSFER_POSITIVE_ADJOINT_DEFECT.md`. No source theorem for finite-rank constructible sheaves is applied to infinite coefficients without the explicit local calculation.

Sheaf derived statements use sheaves of complex vector spaces, and all specified coefficient/current maps are continuous. Kernels and quotients retain their actual topologies; no closed image is inferred from an algebraic quotient. Continuous duals are distinguished from unrestricted algebraic duals.

ADC1. The full source and the particular attaching map

Use the original source spaces and quotient

$$S = \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\},$$

$$A = \{a \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j\}, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu),$$

$$J = \Sigma S, \quad Q = A/J, \quad q : A \rightarrow Q, \quad B = A'_\beta, \quad j = q' : Q'_\beta \hookrightarrow B. \quad (\text{ADC1.1})$$

SSI/ESI give closed J and the continuous inverse of $\Sigma : S \rightarrow J$. SDT/CSD give the strong embedding $j(Q') = J^\perp$. The prime denotes the continuous complex-linear dual, and β denotes uniform convergence on bounded subsets.

The complete original transform is

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta^{-1} F(u) = \frac{u^{-1/2}}{\pi} \int_{\mathbb{R}} F(1/2 + it) u^{-it} dt. \quad (\text{ADC1.2})$$

It identifies J with the entire strip-Schwartz ideal of full vanishing jets at the original nontrivial zeros, with orders m_ρ . The retained divisor is

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \quad (\text{ADC1.3})$$

Its exceptional values remain

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = F_0(2) = \frac{\pi}{24}, \quad F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) \neq 0. \quad (\text{ADC1.4})$$

At $\rho + t$, the full local factor is

$$F_0(\rho + t) = t^{m_\rho} \frac{(\rho + t)(\rho + t - 1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho + t)/2) \frac{\zeta(\rho + t)}{t^{m_\rho}}. \quad (\text{ADC1.5})$$

Every derivative of this factor is retained. The original arithmetic is

$$\zeta(s) = 1 + \sum_{n \geq 2} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{ADC1.6})$$

Thus the unit and all prime-power repetitions remain in the source. No modified zero divisor is substituted.

Let $\mathcal{Z}$ be the distinct original nontrivial zeros, $\rho^\# = 1 - \bar{\rho}$, and

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_\rho m_\rho x_\rho \bar{y}_\rho, \quad (\mathcal{J}y)_\rho = y_{\rho^\#}.$$

The actual continuous map $E : Q \rightarrow H$ is $E(F)_\rho = (\Theta_Q F)(\rho)$. Its image contains every finite value vector and is dense. Its full higher-jet kernel is retained in Q . Put $C = \overline{H}$, with its conjugate complex structure. For a fixed recovered parameter $r > 1$,

$$(T_r y)_\rho = r^\rho y_\rho, \quad U_r = r T_{1/r}, \quad D_r = T_r^* - U_r, \quad d_r(\rho) := e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}). \quad (\text{ADC1.7})$$

Then $(D_r y)_\rho = d_r(\rho) y_\rho$. These are bounded operators by the known open strip. An integer $r = n$ is a geometric cover degree; a noninteger r is only a coefficient parameter.

The precise attaching map is

$$A_H(y)(F) = \langle EF, \mathcal{J}y \rangle_+, \quad \sigma_r : C \rightarrow Q'_\beta, \quad \sigma_r(\bar{y}) = A_H(D_r y), \quad a_r = j\sigma_r : C \rightarrow B. \quad (\text{ADC1.8})$$

A_H is anti-linear, so σ_r is complex-linear. For a bounded test set $K \subset Q$,

$$\sup_{F \in K} |\sigma_r(\bar{y})(F)| \leq \sup_{F \in K} \|EF\|_+ \|D_r\| \|y\|_+, \quad (\text{ADC1.9})$$

proving strong continuity and absolute convergence. Density of $E(Q)$ proves injectivity of A_H ; j is injective. Consequently

$$\ker a_r = \ker \sigma_r = \overline{\ker D_r} =: C_{\text{crit}}, \quad (\text{ADC1.10})$$

where C_{crit} consists exactly of coordinates supported at the actual critical-line zeros. The equality follows from strict monotonicity of r^x in (ADC1.7); it asserts neither existence nor nonexistence of an off-line zero.

For comparison with the original residue realization, if $\zeta(\rho + t) = t^{m_\rho} u_\rho(t)$, the functional in (ADC1.8) is the strong limit of the original- ζ residues with finite numerators

$$h_{B_0}(y) = \sum_{\rho \in B_0} m_\rho (-1)^{m_\rho - 1} u_\rho(0) \overline{(D_r y)_{\rho^\#}} e_{1-\rho, m_\rho - 1}, \quad \mathcal{R}(F, h) = \sum_\rho \text{Res}_{s=\rho} \frac{F(s) h(1-s)}{\zeta(s)} ds. \quad (\text{ADC1.11})$$

The e 's are RTT's full-jet isolators. Its residue calculation gives the exact coefficient of the value functional; (ADC1.9) controls the strong limit. Neither multiplicity nor the leading original-zeta unit is omitted.

ADC2. The current cone, endpoints, and its four local functors

Fix a pole disk with positive coordinate z at zero or $1/z$ at infinity. Let $\mathcal{T}_A$ be the continuous compact-test current complex,

$$\mathcal{T}_A^n(U) = (\mathcal{E}_c^{-n}(U; A))', \quad dT(\phi) = (-1)^{n+1}T(d\phi), \quad -2 \leq n \leq 0.$$

Its constants represent $\underline{B}[2]$. Positive Dirac and angular currents satisfy

$$\delta_p \lambda(f) = \lambda(f(p)), \quad h_p \lambda(\omega) = \lambda\left(\int \frac{d \arg z_p}{2\pi} \wedge \omega\right), \quad dh_p \lambda = \delta_p \lambda. \quad (\text{ADC2.1})$$

The last equality is Stokes with positive small-circle integral one. All A -valued integrals are formed before the functional is applied.

The actual receiver is

$$\mathcal{E}_r = \text{Cone}(-\delta_p a_r)[-1].$$

In degrees $-1, 0, 1$ its terms and differentials are

$$\mathcal{T}_A^{-2}, \quad \mathcal{T}_A^{-1} \oplus i_* C, \quad \mathcal{T}_A^0, \quad d^{-1}b = (-d_T b, 0), \quad d^0(b, x) = -d_T b + \delta_p a_r x. \quad (\text{ADC2.2})$$

It is RGR's actual homotopy pullback; the triangle is

$$\underline{B}[1] \longrightarrow \mathcal{E}_r \longrightarrow i_* C \xrightarrow{\text{Gys}_B a_r} \underline{B}[2]. \quad (\text{ADC2.3})$$

On a disk the constant and angular representatives give the continuous cohomology models

$$i^* \mathcal{E}_r \simeq B[1] \oplus C, \quad R\Gamma(D^*, \mathcal{E}_r) \simeq B[1] \oplus B[\vartheta_p],$$

$$\lambda \mapsto c_\lambda \quad (\deg -1), \quad x \mapsto (h_p a_r x, x) \quad (\deg 0), \quad \text{res} = (1_B, a_r). \quad (\text{ADC2.4})$$

Closedness of the latter representative follows from (ADC2.1); the compact-test contraction removes the remaining current terms. These are continuous comparisons on fixed disks, and their cohomology identifications agree as disks shrink, as in CSD/RGR.

Pulling the punctured constant system to the universal cover and contracting leaves

$$R\psi \mathcal{E}_r = B[1], \quad T = 1, \quad \text{sp} : B[1] \oplus C \longrightarrow B[1] \text{ is projection.} \quad (\text{ADC2.5})$$

Its vanishing cone is $R\phi \mathcal{E}_r = C[1]$. Equivalently the shifted functors $\Psi = R\psi[-1]$, $\Phi = R\phi[-1]$ have

$$\Psi \mathcal{E}_r = B, \quad \Phi \mathcal{E}_r = C, \quad \text{can} = 0, \quad \text{var} = a_r. \quad (\text{ADC2.6})$$

The positive angular convention fixes the plus in variation; it is the second restriction component in (ADC2.4), not a sign deduced from $1 - T = 0$.

The restriction fibre first has coefficient complex

$$B \xrightarrow{b \mapsto (0, b)} C \oplus B \xrightarrow{(x, b) \mapsto a_r x} B \quad \text{in degrees } -1, 0, 1. \quad (\text{ADC2.7})$$

Cancel its identity pair to obtain

$$i^! \mathcal{E}_r \simeq [C \xrightarrow{+a_r} B] \quad (\deg 0, 1). \quad (\text{ADC2.8})$$

The target is positively oriented: in the shifted current differential, $dh_p b = -\delta_p b$, so the fibre differential gives $d(h_p b, 0) = (-\delta_p b, h_p b)$. The positive Dirac class is therefore the positive angular boundary coordinate. This explains why (ADC2.8) has $+a_r$, whereas a literal standard shifted cone on a_r has differential $-a_r$; negating its target gives the displayed comparison.

Both source endpoint lines remain as in RGR10:

$$\mathcal{E}_{r,\text{full}} = \mathcal{E}_r \oplus i_* E'_p[-1]. \quad (\text{ADC2.9})$$

Their stalk and costalk are $E'_p[-1]$, their nearby complex is zero, and their unshifted vanishing complex is $E'_p[0]$. Thus they appear in special and support degree 1, and vanishing degree 0, separately from (ADC2.5)–(ADC2.8).

Here $E_+ = \mathbb{C}c_0 \oplus \mathbb{C}c_1$ and $E_- = \mathbb{C}d_0 \oplus \mathbb{C}d_1$ retain respectively the original value-at-zero and integral channels in the two source charts. Their original coefficient actions are

$$\rho_+(t)|_{E_+} = \text{diag}(1, t), \quad \rho_-(t)|_{E_-} = \text{diag}(t, 1). \quad (\text{ADC2.10})$$

These are the endpoint components of the complete source maps in (LNC5.2); transposition keeps each indicated coordinate label.

ADC3. The exact invariant-cycle cross of this receiver

Write

$$S_r = a_r(C) \subset B, \quad N_r = B/S_r. \quad (\text{ADC3.1})$$

These are the actual image and algebraic quotient with their stated topologies. No closedness is asserted. From (ADC2.8), the full support groups are

$$H_p^0(\mathcal{E}_{r,\text{full}}) = C_{\text{crit}}, \quad H_p^1(\mathcal{E}_{r,\text{full}}) = N_r \oplus E'_p. \quad (\text{ADC3.2})$$

Let $V_i = H^i(R\psi\mathcal{E}_{r,\text{full}})$, $A_i^{\text{sp}} = H^i(i^*\mathcal{E}_{r,\text{full}})$, $B_i^{\text{punct}} = H^i(D^*, \mathcal{E}_{r,\text{full}})$. Define the actual invariants $C_i^{\text{inv}} = V_i^T$, angular coinvariants $K_i = V_{i-1,T}[\vartheta_p]$, and $O_i = H_p^{i+1}$. The only nearby group is $V_{-1} = B$, with monodromy identity. Thus the entire nonzero cross is:

i	A_i^{sp}	K_i	B_i^{punct}	C_i^{inv}	O_i	α_i	∂_i
-1	B	0	B	B	C_{crit}	1	0
0	C	$B[\vartheta_p]$	$B[\vartheta_p]$	0	$N_r \oplus E'_p$	a_r	$b \mapsto ([b], 0)$
1	E'_p	0	0	0	0	0	0

The row $0 \rightarrow K_i \xrightarrow{j_i} B_i^{\text{punct}} \rightarrow C_i^{\text{inv}} \rightarrow 0$ has identity π_{-1} and identity j_0 . The complete remaining localization part is

$$0 \rightarrow C_{\text{crit}} \rightarrow C \xrightarrow{a_r} B \xrightarrow{b \mapsto ([b], 0)} N_r \oplus E'_p \xrightarrow{\text{pr}_{E'_p}} E'_p \rightarrow 0. \quad (\text{ADC3.3})$$

Every sign follows from (ADC2.7)–(ADC2.8).

It follows by these actual maps, without a weight argument, that

$$\boxed{\text{im } \partial_i / \partial_i j_i K_i = 0 \quad \text{for every } i.} \quad (\text{ADC3.4})$$

At degree -1 the boundary is zero. At degree zero the coinvariant injection is identity, so its boundary image is all $N_r \oplus 0$. The quotient of the whole receiver is still

$$O_0 / \partial_0 j_0 K_0 = E'_p, \quad (\text{ADC3.5})$$

not zero when endpoints are retained. This is exactly the remaining specialization kernel in degree one. Equations (ADC3.4)–(ADC3.5) distinguish the obstruction image from the whole quotient receiver.

The vanishing of (ADC3.4) does not imply the attaching map vanished: it holds for the present a_r before any conclusion about D_r . The stronger conclusion of DC8 has a different test here:

$$\ker(\partial_0 j_0) = S_r, \quad K_0 \xrightarrow{\partial_0 j_0} \text{im } \partial_0 = N_r \text{ is injective} \iff a_r = 0 \iff D_r = 0. \quad (\text{ADC3.6})$$

The actual source subquotient in DC5.5 is

$$0 \rightarrow C_{\text{crit}} \rightarrow C \xrightarrow{a_r} S_r \rightarrow 0. \quad (\text{ADC3.7})$$

The map $C/C_{\text{crit}} \rightarrow S_r$ is a continuous algebraic bijection. We do not assume its inverse continuous for the ambient subspace topology of S_r .

There is also an exact calculation of the stronger DC9 candidate:

$$\frac{A_0^{\text{sp}}}{\text{im}(H_p^0 \rightarrow A_0^{\text{sp}})} = C/C_{\text{crit}} \rightarrow C_0^{\text{inv}} = 0. \quad (\text{ADC3.8})$$

Its kernel is the entire displayed source quotient, identified algebraically with S_r . Thus the actual next object that must be controlled after the automatically zero invariant-cycle obstruction is S_r , not the already zero quotient (ADC3.4).

ADC4. The exact relation to the original coefficient quotient

Classical human-source attribution: extension of continuous linear functionals uses Hahn–Banach; see Terence Tao, 245B Notes 6, Theorem 1 and its complex proof. A continuous seminorm and its kernel reduce the locally convex application to that normed-space result. This does not assert a continuous splitting.

Write $S_r^Q = \sigma_r(C) \subset Q'$. Since j is injective, $S_r = jS_r^Q$. The full continuous-transpose coefficient row is

$$0 \rightarrow Q' \xrightarrow{j} B \xrightarrow{k'} J' \rightarrow 0, \quad (\text{ADC4.1})$$

where $k : J \hookrightarrow A$. Exactness is the quotient definition and Hahn–Banach on closed J ; no strong-open mapping theorem is used. It induces the exact row

$$0 \rightarrow Q'/S_r^Q \xrightarrow{[\lambda] \mapsto [j\lambda]} N_r \xrightarrow{[b] \mapsto k'b} J' \rightarrow 0. \quad (\text{ADC4.2})$$

For injectivity, $j\lambda \in S_r = jS_r^Q$ implies $\lambda \in S_r^Q$. The middle kernel consists of classes with $k'b = 0$, hence $b = j\lambda$. Surjectivity is that of k' . These prove every exactness assertion.

The row is exactly the pushout of (ADC4.1) along $Q' \rightarrow Q'/S_r^Q$: send a pushout class $[(b, [\lambda])]$ to $[b + j\lambda]$, and send $[b]$ back to $[(b, 0)]$. Changing λ by an element of S_r^Q or using the relation $(j\mu, -[\mu])$ changes the first formula by an element of S_r . Both compositions are identity. The formulas descend from continuous maps and are continuous for quotient topologies.

The sheaf comparison retaining this row is RGR9's actual coefficient fibre

$$\mathcal{K}_r = \text{Cone}(-\text{Gys}_{Q'} \sigma_r)[-1], \quad \mathcal{K}_r \rightarrow \mathcal{E}_r \rightarrow \underline{J}'[1] \rightarrow \mathcal{K}_r[1]. \quad (\text{ADC4.3})$$

The first map uses j on nearby coefficients and 1_C on vanishing coefficients; the second uses k' and zero. On costalks they are

$$[C \xrightarrow{\sigma_r} Q'] \xrightarrow{(1_C, j)} [C \xrightarrow{a_r} B] \rightarrow J'[-1]. \quad (\text{ADC4.4})$$

These are literal continuous current maps induced by the original q and k . Their cokernel row is (ADC4.2). In particular the source of the arithmetic boundary-image part is the dual of the same original LNC quotient Q , rather than an unrelated coefficient.

The reduced arithmetic receiver $\mathcal{K}_r$ has the endpoint-free part of ADC3's cross, with B replaced by Q' , a_r by σ_r , and boundary image Q'/S_r^Q . Alternatively $\mathcal{K}_{r, \text{full}} = \mathcal{K}_r \oplus i_* E'_p[-1]$ retains exactly the endpoint terms in the full ADC3 table. Its coinvariant-boundary kernel is exactly S_r^Q . Both invariant-cycle quotients still vanish. This isolates the original arithmetic coefficient inside the full A' receiver without changing the test for the attaching map.

Finally, the actual morphism $\mathcal{E}_r \rightarrow D_c P$ of RGR5 has nearby map 1_B and vanishing map $\sigma_r : C \rightarrow Q'$; on costalks it is

$$[C \xrightarrow{a_r} B] \xrightarrow{(\sigma_r, 1_B)} [Q' \xrightarrow{j} B]. \quad (\text{ADC4.5})$$

Its cone is $i_* \text{Cone}(\sigma_r)$, retaining the kernel C_{crit} and quotient Q'/S_r^Q . It connects ADC3 to the original LNC/LVD dual local arrows by an actual sheaf map. No equality of a Gysin Hom class and an M -module Ext class is inferred.

ADC5. The coefficient covariance and the full image map

Write T_t^Q for the actual dilation induced on Q and $U_t^Q = tT_{1/t}^Q$. Their value maps intertwine with the corresponding bounded T_t, U_t on H . GTAH's exact identities are

$$T_t^* = JU_t J, \quad U_t^* = JT_t J, \quad t > 0. \quad (\text{ADC5.1})$$

Direct evaluation gives, for $F \in Q, y \in H$,

$$(T_t^Q)' A_H(y)(F) = \langle T_t EF, Jy \rangle_+ = \langle EF, T_t^* Jy \rangle_+ = A_H(U_t y)(F).$$

The second identity follows by exchanging T_t, U_t . Since D_r is diagonal, it commutes with both. Thus, with linear maps on the conjugate space,

$$\begin{aligned} \sigma_r \bar{U}_t &= (T_t^Q)' \sigma_r, & \sigma_r \bar{T}_t &= (U_t^Q)' \sigma_r, \\ a_r \bar{U}_t &= T_t' a_r, & a_r \bar{T}_t &= U_t' a_r, \quad U_t' = tT_{1/t}'. \end{aligned} \quad (\text{ADC5.2})$$

All maps are continuous; transposes are strong-continuous because continuous primal maps carry bounded sets to bounded sets. Consequently S_r^Q, S_r , their exact quotients in ADC4, and $\ker a_r$ are stable under these actions. Their induced quotient actions are continuous even when an image quotient is not separated.

This gives an exact description on each value coordinate. Let $e_\rho \in H$ be the vector of value one at ρ and zero elsewhere, and let $\delta_{\rho^\#, 0} \in Q'$ be evaluation at $\rho^\#$. Then

$$\sigma_r(\bar{e}_\rho) = m_\rho \overline{d_r(\rho)} \delta_{\rho^\#, 0}, \quad a_r(\bar{e}_\rho) = m_\rho \overline{d_r(\rho)} q' \delta_{\rho^\#, 0}. \quad (\text{ADC5.3})$$

The calculation is substitution into (ADC1.8) and $m_{\rho^\#} = m_\rho$. In particular, every actual off-critical value functional lies in S_r^Q ; it vanishes from this formula exactly when ρ is on the critical line. Finite-coordinate vectors are dense in H , and the image map is continuous, but this does not assert density of finite primal full-jet blocks in Q or closedness of S_r .

On a nonzero image line the two actions have exactly matching source and target eigenvalues:

action	C/C_{crit} at $\bar{e}_\rho$	S_r at $q' \delta_{\rho^\#, 0}$
$\bar{U}_t / T_t'$	$t^{1-\bar{\rho}}$	$t^{\rho^\#}$
$\bar{T}_t / U_t'$	$t^{\bar{\rho}}$	$t^{1-\rho^\#}$

(\text{ADC5.4})

The entries are equal because $\rho^\# = 1 - \bar{\rho}$. This is an intertwining map on the actual entire source, with displayed formulas on its value coordinates; it is not a proposed replacement arithmetic.

The continuous pairing with the original full quotient is equally exact. Define

$$Z_r = \{F \in Q : \sigma_r(x)(F) = 0 \text{ for every } x \in C\}. \quad (\text{ADC5.5})$$

Using (ADC1.8) and the positive inner product gives

$$\begin{aligned} \sigma_r(\bar{y})(F) &= \langle EF, JD_r y \rangle_+ = \langle D_r^* JEF, y \rangle_+, \\ Z_r &= \ker(D_r^* JE) = \{F \in Q : F(\rho) = 0 \text{ at every actual off-critical } \rho\}. \end{aligned} \quad (\text{ADC5.6})$$

For the last equality, the scalar multiplying the reflected value is nonzero exactly at an off-critical zero; reflection preserves that subset. All higher jets remain in Z_r when the values vanish. The pairing descends to

$$(Q/Z_r) \times (C/C_{\text{crit}}) \longrightarrow \mathbb{C}, \quad ([F], [\bar{y}]) \longmapsto \sigma_r(\bar{y})(F). \quad (\text{ADC5.7})$$

It is separately induced by the continuous original pairing. Its left and right radicals are zero by (ADC5.6) and the injectivity calculation (ADC1.10). This proves a concrete relation to the original Q , without asserting that either infinite-dimensional quotient is the entire continuous dual of the other. Topological completions and image closures are additional calculated receivers, not used to change (ADC5.7).

ADC6. What the genuine power-cover maps do to this cross

For a recovered integer $n \geq 1$, the proper-current map over $b_n(z_p) = z_p^n$ is a map $b_{n*}\mathcal{E}_r \rightarrow \mathcal{E}_r$. Its literal nearby source is $B^n[1]$, with cyclic permutation monodromy, and its nearby map is

$$(b_0, \dots, b_{n-1}) \longmapsto T'_n \sum_{j=0}^{n-1} b_j. \quad (\text{ADC6.1})$$

To derive it, pull back the target universal cover: it has n contractible inverse components. On each component the holomorphic local isomorphism preserves the positive orientation and pushes its constant current to $T'_n b_j$; add the branches. This is the same full branch calculation as LNC6.14 and retains all n components.

On nearby invariants the map is nT'_n ; on the angular coinvariants, identified by sum, it is T'_n . A proper map has a single inverse-image pole here, so its actual stalk maps and positive costalk maps are

	first displayed degree	second displayed degree	
$i^*\mathcal{E}_r = B[1] \oplus C$	nT'_n	$\bar{U}_n$	(ADC6.2)
$i^!\mathcal{E}_r = [C \rightarrow B]$	$\bar{U}_n$	T'_n	

For the point-supported coordinate this uses (ADC5.2): $a_r \bar{U}_n = T'_n a_r$. For the current differential, pushforward commutes with d , and Dirac pushforward gives $b_{n*} \delta_p \lambda = \delta_p T'_n \lambda$. The angular current pushes forward with degree one; its n local inverse branches each contribute $1/n$. This proves the entire chain map, not only its induced eigenvalues.

The independently constructed weighted geometric trace on the original primal sheaf has transpose giving the reverse map $\mathcal{E}_r \rightarrow b_{n*}\mathcal{E}_r$. Its full nearby map and invariant/coinvariant factors are

$$b \longmapsto (T'_{1/n} b, \dots, T'_{1/n} b), \quad B_{\text{inv}} : T'_{1/n}, \quad B_{\text{coinv}} : nT'_{1/n} = U'_n. \quad (\text{ADC6.3})$$

The finite-coordinate diagonal has sum $nT'_{1/n}$, which fixes the angular factor. The point map is $\bar{T}_n$, and the costalk identity is exactly $a_r \bar{T}_n = U'_n a_r$. Thus on the cross the resulting maps are

$$i^* : (T'_{1/n}, \bar{T}_n), \quad i^! : (\bar{T}_n, nT'_{1/n}). \quad (\text{ADC6.4})$$

The existence of the entire sheaf trace, including its finite-coordinate nearby/vanishing correction, is constructed in DCA3–DCA7 of CC_ADJOINT_DEFECT_COVER_ACTIONS.md. Its actual remaining coordinates can be displayed here. Put $V_n = B^n$, let $\mathbf{t}(v)_j = v_{j-1}$ with cyclic indices, and let $\Delta b = (b, \dots, b)$, $\Sigma_n v = \sum_j v_j$, $e_0 b = (b, 0, \dots, 0)$. The letter $\mathbf{t}$ denotes sheet monodromy and is distinct from the support τ . Then

$$\begin{aligned} \Phi(b_{n*}\mathcal{E}_r) &= C \oplus (V_n/\Delta B), \quad \text{can}_n(v) = (0, [v]), \quad \text{var}_n(x, [v]) = e_0 a_r x - (\mathbf{t} - 1)v, \\ \mathbf{t}_\Phi(x, [v]) &= (x, [\mathbf{t}v - e_0 a_r x]). \end{aligned} \quad (\text{ADC6.6})$$

These are DCA3.7–DCA3.9's actual cut-cone coordinates. In particular $\text{var}_n \text{can}_n = 1 - \mathbf{t}$ and $\text{can}_n \text{var}_n = 1 - \mathbf{t}_\Phi$. The constant stalk restriction is Δ , and its angular restriction is $e_0 a_r$; applying Σ_n recovers exactly a_r , with no lost degree factor.

Proper-current pushforward on this full vanishing space is $(x, [v]) \mapsto \overline{U}_n x$. For the reverse map define

$$c_x = T'_{1/n} a_r x, \quad w_n(x) = (0, c_x, 2c_x, \dots, (n-1)c_x).$$

Its full vanishing arrow is

$$x \mapsto (\overline{T}_n x, [w_n(x)]), \quad (t-1)w_n(x) = ne_0 c_x - \Delta c_x. \quad (\text{ADC6.7})$$

The zeroth component of the last equality is $(n-1)c_x$, and each remaining component is $-c_x$. Since $a_r \overline{T}_n x = nc_x$, this proves the entire variation square:

$$\text{var}_n(\overline{T}_n x, [w_n(x)]) = \Delta T'_{1/n} a_r x. \quad (\text{ADC6.8})$$

The correction is also the actual gluing homotopy. In the punctured circle complex $[V_n \xrightarrow{t-1} V_n]$ in degrees $(-1, 0)$, the difference between the two stalk-restriction composites is $dw_n + w_n d$. DCA3.5 realizes the sheaf from the homotopy fibre of $Rj_* L \oplus i_* S^{\text{st}} \rightarrow i_* i^* Rj_* L$, $(u, s) \mapsto u - \text{res}(s)$. On its restriction fibre the map is explicitly $(s, v) \mapsto (f_S s, f_N v + w_n(s))$; substitution in $d(s, v) = (ds, \text{res}(s) - dv)$ proves the chain identity. This constructs the sheaf map rather than inferring it from cohomology. All finite-coordinate operations are continuous. Thus (ADC6.3)–(ADC6.4) are the invariant/coinvariant maps of this full arrow, not a holomorphic n -th-root inverse. The additional $V_n/\Delta B$ has been retained.

Both maps descend to $N_r = B/S_r$, by (ADC5.2). On the positive boundary $\partial_0 : B \rightarrow N_r \oplus E'_p$ they commute because the induced quotient actions are precisely the actions of T'_n or $nT'_{1/n}$ on the same representative b . The coefficient covariance also proves commutation with every map in (ADC4.2)–(ADC4.5). The endpoint summands are transported by their original dual pole characters: current pushforward uses the dual endpoint part of $\rho_p(n)$, and the weighted-transfer transpose uses the corresponding $n\rho_p(1/n)'$. Their boundary component remains zero and their inclusion/projection is unchanged.

In the dual coordinate labels of (ADC2.10), these endpoint maps are explicitly

	E'_+	E'_-	
current pushforward	diag(1, n)	diag(n , 1)	
weighted-transfer transpose	diag(n , 1)	diag(1, n).	(ADC6.5)

The second row is obtained by substituting $t = 1/n$ in (ADC2.10) and multiplying the transpose by the retained degree n .

For $t > 0$ not an integer, (ADC5.2) still gives the actual coefficient actions. The maps in this paragraph with geometric degree n are not reinterpreted as noninteger covers.

ADC7. Testing Deligne's two weight directions on the actual boundary

DC6–DC8 use two independent inputs: an upper bound on the special-fibre classes and a lower bound on the coinvariants and supported cohomology, the latter obtained by genuine duality with its retained Tate factor. The full source conclusion is stronger than (ADC3.4): it identifies the low-weight piece with $\ker \partial$, disjoint from the coinvariant summand, and proves $\partial j : K_i \rightarrow \text{im } \partial_i$ injective.

Here all these maps have been computed. In degree zero, special classes map by $a_r : C \rightarrow K_0 = B$. Thus a separation that killed the intersection of special image and coinvariants would kill exactly S_r , by (ADC3.6). We now test the existing actions on that intersection.

For a fixed geometric degree $n > 1$, define the logarithmic weight of a scalar $\lambda \neq 0$ as $w_n(\lambda) = 2 \log |\lambda| / \log n$. This is an explicit modulus calculation, not an assertion that the coefficient space is a finite-dimensional mixed Frobenius module. Under current pushforward, the invariant nearby line indexed by $\rho^\#$ has scalar $n^{1+\rho^\#}$, whereas its angular coinvariant has scalar $n^{\rho^\#}$. Their respective logarithmic weights are

$$2(2 - \Re \rho), \quad 2(1 - \Re \rho). \quad (\text{ADC7.1})$$

The decrease by two is the trace direction of this current pushforward; it is not Deligne's pullback Tate increase.

For the weighted-transfer transpose, the invariant line has scalar $n^{-\rho^\#}$ and the angular line has scalar $n^{1-\rho^\#}$. Their weights are

$$-2(1 - \Re \rho), \quad 2\Re \rho. \quad (\text{ADC7.2})$$

This has the positive angular increase by two. Nevertheless the special source class $\bar{e}_\rho$ has exactly the same angular scalar $n^{\bar{\rho}} = n^{1-\rho^\#}$, by (ADC5.4). Under the pushforward it likewise has exactly the same scalar as the angular target, $n^{\rho^\#}$. Consequently neither actual direction separates the source quotient C/C_{crit} from S_r : on every nonzero image line they have identical eigenvalues. Multiplying both sides of an equivariant map by a common Tate character shifts both weights equally and does not change this equality. Multiplying only one side would require a different map or action; none is inserted here.

The full boundary image N_r has a further exact source component that also prevents assigning it a uniform one-sided weight bound from these formulas. For any s_0 not an original nontrivial zero, evaluation

$$\epsilon_{s_0}(a) = \Theta a(s_0) \in B \quad (\text{ADC7.3})$$

is continuous and has nonzero restriction to J , because $F_0(s_0) \neq 0$. Since $S_r \subset J^\perp$, its class in N_r is nonzero. Under current pushforward angular action T'_n it has scalar n^{s_0} ; under weighted-transfer angular action $nT'_{1/n}$ it has scalar n^{1-s_0} . The resulting weights

$$2\Re s_0, \quad 2(1 - \Re s_0) \quad (\text{ADC7.4})$$

are unbounded in both directions as s_0 ranges outside the original divisor. This uses the actual J' quotient in (ADC4.2), with the full entire F_0 , including its nonzero endpoint and trivial-zero values. It is not a hypothetical zeta spectrum.

Passing to the arithmetic subreceiver $\mathcal{K}_r$ removes precisely this J' quotient by the proven triangle (ADC4.3). It leaves the actual row

$$0 \rightarrow S_r^Q \rightarrow Q' \rightarrow Q'/S_r^Q \rightarrow 0 \quad (\text{ADC7.5})$$

and the special-image map σ_r . Its kernel is still C_{crit} , and its nonzero image lines still have the matching weights (ADC5.4). Thus the next useful attempted reduction has been performed: the full unwanted coefficient quotient is removed by an exact existing map, while the arithmetic intersection that DC's stronger argument would have to kill remains explicitly S_r^Q .

Finally, the normal separator c from GMS/FOD has identity action on the entire original Q ; its transpose is identity on Q' . Hence it acts as identity on S_r^Q , not zero. The actual equality $ce_+ = 0$ for the translated normal extension cannot therefore be reused as $cs_r = 0$. The precise relation is the original dual extension and its pushout in ADC4. This verifies the existing attempt using the full earlier results rather than substituting the already killed normal class for the present attaching map.

ADC8. The continuous transpose and the actual original-quotient cross

Classical human-source attribution: the complex Hilbert-space Riesz representation theorem, with the conjugate-space convention, is recorded in Terence Tao, 245B Notes 5, Theorem 1, Remark 2 and Exercise 15. The explicit maps and signs used here remain displayed below.

The coefficient transpose can be computed without assuming a duality theorem. Identify

$$C' = (\overline{H})' \simeq H, \quad x \mapsto (\bar{y} \mapsto \langle x, y \rangle_+). \quad (\text{ADC8.1})$$

This is the strong topological Riesz isomorphism for the conjugate Hilbert space. Evaluation $A \rightarrow (A'_\beta)'$, followed by a'_r , gives

$$b_r : A \rightarrow H, \quad b_r = D_r^* J E q. \quad (\text{ADC8.2})$$

Indeed, on $\bar{y}$,

$$a'_r(\text{ev}_f)(\bar{y}) = a_r(\bar{y})(f) = \langle E q f, J D_r y \rangle_+ = \langle D_r^* J E q f, y \rangle_+. \quad (\text{ADC8.3})$$

Every map here is continuous, by (ADC1.9). Define also

$$\beta_r : Q \rightarrow H, \quad \beta_r = D_r^* J E, \quad b_r = \beta_r q. \quad (\text{ADC8.4})$$

Its kernel is exactly Z_r from (ADC5.5)–(ADC5.6); it retains all higher jets and all critical-line values.

The actual coefficient cone to which these maps apply is

$$\mathcal{P}_r = \text{Cone}(\underline{A} \xrightarrow{b_r \text{ ev}_p} i_* H) = [\underline{A}^{-1} \xrightarrow{+b_r} i_* H^0]. \quad (\text{ADC8.5})$$

This is constructed by the displayed map on the original source. The following cross uses its literal two-term complex, so no arbitrary exactness of continuous dualization is being presumed.

The required current-level comparison is now explicit. ACD2–ACD4 in CC_ADJOINT_DEFECT_CONTINUOUS_DUAL.md construct the fine resolution

$$\mathcal{L}_r^{-1} = \mathcal{E}_A^0, \quad \mathcal{L}_r^0 = \mathcal{E}_A^1 \oplus i_* H, \quad \mathcal{L}_r^1 = \mathcal{E}_A^2, \quad d^{-1}f = (-df, b_r f(p)), \quad d^0(\omega, h) = -d\omega. \quad (\text{ADC8.5a})$$

Inclusion of locally constant functions and the identity on H compare (ADC8.5) with this resolution. The coefficient-valued Poincaré contraction proves this is a quasi-isomorphism. It does not require the image of b_r to be closed. Use the fixed-support smooth seminorms and their compact-support locally convex inductive limit to form the actual continuous Hom sheaf $\mathcal{D}_r^k(U) = \Gamma_c(U, \mathcal{L}_r^{-k})'$, with $d_{\mathcal{D}}\ell = (-1)^{k+1}\ell d_{\mathcal{L}}$. Its degrees $(-1, 0, 1)$ are the current terms $\mathcal{T}_A^{-2}, \mathcal{T}_A^{-1} \oplus i_* H', \mathcal{T}_A^0$, and

$$d_{\mathcal{D}}^{-1}v = (d_T v, 0), \quad d_{\mathcal{D}}^0(u, \lambda) = d_T u - \delta_p b'_r \lambda. \quad (\text{ADC8.5b})$$

For instance the second formula on a test f is $-u(-df) - \lambda(b_r f(p))$; this fixes both signs. With $R : C \rightarrow H'$, $R(\bar{y})(h) = \langle h, y \rangle_+$, equation (ADC8.3) gives $b'_r R = a_r$. The map

$$\mathcal{D}_r \longrightarrow \mathcal{E}_r, \quad v \mapsto -v \text{ (deg } -1), \quad (u, Rx) \mapsto (u, x) \text{ (deg } 0), \quad w \mapsto -w \text{ (deg } 1) \quad (\text{ADC8.5c})$$

is a continuous chain isomorphism: the differential composites are $(d_T v, 0)$ and $-d_T u + \delta_p a_r x$. This identifies $\mathcal{E}_r$ with the literal continuous support-dual complex of the constructed $\mathcal{P}_r$ resolution. It is not an assertion that continuous dualization is exact on every locally convex complex. The positive compact-form pairing in outer degrees has the minus signs just displayed; the point pairing is $R(x)(h)$. ACD3's compact-support contraction proves the local support comparison, with the same positive Dirac class as (ADC2.8).

Its restriction to the puncture is $\underline{A}[1]$, and its pole complex is $[A^{-1} \xrightarrow{b_r} H^0]$. Therefore

$$R\psi\mathcal{P}_r = A[1], \quad T = 1, \quad H^{-1}(i^*\mathcal{P}_r) = \ker b_r, \quad H^0(i^*\mathcal{P}_r) = H/b_r A. \quad (\text{ADC8.6})$$

The map to $A[1] \oplus A[\vartheta_p]$ is identity on the degree-minus-one A and zero on H . In its restriction fibre the degree-zero coordinate is $H \oplus A$, with preceding differential $a \mapsto (b_r a, a)$. Its quotient coordinate is $(h, a) \mapsto h - b_r a$. This cancels the identity pair and gives

$$i^!\mathcal{P}_r \simeq H[0] \oplus A[-1], \quad \partial_{-1} = -b_r : A \rightarrow H, \quad \partial_0 = +1 : A[\vartheta_p] \rightarrow A. \quad (\text{ADC8.7})$$

The last sign is positive after the same cochain shift as in LVD3/RGR6; it also follows from the retained positive Dirac coordinate in the fibre. The full source includes the endpoint term

$$\mathcal{P}_{r,\text{full}} = \mathcal{P}_r \oplus i_* E_p''[1], \quad (\text{ADC8.8})$$

where $E_p'' \simeq E_p$ by finite-dimensional evaluation, with both original labels retained. This contributes special/support degree -1 , no nearby group, and unshifted vanishing degree -2 .

The resulting entire nonzero cross is:

i	special A_i	K_i	punctured B_i	nearby invariants C_i	support O_i	α_i	∂_i
-2	0	0	0	0	E_p''	0	0
-1	$\ker b_r \oplus E_p''$	0	A	A	H	$(a, e) \mapsto a$	$-b_r$
0	$H/b_r A$	$A[\partial_p]$	$A[\partial_p]$	0	A	0	+1

In particular the genuine invariant-cycle obstruction is now

$$\boxed{\operatorname{im} \partial_{-1}/\partial_{-1} j_{-1} K_{-1} = b_r A = \beta_r Q, \quad \operatorname{coker} \operatorname{sp}_{-1} = A/\ker b_r \simeq Q/Z_r.} \quad (\text{ADC8.9})$$

The first identification of the cokernel with the image uses $[a] \mapsto -b_r a$, retaining the boundary sign. The induced algebraic bijection $Q/Z_r \rightarrow b_r A$ can equivalently use $[F] \mapsto \beta_r F$ when the minus is stated separately. Quotient topologies are retained, and no inverse for its ambient image topology is asserted. All other obstruction images vanish.

This dual direction receives the original LNC quotient through an explicit map of the full local source. Put $P = \mathcal{F}_{\text{red}}[1] \simeq \operatorname{Cone}(q \operatorname{ev}_p)$, with terms $\underline{A}^{-1} \rightarrow i_* Q^0$. Then

$$f_r : P \longrightarrow \mathcal{P}_r, \quad f_r^{-1} = 1_A, \quad f_r^0 = \beta_r \quad (\text{ADC8.10})$$

is a continuous sheaf chain map since $\beta_r q = b_r$. On the full objects add the original evaluation $E_p[1] \rightarrow E_p''[1]$. Its nearby map is identity A , and its vanishing map is $\beta_r : Q \rightarrow H$. On support degree zero it is β_r , and on support degree one it is identity A . Thus the original boundary $-q : A \rightarrow Q$ and the new boundary $-b_r : A \rightarrow H$ satisfy

$$\begin{array}{ccc} A & \xrightarrow{-q} & Q \\ \parallel & & \downarrow \beta_r \\ A & \xrightarrow{-b_r} & H. \end{array} \quad (\text{ADC8.11})$$

It satisfies all four commuting-map identities (DC10.2), with the explicit shifts and coefficients just computed. Here those identities are proved on the displayed continuous complexes; the finite-dimensional Frobenius hypotheses used for DC10's weight conclusions are not asserted for them. Its induced map on obstruction images is the actual surjection

$$Q \xrightarrow{\beta_r} \beta_r Q, \quad \ker \beta_r = Z_r. \quad (\text{ADC8.12})$$

This constructs the precise arithmetic subquotient of the original local defect, rather than naming two different receivers as if they were identical.

The transpose is also the actual old current map (ADC4.5), with its representative sign retained. On the fine resolutions of the plus- q and plus- b_r cones, f_r is identity on forms and β_r on the point term; its transpose is identity on currents and $\beta_r' R = \sigma_r$ on point functionals. The literal dual of the plus- q cone in the sign convention (ADC8.5c) is $\operatorname{Cone}(-\delta_p q')[-1]$. The accepted RGR representative is $\operatorname{Cone}(+\delta_p q')[-1]$; their comparison negates the point coordinate. Therefore the map into that accepted representative is $(u, x) \mapsto (u, -\sigma_r x)$, as proved directly in (ACD4.4). Its positive costalk coordinates are still $(\sigma_r, 1_B)$, by the simultaneous coordinate comparisons in (RGR6.3a). Thus the local source square and the earlier current comparison are related by an actual continuous transpose, rather than by a coincidence of coefficient groups.

Finally $b_r = 0$ if and only if $D_r = 0$: if $b_r = 0$, the bounded map $D_r^* J$ vanishes on dense $E(Q)$, hence on H ; conversely $D_r = 0$ makes every formula zero. Therefore the new cross's surjectivity in degree -1 is equivalent to vanishing of the actual adjoint discrepancy. Unlike (ADC3.4), its obstruction image is the part that detects that discrepancy.

The exact coefficient actions on this new boundary can also be calculated:

$$\beta_r T_t^Q = U_t^* \beta_r, \quad \beta_r U_t^Q = T_t^* \beta_r, \quad b_r T_t^A = U_t^* b_r, \quad b_r U_t^A = T_t^* b_r. \quad (\text{ADC8.13})$$

For the first equality, $ET_t^Q = T_tE$ and $JT_t = U_t^*J$ by (ADC5.1); the diagonal D_r^* commutes with U_t^* . This gives $D_r^*JT_tE = U_t^*D_r^*JE$. The second follows with T_t, U_t exchanged; composing with q gives the remaining formulas. They define actual continuous chain maps on (ADC8.5), and show that the two coefficient directions descend to its invariant-cycle image. For a value class in Q/Z_r supported at an actual ρ , its nonzero image lies at $\rho^\#$ in H . A representative in the complete Q can still have higher-jet Jordan terms; these lie in the recorded kernel Z_r , rather than being deleted from Q . The source and target scalars on the indicated quotient and image are respectively

$$t^\rho = t^{1-\rho^\#}, \quad t^{1-\rho} = t^{\rho^\#}. \quad (\text{ADC8.14})$$

Their weights again agree on every nonzero image line. This proves the compatibility directly on the new cross, rather than inferring a vanished map from the existence of the two actions.

ADC9. Exact outcome and source-use record

The two calculated crosses test different parts of the same actual attaching data:

receiver	invariant-cycle obstruction image	additional retained defect
$\mathcal{E}_r$	0	$S_r = \ker(K_0 \rightarrow \text{im } \partial_0)$
$\mathcal{P}_r$	$b_r A \simeq Q/Z_r$	$H/b_r A$ in special degree 0.

(ADC9.1)

Their source maps are the actual $a_r(\bar{y}) = q'A_H(D_r y)$ and $b_r = D_r^*JEq$, not independent examples. The complete point-source quotient C/C_{crit} and the original arithmetic quotient Q/Z_r are joined by the nondegenerate pairing (ADC5.7). The comparison maps with the original source are (ADC4.5), (ADC8.10), and the boundary square (ADC8.11). The full original dual extension survives as the pushout (ADC4.2), with the source kernel and every endpoint degree retained.

The next attempted step required by WU065 has therefore been carried out. Starting from the automatically vanishing invariant-cycle cokernel of $\mathcal{E}_r$, the calculation first isolates the stronger coinvariant-injectivity defect, then removes its full J' coefficient quotient by the actual source triangle, and then constructs the transpose cone receiving the original invariant-cycle quotient. Its obstruction is precisely the image of the adjoint discrepancy on original classes. The two actual coefficient directions are computed on that image in (ADC8.13)–(ADC8.14); neither has separated source and target weights there. This describes the proved effect of the existing maps, without declaring a finite-field purity theorem for these coefficients or using a new twist to force one.

Reading coverage for this derivation: the connected USER_ARGUMENT_RECONSTRUCTION.md, its correction-precedence table, the verbatim WU062 and WU064–WU065 passages; the complete DC0–DC12 proof; LVD0–LVD9; RGR0–RGR12, including the full original-source, current-cone and positive-adjoint distinctions; GTAH0–GTAH4; ACD0–ACD8, with the literal current-resolution and source-map comparison in ACD2–ACD4 used in ADC8; DCA0–DCA12, including its complete cut-cone, genuine cover/trace and composition homotopies, with DCA3 and DCA6 used explicitly in ADC6; and the accepted LNC0–LNC11 proof, previously authored and independently reviewed here. LNC's current, nearby and support maps are used with their already proved continuous contractions and signs. The original source references and their exact witness status are preserved in DC0, LNC11, LVD0 and RGR0. In particular Deligne's local invariant-cycle proof is read through the identified current French transcription of Weil II; this is not claimed to be Deligne-authored TeX.

This is a calculation on the actual recovered arithmetic and current/sheaf maps. It establishes neither an off-critical zero nor the vanishing of D_r . It supplies the exact subquotients and commuting maps through which that discrepancy enters the stronger Deligne comparison.

Actual power-cover actions on the original adjoint-defect receiver

Complete independent local derivation. Proof locators **DCA0–DCA12**.

DCA0. The object, provenance, and category

This calculation continues the particular receiver of RGR11. The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All coefficient spaces, complex coordinates, positive integers, and operators below occur after the original arithmetic reconstruction. The two original branches retain their separate counters. No numerical coordinate or operation is assigned to the support.

The inputs read for this calculation are `CC_RESIDUE_GYSIN_RECEIVER_INDEPENDENT.md`, `RGR1`, `RGR9–RGR11`; `CC_CONTINUOUS_SUPPORT_DUAL_SHEAF.md`, `CSD3–CSD6` and `CSD9–CSD11`; `CC_LOCAL_NEARBY_CYCLES_AND_SPECIALIZATION.md`, `LNC5–LNC6`; the retained `GTR1–GTR4` proof in `CC_RAMIFIED_TRACE_AND_RESIDUE_ACTION.md`; `CC_ADJOINT_DEFECT_INVARIANT_CYCLE_CROSS.md`, `ADC1–ADC6`; and the continuous-dual comparison in `CC_ADJOINT_DEFECT_CONTINUOUS_DUAL.md`, `ACD1–ACD4`. These are exact programme dependencies, not new primary-source readings. The human-source identity for the original coefficient sheaf remains Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5, with the receiver comparison proved in CSP. The cover actions on the new receiver are proved below.

Derived statements use sheaves of complex vector spaces. Every coefficient operator, finite-coordinate homotopy, and specified current map is continuous. The continuous dual is never replaced by an algebraic dual. The proof does not assume a finite-rank perverse classification, a strong-dual open mapping theorem, or pullback of arbitrary distributions through a branch point. The notation $K[1]$ puts a degree-zero coefficient K in degree -1 .

DCA1. The actual arithmetic map and its two covariances

Retain

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), \ f(0) = 0, \ \int_{\mathbb{R}} f(v) \, dv = 0\}, \\ A &= \{b \in C^\infty(\mathbb{R}_{>0}) : \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j b(u)| < \infty \text{ for every } N, j \geq 0\}, \\ \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), \quad J = \Sigma S, \quad Q = A/J, \quad q : A \rightarrow Q, \quad B = A'_\beta, \quad j = q' : Q'_\beta \hookrightarrow B. \end{aligned} \quad (\text{DCA1.1})$$

The accepted original-image theorem gives closed J and the continuous inverse of $\Sigma : S \rightarrow J$. The original Mellin map and divisor are

$$\begin{aligned} \Theta b(s) &= \frac{1}{2} \int_0^\infty b(u) u^s \frac{du}{u}, \quad \Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1), \\ f_0(v) &= \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \quad (\text{DCA1.2})$$

The latter entire function has values $1/8$ at both $0, 1$, value $\pi/24$ at both $-1, 2$, and

$$\Theta \Sigma f_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) \quad (k \geq 1). \quad (\text{DCA1.3})$$

At an actual nontrivial zero ρ of multiplicity m_ρ , its full local expression is

$$t^{m_\rho} \frac{(\rho+t)(\rho+t-1)}{8} \pi^{-(\rho+t)/2} \Gamma((\rho+t)/2) \frac{\zeta(\rho+t)}{t^{m_\rho}}. \quad (\text{DCA1.4})$$

The local factor and every derivative remain part of the original comparison. The quotient Q retains all vanishing-jet conditions through order $m_\rho - 1$. It is not replaced by its value quotient in the argument below. The original function is still

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks} \quad (\Re s > 1). \quad (\text{DCA1.5})$$

Let $\mathcal{Z}$ be the actual distinct nontrivial zeros, put $\rho^\# = 1 - \bar{\rho}$, and retain

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_{\rho \in \mathcal{Z}} m_\rho x_\rho \overline{y_\rho}, \quad (\mathcal{J}y)_\rho = y_{\rho^\#}. \quad (\text{DCA1.6})$$

The actual value map $E : Q \rightarrow H$ has $(EF)_\rho = (\Theta_Q F)(\rho)$, is continuous, and has dense image by the retained full-jet interpolation theorem. The map

$$A_H(y)(F) = \langle EF, \mathcal{J}y \rangle_+ \quad (\text{DCA1.7})$$

is continuous into Q'_β , anti-linear, and injective. For a bounded subset $K \subset Q$, its strong seminorm is at most $\sup_{F \in K} \|EF\|_+ \|y\|_+$; density of $E(Q)$ proves injectivity.

For a fixed recovered parameter $r > 1$, use the actual operator

$$(T_r y)_\rho = r^\rho y_\rho, \quad U_r = r T_{1/r}, \quad D_r = T_r^* - U_r, \\ (D_r y)_\rho = d_r(\rho) y_\rho, \quad d_r(\rho) = e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}). \quad (\text{DCA1.8})$$

The known open strip makes these bounded. Set

$$C = \overline{H}, \quad \sigma_r(\bar{y}) = A_H(D_r y), \quad a = a_r = j\sigma_r : C \longrightarrow B. \quad (\text{DCA1.9})$$

This is exactly RGR11.3. The parameter r is fixed; the cover degree n below is a separate positive integer.

On A , write $T_t b(u) = b(u/t)$. It preserves J , descends to T_t^Q , and its value action is the displayed diagonal T_t . Directly on the actual zero coordinates,

$$T_t^* = J U_t J, \quad U_t^* = J T_t J. \quad (\text{DCA1.10})$$

For example $(J U_t \mathcal{J} y)_\rho = t^{1-\rho^\#} y_\rho = t^{\bar{\rho}} y_\rho$, the positive Hilbert adjoint of multiplication by t^ρ . Hence

$$(T_t^Q)' A_H(y)(F) = \langle T_t EF, \mathcal{J}y \rangle_+ = \langle EF, T_t^* \mathcal{J}y \rangle_+ = A_H(U_t y)(F). \quad (\text{DCA1.11})$$

Exchanging T_t, U_t proves the other covariance. Since the original diagonal D_r commutes with both operators, for

$$R = T'_n, \quad R_- = T'_{1/n}, \quad S_n = \overline{U}_n : C \rightarrow C, \quad L_n = \overline{T}_n : C \rightarrow C \quad (\text{DCA1.12})$$

one obtains the complete identities

$$\boxed{aS_n = Ra, \quad aL_n = nR_-a, \quad RR_- = R_-R = 1_B, \quad S_n L_n = L_n S_n = n1_C.} \quad (\text{DCA1.13})$$

Here conjugation specifies the linear operators on the conjugate vector space, rather than erasing the anti-linearity of (DCA1.7). All transposes are strong-continuous because their primal maps take bounded sets to bounded sets.

DCA2. The current cone and the positive support coordinate

Fix a positively oriented pole disk D , with coordinate $z_p = z$ at zero or $z_p = 1/z$ at infinity, and $i : \{p\} \hookrightarrow D$. The coefficient current complex is

$$\mathcal{T}_A^k(U) = (\mathcal{E}_c^{-k}(U; A))', \quad dT(\phi) = (-1)^{k+1}T(d\phi), \quad -2 \leq k \leq 0. \quad (\text{DCA2.1})$$

Its positive representatives are

$$c_b(\omega) = b\left(\int \omega\right), \quad h_p b(\eta) = b\left(\int \frac{d \arg z_p}{2\pi} \wedge \eta\right), \quad \delta_p b(f) = b(f(p)). \quad (\text{DCA2.2})$$

The angular coefficient is locally integrable, and Stokes on a punctured disk gives $dh_p b = \delta_p b$. The compact-test contraction in CSD2 identifies $\mathcal{T}_A \simeq \underline{B}[2]$, with the positive orientation above.

The actual receiver is

$$\begin{aligned} \mathcal{E} &= \mathcal{E}_r = \text{Cone}(-\delta_p a)[-1], \\ \mathcal{E}^{-1} &= \mathcal{T}_A^{-2}, \quad \mathcal{E}^0 = \mathcal{T}_A^{-1} \oplus i_* C, \quad \mathcal{E}^1 = \mathcal{T}_A^0, \\ d^{-1}b &= (-d_T b, 0), \quad d^0(b, x) = -d_T b + \delta_p a x. \end{aligned} \quad (\text{DCA2.3})$$

The quotient coordinate is x , so its connecting arrow is $+\text{Gys}_B a$:

$$\underline{B}[1] \longrightarrow \mathcal{E} \longrightarrow i_* C \xrightarrow{+\text{Gys}_B a} \underline{B}[2]. \quad (\text{DCA2.4})$$

The closed stalk representatives $b \mapsto c_b$ and $x \mapsto (h_p a x, x)$ give

$$i^* \mathcal{E} \simeq B[1] \oplus C, \quad R\Gamma(D^*, \mathcal{E}) \simeq B[1] \oplus B[0], \quad \text{res} = (1_B, a). \quad (\text{DCA2.5})$$

The second punctured coefficient is measured by the positive angular current. In the restriction fibre convention $d(s, t) = (ds, \text{res}(s) - dt)$, cancelling its identity pair gives

$$i^! \mathcal{E} \simeq [C \xrightarrow{+a} B] \quad \text{in degrees } 0, 1. \quad (\text{DCA2.6})$$

The sign is also seen directly: $d(h_p b, 0) = (-\delta_p b, h_p b)$, so the positive Dirac class and the positive angular boundary coordinate coincide. For shifted nearby and vanishing functors $\Psi = R\psi[-1]$, $\Phi = R\phi[-1]$,

$$\Psi \mathcal{E} = B, \quad \Phi \mathcal{E} = C, \quad \text{can} = 0, \quad \text{var} = +a. \quad (\text{DCA2.7})$$

These statements are consequences of the current and restriction cones, not applications of a finite-dimensional classification.

DCA3. The full cover and a continuous cut model

Let $b_n : D_s \rightarrow D$, $z_p \mapsto z_p^n$, with the source disk chosen as the inverse image of the target disk. This is a finite proper map. Its direct-image stalk is the finite direct sum of the source stalks above that point: small disjoint neighborhoods of the finite fibre contain the inverse image of some target neighborhood, by properness. This stalk formula proves exactness of b_{n*} on vector-space sheaves without a finite-rank assumption. At the pole there is one inverse-image point.

The inverse image of the target universal cover has exactly n components. Label them $0, \dots, n-1$ in positive cyclic order and put

$$\begin{aligned} V_n &= B^n, \quad \mathbf{t}(v_0, \dots, v_{n-1}) = (v_{n-1}, v_0, \dots, v_{n-2}), \\ \Delta b &= (b, \dots, b), \quad \Sigma_n(v) = \sum_{j=0}^{n-1} v_j, \quad e_0 b = (b, 0, \dots, 0). \end{aligned} \quad (\text{DCA3.1})$$

Thus $\Psi(b_{n*} \mathcal{E}) = V_n$, with monodromy $\mathbf{t}$. In this section Σ_n is a finite sheet sum, different from the original summation source Σ in (DCA1.1).

The circle cochain model of the punctured direct image is

$$N_n = [V_n \xrightarrow{\mathbf{t}-1} V_n] \quad \text{in degrees } -1, 0. \quad (\text{DCA3.2})$$

It follows by cutting a positive circle once, recording its constant value and the difference of its values across the cut. Constant sections are ΔB , and the positive loop class is measured by Σ_n . Explicitly

$$\ker(\mathbf{t}-1) = \Delta B, \quad \text{im}(\mathbf{t}-1) = \ker \Sigma_n, \quad V_n/(\mathbf{t}-1)V_n \xrightarrow{[v] \mapsto \Sigma_n v} B. \quad (\text{DCA3.3})$$

For the middle equality, if $\sum_j w_j = 0$, set $v_0 = 0$ and $v_j = -\sum_{k=1}^j w_k$ for $1 \leq j \leq n-1$. Then $v_{j-1} - v_j = w_j$ and $v_{n-1} - v_0 = w_0$, proving $(\mathbf{t}-1)v = w$. This solution is a finite continuous operator.

The stalk model is $S^{\text{st}} = B[1] \oplus C$. In the cut model its restriction has the representative

$$r_n^{-1} = \Delta : B \rightarrow V_n, \quad r_n^0 = e_0 a : C \rightarrow V_n. \quad (\text{DCA3.4})$$

The first formula is restriction of a constant current to every sheet. The second has positive loop coordinate $\Sigma_n e_0 a x = a x$, exactly the original angular restriction. Any representative produced from a different cut differs by an element of $\ker \Sigma_n$; the explicit solution in (DCA3.3) gives its continuous chain homotopy to (DCA3.4). The cut therefore changes the representative, not the original angular map.

For clarity about what these data construct, if $j_D : D^* \hookrightarrow D$, a complex with punctured restriction L , stalk S^{st} , and restriction arrow $r : S^{\text{st}} \rightarrow i^* Rj_{D*} L$ is represented by the homotopy fibre of

$$Rj_{D*} L \oplus i_* S^{\text{st}} \longrightarrow i_* i^* Rj_{D*} L, \quad (u, s) \longmapsto u - rs. \quad (\text{DCA3.5})$$

On the puncture this fibre is L ; at the pole it is the fibre of $(1, -r) : N \oplus S^{\text{st}} \rightarrow N$, which retracts to S^{st} . The natural comparison with the original complex is therefore a stalkwise quasi-isomorphism. A map of punctured complexes, a map of stalk complexes, and an explicit homotopy between their restriction composites consequently give an actual derived sheaf morphism via this fibre formula. No classification theorem is needed for that construction.

The associated restriction fibre is

$$B \xrightarrow{b \mapsto (0, \Delta b)} C \oplus V_n \xrightarrow{(x, v) \mapsto e_0 a x - (\mathbf{t}-1)v} V_n \quad (\deg -1, 0, 1). \quad (\text{DCA3.6})$$

The first arrow is a split continuous embedding; for example projection to the zeroth component splits Δ . Quotienting this identity pair retains

$$\Phi(b_{n*} \mathcal{E}) = C \oplus D_n, \quad D_n = V_n / \Delta B, \quad \boxed{\mathcal{V}_n(x, [v]) = e_0 a x - (\mathbf{t}-1)v.} \quad (\text{DCA3.7})$$

This is well defined because $(\mathbf{t}-1)\Delta = 0$. The canonical arrow is $\mathcal{C}_n(v) = (0, [v])$. Our positive support-cone convention gives

$$\mathcal{V}_n \mathcal{C}_n = 1 - \mathbf{t}, \quad \mathbf{t}_\Phi(x, [v]) = (x, [\mathbf{t}v - e_0 a x]), \quad \mathcal{C}_n \mathcal{V}_n = 1 - \mathbf{t}_\Phi. \quad (\text{DCA3.8})$$

These formulas fix both signs. Iterating the middle formula gives

$$\mathbf{t}_\Phi^k(x, [v]) = \left(x, \left[\mathbf{t}^k v - \sum_{\ell=0}^{k-1} \mathbf{t}^\ell e_0 a x \right] \right), \quad \mathbf{t}_\Phi^n = 1, \quad (\text{DCA3.9})$$

since $\sum_{\ell=0}^{n-1} \mathbf{t}^\ell e_0 a x = \Delta a x$. This retains the full vanishing object and its monodromy, including the cover contribution D_n .

DCA4. The genuine proper-current map

On target tests define

$$(F_n T)(\phi) = T(T_n b_n^* \phi). \quad (\text{DCA4.1})$$

Properness preserves compact support, and smooth pullback followed by the continuous coefficient operator is continuous on the fixed-support test spaces. Transposition is therefore continuous for both current dual topologies. It commutes with the Hom differential. Its representative factors are

$$F_n c_b = c_{nRb}, \quad F_n h_p b = h_p Rb, \quad F_n \delta_p b = \delta_p Rb. \quad (\text{DCA4.2})$$

For the first equality, the integral of the pulled-back compact two-form is n times its integral. On the puncture each inverse branch contributes $1/n$ of the positive angular form, so the sum of all n branch contributions is that same form. Equality extends as a locally integrable current through the pole. Evaluation at the single inverse-image pole proves the last equality. Thus the three displayed factors have separate geometric proofs.

By (DCA1.13),

$$\boxed{F_n : b_{n*} \mathcal{E} \rightarrow \mathcal{E}, \quad (b, x) \mapsto (F_n b, S_n x)} \quad (\text{DCA4.3})$$

is a current chain map: its attaching term satisfies $F_n \delta_p a x = \delta_p R a x = \delta_p a S_n x$. Its stalk map, in the actual representatives of (DCA2.5), is

$$i^* F_n = (nR, S_n) : B[1] \oplus C \rightarrow B[1] \oplus C. \quad (\text{DCA4.4})$$

The source nearby space is all of V_n , and the map is

$$\boxed{F_{n,\Psi} = R\Sigma_n : V_n \rightarrow B.} \quad (\text{DCA4.5})$$

A single inverse branch is an orientation-preserving local isomorphism and contributes R , with no additional local degree; summing the branches proves the formula. It intertwines $\mathfrak{t}$ with identity.

On $N_n \rightarrow N_1$ the two cochain maps are both $R\Sigma_n$. They commute with restriction, since

$$R\Sigma_n \Delta = nR, \quad R\Sigma_n e_0 a = Ra = aS_n. \quad (\text{DCA4.6})$$

Thus the vanishing and costalk maps are exactly

$$\boxed{F_{n,\Phi}(x, [v]) = S_n x, \quad i^! F_n = (S_n, R) : [C \xrightarrow{a} B] \rightarrow [C \xrightarrow{a} B].} \quad (\text{DCA4.7})$$

The extra summand D_n in the literal vanishing source maps to zero; it remains part of that source. The variation square is the identity

$$R\Sigma_n \mathcal{V}_n(x, [v]) = Rax = aS_n x. \quad (\text{DCA4.8})$$

DCA5. Invariants, coinvariants, and actual Gysin adjunction

On nearby invariants, (DCA4.5) restricts to

$$\Delta B \rightarrow B, \quad \Delta b \mapsto nRb. \quad (\text{DCA5.1})$$

On coinvariants, using the exact positive identification (DCA3.3), it is

$$B \rightarrow B, \quad b \mapsto Rb. \quad (\text{DCA5.2})$$

The diagonal is not the entire nearby space; its composite with the coinvariant quotient is multiplication by n . These two functors therefore see the two different factors of the same genuine map.

The Gysin square is

$$\begin{array}{ccc} i_* C & \xrightarrow{b_{n*}(\text{Gys}_B a)} & b_{n*} \underline{B}[2] \\ \downarrow S_n & & \downarrow F_n \\ i_* C & \xrightarrow{\text{Gys}_B a} & \underline{B}[2]. \end{array} \quad (\text{DCA5.3})$$

At the level of currents this is $F_n \delta_p a = \delta_p a S_n$. Under the oriented support adjunction $i^! \underline{B}[2] = B$, its coefficient equality is

$$Ra = aS_n. \quad (\text{DCA5.4})$$

The ordinary stalk factor nR belongs instead to (DCA5.1). Replacing (DCA5.4) by $nRa = aS_n$ would change the functor and would be a false Gysin square. The distinction is proved by the original Dirac and angular currents, not assigned by a weight label.

DCA6. Construction of the transfer-adjoint direction without branch-point current pullback

There is a derived sheaf map

$$\mathbb{L}_n : \mathcal{E} \longrightarrow b_{n*} \mathcal{E} \quad (\text{DCA6.1})$$

whose generic coefficient is R_- , whose source coefficient on C is L_n , and whose positive supported coefficient is nR_- . We construct its complete gluing data.

Use the constant-sheaf unit with coefficient R_- . On the puncture its map is ΔR_- into the full n -sheet coefficient. On the pole stalk it is R_- . On positive support it is nR_- : a positive boundary circle is mapped with degree n , or equivalently $b_n^*(d \arg z_p / (2\pi)) = n d \arg z_p / (2\pi)$. Its Gysin square is exactly

$$nR_- a = aL_n, \quad (\text{DCA6.2})$$

already proved for the actual arithmetic map in (DCA1.13).

In the explicit cut models use

$$f_{\text{st}} = (R_-, L_n), \quad f_N^{-1} = \Delta R_-, \quad f_N^0 = \Delta R_-. \quad (\text{DCA6.3})$$

For $x \in C$, set

$$z_x = R_- ax, \quad w_n(x) = (0, z_x, 2z_x, \dots, (n-1)z_x) \in V_n. \quad (\text{DCA6.4})$$

All coordinates are retained. Direct componentwise subtraction gives

$$(\mathfrak{t} - 1)w_n(x) = ne_0 z_x - \Delta z_x. \quad (\text{DCA6.5})$$

The zeroth component is $(n-1)z_x$, and each of the other $n-1$ components is $-z_x$. Therefore

$$r_n f_{\text{st}} - f_N r_1 = d_{N_n} w_n + w_n d_{S^{\text{st}}}. \quad (\text{DCA6.6})$$

In degree -1 both sides vanish; in degree zero the equality is (DCA6.2) and (DCA6.5). Formula (DCA3.5) now constructs (DCA6.1) from these continuous maps and this specified homotopy. No pullback of a singular current at the branch point has been used.

For reference, if a restriction-fibre map has coefficient maps f_S, f_N and homotopy k satisfying $r' f_S - f_N r = dk + kd$, its formula is

$$(s, t) \longmapsto (f_S s, f_N t + ks). \quad (\text{DCA6.7})$$

Substitution into $d(s, t) = (ds, rs - dt)$ proves the chain identity. Applying this formula gives

$$\boxed{\mathbb{L}_{n, \Psi} = \Delta R_-, \quad \mathbb{L}_{n, \Phi}(x) = (L_n x, [w_n(x)])}. \quad (\text{DCA6.8})$$

The raw-cut correction $[w_n(x)]$ is necessary: omitting it would give variation $ne_0 R_- ax$ instead of $\Delta R_- ax$. With it,

$$\mathcal{V}_n(L_n x, [w_n(x)]) = e_0 a L_n x - (\mathfrak{t} - 1)w_n(x) = \Delta R_- ax. \quad (\text{DCA6.9})$$

The map also lands in monodromy invariants, since $\mathfrak{t} w_n(x) - e_0 a L_n x - w_n(x) = -\Delta z_x$, whose class in D_n is zero.

The complete local coefficient maps are

$$\boxed{i^* \mathbf{L}_n = (R_-, L_n), \quad i^! \mathbf{L}_n = (L_n, nR_-) : [C \xrightarrow{a} B] \rightarrow [C \xrightarrow{a} B].} \quad (\text{DCA6.10})$$

The positive angular map is nR_- , because its class in (DCA3.3) is $\Sigma_n \Delta R_- = nR_-$. It is consequently also the positive Dirac support factor. These are the generic/angular/Dirac factors

$$(R_-, nR_-, nR_-), \quad (\text{DCA6.11})$$

with their actual domains, rather than three scalar assignments to one functor.

One can display the invariant copy of C inside the full vanishing space without discarding the complement:

$$\iota_n(x) = \left(x, \left[0, \frac{ax}{n}, \frac{2ax}{n}, \dots, \frac{(n-1)ax}{n} \right] \right). \quad (\text{DCA6.12})$$

It satisfies $\mathbf{t}_\Phi \iota_n = \iota_n$ and $\mathcal{V}_n \iota_n = \Delta a/n$, by the same componentwise calculation. Formula (DCA6.8) is exactly $\iota_n L_n$, because $aL_n = nR_- a$. All fractions in this comparison are displayed; (DCA6.4) remains the actual chosen cut correction.

DCA7. Both compositions, including their homotopies

The composite in the target direction is

$$\boxed{\mathbf{F}_n \mathbf{L}_n = n \, 1_{\mathcal{E}}} \quad \text{in the derived sheaf category.} \quad (\text{DCA7.1})$$

On the stalk complexes its factors are $nRR_- = n$ and $S_n L_n = n$. On the punctured cochain complexes they are $R\Sigma_n \Delta R_- = n$. Its gluing homotopy has the remaining value

$$k(x) = R\Sigma_n w_n(x) = \sum_{j=0}^{n-1} j \, ax = \frac{n(n-1)}{2} \, ax. \quad (\text{DCA7.2})$$

This term has not been suppressed. The degree-minus-one stalk homotopy $\ell : C \rightarrow B$, $\ell(x) = \sum_{j=0}^{n-1} j \, ax$, removes it, because the target restriction in degree -1 is $r_1^{-1} = 1_B$. More explicitly, for the restriction fibre, $H(s, t) = (\ell s, 0)$ has $dH + Hd = (0, ks)$. The same homotopy on the supported summand of the reconstruction (DCA3.5) proves (DCA7.1) at the sheaf-cone level.

Let $\mathbf{d}$ be the positively chosen deck generator on $b_{n*} \mathcal{E}$. It is identity on the stalk model and $\mathbf{t}$ in both punctured degrees. Its gluing homotopy is $-e_0 a$, because

$$r_n - \mathbf{t} r_n = -(\mathbf{t} - 1) e_0 a. \quad (\text{DCA7.3})$$

Its induced vanishing map is exactly $\mathbf{t}_\Phi$ in (DCA3.8). The reverse composite is

$$\boxed{\mathbf{L}_n \mathbf{F}_n = \sum_{k=0}^{n-1} \mathbf{d}^k.} \quad (\text{DCA7.4})$$

On full nearby coefficients both sides are

$$\Delta \Sigma_n = \sum_{k=0}^{n-1} \mathbf{t}^k : V_n \rightarrow V_n. \quad (\text{DCA7.5})$$

On the pole stalk they are $n \, 1$. On vanishing coefficients the composite from (DCA4.7) and (DCA6.8) is

$$(x, [v]) \mapsto (nx, [(0, ax, 2ax, \dots, (n-1)ax)]). \quad (\text{DCA7.6})$$

For the deck norm, sum (DCA3.9). The term $\sum_k \mathfrak{t}^k v = \Delta \Sigma_n v$ has zero class in D_n , and the correction is

$$-\sum_{\ell=0}^{n-2} (n-1-\ell) \mathfrak{t}^\ell e_0 a x. \quad (\text{DCA7.7})$$

Its coordinates are $(-(n-1)ax, -(n-2)ax, \dots, -ax, 0)$. Adding $(n-1)\Delta ax$ gives exactly the vector in (DCA7.6), proving equality in the quotient.

At the gluing-homotopy level, the difference between the composite correction and (DCA7.7) is precisely $(n-1)\Delta ax$. The stalk homotopy $\ell(x) = (n-1)ax$ removes it through $r_n^{-1} = \Delta$, exactly as in the proof of (DCA7.1). Thus (DCA7.4) holds on the derived sheaf objects, not merely on their cohomology groups. Likewise $\mathbf{d}^n = 1$: its accumulated homotopy $-\Delta a$ is removed by the stalk homotopy $-a$.

For $n > 1$, the deck norm is not generally multiplication by n on V_n : it takes $(v_0, \dots, v_{n-1})$ to $\Delta(\sum_j v_j)$. The actual nonzero coefficient space $B = A'$ is retained in this assertion. On invariants its value is n , and on the extra cyclic components it vanishes. On costalks both compositions are n in each degree, consistently with (DCA4.7), (DCA6.10), and the identity deck action on the single supported point.

DCA8. The original dual extension and all its comparison arrows

Classical human-source attribution: extension of continuous linear functionals uses Hahn–Banach; see Terence Tao, 245B Notes 6, Theorem 1 and its complex proof. A continuous seminorm and its kernel reduce the locally convex application to that normed-space result. This does not assert a continuous splitting.

The closed original source gives the full exact continuous-transpose row

$$0 \longrightarrow Q' \xrightarrow{j} B \xrightarrow{k'} J' \longrightarrow 0, \quad k : J \hookrightarrow A. \quad (\text{DCA8.1})$$

Its surjectivity is continuous Hahn–Banach extension from the closed subspace J . This does not assert that $k' : B_\beta \rightarrow J'_\beta$ is open.

The original dilation preserves J , so (DCA8.1) commutes with $(T_n^Q)', T'_n, (T_n|_J)'$, and also with their inverse-parameter maps and the displayed factor n . The actual receiver

$$\mathcal{K}_r = \text{Cone}(-\text{Gys}_{Q'} \sigma_r)[-1] \quad (\text{DCA8.2})$$

has the proved triangle

$$\mathcal{K}_r \longrightarrow \mathcal{E}_r \longrightarrow \underline{J}'[1] \longrightarrow \mathcal{K}_r[1]. \quad (\text{DCA8.3})$$

In continuous current representatives its first map is $(v, x) \mapsto (q^\vee v, x)$, and its second is $(b, x) \mapsto k^\vee b$. The coefficient identity $k'j = 0$ proves their composite zero. Cancelling their shared supported C coordinate reduces the cone comparison to the constant-current coefficient row (DCA8.1), proving the triangle as in RGR9.

Every construction DCA3–DCA7 applies with B replaced by Q' , a by σ_r , and with its actual transpose dilation. Naturality of $q^\vee, k^\vee$, of the positive Gysin map, and of the finite sheet sums proves that both cover directions commute with (DCA8.3). In particular the first map uses j on each of the n nearby sheets and identity on the original C component; the raw-cut corrections commute because $j\sigma_r = a$. The quotient J' has the corresponding constant-sheaf trace and unit maps.

There is also the actual map $\mathcal{E}_r \rightarrow D_c P$, where $P = \ker(\underline{A} \rightarrow i_* Q)[1]$. In positive quotient coordinates it uses 1_B on nearby coefficients and σ_r on vanishing coefficients. Its cone is $i_* \text{Cone}(\sigma_r)$, as proved in RGR5. The cover directions commute with this map by

$$\sigma_r S_n = (T_n^Q)' \sigma_r, \quad \sigma_r L_n = n(T_{1/n}^Q)' \sigma_r. \quad (\text{DCA8.4})$$

Its raw-cut correction commutes componentwise by the same two identities. Consequently the cover calculation acts on the receiver of the original dual extension; it has not introduced an unrelated coefficient extension.

DCA9. Both endpoint lines and the actual GTR factors

The original pole source is

$$V_+ = S \oplus \mathbb{C}_{c_0} \oplus \mathbb{C}_{c_1}, \quad V_- = S \oplus \mathbb{C}_{d_0} \oplus \mathbb{C}_{d_1},$$

$$\rho_+(t)(f, c_0, c_1) = (f(\cdot/t), c_0, tc_1), \quad \rho_-(t)(h, d_0, d_1) = (th(t\cdot), td_0, d_1). \quad (\text{DCA9.1})$$

The four labels remain the original value-at-zero and real-integral labels at their respective poles. Their dual coordinate lines are denoted by the corresponding primed labels. The full receiver is

$$\mathcal{E}_{r,\text{full}} = \mathcal{E}_r \oplus i_* E'_p[-1]. \quad (\text{DCA9.2})$$

The endpoint summand has zero nearby object, stalk and costalk $E'_p[-1]$, and shifted vanishing object $E'_p[-1]$. Its unshifted vanishing complex is $E'_p[0]$. Thus the endpoints occur in stalk and support degree 1, separately from the two-term supported complex in (DCA2.6).

The actual CSD/GTR maps on this summand are

$$\mathbf{F}_n|_{E'_p} = \rho_p(n)'|_{E'_p}, \quad \mathbf{L}_n|_{E'_p} = n\rho_p(1/n)'|_{E'_p}. \quad (\text{DCA9.3})$$

In the full retained order (c'_0, c'_1, d'_0, d'_1) , these are

$$\boxed{\mathbf{F}_n : (1, n, n, 1), \quad \mathbf{L}_n : (n, 1, 1, n).} \quad (\text{DCA9.4})$$

They follow by transposing the two explicit diagonal endpoint blocks in (DCA9.1); each product is n . A deck rotation fixes the pole and acts identically on each endpoint coordinate, so its norm there is n , also proving the reverse composition on this retained summand.

For comparison, the full original global coefficient complex has

$$D = [V_+ \oplus V_- \xrightarrow{r_+ - r_-} A \xrightarrow{0} A] \quad \text{in degrees } 0, 1, 2,$$

$$r_+ = \Sigma, \quad r_- = R_A \Sigma = \hat{\Sigma}, \quad R_A b(u) = u^{-1}b(u^{-1}). \quad (\text{DCA9.5})$$

Its positive geometric pullback and weighted ramified trace with inverse coefficients have the exact cochain representatives

$$\mathbf{B}_n = (\rho(n), T_n, nT_n), \quad \mathbf{U}_n = (n\rho(1/n), nT_{1/n}, T_{1/n}). \quad (\text{DCA9.6})$$

The weighted trace sums all interior sheets and is n times identity at each pole, including both endpoint lines. This gives $\text{Tr}_n u_n = n1$ on every stalk; its global factors then follow from the angular and top-degree orientation integrals, as calculated in GTR2–GTR4.

Transposition of these degree-zero maps adds no Hom-differential sign. The full original dual rows are

dual degree	−2	−1	0
$\mathbf{B}'_n$	nT'_n	T'_n	$\rho(n)'$
$(\mathbf{B}_n^{-1})'$	$n^{-1}T'_{1/n}$	$T'_{1/n}$	$\rho(1/n)'$
$\mathbf{U}'_n$	$T'_{1/n}$	$nT'_{1/n}$	$n\rho(1/n)'$

(DCA9.7)

All three rows are retained. The second is the pairing contragredient and differs from the third by the full factor n . The current pushforward direction (DCA4.3) has the first row's geometry; the constructed opposite direction (DCA6.1) has the third row's geometry. Their endpoint maps are exactly (DCA9.3). This establishes the local cover structure on the new cone. The separate literal dual identification is supplied by

ACD1–ACD4: put $b_r = D_r^* JEq : A \rightarrow H$ and $\mathcal{P}_r = [\underline{A}^{-1} \xrightarrow{+b_r \text{ev}_p} i_* H^0]$. The fine resolution ACD2.2 has continuous compact-test dual chain-isomorphic to $\mathcal{E}_r$ by the degree signs $(-1, +1, -1)$ and the Riesz map $\overline{H} \rightarrow H'$. Its full pairing is ACD3.3, and the map $\mathcal{E}_r \rightarrow D_c P$ is the transpose of the actual original-source map ACD4.3, with the point-coordinate minus in ACD4.4. These explicit comparisons identify the primal object and its pairing; the identification is not inferred from the table of cohomology factors.

DCA10. The actual arithmetic image receives matching actions

The conclusions about possible separation can be checked directly on the original arithmetic map rather than on a substitute example. For the actual value vector $e_\rho \in H$, the definition (DCA1.7) gives

$$\sigma_r(\bar{e}_\rho) = m_\rho \overline{d_r(\rho)} \delta_{\rho^\#,0}, \quad a(\bar{e}_\rho) = m_\rho \overline{d_r(\rho)} q' \delta_{\rho^\#,0}. \quad (\text{DCA10.1})$$

Here $\delta_{\eta,0}(F) = (\Theta_Q F)(\eta)$, so the original Mellin factor $1/2$ is part of this functional. Its multiplicity and the full original defect coefficient have both been retained. The equality follows by putting $D_r e_\rho = d_r(\rho) e_\rho$ into the weighted reflected inner product.

The two exact covariance eigenvalue comparisons on this displayed coordinate are

map	source coefficient on $\bar{e}_\rho$	target coefficient on $q' \delta_{\rho^\#,0}$	
$aS_n = Ra$	$n^{1-\bar{\rho}}$	$n^{\rho^\#}$	(DCA10.2)
$aL_n = nR_-a$	$n^{\bar{\rho}}$	$n^{1-\rho^\#}$	

Each row agrees identically because $\rho^\# = 1 - \bar{\rho}$. The coefficients are equal whether or not $d_r(\rho)$ vanishes. When it vanishes, (DCA10.1) is the zero map at that actual coordinate; when it does not, (DCA10.2) is exact equivariance on its nonzero image line. No existence of an off-critical zero is assumed.

The closed source kernel is

$$\ker a = \overline{\ker D_r} = \{\bar{y} : y_\rho = 0 \text{ whenever } \Re \rho \neq 1/2\}. \quad (\text{DCA10.3})$$

The equality follows from injectivity of j, A_H , and from $r^{\Re \rho} = r^{1-\Re \rho}$ if and only if $\Re \rho = 1/2$. Both actions in (DCA1.13) preserve this exact kernel and the actual image $a(C)$. Their induced quotient maps are continuous for quotient topologies, without any conclusion that the image is closed.

Thus the discrepancy between the constant-current factor nR and the angular/support factor R does not impose a zero attaching map. Those factors are the two receivers (DCA5.1)–(DCA5.2) of the full nearby map $R\Sigma_n$. The supported arithmetic map has precisely the matching eigenvalues in (DCA10.2). To demand an incompatible weight on that same supported map would alter its already proved action. In particular, this cover calculation has not proved $a = 0$ or $D_r = 0$.

DCA11. The exact continuation supplied by the calculation

The resulting operation on the actual Gysin obstruction $\mathfrak{o}_r = \text{Gys}_B a$ is now computed, together with its full nearby source, both vanishing maps, costalk maps, endpoint maps, and the connecting morphisms to the dual original extension. Under support adjunction it obeys $Ra = aS_n$ in the proper-current direction and $nR_-a = aL_n$ in the opposite direction. These are identities on the complete source $\bar{H}$, not merely on the displayed finite value coordinates.

The next receiver after the automatically vanishing invariant-cycle image quotient is the actual image $a(C)$, as computed in ADC3. Its source is the exact quotient $C/\ker a$, and DCA10 gives its existing equivariant map into $q'Q' \subset A'$. The concrete cover attempt is complete: its action does not separate this source from this target, because the actions match exactly. The remaining calculation must use additional proved structure of that same image or of its original continuous pairing, while retaining (DCA1.13), (DCA8.3), and all endpoint terms. Replacing the supported target factor by its invariant factor would contradict (DCA5.3), rather than supply such a calculation.

DCA12. Scope and reproducible algebraic checks

Every finite-sheet identity used in the proof can be checked by the following displayed component calculations: $(t-1)\Delta = 0$; $\Sigma_n(t-1) = 0$; the partial-sum inverse in DCA3; $(t-1)(0, z, \dots, (n-1)z) = ne_0 z - \Delta z$; $\sum_k t^k = \Delta \Sigma_n$; and the coordinate difference $(n-1)\Delta ax$ between (DCA7.6) and (DCA7.7). They prove the signs, the retained raw-cut correction, and both derived composition laws for every positive integer n ,

including $n = 1$, when the extra sheet quotient $D_n = B^n/\Delta B$ is zero, the correction vector is zero, and both cover maps are identity. The independently fixed arithmetic parameter remains $r > 1$.

The argument constructs all stated sheaf maps from continuous currents or explicit continuous restriction-cone data. It never infers a unique current chain map from cohomology alone. No conclusion about strong openness of $A'_\beta \rightarrow J'_\beta$, image closure of $a(C)$, finite-dimensionality, arithmetic purity, or vanishing of the actual adjoint defect is inserted. Every original coefficient and endpoint contribution used in the construction remains present in the formulas above.

The actual adjoint-defect source quotient and its continuous dual

Independent review and receiving calculation. Proof locators **AST0–AST9**.

AST0. Construction stage, inputs and review coverage

The supporting datum remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All vector spaces, arithmetic parameters, analytic coordinates and quotients below belong to the already reconstructed coefficient system. The two branches retain their separate recovered counters. No arithmetic, metric, midpoint or vector operation is assigned to the support.

The connected reconstruction, its correction-precedence table, the current construction guide, and verbatim WU062 and WU064–WU065 were reread before this calculation. The new operation is a quotient of the actual ACD specialization source, followed by continuous duality; its prerequisites are proved below. The user's desired weight conclusion is not added as a hypothesis.

The complete written proof GDC0–GDC14, final SHA256 5d0f471b3250e4ee894caca68232c556d64a137a49a9fe25a641db61b94f79ab, was independently read and checked in full. Its earlier version f9e15dff7755b0d9c79c9dd896e55966b87302051fcf8086fad90e56da126589 was also fully read; the final typography repair preserves the proof. The review checked its reflection sign, exact annihilator, strong closure, bounded parameter changes, all restriction and quotient topologies, receiver triangle, positive-residue restriction, finite multiplicities and source scope. A separate bounded independent check of GDC7 and the invoked SDT4 compact-lift construction also found no defect. This is a completed independent audit of the stated final version, in addition to its earlier root verification.

The receiving proof uses the actual maps of ACD0–ACD8, SHA256 1373884c33e74aea9c0d48404152b16812b434b6cc4760227c8596666e919fda, previously read and independently checked in full, and the explicit seminorm, finite-net and compact-lift arguments of SDT1–SDT4, reread for this extension. These are programme derivations. The human coefficient source remains Connes–Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5; the retained author TeX at sources/CC_0903_2024_v3/author_source/announ c3.tex, lines1376–1667, was read in the preceding source reconstruction. No new whole-paper reading or identification with Deligne's étale invariant-cycle cross is claimed.

AST1. The actual coefficient, the full original divisor and the defect

Retain the original spaces and maps

$$\begin{aligned} S &= \{f \in \mathcal{S}(\mathbb{R}; \mathbb{C}) : f(-v) = f(v), f(0) = 0, \int_{\mathbb{R}} f(v) dv = 0\}, \\ A &= \{a \in C^\infty(\mathbb{R}_{>0}) : p_{N,j}(a) = \sup_{u>0} (u^N + u^{-N}) |(u\partial_u)^j a(u)| < \infty \text{ for every } N, j \geq 0\}, \\ \Sigma f(u) &= 2 \sum_{k \geq 1} f(ku), \quad J = \Sigma S, \quad q : A \rightarrow Q = A/J, \quad B = A'_\beta. \end{aligned} \tag{AST1.1}$$

The accepted exact-image theorem makes J closed. The Mellin comparison, with its original factor, is

$$\Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}, \quad \Theta \Sigma f(s) = \zeta(s) \int_0^\infty f(v) v^s \frac{dv}{v} \quad (\Re s > 1). \tag{AST1.2}$$

Its entire rapid-strip quotient is $Q \simeq \mathcal{B}/J$, where J imposes all multiplicities of the actual nontrivial zeros. The original source vector

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}$$

has exactly

$$F_0(s) = \Theta \Sigma f_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

$$F_0(0) = F_0(1) = \frac{1}{8}, \quad F_0(-1) = F_0(2) = \frac{\pi}{24}, \quad F_0(-2k) = \frac{k(2k+1)(-1)^k \pi^k}{2k!} \zeta'(-2k) \quad (k \geq 1). \quad (\text{AST1.3})$$

These values include the cancellations of the original pole and trivial zeros; none creates an extra quotient jet. At an actual nontrivial zero ρ , writing $z = s - \rho$, the retained local expression is

$$F_0(\rho + z) = z^{m_\rho} \frac{(\rho + z)(\rho + z - 1)}{8} \pi^{-(\rho+z)/2} \Gamma((\rho + z)/2) \frac{\zeta(\rho + z)}{z^{m_\rho}}. \quad (\text{AST1.4})$$

Every derivative in this product uses the full Leibniz sum. Explicitly, away from its individually singular factors and then by holomorphic continuation where the whole expression is regular,

$$F_0^{(j)}(s) = \sum_{k_0+k_1+k_2+k_3=j} \frac{j!}{k_0!k_1!k_2!k_3!} \left(\frac{s(s-1)}{8} \right)^{(k_0)} \left(-\frac{\log \pi}{2} \right)^{k_1} \pi^{-s/2} 2^{-k_2} \Gamma^{(k_2)}(s/2) \zeta^{(k_3)}(s). \quad (\text{AST1.5})$$

This entire comparison remains a comparison with original ζ , not its replacement. In its original half-plane the unit term, Euler product and logarithmic derivative remain

$$\zeta(s) = 1 + \sum_{k \geq 2} k^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad -\frac{\zeta'(s)}{\zeta(s)} = \sum_p \sum_{k \geq 1} (\log p) p^{-ks}. \quad (\text{AST1.6})$$

Let $\mathcal{Z}$ be the actual distinct nontrivial zeros, with multiplicities m_ρ , and set $\rho^\# = 1 - \bar{\rho}$. Use

$$H = \ell^2(\mathcal{Z}, m), \quad \langle x, y \rangle_+ = \sum_\rho m_\rho x_\rho \overline{y_\rho}, \quad (\mathcal{J}y)_\rho = y_{\rho^\#}, \quad E : Q \rightarrow H. \quad (\text{AST1.7})$$

Here $(EF)_\rho = (\Theta_Q F)(\rho)$, and the inner product is linear in the first variable. The proved full-jet isolators realize every finite value vector, so E is continuous with dense image. Put $H_L = \ell^2(\mathcal{Z}_L, m)$, $H_O = \ell^2(\mathcal{Z}_O, m)$, where the subsets denote respectively $\Re \rho = 1/2$ and $\Re \rho \neq 1/2$, and write P_L, P_O for their orthogonal projections. These definitions do not assert that either subset is empty or nonempty.

For a coefficient parameter $t > 0$, $(T_t y)_\rho = t^\rho y_\rho$ on H ; on A the original action is $(T_t a)(u) = a(u/t)$, descending to Q and giving the stated value action under E . For the fixed coefficient parameter $r > 1$, ACD uses

$$D_r = T_r^* - r T_{1/r}, \quad d_r(\rho) = e^{-i\Im \rho \log r} (r^{\Re \rho} - r^{1-\Re \rho}), \quad \beta_r = D_r^* \mathcal{J} E : Q \rightarrow H, \quad b_r = \beta_r q. \quad (\text{AST1.8})$$

Thus, with no multiplicity factor absorbed into the vector coordinate,

$$(\beta_r F)_\rho = \overline{d_r(\rho)} F(\rho^\#). \quad (\text{AST1.9})$$

For $C = \overline{H}$, the Riesz map is

$$R : C \rightarrow H', \quad R(\overline{y})(h) = \langle h, y \rangle_+.$$

The exact continuous transposes are

$$\begin{aligned} \sigma_r &= \beta'_r R : C \rightarrow Q'_\beta, \quad a_r = b'_r R = q'_r \sigma_r : C \rightarrow B, \\ \sigma_r(\overline{y})(F) &= \langle \beta_r F, y \rangle_+ = \langle EF, \mathcal{J} D_r y \rangle_+. \end{aligned} \quad (\text{AST1.10})$$

Continuity into strong duals follows from the estimate on every bounded test set K , $\sup_{F \in K} |\langle \beta_r F, y \rangle_+| \leq \sup_{F \in K} \|\beta_r F\|_+ \|y\|_+$.

AST2. The specialization kernel is exactly the GDC source

Because $r > 1$, the scalar $d_r(\rho)$ vanishes exactly on $\mathcal{Z}_L$. Reflection preserves both subsets and their multiplicities. Formula (AST1.9) therefore proves

$$\boxed{Z_r = \ker \beta_r = N_O := \ker(P_O E), \quad \ker b_r = A_O := q^{-1}N_O.} \quad (\text{AST2.1})$$

Both kernels are closed in their specified Fréchet spaces, and they are independent of $r > 1$. The actual common specialization source is consequently

$$\mathcal{R} := Q/N_O \xleftarrow{\sim} A/A_O, \quad [a] \mapsto [q(a)]. \quad (\text{AST2.2})$$

This is a topological quotient isomorphism. Indeed q and the quotient $Q \rightarrow Q/N_O$ are open; their composite has kernel A_O . An open set in A/A_O is the image of an open set in A , whose composite image is open in Q/N_O . The inverse is therefore continuous, without selecting any section.

In the genuine ACD restriction fibre the quotient coordinate is $(h, a) \mapsto h - b_r a$. Write $\vartheta_p = d \arg z_p / (2\pi)$ for the positive angular form in the oriented pole coordinate. Its boundaries and their source comparison are exactly

$$\begin{aligned} \partial_{-1} &= -b_r : A \rightarrow H, & \partial_0 &= +1 : A[\vartheta_p] \rightarrow A, \\ \text{coker } \text{sp}_{-1} &= A/A_O \simeq \mathcal{R} \xrightarrow{-\bar{\beta}_r} b_r A, & \bar{\beta}_r([F]) &= \beta_r F. \end{aligned} \quad (\text{AST2.3})$$

The last arrow is a continuous bijection onto the image with its Hilbert subspace topology. It is a topological isomorphism if that image is instead given its transported quotient topology. No inverse continuity into the Hilbert subspace topology is asserted. The square with the original boundary $-q : A \rightarrow Q$ commutes because $b_r = \beta_r q$. Thus (AST2.2) is a refinement of the actual specialization calculation, not a quotient selected independently of that boundary.

At each actual multiplicity block $\mathbb{C}[z]/z^{m_\rho}$, the kernel N_O retains the whole block on $\mathcal{Z}_L$, and retains $z\mathbb{C}[z]/z^{m_\rho}$ on $\mathcal{Z}_O$. Accordingly $\mathcal{R}$ observes one off-line value coordinate and no line coordinate on that block. All higher jets remain in the explicitly retained kernel. This local description neither replaces the full global quotient by a product nor proves unrestricted interpolation.

AST3. The quotient topology and its full strong dual

We prove the new quotient's bounded-compact property rather than assuming a quotient theorem for general Montel spaces. Put

$$P_n(a) = \max_{0 \leq j \leq n} \sup_{x \in \mathbb{R}} (e^{nx} + e^{-nx}) \left| \frac{d^j}{dx^j} a(e^x) \right|, \quad P_n^O([a]) = \inf_{v \in A_O} P_n(a + v). \quad (\text{AST3.1})$$

The quotient topology on $\mathcal{R} = A/A_O$ is precisely the topology of these increasing seminorms. Closedness of A_O proves Hausdorffness. Completeness follows by the SDT1 successive-lift argument with A_O in place of J : for a Cauchy subsequence choose lifts of successive differences with $P_k < 2^{-k}$; their series converges in A and maps to the required quotient limit.

Let $K \subset \mathcal{R}$ be bounded. For each fixed n , choose representatives of its elements with

$$P_{n+1}(a_y) < 1 + \sup_{z \in K} P_{n+1}^O(z). \quad (\text{AST3.2})$$

SDT2 proves that the set with this single P_{n+1} bound is totally bounded for P_n , using both original exponential weights and one additional derivative. Its finite net maps to a finite P_n^O -net for K . This holds for every n , so K is totally bounded in the complete quotient metric. Its closure is compact. Thus $\mathcal{R}$ is a Montel Fréchet space by this direct calculation.

The strict quotient $\pi : Q \rightarrow \mathcal{R}$ therefore lifts every bounded set to a bounded set in Q . Indeed take the compact closure just proved, apply SDT4's finite-choice compact-lift construction to this strict Fréchet

quotient, and intersect the resulting compact lift with the preimage of the original bounded set. The image of that bounded lift is exactly the original set. The same construction applies to $\tilde{\pi} = \pi q : A \rightarrow \mathcal{R}$.

Continuous descent of functionals gives algebraic bijections onto the indicated annihilators. Their strong topologies agree because for any bounded $K \subset \mathcal{R}$ and an exact bounded lift $\tilde{K} \subset Q$,

$$\sup_{x \in K} |\lambda(x)| = \sup_{F \in \tilde{K}} |(\pi' \lambda)(F)|. \quad (\text{AST3.3})$$

The forward seminorm estimate follows by mapping any bounded set of Q to a bounded set of $\mathcal{R}$. Thus

$$\boxed{\pi' : \mathcal{R}'_\beta \xrightarrow{\sim} S_O = N_O^\perp \subset Q'_\beta, \quad \tilde{\pi}' : \mathcal{R}'_\beta \xrightarrow{\sim} A_O^\perp \subset B.} \quad (\text{AST3.4})$$

These are closed strong topological embeddings, and $\tilde{\pi}' = q' \pi'$. This proves the exact dual of the actual ACD source quotient.

GDC7's independently checked restriction theorem now gives the strict exact rows

$$\begin{aligned} 0 \rightarrow \mathcal{R}'_\beta &\xrightarrow{\pi'} Q'_\beta \xrightarrow{\text{res}} (N_O)'_\beta \rightarrow 0, \\ 0 \rightarrow \mathcal{R}'_\beta &\xrightarrow{\tilde{\pi}'} B \xrightarrow{\text{res}} (A_O)'_\beta \rightarrow 0. \end{aligned} \quad (\text{AST3.5})$$

The restrictions are strongly open, not merely onto as vector maps. Their bounded dual families also have bounded lifts by the common Hahn–Banach seminorm estimate proved in GDC7. The embeddings, openness and bounded lifts used here have separate proofs; no continuous linear section is inferred.

AST4. The actual pairing is strongly dense in this dual

The map $\bar{\beta}_r : \mathcal{R} \rightarrow H_O$ is continuous and injective by (AST2.1). Its image is dense in H_O : for a finite vector h on off-line zeros, prescribe the finite values $F(\rho^\#) = h_\rho / d_r(\rho)$, with zero values elsewhere, using the already proved full-jet isolators. Formula (AST1.9) then gives exactly $\beta_r F = h$. This proves Hilbert density, without claiming density of finite blocks in Q .

Define the continuous linear transpose receiver

$$\gamma_r : \overline{H_O} \rightarrow \mathcal{R}'_\beta, \quad \gamma_r(\bar{y})([F]) = \langle \beta_r F, y \rangle_+. \quad (\text{AST4.1})$$

It is injective by that density and has the exact comparison

$$\pi' \gamma_r = \sigma_r|_{\overline{H_O}}, \quad \tilde{\pi}' \gamma_r = a_r|_{\overline{H_O}}. \quad (\text{AST4.2})$$

The strong topological isomorphism in (AST3.4), together with GDC2's proved strong closure, gives

$$\overline{\gamma_r(\overline{H_O})}^\beta = \mathcal{R}'_\beta. \quad (\text{AST4.3})$$

Thus the nondegenerate ACD pairing on $\mathcal{R} \times \overline{H_O}$ now has its exact dual receiver. It need not exhaust that dual before closure.

Its Hilbert norm on the source is exactly

$$\| [F] \|_r^2 := \|\bar{\beta}_r[F]\|_+^2 = \sum_{\rho \in \mathcal{Z}_O} m_\rho |d_r(\rho)|^2 |F(\rho^\#)|^2 = \sum_{\rho \in \mathcal{Z}_O} m_\rho |d_r(\rho)|^2 |F(\rho)|^2. \quad (\text{AST4.4})$$

The second equality reindexes by $\#$, using equal multiplicities and $d_r(\rho^\#) = -d_r(\rho)$. Completion for this displayed norm is H_O , via $\bar{\beta}_r$. This norm topology is not equated with the original Fréchet quotient topology.

AST5. The split dual receiver is the transpose of an actual primal subcomplex

At a pole, ACD's actual primal object is

$$\mathcal{P}_r = [\underline{A}^{-1} \xrightarrow{+b_r} i_* H^0].$$

Its source kernel constructs the literal subcomplex

$$\mathcal{P}_O = [\underline{A}_O^{-1} \xrightarrow{0} i_* H^0] = \underline{A}_O[1] \oplus i_* H. \quad (\text{AST5.1})$$

The map $\mathcal{P}_O \rightarrow \mathcal{P}_r$ is inclusion on A_O and identity on H . It is a chain map because $b_r|_{A_O} = 0$. Its literal quotient is the constant sheaf $\underline{\mathcal{R}}[1]$, so there is a short exact sequence of the actual complexes

$$0 \rightarrow \mathcal{P}_O \rightarrow \mathcal{P}_r \xrightarrow{(\tilde{\pi}, 0)} \underline{\mathcal{R}}[1] \rightarrow 0. \quad (\text{AST5.2})$$

Each coefficient inclusion/quotient has its proved original topology. The quotient complex itself has zero differential. First define the ordinary short-exact-sequence boundary by differentiating a lifted cocycle with a positive sign. A stalk class $[a] \in H^{-1}(\underline{\mathcal{R}}[1]) = \mathcal{R}$ lifts to $a \in A = \mathcal{P}_r^{-1}$; its differential is $+b_r a \in H = \mathcal{P}_O^0$. Changing the lift by $v \in A_O$ changes this differential by $b_r v = 0$. Hence

$$\delta_{\text{SES}}([a]) = +b_r a = +\bar{\beta}_r([a]). \quad (\text{AST5.2a})$$

There is a separate sign in the retained mapping-cone convention. For the inclusion $i : \mathcal{P}_O \rightarrow \mathcal{P}_r$, take $\text{Cone}(i)^k = \mathcal{P}_r^k \oplus \mathcal{P}_O^{k+1}$, $d(v, u) = (dv + iu, -du)$, and last cone arrow given by positive projection to $\mathcal{P}_O[1]$. The quasi-isomorphism from this cone to $\underline{\mathcal{R}}[1]$ is the positive quotient on its $\mathcal{P}_r$ coordinate. At the pole the cone has $A \oplus H$ in degree -1 , with differential $(a, h) \mapsto b_r a + h$. The cocycle $(a, -b_r a)$ represents $[a]$, and its positive last projection is $-b_r a$. Consequently the distinguished triangle with the literal positive quotient is

$$\mathcal{P}_O \xrightarrow{i} \mathcal{P}_r \xrightarrow{+\tilde{\pi}} \underline{\mathcal{R}}[1] \xrightarrow{\kappa_r} \mathcal{P}_O[1], \quad H^{-1}(i^* \kappa_r) = -\bar{\beta}_r = -\delta_{\text{SES}}. \quad (\text{AST5.2b})$$

Changing the quotient coordinate by $-1_{\mathcal{R}}$ instead gives the isomorphic triangle with middle arrow $-\tilde{\pi}$, last arrow $-\kappa_r$, and positive stalk boundary $+\bar{\beta}_r$. Thus the positive differentiate-a-lift boundary and the stated cone projection have their exact coordinate comparison; their signs have not been identified silently.

Both signs of this map are continuous by the original quotient topology, injective, and dense in H_O by AST4. A lift of an individual class is all the calculation requires; it supplies no continuous section of $A \rightarrow \mathcal{R}$. The actual localization boundary in (AST2.3) is $-b_r$ because its restriction-fibre quotient coordinate is $h - b_r a$. It agrees with $H^{-1}(i^* \kappa_r) \tilde{\pi}$ as a coefficient map, while the positive short-exact-sequence convention gives the negative of that map. These are distinct constructions with the stated exact relation.

Use ACD's fine resolution with the minus de Rham differential. The inclusion of smooth A_O -valued forms into smooth A -valued forms and the identity on the point coefficient give a continuous fine-resolution map. Its literal compact-test transpose, with ACD's degree signs $(-1, +1, -1)$ and Riesz identification, is

$$\mathcal{E}_r \rightarrow \mathcal{T}_{A_O}[-1] \oplus i_* \bar{H}, \quad (u, x) \mapsto (\iota_O^\vee u, x). \quad (\text{AST5.3})$$

Here $\mathcal{E}_r = \text{Cone}(-\delta_p a_r)[-1]$. The target attaching map is zero because the coefficient restriction $B \rightarrow (A_O)'$ kills a_r . Explicitly $\iota_O^\vee \delta_p a_r = \delta_p(a_r|_{A_O}) = 0$; hence the current differential commutes. This is exactly GDC10.3, now identified as the continuous transpose of (AST5.2)'s first map.

The compact-test contraction applies to the complete Fréchet coefficient A_O , giving $\mathcal{T}_{A_O}[-1] \simeq (A_O)'[1]$. On underlying vector-space sheaves, (AST3.5) and cancellation of the common supported term give the exact receiver triangle

$$\boxed{\underline{\mathcal{R}}'[1] \rightarrow \mathcal{E}_r \rightarrow (\underline{A}_O)'[1] \oplus i_* \bar{H} \rightarrow \underline{\mathcal{R}}'[2]}. \quad (\text{AST5.4})$$

Under $\tilde{\pi}'$, its first coefficient is exactly $A_O^\perp$, and the triangle is GDC10.4. For a direct verification of its fibre, form the two current Gysin triangles with common supported coordinate. Their comparison cancels that

coordinate; compact-test contractions reduce the remaining comparison to the actual coefficient restriction $B \rightarrow (A_O)'$, whose kernel is $\mathcal{R}'$ by (AST3.5). The fibre is thus $\mathcal{R}'[1]$. No general exactness claim for strong duals of arbitrary complexes is used.

The positive costalk map is consequently

$$[\overline{H} \xrightarrow{+a_r} B] \longrightarrow [\overline{H} \xrightarrow{0} (A_O)'], \quad (x, b) \mapsto (x, b|_{A_O}). \quad (\text{AST5.5})$$

Its degree-one map is the separated quotient of the actual unseparated cohomology:

$$B/a_r C \longrightarrow B/A_O^\perp \xrightarrow{\sim} (A_O)'_\beta, \quad \ker = A_O^\perp/a_r C. \quad (\text{AST5.6})$$

That kernel is precisely the closure of zero in the original costalk quotient. The retained source $\mathcal{R}$ and its dual kernel in (AST5.4) identify the exact cost of this split receiver.

AST6. Parameter isomorphisms are actual transpose comparisons

For $r, t > 1$, retain GDC3's bounded invertible diagonal map

$$v_{r,t}(\rho) = \begin{cases} e^{-i\Im \rho \log(r/t)} \frac{r^{\Re \rho} - r^{1-\Re \rho}}{t^{\Re \rho} - t^{1-\Re \rho}}, & \rho \in \mathcal{Z}_O, \\ 1, & \rho \in \mathcal{Z}_L, \end{cases} \quad V_{r,t}y = (v_{r,t}(\rho)y)_\rho. \quad (\text{AST6.1})$$

Its explicit off-line modulus bounds are

$$\frac{2\sqrt{r} \log r}{(t+1) \log t} \leq |v_{r,t}(\rho)| \leq \frac{(r+1) \log r}{2\sqrt{t} \log t}; \quad (\text{AST6.2})$$

on line coordinates its value is one. The nonzero constants prove bounded invertibility, even for zeros approaching the line. Put $M_{r,t} = V_{r,t}^*$. Since $D_r = D_t V_{r,t}$, taking Hilbert adjoints gives the exact identities

$$\beta_r = M_{r,t} \beta_t, \quad b_r = M_{r,t} b_t, \quad \sigma_t \overline{V}_{r,t} = \sigma_r, \quad a_t \overline{V}_{r,t} = a_r. \quad (\text{AST6.3})$$

In particular

$$\mathcal{P}_t \xrightarrow{(1_A, M_{r,t})} \mathcal{P}_r, \quad \mathcal{E}_r \xrightarrow{(1_B, \overline{V}_{r,t})} \mathcal{E}_t \quad (\text{AST6.4})$$

are continuous cochain isomorphisms with inverses obtained by exchanging r, t . The first has the same sign on its positive primal differential; the second has the same sign on its positive Gysin attaching term. They are literal continuous transposes under ACD's pairing, because

$$M'_{r,t} R(\overline{y})(h) = \langle M_{r,t} h, y \rangle_+ = \langle h, V_{r,t} y \rangle_+ = R(\overline{V}_{r,t} \overline{y})(h). \quad (\text{AST6.5})$$

The diagonal operators commute and satisfy the GDC cocycle identity, so these maps satisfy the corresponding compositional identities in their indicated directions. They fix $A, A_O, Q, N_O, \mathcal{R}$, the entire nearby dual coefficient B , and the line-supported coefficient H_L or $\overline{H}_L$. They do not identify the numerical defect sizes for different parameters.

The localization boundary squares are now explicit:

$$-b_r = M_{r,t}(-b_t), \quad 1_A = 1_A 1_A, \quad a_r = a_t \overline{V}_{r,t}. \quad (\text{AST6.6})$$

The induced map on the quotient $\text{coker sp}_{-1} = \mathcal{R}$ is identity; its map between the displayed Hilbert images is $M_{r,t}$. Its map on the dual source $\mathcal{R}'$ in (AST5.4) is identity, and $\gamma_t \overline{V}_{r,t} = \gamma_r$ on $\overline{H}_O$.

These parameter maps preserve the actual cover structures. For a recovered integer degree $n \geq 1$, $V_{r,t}$ and $M_{r,t}$ commute with every diagonal T_n, U_n and their Hilbert adjoints. Thus (AST6.4) commutes with the primal unit and weighted trace and the dual DCA maps, including their finite-sheet corrections. In DCA's correction vector every occurrence of $R_{-a_r} x$ equals $R_{-a_t} \overline{V}_{r,t} x$, so all sheet coordinates commute, not merely their sums. The integer cover and its angular degree factors are retained; no geometric cover of noninteger degree is asserted.

AST7. Both poles, the global specialization map and endpoints

At both poles use the same existing coefficient trivialization. The subcomplex construction becomes

$$\mathcal{P}_O^Y = \underline{A}_O[1] \oplus i_+ H \oplus i_- H \longrightarrow \mathcal{P}_r^Y = [\underline{A}^{-1} \xrightarrow{(b_r, b_r)} i_+ H \oplus i_- H], \quad (\text{AST7.1})$$

with quotient $\underline{\mathcal{R}}_Y[1]$. The source and dual maps are identity on each original endpoint summand $E_p[1]$ and $E'_p[-1]$. Both distinct value-at-zero and integral labels remain; none enters the kernel N_O by a relabelling of a zeta endpoint.

In ACD6's proved global finite model the parameter map is

$$[A^{-1} \xrightarrow{(b_t, b_t)} H^0 \oplus H^0 \xrightarrow{0} A^1] \longrightarrow [A^{-1} \xrightarrow{(b_r, b_r)} H^0 \oplus H^0 \xrightarrow{0} A^1],$$

$$(1_A, M_{r,t} \oplus M_{r,t}, 1_A). \quad (\text{AST7.2})$$

It induces identity on $A_O \oplus E_+ \oplus E_-$ in degree -1 , identity on A in degree one, and

$$H \oplus (H/b_t A) \longrightarrow H \oplus (H/b_r A), \quad (h, [k]) \longmapsto (M_{r,t} h, [M_{r,t} k]) \quad (\text{AST7.3})$$

in degree zero. This is well defined with a continuous inverse because $M_{r,t} b_t A = b_r A$, and quotient topology gives continuity even if these quotients are unseparated. The original global source map remains

$$Q \longrightarrow H \oplus (H/b_r A), \quad F \longmapsto (\beta_r F, 0), \quad \ker = N_O. \quad (\text{AST7.4})$$

Equations (AST6.3) and (AST7.3) prove its parameter compatibility on all original classes.

The dual finite global model has B^{-1} , $C^0 \oplus C^0$, B^1 and differential $(x_+, x_-) \mapsto a_r x_+ + a_r x_-$. Its parameter map is identity on both B terms and $\bar{V}_{r,t}$ on each point term. The two positive Dirac contributions remain a sum. Since GDC3 proves equality of the actual images $a_r C' = a_t C$, the degree-one quotient maps are literally identity on $B/a_r C$, and their separated maps are identity on $(A_O)'$. Endpoints extend these statements by their identity maps. This proves preservation of the global receiving maps without declaring an anti-diagonal cancellation to be vanishing of the local attaching map.

AST8. The restricted original extension and the surviving positive residue

The actual kernel source belongs to the strict original pullback row

$$0 \rightarrow J \rightarrow A_O \xrightarrow{q_O} N_O \rightarrow 0. \quad (\text{AST8.1})$$

Its continuous strong dual is, by GDC7,

$$0 \rightarrow (N_O)'_{\beta} \xrightarrow{q'_O} (A_O)'_{\beta} \rightarrow J'_{\beta} \rightarrow 0. \quad (\text{AST8.2})$$

This row retains all of J , in contrast to replacing the coefficient by a zeta value space. It also proves that the split receiver's nearby coefficient still contains the dual of the source carrying all line values and all higher jets.

For completeness, its retained original residue is the proved exact restriction

$$A_H(y)|_{N_O}(F) = \sum_{\rho \in \mathcal{Z}_L} m_{\rho} F(\rho) \overline{y}_{\rho} = \langle EF, P_L y \rangle_+ \quad (F \in N_O). \quad (\text{AST8.3})$$

It injects $\overline{H_L}$ into $(N_O)'$, by testing each actual finite line-value isolator, and then into $(A_O)'$ through q'_O . Precisely, N_O/N_0 , where $N_0 = \ker E$, with norm $\|[F]\| = \|EF\|_+$, completes to H_L by GDC8.3. The injected H_L already is complete in its own Hilbert norm. Neither that norm nor that Hilbert image is identified with the full strong dual $(N_O)'$; the original higher-jet functionals remain present. The full source remains the exact extension $0 \rightarrow N_O \rightarrow Q \rightarrow \mathcal{R} \rightarrow 0$; no source projector splitting this row has been constructed or assumed.

AST9. Exact result and limits of the comparison

The actual ACD specialization source has now been identified as $\mathcal{R} = Q/N_O$, with its original quotient topology. Its full strong dual is the closed GDC defect receiver S_O , and its image in the original dual coefficient is $A_O^\perp$. The original specialization map, its sign, its two-pole map and every parameter comparison commute with these identifications. The GDC split current receiver is the literal continuous transpose of the actual subcomplex (AST5.1); the discarded coefficient is retained as $\underline{\mathcal{R}}'[1]$ in its exact triangle.

No existence or nonexistence of an off-line zero has entered this proof. If an actual off-line zero is present, its full-jet isolator makes its value class nonzero in $\mathcal{R}$; if none is present then $N_O = Q$ and $\mathcal{R} = 0$. These are exact statements about the proved receiver. They do not prove which case holds, impose a common weight, or turn the coefficient quotient into Deligne's geometric inertia cross. What advances the comparison is the exhibited primal subcomplex, its quotient topology, its complete dual and its commuting specialization maps, all on the original arithmetic coefficients.

All continuous positive transfer forms on the original full-jet quotient

25 September 2026. Independent derivation CFP0–CFP12 and the proved analytic lemma CFPA. The inner product convention is linear in the first variable.

CFP0. The actual source and the statement

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. All numbers, coefficient spaces, arithmetic labels and operations in this proof occur after the complete arithmetic reconstruction. No coefficient value, addition, parity, coordinate or metric is assigned to τ . The original two branch records remain distinct.

Retain the source from RD1, RTT0 and PTQ6:

$$\mathcal{B} = \left\{ F \text{ entire} : b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \quad (A, M \in \mathbb{Z}_{\geq 0}) \right\},$$

$$\mathcal{J} = \{F \in \mathcal{B} : F^{(j)}(\rho) = 0 \text{ for every nontrivial zero } \rho \text{ of original } \zeta, \ 0 \leq j < m_\rho\}, \quad \mathcal{Q} = \mathcal{B}/\mathcal{J}. \quad (\text{CFP0.1})$$

The topology is this Fréchet quotient topology. Every multiplicity m_ρ is retained. Write $q : \mathcal{B} \rightarrow \mathcal{Q}$, and retain

$$T_a q(F) = q(a^s F(s)), \quad Lq(F) = q(sF(s)), \quad a > 0; \quad \mathcal{U}_n = nT_{1/n} \quad (n \geq 1). \quad (\text{CFP0.2})$$

The integer n in the transfer is the already constructed cover degree. Its coefficient inverse is not a cover of fractional degree.

Classification. All jointly continuous positive semidefinite Hermitian forms B on $\mathcal{Q}$ satisfying

$$B(T_n F, G) = B(F, \mathcal{U}_n G) \quad \text{for every recovered integer } n \geq 1 \quad (\text{CFP0.3})$$

are exactly

$$B(F, G) = \sum_{\substack{\rho \in \mathcal{Z} \\ \Re \rho = 1/2}} m_\rho c_\rho F(\rho) \overline{G(\rho)}, \quad 0 \leq c_\rho \leq C(1 + |\Im \rho|)^d \quad (\text{CFP0.4})$$

for some finite $C, d \geq 0$ depending on the form. The weights are unique. Here and below $F(\rho)$ means evaluation of any source representative of its class.

The proof does not assume RH, simplicity, boundedness of B for the previous value Hilbert norm, or density of the finite-support subspace in $\mathcal{Q}$. In particular it extends PTQ4 beyond bounded forms on that Hilbert space. The full source, its higher jets, and all actual off-line zeros, if present, remain in the domain; the theorem determines their image in every continuous positive transfer receiver.

CFP1. Source continuity and the positive completion

Multiplication by e^{ts} preserves $\mathcal{B}$, $\mathcal{J}$ and the full vanishing-jet conditions. Its action on an actual jet is multiplication by the complete jet of e^{ts} , rather than the identity. Moreover,

$$b_{A,M}(e^{ts} F) \leq e^{A|t|} b_{A,M}(F), \quad b_{A,M}(sF) \leq (A+1) b_{A,M+1}(F). \quad (\text{CFP1.1})$$

For real t , the identity $e^{hs} - 1 = hs \int_0^1 e^{uhs} du$ proves continuity of $t \mapsto e^{ts} F$ in every seminorm. The analogous second-order integral remainder proves its derivative is $se^{ts} F$ in $\mathcal{B}$. Repetition gives all derivatives. These facts pass through the quotient.

Joint continuity of $B(qF, qG)$, positivity and Cauchy–Schwarz give integers A, M and a constant C_B such that

$$B(qF, qF) \leq C_B b_{A,M}(F)^2. \quad (\text{CFP1.2})$$

Indeed continuity at the origin first bounds the form by a finite maximum of seminorms in each variable; the increasing family $b_{A,M}$ absorbs that finite maximum. Alternatively continuity of the quadratic form supplies the same bound by scaling. Cauchy–Schwarz follows directly by applying positivity to $x + zy$ and minimizing over $z \in \mathbb{C}$.

Let $\mathcal{N}_B = \{F : B(F, F) = 0\}$. Cauchy–Schwarz identifies it with the full radical. Complete $\mathcal{Q}/\mathcal{N}_B$ for the specified inner product B , obtaining a Hilbert space H_B . The canonical map

$$\pi : \mathcal{Q} \longrightarrow H_B \quad \text{has dense image, and} \quad \langle \pi F, \pi G \rangle = B(F, G). \quad (\text{CFP1.3})$$

The map πq is continuous by (CFP1.2), hence π is continuous by the quotient topology. This construction does not identify the original Fréchet topology with the new Hilbert norm.

CFP2. Every integer relation gives the full continuous unitary action

The exact identities $\cup_n T_n = nI$ and (CFP0.3) give

$$B(T_n F, T_n G) = nB(F, G). \quad (\text{CFP2.1})$$

Conversely (CFP2.1), with G replaced by $T_n^{-1}G$, gives (CFP0.3). Invertibility of T_n also proves $B(T_{1/n}F, T_{1/n}G) = n^{-1}B(F, G)$. Composition therefore gives

$$B(T_a F, T_a G) = aB(F, G) \quad (a \in \mathbb{Q}_{>0}). \quad (\text{CFP2.2})$$

For arbitrary real $a > 0$, choose positive rationals tending to a . The source continuity in CFP1 and the continuity of B give exactly the same identity. This uses the entire already recovered arithmetic family and its coefficient operations; it makes no assertion that a finite set of prime labels initiates the arithmetic reconstruction.

For every real t , define the auxiliary Hilbert action on the dense source image by

$$V_t \pi F = e^{-t/2} \pi T_{e^t} F. \quad (\text{CFP2.3})$$

The radical is invariant, and (CFP2.2) shows that this is an isometry with inverse V_{-t} . It extends uniquely to a unitary on H_B . Source continuity proves strong continuity on the dense image; the unitary norm bound and approximation by that image prove it on all of H_B . The group law follows on the dense image and hence everywhere.

The exact original action is still

$$\boxed{\pi T_{e^t} = e^{t/2} V_t \pi.} \quad (\text{CFP2.4})$$

Thus its degree is not removed or replaced. The auxiliary unitary group is the specified comparison of the original action with the positive degree character. In particular, for every source $H \in \mathcal{B}$,

$$\left. \frac{d}{dt} \right|_{t=0} V_t \pi q(H) = \pi q((s - 1/2)H). \quad (\text{CFP2.5})$$

The derivative exists in the Hilbert norm because it already exists in the stronger source topology.

CFP3. The complete original-zeta source integral

Retain the original even Schwartz vector and the complete summation map:

$$f_0(v) = \frac{\pi}{2} v^2 (2\pi v^2 - 3) e^{-\pi v^2}, \quad \Sigma f(u) = 2 \sum_{n \geq 1} f(nu), \quad \Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}. \quad (\text{CFP3.1})$$

The two source conditions are $f_0(0) = 0$ and $\int_{\mathbb{R}} f_0 = 0$. The second follows from $\int v^2 e^{-\pi v^2} dv = 1/(2\pi)$ and $\int v^4 e^{-\pi v^2} dv = 3/(4\pi^2)$.

For the Fourier convention $\hat{f}(y) = \int f(v)e^{-2\pi i v y} dv$, differentiation of the Gaussian transform gives

$$\begin{aligned}\widehat{v^2 e^{-\pi v^2}}(y) &= (1/(2\pi) - y^2)e^{-\pi y^2}, \\ \widehat{v^4 e^{-\pi v^2}}(y) &= (y^4 - 3y^2/\pi + 3/(4\pi^2))e^{-\pi y^2}.\end{aligned}\tag{CFP3.2}$$

Substitution of the two original coefficients proves $\hat{f}_0 = f_0$, including cancellation of the constant coefficient. Poisson summation on this Schwartz vector, with both zero terms zero, yields

$$\Sigma f_0(u) = u^{-1} \Sigma f_0(u^{-1}).\tag{CFP3.3}$$

Here the Poisson identity itself follows by periodizing a Schwartz function, computing its Fourier coefficients by termwise integration, and evaluating its absolutely convergent Fourier series at zero. Thus no unsupported zero term has been silently discarded.

Put

$$k(t) = \frac{1}{2} e^{t/2} \Sigma f_0(e^t) = e^{t/2} \sum_{n \geq 1} \frac{\pi}{2} n^2 e^{2t} (2\pi n^2 e^{2t} - 3) e^{-\pi n^2 e^{2t}}.\tag{CFP3.4}$$

Equation (CFP3.3) gives $k(-t) = k(t)$. For $t \geq 0$, the absolute value of the summand is bounded by $C n^4 e^{(9/2)t} e^{-\pi n^2 e^{2t}}$. Since $n^2 x \geq (n^2 + x)/2$ for $n \geq 1$, $x \geq 1$,

$$|k(t)| \leq C_0 \exp((9/2)|t| - (\pi/2)e^{2|t}|).\tag{CFP3.5}$$

The constant C_0 includes the finite sum $\sum n^4 e^{-\pi n^2/2}$ and the original two polynomial coefficients. This bound proves integrability against $|t|^j e^{R|t|}$ for every finite j, R .

The exact source transform, with original ζ explicit, is

$$\begin{aligned}\int_{\mathbb{R}} k(t) e^{(s-1/2)t} dt &= \frac{1}{2} \int_0^\infty \Sigma f_0(u) u^s \frac{du}{u} \\ &= \zeta(s) \int_0^\infty f_0(v) v^s \frac{dv}{v} \\ &= \left[\frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) = F_0(s) \right].\end{aligned}\tag{CFP3.6}$$

The second identity first holds on $\Re s > 1$ by absolute convergence. The Mellin integral there is

$$\frac{1}{2} \pi^{-s/2} \Gamma((s+4)/2) - \frac{3}{4} \pi^{-s/2} \Gamma((s+2)/2) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2).\tag{CFP3.7}$$

This records both Gaussian summands before their exact Gamma recursion. The left side of (CFP3.6) is entire by (CFP3.5), so the equality continues through every exceptional point. No completed zeta is used as a replacement for original ζ ; F_0 is this proved source multiplier.

All endpoint and trivial-zero values remain in that comparison:

$$\begin{aligned}F_0(0) &= F_0(1) = \frac{1}{8}, \\ F_0(-2r) &= \frac{r(2r+1)(-1)^r \pi^r}{2r!} \zeta'(-2r) \\ &= \frac{(1+2r)(2r)}{8} \pi^{-(1+2r)/2} \Gamma((1+2r)/2) \zeta(1+2r) \neq 0 \quad (r \geq 1).\end{aligned}\tag{CFP3.8}$$

At zero use $\Gamma(s/2) = 2/s + O(1)$ and $\zeta(0) = -1/2$; at one use the original zeta residue one. At $-2r$, use the Gamma residue $2(-1)^r/r!$ and the original simple trivial zero. Its nonzero derivative follows directly from the original functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s).\tag{CFP3.9}$$

Differentiating its simple sine zero supplies the derivative in (CFP3.8); reflection of the full product supplies the displayed value at $1 + 2r$. On the open critical strip, every multiplier in (CFP3.6) is a holomorphic unit. Consequently the zeros of F_0 are exactly the original nontrivial zeta zeros with their unchanged multiplicities.

To retain all derivative terms as well, write only as an auxiliary notation $M_0(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)/8$. On every nonexceptional germ,

$$F_0^{(r)}(s) = \sum_{j=0}^r \binom{r}{j} M_0^{(j)}(s) \zeta^{(r-j)}(s),$$

$$M_0^{(j)}(s) = \sum_{j_0+j_1+j_2=j} \frac{j!}{j_0!j_1!j_2!} \left(\frac{s(s-1)}{8} \right)^{(j_0)} \left(-\frac{\log \pi}{2} \right)^{j_1} \pi^{-s/2} 2^{-j_2} \Gamma^{(j_2)}(s/2). \quad (\text{CFP3.10})$$

The entire continuation at exceptional points is given by the integral, whose exact derivatives are $\int t^r k(t) e^{(s-1/2)t} dt$. No Gamma, endpoint or scale derivative is replaced by a constant.

CFP4. The source annihilator is an actual convergent group integral

For each $F \in \mathcal{B}$,

$$b_{A,M}(k(t)e^{t(s-1/2)}F) \leq |k(t)|e^{(A+1/2)|t|}b_{A,M}(F). \quad (\text{CFP4.1})$$

This bound is integrable by (CFP3.5). Integrating on finite intervals and passing to the limit in every seminorm therefore defines an integral in the complete Fréchet space $\mathcal{B}$. Point evaluation is continuous, so (CFP3.6) identifies that integral exactly:

$$\int_{\mathbb{R}} k(t)e^{t(s-1/2)}F(s) dt = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s) F(s). \quad (\text{CFP4.2})$$

This product belongs to $\mathcal{J}$, since every required zero order occurs in the full F_0 factor. It follows, after the continuous maps q and π , that

$$\int_{\mathbb{R}} k(t)V_t \pi q(F) dt = 0. \quad (\text{CFP4.3})$$

As $k \in L^1(\mathbb{R})$ and V_t is unitary, its Hilbert integral is a bounded operator of norm at most $\|k\|_1$. Equation (CFP4.3) holds on a dense image, hence

$$\boxed{\int_{\mathbb{R}} k(t)V_t dt = 0 \quad \text{on all of } H_B.} \quad (\text{CFP4.4})$$

This is an annihilation of the completed receiver by the original source vector. It neither identifies F_0 with original ζ nor declares that the original source quotient is a quotient by the principal ideal $F_0\mathcal{B}$. Only the proved inclusion $F_0\mathcal{B} \subset \mathcal{J}$ has been used.

CFP5. Spectral support and the complete atomic decomposition

The analytic lemma CFPA below is proved for any strongly continuous unitary group and any L^1 annihilating kernel. Applying it to (CFP4.4) gives spectral support in

$$\Gamma = \{\gamma \in \mathbb{R} : \frac{(1/2 + i\gamma)(-1/2 + i\gamma)}{8} \pi^{-(1/2+i\gamma)/2} \Gamma((1/2 + i\gamma)/2) \zeta(1/2 + i\gamma) = 0\}. \quad (\text{CFP5.1})$$

All factors outside ζ in this formula are nonzero and finite. Thus Γ is exactly the set of ordinates of actual critical-line zeros of original ζ . It is discrete and countable; it has finitely many elements in every compact interval because the entire nonzero source function F_0 has isolated zeros.

There are mutually orthogonal projections P_γ with

$$V_t P_\gamma = e^{it\gamma} P_\gamma, \quad x = \sum_{\gamma \in \Gamma} P_\gamma x, \quad \|x\|^2 = \sum_{\gamma \in \Gamma} \|P_\gamma x\|^2. \quad (\text{CFP5.2})$$

The series converges in norm, and every zero projection is permitted. Explicitly,

$$P_\gamma x = \lim_{R \rightarrow \infty} \frac{1}{2R} \int_{-R}^R e^{-it\gamma} V_t x \, dt. \quad (\text{CFP5.3})$$

The proof in CFPA uses the full source integral and scalar positive measures, not any assumed density of finite original jets. The original degree action on this fibre remains

$$\pi T_{e^t} = e^{t/2} V_t \pi, \quad T_{n,\gamma} = n^{1/2+i\gamma}, \quad \mathbf{U}_{n,\gamma} = n^{1/2-i\gamma}. \quad (\text{CFP5.4})$$

CFP6. Division forces rank one and detects every killed jet

Fix $\gamma \in \Gamma$, put $\rho = 1/2 + i\gamma$, and retain the entire Gaussian

$$g_\rho(s) = e^{(s-\rho)^2}, \quad g_\rho(\rho) = 1. \quad (\text{CFP6.1})$$

It belongs to $\mathcal{B}$. For arbitrary $F \in \mathcal{B}$, define

$$H_{F,\rho}(s) = \frac{F(s) - F(\rho)g_\rho(s)}{s - \rho}. \quad (\text{CFP6.2})$$

The singularity at ρ is removable. This division remains in $\mathcal{B}$: outside a disk about ρ , the denominator is bounded below; inside that disk, the holomorphic quotient is bounded by its value on a larger boundary circle, and each polynomial weight is bounded there. Applying this argument on every strip, with a slightly enlarged strip for that circle, proves all defining seminorms finite. No infinite interpolation assertion is needed.

Differentiate the eigenprojection identity $P_\gamma V_t = e^{it\gamma} P_\gamma$ on the actual smooth vector $\pi q(H_{F,\rho})$. By (CFP2.5),

$$P_\gamma \pi q((s - 1/2)H_{F,\rho}) = i\gamma P_\gamma \pi q(H_{F,\rho}).$$

Therefore

$$\boxed{P_\gamma \pi q(F) = F(\rho)v_\rho, \quad v_\rho = P_\gamma \pi q(g_\rho).} \quad (\text{CFP6.3})$$

Since $\pi q(\mathcal{B})$ is dense, the range of P_γ is the closure of its image on that dense set. Formula (CFP6.3) proves

$$\text{Ran } P_\gamma = \mathbb{C}v_\rho; \quad \dim \text{Ran } P_\gamma \in \{0, 1\}. \quad (\text{CFP6.4})$$

The possible zero dimension is retained. This is the required multiplicity statement for the positive receiver, not a simplicity assertion about zeta.

The original source still has its full local algebra $\mathbb{C}[z]/(z^{m_\rho})$. The exact globally realized block map is

$$\sum_{j=0}^{m_\rho-1} a_j z^j \mapsto \begin{cases} a_0 v_\rho, & \Re \rho = 1/2, \\ 0, & \Re \rho \neq 1/2. \end{cases} \quad (\text{CFP6.5})$$

To verify it globally, use the RD1 isolator $e_{\rho,j}$, whose required jets vanish at every other actual zero. Equation (CFP6.3) and the complete decomposition (CFP5.2) give (CFP6.5). In particular the kernel on a block with $v_\rho \neq 0$ consists of all $m_\rho - 1$ higher-jet directions; when $v_\rho = 0$ or the zero is off-line, it is the entire original m_ρ -dimensional block. No source class is declared zero in $\mathcal{Q}$ merely because this map kills it.

CFP7. The exact weights and their necessary growth

By (CFP5.2) and (CFP6.3), for arbitrary original classes,

$$B(F, G) = \sum_{\rho \in \mathcal{Z}_{\text{line}}} w_\rho F(\rho) \overline{G(\rho)}, \quad w_\rho = \|v_\rho\|^2 \geq 0. \quad (\text{CFP7.1})$$

The sum is absolutely convergent by Cauchy–Schwarz applied to the two norm-convergent orthogonal expansions. Retain the original multiplicity by defining

$$c_\rho = w_\rho/m_\rho. \quad (\text{CFP7.2})$$

This gives exactly (CFP0.4). The weights are unique: testing on the original isolator $e_{\rho,0}$ gives $B(e_{\rho,0}, e_{\rho,0}) = w_\rho$. The multiplicity has not been replaced by one; w_ρ is the total scalar mass of this receiver, while c_ρ is its coefficient relative to the original multiplicity trace.

For the continuity bound (CFP1.2), the Gaussian satisfies

$$|g_\rho(\sigma + iy)| = e^{(\sigma-1/2)^2 - (y-\gamma)^2},$$

$$b_{A,M}(g_\rho) \leq e^{(A+1/2)^2} (1 + |\gamma|)^M \sup_{u \in \mathbb{R}} (1 + |u|)^M e^{-u^2}. \quad (\text{CFP7.3})$$

The last supremum is finite. Since P_γ is an orthogonal projection,

$$0 \leq m_\rho c_\rho = w_\rho \leq \|\pi q(g_\rho)\|^2 \leq C'(1 + |\gamma|)^{2M}. \quad (\text{CFP7.4})$$

As $m_\rho \geq 1$, this proves the necessary polynomial bound on c_ρ . It also proves the stronger polynomial bound on its actual atomic mass $m_\rho c_\rho$. No uniform lower bound for zero spacing, isolator coefficients, or $\zeta'(\rho)$ is involved.

CFP8. Every indicated polynomial family gives a continuous form

Human-source attribution: the unconditional count used here is the classical Riemann–von Mangoldt theorem. The inspected witness already documented at PTQ8 is Alain Connes, *The Riemann Hypothesis: Past, Present and a Letter Through Time*, the Hardy–Littlewood discussion preceding “Zero-free regions and zero-density estimates.” Its asymptotic implies the bound below; this is not a new zero-counting theorem.

Conversely suppose $0 \leq c_\rho \leq C(1 + |\gamma|)^d$. The actual zero count with multiplicities is

$$N(R) = \sum_{\rho \in \mathcal{Z}, |\Im \rho| \leq R} m_\rho = O(R \log(R + 2)). \quad (\text{CFP8.1})$$

This unconditional count is one of the original divisor estimates already used in RTT1. It does not impose RH. By grouping zeros into dyadic height intervals, it implies $\sum_\rho m_\rho (1 + |\Im \rho|)^{-q} < \infty$ for every $q > 1$. Choose an integer M with $2M - d > 2$. Then

$$\sum_{\rho \in \mathcal{Z}_{\text{line}}} m_\rho c_\rho |F(\rho)|^2 \leq C b_{1,M}(F)^2 \sum_{\rho \in \mathcal{Z}} m_\rho (1 + |\Im \rho|)^{d-2M} < \infty. \quad (\text{CFP8.2})$$

Cauchy–Schwarz proves absolute convergence of the two-variable form and joint continuity on $\mathcal{B}$. It annihilates $\mathcal{I}$ in either argument and hence gives a jointly continuous form on the quotient: take the infimum of the displayed seminorm bound over representatives in each argument. Its positivity is termwise.

For every $n \geq 1$, on a retained coordinate $\bar{\rho} = 1 - \rho$, so

$$\overline{n^{1-\rho}} = n^\rho.$$

Substitution in the absolutely convergent sums gives exactly $B(T_n F, G) = B(F, n T_{1/n} G)$. This proves sufficiency and completes the classification.

CFP9. Universal source kernel, without a value-norm hypothesis

For a particular weight family define

$$\mathcal{Z}_c = \{\rho \in \mathcal{Z}_{\text{line}} : c_\rho > 0\}, \quad \mathcal{K}_c = \{[F] \in \mathcal{Q} : F(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}_c\}. \quad (\text{CFP9.1})$$

Equation (CFP0.4) proves that this is exactly the radical. The completion is canonically

$$\overline{\mathcal{Q}/\mathcal{K}_c}^{,B} \simeq \ell^2(\mathcal{Z}_c, m_\rho c_\rho), \quad (\text{CFP9.2})$$

with map $[F] \mapsto (F(\rho))_{\rho \in \mathcal{Z}_c}$. The norm identity follows from the formula. Every finite sequence occurs, by the actual full-jet isolators, and finite sequences are dense in the displayed weighted Hilbert space. This proves surjectivity of the map on completions without any claim about finite-jet density in the original topology.

The intersection of radicals of **all** continuous positive forms satisfying the original transfer identities is therefore exactly

$$\boxed{\mathcal{K}_{\text{all}} = \{[F] : F(\rho) = 0 \quad (\rho \in \mathcal{Z}_{\text{line}})\} = \mathcal{I}_{\mathcal{Z}_{\text{line}}}^{\text{val}} / \mathcal{I}.} \quad (\text{CFP9.3})$$

Every classified form kills this space. Conversely the choice $c_\rho = 1$ is permitted by CFP8 and detects every class outside it. Thus this common kernel is proved in the actual Fréchet category, rather than inferred from boundedness in PTQ's auxiliary value norm.

At an individual original nontrivial zero, the scalar relation responsible for this kernel can also be checked directly: $B(T_n x, T_n x) = nB(x, x)$ forces a pure eigenvector of exponent ρ to have zero norm when $n^{2\Re \rho} \neq n$. The global argument above additionally proves that no continuous residual sector escapes this finite-coordinate calculation and that all higher jets are killed. That global assertion is precisely what finite-support computations alone do not supply.

CFP10. Every continuous positive Hilbert receiver factors through the classified kernel

Let $X : \mathcal{Q} \rightarrow K$ be a continuous linear map into a Hilbert space, with bounded receiving operators A_n, C_n such that

$$XT_n = A_n X, \quad XU_n = C_n X, \quad C_n = A_n^*. \quad (\text{CFP10.1})$$

Its pulled-back form $B_X(F, G) = \langle XF, XG \rangle$ is jointly continuous, positive, and satisfies (CFP0.3). Consequently

$$X\mathcal{K}_{\text{all}} = 0, \quad X : \mathcal{Q} \longrightarrow \mathcal{Q}/\mathcal{K}_{\text{all}} \longrightarrow K \quad (\text{CFP10.2})$$

is a unique continuous factorization in the quotient topology. The classification supplies its exact nonnegative polynomial weights and identifies the completion of its image with (CFP9.2). The original degree relation is retained: on the closed span of $X\mathcal{Q}$, the identities imply $A_n^* A_n = nI$, since it holds on the dense image.

Thus allowing arbitrary continuous Hilbert receivers from the source creates additional polynomial weights, but creates no additional supported zero, off-line value, or nonconstant jet direction compatible with a positive adjoint. This is an exact comparison with PTQ: bounded forms on its specified H correspond to the subfamily with bounded c_ρ ; continuity on the stronger source permits the polynomially growing families of (CFP0.4).

CFP11. The original Weil form and all arithmetic contributions remain

Retain the original source Weil form and its exact zero observation:

$$W(F, G) = \sum_{\rho \in \mathcal{Z}} m_\rho F(\rho) \overline{G(1 - \bar{\rho})}. \quad (\text{CFP11.1})$$

RTT1 and the count (CFP8.1) prove joint continuity; substitution proves the original transfer identity. At a line zero its contribution is the $c_\rho = 1$ contribution in (CFP0.4). For an off-line reflected orbit, retain both actual terms

$$m_\rho (F(\rho) \overline{G(\rho^\#)} + F(\rho^\#) \overline{G(\rho)}), \quad \rho^\# = 1 - \bar{\rho}. \quad (\text{CFP11.2})$$

The globally realized difference $e_{\rho,0} - e_{\rho^\#,0}$ has value $-2m_\rho$ under W . The sum has value $+2m_\rho$. Both statements follow by direct substitution with all other values zero. The classification therefore does not

make this original form positive by a change of source topology: it identifies exactly the space that every positive transfer form would have to kill.

For completeness retain the actual arithmetic receiver from RTT9. Put

$$A_{F,G}(s) = F(s)\overline{G(1-\bar{s})}, \quad A_{F,G}(s) = \int_{\mathbb{R}} h(v)e^{-(s-1/2)v}dv, \quad \hat{h}(t) = \int_{\mathbb{R}} h(v)e^{-itv}dv.$$

Then the unchanged full formula is

$$\begin{aligned} W(F, G) &= A_{F,G}(0) + A_{F,G}(1) + A_{\infty}(h) - P_{\text{hist}}(h), \\ A_{\infty}(h) &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{h}(t) \left(\Re \frac{\Gamma'(1/4 + it/2)}{\Gamma(1/4 + it/2)} - \log \pi \right) dt, \\ P_{\text{hist}}(h) &= \sum_{n \geq 2} \frac{\log L_n - \log L_{n-1}}{\sqrt{n}} (h(\log n) + h(-\log n)), \quad L_n = \text{lcm}(1, \dots, n). \end{aligned} \quad (\text{CFP11.3})$$

The coefficient is the original $\Lambda(n)$, with every prime-power repetition. At each finite trivial-zero cutoff R , retain also

$$\begin{aligned} V_{\zeta,R} &= W(F, G) + \sum_{r=1}^R A_{F,G}(-2r) - A_{F,G}(1), \\ G_R &= A_{\infty}(h) + A_{F,G}(0) + \sum_{r=1}^R A_{F,G}(-2r), \quad V_{\zeta,R} = G_R - P_{\text{hist}}(h). \end{aligned} \quad (\text{CFP11.4})$$

These are the original proved RTT/GIQ identities, retained here without a divergent infinite trivial-zero sum. No weighted form with arbitrary c_{ρ} , and no compression removing (CFP11.2), is asserted to equal their right side. The original supported endpoints and their coefficient labels remain those in that established source formula.

CFP12. The next calculation performed: the exact source-topology kernel

The issue left by PTQ was whether its value-Hilbert boundedness excluded other continuous positive source forms carrying hidden higher jets or off-line data. CFP1–CFP10 answer that question by the full source integral, the actual unitary completion and exact Gaussian division: no such additional continuous directions exist, and the complete additional freedom is the polynomial family of weights.

This does not prove that $\mathcal{K}_{\text{all}}$ consists only of the original higher-jet radical. Their precise difference is retained:

$$\mathcal{N}_0 = \{[F] : F(\rho) = 0 \ \forall \rho \in \mathcal{Z}\} \subset \mathcal{K}_{\text{all}}, \quad \mathcal{K}_{\text{all}}/\mathcal{N}_0 \cong \frac{\mathcal{J}_{\mathcal{Z}}^{\text{val}}}{\mathcal{J}_{\mathcal{Z}}^{\text{line}}}. \quad (\text{CFP12.1})$$

The displayed isomorphism is induced by the identity on representatives; its kernel and surjectivity follow directly from the two quotients. It retains the actual off-line value data. The existing off-line value observation injects this quotient into H_{off} , with dense image by the full original isolators. Thus the exact obstruction exposed by extending PTQ is the same source-topological off-line observation, rather than an unknown class of unbounded positive forms. The receiver of the remaining geometric comparison is now fixed without a value-norm hypothesis.

One further exact consequence records what a faithful positive form would mean. A classified form is positive definite on the **full** $\mathcal{Q}$ exactly when every actual zero lies on the line, every multiplicity is one, and every c_{ρ} is strictly positive. Necessity follows respectively from an off-line isolator, a higher-jet isolator, or a value isolator at a zero-weight coordinate. Sufficiency follows because a nonzero full-jet class then has a nonzero retained value. This statement is not used to demand simplicity for ordinary Weil positivity: the original Weil form is allowed its precisely identified higher-jet radical.

CFPA. Proof of the unitary annihilator lemma used above

Human-source attribution added for publication: the positive Fourier-measure representation is Bochner's classical theorem; see Terence Tao, 245C Notes 2, Theorem 57. The proof below supplies this special case with its displayed Fourier convention and all constants; no novelty is claimed for that theorem.

Here is a proof of exactly the spectral facts used in CFP5. It is included to avoid leaving a new spectral assertion implicit. Basic Fourier inversion for integrable functions, Gaussian Fourier integration, the Riesz representation theorem for finite measures, and Hilbert completion are used with their usual stated domains.

For the classical Fourier inversion theorem, see Tao, 245C Notes 2, equation (11); the Fourier convention and constants below are retained. The weak-star compactness used below is Banach–Alaoglu; see Tao, 245B Notes 11, Theorem 3.

Let $(V_t)_{t \in \mathbb{R}}$ be a strongly continuous unitary group on a Hilbert space H . For $x \in H$ set

$$f_x(t) = \langle V_t x, x \rangle. \quad (\text{CFPA.1})$$

This function is continuous, bounded by $\|x\|^2$, and positive definite: for finite coefficients,

$$\sum_{i,j} a_i \overline{a_j} f_x(t_i - t_j) = \left\| \sum_i a_i V_{t_i} x \right\|^2 \geq 0. \quad (\text{CFPA.2})$$

First prove the positive Fourier-measure representation needed here. If f is continuous positive definite, then $f(0) \geq 0$, $f(-t) = \overline{f(t)}$, and $|f(t)| \leq f(0)$, by the two-point positive matrix. For $\varepsilon > 0$, put $f_\varepsilon(t) = f(t)e^{-\varepsilon t^2}$. The Gaussian is a positive mixture of characters, so integrating (CFPA.2) after multiplying the coefficients by those characters proves that f_ε is positive definite. It is also integrable.

Define

$$p_\varepsilon(\lambda) = \frac{1}{2\pi} \int_{\mathbb{R}} f_\varepsilon(t) e^{-it\lambda} dt. \quad (\text{CFPA.3})$$

This is continuous and nonnegative. To prove the sign, positive definiteness, approximation of integrals by finite sums, and the interval function $e^{-i\lambda t} 1_{[-R,R]}(t)$ show

$$0 \leq \frac{1}{2R} \int_{-R}^R \int_{-R}^R f_\varepsilon(t-u) e^{-i\lambda(t-u)} dt du = \int_{-2R}^{2R} \left(1 - \frac{|v|}{2R}\right) f_\varepsilon(v) e^{-i\lambda v} dv.$$

Dominated convergence as $R \rightarrow \infty$ proves (CFPA.3) is nonnegative. It is real because of the conjugate symmetry.

For $\delta > 0$, Fourier integration and Fubini give

$$\int_{\mathbb{R}} e^{-\delta \lambda^2} p_\varepsilon(\lambda) d\lambda = \frac{1}{2\sqrt{\pi\delta}} \int_{\mathbb{R}} e^{-t^2/(4\delta)} f_\varepsilon(t) dt \rightarrow f(0) \quad (\delta \downarrow 0). \quad (\text{CFPA.4})$$

Monotone convergence on the nonnegative left side proves $\int p_\varepsilon = f(0)$. Fourier inversion, now with both transforms integrable, gives $f_\varepsilon(t) = \int e^{it\lambda} p_\varepsilon(\lambda) d\lambda$.

The measure-representation step uses the classical Riesz representation theorem for positive functionals, not Hilbert-space Riesz representation; see Terence Tao, 245B Notes 12, Theorem 24. The mass and tightness calculation in this application follows explicitly.

The finite positive measures $\mu_\varepsilon = p_\varepsilon d\lambda$, of this fixed total mass, have a weakly convergent subnet as functionals on $C_0(\mathbb{R})$, by weak-star compactness. There is no loss of mass at infinity as $\varepsilon \downarrow 0$: for $a > 0$,

$$\int \left(1 - \frac{\sin(a\lambda)}{a\lambda}\right) d\mu_\varepsilon = f(0) - \frac{1}{2a} \int_{-a}^a f_\varepsilon(t) dt. \quad (\text{CFPA.5})$$

The right side tends uniformly to zero when $a \downarrow 0$ and $0 < \varepsilon \leq 1$, by continuity of f at zero. The integrand is at least $1/2$ when $|\lambda| \geq 2/a$. Thus the measures are uniformly tight. Riesz representation gives a finite positive limiting measure μ , with total mass $f(0)$. Tightness allows the convergence to be tested against the bounded character $e^{it\lambda}$, by a compact cutoff and a uniformly small tail. It follows that

$$f(t) = \int_{\mathbb{R}} e^{it\lambda} d\mu(\lambda), \quad \mu(\mathbb{R}) = f(0). \quad (\text{CFPA.6})$$

Uniqueness follows from Gaussian smoothing: a finite complex measure with identically zero Fourier transform has zero convolution with every Gaussian, by Fubini and the Gaussian inversion formula; those Gaussian convolutions approximate the measure against every continuous compactly supported test, so the measure is zero. This proves the required representation without an unproved spectral decomposition.

Apply this result to f_x , obtaining μ_x with total mass $\|x\|^2$. For $h \in L^1(\mathbb{R})$, the Bochner integral $V(h)x = \int h(t)V_t x dt$ exists and satisfies

$$\|V(h)x\|^2 = \int_{\mathbb{R}} \left| \int_{\mathbb{R}} h(t)e^{it\lambda} dt \right|^2 d\mu_x(\lambda). \quad (\text{CFPA.7})$$

Indeed expand its squared norm as a double integral with integrand $h(t)\overline{h(u)}f_x(t-u)$, then use (CFPA.6) and Fubini. The bound $|f_x| \leq \|x\|^2$ supplies absolute integrability.

If $V(k) = 0$, (CFPA.7) proves that μ_x is supported on the zero set of $\varphi(\lambda) = \int k(t)e^{it\lambda} dt$. More explicitly, its complement is the union of the measurable sets $\{|\varphi| \geq 1/j\}$, each of zero measure. Suppose this zero set is a discrete countable set Γ , as it is in (CFP5.1).

For each $\gamma \in \Gamma$ take

$$h_{R,\gamma}(t) = \frac{1}{2R} e^{-it\gamma} 1_{[-R,R]}(t), \quad \int h_{R,\gamma}(t)e^{it\lambda} dt = \frac{\sin(R(\lambda - \gamma))}{R(\lambda - \gamma)}, \quad (\text{CFPA.8})$$

with value one at $\lambda = \gamma$. These multipliers converge pointwise to the indicator of $\{\gamma\}$ and are bounded by one. Formula (CFPA.7), applied to differences of the $h_{R,\gamma}$, and dominated convergence show that $V(h_{R,\gamma})x$ is Cauchy. Call its limit $P_\gamma x$. Its norm is

$$\|P_\gamma x\|^2 = \mu_x(\{\gamma\}). \quad (\text{CFPA.9})$$

Changing the interval of integration by a fixed translation changes its average in norm by at most $|a|\|x\|/R$. Hence $V_a P_\gamma x = e^{ia\gamma} P_\gamma x$. Conversely if y is an eigenvector with this character, then $V(h_{R,\gamma})y = y$. If x is orthogonal to all such eigenvectors, each average remains orthogonal to them, by unitarity and their eigenvector property; so its limit is zero. These facts prove that P_γ is exactly the orthogonal projection onto that eigenspace.

Different frequencies have orthogonal eigenspaces, because invariance of an inner product multiplies it by $e^{it(\gamma-\eta)}$ for every t . Finally (CFPA.9), countable additivity, and the support assertion give

$$\sum_{\gamma \in \Gamma} \|P_\gamma x\|^2 = \mu_x(\Gamma) = \mu_x(\mathbb{R}) = \|x\|^2. \quad (\text{CFPA.10})$$

Orthogonality now proves $x = \sum P_\gamma x$ in norm. This proves every spectral statement used in CFP5, including possible zero fibres and completeness. In particular there is no undeclared continuous spectral remainder.

Proof sources and reading scope

The full PTQ0–PTQ11 proof was read for the earlier bounded category. The relevant RTT0–RTT2, RTT5–RTT11 and RD1–RD5 were read for the actual Fréchet quotient, exact isolators, full residue and degree maps. PGF5.1–PGF5.8 and the exact Gaussian formula were read for the original Schwartz source; the preceding derivation recomputes its Fourier and Mellin constants. PSC0–PSC2 were read for the precise source kernel

and quotient topology. The arithmetic identity in CFP11 is the retained proved RTT9/GIQ9 identity, not a new independent derivation of the explicit formula.

Original derivation reading record, retained as provenance: the private canonical-index title query for spectral-theorem, unitary-group and positive-definite-function sources returned no relevant exact original source for the unitary-measure argument; unrelated routing hits were not counted as reading. CFPA supplies the needed proof directly. At that derivation stage no newly located human paper was cited as read, and no novelty claim was made for that classical analytic lemma. The Bochner attribution above was added for this publication edition; it does not retrospectively expand the derivation's source-reading claim.

Continuous positivity and the actual specialization obstruction

Complete receiving calculation, CPS0–CPS8. All statements concern the actual original-zeta coefficient source, after its complete arithmetic reconstruction.

CPS0. Definitions, construction stage and source maps

The support remains $\tau\langle Z_1; \text{no } Z_2 \rangle$. Its addition remains retracted. No value, parity, coordinate, midpoint, metric or vector operation is assigned to it. The complete correction chains and the user's global admissibility and quotientability arguments in LATEST_DIRECT_ARGUMENTS.md were consulted before this calculation. The user's request is to derive a common receiver from all established structure, not to stipulate its faithfulness or purity. The two recovered branch histories are kept separate.

Use the actual Fréchet coefficient

$$Q = \mathcal{B}/\mathcal{J}, \quad \mathcal{B} = \{F \text{ entire} : b_{A,M}(F) = \sup_{|\Re s| \leq A} (1 + |\Im s|)^M |F(s)| < \infty \text{ for all } A, M\}, \quad (\text{CPS0.1})$$

where $\mathcal{J}$ requires vanishing through order $m_\rho - 1$ at every actual original nontrivial zero ρ of ζ . The Mellin source is

$$Q \simeq A/J, \quad J = \Sigma S, \quad \Theta a(s) = \frac{1}{2} \int_0^\infty a(u) u^s \frac{du}{u}. \quad (\text{CPS0.2})$$

Its complete proof and source topologies are retained in RTT0–RTT1, ACD1 and AST1–AST3. The coefficient operation

$$T_a[F] = [a^s F(s)], \quad a > 0, \quad (\text{CPS0.3})$$

is continuous, preserves $\mathcal{J}$, and has continuous inverse $T_{1/a}$. Recovered integers n are genuine cover degrees, with geometric transfer $nT_{1/n}$. General positive a is a coefficient parameter, not a noninteger cover.

Write $\mathcal{Z}_L = \{\rho : \Re \rho = 1/2\}$, $\mathcal{Z}_O = \mathcal{Z} \setminus \mathcal{Z}_L$, and retain all multiplicities. Define the closed invariant subspaces

$$\begin{aligned} N_L &= \{F \in Q : F(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}_L\}, \\ N_O &= \{F \in Q : F(\rho) = 0 \text{ for every } \rho \in \mathcal{Z}_O\}, \\ N_0 &= N_L \cap N_O = \ker E, \quad E(F) = (F(\rho))_{\rho \in \mathcal{Z}}. \end{aligned} \quad (\text{CPS0.4})$$

These definitions impose conditions only on values; higher jets stay in the indicated kernels. ACD and AST calculate the actual specialization obstruction quotient as

$$R = Q/N_O \simeq A/A_O, \quad A_O = q^{-1}N_O. \quad (\text{CPS0.5})$$

This is an existing quotient of the source, with its Fréchet quotient topology. No positive form has been selected in defining it.

CPS1. The complete positive family and its precise radical

Let $\mathcal{P}(Q)$ be the family of all continuous positive semidefinite Hermitian forms on Q , linear in the first variable, satisfying

$$B(T_n F, G) = B(F, nT_{1/n} G) \quad (n \geq 1 \text{ a recovered integer; } F, G \in Q). \quad (\text{CPS1.1})$$

The complete CFP proof classifies this family without assuming any finite-support density in Q :

$$B_c(F, G) = \sum_{\rho \in \mathcal{Z}_L} m_\rho c_\rho F(\rho) \overline{G(\rho)}, \quad c_\rho \geq 0, \quad c_\rho \leq C(1 + |\Im \rho|)^d \quad (\text{CPS1.2})$$

for some finite C, d depending on the form. Conversely every such coefficient family defines a continuous form satisfying (CPS1.1). The degree factor n and all multiplicities remain in that theorem.

It follows that

$$\bigcap_{B \in \mathcal{P}(Q)} \text{rad } B = N_L. \quad (\text{CPS1.3})$$

Indeed every form in (CPS1.2) vanishes on N_L . Conversely choose $c_\rho = 1$ for every line zero. A zero diagonal value is then a sum of nonnegative numbers equal to zero, so each value $F(\rho)$ is zero. Positivity implies that zero norm is the full radical by Cauchy–Schwarz. The argument also covers an empty line-zero set, when both sides of (CPS1.3) are all of Q .

This computes a receiver radical, not the zero subspace of the original coefficient. No original jet is identified with zero in Q .

CPS2. All positive seminorms and their complete receiver

For every integer $k \geq 0$ put

$$p_k(F)^2 = \sum_{\rho=1/2+i\gamma \in \mathcal{Z}_L} m_\rho (1+\gamma^2)^k |F(\rho)|^2. \quad (\text{CPS2.1})$$

These are continuous seminorms on Q . To verify convergence and continuity directly, choose an integer M with $2M - 2k > 2$. Then

$$p_k([F])^2 \leq b_{1,M}(F)^2 \sum_{\rho \in \mathcal{Z}_L} m_\rho (1+\gamma^2)^k (1+|\gamma|)^{-2M} < \infty. \quad (\text{CPS2.2})$$

The last convergence follows from the unconditional full zero count $\sum_{|\Im \rho| \leq T} m_\rho = O(T \log(T+2))$, by summing over dyadic intervals. This bound on $\mathcal{B}$ descends continuously through its quotient by $\mathcal{J}$. Every p_k^2 is a form in (CPS1.2).

Conversely, for the coefficient family of any B_c , choose an integer k with $2k \geq d$. Since $(1+|\gamma|)^2 \leq 2(1+\gamma^2)$, one obtains

$$B_c(F, F) \leq C 2^{d/2} p_k(F)^2. \quad (\text{CPS2.3})$$

Thus the topology generated by every positive form in $\mathcal{P}(Q)$ is exactly the topology generated by the countable family p_k . This is a proved comparison with the original Fréchet topology; equality with that original topology is not asserted.

Define the sequence space

$$\mathcal{S}_L = \left\{ x = (x_\rho)_{\rho \in \mathcal{Z}_L} : \|x\|_k^2 = \sum_{\rho=1/2+i\gamma} m_\rho (1+\gamma^2)^k |x_\rho|^2 < \infty \text{ for every } k \geq 0 \right\}. \quad (\text{CPS2.4})$$

It is a complete Fréchet space: a sequence Cauchy in every displayed Hilbert norm has limits in every corresponding weighted ℓ^2 space, and continuity of each coordinate evaluation makes all these limits the same coordinate family. That family has every finite norm and is the simultaneous limit. The usual countable seminorm metric then proves completeness.

The map

$$E_L : Q \longrightarrow \mathcal{S}_L, \quad F \longmapsto (F(\rho))_{\rho \in \mathcal{Z}_L} \quad (\text{CPS2.5})$$

is continuous, has kernel N_L , and has dense image. For the last assertion, the full-jet isolator $e_{\rho,0}$ of RTT0 maps to the individual value coordinate. Hence the image contains all finite sequences. Given $x \in \mathcal{S}_L$, its truncations by $|\gamma| \leq T$ converge in every $\|\cdot\|_k$, since each defining nonnegative series converges. This proves density globally, not by a numerical test or a density assumption about primal jets.

Consequently the Hausdorff completion of Q/N_L for all its positive-form seminorms is exactly $\mathcal{S}_L$, with the explicit map (CPS2.5). The original Fréchet quotient Q/N_L and this specified completion are both retained.

CPS3. Universal Hilbert receiving property

Let $V : Q \rightarrow K$ be any continuous linear map to a Hilbert space for which the pulled-back inner product belongs to $\mathcal{P}(Q)$. The classification gives

$$\|VF\|_K^2 = B_c(F, F) \leq C' p_k(F)^2 \quad (\text{CPS3.1})$$

with finite k, C' . It follows that there is one and only one continuous linear map

$$\widetilde{V} : \mathcal{S}_L \longrightarrow K, \quad V = \widetilde{V} E_L. \quad (\text{CPS3.2})$$

To construct it, first set $\widetilde{V}(E_L F) = VF$. Equality of two source images means their difference is in N_L , hence has zero norm under V , so this is well defined. Inequality (CPS3.1) makes the map continuous for $\|\cdot\|_k$; extend it to the complete limit of each approximating sequence in the dense subspace $E_L Q$. The same bound makes the extension independent of the sequence, and uniqueness follows from density.

The coefficient action on this receiver is explicit:

$$(\widetilde{T}_a x)_\rho = a^{1/2+i\gamma} x_\rho, \quad \|\widetilde{T}_a x\|_k^2 = a \|x\|_k^2. \quad (\text{CPS3.3})$$

It is strongly continuous in $a > 0$ in each norm, by dominated convergence on compact parameter intervals, and $E_L T_a = \widetilde{T}_a E_L$. An equivariance identity for V with continuous Hilbert receiving operators extends through (CPS3.2) by this density argument. The factor $a^{1/2}$ has not been dropped from the original transfer.

CPS4. The existing specialization obstruction has zero positive receiver

The quotient map $\pi : Q \rightarrow R = Q/N_O$ intertwines every T_a , since N_O is invariant. Suppose B_R is any continuous positive Hermitian form on R with the actual integer transfer-adjoint identity. Pulling it back along π gives a form $B \in \mathcal{P}(Q)$. For every actual line zero ρ , the full-jet isolator $e_{\rho,0}$ belongs to N_O , because its value at every off-line zero vanishes. Its image in R is zero. Applying (CPS1.2) yields

$$0 = B_R(\pi e_{\rho,0}, \pi e_{\rho,0}) = B(e_{\rho,0}, e_{\rho,0}) = m_\rho c_\rho. \quad (\text{CPS4.1})$$

Every m_ρ is a positive integer, so every coefficient c_ρ is zero. Therefore $B = 0$. Surjectivity of π gives

$$\boxed{\mathcal{P}(R) = \{0\}}. \quad (\text{CPS4.2})$$

No existence or nonexistence of off-line zeros entered this proof. No finite primal-block density was used. It follows also that every continuous Hilbert receiving map from R whose inner product satisfies that transfer identity is the zero map: its squared norm is zero on each source element.

The original invariant-cycle obstruction remains the actual image

$$\beta_r Q = b_r A, \quad \beta_r = D_r^* J E, \quad b_r = \beta_r q, \quad (\text{CPS4.3})$$

and $[F] \mapsto -\beta_r F$ identifies the original specialization cokernel R algebraically with the signed boundary image. AST proves its original quotient topology and its dense injective Hilbert receiving map. That receiving map does not acquire the positive transfer-adjoint identity from injectivity or density. Equation (CPS4.2) computes the zero positive receiver of this exact obstruction; it does not prove $R = 0$.

CPS5. The source subcomplex still supplies the whole positive completion

Restrict (CPS2.5) to the actual source kernel N_O . Its kernel is $N_O \cap N_L = N_0$. Its image still contains every finite line-value sequence, by the same full-jet isolators. Therefore

$$(\widehat{N_O/N_0})^{(p_k)} \xrightarrow{\sim} \mathcal{S}_L \quad \text{and} \quad (\widehat{Q/N_L})^{(p_k)} \xrightarrow{\sim} \mathcal{S}_L \quad (\text{CPS5.1})$$

are the same completion under the actual inclusion $N_O \hookrightarrow Q$. The notation on the left specifies the seminorms inherited from (CPS2.1). It does not assert that every continuous positive form intrinsic to N_O extends to Q .

This identifies the positive effect of the exact source row and its strong dual:

$$0 \longrightarrow N_O \longrightarrow Q \longrightarrow R \longrightarrow 0, \quad 0 \longrightarrow R'_\beta \longrightarrow Q'_\beta \longrightarrow (N_O)'_\beta \longrightarrow 0. \quad (\text{CPS5.2})$$

The second row is the strict topological row proved in AST3 (AST3.5), not a newly presumed exactness of arbitrary dualization. The embedding of R'_β is exactly $N_O^\perp$. After q' , it is $A_O^\perp$, the strong closure of the Gysin attaching image in GDC. Thus the subcomplex whose transpose yields the split current receiver retains the entire completion (CPS5.1), while its removed coefficient quotient has precisely (CPS4.2). Both sides of the original exact row remain explicit.

CPS6. The original form and its full complementary term

The original Weil form remains

$$W(F, G) = \sum_{\rho \in \mathcal{Z}} m_\rho F(\rho) \overline{G(1 - \bar{\rho})} = W_L(F, G) + W_O(F, G), \quad (\text{CPS6.1})$$

where

$$W_L(F, G) = \sum_{\rho \in \mathcal{Z}_L} m_\rho F(\rho) \overline{G(\rho)}, \quad W_O(F, G) = \sum_{\rho \in \mathcal{Z}_O} m_\rho F(\rho) \overline{G(1 - \bar{\rho})}. \quad (\text{CPS6.2})$$

Absolute convergence follows from the full zero count and rapid strip decay, or from RTT1's Hilbert comparison. The form on $\mathcal{S}_L$ with weights $c_\rho = 1$ pulls back to exactly W_L . Its difference from the original form is exactly W_O , with the displayed reflection and all multiplicities retained. Restriction to N_O kills W_O , because both arguments have every off-line value zero. This is a direct consequence of the source restriction, not a claim that the original off-line term vanished on Q .

For any actual off-line reflection pair $\rho, \rho^\#$, use their original full-jet isolators and let $F = e_{\rho,0} - e_{\rho^\#,0}$. Its exact original value is

$$W(F, F) = -2m_\rho, \quad W_L(F, F) = 0, \quad W_O(F, F) = -2m_\rho. \quad (\text{CPS6.3})$$

The plus combination has value $+2m_\rho$. These formulas apply to each member of the actual off-line divisor, without claiming such a member exists. They prove that the positive quotient has not invisibly reproduced a nonzero complementary term. They are exact global source calculations, not numerical searches.

The multiplier used in CFP's source integral is retained in full:

$$F_0(s) = \frac{s(s-1)}{8} \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F_0(0) = F_0(1) = \frac{1}{8}, \quad (\text{CPS6.4})$$

$$F_0(-2j) = \frac{j(2j+1)(-1)^j \pi^j}{2j!} \zeta'(-2j) \quad (j \geq 1). \quad (\text{CPS6.5})$$

Its full local unit at every nontrivial zero is the one in ADC1.5. These values and factors remain part of the original-zeta comparison; using its vanishing action on Q does not replace original ζ or suppress its trivial-zero and pole data. PSC retains the exact full arithmetic trace decomposition and its endpoint and Gamma terms. Both endpoint lines in the sheaf comparison remain as ACD/AST's separate labelled summands; none is removed by (CPS0.4).

CPS7. What the new calculation adds to the weight argument

PTQ classified bounded forms on the value Hilbert space. CFP and CPS classify every continuous positive transfer-adjoint form on the actual Fréchet source, including polynomially unbounded value weights, and compute their complete common receiver. This closes the earlier restriction to Hilbert-bounded forms. It

also identifies the positive receiver of the actual specialization obstruction as zero by (CPS4.1), using the original quotient and isolators.

The original Deligne comparison is sharper than the existence of a positive receiver: DC6–DC8 derives disjoint weights from geometric inputs and kills the actual boundary image. Here ACD8 and DCA10 retain matching source and target characters on that image. CPS computes what all compatible positive receivers do to the existing image, while AST gives its exact continuous dual. The full source-to-receiver map and its radical have therefore been calculated. They are not a proof that the original boundary is null-homotopic.

The next source-derived comparison is now precise: any proposed positive geometric receiving map must be calculated before completing it, with its kernel compared to N_L and its original trace compared to (CPS6.1), including W_O . This is not an assumed additional theorem. The present calculation already tests the canonical map E_L and the actual AST source restriction: both give (CPS5.1), both have the displayed kernels, and their original trace difference is (CPS6.2). No declaration of purity has been used to close that difference.

CPS8. Proof and source coverage

The entire ADC0–ADC9 and DCA0–DCA12 proofs were read, including the complete nearby-sheet corrections, their cone homotopies, the genuine integer degree maps, the endpoint lines and the exact original specialization cross. The entire ACD0–ACD8 and AST0–AST9 comparisons were read; AST’s updated review of the final GDC source is retained. The complete CFP proof is the analytic input to CPS1. RTT’s original full-jet source and exact residue comparison, PTQ’s bounded-form theorem, PSC’s full arithmetic decomposition and GDC’s strong topology are retained alongside this proof.

Human coefficient source: Alain Connes and Caterina Consani, *Schemes over $\mathbb{F}_1$ and zeta functions*, arXiv:0903.2024v3, §5. Weight comparison: Pierre Deligne, *La conjecture de Weil. II*, IHÉS 52 (1980), §3.6. Existing exact reading ledgers remain authoritative; no new whole-human-paper reading is claimed here. CFP records its own exact analytic source and proof coverage. These receiving theorems do not assert a proof of RH.

Verification of the continuous positive-form classification

25 September 2026. Complete root verification of CFP0–CFP12 and CFPA, with the exact accepted source hash recorded in CONTINUOUS_POSITIVE_RECEIPT.json. The independent derivation and this verification are distinct review scopes.

The entire proof was read. CFP1’s seminorm bounds give actual differentiable source dilation; no density of finite primal jets is presumed. CFP2 derives the full positive-rational scaling and then the real coefficient action by source continuity. Its comparison unitary action retains the original positive degree factor explicitly. The form radical is invariant under both parameter directions, so the completion action is well defined and invertible.

CFP3’s original Gaussian constants were recalculated. Fourier transformation of the two terms gives the original polynomial again, with both constant contributions cancelling. The zero value and real integral are both zero for that specific source vector. Poisson summation therefore yields the exact inversion identity, making the displayed kernel even. The bound on the two tails is integrable against every finite exponential and polynomial weight. The Mellin transform of each of the two original terms is the one in CFP3.7, with their stated signs, coefficients and Gamma arguments. Their Gamma recursion gives the full $s(s-1)/8$ multiplier of original zeta. The endpoint values, all negative even values and every displayed derivative factor agree with the original functional equation and its simple trivial-zero derivative. No factor was treated as one.

CFP4’s source-valued integral is convergent in each actual Fréchet seminorm. It equals multiplication by that full multiplier and lies in the full vanishing-jet ideal. Passing through the continuous quotient and Hilbert map is justified. The resulting Hilbert operator is bounded by the kernel’s L1 norm, so its vanishing on the dense source image gives vanishing on the whole completion. This requires only the proved inclusion of the multiplier product in the ideal, not a principal-ideal identification.

CFPA was checked in full. The two-point matrix gives boundedness and conjugate symmetry of a positive definite function. Gaussian regularization preserves positive definiteness. The interval double integral has the exact triangular convolution factor and proves nonnegativity of the Fourier density. Gaussian inversion retains the factor $1/(2\sqrt{\pi\delta})$. Monotone convergence proves its full finite mass. The sinc identity gives uniform tightness, so the weak-star limit retains the total mass and each character integral. Gaussian convolution proves uniqueness. The double-integral norm identity uses the correct sign for an inner product linear in its first variable. Applied to the annihilating kernel, it removes spectral measure outside the exact zero set. The time averages converge in norm by that identity; their ranges are the stated character eigenspaces and their orthogonal sum is all of the Hilbert space by countable mass additivity. This proves that no continuous spectral remainder was silently dropped.

CFP5 applies this result to the actual full source multiplier, whose real-frequency zeros are exactly the original line-zero ordinates. CFP6’s entire division is valid on every strip and its derivative argument applies to genuine source-smooth vectors. Each spectral fibre is consequently zero or one dimensional; this is a claim about the receiver, not zeta multiplicity. The full-jet isolators verify every local block map. CFP7’s translated Gaussian bound is uniform in the zero ordinate apart from the displayed polynomial, so source continuity proves a polynomial bound on each actual atomic mass. Division by the retained positive integer multiplicity gives the announced coefficients. CFP8 proves the converse by the unconditional full zero count and quotient continuity. The multiplication characters verify the exact integer adjoint identity.

CFP9–CFP10 correctly identify the form radical, the weighted Hilbert completion and factorization of each continuous positive Hilbert receiver. Their density assertion is solely density of finite value sequences in the target; it is not finite-jet density in the original Fréchet space. CFP11 retains the original reflected Weil form and the previously proved original arithmetic terms. The two signs on each actual off-line pair are obtained by full isolators; no off-line zero is assumed to exist. CFP12 distinguishes positivity on the value quotient from positive definiteness on all original jets. Its higher-jet and off-line kernels agree with the classification.

The complete received ACD, ADC, DCA and AST texts were also read for the CPS receiving calculation. The root did not newly independently rederive all their previous source-image theorems. Their exact current-level signs, finite-sheet corrections, cover degrees, boundary source, quotient topologies and endpoints are

retained as prior proved inputs. AST contains an independent full review of final GDC hash 5d0f471b3250e4ee894caca68232c556d64a137a49a9fe25a641db61b94f79ab, including its final typography repair. That later review strengthens the earlier review scope without changing the retained earlier receipt.

No mathematical defect was found in the complete CFP proof. CPS was independently reviewed by the CFP derivation agent; its separate review specifies the exact scope. The original weight-separated vanishing target remains active. No numerical zero sample, assumed RH, or assumed geometric purity was used in either new derivation.

Independent mathematical review of the continuous positive receiver

25 September 2026. Reviewed the entire POSITIVE_SOURCE_COMPLETION_AND_SPECIALIZATION.md, CPS0–CPS8, SHA256 d6eed5950b21e72365741b7da81955f41f89beff67d6004f9d5c7b3609fd79b0.

The source used for its classification is CONTINUOUS_POSITIVE_TRANSFER_FORMS.md, CFP0–CFP12 and CFPA, SHA256 ce8da36c1c6a61acbd2a083751cd86597b4dbb8d2940892592a458f242034d1e. That proof was independently derived in the current task, with the complete original multiplier and source integral. It proves all continuous source forms, not only forms bounded for an earlier value norm. This hash includes the clarification that the full vanishing-jet conditions are preserved while each actual jet transforms by the full jet of e^{ts} .

The following checks cover the receiving arguments in CPS. No mathematical correction was found.

1. **CPS1, common radical.** Every classified form vanishes on the actual line-value-zero source subspace N_L , and the permitted coefficient family $c_\rho = 1$ has precisely that radical. Positivity and Cauchy–Schwarz identify zero norm with the full Hermitian radical. This includes an empty index set without a new existence assertion.
2. **CPS2, topology generated by the entire family.** Every polynomial coefficient family is bounded by a single p_k^2 , with its constant retained, and every p_k^2 is itself in the family. This proves equality of these two generated topologies. Continuity in the original source follows from the exact zero count and the displayed strip bound; it does not identify the original source topology with the generated one. Completeness of $\mathcal{S}_L$ follows by consistent weighted Hilbert limits, and its finite truncations converge in every weight. Original global isolators give those finite coordinate vectors in the image. This is density in $\mathcal{S}_L$, not a primal density assertion.
3. **CPS3, Hilbert receiving map.** Its single-weight estimate gives a well-defined map on $E_L Q$, since the kernel has zero image norm. The estimate supplies a unique extension to the completion and retains every original degree coefficient. Any receiving equivariance relation extends by density with the stated continuity hypothesis.
4. **CPS4, actual specialization quotient.** Pullback of a continuous positive transfer form on $R = Q/N_O$ is a form classified by CFP. Each original line isolator lies in N_O , so its squared norm is zero. The exact isolator value is $m_\rho c_\rho$; all original multiplicities are positive. Every coefficient therefore vanishes. Surjectivity then gives $\mathcal{P}(R) = \{0\}$. This proves a zero positive receiver without asserting that R itself is zero or that off-line zeros exist.
5. **CPS5, inherited positive completion.** The restriction $E_L|_{N_O}$ has kernel $N_O \cap N_L = N_0$, and its image contains all finite line sequences. Its completion for the inherited p_k is consequently the same $\mathcal{S}_L$. The proof correctly does not claim that every form continuous for the intrinsic topology on N_O extends to Q .
6. **CPS6–CPS7, retained complement.** The Weil decomposition keeps both terms of every reflected off-line pair and all multiplicities. The two exact isolator combinations have values $\pm 2m_\rho$. Restricting to N_O removes this term from that restriction only. The displayed original-zeta multiplier, endpoint values, and trivial-zero derivative coefficients match the full source comparison in CFP3. No source-boundary vanishing or purity claim follows from the zero positive receiver, and none is asserted.

The AST identification of the specialization quotient and its strict strong-dual exact sequence is explicitly an earlier proof dependency in CPS0/CPS5. This review checked its use and the receiving calculation, but did not independently re-prove the entire AST source. The preceding review is therefore not a claim of new full reading or acceptance of every ancestor proof. No publication or numerical RH verification is part of this review.